the diagnostic approaches to RLS.

Table 15-15
Medications and Drugs Causing or Worsening Certain Parasomnias

REM Sleep Behavior Disorder
Biperiden
Caffeine
Monoamine oxidase inhibitors (MAOIs)
Selegiline
Serotonin agonists
Tricyclic antidepressants
Venlafaxine
Withdrawal syndromes
Alcohol
Meprobamate
Pentazocine
Nitrazepam
Nightmares
Alcohol abuse or withdrawal
Drug-induced REM sleep rebound (e.g., withdrawal from REM-suppressing drugs such as methamphetamine)
l-DOPA
β-Blockers

A variety of medical conditions can cause secondary RLS, and we list these in Table 15-17. A detailed history and physical examination are essential parts of the RLS workup. Also, we should check a ferritin level in every patient with symptoms consistent with RLS.

Periodic Limb Movement Disorder. Periodic limb movement disorder (PLMD), previously called *nocturnal myoclonus,* involves brief, stereotypic, repetitive, nonepileptiform movements of the limbs, usually the legs. It

occurs primarily in NREM sleep and involves an extension of the big toe. A partial flexion of the ankle, knee, and hip may also occur. These movements range from 0.5 to 5 seconds in duration and occur every 20 to 40 seconds. The leg movements are frequently associated with brief arousals from sleep and, as a result, can (but do not always) disturb sleep architecture. The prevalence of PLMD increases with aging and can occur in association with folate deficiency, renal disease, anemia, and the use of antidepressants.

Sleep-Related Leg Cramps. Nocturnal leg cramps are much like leg cramps that occur during wakefulness. They usually affect the calf and are painful muscle contractions. The pain awakens the sleeper and thereby disrupts sleep. Metabolic disorders, mineral deficiencies, electrolytic imbalances, diabetes, and pregnancy are known precipitators. The reason that some individuals have repeated leg cramps during sleep and not during the day is not known.

Sleep-Related Bruxism. Sleep-related bruxism occurs when an individual grinds or clenches the teeth during sleep. Formerly classified as a parasomnia, sleep bruxism can produce abnormal wear and damage to teeth, provoke tooth and jaw pain, or make loud, unpleasant sounds that disturb the bed partner. Sometimes atypical facial pain and headache also result. More than 85 percent of the population may brux at one time or another; however, it is clinically significant in only about 5 percent. Teeth grinding occurs in any sleep stage but is most common at the transition to sleep, in stage 2 sleep, and during REM sleep. Some evidence indicates that teeth grinding during REM sleep is more commonly associated with dental wear or damage. Sleep bruxism does not appear to be exacerbated by dental malocclusion. It worsens during periods of stress. Researchers studying sleep bruxism have found that many patients seem to have less frequent teeth grinding when sleeping in the laboratory; therefore, we may need repeat studies to document the disorder. By contrast, bruxism frequently appears on polysomnographic recordings made for other purposes. Sleep bruxism may occur secondary to sleep-related breathing disorders, the use of psychostimulants (e.g., amphetamine, cocaine), alcohol ingestion, and treatment with some SSRIs. Sleep bruxism can infrequently occur (monthly), regularly (weekly), or frequently (nightly). We judge the severity by considering the degree of sleep

disruption, consequent pain, and dental damage.

Table 15-16
Restless Legs Syndrome

	DSM-5	ICSD-3	ICD-10
Name	Restless legs syndrome	Restless legs syndrome	Other specified extrapyramida and movement disorders **Restless legs syndrome**
Duration	≥3×/wk for ≥3 mo		
Symptoms	Urge to move legs Worse at night Urge relieved by movement	Urge to move legs Uncomfortable sensations occurring during rest/inactivity At least partially relieved by movement Tend to have circadian pattern (occurring primarily at night)	Experiencing a strong urge to move legs to relieve discomfort often occurs while lying down or sitting and tends to be worse at night
# Symptoms needed	All the above	All the above	

Exclusion (not the result of):	Akathisia Effects of a substance or medication Another mental disorder or medical condition Behavioral condition (i.e., habitual movements)	
Psychosocial impact	Marked distress and/or impairment in areas of functioning present	Marked distress and/or impairment in areas of functioning present

Sleep-Related Rhythmic Movement Disorder. This sleep disorder consists of repetitive, rhythmic movements, usually involving the head and neck. Most often occurring at the transition from wakefulness to sleep, this movement disorder may also continue during light sleep. Formerly classified as a parasomnia, sleep-related rhythmic movement disorder has many names, including *jactatio capitis nocturna,* headbanging, head rolling, body rocking, and *rhythmie du sommeil.* Most infants body rock. Some clinicians believe that body rocking develops from the soothing effect of vestibular stimulation. If the rhythmic movement persists into childhood, the risk of injury increases. This is particularly true if the movement includes headbanging. The male to female ratio is 4:1. Severity ranges from less than one episode weekly to nightly episodes producing injury.

Table 15-17
Medical Disorders That Can Cause Secondary Restless Legs Syndrome

COPD
Diabetes
Fibromyalgia
Iron and folic acid deficiency anemias
Neuropathies
Rheumatoid arthritis
Thyroid diseases
Uremia

Benign Sleep Myoclonus of Infancy. Previously called benign neonatal sleep myoclonus, this disorder is characterized by asynchronous jerking of limbs and trunk during quiet sleep in neonates. This benign, apparently rare, parasomnia usually begins within the first week of life and may last a few days or several months. It does not require any treatment and should resolve with time.

Propriospinal Myoclonus at Sleep Onset. Propriospinal myoclonus at sleep onset is a spinal cord–mediated movement disorder, sometimes associated with spinal cord lesions. Movements appear during times of wakeful relaxation and may interfere with sleep onset. They start in the abdominal and truncal muscles and then progress to the neck and proximal muscles of the limbs.

Sleep-Related Movement Disorder due to Medication or Substance Use and Sleep-Related Movement Disorder due to Medical Condition. A variety of medications, substances, and comorbid conditions can produce or exacerbate sleep-related movement disorders. Table 15-18 lists some example medications. Neurologic diseases that are associated with daytime movement disorders can also be associated with sleep-related movement disorders. Stress, anxiety, and sleep deprivation may contribute to bruxism.

Table 15-18
Medications Causing Sleep-Related Movement Disorders

Antidepressants (including most tricyclics and SSRIs)
Antiemetics
Antihistamines
Antipsychotics (restless legs symptoms, periodic limb movement disorder)
Calcium channel blockers
Lithium
Stimulants (rhythmic movement disorders, bruxism)

Isolated Symptoms and Normal Variants

Sleep Talking. As the name implies, sleep talking in classic form involves unconscious speech during sleep. Unless it annoys the bed partner, it is seldom recognized. Fever, stress, or talking with the sleeper can all induce the behavior. Somniloquy may accompany sleep terror, sleepwalking, confusional arousals, OSAs, and RBD.

Sleep Starts (Hypnic Jerks). Sleep starts are sudden, brief muscle contractions that occur at the transition between wakefulness and sleep in 60 to 70 percent of adults. The contractions commonly involve the legs; however, sometimes there is movement in the arms and head. Sometimes called a “hypnic jerk,” it is usually benign. The sleep start, however, can interfere with the ability to fall asleep and may be accompanied by sensations of falling, a hallucinated flash of light, or a loud crackling sound. In severe cases, the sleep start produces profound sleep-onset insomnia.

CLINICAL TOOLS IN SLEEP MEDICINE

Clinical Interview

A careful and thorough clinical interview is one of the most informative parts of a workup for sleep disorders. The habitual bedtime and arising times for both weekdays and weekends, the frequency, duration, and restorative nature

of naps, and overall level of sleepiness are good places to begin. Specific sleep problems relating to difficulty initiating and maintaining sleep are informative, including whether there is rumination at bedtime, fear of not being able to sleep, or excessive worry when attempting to sleep. We should assess for movements, including leg movement, leg sensations, leg cramps, teeth grinding, and dream enactments (with or without injury). We should also review for morning headaches, morning dry mouth, nocturnal reflux, hyperhidrosis (excess sweating), nocturia, enuresis, nocturnal tongue biting, nightmares, sleep terrors, and other sleep-related problems. In some cases, it can be crucial to ask about the presence of family pets and whether they sleep in the bedroom (or bed). A sleep history questionnaire is often helpful in diagnosing a patient's sleep disorder (Table 15-19).

Clinical Tests

Polysomnography. Polysomnography is the continuous, attended, comprehensive recording of the physiologic changes that occur during sleep. A polysomnogram is typically recorded at night and lasts between 6 and 8 hours. Brain wave activity, eye movements, submentalis electromyography activity, nasal–oral airflow, nasal pressure, respiratory effort, oxyhemoglobin saturation, heart rhythm, and leg movements during sleep are measured. Usually, the laboratory records body position and snoring sounds. Brain wave activity, eye movements, and submentalis electromyogram are essential for identifying sleep stages and CNS arousals. Muscle tension and movements subside with deeper sleep and can also be useful in the diagnosis of PLMD and RLS. Nasal airflow, respiratory effort, and oxyhemoglobin saturation are instrumental in diagnosing sleep apnea and other sleep-related breathing disorders.

Table 15-20 lists the most common reasons for ordering polysomnography. A sleep study is not needed to diagnose RLS.

Table 15-21 lists standard polysomnographic measures.

Home Sleep Testing. Home sleep testing involves recording a limited number of cardiopulmonary parameters to assess patients for sleep-related breathing disorders. Home sleep testing is much less expensive than polysomnography. The test can record airflow, respiratory effort, heart

rhythm, snoring sounds, and oximetry. Several devices are commercially available that are capable of detecting sleep apnea in patients with moderate to severe pathophysiology. Negative studies are problematic because home sleep testing is less sensitive than full laboratory polysomnography. Patients with negative tests, notwithstanding obvious symptoms or comorbidity, should be scheduled for laboratory polysomnography. Also, home sleep testing does not check for the full spectrum of sleep disorders; therefore, the presence of residual symptoms after a breathing disorder is diagnosed in this manner and treated indicates a need for careful follow-up.

Multiple Sleep Latency Test. The multiple sleep latency test (MSLT) is essential for the workup of narcolepsy. Every 2 hours, beginning 2 hours after morning awakening, the patient is provided with 20-minute nap opportunities and instructed not to resist falling asleep. During sleep, the laboratory records electroencephalographic, electrooculographic, and submentalis electromyography activity to determine the sleep stage. The latency to sleep helps assess hypersomnolence, and the appearance of REM sleep on two or more nap opportunities confirms narcolepsy (or on one nap if the previous night's polysomnography revealed REM sleep within 15 minutes of sleep onset). If the patient falls asleep on a given nap opportunity, the laboratory will wake the patient 15 minutes after initial sleep onset. If the patient does not fall asleep after 20 minutes, the laboratory will terminate the session. The laboratory gives the patient five nap opportunities during 2-hour intervals across the day.

Maintenance of Wakefulness Test. Similar to the MSLT, the maintenance of wakefulness test (MWT) provides 40-minute test sessions at 2-hour intervals across the day. However, in this test, the laboratory instructs the patient to try to remain awake. This technique is used to assess treatment outcomes and is sometimes part of "fit for duty" testing. Patients are recorded in a darkened room while sitting in a comfortable chair or a bed with a bolster pillow. The first epoch of stage 2, 3, or 4 sleep or REM or three consecutive epochs of stage 1 mark unequivocal sleep onset. Falling asleep on MWT indicates some level of sleepiness. Any sleep latency of fewer than 8 minutes is abnormal. Sleep latency ranging from 8 to 40 minutes is of unknown significance.

Actigraphy. An actigraph is a device that measures and records movement. It is usually worn on the wrist (like a watch) and is a rough measure of the sleep–wake cycle. Depending on the model and the settings, it can make continuous recordings for days or weeks. It can be especially useful for assessing insomnia, circadian rhythm disorders, movement disorders, and an assortment of rare events.

Common Test Findings in Sleep–Wake Disorders

Hypersomnolence Disorder. The EEG sleep pattern is essentially the same as that found in healthy individuals who are sleep deprived (Fig. 15-2). Unlike a sleep-deprived individual, the sleep pattern continues in this profile even after several nights of extended sleep. Also, we may see elevated slow-wave sleep.

Narcolepsy. When the diagnosis is not clinically apparent, a nighttime polysomnographic recording reveals a characteristic sleep-onset REM period (Fig. 15-3). A test of daytime MSLT shows rapid sleep onset and usually one or more sleep-onset REM periods.



Table 15-19
Example Sleep History Questionnaire

Patient name _____	
Date _____	
Please check the appropriate box or give short answers for the followi	
	Ye
1. Do you feel sleepy or have sleep attacks during the day?	☐
2. Do you nap during the day?	☐
3. Do you have trouble concentrating during the day?	☐
4. Do you have trouble falling asleep when you first go to bed?	☐

5. Do you awaken during the night? ☐

6. Do you awaken more than once? ☐

7. Do you awaken too early in the morning? ☐

8. How long have you had trouble sleeping? ☐

What do you think precipitated the problem?

9. How would you describe your usual night's sleep (hours of sleep, quality of sleep, etc.)?

10. Does your schedule for sleep and rising on the weekend differ from what it is during the week? ☐

11. Do others live at home who interrupt your sleep? ☐

12. Are you regularly awakened at night by pain or the need to use the bathroom? ☐

13. Does your job require shift changes or travel? ☐

14. Do you drink caffeinated beverages (coffee, tea, or soft drinks)? ☐

15. Apart from difficulty in sleeping, what, if any, other medical problems do you have?

16. What sleep medications, prescription or nonprescription, do you take? (Please include the dosage, how often you take it, and for how many months or years you have taken it.)

17. What other prescription and over-the-counter medications do you regularly use? (Again, please include the dosage, the frequency, and the duration.)

_____
_____
_____

18. Have you ever suffered from depression, anxiety, or similar problems? ☐

19. Do you snore? ☐

Questions for the Sleep Partner

1. Does your sleep partner snore? ☐

2. Does your sleep partner seem to stop breathing repeatedly during the night? ☐

3. Does your sleep partner jerk his or her legs or kick you while he or she is sleeping? ☐

4. Have you ever experienced trouble sleeping? Please explain. ☐

_____
_____
_____



Table 15-20
Indications for Polysomnography

Common Reasons

Diagnosis of sleep-related breathing disorders

Positive airway pressure (PAP) titration and assessment of treatment efficacy

Evaluation of sleep-related behaviors that are violent or may potentially harm the patient or bed partner

Differentiating narcolepsy from other hypersomnolence disorders

Differentiating parasomnias from nocturnal seizures

Other Possible Uses

Diagnose atypical parasomnias
Sleep-related problems secondary to neuromuscular disorders
Periodic limb movement disorder
Arousals secondary to seizure disorder
Excessive daytime sleepiness
Waking up gasping or choking
Narcolepsy (to assess sleep quality and quantity prior to a multiple sleep latency test)

Obstructive Sleep Apnea. On the polysomnogram, episodes of OSA appear as multiple periods of at least 10 seconds' duration in which nasal and oral airflow ceases either wholly or partially. Meanwhile, the abdominal and chest expansion leads indicate the continuing efforts of the diaphragm and accessory muscles of respiration to move air through the obstruction (Fig. 15-4). The arterial oxygen saturation drops, and often we see bradycardia, along with possible arrhythmias, such as premature ventricular contractions. In the end, an arousal reflex takes place, seen as a waking signal, and possibly as a motor artifact on the EEG channels. At this moment, sometimes called the *breakthrough,* the patient can make brief restless movements in bed.



Table 15-21
Common Polysomnographic Measures

Sleep latency: Period of time from turning out the lights until the appearance of stage 2 sleep

Early morning awakening: Time of being continuously awake from the last stage of the sleep until the end of the sleep record (usually at 7 am)

Sleep efficiency: Total sleep time or total time of the sleep record × 100

Apnea index: Number of apneas longer than 10 s/h of sleep

Nocturnal myoclonus index: Number of periodic leg movements per hour

Rapid eye movement (REM) latency: Period of time from the onset of sleep until the first REM period of the night

Sleep-onset REM period: REM sleep within the first 10 min of sleep

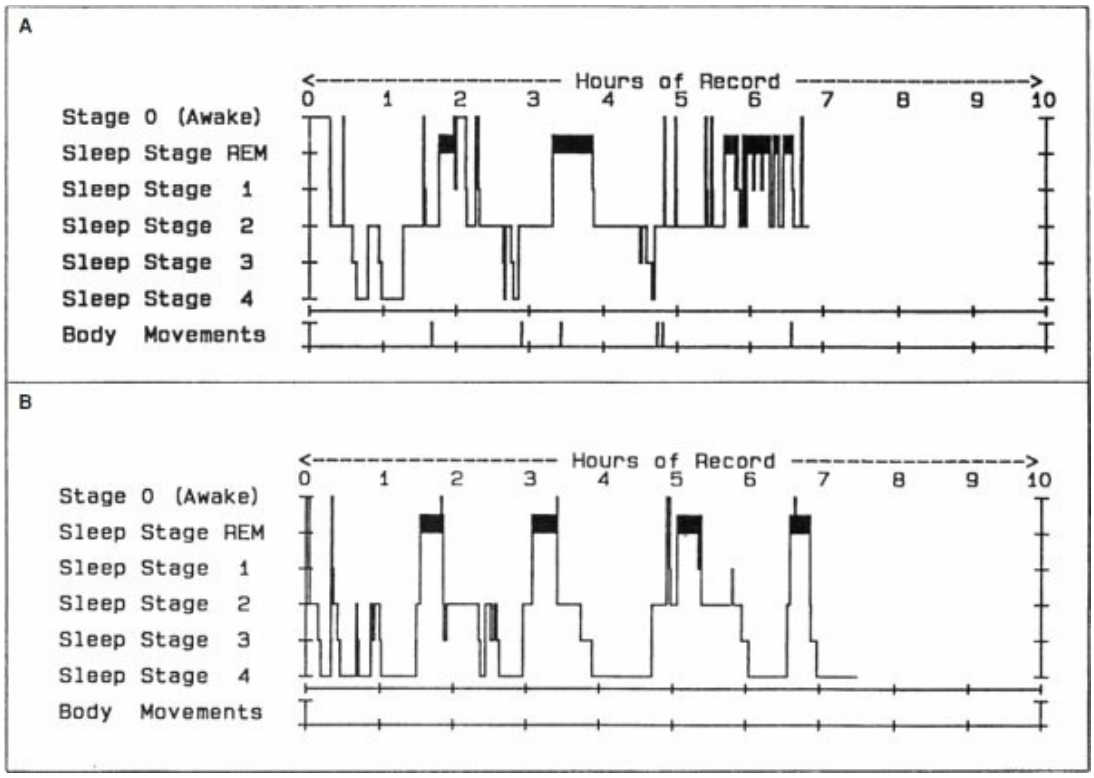


FIGURE 15-2
Sleep stage histograms comparing a normal sleep profile with that of a patient with hypersomnolence. **Panel A** illustrates the sleep pattern recorded in a healthy young adult. His sleep was interrupted several times by awakenings associated with other concurrent laboratory procedures; however, all parameters were within normal limits. **Panel B** depicts the sleep pattern of a patient with severe daytime sleepiness. The patient did not have narcolepsy or apnea. In the laboratory, sleep onset was rapid, sleep continued virtually without interruption, and he was extremely difficult to arouse. This recording was terminated after 7.5 hours to begin daytime testing; however, the patient's sleep revealed little indication of progressing toward a lighter state. In contrast to the normal sleep pattern, slow-wave sleep duration was elevated and it continued to emerge in alternation with REM sleep throughout the night. The patient awoke feeling unrefreshed and not satisfied with his sleep.

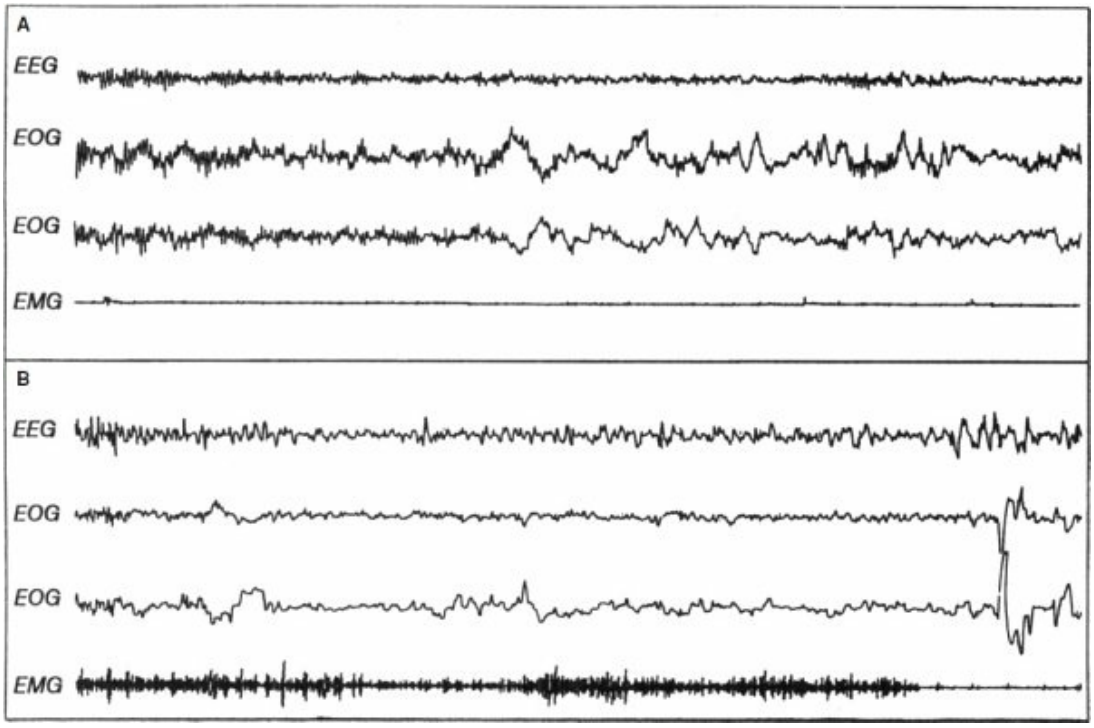


FIGURE 15-3
Polysomnographic tracing comparing normal sleep onset with that of a patient with narcolepsy. Each panel illustrates approximately 30 seconds of polysomnographic recording beginning with relaxed wakefulness. **Panel A** (normal sleep progression) shows a reduction of EEG α activity and development of slow rolling eye movements. **Panel B** shows the normally expected abatement of EEG α activity associated with increased θ activity and the appearance of a few slow eye movements. However, within 25 seconds (far right of figure) a swift loss of muscle tone occurs accompanied by rapid eye movements. This appearance of sleep-onset REM sleep (SOREM) characterizes narcolepsy and is part of the diagnostic criteria.

Central Sleep Apnea. The polysomnographic features of CSA are similar to those of OSA. However, during the periods of apnea, cessation of respiratory effort is seen in the abdominal and chest expansion leads. In Cheyne–Stokes breathing, polysomnography reveals five or more central apnea and hypopnea episodes per hour of sleep. In CSA, due to a medical disorder without Cheyne–Stokes breathing, diagnostic criteria require a polysomnographically verified rate of 5 or more central apneas and hypopneas per hour of sleep. Also, CSA due to medication or substance requires five or more central apneas per hour.

Sleep-Related Hypoventilation. Polysomnography demonstrates either episodes with elevated CO_2 levels and decreased respirations or, if not

monitoring CO_2, episodes of persistently low oxyhemoglobin levels that are not provoked by apnea or hypopnea.

Parasomnias. Sleep studies help to develop a differential diagnosis and rule out that the unusual behavior is secondary to a seizure, sleep-related breathing disorder, or another sleep disorder. Figures 15-5 to 15-7 give examples of polysomnograms for sleep terrors, RLS, and sleep-related bruxism.

DIFFERENTIAL DIAGNOSIS

In general, the differential diagnosis of sleep–wake disorders should include other individual sleep–wake disorders, psychiatric disorders, medical conditions, medications and substance use, and normal variants.

Insomnia Disorder

Many psychiatric disorders can cause a person to experience insomnia. For example, MDD, bipolar disorder, and PTSD can all present with sleep disruption. Many medications and substances also affect sleep, some of which we listed in Table 15-7. Other sleep–wake disorders should also be considered, including circadian rhythm sleep–wake disorder, RLS, sleep-related breathing disorders, narcolepsy, and parasomnias. Finally, one should consider whether the sleep pattern is a normal variant in the population (e.g., a “short sleeper”) or is occurring in the context of an acute stressor.

Hypersomnolence Disorder

Psychiatric disorders associated with increased sleep or fatigue, such as MDD or bipolar depression, should be considered part of the differential diagnosis. Certain medications or substances can also cause sedation, such as sedative–hypnotics and opioids. Other sleep–wake disorders should be considered, including circadian rhythm sleep–wake disorder, sleep-related breathing disorders, and parasomnias. Also, poor sleep quality can lead to excessive daytime sleepiness, and there are some persons for whom it is a normal variant to require more sleep (e.g., “long sleepers”).

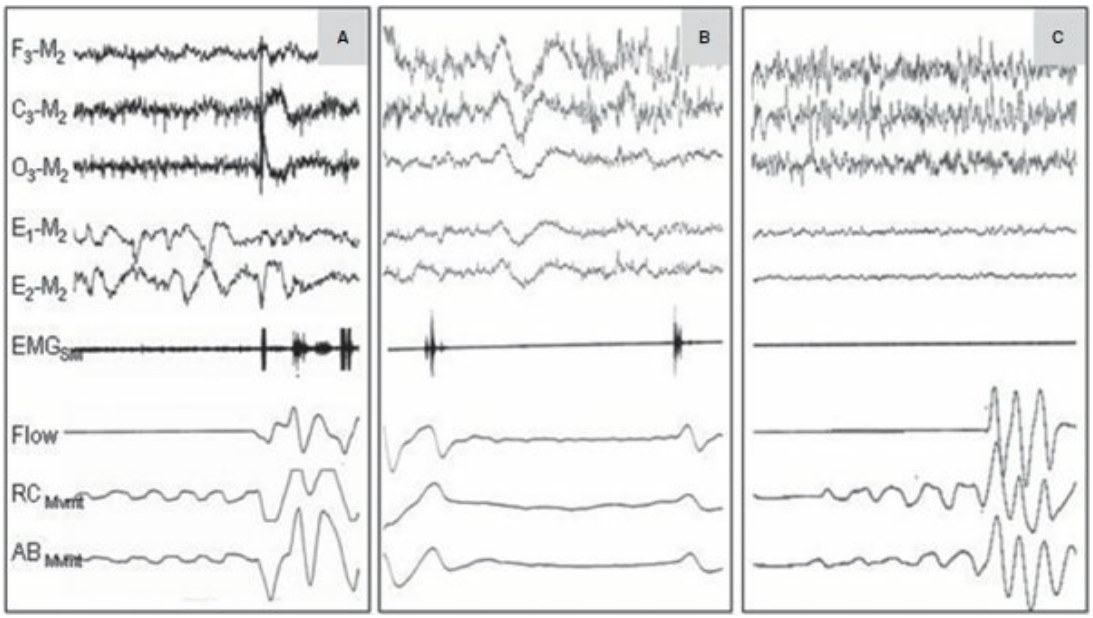


FIGURE 15-4

Polysomnographic tracings illustrating sleep apnea episodes. **Panel A** illustrates an obstructive sleep apnea, **Panel B** shows a central sleep apnea, and **Panel C** depicts a mixed sleep apnea. Note the presence of respiratory effort (rib cage and abdominal movement), notwithstanding cessation of airflow, throughout the obstructive apnea and during the latter portion of the mixed apnea. By contrast, respiratory effort is absent during the flow cessation throughout the central apnea and during the early portion of the mixed apnea. Notation: F_3 (left frontal scalp derivation); M_2 (right mastoid electrode location); C_3 (left central scalp derivation); O_3 (left occipital scalp derivation); E_1 (left outer canthus eye electrode location); E_2 (right outer canthus eye electrode location); EMG_{SM} (surface electrode electromyogram from submentalis muscle); Flow (airflow measured with nasal–nasal–oral thermistors); RC Mvmt (rib cage movement measured with inductive plethysmography); AB Mvmt (abdominal movement measured with inductive plethysmography). (From Hirshkowitz M, Sharafkhaneh A. Diagnostic assessment methods in adults. In: Kryger MH, Avidan AY, Berry RB, eds. *Atlas of Clinical Sleep Medicine*. 2nd ed. Elsevier Saunders: Philadelphia, PA; 2014:378.)

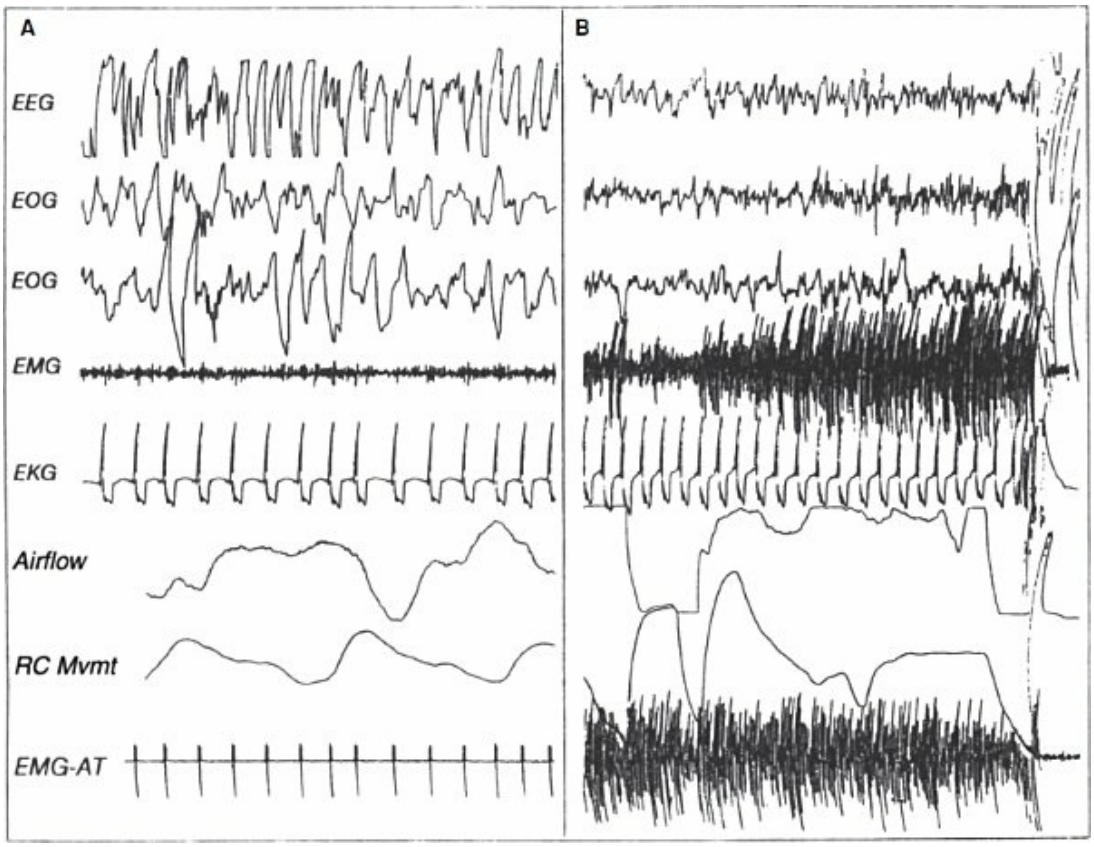


FIGURE 15-5

Polysomnographic tracing illustrating a sleep terror. **Panel A** shows approximately 14 seconds of tracing occurring immediately before the sleep terror. Prominent EEG slow-wave activity and other characteristics of stage 4 sleep are seen. **Panel B** shows the awakening, accompanied by tachycardia and movement. EEG activity is ambiguous and the patient eventually disconnected his electrodes as he thrashed about in bed (visible at far right of figure). Although the patient was screaming and greatly agitated, there was no report of dreaming. In the morning, there was little recollection of anything having occurred during the night.

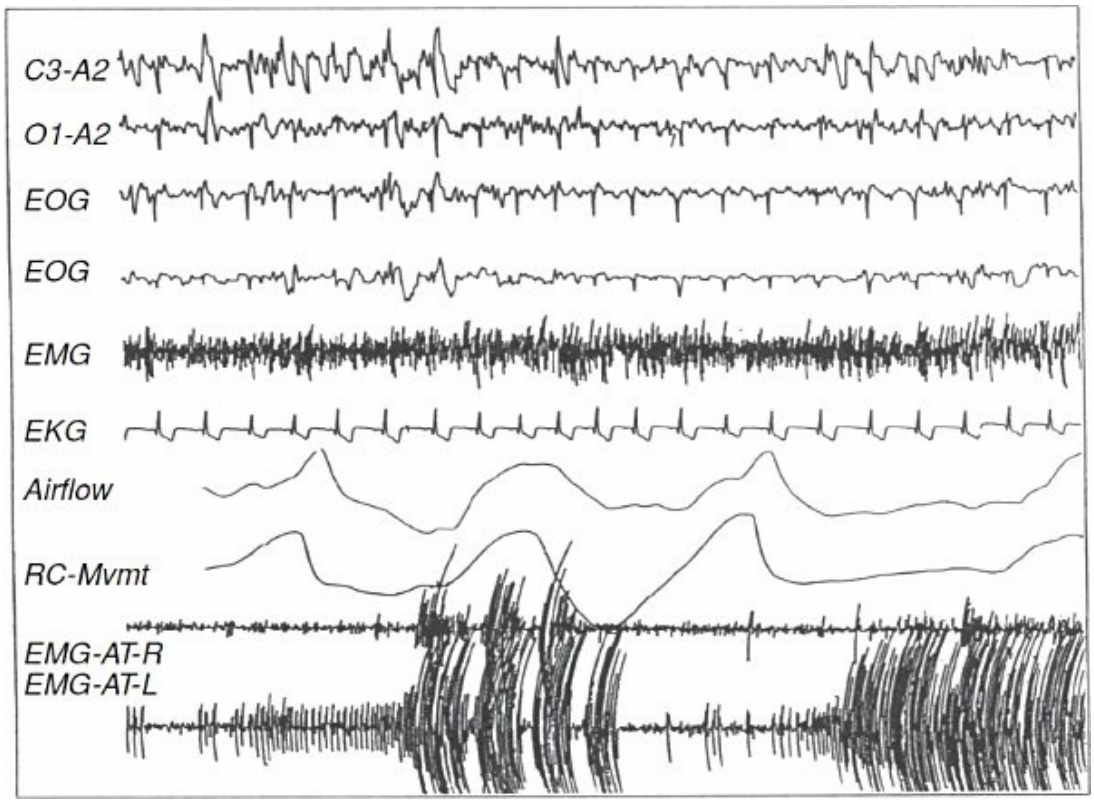


FIGURE 15-6

Restless legs syndrome. This patient presented with complaints of uncomfortable, crawling sensations in the legs when trying to fall asleep. Patients commonly report an urge to move the leg in order to dispel the sensation. This figure shows a bilateral pattern of leg EMG activity; however, the discharge is more pronounced in the left anterior tibialis (EMG-AT-L) than the right (EMG-AT-R). This pattern continued for more than an hour as the patient attempted to fall asleep; note that the sharp activity in central and occipital EEG (C3-A2 and O1-A2, respectively) and EOG is EKG artifact and not an EEG abnormality.

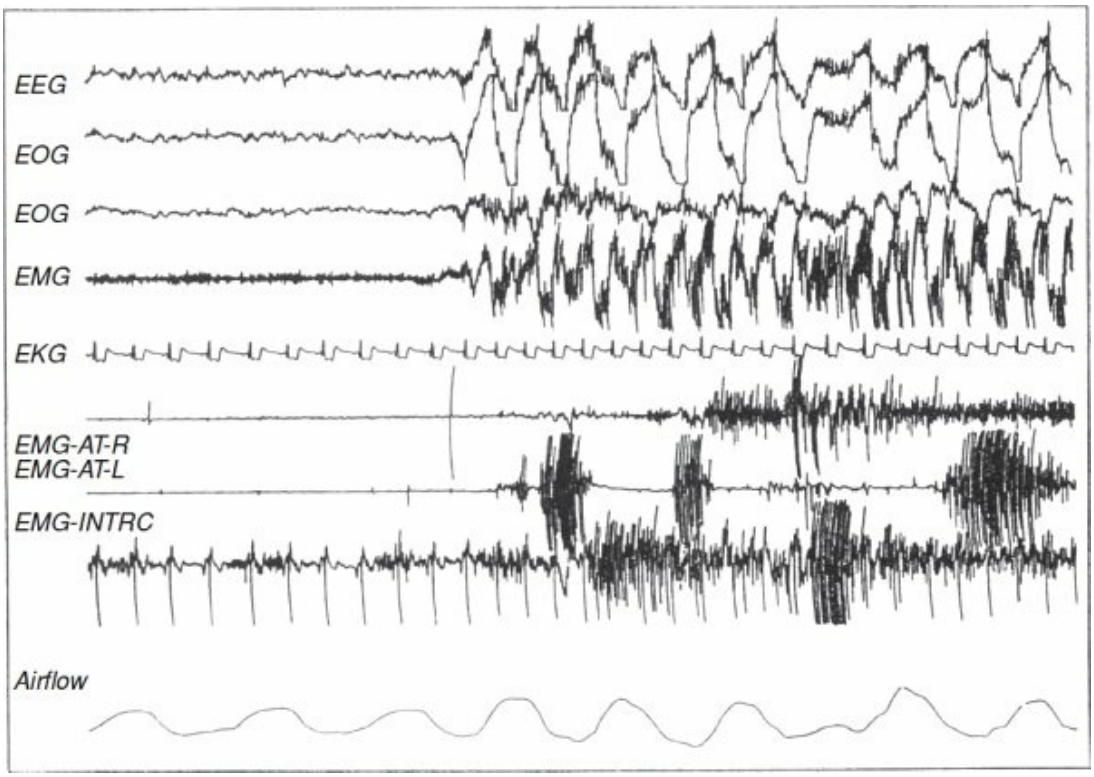


FIGURE 15-7
Polysomnographic tracing showing sleep-related bruxism. This figure showing approximately 25 seconds of tracing was obtained from a patient during an episode of bruxism. Bruxism can occur during any stage of sleep or wakefulness. The interference pattern of EEG-EOG-EMG channels is typical and reflects the rhythmic jaw movement and grinding of the teeth. This patient had many such episodes, some that caused awakening. Readily observable tooth damage and jaw pain were noted.

Table 15-22
Medical Disorders Causing Cataplexy

Disorders affecting the hypothalamus
Tumors
Sarcoidosis
Multiple sclerosis plaques
Paraneoplastic syndrome
Neimann–Pick type C disease
Coffin–Lowry syndrome

Narcolepsy

The differential diagnosis of narcolepsy includes many of the same disorders as the differential diagnosis of hypersomnolence disorder. Also, cataplexy can be confused for seizures or neurologic movement disorders (e.g., chorea). Many disorders can precipitate cataplexy, and we list some of the more common in Table 15-22. All of these disorders would preclude a diagnosis of narcolepsy. Cataplexy is usually absent in narcolepsy associated with multiple sclerosis, myotonic dystrophy, Prader–Willi syndrome, Parkinson disease, and multisystem atrophy.

Sleep-Related Breathing Disorders

The differential diagnosis of sleep-related breathing disorders includes many of the same disorders as the differential diagnosis of insomnia and hypersomnolence disorders, given the clinical presentation of fatigue and daytime sleepiness. Other possible conditions include panic attacks and attention-deficit/hyperactivity disorder (as repeated apneic episodes may lead to problems with focus, attention, and school- or job-related performance).

Circadian Rhythm Sleep–Wake Disorders

The differential diagnosis of circadian rhythm sleep–wake disorders includes insomnia disorder, other sleep–wake disorders, and mood disorders. One should also consider the possibility that the sleep pattern represents a normal variant.

Parasomnias and Sleep-Related Movement Disorders

The differential diagnosis of parasomnias can be complicated and should include other individual parasomnias, sleep-related breathing disorders, nocturnal seizures, panic attacks, other sleep–wake disorders, amnestic and dissociative syndromes, and numerous medical conditions. The differential diagnosis of RLS includes multiple medical conditions, such as arthritis, neuropathy, positional ischemia, and peripheral artery disease, to name a few, while the differential diagnosis of sleep-related bruxism should include nocturnal seizures.

COMORBIDITY

Insomnia Disorder

Several psychiatric disorders are often comorbid with insomnia. We list some common ones in Table 15-23. Nearly 90 percent of patients with MDD have insomnia. Of the most significant concern, insomnia is an independent risk factor for suicide in patients with MDD. Insomnia also predicts who will develop posttraumatic stress disorder following a trauma. For bipolar patients, lack of sleep may precipitate a manic episode. Patients with schizophrenia may deny having slept despite objective measures showing normal sleep.

Hypersomnolence Disorder

Many disorders also occur with hypersomnolence disorder, and we list some of these in Table 15-23. Medication use disorders are also common, particularly the stimulants, as the individual attempts to stay awake.

Narcolepsy

Narcolepsy is associated with psychiatric disorders in general, as well as many medical disorders. We list some of these in Table 15-23.

Sleep-Related Breathing Disorders

Table 15-23 lists several disorders associated with sleep-related breathing disorders. The section on diagnosis discusses conditions associated with central sleep apnea and sleep-related hypoventilation.

Circadian Rhythm Sleep–Wake Disorders

The circadian rhythm sleep–wake disorders can be associated with psychiatric and medical comorbidities; we list some of these in Table 15-23. The shift work type is especially associated with many psychiatric and medical disorders.

Parasomnias and Sleep-Related Movement Disorders

Each parasomnia has its own set of common comorbid diagnoses, and we list some examples in Table 15-23. Many individuals who present with RBD will go on to develop a neurodegenerative disorder.

TREATMENT APPROACH

Insomnia

Pharmacologic Treatment. For many years, benzodiazepines were the most commonly prescribed sedative–hypnotic medications for treating insomnia. Other benzodiazepine-receptor agonists (e.g., zolpidem and similar drugs) then emerged as an option to treat insomnia. The subsequent development of a low-dose antihistamine (doxepin), a melatonin receptor agonist (ramelteon), and an orexin antagonist (suvorexant) have more recently augmented the armamentarium of medications approved for treating insomnia. However, unapproved off-label (and not recommended) use of sedating antidepressants is commonplace. Additionally, a variety of over-the-counter (OTC) sleep aids is available. Nonprescription formulas include sedating antihistamines, protein precursors, and other substances. L-tryptophan was trendy and readily available at health food stores until an outbreak of eosinophilia led to pulling it off the shelves. However, the cause was a single manufacturer, and L-tryptophan subsequently became available OTC again. Melatonin is the leader among self-administered food additives believed by some to alleviate sleeplessness. Low-dose melatonin can improve sleep when used to align or reentrain the circadian rhythm with the desired sleep–wake schedule.



Table 15-23
Disorders That Are Commonly Comorbid with Various Sleep Disorders

Insomnia Disorder
Bipolar disorder
Generalized anxiety disorder
Major depressive disorder
Posttraumatic stress disorder
Schizophrenia
Hypersomnolence Disorder
Bipolar disorder

Depressive disorders (particularly seasonal depression)
Neurodegenerative disorders
Substance use disorders (stimulants)

Narcolepsy
Anxiety
Depressive disorders
Obstructive sleep apnea
Headache
Hyperlipidemia
Hypertension
Obesity
Peripheral neuropathy
Thyroid disease, other endocrinopathies

Sleep-Related Breathing Disorders
Anxiety disorders
Dementia
Mood disorders
Post-traumatic stress disorder
Psychotic disorders
Cardiovascular disease
Diabetes mellitus
Heart failure
Hypertension
Obesity

Circadian Rhythm Sleep-Wake Disorders
Depression
Neurodegenerative disorders
Neurodevelopmental disorders
Personality disorder
Somatic symptom disorders
Blindness
Cardiovascular disease

Diabetes mellitus
Gastrointestinal disease
Substance use disorders
Parasomnias and Sleep-Related Movement Disorders
Sleepwalking
Obsessive compulsive disorder
Major depression
Nightmare disorder
Posttraumatic stress disorder
Acute stress disorder
Schizophrenia
Anxiety disorders
Personality disorders
Bereavement
Pain
Neurodegenerative disorders
Cardiovascular disease
REM sleep behavior disorder
Narcolepsy
Neurodegenerative disorders
Restless legs syndrome
Depression
Anxiety
Attentional disorders
Iron deficiency

Drug manufacturers must rigorously test prescription medications in clinical trials; therefore, they hold an advantage over the virtually untested OTCs. Medication must be proven safe and effective to become FDA approved.

Most hypnotic medications are approved for short-, not long-term use. Exceptions include zolpidem-modified release, eszopiclone, and ramelteon, which the FDA approved for long-term therapy. When properly used, hypnotics can provide immediate and adequate relief from sleeplessness.

Insomnia, however, usually returns upon discontinuation of dosing.

The clinical practice guideline from the American Academy of Sleep Medicine recommends the use of suvorexant, eszopiclone, zaleplon, zolpidem, triazolam, temazepam, ramelteon, and doxepin for the treatment of insomnia (with specific agents having different indications for sleep-onset or sleep-maintenance insomnia). They recommend against the use of trazodone, tiagabine, diphenhydramine, melatonin, tryptophan, and valerian for insomnia treatment.

Cognitive-Behavioral Therapy for Insomnia. Cognitive-behavioral therapy for insomnia (CBTi), as a treatment modality, uses a combination of behavioral and cognitive techniques to overcome dysfunctional sleep behaviors, misperceptions, and distorted, disruptive thoughts about sleep. Behavioral techniques include universal sleep hygiene, stimulus control therapy, sleep restriction therapy, relaxation therapies, and biofeedback.

Studies repeatedly show significant, sustained improvement in sleep symptoms, including number and duration of awakenings and sleep latency, from CBTi. Short-term benefits are similar to that of medication, but CBTi tends to have lasting benefits, even 36 months after treatment. With the cessation of the medication, insomnia frequently returns and is sometimes accompanied by rebound insomnia. CBTi has no apparent adverse effects. There are no established “best practice” guidelines for length or quantity of sessions. CBTi, however, is not without limitations. Most data do not compare the efficacy of the individual components of CBTi. However, sleep hygiene education alone produces an insignificant effect on sleep. Intuitively, the multicomponent approach addresses many of the variables contributing to insomnia. Other limitations of CBTi include lack of or unavailability of trained specialists, cost of sessions, and noncoverage by insurance carriers.

The effects of CBTi take longer to emerge than the effects of medications. Usually, when patients finally come for treatment of their insomnia, they are desperate. It is challenging to convince them to try a therapy that may take several weeks before it will provide relief. Furthermore, patients do not assume a passive role in this type of therapy; they must be active participants. Many individuals not only want a “quick fix,” but they also want to undergo a procedure or have something administered rather than be involved in the therapeutic process. For optimal CBTi, patients must commit to come to

multiple sessions and also be open to the idea that modifying thoughts and behaviors about sleep can improve the symptoms of insomnia.

Although firmly focused on cognitive and behavioral issues, it helps to extend CBT just slightly into the psychodynamic sphere. For some patients with long-standing difficulty sleeping, being an insomniac becomes an essential part of their identity. There may be primary or secondary gain to such identification. It is the negative emotional response (i.e., anger at the inability to control one's sleep, feeling like a failure because one cannot sleep) to insomnia that contributes to its chronicity. In general, these individuals tend to internalize rather than express emotion, feel a heightened need for control, experience interpersonal difficulties and have significant discontent with past events. For this subset of people, neglecting the emotional response can limit the response to treatment and cause a relapse. The clinician will be better able to intercept barriers to treatment when they attune themselves to a patient's tendency to view something as a failure rather than a challenge.

Universal Sleep Hygiene. A patient's lifestyle can lead to a sleep disturbance. We usually call this *inadequate sleep hygiene*, referring to a problem in following generally accepted practices to aid sleep. These include, for instance, keeping regular hours of bedtime and arousal, avoiding excessive caffeine, not eating heavy meals before bedtime, and getting adequate exercise. Many behaviors can interfere with sleep and may do so by increasing nervous system arousal near bedtime or by altering circadian rhythms.

The focus of universal sleep hygiene is on modifiable environmental and lifestyle components that may interfere with sleep, as well as behaviors that may improve sleep. Because some of these behaviors are difficult to change, we should concentrate on only one or two items, chosen in collaboration with the patient, to address. The patient then has the best chance at a successful intervention. We list some sleep-enhancing directives in Table 15-24. Often a few simple alterations in a patient's habits or sleep environment are useful. The clinician, however, needs to spend time reviewing both the patient's routine and its irregularity. In some respects, the essence of insomnia is its variability. The day-to-day changes in behavior and the changing severity of sleeplessness can obscure the factors responsible for the problem. A carefully

explained program of sleep hygiene, with follow-up, represents a reasonably inexpensive intervention. Furthermore, improving sleep habits can enhance sleep, even when the primary cause of insomnia is physical. However, sleep hygiene education does not appear to be as effective as CBTi.

Stimulus Control Therapy. Stimulus control therapy is a deconditioning paradigm that aims to break the cycle of problems commonly associated with difficulty initiating sleep. By attempting to undo conditioning that undermines sleep, stimulus control therapy helps reduce both primary and reactive factors involved in insomnia. The rules attempt to enhance stimulus cues for sleeping and diminish associations with sleeplessness. The instructions are simple; however, the patient must consistently follow them. We list an example of these instructions in Table 15-25. Stimulus control therapy does work; however, we might not see results for weeks or even a month. If continually practiced, the bouts of insomnia lessen in both frequency and severity.

Sleep Restriction Therapy. Sleep restriction therapy is a strategy designed to increase sleep efficiency by decreasing the amount of time spent awake while lying in bed. This therapy targets explicitly those patients who lie awake in bed, unable to sleep. Restricting time in bed can help to consolidate sleep. If the patient reports sleeping only 5 hours of an 8-hour scheduled time in bed, reduce the time in bed. It is advised, however, not to reduce bedtime to less than 4 hours per night and to warn the patient about the hazards of daytime sleepiness. Sleep at other times during the day must be avoided, except in the elderly, who may take a 30-minute nap. The clinician then monitors sleep efficiency (time asleep as a percentage of the time in bed). When sleep efficiency reaches 85 percent (averaged over five nights), we should increase the time in bed by 15 minutes. Sleep restriction therapy produces a gradual and steady decline in nocturnal wakefulness.



Table 15-24
Do's and Don'ts for Good Sleep Hygiene

	DO	DON'T
Maintain regular hours of bedtime and arising	✓	

If you are hungry, have a light snack before bedtime	✓	
Maintain a regular exercise schedule	✓	
Give yourself approximately an hour to wind down before going to bed	✓	
If you are preoccupied or worried about something at bedtime, write it down and deal with it in the morning	✓	
Keep the bedroom cool	✓	
Keep the bedroom dark	✓	
Keep the bedroom quiet	✓	
Take naps		✓
Watch the clock so you know how bad your insomnia actually is		✓
Exercise right before going to bed in order wear yourself out		✓
Watch television in bed when you cannot sleep		✓
Eat a heavy meal before bedtime to help you sleep		✓
Drink coffee in the afternoon and evening		✓
If you cannot sleep, smoke a cigarette		✓
Use alcohol to help in going to sleep		✓
Read in bed when you cannot sleep		✓
Eat in bed		✓
Exercise in bed		✓
Talk on the phone in bed		✓

Relaxation Therapy and Biofeedback. The most critical aspect of relaxation therapy is to perform it properly. Self-hypnosis, progressive

relaxation, guided imagery, and deep breathing exercises are all useful if they produce relaxation. The goal is to find the optimal technique for each patient, but not all patients need help in relaxing. *Progressive muscle relaxation* is especially useful for patients who experience muscle tension. The patients should purposefully tense (5 to 6 seconds) and then relax (20 to 30 seconds) muscle groups, beginning at the head and ending at the feet. The patient should appreciate the difference between tension and relaxation. *Guided imagery* has the patient visualize a pleasant, restful scene, engaging all of his or her senses.



Table 15-25
Stimulus Control Instructions

1. Go to bed only when sleepy to maximize success.
2. Use the bed only for sleeping. Do not watch television in bed, do not read, do not eat, and do not talk on the telephone while in bed.
3. Do not lie in bed and become frustrated if unable to sleep. After a few minutes (do not watch the clock), get up, go to another room, and do something nonarousing until sleepiness returns. Repeat as often as needed.
4. Awaken at the same time every morning (regardless of bedtime, total sleep time, or day of week) and totally avoid napping.



Table 15-26
Abdominal Breathing

1. In the supine position, the patient should breathe normally through his or her mouth or nose, whichever is more comfortable, and attend to his or her breathing pattern.
2. While maintaining that rhythm, the patient should begin to breathe more with his or her abdomen and less with his or her chest.
3. The patient should pause for a half second after each breath cycle (in and out) and evaluate the breath. How did it feel? Was it

smooth? Eventually each breath will become uniform and smooth.
4. The patient should find a place where he or she can best feel the air move in and out. Concentrate on that spot and on the air moving in and out.
5. The patient should visualize intrusive thoughts as floating away; if there are too many thoughts, stop practicing and try again later.

Breathing exercises should be practiced for at least 20 min/day for 2 weeks. Once mastered, the technique should be used once at bedtime for 30 minutes. If it does not work, the patient should try a different night again. It is essential that the technique not become associated with failure to fall asleep. Table 15-26 gives example instructions for *abdominal breathing,* an important technique. The patient should become comfortable with each step before moving on to the next.

Biofeedback provides stimulus cues for physiologic markers of relaxation and can increase self-awareness. Muscle tension in the forehead or finger temperature is measured. Finger temperature rises when a person becomes more relaxed. Patients require thorough and adequate training; merely giving them an instruction tape is not especially helpful. Techniques are ideally mastered during the day for several weeks before application to the sleep problem; the patient preferably does this outside of the bed. By the time the patient uses the techniques in bed, the skill should be automatic. Relaxation techniques readily lend themselves to being combined with sleep hygiene and stimulus control therapies. Sometimes they make for good distractions from thinking about the inability to sleep. The ruminations fuel insomnia, and if the ruminator can be distracted, then the person may sleep better.

Cognitive Therapy. This practical, validated treatment for a variety of psychiatric conditions, including major depression and generalized anxiety, has been adapted for use with insomnia. The cognitive aspect of insomnia treatment targets the negative emotional response to an appraisal of a sleep-related situation. The negative emotional response produces emotional arousal, which in turn contributes to or perpetuates insomnia. People who have maladaptive cognitions tend to exaggerate the negative consequences of insomnia: “There must be something wrong with me if I cannot fall asleep in

40 minutes." They also tend to have unrealistic expectations about their sleep requirements: "If I do not sleep 8 hours a night, then my whole day will be ruined." The first step is to identify these cognitions, then challenge their validity, and finally, substitute them with more adaptive cognitions.

DH was a 42-year-old man with a 5-year history of insomnia. He identified losing his job and the birth of a colicky baby as precipitating factors in his inability to sleep. However, even after he found a new position with better hours and pay and with the child sleeping through the night, DH continued to experience difficulty falling and staying asleep. Perpetuating factors included low back pain and a spouse with periodic limb movement disorder. He reported spending 8 to 9 hours in bed each night and sleeping only 4 to 5 hours intermittently. He watched 1 hour of television in bed before turning out the light for bedtime. He spent hours watching the minutes tick away. He did not wake feeling rested, and when his alarm went off, he was frequently already awake and had thoughts such as, "I hardly slept at all last night. I should be able to get more sleep. There must be something wrong. Great, I'll be too tired to concentrate on anything today."

Examples of maladaptive thoughts: *"I should be able to get more sleep."* This is a faulty appraisal of sleeping ability and may relate to a need for control over sleep. This need for control interferes with having a more laissez-faire attitude about a few missed hours of rest. Such thoughts can also lead to feelings of frustration and anger. *"Great, I'll be too tired to concentrate on anything today."* This is a misattribution of daytime impairment due to poor sleep. DH was also magnifying the negative and discounting the positive with his black-and-white or all-or-nothing thinking. Could DH be too tired to concentrate on some things, but not all things? Might his inability to concentrate be due to a myriad of other factors? "*There must be something wrong with me (if I can't get enough sleep).*" This is catastrophizing and emotional reasoning: just because a person had a feeling does not mean that the thought or feeling is valid. A firmly held belief that sleeplessness negatively affects physical and mental health can set off catastrophizing. (Courtesy of Max Hirshkowitz, Ph.D., Rhoda G. Seplowitz-Hafkin, M.D., and Amir

Sharafkhaneh, M.D., Ph.D.)

Paradoxical Intention. This is a cognitive technique with conflicting evidence regarding its efficacy. In clinical practice, compliance is often a barrier, but it does work for a limited number of patients. The theory is that performance anxiety interferes with sleep onset. Thus, when the patient tries to stay awake for as long as possible rather than trying to fall asleep, performance anxiety will be reduced, and sleep latency will improve.

Hypersomnolence Disorder

We usually treat hypersomnolence disorder with medication. Although there are no medications approved for its treatment, physicians commonly use stimulants, with varying degrees of success. The wake-promoting substance modafinil or traditional psychostimulants (amphetamines and their derivatives) sometimes help manage the sleepiness. In addition to drug therapies, the overall therapeutic approach usually includes lifestyle adjustment, psychological counseling, drug holidays to reduce tolerance (if using stimulants), and careful monitoring of refills, general health, and cardiac status.

Narcolepsy

Currently, we cannot cure narcolepsy; however, the symptoms can be managed with either the wake-promoting substances modafinil or armodafinil, or traditional psychostimulants like amphetamines and their derivatives. The FDA has approved two medications for the treatment of narcolepsy as wake-promoting agents: pitolisant, which is a selective H_3 receptor antagonist/inverse agonist, and solriamfetol, a selective norepinephrine–dopamine reuptake inhibitor. For managing cataplexy, clinicians often administer REM sleep–suppressing drugs (e.g., many antidepressants). This approach capitalizes on the REM sleep-suppressant properties of these drugs. Because cataplexy is presumably an intrusion of REM sleep phenomena into the awake state, the rationale is clear. Many reports indicate that imipramine and protriptyline are quite effective in reducing or eliminating cataplexy. Selective serotonin reuptake inhibitors

have gained popularity because they are associated with fewer side effects than the tricyclic antidepressants. Sodium oxybate has proven extremely useful for reducing cataplexy, even in cases where clinicians thought that the cataplexy was intractable. Studies also suggest that sodium oxybate helps improve sleep and relieves some of the sleepiness associated with narcolepsy. Although drug therapy is the treatment of choice, the overall therapeutic approach should include scheduled naps, lifestyle adjustment, psychological counseling, drug holidays to reduce tolerance, and careful monitoring of drug refills, general health, and cardiac status.

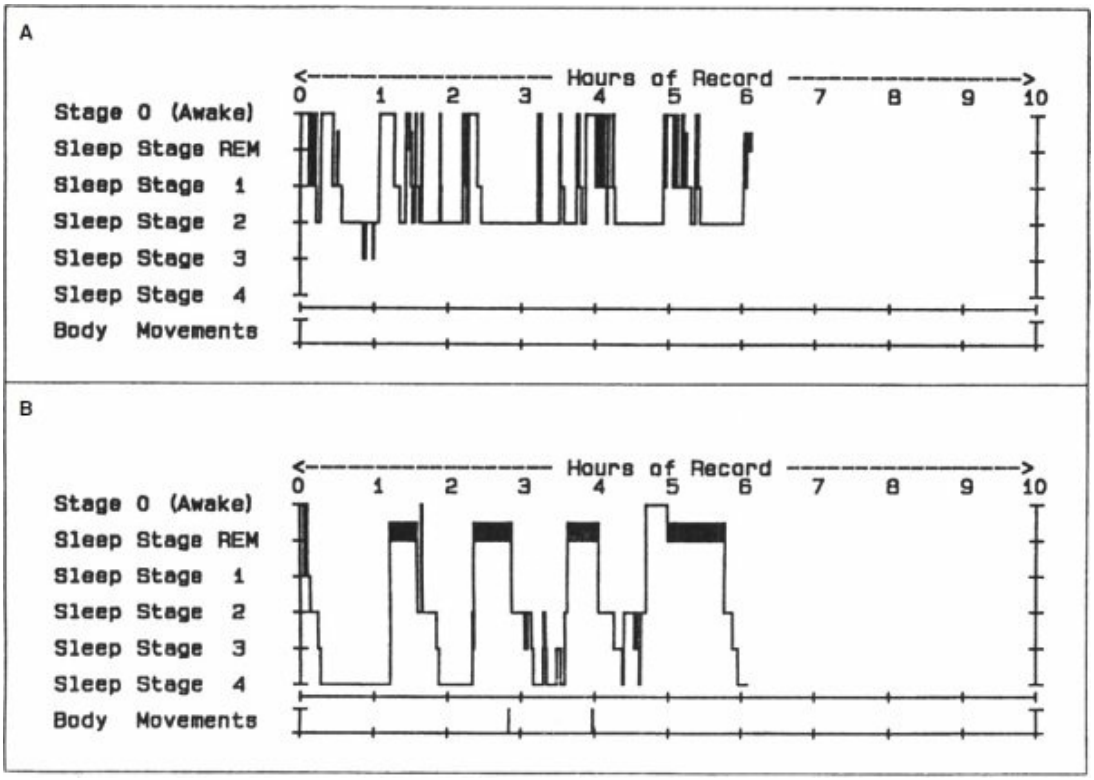


FIGURE 15-8
Sleep histogram showing CPAP-related sleep improvement. Sleep stage histogram illustrating the immediate, dramatic improvement in sleep architecture produced by treating obstructive sleep apnea with continuous positive airway pressure (CPAP) therapy. **Panel A** illustrates the abnormal sleep pattern on a night when the patient had more than 200 episodes of obstructive sleep apnea. Sleep is disturbed by frequent awakenings while REM and slow-wave (stages 3 and 4) sleep are nearly absent. **Panel B** shows data from the same patient being treated with CPAP on the next night. Normalization of sleep continuity with a massive rebound in REM and slow-wave sleep is evident.

Breathing-Related Sleep Disorders

Obstructive Sleep Apnea-Hypopnea. A number of treatments are available for OSA, including PAP, oral appliances, positional therapy, surgical intervention, and weight loss. Medications have not been useful for treating OSA. Wakefulness promoting agents, such as modafinil, solriamfetol, and pitolisant, may help with the residual sleepiness that may persist despite the use of other OSA treatments such as PAP. However, these medications do not address the pathophysiology of airway occlusion.

POSITIVE AIRWAY PRESSURE. PAP is the preferred treatment for OSA (Fig. 15-8). The PAP apparatus consists of a fan-driven blower, a nasal mask or oronasal interface, and tubing connecting the two. The airflow through the interface (usually a mask or nasal pillows) provides a positive pressure that offsets oropharyngeal collapse produced by inspiratory negative thoracic pressure. In this manner, it acts as a pneumatic splint, thereby maintaining the airway. When appropriately titrating the pressure, even the most severe sleep apnea can be alleviated. Results are often dramatic. PAP devices come in several varieties. The most common are systems that provide a preset continuous positive airway pressure (CPAP). For individuals who find it difficult to exhale against a continuous pressure, bilevel positive airway pressure (BPAP) may provide a solution. BPAP devices have different inspiratory and expiratory pressure settings. More recently, systems that sense the patient's changes in airway resistance and automatically adjust the positive airway pressure have been gaining popularity. Such automatic positive airway pressure (APAP) systems should theoretically be able to adapt to changes in pressure requirements produced by sleep deprivation, medications, weight change, sleep stage, illness, and aging. Finally, there are timed bilevel and servo-ventilation systems, but these fall into the category of noninvasive positive pressure ventilation (NIPPV) systems, which are more appropriate for treating other pulmonary diseases and breathing problems in neuromuscular diseases. The tremendous efficacy and remarkable safety of PAP therapies have made them the standard of care for patients who can tolerate sleeping with the machine. The principal therapeutic challenge is utilization. Patient education and systematic follow-up are crucial. When problems with the interface, the pressure, nasal stuffiness, and other barriers to routine, nightly use arise, we should remedy these quickly to ensure therapeutic adherence. When used properly, the success of PAP therapy has

rendered surgical intervention a secondary option, resorted to mainly after PAP failure, rejection, or nonadherence.

ORAL APPLIANCE. Oral appliances represent another therapeutic option that is gaining popularity. A variety of oral appliances have also been developed to treat snoring and can provide benefit for mild to moderate OSA cases. The general approach is to manipulate the position of the mandible, lift the palate, or retain the tongue. Randomized trials indicate that some oral appliances improve airway patency sufficiently. However, in patients with severe OSA, improvement does not always reach satisfactory levels; therefore, follow-up evaluations are critically important.

POSITIONAL THERAPY. In some patients, sleep-disordered breathing occurs only in the supine position. In such situations, preventing patients from sleeping on their backs may produce beneficial results. Tennis balls sewn onto or placed into pockets on the back of the nightshirt or foam wedges may accomplish this goal. Although such interventions seem to be useful in clinical practice, large-scale systematic clinical trials are needed.

SURGICAL INTERVENTION. Aggressive surgical treatments evolved soon after OSA's pathophysiologic and potentially life-threatening consequences were recognized. The earliest surgical intervention aimed to create a patent airway; thus, in the late 1970s, tracheostomies were performed on individuals with severe apnea. Although no longer the preferred treatment, it remains a standard against which we judge newer, more refined therapies. Second-generation surgical approaches attempt to correct airway obstructions and malformations. Early studies of uvulopalatopharyngoplasty (UPPP) suggested that modification of the soft palate effectively relieved most sleep apnea. Later follow-up results were less impressive. Approximately 30 to 50 percent of patients with sleep apnea benefit from UPPP. These patients are likely those with oropharyngeal obstruction; thus, careful attention to selection criteria presumably improves outcomes. However, if obstruction occurs in the posterior airway space (PAS), maxillomandibular surgery may be appropriate. In retrognathic patients or patients with cephalometrics revealing compromised PAS, moving the jaw forward can achieve impressive normalization of breathing during sleep.

WEIGHT LOSS. Weight loss is known to help many patients. However, because losing weight and keeping it off is difficult and unreliably achieved, the prudent clinician should recommend weight loss but also rely on other therapies.

CSA due to High-Altitude Periodic Breathing. We can treat CSA due to high-altitude periodic breathing with acetazolamide, which lowers serum pH and increases the respiratory drive. Acetazolamide side effects include metabolic acidosis, electrolyte imbalance, anaphylaxis, Stevens–Johnson syndrome, toxic epidermal necrolysis, and agranulocytosis. Common reactions include but are not limited to fatigue, anorexia, taste changes, polyuria, diarrhea, melena, tinnitus, and photosensitivity.

Circadian Rhythm Sleep–Wake Disorders

Light Therapy. Research indicates that exposing an individual to bright lights can reset the circadian pacemaker. This seems especially the case when the light is bright (higher than 10,000 lux) or in the blue spectrum. With the precise timing of bright light exposure, we can reset the biologic clock. Exposure to light modifies the set-point of the biologic clock. Using core body temperature as a physiologic marker, one can use bright lights to produce a phase delay when presented before the temperature nadir. By contrast, light exposure after the temperature nadir evokes phase advance. The closer one presents light to the point of inflection (temperature nadir), the more robust is the response in altering the cycle. Thus, we can use early-morning bright-light therapy to phase advance individuals with delayed sleep phase syndrome.

Similarly, exposure to bright light in the evening can help patients with advanced sleep phase syndrome. The blue part of the light spectrum is the crucial ingredient in phase setting and shifting. Therefore, blue light exposure (or limitation with blue-blocking eyewear) at specific times can provide therapeutic benefit. Clinicians are using light therapy to reset the circadian rhythm of shift workers, astronauts, and individuals experiencing jet lag.

Medication. Researchers posit that melatonin secretion acts as the biologic substrate for the internal circadian oscillator. Under normal circumstances,

melatonin levels begin to rise at dusk and remain elevated until dawn. Bright light suppresses the release of melatonin. Melatonin, in a sense, is the signal of darkness in the brain. As such, we can use it clinically to manage sighted patients with disturbed sleep–wake cycles. Melatonin is available over the counter in the United States. A prescription synthetic melatonin agonist (ramelteon) is also available. Ramelteon is FDA approved for treating patients with sleep-onset insomnia but is used off label for the entire spectrum of circadian rhythm sleep disorders. Outside the United States, a sustained-release prescription form of melatonin is available. Of interest, the only medication approved for shift work sleep disorder is the wake-promoting compound modafinil. Modafinil treats sleepiness occurring during night shift work.

Chronotherapy. Chronotherapy is one technique used to reset the biologic clock. It involves progressively phase delaying a person until synchronizing the circadian oscillator with the desired sleep–wake schedule. Most young and middle-aged adults have a propensity to phase delay the circadian sleep–wake rhythm progressively anyhow. Thus, phase delaying each night by 2 to 3 hours is simpler than phase advancing because it capitalizes on a natural tendency. Halting the phase delay at the appropriate moment and maintaining the desired synchrony can be a challenge. The patient also has to cope with an odd sleep–wake schedule for the better part of a week during therapy (which can interfere with school or work). For these reasons, the development of light therapy and pharmacologic interventions has overshadowed chronotherapy.

Parasomnias and Sleep-Related Movement Disorders

Nightmare Disorder. Treatment using behavioral techniques can be helpful. Desensitization and exposure therapy, image rehearsal therapy, lucid dream therapy, and cognitive therapy can help address nightmare disorder. Additionally, evidence for the use of prazosin, a CNS α-1 receptor antagonist, in the treatment of PTSD-related nightmares is growing. Nitrazepam and triazolam may also be helpful.

Recurrent Isolated Sleep Paralysis. Improved sleep hygiene and

assuring sufficient sleep are first-line therapies. Sometimes, if the individual voluntarily makes very–rapid eye movements or is touched by another person, the episode will usually terminate.

Sleep Enuresis. Treating sleep enuresis involves medication, behavioral interventions, or both. Among medications, desmopressin acetate is the preferred choice. Other medications used to treat sleep enuresis include imipramine and oxybutynin chloride. Behavioral treatments, including bladder training, using conditioning devices (bell and pad), diet modification, and fluid restriction, have reported success when properly administered. Other treatments include psychotherapy, motivational strategies, and hypnotherapy.

Restless Legs Syndrome. Pharmacologically, the dopaminergic agonists pramipexole, rotigotine, and ropinirole are FDA approved and represent the treatment of choice. Other agents used to treat RLS include dopamine precursors (e.g., levodopa), benzodiazepines, opiates (including prolonged-release oxycodone–naloxone), and antiepileptic drugs (e.g., gabapentin, pregabalin). When iron deficiency is present, or when ferritin levels are low, iron supplementation can also be helpful. Nonpharmacologic treatment includes avoiding alcohol or nicotine use close to bedtime, massaging the affected parts of the legs, taking hot baths, applying hot or cold to the affected areas, magnetic or electric stimulation therapy, and moderate exercise.

Periodic Limb Movement Disorder Treatment. Pharmacotherapy for PLMD associated with RLS is the same as for RLS. Clinical trials of pharmacotherapy for other forms of PLMD are lacking. However, benzodiazepines, especially clonazepam, reportedly improve sleep in patients with PLMD.

Sleep-Related Bruxism. The usual treatment involves having the patient wear an oral appliance to protect the teeth during sleep. There are two basic types of appliances: the soft one (mouth guard) is typically used in the short term, whereas the hard acrylic one (bite splint) is used long term and requires regular follow-up. Relaxation, biofeedback, hypnosis, physical therapy, and

stress management can also treat sleep bruxism. A variety of drug therapies have been tried (benzodiazepines, muscle relaxants, dopaminergic agonists, and propranolol); however, reliable outcome data are not available. Botulinum toxin can also improve the frequency of bruxism episodes.

Benign Sleep Myoclonus of Infancy. This benign condition typically lasts a few days to several months and does not need to be treated.

Propriospinal Myoclonus at Sleep Onset. Treatment with clonazepam or anticonvulsants may be useful.

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16 Human Sexuality and Sexual Dysfunctions

Sexuality is a broad topic, and no brief synopsis can do it justice. For the psychiatrist, it loosely divides into the subjects of normal sexuality, sexual dysfunction, paraphilias, and gender dysphoria. As with other chapters, we are focusing on the disorders that might bring a patient to a psychiatrist. There are two broad categories of disorders that we loosely group under sexual disorders: sexual dysfunction and paraphilias. By sexual dysfunctions, we are referring to disorders of normal sexual functioning. The paraphilias are unusual expressions of sexuality that are considered aberrant or even illegal. As these represent fundamentally different problems that require very different approaches, we separate these into two sections. We then discuss normal sexuality as part of our consideration of the pathology and etiology of these disorders. We will discuss gender dysphoria in the next section.

For a fuller discussion, we would refer the reader to Virginia Sadock's excellent review of the subject in the *Comprehensive Textbook.*

SEXUAL DYSFUNCTIONS

The essential features of sexual dysfunctions are an inability to respond to sexual stimulation, or the experience of pain during the sexual act. Dysfunction can be defined by a disturbance in the subjective sense of pleasure or desire usually associated with sex, or by the objective performance. According to ICD-10, sexual dysfunction refers to a person's inability "to participate in a sexual relationship as he or she would wish."

In DSM-5, the sexual dysfunctions include male hypoactive sexual desire disorder, female sexual interest/arousal disorder, erectile disorder, female

orgasmic disorder, delayed ejaculation, premature (early) ejaculation, genito-pelvic pain/penetration disorder, substance/medication-induced sexual dysfunction, other specified sexual dysfunction, and unspecified sexual dysfunction. We diagnose sexual dysfunctions only when they are a significant part of the clinical picture. If more than one dysfunction exists, we should diagnose them all. Sexual dysfunctions can be lifelong or acquired, generalized or situational, and result from psychological factors, physiologic factors, combined factors, and numerous stressors, including prohibitive cultural mores, health and partner issues, and relationship conflicts. If the dysfunction is attributable entirely to a general medical condition, substance use, or adverse effects of medication, then sexual dysfunction due to a general medical condition or substance-induced sexual dysfunction is diagnosed. DSM-5 indicates the severity of the dysfunction by whether the patient's distress is mild, moderate, or severe.

Sexual dysfunctions are frequently associated with other mental disorders, such as depressive disorders, anxiety disorders, personality disorders, and schizophrenia. In many instances, sexual dysfunction is comorbid with another psychiatric disorder. If the dysfunction is mostly attributable to an underlying psychiatric disorder, we should only diagnose the underlying disorder. Sexual dysfunctions are usually self-perpetuating, with the patients increasingly subjected to ongoing performance anxiety and a concomitant inability to experience pleasure. In relationships, the sexually functional partner often reacts with distress or anger due to feelings of deprivation or a sense that he or she is an insufficiently attractive or adequate sexual partner. In such cases, the clinician must consider whether the sexual problem preceded or arose from relationship difficulties and weigh whether a diagnosis of sexual dysfunction relevant to relationship issues is appropriate.

A sex history provides essential information about patients, regardless of the presence of a sexual disorder or whether that is the patient's chief complaint. The information can be obtained gradually through open-ended questions. The outline in Table 16-1 provides a guide to the topics to be covered.

Desire, Interest, and Arousal Disorders

Male Hypoactive Sexual Desire Disorder. Male hypoactive sexual

desire disorder is a deficiency or absence of sexual fantasies and desire for sexual activity for a minimum duration of approximately 6 months (Table 16-2). Men for whom this is a lifelong condition have never experienced many spontaneous erotic/sexual thoughts. Minimal spontaneous sexual thinking or minimal desire for sex ahead of sexual experiences is not a diagnosable disorder in women, primarily if the sexual encounter then triggers the desire. The reported prevalence of low desire is highest at the younger and older ends of the age spectrum, with only 2 percent of men ages 16 to 44 affected by this disorder. A reported 6 percent of men ages 18 to 24, and 40 percent of men ages 66 to 74, have problems with sexual desire. Some men may confuse decreased desire with decreased activity. Their erotic thoughts and fantasies are undiminished, but they no longer act on them due to health issues, unavailability of a partner, or another sexual dysfunction such as erectile disorder.

A variety of causative factors are associated with low sexual desire. Patients with desire problems often use inhibition of desire defensively, to protect against unconscious fears about sex.

Abstinence from sex for a prolonged period sometimes results in suppression of sexual impulses. Loss of desire may also be an expression of hostility to a partner or the sign of a deteriorating relationship. The presence of desire depends on several factors: biologic drive, adequate self-esteem, the ability to accept oneself as a sexual person, the availability of an appropriate partner, and a good relationship in nonsexual areas with a partner. Damage to, or absence of, any of these factors can diminish desire.

In making the diagnosis, clinicians must evaluate a patient’s age, general health, any medication regimen, and life stresses. The clinician must attempt to establish a baseline of sexual interest before the disorder began. The need for sexual contact and satisfaction varies among persons. We should not make a diagnosis unless the lack of desire is a source of distress to a patient.



Table 16-1
Taking a Sex History

I. Identifying data
A. Age
B. Sex

C. Occupation
D. Relationship status—single, married, number of times previously married, separated, divorced, cohabiting, serious involvement, casual dating (difficulty forming or keeping relationships should be assessed throughout the interview)
E. Sexual orientation—heterosexual, homosexual, or bisexual (this may also be ascertained later in the interview)

II. Current functioning
A. Unsatisfactory to highly satisfactory
B. If unsatisfactory, why?
C. Feeling about partner satisfaction
D. Dysfunctions?—e.g., lack of desire, erectile disorder, inhibited female interest/arousal, anorgasmia, premature ejaculation, retarded ejaculation, pain associated with intercourse (dysfunction discussed below)
1. Onset—lifelong or acquired
a. If acquired, when?
b. Did onset coincide with drug use (medications or illegal recreational drugs), life stresses (e.g., loss of job, birth of child), interpersonal difficulties
2. Generalized—occurs in most situations or with most partners
3. Situational
a. Only with current partner
b. In any committed relationship
c. Only with masturbation
d. In socially proscribed circumstance (e.g., affair)
e. In definable circumstance (e.g., very late at night, in parental home, when partner initiated sex play)
E. Frequency—partnered sex (coital and noncoital sex play)
F. Desire/libido—how often are sexual feelings, thoughts, fantasies, dreams, experienced? (per day, week, etc.)
G. Description of typical sexual interaction
1. Manner of initiation or invitation (e.g., verbal or physical? Does same person always initiate?)

2. Presence, type, and extent of foreplay (e.g., kissing, caressing, manual or oral genital stimulation)
3. Coitus? positions used?
4. Verbalization during sex? if so, what kind?
5. Afterplay? (whether sex act is completed or disrupted by dysfunction); typical activities (e.g., holding, talking, return to daily activities, sleeping)
6. Feeling after sex: relaxed, tense, angry, loving

H. Sexual compulsivity?—intrusion of sexual thoughts or participation in sexual activities to a degree that interferes with relationships or work, requires deception and may endanger the patient

III. Past sexual history

A. Childhood sexuality

1. Parental attitudes about sex—degree of openness or reserve (assess unusual prudery or seductiveness)
2. Parents' attitudes about nudity and modesty
3. Learning about sex
   a. From parents? (initiated by child's questions or parent volunteering information? which parent? what was child's age?) subjects covered (e.g., pregnancy, birth, intercourse, menstruation, nocturnal emission, masturbation)
   b. From books, magazines, or friends at school or through religious group?
   c. Significant misinformation
   d. Feeling about information
4. Viewing or hearing primal scene—reaction?
5. Viewing sex play or intercourse of person other than parent
6. Viewing sex between pets or other animals

B. Childhood sex activities

1. Genital self-stimulation before adolescence; age? reaction if apprehended?
2. Awareness of self as boy or girl; bathroom sensual activities? (regarding urine, feces, odor, enemas)

3. Sexual play or exploration with another child (playing doctor)—type of activity (e.g., looking, manual touching, genital touching); reactions or consequences if apprehended (by whom?)

IV. Adolescence

A. Age of onset of puberty—development of secondary sex characteristics, age of menarche for girl, wet dreams or first ejaculation for boy (preparation for and reaction to)

B. Sense of self as feminine or masculine—body image, acceptance by peers (opposite sex and same sex), sense of sexual desirability, onset of coital fantasies

C. Sex activities

1. Masturbation—age begun; ever punished or prohibited? method used, accompanying fantasies, frequency (questions about masturbation and fantasies are among the most sensitive for patients to answer)
2. Homosexual activities—ongoing or rare and experimental episodes, approached by others? If homosexual, has there been any heterosexual experimentation?
3. Dating—casual or steady, description of first crush, infatuation, or first love
4. Experiences of kissing, necking, petting (“making out” or “fooling around”), age begun, frequency, number of partners, circumstances, type(s) of activity
5. Orgasm—when first experienced? (may not be experienced during adolescence), with masturbation, during sleep, or with partner? with intercourse or other sex play? frequency?
6. First coitus—age, circumstances, partner, reactions (may not be experienced during adolescence); contraception and/or safe sex precautions used

V. Adult sexual activities (may be experienced by some adolescents)

A. Premarital sex

1. Types of sex play experiences—frequency of sexual interactions, types and number of partners

2. Contraception and/or safe sex precautions used
3. First coitus (if not experienced in adolescence) age, circumstances, partner
4. Cohabitation—age begun, duration, description of partner, sexual fidelity, types of sexual activity, frequency, satisfaction, number of cohabiting relationships, reasons for breakup(s)
5. Engagement—age, activity during engagement period with fiancé(e), with others; length of engagement

B. Marriage (if multiple marriages have occurred, explore sexual activity, reasons for marriage, and reasons for divorce in each marriage)
1. Types and frequency of sexual interaction—describe typical sexual interaction (see above), satisfaction with sex life? view of partner's feeling
2. First sexual experience with spouse—when? what were the circumstances? was it satisfying? disappointing?
3. Honeymoon—setting, duration, pleasant or unpleasant, sexually active, frequency? problems? compatibility?
4. Effect of pregnancies and children on marital sex
5. Extramarital sex—number of incidents, partner; emotional attachment to extramarital partners? feelings about extramarital sex
6. Postmarital masturbation—frequency? effect on marital sex?
7. Extramarital sex by partner—effect on interviewee
8. Ménage à trois or multiple sex (swinging)
9. Areas of conflict in marriage (e.g., parenting, finances, division of responsibilities, priorities)

VI. Sex after widowhood, separation, divorce—celibacy, orgasms in sleep, masturbation, noncoital sex play, intercourse (number of and relationship to partners), other

VII. Special issues
A. History of rape, incest, sexual or physical abuse
B. Spousal abuse (current)

C. Chronic illness (physical or psychiatric)
D. History or presence of sexually transmitted diseases
E. Fertility problems
F. Abortions, miscarriages, or unwanted or illegitimate pregnancies
G. Gender identity conflict—(e.g., transsexualism, wearing clothes of opposite sex)
H. Paraphilias—(e.g., fetishes, voyeurism, sadomasochism)



Table 16-2
Male Hypoactive Sexual Desire Disorder

	DSM-5	ICD-10
Name	Male Hypoactive Sexual Desire Disorder	Lack or loss of sexual desire Sexual aversion and lack of sexual enjoyment
Duration	≥6 mo	
Symptoms	Decreased thoughts or interest in sex	Lack or loss of sexual desire not attributable to physiologic difficulties related to intercourse Sexual aversion due to anxiety and subsequent avoidance of sexual activity, or lack of enjoyment given that orgasm is not accompanied by appropriate pleasure
# Symptoms Needed	Symptoms should be recurrent	
Exclusions (not	Sociocultural	

result of):	influences Age Relationship problem Substance use (including prescribed meds) Another medical condition Another psychiatric condition	
Psychosocial Impact	Marked distress	
Symptom Specifiers	Generalized vs. Situational	
Severity Specifiers	**Mild:** causes mild distress **Moderate:** causes moderate distress **Severe:** causes severe distress	
Course Specifiers	Lifelong vs. Acquired	Note: ICD-10 classifies this diagnosis as including male hypoactive sexual desire disorder as well as female sexual arousal disorder (Table 16-3)

Table 16-3
Female Sexual Interest/Arousal Disorder

	DSM-5	ICD-10
Name	Female Sexual Interest/Arousal Disorder	Lack or loss of sexual desire Sexual aversion and lack of sexual enjoyment
Duration	≥6 mo	
Symptoms	Decreased arousal or interest in sex: Reduced interest in sex Reduced sexual thoughts/fantasies Reduced initiation/receptiveness to sexual activity Absent or reduced sexual pleasure Absent or reduced pleasure from sexual cues Absent or reduced sensation during sex	Lack or loss of sexual desire not attributable to physiologic difficulties related to intercourse Sexual aversion due to anxiety and subsequent avoidance of sexual activity, or lack of enjoyment given that orgasm is not accompanied by appropriate pleasure
# Symptoms Needed	≥3 during most sexual experiences	
Exclusions (not result of):	Relationship problem Substance use (including prescribed meds) Another medical condition Another psychiatric	

	condition
Psychosocial Impact	Marked distress
Symptom Specifiers	Generalized vs. Situational
Severity Specifiers	**Mild:** causes mild distress **Moderate:** causes moderate distress **Severe:** causes severe distress
Course Specifiers	Lifelong vs. Acquired

Female Sexual Interest/Arousal Disorder. The combination of interest (or desire) and arousal into one dysfunction category reflects the recognition that women do not necessarily move stepwise from desire to arousal, but often experience desire synchronously with, or even following, beginning feelings of arousal. This is particularly true for women in long-term relationships. As a corollary, women experiencing sexual dysfunction may experience either inability to feel interest or arousal, and they may also have difficulty achieving orgasm or experience pain. Some may experience dysfunction across the entire range of sexual response. Complaints in this dysfunction category present variously as a decrease or paucity of erotic feelings, thoughts, or fantasies; a decreased impulse to initiate sex; a decreased or absent receptivity to partner overtures; or an inability to respond to partner stimulation (Table 16-3).

A complicating factor in this diagnosis is that a subjective sense of arousal is often poorly correlated with genital lubrication in both normal and dysfunctional women. Therefore, complaints of lack of pleasure are sufficient for this diagnosis even when vaginal lubrication and congestion are present. A woman complaining of lack of arousal may lubricate vaginally, but may not experience a subjective sense of excitement. Alterations in testosterone, estrogen, prolactin, and thyroxin levels occur in female sexual arousal

disorder. Also, medications with antihistaminic or anticholinergic properties cause a decrease in vaginal lubrication.

We should evaluate factors such as life stresses, aging, menopause, adequate sexual stimulation, and general health before making this diagnosis. Relationship problems are particularly relevant to acquired interest/arousal disorder. In one study of couples with markedly decreased sexual interaction, the most prevalent etiology was marital discord.

Male Erectile Disorder. Male erectile disorder was historically called *impotence.* The term was dropped for a more medical designation, but also because it was considered derogatory and had negative connotations for the man with the problem. However, it describes with accuracy the feelings of powerlessness, helplessness, and resultant low self-esteem men with this dysfunction frequently suffer (Table 16-4). A man with a lifelong male erectile disorder has never been able to obtain an erection sufficient for insertion. In acquired male erectile disorder, a man has successfully achieved penetration at some time in his sexual life but is later unable to do so. In situational male erectile disorder, a man can have coitus in certain circumstances but not in others; for example, he may function effectively with a sex worker but be unable to have an erection when with his partner.

Acquired male erectile disorder occurs in 10 to 20 percent of all men. Erectile disorder is the chief complaint of more than 50 percent of all men treated for sexual disorders. Lifelong male erectile disorder is rare; it occurs in about 1 percent of men younger than age 35. The incidence of erectile disorder increases with age. It has been reported variously as 2 to 8 percent of the young adult population, and much more common in older men. Male erectile disorder, however, is not universal in aging men; having an available sex partner is related to continuing potency, as is a history of consistent sexual activity and the absence of vascular, neurologic, or endocrine disease. Twenty percent of men fear erectile dysfunction before their first coitus; the reported incidence of actual erectile dysfunction during the first coitus is 8 percent.

Male erectile disorder can be organic or psychological or a combination of both, but in young and middle-aged men, the cause is usually psychological. A good history is of primary importance in determining the cause of the dysfunction. If a man reports having spontaneous erections at times when he

does not plan to have intercourse, having morning erections, or having good erections with masturbation or with partners other than his usual one, the physiologic causes of his erectile disorder can be considered negligible, and he can avoid costly diagnostic procedures. Male erectile disorder caused by a general medical condition or a pharmacologic substance is discussed later in this section.



Table 16-4
Erectile Disorder or Failure of Genital Response

	DSM-5	ICD-10
Name	Erectile Disorder	Failure of genital response
Duration	≥6 mo	
Symptoms	Difficulties with erections: Achieving Maintaining Rigidity	**Male** Difficulty with erections: developing maintaining **Female** Difficulty with: Vaginal dryness Lack of adequate lubrication
# Symptoms Needed	≥1 during most sexual encounters	
Exclusions (not result of):	Relationship problem Substance use (including prescribed meds) Another medical	

	condition Another psychiatric condition	
Psychosocial Impact	Marked distress	
Symptom Specifiers	Generalized vs. Situational	
Severity Specifiers	**Mild:** causes mild distress **Moderate:** causes moderate distress **Severe:** causes severe distress	
Course Specifiers	Lifelong vs. Acquired	
Comments		Note: ICD-10 classifies this diagnosis as including male erectile disorder as well as female sexual arousal disorder

A man may be unable to express a sexual impulse because of fear, anxiety, anger, or moral prohibition. In an ongoing relationship, the disorder may reflect difficulties between the partners, mainly when a man cannot communicate his needs or his anger directly and constructively. Also, episodes of erectile disorder are reinforcing, with the man becoming increasingly anxious before each sexual encounter.

Mr. Y came for therapy after his wife complained about their lack of sexual interaction. The patient avoided sex because of his frequent erectile dysfunction and the painful feelings of inadequacy he suffered after his "failures." He presented as an articulate, gentle, and self-blaming man.

He was faithful to his wife but masturbated frequently. His fantasies involved explicit sadistic components, including hanging and biting women. The contrast between his angry, aggressive fantasies and his loving, considerate behavior toward his wife symbolized his conflicts about his sexuality, his masculinity, and his mixed feelings about women. The physician diagnosed him with erectile disorder, situational type.

Orgasm Disorders

Female Orgasmic Disorder. Female orgasmic disorder, sometimes called *inhibited female orgasm* or *anorgasmia*, is defined as the recurrent or persistent inhibition of female orgasm, as manifested by the recurrent delay in, or absence of, orgasm after a normal sexual excitement phase that a clinician judges to be adequate in focus, intensity, and duration—in short, a woman's inability to achieve orgasm by masturbation or coitus (Table 16-5). Women who can achieve orgasm by one of these methods are not necessarily categorized as anorgasmic, although there may be some sexual inhibition. Some anorgasmic women are not distressed by the lack of climax and derive pleasure from sexual activity. However, the woman may still present for treatment as her partner is distressed by her apparent disinterest.

Research on the physiology of the female sexual response has shown that orgasms caused by clitoral stimulation and those caused by vaginal stimulation are physiologically identical. Many women achieve orgasm during coitus by a combination of manual clitoral stimulation and penile vaginal stimulation.

A woman with a lifelong female orgasmic disorder has never experienced orgasm by any kind of stimulation. A woman with an acquired orgasmic disorder has previously experienced at least one orgasm, regardless of the circumstances or means of stimulation, whether by masturbation or while

dreaming during sleep. Studies have shown that women achieve orgasm more consistently with masturbation than with partnered sex. Lifelong female orgasmic disorder is more common among unmarried women than married women. Increased orgasmic potential in women older than 35 years of age may relate to less psychologic inhibition, greater sexual experience, or both.

Acquired female orgasmic disorder is a common complaint in clinical populations. One clinical treatment facility reported having about four times as many nonorgasmic women in its practice as female patients with all other sexual disorders. In another study, 46 percent of women complained of difficulty reaching orgasm. Inhibition of arousal and orgasmic problems often occur together. The overall prevalence of female orgasmic disorder from all causes is estimated to be 30 percent. Orgasmic dysfunction in some females may have a genetic basis and cannot be attributed solely to psychologic differences. That study demonstrated an estimated heritability for difficulty reaching orgasm with intercourse of 34 percent and an estimated heritability in women who could not climax with masturbation of 45 percent.



Table 16-5
Female Orgasmic Disorder

	DSM-5	ICD-10
Name	Female Orgasmic Disorder	Orgasmic dysfunction
Duration	≥6 mo	
Symptoms	Difficulties with orgasms Delayed/reduced frequency Reduced intensity	Marked delay or absence of orgasm
# Symptoms Needed	≥1 during most sexual encounters	
Exclusions (not result of):	Relationship problem Substance use	Another medical condition

	(including prescribed meds) Another medical condition Another psychiatric condition	
Psychosocial Impact	Marked distress	
Symptom Specifiers	Generalized vs. Situational	
Severity Specifiers	**Mild:** causes mild distress **Moderate:** causes moderate distress **Severe:** causes severe distress	
Course Specifiers	Lifelong vs. Acquired Never experienced an orgasm under any conditions	
Comments		Note: ICD 10 classifies orgasmic dysfunction as occurring in men and women, including delayed ejaculation in this category as per Table 16-6

Numerous psychological factors are associated with female orgasmic disorder. They include fears of impregnation, rejection by a sex partner, and damage to the vagina; hostility toward men; poor body image; and feelings of guilt about sexual impulses. Some women equate orgasm with loss of control or with aggressive, destructive, or violent impulses; they may express their

fear of these impulses through inhibition of arousal or orgasm. Cultural expectations and social restrictions on women are also relevant. Despite the sexual revolution of the 1960s, many women in the United States still grow to believe that sexual pleasure is not a natural entitlement for so-called decent women. Nonorgasmic women may be otherwise symptom-free or may experience frustration in a variety of ways; they may have such pelvic complaints as lower abdominal pain, itching, and vaginal discharge, as well as increased tension, irritability, and fatigue.

Delayed Ejaculation. In male delayed ejaculation, sometimes called *retarded ejaculation*, a man achieves ejaculation during coitus with great difficulty, if at all (Table 16-6). The problem is rarely present with masturbation but appears as a problem during partnered sex. A man with lifelong delayed ejaculation has never been able to ejaculate during partnered sexual activity. The problem is usually most pronounced during coital activity. We would consider the disorder to be acquired if it develops after previously normal functioning. Some researchers think that orgasm and ejaculation should be differentiated, especially in the case of men who ejaculate but complain of a decreased or absent subjective sense of pleasure during the orgasmic experience (orgasmic anhedonia).

The incidence of male orgasmic disorder is much lower than the incidence of premature ejaculation or erectile disorder. The prevalence is about 5 percent. However, there may be an increase in this disorder presenting to sex therapy programs. An increase in antidepressant use may be one explanation for this. Another is the high use of Internet pornography. These sites offer a level of stimulation involving such a variety of people and acts that they may inure the man to the stimulation of more typical partnered activity. Some studies of adolescent males who use these sites frequently, before live sexual interaction, have reported that these teens do not develop neuronal synapses that will enable them to respond to usual partnered interactions with sufficient pleasure to allow them to achieve climax.

Lifelong delayed ejaculation indicates severe psychopathology. A man may come from a rigid, puritanical background; he may perceive sex as sinful and the genitals as dirty, and he may have conscious or unconscious incest wishes and guilt. He usually has difficulty with closeness in areas beyond those of sexual relations. In a few cases, an attention-deficit/hyperactivity

disorder may aggravate the condition. A man's distractibility prevents sufficient arousal for climax to occur.

In an ongoing relationship, acquired male delayed ejaculation disorder frequently reflects interpersonal difficulties. The disorder may be a man's way of coping with real or fantasized changes in a relationship, such as plans for pregnancy about which the man is ambivalent, the loss of sexual attraction to the partner, or demands by the partner for greater commitment as expressed by sexual performance. In some men, the inability to ejaculate reflects unexpressed hostility toward a woman. The problem is more common among men with obsessive-compulsive disorder (OCD) than among others.



Table 16-6
Delayed Ejaculation

	DSM-5	ICD-10
Name	Delayed Ejaculation	Orgasmic dysfunction
Duration	≥6 mo	
Symptoms	Difficulty with ejaculations: Delayed Absent or reduced frequency	Marked delay or absence of orgasm
# Symptoms Needed	≥1	
Exclusions (not result of):	Relationship problem Substance use (including prescribed meds) Another	Another medical condition

	medical condition Another psychiatric condition	
Psychosocial Impact	Marked distress	
Symptom Specifiers	Generalized vs. Situational	
Severity Specifiers	**Mild:** causes mild distress **Moderate:** causes moderate distress **Severe:** causes severe distress	
Course Specifiers	Lifelong vs. Acquired	
Comments		Note: ICD 10 classifies orgasmic dysfunction as occurring in men and women, including delayed ejaculation in this category as per Table 16-5

A couple presented with the man as the identified patient; he was unable to ejaculate with intercourse. He had always had difficulty reaching climax, except in rare circumstances. He ejaculated once when he was with two women at the same time and once when he was experimenting with cocaine. He currently was not using any substances except for

moderate use of alcohol. This patient was committed to his marriage, although he had extramarital sexual experiences. He did not ejaculate with coitus in those situations either, although he could climax with oral sex. He stated he was more interested in "the conquest" than in the sex itself. He could climax with masturbation, although he rarely masturbated himself, but went to massage parlors. He had issues with anger at women and considered his wife to be excessively critical.

He had difficulty doing any of the exercises that required him to pleasure his wife. His difficulty giving also made it hard for him to enjoy mutual pleasuring. It was easier for him to be the recipient of stimulation. Because of this patient's problems with impulsiveness, narcissism, and dependency, it was necessary to combine introspective psychotherapy with a regimen of behavioral exercises.

The psychiatrist diagnosed the patient with delayed ejaculation, lifelong type.

Premature (Early) Ejaculation. In premature ejaculation, men persistently or recurrently achieve orgasm and ejaculation before they wish to. The diagnosis occurs when a man regularly ejaculates before or within approximately 1 minute after penetration. DSM-5 refers only to "vaginal penetration" in its diagnostic criteria, even though the disorder can occur in gay men or those who do not engage in vaginal penetration for other reasons. DSM-5 defines the disorder as mild if ejaculation occurs within approximately 30 seconds to 1 minute of vaginal penetration, moderate if ejaculation occurs within approximately 15 to 30 seconds of vaginal penetration, and severe when ejaculation occurs at the start of sexual activity or within approximately 15 seconds of vaginal penetration. A difficulty with these specifiers involves time distortions, which patients make in both overestimating and underestimating time from penetration to climax. Clinicians need to consider factors that affect the duration of the excitement phase of the sexual response, such as age, the novelty of the sex partner, and the frequency of coitus (Table 16-7). As with the other sexual dysfunctions, we would not diagnose premature ejaculation when it is caused exclusively by organic factors or when it is symptomatic of another clinical psychiatric syndrome.

Premature ejaculation appears to be more common among college-educated men than among men with less education. The complaint is likely related to their concern for partner satisfaction, although data is limited, and we do not know for sure. Premature ejaculation is the chief complaint of about 35 to 40 percent of men treated for sexual disorders. In DSM-5, the writers state that the disorder, with its newly defined time parameter, would now be an accurate diagnosis for only 1 to 3 percent of men. Some researchers divide men who experience premature ejaculation into two groups: those who are physiologically predisposed to climax quickly because of shorter nerve latency time and those with a psychogenic or behaviorally conditioned cause. Difficulty in ejaculatory control can be associated with anxiety regarding the sex act, with unconscious fears about the vagina, or with negative cultural conditioning. Men whose early sexual contacts occurred mainly with sex workers who demanded that the sex act proceed quickly or whose sexual contacts took place in situations in which discovery would be embarrassing (e.g., in a shared dormitory room or the parental home) may be conditioned to orgasm rapidly. With young, inexperienced men, who have the problem, it may resolve in time. In ongoing relationships, the partner has a significant influence on a premature ejaculator, and a stressful marriage exacerbates the disorder. The developmental background and the psychodynamics found in premature ejaculation and erectile disorder are similar.



Table 16-7
Premature (Early) Ejaculation

	DSM-5	ICD-10
Name	Premature (Early) Ejaculation	Premature ejaculation
Duration	≥6 mo	
Symptoms	Ejaculation occurs Within 1 min of penetration Before it was	Difficulty controlling ejaculation

	intended	
# Symptoms Needed	Above occurs in most sexual encounters	
Exclusions (not result of):	Relationship problem Substance use (including prescribed meds) Another medical condition Another psychiatric condition	
Psychosocial Impact	Marked distress	Both partners do not sufficiently enjoy the sexual encounter
Symptom Specifiers	Generalized vs. Situational	
Severity Specifiers	Defined by time from penetration until ejaculation: **Mild:** 30 s–1 min **Moderate:** 15–30 s **Severe:** <15 s	
Course Specifiers	Lifelong vs. Acquired	

The focus of this diagnosis is on males, and data on premature female orgasm are lacking. There are some case reports of, for example, multiple spontaneous orgasms without sexual stimulation. In some cases, an epileptogenic focus may be the cause. There are also some rare case reports of spontaneous orgasms in women taking serotonergic antidepressants.

Orgasmic Anhedonia. Orgasmic anhedonia is a condition in which a

person has no physical sensation of orgasm, even though the physiologic component (e.g., ejaculation) remains intact. Organic causes, such as sacral and cephalic lesions that interfere with afferent pathways from the genitalia to the cortex, must be ruled out. Psychiatric causes usually relate to extreme guilt about experiencing sexual pleasure. These feelings produce a dissociative response that isolates the affective component of the orgasmic experience from consciousness. In DSM-5 this would be diagnosed as an "other specified sexual dysfunction."

Sexual Pain Disorders

Genito-Pelvic Pain/Penetration Disorder. In DSM-5, this disorder refers to one or more of the following complaints, of which any two or more may occur together: difficulty having intercourse, genito-pelvic pain, fear of pain or penetration, and tension of the pelvic floor muscles (Table 16-8). Previously, we diagnosed these pain disorders as *dyspareunia* or *vaginismus.* These former diagnoses could coexist, or one could lead to the other and could understandably lead to fear of pain with sex. Thus, it is reasonable to gather these diagnoses into one diagnostic category. For this clinical discussion, however, the distinct categories of dyspareunia and vaginismus remain clinically useful.

DYSPAREUNIA. Dyspareunia is recurrent or persistent genital pain occurring before, during, or after intercourse. Dyspareunia is related to and often coincides with vaginismus. Repeated episodes of vaginismus can lead to dyspareunia and vice versa; in either case, we should rule out somatic causes, or lack of lubrication. Perhaps 5 percent of women in North America report recurrent pain during intercourse.

In most cases, dynamic factors are considered causative. Chronic pelvic pain is a common complaint in women with a history of rape or childhood sexual abuse. Painful coitus can result from tension and anxiety about the sex act that causes women to contract their pelvic floor muscles involuntarily. The pain is real and makes intercourse unpleasant or unbearable. The anticipation of further pain may cause women to avoid coitus altogether. If a partner proceeds with intercourse regardless of a woman's state of readiness, this aggravates the condition. There is an increase in reported dyspareunia

postmenopausally due to hormonally induced physiologic changes in the vagina; however, specific complaints of difficulty having intercourse occur more often in premenopausal women. There is some increase in dyspareunia in the immediate postpartum population, but it is usually temporary. Dyspareunia may present as any of the four complaints listed under genito-pelvic pain/penetration disorder, and we would diagnose this as a genito-pelvic pain/penetration disorder.

VAGINISMUS. Vaginismus is a constriction of the outer third of the vagina due to involuntary pelvic floor muscle tightening or spasm. It then interferes with penile insertion and intercourse. This response may occur during a gynecologic examination when involuntary vaginal constriction prevents the introduction of the speculum into the vagina. We would not make the diagnosis when the dysfunction is caused exclusively by organic factors or when it is symptomatic of another mental disorder.

Vaginismus may be complete; that is, no penetration of the vagina is possible, whether by the penis, fingers, a speculum during a gynecologic examination, or even if the woman tries to use the smallest size tampon. Many women who discover this when they become sexually active have avoided the use of tampons previously. In a less severe form of vaginismus, pelvic floor muscle tightening due to pain or fear of pain makes penetration difficult, but not impossible. Penetration may be achieved with the smallest size speculum or little fingers. In mild cases, the muscles relax after the initial difficulty with penetration, and the woman can continue with sexual play, sometimes even with coitus.

Miss B was a 27-year-old single woman who presented for therapy because of an inability to have intercourse. She described episodes with a recent boyfriend in which he had tried vaginal penetration but had been unable to enter. The boyfriend did not have erectile dysfunction. Miss B experienced desire and was able to achieve orgasm through manual or oral stimulation. For almost a year, she and her boyfriend had sex play without intercourse. However, he complained increasingly about his frustration at the lack of coitus, which he had enjoyed in previous relationships. Miss B had a conscious fear of penetration and dreaded

going to the gynecologist, although she was able to use tampons when she menstruated. Her therapist diagnosed her with genito-pelvic pain/penetration disorder, lifelong type.



Table 16-8
Genito-Pelvic Pain/Penetration Disorder

	DSM-5	ICD-10
Name	**Genito-Pelvic Pain/Penetration Disorder**	**Nonorganic vaginismus** **Nonorganic dyspareunia**
Duration	≥6 mo	
Symptoms	During vaginal penetration, the woman experiences Difficulties with penetration Pain Associated symptoms during or in anticipation of penetration: Fear or anxiety Tension or tightening of pelvic muscles	*Vaginismus*: spasm of pelvic floor muscles around the vagina, causing pain and difficulties with penetration Dyspareunia: pain or discomfort related to penile penetration or intercourse
# Symptoms Needed	One or more of the above	
Exclusions (not result of):	Relationship problem Substance use	Another medical condition

	(including prescribed meds) Another medical condition Another psychiatric condition
Psychosocial Impact	Marked distress
Symptom Specifiers	
Severity Specifiers	**Mild:** causes mild distress **Moderate:** causes moderate distress **Severe:** causes severe distress
Course Specifiers	Lifelong vs. Acquired

Vaginismus is less prevalent than female orgasmic disorder. It most often afflicts highly educated women and those in high socioeconomic groups. Women with vaginismus may consciously wish to have coitus, but unconsciously wish to keep a penis from entering their bodies. Sexual trauma, such as rape, may cause vaginismus. The anticipation of pain at the first coital experience may cause vaginismus. Clinicians have noted that a strict religious upbringing in which the patient associates sex with sin is frequent in these patients. Other women have problems in dyadic relationships; if women feel emotionally abused by their partners, they may protest in this nonverbal fashion. Some women who have experienced significant pain in childhood due to surgical or dental interventions become guarded about any breach of body integrity and develop vaginismus. Vaginismus may present as any of the four complaints under genito-pelvic

pain/penetration disorder, and we should diagnose this as genito-pelvic pain/penetration disorder.

Postcoital Headache. Postcoital headache, characterized by headache immediately after coitus, may last for several hours. It is usually described as throbbing and localized to the occipital or frontal area. The cause is unknown. There may be vascular, muscle contraction (tension), or psychogenic causes. Coitus may precipitate migraine or cluster headaches in predisposed persons. In DSM-5 this disorder would not be included among the above pain disorders as the pain is not directly associated with the sexual act. When it is a source of distress or dysfunction, we list it as an "other specified sexual dysfunction."

Sexual Dysfunction due to a General Medical Condition

Male Erectile Disorder due to a General Medical Condition. The incidence of psychological male erectile disorder has been the focus of many studies. Statistics indicate that 20 to 50 percent of men with erectile disorder have an organic basis for the disorder. A physiologic etiology is more likely in men older than 50 and the most likely cause in men older than age 60. Table 16-9 lists the medical causes of male erectile disorder. Side effects of medication can impair male sexual functioning in a variety of ways (Table 16-10). Castration (removal of the testes) does not always lead to sexual dysfunction because erection may still occur. A reflex arc, fired when stimulating the inner thigh, passes through the sacral cord erectile center to account for the phenomenon.

Several procedures, benign and invasive, are used to help differentiate the causes of erectile disorder. The procedures include monitoring nocturnal penile tumescence (erections that occur during sleep), generally associated with rapid eye movement; monitoring tumescence with a strain gauge; measuring blood pressure in the penis with a penile plethysmograph or an ultrasound (Doppler) flowmeter, both of which assess blood flow in the internal pudendal artery; and measuring pudendal nerve latency time. Other diagnostic tests that delineate physiologic bases for impotence include glucose tolerance tests, plasma hormone assays, liver and thyroid function tests, prolactin and follicle-stimulating hormone (FSH) determinations, and

cystometric examinations. Invasive diagnostic studies include penile arteriography, infusion cavernosonography, and radioactive xenon penography. Invasive procedures require expert interpretation and are used only for patients who are candidates for vascular reconstructive procedures.

Dyspareunia due to a General Medical Condition. An estimated 30 percent of all surgical procedures on the female genital area result in temporary dyspareunia. Also, 30 to 40 percent of women with the complaint seen in sex therapy clinics have pelvic pathology. Organic abnormalities leading to dyspareunia and vaginismus include irritated or infected hymenal remnants, episiotomy scars, Bartholin gland infection, various forms of vaginitis and cervicitis, endometriosis, and adenomyosis. Women with myomata, endometriosis, and adenomyosis may report postcoital pain owing to the uterine contractions during orgasm. Postmenopausal women may have dyspareunia resulting from thinning of the vaginal mucosa and reduced lubrication.



Table 16-9
Diseases and Other Medical Conditions Implicated in Male Erectile Disorder

Infectious and parasitic diseases	Neurologic disorders
Elephantiasis	Multiple sclerosis
Mumps	Transverse myelitis
Cardiovascular disease[a]	Parkinson disease
Atherosclerotic disease	Temporal lobe epilepsy
Aortic aneurysm	Traumatic and neoplastic spinal cord diseases[a]
Leriche syndrome	Central nervous system tumor
Cardiac failure	Amyotrophic lateral sclerosis
Renal and urologic disorders	Peripheral neuropathy
Peyronie disease	General paresis
	Tabes dorsalis
	Pharmacologic factors
	Alcohol and other dependence-inducing substances (heroin, methadone, morphine,

- Chronic renal failure
- Hydrocele and varicocele

Hepatic disorders
- Cirrhosis (usually associated with alcohol dependence)

Pulmonary disorders
- Respiratory failure

Genetics
- Klinefelter syndrome
- Congenital penile vascular and structural abnormalities

Nutritional disorders
- Malnutrition
- Vitamin deficiencies
- Obesity

Endocrine disorders[a]
- Diabetes mellitus
- Dysfunction of the pituitary–adrenal–testis axis
- Acromegaly
- Addison disease
- Chromophobe adenoma

cocaine, amphetamines, and barbiturates)
- Prescribed drugs (psychotropic drugs, antihypertensive drugs, estrogens, and antiandrogens)

Poisoning
- Lead (plumbism)
- Herbicides

Surgical procedures[a]
- Perineal prostatectomy
- Abdominal–perineal colon resection
- Sympathectomy (frequently interferes with ejaculation)
- Aortoiliac surgery
- Radical cystectomy
- Retroperitoneal lymphadenectomy

Miscellaneous
- Radiation therapy
- Pelvic fracture
- Any severe systemic disease or debilitating condition

Adrenal neoplasia
Myxedema
Hyperthyroidism

[a]In the United States an estimated 2 million men are impotent because they have diabetes mellitus; an additional 300,000 are impotent because of other endocrine diseases; 1.5 million are impotent as a result of vascular disease; 180,000 because of multiple sclerosis; 400,000 because of traumas and fractures leading to pelvic fractures or spinal cord injuries; and another 650,000 are impotent as a result of radical surgery, including prostatectomies, colostomies, and cystectomies.

Two conditions not readily apparent on physical examination that produce dyspareunia are vulvar vestibulitis and interstitial cystitis. The former may present with chronic vulvar pain, and the latter produces pain most intensely following orgasm. Dyspareunia can also occur in men, but it is uncommon and is usually associated with an organic condition, such as Peyronie disease, which consists of sclerotic plaques on the penis that cause penile curvature.

Male Hypoactive Sexual Desire Disorder and Female Interest/Arousal Disorder due to a General Medical Condition. Sexual desire commonly decreases after major illness or surgery, particularly when the body image is affected after such procedures as mastectomy, ileostomy, hysterectomy, and prostatectomy. Illnesses that deplete a person's energy, chronic conditions that require physical and psychological adaptation, and serious illnesses that can cause a person to become depressed can all markedly lessen sexual desire.

In some cases, biochemical correlates are associated with hypoactive sexual desire disorder (Table 16-11). A recent study found markedly lower levels of serum testosterone in men complaining of low desire than in normal controls in a sleep-laboratory situation. Drugs that depress the central nervous system (CNS) or decrease testosterone production can decrease desire.

Other Male Sexual Dysfunction due to a General Medical Condition. Delayed ejaculation can have physiologic causes and can occur after surgery on the genitourinary tract, such as prostatectomy. It may also be associated with Parkinson disease and other neurologic disorders involving the lumbar or sacral sections of the spinal cord. The antihypertensive drug guanethidine monosulfate, methyldopa, the phenothiazines, the tricyclic

drugs, and the selective serotonin reuptake inhibitors (SSRIs), among others, have been implicated in retarded ejaculation. Also, we should differentiate delayed ejaculation from retrograde ejaculation, in which ejaculation occurs but the seminal fluid passes back into the bladder. Retrograde ejaculation always has an organic cause. It can develop after genitourinary surgery, and it is also associated with medications that have anticholinergic adverse effects, such as the phenothiazines.

Other Female Sexual Dysfunction due to a General Medical Condition. Some medical conditions—specifically, endocrine diseases such as hypothyroidism, diabetes mellitus, and primary hyperprolactinemia—can affect a woman's ability to have orgasms. Several drugs also affect some women's capacity to have orgasms (Table 16-12). Antihypertensive medications, CNS stimulants, tricyclic drugs, SSRIs, and, frequently, monoamine oxidase inhibitors (MAOIs) have interfered with female orgasmic capacity. One study of women taking MAOIs, however, found that after 16 to 18 weeks of pharmacotherapy, the adverse effect of the medication disappeared, and the women were able to reexperience orgasms, although they continued taking an undiminished dose of the drug.



Table 16-10
Some Pharmacologic Agents Implicated in Male Sexual Dysfunctions

Drug	Impairs Erection	Impairs Ejaculation
Psychiatric drugs		
Cyclic drugs[a]		
Imipramine (Tofranil)	+	+
Protriptyline (Vivactil)	+	+
Desipramine (Pertofrane)	+	+
Clomipramine (Anafranil)	+	+
Amitriptyline (Elavil)	+	+

Trazodone (Desyrel)[b]	–	–
Monoamine oxidase inhibitors		
Tranylcypromine (Parnate)	+	+
Phenelzine (Nardil)	+	+
Pargyline (Eutonyl)	–	+
Isocarboxazid (Marplan)	–	+
Other mood-active drugs		
Lithium (Eskalith)	+	+
Amphetamines	+	+
Fluoxetine (Prozac)[c]	–	+
Antipsychotics[d]		
Fluphenazine (Prolixin)	+	+
Thioridazine (Mellaril)	+	+
Chlorprothixene (Taractan)	–	+
Mesoridazine (Serentil)	–	+
Perphenazine (Trilafon)	–	+
Trifluoperazine (Stelazine)	–	+
Reserpine (Serpasil)	+	+
Haloperidol (Haldol)	–	+
Antianxiety agent[e]		
Chlordiazepoxide (Librium)	–	+
Antihypertensive drugs		

Clonidine (Catapres)	+	+
Methyldopa (Aldomet)	+	+
Spironolactone (Aldactone)	+	–
Hydrochlorothiazide	+	–
Guanethidine (Ismelin)	+	+
Commonly abused substances		
Alcohol	+	+
Barbiturates	+	+
Cannabis	+	–
Cocaine	+	+
Heroin	+	+
Methadone	+	–
Morphine	+	+
Miscellaneous drugs		
Antiparkinsonian agents	+	+
Clofibrate (Atromid-S)	+	–
Digoxin (Lanoxin)	+	–
Glutethimide (Doriden)	+	+
Indomethacin (Indocin)	+	–
Phentolamine (Regitine)	–	+
Propranolol (Inderal)	+	–

[a]The incidence of male erectile disorder associated with the use of tricyclic drugs is low.

[b]Trazodone has been causative in some cases of priapism.

[c]All SSRIs can produce sexual dysfunction, more commonly, in men.

[d]Impairment of sexual function is not a common complication of the use of antipsychotics.

Priapism has occasionally occurred in association with the use of antipsychotics.

[e]Benzodiazepines have been reported to decrease libido, but in some patients the diminution of anxiety caused by those drugs enhances sexual function.

Substance/Medication-Induced Sexual Dysfunction. We diagnose substance-induced sexual dysfunction when evidence of substance intoxication or withdrawal is apparent from the history, physical examination, or laboratory findings. The disturbance in sexual function must be predominant in the clinical picture. Distressing sexual dysfunction occurs soon after significant substance intoxication or withdrawal, or after exposure to a medication or a change in medication use. Specified substances include alcohol, amphetamines or related substances, cocaine, opioids, sedatives, hypnotics, or anxiolytics, and other or unknown substances.

Abused recreational substances affect sexual function in various ways. In small doses, many substances enhance sexual performance by decreasing inhibition or anxiety or by causing a temporary elevation of mood. With continued use, however, erectile engorgement and orgasmic and ejaculatory capacities become impaired. The abuse of sedatives, anxiolytics, hypnotics, and particularly opiates and opioids nearly always depresses desire. Alcohol may foster the initiation of sexual activity by removing inhibition, but it also impairs performance. Cocaine and amphetamines produce the similar effects. Although no direct evidence indicates that sexual drive is enhanced, users initially have feelings of increased energy and may become sexually active; ultimately, dysfunction occurs. Men usually go through two stages: an experience of prolonged erection without ejaculation, and then a gradual loss of erectile capability.

Patients recovering from substance dependency may need therapy to regain sexual function, partly because of psychological readjustment to a nondependent state. Many substance abusers have always had difficulty with intimate interactions. Others who spent their crucial developmental years under the influence of a substance have missed the experiences that would have enabled them to learn social and sexual skills.

Almost every pharmacologic agent, particularly those used in psychiatry, has been associated with an effect on sexuality. In men, these effects include decreased sex drive, erectile failure, decreased volume of ejaculate, and delayed or retrograde ejaculation. In women, decreased sex drive, decreased

vaginal lubrication, inhibited or delayed orgasm, and decreased or absent vaginal contractions may occur. Drugs may also enhance sexual responses and increase the sex drive, but this is less common than adverse effects.

ANTIPSYCHOTIC DRUGS. Most antipsychotic drugs are dopamine receptor antagonists that also block adrenergic and cholinergic receptors, thus accounting for the adverse sexual effects (Table 16-13). Chlorpromazine and trifluoperazine are potent anticholinergics, and they impair erection and ejaculation. With some drugs, the seminal fluid backs up into the bladder rather than being propelled through the penile urethra. Patients still have a pleasurable sensation, but the orgasm is dry. When urinating after orgasm, the urine may be milky white because it contains the ejaculate. The condition is startling but harmless. Paradoxically, antipsychotics may rarely cause priapism.



Table 16-11
Neurophysiology of Sexual Dysfunction

	DA	5-HT	NE	ACh	Clinical Correlation
Erection	↑	°	α_2, β ↓ ↑	M	Antipsychotics may lead to erectile dysfunction (DA block): DA agonists may lead to enhanced erection and libido; priapism with trazodone (α_1 block); β-blockers may lead to impotence
Ejaculation and orgasm	°	± ↓	α_1 ↑	M	α-Blockers (tricyclic drugs, MAOIs, thioridazine) may lead to impaired ejaculation; 5-HT agents may inhibit orgasm

↑, facilities; ↓, inhibits or decreases; ±, some; ACh, acetylcholine; DA, dopamine; 5-HT, serotonin; M, modulates; NE, norepinephrine; °, minimal.
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ANTIDEPRESSANT DRUGS. The tricyclic and tetracyclic antidepressants have anticholinergic effects that interfere with erection and delay ejaculation. Because the anticholinergic effects vary among the cyclic antidepressants, those with the fewest effects (e.g., desipramine) produce the fewest sexual adverse effects.

Some men report increased sensitivity of the glans that is pleasurable, and that does not interfere with erection, although it delays ejaculation. In some cases, however, the tricyclic causes painful ejaculation, perhaps as the result of interference with seminal propulsion caused by interference with, in turn, urethral, prostatic, vas, and epididymal smooth muscle contractions. Clomipramine may increase sex drive in some persons. Selegiline, a selective MAO type B (MAO_B) inhibitor, and bupropion may also increase sex drive,

possibly by dopaminergic activity and increased production of norepinephrine.

Venlafaxine and the SSRIs most often have adverse effects because of the rise in serotonin levels. A lowering of the sex drive and difficulty reaching orgasm occur in both sexes. Cyproheptadine, an antihistamine with antiserotonergic effects, and methylphenidate, with adrenergic effects, may reverse those adverse effects. Trazodone is associated with the rare occurrence of priapism, the symptom of prolonged erection in the absence of sexual stimuli. That symptom appears to result from the α_1-adrenergic antagonism of trazodone.



Table 16-12
Some Pharmacologic Agents Implicated in Inhibited Female Orgasm[a]

Tricyclic antidepressants
Imipramine (Tofranil)
Clomipramine (Anafranil)
Nortriptyline (Aventyl)
Monoamine oxidase inhibitors
Tranylcypromine (Parnate)
Phenelzine (Nardil)
Isocarboxazid (Marplan)
Dopamine receptor antagonists
Thioridazine (Mellaril)
Trifluoperazine (Stelazine)
Selective serotonin reuptake inhibitors
Fluoxetine (Prozac)
Paroxetine (Paxil)
Sertraline (Zoloft)
Fluvoxamine (Luvox)
Citalopram (Celexa)

[a]The interrelation between female sexual dysfunction and pharmacologic agents has been less extensively evaluated than male reactions. Oral contraceptives are reported to decrease

libido in some women, and some drugs with anticholinergic side effects may impair arousal as well as orgasm. Prolonged use of oral contraceptives may also cause physiologic menopausal-like changes resulting in genito-pelvic pain/penetration disorder. Benzodiazepines have been reported to decrease libido, but in some patients the diminution of anxiety caused by those drugs enhances sexual function. Both increase and decrease in libido have been reported with psychoactive agents. It is difficult to separate those effects from the underlying condition or from improvement of the condition. Sexual dysfunction associated with the use of a drug disappears when use of the drug is discontinued.

The MAOIs affect biogenic amines broadly. Accordingly, they produce impaired erection, delayed or retrograde ejaculation, vaginal dryness, and inhibited orgasm. Tranylcypromine has a paradoxical sexually stimulating effect in some persons, possibly as a result of its amphetamine-like properties.

Mr. W presented with the complaint of inability to achieve orgasm. His problem dated from the time, 18 months previously, when he started fluoxetine. Before that time, he had been able to achieve orgasm through masturbation and coitus with his wife.

Mr. W tried several other SSRIs, as well as venlafaxine, but the side effect of delayed ejaculation persisted. None of the usual antidotes to SSRI-induced anorgasmia proved effective, and the patient then tried antidepressants of other categories. Mr. W was able to respond to bupropion and clonazepam. This combination treated his depression and anxiety, and his delayed ejaculation resolved.

His psychiatrist diagnosed pharmacologically induced delayed ejaculation.



Table 16-13
Diagnostic Issues with Sex and Some Antipsychotic Drugs

Differential diagnosis of drug-induced sexual dysfunction	Problem after drug therapy started or drug overdose Problem not situation or partner specific Not a lifelong or recurrent

	problem
	No obvious nonpharmacologic precipitant
	Dissipates with drug discontinuation
Antipsychotic drugs and ejaculatory problems	Perphenazine
	Chlorpromazine
	Trifluoperazine
	Haloperidol
	Mesoridazine
	Chlorprothixene
Antipsychotic drugs and priapism	Perphenazine
	Mesoridazine
	Chlorpromazine
	Thioridazine
	Fluphenazine
	Molindone
	Risperidone
	Clozapine

Table by R. T. Seagraves, M.D.

Because depression is associated with decreased libido, varying levels of sexual dysfunction and anhedonia are part of the disease process. Some patients report improved sexual functioning as their depression improves because of antidepressant medication. The phenomenon makes evaluating sexual side effects difficult; also, the side effects may disappear with time, perhaps because a biogenic amine homeostatic mechanism comes into play.

LITHIUM. Lithium regulates mood and, in the manic state, may reduce hypersexuality, possibly by a dopamine antagonist activity. Some patients may have impaired erections.

SYMPATHOMIMETICS. Psychostimulants, including amphetamines, methylphenidate, and pemoline, raise the plasma levels of norepinephrine and dopamine. They can increase libido. However, with prolonged use, men may experience a loss of desire and erections.

α-ADRENERGIC AND β-ADRENERGIC RECEPTOR ANTAGONISTS. α-Adrenergic and β-adrenergic receptor antagonists, used in the treatment of hypertension, angina, and certain cardiac arrhythmias, diminish tonic sympathetic nerve outflow from vasomotor centers in the brain. As a result, they can cause impotence, decrease the volume of ejaculate, and produce retrograde ejaculation. They can affect libido in both sexes.

ANTICHOLINERGICS. The anticholinergics block cholinergic receptors and include such drugs as amantadine and benztropine. They produce dryness of the mucous membranes (including those of the vagina) and erectile disorder. However, amantadine may reverse SSRI-induced orgasmic dysfunction through its dopaminergic effect.

ANTIHISTAMINES. Drugs such as diphenhydramine have anticholinergic activity and are mildly hypnotic. They may inhibit sexual function as a result. Cyproheptadine, although an antihistamine, also has potent activity as a serotonin antagonist. It can block the serotonergic sexual adverse effects produced by SSRIs, such as delayed orgasm.

ANTIANXIETY AGENTS. The dominant class of anxiolytics is the benzodiazepines. They act on the γ-aminobutyric acid (GABA) receptors, which are involved in cognition, memory, and motor control. Because benzodiazepines decrease plasma epinephrine concentrations, they diminish anxiety, and as a result, they improve sexual function in persons inhibited by anxiety.

ALCOHOL. Alcohol suppresses CNS activity generally and can produce

erectile disorders in men as a result. Alcohol has a direct gonadal effect that decreases testosterone levels in men; paradoxically, it can produce a slight rise in testosterone levels in women. The latter finding may account for women who report increased libido after drinking small amounts of alcohol. The long-term use of alcohol reduces the ability of the liver to metabolize estrogenic compounds in men, which produces signs of feminization (such as gynecomastia as a result of testicular atrophy).

OPIOIDS. Opioids, such as heroin, have adverse sexual effects, such as erectile failure and decreased libido. The alteration of consciousness may enhance the sexual experience of occasional users.

HALLUCINOGENS. The hallucinogens include lysergic acid diethylamide (LSD), phencyclidine (PCP), psilocybin (from some mushrooms), and mescaline (from peyote cactus). In addition to inducing hallucinations, the drugs cause loss of contact with reality and an expanding and heightening of consciousness. Some users report that the sexual experience is similarly enhanced, but others experience anxiety, delirium, or psychosis, which interfere with sexual function.

CANNABIS. The altered state of consciousness produced by cannabis may enhance sexual pleasure for some persons. Its prolonged use depresses testosterone levels.

BARBITURATES AND SIMILARLY ACTING DRUGS. Barbiturates and similarly acting sedative-hypnotic drugs may enhance sexual responsiveness in persons who are sexually unresponsive as a result of anxiety. They have no direct effect on the sex organs; however, they do produce an alteration in consciousness that some persons find pleasurable. These drugs are subject to abuse, and use can be fatal when combined with alcohol or other CNS depressants.

Methaqualone acquired a reputation as a sexual enhancer, a reputation without basis. It is no longer legally sold in the United States.

Treatment

Before 1970, the most common treatment for sexual dysfunctions was

individual psychotherapy. Classic psychodynamic theory holds that sexual inadequacy has its roots in early developmental conflicts, and the therapist treated the sexual disorder as part of a pervasive emotional disturbance. The symptoms of sexual dysfunctions, however, frequently become secondarily autonomous and continue to persist, even when other problems evolving from the patients' pathology have been resolved. The addition of behavioral techniques is often necessary to cure a sexual problem. Depending on the disorder, medications can be a first-line treatment or an adjunct to therapy.

Somatic Treatments. Biologic treatments, including pharmacotherapy, surgery, and mechanical devices, can treat specific cases of sexual disorder. Most of the recent advances involve male sexual dysfunction.

PHARMACOTHERAPY. Medications for erectile disorder include sildenafil and its congeners (Table 16-14), oral phentolamine, alprostadil, and injectable medications. Other options include papaverine, prostaglandin E1, phentolamine, or some combination of these; and transurethral alprostadil (MUSE).

Sildenafil is a nitric oxide enhancer that facilitates the inflow of blood to the penis necessary for an erection. The drug takes effect about 1 hour after ingestion, and its effect can last up to 4 hours. Sildenafil is not effective in the absence of sexual stimulation. The most common adverse events associated with its use are headaches, flushing, and dyspepsia. Persons taking organic nitrates should not take sildenafil, as the combined action of the two drugs can cause sudden, and sometimes fatal, drops in systemic blood pressure. Sildenafil is not effective in all cases of erectile dysfunction. It fails to produce an erection that is sufficiently rigid for penetration in about 50 percent of men who have had radical prostate surgery or in those with long-standing insulin-dependent diabetes. It is also ineffective in some instances of nerve damage.

A small number of patients developed nonarteritic ischemic optic neuropathy (NAION) soon after using sildenafil. This side effect may be more common in patients with cardiovascular risk factors. Although rare, sildenafil may provoke NAION in individuals with an arteriosclerotic risk profile. Exceedingly rare cases of hearing loss may also occur.

Sildenafil use in women results in vaginal lubrication, but not in increased

desire. Anecdotal reports, however, describe individual women who have experienced intensified excitement with sildenafil.

Oral phentolamine and apomorphine are not FDA approved at present but have proved useful as potency enhancers in men with minimal erectile dysfunction. Phentolamine reduces sympathetic tone and relaxes corporeal smooth muscle. Adverse events include hypotension, tachycardia, and dizziness. Apomorphine effects are mediated by the autonomic nervous system and result in vasodilation that facilitates the inflow of blood to the penis. Adverse events include nausea and sweating.



Table 16-14
Pharmacokinetics of the PDE-5 Inhibitors

		Sildenafil (Viagra)	Vardenafil (Levitra)	Vardenafil ODT (Staxyn)	Tadalafil (Cialis)	Avanafil (Stendra)
Recommended dose (mg/day)		25–100	5–20	10	5–20	100–200
Time to peak concentration (hr)		0.5–2	0.5–2	0.75–2.5	0.5–6	0.5–0.75
Half-life (hr)		4	4–5	4–6	17.5	3–5
Usual duration of action (hr)		12	12		36	6
Delay after high fat meal (hr)		1	1		0	1–1.25
Metabolism	Major enzyme	CYP3A4				
	Additional enzymes	CYP2C9	CYP3A5, CYP2C		CYP3A5, CYP2C	CYP2C
Elimination (%)	Feces	80	91–95		61	62
	Urine	13	2–6		36	21

Sources: Prescriber's Digital Reference, https://www.pdr.net/; Huang SA, Lie JD. Phosphodiesterase-5 (PDE5) Inhibitors in the management of erectile dysfunction. *P T*. 2013;38(7):407–419; Katz EG, Tan RB, Rittenberg D, Hellstrom WJ. Avanafil for erectile dysfunction in elderly and younger adults: differential pharmacology and clinical utility. *Ther Clin Risk Manag*. 2014;10:701–711.

In contrast to the oral medications, injectable and transurethral forms of alprostadil act locally on the penis and can produce erections in the absence of sexual stimulation. Alprostadil contains a naturally occurring form of prostaglandin E, a vasodilating agent. Alprostadil may be administered by direct injection into the corpora cavernosa or by intraurethral insertion of a pellet through a cannula. The firm erection produced within 2 to 3 minutes after administration of the drug may last as long as 1 hour. Infrequent and reversible adverse effects of injections include penile bruising and changes in liver function test results. Possible hazardous sequelae exist, including priapism and sclerosis of the small veins of the penis. Users of transurethral alprostadil sometimes complain of burning sensations in the penis.

Two small trials found different topical agents effective in alleviating erectile dysfunction. One cream consists of three vasoactive substances

absorbed through the skin: aminophylline, isosorbide dinitrate, and co-dergocrine mesylate, which is a mixture of ergot alkaloids. The other is a gel containing alprostadil and an additional ingredient, which temporarily makes the outer layer of the skin more permeable.

Also, a cream incorporating alprostadil exists for female sexual arousal disorder. Vaginally applied phentolamine mesylate, an α-receptor antagonist, can increase vasocongestion and a subjective sense of arousal. Flibanserin is also FDA-approved to increase desire in women.

The pharmacologic treatments described in the preceding text are useful in the treatment of arousal dysfunction of various causes: neurogenic, arterial insufficiency, venous leakage, psychogenic, and mixed. When coupled with insight-oriented or behavioral sex therapy, the use of medications can reverse psychogenic arousal disorder that is resistant to psychotherapy alone, the ultimate goal being pharmacologically unassisted sexual functioning.

Other Pharmacologic Agents. Numerous other pharmacologic agents have been used to treat various sexual disorders. Intravenous methohexital sodium assists with desensitization therapy. Antianxiety agents may have some application for tense patients. The side effects of antidepressants, in particular the SSRIs and tricyclic drugs, have been used to prolong the sexual response in patients with premature ejaculation. This approach is particularly useful in patients who are refractory to behavioral techniques. Topical anesthetic creams may help decrease the intravaginal ejaculation latency time (IELT) in cases of premature ejaculation. Antidepressants can help patients who are phobic of sex and in those with posttraumatic stress disorder following rape. Trazodone is an antidepressant that improves nocturnal erections. We should always weigh the risks of taking such medications against their possible benefits. Bromocriptine treats hyperprolactinemia, which is frequently associated with hypogonadism and any associated sexual dysfunction. In such patients, it is necessary to rule out pituitary tumors.

Several substances have popular standing as aphrodisiacs; for example, ginseng root and yohimbine. Studies, however, have not confirmed any aphrodisiac properties. Yohimbine, an α-receptor antagonist, may cause dilation of the penile artery; however, the American Urologic Association does not recommend its use to treat organic erectile dysfunction. Many

recreational drugs, including cocaine, amphetamines, alcohol, and cannabis, are considered enhancers of sexual performance. Although they may provide the user with an initial benefit because of their tranquilizing, disinhibiting, or mood-elevating effects, consistent or prolonged use of any of these substances impairs sexual functioning.

Dopaminergic agents may increase libido and improve sex function. Those drugs include L-dopa, a dopamine precursor, and bromocriptine, a dopamine agonist. The antidepressant bupropion has dopaminergic effects and has increased sex drive in some patients. Selegiline, an MAOI, is selective for MAO_B and is dopaminergic. It improves sexual functioning in older persons.

HORMONE THERAPY. Androgens increase the sex drive in women and men with low testosterone concentrations. Women may experience virilizing effects, some of which are irreversible (e.g., deepening of the voice). In men, prolonged use of androgens produces hypertension and prostatic enlargement. Testosterone is most effective when given parenterally; however, effective oral and transdermal preparations are available.

Women who use estrogens for replacement therapy or contraception may report decreased libido; in such cases, a combined preparation of estrogen and testosterone may help. Estrogen itself prevents thinning of the vaginal mucous membrane and facilitates lubrication. Several forms of locally delivered estrogen—vaginal rings, vaginal creams, and vaginal tablets—provide alternate administration routes to treat women with arousal problems or genital atrophy. Because tablets, creams, and rings do not significantly increase circulating estrogen levels, we should consider these devices for patients with breast cancer with arousal problems.

ANTIANDROGENS AND ANTIESTROGENS. Estrogens and progesterone are antiandrogens that have been used to treat compulsive sexual behavior in men, usually in sex offenders. Clomiphene and tamoxifen are both antiestrogens, and both stimulate gonadotropin-releasing hormone (GnRH) secretion and increase testosterone concentrations, thereby increasing libido. Women treated for breast cancer with tamoxifen report an increased libido. However, tamoxifen may cause uterine cancer.

MECHANICAL TREATMENT APPROACHES. Male patients with

arteriosclerosis (especially of the distal aorta, known as Leriche syndrome), may lose their erection during active pelvic thrusting. The need for increased blood in the gluteal muscles and others served by the ilial or hypogastric arteries takes blood away (steals) from the pudendal artery and, thus, interferes with penile blood flow. They can obtain relief by decreasing pelvic thrusting, which is also aided by the woman's superior coital position.

VACUUM PUMP. Vacuum pumps are mechanical devices that patients without a vascular disease can use to obtain erections. The vacuum draws blood to the penis and kept there by a ring at the base of the penis. This device has no adverse effects, but it is cumbersome, and partners must be willing to accept its use. Some women complain that the penis is redder and cooler than a natural erection or find the process objectionable.

A similar device, called EROS, has been developed to create clitoral erections in women. EROS is a small suction cup that fits over the clitoral region and draws blood into the clitoris. Studies have reported its success in treating female sexual arousal disorder. Vibrators used to stimulate the clitoral area have been successful in treating anorgasmic women.

OTHER SOMATIC TREATMENTS

Male Prostheses. Surgical treatment is rarely indicated, but penile prosthetic devices are available for men with inadequate erectile responses who are resistant to other treatment methods or who have medically caused deficiencies. The two main types of prostheses are (1) a semi-rigid rod prosthesis that produces a permanent erection that can be positioned close to the body for concealment and (2) an inflatable type that is implanted with its reservoir and pump for inflation and deflation. The latter type can mimic normal physiologic functioning.

Vascular Surgery. When vascular insufficiency is present due to atherosclerosis or other blockages, bypass surgery of penile arteries may help in selected cases.

Psychosocial Treatments

DUAL-SEX THERAPY. The theoretical basis of dual-sex therapy is the concept of the marital unit or dyad as the object of therapy; the approach represented the

significant advance in the diagnosis and treatment of sexual disorders in the 20th century. The methodology was originated and developed by Masters and Johnson. In dual-sex therapy, when a dysfunctional person is in a relationship, the couple must be treated. Because both are involved in a sexually distressing situation, both must participate in the therapy program. The sexual problem often reflects other areas of disharmony or misunderstanding in the relationship so that the entire relationship is treated, with emphasis on the sexual functioning of the partners.

The keystone of the program is the roundtable session in which a male and female therapy team clarifies, discusses, and works through problems with the couple. The four-way sessions require active participation by the patients. Therapists and patients discuss the psychological and physiologic aspects of sexual functioning, and therapists have an educative attitude. Therapists suggest specific sexual activities for the couple to follow in the privacy of their home. The therapy aims to establish or reestablish communication within the partner unit. Sex is emphasized as a natural function that flourishes in the appropriate domestic climate, and improved communication is encouraged toward that end. In a variation of this therapy that has proved useful, one therapist may treat the couple. Treatment is short term and is behaviorally oriented. The therapists attempt to reflect the situation as they see it, rather than interpret underlying dynamics. An undistorted picture of the relationship presented by the therapists often corrects the myopic, narrow view held by each partner. This new perspective can interrupt the couple’s destructive pattern of relating and can encourage improved, more effective communication. Specific exercises are prescribed for the couple to treat their particular problems. Sexual inadequacy often involves a lack of information, misinformation, and performance fear. Therefore, the therapist prohibits any sexual play other than what was prescribed. Beginning exercises usually focus on heightening sensory awareness to touch, sight, sound, and smell. Initially, the therapist also prohibits intercourse, and the couple learns to give and receive bodily pleasure without the pressure of performance or penetration. At the same time, they learn how to communicate nonverbally in a mutually satisfactory way, and they learn that sexual foreplay is an enjoyable alternative to intercourse and orgasm.

During the sensate focus exercises, the couple receives much reinforcement to reduce anxiety. They can use fantasies to distract them from

obsessive concerns about performance (spectatoring). They should consider the needs of both the dysfunctional partner and the nondysfunctional partner. If either partner becomes sexually excited by the exercises, the other is encouraged to bring them to orgasm by manual or oral means. Open communication and the expression of mutual needs are encouraged. Resistances, such as claims of fatigue or not enough time to complete the exercises, are common and must be dealt with by the therapists. Issues of body image, fear of being touched, and difficulty touching oneself arise frequently. Eventually, the therapy allows genital stimulation in addition to general body stimulation. The couple should sequentially to try various positions for intercourse, without necessarily completing the act, and to use varieties of stimulating techniques before they proceed with intercourse.

Psychotherapy sessions follow each new exercise period, and problems and satisfactions, both sexual and in other areas of the couple's lives, are discussed. The therapist reviews specific instructions and introduces new exercises geared to the individual couple's progress. Gradually, the couple gains confidence and learns to communicate, verbally and sexually. Dual-sex therapy is most effective when sexual dysfunction exists apart from other psychopathology. The more difficult treatment cases involve couples with severe marital discord. Desire disorders are particularly challenging to treat. They require more extended and intensive therapy than some other disorders, and their outcomes vary greatly.

SPECIFIC TECHNIQUES AND EXERCISES. Various techniques can treat various sexual dysfunctions. In cases of vaginismus, a woman is advised to dilate her vaginal opening with her fingers or with size-graduated dilators. Dilators can also treat cases of dyspareunia. Sometimes, specially trained physiotherapists can assist the treatment by helping the patients to relax their perineal muscles.

In cases of premature ejaculation, an exercise known as the squeeze technique can raise the threshold of penile excitability. In this exercise, the man or the woman stimulates the erect penis until feeling the earliest sensations of impending ejaculation. At this point, the partner forcefully squeezes the coronal ridge of the glans, diminishing the erection and inhibiting ejaculation. The exercise program eventually raises the threshold of the sensation of ejaculatory inevitability and allows the man to focus on the sensations of arousal without anxiety and develop confidence in his sexual

performance. A variant of the exercise is the stop-start technique developed by James H. Semans, in which the woman stops all stimulation of the penis when the man first senses an impending ejaculation without squeezing. Research has shown that the presence or absence of circumcision has no bearing on a man's ejaculatory control; the glans is equally sensitive in the two states. Sex therapy has been most successful in the treatment of premature ejaculation.

A man with a sexual desire disorder or male erectile disorder can masturbate to prove that full erection and ejaculation are possible. Delayed ejaculation is managed initially by extravaginal ejaculation and then by gradual vaginal entry after stimulation to a point near ejaculation. Most importantly, the early exercises forbid ejaculation to remove the pressure to climax and allow the man to immerse himself in sexual pleasuring.

In cases of lifelong female orgasmic disorder, the woman is directed to masturbate, sometimes using a vibrator. The shaft of the clitoris is the masturbatory site most preferred by women, and orgasm depends on adequate clitoral stimulation. An area on the anterior wall of the vagina exists in some women as a site of sexual excitation, known as the *G-spot*, but reports of an ejaculatory phenomenon at orgasm in women following the stimulation of the G-spot is only anecdotal.

BEHAVIOR THERAPY. Behavioral approaches were initially designed for the treatment of phobias but are now used to treat other problems as well. Behavior therapists assume that the patient learns sexual dysfunction as maladaptive behavior, which causes them to fear sexual interaction. Using traditional techniques, therapists set up a hierarchy of anxiety-provoking situations, ranging from least threatening (e.g., the thought of kissing) to most threatening (e.g., the thought of penile penetration). The behavior therapist enables the patient to master the anxiety through a standard program of systematic desensitization, which inhibits the learned anxious response by encouraging behaviors antithetical to anxiety. The patient first deals with the least anxiety-producing situation in fantasy and progresses by steps to the most anxiety-producing situation. Medication, hypnosis, and specialized training in deep muscle relaxation can help with the initial mastery of anxiety.

Assertiveness training can help teach patients to express sexual needs

openly and without fear. The patient receives exercises in assertiveness, given in conjunction with sex therapy, and they are encouraged to make sexual requests and to refuse to comply with requests perceived as unreasonable. Sexual exercises may be prescribed for patients to perform at home, and the therapist helps the patient to establish a hierarchy, starting with those activities that have proved most pleasurable and successful in the past.

One treatment variation involves the participation of the patient's sexual partner in the desensitization program. The partner, rather than the therapist, presents items of increasing stimulation value to the patient. A cooperative partner is necessary to help the patient carry gains made during treatment sessions to sexual activity at home.

Couples who regularly practice assigned exercises appear to have a much higher likelihood of success than do more resistant couples or those whose interaction involves sadomasochistic or depressive features or mechanisms of blame and projection. Attitude flexibility is also a favorable prognostic factor. Overall, younger couples tend to complete sex therapy more often than older couples. Couples whose interactional difficulties center on their sex problems, such as inhibition, frustration, or fear of performance failure, are also likely to respond well to therapy.

MINDFULNESS. Mindfulness is a cognitive technique that has been helpful in the treatment of sexual dysfunction. The patient focuses on the moment and maintains an awareness of sensations—visual, tactile, auditory, and olfactory—that he or she experiences in the moment. The aim is to distract the patient from "spectatoring" (watching him or herself) and center the person on the sensations that lead to arousal or orgasm. Hopefully, this shift in focus allows patients to become immersed in the pleasure of the experience and remove themselves from self-judgment and performance anxiety.

GROUP THERAPY. Group therapy helps to examine both intrapsychic and interpersonal problems in patients with sexual disorders. A therapy group provides a strong support system for a patient who feels ashamed, anxious, or guilty about a particular sexual problem. It is a useful forum in which to counteract sexual myths, correct misconceptions, and provide accurate information about sexual anatomy, physiology, and varieties of behavior.

One can organize the groups in several ways. Members may all share the

same problem, such as premature ejaculation; members may all be of the same sex with different sexual problems, or groups may be composed of both men and women who are experiencing a variety of sexual problems. Group therapy can be an adjunct to other forms of therapy or the prime mode of treatment. Groups organized to treat a particular dysfunction are usually behavioral in approach.

Groups composed of married couples with sexual dysfunctions are also useful. A group provides the opportunity to gather accurate information, offers consensual validation of individual preferences, and enhances self-esteem and self-acceptance. Techniques, such as role-playing and psychodrama, may be used in treatment. Such groups are not indicated for couples when one partner is uncooperative, when a patient has a severe depressive disorder or psychosis, when a patient finds explicit sexual audiovisual material repugnant, or when a patient fears or dislikes groups.

HYPNOTHERAPY. Hypnotherapists focus specifically on the anxiety-producing situation—that is, the sexual interaction that results in dysfunction. The successful use of hypnosis enables patients to gain control over the symptom that has been lowering self-esteem and disrupting psychological homeostasis. The patient's cooperation is first obtained and encouraged during a series of nonhypnotic sessions with the therapist. Those discussions permit the development of a secure doctor–patient relationship, a sense of physical and psychological comfort on the part of the patient, and the establishment of mutually desired treatment goals. During this time, the therapist assesses the patient's capacity for the trance experience. The nonhypnotic sessions also permit the clinician to take a psychiatric history and perform a mental status examination before beginning hypnotherapy. The focus of treatment is on symptom removal and attitude alteration. The patient develops alternative means of dealing with the anxiety-provoking situation, the sexual encounter.

Also, the therapist teaches relaxation techniques for use before sexual relations. With these methods to alleviate anxiety, the physiologic responses to sexual stimulation can more readily result in pleasurable excitation and discharge. The therapy helps to remove psychological impediments to vaginal lubrication, erection, and orgasms so that normal sexual functioning ensues. Hypnosis may be added to a basic individual psychotherapy program to accelerate the effects of psychotherapeutic intervention.

ANALYTICALLY ORIENTED SEX THERAPY. Some therapists combine sex therapy with psychodynamic psychotherapy. The therapy occurs over a more extended period than usual, which allows learning or relearning of sexual satisfaction under the realities of patients' day-to-day lives. The addition of psychodynamic conceptualizations to behavioral techniques used to treat sexual dysfunctions allows the treatment of patients with sexual disorders associated with other psychopathology.

The material and dynamics that emerge in patients in analytically oriented sex therapy are the same as those in psychoanalytic therapy, such as dreams, fear of punishment, aggressive feelings, difficulty trusting a partner, fear of intimacy, oedipal feelings, and fear of genital mutilation. The combined approach of analytically oriented sex therapy is used by the general psychiatrist, who carefully judges the optimal timing of sex therapy and the ability of patients to tolerate the directive approach that focuses on their sexual difficulties.

PARAPHILIC DISORDERS

Paraphilias or perversions are sexual stimuli or acts that are deviations from normal sexual behaviors but are necessary for some persons to experience arousal and orgasm. DSM-5 only considers this a disorder when a person has acted on the fantasy or impulse, or if the fantasies or impulses lead to significant distress. Individuals with paraphilic interests can experience sexual pleasure, but they do not respond to stimuli that are typically considered erotic. The paraphiliac person's sexuality is mainly restricted to specific deviant stimuli or acts. Persons who occasionally experiment with paraphilic behavior (e.g., infrequent episode of bondage or dressing in costumes), but can respond to more typical erotic stimuli, do not have paraphilic disorders.

Paraphilic disorders can range from nearly normal behavior to behavior that is destructive or hurtful only to a person's self or to a person's self and partner, and finally to behavior that is deemed destructive or threatening to the community at large. DSM-5 lists pedophilia, frotteurism, voyeurism, exhibitionism, sexual sadism, sexual masochism, fetishism, and transvestism with explicit diagnostic criteria because of their threat to others or because they are relatively common paraphilias. Many other paraphilias exist.

A particular fantasy with its unconscious and conscious components is the pathognomonic element of the paraphilia, with sexual arousal and orgasm being associated phenomena that *reinforce the fantasy or impulse.* The influence of these fantasies and their behavioral manifestations often extend beyond the sexual sphere to pervade people's lives.

The primary functions of human sexual behavior are to assist in bonding, to create mutual pleasure in cooperation with a partner, to express and enhance love between two persons, and to procreate. Paraphilic disorders entail divergent behaviors in that those acts involve aggression, victimization, and extreme one-sidedness. The behaviors exclude or harm others and disrupt the potential for bonding between persons. Moreover, paraphilic sexual scripts often serve other vital psychic functions. They may assuage anxiety, bind aggression, or stabilize identity.

The Clinical Presentation and Diagnosis

In DSM-5, the criteria for paraphilic disorder require the patient to experience intense and recurrent arousal from their deviant fantasy for at least 6 months and to either act on the paraphilic impulse or experience significant distress as a result of the impulse. The fantasy distressing the patient contains unusual sexual material that is relatively fixed and shows only minor variations. Arousal and orgasm depend on the mental elaboration, if not the behavioral playing out of the fantasy. Sexual activity is ritualized or stereotyped and makes use of degraded, reduced, or dehumanized objects.

Exhibitionism

Exhibitionism is the recurrent urge to expose the genitals to a stranger or an unsuspecting person. Sexual excitement occurs in anticipation of the exposure, and orgasm is brought about by masturbation during or after the event. In almost 100 percent of cases, those with exhibitionism are men exposing themselves to women. The dynamic of men with exhibitionism is to assert their masculinity by showing their penises and by watching the victims' reactions—fright, surprise, and disgust. In other related paraphilias, the central themes involve derivatives of looking or showing.

A substance-abusing professional was finally able to attain sobriety at age

33 years. With this accomplishment, he met a woman and got married, began to work steadily for the first time in his life, and was able to impregnate his new wife. His preferred sexual activity had been masturbation in semi-public places. The patient had a strong sense that his mother had always thought him to be inadequate, did not like to spend time with him, and constantly made negative comparisons between him and his "all-boy" younger brother. He recalled several times when his father had tried to explain his mother's antipathy: "It is just one of those things, son: your mother does not seem to like you." Without substance abuse, he gave up his exhibitionism, but he quickly developed sexual incapacity with his wife and became "addicted" to phone sex. (Courtesy of Stephen B. Levine, M.D.)

Specifiers added to exhibitionistic disorder by DSM-5 differentiate between arousal from exposing genitals to prepubertal children, to physically mature individuals, or both (Table 16-15).

Fetishism

In fetishism, the sexual focus is on objects (e.g., shoes, gloves, pantyhose, and stockings) that are intimately associated with the human body, or on nongenital body parts. The latter focus is sometimes called *partialism*. DSM-5 applies the diagnosis of fetishistic disorder to partialism and attaches the following specifiers to fetishistic disorder: body part(s); nonliving parts; other (Table 16-16). The particular fetish used is linked to someone closely involved with a patient during childhood and has a quality associated with this loved, needed, or even traumatizing person. Usually, the disorder begins by adolescence, although the fetish may begin in childhood. Once established, the disorder tends to be chronic.



Table 16-15
Exhibitionism

	DSM-5	ICD-10
Name	Exhibitionistic Disorder	Exhibitionism

Duration	≥6 mo	
Symptoms	Sexual arousal/fantasies/urges/behaviors that result from exposing one's genitals to unsuspecting individuals	Exposing geni public, often of the sexual excitemer associate
Psychosocial Impact	Marked distress and/or psychosocial impairment	
Symptom Specifiers	Sexually aroused by: exposing genitals to prepubertal children exposing genitals to physically mature individuals both	
Severity Specifiers	**In a controlled environment** lives in institution/other controlled environment	
Course Specifiers	**In full remission:** no symptoms or distress for ≥5 yr	



Table 16-16
Fetishism

	DSM-5	ICD-10
Name	Fetishistic Disorder	F65.0 Fetishis
Duration	≥6 mo	
Symptoms	Sexual arousal/fantasies/urges/behaviors that result from objects/nongenital body parts	Sexual arousal/gra from nonliv objects

Psychosocial Impact	Marked distress and/or psychosocial impairment
Exclusions (not result of):	Transvestic disorder Using genital stimulation devices
Symptom Specifiers	Body Part(s) Nonliving object(s) Other
Severity Specifiers	**In a controlled environment** (lives in institution/other controlled environment)
Course Specifiers	**In full remission:** no symptoms or distress for ≥5 yr

The patient may direct sexual activity at the fetish itself (e.g., masturbation with or into a shoe), or may incorporate the fetish into sexual intercourse (e.g., the demand that high-heeled shoes be worn). The disorder is almost exclusively found in men. Learning theorists believe that the object was associated with sexual stimulation at an early age.

A 50-year-old man entered treatment with a chief complaint of erectile disorder experienced primarily with his wife. He was suffering from a moderate depression that related to both his marital issues and business problems. He had no erectile problems with women he picked up in bars or knew and arranged to meet in bars, partly because where he lived, smoking was allowed in bars. The woman's act of smoking a cigarette was necessary to his sexual arousal. His family history included an alcoholic mother and an emotionally abusive father who was a chain smoker. On family car trips, the father would smoke, with all the car windows up. If the patient complained of feeling nauseous, the father would tell him to "shut up." He recalled being very attracted to a Sunday school teacher who smoked when he was six years old. He first smoked when he was 13, sneaking and hiding behind his house. His first cigarette was one he stole from a pack on his mother's night table.

Frotteurism

Frotteurism is usually characterized by a man's rubbing his penis against the buttocks or other body parts of a fully clothed woman to achieve orgasm (Table 16-17). At other times, he may use his hands to rub an unsuspecting victim. The acts usually occur in crowded places, particularly in subways and buses. Those with frotteurism are passive and isolated, and frottage is often their only source of sexual gratification. The expression of aggression in this paraphilia is readily apparent.

Pedophilia

Pedophilia involves recurrent intense sexual urges toward, or arousal by, children. Per DSM-5, the child must be under 14 years old, and the patient must be at least 16 years old and at least five years older than the victims.

Most child molestations involve genital fondling or oral sex. Vaginal or anal penetration of children infrequently occurs, except in cases of incest. Although most child victims coming to public attention are girls, this finding may be a product of the referral process. Offenders report that when they touch a child, most (60 percent) of the victims are boys. This figure is in sharp contrast to the figure for nontouching victimization of children, such as window peeping and exhibitionism; 99 percent of all such cases occur against girls. DSM-5 includes criteria to specify the genders of the patient's target, and Table 16-18 compares the criteria for this disorder.

Of persons with pedophilia, 95 percent are heterosexual, and 50 percent have consumed alcohol to excess at the time of the incident. Many perpetrators concomitantly or have previously committed exhibitionism, voyeurism, or rape.

Incest is related to pedophilia by the frequent selection of an immature child as a sex object, the subtle or overt element of coercion, and occasionally the preferential nature of the adult–child liaison.



Table 16-17
Frotteurism

DSM-5	ICD-10

Name	Frotteuristic Disorder	Other disorder of sexual preference *Frotteurism*
Duration	≥6 mo	
Symptoms	Sexual arousal/fantasies/urges/behaviors that result from touching/rubbing against a nonconsenting individual	Sexual arousa from rubbin against people in public spaces
Psychosocial Impact	Marked distress and/or psychosocial impairment	
Severity Specifiers	**In a controlled environment** (lives in institution/other controlled environment)	
Course Specifiers	**In full remission:** no symptoms or distress for ≥5 yr	
Comments		This category also includes *Necrophilia* in addition t behaviors such as making obscene phone calls engaging in sexual activity with animals, an

use of strangulatic for sexual excitement

A 62-year-old married janitor had worked as a fourth-grade school teacher for 26 years before he transferred school districts, and several years later mysteriously lost his second job. He was referred for help after his family discovered that he had repeatedly fondled the genitals of his 4- and 6-year-old granddaughters. A father of five who had not had sex with his wife for 30 years after strenuously objecting to her cigarette smoking, he was generous, helpful, and cooperative with his children and grandchildren. Intellectually slow, he preferred comic books and had a charming manner of playing with young children "like he was one himself." By his estimate, he had touched the buttocks and genitals of at least 300 girl students, thinking only of how they did not know what he was doing because he was affectionate, and they were too young to realize what was happening. He loved the anticipation and excitement of this behavior. His teaching career ended when parents complained to a principal. The principal discovered that the man had lost his last job for the same reason. (Courtesy of Stephen B. Levine, M.D.)

Sexual Masochism

Masochism takes its name from the 19th-century novelist Sacher-Masoch, whose characters derived sexual pleasure from being abused and dominated by women. According to the DSM-5, persons with sexual masochism have a recurrent preoccupation with sexual urges and fantasies involving the act of being humiliated, beaten, bound, or otherwise made to suffer (Table 16-19). The diagnosis includes asphyxiophilia or autoerotic asphyxiation as a subtype. It is the practice of achieving or heightening sexual arousal with restriction of breathing. Sexual masochistic practices are more common among men than among women. Persons with sexual masochism may have had childhood experiences that convinced them that pain is a prerequisite for sexual pleasure. About 30 percent of those with sexual masochism also have

sadistic fantasies. Moral masochism involves a need to suffer but does not include sexual fantasies.



Table 16-18
Pedophilia

	DSM-5	ICD-10
Name	Pedophilic Disorder	Pedophilia
Duration	≥6 mo	
Symptoms	Sexual arousal/fantasies/urges/behaviors that result from sexual activity with prepubescent children Individual must be 16 yr or older at time of diagnosis, and must also be a minimum of 5 yr older than the child of sexual interest	Sexually preferring young boys and girls Often preferring prepubertal or early pubertal age
# Symptoms Needed	Both of the above	
Exclusion	≤16 yr old Adolescent in relationship with child aged 12 or 13	
Psychosocial Impact	Marked distress and/or psychosocial impairment	
Symptom Specifiers	**Exclusive type** (attracted only to children) **Nonexclusive type** **Sexually attracted to males** **Sexually attracted to females**	

	Sexually attracted to both males and females **Limited to incest**



Table 16-19
Sexual Masochism

	DSM-5	ICD-10
Name	Sexual Masochism Disorder	Sadomasochis
Duration	≥6 mo	
Symptoms	Sexual arousal/fantasies/urges/behaviors that result from being humiliated/hit/bound/made to suffer	Preference for sexual activity to include: Bondage Infliction of pain or humiliatio (as the recipient)
Psychosocial Impact	Marked distress and/or psychosocial impairment	
Symptom Specifiers	**With asphyxiophilia:** arousal from restriction of breathing	
Severity Specifiers	**In a controlled environment:** lives in institution/other controlled environment	
Course Specifiers	**In full remission:** no symptoms or distress for ≥5 yr	
Comments		ICD combined masochism

and sadism into one diagnosis

A 27-year-old woman presented for a job interview. She was accompanied by a man whom she introduced, saying, “This is my lover.” When asked about this, the woman said that her companion had ordered her to bring him and introduce him in that way. She further explained that she was part of a group that utilized sadomasochistic techniques in their sexual play.

Sexual Sadism

Sexual sadism is recurrent and intense sexual arousal from the physical and psychological suffering of another person. The name is from the Marquis de Sade, an 18th-century French author who wrote novels detailing violent sexual acts against women, and was himself imprisoned for such acts. A person must have experienced these feelings for at least 6 months and must have either acted on sadistic fantasies or experienced significant distress as a result of these fantasies to receive a diagnosis of sexual sadism disorder (Table 16-20).

The onset of the disorder is usually before the age of 18 years, and most persons with sexual sadism are male. Sexual sadism likely derives from early abusive life experiences. However, there is little data about how the disorder develops. Sexual sadism is related to rape, although rape is more aptly considered an expression of power. Some sadistic rapists, however, kill their victims after having sex (so-called lust murders). In many cases, these persons have underlying schizophrenia, and they may also suffer from a dissociative disorder or have a history of head trauma.

Voyeurism

Voyeurism, also known as *scopophilia*, is the recurrent preoccupation with fantasies and acts that involve observing unsuspecting persons who are naked or engaged in grooming or sexual activity (Table 16-21). Masturbation to orgasm usually accompanies or follows the event. The first voyeuristic act

usually occurs during childhood, and the paraphilia is most common in men. When persons with voyeurism are apprehended, the charge is usually loitering.

Transvestism

Transvestism, formerly called transvestic fetishism, is described as fantasies and sexual urges to dress in opposite gender clothing as a means of arousal and as an adjunct to masturbation or coitus. DSM also has various specifiers depending on the exact nature of the fantasy, and Table 16-22 compares the criteria for this disorder.



Table 16-20
Sexual Sadism

	DSM-5	ICD-10
Name	Sexual Sadism Disorder	Sadomasochis
Duration	≥6 mo	
Symptoms	Sexual arousal/fantasies/urges/behaviors that result from inflicting physical or psychological suffering/humiliation on others	Preference for sexual activity to include: Bondage Infliction of pain or humiliatio (as the provider)
Psychosocial Impact	Marked distress and/or psychosocial impairment or engaging with nonconsenting person	
Severity Specifiers	**In a controlled environment:** lives in institution/other controlled	

	environment	
Course Specifiers	**In full remission:** no symptoms or distress for ≥5 yr	
Comments		ICD combined masochism and sadism into one diagnosis



Table 16-21
Voyeurism

	DSM-5	ICD-10
Name	Voyeuristic Disorder	Voyeurism
Duration	≥6 mo	
Symptoms	Sexual arousal/fantasies/urges/behaviors that result from observing an unaware person as they undress/have sex	Experiencing sexual pleasure from watching unaware people engage in: sex other intimat activities (undressin
# Symptoms Needed		
Exclusion	≤18 yr old	
Psychosocial	Marked distress and or impairment	

Impact	
Symptom Specifiers	
Severity Specifiers	**In a controlled environment:** lives in institution/other controlled environment
Course Specifiers	**In full remission:** no symptoms or distress for ≥5 yr

Transvestism typically begins in childhood or early adolescence. With time, some men want to dress and live permanently as women. Very rarely, women want to dress and live as men. DSM-5 considers this to be a combination of transvestic disorder and gender dysphoria. Usually, a person wears more than one article of opposite sex clothing; frequently, this involves an entire wardrobe. When a man with transvestism cross-dresses, the appearance of femininity may be striking, although not usually to the degree found in transsexualism. When not dressed in women's clothes, men with transvestism may be hypermasculine in appearance and occupation. Cross-dressing can range from solitary, guilt-ridden behaviors to ego-syntonic, social membership in a subculture.

The overt clinical syndrome of transvestism may begin in latency but is more often seen around pubescence or in adolescence. Frank dressing in opposite-sex clothing usually does not begin until the person is relatively independent of their parents.

Other Specified Paraphilic Disorder

DSM-5 includes a generalized category for paraphilias that cause distress but do not fit into one of the previous categories. We describe some examples of these next.

In some cases, we may not wish to specify the exact paraphilia, such as when we do not yet have enough information; in these cases, we would diagnose an unspecified paraphilic disorder.

Telephone and Computer Scatologia. Telephone scatologia involves

obscene phone calling to an unsuspecting recipient. Tension and arousal begin in anticipation of phoning; the recipient of the call listens while the telephoner (usually male) verbally exposes his preoccupations or induces her to talk about her sexual activity. The conversation is accompanied by masturbation.

Persons also use interactive computer networks, sometimes compulsively, to send obscene messages by electronic mail and to transmit sexually explicit messages and video images. Because of the anonymity of the users in chat rooms who use aliases, on-line or computer sex (cybersex) allows some persons to play the role of the opposite sex ("genderbending"), which represents an alternative method of expressing transvestic or other fantasies. A danger of on-line cybersex is that pedophiles often make contact with children or adolescents who lure children into meeting and then molest them. Many on-line contacts develop into off-line liaisons. Although some persons report that the off-line encounters develop into meaningful relationships, most such meetings are filled with disappointment and disillusionment, as the fantasized person fails to meet the unconscious expectations of the ideal partner. In other situations, when adults meet, rape or even homicide may occur.



Table 16-22
Transvestism

	DSM-5	ICD-10
Name	Transvestic Disorder	Fetishistic transvestis
Duration	≥6 mo	
Symptoms	Sexual arousal/fantasies/urges/behaviors that result from cross-dressing	Sexual excitement from wearing opposite se clothes Desire to

		remove afte arousal declines
Psychosocial Impact	Marked distress and/or impairment	
Symptom Specifiers	**With fetishism** (sexually aroused by specific fabrics or garments) **With autogynephilia** (sexually aroused by imagining self as female)	
Severity Specifiers	**In a controlled environment:** lives in institution/other controlled environment	
Course Specifiers	**In full remission:** no symptoms or distress for ≥5 yr	

Necrophilia. Necrophilia is an obsession with obtaining sexual gratification from cadavers. Most persons with this disorder find corpses in morgues, but some may rob graves or even murder to satisfy their sexual urges. In the few cases studied, those with necrophilia believed that they were inflicting the greatest conceivable humiliation on their lifeless victims. Some experts consider this a type of psychosis.

Partialism. Persons with the disorder of partialism concentrate their sexual activity on one part of the body to the exclusion of all others. Mouth–genital contacts—such as cunnilingus (oral contact with a woman's external genitals), fellatio (oral contact with the penis), and anilingus (oral contact with the anus)—is typically associated with foreplay. However, when a person uses these activities as the sole source of sexual gratification and cannot have or refuses to have coitus, a paraphilia exists. It is also known as *oralism.* As described earlier, transvestism is a type of partialism but differs in that the focus is not on a sexual organ.

Zoophilia. In zoophilia, persons incorporate animals into arousal fantasies

or sexual activities, including intercourse, masturbation, and oral–genital contact. Zoophilia, as an organized paraphilia, is rare. For many persons, animals are the primary source of relatedness, so it is not surprising that some may use a broad variety of domestic animals sensually or sexually.

Sexual relations with animals may occasionally be an outgrowth of availability or convenience, especially in parts of the world where rigid convention precludes premarital sexuality and in situations of enforced isolation. Because masturbation is also available in such situations, however, a predilection for animal contact is probably present in opportunistic zoophilia.

Coprophilia, Urophilia, and Klismaphilia. Coprophilia is sexual pleasure associated with the desire to defecate on a partner, to be defecated on, or to eat feces (coprophagia). A variant is the compulsive utterance of obscene words (coprolalia). Urophilia, a form of urethral eroticism, is an interest in sexual pleasure associated with the desire to urinate on a partner or to be urinated on. The disorder may be associated with masturbatory techniques involving the insertion of foreign objects into the urethra for sexual stimulation. Similarly, klismaphilia is the use of enemas for sexual stimulation.

Hypoxyphilia. Hypoxyphilia is the desire to achieve an altered state of consciousness secondary to hypoxia while experiencing orgasm. Persons may use a drug (e.g., a volatile nitrite or nitrous oxide) to produce hypoxia. Autoerotic asphyxiation is related and is a form of sexual masochism.

Differential Diagnosis

Clinicians must differentiate a paraphilia from an experimental act that is not recurrent or compulsive and done for its novelty. Paraphilic activity most likely begins during adolescence. Some paraphilias (especially the bizarre types) are associated with other mental disorders, such as schizophrenia. Brain diseases can also release perverse impulses.

Course and Prognosis

The difficulty in controlling or curing paraphilic disorders rests in the fact

that it is hard for people to give up sexual pleasure with no assurance that alternative approaches will be as sexually gratifying. The prognosis is poor when it begins at an early age, and when it occurs frequently. Also, a lack of guilt is a bad predictor of outcome, as is a substance use disorder. The course and the prognosis are better when patients also engage in regular sexual activity. It may also be better when patients self-refer than are forced to present for legal reasons.

Treatment

There are five types of psychiatric interventions used to treat persons with paraphilic disorder: external control, reduction of sexual drives, treatment of comorbid conditions (e.g., depression or anxiety), cognitive-behavioral therapy, and dynamic psychotherapy.

Prison is an external control mechanism for sexual crimes that usually does not contain a treatment element. When victimization occurs in a family or work setting, the external control comes from informing supervisors, peers, or other adult family members of the problem and advising them about eliminating opportunities for the perpetrator to act on urges.

Drug therapy, including antipsychotic or antidepressant medication, is indicated for the treatment of schizophrenia or depressive disorders if the paraphilia is associated with these disorders. Antiandrogens, such as cyproterone acetate in Europe and medroxyprogesterone acetate (Depo-Provera) in the United States, may reduce the drive to behave sexually by decreasing serum testosterone levels to subnormal concentrations. Serotonergic agents, such as fluoxetine, have been used with limited success in some patients with paraphilia.

Cognitive-behavioral therapy can disrupt learned paraphilic patterns and modify behavior to make it socially acceptable. The interventions include social skills training, sex education, cognitive restructuring (confronting and destroying the rationalizations used to support the victimization of others), and development of victim empathy. The therapist can also teach the patient imaginal desensitization and relaxation technique. The patient can also learn what triggers the paraphilic impulse so that they can avoid these stimuli. In modified aversive behavior rehearsal, the therapist videotapes the perpetrators acting out their paraphilia with a mannequin. Then the therapist

or a peer group may confront the patient with questions about feelings, thoughts, motives associated with the act and repeatedly try to correct cognitive distortions and point out the lack of victim empathy to the patient.

Insight-oriented psychotherapy is a long-standing treatment approach. Patients have the opportunity to understand their dynamics and the events that caused the paraphilia to develop. In particular, they become aware of the daily events that cause them to act on their impulses (e.g., a real or fantasized rejection). Treatment helps them deal more effectively with life stresses and enhances their capacity to relate to a life partner. Also, psychotherapy allows patients to regain self-esteem, which in turn allows them to approach a partner in a more normal sexual manner. Sex therapy is an appropriate adjunct to the treatment of patients with specific sexual dysfunctions when they attempt nondeviant sexual activities.

Good treatment predictors include the presence of a single paraphilia, normal intelligence, the absence of substance abuse, the absence of nonsexual antisocial personality traits, and the presence of a successful adult attachment. Paraphilic disorders, however, remain significant treatment challenges even under these circumstances.

Epidemiology

Paraphilias are rare, but the insistent, repetitive nature of the disorders results in a high frequency of such acts. Thus, many people are victims of persons with paraphilic disorders. It may be that the prevalence of paraphilias is much higher than that found in clinical care, given the large commercial market in paraphilic pornography and paraphernalia. We do not know how many consumers of these materials act on their fantasies or cannot respond to typical erotic stimuli.

Among legally identified cases of paraphilic disorders, pedophilia is most common. Of all children, 10 to 20 percent have been molested by age 18. Because the act involves a child, it has gotten more attention than some other paraphilic disorders. Persons with exhibitionism who publicly display themselves to young children are also commonly apprehended. Those with voyeurism may be apprehended, but their risk is not considerable. Of adult females, 20 percent have been the targets of persons with exhibitionism and voyeurism. Sexual masochism and sexual sadism are underrepresented in any

prevalence estimates. Sexual sadism usually comes to attention only in sensational cases of rape, brutality, and lust murder. The excretory paraphilic disorders rarely come to clinical attention, as it usually takes place between consenting adults or between a sex worker and client. Persons with fetishism rarely become entangled in the legal system. Those with transvestism may be arrested occasionally for disturbing the peace or on other misdemeanor charges if they are men dressed in women's clothes, but an arrest is more common among those with gender identity disorders. Zoophilia, as an actual paraphilic disorder, is rare (Table 16-23).

As usually defined, the paraphilias seem to be mostly male conditions. Fetishism almost always occurs in men. More than 50 percent of all paraphilias have their onset before age 18. Patients with paraphilia frequently have three to five paraphilias, either concurrently or at different times in their lives. This pattern of occurrence is especially the case with exhibitionism, fetishism, sexual masochism, sexual sadism, transvestic fetishism, voyeurism, and zoophilia (see Table 16-23). The occurrence of paraphilic behavior peaks between ages 15 and 25 and gradually declines. DSM-5 suggests the paraphilia designation be reserved for those ages 18 and older to avoid pathologizing healthy sexual curiosity and occasional experimentation in adolescence. In men older than 50, criminal paraphilic acts are rare. Those that occur are practiced in isolation or with a cooperative partner.

Table 16-23
Frequency of Paraphilic Acts Committed by Patients with Paraphilia Seeking Outpatient Treatment

Diagnostic Category	Patients with Paraphilia Seeking Outpatient Treatment (%)	Paraphilic Acts per Patient with Paraphilia
Pedophilia	45	5
Exhibitionism	25	50
Voyeurism	12	17
Frotteurism	6	30

Sexual masochism	3	36
Transvestic fetishism	3	25
Sexual sadism	3	3
Fetishism	2	3
Zoophilia	1	2

[a]Median number.

Courtesy of Gene G. Abel, M.D.

Etiology

Psychosocial Factors. Many paraphilias can be traced back to childhood experiences that condition or socialize children into committing a paraphilic act. The first shared sexual experience can be significant in that regard. Molestation as a child can predispose a person to accept continued abuse as an adult or, conversely, to become an abuser of others. Also, early experiences of abuse that are not explicitly sexual, such as spanking, enemas, or verbal humiliation, can be sexualized by a child and can form the basis for a paraphilia. Such experiences can result in the development of an *eroticized child.*

A 34-year-old man presented for the treatment of an erectile disorder. He was frequently unable to obtain an erection sufficient for coitus with his wife. The problem disappeared whenever she was willing to act out his bondage fantasy and tie him up with ropes, a scenario he intensely desired. He explained that he felt free to be sexual when tied up because it reassured him that he could move vigorously and not hurt the woman. Also, he gave a history of being tied up “in fun” when he was a child by a babysitter who would then tickle him until he begged her to stop.

The onset of paraphilic acts can result from persons’ modeling their behavior on the behavior of others who have carried out paraphilic acts,

mimicking sexual behavior depicted in the media, or recalling emotionally laden events from the past, such as their molestation. Learning theory indicates that because the fantasizing of paraphilic interests begins at an early age and because children are likely to hide their fantasies from others (who might discourage them), the use and misuse of paraphilic fantasies and urges continue uninhibited until late in life. Only then do persons begin to realize that such paraphilic interests and urges are inconsistent with societal norms. By that time, however, the repetitive use of such fantasies has become ingrained, and sexual thoughts and behaviors have become associated with or conditioned to paraphilic fantasies.

In the classic psychoanalytic model, persons with a paraphilia have failed to complete the normal developmental process toward sexual adjustment, but new psychoanalytic approaches have modified the model. What distinguishes one paraphilia from another is the method chosen by a person (usually male) to cope with the anxiety caused by the threat of castration by the father and separation from the mother. However bizarre its manifestation, the resulting behavior provides an outlet for the sexual and aggressive drives that would otherwise channel into normal sexual behavior.

Biologic Factors. Several studies have identified abnormal biologic findings in persons with paraphilias. None has used random samples of such persons; instead, they have extensively investigated patients with paraphilia at large medical centers. Among these patients, those with positive physical findings included 74 percent with abnormal hormone levels, 27 percent with hard or soft neurologic signs, 24 percent with chromosomal abnormalities, 9 percent with seizures, 9 percent with dyslexia, 4 percent with abnormal electroencephalography (EEG) studies, 4 percent with major mental disorders, and 4 percent with an intellectual disorder. It is not clear whether these abnormalities are causally related or incidental to the paraphilia.

Psychophysiological tests can measure the penile volumetric size in response to paraphilic and nonparaphilic stimuli. The procedures may be of use in diagnosis and treatment but are of questionable diagnostic validity because some men can suppress their erectile responses.

SEX ADDICTION AND COMPULSIVITY

The concept of sex addiction developed over the last two decades to refer to persons who compulsively seek out sexual experiences and whose behavior becomes impaired if they are unable to gratify their sexual impulses. This concept derives from the model of substance addiction or such addictive behaviors as gambling. Addiction implies psychological dependence, physical dependence, and the presence of a withdrawal syndrome if the substance or behavior is not available.

DSM-5 does not contain the diagnosis of sex addiction or compulsive sexuality, and the concept remains controversial. Nevertheless, the phenomenon of a person whose life revolves around sex-seeking behavior and activities, who spends an excessive amount of time in such behavior, and who often tries to stop such behavior but is unable to do so is well known to clinicians. Such persons show repeated and increasingly frequent attempts to have a sexual experience, deprivation of which gives rise to symptoms of distress. When encountering such a story, this should alert the clinician to find an underlying cause for the behavior.

Sex addicts cannot control their sexual impulses, which can involve the entire spectrum of sexual fantasy or behavior. Eventually, the need for sexual activity increases, and the person's behavior is mainly motivated by the persistent desire to experience the sex act. The history usually reveals a long-standing pattern of such behavior, which the person repeatedly has tried to stop, but without success. Although a patient may have feelings of guilt and remorse after the act, these feelings do not suffice to prevent its recurrence. The patient may report that their need to act out is most severe during stressful periods or when angry, depressed, anxious, or otherwise dysphoric. Most acts culminate in sexual orgasm. Eventually, the sexual activity interferes with the person's social, vocational, or marital life, which begins to deteriorate. Table 16-24 lists the signs of sexual addiction.



Table 16-24
Signs of Sexual Addiction

1. Out-of-control behavior
2. Severe adverse consequences (medical, legal, interpersonal) due to sexual behavior

3. Persistent pursuit of self-destructive or high-risk sexual behavior
4. Repeated attempts to limit or stop sexual behavior
5. Sexual obsession and fantasy as a primary coping mechanism
6. The need for increasing amounts of sexual activity
7. Severe mood changes related to sexual activity (e.g., depression, euphoria)
8. Inordinate amount of time spent in obtaining sex, being sexual, or recovering from sexual experience
9. Interference of sexual behavior in social, occupational, or recreational activities

Some men who appear to be hypersexual, as manifested by their need to have many sexual encounters or conquests, use their sexual activities to mask deep feelings of inferiority, this is sometimes called *Don Juanism*. Some are unconsciously attracted to other men, which they deny by compulsive sexual contact with women. After having sex, most Don Juans are no longer interested in the woman. The condition is also called *satyriasis*.

In a woman, a similar condition is called nymphomania. Of the few scientific studies of the condition, those patients who were studied usually have had one or more sexual disorders, often including female orgasmic disorder. The woman often has an intense fear of losing love and, through her actions, attempts to satisfy her dependence needs rather than gratify her sexual impulses.

Comorbidity

Many sex addicts have an associated psychiatric disorder. The comorbid diagnosis may be challenging to recognize, given that the addictive disorder produces considerable stress and anxiety on its own. Most common are substance use disorders (up to 80 percent in some studies), which not only complicates the task of diagnosis, but also complicates treatment.

Treatment

Self-help groups based on the 12-step concept used in Alcoholics Anonymous (AA) have been used successfully with many sex addicts. They include such groups as Sexaholics Anonymous (SA), Sex and Love Addicts Anonymous (SLAA), and Sex Addicts Anonymous (SAA). The groups differ in that some are for men or women, or married persons or couples. All advocate some abstinence from either the addictive behavior or sex in general. Should a substance use disorder also be present, the patient often requires referral to AA or Narcotics Anonymous (NA) as well. Patients may enter an inpatient treatment unit when they lack sufficient motivation to control their behavior on an outpatient basis or pose a danger to themselves or others. Also, severe medical or psychiatric symptoms may require careful supervision and treatment best carried out in a hospital.

A 42-year-old married businessman with two children was considered a model of virtue in his community. He was active in his church and on the boards of several charitable organizations. He was living a secret life, however, and would lie to his wife, telling her that he was at a board meeting when he was visiting massage parlors for paid sex. He eventually was engaging in the behavior four to five times a day, and although he tried to quit many times, he was unable to do so. He knew that he was harming himself by putting his reputation and marriage at risk.

The patient presented himself to the psychiatric emergency room, stating that he would prefer to be dead rather than continue the behavior described. He was admitted with a diagnosis of major depressive disorder and started on a daily dose of 20 mg of fluoxetine. He also received 100 mg of medroxyprogesterone intramuscularly once a day. His need to masturbate diminished markedly and ceased entirely on the third hospital day, as did his mental preoccupation with sex. His clinicians discontinued the medroxyprogesterone on the sixth day and discharged him. He continued to take fluoxetine, enrolled in a local SA group, and entered individual and couples psychotherapy. His addictive behavior eventually stopped, he was having satisfactory sexual relations with his wife, and he was no longer suicidal or depressed.

Pharmacotherapy. Certain medications may be of use in treating sex

addiction, however, because of their specific effects on reducing the sex drive. SSRIs reduce libido in some persons. Compulsive masturbation is an example of a behavioral pattern that may benefit from such medication. Medroxyprogesterone acetate diminishes libido in men and, thus, makes it easier to control sexually addictive behavior.

There is little information on the use of antiandrogens in women to control hypersexuality; however, they might be of benefit. Antiandrogenic agents (e.g., cyproterone acetate) are not available in the United States, but in Europe, they have had varying success. Antiandrogenic medications remain controversial, and some clinicians see it as chemical castration.

Psychotherapy. Insight-oriented psychotherapy may help patients understand the dynamics of their behavioral patterns. Supportive psychotherapy can help repair the interpersonal, social, or occupational damage that occurs. Cognitive behavioral therapy helps the patient recognize dysphoric states that precipitate sexual acting out. Marital therapy or couples therapy can help the patient regain self-esteem. It is also helpful to the partners who need assistance in understanding the disease and dealing with their complex reactions to the situation. Finally, psychotherapy may help in the treatment of any associated psychiatric disorder.

NORMAL SEXUALITY

Sexuality is determined by anatomy, physiology, the culture in which a person lives, relationships with others, and developmental experiences throughout the life cycle. It includes the perception of being male or female and private thoughts and fantasies as well as behavior. To the average person, sexual attraction to another person and the passion and love that follow are deeply associated with feelings of intimate happiness.

Healthy sexual behavior brings pleasure to oneself and one’s partner and involves stimulation of the primary sex organs, including coitus; it is devoid of inappropriate feelings of guilt or anxiety and is not compulsive. Societal understanding of what defines normal sexual behavior is inconstant and varies from era to era, reflecting the cultural mores of the time.

Terms

Sexuality and personality cannot be separated as they are related; hence the term *psychosexual,* to describe personality development and functioning as affected by sexuality. The term *psychosexual* applies to more than sexual feelings and behavior, and it is not synonymous with *libido* in the Freudian sense.

Childhood Sexuality

Before Freud described the effects of childhood experiences on the personalities of adults, the universality of sexual activity and sexual learning in children was unrecognized. Most sexual learning experiences in childhood occur without the parents' knowledge, but awareness of a child's sex does influence parental behavior. Male infants, for instance, tend to be handled more vigorously, and female infants cuddled more. Fathers spend more time with their infant sons than with their daughters, and they also tend to be more aware of their sons' adolescent concerns than of their daughters' anxieties. Boys are more likely than girls to be physically disciplined. A child's sex affects parental tolerance for aggression and reinforcement or extinction of activity and intellectual, aesthetic, and athletic interests.

Observation of children reveals that genital play in infants is part of healthy development. According to Harry Harlow, interaction with mothers and peers is necessary for the development of active adult sexual behavior in monkeys, a finding that has relevance to the socialization of children. During a critical period in development, infants are especially susceptible to certain stimuli; later, they may be immune to these stimuli. We do not entirely understand the detailed relation of critical periods to psychosexual development; Freud's stages of psychosexual development—oral, anal, phallic, latent, and genital—presumably provide a broad framework.

Psychosexual Factors

Sexuality depends on four interrelated psychosexual factors: sexual identity, gender identity, sexual orientation, and sexual behavior. These factors affect personality, growth, development, and functioning. Sexuality is something more than physical sex, coital or noncoital, and something less than all behaviors directed toward attaining pleasure.

Sexual Identity, Gender Identity, and Sexual Orientation

Sexual identity is the pattern of a person's biologic sexual characteristics: chromosomes, external genitalia, internal genitalia, hormonal composition, gonads, and secondary sex characteristics. In healthy development, these characteristics form a cohesive pattern that leaves a person in no doubt about his or her sex. Gender identity is a person's sense of maleness or femaleness. Sexual identity and gender identity are interactive. Genetic influences and hormones affect behavior, and the environment affects hormonal production and gene expression (Table 16-25).

Sexual Identity. Modern embryologic studies have shown that all mammalian embryos, whether genetically male (XY genotype) or genetically female (XX genotype), are anatomically female during the early stages of fetal life. Differentiation of the male from the female results from the action of fetal androgens; the action begins about the sixth week of embryonic life and finishes by the end of the third month. Recent research has focused on the possible roles of critical genes in fetal sexual development. A testis develops because of SRY and SOX9 action, and an ovary develops in the absence of such action. DAX1 plays a part in the fetal development of both sexes, and WNT4 action is needed for the development of the müllerian ducts in the female fetus. Other studies have explained the effects of fetal hormones on the masculinization or feminization of the brain. In animals, prenatal hormonal stimulation of the brain is necessary for male and female reproductive and copulatory behavior. The fetus is also vulnerable to exogenously administered androgens during that period. For instance, if a pregnant woman receives sufficient exogenous androgens, her female fetus that possesses ovaries can develop external genitalia resembling those of a male fetus.

In the past, newborns with ambiguous genitalia were assigned their sexual identity at birth. The theory underlying this action was that parents and child would feel less confusion and that the child would accept the assigned sex and more easily develop a stable sense of being male or female. Although this worked for some children, others developed a gender identity at odds with their assigned sex. For example, an infant designated female at birth could instead feel male throughout childhood, and more emphatically at

puberty. In some cases, this conflict led to depression and even suicide. Current practice usually allows the child to develop with the ambiguity, which permits a sense of gender identity to evolve as the child grows. The gender identity is then more congruent with the child's emotional sense of maleness or femaleness. Ideally, the family receives support from a medical team composed of a pediatrician, an endocrinologist, and a psychiatrist throughout this developmental process.



Table 16-25
Classification of Intersexual Disorders[a]

Syndrome	Description
Virilizing adrenal hyperplasia (adrenogenital syndrome)	Results from excess androgens in fetus with XX genotype; most common female intersex disorder; associated with enlarged clitoris, fused labia, hirsutism in adolescence
Turner syndrome	Results from absence of second female sex chromosome (XO); associated with web neck, dwarfism, cubitus valgus; no sex hormones produced; infertile
Klinefelter syndrome	Genotype is XXY; male habitus present with small penis and rudimentary testes because of low androgen production; weak libido; usually assigned as male
Androgen insensitivity syndrome (testicular-feminizing syndrome)	Congenital X-linked recessive disorder that results in inability of tissues to respond to androgens; external genitals look female and cryptorchid testes present; in extreme form patient has breasts, normal external genitals, short blind vagina, and absence of pubic and axillary hair

Enzymatic defects in XY genotype (e.g., 5-α-reductase deficiency, 17-hydroxy-steroid deficiency)	Congenital interruption in production of testosterone that produces ambiguous genitals and female habitus
Hermaphroditism	True hermaphrodite is rare and characterized by both testes and ovaries in same person (may be 46 XX or 46 XY)
Pseudohermaphroditism	Usually the result of endocrine or enzymatic defect (e.g., adrenal hyperplasia) in persons with normal chromosomes; female pseudohermaphrodites have masculine-looking genitals but are XX; male pseudohermaphrodites have rudimentary testes and external genitals and are XY

[a]Intersexual disorders include a variety of syndromes that produce persons with gross anatomical or physiologic aspects of the opposite sex.

Gender Identity. In infants with an unambiguous sexual identity, almost everyone has a firm conviction that "I am male" or "I am female" by 2 to 3 years of age. However, even if maleness and femaleness develop typically, persons must still develop a sense of masculinity or femininity.

Gender identity, according to Robert Stoller, "connotes psychological aspects of behavior related to masculinity and femininity." Stoller considers gender social and sex biologic: "Most often, the two are relatively congruent, that is, males tend to be manly and females womanly." Nevertheless, sex and gender can develop in conflicting or even opposite ways. Gender identity results from an almost infinite series of cues derived from experiences with family members, teachers, friends, and coworkers, and from cultural phenomena. Physical characteristics derived from a person's biologic sex—such as physique, body shape, and physical dimensions—interrelate with an intricate system of stimuli, including rewards and punishment and parental

gender labels, to establish gender identity.

Thus, the formation of gender identity arises from parental and cultural attitudes, the infant's external genitalia, and genetic influence, which is physiologically active by the sixth week of fetal life. Although family, cultural, and biological influences may complicate the establishment of a sense of masculinity or femininity, persons usually develop a relatively secure sense of identification with their biologic sex—a stable gender identity.

GENDER ROLE. Related to, and in part derived from, gender identity is gender role behavior. John Money and Anke Ehrhardt described gender role behavior as all those things that a person says or does to disclose himself or herself as having the status of boy or man, girl or woman, respectively. A gender role is not established at birth, but builds gradually through (1) experiences encountered and transacted through casual and unplanned learning, (2) explicit instruction and inculcation, and (3) spontaneously putting two and two together to sometimes make four and sometimes five. The usual outcome is a congruence of gender identity and gender roles. Although biologic attributes are significant, the primary factor in achieving the role appropriate to a person's sex is learning.

Research on sex differences in children's behavior reveals more psychological similarities than differences. Girls, however, are found to be less susceptible to tantrums after the age of 18 months than are boys, and boys generally are more physically and verbally aggressive than are girls from age two onward. Little girls and little boys are similarly active, but boys are more easily stimulated to sudden bursts of activity when they are in groups. Some researchers speculate that, although aggression is a learned behavior, male hormones may have sensitized boys' neural organizations to absorb these lessons more readily than do girls.

Persons' gender roles can seem to be opposed to their gender identities. They may identify with their sex and yet adopt the dress, hairstyle, or other characteristics of the opposite sex. Alternatively, they may identify with the opposite sex and yet for expediency adopt many behavioral characteristics of their sex. A further discussion of gender issues appears in Chapter 17.

Sexual Orientation. Sexual orientation describes the object of a person's

sexual impulses: opposite sex, same-sex, or both sexes. A group of people have defined themselves as “asexual” and assert this as a positive identity. Some researchers believe this lack of attraction to any object is a manifestation of a desire disorder. Other people wish not to define their sexual orientation at all and avoid labels. Still, others describe themselves as polysexual or pansexual.

The Biology of Sexual Behavior

The Brain

CORTEX. The cortex is involved both in controlling sexual impulses and in processing sexual stimuli that may lead to sexual activity. In studies of young men, some areas of the brain are more active during sexual stimulation than others. These include the orbitofrontal cortex, which is involved in emotions; the left anterior cingulate cortex, which is involved in hormone control and sexual arousal; and the right caudate nucleus, whose activity is a factor in whether sexual activity follows arousal.

LIMBIC SYSTEM. In all mammals, the limbic system is directly involved with elements of sexual functioning. Chemical or electrical stimulation of the lower part of the septum and the contiguous preoptic area, the fimbria of the hippocampus, the mammillary bodies, and the anterior thalamic nuclei have all elicited penile erections.

Studies of the brain in women have revealed that those areas activated by emotions of fear or anxiety are notably quiescent when the woman experiences an orgasm.

BRAINSTEM. Brainstem sites exert inhibitory and excitatory control over spinal sexual reflexes. The nucleus paragigantocellularis projects directly to pelvic efferent neurons in the lumbosacral spinal cord, apparently causing them to secrete serotonin, which is known to inhibit orgasms. The lumbosacral cord also receives projections from other serotonergic nuclei in the brainstem.

BRAIN NEUROTRANSMITTERS. Many neurotransmitters, including dopamine, epinephrine, norepinephrine, and serotonin, are produced in the brain and affect sexual function. For example, an increase in dopamine likely increases

libido. Serotonin, produced in the upper pons and midbrain, exerts an inhibitory effect on sexual function. Oxytocin is released with orgasm and may reinforce pleasurable activities.

Spinal Cord. Ultimately, sexual arousal and climax are organized at the spinal level. Afferents from the pudendal, pelvic, and hypogastric nerves convey the sensory stimuli related to sexual function. Several separate experiments suggest that spinal neurons mediate sexual reflexes in the central gray region of the lumbosacral segments.

Physiologic Responses. Sexual response is a psychophysiological experience. Both psychological and physical stimuli trigger arousal and an aroused person experiences both physiologic and emotional tension. With orgasm, usually, subjective perception of a peak of physical reaction and release occurs along with a feeling of well-being. Psychosexual development, psychological attitudes toward sexuality, and attitudes toward one's sexual partner are directly involved with, and affect, the physiology of human sexual response.

Usually, men and women experience a sequence of physiologic responses to sexual stimulation. In the first detailed description of these responses, Masters and Johnson observed that the physiologic process involves increasing levels of vasocongestion and myotonia (tumescence) and the subsequent release of the vascular activity and muscle tone as a result of orgasm (detumescence). Tables 16-26 and 16-27 describe the physiologic male and female sexual response cycles. It is important to remember that the sequence of responses can overlap and fluctuate. A sexual fantasy or the desire to have sex frequently precedes the physiologic responses of excitement, orgasm, and resolution, particularly in the male. Also, a person's subjective experiences are as relevant to sexual satisfaction as the objective physiologic response. Figures 16-1 and 16-2 illustrate several possible patterns in the phases of the male sexual response and female sexual response, respectively.

Hormones and Sexual Behavior. In general, substances that increase dopamine levels in the brain increase desire, whereas substances that augment serotonin decrease desire. Testosterone increases libido in both men

and women, although estrogen is a crucial factor in the lubrication involved in female arousal and may increase sensitivity in the woman to stimulation. Recent studies indicate that estrogen is also a factor in the male sexual response and that a decrease in estrogen in the middle-aged male results in higher fat accumulation just as it does in women. Progesterone mildly depresses desire in men and women as do excessive prolactin and cortisol. Oxytocin is involved in pleasurable sensations during sex and increases in men and women after orgasm.

Gender Differences in Desire and Erotic Stimuli

Sexual impulses and desire exist in men and women. In measuring desire by the frequency of spontaneous sexual thoughts, interest in participating in sexual activity, and alertness to sexual cues, males generally possess a higher baseline level of desire than do women, which may be biologically determined. Motivations for having sex, other than desire, exist in both men and women but seem to be more varied and prevalent in women. In women, they may include a wish to reinforce the pair bond, the need for a feeling of closeness, a way of preventing the man from straying, or a desire to please the partner.

Although explicit sexual fantasies are common to both sexes, the external stimuli for the fantasies frequently differ for men and women. Many men respond sexually to visual stimuli of nude or barely dressed women. Women report responding sexually to romantic stories such as a demonstrative hero whose passion for the heroine impels him toward a lifetime commitment to her. A complicating factor is that a woman’s subjective sense of arousal is not always congruent with her physiologic state of arousal. Her sense of excitement may reflect a readiness to be aroused rather than physiologic lubrication. Conversely, she may experience signs of arousal, including vaginal lubrication, without being aware of them. This situation rarely occurs in men.

Masturbation

Masturbation is usually a standard precursor of object-related sexual behavior. To paraphrase MS Patton, no other form of sexual activity is more discussed, more condemned, and more universally practiced. Research by

Kinsey into the prevalence of masturbation indicated that nearly all men and three-fourths of all women masturbate sometime during their lives.

Longitudinal studies of development show that sexual self-stimulation is common in infancy and childhood. Just as infants learn to explore the functions of their fingers and mouths, they learn to do the same with their genitalia. At about 15 to 19 months of age, both sexes begin genital self-stimulation. Pleasurable sensations result from any gentle touch to the genital region. Those sensations, coupled with the ordinary desire for exploration of the body, produce a healthy interest in masturbatory pleasure at that time. Children also develop an increased interest in the genitalia of others—parents, children, and even animals. As youngsters acquire playmates, the curiosity about their own and others' genitalia motivates episodes of exhibitionism or genital exploration. Such experiences, unless blocked by guilty fear, contribute to continued pleasure from sexual stimulation.

With the approach of puberty, the upsurge of sex hormones, and the development of secondary sex characteristics, sexual curiosity intensifies, and masturbation increases. Adolescents are physically capable of coitus and orgasm but are usually inhibited by social restraints. The dual and often conflicting pressures of establishing their sexual identities and controlling their sexual impulses produce a strong physiologic sexual tension in teenagers that demands release, and masturbation is a healthy way to reduce sexual tensions. In general, males learn to masturbate to orgasm earlier than females and masturbate more frequently. An important emotional difference between the adolescent and the youngster of earlier years is the presence of coital fantasies during masturbation in the adolescent. These fantasies are an essential adjunct to the development of sexual identity; in the comparative safety of the imagination, the adolescent learns to perform the adult sex role. This autoerotic activity continues into the young adult years when coitus becomes more common.



Table 16-26
Male Sexual Response Cycle[a]

Organ	Excitement Phase	Orgasmic Phase	Resolution Phase

	Lasts several minutes to several hours; heightened excitement before orgasm, 30 s to 3 min	3–5 s	10–15 min; if no orgasm, ½–1 day
Skin	Just before orgasm: sexual flush inconsistently appears; maculopapular rash originates on abdomen and spreads to anterior chest wall, face, and neck and can include shoulders and forearms	Well-developed flush	Flush disappears in reverse order of appearance; inconsistently appearing film of perspiration on soles of feet and palms of hands
Penis	Erection in 10–30 s caused by vasocongestion of erectile bodies of corpus cavernosa of shaft; loss of erection may occur with introduction of asexual stimulus, loud	Ejaculation; emission phase marked by three to four 0.8-s contractions of vas, seminal vesicles, prostate; ejaculation proper	Erection: partial involution in 5–10 s with variable refractory period; full detumescence in 5–30 min

	noise; with heightened excitement, size of glands and diameter of penile shaft increase further	marked by 0.8-s contractions of urethra and ejaculatory spurt of 12–20 in at age 18, decreasing with age to seepage at 70	
Scrotum and testes	Tightening and lifting of scrotal sac and elevation of testes; with heightened excitement, 50% increase in size of testes over unstimulated state and flattening against perineum, signaling impending ejaculation	No change	Decrease to baseline size because of loss of vasocongestion; testicular and scrotal descent within 5–30 min after orgasm; involution may take several hours if no orgasmic release takes place
Cowper glands	2–3 drops of mucoid fluid that contain	No change	No change

	viable sperm are secreted during heightened excitement		
Other	Breasts: inconsistent nipple erection with heightened excitement before orgasm Myotonia: semispastic contractions of facial, abdominal, and intercostal muscles Tachycardia: up to 175 beats a minute Blood pressure: rise in systolic 20–80 mm; in diastolic 10–40 mm Respiration: increased	Loss of voluntary muscular control Rectum: rhythmical contractions of sphincter Heart rate: up to 180 beats a minute Blood pressure: up to 40–100 mm systolic; 20–50 mm diastolic Respiration: up to 40 respirations a minute	Return to baseline state in 5–10 min A refractory period follows orgasm, during which time the male cannot be rearoused to erection and is unresponsive to stimulation. The length of the refractory period is age and situation dependent

[a]A desire phase consisting of sex fantasies and desire to have sex precedes excitement phase.

Table by Virginia Sadock, M.D.

Couples in a sexual relationship do not abandon masturbation entirely. When coitus is unsatisfactory or is unavailable because of illness or the

absence of the partner, self-stimulation often serves an adaptive purpose, combining sensual pleasure and tension release.

Kinsey reported that when women masturbate, most prefer clitoral stimulation. Masters and Johnson stated that women prefer the shaft of the clitoris to the glans because the glans is hypersensitive to intense stimulation. Most men masturbate by vigorously stroking the penile shaft and glans.

Several studies found that in men, orgasm from masturbation raised the serum prostate-specific antigen (PSA) significantly. Male patients scheduled for PSA tests should be advised not to masturbate (or have coitus) for at least seven days before the examination.

Moral taboos against masturbation have generated myths that masturbation causes mental illness or decreased sexual potency. No scientific evidence supports such claims. Masturbation is a psychopathological symptom only when it becomes a compulsion beyond a person's willful control. Then, it is a symptom of emotional disturbance, not because it is sexual, but because it is compulsive. Masturbation is probably a universal aspect of psychosexual development, and, in most cases, it is adaptive.

Coitus

The first coitus is a rite of passage for both men and women. In the United States, most people have experienced coitus by young adulthood. In a study of people ages 18 to 59, over 95 percent had included coitus in their last sexual interaction.



Table 16-27
Female Sexual Response Cycle[a]

Organ	Excitement Phase	Orgasmic Phase	Resolution Phase
	Lasts several minutes to several hours; heightened excitement before orgasm,	3–15 seconds	10–15 minutes; if no orgasm, ½–1 day

	30 s to 3 min		
Skin	Just before orgasm: sexual flush inconsistently appears; maculopapular rash originates on abdomen and spreads to anterior chest wall, face, and neck; can include shoulders and forearms	Well-developed flush	Flush disappears in reverse order of appearance; inconsistently appearing film of perspiration on soles of feet and palms of hands
Breasts	Nipple erection in two-thirds of women, venous congestion and areolar enlargement; size increases to one-fourth over normal	Breasts may become tremulous	Return to normal in about 30 min
Clitoris	Enlargement in diameter of glans and shaft; just before orgasm, shaft retracts into prepuce	No change	Shaft returns to normal position in 5–10 s; detumescence in 5–30 min; if no orgasm, detumescence

			takes several hours
Labia majora	Nullipara: elevate and flatten against perineum	No change	Nullipara: decrease to normal size in 1–2 min
	Multipara: congestion and edema		Multipara: decrease to normal size in 10–15 min
Labia minora	Size increased two to three times over normal; change to pink, red, deep red before orgasm	Contractions of proximal labia minora	Return to normal within 5 min
Vagina	Color change to dark purple; vaginal transudate appears 10–30 s after arousal; elongation and ballooning of vagina; lower third of vagina constricts before orgasm	3–15 contractions of lower third of vagina at intervals of 0.8 s	Ejaculate forms seminal pool in upper two-thirds of vagina; congestion disappears in seconds or, if no orgasm, in 20–30 min
Uterus	Ascends into false pelvis; labor-like contractions	Contractions throughout orgasm	Contractions cease, and uterus descends to

	begin in heightened excitement just before orgasm		normal position
Other	Myotonia A few drops of mucoid secretion from Bartholin glands during heightened excitement Cervix swells slightly and is passively elevated with uterus	Loss of voluntary muscular control Rectum: rhythmical contractions of sphincter Hyperventilation and tachycardia	Return to baseline status in seconds to minutes Cervix color and size return to normal, and cervix descends into seminal pool

[a]A desire phase consisting of sex fantasies and desire to have sex may precede or overlap with the excitement phase.

Table by Virginia Sadock, M.D.

The young man experiencing intercourse for the first time is vulnerable in his pride and self-esteem. Cultural myths still perpetuate the idea that he should be able to have an erection with no, or little, stimulation, and that he should have an effortless mastery over the situation, even though it is an act that he has never before experienced. Cultural pressure on the woman with her first coitus reflects remaining cultural ambivalence about her loss of virginity, despite the current era of sexual liberality. This reality is illustrated by a survey that found only 50 percent of young women use contraception during their first coitus, and of that 50 percent, an even smaller number use it consistently after that. Young women with a history of masturbation are more likely to approach intercourse with positive anticipation and confidence.

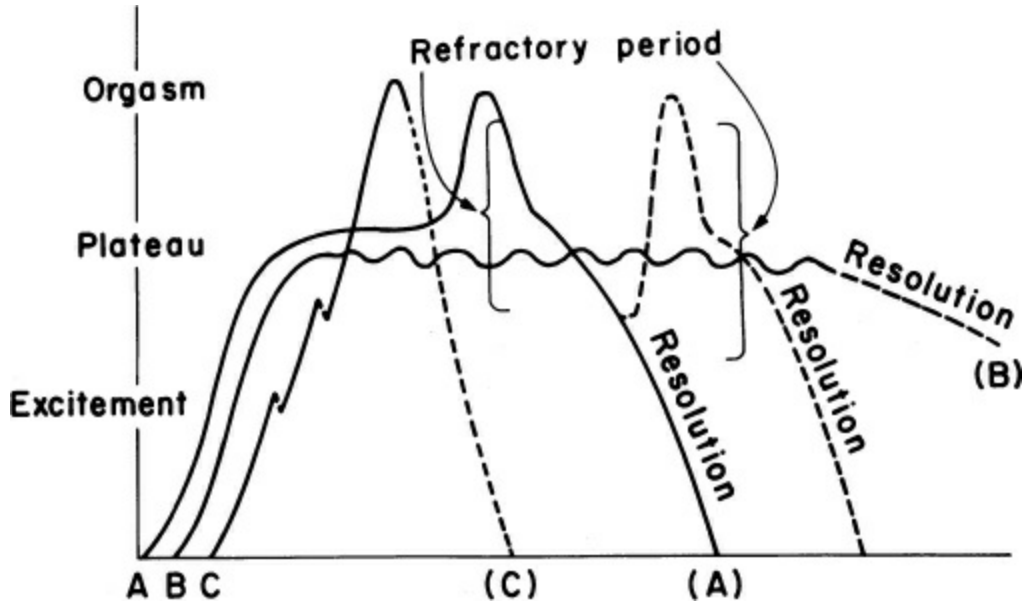


FIGURE 16-1
Male sexual response. An individual man may experience any of these three patterns (**A, B,** or **C**) during a particular sexual experience. (From Walker JI, ed. *Essentials of Clinical Psychiatry*. Philadelphia, PA: JB Lippincott; 1985:276, with permission.)

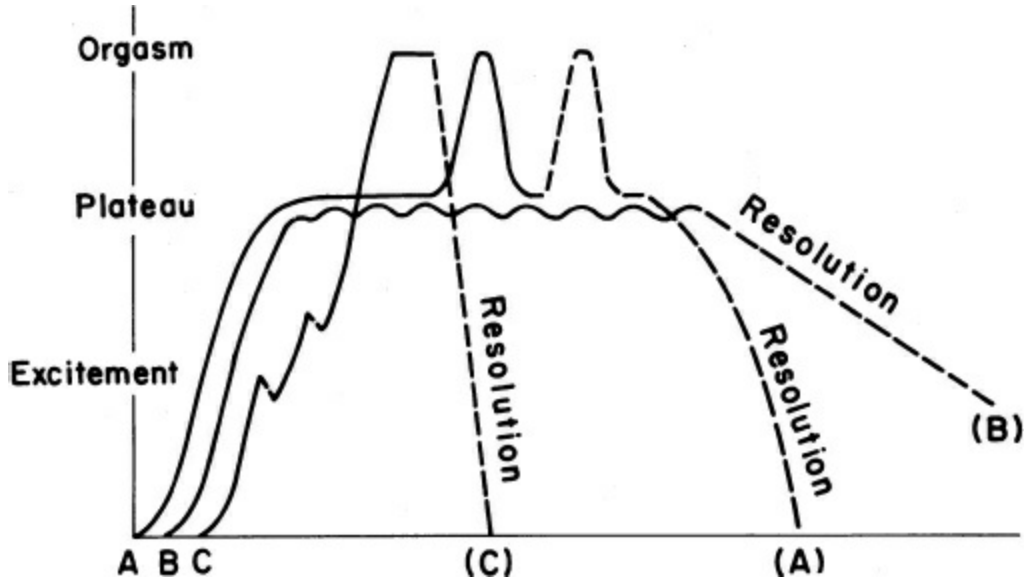


FIGURE 16-2
Female sexual response. An individual woman may experience any of these three patterns (**A, B,** or **C**) during a particular sexual experience. (From Walker JI, ed. *Essentials of Clinical Psychiatry*, Philadelphia, PA: JB Lippincott; 1985:276. with permission.)

In the last decade, coitus has also been part of the sexual repertoire of elderly adults, due to the development of sildenafil type drugs, which

facilitate erections in men, and hormonally enhanced creams or hormonal pills, which counteract vaginal atrophy in postmenopausal women. Before the development of these drugs, many elderly adults enjoyed gratifying sex play, exclusive of coitus.

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17 Gender Dysphoria, Gender Identity, and Related Conditions

INTRODUCTION

In this chapter, we discuss gender identity and associated disorders. We start with gender identity and related conditions.

Gender identity refers to the sense one has of being male or female, which corresponds most often to the person's anatomical sex. Persons with gender dysphoria express their discontent with their assigned sex as a desire to have the body of the other sex or to be regarded socially as a person of the other sex.

The term gender dysphoria appears as a diagnosis for the first time in DSM-5 to refer to those persons with a marked incongruence between their experienced or expressed gender and the one assigned at birth. It was known as gender identity disorder in the previous edition of DSM.

In ICD-10, gender identity issues appear under Disorders of Adult Behavior and Personality in the category Gender Identity Disorders and include five diagnoses: transsexualism, dual-role transvestic, gender identity disorder of childhood, other gender identity disorders, and gender identity disorder, unspecified.

The ICD Working Group on the Classification of Sexual Disorders and Sexual Health is recommending that for ICD-11, gender identity concerns be moved from the psychological sections and is considering options that would list these concerns in a separate chapter, as medical diagnoses, or as part of a new chapter on sexual health and sexual disorders.

The term *transgender* is a general term used to refer to those who identify

with a gender different from the one they were born with (sometimes referred to as their assigned gender). Transgender people are a diverse group: those who want to have the body of another sex known as transsexuals; those who feel they are between genders, of both genders, or neither gender known as genderqueer; and those who wear clothing traditionally associated with another gender, but who maintain a gender identity that is the same as their birth-assigned gender known as cross-dressers. Contrary to popular belief, most transgender people do not have genital surgery. Some do not desire it, and others who do may be unable to afford it. Transgender people may be of any sexual orientation. For example, a transgender man, assigned female at birth, may identify as gay (attracted to other men), straight (attracted to women), or bisexual (attracted to both men and women).

DIAGNOSIS AND CLINICAL FEATURES

Children

The DSM-5 defines gender dysphoria in children as incongruence between expressed and assigned gender, with the most critical criterion being a desire to be another gender or insistence that one is another gender. By emphasizing the importance of the child's self-perception, the creators of the diagnosis attempt to limit its use to those children who clearly state their wishes to be another gender, rather than encompassing a broader group of children who might be considered by adults to be gender nonconforming. However, a child's behavior may also lead to this diagnosis. Table 17-1 compares the approaches to diagnosing gender dysphoria or gender identity disorder.

Many children with gender dysphoria prefer clothing typical of another gender, preferentially choose playmates of another gender, enjoy games and toys associated with another gender, and take on the roles of another gender during play. In gender dysphoria, these social characteristics accompany other traits that are less likely to be socially influenced, such as a strong desire to be the other gender, dislike of one's sexual anatomy, or desire for primary or secondary sexual characteristics of the desired gender. Children may express a desire to have different genitals, state that their genitals are going to change, or urinate in the position (standing or sitting) typical of another gender. Also, although the caregivers may be distressed, it is the

child who must feel clinically significant distress or impairment because of the condition for there to be a diagnosis.

Adolescents and Adults

Adolescents and adults diagnosed with gender dysphoria must also show an incongruence between expressed and assigned gender. Also, they must meet at least two of six criteria, half of which are related to their current (or in the cases of early adolescents, future) secondary sex characteristics or desired secondary sex characteristics. Other criteria include a strong desire to be another gender, be treated as another gender, or the belief that one has the typical feelings and reactions of another gender (see Table 17-1).

In practice, most adults who present to mental health practitioners with reports of gender-related concerns are aware of the concept of transgender identity. They may be interested in therapy to explore gender issues. They may also make contact to request a letter recommending hormone treatment or surgery. The cultural trope of being "trapped in the wrong body" does not apply to all, or even most, people who identify as transgender, so clinicians should be aware of using open and affirming approaches, taking language cues from their patients.

The DSM-5 criteria are noticeably open to the idea that some people do not fit into the traditional gender binary and might desire to be alternative genders, such as genderqueer. As with children, adolescents and adults should be personally distressed or impaired by their feelings. The adolescent and adult criteria also contain a posttransition specifier, which is for people living in their affirmed genders. For this specifier, they have to have had at least one medical or surgical procedure or be preparing for one.



Table 17-1
Gender Dysphoria

	DSM-5	ICD-10
Name	**Gender Dysphoria**	**Gender Identity Disorder (Several Disorders)**
Duration	≥6 mo	

Symptoms	Conflict between experienced and assigned gender: **Children** • Desire or insistence of being another gender • Dressing like the other gender • Preference for other gender roles during play • Preference for other gender toys, games, or stereotypes • Rejection of same gender toys, games, or stereotypes • Preferences for other gender playmates • Dislike of one's anatomy • Desire for physical characteristics of other gender **Adults** • Conflict between experienced gender and primary/secondary sex characteristics • Desire to remove	**Transsexualism:** desire to live as member of the opposite sex **Dual-role transvestism:** temporarily wearing clothes of opposite sex to experience the other gender, but not desiring sex change. **Gender identity disorder of childhood:** • Distress about assigned sex • Desire to be other sex • Dressing like other sex • Disliking one's assigned sex

	one's primary/secondary sex characteristics • Desire for other another gender's primary/secondary sex characteristics • Desire to be another gender • Feeling one has the psychological characteristics of another gender	
# Symptoms needed	Children: ≥6 Adults: ≥2	
Exclusions (not result of):		Not motivated by sexual excitement **In children:** Normal variants ("tomboyishness" or effeminate behavior) Ego-dystonic sexual orientation Sexual maturation disorder
Psychosocial impact	Distress or functional impairment	Distress
Symptom specifiers	With a disorder of sex development	
Course specifiers	Post transition: ≥1 procedure/treatment for gender change	

and is living as the other gender

Other Specified

The category *other specified gender dysphoria* is for cases where the presentation causes clinically significant distress or impairment but does not meet the full criteria for gender dysphoria. When using this diagnosis, the clinician should specify why the full criteria were not met.

Unspecified

The category *unspecified gender dysphoria* can be applied when full criteria are not met, and the clinician chooses not to specify the reason.

A 27-year-old assigned female at birth was referred to a gender identity clinic reporting having felt different as a child from other girls, although unable then to identify the source. As a young girl, she enjoyed playing sports with girls and boys but generally preferred the companionship of boys. She preferred wearing unisex or boyish clothes and resisted wearing a skirt or dress. Everyone referred to her as a tomboy. She tried to hide her breast development by wearing loose-fitting tops and stooping forward. Menses were embarrassing and poignantly reminded her of her femaleness, which was becoming increasingly alienating. As sexual attractions evolved, they were directed exclusively to female partners. In her late teens, she had one sexual experience with a man, and it was aversive. She began socializing in lesbian circles, but did not feel comfortable there and did not consider herself lesbian, but more a man. For sexual partners, she wanted heterosexual women and wanted to be considered by the partner as a man. As gender dysphoric feelings became increasingly pronounced, she consulted transsexual sites on the Internet and contacted a female-to-male transsexual community support group. She then set into motion the process of clinical referral. She transitioned to living as a man, had a name change, and received androgen injections. The patient's voice deepened, facial and body hair grew, menses stopped, and sex drive increased, along with clitoral hypertrophy. After 2 years,

the patient underwent a bilateral mastectomy and was on the waitlist for phalloplasty and hysterectomy–oophorectomy. Employment as a man continues, as does a 3-year relationship with a female partner. The partner has a child from a previous marriage. (Adapted from a case of Richard Green, M.D.)

DIFFERENTIAL DIAGNOSIS

Differential Diagnosis of Children

Children diagnosed with gender dysphoria predicted to be more likely than others to identify as transgender as adults are differentiated from other gender-nonconforming children by statements about desired anatomical changes, as well as the persistence of the diagnosis over time. Children whose gender dysphoria persists over time may make repeated statements about a desire to be or believe that they are another gender. Other gender-nonconforming children may make these statements for short periods but not repeatedly, or may not make these types of statements, and may instead prefer clothing and behaviors associated with another gender but show contentment with their birth-assigned gender.

The diagnosis of gender dysphoria no longer excludes intersex people and instead is coded with a specifier in the cases where intersex people are gender dysphoric about their birth-assigned gender. Medical history is important to distinguish between children with or without intersex conditions. The standards of care for intersex children have changed dramatically over the last few decades due to activism by intersex adults and supportive medical and mental health professionals. Historically, intersex babies were often subjected to early surgical procedures to create a more standard male or female appearance. These procedures had the potential to cause sexual dysfunction, such as inability to orgasm, and permanent sterility. Recently, these practices have changed considerably so that more intersex people can make decisions about their bodies later in life.

Differential Diagnosis of Adolescents and Adults

Those who meet the criteria for a diagnosis of gender dysphoria must

experience clinical distress or impairment related to their gender identity. This requirement excludes the diagnosis of transgender or gender-nonconforming people who are not clinically distressed by their gender identities. There are certain mental illnesses in which transgender identity may be a component of delusional thinking, such as in schizophrenia. However, this is extremely rare and is different from transgender identity or gender dysphoria as the gender identity issues diminish with the treatment of the psychosis. Body dysmorphic disorder may be a differential diagnosis for some patients who present with a desire to change gendered body parts. However, with body dysmorphic disorder, the focus is on the body part looking abnormal, not on gender. Transvestic disorder, discussed in the previous chapter, is a recurrent and intense sexual arousal from cross-dressing that causes clinically significant distress or impairment. This diagnosis is differentiated from gender dysphoria by the patient's gender identity being consistent with their gender assigned at birth, and by sexual excitement linked to cross-dressing coming to interfere with the person's life.

COURSE AND PROGNOSIS

Children

Children typically begin to develop a sense of their gender identity around age 3. At this point, they may develop gendered behaviors and interests, and some may begin to express a desire to be another gender. It is often around school age that children are first brought for clinical consultations, as this is when they begin to interact with classmates closely and to be scrutinized by adults other than their caregivers. Some adults who identify as transgender recall that as children, they did not show behaviors consistent with another gender then. Some say later that they worked hard to appear stereotypical to their assigned gender, whereas others deny being able to recall gender identity concerns. Approaching puberty, many children diagnosed with gender dysphoria begin to show increased levels of anxiety related to anticipated changes to their bodies.

Children diagnosed with gender dysphoria do not necessarily grow up to identify as transgender adults. Several studies have demonstrated that more than half of those diagnosed with gender identity disorder, based on the

DSM-IV, later identify with their birth-assigned gender once they reach adulthood. Those children who do identify as transgender as adults tend to have more extreme gender dysphoria as children. Many studies show increased rates of gay and bisexual identity among those who were gender nonconforming as children.

COMORBIDITY

Comorbidity in Children

Children diagnosed with gender dysphoria show higher rates than other children of depressive disorders, anxiety disorders, and impulse control disorders. This high rate is likely related to the stigma faced by these children related to their gendered behaviors and identities. There are also reports that those diagnosed with gender dysphoria are more likely than others to fall on the autism spectrum. Some researchers posit that this may be related to intrauterine hormone exposure.

Comorbidity in Adults

Adults diagnosed with gender dysphoria show higher rates than other adults of depressive disorders, anxiety disorders, suicidality, self-harming behaviors, and substance abuse. The lifetime rate of suicidal thoughts in transgender people is about 40 percent. The minority stress model predicts increases in mental illness in groups that are stigmatized, discriminated against, harassed, and abused at higher rates than others. DSM-5 reports that persons with late-onset gender dysphoria may have more significant fluctuations in the extent of their distress and more ambivalence about and less satisfaction after sex reassignment surgery.

TREATMENT

Children

Treatment of gender identity issues in children typically consists of individual, family, and group therapy that guides children in exploring their gendered interests and identities. Some providers practice reparative, or conversion therapy, which attempts to change a person’s gender identity or

sexual orientation. This type of therapy is contrary to position statements by the American Psychiatric Association and practice guidelines of the American Academy of Child and Adolescent Psychiatry, and we consider this to treatment to be unethical.

Adolescents

As gender-nonconforming children approach puberty, some show intense fear and preoccupation related to the physical changes they anticipate or are beginning to experience. In addition to providing psychotherapy, many clinicians use these adolescents' reactions to the first signs of puberty as a compass to determine if puberty-blocking medications should be a consideration. Puberty-blocking medications are gonadotropin-releasing hormone (GnRH) agonists that can temporarily block the release of hormones that lead to secondary sex characteristics, giving adolescents and their families time to reflect on the best options moving forward. GnRH agonists are used in other populations (e.g., children with precocious puberty) and thought safe. However, we should consider such steps carefully.

Adults

Treatment of adults who identify as transgender may include psychotherapy to explore gender issues, hormonal treatment, and surgical treatment. Hormonal and surgical interventions may decrease depression and improve the quality of life for such persons.

Mental Health Treatment

The history of poor treatment and medicalization of transgender people by mental health providers has led to a decreased interest on the part of trans-identified people in engaging in mental health care. Many surgeons, and some physicians who prescribe transition-related hormones, require a letter from a mental health provider, so for many transgender people, the mental health worker is a gatekeeper. Many community clinics are now using informed consent models for hormone treatment, thereby decreasing the need for mental health providers to play the role of gatekeepers. The World Professional Association for Transgender Health (WPATH) Standards of Care (SOC) for the health of transsexual, transgender, and gender-

nonconforming people have recently become more flexible and open to informed consent models. Some mental health providers are specializing in working with transgender populations, and this is increasing the rate at which transgender people engage in psychotherapy.

Hormones

Hormone treatment of transgender men most often involves testosterone, usually taken by injection every week or every other week. Initial changes with testosterone therapy include increased acne, muscle mass, and libido, as well as a cessation of menses, usually within the first few months. Subsequent and more permanent changes include deepening of the voice, increased body hair, and enlargement of the clitoris. Monitoring includes hemoglobin/hematocrit levels, as testosterone can rarely cause an increase in red blood cell counts that can lead to stroke. Like all steroid hormones, the liver metabolizes testosterone, so we should obtain routine liver function tests. Clinicians also want to monitor cholesterol and screen for diabetes, as testosterone treatment may increase the likelihood of lipid abnormalities and diabetes. Those beginning hormone treatments are routinely counseled on fertility, as future fertility may be affected by testosterone.

Transgender women may take estrogen, testosterone-blockers, or progesterone, often in combination. These hormones can cause softening of the skin and redistribution of fat, as well as breast growth. Breast development varies between people but does not generally exceed bra cup size B. Experts recommend for patients to be on hormones for 18 to 24 months before having breast augmentation, allowing the breasts to develop to their final size. Sex drive can decrease, as well as erections and ejaculation. Body hair can decrease somewhat, but often not as much as desired, prompting many women to obtain electrolysis. There is no change in voice, as testosterone has permanently altered the vocal cords, and many women seek out voice coaching. Those on estrogen should avoid cigarette smoking, as the combination can lead to an increased risk of blood clots. We should monitor blood pressure, as well as liver function and cholesterol. Also, providers routinely test prolactin as this hormone can increase with estrogen therapy, and in rare cases, transgender women may develop prolactinomas. Reproductive counseling is essential before beginning estrogen treatment

because permanent sterility is almost always the outcome.

Surgery

Many fewer people undergo gender-related surgeries than take hormones. Some people do not desire gender-related surgeries. Others cannot afford them or are not convinced that they will be satisfied with the currently available results.

The most common type of surgery for both trans men and trans women is “top surgery,” or chest surgery. Transgender men may have surgery to construct a male-contoured chest. Trans-women may have breast augmentation.

“Bottom surgery” is less common. Transgender men may have a metoidioplasty, in which the surgeon frees the clitoris from the ligament, attaching it to the body, and adds tissue increasing its length and girth. Scrotoplasty, the placement of testicular implants, is another way to create male-appearing genitalia. Phalloplasty, the creation of a penis, is less commonly performed because it is expensive, involves multiple procedures, requires donor skin from another part of the body, and has limited functionality. Bottom surgery for women is typically vaginoplasty, also commonly known as Sex Reassignment Surgery (SRS). Here, the surgeon removes the testicles, reconstructs the penis to form a clitoris, and creates a vagina. Techniques for vaginoplasty are becoming very good, but the procedure remains expensive. Because of this, some women, especially those with less money, may have orchiectomies, where the surgeon removes the testicles. These can be in-office procedures with a local anesthetic. The procedure is useful for substantially decreasing the body’s production of androgens like testosterone. Less widely discussed, but essential to many women, are facial feminization surgeries that alter the cheeks, forehead, nose, and lips to create a more feminine facial appearance. The face is often used by persons to recognize gender in another person, and having facial features that match the affirmed gender can facilitate social interaction and provide safety from harassment and violence. Transgender men rarely undergo facial surgeries, as testosterone typically causes the face to appear more masculine.

Because surgery is inaccessible to many, there are rare cases of self-surgery, and some people have surgeries performed under unsafe conditions.

Women may inject industrial-grade silicone to produce body curves. Silicone injection done without the supervision of a medical professional can result in body mutilation, infection, and even silicone blood clots that can lead to embolism and death.

EPIDEMIOLOGY

Children

Most children with gender dysphoria come to clinical attention in early grade school years. Parents, however, typically report that the cross-gender behaviors were apparent before 3 years of age. Among a sample of boys younger than age 12 referred for a range of clinical problems, the reported desire to be the other sex was 10 percent. For clinically referred girls younger than age 12, the reported desire to be the other sex was 5 percent. The sex ratio of children referred for gender dysphoria is 4 to 5 boys for each girl, which is likely due in part to societal stigma directed toward feminine boys. The sex ratio is equal in adolescents referred for gender dysphoria. Researchers have observed that many children considered to have shown gender-nonconforming behavior do not grow up to be transgender adults; conversely, many people who later come out as transgender adults report that they were not identified as gender nonconforming during childhood.

Adults

The estimates of gender dysphoria in adults emanate from European hormonal/surgical clinics with a prevalence of 1 in 11,000 male- and 1 in 30,000 female-assigned people. DSM-5 reports a prevalence rate ranging from 0.005 to 0.014 percent for male- and 0.002 to 0.003 percent for female-assigned people. Most clinical centers report a sex ratio of three to five male patients for each female patient. Most adults with gender dysphoria report having felt different from other children of their same sex, although, in retrospect, many could not identify the source of that difference. Many report feeling extensively cross-gender identified from the earliest years, with the cross-gender identification becoming more profound in adolescence and young adulthood. Overall the prevalence of male to female dysphoria is higher than female to male dysphoria. An important factor in diagnosis is that

there is greater social acceptance of birth-assigned females dressing and behaving as boys (so-called tomboys) than there is of birth-assigned males acting as females (so-called sissies). Some researchers speculate that 1 in 500 adults may fall somewhere on a transgender spectrum, based on population data rather than clinical data.

ETIOLOGY

Biologic Factors

For mammals, the resting state of tissue is initially female. Males only develop if the body introduces androgen, which is set off by the Y chromosome. The result is testicular development. Without testes and androgen, female external genitalia develop. Thus, maleness and masculinity depend on fetal and perinatal androgens. Sex steroids govern sexual behavior in lower animals, but this effect diminishes as the evolutionary tree is scaled. Sex steroids influence the expression of sexual behavior in mature men or women; that is, testosterone can increase libido and aggressiveness in women, and estrogen can decrease libido and aggressiveness in men. However, masculinity, femininity, and gender identity are more a product of postnatal life events than the prenatal hormonal organization.

Brain organization theory refers to the masculinization or feminization of the brain in utero. Testosterone affects brain neurons that contribute to the masculinization of the brain in such areas as the hypothalamus. Whether testosterone contributes to so-called masculine or feminine behavioral patterns remains a controversial issue.

Genetic causes of gender dysphoria are under study, but we have not identified candidate genes, and chromosomal variations are uncommon in transgender populations. Case reports of identical twins have shown some pairs that are concordant for transgender issues and others that are not.

A variety of other approaches to understanding gender dysphoria are underway. These include imaging studies that have shown changes in white matter tracts, cerebral blood flow, and cerebral activation patterns in patients with gender dysphoria, but this is preliminary work. An incidental finding is that transgender persons are likely to be left-handed, the significance of which is unknown.

Psychosocial Factors

Children usually develop a gender identity consonant with their assigned sex. Many factors influence the formation of gender identity, including children's temperament and the interaction of that with parents' qualities and attitudes. Culturally acceptable gender roles exist: Society does not expect boys to be effeminate, or girls to be masculine. There are boys' games (e.g., cops and robbers) and girls' toys (e.g., dolls and dollhouses). Children learn these roles, although some investigators believe that some boys are temperamentally delicate and sensitive and that some girls are aggressive and energized—traits that are stereotypically known in today's culture as feminine and masculine, respectively. However, in Western society, we have a higher tolerance for "nontraditional" gender role behaviors than we once did.

Sigmund Freud believed that gender identity problems resulted from conflicts experienced by children within the oedipal triangle. In his view, both real family events and children's fantasies fuel these conflicts. Whatever interferes with a child loving the opposite-sex parent and identifying with the same-sex parent interferes with healthy gender identity development.

Since Freud, psychoanalysts have postulated that the quality of the mother–child relationship in the first years of life is paramount in establishing gender identity. During this period, mothers usually facilitate their children's awareness of, and pride in, their gender: they value their children as little boys and girls. Analysts argue that devaluing, hostile mothering can result in gender problems. At the same time, the separation–individuation process is unfolding. When gender problems become associated with separation–individuation problems, the result can be the use of sexuality to remain in relationships characterized by shifts between a desperate infantile closeness and a hostile, devaluing distance.

Some children receive the message that they would be more valued if they adopted the gender identity of the opposite sex. Rejected or abused children may act on such a belief. A mother's death, extended absence, or depression can influence the process, and a young boy may react by totally identifying with her—that is, by becoming a mother to replace her.

The father's role is also vital in the early years, and his presence can help the separation–individuation process. Without a father, the mother and child

may remain overly close. For a girl, the father usually is the prototype of future love objects; for a boy, the father is a model for male identification.

Learning theory postulates that children may be rewarded or punished by parents and teachers based on gendered behavior, thus influencing the way children express their gender identities. Children also learn how to label people according to gender. Eventually, they also learn that we should not dictate gender based on surface appearances such as clothing or hairstyle.

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18 Disruptive, Impulse-Control, and Conduct Disorders

Six conditions comprise the category of *disruptive, impulse-control, and conduct disorders.* They include two that are associated with childhood: (1) oppositional defiant disorder and (2) conduct disorder; we discuss both in Chapter 2. DSM-5 includes antisocial personality disorder in both this section and in the personality disorders section; we will consider that with the personality disorders (Chapter 19). The remaining three disorders are intermittent explosive disorder, kleptomania, and pyromania, which are discussed here as well as other specified and unspecified disorders that relate to this category.

THE CLINICAL PRESENTATION

Persons suffering from one of these disorders have either an inability to resist an intense impulse or drive or are tempted to perform a particular act that is clearly harmful to themselves or others, or both. Before the event, the individual usually experiences mounting tension and arousal, sometimes—but not consistently—mingled with conscious anticipatory pleasure. Completing the action brings immediate gratification and relief. Within a variable time afterward, the individual experiences a conflation of remorse, guilt, self-reproach, and dread. Shameful secretiveness about the repeated impulsive activity frequently expands to permeate the individual's entire life, often delaying treatment.

Intermittent Explosive Disorder

Patients with intermittent explosive disorder have discrete episodes of losing control of their aggressive impulses; these episodes can result in severe assault or the destruction of property. The aggressiveness expressed is grossly out of proportion to any stressors that may have helped elicit the events. The episodes appear within minutes or hours and, regardless of duration, remit spontaneously and quickly. After each episode, patients usually show genuine regret or guilt, and signs of generalized impulsivity or aggressiveness are absent between events. Clinicians should not diagnose intermittent explosive disorder if they can better explain the symptoms with another disorder: one can imagine a long list of possible disorders, and we discuss some differentials in that section.

A 36-year-old real estate agent sought assistance for difficulty with his anger. He was quite competent at his job, although he frequently lost clients when he became enraged over their indecisiveness. On several occasions, he became verbally abusive, leading clients to find ways out of escrow closings. The impulsive aggression also led to the termination of multiple relationships because sudden angry outbursts contained demeaning accusations toward his girlfriends. These outbursts frequently occurred in the absence of any apparent conflict. On numerous occasions, the patient became so uncontrollably enraged that he threw things across the room, including books, his desk, and the contents of the refrigerator. Between episodes, he was a kind and likable individual with many friends. He enjoyed drinking on the weekends and had a history of two arrests for driving while intoxicated. On one of these occasions, he became involved in a verbal altercation with a police officer. He had a history of drug experimentation in college that included cocaine and marijuana.

A mental status examination revealed a generally cooperative patient. However, he became quite defensive when questioned about his anger and quickly felt accused and blamed by the interviewer for his past behaviors. He had no significant medical history and no signs of neurologic problems. He had never been in psychiatric treatment before this evaluation. He was on no medications. He denied any symptoms of a mood disorder or any other antisocial activity.

Treatment included the use of carbamazepine and a combination of supportive and cognitive-behavioral psychotherapy. The patient's angry outbursts improved as he became aware of the early signs that he was about to lose control. He learned techniques to avoid confrontation when such warning signs occurred. (Courtesy of Vivien K. Burt, M.D., Ph.D. and Jeffrey William Katzman, M.D.)

Kleptomania

Patients with kleptomania (from the Greek *kleptēs,* "thief") describe a recurrent failure to resist the impulse to steal objects, even though they have no need for the objects—often it has no value to the person, monetary or otherwise, and the patient could have easily afforded it. For patients with this disorder, the goal is not to obtain a specific item, it is the act of stealing itself. After taking the objects, the patient often gives them away, returns them secretly, or hides them.

As with other impulse-control disorders, patients with kleptomania experience mounting tension before the act, followed by gratification and lessening of tension after the action. They may or may not feel guilt or remorse and might feel depressed about their behavior. The theft is not planned and does not involve others. Although patients are unlikely to take something when immediate arrest is probable (e.g., in front of a security guard at a store), they may not rationally consider the risks involved even though repeated arrests lead to pain and humiliation. Patients may feel guilt and anxiety after the theft, but they do not feel anger or vengeance.

Jane was a 42-year-old, highly successful, single executive from a wealthy background. She called herself a "shop-'til-you-drop type" and had always been able to afford the expensive designer clothing that she loved. Since college, her "legit" shopping had been paralleled by "boosting" cheap panties and brassieres from discount stores. She did not wear the stolen items; indeed, she considered them "sleazy." She could never bring herself to get rid of them either and kept boxes filled with stolen lingerie in a storage facility.

Jane talked or bought her way out of trouble until her 30s when she

was arrested while stealing pantyhose from the same K-Mart for the third time in as many months. As a condition of probation, she was ordered to see a psychiatrist. Her attendance was sporadic, and several more thefts occurred over the next 2 years. She also experienced substantial depression, which she tried to alleviate by heavy drinking.

Jane finally began taking her problem seriously after yet another arrest precipitated a suicidal gesture. She started keeping appointments regularly and consented to take citalopram and naltrexone. She believes that her participation in an Alcoholics Anonymous (AA) group for high-pressured executives has been at least as effective—if not more so—in controlling her stealing. (Courtesy of Harvey Roy Greenberg, M.D.)



Table 18-1
Intermittent Explosive Disorder

	Intermittent Explosive Disorder	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Intermittent explosive disorder	Other habit and impulse disorders (intermittent explosive disorder)
Duration	If symptom 1, average of 2×/wk for 3 mo If symptom 2, occurs for at least 1 y	
Symptoms	Nonpremeditated, unmotivated verbal or physical aggression, out of proportion to the provocation 1. Not causing damage 2. Causing damage	Persistently repeated maladaptive behavior Failure to resist the impulse Preceded by a period of tension Followed by a period

		of release
Required number of symptoms	Either 1 or 2 above If symptom 2, has to occur at least 3 times within the year	All of above
Psychosocial consequences of symptoms	Distress or impaired functioning (social, occupational, or other significant areas) Or, financial or legal consequences	
Exclusions (not result of):	Medical illness Substance Other psychiatric disorder Age <6	Not secondary to another psychiatric syndrome
Comments		This is diagnosed as a habit and impulse disorder, "other," and not specifically defined



Table 18-2
Kleptomania Diagnoses

	Kleptomania	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Kleptomania	Kleptomania
Symptoms	Recurrent stealing • Failure to resist the impulse to steal • Items stolen are not	Persistently repeated maladaptive behavior • Failure to resist the impulse

	needed • Preceded by a period of tension • Followed by a period of gratification or release	• Preceded by a period of tension • Followed by a period of release
Required number of symptoms	All of above	All of above
Exclusions (not result of):	Anger Vengeance Psychosis Conduct disorder Mania Antisocial personality disorder	Not secondary to another psychiatric syndrome
Comments		Included among the habit and impulse disorders

Pyromania

Patients with pyromania repeatedly and deliberately set fires. They feel an urge to do this and relief or pleasure after they do. They typically are fascinated with all aspects of fires and may set off false alarms. They often are attracted to firefighting, may spend time at their local fire department, volunteer, or even become professional firefighters. Their curiosity is evident, but they show no remorse and may be indifferent to the consequences for life or property. Firesetters may gain satisfaction from the resulting destruction; frequently, they leave apparent clues.

DIAGNOSIS

Intermittent Explosive Disorder

Table 18-1 lists the diagnostic approaches to intermittent explosive disorder in DSM-5 and ICD-10. The key feature of this disorder is an aggressive outburst that has a rapid onset, typically without warning, lasting for less than 30 minutes. There is usually some provocation, although the response is disproportionate to that provocation. A less severe episode may occur in between the more severe ones.

As with most psychiatric disorders, the disorders cause significant distress or impairment in psychosocial functioning and are not better accounted for by another psychiatric, substance use, or other medical disorder. Relatively unique to this disorder is an age requirement, in which the individual has to be above the age of 6 years (or of comparable developmental level).

The diagnosis of intermittent explosive disorder should be the result of history-taking that reveals several episodes of loss of control associated with aggressive outbursts. One discrete episode does not justify the diagnosis. The histories typically describe a childhood in an atmosphere of an alcohol use disorder, violence, and emotional instability. Patients' work histories are poor; they report job losses, marital difficulties, and trouble with the law. Most patients have sought psychiatric help in the past but to no avail. Anxiety, guilt, and depression usually follow an outburst, but this is not a constant finding.

Kleptomania

Table 18-2 lists the diagnostic approaches to kleptomania in DSM-5 and ICD-10. The essential feature of kleptomania is recurrent, intrusive, and irresistible urges or impulses to steal unneeded objects. The act is accompanied by tension before the action and pleasure, gratification, or relief after it. It is not associated with anger or vengeance and is not better explained by some other disorder, as discussed in the Differential section.

Pyromania

Table 18-3 lists the diagnostic approaches to kleptomania in DSM-5 and ICD-10. To be diagnosed with pyromania, patients must set fires, not just desire or imagine them. This action is repetitive, deliberate, and purposeful. They feel tension or arousal before setting the fires as well as a fascination

and attraction to fires. They may make considerable preparations before starting the fire. After setting the fire, they feel pleasure, gratification or relief when witnessing or participating in the aftermath of the fire. Perhaps related to that gratification, they also feel a similar attraction and interest with firefighting, including the activities and equipment associated with that profession. Like the discussion of kleptomania, they do not personally benefit, financially or otherwise, from the fire, hence the distinction between pyromania and arson, and the disorder is not better explained by another cause (see the Differential section).

Other Specified or Unspecified Disorders

DSM includes “other specified or unspecified disorders” for most of the diagnostic categories. This is usually a residual category for disorders that do not meet the criteria for the major disorders in the class. In this case, this group includes some unique disorders, some of which stand at the border between impulsive and compulsive disorders (see the section on Differential Diagnosis for a discussion of the difference). Many of these, such as Internet gaming disorder and related, are better considered as addictive disorders and included in our substance use disorders chapter. This is because even though not due to substances, the phenomenology, pathology, course, and treatment approach of disorders such as Internet gaming disorder appears to be very similar.

OBJECTIVE TESTS FOR THE DISORDERS

Physical Findings and Laboratory Examination

Blood chemistry (liver and thyroid function tests, fasting blood glucose, electrolytes), urinalysis (including drug toxicology), and syphilis serology may help rule out other causes of aggression or impulsive behavior. Magnetic resonance imaging (MRI) may reveal changes in the prefrontal cortex, which is associated with loss of impulse control.

DIFFERENTIAL DIAGNOSIS FOR THE DISRUPTIVE AND IMPULSE-CONTROL DISORDERS

It can be challenging to distinguish between impulse disorders and compulsive disorders, and the terms in practice are sometimes vague and overlapping. Technically, an *impulse* is a tension state that can exist without an action; a *compulsion* is a tension state that always has an action component. In the case of compulsive disorders (such as obsessive-compulsive disorder [OCD] or trichotillomania), the patients feel "compelled" to act out their pathologic behavior; they cannot resist the impulse to do so. Impulses and compulsions also differ by their expected result: people acting on an impulse expect to receive some pleasure, whereas compulsions are often ego-dystonic, meaning that a person does not desire or enjoy the act despite feeling compelled to do it. Even this is complicated, as in some cases impulse-control disorders also elicit guilt, which disturbs the sense of pleasure. Similarly, not all compulsions are ego-dystonic; for example, compulsive video game playing may have a pleasurable component. Both impulsive and compulsive behaviors are repetitive; however, the repeated acting out of impulses leads to psychosocial impairment, whereas compulsive behavior does not always carry that risk.

Intermittent Explosive Disorder

One should only diagnose an intermittent explosive disorder after other disorders associated with the occasional loss of control of aggressive impulses have been ruled out as the primary cause. These other disorders include psychotic disorders, personality change because of a general medical condition, antisocial or borderline personality disorder, substance intoxication (e.g., alcohol, barbiturates, hallucinogens, amphetamines), epilepsy, brain tumors, degenerative diseases, and endocrine disorders.



Table 18-3
Pyromania

	Pyromania	
Disorder	**DSM-5**	**ICD-10**
Diagnostic name	Pyromania	Pyromania

Symptoms	Fire setting • 1+ incidents • Deliberate and purposeful • Preoccupied by fire, burning, and related phenomena • Followed by a period of gratification or release	Fire setting • Multiple acts or attempts • No apparent motive • Preoccupied by fire, burning, and related phenomena • Preceded by a period of tension • Followed by a period of excitement
Required number of symptoms	All of above	All of above
Exclusions (not result of):	• Conduct disorder • Mania • Antisocial personality disorder • Psychosis • Impaired cognition or judgment Motivated by: • Desire for a perceived financial or social benefit • Sociopolitical ideology • Need to conceal criminal activity • Anger • Vengeance	• Dissocial personality disorder • Alcohol or substance disorder • Feigned illness • Conduct disorder • Organic mental disorder • Schizophrenia
Comments		Included among the habit and impulse disorders

The episodic and discrete nature of intermittent explosive disorder differentiates it from conduct disorder and antisocial personality disorder, which are more pervasive patterns of behavior in which aggression is a consistent feature, and not just associated with discrete outbursts.

In schizophrenia, patients may display violent behavior in response to delusions and hallucinations, and they show gross impairments in reality testing. Hostile patients with mania may be impulsively aggressive, but the underlying diagnosis is generally apparent from their mental status examinations and clinical presentations.

Amok is a culture-bound syndrome usually associated with southeastern Asia (but also seen elsewhere), which consists of episodes of acute violent behavior for which the person claims amnesia. It is distinguished from intermittent explosive disorder by a single episode and prominent dissociative features.

Among medical disorders, certain neurologic disorders can resemble impulse-control disorders. The best-documented examples are with Parkinson disease, which can resemble both impulse-control disorders and addictive disorder.

Kleptomania

Shoplifting has become a national epidemic. Few shoplifters have true kleptomania; most are teenagers and young adults who steal in pairs or small groups for “kicks” as well as goods and do not have a psychiatric disorder.

Individuals with an antisocial personality disorder steal for personal gain, with some degree of premeditation and planning. Often they convince others to rob for them. Antisocial stealing regularly involves the threat of harm or actual violence, mainly to elude capture. Guilt and remorse are distinctively lacking, or patients are patently insincere.

Episodes of theft occasionally occur during psychotic illness, for example, acute mania, major depression with psychotic features, or schizophrenia. Psychotic stealing is a product of pathologic elevation or depression of mood or command hallucinations or delusions.

Acute intoxication with drugs or alcohol may precipitate theft in an individual with another psychiatric disorder or without significant psychopathology. Patients with Alzheimer disease or other dementing

organic illness may leave a store without paying, owing to forgetfulness rather than larcenous intent.

Some individuals with an antisocial personality disorder or no disorder may steal and then claim to suffer from kleptomania. This would be an example of malingering, not kleptomania, although with sufficiently intelligent perpetrators it may be challenging to tell the difference.

Pyromania

Clinicians should have little trouble distinguishing between pyromania and the fascination of many young children with matches, lighters, and fire as part of the normal investigation of their environments. One should also separate pyromania from incendiary acts of sabotage carried out by dissident political extremists or by arsonists.

When a fire setting occurs in conduct disorder and antisocial personality disorder, it is a deliberate act, not a failure to resist an impulse. Such individuals may set fires for profit, sabotage, or retaliation. Patients with schizophrenia or mania may set fires in response to delusions or hallucinations. Patients with brain dysfunction (e.g., dementia), mental retardation, or substance intoxication may set fires because of a failure to appreciate the consequences of the act.

COMORBIDITY

Intermittent Explosive Disorder

More than 80 percent of individuals with intermittent explosive disorder meet the criteria for another psychiatric disorder. These include other disorders of impulse-control and substance use, and mood, anxiety, personality (antisocial and borderline), posttraumatic stress, and eating disorders. Also, individuals with intermittent explosive disorder have been reported to be at increased risk of self-harm.

Kleptomania

Patients with kleptomania are said to have high lifetime comorbidity of major mood disorders (usually, but not exclusively, depressive) and various anxiety disorders. Associated conditions also include other disorders such as

pathologic gambling and compulsive shopping, eating disorders (especially bulimia nervosa), personality disorders, OCD, and substance use disorders, especially alcohol use disorder. Individuals with kleptomania are also reported to have higher rates of suicide.

Pyromania

Pyromania is significantly associated with substance use disorders (especially alcohol and marijuana use disorders), affective disorders, depressive or bipolar, and other impulse-control disorders. They are also reported to have increased suicidal thoughts.

COURSE AND PROGNOSIS

Intermittent Explosive Disorder

Intermittent explosive disorder may begin at any stage of life but usually appears between late adolescence and early adulthood. The onset can be sudden or insidious, and the course can be episodic or chronic. In most cases, the disorder decreases in severity with the start of middle age, but heightened organic impairment can lead to frequent and severe episodes. Intermittent explosive disorder is more common in men than women.

Kleptomania

Kleptomania may begin in childhood, although most children and adolescents who steal do not become kleptomaniac adults. The onset of the disorder generally is adolescence. In quiescent cases, loss or disappointment may precipitate new bouts of the disorder.

The course of the disorder waxes and wanes but tends to be chronic—the frequency of stealing ranges from less than one to multiple episodes per month. Most patients with kleptomania steal from retail stores, but they may also take from family members in their households. People with the disorder sometimes have bouts of being unable to resist the impulse to steal, followed by free periods that last for weeks or months. The spontaneous recovery rate of kleptomania is unknown.

Severe impairment and complications are usually secondary to being caught, particularly to being arrested. Many persons seem never to have

consciously considered the possibility of facing the consequences of their acts. Often, the disorder in no way impairs a person's social or work functioning.

The prognosis with treatment can be good, but few patients come for help of their own accord.

Pyromania

It should not be too surprising that we know very little about the development and course of pyromania. Most individuals with pyromania start setting fires in adolescence or young adulthood. The frequency and intensity of fire setting may increase over time or may wax and wane. The course of pyromania is unknown, but smaller studies suggest it may be chronic. Pyromania occurs more frequently in males with weaker social skills and learning difficulties.

TREATMENT APPROACH

Intermittent Explosive Disorder

A combined pharmacologic and psychotherapeutic approach has the best chance of success. Psychotherapy with patients who have an intermittent explosive disorder is difficult, however, because of their angry outbursts. Therapists may have problems with countertransference and limit-setting. Group psychotherapy may be helpful, and family therapy is useful, especially when the explosive patient is an adolescent or a young adult. A goal of treatment is to have the patient recognize and verbalize the thoughts or feelings that precede the explosive outbursts instead of acting them out. Cognitive-behavioral therapy and contingency management may be helpful.

Anticonvulsants have long been used, with mixed results, in treating explosive patients. Lithium has been reported useful in generally lessening aggressive behavior, and carbamazepine, valproate, and phenytoin have been reported helpful. Some clinicians have also used other anticonvulsants (e.g., topiramate). Clinicians occasionally use benzodiazepines but the meds can cause disinhibition, thus worsening the behaviors.

Antipsychotics and tricyclic drugs have been effective in some cases, but clinicians must then question whether schizophrenia or a mood disorder is the

correct diagnosis. With a likelihood of subcortical seizure-like activity, medications that lower the seizure threshold can aggravate the situation. Selective serotonin reuptake inhibitors (SSRIs), trazodone, and buspirone are useful in reducing impulsivity and aggression. Propranolol and other β-adrenergic receptor antagonists and calcium channel inhibitors have also been effective in some cases.

Some neurosurgeons have performed operative treatments for intractable violence and aggression. No evidence indicates that such treatment is effective.

Kleptomania

Because true kleptomania is rare, reports of treatment tend to be individual case descriptions or a short series of cases. Insight-oriented psychotherapy and psychoanalysis have been successful but depend on patients' motivations. Those who feel guilt and shame may be helped by insight-oriented psychotherapy because of their increased motivation to change their behavior.

Case reports and case series suggest that cognitive-behavioral therapy may be helpful. Specific behavior therapies, including systematic desensitization, aversive conditioning, and a combination of aversive conditioning and altered social contingencies, have been reported successful, even when motivation was lacking. SSRIs, such as fluoxetine and fluvoxamine, appear to be effective in some patients with kleptomania; however, there are also reports of fluoxetine having no benefit in kleptomania. Case reports indicated successful treatment with tricyclic drugs, trazodone, lithium, valproate, topiramate, naltrexone, methylphenidate, and electroconvulsive therapy. In a controlled trial of antidepressants, escitalopram was found to have no benefit in treating kleptomania. Though limited by small sample size and open-label design, high-dose naltrexone was reported to be effective in reducing the urge to steal in patients with kleptomania.

Pyromania

Little has been written about the treatment of pyromania, and treating firesetters has been difficult because of their lack of motivation. No single treatment has been proved effective; as a result, most therapists try various

modalities, including behavioral approaches. Among the treatments attempted, CBT appears to be the most promising.

Because of the recurrent nature of pyromania, any treatment program should include the supervision of patients to prevent a repeated episode of fire setting. Incarceration may be the only method of avoiding a recurrence while their therapy can commence.

We should treat firesetting by children with the utmost seriousness. When possible we should consider intensive interventions, but as therapeutic and preventive measures, not as punishment. In the case of children and adolescents, treatment of pyromania or fire setting should include family therapy.

Several case reports have been published using SSRIs, lithium, naltrexone, stimulants, topiramate, valproate, carbamazepine, antiandrogen agents, clonazepam, and olanzapine. Unfortunately, many have reported no benefit from pharmacotherapy.

EPIDEMIOLOGY

For all these disorders, we do not know much about their epidemiology, as they are all underreported. This is understandable given the potential ramifications of self-reporting what is usually illegal behavior.

The lifetime prevalence rate for intermittent explosive disorder is estimated to be around 5 to 8 percent, and it appears to be more common in men than in women. The men are likely to be found in correctional institutions and the women in psychiatric facilities. In one study, about 2 percent of all persons admitted to a university hospital psychiatric service had an intermittent explosive disorder; 80 percent were men.

The prevalence of kleptomania is not known, but it is estimated to be about 0.6 percent. The range varies from 4 to 24 percent of those arrested for shoplifting. The male-to-female ratio is 1:3 in clinical samples.

No information is available on the prevalence of pyromania, but only a small percentage of adults who set fires have pyromania. The disorder is found far more often in men than in women. In the United States, 60 percent of individuals who set fires are adolescents; most, however, would not meet the criteria for pyromania.

ETIOLOGY

For all of these disorders, the cause is not known. Psychodynamic, psychosocial, and biologic factors all undoubtedly play an essential role. Some disruptive and impulse-control disorders may have common underlying neurobiologic mechanisms. Fatigue, constant stimulation, and psychic trauma can lower a person's resistance to control impulses.

Intermittent Explosive Disorder

Other factors associated with impulsive aggression include childhood exposure to violence, maltreatment, and neglect. One study found that there was significant association between intermittent explosive disorder and aversive childhood parenting environment comparable to other psychopathologies. Another study found that physical abuse in childhood was independently associated with intermittent explosive disorder, with impulsivity and aggression as the mediators of the relationship between physical abuse and intermittent explosive disorder. Others have shown that intermittent explosive disorder is associated with childhood exposure to interpersonal traumatic events.

Prefrontal cortical dysfunction has been associated with impulsive aggression. Some investigators suggest that disordered brain physiology, particularly in the limbic system, is involved in most cases of episodic violence. Compelling evidence indicates that serotonergic neurons mediate behavioral inhibition. Decreased serotonergic transmission, which can be induced by inhibiting serotonin synthesis or by antagonizing its effects, reduces the impact of punishment as a deterrent to behavior. The restoration of serotonin activity (e.g., by administering L-tryptophan, a serotonin precursor) restores the behavioral effect of punishment and may aid the control of violent tendencies. Low levels of CSF 5-hydroxyindoleacetic acid (5-HIAA) correlate with violent suicide attempts and impulsive aggression in some studies. High CSF testosterone concentrations are associated with aggressiveness and interpersonal violence in men. Antiandrogenic agents have been shown to decrease aggression.

Evidence indicates that intermittent explosive disorder is more common in first-degree biologic relatives of persons with the disorder than in the general population. This association does not imply a genetic etiology, as many other

familial and social factors are surely influential. A gene–environment interaction may exist where individuals who have specific serotonin gene polymorphisms may be more prone to impulsive aggression behavior after exposure to childhood maltreatment.

Kleptomania

We know little about the etiology of kleptomania. There have been case reports of kleptomania associated with head trauma, frontal lobe lesions, cortical atrophy, dementia, and hypoglycemia secondary to insulinoma. There are implications for subtle white matter pathology based on diffusion tensor imaging results. Another study found that although, as a group, patients with kleptomania did not show deficits on neuropsychological testing, the severity of illness was associated with impairment in executive functioning.

Data are lacking and limited regarding the genetics of kleptomania. A study found that individuals with kleptomania had more first-degree relatives with alcohol use disorders compared to controls. Serotonin and dopamine, as well as the opioid and glutamatergic neurotransmitter systems, may play a role in kleptomania. Additionally, there are higher rates of OCD in relatives of individuals with kleptomania compared to the general population.

Pyromania

The etiology of pyromania is also poorly understood. The literature on the origins of pyromania has suggested factors ranging from temperament to environmental factors and parental psychopathology. Other authors have suggested pyromania, and the other impulse-control disorders, may share a neurobiologic link with substance use disorders. Limited studies suggest that pyromania may be associated with lower concentration of norepinephrine and serotonin metabolites 3-methoxy-4-hydroxyphenylglycol (MHPG) and 5-hydroxyindoleacetic acid (5-HIAA).

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19

Personality Disorders

The person-centered approach to understanding personality and its disorders distinguish psychiatry fundamentally from all other branches of medicine. A person is a self-aware human being, as C. Robert Cloninger said, not "a machine-like object that lacks self-awareness." Personality refers to all of the characteristics that distinguish a continually developing, self-organizing human being from a predictable machine-like object. In other words, personality refers to all the ways someone shapes and adapts in a unique way to an ever-changing internal and external environment.

Persons with personality disorders are far more likely to refuse psychiatric help and to deny their problems than persons with anxiety disorders, depressive disorders, or obsessive-compulsive disorder. In general, personality disorder symptoms are ego-syntonic (i.e., acceptable to the ego, as opposed to ego-dystonic) and alloplastic (i.e., adapt by trying to alter the external environment rather than themselves). Persons with personality disorders do not feel anxiety about their maladaptive behavior. Because they do not routinely acknowledge pain from what others perceive as their symptoms, they often seem disinterested in treatment and impervious to recovery.

THE CLINICAL PRESENTATION

Persons suffering from one of these disorders have long-standing, maladaptive behavior traits that are pervasive and apparent in a wide range of personal and social contexts.

Paranoid Personality Disorder

Persons with paranoid personality disorder have long-standing suspiciousness and mistrust of persons in general. They refuse responsibility for their feelings and assign responsibility to others. They are often hostile, irritable, and angry.

Schizoid Personality Disorder

Schizoid personality disorder is characterized by a lifelong pattern of social withdrawal. Others often see persons with schizoid personality disorder as eccentric, isolated, or lonely. Their discomfort with human interaction, their introversion, and their bland, constricted affect are noteworthy.

Schizotypal Personality Disorder

Persons with schizotypal personality disorder are strikingly odd or strange, even to laypersons. Magical thinking, peculiar notions, ideas of reference, illusions, and derealization are commonplace.

Antisocial Personality Disorder

Antisocial personality disorder is an inability to conform to the social norms that ordinarily govern many aspects of a person’s adolescent and adult behavior. Although often characterized by continual antisocial or criminal acts, the disorder is not synonymous with criminality.

Borderline Personality Disorder

Patients with borderline personality disorder stand on the border between neurosis and psychosis, and they have extraordinarily unstable affect, mood, behavior, object relations, and self-image.

Histrionic Personality Disorder

Persons with histrionic personality disorder are excitable and emotional and behave in a colorful, dramatic, extroverted fashion. Accompanying their flamboyant aspects, however, is often an inability to maintain sincere, long-lasting attachments.

Narcissistic Personality Disorder

Individuals with narcissistic personality disorder have a heightened sense of self-importance, lack of empathy, and grandiose feelings of uniqueness. Underneath, however, their self-esteem is fragile and vulnerable to even minor criticism.

Avoidant Personality Disorder

Persons with avoidant personality disorder show extreme sensitivity to rejection and may lead socially withdrawn lives. Although shy, they are not asocial and show a great desire for companionship, but they need unusually strong guarantees of uncritical acceptance. We often describe this group as having an inferiority complex.

Dependent Personality Disorder

Patients with dependent personality disorder subordinate their own needs to those of others, get others to assume responsibility for significant areas of their lives, lack self-confidence, and may experience intense discomfort when alone for more than a brief period.

Obsessive-Compulsive Personality Disorder

Persons with obsessive-compulsive personality disorder typically are emotionally constricted, orderly, perseverative, stubborn, and indecisive. The essential feature of the disorder is a pervasive pattern of perfectionism and inflexibility.

Personality Change due to a General Medical Condition

Personality change due to a general medical condition is a significant occurrence. ICD-10 includes the category personality and behavioral disorders due to brain disease, damage, and dysfunction, which includes organic personality disorder, postencephalitic syndrome, and postconcussional syndrome. Persons with this disorder have a marked change in personality style and traits from their previous level of functioning. Patients must show evidence of a causative medical factor antedating the onset of personality change.

DIAGNOSIS

DSM-5's categorical approach to personality disorders is problematic. It seems counterintuitive to think that we can boil down disturbed personality into a few symptoms. Experts in the field have argued for a more dimensional approach that is grounded in research on temperament and personality, some of which we later describe. The DSM-5 Personality Disorders Workgroup did work on such a change, in which the clinician would focus on the impairments in a person's personality functioning, using an approach based on a widely accepted model of personality. Although the authors of DSM-5 did consider such a change, they felt it too radical a leap for clinical use. Instead, they included the approach that the workgroup recommended in a later section of the book, "Emerging Measures and Models." For now, the categorical approach remains. However, this remains a controversial area of psychiatry, and many rightly question the validity of this approach.

ICD-10, with its more descriptive approach, is a little less controversial. However, despite some wording differences, its approach is similar to DSM-5's, although it lists more disorders than does DSM-5.

Table 19-1 compares the general approaches to diagnosing personality disorders.

DSM-5 divides the disorders into three categories, or clusters: A, B, and C. Cluster A includes three personality disorders with odd, aloof features (paranoid, schizoid, and schizotypal). Cluster B includes four personality disorders with dramatic, impulsive, exploitative, and erratic features (borderline, antisocial, narcissistic, and histrionic). Cluster C includes three personality disorders sharing anxious and fearful features (avoidant, dependent, and obsessive-compulsive). Individuals frequently exhibit traits that are not limited to a single personality disorder. When a patient meets the criteria for more than one personality disorder, clinicians should diagnose each.

Many of the features of various personality disorders can occur during an episode of another mental disorder. We should only use a diagnosis of personality disorder when the features are typical of long-term functioning and are not limited to a discrete episode of another mental disorder. Likewise, when maladaptive behavior features are due to the direct psychological effects of another mental disorder, including substance use, a diagnosis of

personality disorder is not warranted. However, it is possible to diagnose personality change due to another medical condition such as temporal lobe epilepsy; in this case, we must specify the medical condition along with the associated diagnosis of personality change.



Table 19-1
Personality Disorders

	DSM-5	ICD-10
Name	**General Personality Disorder**	**Disorders of Adult Personality and Behavior**
Duration	Persistent, long duration, usually since teens or early adulthood	Persistent
Symptoms	The symptoms comprise a fixed pattern of behavior. The domains include: • Cognitive and perceptual • Mood and affect • Interpersonal • Behavioral	Consistent, fixed, and inflexible pattern of behaviors, involving • Dysfunctional relationships • Distorted view of oneself
# Symptoms Needed	2/4 of the domains above	
Exclusions (not	Another mental	

result of):	disorder Substance use Another medical illness	
Psychosocial Impact	Distress, functional impairment	Usually causes distress Difficulties with social performance
Specific Disorders	• **Cluster A:** paranoid personality disorder, schizoid personality disorder, schizotypal personality disorder • **Cluster B:** antisocial personality disorder, borderline personality disorder, histrionic personality disorder, narcissistic personality disorder • **Cluster C:** avoidant personality disorder,	• **Specific personality disorder:** Paranoid personality disorder, schizoid personality disorder, dissocial personality disorder, emotionally unstable personality disorder, histrionic personality disorder, anankastic personality disorder, anxious [avoidant] personality disorder, dependent personality disorder, other specific personality disorder (eccentric, haltlose, immature, narcissistic, passive-aggressive, psychoneurotic), personality disorder unspecified • **Mixed and other personality disorders** • **Enduring personality changes:** due to brain damage and disease, after catastrophic experience, after psychiatric illness, chronic pain personality syndrome, enduring personality change

	dependent personality disorder, obsessive-compulsive personality disorder	unspecified

This compares the general criteria for all the disorders. Please see additional tables for specific personality disorders.



Table 19-2
Paranoid Personality Disorder: Specific Symptoms

DSM-5		ICD-10	
Paranoid Personality Disorder		**Paranoid Personality D**	
Suspiciousness, evident from the following (≥4): • Suspicion/persecutory thoughts • Doubting others' loyalty • Distrust, inability to confide in others • Interpreting experiences as threatening or hostile • Holds grudges • Feeling attacked by others • Distrusting partner's faithfulness	Exclusions (not result of): • *Schizophrenia* • *Depression with psychotic features* • *Other psychoses* • *Substance use* • *A medical condition*	• Excessive sensitivity to setbacks or insults • Suspiciousness • Interpreting experiences as threatening or hostile • Distrusting partner's faithfulness • May also be self-important or overall self-referential	Exc • I • S • I

Cluster A: Odd and Eccentric

Paranoid Personality Disorder. The hallmarks of paranoid personality disorder are excessive suspiciousness and distrust of others expressed as a pervasive tendency to interpret the actions of others as deliberately demeaning, malevolent, threatening, exploiting, or deceiving. This tendency begins in early adulthood and appears in a variety of contexts. Almost invariably, those with the disorder expect to be exploited or harmed by others in some way. They frequently dispute, without any justification, friends' or associates' loyalty or trustworthiness. Such persons are often pathologically jealous and, for no reason, question the fidelity of their spouses or sexual partners. Persons with this disorder externalize their own emotions and use the defense of projection; they attribute to others the impulses and thoughts that they cannot accept in themselves. Ideas of reference and logically defended illusions are common. Table 19-2 compares the different approaches to diagnosing paranoid personality disorder.

On psychiatric examination, patients with paranoid personality disorder may be formal in their style and act baffled about having to seek psychiatric help. Muscular tension, an inability to relax, and a need to scan the environment for clues may be evident, and the patient's manner is often severe and humorless. Although some premises of their arguments may be false, their speech is goal-directed and logical. Their thought content shows evidence of projection, prejudice, and occasional ideas of reference.

Schizoid Personality Disorder. Persons with schizoid personality disorder seem to be cold and aloof; they display a remote reserve and show no involvement with everyday events and the concerns of others. They appear quiet, distant, seclusive, and unsociable. They may pursue their own lives with remarkably little need or longing for emotional ties, and they are the last to be aware of changes in popular fashion. Table 19-3 compares the different approaches to diagnosing schizoid personality disorder.

The life histories of such persons reflect solitary interests and success at noncompetitive, lonely jobs that others find difficult to tolerate. Their sexual lives may exist exclusively in fantasy, and they may postpone mature sexuality indefinitely. Persons with schizoid personality disorder usually reveal a lifelong inability to express anger directly. They often lack close friends or confidants and are indifferent to praise and criticism. Although persons with schizoid personality disorder appear self-absorbed and lost in

daydreams, they have an average capacity to recognize reality.

On an initial psychiatric examination, patients with schizoid personality disorder may appear ill at ease. They rarely tolerate eye contact, and interviewers may surmise that such patients are eager for the interview to end. Their affect may be constricted, aloof, or inappropriately severe, but underneath the aloofness, sensitive clinicians can recognize fear. These patients find it challenging to be lighthearted: their efforts at humor may seem adolescent and off the mark. Their speech is goal-directed, but they are likely to give short answers to questions and to avoid spontaneous conversation. They may occasionally use unusual figures of speech, such as an odd metaphor, and may be fascinated with inanimate objects or metaphysical constructs. Their sensorium is intact, their memory functions well, and their proverb interpretations are abstract.



Table 19-3
Schizoid Personality Disorder: Specific Symptoms

DSM-5		ICD-10	
Schizoid Personality Disorder		**Schizoid Personality Disorder**	
Detachment from interpersonal relationships (≥4): • Doesn't enjoy relationships • Prefers doing things alone • Not interested in sexual relations • Doesn't enjoy activities • Few if any friends or	Exclusions (not result of): • *Schizophrenia* • *Depression with psychotic features* • *Bipolar disorder with psychotic features* • *Autism spectrum disorder*	• Retreats from close contact or emotional encounters • Prefers fantasy, isolation, and introspection • Difficulty expressing emotions • Difficulty experiencing pleasure	Exclusions (nc result of): • *Asperger syndrome* • *Delusional disorder* • *Schizophr* • *Schizotypa disorder*

acquaintances • Doesn't care what others think of them • Emotionally constricted	• *Another psychotic disorder* • *Another medical condition*



Table 19-4
Schizotypal Personality Disorder: Specific Symptoms

DSM-5		ICD-10	
Schizotypal Personality Disorder		**Schizotypal Disorder (class under the psychotic disord ICD-10, not personality dis**	
Social difficulties and perceptual disturbances (≥5): • Ideas of reference • Magical/strange thinking • Odd perceptions • Odd speech/thoughts • Odd affect • Odd behavior • Suspicious or persecutory thoughts • Few if any friends or acquaintances • Socially anxious	Exclusions (not result of): • Schizophrenia • Depression with psychotic features • Bipolar disorder with psychotic features • Autism spectrum disorder • Another psychotic disorder	• Eccentric behavior • Abnormal thinking • Cold affect • Odd behavior • Social isolation • Perceptual disturbances: Quasi-psychotic episodes Illusions Delusional-like ideas Hallucinations • Starts suddenly, course is	Exclusion result • Schizo other psych disor • Asper syndr • Schizo perso disor

(because of thoughts)	more like that for personality disorders

Schizotypal Personality Disorder. Schizotypal personality disorder is characterized by pervasive discomfort with and inability to maintain close relationships, as well as eccentric behavior. These individuals demonstrate peculiarities of thinking, behavior, and appearance. Taking a history may be difficult because of the patients' unusual way of communicating.

Patients with schizotypal personality disorder exhibit disturbed thinking and communicating. Although frank thought disorder is absent, their speech may be distinctive or peculiar, may have meaning only to them, and often needs interpretation. These patients may be superstitious or claim powers of clairvoyance and may believe that they have other special powers of thought and insight. Their inner world may contain vivid imaginary relationships and child-like fears and fantasies, and they may experience perceptual illusions, as well. Table 19-4 compares the different approaches to diagnosing schizotypal personality disorder.

Because persons with schizotypal personality disorder have poor interpersonal relationships and may act inappropriately, they are isolated and have few, if any, friends. Under stress, patients with schizotypal personality disorder may decompensate and have psychotic symptoms, but these are usually brief. Patients with severe cases of the disorder may exhibit anhedonia and severe depression.

Cluster B: Dramatic

Antisocial Personality Disorder. The hallmarks of antisocial personality disorder are pervasive disrespect for and infringement on the rights of others. A person has to be 18 years of age or older, has to have demonstrated this pattern of behavior since age 15, and must have demonstrated evidence of conduct disorder before the age of 15 years (conduct disorder involves a repetitive and persistent pattern of behavior in which they violate the fundamental rights of others or major age-appropriate social rules). Table 19-

5 compares the different approaches to diagnosing antisocial personality disorder.

Patients with antisocial personality disorder can often seem to be normal and even charming and ingratiating. Their histories, however, reveal many areas of disordered life functioning. Lying, truancy, running away from home, thefts, fights, substance abuse, and illegal activities are typical experiences that patients report as beginning in childhood. These patients often impress clinicians with the colorful, seductive aspects of their personalities, but the clinician may also regard them as manipulative and demanding. They frequently have a heightened sense of reality testing and often impress observers as having excellent verbal intelligence.



Table 19-5
Antisocial Personality Disorder: Specific Symptoms

DSM-5		ICD-10	
Antisocial Personality Disorder		**Dissocial Personality Disor**	
≥age 18 Began by age 15 Conduct disorder before age 15 (diagnosed or evidence for) (≥3): • Disregards legal and social rules/norms • Lies • Impulsive • Irritable or	Exclusions (not result of): • *Schizophrenia* • *Bipolar disorder*	• Disregards social norms • Unconcerned by others' feelings • Easily frustrated/distressed • Easily becomes aggressive • Blames others	Exclusic resul • Conc disc • Emo unst pers disc

aggressive
- Neglects safety (self/others)
- Irresponsible
- No remorse

Persons with antisocial personality disorder are highly representative of the so-called con men. They are incredibly manipulative and can frequently talk others into participating in schemes for easy ways to make money or to achieve fame or notoriety. These schemes may eventually lead the unwary to financial ruin, social embarrassment, or both. Those with this disorder do not tell the truth and cannot be trusted to carry out any task or adhere to any conventional standard of morality. Promiscuity, spousal abuse, child abuse, and drunk driving are frequent events in their lives. A notable finding is a lack of remorse for these actions; that is, they appear to lack a conscience.

Patients with antisocial personality disorder can fool even the most experienced clinicians. In an interview, patients can appear composed and credible, but beneath the veneer lurks tension, hostility, irritability, and rage. A stress interview, in which the clinician vigorously confronts the patient with inconsistencies in their histories, may be necessary to reveal the pathology. A diagnostic workup should include a thorough neurologic examination. Because patients often show abnormal EEG results and soft neurologic signs suggesting minimal brain damage in childhood, these findings may support the clinical impression.

Borderline Personality Disorder. Persons with borderline personality disorder almost always appear to be in a state of crisis. Mood swings are frequent. Patients can be argumentative at one moment, depressed the next, and later complain of having no feelings. Patients can have short-lived psychotic episodes rather than full-blown psychotic breaks, and the symptoms are circumscribed, fleeting, or questionable. The behavior of patients with borderline personality disorder is highly unpredictable, and their achievements are rarely at the level of their abilities. The painful nature of their lives reflects in their repetitive self-destructive acts. Such patients may

slash their wrists and perform other self-mutilation to elicit help from others, to express anger, or to numb themselves to overwhelming affect. Table 19-6 compares the different approaches to diagnosing borderline personality disorder.

Because they feel both dependent and hostile, persons with this disorder have tumultuous interpersonal relationships. They can be dependent on those with whom they are close and, when frustrated, can express enormous anger toward their intimate friends. Patients with borderline personality disorder cannot tolerate being alone, and they prefer a frantic search for companionship, no matter how unsatisfactory, to their own company. To assuage loneliness, if only for brief periods, they accept a stranger as a friend or behave promiscuously. They often complain about chronic feelings of emptiness and boredom and the lack of a consistent sense of identity (identity diffusion); when pressed, they may complain about how depressed they usually feel, despite the flurry of other affects.

Functionally, patients with borderline personality disorder distort their relationships by considering each person to be either all good or all bad. They see persons as either nurturing attachment figures or as hateful, sadistic figures who deprive them of security needs and threaten them with abandonment whenever they feel dependent. As a result of this splitting, the good person is idealized, and the bad person devalued. Shifts of allegiance from one person or group to another are frequent. Some clinicians use the concepts of panphobia, pananxiety, panambivalence, and chaotic sexuality to delineate these patients' characteristics.

Projective identification is a defense mechanism that often occurs in patients with borderline personality disorder. In this primitive defense mechanism, the patient projects intolerable aspects of themselves onto another person, inducing them to play the projected role, and the two persons act in unison. Therapists must be aware of this process so they can act neutrally toward such patients.

Histrionic Personality Disorder. Persons with histrionic personality disorder show a high degree of attention-seeking behavior. They tend to exaggerate their thoughts and feelings and make everything sound more vital than it is. They display temper tantrums, tears, and accusations when they are not the center of attention or are not receiving praise or approval. Table 19-7

compares the different approaches to diagnosing histrionic personality disorder.

Seductive behavior is typical in both sexes. Sexual fantasies about persons with whom patients are involved are common, but patients are inconsistent about verbalizing these fantasies and may be coy or flirtatious rather than sexually aggressive. Despite these behaviors, they may have a psychosexual dysfunction, such as anorgasmia. They may act on their sexual impulses to reassure themselves that they are attractive to the other sex. Their relationships tend to be superficial, however, and they can be vain, self-absorbed, and fickle. Their deep dependence needs make them overly trusting and gullible.



Table 19-6
Borderline Personality Disorder: Specific Symptoms

DSM-5	ICD-10	
Borderline Personality Disorder	**Emotionally Unstable Personality Disorder**	
Conflict/impulsivity (≥5): • Avoids abandonment • Intense, unstable relations marked by splitting • Unstable self-image • Self-harm, other impulsive behaviors • Suicidal ideation or behavior • Labile affect • Feeling empty inside • Poor anger management • Paranoia/dissociation, usually due to stress	• Impulsiveness • Labile mood • Outbursts • Interpersonal conflict Two subtypes: (1) Impulsive type (emotional lability and lack of impulse control) (2) Borderline type (poor self-image, self-worth, relationships, with associated self-harm)	Exclusions (not result of): • Dissocial personality disorder

Table 19-7
Histrionic Personality Disorder: Specific Symptoms

DSM-5	ICD-10
Histrionic Personality Disorder	**Histrionic Personality Disorder**
Needing attention, very emotional (≥5): • Uncomfortable when not the center of attention • Flirtatious, provocatively sexual • Emotionally labile • Physically provocative or flamboyant • Speech is vague • Speech is exaggerated/dramatic • Suggestible • Overestimates the intimacy of relationships	• Affect is shallow/labile • Self-dramatization • Speech and emotions are exaggerated • Suggestible • Egocentric • Not concerned with others • Attention seeking • Easily offended or hurt

The major defenses of patients with histrionic personality disorder are repression and dissociation. Accordingly, such patients are unaware of their true feelings and cannot explain their motivations. Under stress, reality testing quickly becomes impaired.

In interviews, patients with histrionic personality disorder are generally cooperative and eager to give a detailed history. Gestures and dramatic punctuation in their conversations are commonplace; they may make frequent slips of the tongue, and their language is colorful. Affective display is typical, but when pressed to acknowledge certain feelings (e.g., anger, sadness, and sexual wishes), they may respond with surprise, indignation, or denial. The results of the cognitive examination are usually normal, although they may show a lack of perseverance on arithmetic or concentration tasks, and the patients' forgetfulness of affect-laden material may be astonishing.

Narcissistic Personality Disorder. Persons with narcissistic personality disorder have a grandiose sense of self-importance; they consider themselves special and expect special treatment. Their sense of entitlement is striking. They handle criticism poorly and may become enraged when someone dares to criticize them, or they may appear utterly indifferent to criticism. Persons with this disorder want their way and are frequently ambitious to achieve fame and fortune. Their relationships are tenuous, and they can make others furious by their refusal to obey conventional rules of behavior. Interpersonal exploitativeness is commonplace. They cannot show empathy, and they feign sympathy only to achieve their selfish ends. Because of their fragile self-esteem, they are susceptible to depression. Interpersonal difficulties, occupational problems, rejection, and loss are among the stresses that narcissists commonly produce by their behavior—stresses they are least able to handle. Table 19-8 compares the different approaches to diagnosing narcissistic personality disorder.



Table 19-8
Narcissistic Personality Disorder: Specific Symptoms

DSM-5	ICD-10
Narcissistic Personality Disorder	**Narcissistic Personality**
Self-important, lacking empathy for others (≥5): • Grandiose • Preoccupied with fantasies about success • Feeling special/unique • Needing others to admire them for validation • Expecting special treatment • Exploiting others • Lacking empathy • Jealous of others' success,	Not described. Listed as an "other specific personality disorder"

assuming others are jealous of them
- Arrogance

Cluster C: Anxious

Avoidant Personality Disorder. Hypersensitivity to rejection by others is the central feature of avoidant personality disorder, and timidity is the primary personality trait that is displayed. These persons desire the warmth and security of human companionship but justify their avoidance of relationships by their fear of rejection. When talking with someone, they express uncertainty, show a lack of self-confidence, and may speak in a self-effacing manner. Because they are hypervigilant about rejection, they are afraid to speak up in public or to make requests of others. They are apt to misinterpret others' comments as derogatory or ridiculing. The refusal of any request leads them to withdraw from others and to feel hurt. Table 19-9 compares the different approaches to diagnosing avoidant personality disorder.

In the vocational sphere, patients with avoidant personality disorder often take jobs on the sidelines. They rarely attain much personal advancement or exercise much authority but seem shy and eager to please. These persons are generally unwilling to enter relationships unless they have an unusually strong guarantee of uncritical acceptance. Consequently, they often have no close friends or confidants.

In clinical interviews, patients' most striking aspect is anxiety about talking with an interviewer. Their nervous and tense manner may wax and wane depending on whether they think the interviewer likes them. They may also seem vulnerable to the interviewer's comments and suggestions and may regard a clarification or interpretation as criticism.

Dependent Personality Disorder. Dependent personality disorder is a pervasive pattern of dependent and submissive behavior. Persons with the disorder cannot make decisions without significant and unwarranted advice and encouragement from others. They avoid positions of responsibility and become anxious if asked to assume a leadership role. They prefer to be

submissive. When on their own, they find it difficult to persevere at tasks but may find it easy to perform these tasks for someone else. Table 19-10 compares the different approaches to diagnosing dependent personality disorder.



Table 19-9
Avoidant Personality Disorder: Specific Symptoms

DSM-5	ICD-10
Avoidant Personality Disorder	**Anxious [Avoidant] Personality Disorder**
Hypersensitive, lacking confidence (≥4): • Avoids others • Fears being disliked • Avoids relationships for fear of shame • Fears rejection/criticism • Inhibited in relationships • Avoids novelty for fear of embarrassment	• Fearful, insecure, feels inferior to others • Craves acceptance • Fears or is sensitive to rejection • Relationships are superficial • Avoids activities/situations perceived as risky



Table 19-10
Dependent Personality Disorder: Specific Symptoms

DSM-5	ICD-10
Dependent Personality Disorder	**Dependent Personality Disorder**
Fears separation, needs others to care for them (≥5): • Cannot make decisions alone • Avoids taking on responsibility for important things	• Relies on others and needs them for validation • Cannot make decisions without another's approval • Fears abandonment

- Cannot disagree with others for fear they will disapprove of them
- Lacks confidence, cannot initiate new things
- Seeks acceptance of others
- Fears being alone/independence
- When relationship ends, quickly seeks new one
- Fears having to take care of self
- Fears helplessness/incompetence
- Passive, lets another make decisions
- Avoids responsibility

Because persons with the disorder do not like to be alone, they seek out others on whom they can depend; their relationships, thus, are distorted by their need to be attached to another person. In a shared psychotic disorder (folie à deux), one member of the pair usually has dependent personality disorder; the submissive partner takes on the delusional system of the more aggressive, assertive partner on whom he or she depends.

Pessimism, self-doubt, passivity, and fears of expressing sexual and aggressive feelings all typify the behavior of persons with a dependent personality disorder. An abusive, unfaithful, or alcoholic spouse may be tolerated for long periods to avoid disturbing the sense of attachment.

In interviews, patients appear compliant. They try to cooperate, welcome specific questions and look for guidance.

Obsessive-Compulsive Personality Disorder. Persons with obsessive-compulsive personality disorder are preoccupied with rules, regulations, orderliness, neatness, details, and the achievement of perfection. These traits account for the general constriction of the entire personality. They insist that rules be followed rigidly and cannot tolerate what they consider infractions. Accordingly, they lack flexibility and are intolerant. They are capable of prolonged work, provided it is routinized and does not require changes to which they cannot adapt. Table 19-11 compares the different approaches to diagnosing obsessive-compulsive personality disorder.

Persons with obsessive-compulsive personality disorder have limited interpersonal skills. They are formal and severe and often lack a sense of humor. They alienate those around them, are unable to compromise, and

insist that others submit to their needs. They are eager to please those whom they see as more powerful, however, and they carry out these persons' wishes in an authoritarian manner. Because they fear to make mistakes, they are indecisive and ruminate about making decisions. Although a stable marriage and occupational adequacy are common, persons with obsessive-compulsive personality disorder have few friends. Anything that threatens to upset their perceived stability or the routine of their lives can precipitate much anxiety otherwise bound up in the rituals that they impose on their lives and try to impose on others.

In interviews, patients with an obsessive-compulsive personality disorder may have a stiff, formal, and rigid demeanor. Their affect is not blunted or flat but often constricted. They lack spontaneity, and their mood is usually serious. Such patients may be anxious about not being in control of the interview. Their answers to questions are unusually detailed. The defense mechanisms they use are rationalization, isolation, intellectualization, reaction formation, and undoing.

Other Personality Disorders

Personality Change due to Another Medical Condition. This diagnosis involves a change in personality from previous patterns of behavior or an exacerbation of previous personality characteristics. Impaired control of the expression of emotions and impulses is a cardinal feature. Emotions are characteristically labile and shallow, although euphoria or apathy may be prominent. The euphoria may mimic hypomania, but genuine elation is absent, and patients may admit to not feeling happy. There is a hollow and silly ring to their excitement and facile jocularity, particularly when the frontal lobes are involved. Also associated with damage to the frontal lobes, the so-called frontal lobe syndrome, consists of prominent indifference and apathy, characterized by a lack of concern for events in the immediate environment. Temper outbursts, which can occur with little or no provocation, especially after alcohol ingestion, can result in violent behavior. The expression of impulses may include inappropriate jokes, a coarse manner, improper sexual advances, and antisocial conduct resulting in conflicts with the law, such as assaults on others, sexual misdemeanors, and shoplifting. They typically have diminished foresight, and cannot anticipate

the social or legal consequences of actions. Persons with temporal lobe epilepsy characteristically show humorlessness, hypergraphia, hyperreligiosity, and marked aggressiveness during seizures.



Table 19-11
Obsessive-Compulsive Personality Disorder: Specific Symptoms

Obsessive-Compulsive Personality Disorder		Anankastic Personality Dis	
Orderliness, perfectionism, self-control (≥4): • Attention to rules/details/order • Cannot complete things because of needing it to be perfect • Relationships neglected because of devotion to work • Inflexible thinking • Cannot part with things	• Cannot delegate • Stingy • Stubborn/rigid	• Plagued by doubt • Perfectionism, checking, attention to detail • Stubborn/rigid • Overly cautious • Unwanted thoughts or impulses, but not to the point of obsessions or compulsions	Exclusion result • Obses comp disord

Persons with personality change due to a general medical condition have a clear sensorium. Mild disorders of cognitive function often coexist but do not amount to intellectual deterioration. Patients may be inattentive, which may account for disorders of recent memory. With some prodding, however, patients are likely to recall what they claim to have forgotten. We should suspect the diagnosis in patients who show marked changes in behavior or personality involving emotional lability and impaired impulse control, who have no history of mental disorder, and whose personality changes occur

abruptly or over a relatively brief time.

Other Specified Personality Disorder. DSM-5 reserves the category of other specified personality disorder for situations that meet the general criteria for a personality disorder, but the presentation does not fit into any of the personality disorder categories described above. Passive-aggressive personality and depressive personality are examples. A narrow spectrum of behavior or a particular trait—such as oppositionalism, sadism, or masochism—can also be classified in this category. A patient with features of more than one personality disorder but without the complete criteria of any one disorder can be assigned this classification.

Unspecified Personality Disorder. This diagnosis is like the above Other Specified Personality Disorder but is used in situations when we may choose not to indicate why the patient does not meet the criteria for a specific personality disorder. A typical example is a case in which there is not enough information to make a specific personality disorder diagnosis.

DIFFERENTIAL DIAGNOSIS

Many of the features of various personality disorders may occur during an episode of another mental disorder. Some of the personality disorders, particularly within a given cluster, may also share common features.

Paranoid Personality Disorder

Paranoid personality disorder can usually be differentiated from delusional disorder by the absence of fixed delusions. Unlike persons with schizophrenia, those with personality disorders typically lack hallucinations or formal thought disorder. However, when a brief reactive psychosis with delusions complicates the clinical picture of paranoid personality disorder, this distinction is far more complicated. Paranoid personality disorder can be distinguished from a borderline personality disorder because patients who are paranoid are rarely capable of overly involved, tumultuous relationships with others. Patients with paranoia lack the long history of antisocial behavior of persons with antisocial character. Persons with schizoid personality disorder are withdrawn and aloof and do not have paranoid ideation.

Schizoid Personality Disorder

Schizoid personality disorder differs from schizophrenia, delusional disorder, and affective disorder with psychotic features based on periods with positive psychotic symptoms, such as delusions and hallucinations, in the latter. Although patients with paranoid personality disorder share many traits with those with schizoid personality disorder, the former exhibit more social engagement, a history of aggressive verbal behavior, and a greater tendency to project their feelings onto others. If just as emotionally constricted, patients with obsessive-compulsive and avoidant personality disorders experience loneliness as dysphoric, possess a more abundant history of past object relations, and do not engage as much in autistic reverie. The chief distinction between a patient with schizotypal personality disorder and one with schizoid personality disorder is that the patient who is schizotypal is more similar to a patient with schizophrenia in oddities of perception, thought, behavior, and communication. Patients with avoidant personality disorder are isolated but do wish to participate in activities, a characteristic absent in those with schizoid personality disorder. Schizoid personality disorder is distinguished from autism spectrum disorder by more severely impaired social interactions and stereotypical behaviors and interests with autism.

Schizotypal Personality Disorder

Persons with schizotypal personality disorder can be distinguished from those with schizoid and avoidant personality disorders by the presence of oddities in their behavior, thinking, perception, and communication. Schizotypal personality disorder is difficult to distinguish from the heterogeneous group of solitary, odd children whose behavior is characterized by social isolation, eccentricity, and peculiarities in language seen in autism spectrum disorder and communication disorders. We distinguish the latter by the primacy and severity of the language disorder and accompanying compensatory efforts by the patient. Also, individuals with autism often have more severely impaired social interactions and restricted behaviors and interests than those with schizotypal disorder. Patients with schizotypal personality disorder can be distinguished from those with schizophrenia by the absence of psychosis. If psychotic symptoms do appear, they are brief and fragmentary. Patients with

paranoid personality disorder are suspicious but lack the odd behavior of patients with schizotypal personality disorder.

Antisocial Personality Disorder

Antisocial personality disorder differs from mere illegal behavior in that antisocial personality disorder involves many areas of a person's life. When illegal behavior is only for gain and does not include rigid, maladaptive, and persistent personality traits, we classify it as criminal behavior not associated with a personality disorder in DSM-5.

Many of these persons have a neurologic or mental disorder that has been either overlooked or undiagnosed. More difficult is the differentiation of antisocial personality disorder from substance use disorders. When both substance use disorders and antisocial behavior begin in childhood and continue into adult life, we should diagnose both disorders. When, however, the antisocial behavior is secondary to premorbid alcohol use disorder or another substance use disorder, the diagnosis of antisocial personality disorder is not warranted.

In diagnosing antisocial personality disorder, clinicians must adjust for the distorting effects of socioeconomic status, cultural background, and sex. Furthermore, the diagnosis of antisocial personality disorder is not warranted when intellectual disability, schizophrenia, or mania can explain the symptoms. Other personality disorders, such as narcissistic, histrionic, and paranoid, can be differentiated from antisocial personality disorder as the former personality disorders rarely include serious criminality and aggressiveness. Borderline personality disorder can sometimes be associated with criminality. However, individuals with borderline personality who commit crimes tend to display high novelty seeking and high harm avoidance behaviors, whereas those with antisocial personality tend to display high novelty seeking and low harm avoidance behaviors.

Borderline Personality Disorder

Borderline personality disorder differs from major depressive disorder, bipolar disorder, dysthymic disorder, and cyclothymia based on the presence of core borderline personality symptoms that are not typically present in mood disorders (e.g., fear of abandonment, highly unpredictable behavior,

tumultuous interpersonal relationships, seeing others as either all good or all bad, complaints of being numb or empty, and the lack of a consistent sense of identity). It differs from identity problems, which are typically limited to a developmental stage.

Borderline personality disorder differs from schizophrenia on the basis that the patient with borderline personality lacks prolonged psychotic episodes, thought disorder, and other classic schizophrenic signs. Patients with schizotypal personality disorder show marked peculiarities of thinking, strange ideation, and recurrent ideas of reference. Those with paranoid personality disorder have extreme suspiciousness. In contrast, patients with borderline personality disorder generally have short-lived psychotic episodes, when present.

Histrionic Personality Disorder

Distinguishing between histrionic personality disorder and borderline personality disorder is challenging, but in borderline personality disorder, suicide attempts, identity diffusion, and brief psychotic episodes are more likely. Although both conditions can exist in the same patient, clinicians should separate the two. Somatic symptom disorder may occur in conjunction with histrionic personality disorder, and we should diagnose both as well. Patients with brief psychotic disorder and dissociative disorders may warrant a coexisting diagnosis of histrionic personality disorder.

Narcissistic Personality Disorder

Borderline, histrionic, and antisocial personality disorders often accompany narcissistic personality disorder, so differential diagnosis is difficult. Patients with narcissistic personality disorder have less anxiety than those with borderline personality disorder; their lives tend to be less chaotic, and they are less likely to attempt suicide. Patients with antisocial personality disorder have a history of impulsive behavior, often associated with alcohol or other substance use, which frequently gets them into trouble with the law. Patients with histrionic personality disorder show features of exhibitionism and interpersonal manipulativeness that resemble those of patients with narcissistic personality disorder. Narcissistic personality disorder can be distinguished from the grandiosity of mania by the episodic course,

associated euphoria, and functional impairment in a manic or hypomanic episode.

Avoidant Personality Disorder

Avoidant personality disorder is complicated to distinguish from social anxiety disorder. In social anxiety disorder, specific situations, rather than interpersonal contact in general, are avoided. They may co-occur. Panic disorder with agoraphobia also manifests avoidance, but usually after the onset of panic attacks.

Patients with schizoid and schizotypal personality disorders may be indistinguishable from those with avoidant personality disorder. However, patients with avoidant personality disorder desire social interaction, unlike patients with schizoid personality disorder, who want to be alone. Patients with avoidant personality disorder are not as demanding, irritable, or unpredictable as those with borderline and histrionic personality disorders. Avoidant personality disorder and dependent personality disorder are similar. Patients with dependent personality disorder are presumed to have a greater fear of being abandoned or unloved than those with avoidant personality disorder, but the clinical picture may be indistinguishable.

Dependent Personality Disorder

The traits of dependence occur in many psychiatric disorders, so the differential diagnosis is difficult. Dependence is a prominent factor in patients with histrionic and borderline personality disorders. However, those with dependent personality disorder usually have a long-term relationship with one person rather than a series of persons on whom they are dependent, and they do not tend to be overly manipulative. Dependent behavior can also occur in patients with agoraphobia, panic disorder, and depressive disorders, but these patients tend to have a high level of overt anxiety, panic, or depression, respectively.

Obsessive-Compulsive Personality Disorder

When recurrent obsessions or compulsions are present, we should diagnose an obsessive-compulsive disorder. Perhaps the most challenging distinction is between outpatients with some obsessive-compulsive traits and those with an

obsessive-compulsive personality disorder. We should reserve the personality disorder diagnosis for those with significant impairments in their occupational or social effectiveness. In some cases, delusional disorder coexists with personality disorders, and we should diagnose both.

Personality Change due to Another Medical Condition

Dementia involves global deterioration in intellectual and behavioral capacities, of which personality change is just one category. A personality change may herald a cognitive disorder that eventually will deteriorate into dementia. In these cases, as deterioration begins to encompass significant memory and cognitive deficits, the diagnosis of the disorder changes from personality change caused by a general medical condition to dementia. In differentiating the specific syndrome from other disorders in which personality change may occur—such as schizophrenia, delusional disorder, mood disorders, and impulse control disorders—physicians must consider the most critical factor, the presence of a personality change due to a specific organic causative factor.

COMORBIDITY

Personality disorders occur in 10 to 20 percent of the general population and about half of psychiatric inpatients or outpatients. Personality disorders are frequently comorbid with other clinical syndromes and are predisposing factors for other psychiatric disorders and symptoms (e.g., substance use, suicide, affective disorders, impulse-control disorders, eating disorders, and anxiety disorders).

Paranoid Personality Disorder

These individuals are at increased risk for major depression, obsessive-compulsive disorder, agoraphobia, and substance use disorders. The most common co-occurring personality disorders are schizotypal, schizoid, narcissistic, avoidant, and borderline. Paranoid personality disorder may be a premorbid antecedent of delusional disorder, persecutory type.

Schizoid Personality Disorder

This personality disorder sometimes appears as the premorbid antecedent of delusional disorder, schizophrenia, or rarely major depression. Paranoid, schizotypal, and avoidant are the most common co-occurring personality disorders.

Schizotypal Personality Disorder

More than half of these patients have had at least one episode of major depression, and 30 to 50 percent have major depression concurrent with this personality disorder. The most common co-occurring personality disorders are schizoid, paranoid, avoidant, and borderline.

Antisocial Personality Disorder

These patients are at increased risk for impulse control disorders, major depression, substance use disorders, pathologic gambling, anxiety disorders, and somatic symptom disorder. Narcissistic, borderline, and histrionic are the most common co-occurring personality disorders.

Borderline Personality Disorder

These individuals are at increased risk for major depression, substance use disorders, eating disorders (notably bulimia), posttraumatic stress disorder, attention-deficit hyperactivity disorder, and somatic symptom disorder. Borderline personality disorder can co-occur with most other personality disorders.

Histrionic Personality Disorder

Patients with histrionic personality disorder are at increased risk for major depression, somatic symptom disorder, and conversion disorder. Narcissistic, borderline, antisocial, and dependent are the most common co-occurring personality disorders.

Narcissistic Personality Disorder

These patients are at increased risk for major depression and substance use disorders (especially cocaine use). The most common co-occurring personality disorders are borderline, antisocial, histrionic, and paranoid.

Avoidant Personality Disorder

Patients with avoidant personality disorder are at increased risk for mood and anxiety disorders (especially social anxiety disorder). The most common co-occurring personality disorders are schizotypal, schizoid, paranoid, dependent, and borderline. Somatic symptom disorder may be comorbid.

Dependent Personality Disorder

These patients are at increased risk for major depression, anxiety disorders, and adjustment disorder. Somatic symptom disorder may be comorbid. The personality disorders that co-occur most commonly are histrionic, avoidant, and borderline.

Obsessive-Compulsive Personality Disorder

These individuals are at increased risk for major depression and anxiety disorder. Equivocal evidence exists for an increased risk of obsessive-compulsive disorder.

COURSE AND PROGNOSIS

Personality disorders are chronic, and they last over decades. They often interfere with the treatment outcomes of comorbid psychiatric disorders and lead to personal incapacitation, morbidity, and mortality in these patients. People with personality disorders have chronic impairments in their ability to work and love, tend to be less educated, drug dependent, single, unemployed, and to have marital difficulties. They consume a large portion of community services, social welfare benefits, public health, and prison resources.

Paranoid Personality Disorder

There are no adequate, systematic long-term studies of paranoid personality disorder. In some, paranoid personality disorder is lifelong; in others, it is a harbinger of schizophrenia. In still others, paranoid traits give way to reaction formation, appropriate concern with morality, and altruistic concerns as they mature or as stress diminishes. In general, however, those with paranoid personality disorder have lifelong problems working and living with others.

Occupational and marital problems are common. Complications may

include brief reactive psychosis, particularly in response to stress.

Schizoid Personality Disorder

The onset of schizoid personality disorder usually occurs in early childhood or adolescence. As with all personality disorders, schizoid personality disorder is long-lasting but not necessarily lifelong. The proportion of patients who incur schizophrenia is unknown. These individuals frequently have severe problems in social relations, and occupational problems develop when interpersonal involvement is required. Solitary work sometimes favorably affects overall performance. Complications may include very brief reactive psychosis, particularly in response to stress.

Schizotypal Personality Disorder

This disorder can be the premorbid personality of the patient with schizophrenia. Some, however, maintain a stable schizotypal personality throughout their lives and marry and work, despite their oddities. Complications may include transient psychotic episodes, particularly in response to stress. Symptoms sometimes become so significant that individuals meet the criteria for schizophreniform disorder, delusional disorder, and brief psychotic disorder. One long-term study reported that 10 percent of those with schizotypal personality disorder eventually committed suicide.

Antisocial Personality Disorder

When an antisocial personality disorder develops, it runs an unremitting course, with the height of antisocial behavior usually occurring in late adolescence. The prognosis varies. Some reports indicate that symptoms decrease as a person ages. Even after the severe antisocial behavior “burns out,” people diagnosed with antisocial personality disorder usually continue to be irritable, impulsive, and detached. Complications may include dysphoria, tension, low tolerance for boredom, depressed mood, and premature, violent death.

Borderline Personality Disorder

The course of borderline personality disorder is variable and most commonly

follows a pattern of chronic instability in early adulthood, with episodes of severe affective and impulsive dyscontrol. The impairment and the risk of suicide are highest in the young adult years and gradually wane with advancing age. In the fourth and fifth decades, these individuals tend to attain greater stability in their relationships and functioning. Impairment typically involves frequent job losses, interrupted education, and broken marriages. Complications may include psychotic-like symptoms (hallucinations, body image distortions, hypnagogic phenomena, ideas of reference) in response to stress, as well as premature death or physical handicaps from suicide and suicidal gestures, failed suicide, and self-injurious behavior.

Histrionic Personality Disorder

With age, persons with histrionic personality disorder show fewer symptoms, but because they lack the energy of earlier years, the difference in the number of symptoms may be more apparent than real. Persons with this disorder are sensation seekers, and they may get into trouble with the law, abuse substances, and act promiscuously. Complications may include frequent suicidal gestures and threats to coerce better caregiving; interpersonal relations that are unstable, shallow, and generally ungratifying; and frequent marital problems secondary to the tendency to neglect long-term relationships for the excitement of new relationships.

Narcissistic Personality Disorder

Narcissistic personality disorder is chronic and difficult to treat. Patients with the disorder must continuously deal with blows to their narcissism resulting from their behavior or life experience. Patients handle aging poorly; they value beauty, strength, and youthful attributes, to which they cling inappropriately. They may be more vulnerable, therefore, to midlife crises than are other groups. However, narcissistic symptoms tend to diminish after the age of 40 years, when pessimism usually develops. Impairment is frequently severe and includes marital problems and interpersonal relationships in general. Complications may include social withdrawal, depressed mood, and dysthymic or major depressive disorder in reaction to criticism or failure.

Avoidant Personality Disorder

Avoidant personality disorder frequently begins in childhood with shyness and fear of strangers and new situations. Impairment can be severe and typically includes occupational and social difficulties. However, many persons with avoidant personality disorder can function in a protected environment. Some marry, have children, and live their lives surrounded only by family members. If their support system fails, however, they are subject to depression, anxiety, and anger. Phobic avoidance is common, and patients with the disorder may give histories of social anxiety disorder or incur social anxiety disorder in the course of their illness.

Dependent Personality Disorder

We know little about the course of dependent personality disorder. Occupational functioning tends to be impaired because persons with the disorder cannot act independently and without close supervision. Social relationships are limited to those on whom they can depend, and many suffer physical or mental abuse because they cannot assert themselves. They risk major depressive disorder if they lose the person on whom they depend, but with treatment, the prognosis is favorable. Complications may include low socioeconomic status and inadequate family and marital functioning, as well as mood disorders, anxiety disorders, adjustment disorder, and social anxiety disorder.

Obsessive-Compulsive Personality Disorder

The course of obsessive-compulsive personality disorder is variable and unpredictable. Occupational and social difficulties are typical. From time to time, persons may develop obsessions or compulsions in the course of their disorder. Some adolescents with obsessive-compulsive personality disorder evolve into warm, open, and loving adults; in others, the disorder can be either the harbinger of schizophrenia or—decades later and exacerbated by the aging process—major depressive disorder.

Persons with obsessive-compulsive personality disorder may flourish in positions demanding methodical, deductive, or detailed work, but they are vulnerable to unexpected changes, and their personal lives may remain barren. Depressive disorders, especially those of late-onset, are common.

Complications may include distress and difficulties when confronted with new situations that require flexibility and compromise, as well as myocardial infarction secondary to features typical of type A personalities, such as time urgency, hostility, and competitiveness.

Personality Change due to Another Medical Condition

Both the course and the prognosis of personality change due to a general medical condition depend on its cause. If the disorder results from structural damage to the brain, the disorder tends to persist. In cases of head trauma or vascular accident, the disorder may follow a period of coma and delirium, and it may be permanent. The personality change can evolve into dementia in cases of brain tumors, multiple sclerosis, and Huntington disease. Personality changes produced by chronic intoxication, medical illness, or drug therapy (such as levodopa for parkinsonism) may reverse with appropriate treatment of the cause. Some patients require custodial care or at least close supervision to meet their basic needs, avoid repeated conflicts with the law, and protect themselves and their families from the hostility of others and destitution resulting from impulsive and ill-considered actions.

TREATMENT APPROACH

Individuals with personality disorders do not recognize that they are ill and seldom seek help unless others (such as a spouse or parents) are insistent. This usually happens when maladaptive behaviors create marital, family, and career problems, or when other disorders (e.g., anxiety, depression, substance use) or somatic symptoms (e.g., obesity, heart disease, COPD) complicate their clinical picture. In general, patients with personality disorders require a multifaceted treatment plan that often combines psychotherapy and pharmacotherapy.

Combining Psychotherapy and Pharmacotherapy

Most patients with a personality disorder will require a combination of psychotherapy and pharmacotherapy. Psychotherapy is the primary treatment, as it promotes the maturation of character and, ultimately, the encouragement of the patient’s capacity to develop better adaptive solutions. However,

psychotherapy is very hard to implement in the setting of unstable affect and risky or self-destructive behaviors, all very common in the initial stages of treatment. During these early stages, pharmacotherapy achieves a relatively prompt control of affect and behavior. Most importantly, this nonspecific improvement sets a more suitable platform for the work in psychotherapy.

Pharmacotherapy. In the treatment of patients with personality disorders, medications can target specific symptoms of their disorders with the goals of relieving subjective distress, risky or self-destructive behaviors, or conflict with others, thereby preparing them for or allowing them to participate in psychotherapeutic approaches. We should identify target symptoms, in order to choose agents that effect those symptoms. Table 19-12 summarizes drug choices for various target symptoms of personality disorders.

Paranoid Personality Disorder

Psychotherapy. Psychotherapy is the treatment of choice for those with paranoid personality disorder. Therapists should be straightforward in all their dealings with these patients. If the patient accuses the therapist of an inconsistency or a fault, such as lateness for an appointment, honesty and an apology are preferable to a defensive explanation. Therapists must remember that trust and tolerance of intimacy are troubled areas for patients with this disorder. Individual psychotherapy thus requires a professional and not overly warm style from therapists. Clinicians' overzealous use of interpretation—especially interpretation about deep feelings of dependence, sexual concerns, and wishes for intimacy—increases patients' mistrust significantly. Generally, patients who are paranoid do not do well in group psychotherapy, although it can be useful for improving social skills and diminishing suspiciousness through role-playing. Many cannot tolerate the intrusiveness of behavior therapy, also used for social skills training.

At times, patients with paranoid personality disorder behave so threateningly that therapists must control or set limits on their actions. Delusional accusations should be realistically handled, but gently and without humiliating patients. Patients who are paranoid are profoundly frightened when they feel that those trying to help them are weak and helpless; therefore, therapists should never offer to take control unless they are willing

and able to do so.

Pharmacotherapy. There is little evidence to guide the use of pharmacotherapy in paranoid personality disorder. Treatment selection should be tailored to the individual patient and guided by target symptoms, such as the use of low-dose novel antipsychotics for psychotic symptoms or the use of anticonvulsants for irritability.

Schizoid Personality Disorder

Psychotherapy. The treatment of patients with schizoid personality disorder is similar to that of those with paranoid personality disorder. Patients who are schizoid tend toward introspection; however, these tendencies are consistent with psychotherapists' expectations, and such patients may become devoted if distant patients. As trust develops, patients who are schizoid may, with great trepidation, reveal a plethora of fantasies, imaginary friends, and fears of unbearable dependence—even of merging with the therapist.

In group therapy settings, patients with schizoid personality disorder may be silent for long periods; nonetheless, they do become involved. The patients should be protected against aggressive attack by group members for their proclivity to be silent. With time, the group members become important to patients who are schizoid and may provide the only social contact in their otherwise isolated existence.

Pharmacotherapy. Limited evidence exists to guide the psychopharmacologic treatment of patients with schizoid personality disorder. The use of psychotropics to target specific symptoms, such as social and emotional detachment, may be appropriate.

Schizotypal Personality Disorder

Psychotherapy. The principles of treatment of schizotypal personality disorder do not differ from those of schizoid personality disorder, but clinicians must deal sensitively with the former, mainly using care to avoid ridiculing or judging patients' odd beliefs. Schizotypal patients have peculiar

patterns of thinking, and some are involved in cults, strange religious practices, or the occult.

Pharmacotherapy. Antipsychotic medication may be useful for dealing with ideas of reference, illusions, and other symptoms of the disorder and can be an adjunct to psychotherapy. Antidepressants are useful when a depressive component of the personality is present.

Antisocial Personality Disorder

Psychotherapy. There is limited evidence to guide the use of psychotherapeutic approaches in antisocial personality disorder. Overall, the limited evidence suggests that these individuals seem to respond better to contingency management and other reward-based interventions than they do to cognitive behavioral therapy.

Pharmacotherapy. Pharmacotherapy can deal with incapacitating symptoms such as anxiety, rage, and depression, but because patients often misuse substances, we should use medications judiciously. One might use anticonvulsants to treat aggressive behaviors, especially if there are abnormal waveforms on an EEG. β-Adrenergic receptor antagonists, lithium, and antipsychotics may also reduce aggression.

Borderline Personality Disorder

Psychotherapy. Psychotherapy for patients with borderline personality disorder is an area of intensive investigation and has been the treatment of choice. It seems to be most successful in combination with pharmacotherapy.

Psychotherapy is difficult for the patient and therapist alike. Patients regress quickly, act out their impulses, and show labile or fixed negative or positive transferences, which are difficult to analyze. Projective identification may also cause countertransference problems when therapists are unaware that patients are unconsciously trying to coerce them to act out a particular behavior. The splitting defense mechanism causes patients to love and hate therapists and others in the environment alternately. A reality-oriented approach is more effective than in-depth interpretations of the unconscious.

Table 19-12
Pharmacotherapy of Target Symptom Domains of Personality Disorders

Choice of Drugs According to Target Symptoms of Personality Disorders		
Target Symptom	**Drug/Treatment of Choice**	**Not Recommended**
I. Mood Dysregulation and Anxiety		
Anxiety		
Chronic cognitive	Psychotherapy, SSRIs, SNRIs, MAOIs, Low-Dose Novel Psychotropics (aripiprazole, quetiapine) Valproate and other GABA analogs (clonazepam and buspirone)	Benzodiazepines
Chronic somatic	MAOIs, SNRIs (duloxetine, milnacipran) Pregabalin and other GABA analogs, TCAs, beta-blockers	If used—benzodiazepir with long half- and short trial preferred
Obsessions	SSRIs, Antipsychotics (quetiapine), TCAs (clomipramine), Mild NMDA antagonists	

	(riluzole, memantine)	
Acute and severe	Mirtazapine, Novel Psychotropics (quetiapine, aripiprazole, clozapine), TCAs, clonazepam, valproates, lithium	
Depression		
Atypical depression/dysphoria Classical depression	MAOIs, SSRIs, SNRIs, Aripiprazole, Lurasidone, ziprasidone, quetiapine, Atypical antipsychotics (as monotherapy or augmentation)	TCAs
Emotional lability/rapid cycling	Lithium, Lamotrigine, Valproate, Lower-dose antipsychotics (olanzapine, aripiprazole, clozapine, ziprasidone)	TCAs (“catecholamin stress”), Standard antidepressan (risk of switchi to mania)
II. Behavior Dyscontrol		
Aggression/impulsivity		
Affective aggression “Hot temper” with normal	Lithium, SSRIs, Anticonvulsants,	Benzodiazepines (disinhibition)

EEG	Low-dose antipsychotics	
Predatory aggression (cold blooded revenge/cruelty)	No Effective Pharmacologic Tx	Benzodiazepines (disinhibition)
Organic-like aggression (traumatic brain injury)	Beta-Blockers, Valproates, Quetiapine, Carbamazepine (CBMZ), TCAs, cholinesterase inhibitors (donepezil)	Benzodiazepines (disinhibition, delirium)
Ictal aggression (abnormal EEG)	CBMZ, Diphenylhydantoin, Valproate Benzodiazepines (clonazepam)	TCAs, Low-poten typical antipsychotics (both increase risk of seizure
III. Social and Emotional Detachment		
Chronic asociality and disinterest		
Blunted affect	Low-Dose Antipsychotics (aripiprazole, olanzapine, low-dose clozapine, sulpiride)	
IV. Cognitive-Perceptual Distortions/Psychotic Symptoms		
Acute and brief psychotic episodes	Novel Antipsychotics (Risperdal, olanzapine) Typical neuroleptics (for the duration of	

	psychosis)
Chronic and low-level psychotic-like symptoms	Novel Antipsychotics

Adapted with permission of the Center for Well-Being at Washington University.

Therapists have used behavior therapy to help patients manage their impulses and angry outbursts and to reduce their sensitivity to criticism and rejection. Social skills training, especially with videotape playback, helps patients to see how their actions affect others and thereby improve their interpersonal behavior.

Patients with borderline personality disorder often do well in a hospital setting in which they receive intensive psychotherapy on both an individual and a group basis. In a hospital, they can also interact with trained staff members from a variety of disciplines and receive occupational, recreational, and vocational therapy. Such programs are especially helpful when the home environment is detrimental to a patient's rehabilitation because of intrafamilial conflicts or other stresses, such as abuse. Within the protected environment of the hospital, we can give limits to patients who are excessively impulsive, self-destructive, or self-mutilating, and observe their actions. Under ideal circumstances, patients remain in the hospital until they show marked improvement, up to 1 year in some cases. We can then discharge patients to specialized support systems, such as day hospitals, night hospitals, and halfway houses.

DIALECTICAL BEHAVIOR THERAPY. Dialectical behavior therapy (DBT) is the psychosocial treatment that has received the most empirical support for patients with borderline personality disorder. The method is eclectic, drawing on concepts derived from supportive, cognitive, and behavioral therapies. There are four primary modes of treatment in DBT: group skill training, individual therapy, phone consultation, and consultation team. Patients are seen weekly, with the goal of improving interpersonal skills and decreasing self-destructive behavior. Patients with borderline personality disorder receive help in dealing with the ambivalent feelings that are characteristic of the disorder. As with other behavioral approaches, DBT assumes all behavior

(including thoughts and feelings) is learned and that patients with borderline personality disorder behave in ways that reinforce or even reward their behavior, regardless of how maladaptive it is. See Chapter 23 for a more detailed discussion of DBT.

MENTALIZATION-BASED TREATMENT. Another type of psychotherapy for borderline personality disorder is called mentalization-based therapy (MBT). Mentalization is a social construct that allows a person to be attentive to the mental states of oneself and of others; it comes from a person's awareness of mental processes and subjective states that arise in interpersonal interactions. MBT is based on a theory that borderline personality symptoms, such as difficulty regulating emotions and managing impulsivity, are a result of patients' reduced capacities to mentalize. Thus, recovery of mentalization should help patients build relationship skills as they learn to regulate their thoughts and feelings better. MBT was effective for borderline personality disorder in several randomized, controlled research trials.

TRANSFERENCE-FOCUSED PSYCHOTHERAPY. Transference-focused psychotherapy (TFP) is a modified form of psychodynamic psychotherapy used for the treatment of borderline personality disorder grounded in object relations theory. The therapist relies on two major processes in working with the patient: the first is clarification, in which the transference is analyzed more directly than in traditional psychotherapy so that the patient becomes quickly aware of his or her distortions about the therapist, and the second is confrontation, whereby the therapist points out how these transferential distortions interfere with interpersonal relations toward others (objects). The mechanism of splitting used by borderline patients is characterized by their having a good object and a bad object, and they use this as a defense against anxiety. If therapy is successful, then the need for splitting diminishes, object relations are improved, and the patient achieves a more normal level of functioning. Studies comparing TFP, DBT, psychodynamic psychotherapy, and supportive psychotherapy show that all are useful and all show varying degrees of success.

Pharmacotherapy. Pharmacotherapy is useful to deal with specific personality features that interfere with patients' overall functioning.

Antipsychotics may help control anger, hostility, and brief psychotic episodes. Antidepressants improve the depressed mood common in patients with borderline personality disorder. Benzodiazepines should be avoided not only due to the risk of abuse, but also because patients may become disinhibited with this class of drugs. Anticonvulsants, such as carbamazepine, may improve functioning for some patients. Serotonergic agents, such as selective serotonin reuptake inhibitors (SSRIs), have been helpful in some cases.

Histrionic Personality Disorder

Psychotherapy. Patients with histrionic personality disorder are often unaware of their real feelings; clarification of their inner feelings is a necessary therapeutic process. Psychoanalytically oriented psychotherapy is probably the treatment of choice for histrionic personality disorder.

Pharmacotherapy. Pharmacotherapy can be adjunctive to target symptoms (e.g., the use of antidepressants for depression and somatic complaints, antianxiety agents for anxiety, and antipsychotics for derealization and illusions).

Narcissistic Personality Disorder

Psychotherapy. Because patients must renounce their narcissism to make progress, the treatment of narcissistic personality disorder is challenging. Some experts have advocated using psychoanalytic approaches to effect change, but much research is required to validate the diagnosis and to determine the best treatment. Some clinicians advocate group therapy for their patients so they can learn how to share with others and, under ideal circumstances, can develop an empathic response to others.

Pharmacotherapy. Lithium can help patients whose clinical picture includes mood swings. Because patients with narcissistic personality disorder tolerate rejection poorly and are susceptible to depression, antidepressants, especially serotonergic drugs, may also be of use.

Avoidant Personality Disorder

Psychotherapy. Psychotherapeutic treatment depends on solidifying an alliance with patients. As trust develops, a therapist must convey an accepting attitude toward the patient's fears, especially the fear of rejection. The therapist eventually encourages a patient to move out into the world to take what they perceive as significant risks of humiliation, rejection, and failure. However, therapists should be cautious when giving assignments to exercise new social skills outside therapy; failure can reinforce the patient's already weak self-esteem. Group therapy may help patients understand how their sensitivity to rejection affects them and others. Assertiveness training is a form of behavior therapy that may teach patients to express their needs openly and to enlarge their self-esteem.

Pharmacotherapy. Pharmacotherapy can help manage anxiety and depression when they are associated with the disorder. Some patients are helped by β-adrenergic receptor antagonists, such as atenolol, to manage autonomic nervous system hyperactivity, which tends to be high in patients with avoidant personality disorder, especially when they approach feared situations. Serotonergic agents may help with rejection sensitivity. Theoretically, dopaminergic drugs might engender novelty-seeking behavior in these patients; however, the patient must be psychologically prepared for any new experience that might result.

Dependent Personality Disorder

Psychotherapy. The treatment of dependent personality disorder is often successful. Insight-oriented therapies enable patients to understand the antecedents of their behavior, and with the support of a therapist, patients can become more independent, assertive, and self-reliant. Behavioral therapy, assertiveness training, family therapy, and group therapy have all been used, with successful outcomes in many cases.

A pitfall may arise in treatment when a therapist encourages a patient to change the dynamics of a pathologic relationship (e.g., supports a physically abused wife in seeking help from the police). At this point, patients may become anxious and unable to cooperate in therapy; they may feel torn

between complying with the therapist and losing an external pathologic relationship. Therapists must show great respect for these patients' feelings of attachment, no matter how pathologic these feelings may seem.

Pharmacotherapy. Pharmacotherapy can help deal with specific symptoms, such as anxiety and depression, which are commonly associated features of dependent personality disorder. Patients who experience panic attacks or who have high levels of separation anxiety may benefit from symptom-targeted pharmacotherapy, such as antidepressant agents.

Obsessive-Compulsive Personality Disorder

Psychotherapy. Unlike patients with other personality disorders, some individuals with obsessive-compulsive personality disorder are often aware of their suffering and seek treatment on their own. Though limited evidence exists to guide treatment, there have been several studies that support cognitive therapy or cognitive behavioral therapy as an effective treatment for individuals with this disorder, delivered either individually or in a group. Additionally, interpersonal psychotherapy may improve depressive symptoms in individuals with obsessive-compulsive personality disorder.

Pharmacotherapy. There is limited evidence for the pharmacologic treatment of obsessive-compulsive personality disorder. Some studies have suggested benefits from fluvoxamine and carbamazepine, and in those patients with comorbid depression, citalopram may be helpful. In one study comparing citalopram and sertraline, both medications reduced the number of obsessive-compulsive personality disorder traits, though citalopram performed better than sertraline.

Personality Change due to Another Medical Condition

Management of personality change disorder involves the treatment of the underlying medical condition when possible. Psychopharmacological treatment of specific symptoms may help in some cases, such as antidepressants for depression or anticonvulsants for irritability, aggression, or impulsivity.

Patients with severe cognitive impairment or weakened behavioral controls

may need counseling to help avoid difficulties at work or to prevent social embarrassment. As a rule, patients' families need emotional support and concrete advice on how to help minimize patients' undesirable conduct. Patients should avoid alcohol, and we should curtail social engagements when patients tend to act in a grossly offensive manner.

EPIDEMIOLOGY

Personality disorders are common. They occur in 10 to 20 percent of the general population, and approximately 50 percent of all psychiatric patients.

Paranoid Personality Disorder

Data suggest that the prevalence of paranoid personality disorder is 0.5 to 4.4 percent of the general population. Those with the disorder rarely seek treatment themselves; when referred to treatment by a spouse or an employer, they can often pull themselves together and appear undistressed. Relatives of patients with schizophrenia show a higher incidence of paranoid personality disorder than control participants. Some evidence suggests a more specific familial relationship with delusional disorder, persecutory type. The diagnosis is more common in men.

Schizoid Personality Disorder

The prevalence of schizoid personality disorder is estimated to be 3.1 to 4.9 percent of the general population, according to DSM-5. This diagnosis is more common in men. Persons with the disorder tend to gravitate toward solitary jobs that involve little or no contact with others. Many prefer night work to day work so that they need not deal with many persons. There appears to be an increased prevalence among the relatives of individuals with schizophrenia or schizotypal personality disorder.

Schizotypal Personality Disorder

Schizotypal personality disorder occurs in 3.9 to 4.6 percent of the population, according to DSM-5. The sex ratio is unknown; however, it is frequent in females with fragile X syndrome. A higher association of cases exists among the biologic relatives of patients with schizophrenia than among

control participants and a higher incidence among monozygotic twins than among dizygotic twins (33 percent vs. 4 percent in one study).

Antisocial Personality Disorder

According to DSM-5, prevalence rates for antisocial personality disorder are 3 percent for males and 1 percent for females in the general population. It is frequent among the first-degree biologic relatives of individuals with this disorder. Biologic relatives of females with antisocial personality disorder are at increased risk for the same disorder compared to biologic relatives of males with antisocial personality disorder. Genetic studies have suggested familial transmission of antisocial personality disorder, substance use, and somatic symptom disorder (most studies used the previous criteria for the related somatization disorder), the former two being characteristic of males, and the latter of females in the same family. Adoption studies have shown that both genetic and environmental factors contribute to the risk of this disorder. Both adopted and biologic children of parents with antisocial personality disorder are at increased risk for this disorder. Conduct disorder (before the age of 10 years) and accompanying attention-deficit hyperactivity disorder increases the likelihood of developing antisocial personality in adult life. Conduct disorder is more likely to develop into antisocial personality disorder with erratic parenting, neglect, or inconsistent parental discipline.

Borderline Personality Disorder

Borderline personality disorder is thought to be present in about 2 percent of the general population and is more common in women than in men. The disorder is more common in younger than in older samples, suggesting a natural tendency toward maturation and remission. Physical and sexual abuse, neglect, hostile conduct, and early parental loss or separation are common in the childhood histories of patients with this disorder. Borderline personality disorder is five times more common among relatives of individuals with the same disorder than the general population. An increased prevalence of mood disorders, antisocial personality disorder, and substance use disorder is common in first-degree relatives of persons with borderline personality disorder.

Histrionic Personality Disorder

Limited data from general population studies suggest a prevalence of histrionic personality disorder of about 2 percent. Although traditionally thought to occur more frequently in women, there is some evidence to suggest that this disorder might be equally frequent among men and women. Histrionic personality disorder tends to run in families. Some experts suggest that there is a genetic link between histrionic and antisocial personality disorder and alcohol use disorder.

Narcissistic Personality Disorder

According to DSM-5, estimates of the prevalence of narcissistic personality disorder range from less than 1 percent in the general population to 2 to 16 percent in the clinical population; this disorder is more common in men.

Avoidant Personality Disorder

The prevalence of the disorder is about 0.5 to 2 percent of the general population, according to DSM-5. It occurs in men and women equally.

Dependent Personality Disorder

Dependent personality disorder is equally common in women and men. DSM-5 reports an estimated prevalence of 0.5 to 0.6 percent. Persons with chronic physical illness in childhood or separation anxiety disorder may be most susceptible to dependent personality disorder.

Obsessive-Compulsive Personality Disorder

DSM-5 reports an estimated prevalence ranging from 2 to 8 percent of the general population. It is more common in men than in women. Some studies have demonstrated a familial aggregation of this disorder.

ETIOLOGY

Genetic Factors

The best evidence that genetic factors contribute to personality disorders comes from investigations of more than 15,000 pairs of twins in the United

States. The concordance for personality disorders among monozygotic twins was several times that among dizygotic twins. Moreover, according to one study, monozygotic twins reared apart are about as similar as monozygotic twins reared together. Similarities include multiple measures of personality and temperament, occupational and leisure-time interests, and social attitudes.

Cluster A personality disorders are more common in the biologic relatives of patients with schizophrenia than in control groups. Adoption, family, and twin studies demonstrate an increased prevalence of schizotypal features in the families of schizophrenic patients, mainly when schizotypal features were not associated with comorbid affective symptoms. Less correlation exists between paranoid or schizoid personality disorder and schizophrenia.

Cluster B personality disorders have a genetic basis. Antisocial personality disorder is associated with alcohol use disorders. Depression is common in the family backgrounds of patients with borderline personality disorder. These patients have more relatives with mood disorders than do control groups, and persons with borderline personality disorder often have a mood disorder as well. There is a strong association between histrionic personality disorder and somatic symptom disorder; patients with each disorder show an overlap of symptoms.

Cluster C personality disorders may also have a genetic base. Patients with avoidant personality disorder often have high anxiety levels. Obsessive-compulsive traits are more common in monozygotic twins than in dizygotic twins, and patients with obsessive-compulsive personality disorder show some signs associated with depression—for example, shortened rapid eye movement (REM) latency period and abnormal dexamethasone-suppression test (DST) results.

Biologic Factors

Hormones. Persons who exhibit impulsive traits also often show high levels of testosterone, 17-estradiol, and estrone. In nonhuman primates, androgens increase the likelihood of aggression and sexual behavior, but the role of testosterone in human aggression is unclear. DST results are abnormal in some patients with borderline personality disorder who also have depressive symptoms.

Platelet Monoamine Oxidase. Low platelet monoamine oxidase (MAO) levels are associated with activity and sociability in monkeys. College students with low platelet MAO levels report spending more time in social activities than students with high platelet MAO levels. Low platelet MAO levels occur in some patients with schizotypal disorders.

Smooth Pursuit Eye Movements. Smooth pursuit eye movements are saccadic (i.e., jumpy) in persons who are introverted, who have low self-esteem and tend to withdraw, and who have schizotypal personality disorder. These findings have no clinical application, but they do indicate the role of inheritance.

Neurotransmitters. Studies of personality traits and the dopaminergic and serotonergic systems indicate an arousal-activating function for these neurotransmitters. Levels of 5-hydroxyindoleacetic acid (5-HIAA), a metabolite of serotonin, are low in persons who attempt suicide and in patients who are impulsive and aggressive. Raising serotonin levels with serotonergic agents such as fluoxetine can produce dramatic changes in some character traits of personality. In many persons, serotonin reduces depression, impulsiveness, and rumination and can produce a sense of general well-being. Increased dopamine concentrations in the central nervous system produced by certain psychostimulants (e.g., amphetamines) can induce euphoria. The effects of neurotransmitters on personality traits have generated much interest and controversy about whether personality traits are inborn or acquired.



Table 19-13
Medical Conditions Associated with Personality Change

Head trauma
Cerebrovascular diseases
Cerebral tumors
Epilepsy (particularly, complex partial epilepsy)
Huntington disease
Multiple sclerosis
Endocrine disorders

Heavy metal poisoning (manganese, mercury)
Neurosyphilis
Acquired immune deficiency syndrome (AIDS)

Electrophysiology. Changes in electrical conductance on the electroencephalogram (EEG) occur in some patients with personality disorders, most commonly antisocial and borderline types; these changes appear as slow-wave activity on EEGs.

Other Biologic Factors. In the case of personality change due to another medical condition, structural damage to the brain is usually the cause, and head trauma is probably the most common cause. Cerebral neoplasms and vascular accidents, particularly of the temporal and frontal lobes, are also common causes. The conditions most often associated with personality change are listed in Table 19-13.

Psychoanalytic Factors

Sigmund Freud suggested that personality traits are related to a fixation at one psychosexual stage of development. For example, those with an oral character are passive and dependent because they are fixated at the oral stage when the dependence on others for food is prominent. Those with an anal character are stubborn, parsimonious, and highly conscientious because of struggles over toilet training during the anal period.

Wilhelm Reich subsequently coined the term *character armor* to describe individuals' characteristic defensive styles for protecting themselves from internal impulses and interpersonal anxiety in significant relationships. Reich's theory has had a broad influence on contemporary concepts of personality and personality disorders. For example, each human being's unique stamp of personality depends on their characteristic defense mechanisms. Each personality disorder has a cluster of defenses that help psychodynamic clinicians recognize the type of character pathology present. Persons with paranoid personality disorder, for instance, use projection, whereas schizoid personality disorder is associated with withdrawal.

When defenses work effectively, persons with personality disorders master feelings of anxiety, depression, anger, shame, guilt, and other affects. Their

behavior is ego-syntonic; that is, it creates no distress for them even though it may adversely affect others. They may also be reluctant to engage in a treatment process; because their defenses are essential to controlling unpleasant affects, they are not interested in surrendering them.

In addition to characteristic defenses in personality disorders, another central feature is internal object relations. During development, a person develops particular patterns of the self in relation to others. Through introjection, children internalize a parent or another significant person as an internal presence that continues to feel like an object rather than a self. Through identification, children internalize parents and others in such a way that they incorporate the traits of the external object into the self, and the child "owns" the traits. These internal self-representations and object representations are crucial in developing the personality and, through externalization and projective identification, are played out in interpersonal scenarios in which they coerce others into playing a role in the person's internal life. Hence, persons with personality disorders are also identified by particular patterns of interpersonal relatedness that stem from these internal object relations patterns.

Defense Mechanisms. To help those with personality disorders, psychiatrists must appreciate patients' underlying defenses, the unconscious mental processes that the ego uses to resolve conflicts among the four lodestars of the inner life: instinct (wish or need), reality, important persons, and conscience. When defenses are most effective, especially in those with personality disorders, they can abolish anxiety and depression at the conscious level. Thus, abandoning a defense increases conscious awareness of anxiety and depression—a primary reason that those with personality disorders are reluctant to alter their behavior.

Although we tend to characterize patients with personality disorders by their most dominant or rigid mechanism, each patient uses several defenses. Therefore, the management of defense mechanisms used by patients with personality disorders is discussed here as a general topic and not as an aspect of the specific disorders. While many formulations are presented here in the language of psychoanalytic psychiatry, it is possible to translate them into principles consistent with cognitive and behavioral approaches.

FANTASY. Many people who are labeled schizoid—those who are eccentric, lonely, or frightened—seek solace and satisfaction within themselves by creating imaginary lives, especially imaginary friends. In their extensive dependence on fantasy, these persons often seem to be strikingly aloof. Therapists must understand that the unsociableness of these patients rests on a fear of intimacy. Rather than criticizing them or feeling rebuffed by their rejection, therapists should maintain a quiet, reassuring, and considerate interest without insisting on reciprocal responses. Recognition of patients' fear of closeness and respect for their eccentric ways are both therapeutic and useful.

DISSOCIATION. Dissociation is an unconscious defense mechanism that involves separating a mental or behavioral process (e.g., thought, feeling) from the rest of the person's psychic activity. Patients with borderline personality disorder may demonstrate dissociation during times of increased stress, including derealization or depersonalization.

ISOLATION. Isolation involves the separation of an idea or memory from its attached emotion. It is characteristic of controlled, orderly persons who are often labeled obsessive-compulsive personalities. These individuals remember the truth in fine detail but without affect. In a crisis, patients may show intensified self-restraint, overly formal social behavior, and obstinacy. Patients' quests for control may annoy clinicians or make them anxious. Often, such patients respond well to precise, systematic, and rational explanations and value efficiency, cleanliness, and punctuality as much as they do clinicians' effective responsiveness. Whenever possible, therapists should allow such patients to control their own care and should not engage in a battle of wills.

PROJECTION. In projection, patients attribute their unacknowledged feelings to others. Patients' excessive faultfinding and sensitivity to criticism may appear to therapists as prejudiced, hypervigilant, and injustice-collecting, but the therapist should not respond with defensiveness and argument. Instead, clinicians should frankly acknowledge even minor mistakes on their part and should discuss the possibility of future difficulties. Strict honesty, concern for patients' rights, and maintaining the same formal, concerning distance as

used with patients who use fantasy defenses are all helpful. Confrontation guarantees a lasting enemy and early termination of the interview. Therapists need not agree with patients' injustice-collecting, but they should ask whether both can agree to disagree.

The technique of counterprojection is especially helpful. Clinicians acknowledge and give paranoid patients full credit for their feelings and perceptions; they neither dispute patients' complaints nor reinforce them but agree that the world described by patients is conceivable. Interviewers can then talk about real motives and feelings, misattributed to someone else, and begin to cement an alliance with patients.

SPLITTING. In splitting, the patient divides persons toward whom they are, or have been, ambivalent into good and bad. For example, in an inpatient setting, a patient may idealize some staff members and uniformly disparage others. This defense behavior can be highly disruptive in a hospital ward and can ultimately provoke the staff to turn against the patient. When staff members anticipate the process, discuss it at staff meetings, and gently confront the patient with the fact that no one is all good or all bad, the phenomenon of splitting can be dealt with effectively.

ACTING OUT. In acting out, patients directly express unconscious wishes or conflicts through action to avoid being conscious of either the accompanying idea or the affect. Tantrums, apparently motiveless assaults and pleasureless promiscuity are typical examples. Because the behavior occurs outside reflective awareness, acting out appears to observers to be unaccompanied by guilt, but when acting out is impossible, the conflict behind the defense may be accessible. The clinician faced with acting out, either aggressive or sexual, in an interview situation, must recognize that the patient has lost control, that the patient will mishear anything the interviewer says, and that getting the patient's attention is of paramount importance. Depending on the circumstances, a clinician's response may be, "How can I help you if you keep screaming?" Alternatively, if the patient's loss of control seems to be escalating, say, "If you continue screaming, I will leave." An interviewer who feels genuinely frightened of the patient can simply leave and, if necessary, ask for help from ward attendants or the police.

PROJECTIVE IDENTIFICATION. The defense mechanism of projective identification appears mainly in borderline personality disorder and consists of three steps. First, the patient projects an aspect of the self onto someone else. The projector then tries to coerce the other person into identifying what they projected. Finally, the recipient of the projection and the projector feel a sense of oneness or union.

Temperament

Temperament refers to the body's biases in the modulation of conditioned behavioral responses to prescriptive physical stimuli. Behavioral conditioning (i.e., procedural learning) involves presemantic sensations that elicit basic emotions, such as fear or anger, independent of conscious recognition, descriptive observation, reflection, or reasoning. Pioneering work by A. Thomas and S. Chess conceptualized temperament as the stylistic component ("how") of behavior, as differentiated from the motivation ("why") and the content ("what") of behavior. Modern concepts of temperament emphasize its emotional, motivational, and adaptive aspects. Specifically, four major temperament traits have been identified and subjected to extensive neurobiologic, psychosocial, and clinical investigation: harm avoidance, novelty seeking, reward dependence, and persistence. These four temperaments are now understood to be genetically independent dimensions that occur in all possible combinations within the same individual rather than as mutually exclusive categories.

Psychobiology of Temperament. Temperament traits of harm avoidance, novelty seeking, reward dependence, and persistence are heritable differences underlying one's automatic response to danger, novelty, social approval, and intermittent reward, respectively. These four temperaments are closely associated with the four basic emotions of fear (harm avoidance), anger (novelty seeking), attachment (reward dependence), and ambition (persistence).

Individual differences in temperament and basic emotions modify the processing of sensory information and shape early learning characteristics, especially associative conditioning of unconscious behavior responses and preattentive components of perception. Temperament occurs in terms of

heritable biases in emotionality and learning that underlie the acquisition of emotion-based, automatic behavioral traits and habits observable early in life and relatively stable over one's lifespan.

Each of the four major dimensions is a normally distributed quantitative trait, moderately heritable, observable early in childhood, relatively stable in time, and moderately predictive of adolescent and adult behavior. The four dimensions are genetically homogeneous and independently inherited from one another, as per several extensive international twin studies. Temperamental differences, which are not very stable initially, tend to stabilize during the second and third years of life. Accordingly, ratings of these four temperament traits at ages 10 to 11 years were moderately predictive of personality traits at ages 15, 18, and 27 years in a large sample of Swedish children.

The four dimensions are universal across different cultures, ethnic groups, and political systems on every inhabited continent. We call these aspects of personality temperaments because they are heritable, manifest early in life, are developmentally stable, and are consistent in different cultures. Temperament traits are similar to crystallized intelligence in that they do not show the rapid changes with increasing age or across birth cohorts found for fluid intelligence and character traits. Table 19-14 summarizes contrasting sets of behaviors that distinguish extreme scorers on the four dimensions of temperament. Note that each extreme of these dimensions has specific adaptive advantages and disadvantages so that neither high nor low scores inherently mean better adaptation.

Each of the four temperament dimensions has unique genetic determinants according to family and twin studies, as well as studies of genetic associations with specific DNA markers.

HARM AVOIDANCE. Harm avoidance involves a heritable bias in the inhibition of behavior in response to signals of punishment and frustrative non-reward. High harm avoidance is a fear of uncertainty, social inhibition, shyness, passive avoidance of problems/danger, rapid fatigability, and pessimistic worry in anticipation of problems even in situations that do not worry others. Those low in harm avoidance are carefree, courageous, energetic, outgoing, and optimistic, even in situations that worry most people.

The psychobiology of harm avoidance is complex. Benzodiazepines

disinhibit avoidance by γ-aminobutyric acid (GABA)-ergic inhibition of serotonergic neurons originating in the dorsal raphe nuclei. The anterior serotonergic cells in the dorsal raphe nucleus intermingle with the dopaminergic cells of the ventral tegmental area, and both groups innervate the same structures (e.g., basal ganglia, accumbens, amygdala), providing opposing dopaminergic-serotonergic influences in the modulation of approach and avoidance behavior. Persons given serotonin drugs show decreased harm avoidance behavior.

Table 19-14
Descriptors of Individuals Who Score High or Low on the Four Temperament Dimensions

	Descriptors of Extreme Variants	
Temperament Dimension	**High**	**Low**
Harm avoidance	Pessimistic	Optimistic
	Fearful	Daring
	Shy	Outgoing
	Fatigable	Energetic
Novelty seeking	Exploratory	Reserved
	Impulsive	Deliberate
	Extravagant	Thrifty
	Disorderly	Orderly
Reward dependence	Sentimental	Detached
	Open	Aloof
	Warm	Cold
	Affectionate	Independent
Persistence	Industrious	Lazy

	Determined	Spoiled
	Ambitious	Underachiever
	Perfectionist	Pragmatist

NOVELTY SEEKING. Novelty seeking reflects a heritable bias in the initiation or activation of appetitive approach in response to novelty, approach to signals of reward, active avoidance of conditioned signals of punishment, and escape from unconditioned punishment. Novelty seeking is an exploratory activity in response to novelty, impulsiveness, extravagance in approach to cues of reward, and active avoidance of frustration. Individuals high in novelty seeking are quick-tempered, curious, easily bored, impulsive, extravagant, and disorderly. Those low in novelty seeking are slow tempered, uninquiring, stoical, reflective, frugal, reserved, tolerant of monotony, and orderly.

Dopaminergic projections have a crucial role in novelty seeking. Novelty seeking seems to involve increased reuptake of dopamine at presynaptic terminals, thereby requiring frequent stimulation to maintain optimal levels of postsynaptic dopaminergic stimulation. Studies of candidate genes involved in dopamine neurotransmission, have provided evidence of association with novelty seeking and no other dimension of temperament.

REWARD DEPENDENCE. Reward dependence reflects the maintenance of behavior in response to cues of social reward. Individuals high in reward dependence are tender-hearted, sensitive, dedicated, dependent, and warmly sociable. Those low in reward dependence are practical, tough-minded, cold, socially insensitive, irresolute, and indifferent if alone.

Noradrenergic projections from the locus coeruleus and serotonergic projections from the median raphe likely influence such reward conditioning. The “love hormone” oxytocin’s plasma level is positively correlated ($r = 0.5$) with individual differences in reward dependence but not other dimensions of personality.

PERSISTENCE. Persistence reflects the maintenance of behavior despite frustration, fatigue, and intermittent reinforcement. It manifests as industriousness, determination, ambitiousness, and perfectionism. Highly persistent individuals are hard-working, perseverant, and ambitious

overachievers who tend to intensify their effort in response to anticipated rewards and view frustration and fatigue as personal challenges. Those low in persistence are indolent, inactive, unstable, and erratic; they tend to give up easily when faced with frustration, rarely strive for higher accomplishments, and manifest little perseverance even in response to intermittent reward.

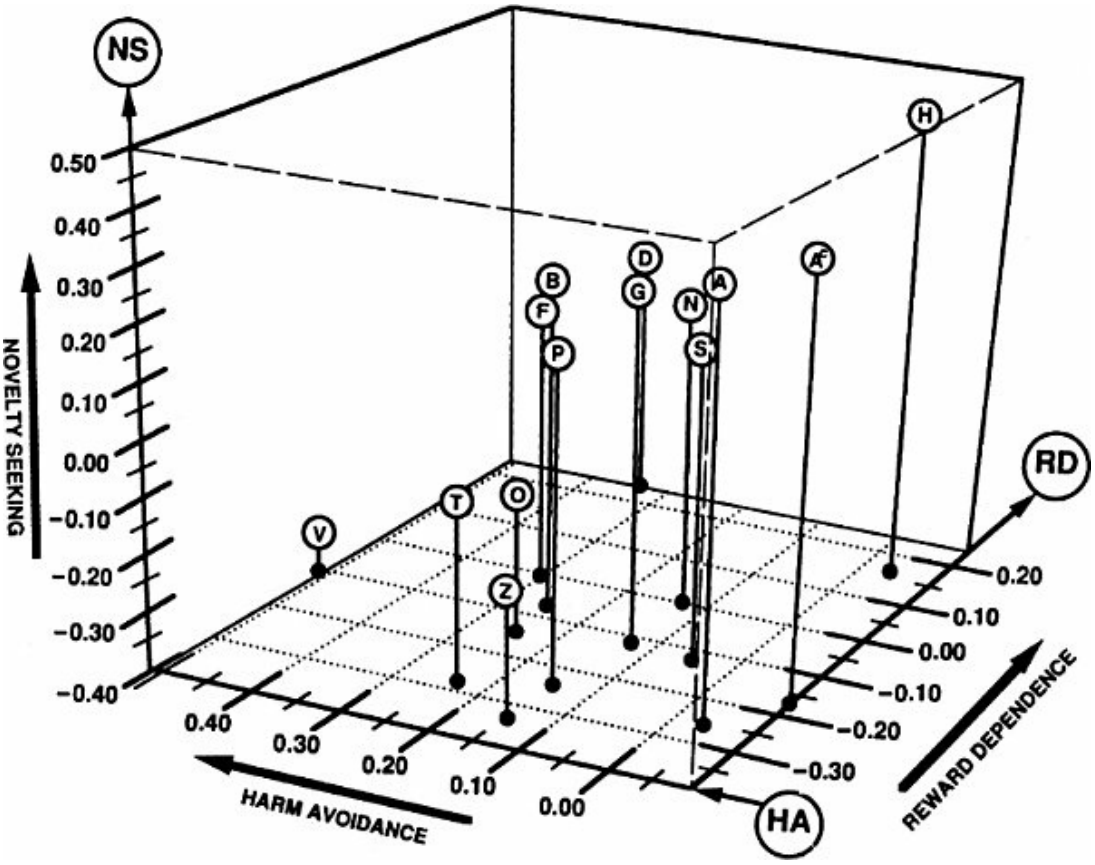


FIGURE 19-1
Correlations between individual personality disorder subtypes and temperament traits of harm avoidance (HA), novelty seeking (NS), and reward dependence (RD). A, antisocial; Ac, antisocial children; B, borderline; D, dependent; F, self-defeating; G, passive aggressive; H, histrionic; N, narcissistic; O, obsessive; P, paranoid; S, sadistic; GT, schizotypal; V, avoidant; Z, schizoid.

Individual differences in persistence have been strongly correlated ($r = 0.8$) with responses measured by fMRI in a circuit involving the ventral striatum, orbitofrontal cortex/rostral insula, and dorsal anterior cingulate cortex.

Figure 19-1 illustrates the interrelationships between the personality disorders and the temperament traits of harm avoidance, novelty seeking, and reward dependence.

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20

Other Conditions that May be a Focus of Clinical Attention

INTRODUCTION

DSM-5 includes a section called Other Conditions that May be a Focus of Clinical Attention. ICD-10 has a similar section called Factors Influencing Health Status and Contact with Health Services. These sections are for conditions that are not mental disorders but that have led to contact with the mental health care system. In some instances, one of these conditions will be noted during a psychiatric evaluation (e.g., divorce), although there is no mental disorder. In other instances, the diagnostic evaluation reveals no mental disorder, but there is a need to note the primary reason for contact with the mental health care system (e.g., homelessness).

In some cases, we may eventually find a mental disorder, but the focus of attention or treatment is on a condition that is not caused by a mental disorder. For example, a patient with an anxiety disorder may receive treatment for a marital problem that is unrelated to the anxiety disorder itself.

Table 20-1 includes the many conditions DSM-5 lists that may be a focus of clinical attention, or that may influence the diagnosis, treatment, or course of a mental disorder contained in DSM-5. The list of conditions that make up this category covers the entire life cycle from infancy through childhood, adolescence, adulthood, and old age. The list of conditions covers almost every conceivable life circumstance, from divorce to problems related to being in military service. Each of these conditions or circumstances is capable of having a profound input on a particular mental illness or the human experience in general.

MALINGERING

Malingering is the deliberate falsification of physical or psychological symptoms in an attempt to achieve a secondary gain such as avoiding military duty, avoiding work, obtaining financial compensation, evading criminal prosecution, or obtaining drugs. Under some circumstances, malingering may represent adaptive behavior—for example, as mentioned below, feigning illness while a captive of the enemy during wartime.

The clinician should consider possible malingering when encountering any combination of the following: (1) medicolegal context of presentation (e.g., the person is referred by an attorney to the clinician for examination), (2) evident discrepancy between the individual's claimed stress or disability and the objective findings, (3) lack of cooperation during the diagnostic evaluation and in complying with the prescribed treatment regimen, and (4) the presence of antisocial personality disorder.

Diagnosis and Clinical Features

Avoidance of Criminal Responsibility, Trial, and Punishment. Criminals may pretend to be incompetent to avoid standing trial; they may feign insanity at the time of the perpetration of the crime, malinger symptoms to receive a less harsh penalty, or attempt to act too incapacitated (incompetent) for punishment.

Avoidance of Military Service or Particularly Hazardous Duties. Persons may malinger to avoid conscription into the armed forces, and, after being conscripted, they may feign illness to escape from particularly onerous or hazardous duties.

Financial Gain. Malingerers may seek financial gain in the form of undeserved disability insurance, veterans' benefits, workers' compensation, or tort damages for purported psychological injury.

Avoidance of Work, Social Responsibility, and Social Consequences. Individuals may malinger to escape from unpleasant vocational or social circumstances or to avoid the social and litigation-related

consequences of vocational or social improprieties.

An owner of a previously successful photographic equipment supplier declared bankruptcy in a way that the government maintained was illegal. Subsequently, the government indicted the defendant on various counts of fraud. The defendant's counsel maintained that the defendant was too depressed to cooperate with him and that, because of that depression, he experienced memory loss that made it impossible to understand what had occurred and, therefore, impossible to provide a meaningful defense. The government's forensic psychiatrist evaluated the defendant to ascertain the nature of his depression and to determine whether it was causing cognitive problems.

When asked early in his evaluation when his birthday was, he responded, "Oh, what does it matter? It was in the 40s or 50s." Similarly, when queried about where he was born, he said, "Somewhere in Hungary." Even when pressed for more specifics, he refused to elaborate. Nevertheless, at many points later in his evaluation, he responded with complete, often detailed information about transactions not related to those for which he was under indictment. It was the impression of the evaluator that the defendant was malingering grossly and inconsistently, incompatible with the kinds of decreases in cognitive skills that occasionally attend major depression. (Adapted from a case of Mark J. Mills, J.D., M.D. and Mark S. Lipian, M.D., Ph.D.)



Table 20-1
Conditions that May be a Focus of Clinical Attention

Relational Problems

- Problems Related to Family Upbringing
  - Parent–Child Problem
  - Sibling Problem
  - Upbringing Away from Parents
  - Parental Relationship Distress
- Other Problems Related to Primary Support System

- Relationship Distress with Spouse or Partner
- Separation or Divorce
- Emotional Distress within Family
- Uncomplicated Bereavement

Abuse and Neglect

- Problems of Child Maltreatment and Neglect
  - Child Physical Abuse
  - Child Sexual Abuse
  - Child Neglect
  - Child Psychological Abuse
- Problems of Adult Maltreatment and Neglect
  - Physical Violence of Spouse or Partner
  - Sexual Violence of Spouse or Partner
  - Spouse or Partner Neglect
  - Psychological Spouse or Partner Abuse
  - Adult Abuse by Nonspouse or Nonpartner (e.g., Physical, Sexual, Psychological)

Educational and Occupational Problems

- Educational Problems
- Occupational Problems
  - Related to Current Military Deployment Status
  - Other (e.g., Job change, Loss of job, Stress)

Housing and Economic Problems

- Housing Problems
  - Homelessness
  - Inadequate Housing (e.g., Lack of Heat or Electricity, Insect or Rodent Infestation)
  - Neighbor, Lodger, or Landlord Discord
  - Residential Institution (Does Not Include Psychological Reaction to Change in Living Situation; see Adjustment Disorders, Chapter 10)
- Economic Problems
  - Lack of Adequate Food or Safe Drinking Water

- Extreme Poverty
- Low Income

Other Problems Related to the Social Environment
- Phase of Life Problem
- Acculturation Difficulty
- Social Exclusion or Rejection
- Discrimination or Persecution

Problems Related to Crime or Interaction with the Legal System (e.g., Victim of Crime, Imprisonment, Release from Prison)

Other Health Service Encounters for Counseling and Medical Advice (e.g., Sexual Counseling)

Problems Related to Other Psychological, Personal, and Environmental Circumstances
- Religious or Spiritual Problem
- Victim of Terrorism or Torture
- Exposure to Disaster or War

Other Circumstances of Personal History
- Adult Antisocial Behavior
- Child or Adolescent Antisocial Behavior
- Problems Related to Access to Medical and Other Health Care
- Nonadherence to Medical Treatment (e.g., Overweight or Obesity, Malingering, Wandering Associated with a Mental Disorder)
- Borderline Intellectual Functioning

Adapted from *Diagnostic and Statistical Manual of Mental Disorders*. 5th ed., American Psychiatric Association; 2013.

Facilitation of Transfer from Prison to Hospital. Prisoners may malinger (fake bad) to obtain a transfer to a psychiatric hospital from which they may hope to escape or in which they expect to do "easier time." The prison context may also give rise to dissimulation (faking good). However, the prospect of an indeterminate number of days on a mental health ward may prompt an inmate with actual psychiatric symptoms to make every effort to

conceal them.

Admission to a Hospital. In this era of deinstitutionalization and homelessness, individuals may malinger to gain admission to a psychiatric hospital. The individual may see such institutions as providing free room and board, a haven from the police, or refuge from rival gang members or disgruntled drug cronies who have made street life even more unbearable and hazardous than it usually is.

> A robust, neatly attired man presented to the psychiatric emergency department in the early-morning hours. He stated that "the voices" were worse than before and that he wished readmission to the hospital. When the psychiatrist challenged him, observing that his discharge was just that afternoon, that he routinely left the hospital in the morning and demanded rehospitalization at night, and that, despite multiple hospitalizations, clinicians increasingly doubted his reported history of hallucinations, the man became belligerent. When the psychiatrist still refused to admit him, the patient grabbed the psychiatrist's clothes, threatening him but inflicting no harm. The psychiatrist asked the hospital police to escort him off the grounds. The treatment team told the patient that he could seek readmission to his regular ward during the day. Subsequent contact with the patient's ward revealed that their diagnoses were substance abuse and homelessness; his apparent schizophrenia appeared never to have been an actual issue in his treatment. (Courtesy of Mark J. Mills, J.D., M.D. and Mark S. Lipian, M.D., Ph.D.)

Drug Seeking. Malingerers may feign illness to obtain favored medications, either for personal use or, in a prison setting, as currency to barter for cigarettes, protection, or other inmate-provided favors.

> The plaintiff, a woman in her late 20s, was injured while dancing at a club. Although her claim initially appeared bona fide, subsequent investigation cast doubt on the mechanism of injury that she claimed—namely, that a misplaced electrical cord under a carpet caused her to slip.

This was true, she claimed, even though she had been dancing in a particularly jerky manner that could have easily caused problems without tripping.

Subsequently, she sought medical and surgical treatment for torn cartilage in her injured knee. Even though the initial surgery went well, she kept reinjuring the knee with various "slips." As a result, she requested narcotic analgesics. A careful medical record review revealed that she was obtaining such medications from multiple practitioners and that she had forged at least one prescription.

In reviewing the case before binding arbitration, it was the opinion of the orthopedic and psychiatric consultants that, although the initial injury and reported pain were real, the plaintiff consciously elaborated her injuries to obtain the desired narcotic analgesics. (Courtesy of Mark J. Mills, J.D., M.D. and Mark S. Lipian, M.D., Ph.D.)

Child Custody. Minimizing difficulties or faking good for the sake of obtaining child custody can occur when one party accurately accuses the other of being an unfit parent because of psychological conditions. The accused party may feel compelled to minimize symptoms or to portray him- or herself in a positive light to reduce the chances of being deemed unfit and losing custody.

Differential Diagnosis

The challenge is to differentiate malingering from an actual physical or psychiatric illness. Furthermore, the possibility of partial malingering, which is an exaggeration of existing symptoms, must be entertained. Also, the possibility exists of unintentional, dynamically driven misattribution of genuine symptoms (e.g., of depression) to an incorrect environmental cause (e.g., to sexual harassment rather than to narcissistic injury).

We should also remember that a real psychiatric disorder and malingering are not mutually exclusive.

Factitious disorder is distinguished from malingering by motivation (sick role vs. tangible pain), whereas the somatoform disorders involve no conscious volition. In conversion disorder, as in malingering, objective signs cannot account for subjective experience, and differentiation between the two

disorders can be difficult. Table 20-2 lists some variables that may aid in distinguishing between these two conditions.

Epidemiology

There is perhaps a 1 percent prevalence of malingering among mental health patients in civilian clinical practice, with the estimate rising to 5 percent in the military. In a litigious context, during interviews of criminal defendants, the estimated prevalence of malingering is much higher—between 10 and 20 percent. Approximately 50 percent of children presenting with conduct disorders have reported lying-related issues.

Although there is no apparent familial or genetic patterns, and no clear-cut sex or age bias at onset, malingering does appear to be highly prevalent in some military, prison, and litigious populations and, in Western society, in men from youth through middle age. Associated disorders include conduct and anxiety disorders in children and antisocial, borderline, and narcissistic personality disorders in adults.

Etiology

Although there do not seem to be any biologic factors causally related to malingering, its frequent association with antisocial personality disorder raises the possibility that hypoarousability may be an underlying metabolic factor. Still, no predisposing genetic, neurophysiologic, neurochemical, or neuroendocrinologic forces are presently known.



Table 20-2
Factors Aiding in the Differentiation between Malingering and Conversion Disorder

1. Malingerers are more likely to be suspicious, uncooperative, aloof, and unfriendly; patients with conversion disorder are likely to be friendly, cooperative, appealing, dependent, and clinging.
2. Malingerers may try to avoid diagnostic evaluations and refuse recommended treatment; patients with conversion disorder likely welcome evaluation and treatment, “searching for an answer.”
3. Malingerers likely refuse employment opportunities designed to

circumvent their disability; patients with conversion disorder likely accept such opportunities.

4. Malingerers are more likely to provide extremely detailed and exacting descriptions of events precipitating their “illness”; patients with conversion disorder are more likely to report historical gaps, inaccuracies, and vagaries.

Course and Prognosis

Malingering persists as long as the malingerer believes it will likely produce the desired rewards. In the absence of concurrent diagnoses, after the individual has attained the goal, the feigned symptoms disappear. In some structured settings, such as the military or prison units, ignoring the malingered behavior may result in its disappearance, mainly if an expectation of continued productive performance, despite complaints, is made clear. In children, malingering is most likely associated with a predisposing anxiety or conduct disorder; proper attention to this developing problem may alleviate the child’s propensity to malinger.

Treatment

The appropriate stance for the psychiatrist is clinical neutrality. If malingering is suspected, a careful differential investigation should ensue. If, after the diagnostic evaluation, malingering seems most likely, the clinician should tactfully but firmly confront the patient with the apparent outcome. The clinician should also try to understand the underlying reasons for the ruse, and explore alternative, more acceptable pathways to the desired outcome. Coexisting psychiatric disorders should be thoroughly assessed. Only if the patient is utterly unwilling to interact with the physician under any terms other than manipulation should the physician abandon the interaction.

BEREAVEMENT

Normal bereavement begins immediately after or within a few months of the loss of a loved one. Typical signs and symptoms include feelings of sadness, preoccupation with thoughts about the deceased, tearfulness, irritability,

insomnia, and difficulties concentrating and carrying out daily activities. Based on the cultural group, bereavement is limited to a varying time, usually 6 months, but it can be longer. Normal bereavement, however, can lead to a full depressive disorder that requires treatment. Some grieving individuals present with symptoms characteristic of a major depressive episode such as depressed mood, insomnia, anorexia, and weight loss. The duration of grief and bereavement varies considerably among different cultural groups and within the same cultural group. The diagnosis of depressive disorder is generally not given unless the symptoms are still present 2 months after the loss. However, the presence of specific symptoms that are not characteristic of a "normal" grief reaction may help differentiate bereavement from depression. These include (1) guilt about things other than actions taken or not taken by the survivor at the time of the death, (2) thoughts of death other than the survivor feeling that he or she would be better off dead or should have died with the deceased person, (3) morbid preoccupation with worthlessness, (4) marked psychomotor retardation, (5) prolonged and marked functional impairment, and (6) hallucinatory experiences other than thinking that he or she hears the voice of or transiently sees the image of the deceased person.

OCCUPATIONAL PROBLEMS

Occupational problems often arise during stressful changes in work, namely, at initial entry into the workforce or when making job changes within the same organization to a higher position because of excellent quality performance or to a parallel position because of corporate need. Distress often occurs when not choosing these changes or if the person feels unprepared for the change. It is also true during layoffs and at retirement, especially if retirement is mandatory, and the person is unprepared for this event. Work distress can result if one's work situation changes to work overload, or conversely, a lack of challenge. Also, an individual may feel unable to fulfill work expectations, perhaps because they lack sufficient authority to enact any change or meet with conflicting expectations. Working with harsh or unreasonable superiors can be another common source of work distress.

Work Choices and Changes

Young adults without role models or guidance from families, mentors, or others in their communities too often underestimate their potential lifetime abilities to learn a trade or earn a college or postgraduate degree. Also, women and members of minority groups often feel less prepared to accept work challenges, fear rejection, and do not apply for jobs for which they are qualified. On the other hand, men, in fields in which they are underrepresented, often and confidently move up the career ladder faster (glass elevator). As part of initial interviews for evaluation of occupational problems, patients should be encouraged to consider their heretofore unrecognized, unadmitted talents; long-held, yet unexpressed, dreams and goals regarding work; actual successes in work and school; and motivation to risk learning what they would find satisfying.

Minorities and those in low-paying and low-skilled jobs too often have less job security. Business and institutional reorganization and consequent downsizing, factory closings, and moves affect many, often leaving these workers feeling hopeless and helpless about future employment, on welfare, angry, and depressed.

With ongoing and often sudden downsizing of corporations and businesses, men and women continue to struggle with unexpected job loss and premature retirement even when finances are not an issue. Also, men, in particular, define themselves by their work roles and thus experience more occupational distress from these changes. Women may adjust faster to retirement, but they often have less financial security than men. Even today, white women earn approximately 80 cents on the dollar, and African American and Hispanic women earn even less for comparable work. In addition to lower pay, women may have lower status, find themselves widowed more often than men, and are more likely to be caring for children, grandchildren, and elderly relatives. Women represent more of the single working parent group and the working poor.

Stress and the Workplace

More than 30 percent of workers report that they are under stress at work. At least 15 percent of occupational disability claims involve workplace distress. Expected distress follows recognized and uncontrollable work changes—

downsizing, mergers and acquisitions; work overload; and chronic physical strains, including work noise, temperature, bodily injuries, and strain from performing computer work.

Work frustration can also arise from an individual worker's unrecognized (and therefore unresolved) psychodynamic issues, such as working appropriately with superiors and not relating to one's supervisor as a parent figure. Other developmental issues include unresolved problems with competition, assertiveness, envy, fear of success, and the inability to communicate verbally in a constructive manner.

After the September 11, 2001, World Trade Center tragedy, a 32-year-old married male firefighter, who had been away on vacation that day with his wife and children, began to exhibit changed behaviors at home and work. At home, he appeared not to listen to his two children and, instead, focused his attention on television sporting events. At work, he also appeared to be more focused on cooking the same dinners for his peers and watching television than on interacting verbally with his remaining peers and the new chief. In several months, a chaplain visited the station several times and talked to the firefighters about survivor guilt and the 9/11 tragedy, and the firefighter began to return somewhat to his former healthier behaviors. (Courtesy of Leah J. Dickstein, M.D.)

Often, work conflicts reflect similar conflicts in the worker's personal life, and referral for treatment, unless there is insight, is in order. Some studies have found that massage therapy, meditation, and yoga at intervals during the workday relieve stress when used regularly. Approaches using cognitive therapy have also helped people reduce work pressure.

Suicide Risk

Some occupations—health professionals, financial service workers, and police, the first and latter groups because of more access to lethal drugs and weapons—both attract persons with high suicide risk and involve increased chronic distress that may lead to higher suicide rates.

Career and Job Problems of Women

Most women work outside the home out of necessity to support themselves or their dependents (whether children or adults) or as part of a working couple. With the divorce rate remaining at the 50 percent level, many women find themselves economically more impoverished after a divorce than when married, although divorced men usually find their economic status improved. Despite more than four decades of increasing knowledge about and concern for women's status in the workplace, unique gender issues, bias, and lack of accommodation to their unique needs at certain life stages (i.e., pregnancy and postpartum, the primary responsibility for young healthy and ill children) continue. However, women were the largest group establishing new small businesses in the 1990s. Many have left large corporations where they felt undervalued because of their gender. Women experience problems when they are the sole woman in a man's field. Despite increasing recognition of the need for men in relationships with women to assume home and family responsibilities, fewer than 25 percent of men do so equitably.

Women of childbearing and child-rearing ages continue to find themselves in conflict with job expectations, opportunities, and personal responsibilities. High-quality, on-site, dependent-care facilities with extended hours are rare and often out of range financially. Major unresolved work issues that are unique to women at certain life stages include flextime and paid and unpaid dependent leave options. Beyond dependent care issues, women in the workforce continue to experience distress after chronic and repeated sexual harassment, despite its illegality and media attention. Increasingly, more women have travel responsibilities, work long hours, work shifts beyond daylight hours, and experience personal workplace violence.

Among dual-career families and partners, the woman is more likely to move when the man chooses to move for a work opportunity than vice versa. Consequently, a woman's career is interrupted more often. There is now less hesitation to have the two members of a relationship work for the same organization than previously, albeit usually in different departments. Work distress may also stem from continuous miscommunication, especially that based on gender.

Working Teenagers

With unemployment increasing, many teenagers work part time while attending high school. Consequently, stress can arise because of less parent–teenager interaction and constructive parental control issues about teens' use of earnings, time spent away from home, and consequent behaviors both in and outside the home. When both parents or a single parent, as well as the teenager, work outside the home, often on different schedules, parent–teen verbal communication must be proactive, transparent, and ongoing.

Working within the Home

Although most women with children of all ages must work outside the home, at times, they may be home full time or part time or may work at home. When their husbands or partners work full time outside the home, problems may develop from each one's perceived expectations of the other. Women who care for children and their home exclusively may be seen by their partners as not only economically dependent and inferior but also not as competent and not understanding of the man's stressors and needs. Ongoing respectful listening and verbal communication must be encouraged.

People in organizations are increasingly taking work home as their work expectations increase. This work-at-home experience can and does interfere with personal lives and satisfaction, which can then have further repercussions at work.

Chronic Illness

As general and other medical and psychiatric treatments for chronic diseases improve, employers have been increasingly concerned about accommodating patients with acquired immunodeficiency syndrome (AIDS), diabetes mellitus, and other disorders. The issue of mandatory testing for AIDS and substance abuse (alcohol and other illegal substances) continues to be of concern. Employee assistance programs offering education about general and mental health topics have proved timely and cost-effective.

Domestic Violence

Although occurring in the home, signs and symptoms that interfere with work may help identify victims of domestic violence. Trained professionals must question all employees experiencing work distress about domestic violence

and, when indicated, refer individuals for assistance, which includes safety in the workplace.

Job Loss

Regardless of the reason for job loss, most people experience distress, at least temporarily, including symptoms of normal grief, loss of self-esteem, anger, and reactive depressive and anxiety symptoms, as well as somatic symptoms and possibly the onset of or increase in substance abuse or domestic violence. Timely education, support programs, and vocational guidance can be helpful, as is access to treatment.

Vocational Rehabilitation

Rehabilitation is often necessary for those traumatized by stresses in the workplace, those who had to take a leave for medical or psychiatric reasons, and those who lose their jobs. Individual or group counseling enables persons to improve personal relationships, raise self-esteem, or learn new work skills. Patients with schizophrenia may benefit from sheltered workshops in which they perform work geared to their level of function. Some patients with schizophrenia or autism do well in tasks that are repetitive or require obsessive concern with details.

ADULT ANTISOCIAL BEHAVIOR

Characterized by activities that are illegal, immoral, or both, antisocial behavior usually begins in childhood and often persists throughout life. The term *antisocial behavior* somewhat confusingly applies both to persons' actions that are not due to a mental disorder and to actions by those who never received a neuropsychiatric workup to determine the presence or absence of a mental disorder. As Dorothy Lewis noted, the term can apply to behavior by otherwise normal persons who "struggle to make a dishonest living."

Diagnosis and Clinical Features

The diagnosis of adult antisocial behavior is one of exclusion. Substance dependence in such behavior often makes it difficult to separate the antisocial

behavior related primarily to substance dependence from disordered behaviors that occurred either before substance use or during episodes unrelated to substance dependence.

It is also essential that we differentiate this from an antisocial personality disorder, and when a patient meets the criteria for the personality disorder, that diagnosis would supersede this one.

During the manic phases of bipolar I disorder, certain aspects of behavior, such as wanderlust, sexual promiscuity, and financial difficulties can be similar to adult antisocial behavior. Patients with schizophrenia may have episodes of adult antisocial behavior, but the symptom picture is usually clear, especially regarding thought disorder, delusions, and hallucinations on the mental status examination.

Neurologic conditions can be associated with adult antisocial behavior, and, when indicated, a full workup, including neuroimaging, can help identify a cause of the behavior. We should consider complex partial seizures in the differential diagnosis. When making a clear-cut diagnosis of epilepsy or encephalitis, we should then consider whether the disorder is contributing to antisocial behavior. Abnormal EEG findings are prevalent among violent offenders: an estimated 50 percent of aggressive criminals have abnormal EEG findings.

Persons with adult antisocial behavior have difficulties in work, marriage, and money matters and conflicts with various authorities. Table 20-3 summarizes the symptoms of adult antisocial behavior.

Epidemiology

Depending on the specifics of the case finding, adult antisocial behavior may range from 5 to 15 percent of the population. Within prison populations, investigators report prevalence figures between 20 and 80 percent. Men account for more adult antisocial behavior than do women.



Table 20-3
Symptoms of Adult Antisocial Behavior

Life Area	Antisocial Patients with Significant Problems in Area (%)

Work problems	85
Marital problems	81
Financial dependence	79
Arrests	75
Alcohol abuse	72
School problems	71
Impulsiveness	67
Sexual behavior	64
Wild adolescence	62
Vagrancy	60
Belligerence	58
Social isolation	56
Military record (of those serving)	53
Lack of guilt	40
Somatic complaints	31
Use of aliases	29
Pathologic lying	16
Drug abuse	15
Suicide attempts	11

Data from Robins L. *Deviant Children Grown Up: A Sociological and Psychiatric Study of Sociopathic Personality*. Baltimore, MD: Williams & Wilkins; 1966.

Etiology

Antisocial behaviors in adulthood are characteristic of a variety of persons, ranging from those with no demonstrable psychopathology to those who are

severely impaired and have psychotic disorders, cognitive disorders, and retardation, among other conditions. A comprehensive neuropsychiatric assessment of antisocial adults is indicated and may reveal potentially treatable psychiatric and neurologic impairments. Only in the absence of mental disorders can patients be categorized as displaying adult antisocial behavior. Genetic and social factors may influence adult antisocial behavior.

Genetic Factors. Some studies found a 60 percent concordance rate in monozygotic twins and about a 30 percent concordance rate in dizygotic twins. Adoption studies show a high rate of antisocial behavior in the biologic relatives of adoptees identified with antisocial behavior and a high incidence of antisocial behavior in the adopted-away offspring of those with antisocial behavior. The prenatal and perinatal periods of those who subsequently display antisocial behavior often are associated with low birth weight, mental retardation, and prenatal exposure to alcohol and other drugs of abuse.

Social Factors. Studies have shown that in neighborhoods in which families with low socioeconomic status (SES) predominate, the sons of unskilled workers are more likely to commit more offenses and more serious criminal offenses than do the sons of middle-class and skilled workers, at least during adolescence and early adulthood. These data are not as precise for women, but the findings are generally similar in studies from many countries. Areas of family training differ by SES group. Middle-SES parents use love-oriented techniques in discipline. They withdraw affection rather than impose physical punishment, as is done in low-SES groups. Negative parental attitudes toward aggressive behavior attempt to curb that behavior, and the ability to communicate parental values are more characteristic of middle- and high-SES groups than of low ones. Adult antisocial behavior is associated with the use and abuse of alcohol and other substances and with the easy availability of handguns.

Treatment

In general, therapists are pessimistic about treating adult antisocial behavior. They have little hope of changing a pattern that has been present almost continuously throughout a person’s life. Psychotherapy has not been

effective, and no breakthroughs with biologic treatments, including medications, have occurred.

Therapists show more enthusiasm for the use of therapeutic communities and other forms of group treatment, although the data provide little basis for optimism. Many adult criminals, incarcerated in institutional settings, have shown some response to group therapy approaches. The history of violence, criminality, and antisocial behavior has shown that such behaviors seem to decrease after age 40 years. Recidivism in criminals, which can reach 90 percent in some studies, also decreases in middle age.

Prevention. Because antisocial behavior often begins during childhood, the primary focus must be on delinquency prevention. Any measures that improve the physical and mental health of socioeconomically disadvantaged children and their families are likely to reduce delinquency and violent crime. Often, recurrently violent persons have sustained many insults to the central nervous system (CNS) prenatally and throughout childhood and adolescence. Consequently, we need to develop programs to educate parents about the dangers to their children of CNS injury from maltreatment, including the effects of psychoactive substances on the brains of growing fetuses. Public education about the releasing effect of alcohol on violent behaviors (as well as its contribution to vehicular homicide) may also reduce crime.

In a Surgeon General's Workshop on Violence and Public Health (1985), the work group on preventing assault and homicide emphasized the importance of discouraging corporal punishment in the home, forbidding it in the schools, and even abolishing capital punishment by the state, saying that all are models and sanctions for violence. No evidence indicates that capital punishment reduces crime in the states that have it. Opponents of capital punishment see it as "vengeance," not punishment.

Although persons disagree about the contribution of violence in the media to violent crime, the propaganda potential of the media is universally recognized. We have not fully realized the extent to which the media, such as television, can be used to transmit positive social values. The guidelines issued by the television industry to indicate the amount of sex and violence in programs is an attempt to deal with the issue; however, program content that espouses traditional societal values would be beneficial.

The most successful preventive measures within the field of medicine have

come from community-wide public health programs (e.g., campaigns against smoking) and from programs that detect individual vulnerabilities (e.g., individual monitoring of blood pressure). Studies of adult antisocial behavior reveal the contribution of broad cultural factors and constellations of individual biopsychosocial vulnerabilities. Prevention programs must recognize and address both kinds of factors.

RELIGIOUS OR SPIRITUAL PROBLEM

A religious or spiritual problem can bring the person to the psychiatrist under one of several circumstances. For example, a person may begin to question his or her faith and choose not to discuss the problem with a spiritual advisor. Alternatively, a person may wish to convert to a new faith, to marry or improve a marriage when the partner is of a different faith.

Psychiatrists must enable and assist patients in distinguishing religious thought or experience from psychopathology and, if this is a problem, encourage patients to work through the issues independently or with assistance. Religious imagery may be recognized in mental illness when persons state they believe that they have been commanded by God to take a dangerous or grandiose action.

Religious experiences may factor into a person's life in unexpected ways, as in the following case. A midcareer male surgeon who was very successful but long overcommitted to his private practice and his academic responsibilities revealed to his often-neglected wife that, at age 9 years, he was approached by his religious leader to get close physically and ultimately engaged in sexual acts over several years. Believing it was his fault, he never told anyone and decided never to have children. After telling his wife about the experience, they engaged in family therapy to work through the stresses the confession produced in their marriage.

Cults

Recently, cults have appeared to be less popular and less attractive to naïve late adolescents and young adults seeking assistance in discovering who they are as they struggle to develop more mature relationships with their parents.

Cults are led by charismatic leaders, often out of control themselves, with inappropriate and often unethical values but purporting to offer acceptance and guidance to troubled followers. Cult members are actively controlled and forced to dissolve allegiance to family and others to serve the cult leader's directives and personal needs. These young members often come from educated families who then seek professional help in persuading their children to leave the cult and enter deprogramming therapy to restore psychological stability to the former cult members. Deprogramming and adjustment back into the family, society, and an independent life are time-intensive and long term with resultant posttraumatic stress disorder (PTSD), which must be recognized and treated.

ACCULTURATION PROBLEM

Acculturation is the process whereby a person from one culture changes in their manner, custom, and dress to adapt to a different culture. It leads to assimilation in which the person has identified with the new culture, usually without conflict or ambivalence. In some cases, however, significant cultural change can evoke severe distress, termed *culture shock.* This condition arises when individuals suddenly find themselves in a new culture in which they feel entirely alien. They may also feel conflict over which lifestyles to maintain, change, or adopt. Children and young adult immigrants often adapt more quickly than do middle-aged and elderly immigrants. Younger immigrants often learn the new language more efficiently and continue to mature in the new culture, but those who are more senior, having had more stability and unchanging routines in their former culture, struggle more to adapt. Culture shock from immigration differs from the restless and continuous moving of psychiatric patients secondary to their illness.

Culture shock can occur within a person's own country with geographic, school, and work changes, such as joining the military, experiencing school busing, moving across the country, or moving to a vastly different neighborhood or from a rural area to a metropolis. Reactive symptoms, which are understandable, include anxiety, depression, isolation, fear, and a sense of loss of identity as the person adjusts. If the person is part of a family or group making this transition and the move is positive and planned, stress can be lower. Furthermore, if one can maintain selected cultural mores safely as they

integrate into the new culture, it may be less stressful.

Constant geographic moves because of chosen work opportunities or necessity involve a large proportion of workers in the United States. Joining activities in the new community and actively trying to meet neighbors and coworkers can lessen the culture shock.

An 18-year-old, first-year female college student offered an academic scholarship by a small Southern college with a major in her field of interest, realized on her return home to the Midwest for winter break that she felt like a misfit among her dorm peers. They were friendly yet generally kept their distance from her after class. At home, she discussed her experiences with high school friends, who replied that they had heard about such cultural dissonance from peers at their Midwestern colleges. The student returned to college feeling that it was not her fault or imagination and slowly began to reach out more assertively to her peers so they could get to know her beyond stereotypical beliefs and so she could do the same.

Brainwashing

First practiced by the Chinese Communists on American prisoners during the Korean War, *brainwashing* is the deliberate creation of culture shock. Individuals are isolated, intimidated, and made to feel different and out of place to break their spirits and destroy their coping skills. When a person appears mentally weak and helpless, the aggressors impose new ideas on them that they would never have accepted in their normal state. As with those involved in cults, on release and return to their homes, brainwashed individuals with PTSD require deprogramming treatment, including re-education and ongoing supportive psychotherapy, both on an individual and group basis. Treatment is usually long-term to rebuild healthy self-esteem and coping skills.

Prisoners of War and Torture Victims

Prisoners who survive war or torture experiences do so because of personal inner strengths developed in their earlier lives, beginning within their

emotionally healthy and caring families; if they come from troubled families, they are more likely to commit suicide during imprisonment and torture. Prisoners must continuously cope with ongoing anxiety, fear, isolation from known lives, and complete loss of all control over their lives. Those who appear to cope best believe they must survive for a reason (e.g., to tell others what they experienced or to find and return to loved ones). Prisoners who cope best describe living simultaneously on two levels—coping in the here and now to survive the situation while maintaining constant mental connections to their past values and experiences and those important to them.

Beyond the surviving prisoner's personal difficulties, including PTSD, if and when his or her survival behavior continues, his or her family may be affected by the surviving prisoner's inordinate fear of police and strangers, overprotection and overburdening of children to replace those significant others lost, lack of sharing of the past, continued isolation from current communities, or inappropriately expressed anger. Thus, another generation (i.e., children of survivors) can be affected in their personal development and psychological functioning and may require psychiatric evaluation and treatment. (See also Chapter 10, Trauma and Stressor-Related Disorders, for further discussion of these topics.)

A 75-year-old Catholic female survivor of the Pawiak prison in Warsaw, Poland, and then of a concentration camp after her capture as a member of the underground in World War II stated that she had wanted to become a painter. In camp, she carved the Madonna and Child on her toothbrush and sent it home to her mother. She made other clandestine carvings for several women in her barracks to send home to their families, which pleased everyone. After the war, she became a well-known sculptress with exhibits throughout Europe. Many of her art pieces taught people about suffering and respect for others who are of different religions and cultures.

PHASE OF LIFE PROBLEM

Phase of life problems may occur at any point along the life cycle: the first day of school as a child, the divorce of a parent during adolescence, starting

college as a young adult, marriage, having children, illness, caring for aged parents, and many others. Although, on some level, adults recognize that life events will intrude on expected plans in the course of a lifetime, unexpected, multiple, significant negative occurrences, especially if they are chronic, overwhelm a person's ability to recover and function constructively. The typical phase of life problems include relationship changes, such as a changed significant personal relationship or loss, job crises, and parenthood.

Because of sex-role socialization and consequent cultural expectations, where men appear externally better able to handle these phases of life problems, women, people with lower SES, and minority group members appear more vulnerable to negative experiences, perhaps because they feel less empowered psychologically. Significant life changes precipitate distress in the form of anxiety and depressive symptoms, an inability to express reactive emotions directly, and often difficulties in coping with ongoing or changed life responsibilities.

Individuals with positive attitudes, healthy family and personal relationships, and mature defense mechanisms and coping styles, including essential trust in self and others, good verbal communication skills, a capacity for creative and positive thinking, and the ability to be flexible, reliable, and energetic, appear to be best able to cope with phase of life problems. Furthermore, a capacity for sublimation; adequate financial and work status; solid values; and healthy, feasible goals can enable people to face, accept, and deal realistically with expected and unexpected life problems and changes.

NONCOMPLIANCE WITH TREATMENT

Compliance is the degree to which a patient carries out the recommendations of the treating physician. A positive doctor–patient relationship can help foster compliance, but even in those circumstances, the patient may be reluctant to comply with a physician's advice. In psychiatry, a significant concern is medication noncompliance, which may result from discomforting side effects, expense, personal value judgments, and denial of illness, among many others. We should only use this category when the problem is sufficiently severe to warrant independent clinical attention.

RELATIONAL PROBLEMS

An adult's psychological health and sense of well-being depend, to a significant degree, on the quality of his or her critical relationships—that is, on patterns of interaction with a partner and children, parents and siblings, and friends and colleagues. Problems in the interaction between any of these significant others can lead to clinical symptoms and impaired functioning among one or more members of the relational unit. Relational problems may be a focus of clinical attention (1) when a relational unit is distressed and dysfunctional or threatened with dissolution and (2) when the relational problems precede, accompany, or follow other psychiatric or medical disorders. Indeed, the relational context of the patient influences other medical or psychiatric symptoms. Conversely, the functioning of a relational unit is affected by a member's general and other medical or psychiatric illness. Relational disorders require a different clinical approach than other disorders. Instead of focusing primarily on the link between symptoms, signs, and the workings of the individual mind, the clinician must also focus on interactions between the individuals involved. Also, the clinician should consider how these interactions are related to the general and other medical or psychiatric symptoms in a meaningful way.

Definition

Relational problems are patterns of interaction between members of a relational unit that are associated with significantly impaired functioning in one or more individual members. Thus one may have parent–child problems, sibling-related problems, or other dyad or triad impairments. At times the entire unit, such as the family itself, may be dysfunctional.

Epidemiology

No reliable figures are available on the prevalence of relational problems. They can be assumed to be ubiquitous; however, most relational problems resolve without professional intervention. We should consider the nature, frequency, and effects of the problem on those involved before using this diagnosis. For example, divorce, which occurs in just under 50 percent of marriages, is a problem that partners resolve through the remedy of divorce.

We do not need to use this diagnosis for most divorces. If the persons cannot resolve their disputation and continue to live together in a sadomasochistic or pathologically depressed relationship with unhappiness and abuse, then they should be so labeled. Relationship problems between involved persons that cannot be resolved by friends, family, or clergy require professional intervention by psychiatrists, clinical psychologists, social workers, and other mental health professionals.

Relational Problem Related to a Mental Disorder or General Medical Condition

When a family member is ill either from a psychiatric or medical illness, there are reverberations throughout the family unit. Studies indicate that whereas satisfying relationships may have a health-protective influence, relationship distress tends to be associated with an increased incidence of illness. Psychophysiologic mechanisms help to explain how relational systems affect physical or mental health—for example, the intense emotions generated in human attachment systems can affect vascular reactivity and immune processes. Thus, stress-related psychological or physical symptoms can be an expression of family dysfunction.

Adults must often assume responsibility for caring for aging parents while they are still caring for their children, and this dual obligation can create stress. When adults take care of their parents, both parties must adapt to a reversal of their former roles, and the caretakers not only face the potential loss of their parents but also must cope with evidence of their mortality.

Some caretakers abuse their aging parents, a problem that is now receiving attention. Abuse is most likely to occur when the caretaking offspring have substance abuse problems, are under economic stress, and have no relief from their caretaking duties or when the parent is bedridden or has a chronic illness requiring constant nursing attention. Women suffer abuse more often than men, and most abuse occurs in persons older than age 75 years.

The development of chronic illness in a family member stresses the family system and requires adaptation by both the sick person and the other family members. The person who has become sick must frequently face a loss of autonomy, an increased sense of vulnerability, and sometimes a taxing medical regimen. The other family members must experience the loss of the

person as he or she was before the illness, and they usually have substantial caretaking responsibility—for example, in debilitating neurologic diseases, including dementia of the Alzheimer's type, and diseases such as AIDS and cancer. In these cases, the whole family must deal with the stress of prospective death as well as the current illness. Some families use the anger engendered by such situations to create support organizations, increase public awareness of the disease, and rally around the sick member. However, chronic illness frequently produces depression in family members and can cause them to withdraw from or attack one another. The burden of caring for ill family members falls disproportionately on the women in a family—mothers, daughters, and daughters-in-law.

Chronic emotional illness also requires major adaptations by families. For instance, family members may react with chaos or fear of the psychotic productions of a family member with schizophrenia. The regression, exaggerated emotions, frequent hospitalizations, and economic and social dependence of a person with schizophrenia can stress the family system. Family members may react with hostile feelings (referred to as expressed emotion) that are associated with a poor prognosis for the person who is sick. Similarly, a family member with bipolar I disorder can disrupt a family, particularly during manic episodes.

Family devastation can occur when illness (1) suddenly strikes a previously healthy person, (2) occurs earlier than expected in the life cycle (some impairment of physical capacities is expected in old age, although many older persons are healthy), (3) affects the economic stability of the family, and (4) when there are few options for improving the condition of the sick family member.

Parent–Child Relational Problem

Parents differ widely in sensing the needs of their infants. Some quickly note their child's moods and needs; others are slow to respond. Parental responsiveness interacts with the children's temperament to affect the quality of the attachment between child and parent. The diagnosis of the parent–child relational problem applies when the focus of clinical attention is a pattern of interaction between parent and child that is associated with clinically significant impairment in individual or family functioning or with clinically

significant symptoms. Examples include impaired communication, overprotection, and inadequate discipline.

Research on parenting skills has isolated two primary dimensions: (1) a permissive–restrictive dimension and (2) a warm and accepting versus a cold and hostile dimension. A typology that separates parents on these dimensions distinguishes among *authoritarian* (restrictive and cold), *permissive* (minimally restrictive and accepting), and *authoritative* (restrictive as needed but also warm and accepting) parenting styles. Children of authoritarian parents tend to be withdrawn or conflicted; those of permissive parents are likely to be more aggressive, impulsive, and low achievers; and children of authoritative parents seem to function at the highest level, socially and cognitively. However, switching from an authoritarian to a permissive mode may create a negative reinforcement pattern.

Difficulties in many situations stress the usual parent–child interaction. Substantial evidence indicates that marital discord leads to problems in children, from depression and withdrawal to conduct disorder and poor performance at school. Parents may resort to *triangulation*, where they recruit a child as an ally in the struggle with the partner. Divorces and remarriages stress the parent–child relationship and may create painful loyalty conflicts. Stepparents often find it challenging to assume a parental role and may resent the special relationship that exists between their new marital partner and the children from that partner's previous marriages. The resentment of a stepparent by a stepchild and the favoring of a natural child are usual reactions in a new family's initial phases of adjustment. When a second child is born, both familial stress and happiness may result, although happiness is the dominant emotion in most families. The birth of a child can also be troublesome when parents had adopted a child in the belief that they were infertile. Single-parent families usually consist of a mother and children, and their relationship is often affected by financial and emotional problems.

Other situations that can produce a parent–child problem are the development of fatal, disabling, or chronic illness, such as leukemia, epilepsy, sickle-cell anemia, or spinal cord injury, in either the parent or child. The birth of a child with congenital disabilities, such as cerebral palsy, blindness, or deafness, may also produce parent–child problems. These situations, which are not rare, challenge the emotional resources of those involved. Parents and the child must face present and potential loss and must

adjust their day-to-day lives physically, economically, and emotionally. These situations can strain the healthiest families and produce parent–child problems not only with the sick person but also with the unaffected family members. In a family with a severely sick child, parents may resent, prefer, or neglect the other children because the ill child requires much time and attention.

Parents with children who have emotional disorders face particular problems, depending on the child's illness. In families with a child with schizophrenia, family treatment is beneficial and improves the social adjustment of the patient. Similarly, family therapy is useful when a child has a mood disorder. In families with a substance-abusing child or adolescent, family involvement is crucial to help control the drug-seeking behavior and to allow family members to verbalize the feelings of frustration and anger that are invariably present.

Normal developmental crises can also be related to parent–child problems. For instance, adolescence is a time of frequent conflict as the adolescent resists rules and demands, increasing autonomy and, at the same time, elicits protective control by displaying immature and dangerous behavior.

The parents of sons ages 18, 15, and 11 years presented with distress about the behavior of their middle child. The family had been cohesive with satisfactory relationships among all members until 6 months before this consultation. At that time, the 15-year-old son began seeing a girl from a comparatively unsupervised household. Frequent arguments had developed between parents and son regarding going out on school nights, curfews, and neglect of schoolwork. The son's combativeness and lowered academic achievement upset his parents to a great deal. They had not experienced similar conflicts with their oldest child. The adolescent, however, maintained a good relationship with his siblings and friends, did not have behavior problems at school, continued to participate in the school basketball team, and was not a substance user.

Daycare Centers. Quality of care during the first 3 years of life is crucial to neuropsychologic development. The National Institute of Child Health and Human Development does not consider daycare harmful to children,

especially when the caregivers and daycare teachers provide consistent, empathetic, nurturing care. Not all daycare centers can meet that level of care, however, especially those located in impoverished urban areas. Children receiving less than optimal caring exhibit decreased intellectual and verbal skills that indicate delayed neurocognitive development. They may also become irritable, anxious, or depressed, which interferes with the parent–child bonding experience, and they are less assertive and less effectively toilet trained by the age of 5 years.

Currently, more than 55 percent of women are in the workforce, many of whom have no choice but to place their children in daycare centers. Close to 50 percent of entering medical students are women; few medical centers, however, make adequate provisions for on-site daycare centers for their students or staff. Similarly, corporations need to provide on-site, high-quality care for the children of their employees. Not only will that approach benefit the children, but also corporate economic benefits will accrue as a result of reduced absenteeism, increased productivity, and happier working mothers. Such programs have the added benefit of decreasing stresses on marriages.

Partner Relational Problem

Partner relational problems may include negative communication (e.g., criticisms), distorted communication (e.g., unrealistic expectations), or noncommunication (e.g., withdrawal) associated with clinically significant impairment in individual or family functioning or symptoms in one or both partners.

When persons have partner relational problems, psychiatrists must assess whether a patient's distress arises from the relationship or a mental disorder. Mental disorders are more common in single persons—those who never married or who are widowed, separated, or divorced—than among married persons. Clinicians should evaluate developmental, sexual, and occupational and relationship histories, for purposes of diagnosis.

Marriage demands a sustained level of adaptation from both partners. In a troubled marriage, a therapist can encourage the partners to explore areas such as the extent of communication between the partners, their ways of solving disputes, their attitudes toward childbearing and child-rearing, their relationships with their in-laws, their attitudes toward social life, their

handling of finances, and their sexual interaction. The birth of a child, abortion or miscarriage, economic stresses, moves to new areas, episodes of illness, significant career changes, and any situations that involve a significant change in marital roles can precipitate stressful periods in a relationship. Illness in a child exerts the most significant strain on a marriage, and marriages in which a child has died through illness or accident more often than not end in divorce. Complaints of lifelong anorgasmia or impotence by marital partners usually indicate intrapsychic problems, although sexual dissatisfaction is involved in many cases of marital maladjustment.

Adjustment to marital roles can be a problem when partners are from different backgrounds and have grown up with different value systems. For example, members of low SES groups perceive a wife as making most of the decisions in the family, and they accept physical punishment as a way to discipline children. Middle-class persons perceive family decision-making processes as shared, with the husband often being the final arbiter, and they prefer to discipline children verbally. Therapists and partners can handle the problems involving conflicts in values, adjustment to new roles, and poor communication by examining the couple's relationship, as in marital therapy.

Epidemiologic surveys show that unhappy marriages are a risk factor for major depressive disorders. Marital discord also affects physical health. For example, in a study of women age 30 to 65 years with coronary artery disease, marital stress worsened the prognosis 2.9 times for recurrent coronary events. Marital conflict was also associated with a 46 percent higher relative death risk among female patients having hemodialysis and with elevations in serum epinephrine, norepinephrine, and corticotrophin levels in both men and women. In one study, patients with high levels of hostile marital behavior had slower healing wounds, lower production of proinflammatory cytokines, and higher cytokine production in peripheral blood. Overall, women show better psychological and physiologic responsiveness to conflict than men.

Physician Marriages. Physicians have a higher risk of divorce than other occupational groups. The incidence of divorce among physicians is about 25 to 30 percent. Specialty choice influenced divorce. The highest rate of divorce occurred in psychiatrists (50 percent) followed by surgeons (33

percent) and internists, pediatricians, and pathologists (31 percent). The average age at first marriage was 26 years among all groups.

It is not clear why physicians are at high risk for divorce. Factors implicated include the stresses of dealing with dying patients, making life-and-death decisions, working long hours, and the constant risk of malpractice litigation. Such stressors may predispose physicians to a variety of emotional ills, with the most common being depression and substance abuse, including alcoholism. Such persons generally cannot deal with the complex interactions required to maintain successful long-term relationships of any kind, and marriage requires the most interpersonal skills of all.

Sibling Relational Problem

Sibling relationships tend to be characterized by competition, comparison, and cooperation. Intense sibling rivalry can occur with the birth of a child and can persist as the children grow up, compete for parental approval, and measure their accomplishments against one another. Alliances between siblings are equally common. Siblings may learn to protect one another against parental control or aggression. In households with three children, one pair tends to become tightly involved with one another, leaving the extra child in the position of an outsider.

Relational problems can arise when a family treats siblings unequally; for instance, idealizing one child while casting another in the role of the family scapegoat. Differences in gender roles and expectations expressed by the parents can underlie sibling rivalry. Parent–child relationships also are dependent on personality interactions. A child's resentment directed at a parental figure or a child's own disavowed dark emotions can be projected onto a sibling and can fuel an intense hate relationship.

An ill child creates stress for the child's siblings. Parental concern and attention to the sick child can elicit envy in the siblings. Also, chronic disability can leave the sick child feeling devalued and rejected by siblings, and the latter may develop a sense of superiority and may feel embarrassed about having a disabled sister or brother. Twin relationships have become an area of increasing study. Preliminary data show that twins are more likely to be cooperative than competitive. Whether or not identical twins should be dressed differently during their toddler years to ensure a separate identity is

open to question as is the issue of whether or not they should be in separate classrooms when they begin school.

Other Relational Problems

People, across the life cycle, may become involved in relational problems with leaders and others in their communities at large. In such relationships, conflicts are frequent and can bring about stress-related symptoms. Many relational problems of children occur in the school setting and involve peers. Impaired peer relationships can be the chief complaint in attention-deficit or conduct disorders, as well as in depressive and other psychiatric disorders of childhood, adolescence, and adulthood.

Racial, ethnic, and religious prejudices and ignorance cause problems in interpersonal relationships. In the workplace and communities at large, sexual harassment is often a combination of inappropriate sexual interactions, inappropriate displays of abuse of power and dominance, and expressions of negative gender stereotypes. This harassment most often victimizes women and gay men. However, it can also affect children and adolescents of both sexes.

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B Treatment Across the Lifespan



21 Psychopharmacology

Psychopharmacologic advances continue to expand the parameters of psychiatric treatments dramatically. Our increasing understanding of the brain has led to more effective, less toxic, better-tolerated, and more specific therapeutic agents. With the ever-increasing sophistication and array of treatment options, clinicians, however, must remain aware of potential adverse effects, drug–drug (and drug–food or drug–supplement) interactions, and how to manage the emergence of unwanted or unintended consequences. Newer drugs often cause side effects that are not recognized initially. Keeping up with the latest research findings is important as these findings proliferate. A thorough understanding of the management of medication-induced side effects (either through treating the effect with another agent or substituting another primary agent) is necessary.

Medications used to treat psychiatric disorders are sometimes referred to as *psychotropic drugs*. These drugs are commonly described by their primary clinical application, for example, *antidepressants, antipsychotics, mood stabilizers, anxiolytics, hypnotics, cognitive enhancers,* and *stimulants*. A problem with this approach is that, in many instances, drugs have multiple indications. For example, drugs such as the selective serotonin reuptake inhibitors (SSRIs) are both antidepressants and anxiolytics, and the serotonin-dopamine antagonists (SDAs) are both antipsychotics and mood stabilizers. The term is largely synonymous with simply “psychiatric drugs” and we use both terms synonymously in this book.

Psychotropic drugs have also been organized according to structure (e.g., tricyclic), mechanism (e.g., monoamine oxidase inhibitor [MAOI]), history (e.g., first-generation, traditional), uniqueness (e.g., atypical), or indication (e.g., antidepressant). A further source of confusion is that many drugs used

to treat medical and neurologic conditions are routinely used to treat psychiatric disorders.

Also, psychotropic drug terminology can be confusing. The first pharmaceutical agents used to treat schizophrenia were termed *tranquilizers*. When newer drugs emerged as therapies for anxiety, a distinction was drawn between *major* and *minor tranquilizers*. At first, antidepressants were tricyclic antidepressants (TCAs) or MAOIs. In the 1970s and 1980s, as newer antidepressant drugs emerged, they were labeled as *second-* or *third-generation antidepressants*. More recently, older agents used as treatments for psychosis became known as *typical, conventional,* or first-generation *traditional* neuroleptics. Newer ones became *atypical neuroleptics* or second-generation antipsychotics.

In this chapter, we still start with a general review of psychopharmacology, as it applies to psychotropic medications. We will then review the major classes of drugs. For the purposes here, we will use the first system of organization described above; that is, we will classify them by their primary application. Also, we will divert from our usual practice of mainly using generic names of the drugs to, in many cases, include brand names as well. However, in practice, most psychotropic medications are now available through generic preparations, and few patients receive the brand version of a medication. Also, the brand names here reflect US brands. Drug companies in other countries may produce medications under a different brand name.

DRUG SELECTION

Although all U.S. Food and Drug Administration (FDA)-approved psychotropics are similar in overall effectiveness for their indicated disorder, they differ considerably in their pharmacology and their efficacy and adverse effects on individual patients. The ability of a drug to prove effective, thus, is only partially predictable and is dependent on poorly understood patient variables. Nevertheless, some drugs may have a niche in which they can be uniquely helpful for a subgroup of patients, without demonstrating any overall superiority in efficacy. No drug is universally effective, and no evidence indicates the unambiguous superiority of any single agent as a treatment for any major psychiatric disorders. The only exception, clozapine (Clozaril), has been approved by the FDA as a treatment for cases of

treatment-refractory schizophrenia.

Decisions about drug selection and use are made on a case-by-case basis, relying on individual judgment by the physician. Other factors in drug selection are the characteristics of the drug and the nature of the patient's illness. Each of these components affects the probability of a successful outcome.

DRUG FACTORS

Pharmacodynamics

Both genetic and environmental factors influence individual response to and tolerability of psychotropic agents. Thus, a drug that may not prove effective in many patients with a disorder can dramatically improve symptoms in others. In these cases, identification of characteristics that might predict potential candidates for that drug becomes important but often remains elusive.

Drugs, even within the same class, are distinguished from one another by often subtle differences in molecular structure, types of interactions with neurotransmitter systems, differences in pharmacokinetics, the presence or absence of active metabolites, and protein binding. These differences, along with patient biochemical variations, account for the profile of efficacy, tolerability, and safety and the risk-to-benefit ratio for the individual. These multiple variables, some poorly understood, make it difficult to predict a drug's effect with certainty. Nevertheless, knowledge of the nature of each property increases the likelihood of successful treatment. The clinical effects of drugs are best understood in terms of *pharmacokinetics* and *pharmacodynamics*. Pharmacokinetics describes *what the body does to a drug*, whereas pharmacodynamics is *what the drug does to the body*.

Pharmacokinetics and pharmacodynamics need to be seen in the context of the underlying variability among patients for how drug effects are expressed clinically. Patients differ in their therapeutic response to a drug, and the experience of side effects. It is increasingly clear that these differences have a strong genetic basis. Pharmacogenetics research is attempting to identify the role of genetics in drug response.

Primary pharmacodynamic considerations include receptor mechanisms,

the dose–response curve, the therapeutic index, and the development of tolerance, dependence, and withdrawal phenomena. The drug's mechanism of action is subsumed under pharmacodynamics. The clinical response to a drug, including adverse reactions, results from an interaction between that drug and a patient's susceptibility to those actions. Pharmacogenetic studies are beginning to identify genetic polymorphisms linked to individual differences in treatment response and sensitivity to side effects.

Mechanisms

The mechanisms through which most psychotropic drugs produce their therapeutic effects remain poorly understood. Standard explanations focus on ways that drugs alter synaptic concentrations of dopamine, serotonin, norepinephrine, histamine, γ-aminobutyric acid (GABA), or acetylcholine. These changes are said to result from receptor antagonists or agonists, interference with neurotransmitter reuptake, enhancement of neurotransmitter release, or inhibition of enzymes. Specific drugs are associated with permutations or combinations of these actions. For example, a drug can be an agonist for a receptor, thus stimulating the specific biologic activity of the receptor, or an antagonist, thus inhibiting the biologic activity. Some drugs are partial agonists because they are not capable of fully activating a specific receptor. Some psychotropic drugs also produce clinical effects through mechanisms other than receptor interactions. For example, lithium (Eskalith) can act by directly inhibiting the enzyme inositol-1-phosphatase. Some effects are closely linked to a specific synaptic effect. For example, most medications that treat psychosis share the ability to block the dopamine type 2 (D_2) receptor. Similarly, benzodiazepine agonists bind a receptor complex that contains benzodiazepine and GABA receptors.

Additionally, illustrating the fact that the mechanisms of action of psychotropic drugs remain only partially understood are observations that medications that do not directly target monoamine neurotransmitters can be remarkably effective in treating some psychiatric disorders. For example, ketamine (Ketalar), an anesthetic agent that targets glutamate, can rapidly and dramatically alleviate symptoms of depression when given as a slow infusion. Another example involves the antibiotic minocycline (Solodyn), which has been shown to have antidepressant effects. Along with other findings, this

suggests that the immune system and inflammatory responses may underlie some mood disorders.

Accounts of the so-called mechanisms of action should nevertheless be kept in perspective. Explanations of how psychotropic drugs work that focus on synaptic elements represent an oversimplification of a complex series of events. If merely raising or lowering levels of neurotransmitter activity is associated with the clinical effects of a drug, then all drugs that cause these changes should produce equivalent benefits. This predictability is not the case. Multiple obscure actions, several steps removed from events at neuronal receptor sites, are probably responsible for the therapeutic effects of psychotropic drugs. These *downstream* elements are postulated to represent the actual reasons that these drugs produce clinical improvement. A glossary of terms related to receptor drug interactions is given in Table 21-1.

SIDE EFFECTS

Side effects are an unavoidable risk of medication treatment. Although it is impossible to have an encyclopedic knowledge of all possible adverse drug effects, prescribing clinicians should be familiar with the more common adverse effects, as well as those with serious medical consequences. No single text or document, including the product information, contains a complete list of possible treatment-emergent events.

Side effect considerations include the probability of its occurrence, its impact on a patient's quality of life, its time course, and its cause. Just as no one drug is sure to produce clinical improvement in all patients, no side effect, no matter how common, occurs in every patient. When concurrent medical disorders or a history of a similar adverse reaction puts a patient at increased risk for a side effect, it is logical to consider prescribing a compound not typically associated with that adverse reaction.

Side effects can result from the same pharmacologic action that is responsible for a drug's therapeutic activity or from an unrelated property. In examples of the latter, some of the most common adverse effects of the TCAs are caused by blockade of muscarinic acetylcholine receptors or histamine receptors. If a patient is sensitive to these effects, alternative agents without these properties should be prescribed. When side effects are manifestations of the drug's presumed mechanism of action, side effects may be unavoidable.

Thus, blockade of serotonin reuptake by SSRIs can cause nausea and sexual dysfunction. The D_2 blockade of drugs used to treat psychosis can cause extrapyramidal side effects. Agonist action of benzodiazepine receptors can cause ataxia and daytime sleepiness. In these cases, additional medications are frequently used to make the primary agent better tolerated.

Time Course

Adverse effects differ in terms of their onset and duration. Some side effects appear at the outset of treatment and then rapidly diminish. Nausea occurring with SSRIs or venlafaxine (Effexor) and sedation occurring with mirtazapine (Remeron) are good examples of early, time-limited side effects. Early-onset, but persistent, side effects include a dry mouth that is associated with noradrenergic reuptake inhibition or antimuscarinic activity. Some side effects appear later in treatment (*late-appearing side effects*) and, sometimes may be just the opposite of adverse events early in treatment. For example, patients may typically lose weight during early treatment with SSRIs, only to find, over time, a reversal occurs, so that they gain weight.

Similarly, early activation or agitation may be followed by constant fatigue or apathy. Because most data about new drugs come from short-term studies, generally 8 weeks in duration, early-onset side effects are overrepresented in product information and descriptions of newly marketed information. Clinicians must follow the letters to the editor sections of journals and other sources of information to update their understanding of the real side effect profile of a drug.

Adverse effects differ in their impact on compliance and the potential to cause harm. Depending on a patient's threshold of tolerance for a side effect and the impact on the quality of life, side effects can lead to drug discontinuation. Examples of serious side effects include agranulocytosis (clozapine), Stevens–Johnson syndrome (lamotrigine [Lamictal]), hepatic failure (nefazodone [Serzone]), stroke (phenelzine [Nardil]), and heart block (thioridazine [Mellaril]). Overall, the risk of life-threatening side effects with psychotropics is low. Drugs that carry such a risk should be monitored more closely, and the prescribing physician should take into account whether the potential clinical benefits justify the additional risk. Any drug with a serious risk, as reflected in a black box warning, is generally used less extensively

than would otherwise be the case.



Table 21-1
Glossary of Receptor Drug Interactions

Receptor Interaction	Definition	Examples and Comments
Agonist (full agonist)	A drug or medication that binds to a specific receptor producing an effect identical to that usually produced by the neurotransmitter affecting that receptor. Drugs are often designed as receptor agonists to treat a variety of diseases and disorders in which the original neurotransmitter is missing or diminished.	Full agonists include opioids such as morphine, methadone, oxycodone, hydrocodone, heroin, codeine, meperidine, propoxyphene, and fentanyl. Benzodiazepines act as agonists at the GABA receptor complex.
Antagonist	A compound that binds to a receptor that blocks or reduces the action of another substance (agonist) at the receptor site involved. Antagonists that compete with an agonist for a receptor are *competitive antagonists* Those that antagonize by other means are *noncompetitive antagonists*	Flumazenil is a competitive benzodiazepine receptor antagonist. It competitively inhibits the activity at the benzodiazepine recognition site on the GABA/benzodiazepine receptor complex. It is the purest antagonist synthesized. Drugs used in the treatment of schizophrenia block dopamine 2 receptors. Examples of opioid

		antagonists include naltrexone and naloxone.
Partial agonist (mixed agonist)	A compound that (even when fully occupying a receptor) possesses affinity for a receptor, but elicits a partial pharmacologic response at the receptor involved. Partial agonists are often structural analogs of agonist molecules. If neurotransmitter concentrations are low, partial agonists may behave as an agonist. This is why these medications are sometimes called mixed agonists.	Buprenorphine is a partial agonist that produces typical opioid agonist effects and side effects, such as euphoria and respiratory depression, but its maximal effects are less than those of full agonists like heroin and methadone. When used at low doses buprenorphine produces sufficient agonist effect to enable opioid-addicted individuals to discontinue the drugs with fewer withdrawal symptoms.
Inverse agonist	An inverse agonist is an agent that binds to the same receptor as an agonist for that receptor but produces the opposite pharmacologic effect.	Several inverse agonists are currently in clinical development. One particular example is R015-4513, which is the inverse agonist of the benzodiazepine class of drugs. R015-4513 and the benzodiazepines both utilize the same GABA binding site on neurons, yet R015-4513 has the opposite effect, producing severe anxiety rather than the sedative and anxiolytic

	effects associated with benzodiazepines.
	Cannabinoid inverse agonists have been found to reduce appetite, the opposite of the craving effect associated with cannabis.

GABA, γ-aminobutyric acid.
Table by Norman Sussman, M.D.

In the case of haloperidol (Haldol) and other dopamine receptor antagonists (DRAs), long-term complications, such as tardive dyskinesia, have been well documented. Emerging evidence also suggests that the use of dopamine antagonists is associated with a small increase in the risk of breast cancer and that this is related to larger cumulative doses. In cases in which serious risk is associated with a drug, closer medical monitoring of medication treatment is warranted. Because the most widely used psychotropics, such as the SSRIs and SDAs, have only been in use since the 1980s or 1990s, there is less certainty about long-term effects, but no evidence indicates that side effects are not merely extensions of those already evident during initial therapy. It should also be kept in mind that most drugs used in the treatment of chronic medical disorders have not been in use sufficiently long to provide assurances about unintended long-term adverse effects.

Suicidal Ideation and Antidepressant Treatment

The issue of antidepressant-associated suicide has become front-page news, the result of an analysis suggesting a link between medication use and suicidal ideation among children, adolescents, and adults up to age 24 in short-term (4 to 16 weeks), placebo-controlled trials of nine newer antidepressant drugs. The data from trials involving more than 4,400 patients suggested that the average risk of suicidal thinking or behavior (suicidality) during the first few months of treatment in those receiving antidepressants was 4 percent, twice the placebo risk of 2 percent. No suicides occurred in

these trials. The analysis also showed no increase in suicide risk among the 25- to 65-age group. Antidepressants reduced suicidality among those over age 65.

Following public hearings on the subject, in October 2004, the FDA requested the addition of black box warnings—the most severe warning placed on the labeling of prescription medication—to all antidepressant drugs, old and new. This action raised the alarm among parents and physicians and prompted an explosion of advertisements by malpractice attorneys. Most important, antidepressant prescriptions written for adolescents declined, whereas those for adults flattened, after years of growth.

A large study of real-world patients published in the January 2006 issue of the *American Journal of Psychiatry* raised severe doubt about antidepressants and actual suicidality and about the wisdom of the FDA's decision to change the labeling. The study examined suicides and hospitalizations for suicide attempts in the medical records of 65,103 members of a nonprofit insurer in the Pacific Northwest that covers about 500,000 people who received antidepressants from 1992 to 2003. It found that (1) newer antidepressants were associated with a more rapid and more significant reduction in risk than older types of antidepressants, and (2) patients were significantly more likely to attempt or commit suicide in the month before they began drug therapy than in the 6 months after starting it.

This controversy is not the first time credible evidence has contradicted a significant link between antidepressant use and increased risk of suicide. At the hearings that led to the black box warning, John Mann of Columbia University presented population data showing that since 1987, the year before fluoxetine (Prozac) became the first marketed SSRI, suicide rates in the United States began dropping and that areas in the United States with the highest SSRI prescription rates had the most significant decline in suicides. For every 10 percent increase in prescription rates, the US suicide rate declined 3 percent.

Another study, a review of 588 case files of patients aged 10 to 19, found that a 1 percent increase in antidepressant use was associated with a decrease of 0.23 suicides per 100,000 adolescents per year.

A more critical question, given how slight the risk may be, if indeed it exists, is whether as a result of the FDA's ill-considered actions, some

depressed patients are not getting potentially life-saving treatment. Epidemiologic findings from several countries, including the United States, have shown that decreased prescribing of antidepressants for depressed children and adolescents resulted in an increase in suicide rates in those populations.

Medication-Induced Movement Disorders

Medication-induced movement disorders are commonly associated with the use of psychotropic drugs. Although most frequently associated with drugs that block dopamine type 2 (D_2) receptors, abnormal motor activity may occur with other types of medications as well. Sometimes it can be challenging to determine if abnormal motor movements are an adverse event or a symptom of an underlying disorder. For example, anxiety can resemble akathisia, and alcohol or benzodiazepine withdrawal can cause tremors. The American Psychiatric Association has decided to retain the term *neuroleptic* when discussing side effects associated with drugs used to treat psychosis—the DRAs and second-generation antipsychotics (SGAs). The rationale for continued use of the term is that it was initially used to describe the tendency of these drugs to cause abnormal movements. Note that SGA and SDA are often used interchangeably when referring to the second-generation antipsychotics.

The most common neuroleptic-related movement disorders are parkinsonism, acute dystonia, and acute akathisia. The neuroleptic malignant syndrome is a life-threatening and often misdiagnosed condition. Neuroleptic-induced tardive dyskinesia is a late-appearing adverse effect of neuroleptic drugs and can be irreversible; recent data, however, indicate that the syndrome, although still severe and potentially disabling, is less pernicious than was previously thought in patients taking DRAs. The newer antipsychotics, the SDAs, block binding to dopamine receptors to a much lesser degree and thereby are presumed to be less likely to produce such movement disorders. Nevertheless, this risk remains, and vigilance is still required when these drugs are prescribed.

Table 21-2 lists the selected medications associated with movement disorders and their impact on relevant neuroreceptors.

Neuroleptic-Induced Parkinsonism and Other Medication-Induced Parkinsonism

DIAGNOSIS, SIGNS, AND SYMPTOMS. Symptoms of neuroleptic-induced parkinsonism and other medication-induced parkinsonism include muscle stiffness (lead pipe rigidity), cogwheel rigidity, shuffling gait, stooped posture, and drooling. The pill-
rolling tremor of idiopathic parkinsonism is rare, but a regular, coarse tremor similar to essential tremor may be present. The so-called *rabbit syndrome*, a tremor affecting the lips and perioral muscles, is another parkinsonian effect seen with antipsychotics, although perioral tremor is more likely than other tremors to occur late in the course of treatment.

EPIDEMIOLOGY. Parkinsonian adverse effects typically occur within 5 to 90 days of the initiation of treatment. Patients who are elderly and female are at the highest risk for neuroleptic-induced parkinsonism, although the disorder can occur at all ages.

ETIOLOGY. The blockade of D_2 receptors causes neuroleptic-induced parkinsonism in the caudate at the termination of the nigrostriatal dopamine neurons. All antipsychotics can cause the symptoms, especially high-potency drugs with low levels of anticholinergic activity, most notably haloperidol (Haldol).

DIFFERENTIAL DIAGNOSIS. Included in the differential diagnosis are idiopathic parkinsonism, other organic causes of parkinsonism, and depression, which can also be associated with parkinsonian symptoms. Decreased psychomotor activity and blunted facial expression are symptoms of depression and idiopathic parkinsonism.

TREATMENT. Parkinsonism can be treated with anticholinergic agents, benztropine (Cogentin), amantadine (Symmetrel), or diphenhydramine (Benadryl) (Table 21-3). Anticholinergics should be withdrawn after 4 to 6 weeks to assess whether tolerance to the parkinsonian effects has developed; about half of patients with neuroleptic-induced parkinsonism require continued treatment. Even after the antipsychotics are withdrawn, parkinsonian symptoms can last up to 2 weeks. In elderly patients, this may

last as long as 3 months. With such patients, the clinician may continue the anticholinergic drug after the antipsychotic has been stopped until the parkinsonian symptoms resolve completely.

Neuroleptic Malignant Syndrome

DIAGNOSIS, SIGNS, AND SYMPTOMS. *Neuroleptic malignant syndrome* is a life-threatening complication that can occur anytime during antipsychotic treatment. The motor and behavioral symptoms include muscular rigidity and dystonia, akinesia, mutism, obtundation, and agitation. The autonomic symptoms include hyperthermia, diaphoresis, and increased pulse and blood pressure (BP). Laboratory findings include an increased white blood cell (WBC) count and increased levels of creatinine phosphokinase, liver enzymes, plasma myoglobin, and myoglobinuria, occasionally associated with renal failure.

Table 21-2
Selected Medications Associated with Movement Disorders: Impact on Relevant Neuroreceptors

Type (Subtype)	Name (Brand)	D_2 Blockade	5-HT_2 Blockade	mACh Blockade
Antipsychotics				
Phenothiazine (Aliphatic)	Chlorpromazine (Thorazine)	Low	High	High
Phenothiazine (Piperidines)	Thioridazine (Mellaril)	Low	Med	High
	Mesoridazine (Serentil)	Low	Med	High
Phenothiazine (Piperazines)	Trifluoperazine (Stelazine)	Med	Med	Med
	Fluphenazine (Prolixin)	High	Low	Low
	Perphenazine (Trilafon)	High	Med	Low
Thioxanthenes	Thiothixene (Navane)	High	Med	Low
	Chlorprothivene (Taractan)	Med	High	Med
Dibenzoxazepines	Loxapine (Loxitane)	Med	High	Low
Butyrophenones	Haloperidol (Haldol)	High	Low	Low
	Droperidol (Inapsine)	High	Med	—
Diphenyl-butylpiperidines	Pimozide (Orap)	High	Med	Low
Dihydroindolones	Molindone (Moban)	Med	Low	Low
Dibenzodiazepines	Clozapine (Clozaril)	Low	High	High
Benzisoxazole	Risperidone (Risperdal)	High	High	Low
Thienobenzodiazepines	Olanzapine (Zyprexa)	Low	High	High
Dibenzothiazepines	Quetiapine (Seroquel)	Low/med	Low/med	Low
Benzisothiazolvils	Ziprasidone (Geodon)	Med	High	Low
Quinolones	Aripiprazole (Abilify)	High (as partial agonist)	High	Low
Nonantipsychotic psychotropics	Lithium (Eskalith)	N/A	N/A	N/A
Anticonvulsants		Low	Low	Low
Antidepressants		Low (except amoxapine)	(Varies)	(Varies)
Nonpsychotropics	Prochlorperazine (Compazine)	High	Med	Low
	Metoclopramide (Reglan)	High	High	—

D_2, dopamine type 2; 5-HT_2, 5-hydroxytryptomine type 2; mACh, muscarinic acetylcholine; N/A, not applicable.
Adapted from Janicak PG, Davis JM, Preskorn SH, Ayd FJ Jr. *Principles and Practice of Psychopharmacotherapy*. 3rd ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2001, with permission.

EPIDEMIOLOGY. About 0.01 to 0.02 percent of patients treated with antipsychotics develop neuroleptic malignant syndrome. Men are affected more frequently than women, and young patients are affected more commonly than elderly patients. The mortality rate can reach 10 to 20 percent or even higher when depot antipsychotic medications are involved.

COURSE AND PROGNOSIS. The symptoms usually evolve over 24 to 72 hours, and the untreated syndrome lasts 10 to 14 days. The diagnosis is often missed in the early stages, and the withdrawal or agitation may mistakenly be considered to reflect an exacerbation of the psychosis.

TREATMENT. In addition to supportive medical treatment, the most commonly used medications for the condition are dantrolene (Dantrium) and bromocriptine (Parlodel), although amantadine (Symmetrel) is sometimes used (Table 21-4). Bromocriptine and amantadine pose direct DRA effects and may overcome the antipsychotic-induced dopamine receptor blockade.

We should reduce the antipsychotic dose to the lowest effective dose possible. High-potency drugs, such as haloperidol, pose the greatest risk. Antipsychotic drugs with anticholinergic effects seem less likely to cause neuroleptic malignant syndrome. Electroconvulsive therapy (ECT) has been used.

Medication-Induced Acute Dystonia

DIAGNOSIS, SIGNS, AND SYMPTOMS. *Dystonias* are brief or prolonged contractions of muscles that result in obviously abnormal movements or postures, including oculogyric crises, tongue protrusion, trismus, torticollis, laryngeal–pharyngeal dystonias, and dystonic postures of the limbs and trunk. Other dystonias include blepharospasm and glossopharyngeal dystonia; the latter results in dysarthria, dysphagia, and even difficulty in breathing, which can cause cyanosis. Children are particularly likely to evidence opisthotonos, scoliosis, lordosis, and writhing movements. Dystonia can be painful and frightening and often results in noncompliance with future drug treatment regimens.

EPIDEMIOLOGY. Their early-onset characterizes the development of acute dystonic symptoms during treatment with neuroleptics. There is a higher incidence of acute dystonia in men, in patients younger than age 30 years, and patients who were given high dosages of high-potency medications.

ETIOLOGY. Although it is most frequent with intramuscular doses of high-potency antipsychotics, dystonia can occur with any antipsychotic. The mechanism of action is thought to be dopaminergic hyperactivity in the basal ganglia that occurs when the central nervous system (CNS) levels of the antipsychotic drug begin to fall between doses.



Table 21-3
Drug Treatment of Extrapyramidal Disorders

Generic Name	Trade Name	Usual Daily Dosage	Indications
Anticholinergics			

Benztropine	Cogentin	PO 0.5–2 mg tid; IM or IV 1–2 mg	Acute dystonia, parkinsonism, akinesia, akathisia
Biperiden	Akineton	PO 2–6 mg tid; IM or IV 2 mg	
Procyclidine	Kemadrin	PO 2.5–5 mg bid-qid	
Trihexyphenidyl	Artane, Tremin	PO 2–5 mg tid	
Orphenadrine	Norflex, Disipal	PO 50–100 mg bid-qid; IV 60 mg	Rabbit syndrome
Antihistamine			
Diphenhydramine	Benadryl	PO 25 mg qid; IM or IV 25 mg	Acute dystonia, parkinsonism, akinesia, rabbit syndrome
Amantadine	Symmetrel	PO 100–200 mg bid	Parkinsonism, akinesia, rabbit syndrome
β-Adrenergic Antagonist			
Propranolol	Inderal	PO 20–40 mg tid	Akathisia, tremor
α-Adrenergic Antagonist			
Clonidine	Catapres	PO 0.1 mg tid	Akathisia

Benzodiazepines			
Clonazepam	Klonopin	PO 1 mg bid	Akathisia, acute dystonia
Lorazepam	Ativan	PO 1 mg tid	
Buspirone	BuSpar	PO 20–40 mg qid	Tardive dyskinesia
Vitamin E	—	PO 1,200–1,600 IU/day	Tardive dyskinesia

PO, orally; IM, intramuscularly; IV, intravenously; bid, twice a day; tid, three times a day; qid; four times a day.

DIFFERENTIAL DIAGNOSIS. The differential diagnosis includes seizures and tardive dyskinesia.

COURSE AND PROGNOSIS. Dystonia can fluctuate spontaneously and respond to reassurance, so the clinician gets the false impression that the movement is hysterical or entirely under conscious control.



Table 21-4
Treatment of Neuroleptic Malignant Syndrome

Intervention	Dosing	Effectiveness
Amantadine	200–400 mg PO/day in divided doses	Beneficial as monotherapy or in combination; decrease in death rate
Bromocriptine	2.5 mg PO bid or tid, may increase to a total of 45 mg/day	Mortality reduced as a single or combined agent
Levodopa/carbidopa	Levodopa 50–100 mg/day	Case reports of

	IV as continuous infusion	dramatic improvement
Electroconvulsive therapy	Reports of good outcome with both unilateral and bilateral treatments; response may occur in as few as three treatments	Effective when medications have failed; also may treat underlying psychiatric disorder
Dantrolene	1 mg/kg/day for 8 days, then continue as PO for 7 additional days	Benefits may occur in minutes or hours as a single agent or in combination
Benzodiazepines	1–2 mg IM as test dose; if effective, switch to PO; consider use if underlying disorder has catatonic symptoms	Has been reported effective when other agents have failed
Supportive measures	IV hydration, cooling blankets, ice packs, ice-water enema, oxygenation, antipyretics	Often effective as initial approach early in the episode

PO, orally; bid, twice a day; tid, three times a day; IV, intravenously; IM, intramuscularly.
Adapted from Davis JM, Caroif SN, Mann SC. Treatment of neuroleptic malignant syndrome. *Psychiatr Ann*. 2000;30:325–331, with permission.

TREATMENT. Prophylaxis with anticholinergics or related drugs (outlined in Table 21-3) usually prevents dystonia, although the risks of prophylactic treatment weigh against that benefit. Treatment with intramuscular anticholinergics or intravenous or intramuscular diphenhydramine (Benadryl) (50 mg) almost always relieves the symptoms. Diazepam (Valium) (10 mg intravenously), amobarbital (Amytal), caffeine sodium benzoate, and hypnosis have also been reported to be effective. Although tolerance for the

adverse effects usually develops, it is prudent to change the antipsychotic if the patient is particularly concerned that the reaction may recur.

Medication-Induced Acute Akathisia

DIAGNOSIS, SIGNS, AND SYMPTOMS. *Akathisia* is subjective feelings of restlessness, objective signs of restlessness, or both. Examples include a sense of anxiety, inability to relax, jitteriness, pacing, rocking motions while sitting, and rapid alternation of sitting and standing. Akathisia has been associated with the use of a wide range of psychiatric drugs, including antipsychotics, antidepressants, and sympathomimetics. Once akathisia is recognized and diagnosed, the antipsychotic dose should be reduced to a minimally adequate level. Akathisia may be associated with a poor treatment outcome.

EPIDEMIOLOGY. Middle-aged women are at increased risk of akathisia, and the time course is similar to that for neuroleptic-induced parkinsonism.

TREATMENT. Three necessary steps in the treatment of akathisia are reducing medication dosage, attempting treatment with appropriate drugs, and considering changing the neuroleptic. The most efficacious drugs are β-adrenergic receptor antagonists, although anticholinergic drugs, benzodiazepines, and cyproheptadine (Periactin) may benefit some patients. In some cases of akathisia, no treatment seems to be effective.

Tardive Dyskinesia

DIAGNOSIS, SIGNS, AND SYMPTOMS. *Tardive dyskinesia* is a delayed effect of antipsychotics; it rarely occurs until after 6 months of treatment. The disorder consists of abnormal, involuntary, irregular choreoathetoid movements of the muscles of the head, limbs, and trunk. The severity of the movements ranges from minimal—often missed by patients and their families—to grossly incapacitating. Perioral movements are the most common and include darting, twisting, and protruding movements of the tongue; chewing and lateral jaw movements; lip-puckering; and facial grimacing. Finger movements and hand clenching are also typical. Torticollis, retrocollis, trunk twisting, and pelvic thrusting occur in severe cases. In the most severe cases,

patients may have breathing and swallowing irregularities that result in aerophagia, belching, and grunting. Respiratory dyskinesia has also been reported. Dyskinesia is exacerbated by stress and disappears during sleep.



Table 21-5
Abnormal Involuntary Movement Scale (AIMS) Examination Procedure

Patient Identification	Date

Rated by

Either before or after completing the examination procedure, observe the patient unobtrusively at rest (e.g., in waiting room).

The chair to be used in this examination should be a hard, firm one without arms.

After observing the patient, rate him or her on a scale of 0 (none), 1 (minimal), 2 (mild), 3 (moderate), and 4 (severe), according to the severity of the symptoms.

Ask patient whether there is anything in his or her mouth (e.g., gum, candy) and, if so, to remove it.

Ask patient about the current condition of his or her teeth. Ask patient if he or she wears dentures. Do teeth or dentures bother patient now?

Ask patient whether he or she notices movement in mouth, face, hands, or feet. If yes, ask patient to describe and indicate to what extent movements currently bother patient or interfere with his or her activities.

0 1 2 3 4 Have patient sit in chair with hands on knees, legs slightly apart, and feet flat on floor. (Look at entire body for movement while in this position.)

0 1 2 3 4 Ask patient to sit with hands hanging unsupported—If male, between legs; if female and wearing a dress, hanging over knees. (Observe hands and other body areas.)

0 1 2 3 4 Ask patient to open mouth. (Observe tongue at rest within mouth.) Do this twice.

0 1 2 3 4 Ask patient to protrude tongue. (Observe abnormalities of tongue movement.) Do this twice.
0 1 2 3 4 Ask patient to tap thumb, with each finger, as rapidly as possible for 10–15 seconds; separately with right hand, then with left hand. (Observe facial and leg movements.)
0 1 2 3 4 Flex and extend patient's left and right arms. (One at a time.)
0 1 2 3 4 Ask patient to stand up. (Observe in profile. Observe all body areas again, hips included.)
0 1 2 3 4 [a]Ask patient to extend both arms outstretched in front with palms down. (Observe trunk, legs, and mouth.)
0 1 2 3 4 [a]Have patient walk a few paces, turn, and walk back to chair. (Observe hands and gait.)
Do this twice.

[a]Activated movements.

EPIDEMIOLOGY. Tardive dyskinesia develops in about 10 to 20 percent of patients who are treated for more than a year. About 20 to 40 percent of patients who require long-term hospitalization have tardive dyskinesia. Women are more likely to be affected than men. Children, patients who are more than 50 years of age, and patients with brain damage or mood disorders are also at high risk.

COURSE AND PROGNOSIS. Between 5 and 40 percent of all cases of tardive dyskinesia eventually remit, and between 50 and 90 percent of all mild cases remit. Tardive dyskinesia is less likely to remit in elderly patients than in young patients, however.

TREATMENT. The three basic approaches to tardive dyskinesia are prevention, diagnosis, and management. Prevention is best achieved by using antipsychotic medications only when indicated and in the lowest effective doses. The atypical antipsychotics are associated with less tardive dyskinesia than the older antipsychotics. Clozapine (Clozaril) is the only antipsychotic to have minimal risk of tardive dyskinesia and can even help improve preexisting symptoms of tardive dyskinesia. This quality of the drug may be

due to its low affinity for D_2 receptors and high affinity for 5-hydroxytryptamine (5-HT) receptor antagonism. Patients who are receiving antipsychotics should be examined regularly for the appearance of abnormal movements, preferably with the use of a standardized rating scale (Table 21-5). Patients frequently experience an exacerbation of their symptoms when the DRA is withheld, whereas substitution of an SDA may limit the abnormal movements without worsening the progression of the dyskinesia.

Once tardive dyskinesia is recognized, the clinician should consider reducing the dose of the antipsychotic or even stopping the medication altogether. Alternatively, the clinician may switch the patient to clozapine or one of the new SDAs. In patients who cannot continue taking any antipsychotic medication, lithium (Eskalith), carbamazepine (Tegretol), or benzodiazepines may effectively reduce the symptoms of both the movement disorder and the psychosis. In 2017, the FDA-approved valbenazine (Ingrezza) for the treatment of adults with tardive dyskinesia, making this the first approved medication for tardive dyskinesia.

Tardive Dystonia and Tardive Akathisia. On occasion, dystonia and akathisia emerge late in the course of treatment. These symptoms may persist for months or years despite drug discontinuation or dose reduction.

Medication-Induced Postural Tremor

DIAGNOSIS, SIGNS, AND SYMPTOMS. *Tremor* is a rhythmic alteration in a movement that is usually faster than one beat per second. Fine tremor (8 to 12 Hz) is most common.

EPIDEMIOLOGY. Typically, tremors decrease during periods of relaxation and sleep and increase with stress or anxiety.

ETIOLOGY. Whereas all the above diagnoses specifically include an association with a neuroleptic, a range of psychiatric medications can produce tremors—most notably, lithium, stimulants, antidepressants, caffeine, and valproate (Depakene).

TREATMENT. The treatment involves four principles:

1. The lowest possible dose of the psychiatric drug should be taken.
2. Patients should minimize caffeine consumption.
3. The psychiatric drug should be taken at bedtime to minimize the amount of daytime tremor.
4. β-Adrenergic receptor antagonists (e.g., propranolol [Inderal]) can be given to treat drug-induced tremors.

Other Medication-Induced Movement Disorders

NOCTURNAL MYOCLONUS. *Nocturnal myoclonus* consists of highly stereotyped, abrupt contractions of specific leg muscles during sleep. Patients lack any subjective awareness of the leg jerks. The condition may be present in about 40 percent of persons over 65 years of age. The cause is unknown, but it is a rare side effect of SSRIs.

The repetitive leg movements occur every 20 to 60 seconds, with extensions of the large toe and flexion of the ankle, the knee, and the hips. Frequent awakenings, unrefreshing sleep, and daytime sleepiness are significant symptoms. No treatment for nocturnal myoclonus is universally effective. Benzodiazepines, levodopa (Larodopa), quinine, and, in rare cases, opioids may be useful.

RESTLESS LEG SYNDROME. In *restless leg syndrome*, persons feel deep sensations of creeping inside the calves whenever sitting or lying down. The dysesthesias are rarely painful but are agonizingly relentless and cause an almost irresistible urge to move the legs; thus, this syndrome interferes with sleep and with falling asleep. It peaks in middle age and occurs in 5 percent of the population. The cause is unknown, but it is a rare side effect of SSRIs.

Symptoms are relieved by movement and by leg massage. The dopamine receptor agonists ropinirole (Requip) and pramipexole (Mirapex) are effective in treating this syndrome. Other treatments include the benzodiazepines, levodopa, quinine, opioids, propranolol, valproate, and carbamazepine.

Hyperthermic Syndromes. All the medication-induced movement disorders may be associated with hyperthermia. Table 21-6 lists the various conditions associated with hyperthermia.

Side Effects Associated with Newer Medications

All medications are associated with side effects. The clinician should be aware of these, be able to recognize them, and take appropriate measures to treat them.

Somnolence. Sedation is often an intended effect of many psychotropic drugs, mainly when used to treat insomnia, anxiety, or agitation. Daytime sleepiness, or somnolence, is also an unwanted adverse event, however. It is essential for the clinician to alert patients to the possibility of sedation and to document that the person was advised to exercise caution when operating any type of vehicle or mechanical equipment. Some somnolence results from a carryover of nighttime use of drugs as hypnotics. Even with drugs such as the SSRIs, which are activating for many patients, somnolence can be problematic. In some instances, it results from impairment of sleep quality. Chronic use of SSRIs can cause some patients to experience a subjective sense of fatigue, exhaustion, or yawning, even with adequate amounts of sleep. Management of unwanted somnolence includes adjustment of dose or timing of administration, switching to alternative medications, the addition of small doses of stimulants, or the addition of modafinil (Provigil).

Gastrointestinal Disturbances. The significant gastrointestinal (GI) side effects of the older antidepressant and antipsychotic drugs consisted primarily of constipation and dry mouth, a consequence of their antimuscarinic activity. Most of the newer drugs have little antimuscarinic activity but do have effects on the serotonin system. Most of the body's serotonin is in the GI tract, and serotonergic drugs often cause varying degrees of stomach pain, nausea, flatulence, and diarrhea. In most cases, these side effects are transient, but some persons never accommodate and must switch to another class of drugs. The initial use of lower doses or the use of delayed-release preparations are the most effective strategies for minimizing GI side effects.

Movement Disorders. The introduction of SDAs has dramatically reduced the incidence of medication-induced movement disorders, but varying degrees of dose-related parkinsonism, akathisia, and dystonia still occur. Risperidone (Risperdal) most closely resembles the older agents in terms of these side effects. Olanzapine (Zyprexa) also causes more extrapyramidal

effects than clinical trials suggested. Aripiprazole (Abilify) causes severe akathisia. There have been rare reports of SSRI-induced movement disorders, ranging from akathisia to tardive dyskinesia.

Sexual Dysfunction. The use of psychiatric drugs can be associated with sexual dysfunction—decreased libido, impaired ejaculation and erection, and inhibition of female orgasm. In clinical trials with the SSRIs, the extent of sexual side effects was grossly underestimated because data were based on spontaneous reports by patients. The rate of sexual dysfunction in the original fluoxetine product information, for example, was less than 5 percent. In subsequent studies in which specific questions elicited information about sexual side effects, the rate of SSRI-associated sexual dysfunction was found to be between 35 and 75 percent. In clinical practice, patients are not likely to report sexual dysfunction spontaneously to the physician, so it is essential to ask about this side effect. Also, some sexual dysfunctions may be related to the primary psychiatric disorder.



Table 21-6
Drug-Induced Central Hyperthermic Syndromes[a]

Condition (and Mechanism)	Common Drug Causes	Frequent Symptoms	Possible Treatment[a]	Clinical Course
Hyperthermia (↓ heat dissipation, ↑ heat production)	Atropine, lidocaine, meperidine NSAID toxicity, pheochromocytoma, thyrotoxicosis	Hyperthermia, diaphoresis, malaise	Acetaminophen per rectum (325 mg every 4 hr), diazepam oral or per rectum (5 mg every 8 hr) for febrile seizures	Benign, febrile seizures in children
Malignant hyperthermia (↑ heat production)	NMJ blockers (succinylcholine), halothane	Hyperthermia, muscle rigidity, arrhythmias, ischemia,[c] hypotension, rhabdomyolysis; disseminated intravascular coagulation	Dantrolene sodium (1–2 mg/kg/min IV infusion)[d]	Familial, 10% mortality if untreated
Tricyclic overdose (↑ heat production)	Tricyclic antidepressants, cocaine	Hyperthermia, confusion, visual hallucinations, agitation, hyperreflexia, muscle relaxation, anticholinergic effects (dry skin, pupil dilation), arrhythmias	Sodium bicarbonate (1 mEq/kg IV bolus) if arrhythmia is present, physostigmine (1–3 mg IV) with cardiac monitoring	Fatalities have occurred if untreated
Autonomic hyperreflexia (↑ heat production)	CNS stimulants (amphetamines)	Hyperthermia, excitement, hyperreflexia	Trimethaphan (0.3–7 mg/min IV infusion)	Reversible
Lethal catatonia (↓ heat dissipation)	Lead poisoning	Hyperthermia, intense anxiety, destructive behavior, psychosis	Lorazepam (1–2 mg IV every 4 hr), antipsychotics may be contraindicated	High mortality if untreated
Neuroleptic malignant syndrome (mixed; hypothalamic, ↓ heat dissipation, ↑ heat production)	Antipsychotics (neuroleptics), methyldopa, reserpine	Hyperthermia, muscle rigidity, diaphoresis (60%), leukocytosis, delirium, rhabdomyolysis, elevated CPK, autonomic deregulation, extrapyramidal symptoms	Bromocriptine (2–10 mg every 8 hr orally or nasogastric tube), lysuride (0.02–0.1 mg/hr IV infusion), carbidopa-levodopa (Sinemet) (25/100 PO every 8 hr), dantrolene sodium (0.3–1 mg/kg IV every 6 hr)	Rapid onset, 20% mortality if untreated

NSAID, nonsteroidal anti-inflammatory drug; MAOI, monoamine oxidase inhibitor; NMJ, neuromuscular junction; CNS, central nervous system; CPK, creatine phosphokinase; PO, orally; IV, intravenously.
[a]Boldface indicates features that may be used to distinguish one syndrome from another.
[b]Gastric lavage and supportive measures, including cooling, are required in most cases.
[c]Oxygen consumption increases by 7% for every 1°F increase in body temperature.
[d]Has been associated with idiosyncratic hepatocellular injury, as well as severe hypotension in one case.
From Theoharides TC, Harris RS, Weckstein D. Neuroleptic malignant-like syndrome due to cyclobenzaprine? [letter]. *J Clin Psychopharmacol.* 1995;15:80, with permission.

Nevertheless, if sexual dysfunction emerges after pharmacotherapy has begun and the primary response to treatment has been positive, it may be worthwhile to attempt to treat the symptoms. Long lists of possible antidotes to these side effects have evolved, but few interventions are consistently effective, and few have more than anecdotal evidence to support their use. The clinician and patient should consider the possibility of sexual side effects when selecting a drug and switching treatment to another drug that is less or not at all associated with sexual dysfunction if this adverse effect is not acceptable to the patient.

Weight Gain. Weight gain accompanies the use of many psychotropic drugs as a result of retained fluid, increased caloric intake, decreased exercise, or altered metabolism. Weight gain can also occur as a symptom of a disorder, as in bulimia or atypical depression. In these cases, it may be a sign of

recovery. Treatment-emergent increase in body weight is a common reason for noncompliance with a drug regimen. No specific mechanisms have been identified as causing weight gain, and it appears that the histamine and serotonin systems mediate changes in weight associated with many drugs used to treat depression and psychosis. Metformin (Glucophage) has been reported to facilitate weight loss among patients whose weight gain is attributed to the use of serotonin-dopamine reuptake inhibitors and valproic acid (Depakene). Valproate (Depacon), as well as olanzapine, has been linked to the development of insulin resistance, which could induce appetite increase, with the subsequent weight increase. Weight gain is a noteworthy side effect of clozapine (Clozaril) and olanzapine. Genetic factors that regulate body weight, as well as the related problem of diabetes mellitus, seem to involve the 5-HT_{2C} receptor. There is a genetic polymorphism of the promoter region of this receptor, and patients with a variant allele have less weight gain. Drugs with a strong 5-HT_{2C} affinity would be expected to have a more significant impact on body weight of patients with a polymorphism of the 5-HT_{2C} receptor promoter region.

Weight Loss. Initial weight loss is associated with SSRI treatment but is usually transient, with most weight being regained within the first few months. Bupropion (Wellbutrin) has been shown to cause modest weight loss that is sustained. When combined with diet and lifestyle changes, bupropion can facilitate more significant weight loss. Topiramate (Topamax) and zonisamide (Zonegran), marketed as treatments for epilepsy, sometimes produce substantial, sustained loss of weight.

Glucose Changes. Increased risk of glucose abnormalities, including diabetes mellitus, is associated with weight increase during psychotropic drug therapy. Clozapine and olanzapine are associated with a greater risk than other SDAs of abnormalities in fasting glucose levels, as well as hyperosmolar diabetes and ketoacidosis. This dysregulation of glucose homeostasis appears to be drug-induced and increases glucagon.

Hyponatremia. Hyponatremia is associated with oxcarbazepine (Trileptal) and SSRI treatment, especially in elderly patients. Confusion, agitation, and lethargy are common symptoms.

Cognitive Impairment. Cognitive impairment means a disturbance in the capacity to think. Some agents, such as the benzodiazepine agonists, are recognized as causes of cognitive impairment. Other widely used psychotropics, such as the SSRIs, lamotrigine (Lamictal), topiramate (Topamax), gabapentin (Neurontin), lithium, TCAs, and bupropion, however, are also associated with varying degrees of memory impairment and word-finding difficulties. In contrast to the benzodiazepine-induced anterograde amnesia, these agents cause a more subtle type of absent-mindedness. Drugs with anticholinergic properties are likely to worsen memory performance.

Sweating. Severe perspiration unrelated to ambient temperature is associated with TCAs, SSRIs, and venlafaxine. This side effect is often socially disabling. Attempts can be made to treat this side effect with alpha agents, such as terazosin (Hytrin) and oxybutynin (Ditropan).

Cardiovascular Disturbances. Newer agents are less likely to have direct cardiac effects. Many older agents, such as TCAs and phenothiazines, affected BP and cardiac conduction. Thioridazine (Mellaril), which has been in use for decades, has been shown to prolong the QTc interval in a dose-related manner and may increase the risk of sudden death by delaying ventricular repolarization and causing torsades de pointes. Newer drugs are now routinely scrutinized for evidence of cardiac effects. A promising treatment for psychosis, sertindole (Serlect), was not marketed because the FDA would have required a black box warning. Slight QTc effects noted with ziprasidone (Geodon) delayed the marketing of that drug. Clozapine can cause myocarditis in rare cases.

Rash. Any medication is a potential source of a drug rash. Some psychotropics, such as carbamazepine (Equetro, Tegretol) and lamotrigine, have been linked to an increased risk of severe exfoliative dermatitis. Commonly referred to as Stevens–Johnson syndrome, this condition is a systemic, immune-mediated reaction that can prove fatal or result in permanent scarring or blindness. All patients should be informed about the potential seriousness of widespread lesions that occur above the neck, that involve the mucous membranes, and that may be associated with fever and

lymphadenopathy. If such symptoms manifest, a patient should be instructed at the time that the medication is prescribed to go immediately to an emergency department.

Idiosyncratic and Paradoxical Drug Responses

Idiosyncratic reactions occur in a tiny percentage of patients taking a drug. The reactions are not related to the known pharmacologic properties, and most likely represent a genetically based abnormal sensitivity to a drug. A paradoxical response represents the manifestation of a clinical effect the opposite of what is expected. In March 2007, the FDA reported dissociative-like states associated with certain sedative-hypnotics. These included behaviors such as sleepwalking, binge eating, aggressive outbursts, and night driving, of which the patient was unaware. Table 21-7 lists the drugs required to have warning labels for that effect.

Therapeutic Index

The therapeutic index is a relative measure of the toxicity or safety of a drug and is defined as the ratio of the median toxic dose to the median effective dose. The median toxic dose is the dose at which 50 percent of patients experience a specific toxic effect, and the median effective dose is the dose at which 50 percent of patients have a specified therapeutic effect. When the therapeutic index is high, as it is for haloperidol, it is reflected by the wide range of dosages in which that drug is prescribed. Conversely, the therapeutic index for lithium is relatively low, thus requiring careful monitoring of serum lithium levels in patients for whom the drug is prescribed.

Overdose

Safety in overdose is always a consideration in drug selection. Almost all of the newer agents, however, have a wide margin of safety when taken in overdose. By contrast, a 1-month supply of TCAs could be fatal. The depressed patients they were used to treat are the group most at risk of attempting suicide. Because even the safest drugs can sometimes produce severe medical complications, especially when combined with other agents, clinicians must recognize that the prescribed medication can be used in an attempt to commit suicide. Although it is prudent to write nonrefillable

prescriptions for small quantities, this practice passes along increased copay costs to the patient. Many pharmacy benefit management programs encourage the prescribing of a 3-month supply of medication.



Table 21-7
Sedative-Hypnotics Cited by the U.S. Food and Drug Administration

Drug	Manufacturer
Zolpidem (Ambien/Ambien CR)	Sanofi Aventis
Butabarbital (Butisol Sodium)	MedPointe Pharmaceuticals
Pentobarbital and carbromal (Carbrital)	Parke-Davis
Flurazepam (Dalmane)	Valeant Pharmaceuticals
Quazepam (Doral)	Questcor Pharmaceuticals
Triazolam (Halcion)	Pfizer
Eszopiclone (Lunesta)	Sepracor
Ethchlorvynol (Placidyl)	Abbott
Estazolam (Prosom)	Abbott
Temazepam (Restoril)	Tyco Healthcare
Ramelteon (Rozerem)	Takeda
Secobarbital (Seconal)	Lilly
Zaleplon (Sonata)	King Pharmaceuticals

In cases in which suicide is a significant concern, an attempt should be made to verify that the medication is not being hoarded for a later overdose attempt. Random pill counts or asking a family member to dispense daily doses may be helpful. Some patients attempt suicide just as they are beginning to recover. Large quantities of medications with a low therapeutic index should be prescribed judiciously. Another reason to limit the number of

pills prescribed is the possibility of accidental ingestion of medications by children in the household. Psychotherapeutic medications should be kept in a safe place.

Physicians who work in emergency departments should know which drugs can be hemodialyzed. The issues involved are complex and are not based on any single chemical property of the drug. For example, it is generally presumed that drugs with low protein binding are good candidates for dialysis. Venlafaxine, however, is only 27 percent protein-bound and is too large as a molecule dialyzed. Hemodialysis is useful for treating overdose of lithium and valproic acid.

Pharmacokinetics

Pharmacokinetic drug interactions are the effects of drugs on the plasma concentrations of each other, and *pharmacodynamic drug interactions* are the effects of drugs on the biologic activities of each other. Pharmacokinetic concepts are used to describe and predict the time course of drug concentrations in different parts of the body, such as plasma, adipose tissue, and the CNS. From a clinical perspective, pharmacokinetic methods help explain or predict the onset and duration of drug activity and interactions between drugs that alter their metabolism or excretion.

Pharmacogenetic research focuses on finding variant alleles that alter drug pharmacokinetics and pharmacodynamics. Researchers are attempting to identify genetic differences in how enzymes metabolize psychotropics, as well as CNS proteins directly involved in drug action. Likely, identification of patient genotypes will facilitate the prediction of clinical response to different types of drugs.

Most clinicians need to consult charts or computer programs to determine when potential interactions may occur and, if so, how clinically relevant they may be. Whenever possible, it is preferable to use a medication that produces the minimal risk of drug interactions. Also, it is recommended that prescribers know the interaction profiles of the drugs they most commonly prescribe.

Examples of pharmacokinetic interactions include one drug increasing or decreasing the concentrations of a coadministered compound. These types of interactions can also lead to altered concentrations of metabolites. In some

cases, there may also be interference with the conversion of a drug to its active metabolite. Enormous variability exists among patients concerning pharmacokinetic parameters, such as drug absorption and metabolism. Another type of interaction is represented by interactions involving the kidney. Commonly used medications, such as angiotensin-converting enzyme inhibitors (ACEIs), nonsteroidal anti-inflammatory drugs (NSAIDs), and thiazides, decrease the renal clearance of lithium, increasing the likelihood of severe elevations of lithium. Drug interactions can occur pharmacokinetically or pharmacodynamically.

Pharmacogenetics is being used to study why patients differ in the way they metabolize drugs. In patients who are ultrarapid or extensive metabolizers, the concentrations of a drug may be lower than expected.

PATIENT-RELATED FACTORS

Response to medication and sensitivity to side effects are influenced by factors related to the patient. This response is why there is no one-size-fits-all approach to pharmacologic treatment. Patient-related variables include diagnosis, genetic factors, lifestyle, overall medical status, concurrent disorders, and history of drug response. A patient's attitude toward medication in general, aversion to certain types of side effects, and preference for a specific agent also need to be considered.

Diagnosis

Failure to correctly diagnose a disorder diminishes the likelihood of optimal drug selection. Misdiagnosis not only can result in a missed opportunity, but it also can, at times, produce worsening of symptoms. Inadvertently diagnosing a patient in the depressed phase of bipolar disorder as having unipolar depression can induce mania or rapid cycling. Treatment failure or exacerbation of symptoms should prompt a reassessment of the working diagnosis.

Past Treatment Response

A specific drug should be selected according to the patient's history of drug response (compliance, therapeutic response, adverse effects), the patient's

family history of drug response, the profile of adverse effects for that drug concerning the particular patient, and the prescribing clinician's usual practice. If a drug has previously been effective in treating a patient or a family member, the same drug should be used again. For reasons that are not understood, however, some patients fail to respond to a previously effective agent when it is resumed. A history of severe adverse effects from a specific drug is a strong indicator that the patient would not be compliant with that particular drug.

It is helpful if patients can recall the details of past psychotropic drug treatment: the drugs prescribed, in what dosages, for how long, and in what combinations. Because of their psychiatric disorders, many patients, however, are poor historians. If possible, patients' medical records should be obtained to confirm their reports. Family members are a good source of collateral information.

Response in Family Members

It is widely held that drug responses cluster in families. Response to a drug in a relative may predict a patient's response. Although no conclusive evidence supports this as a consideration in drug selection, existing studies do confirm that a history of positive response to treatment with a drug should be considered in making treatment decisions. Of course, certain drugs may not have been available to a family member depending on the era in which they were treated.

Concurrent Medical or Psychiatric Disorders

The initial assessment should elicit information about coexisting medical disorders. In some cases, a medical disorder may be responsible for the symptoms. Patients with thyroid disease who are not adequately treated may appear depressed. Sleep apnea produces depression and cognitive impairment. Rare conditions, such as Kleine–Levin syndrome, can mimic bipolar disorder. A drug should be selected that minimally exacerbates any preexisting medical problems that a particular patient may have.

Recreational drug use, excessive consumption of alcohol, and frequent ingestion of caffeine-containing beverages can complicate and even undermine psychotropic drug treatment. These compounds possess

significant psychoactive properties and, in some cases, may represent the source of the patient's symptoms. It is reasonable to ask patients to abstain from the use of these substances, at least until the benefits of psychotropic drug treatment have been unequivocally established. Gradual reintroduction of moderate amounts of alcohol, tea, and coffee can then take place. Patients can then observe for themselves whether there are any untoward effects on their clinical status.

INFORMED CONSENT AND PATIENT EDUCATION

Establishing trust and motivating the patient to comply with the medication regimen are essential components of successful treatment. Patients should be informed about treatment options and the probable side effects and unique benefits of each treatment. Patient preference should be respected unless a compelling advantage exists involving efficacy, tolerability, or safety with an alternative agent. If a particular medication is being recommended, the reasons for this recommendation should be explained. Patients are more likely to continue taking their medication if they fully understand the reasons why it is being prescribed.

A strong therapeutic alliance between a clinician and a patient is always helpful. Given the unpredictability of medication response, the frequent occurrence of side effects, and underlying ambivalence about, or fear of taking, medication, a constructive, trusting relationship serves to improve patient compliance. Repeated failed trials may be needed before a response is seen. A patient's confidence in the physician's knowledge and judgment enables medication trials and more complex regimens, such as the use of multiple medications.

Discussions about drug selection should be documented in notes, but a signed informed consent is not needed. Surprisingly, patients who are informed of potential adverse effects report a higher incidence of side effects but do not have higher rates of premature discontinuation.

How the patient and family are engaged in the treatment plan can determine the success of treatment. The psychodynamic meaning of pharmacotherapy to the patient and family and environmental influences, psychosocial stressors, and support should be explored. Some patients may view drug treatment as a panacea, and others may view it as the enemy. With

the patient's consent, relatives and other clinicians should be instructed about the reasons for the drug treatment, as well as the expected benefits and potential risks.

DOSING, DURATION, AND MONITORING

Dosing

The clinically effective dose for treatment depends on the characteristics of the drug and patient factors, such as inherited sensitivity and ability to metabolize a drug, concurrent medical disorders, use of concurrent medications, and history of exposure to previous medications.

Plasma concentrations of many psychotropics can vary up to 10-fold. Thus, to some extent, the optimal dose for an individual is ultimately determined by trial and error, guided by the empirical evidence of the usual dose range for that drug. In some cases, it may prove helpful to test patients for genetic polymorphisms involving hepatic enzymes. Patients who are ultrarapid metabolizers of certain drugs may require higher than standard dosing. Slow metabolizers might demonstrate side effects and even toxicity at very low doses.

Some drugs demonstrate a clear relationship between increases in the dose and clinical response. This dose–response curve plots the drug concentration against the effects of the drug.

The *potency* of a drug refers to the relative dose required to achieve specific effects, not to its efficacy. Haloperidol, for example, is more potent than chlorpromazine (Thorazine) because approximately 5 mg of haloperidol is required to achieve the same therapeutic effect as 100 mg of chlorpromazine. These drugs, however, are equal in their clinical efficacy—that is, the maximal clinical response achievable by the administration of a drug.

Drugs must be used in effective dosages for sufficient periods. Although drug tolerability and safety are always considerations, subtherapeutic doses and incomplete therapeutic trials should be avoided. The use of inadequate doses merely exposes the patient to the risk of side effects without providing the probability of therapeutic benefit. Because of the wide margin of safety associated with most currently prescribed medications, more risk exists in

underdosing than in overshooting the recommended dose range.

The time of dosing is usually based on the plasma half-life of a drug and its side effect profile. Sedating drugs are given all at night or with disproportionate daily doses at night. The opposite is true with activating drugs. The frequency of dosing is not as straightforward as with sedating drugs. Most dosing regimens of psychotropic drugs, such as once-a-day versus divided doses, are based on measurements of plasma concentrations rather than receptor occupancy in the brain. Evidence suggests a significant dissociation exists between the brain and plasma kinetics. Reliance on plasma kinetics as the basis for dosing regimens leads to a misunderstanding of necessary schedules.

As a rule, psychotropic drugs should be used continuously. There are some possible exceptions, such as when treating insomnia, agitation, or situational anxiety. A common mistake is the use of high-potency benzodiazepines, such as alprazolam (Xanax) and clonazepam (Klonopin), only after an attack has begun. These drugs should be used as part of a regular schedule to prevent attacks.

Some patients who experience sexual dysfunction while being treated with SSRIs take a drug holiday; that is, they skip a daily dose from time to time to facilitate sexual performance.

Intermittent dosing regimens of SSRIs are effective as a treatment for premenstrual dysphoric disorder (PMDD). The drugs are taken daily during the 2-week luteal phase of the menstrual cycle.

Duration of Treatment

A common question from a patient is, “How long do I need to take the medication?” The answer depends on multiple variables. These include the specific disorder, the symptom duration, and family history. We should also consider the patient’s tolerance of and benefit from the medication. Patients can be given a reasonable explanation of the probabilities but should be told that it is first best to see if the medication works for them and whether any side effects are acceptable. Any more definitive discussion of treatment duration can be held once we can assess the degree of success. Even patients with a philosophical aversion to the use of psychotropic drugs may elect to stay on the medication indefinitely if the magnitude of improvement is

excellent. Most psychiatric disorders have high rates of chronicity and relapse. Because of this, long-term treatment is often needed to prevent a recurrence. Nevertheless, the fact remains that psychotropic drugs are not said to cure the disorders they treat but rather to help control the symptoms.

Treatment is conceptually broken down into three phases: the initial therapeutic trial, the continuation, and the maintenance phase. The initial period of treatment should last at least several weeks because of the delay in therapeutic effects that characterize most classes of psychotropic drugs. The required duration of a *therapeutic trial* of a drug should be discussed at the outset of treatment so that the patient does not have unrealistic expectations of an immediate improvement in symptoms. Patients are more likely to experience side effects early in the course of pharmacotherapy than any relief from their disorder. In some cases, medication may even exacerbate some symptoms. Patients should be counseled that a meager initial reaction to medication is not an indicator of the outcome of treatment. For instance, many patients with panic disorder develop jitteriness or an increase in panic attacks after starting on tricyclic or SSRI treatment. Benzodiazepine agonists are an exception to the rule that clinical onset is delayed. In most cases, their hypnotic and antianxiety effects are evident immediately.

Ongoing use of medication, however, does not provide absolute protection against relapse. Continuation therapy provides clinically and statistically significant protective effects against relapse. The optimal duration of continuation or maintenance therapy is variable and dependent on the clinical history of the patient. Early-onset chronic major depression, for example, has a more severe course and more significant comorbidity than late-onset chronic major depression. In addition to early-onset, a history of multiple past episodes and the severity and length of a current episode would suggest that longer, even indefinite, treatment is appropriate.

Frequency of Visits

Until an unequivocal response to treatment occurs, patients should be seen as frequently as circumstances warrant. The frequency of follow-up or monitoring visits is determined by clinical judgment. In severely ill patients, this might mean several times a week. Patients on maintenance therapy, even when stable, need monitoring, but no consensus exists on the frequency of

follow-up therapy. Three months is a reasonable interval between visits, but 6 months may be adequate after long-standing treatment.

LABORATORY TESTS AND THERAPEUTIC BLOOD MONITORING

Laboratory testing and therapeutic blood monitoring should be based on clinical circumstances and the drugs being used. For most commonly used psychotropic drugs, routine testing is not required.

Pretreatment tests are routine as part of a workup to establish baseline values and to rule out underlying medical problems that may be causing the psychiatric symptoms or that might complicate treatment with drugs. Results of recently performed tests should be obtained. With agents known to cause cardiac conduction changes, a pretreatment electrocardiogram (ECG) should be obtained before initiating treatment. With lithium and clozapine, the possibility of severe changes in thyroid, renal, hepatic, or hematologic functions requires pretreatment and ongoing monitoring with appropriate laboratory tests.

As a result of both anecdotal and research findings of sometimes severe glucose dysregulation during treatment primarily with SDAs, the FDA has suggested that patients being treated with any atypical antipsychotic be monitored for the emergence of diabetes.

Certain circumstances present in which it is necessary or useful to use plasma concentrations to monitor a patient’s condition. The most common is the monitoring of drugs with narrow therapeutic indexes, such as lithium. Also, we may monitor drugs with a therapeutic window, which is the optimal dose range for a therapeutic response; drug combinations that can lead to interactions that raise drug concentrations of medications or their metabolites, which can cause toxicity; unexplained toxicity at average therapeutic doses; and failure to respond in a patient who may be noncompliant. A clinician should have no reservations about requesting random urine toxicologic tests in a patient who abuses substances.

TREATMENT OUTCOMES

The goal of psychotropic treatment is to eliminate all manifestations of a

disorder, thus enabling the patient to regain the ability to function as well and to enjoy life as fully as before they became ill. This degree of improvement to below the syndromal threshold is defined as *remission*.

Response and Remission

Remission is the preferred outcome of treatment, not only because of the immediate impact on functioning and state of mind but also because emerging evidence suggests that patients in remission are less likely to experience relapse and recurrence of their disorder.

Patients who improve but do not experience a full resolution are considered to be responders. They may exhibit significant improvement but continue to experience symptoms. In depression studies, the *response* is usually defined as a 50 percent or more significant decrease from baseline on a standard rating scale, such as the Hamilton Depression (HAM-D) Scale or the Montgomery-Asberg Depression Rating Scale (MADRS). The scales define r*emission* as an absolute score of 7 or less on the HAM-D or 10 or less on the MADRS.

Our expectations around improvement derive from what we know about the disorder and its response to treatment. Obsessive-compulsive disorder (OCD) and schizophrenia, for example, are more likely to be associated with residual manifestations of illness than major depression or panic disorder. The probability of full remission from OCD with SSRI treatment alone over 2 years is less than 12 percent, and the probability of partial remission is approximately 47 percent.

Treatment Failure

The initial treatment plan should anticipate that the medication may be ineffective. A next-step strategy should be in place at the initiation of treatment. Repeated drug failures should prompt a reassessment of the patient. First, was the original diagnosis correct? In answering this question, the clinician should include the possibility of an undiagnosed medical condition or recreational drug use as the cause of the psychiatric symptoms.

Second, are the observed symptoms related to the original disorder, or are they adverse effects of the drug treatment? Some antipsychotic drugs, for example, can produce akinesia, which resembles psychotic withdrawal, or

akathisia and neuroleptic malignant syndrome, which resemble increased psychotic agitation. Long-term use of SSRIs can produce emotional blunting, which can mimic depression. Intolerance of side effects may be the most common reason for treatment failure.

Third, was the drug administered at an appropriate dosage for a sufficient length of time? Because absorption and metabolism of drugs can vary significantly in patients, the clinician may need to measure plasma levels of a drug to ensure a sufficient dose of the drug.

Fourth, did a pharmacokinetic or pharmacodynamic interaction with another drug that the patient was taking reduce the efficacy of the newly prescribed drug?

Fifth, did the patient take the drug as directed? Drug noncompliance is a common clinical problem that arises as a result of complicated drug regimens (more than one drug in more than one daily dosage), adverse effects (mainly if unnoticed by the clinician), and poor patient education about the drug treatment plan. Patients may discontinue medication when they recover, thinking that they are cured and no longer benefiting from the medication.

Treatment Resistance

Some patients fail to respond to repeated trials of medication. No single factor can explain the ineffectiveness of the various interventions in these cases. Strategies in these cases include the use of drug combinations, high-dose therapy, and the use of unconventional drugs. Limited evidence is available on the comparative success rates associated with any given strategy.

Tolerance

The development of tolerance is marked by a need, over time, to use increased doses of a drug for it to maintain a clinical effect. This decreased responsiveness to a drug occurs after repeated doses. Tolerance also describes decreased sensitivity to adverse effects of the drug, such as nausea. This phenomenon is used as the basis for starting some drugs at subtherapeutic doses, with the plan to adjust the schedule once the patient can tolerate higher doses. Clinical tolerance appears to represent changes in the CNS, such as altered receptor configuration or density. Drugs with similar pharmacologic actions often exhibit cross-tolerance.

Sensitization

Clinically manifested as the reverse of tolerance, sensitization is said to occur when sensitivity to a drug effect increases over time. In these cases, the same dose typically produces more pronounced effects as treatment progresses.

Withdrawal

The development of physiologic adaptation to a drug, with a subsequent risk of withdrawal symptoms, has been reported for many classes of psychotropic drugs. Technically, withdrawal should be considered a side effect. The probability and severity of these reactions are remote with most drugs and more frequent with others. As a general rule, the more abruptly a drug is stopped and the shorter its elimination half-life, the more likely it is that clinically significant withdrawal symptoms will occur. When using some short-acting drugs, withdrawal reactions can result from missed doses and during daily intervals between doses. Gradual tapering of medications after prolonged use is recommended whenever possible. Although this reduces the risk of withdrawal reactions, it does not ensure they will not occur. The so-called sedative-hypnotics and opiates are the agents most often associated with mentally and physically distressing discontinuation reactions. In some cases, such as barbiturate use, withdrawal can be fatal.

Marked differences are found among agents, even within a given class, concerning the probability and severity of discontinuation effects. For example, the benzodiazepines alprazolam and triazolam (Halcion) commonly produce more immediate and intense withdrawal symptoms than other compounds. Among the SSRIs, there is a well-described withdrawal syndrome that appears to be more frequent and severe with paroxetine (Paxil). It can, however, occur with any SSRI. Even fluoxetine can be associated with discontinuation symptoms, but the symptoms may be delayed and attenuated because of the long elimination half-life of its active metabolite. These manifestations are subtle and are delayed for weeks after the last dose. Venlafaxine also produces severe SSRI-like withdrawal syndrome.

In addition to half-life, many variables can influence the likelihood and degree of discontinuation symptoms. Changes in the rate of drug metabolism, as an example, can play a role. Paroxetine is primarily metabolized by the

cytochrome P450 (CYP) 2D6 isoenzyme. However, it is also a potent inhibitor of CYP 2D6. This results in *autoinhibition,* dose-dependent inhibition of its metabolism, with a subsequent increase in plasma concentrations of paroxetine. If the dose of paroxetine is decreased or the drug is stopped, the decline in its plasma concentrations can be steep, causing withdrawal to occur. Withdrawal can occur in rare cases in which the dose of a drug is not decreased, but a second agent, which had been inhibiting its metabolism, was stopped. For example, alprazolam is metabolized via the CYP 3A3/4 enzyme system. Nefazodone inhibits that enzyme. If a patient taking both agents for several weeks discontinues the nefazodone, it could result in a rapid increase in the rate of alprazolam metabolism and a consequent drop in plasma concentrations.

The development of sustained-release versions of drugs, such as alprazolam, paroxetine, and venlafaxine, has not reduced the severity of their withdrawal reactions. The prolonged half-life of those agents results from delayed absorption rather than prolongation of the elimination phase. The frequency of drug dosing is reduced but not the rate of the falloff in plasma concentrations.

Low bioavailability with a generic agent may account for the unexpected loss of clinical effect in the emergence of withdrawal symptoms. The occurrence of these events soon after refilling a prescription should prompt examination of the new medication. It should be confirmed whether the dispensed medication and dose are both correct. It is difficult to ascertain whether generic medications are genuinely equivalent, so the possibility exists that differences in potency may underlie adverse changes in clinical status.

Withdrawal symptoms invariably occur hours or days after dose reduction or discontinuation. Symptoms resolve within a few weeks, so the persistence of symptoms argues against withdrawal. Although depletion studies have been shown to provoke the rapid return of symptoms, in clinical practice, psychotic and mood symptoms do not usually reappear abruptly after long-term treatment.

COMBINATION OF DRUGS

According to the American Psychiatric Association Practice Guidelines for

the Treatment of Psychiatric Disorders, “the use of multiple agents should be avoided if possible” in the treatment of psychiatric disorders. Although *monotherapy* represents the ideal, *polypharmacy*, the simultaneous use of psychotropic medications, has been commonplace since chlorpromazine was combined with reserpine (Diupres) in the early 1950s. The practice of combining drugs and the merits of various *augmentation* or *combination* strategies are routinely discussed in the literature and at scientific meetings. The mean number of simultaneously prescribed medications has increased in recent decades. Among psychiatric inpatients, the mean number of psychotropics prescribed is approximately three. Fixed combinations—drugs that contain more than one active ingredient—have been successfully marketed in the past, and research on new combinations is ongoing. A fluoxetine-olanzapine fixed combination has been approved as a treatment for bipolar disorder. The use of such drugs may increase the patients’ compliance by simplifying the drug regimen. A problem with combination drugs, however, is that the clinician has less flexibility in adjusting the dosage of one of the components; that is, the use of combination drugs can cause two drugs to be administered when only one drug continues to be necessary for therapeutic efficacy (Table 21-8).



Table 21-8
Selected Combination Drugs Used in Psychiatry

Ingredients	Preparation	Amount of Each	Recommended Dosage	Indications
Perphenazine and amitriptyline	—	Tablet: 2:25, 4:25, 4:50, 2:10, 4:10	Initial therapy: tablet of 2:25 or 4:25 qid Maintenance therapy: tablet 2:25 or 4:25 bid or qid.	Depression and associated anxiety
Dextroamphetamine and amphetamine	Adderall	Tablet: 5, 7.5, 10.0, 12.5, 15.0, 20.0, 30.0 mg	3–5 yr: 2.5 mg/day; 6 yr and older: 5 mg/day	Attention-deficit/hyperactivity disorder
	Adderall XR	Capsule: 5, 10, 15, 20, 25, 30 mg	—	—
Chlordiazepoxide and clidinium bromide	—	Capsule: 5:25	One or two capsules tid or qid before meals and at bedtime	Peptic ulcer, gastritis, duodenitis, irritable bowel syndrome, spastic colitis, and mild ulcerative colitis
Chlordiazepoxide and amitriptyline	—	Tablet: 5.0:12.5, 10:25	Tablet of 5:12.5 tid or qid; tablet of 10:25 tid or qid, initially, then may increase to six tablets daily as required	Depression and associated anxiety
Olanzapine and fluoxetine	Symbyax	Capsule: 6:25, 6:50, 12:25, 12:50	Once daily in the evening in a dose range of olanzapine 6–12 mg and fluoxetine 25–50 mg	Depressive episodes associated with bipolar I disorder

qid, four times daily; bid, twice daily; tid, three times daily.

Sometimes distinctions are made between augmentation and combination

therapy. When two psychotropics with the same approved indications are used concurrently, this is termed *combination therapy*. Adding a drug with another indication is termed *augmentation*. Augmentation often entails the use of a drug that is not primarily considered a psychotropic. For example, in treating depression, it is not common to add thyroid hormone to an approved antidepressant.

Almost all patients with bipolar disorder are taking more than one psychotropic agent. Combination treatment with drugs that treat depression and DRA or SDA has long been held as preferable in patients with psychotic depression. Similarly, SSRIs typically produce partial improvement in patients with OCD, so the addition of an SDA may be helpful.

Medications also can be combined to counteract side effects, to treat specific symptoms, and as a temporary measure to transition from one drug to another. It is common practice to add a new medication without the discontinuation of a prior drug, mainly when the first drug has provided partial benefit. This addition can be part of a plan to transition from an agent that is not producing a satisfactory response or an attempt to maintain the patient on combined therapy.

The advantages of combining drugs include building on the existing response, which may be less demoralizing, and the possibility that combinations produce new mechanisms that no single agent can provide. One limitation is that noncompliance and adverse effects increase, and the clinician may not be able to determine whether it was the second drug alone or the combination of drugs that resulted in a therapeutic success or a particular adverse effect. Combining drugs can create a broad spectrum effect and also changes the ratio of metabolites.

COMBINED PSYCHOTHERAPY AND PHARMACOTHERAPY

Many psychiatrists believe that patients are best treated with a combination of medication and psychotherapy. Studies have demonstrated that the results of combined therapy are superior to those of either type of therapy alone. When pharmacotherapy and psychotherapy are used together, the approach should be coordinated, integrated, and synergistic. If two separate clinicians direct psychotherapy and pharmacotherapy, the clinicians must communicate with

each other clearly and often.

SPECIAL POPULATIONS

Although every patient brings a unique combination of demographic and clinical variables to the clinical setting, specific patient populations require special consideration. When treating the young, the elderly, those with medical disorders, and women who want to conceive or are pregnant or nursing, awareness of risks associated with medication assumes increased importance. Data derived from clinical trials are of limited value in guiding many decisions because populations in these studies consisted of healthy young adults and, until recently, excluded many women of child-bearing age. Studies of children and adolescents have become more common, so understanding of treatment effects in this population has grown.

Children

Understanding of the safety and efficacy of most psychotropic drugs, when used to treat children, is based more on clinical experience than on evidence from extensive clinical trial data. Often, so results from adult studies are extrapolated to children. This extrapolation is not necessarily appropriate because of developmental differences in pharmacokinetics and pharmacodynamics. Dosing is another special consideration in drug use with children. Although the small volume of distribution suggests the use of lower doses than those used in adults, a child's higher rate of metabolism suggests that a higher ratio of milligrams of a drug to kilograms of body weight should be used. In practice, beginning with a small dose is best, then increasing it until clinical effects are observed. The clinician should not hesitate, however, to use adult dosages in children if these dosages are effective and the adverse effects are acceptable.

The paucity of research data is a legacy of many years in which manufacturers avoided conducting trials in children because of liability concerns, small market share, and, hence, limited profit potential represented by this population. To correct this problem, the FDA Modernization Act (FDAMA) of 1997 provided for special encouragement and incentives to study drugs for pediatric use.

Pregnant and Nursing Women

No definitive assurances exist that any drug is entirely without risk during pregnancy and lactation. No psychotropic medication is absolutely contraindicated during pregnancy, although drugs with known risks of congenital disabilities, premature birth, or neonatal complications should be avoided if acceptable alternatives are available.

Women who are pregnant or lactating are excluded from clinical trials, and it is only recently that women of child-bearing age have been able to participate in these studies. As a result, there are large gaps in knowledge of the effects of psychotropic agents on the developing fetus and the neonate. Most of what is known are the result of anecdotal reports or data from registries. The basic rule is to avoid administering any drug to a woman who is pregnant (particularly during the first trimester) or who is breastfeeding a child unless the mother's psychiatric disorder is severe, and the therapeutic value of the drug outweighs the theoretical adverse effects on the fetus or newborn. A woman may elect to continue on medication because she does not want to chance a possible recurrence of painful or disabling symptoms.

Among the newer antidepressants, paroxetine is the only one to carry a warning from the FDA, the result of an increased risk of cardiac malformation. The agents with the most well-documented risk of specific congenital disabilities are lithium, carbamazepine, and valproate. Lithium administration during pregnancy is associated with Ebstein anomaly, a severe abnormality of cardiac development, although recent evidence suggests that the risk is not as significant as previously believed. Carbamazepine and valproic acid are associated with neural tube defects, which can be prevented by the use of folate during pregnancy. Lamotrigine may cause oral clefts when used during the first trimester. Some experts advise that all women of child-bearing age who are treated with psychotropics take supplemental folate.

The administration of psychotherapeutic drugs at or near delivery can cause the baby to be overly sedated at delivery and require a respirator. They may also be physically dependent on the drug, requiring detoxification and the treatment of a withdrawal syndrome. Reports exist of a neonatal withdrawal syndrome associated with third-trimester use of SSRIs in pregnant women. They have also been implicated in producing pulmonary

hypertension in newborns.

Virtually all psychiatric drugs are secreted in the milk of a nursing mother; therefore, mothers on those agents should be advised not to breastfeed their infants.

Elderly Patients

The two major concerns when treating geriatric patients with psychotherapeutic drugs are that elderly persons may be more susceptible to adverse effects (particularly cardiac effects) and may metabolize and excrete drugs more slowly, thus requiring lower dosages of medication. In practice, clinicians should begin treating geriatric patients with a small dose, usually approximately half of the usual starting dose. The dose should be raised in small increments, more slowly than for middle-aged adults, until a clinical benefit is achieved or unacceptable adverse effects appear. Although many geriatric patients require a small dose of medication, many others require a full therapeutic dose.

Elderly patients account for approximately one-third of all prescription drug use and a substantial percentage of over-the-counter preparations as well. Even more significant is the incidence of polypharmacy. Recent surveys have found that elderly patients in the community are taking between three and five medications, and that hospitalized elderly patients are treated with an average of ten drugs. Nearly half of all patients in long-term care facilities are prescribed one or more psychotropic agents. Because of these statistics, clinicians need to consider potential types and likelihood of drug interactions when selecting medications.

Psychotropic drugs are causally related to falls in the elderly. Discontinuation of psychotropic drugs results in an estimated 40 percent risk reduction for falls. This association between psychotropics and falls and hip fractures may weaken as newer agents become widely used. As a rule, new-generation compounds produce less unwanted sedation, dizziness, parkinsonism, and postural hypotension.

Age-related changes in renal clearance and hepatic metabolism make it more important to be conservative with the starting doses of medication as well as the rate of dose titration. Within any class of psychotropic agents, those with potentially serious consequences, such as hypotension, cardiac

conduction abnormalities, anticholinergic activity, and respiratory depression, are not suitable choices. Drugs that cause cognitive impairment, such as benzodiazepines and anticholinergics, can mimic or exacerbate symptoms of dementia. Similarly, DRAs can worsen or induce Parkinson disease, another age-related disorder. Some side effects, such as SSRI-associated syndrome of inappropriate secretion of antidiuretic hormone (SIADH) and oxcarbazepine-associated hyponatremia, occur more commonly in older patients.

A common ethical dilemma with the medically ill elderly or those with dementia is the question of their capacity to give informed consent before treatment with psychotropic drugs or ECT.

Medically Ill Patients

There are special considerations, diagnostic and therapeutic, when administering psychiatric drugs to medically ill patients. The medical disorder should be ruled out as a cause of psychiatric symptoms. For example, patients with neurologic or endocrine disorders or those infected with human immunodeficiency virus (HIV) may experience disturbances of mood and cognition. Common medications, such as corticosteroids and L-dopa, are associated with the induction of mania.

A patient with diabetes mellitus is better treated with an agent without the risk of weight gain or glucose dysregulation. Depending on the diagnosis, drugs that might treat the primary psychiatric disorder and also cause weight loss, drugs such as bupropion, topiramate, and zonisamide, should be prescribed for these patients. Patients with obstructive pulmonary disease should not be given sedating drugs, which raise the arousal threshold and suppress respiration. Patients with medical disorders are also taking other medications, which can result in pharmacodynamic and pharmacokinetic interactions. Combined treatment with an inducer of multiple CYP enzymes and a drug that is a substrate for those enzymes could result in subtherapeutic levels, leading to inadequate symptom control. The use of the tuberculosis treatment rifampicin (Rifadin) with carbamazepine is an example of this. The use of drugs that inhibit CYP 2D6, agents such as paroxetine and fluoxetine, can prevent the conversion of hydrocodone (Robidone) and other opiates into an active analgesic form. NSAIDs are also a rare cause of perceptual

disturbances and psychotic symptoms.

Other issues include a potentially increased sensitivity to adverse effects, including increased or decreased metabolism and excretion of the drug, and interactions with other medications. Drug interactions are an obvious concern when drugs with a narrow therapeutic range are being used. Any change in the rate of metabolism or interference with the formation and elimination of metabolites can profoundly influence the activity of that drug. Similarly, interactions that interfere with drug metabolism can produce an increase in side effects and toxicity.

As with children and geriatric patients, the most reasonable clinical practice is to begin with a small dosage, to increase it slowly, and to watch for clinical benefit and adverse effects. Determining the plasma drug concentrations may be helpful for such patients, but therapeutic blood concentrations for most psychotropic drugs are neither necessary nor routinely available.

Substance Abuse

Many patients who seek or need treatment for a psychiatric disorder chronically use illicit substances or drink excessive amounts of alcohol. Discontinuation of chronic drug or alcohol use can result not only in craving but also in clinically significant psychiatric and physiologic withdrawal symptoms. For many patients, successful treatment of their underlying psychiatric disorder may not be possible in the presence of ongoing marijuana, cocaine, and alcohol use. If several trials of medications fail, hospitalization for detoxification may be necessary. Little research and no consensus exist about how to use psychotropic agents in patients who are regular users of cocaine, marijuana, or other recreational drugs.

REGULATORY ISSUES

The FDA has the authority to approve a drug for clinical use and to ensure that product labeling is truthful and contains all information pertinent to the safe and effective use of that drug.

Product information that is FDA approved for marketed drugs appears as a package insert that lists potential side effects, drug interactions, the need for special monitoring, and restrictions for use. In some cases, these adverse

reactions and potential safety hazards warrant a warning label called a *black box label*. The FDA typically negotiates final labeling language with the company; however, in cases where a company refuses to satisfy the FDA, the agency may initiate proceedings to remove the drug from clinical use. In recent years, warning labels have been applied to entire classes of psychotropic drugs, including the SDAs and antidepressants such as the SSRIs.

The product information may also contain a "Contraindications" heading. This section describes instances in which the drug should not be used because the risk of using it outweighs the benefit. If no contraindications are known, this section of the labeling will state "None known."

A Precautions section may contain precautions for most individuals taking the drug, as well as for specific groups, such as pregnant women, nursing mothers, or children. In this section, one will find recommendations for patients to ensure the safe and effective use of the drug. For example, there may be precautions about driving when taking the medication or using substances such as other drugs, food, or alcohol that may have harmful effects if taken while using the medication. The precautions section also provides information about laboratory tests needed to track responses or to identify adverse reactions to the drug or about known interactions with other drugs, foods, or ingredients.

Every product label has an Adverse Reaction section that lists the frequency of undesirable effects that may be associated with the use of a drug. Causes of adverse reactions can include medication errors, such as overdosage, or interactions between different drugs or between drugs and certain foods.

NONAPPROVED DOSAGES AND USES

It is now common practice to treat psychiatric disorders with drugs that are approved for nonpsychiatric conditions. Some examples include propranolol (Inderal) for social anxiety and treatment of lithium-induced tremor; verapamil (Calan, Isoptin) for mania and treatment of MAOI-induced hypertensive crisis; levothyroxine (Levoxyl) for antidepressant augmentation; clonidine (Catapres) and guanfacine (Tenex) for posttraumatic stress disorder (PTSD); dextroamphetamine (Dexedrine) for antidepressant augmentation;

and riluzole (Rilutek) for self-injurious behavior. Off-label use of a drug is not a violation of law or a departure from good medical practice. The FDA does not limit how a physician may use an approved drug. Medications can be prescribed for any reason shown to be medically indicated for the welfare of the patient. Once a drug is approved for commercial use, a physician can, as part of the practice of medicine, lawfully prescribe a different dosage for a patient or may otherwise vary the conditions of use from what is approved in the package labeling without notifying the FDA or obtaining its approval.

Failure to follow the information on the drug label does not in itself impose liability and should not preclude a physician from using sound clinical judgment in the service of the patient. Physicians are permitted to use a drug for indications not included on the drug's official labeling without violating the FDA rules. This fact, however, does not absolve the physician of responsibility for an untoward result from treatment. Patients can still sue for possible medical malpractice with the reasoning that the failure to follow the FDA-approved label can be interpreted as deviating from the prevailing standard of care.

When using a drug off-label or in doses outside the usual range, we should explain to the patient the reasons for this approach. We should also document it in the chart. When doubting the plan to use a drug off label, a consultation with a colleague should be obtained.

In some cases, a drug has obtained limited approval for an indication. Divalproex (Depakote) and risperidone, for example, are approved by the FDA for the acute, but not long-term, treatment of mania. Nevertheless, these drugs are routinely used for long-term prevention of recurrences of mania and bipolar disorder. In the case of lamotrigine, it was accepted as a first-choice agent for the treatment of bipolar disorder long before the FDA approved that indication.

PLACEBOS

Pharmacologically inactive substances have long been known sometimes to produce significant clinical benefits. A patient who believes that a compound is helpful may often derive considerable benefit from taking that substance, whether it is known to be pharmacologically active or not. For many psychiatric disorders, including mild to moderate depression and some

anxiety disorders, well over 30 percent of patients can exhibit significant improvement or remission of symptoms on a placebo. For other conditions, such as schizophrenia, manic episodes, and psychotic depression, the placebo response rate is meager. Whereas suggestion is undoubtedly crucial in the efficacy of placebos (and active drugs), placebos can produce biologic effects. For example, placebo-induced analgesia may sometimes be blocked by naloxone (Narcan), which suggests that endorphins may mediate the analgesia derived from taking a placebo. It is conceivable that placebos may also stimulate endogenous anxiolytic and antidepressant factors, resulting in clinical improvement in patients with depression and anxiety disorders.

Just as placebos can produce benefits, they can also have adverse effects. In many studies, some adverse effects are likely to be more frequent with placebos than with the active drug. Some patients will not tolerate placebos, although they are supposedly inert, and they exhibit adverse effects (called the *nocebo phenomenon*). It is easy to discount such patients as overly suggestible; however, if placebos can stimulate beneficial endogenous factors, perhaps toxic endogenous factors can also be produced.

Prudence is needed in contemplating the use of a placebo in clinical practice. Treating a patient with a placebo without consent can seriously undermine a patient's confidence in the physician if, and when, it is discovered.

21.1 Antipsychotics

SEROTONIN–DOPAMINE ANTAGONISTS AND SIMILARLY ACTING DRUGS (SECOND-GENERATION OR ATYPICAL ANTIPSYCHOTICS)

The SGAs, also known as second-generation or atypical antipsychotic drugs, are a group of pharmacologically diverse drugs that have largely supplanted the older DRAs. The term *atypical* is used because these drugs differ in their side effect profiles, most notably lower risk of extrapyramidal side effects (EPS), and have spectra of action that are broader than those of the DRAs. However, this term is becoming something of a misnomer, as the drugs are

now more commonly used than the “typical” antipsychotics. In contrast to the earlier antipsychotic drugs, the SGAs have significant effects on both the dopamine and serotonin systems. Their pharmacology is complex, with individual drugs in this group having multiple neurotransmitter effects. All SGAs are indicated for the treatment of schizophrenia. Most of these SGA drugs have also received approval as monotherapy or adjunctive therapy in the treatment of bipolar disorder. Some have also been approved as adjuncts for the treatment of major depression.

As of 2020, 13 SGA drugs were approved by the FDA. These include the following:

Aripiprazole (marketed as Abilify)
Asenapine Maleate (marketed as Saphris)
Brexpiprazole (marketed as Rexulti)
Cariprazine (marketed as Vraylar)
Clozapine (marketed as Clozaril)
Iloperidone (marketed as Fanapt)
Lumateperone (marketed as Caplyta)
Lurasidone (marketed as Latuda)
Olanzapine (marketed as Zyprexa, and as Symbyax in combination with fluoxetine)
Paliperidone (marketed as Invega)
Quetiapine (marketed as Seroquel)
Risperidone (marketed as Risperdal)
Ziprasidone (marketed as Geodon)

It is arguable whether the SGAs represent an improvement in overall tolerability than the DRAs. Although there is an improvement concerning a lowered, not absent, risk of EPS, most of the drugs in this group often produce substantial weight gain, which in turn increases the potential for the development of diabetes mellitus. Olanzapine and clozapine appear to account for most cases of weight gain and drug-induced diabetes mellitus. The other agents pose a smaller risk of these side effects; nevertheless, the FDA has requested that all SGAs carry a warning label that patients taking the drugs need to be monitored closely, and has recommended the following factors be considered for all patients prescribed SGAs:

1. Personal and family history of obesity, diabetes, dyslipidemia, hypertension, and cardiovascular disease
2. Weight and height (so that body mass index can be calculated)
3. Waist circumference (at the level of the umbilicus)
4. BP
5. Fasting plasma glucose level
6. Fasting lipid profile

Patients with preexisting diabetes should have regular monitoring, including hemoglobin A1C (HbA1C) and, in some cases, insulin levels. Among these drugs, clozapine sits apart. It is not considered a first-line agent because of side effects (hematologic) and the need for weekly blood tests. Although highly effective in treating both mania and depression, clozapine does not have an FDA indication for these conditions.

Mechanisms of Action

The presumed antipsychotic effects of the SDAs are blockade of D_2 dopamine receptors. Where the SDAs differ from older antipsychotic drugs is their higher ratio interactions with serotonin receptor subtypes, most notably the 5-HT_{2A} subtype, as well as with other neurotransmitter systems. It is hypothesized that these properties account for the distinct tolerability profiles associated with each of the SDAs. All SDAs have different chemical structures, receptor affinities, and side effect profiles. No SDA is identical in its combination of receptor affinities, and the relative contribution of each receptor interaction to the clinical effects is unknown.

Therapeutic Indications

Although initially approved for the treatment of schizophrenia and acute mania, some of these drugs have also been approved as adjunctive therapy in treatment-resistant depression and as adjunctive therapy in major depressive disorder (MDD). They are also useful in PTSD and anxiety disorders. Although clinicians tend to use them in behavioral disturbances associated with dementia, all SDAs carry an FDA boxed warning regarding adverse effects when used in elderly persons with dementia-related psychoses, because elderly patients with dementia-related psychoses are at an increased

risk (1.6 to 1.7 times) of death compared with placebo. All of these agents are considered first-line drugs for schizophrenia except clozapine, which may cause adverse hematologic effects that require weekly blood sampling.

Schizophrenia and Schizoaffective Disorder. The SDAs are useful for treating acute and chronic psychoses such as schizophrenia and schizoaffective disorder, in both adults and adolescents. SDAs are as good as or better than typical antipsychotics (DRAs) for the treatment of positive symptoms in schizophrenia and superior to DRAs for the treatment of negative symptoms. Compared with persons treated with DRAs, persons treated with SDAs have fewer relapses and require less frequent hospitalization, fewer emergency department visits, less phone contact with mental health professionals, and less treatment in day programs.

Because clozapine has potentially life-threatening adverse effects, it is appropriate only for patients with schizophrenia who are resistant to all other antipsychotics. Other indications for clozapine include treatment of persons with severe tardive dyskinesia—which can be reversed with high dosages in some cases—and those with a low threshold for EPS. Persons who tolerate clozapine have done well on long-term therapy. The effectiveness of clozapine may be increased by augmentation with risperidone, which raises clozapine concentrations and sometimes results in dramatic clinical improvement.

Mood Disorders. Most of the SDAs (clozapine being a notable exception) are FDA approved for the treatment of acute mania. Some of these agents, including aripiprazole, olanzapine, quetiapine, and quetiapine XR, are also approved for the maintenance treatment in bipolar disorder as monotherapy or adjunctive therapy. The SDAs improve depressive symptoms in schizophrenia, and both clinical experience and clinical trials show that all of the SDAs augment antidepressants in the acute management of major depression. At this time, olanzapine, in combination with fluoxetine, has been approved for treatment-resistant depression, and aripiprazole and quetiapine XR are indicated for adjunctive therapy to antidepressants in MDDs. Quetiapine and quetiapine XR are also approved in bipolar depression. A fixed combination of olanzapine and fluoxetine (Symbyax) is approved as a treatment for acute bipolar depression.

Other Indications. About 10 percent of patients with schizophrenia exhibit outwardly aggressive or violent behavior, and the SDAs are effective for the treatment of such aggression. Other off-label indications include acquired immunodeficiency syndrome (AIDS) dementia, autism spectrum disorders, Tourette disorder, Huntington disease, and Lesch–Nyhan syndrome. Risperidone and olanzapine have been used to control aggression and self-injury in children. These drugs are sometimes coadministered with stimulants to children with ADHD who are comorbid for either oppositional–defiant disorder or conduct disorder. SDAs—especially olanzapine, quetiapine, and clozapine—are useful in persons who have severe tardive dyskinesia. The SDAs are also effective for treating psychotic depression and for psychosis secondary to head trauma, dementia, or treatment drugs.

Treatment with SDAs decreases the risk of suicide and water intoxication in patients with schizophrenia. Patients with treatment-resistant OCD have responded to the SDAs; however, a few persons treated with the SDAs have been noted to develop treatment-emergent symptoms of OCD. Some patients with borderline personality disorder may improve with the SDAs.

Some data suggest that treatment with conventional DRAs has protective effects against the progression of schizophrenia when used during the first episode of psychosis. Ongoing studies are looking at whether the use of SDAs in at-risk patients with early evidence of disease prevents deterioration, thus improving long-term outcomes.

Adverse Effects

The SDAs share a similar spectrum of adverse reactions but differ considerably in terms of frequency or severity of their occurrence. Specific side effects that are more common with an individual SDA are emphasized in the discussion of each drug in subsequent text.

Risperidone (Risperdal)

Indications. Risperidone is indicated for the acute and maintenance treatment of schizophrenia in adults and the treatment of schizophrenia in adolescents ages 13 to 17 years. Risperidone is also indicated for the short-term treatment of acute manic or mixed episodes associated with bipolar I disorder in adults and children and adolescents ages 10 to 17 years. The

combination of risperidone with lithium or valproate is indicated for the short-term treatment of acute manic or mixed episodes associated with bipolar I disorder.

Risperidone is also indicated for the treatment of irritability associated with an autistic spectrum disorder in children and adolescents ages 5 to 16 years, including symptoms of aggression toward others, deliberate self-injuriousness, temper tantrums, and quickly changing moods.

Pharmacology. Risperidone is a benzisoxazole. It undergoes extensive first-pass hepatic metabolism to 9-hydroxy risperidone, a metabolite with equivalent antipsychotic activity. Peak plasma levels of the parent compound occur within 1 hour for the parent compound and 3 hours for the metabolite. Risperidone has bioactivity of 70 percent. The combined half-life of risperidone and 9-hydroxy risperidone averages 20 hours, so it is useful in once-daily dosing. Risperidone is an antagonist of the serotonin 5-HT_{2A}, dopamine D_2, α_1-adrenergic and α_2-adrenergic, and histamine H_1 receptors. It has a low affinity for α-adrenergic and muscarinic cholinergic receptors. Risperidone is a potent antagonist of D_2 receptors, as potent as haloperidol, however unlike haloperidol it causes less EPS.

Dosages. The recommended dose range and frequency of risperidone dosing have changed since the drug first came into clinical use. Risperidone is available in 0.25-, 0.5-, 1-, 2-, 3-, and 4-mg tablets and a 1-mg/mL oral solution. The initial dosage is usually 1 to 2 mg at night, which can then be increased to 4 mg/day. Positron emission tomography (PET) studies have shown that dosages of 1 to 4 mg/day provide the required D_2 blockade for a therapeutic effect. At first, it was believed that because of its short elimination half-life, risperidone should be given twice a day, but studies have shown equal efficacy with once-a-day dosing. Dosages above 6 mg a day are associated with a higher incidence of adverse effects, particularly EPS. There is no correlation between plasma concentrations and therapeutic effects. Dosing guidelines for adolescents and children are different from those for adults, requiring lower starting dosages; higher dosages are associated with more adverse effects.

Side Effects. The EPS of risperidone is largely dosage dependent, and there has been a trend to using lower doses than initially recommended. Weight gain, anxiety, nausea and vomiting, rhinitis, erectile dysfunction, orgasmic dysfunction, and increased pigmentation are associated with risperidone use. The most common drug-related reasons for discontinuation of risperidone use are EPS, dizziness, hyperkinesias, somnolence, and nausea. Marked elevation of prolactin may occur. Weight gain occurs more commonly with risperidone use in children than in adults.

Risperidone is also available as an orally disintegrating tablet (Risperdal M-Tab), which is available in 0.5, 1, and 2 mg strengths, and in a depot formulation (Risperdal Consta), which is given as an intramuscular (IM) injection formulation every 2 weeks. The dose may be 25, 50, or 75 mg. Oral risperidone should be coadministered with Risperdal Consta for the first 3 weeks before being discontinued.

Drug Interactions. Inhibition of CYP2D6 by drugs such as paroxetine and fluoxetine can block the formation of risperidone's active metabolite. Risperidone is a weak inhibitor of CYP2D6 and has little effect on other drugs. Combined use of risperidone and SSRIs may result in significant elevation of prolactin, with associated galactorrhea and breast enlargement.

Paliperidone (Invega)

Indications. Paliperidone is indicated for the acute and maintenance treatment of schizophrenia. Paliperidone is also indicated for the acute treatment of schizoaffective disorder as monotherapy or as an adjunct to mood stabilizers or antidepressants.

Pharmacology. Paliperidone is a benzisoxazole derivative and is the primary active metabolite of risperidone. Peak plasma concentrations (C_{max}) are achieved approximately 24 hours after dosing, and steady-state concentrations of paliperidone are attained within 4 or 5 days. The hepatic isoenzymes CYP2D6 and CYP3A4 play a limited role in the metabolism and elimination of paliperidone, so no dose adjustment is required in patients with mild or moderate hepatic impairment.

Dosage. Paliperidone is available in 3-, 6-, and 9-mg tablets. The recommended dosage is 6 mg once daily administered in the morning. It can be taken with or without food. It is also available as extended-release tablets, which are also available in 3-, 6-, and 9-mg tablets administered once daily. It is recommended that no more than 12 mg should be administered per day. A long-acting formulation of paliperidone (Invega Sustenna) is given by injection once a month. Invega Sustenna is available as a white to off-white sterile aqueous extended-release suspension for intramuscular injection in dose strengths of 39-, 78-, 117-, 156-, and 234-mg paliperidone palmitate. The drug product hydrolyzes to the active moiety, paliperidone, resulting in dose strengths of 25, 50, 75, 100, and 150 mg of paliperidone, respectively.

Invega Sustenna is provided in a prefilled syringe with a plunger stopper and tip cap. The kit also contains two safety needles (a 1½-in 22-gauge safety needle and a 1-in 23-gauge safety needle). It has a half-life of 25 to 49 days. Monthly injections of 117 mg are recommended, although higher or lower dosages can be used depending on the clinical situation. The first two injections should be in the deltoid muscle because plasma concentrations are 28 percent higher with deltoid versus gluteal administration. Subsequent injections can alternate between gluteal and deltoid sites.

Side Effects. The dose of paliperidone should be reduced in patients with renal impairment. It may cause more sensitivity to temperature extremes, such as very hot or cold conditions. Paliperidone may cause an increase in QT (QTc) interval and should be avoided in combination with other drugs that cause prolongation of QT interval. It may cause orthostatic hypotension, tachycardia, somnolence, akathisia, dystonia, EPS, and parkinsonism.

Olanzapine (Zyprexa)

Indications. Olanzapine is indicated for the treatment of schizophrenia. Oral olanzapine is indicated for use as monotherapy for the acute treatment of manic or mixed episodes associated with bipolar I disorder and maintenance treatment of bipolar I disorder. Oral olanzapine is also indicated for the treatment of manic or mixed episodes associated with bipolar I disorder as an adjunct to lithium or valproate, and olanzapine can also be used in combination with fluoxetine (Symbyax) for the treatment of depressive

episodes associated with bipolar I disorder.

Oral olanzapine and fluoxetine in combination (Symbyax) is indicated for the treatment of treatment-resistant depression. Olanzapine monotherapy is not indicated for the treatment of treatment-resistant depression.

Pharmacology. Approximately 85 percent of olanzapine is absorbed from the GI tract, and about 40 percent of the dosage is inactivated by first-pass hepatic metabolism. Peak concentrations are reached in 5 hours, and the half-life averages 31 hours (range 21 to 54 hours). It is given in once-daily dosing. In addition to 5-HT_{2A} and D_2 antagonism, olanzapine is an antagonist of the D_1, D_4, α_1, 5-HT_{1A}, muscarinic M_1 to M_5, and H_1 receptors.

Dosages. Olanzapine is available in 2.5-, 5-, 7.5-, 10-, 15-, and 20-mg oral and Zydis form (orally disintegrating) tablets. The initial dosage for the treatment of psychosis is usually 5 or 10 mg. For acute mania, it is usually 10 or 15 mg given once daily. It is also available as 5-, 10-, 15-, and 20-mg orally disintegrating tablets that might be useful for patients who have difficulty swallowing pills or who “cheek” their medication.

A starting daily dose of 5 to 10 mg is recommended. After 1 week, the dosage can be raised to 10 mg a day. Given the long half-life, 1 week must be allowed to achieve each new steady-state blood level. Dosages in clinical use ranges vary, with 5 to 20 mg a day is most commonly used, but 30 to 40 mg a day is needed in treatment-resistant patients. A word of caution, however, is that the higher dosages are associated with increased EPS and other adverse effects, and dosages above 20 mg a day were not studied in the pivotal trials that led to the approval of olanzapine. The parenteral form of olanzapine is indicated for the treatment of acute agitation associated with schizophrenia and bipolar disorder, and the IM dosage is 10 mg. Coadministration with benzodiazepines is not approved.

Other Formulations. Olanzapine is available as an extended-release injectable suspension (Relprevv), which is a long-acting atypical IM injection indicated for the treatment of schizophrenia. It is injected deep in the gluteal region and should not be administered intravenously or subcutaneously, nor is it approved for deltoid administration. Before administering the injection, the administrator should aspirate the syringe for several seconds to ensure

that no blood is visible. It carries a boxed warning for postinjection delirium sedation syndrome (PDSS). Patients are at risk for severe sedation (including coma) and must be observed for 3 hours after each injection in a registered facility. In controlled studies, all patients with PDSS recovered, and there were no deaths reported. It is postulated that PDSS is secondary to increased levels of olanzapine secondary to accidental rupture of a blood vessel, causing extreme sedation or delirium. Patients should be managed as clinically appropriate and, if necessary, monitored in a facility capable of resuscitation. The injection can be given every 2 or 4 weeks, depending on the dosing guidelines.

Drug Interactions. Fluvoxamine (Luvox) and cimetidine (Tagamet) increase, and carbamazepine and phenytoin decrease, serum concentrations of olanzapine. Ethanol increases olanzapine absorption by more than 25 percent, leading to increased sedation. Olanzapine has little effect on the metabolism of other drugs.

Side Effects. Other than clozapine, olanzapine consistently causes a more significant amount and more frequent weight gain than other atypicals. This effect is not dose-related and continues over time. Clinical trial data suggest it peaks after 9 months, after which it may continue to increase more slowly. Somnolence, dry mouth, dizziness, constipation, dyspepsia, increased appetite, akathisia, and tremor are associated with olanzapine use. A small number of patients (2 percent) may need to discontinue the use of the drug because of transaminase elevation. There is a dose-related risk of EPS. The manufacturer recommends a “periodic” assessment of blood sugar and transaminases during treatment with olanzapine. There is an FDA-mandated warning about an increased risk of stroke among patients with dementia treated with SDAs, but this risk is small and is outweighed by improved behavioral control that treatment may produce.

Quetiapine (Seroquel)

Indications. Quetiapine is indicated for the treatment of schizophrenia. It is also indicated for the acute treatment of manic episodes associated with bipolar I disorder, both as monotherapy and as an adjunct to lithium or

sodium valproate. It is also indicated as monotherapy for the acute treatment of depressive episodes associated with bipolar disorder and maintenance treatment of bipolar I disorder as an adjunct to lithium or divalproex. It also is indicated as adjunctive therapy to antidepressants for the treatment of MDD.

Pharmacology. Quetiapine is a dibenzothiazepine structurally related to clozapine, but it differs markedly from that agent in biochemical effects. It is rapidly absorbed from the GI tract, with peak plasma concentrations reached in 1 to 2 hours. The steady-state half-life is about 7 hours, and optimal dosing is two or three times per day. Quetiapine, in addition to being an antagonist of D_2 and 5-HT_2, also blocks 5-HT_6, D_1 and H_1, and α_1 and α_2 receptors. It does not block muscarinic or benzodiazepine receptors. The receptor antagonism for quetiapine is generally lower than that for other antipsychotic drugs, and it is not associated with EPS.

Dosages. Quetiapine is available in 25-, 50-, 100-, 200-, 300-, and 400-mg tablets. Quetiapine dosing should begin at 25 mg twice daily, with doses then increased by 25 to 50 mg/dose every 2 to 3 days, up to a target of 300 to 400 mg a day. Studies have shown efficacy in the range of 300 to 800 mg a day. In reality, more aggressive dosing is both tolerated and more effective. It has become evident that the target dose can be achieved more rapidly and that some patients benefit from dosages of as much as 1,200 to 1,600 mg a day. When used at higher doses, serial ECG studies are required. Despite its short elimination half-life, quetiapine can be given to many patients once a day. This dosing is consistent with the observation that quetiapine receptor occupancy remains even when concentrations in the blood have markedly declined. Quetiapine in doses of 25 to 300 mg at night has been used for insomnia.

Other Formulations. Quetiapine XR has comparable bioavailability to an equivalent dose of quetiapine administered two or three times daily. Quetiapine XR is given once daily, preferably in the evening, 3 to 4 hours before bedtime without food or a light meal to prevent an increase in C_{max}. The usual starting dose is 300 mg, and it may be increased to 400 to 800 mg.

Drug Interactions. The potential interactions between quetiapine and other

drugs have been well studied. Phenytoin increases quetiapine clearance fivefold; no significant pharmacokinetic interactions have been noted. Avoid the use of quetiapine with drugs that increase the QT interval and in patients with risk factors for prolonged QT interval. The FDA has added a new warning about quetiapine, cautioning prescribers about potential prolongation of the QT interval when above-recommended amounts of quetiapine are combined with specific drugs. Quetiapine should not be combined with other drugs that are known to prolong QTc. This includes class 1A antiarrhythmics (e.g., quinidine, procainamide) or class III antiarrhythmics (e.g., amiodarone, sotalol), antipsychotic medications (e.g., ziprasidone, chlorpromazine, thioridazine), antibiotics (e.g., gatifloxacin, moxifloxacin), or any other class of medications known to prolong the QTc interval (e.g., pentamidine, levomethadyl acetate, methadone). Quetiapine should also be avoided in circumstances that may increase the risk of occurrence of torsade de pointes or sudden death, including (1) a history of cardiac arrhythmias such as bradycardia; (2) hypokalemia or hypomagnesemia; (3) concomitant use of other drugs that prolong the QTc interval; and (4) presence of congenital prolongation of the QT interval. Postmarketing cases also show increases in QT interval in patients who overdose on quetiapine.

Side Effects. Somnolence, postural hypotension, and dizziness are the most common adverse effects of quetiapine. These are usually transient and are best managed with initial gradual upward titration of the dosage. Quetiapine is the SDA least likely to cause EPS, regardless of dose. This makes it particularly useful in treating patients with Parkinson disease who develop dopamine agonist-induced psychosis. Prolactin elevation is rare and both transient and mild when it occurs. Quetiapine is associated with modest transient weight gain in some persons, but some patients occasionally gain a considerable amount of weight. The relationship between quetiapine and the development of diabetes is not as clearly established as are the cases involving the use of olanzapine. Small increases in heart rate, constipation, and a transient increase in liver transaminases may also occur. Initial concerns about cataract formation, based on animal studies, have not been borne out since the drug has been in clinical use. Nevertheless, it might be prudent to test for lens abnormalities early in treatment and periodically after that.

Ziprasidone (Geodon)

Indications. Ziprasidone is indicated for the treatment of schizophrenia. Ziprasidone is also indicated as monotherapy for the acute treatment of manic or mixed episodes associated with bipolar I disorder and as an adjunct to lithium or valproate for the maintenance treatment of bipolar I disorder.

Pharmacology. Ziprasidone is a benzisothiazole piperazine. Peak plasma concentrations of ziprasidone are reached in 2 to 6 hours. Steady-state levels ranging from 5 to 10 hours are reached between the first and the third days of treatment. The mean terminal half-life at steady state ranges from 5 to 10 hours, which accounts for the recommendation that twice-daily dosing is necessary. Bioavailability doubles when ziprasidone is taken with food, and therefore it should be taken with food.

Peak serum concentrations of IM ziprasidone occur after approximately 1 hour, with a half-life of 2 to 5 hours.

Ziprasidone, similar to the other SDAs, blocks 5-HT_{2A} and D_2 receptors. It is also an antagonist of 5-HT_{1D}, 5-HT_{2C}, D_3, D_4, α_1, and H_1 receptors. It has a very low affinity for D_1, M_1, and α_2 receptors. Also, ziprasidone has agonist activity at the serotonin 5-HT_{1A} receptors and is an SSRI and a norepinephrine reuptake inhibitor. This action is consistent with clinical reports that ziprasidone has antidepressant-like effects in nonschizophrenic patients.

Dosages. Ziprasidone is available in 20-, 40-, 60-, and 80-mg capsules. Ziprasidone for IM use comes as a single-use 20 mg/mL vial. Oral ziprasidone dosing should be initiated at 40 mg a day, divided into two daily doses. Studies have shown efficacy in the range of 80 to 160 mg a day, divided twice daily. In clinical practice, doses as high as 240 mg a day are being used. The recommended IM dosage is 10 to 20 mg every 2 hours for the 10-mg dose and every 4 hours for the 20-mg dose. The maximum total daily dose of IM ziprasidone is 40 mg.

Other than interactions with other drugs that prolong the QTc complex, ziprasidone appears to have a low potential for clinically significant drug interactions.

Side Effects. Somnolence, headache, dizziness, nausea, and lightheadedness are the most common adverse effects in patients taking ziprasidone. The drug has almost no significant effects outside the CNS, is associated with almost no weight gain, and does not cause sustained prolactin elevation. Concerns about prolongation of the QTc complex have deterred some clinicians from using ziprasidone as a first choice. The QTc interval has been shown to increase in patients treated with 40 and 120 mg/day. Ziprasidone is contraindicated in combination with other drugs known to prolong the QTc interval. These include, but are not limited to, dofetilide, sotalol, quinidine, other class IA and III antiarrhythmics, mesoridazine, thioridazine, chlorpromazine, droperidol, pimozide, sparfloxacin, gatifloxacin, moxifloxacin, halofantrine, mefloquine, pentamidine, arsenic trioxide, levomethadyl acetate, dolasetron mesylate, probucol, and tacrolimus. Ziprasidone should be avoided in patients with congenital long QT syndrome and patients with a history of cardiac arrhythmias.

Aripiprazole (Abilify)

Aripiprazole is a potent 5-HT_{2A} antagonist and is indicated for the treatment of both schizophrenia and acute mania. It is also approved for the augmentation of antidepressant agents in MDD. Aripiprazole is a D_2 antagonist, but can also act as a partial D_2 agonist. Partial D_2 agonists compete at D_2 receptors for endogenous dopamine, thereby producing a functional reduction of dopamine activity.

Indications. Aripiprazole is indicated for the treatment of schizophrenia. Short-term, 4- to 6-week studies comparing aripiprazole with haloperidol and risperidone in patients with schizophrenia and schizoaffective disorder have shown comparable efficacy. Dosages of 15, 20, and 30 mg a day were found to be effective. Long-term studies suggest that aripiprazole is effective as a maintenance treatment at a daily dose of 15 to 30 mg.

Aripiprazole is also indicated for the acute and maintenance treatment of manic and mixed episodes associated with bipolar I disorder. It is also used as adjunctive therapy to either lithium or valproate for the acute treatment of manic and mixed episodes associated with bipolar I disorder.

Aripiprazole is indicated for use as adjunctive therapy to antidepressants

for the treatment of MDD. Aripiprazole is also indicated for the treatment of irritability associated with autistic disorder.

Pharmacology. Aripiprazole is well absorbed, reaching peak plasma concentrations after 3 to 5 hours. Absorption is not affected by food. The mean elimination half-life of aripiprazole is about 75 hours. It has a weakly active metabolite with a half-life of 96 hours. These relatively long half-lives make aripiprazole suitable for once-daily dosing. Clearance is reduced in elderly persons. Aripiprazole exhibits linear pharmacokinetics and is primarily metabolized by CYP3A4 and CYP2D6 enzymes. It is 99 percent protein-bound. Aripiprazole is excreted in breast milk in lactating rats.

Mechanistically, aripiprazole acts as a modulator, rather than a blocker, and acts on both postsynaptic D_2 receptors and presynaptic autoreceptors. In theory, this mechanism addresses excessive limbic dopamine (hyperdopaminergic) activity and decreased dopamine (hypodopaminergic) activity in frontal and prefrontal areas—abnormalities that are thought to be present in schizophrenia. The absence of a complete D_2 blockade in the striatal areas would be expected to minimize EPS. Aripiprazole is an α_1-adrenergic receptor antagonist, which may cause some patients to experience orthostatic hypotension. Similar to the so-called atypical antipsychotic agents, aripiprazole is a 5-HT_{2A} antagonist.

Other Uses. A study of aggressive children and adolescents with oppositional defiant disorder or conduct disorder found that there was a positive response in about 60 percent of the subjects. In this study, vomiting and somnolence led to a reduction in initial aripiprazole dosage.

Drug Interactions. Whereas carbamazepine and valproate reduce serum concentrations, ketoconazole, fluoxetine, paroxetine, and quinidine increase aripiprazole serum concentrations. Lithium and valproic acid, two drugs likely to be combined with aripiprazole when treating bipolar disorder, do not affect the steady-state concentrations of aripiprazole. Combined use with antihypertensives may cause hypotension. Drugs that inhibit CYP2D6 activity reduce aripiprazole elimination.

Dosage and Clinical Guidelines. Aripiprazole is available as 5-, 10-, 15-

, 20-, and 30-mg tablets. The effective dosage range is 10 to 30 mg/day. Although the starting dosage is 10 to 15 mg/day, problems with nausea, insomnia, and akathisia have led to the use of lower than recommended starting dosages of aripiprazole. Many clinicians find that an initial dose of 5 mg increases tolerability.

Side Effects. The most commonly reported side effects of aripiprazole are headache, somnolence, agitation, dyspepsia, anxiety, and nausea. Although it is not a frequent cause of EPS, aripiprazole does cause akathisia-like activation. Described as restlessness or agitation, it can be highly distressing and often leads to discontinuation of the medication. Insomnia is another common complaint. Data so far do not indicate that weight gain or diabetes mellitus have an increased incidence with aripiprazole. Prolactin elevation does not typically occur. Aripiprazole does not cause significant QTc interval changes. There have been reports of seizures.

Asenapine (Saphris)

Indications. Asenapine is approved for the acute treatment of adults with schizophrenia and acute treatment of manic or mixed episodes associated with bipolar I disorder with or without psychotic features in adults.

Pharmacology. Asenapine has an affinity for several receptors, including serotonin (5-HT_{2A} and 5-HT_{2C}), noradrenergic (α_2, and α_1), dopaminergic (D_3 and D_4 receptors is higher than its affinity for D_2 receptors), and histamine (H_1). It has a negligible affinity for muscarinic-1 cholinergic receptors and hence less incidence of dry mouth, blurred vision, constipation, and urinary retention. The bioavailability is 35 percent via the sublingual (preferred) route, and it achieves peak plasma concentration in 1 hour. Asenapine is metabolized through glucuronidation and oxidative metabolism by CYP1A2, so coadministration with fluvoxamine and other CYP1A2 inhibitors should be done cautiously.

Dosage. Asenapine is available as 5- and 10-mg sublingual tablets and should be placed under the tongue. This method is because the bioavailability of asenapine is less than 2 percent when swallowed, but is 35 percent when

absorbed sublingually. The agent dissolves in saliva within seconds and is absorbed through the oral mucosa. Sublingual administration avoids first-pass hepatic metabolism. Patients should be advised to avoid drinking or eating for 10 minutes after taking asenapine because this may lower the blood levels. The recommended starting and target dose for schizophrenia is 5 mg twice a day. In bipolar disorder, the patient may be started on 10 mg twice a day, and if necessary, the dosage may be lowered to 5 mg twice a day, depending on the tolerability issues. In acute schizophrenia treatment, there is no evidence of added benefit with a 10 mg twice-daily dose, but there is an apparent increase in specific adverse reactions. In both bipolar I disorder and schizophrenia, the maximum dose should not exceed 10 mg two times a day. The safety of doses above 10 mg twice a day has not been evaluated in clinical studies.

Side Effects. The most common side effects observed in schizophrenic and bipolar disorders are somnolence, dizziness, EPS other than akathisia, and increased weight. In clinical trials, the mean weight gain after 52 weeks is 0.9 kg, and there were no clinically relevant differences in lipid profile and blood glucose after 52 weeks. In clinical trials, asenapine was found to increase the QTc interval in a range of 2 to 5 milliseconds compared to placebo. No patients treated with asenapine experienced QTc increases 60 milliseconds or greater from baseline measurements, nor did any experience a QTc of 500 milliseconds or more. Nevertheless, asenapine should be avoided in combination with other drugs known to prolong QTc interval in patients with congenital prolongation of QT interval or a history of cardiac arrhythmias, and in circumstances that may increase the occurrence of torsades de pointes. Asenapine can elevate prolactin levels, and the elevation can persist during chronic administration. Galactorrhea, amenorrhea, gynecomastia, and impotence may occur.

Clozapine (Clozaril)

Indications. In addition to being the most effective drug treatment for patients who have failed to respond to standard therapies, clozapine has been shown to benefit patients with severe tardive dyskinesia. Clozapine suppresses these dyskinesias, but the abnormal movements return when

clozapine is discontinued. This suppression is true even though clozapine, on rare occasions, may cause tardive dyskinesia. Other clinical situations in which clozapine may be used include the treatment of psychotic patients intolerant of EPS caused by other agents, treatment-resistant mania, severe psychotic depression, idiopathic Parkinson disease, Huntington disease, and suicidal patients with schizophrenia or schizoaffective disorder. Other treatment-resistant disorders that have demonstrated response to clozapine include pervasive developmental disorder, autism of childhood, and OCD (either alone or in combination with an SSRI). Used by itself, clozapine may very rarely induce obsessive-compulsive symptoms.

Pharmacology. Clozapine is a dibenzothiazepine. It is rapidly absorbed, with peak plasma levels reached in about 2 hours. Steady state is achieved in less than 1 week if twice-daily dosing is used. The elimination half-life is about 12 hours. Clozapine has two significant metabolites, one of which, *N*-dimethyl clozapine, may have some pharmacologic activities. Clozapine is an antagonist of 5-HT_{2A}, D_1, D_3, D_4, and α (especially α_1) receptors. It has relatively low potency as a D_2 receptor antagonist. Data from PET scanning show that whereas 10 mg of haloperidol produces 80 percent occupancy of striatal D_2 receptors, clinically significant dosages of clozapine occupy only 40 to 50 percent of striatal D_2 receptors. This difference in D_2 receptor occupancy is probably why clozapine does not cause EPS. It has also been postulated that clozapine and other SDAs bind more loosely to the D_2 receptor, and because of this "fast dissociation," more normal dopamine neurotransmission is possible.

Dosages. Clozapine is available in 25- and 100-mg tablets. The initial dosage is usually 25 mg one or two times daily, although a conservative initial dosage is 12.5 mg twice daily. The dosage can then be increased gradually (25 mg a day every 2 or 3 days) to 300 mg a day in divided doses, usually two or three times a day. Dosages up to 900 mg a day can be used. Testing for blood concentrations of clozapine may be helpful in patients who fail to respond. Studies have found that plasma concentrations greater than 350 μg/mL are associated with a better likelihood of response.

Drug Interactions. Clozapine should not be used with any other drug that is associated with the development of agranulocytosis or bone marrow suppression. Such drugs include carbamazepine, phenytoin, propylthiouracil, sulfonamides, and captopril (Capoten). Lithium combined with clozapine may increase the risk of seizures, confusion, and movement disorders. Lithium should not be used in combination with clozapine by persons who have experienced an episode of neuroleptic malignant syndrome. Clomipramine (Anafranil) can increase the risk of seizure by lowering the seizure threshold and by increasing clozapine plasma concentrations. Risperidone, fluoxetine, paroxetine, and fluvoxamine increase serum concentrations of clozapine. The addition of paroxetine may precipitate clozapine-associated neutropenia.

Side Effects. The most common drug-related adverse effects are sedation, dizziness, syncope, tachycardia, hypotension, electrocardiography (ECG) changes, nausea, and vomiting. Other common adverse effects include fatigue, weight gain, various GI symptoms (most commonly constipation), anticholinergic effects, and subjective muscle weakness. Sialorrhea, or hypersalivation, is a side effect that begins early in treatment and is most evident at night. Patients report that their pillows are drenched with saliva. This side effect is most likely the result of impairment of swallowing. Although there are reports that clonidine or amitriptyline may help reduce hypersalivation, the most practical solution is to put a towel over the pillow.

The risk of seizures is about 4 percent in patients taking dosages greater than 600 mg a day. Leukopenia, granulocytopenia, agranulocytosis, and fever occur in about 1 percent of patients. During the first year of treatment, there is a 0.73 percent risk of clozapine-
induced agranulocytosis. The risk during the second year is 0.07 percent. For neutropenia, the risk is 2.32 percent and 0.69 percent, respectively, during the first and second years of treatment. The only contraindications to the use of clozapine are a WBC count below 3,500 cells/mm^3; a previous bone marrow disorder; a history of agranulocytosis during clozapine treatment; or the use of another drug that is known to suppress the bone marrow, such as carbamazepine (Tegretol).

During the first 6 months of treatment, weekly WBC counts are indicated to monitor the patient for the development of agranulocytosis. If the WBC

count remains normal, the frequency of testing can be decreased to every 2 weeks. Although monitoring is expensive, an early indication of agranulocytosis can prevent a fatal outcome. Clozapine should be discontinued if the WBC count is below 3,000 cells/mm^3 or the granulocyte count is below 1,500/mm^3. Also, a hematologic consultation should be obtained, and obtaining a bone marrow sample should be considered. Persons with agranulocytosis should not be reexposed to the drug. Clozapine cannot be dispensed without proof of monitoring.

Patients exhibiting symptoms of chest pain, shortness of breath, fever, or tachypnea should be immediately evaluated for myocarditis or cardiomyopathy, an infrequent but serious adverse effect ending in death. Serial CPK-MB (creatine phosphokinase with myocardial band fractions), troponin levels, and ECG studies are recommended, with immediate discontinuation of clozapine.

Iloperidone (Fanapt)

Indications. Iloperidone (Fanapt) is indicated for the acute treatment of schizophrenia in adults. The safety and efficacy of iloperidone in children and adolescents has not been established.

Pharmacology. Iloperidone is not a derivative of another antipsychotic agent. It has complex multiple antagonist effects on several neurotransmitter systems. Iloperidone has a strong affinity for dopamine D_3 receptors, followed by decreasing affinities of α_{2c}-noradrenergic, 5-HT_{1a}, D_{2a}, and 5-HT_6 receptors. Iloperidone has a low affinity for histaminergic receptors. As with other antipsychotics, the clinical significance of this receptor binding affinity is unknown.

Iloperidone has a peak concentration of 2 to 4 hours and a half-life that is dependent on hepatic isoenzyme metabolism. It is metabolized primarily through CYP2D6 and CYP3A4, and the dosage should be reduced by half when administered concomitantly with potent inhibitors of these two isoenzymes. The half-life is 18 to 26 hours in CYP2D6 extensive metabolizers and is 31 to 37 hours in CYP2D6 poor metabolizers. Of note, approximately 7 to 10 percent of whites and 3 to 8 percent of African Americans cannot metabolize CYP2D6 substrates; hence, dosing should be

determined with this caveat in mind. Iloperidone should be used with caution in persons with severe hepatic impairment.

Side Effects. Iloperidone prolongs the QT interval and may be associated with arrhythmia and sudden death. Iloperidone prolongs the QTc interval by 9 milliseconds at dosages of 12 mg twice daily. Concurrent use with other agents that prolong the QTc interval may result in additive effects on the QTc interval. The concurrent use of iloperidone with agents that prolong the QTc interval may result in potentially life-threatening cardiac arrhythmias, including torsades de pointes. Concurrent administration of other drugs that are known to prolong the QTc interval should be avoided. Cardiovascular disease, hypokalemia, hypomagnesemia, bradycardia, congenital prolongation of the QT interval, and concurrent use of inhibitors of CYP3A4 or CYP2D6, which metabolize iloperidone, may increase the risk of QT prolongation.

The most common adverse effects reported are dizziness, dry mouth, fatigue, sedation, tachycardia, and orthostatic hypotension (depending on dosing and titration). Despite being a robust D_2 antagonist, the rates of EPS and akathisia are similar to those of placebo. The mean weight gain in short- and long-term trials is 2.1 kg. Due to its relatively limited use, there is no accurate understanding of iloperidone's effects on weight and lipids. Some patients exhibit elevated prolactin levels. Three cases of priapism have been reported in the premarketing phase.

Dosing. Iloperidone must be titrated slowly to avoid orthostatic hypotension. It is available in a titration pack, and the effective dose (12 mg) should be reached in approximately 4 days based on a twice-a-day dosing schedule. It is usually started on day 1 at 1 mg twice a day and increased daily on a twice-a-day schedule to reach 12 mg by day 4. The maximum recommended dose is 12 mg twice a day (24 mg a day), and it can be administered without regard to food.

Lurasidone HCL (Latuda)

Indications. Lurasidone hydrochloride is an oral, once-daily atypical antipsychotic indicated for the treatment of patients with schizophrenia. To

date, there has not been extensive clinical experience with lurasidone.

Side Effects. The most commonly observed adverse reactions associated with the use of lurasidone are similar to those seen with other new-generation antipsychotics. These include, but are not limited to, somnolence, akathisia, nausea, parkinsonism, and agitation. Based on clinical trial data, lurasidone appears to cause less weight gain and metabolic changes than the two other most recently approved SDAs, asenapine and iloperidone. More extensive clinical experience with the drug is required to determine whether this is, in fact, the case.

Drug Interactions. When coadministration of lurasidone with a moderate CYP3A4 inhibitor such as diltiazem is considered, the dose should not exceed 40 mg/day. Lurasidone should not be used in combination with a potent CYP3A4 inhibitor (e.g., ketoconazole). Lurasidone also should not be used in combination with a potent CYP3A4 inducer (e.g., rifampin).

Dosages. Lurasidone is available as 20-, 40-, 80-, and 120-mg tablets. Initial dose titration is not required. The recommended starting dose is 40 mg once daily, and the medication should be taken with food. It is effective in a dose range of 40 to 120 mg/day. Although there is no proven added benefit with the 120 mg/day dose, there may be a dose-related increase in adverse reactions. Still, some patients may benefit from the maximum recommended dose of 160 mg/day. Dose adjustment is recommended in patients with renal impairment. The dose in moderate to severe renal impairment should not exceed 80 mg/day. The dose in severe hepatic impairment patients should not exceed 40 mg/day.

Lumateperone (Caplyta)

Lumateperone was approved by the FDA for the treatment of schizophrenia in adults in December 2019 and became available in February 2020. It antagonizes several receptors, including 5-HT_{2A} receptor, D_1, D_2, and D_4. Although effective for schizophrenia, several studies of its effectiveness in bipolar depression had disappointing results. It is available in several doses, although the highest FDA-approved dose is 42 mg. As this is a new drug, experience with it in post study clinical samples is very limited.

Clinical Guidelines for SDAs

All SDAs are appropriate for the management of an initial psychotic episode, but clozapine is reserved for persons who are refractory to all other antipsychotic drugs. If a person does not respond to the first SDA, other SDAs should be tried. The choice of drug should be based on the patient's clinical status and history of response to medication. Recent studies have challenged the notion that SDAs require 4 to 6 weeks to reach full effectiveness, and it may take up to 8 weeks for the full clinical effects of an SDA to become apparent. The newer meta-analyses suggest that the apparent benefits may be seen as early as 2 to 3 weeks, and early response or failure is an indicator of subsequent response or failure.

Nevertheless, it is acceptable practice to augment an SDA with a high-potency DRA or benzodiazepine in the first few weeks of use. Lorazepam (Ativan) 1 to 2 mg orally or IM can be used as needed for acute agitation. Once effective, dosages can be lowered as tolerated. Clinical improvement may take 6 months of treatment with SDAs in some, particularly treatment-refractory persons.

The use of all SDAs must be initiated at low dosages and gradually tapered upward to therapeutic dosages. The potential development of adverse effects necessitates the gradual increase in dosage. If a person stops taking an SDA for more than 36 hours, drug use should be resumed at the initial titration schedule. After the decision to terminate olanzapine or clozapine use, dosages should be tapered whenever possible to avoid cholinergic rebound symptoms such as diaphoresis, flushing, diarrhea, and hyperactivity.

After assuring themselves of the appropriateness of SDA treatment, the clinician should explain the risks and benefits of SDA treatment to the person and the family. In the case of clozapine, an informed consent procedure should be documented in the person's chart. The patient's history should include information about blood disorders, epilepsy, cardiovascular disease, hepatic and renal diseases, and drug abuse. The presence of a hepatic or renal disease necessitates using low starting dosages of the drug. The physical examination should include supine and standing BP measurements to screen for orthostatic hypotension. The laboratory examination should include an ECG and several complete blood counts (CBCs) with WBC counts, which can then be averaged; and liver and renal function tests. Periodic monitoring

of blood glucose, lipids, and body weight is recommended.

Although the transition from a DRA to an SDA may be made abruptly, it is wiser to taper off the DRA slowly while titrating up the SDA. Clozapine and olanzapine both have anticholinergic effects, and the transition from one to the other can usually be accomplished with little risk of cholinergic rebound. The transition from risperidone to olanzapine is best accomplished by tapering the risperidone off over 3 weeks while simultaneously beginning olanzapine at 10 mg a day. Risperidone, quetiapine, and ziprasidone lack anticholinergic effects, and the abrupt transition from a DRA, olanzapine, or clozapine to one of these agents may cause cholinergic rebound, which consists of excessive salivation, nausea, vomiting, and diarrhea. The risk of cholinergic rebound can be mitigated by initially augmenting risperidone, quetiapine, or ziprasidone with an anticholinergic drug, which is then tapered off slowly. Any initiation and termination of SDA use should be accomplished gradually.

It is wise to overlap the administration of the new drug with the old drug. Of interest, some people have a more robust clinical response while taking the two agents during the transition and then regressing on monotherapy with the newer drug. We know little about the safety of combination antipsychotic treatment.

Persons receiving regular injections of depot formulations of a DRA who are to switch to SDA use are given the first dose of the SDA on the day the next injection is due.

Persons who developed agranulocytosis while taking clozapine can safely switch to olanzapine use, although initiation of olanzapine use amid clozapine-induced agranulocytosis can prolong the time of recovery from the usual 3 to 4 days up to 11 to 12 days. It is prudent to wait for the resolution of agranulocytosis before initiating olanzapine use. Emergence or recurrence of agranulocytosis has not been reported with olanzapine, even in persons who developed it while taking clozapine.

SDA use by pregnant women has not been studied, but consideration should be given to the potential of risperidone to raise prolactin concentrations, sometimes up to three to four times the upper limit of the normal range. Because the drugs can be excreted in breast milk, they should not be taken by nursing mothers. The dosages for selected SDAs are given in Table 21-9.

DOPAMINE RECEPTOR ANTAGONISTS (FIRST-GENERATION ANTIPSYCHOTICS)

The DRAs represent the first group of effective agents for schizophrenia and other psychotic illnesses. The first of these drugs, the phenothiazine chlorpromazine (Thorazine), was introduced in the early 1950s. Other DRAs include all of the antipsychotics in the following groups: phenothiazines, butyrophenones, thioxanthenes, dibenzoxazepines, dihydroindoles, and diphenylbutylpiperidines. Because these agents are associated with extrapyramidal side effects (EPS) at clinically effective dosages, newer antipsychotic drugs—the SDAs—have gradually replaced the older agents in the United States. The SDAs are differentiated from earlier drugs by their lower liability to cause extrapyramidal side effects. These newer drugs have other liabilities, most notably a propensity to cause weight gain, lipid elevations, and diabetes. Therefore, a reason to still consider the use of the DRAs is their lower risk of causing significant metabolic abnormalities. Intermediate-potency DRAs, such as perphenazine (Trilafon), are as effective and well tolerated as the SDAs. Manufacturing of molindone (Moban), the DRA with the lowest risk of weight gain and metabolic side effects, was discontinued in the United States.

Table 21-9
Comparison of Usual Dosing[a] for Some Available Second-Generation Antipsychotics in Schizophrenia

Antipsychotic	Typical Starting Dosage	Maintenance Therapy Dose Range	Titration	Maximum Recommended Dosage
Aripiprazole (Abilify)	10–15 mg tablets once a day	10–30 mg/day	Dosage increases should not be made before 2 wk	30 mg/day
Asenapine (Saphris)	5 mg twice a day	10 mg twice a day	Titration not necessary	20 mg/day
Clozapine (Clozaril)	12.5-mg tablets once or twice a day	150–300 mg/day in divided doses or 200 mg as a single dose in the evening	The dosage should be increased to 25–50 mg on the second day. Further increases may be made in daily increments of 25–50 mg to a target dosage of 300–450 mg/day. Subsequent dosage increases should be made no more than once or twice weekly in increments of no more than 100 mg	900 mg/day
Iloperidone (Fanapt)	1 mg twice a day	12–24 mg a day in divided dose	Start at 1 mg twice a day then move to 2, 4, 6, 8 and 12 mg twice a day. Do this over the course of 7 days	24 mg/day
Lurasidone (Latuda)	40 mg/day	40–80 mg/day	Titration not necessary	120 mg/day
Olanzapine (Zyprexa)	5–10 mg/day tablets or orally disintegrating tablets	10–20 mg/day	Dosage increments of 5 mg once a day are recommended when required at intervals of not less than 1 wk	20 mg/day
Paliperidone (Invega)	3–9 mg extended-release tablets once a day	3–6 mg/day	Plasma concentration rises to a peak approximately 24 hr after dosing	12 mg/day
Quetiapine (Seroquel)	25-mg tablets twice a day	Lowest dose needed to maintain remission	Increase in increments of 25–50 mg two or three times a day on the second and the third day, as tolerated, to a target dosage of 500 mg daily by the fourth day (given in two or three doses/day). Further dosage adjustments, if required, should be of 25–50 mg twice a day and occur at intervals of not fewer than 2 days	800 mg/day
Risperidone (Risperdal)	1-mg tablet and oral solution once a day	2–6 mg once a day	Increase to 2 mg once a day on the second day and 4 mg once a day on the third day. In some patients, a slower titration may be appropriate. When dosage adjustments are necessary, further dosage increments of 1–2 mg/day at intervals of not less than 1 wk are recommended	1–6 mg/day
Risperidone IM long-acting (Consta)	25–50 mg IM injection every 2 wk	Start with oral risperidone for 3 wk	Starting dose: 25 mg every 2 wk	50 mg for 2 wk
Ziprasidone (Geodon)	20-mg capsules twice a day with food	20–80 mg twice a day	Dosage adjustments based on individual clinical status may be made at intervals of not fewer than 2 days	80 mg twice a day
Ziprasidone (IM)	For acute agitation: 10–20 mg, as required, up to a maximum of 40 mg/day	Not applicable	For acute agitation: Doses of 10 mg may be administered every 2 hr, and doses of 20 mg may be administered every 4 hr up to a maximum of 40 mg/day	For acute agitation: 40 mg/day, for not more than 3 consecutive days

Note: Information taken from U.S. Prescribing Information for individual agents.
[a]Dosage adjustments may be required in special populations.
IM, intramuscular.

Pharmacologic Actions

All of the DRAs are well absorbed after oral administration, with liquid preparations being absorbed more efficiently than tablets or capsules. Peak plasma concentrations are usually reached 1 to 4 hours after oral administration and 30 to 60 minutes after parenteral administration. Smoking, coffee, antacids, and food interfere with the absorption of these drugs. Steady-state levels are reached in approximately 3 to 5 days. The half-lives of these drugs are approximately 24 hours. All can be given in one daily oral

dose if tolerated after the patient is in a stable condition. Most DRAs are highly protein-bound. Parenteral formulation of the DRAs results in a more rapid and more reliable onset of action. Bioavailability is also up to 10-fold higher with parenteral administration. Most DRAs are metabolized by cytochrome P450 (CYP) CYP2D6 and 3A isozymes. However, there are differences among the specific agents.



Table 21-10
Factors Influencing the Pharmacokinetics of Antipsychotics

Age	Elderly patients may demonstrate reduced clearance rates.
Medical condition	Decreased hepatic blood flow can reduce clearance.
	Hepatic disease can decrease clearance.
Enzyme inducers	Carbamazepine, phenytoin, ethambutol, barbiturates.
Clearance inhibitors	Include SSRIs, TCAs, cimetidine, β-blockers, isoniazid, methylphenidate, erythromycin, triazolobenzodiazepines, ciprofloxacin, and ketoconazole.
Changes in binding protein	Hypoalbuminemia can occur with malnutrition or hepatic failure.

SSRI, selective serotonin reuptake inhibitor; TCA, tricyclic antidepressant.

Adapted from Ereshefsky L. Pharmacokinetics and drug interactions: update for new antipsychotics. *J Clin Psychiatry*. 1996;57(Suppl 11):12–25.

Long-acting depot parenteral formulations of haloperidol (Haldol Decanoate) and fluphenazine are available in the United States. These agents are usually administered once every 1 to 4 weeks, depending on the dose and the patient. It can take up to 6 months of treatment with depot formulations to reach steady-state plasma levels, indicating that oral therapy should be

continued during the first month or so of depot antipsychotic treatment.

Antipsychotic activity derives from the inhibition of dopaminergic neurotransmission. The DRAs are effective when approximately 72 percent of dopamine D_2 receptors in the brain are occupied. The DRAs also block noradrenergic, cholinergic, and histaminergic receptors, with different drugs having different effects on these receptor systems.

Some generalizations can be made about the DRAs based on their potency. Potency refers to the amount of drug that is required to achieve therapeutic effects. Low-potency drugs such as chlorpromazine and thioridazine (Mellaril), given in doses of several 100 mg/day, typically produce more weight gain and sedation than high-potency agents such as haloperidol and fluphenazine, usually given in doses of less than 10 mg/day. High-potency agents are also more likely to cause EPS. Some factors influencing the pharmacologic actions of DRAs are listed in Table 21-10.

Therapeutic Indications

Many types of psychiatric and neurologic disorders may benefit from treatment with DRAs. Some of these indications are shown in Table 21-11.

Schizophrenia and Schizoaffective Disorder. The DRAs are useful in both the short- and long-term management of schizophrenia and schizoaffective disorder. They both reduce acute symptoms and prevent future exacerbations. These agents produce their most dramatic effects on the positive symptoms of schizophrenia (e.g., hallucinations, delusions, and agitation). Negative symptoms (e.g., emotional withdrawal and ambivalence) are less likely to improve significantly, and they may appear to worsen because these drugs produce constriction of facial expression and akinesia, side effects that mimic negative symptoms.



Table 21-11
Indications for Dopamine Receptor Antagonists

Acute psychotic episodes in schizophrenia and schizoaffective disorder
Maintenance treatment in schizophrenia and schizoaffective

disorders
Mania
Depression with psychotic symptoms
Delusional disorder
Borderline personality disorder
Substance-induced psychotic disorder
Delirium and dementia
Mental disorders caused by a medical condition
Childhood schizophrenia
Autism spectrum disorder
Tourette disorder
Huntington disease

Schizophrenia and schizoaffective disorder are characterized by remission and relapse. DRAs decrease the risk of reemergence of psychosis in patients who have recovered while on medication. After the first episode of psychosis, patients should be maintained on medication for 1 to 2 years; after multiple episodes, for 2 to 5 years.

Mania. DRAs are useful for treating psychotic symptoms of acute mania. Because antimanic agents (e.g., lithium) generally have a slower onset of action than do antipsychotics in the treatment of acute symptoms, it is standard practice to initially combine either a DRA or an SDA with lithium (Eskalith), sodium valproate (Depakote), lamotrigine (Lamictal), or carbamazepine (Tegretol) and then to withdraw the antipsychotic gradually.

Depression with Psychotic Symptoms. Combination treatment with an antipsychotic and an antidepressant is one of the treatments of choice for MDD with psychotic features; the other is ECT.

Delusional Disorder. Patients with delusional disorder often respond favorably to treatment with these drugs. Some persons with borderline personality disorder who develop paranoid thinking may respond to antipsychotic drugs.

Severe Agitation and Violent Behavior. Severely agitated and violent patients, regardless of diagnosis, may be treated with DRAs. Symptoms such as extreme irritability, lack of impulse control, severe hostility, gross hyperactivity, and agitation respond to short-term treatment with these drugs. Children with mental disabilities, especially those with profound intellectual disabilities and autistic disorder, often have associated episodes of violence, aggression, and agitation that respond to treatment with antipsychotic drugs; however, the repeated administration of antipsychotics to control disruptive behavior in children is controversial.

Tourette Disorder. DRAs are used to treat Tourette's disorder, a neurobehavioral disorder marked by motor and vocal tics. Haloperidol and pimozide (Orap) are the drugs most frequently used, but other DRAs are also useful. Some clinicians prefer to use clonidine (Catapres) for this disorder because of its lower risk of neurologic side effects.

Borderline Personality Disorder. Patients with borderline personality disorder who experience transient psychotic symptoms, such as perceptual disturbances, suspiciousness, ideas of reference, and aggression, may need to be treated with a DRA. This disorder is also associated with mood instability, so patients should be evaluated for possible treatment with mood-stabilizing agents.

Dementia and Delirium. About two-thirds of agitated, elderly patients with various forms of dementia improve when given a DRA. Low doses of high-potency drugs (e.g., 0.5 to 1 mg a day of haloperidol) are recommended. DRAs are also used to treat psychotic symptoms and agitation associated with delirium. The cause of the delirium needs to be determined because toxic deliriums caused by anticholinergic agents can be exacerbated by low-potency DRAs, which often have significant antimuscarinic activity. Orthostasis, parkinsonism, and worsened cognition are the most problematic side effects in this elderly population.

Substance-Induced Psychotic Disorder. Intoxication with cocaine, amphetamines, alcohol, phencyclidine, or other drugs can cause psychotic symptoms. Because these symptoms tend to be time-limited, it is preferable

to avoid the use of a DRA unless the patient is severely agitated and aggressive. Usually, benzodiazepines can be used to calm the patient. Benzodiazepines should be used instead of DRAs in cases of phencyclidine intoxication. When a patient is experiencing hallucinations or delusions as a result of alcohol withdrawal, DRAs may increase the risk of seizure.

Childhood Schizophrenia. Children with schizophrenia benefit from treatment with antipsychotic medication, although considerably less research has been devoted to this population. Studies are underway to determine if intervention with medication at the very earliest signs of disturbance in children genetically at risk for schizophrenia can prevent the emergence of more florid symptoms. Careful consideration needs to be given to side effects, especially those involving cognition and alertness.



Table 21-12
Dopamine Receptor Antagonists: Potency and Adverse Effects

Drug Name	Chemical Classification	Therapeutically Equivalent Oral	Side Effects		
			Sedation	Autonomic[a]	Extrapyramidal Reactions[b]
Pimozide[c]	Diphenylbutyl-piperidine	1.5	+	+	+++
Fluphenazine	Phenothiazine: piperazine compound	2	+	+	+++
Haloperidol	Butyrophenone	2	+	+	+++
Thiothixene	Thioxanthene	4	+	+	+++
Trifluoperazine	Phenothiazine: piperazine compound	5	++	+	+++
Perphenazine	Phenothiazine: piperazine compound	8	++	+	++/+++
Molindone	Dihydroindolone	10	++	+	+
Loxapine	Dibenzoxazepine	10	++	+/++	++/+++
Prochlor-perazine[c]	Phenothiazine: piperazine compound	15	++	+	+++
Aceto-phenazine	Phenothiazine: piperazine compound	20	++	+	++/+++
Triflupromazine	Phenothiazine: aliphatic compound	25	+++	++/+++	++
Mesoridazine	Phenothiazine: piperidine compound	50	+++	++	+
Chlorpromazine	Phenothiazine: aliphatic compound	100	+++	+++	++
Chlorprothixene	Thioxanthene	100	+++	+++	+/++
Thioridazine	Phenothiazine: piperidine compound	100	+++	+++	+

[a]Anti–α-adrenergic and anticholinergic effects.
[b]Excluding tardive dyskinesia, which appears to be produced to the same degree and frequency by all agents with equieffective antipsychotic dosages.
[c]Pimozide is used principally in the treatment of Tourette disorder; prochlorperazine is used rarely, if ever, as an antipsychotic agent.
Adapted from American Medical Association. *AMA Drug Evaluations: Annual 1992*. Chicago: American Medical Association, 1992.

Other Psychiatric and Nonpsychiatric Indications. The DRAs reduce the chorea in the early stages of Huntington disease. Patients with this disease

may develop hallucinations, delusions, mania, or hypomania. These and other psychiatric symptoms respond to DRAs. High-potency DRAs should be used. However, clinicians should be aware that patients with a rigid form of this disorder may experience acute EPS. The use of DRAs to treat impulse control disorders should be reserved for patients in whom other interventions have failed. Patients with autism spectrum disorder may exhibit hyperactivity, screaming, and agitation with combativeness. Some of these symptoms respond to high-potency DRAs, but there is little research evidence supporting benefits in these patients.

The rare neurologic disorders ballismus and hemiballismus (which affect only one side of the body), characterized by propulsive movements of the limbs away from the body, also respond to treatment with antipsychotic agents. Other miscellaneous indications for the use of DRAs include the treatment of nausea, emesis, intractable hiccups, and pruritus. Endocrine disorders and temporal lobe epilepsy may be associated with psychosis that responds to antipsychotic treatment.

The most common side effects of DRAs are neurologic. As a rule, low-potency drugs cause the most nonneurologic adverse effects, and high-potency drugs cause the most neurologic adverse effects.

Precautions and Adverse Reactions

Table 21-12 summarizes the most common adverse events associated with the use of DRAs.

Neuroleptic Malignant Syndrome. A potentially fatal side effect of DRA treatment, neuroleptic malignant syndrome, can occur at any time during DRA treatment. Symptoms include extreme hyperthermia, severe muscular rigidity and dystonia, akinesia, mutism, confusion, agitation, and increased pulse rate and BP. Laboratory findings include increased WBC count and levels of creatinine phosphokinase, liver enzymes, plasma myoglobin, and myoglobinuria, occasionally associated with renal failure. The symptoms usually evolve over 24 to 72 hours, and the untreated syndrome lasts 10 to 14 days. The diagnosis is often missed in the early stages, and the withdrawal or agitation may mistakenly be considered to reflect increased psychosis. Men are affected more frequently than are

women, and young persons are affected more commonly than are elderly persons. The mortality rate can reach 20 to 30 percent, or even higher, when depot medications are involved. Rates are also increased when high doses of high-potency agents are used.

If the neuroleptic malignant syndrome is suspected, the DRA should be stopped immediately, and the following done: medical support to cool the person; monitoring of vital signs, electrolytes, fluid balance, and renal output; and symptomatic treatment of fever. Antiparkinsonian medications may reduce some of the muscle rigidity. Dantrolene (Dantrium), a skeletal muscle relaxant (0.8 to 2.5 mg/kg every 6 hours, up to a total dosage of 10 mg a day), may be useful in the treatment of this disorder. When the person can take oral medications, dantrolene can be given in doses of 100 to 200 mg a day. Bromocriptine (20 to 30 mg a day in four divided doses) or amantadine can be added to the regimen. Treatment should usually be continued for 5 to 10 days. When drug treatment is restarted, the clinician should consider switching to a low-potency drug or an SDA, although these agents—including clozapine—may also cause the neuroleptic malignant syndrome.

Seizure Threshold. DRAs may lower the seizure threshold. Chlorpromazine, thioridazine, and other low-potency drugs are thought to be more epileptogenic than are high-potency drugs. The risk of inducing a seizure by drug administration warrants consideration when the person already has a seizure disorder or brain lesion.

Sedation. The blockade of histamine H_1 receptors is the usual cause of sedation associated with DRAs. Chlorpromazine is the most sedating typical antipsychotic. The relative sedative properties of the drugs are summarized in Table 21-12. Giving the entire daily dose at bedtime usually eliminates any problems from sedation, and tolerance for this adverse effect often develops.

Central Anticholinergic Effects. The symptoms of central anticholinergic activity include severe agitation; disorientation to time, person, and place; hallucinations; seizures; high fever; and dilated pupils. Stupor and coma may ensue. The treatment of anticholinergic toxicity consists of discontinuing the causal agent or agents, close medical supervision, and physostigmine (Antilirium, Eserine), 2 mg by slow

intravenous (IV) infusion, repeated within 1 hour as necessary. Too much physostigmine is dangerous, and symptoms of physostigmine toxicity include hypersalivation and sweating. Atropine sulfate (0.5 mg) can reverse the effects of physostigmine toxicity.

Cardiac Effects. The DRAs decrease cardiac contractility, disrupt enzyme contractility in cardiac cells, increase circulating levels of catecholamines, and prolong atrial and ventricular conduction time and refractory periods. Low-potency DRAs, particularly the phenothiazines, are usually more cardiotoxic than are high-potency drugs. One exception is haloperidol, which has been linked to abnormal heart rhythm, ventricular arrhythmias, torsades de pointes, and sudden death when injected IV. Pimozide, sulpiride, and droperidol (a butyrophenone) also prolong the QTc interval and have been associated with torsades de pointes and sudden death. In one study, thioridazine was responsible for 28 (61 percent) of the 46 sudden antipsychotic deaths. In 15 of these cases, it was the only drug ingested. Chlorpromazine also causes prolongation of the QT and PR intervals, blunting of the T waves, and depression of the ST segment. These drugs are thus indicated only when other agents have been ineffective.

Sudden Death. Occasional reports of sudden cardiac death during treatment with DRAs may be the result of cardiac arrhythmias. Other causes may include seizure, asphyxiation, malignant hyperthermia, heatstroke, and neuroleptic malignant syndrome. However, there does seem to be an overall increased incidence of sudden death with antipsychotics.

Orthostatic (Postural) Hypotension. Orthostatic (postural) hypotension is most frequent with low-potency drugs, particularly chlorpromazine, thioridazine, and chlorprothixene. When using intramuscular (IM) low-potency DRAs, the clinician should measure the patient’s BP (lying and standing) before and after the first dose and during the first few days of treatment.

Orthostatic hypotension is mediated by adrenergic blockade and occurs most frequently during the first few days of treatment. Tolerance often develops for this side effect, which is why the initial dosing of these drugs is lower than the usual therapeutic dose. Fainting or falls, although uncommon,

may lead to injury. Patients should be warned of this side effect and instructed to rise slowly after sitting and reclining. Patients should avoid all caffeine and alcohol; should drink at least 2 L of fluid a day; and if not under treatment for hypertension, should add liberal amounts of salt to their diet. Support hose may help some persons.

Hypotension can usually be managed by having patients lie down with their feet higher than their heads and pump their legs as if bicycling. Volume expansion or vasopressor agents, such as norepinephrine (Levophed), may be indicated in severe cases. Because hypotension is produced by α-adrenergic blockade, the drugs also block the α-adrenergic stimulating properties of epinephrine, leaving the β-adrenergic stimulating effects untouched. Therefore, the administration of epinephrine results in a paradoxical worsening of hypotension and is contraindicated in cases of antipsychotic-induced hypotension. Pure α-adrenergic pressor agents, such as metaraminol (Aramine) and norepinephrine, are the drugs of choice in the treatment of the disorder.

Hematologic Effects. Temporary leukopenia with a WBC count of about 3,500 is a common but not a serious problem. Agranulocytosis, a life-threatening hematologic problem, occurs in about 1 in 10,000 persons treated with DRAs. Thrombocytopenic or nonthrombocytopenic purpura, hemolytic anemias, and pancytopenia may rarely occur in persons treated with DRAs. Although routine CBCs are not indicated, if a person reports a sore throat and fever, a CBC should be done immediately to check for the possibility of a severe blood dyscrasia. If blood index values are low, the administration of DRAs should be stopped, and the patient should be transferred to a medical facility. The mortality rate for the complication may be as high as 30 percent.

Peripheral Anticholinergic Effects. Peripheral anticholinergic effects, such as dry mouth and nose, blurred vision, constipation, urinary retention, and mydriasis, are common. This is especially true with low-potency DRAs, for example, chlorpromazine, thioridazine, mesoridazine (Serentil). Some persons may also have nausea and vomiting.

Constipation should be treated with the usual laxative preparations, but severe constipation can progress to paralytic ileus. A decrease in the DRA dosage is warranted in such cases. Pilocarpine (Salagen) may be used to treat

paralytic ileus, although the relief is only transitory. Bethanechol (Urecholine) (20 to 40 mg a day) may be useful in some persons with urinary retention.

Weight gain is associated with increased mortality and morbidity and with medication noncompliance. Low-potency DRAs may cause significant weight gain but not as much as is seen with the SDAs olanzapine (Zyprexa) and clozapine (Clozaril). Molindone and perhaps loxapine (Loxitane) appear to be least likely to cause weight gain.

Endocrine Effects. Blockade of the dopamine receptors in the tuberoinfundibular tract results in the increased secretion of prolactin, which can result in breast enlargement, galactorrhea, amenorrhea, and inhibited orgasm in women and impotence in men. The SDAs, except risperidone (Risperdal), are not significantly associated with an increase in prolactin levels and may be the drugs of choice for persons experiencing disturbing side effects from increased prolactin release.

Sexual Adverse Effects. Both men and women taking DRAs can experience anorgasmia and decreased libido. Up to 50 percent of men who take antipsychotics report ejaculatory and erectile disturbances. Sildenafil (Viagra), vardenafil (Levitra), and tadalafil (Cialis) are often used to treat psychotropic-induced orgasmic dysfunction, but they have not been studied in combination with the DRAs. Thioridazine is mainly associated with decreased libido and retrograde ejaculation in men. Priapism and reports of painful orgasms have also been described, both possibly resulting from α_1-adrenergic antagonist activity.

Skin and Eye Effects. Allergic dermatitis and photosensitivity may occur, especially with low-potency agents. Urticarial, maculopapular, petechial, and edematous eruptions may occur early in treatment, generally in the first few weeks, and remit spontaneously. A photosensitivity reaction that resembles a severe sunburn also occurs in some persons taking chlorpromazine. Persons should be warned of this adverse effect, should spend no more than 30 to 60 minutes in the sun, and should use sunscreens. Long-term chlorpromazine use is associated with blue-gray discoloration of skin areas exposed to sunlight. The skin changes often begin with a tan or golden brown color and

progress to such colors as slate gray, metallic blue, and purple. These discolorations resolve when the patient is switched to another medication.

Irreversible retinal pigmentation is associated with the use of thioridazine at dosages above 1,000 mg a day. An early symptom of the side effect can sometimes be nocturnal confusion related to difficulty with night vision. The pigmentation can progress even after thioridazine administration is stopped, finally resulting in blindness. It is for this reason that the maximum recommended dosage of thioridazine is 800 mg/day.

Patients taking chlorpromazine may develop a relatively benign pigmentation of the eyes, characterized by whitish brown granular deposits concentrated in the anterior lens and posterior cornea and visible only by slit-lens examination. The deposits can progress to opaque white and yellow-brown granules, often stellate. Occasionally, the conjunctiva is discolored by a brown pigment. No retinal damage is seen, and vision is rarely impaired. This condition gradually resolves when chlorpromazine is discontinued.

Jaundice. Elevations of liver enzymes during treatment with a DRA tend to be transient and not clinically significant. When chlorpromazine first came into use, cases of obstructive or cholestatic jaundice were reported, usually in the first month of treatment and heralded by symptoms of upper abdominal pain, nausea, and vomiting. This adverse effect was followed by fever, rash, eosinophilia, bilirubin in the urine, and increases in levels of serum bilirubin, alkaline phosphatase, and hepatic transaminases. Reported cases are now sporadic, but if jaundice occurs, the medication should be discontinued.

Overdoses. Overdoses typically consist of exaggerated DRA side effects. Symptoms and signs include CNS depression, EPS, mydriasis, rigidity, restlessness, decreased deep tendon reflexes, tachycardia, and hypotension. The severe symptoms of overdose include delirium, coma, respiratory depression, and seizures. Haloperidol may be among the safest typical antipsychotics in overdose. After an overdose, electroencephalography (EEG) shows diffuse slowing and low voltage. An extreme overdose may lead to delirium and coma, with respiratory depression and hypotension. Life-threatening overdose usually involves the ingestion of other CNS depressants, such as alcohol or benzodiazepines.

Activated charcoal, if possible, and gastric lavage should be administered

if the overdose is recent. Emetics are not indicated because the antiemetic actions of the DRAs inhibit their efficacy. Seizures can be treated with IV diazepam or phenytoin. Hypotension can be treated with either norepinephrine or dopamine but not epinephrine.

Pregnancy and Lactation. There is a low correlation between the use of antipsychotics during pregnancy and congenital malformations. Nevertheless, antipsychotics should be avoided during pregnancy, particularly in the first trimester, unless the benefit outweighs the risk. High-potency drugs are preferable to low-potency drugs because low-potency drugs are associated with hypotension.

DRAs are secreted in the breast milk, although concentrations are low. Women taking these agents should be advised against breastfeeding.

Drug Interactions

Many pharmacokinetic and pharmacodynamic drug interactions are associated with these drugs (Table 21-13). CYP2D6 is the most common hepatic isozyme involved in DRA pharmacokinetic interactions. Other common drug interactions affect the absorption of the DRAs.

Antacids, activated charcoal, cholestyramine (Questran), kaolin, pectin, and cimetidine (Tagamet) taken within 2 hours of antipsychotic administration can reduce the absorption of these drugs. Anticholinergics may decrease the absorption of the DRAs. The additive anticholinergic activity of the DRAs, anticholinergics, and tricyclic drugs may result in anticholinergic toxicity. Digoxin (Lanoxin) and steroids, both of which decrease gastric motility, can increase DRA absorption.



Table 21-13
Antipsychotic Drug Interactions

Interacting Medication	Mechanism	Clinical Effect
Drug Interactions Assessed to Have Major Severity		
β-Adrenergic receptor	Synergistic pharmacologic effect; antipsychotic	Severe hypotension

antagonists	inhibits metabolism of propranolol; antipsychotic increases plasma concentrations	
Anticholinergics	Pharmacodynamic effects	Decreased antipsychotic effect
	Additive anticholinergic effect	Anticholinergic toxicity
Barbiturates	Phenobarbital induces antipsychotic metabolism	Decreased antipsychotic concentrations
Carbamazepine	Induces antipsychotic metabolism	Up to 50% reduction in antipsychotic concentrations
Charcoal	Reduces GI absorption of antipsychotic and adsorbs drug during enterohepatic circulation	May reduce antipsychotic effect or cause toxicity when used to treat overdose or for GI disturbances
Cigarette smoking	Induction of microsomal enzymes	Reduced plasma concentrations of antipsychotic agents
Epinephrine, norepinephrine	Antipsychotic antagonizes pressor effect	Hypotension
Ethanol	Additive CNS depression	Impaired psychomotor status

Fluvoxamine	Fluvoxamine inhibits metabolism of haloperidol and clozapine	Increased concentrations of haloperidol and clozapine
Guanethidine	Antipsychotic antagonizes guanethidine reuptake	Impaired antihypertensive effect
Lithium	Unknown	Rare reports of neurotoxicity
Meperidine	Additive CNS depression	Hypotension and sedation
Drug Interactions Assessed to Have Minor or Moderate Severity		
Amphetamines, anorexiants	Decreased pharmacologic effect of amphetamine	Diminished weight loss effect; amphetamines may exacerbate psychosis
ACEIs	Additive hypotensive crisis	Hypotension, postural intolerance
Antacids containing aluminum	Insoluble complex formed in GI tract	Possible reduced antipsychotic effect
AD nonspecific	Decreased metabolism of AD through competitive inhibition	Increased AD concentration
Benzodiazepines	Increased pharmacologic effect of the benzodiazepine	Respiratory depression, stupor, hypotension

Bromocriptine	Antipsychotic antagonizes dopamine receptor stimulation	Increased prolactin
Caffeinated beverages	Form precipitate with antipsychotic solutions	Possible diminished antipsychotic effect
Cimetidine	Reduced antipsychotic absorption and clearance	Decreased antipsychotic effect
Clonidine	Antipsychotic potentiates α-adrenergic hypotensive effect	Hypotension or hypertension
Disulfiram	Impairs antipsychotic metabolism	Increased antipsychotic concentrations
Methyldopa	Unknown	BP elevations
Phenytoin	Induction of antipsychotic metabolism; decreased phenytoin metabolism	Decreased antipsychotic concentrations: increased phenytoin levels
SSRIs	Impair antipsychotic metabolism; pharmacodynamic interaction	Sudden onset of extrapyramidal symptoms
Valproic acid	Antipsychotic inhibits valproic acid metabolism	Increased valproic acid half-life and levels

ACEI, angiotensin-converting enzyme inhibitor; AD, antidepressant; BP, blood pressure; CNS, central nervous system; GI, gastrointestinal; SSRI, selective serotonin reuptake inhibitor.

From Ereshosky L, Overman GP, Karp JK. Current psychotropic dosing and monitoring guidelines. *Prim Psychiatry*. 1996;3:21, with permission.

Phenothiazines, especially thioridazine, may decrease the metabolism of and cause toxic concentrations of phenytoin. Barbiturates may increase the metabolism of DRAs.

Tricyclic drugs and SSRIs that inhibit CYP2D6—paroxetine (Paxil), fluoxetine (Prozac), and fluvoxamine (Luvox)—interact with DRAs, resulting in increased plasma concentrations of both drugs. The anticholinergic, sedative, and hypotensive effects of the drugs may also be cumulative.

Typical antipsychotics may inhibit the hypotensive effects of α-methyldopa (Aldomet). Conversely, typical antipsychotics may have an additive effect on some hypotensive drugs. Antipsychotic drugs have a variable effect on the hypotensive effects of clonidine. Propranolol (Inderal) coadministration increases the blood concentrations of both drugs.

The DRAs potentiate the CNS-depressant effects of the sedatives, antihistamines, opiates, opioids, and alcohol, particularly in persons with impaired respiratory status. When these agents are taken with alcohol, the risk for heatstroke may be increased.

Cigarette smoking may decrease the plasma levels of the typical antipsychotic drugs. Epinephrine has a paradoxical hypotensive effect in persons taking typical antipsychotics. These drugs may decrease the blood concentration of warfarin (Coumadin), resulting in decreased bleeding time. The phenothiazines, thioridazine, and pimozide should not be coadministered with other agents that prolong the QT interval. Thioridazine is contraindicated in patients taking drugs that inhibit the CYP2D6 isoenzyme or in patients with reduced levels of CYP2D6.

Laboratory Interferences

Chlorpromazine and perphenazine (Trilafon) may cause both false-positive and false-negative results in immunologic pregnancy tests and falsely elevated bilirubin (with reagent test strips) and urobilinogen (with Ehrlich reagent test) values. These drugs are associated with an abnormal shift in the results of the glucose tolerance test. However, that shift may reflect the effects of the drugs on the glucose-regulating system. Phenothiazines have been reported to interfere with the measurement of 17-ketosteroids and 17-hydroxycorticosteroids and produce false-positive results in tests for

phenylketonuria.

Dosage and Clinical Guidelines

Contraindications to the use of DRAs include the following: (1) a history of a severe allergic response; (2) the possible ingestion of a substance that will interact with the antipsychotic to induce CNS depression (e.g., alcohol, opioids, barbiturates, and benzodiazepines) or anticholinergic delirium (e.g., scopolamine and possibly phencyclidine [PCP]); (3) the presence of a severe cardiac abnormality; (4) a high risk for seizures; (5) the presence of narrow-angle glaucoma or prostatic hypertrophy if a drug with high anticholinergic activity is to be used; and (6) the presence or a history of tardive dyskinesia. Antipsychotics should be administered with caution in persons with hepatic disease because impaired hepatic metabolism may result in high plasma concentrations. The usual assessment should include a CBC with WBC indexes, liver function tests, and electrocardiography (ECG), especially in women older than 40 years of age and men older than 30 years of age. Elderly persons and children are more sensitive to side effects than are young adults, so the dosage of the drug should be adjusted accordingly.

Various patients may respond to widely different dosages of antipsychotics; therefore, there is no set dosage for any given antipsychotic drug. Because of side effects, it is a reasonable clinical practice to begin at a low dosage and increase as necessary. It is important to remember that the maximal effects of a particular dosage may not be evident for 4 to 6 weeks. Available preparations and dosages of the DRAs are given in Table 21-14.

Short-Term Treatment. The equivalent of 5 to 20 mg of haloperidol is a reasonable dose for an adult in an acute state. An older adult may benefit from as little as 1 mg of haloperidol. The administration of more than 25 mg of chlorpromazine in one injection may result in profound hypotension. IM administration results in peak plasma levels in about 30 minutes versus 90 minutes using the oral route. Doses of drugs for IM administration are about half those given by the oral route. In a short-term treatment setting, the person should be observed for 1 hour after the first dose of medication. After that time, most clinicians administer a second dose or a sedative agent (e.g., a benzodiazepine) to achieve effective behavioral control. Possible sedatives

include lorazepam (Ativan) (2 mg IM) and amobarbital (50 to 250 mg IM).

Rapid Neuroleptization. Rapid neuroleptization (also called psychotolysis) is the practice of administering hourly IM doses of antipsychotic medications until marked sedation is achieved. However, several research studies have shown that merely waiting several more hours after one dose yields the same clinical improvement as is seen with repeated doses. Nevertheless, clinicians must be careful to keep patients from becoming violent while they are psychotic. Clinicians can help prevent violent episodes by using adjuvant sedatives or by temporarily using physical restraints until the persons can control their behavior.

Early Treatment. A full 6 weeks may be necessary to evaluate the extent of the improvement in psychotic symptoms. However, agitation and excitement usually improve quickly with antipsychotic treatment. About 75 percent of persons with a short history of illness show significant improvement in their psychosis. Psychotic symptoms, both positive and negative, usually continue to improve 3 to 12 months after the initiation of treatment.

About 5 mg of haloperidol or 300 mg of chlorpromazine is a usual effective daily dose. In the past, much higher doses were used, but evidence suggests that it resulted in more side effects without additional benefits. A single daily dose is usually given at bedtime to help induce sleep and to reduce the incidence of adverse effects. However, bedtime dosing for elderly persons may increase their risk of falling if they get out of bed during the night. The sedative effects of typical antipsychotics last only a few hours, in contrast to the antipsychotic effects, which last for 1 to 3 days.

Intermittent Medications. It is a common clinical practice to order medications to be given intermittently as needed (PRN). Although this practice may be reasonable during the first few days that a person is hospitalized, the amount of time the person takes antipsychotic drugs, rather than an increase in dosage, is what produces therapeutic improvement. Clinicians on inpatient services may feel pressured by staff members to write PRN antipsychotic orders; such orders should include specific symptoms, how often the drugs should be given, and how many doses can be given each day. Clinicians may choose to use small doses for the PRN doses (e.g., 2 mg

of haloperidol) or use a benzodiazepine instead (e.g., 2 mg of lorazepam IM). If PRN doses of an antipsychotic are necessary after the first week of treatment, the clinician may want to consider increasing the standing daily dose of the drug.

Maintenance Treatment. The first 3 to 6 months after a psychotic episode are usually considered a period of stabilization. After that time, the dosage can be decreased by about 20 percent every 6 months until the minimum effective dosage is found. A person is usually maintained on antipsychotic medications for 1 to 2 years after the first psychotic episode. Antipsychotic treatment is often continued for 5 years after a second psychotic episode, and lifetime maintenance is considered after the third psychotic episode, although attempts to reduce the daily dosage can be made every 6 to 12 months.



Table 21-14
Dopamine Receptor Antagonists

Name						Rectal Suppositories (mg)	Adult Dose Range (mg/day)	
Generic or Chemical	Trade	Tablets (mg)	Capsules (mg)	Solution	Parenteral		Acute	Maintenance
Chlorpromazine	Thorazine	10, 25, 50, 100, 200	30, 75, 150, 200, 300	10 mg/5 mL, 30 mg/mL, 100 mg/mL	25 mg/mL	25, 100	100–1,600 PO, 25–400 IM	50–400 PO
Prochlorperazine	Compazine	5, 10, 25	10, 15, 30	5 mg/5 mL	5 mg/mL	2.5, 5, 25	15–200 PO, 40–80 IM	15–60 PO
Perphenazine	Trilafon	2, 4, 8, 16	—	16 mg/5 mL	5 mg/mL	—	12–64 PO, 15–30 IM	8–24 PO
Trifluoperazine	Stelazine	1, 2, 5, 10	—	10 mg/mL	2 mg/mL	—	4–40 PO, 4–10 IM	5–20 PO
Fluphenazine	Prolixin	1, 2.5, 5, 10	—	2.5 mg/5 mL, 5 mg/mL	2.5 mg/mL (IM only)	—	2.5–40.0 PO, 5–20 IM	1.0–15.0 PO, 12.5–50.0 IM (decanoate or enanthate, weekly or biweekly)
Fluphenazine decanoate	—	—	—	—	2.5 mg/mL	—	—	—
Fluphenazine enanthate	—	—	—	2.5 mg/mL	—	—	—	—
Thioridazine	Mellaril	10, 15, 25, 50, 100, 150, 200	—	25 mg/5 mL, 100 mg/5 mL, 30 mg/mL, 100 mg/mL	—	—	200–800 PO	100–300 PO
Mesoridazine	Serentil	10, 25, 50, 100	—	25 mg/mL	25 mg/mL	—	100–400 PO, 25–200 IM	30–150 PO
Haloperidol	Haldol	0.5, 1, 2, 5, 10, 20	—	2 mg/5 mL	5 mg/mL (IM only)	—	5–20 PO, 12.5–25 IM	1–10 PO
Haloperidol decanoate		—	—	—	50 mg/mL, 100 mg/mL (IM only)	—		25–200 IM (decanoate, monthly)
Chlorprothixene	Taractan	10, 25, 50, 100	—	100 mg/5 mL (suspension)	12.5 mg/mL	—	75–600 PO, 75–200 IM	50–400 PO
Thiothixene	Navane	—	1, 2, 5, 10, 20	5 mg/mL	5 mg/mL (IM only), 20 mg/mL (IM only)	—	60–100 PO, 8–30 IM	6–30 PO
Loxapine	Loxitane	—	5, 10, 25, 50	25 mg/5 mL	50 mg/mL	—	20–250 PO, 20–75 IM	20–100 PO
Molindone	Moban	5, 10, 25, 50, 100	—	20 mg/mL	—	—	50–225 PO	5–150 PO
Pimozide	Orap	2	—	—	—	—	0.5–20 PO	0.5–5.0 PO

IM, intramuscular; PO, oral.

Antipsychotic drugs are effective in controlling psychotic symptoms, but persons may report that they prefer being off the drugs because they feel better without them. The clinician must discuss maintenance medication with

patients and take into account their wishes, the severity of their illnesses, and the quality of their support systems. The clinician needs to know enough about the patient's life to try to predict upcoming stressors that might require increasing the dosage or closely monitoring compliance.

Long-Acting Depot Medications. Long-acting depot preparations may be needed to overcome problems with compliance. IM preparations are typically given once every 1 to 4 weeks.

Two depot preparations, a decanoate and an enanthate of fluphenazine, and a decanoate preparation of haloperidol, are available in the United States. The preparations are injected IM into an area of extensive muscle tissue, from which they are absorbed slowly into the blood. Decanoate preparations can be given less frequently than enanthate preparations because they are absorbed more slowly. Although stabilizing a person on the oral preparation of the specific drugs is not necessary before initiating the depot form, it is good practice to give at least one oral dose of the drug to assess the possibility of an adverse effect, such as severe EPS or an allergic reaction.

It is reasonable to begin with either 12.5 mg (0.5 mL) of fluphenazine preparation or 25 mg (0.5 mL) of haloperidol decanoate. If symptoms emerge in the next 2 to 4 weeks, the person can be treated temporarily with additional oral medications or with additional small depot injections. After 3 to 4 weeks, the depot injection can be increased to a single dose equal to the total of the doses given during the initial period.

A good reason to initiate depot treatment with low doses is that the absorption of the preparations may be faster than usual at the onset of treatment, resulting in frightening episodes of dystonia that eventually discourage compliance with the medication. Some clinicians keep patients drug free for 3 to 7 days before initiating depot treatment and give small doses of the depot preparations (3.125 mg of fluphenazine or 6.25 mg of haloperidol) every few days to avoid those initial problems.

Plasma Concentrations

Genetic differences among persons and pharmacokinetic interactions with other drugs influence the metabolism of the antipsychotics. If a person has not improved after 4 to 6 weeks of treatment, the plasma concentration of the

drug should be determined if feasible. After a patient has been on a particular dosage for at least five times the half-life of the drug and thus approaches steady-state concentrations, blood levels may be helpful. It is standard practice to obtain plasma samples at trough levels—just before the daily dose is given, usually at least 12 hours after the previous dose, and most commonly 20 to 24 hours after the previous dose. Most antipsychotics have no well-defined dose–response curve. The best-studied drug is haloperidol, which may have a therapeutic window ranging from 2 to 15 ng/mL. Other therapeutic ranges that have been reasonably well documented are 30 to 100 ng/mL for chlorpromazine and 0.8 to 2.4 ng/mL for perphenazine.

Treatment-Resistant Persons. Unfortunately, 10 to 35 percent of persons with schizophrenia do not obtain significant benefit from the antipsychotic drugs. Treatment resistance is a failure on at least two adequate trials of antipsychotics from two pharmacologic classes. It is useful to determine plasma concentrations for such persons because it is possible that they are slow or rapid metabolizers or are not taking their medication. Clozapine has been conclusively shown to be effective when given to patients who have failed multiple trials of DRAs.

Adjunctive Medications. It is common practice to use DRAs in conjunction with other psychotropic agents, either to treat side effects or to improve symptoms further. Most commonly, this involves the use of lithium or other mood-stabilizing agents, SSRIs, or benzodiazepines. It was once held that antidepressant drugs exacerbated psychosis in patients with schizophrenia. In all likelihood, this observation involved patients with bipolar disorder who were misdiagnosed as having schizophrenia. Abundant evidence suggests that antidepressants improve symptoms of depression in patients with schizophrenia. In some cases, amphetamines can be added to DRAs if patients remain withdrawn and apathetic.

Choice of Drug

Given their proven efficacy in managing acute psychotic symptoms and the fact that prophylactic administration of antiparkinsonian medication prevents or minimizes acute motor abnormalities, DRAs are still valuable, especially for short-term therapy. There is a considerable cost advantage to a DRA

antiparkinsonian regimen compared with monotherapy with a newer antipsychotic agent. Concern about the development of DRA-induced tardive dyskinesia is the primary deterrent to long-term use of these drugs, yet it is not clear that SDAs are entirely free of this complication. Thus, DRAs still occupy an important role in psychiatric treatment. DRAs are not predictably interchangeable. For reasons that cannot be explained, some patients do better on one drug than another. The choice of a particular DRA should be based on the known adverse effect profile of the drugs. Other than a significant advantage in terms of medication cost, the choice currently would be an SDA. If a DRA is thought to be preferable, a high-potency antipsychotic is favored, even though it may be associated with more neurologic adverse effects, mainly because there is a higher incidence of other adverse effects (e.g., cardiac, hypotensive, epileptogenic, sexual, and allergic) with the low-potency drugs. If sedation is the desired goal, either a low-potency antipsychotic can be given in divided doses, or a benzodiazepine can be coadministered.

An unpleasant or dysphoric reaction (a subjective sense of restlessness, oversedation, and acute dystonia) to the first dose of an antipsychotic predicts future inadequate response and noncompliance. Prophylactic use of antiparkinsonian medications may prevent this reaction. In general, clinicians should be vigilant about serious side effects and adverse events (described above) regardless of which drug is used.

21.2 Antidepressants

SELECTIVE SEROTONIN REUPTAKE INHIBITORS

Fluoxetine (Prozac), the first SSRI marketed in the United States, rapidly captured the favor of both clinicians and the general public as reports emerged of dramatic patient responses to treatment of depression. Patients no longer experienced such side effects as dry mouth, constipation, sedation, orthostatic hypotension, and tachycardia, typical side effects associated with the earlier antidepressant drugs—the TCAs and MAOIs. It was also significantly safer when taken in overdose than any previously available antidepressant. A significant effect of fluoxetine's popularity was that it helped ameliorate the long-standing stigma of depression and its treatment.

Table 21-15
Currently Approved Indications of the Selective Serotonin Reuptake Inhibitors in the United States for Adult and Pediatric Populations

	Citalopram (Celexa)	Escitalopram (Lexapro)	Fluoxetine (Prozac)	Fluvoxamine (Luvox)	Paroxetine (Paxil)	Sertraline (Zoloft)	Vilazodone (Viibryd)
Major depressive disorder	Adult	Adult	Adult[a] and pediatric	—	Adult[b]	Adult	Adult
Generalized anxiety disorder	—	Adult	—	—	Adult	—	—
OCD	—	—	Adult and pediatric	Adult and pediatric	Adult	Adult and pediatric	—
Panic disorder	—	—	Adult	—	Adult[b]	Adult	—
PTSD	—	—	—	—	Adult	Adult	—
Social anxiety disorder	—	—	—	—	Adult[b]	Adult	—
Bulimia nervosa	—	—	Adult	—	—	—	—
Premenstrual dysphoric disorder	—	—	Adult[c]	—	Adult[d]	Adult	—

OCD, obsessive-compulsive disorder; PTSD, posttraumatic stress disorder.
[a]Weekly fluoxetine is approved for continuation and maintenance therapy in adults.
[b]Paroxetine and paroxetine controlled-release.
[c]Marketed as Sarafem.
[d]Paroxetine-controlled release is approved for premenstrual dysphoric disorder.

Other SSRIs followed fluoxetine. These include sertraline (Zoloft), paroxetine (Paxil), fluvoxamine (Luvox), citalopram (Celexa), escitalopram (Lexapro), and vilazodone (Viibryd). These drugs are all equally effective in

treating depression. However, some are approved by the U.S. FDA for multiple indications, such as major depression, OCD, PTSD, PMDD, panic disorder, and social phobia (social anxiety disorder) (Table 21-15). Note that fluvoxamine is not FDA approved as an antidepressant, a fact that is due to a marketing decision. It is considered an antidepressant in other countries.

Although all SSRIs are equally effective, there are meaningful differences in pharmacodynamics, pharmacokinetics, and side effects, differences that might affect clinical responses among individual patients. This would explain why some patients have better clinical responses to a particular SSRI than another. The SSRIs have proven more problematic in terms of some side effects than the original clinical trials suggested. Quality-of-life–associated adverse effects such as nausea, sexual dysfunction, and weight gain sometimes mitigate the therapeutic benefits of the SSRIs. There can also be distressing withdrawal symptoms when SSRIs are stopped abruptly. This withdrawal is especially true with paroxetine but also occurs when other SSRIs with short half-lives are stopped.

Pharmacologic Actions

Pharmacokinetics. A significant difference among the SSRIs is their broad range of serum half-lives. Fluoxetine has the most prolonged half-life: 4 to 6 days; its active metabolite has a half-life of 7 to 9 days. The half-life of sertraline is 26 hours, and its less active metabolite has a half-life of 3 to 5 days. The half-lives of the other three, which do not have metabolites with significant pharmacologic activity, are 35 hours for citalopram, 27 to 32 hours for escitalopram, 21 hours for paroxetine, and 15 hours for fluvoxamine. As a rule, the SSRIs are well absorbed after oral administration and have their peak effects in the range of 3 to 8 hours. Absorption of sertraline may be slightly enhanced by food.

There are also differences in plasma protein–binding percentages among the SSRIs, with sertraline, fluoxetine, and paroxetine being the most highly bound and escitalopram being the least bound.

All SSRIs are metabolized in the liver by the CYP450 enzymes. Because the SSRIs have such a broad therapeutic index, other drugs rarely produce problematic increases in SSRI concentrations. The most critical drug–drug interactions involving the SSRIs occur as a result of the SSRIs inhibiting the

metabolism of the coadministered medication. Each of the SSRIs possesses a potential for slowing or blocking the metabolism of many drugs (Table 21-16). Fluvoxamine is the most problematic of the drugs in this respect. It has a marked effect on several of the CYP enzymes. Examples of clinically significant interactions include fluvoxamine and theophylline (Slo-Bid, Theo-Dur) through CYP1A2 interaction, fluvoxamine and clozapine (Clozaril) through CYP1A2 inhibition; and fluvoxamine with alprazolam (Xanax) or clonazepam (Klonopin) through CYP3A4 inhibition. Fluoxetine and paroxetine also possess significant effects on the CYP2D6 isozyme, which may interfere with the efficacy of opiate analogs, such as codeine and hydrocodone, by blocking the conversion of these agents to their active form. Thus, coadministration of fluoxetine and paroxetine with an opiate interferes with its analgesic effects. Sertraline, citalopram, and escitalopram are least likely to complicate treatment because of interactions.

The pharmacokinetics of vilazodone (5 to 80 mg) is dose-proportional. Steady-state plasma levels are achieved in about 3 days. The elimination of vilazodone is primarily by hepatic metabolism with a terminal half-life of approximately 25 hours.

Pharmacodynamics. The SSRIs are believed to exert their therapeutic effects through serotonin reuptake inhibition. They derive their name because they have little effect on the reuptake of norepinephrine or dopamine. Often, adequate clinical activity and saturation of the 5-HT transporters are achieved at starting dosages. As a rule, higher dosages do not increase antidepressant efficacy but may increase the risk of adverse effects.



Table 21-16
CYP450 Inhibitory Potential of Commonly Prescribed Antidepressants

Relative Rank	CYP1A2	CYP2C	CYP2D6	CYP3A
Higher	Fluvoxamine (Luvox)	Fluoxetine Fluvoxamine	Bupropion Fluoxetine Paroxetine	Fluvoxamine Nefazodone Tricyclics
Moderate	Tertiary amine tricyclics Fluoxetine (Prozac)	Sertraline	Secondary amine tricyclics Citalopram (Celexa) Escitalopram (Lexapro) Sertraline	Fluoxetine Sertraline
Low or minimal	Bupropion (Wellbutrin) Mirtazapine (Remeron) Nefazodone (Serzone) Paroxetine (Paxil) Sertraline (Zoloft) Venlafaxine (Effexor)	Paroxetine Venlafaxine	Fluvoxamine Mirtazapine Nefazodone Venlafaxine	Citalopram Escitalopram Mirtazapine Paroxetine Venlafaxine

CYP, cytochrome P450.

Citalopram and escitalopram are the most selective inhibitors of serotonin reuptake, with very little inhibition of norepinephrine or dopamine reuptake and very low affinities for histamine H_1, GABA, or benzodiazepine receptors. The other SSRIs have a similar profile except that fluoxetine weakly inhibits norepinephrine reuptake and binds to 5-HT_{2C} receptors, sertraline weakly inhibits norepinephrine and dopamine reuptake, and paroxetine has significant anticholinergic activity at higher dosages and binds to nitric oxide synthase. The SSRI vilazodone has 5-HT_{1A} receptor agonist properties. The clinical implications of the 5-HT_{1A} receptor agonist effects are not yet evident.

A pharmacodynamic interaction appears to underlie the antidepressant effects of combined fluoxetine–olanzapine. When taken together, these drugs increase brain concentrations of norepinephrine. Concomitant use of SSRIs and drugs in the triptan class (sumatriptan [Imitrex], naratriptan [Amerge], rizatriptan [Maxalt], and zolmitriptan [Zomig]) may result in a severe pharmacodynamic interaction—the development of serotonin syndrome (see “Precautions and Adverse Reactions”). However, many people use triptans while taking low doses of an SSRI for headache prophylaxis without adverse reaction. A similar reaction may occur when SSRIs are combined with tramadol (Ultram).

Therapeutic Indications

Depression. In the United States, all SSRIs other than fluvoxamine have been approved by the FDA for the treatment of depression. Several studies

have found that antidepressants with serotonin-norepinephrine activity—drugs such as the MAOIs, TCAs, venlafaxine (Effexor), and mirtazapine (Remeron)—may produce higher rates of remission than SSRIs in head-to-head studies. The continued role of SSRIs as first-line treatment thus reflects their simplicity of use, safety, and a broad spectrum of action.

Direct comparisons of individual SSRIs have not revealed any to be consistently superior to another. There nevertheless can be considerable diversity in response to the various SSRIs among individuals. For example, more than 50 percent of people who respond poorly to one SSRI will respond favorably to another. Thus, before shifting to non-SSRI antidepressants, it is most reasonable to try other agents in the SSRI class for persons who did not respond to the first SSRI.

Some clinicians have attempted to select a particular SSRI for a specific person based on the drug's unique adverse effect profile. For example, thinking that fluoxetine is an activating and stimulating SSRI, they may assume it is a better choice for an abulic person than paroxetine, which is presumed to be a sedating SSRI. These differences, however, usually vary from person to person. Analyses of clinical trial data show that SSRIs are more effective in patients with more severe symptoms of major depression than those with milder symptoms.

SUICIDE. The FDA has issued a black box warning for antidepressants and suicidal thoughts and behavior in children and young adults. This warning is based on a decade-old analysis of clinical trial data. More recent, comprehensive reanalysis of data has shown that suicidal thoughts and behavior decreased over time for adult and geriatric patients treated with antidepressants as compared with placebo. No differences were found for youths. In adults, reduction in suicide ideation and attempts occurred through a reduction in depressive symptoms. In all age groups, the severity of depression improved with medication and was significantly related to suicide ideation or behavior. It appears that SSRIs, as well as serotonin-norepinephrine reuptake inhibitors (SNRIs), have a protective effect against suicide that is mediated by decreases in depressive symptoms with treatment. For youths, no significant effects of treatment on suicidal thoughts and behavior were found, although depression responded to treatment. No evidence of increased suicide risk was observed in youths receiving active

medication. It is crucial to keep in mind that SSRIs, like all antidepressants, prevent potential suicides as a result of their primary action, the shortening and prevention of depressive episodes. In clinical practice, a few patients become anxious and agitated when started on an SSRI. The appearance of these symptoms could conceivably provoke or aggravate suicidal ideation. Thus, all depressed patients should be closely monitored during the period of maximum risk, the first few days and weeks they are taking SSRIs.

DEPRESSION DURING PREGNANCY AND POSTPARTUM. Rates of relapse of major depression during pregnancy among women who discontinue, attempt to discontinue, or modify their antidepressant regimens are very high. Rates range from 68 to 100 percent of patients. Thus, many women need to continue taking their medication during pregnancy and postpartum. The impact of maternal depression on infant development is unknown. There is no significantly increased risk for major congenital malformations after exposure to SSRIs during pregnancy (with the exception of paroxetine, discussed below). Thus, the risk of relapse into depression when a newly pregnant mother is taken off SSRIs is severalfold higher than the risk to the fetus of exposure to SSRIs.

There is some evidence suggesting increased rates of special care nursery admissions after delivery for children of mothers taking SSRIs. There is also a potential for a discontinuation syndrome with paroxetine. However, there is an absence of clinically significant neonatal complications associated with SSRI use.

Studies that have followed children into their early school years have failed to find any perinatal complications, congenital fetal anomalies, decreases in global intelligence quotient (IQ), language delays, or specific behavioral problems attributable to the use of fluoxetine during pregnancy.

Postpartum depression (with or without psychotic features) affects a small percentage of mothers. Some clinicians start administering SSRIs if the postpartum blues extend beyond a few weeks or if a woman becomes depressed during pregnancy. The head start afforded by starting SSRI administration during pregnancy if a woman is at risk for postpartum depression also protects the newborn, toward whom the woman may have harmful thoughts after parturition.

Babies whose mothers are taking an SSRI in the latter part of pregnancy

may be at a slight risk of developing pulmonary hypertension. Data about the risk of this side effect are inconclusive, but it is estimated to involve 1 to 2 babies for 1,000 births. Paroxetine should be avoided during pregnancy.

The FDA classified paroxetine as a pregnancy Category D medication in the former FDA classification system. In 2005, the FDA issued an alert that paroxetine increases the risk of congenital disabilities, particularly heart defects, when women take it during the first 3 months of pregnancy. Paroxetine should usually not be taken during pregnancy, but for some women who have already been taking paroxetine, the benefits of continuing paroxetine may be greater than the potential risk to the baby. Women taking paroxetine who are pregnant, think they may be pregnant, or plan to become pregnant should talk to their physicians about the potential risks of taking paroxetine during pregnancy.

The FDA alert was based on the findings of studies that showed that women who took paroxetine during the first 3 months of pregnancy were about one-and-a-half to two times as likely to have a baby with a heart defect as women who received other antidepressants or women in the general population. Most of the heart defects in these studies were not life-threatening and happened mainly in the inside walls of the heart muscle, where repairs can be made if needed (atrial and ventricular septal defects). Sometimes these septal defects resolve without treatment. In one of the studies, the risk of heart defects in babies whose mothers had taken paroxetine early in pregnancy was 2 percent, compared with a 1 percent risk in the whole population. In the other study, the risk of heart defects in babies whose mothers had taken paroxetine in the first 3 months of pregnancy was 1.5 percent, compared with 1 percent in babies whose mothers had taken other antidepressants in the first 3 months of pregnancy. This study also showed that women who took paroxetine in the first 3 months of pregnancy were about twice as likely to have a baby with any congenital disability as women who took other antidepressants.

Minimal amounts of SSRIs are found in breast milk, and no harmful effects have been found in breastfed babies. Concentrations of sertraline and escitalopram are exceptionally low in breast milk. However, in some cases, reported concentrations might be higher than average. No decision regarding the use of an SSRI is risk-free. It is thus essential to document that communication of potential risks to the patient has taken place.

DEPRESSION IN ELDERLY AND MEDICALLY ILL PERSONS. The SSRIs are safe and well tolerated when used to treat elderly and medically ill persons. As a class, they have little or no cardiotoxic, anticholinergic, antihistaminergic, or α-adrenergic adverse effects. Paroxetine does have some anticholinergic activity, which may lead to constipation and worsening of cognition. The SSRIs can produce subtle cognitive deficits, prolonged bleeding time, and hyponatremia, all of which may impact the health of this population. The SSRIs are effective in poststroke depression and dramatically reduce the symptom of crying.

DEPRESSION IN CHILDREN. The use of SSRI antidepressants in children and adolescents has been controversial. Few studies have shown clear-cut benefits from the use of these drugs, and studies show that there may be an increase in suicidal or aggressive impulses. However, some children and adolescents do exhibit dramatic responses to these drugs in terms of depression and anxiety. Fluoxetine has most consistently demonstrated effectiveness in reducing symptoms of depressive disorder in both children and adolescents. This fact may be a function of the quality of the clinical trials involved. Sertraline is effective in treating social anxiety disorder in this population, especially when combined with cognitive-behavioral therapy. Given the potential adverse effect of untreated depression and anxiety in a young population and the uncertainty about many aspects of how children and adolescents might react to medication, any use of SSRIs should be undertaken only within the context of comprehensive management of the patient.

Anxiety Disorders

OBSESSIVE-COMPULSIVE DISORDER. Fluvoxamine, paroxetine, sertraline, and fluoxetine are indicated for the treatment of OCD in persons older than the age of 18 years. Fluvoxamine, fluoxetine, and sertraline have also been approved for the treatment of children with OCD (ages 6 to 17 years). About 50 percent of persons with OCD begin to show symptoms in childhood or adolescence, and more than half of these respond favorably to medication. Beneficial responses can be dramatic. Long-term data support the model of

OCD as a genetically determined, lifelong condition that is best treated continuously with drugs and cognitive-behavioral therapy from the onset of symptoms in childhood throughout the lifespan.

SSRI dosages for OCD may need to be higher than those required to treat depression. Although some responses can be seen in the first few weeks of treatment, it may take several months for the maximum effects to become evident. Patients who fail to obtain adequate relief of their OCD symptoms with an SSRI will usually benefit from the addition of a small dose of risperidone (Risperdal). Apart from the extrapyramidal side effects of risperidone, patients should be monitored for increases in prolactin levels when this combination is used. Clinically, hyperprolactinemia may manifest as gynecomastia and galactorrhea (in both men and women) and loss of menses.

Several disorders are now considered to be within the OCD spectrum. This spectrum includes several conditions and symptoms characterized by nonsuicidal self-mutilation, such as trichotillomania, eyebrow picking, nose picking, nail-biting, compulsive picking of skin blemishes, and cutting. Patients with these behaviors benefit from treatment with SSRIs. Other spectrum disorders include compulsive gambling, compulsive shopping, hypochondriasis, and body dysmorphic disorder.

PANIC DISORDER. Paroxetine, fluoxetine, and sertraline are indicated for the treatment of panic disorder, with or without agoraphobia. These agents work less rapidly than do benzodiazepines but are far superior to them for the treatment of panic disorder with comorbid depression. Citalopram and fluvoxamine also may reduce spontaneous or induced panic attacks. Because fluoxetine can initially heighten anxiety symptoms, persons with panic disorder must begin taking small dosages (5 mg a day) and increase the dosage slowly. Low doses of benzodiazepines may be given to manage this side effect.

SOCIAL ANXIETY DISORDER. SSRIs are effective agents in the treatment of social phobia. They reduce both symptoms and disability. The response rate is comparable to that seen with the MAOI phenelzine (Nardil), the previous standard treatment. The SSRIs are safer to use than MAOIs or benzodiazepines.

POSTTRAUMATIC STRESS DISORDER. Pharmacotherapy for PTSD must target specific symptoms in four clusters: reexperiencing, avoidance, negative changes in mood and thinking, and arousal. For long-term treatment, SSRIs appear to have a broader spectrum of therapeutic effects on specific PTSD symptom clusters than do TCAs and MAOIs. Benzodiazepine augmentation is useful in the acute symptomatic state. The SSRIs are associated with marked improvement of both intrusive and avoidant symptoms.

GENERALIZED ANXIETY DISORDER. The SSRIs may be useful for the treatment of specific phobias, generalized anxiety disorder (GAD), and separation anxiety disorder. A thorough, individualized evaluation is the first approach, with particular attention to identifying conditions amenable to drug therapy. Also, cognitive-behavioral or other psychotherapies can be added for greater efficacy.

Bulimia Nervosa and Other Eating Disorders. Fluoxetine is indicated for the treatment of bulimia, which is best done in the context of psychotherapy. Dosages of 60 mg a day are significantly more effective than 20 mg a day. In several well-controlled studies, fluoxetine in dosages of 60 mg a day was superior to placebo in reducing binge eating and induced vomiting. Some experts recommend an initial course of cognitive-behavioral therapy alone. If there is no response in 3 to 6 weeks, then fluoxetine administration is added. The appropriate duration of treatment with fluoxetine and psychotherapy has not been determined.

Fluvoxamine was not effective at a statistically significant level in one double-blind, placebo-controlled trial for inpatients with bulimia.

ANOREXIA NERVOSA. Fluoxetine has been used in inpatient treatment of anorexia nervosa to attempt to control comorbid mood disturbances and obsessive-compulsive symptoms. However, at least two careful studies, one of 7 months' and one of 24 months' duration, failed to find that fluoxetine affected the overall outcome and the maintenance of weight. Effective treatments for anorexia include cognitive-behavioral, interpersonal, psychodynamic, and family therapies in addition to a trial with SSRIs.

OBESITY. Fluoxetine, in combination with a behavioral program, is only

modestly beneficial for weight loss. A significant percentage of all persons who take SSRIs, including fluoxetine, lose weight initially but later may gain weight. However, all SSRIs may cause initial weight gain.

PREMENSTRUAL DYSPHORIC DISORDER. PMDD is characterized by debilitating mood and behavioral changes in the week preceding menstruation that interferes with normal functioning. Sertraline, paroxetine, fluoxetine, and fluvoxamine have been reported to reduce the symptoms of PMDD. Controlled trials of fluoxetine and sertraline administered either throughout the cycle or only during the luteal phase (the 2 weeks between ovulation and menstruation) showed both schedules to be equally effective.

An additional observation of unclear significance was that fluoxetine was associated with changing the duration of the menstrual period by more than 4 days, either lengthening or shortening it. The effects of SSRIs on menstrual cycle length are mostly unknown and may warrant careful monitoring in women of reproductive age.

Off-Label Uses

PREMATURE EJACULATION. The anorgasmic effects of SSRIs make them useful as a treatment for men with premature ejaculation. The SSRIs permit intercourse for a significantly more extended period and are reported to improve sexual satisfaction in couples in which the man has premature ejaculation. Fluoxetine and sertraline are useful for this purpose.

PARAPHILIAS. The SSRIs may reduce obsessive-compulsive behavior in people with paraphilias. The SSRIs diminish the average time per day spent in unconventional sexual fantasies, urges, and activities. Evidence suggests a more significant response to sexual obsessions than for paraphilic behavior.

AUTISM. Obsessive-compulsive behavior, poor social relatedness, and aggression are prominent autistic features that may respond to serotonergic agents such as SSRIs and clomipramine (Anafranil). Sertraline and fluvoxamine have been shown in controlled and open-label trials to mitigate aggressiveness, self-injurious behavior, repetitive behaviors, some degree of language delay, and (rarely) lack of social relatedness in adults with autistic

spectrum disorders. Fluoxetine has been reported to be useful for features of autism in children, adolescents, and adults.

Precautions and Adverse Reactions. SSRI side effects need to be considered in terms of their onset, duration, and severity. For example, nausea and jitteriness are early, generally mild, and time-limited side effects. Although SSRIs share similar side effect profiles, individual drugs in this class may cause a higher rate or carry a more severe risk of specific side effects depending on the patient.

Sexual Dysfunction. All SSRIs cause sexual dysfunction, and it is the most common adverse effect of SSRIs associated with long-term treatment. It has an estimated incidence of between 50 and 80 percent. The most common complaints are anorgasmia, inhibited orgasm, and decreased libido. Some studies suggest that sexual dysfunction is dose-related, but this has not been established. Unlike most of the other adverse effects of SSRIs, sexual inhibition rarely resolves in the first few weeks of use but usually continues as long as the drug is taken. In some cases, there may be an improvement over time.

Strategies to counteract SSRI-induced sexual dysfunction are numerous, and none has been proven to be very useful. Some reports suggest decreasing the dosage or adding bupropion (Wellbutrin) or amphetamine. Reports have described successful treatment of SSRI-induced sexual dysfunction with agents such as sildenafil (Viagra), which are used to treat erectile dysfunction. Ultimately, patients may need to be switched to antidepressants that do not interfere with sexual functioning, drugs such as mirtazapine or bupropion.

Gastrointestinal Adverse Effects. GI side effects are widespread and are mediated mainly through effects on the serotonin 5-HT_3 receptor. The most frequent GI complaints are nausea, diarrhea, anorexia, vomiting, flatulence, and dyspepsia. Sertraline and fluvoxamine produce the most intense GI symptoms. Delayed-release paroxetine, compared with the immediate-release preparation of paroxetine, has less intense GI side effects during the first week of treatment. However, paroxetine, because of its anticholinergic activity, frequently causes constipation. Nausea and loose

stools are usually dose-related and transient, usually resolving within a few weeks. Sometimes flatulence and diarrhea persist, especially during sertraline treatment. Initial anorexia may also occur and is most frequent with fluoxetine. SSRI-induced appetite and weight loss begin as soon as the drug is taken and peak at 20 weeks, after which weight often returns to baseline. Up to one-third of persons taking SSRIs will gain weight, sometimes more than 20 lb. This effect is mediated through a metabolic mechanism, an increase in appetite, or both. It happens gradually and is usually resistant to diet and exercise regimens. Paroxetine is associated with more frequent, rapid, and pronounced weight gain than the other SSRIs, especially among young women.

Cardiovascular Effects. All SSRIs can lengthen the QT interval in otherwise healthy people and cause drug-induced long QT syndrome, especially when taken in overdose. The risk of QTc prolongation increases when an antidepressant and an antipsychotic are used in combination, an increasingly common practice. Citalopram stands out as the SSRI with the most pronounced effect on QT intervals. A QT study to assess the effects of 20- and 60-mg doses of citalopram on the QT interval in adults, compared with placebo, found a maximum mean prolongation in the individually corrected QT intervals were 8.5 milliseconds for 20-mg citalopram and 18.5 milliseconds for 60 mg. For 40 mg, prolongation of the corrected QT interval was estimated to be 12.6 milliseconds. Based on these findings, the FDA has issued the following recommendation regarding citalopram use:

- 20 mg a day is the maximum recommended dose for patients with hepatic impairment, who are older than 60 years of age, who are CYP2C19 poor metabolizers, or who are taking concomitant cimetidine (Tagamet).
- No longer prescribe at doses greater than 40 mg a day.
- Do not use in patients with congenital long QT syndrome.
- Correct hypokalemia and hypomagnesemia before administering citalopram.
- Monitor electrolytes as clinically indicated.

Consider more frequent electrocardiograms in patients with congestive heart failure, bradyarrhythmias, or patients on concomitant medications that

prolong the QT interval.

The fact that citalopram carries a greater risk of causing fatal rhythm abnormalities was confirmed in a review of 469 SSRI poisoning admissions. Accordingly, patients should be advised to contact their prescriber immediately if they experience signs and symptoms of an abnormal heart rate or rhythm while taking citalopram.

The effect of vilazodone (20, 40, 60, and 80 mg) on the QTc interval was evaluated, and a small effect was observed. The upper bound of the 90 percent confidence interval for the largest placebo-adjusted, baseline-corrected QTc interval was below 10 milliseconds, based on the individual correction method (QTcI). This interval is below the threshold for clinical concern. However, it is unknown whether 80 mg is adequate to represent a high clinical exposure condition.

Physicians should consider whether the benefits of androgen deprivation therapy outweigh the potential risks in SSRI treated patients with prostate cancer, as reductions in androgen levels can cause QTc interval prolongation.

Dextromethorphan/Quinidine (Nuedexta) is available as a treatment for pseudobulbar affect, which is defined by involuntary, sudden, and frequent episodes of laughing or crying that are generally out of proportion or inappropriate to the situation. Quinidine can prolong the QT interval and is a potent inhibitor of CYP2D6. It should not be used with other medications that prolong the QT interval and are metabolized by CYP2D6. This drug should be used with caution with any medications that can prolong the QT interval and inhibit CYP3A4, particularly in patients with cardiac disease.

Antepartum use of SSRIs is sometimes associated with QTc interval prolongation in exposed neonates. In a review of 52 newborns exposed to SSRIs in the immediate antepartum period and 52 matched control subjects, the mean QTc was significantly longer in the group of newborns exposed to antidepressants as compared with control subjects. Five (10 percent) newborns exposed to SSRIs had a markedly prolonged QTc interval (greater than 460 milliseconds) compared with none of the unexposed newborns. The most extended QTc interval observed among exposed newborns was 543 milliseconds. All of the drug-associated repolarization abnormalities normalized in subsequent electrocardiographic tracings.

Headaches. The incidence of headache in SSRI trials was 18 to 20 percent,

only one percentage point higher than the placebo rate. Fluoxetine is the most likely to cause headaches. On the other hand, all SSRIs are effective prophylaxis against both migraine and tension-type headaches in many persons.

Central Nervous System Adverse Effects

ANXIETY. Fluoxetine may cause anxiety, particularly in the first few weeks of treatment. However, these initial effects usually give way to an overall reduction in anxiety after a few weeks. Increased anxiety is caused considerably less frequently by paroxetine and escitalopram, which may be better choices if sedation is desired, as in mixed anxiety and depressive disorders.

INSOMNIA AND SEDATION. The primary effect SSRIs exert in the area of insomnia and sedation is improved sleep resulting from the treatment of depression and anxiety. However, as many as 25 percent of persons taking SSRIs note trouble sleeping, excessive somnolence, or overwhelming fatigue. Fluoxetine is the most likely to cause insomnia, for which reason it is often taken in the morning. Sertraline and fluvoxamine are about equally likely to cause insomnia as somnolence, and citalopram and especially paroxetine often cause somnolence. Escitalopram is more likely to interfere with sleep than its isomer, citalopram. Some persons benefit from taking their SSRI dose before going to bed, but others prefer to take it the morning. SSRI-induced insomnia can be treated with benzodiazepines, trazodone (Desyrel) (clinicians must explain the risk of priapism), or other sedating medicines. Significant SSRI-induced somnolence often requires switching to the use of another SSRI or bupropion.

OTHER SLEEP EFFECTS. Many persons taking SSRIs report recalling extremely vivid dreams or nightmares. They describe sleep as “busy.” Other sleep effects of the SSRIs include bruxism, restless legs, nocturnal myoclonus, and sweating.

EMOTIONAL BLUNTING. Emotional blunting is a largely overlooked but frequent side effect associated with chronic SSRI use. Patients report an

inability to cry in response to emotional situations, a feeling of apathy or indifference, or a restriction in the intensity of emotional experiences. This side effect often leads to treatment discontinuation, even when the drugs provide relief from depression or anxiety.

YAWNING. Close clinical observation of patients taking SSRIs reveals an increase in yawning. This side effect is not a reflection of fatigue or poor nocturnal sleep but is the result of SSRI effects on the hypothalamus.

SEIZURES. Seizures have been reported in 0.1 to 0.2 percent of all patients treated with SSRIs, an incidence comparable to that reported with other antidepressants and not significantly different from that with placebo. Seizures are more frequent at the highest doses of SSRIs (e.g., fluoxetine 100 mg a day or higher) than at standard doses.

EXTRAPYRAMIDAL SYMPTOMS. The SSRIs may rarely cause akathisia, dystonia, tremor, cogwheel rigidity, torticollis, opisthotonos, gait disorders, and bradykinesia. Rare cases of tardive dyskinesia have been reported. Some people with well-controlled Parkinson disease may experience acute worsening of their motor symptoms when they take SSRIs.

Anticholinergic Effects. Paroxetine has a mild anticholinergic activity that causes dry mouth, constipation, and sedation in a dose-dependent fashion. Nevertheless, most persons taking paroxetine do not experience cholinergic adverse effects. Other SSRIs are associated with dry mouth, but this effect is not mediated by muscarinic activity.

Hematologic Adverse Effects. The SSRIs can cause functional impairment of platelet aggregation but not a reduction in platelet number. This pharmacologic effect can manifest through easy bruising and excessive or prolonged bleeding. When patients exhibit these signs, a test for bleeding time should be performed. Special monitoring is suggested when patients use SSRIs in conjunction with anticoagulants or aspirin. Concurrent use of SSRIs and NSAIDs is associated with a significantly increased risk of gastric bleeding. In cases where this combination is necessary, we should consider using proton pump inhibitors.

Electrolyte and Glucose Disturbances. The SSRIs may acutely decrease glucose concentrations; therefore, diabetic patients should be carefully monitored. Long-term use may be associated with increased glucose levels, although it remains to be proven whether this is the result of a pharmacologic effect. Antidepressant users may have other characteristics that raise their odds of developing diabetes, or may be more likely to be diagnosed with diabetes or other medical conditions as a result of being in treatment for depression.

Cases of SSRI-associated hyponatremia and the syndrome of inappropriate antidiuretic hormone have been seen in some patients, especially those who are older or treated with diuretics.

Endocrine and Allergic Reactions. The SSRIs can increase prolactin levels and cause mammoplasia and galactorrhea in both men and women. Breast changes are reversible upon discontinuation of the drug, but this may take several months to occur.

Various types of rashes appear in about 4 percent of all patients; in a small subset of these patients, the allergic reaction may generalize and involve the pulmonary system, rarely resulting in fibrotic damage and dyspnea. SSRI treatment may have to be discontinued in patients with drug-related rashes.

Serotonin Syndrome. Concurrent administration of an SSRI with an MAOI, L-tryptophan, or lithium (Eskalith) can raise plasma serotonin concentrations to toxic levels, producing a constellation of symptoms called *serotonin syndrome*. This severe and possibly fatal syndrome of serotonin overstimulation comprises, in order of appearance as the condition worsens, (1) diarrhea; (2) restlessness; (3) extreme agitation, hyperreflexia, and autonomic instability with possible rapid fluctuations in vital signs; (4) myoclonus, seizures, hyperthermia, uncontrollable shivering, and rigidity; and (5) delirium, coma, status epilepticus, cardiovascular collapse, and death.

Treatment of serotonin syndrome consists of removing the offending agents and promptly instituting comprehensive supportive care. Supportive care and treatment may include nitroglycerine, cyproheptadine (Periactin), methysergide (Sansert), cooling blankets, chlorpromazine (Thorazine), dantrolene (Dantrium), benzodiazepines, anticonvulsants, mechanical ventilation, and paralyzing agents.

Sweating. Some patients experience sweating while being treated with SSRIs. This sweating is independent of the temperature in the environment. Nocturnal sweating may drench bed sheets and require a change of nightclothes. Terazosin (Hytrin), 1 or 2 mg/day, is often dramatically effective in counteracting sweating.

Overdose. The adverse reactions associated with overdose of vilazodone at doses of 200 to 280 mg, as observed in clinical trials, included serotonin syndrome, lethargy, restlessness, hallucinations, and disorientation.

Selective Serotonin Reuptake Inhibitor Withdrawal. The abrupt discontinuance of SSRI use, especially one with a shorter half-life such as paroxetine or fluvoxamine, has been associated with a withdrawal syndrome that may include dizziness, weakness, nausea, headache, rebound depression, anxiety, insomnia, poor concentration, upper respiratory symptoms, paresthesias, and migraine-like symptoms. It usually does not appear until after at least 6 weeks of treatment and usually resolves spontaneously in 3 weeks. Persons who experienced transient adverse effects in the first weeks of taking an SSRI are more likely to experience discontinuation symptoms.

Fluoxetine is the SSRI least likely to be associated with this syndrome because the half-life of its metabolite is more than 1 week, and it effectively tapers itself. Fluoxetine has therefore been used in some cases to treat the discontinuation syndrome caused by the termination of other SSRIs. Nevertheless, a delayed and attenuated withdrawal syndrome occurs with fluoxetine as well.

Drug Interactions

The SSRIs do not interfere with most other drugs. A serotonin syndrome (Table 21-17) can develop with concurrent administration of MAOIs, L-tryptophan, lithium, or other antidepressants that inhibit the reuptake of serotonin. Fluoxetine, sertraline, and paroxetine can raise plasma concentrations of TCAs, which can cause clinical toxicity. Several potential pharmacokinetic interactions have been described based on in vitro analyses of the CYP enzymes, but clinically relevant interactions are rare. SSRIs that inhibit CYP2D6 may interfere with the analgesic effects of hydrocodone and oxycodone. These drugs can also reduce the effectiveness of tamoxifen

(Nolvadex, Soltamox). The combined use of SSRIs and NSAIDs increases the risk of gastric bleeding.



Table 21-17
Serotonin Syndrome Symptoms

Diarrhea	Myoclonus
Diaphoresis	Hyperactive reflexes
Tremor	Disorientation
Ataxia	Lability of mood

The SSRIs, particularly fluvoxamine, should not be used with clozapine because it raises clozapine concentrations, increasing the risk of seizure. The SSRIs may increase the duration and severity of zolpidem (Ambien)-induced side effects, including hallucinations.

Fluoxetine. Fluoxetine can be administered with tricyclic drugs, but the clinician should use low dosages of the tricyclic drug. Because the hepatic enzyme CYP2D6 metabolizes it, fluoxetine may interfere with the metabolism of other drugs in the 7 percent of the population who have an inefficient isoform of this enzyme, the so-called poor metabolizers. Fluoxetine may slow down the metabolism of carbamazepine (Tegretol), antineoplastic agents, diazepam (Valium), and phenytoin (Dilantin). Drug interactions have been described for fluoxetine that may affect the plasma levels of benzodiazepines, antipsychotics, and lithium. Fluoxetine and other SSRIs may interact with warfarin (Coumadin), increasing the risk of bleeding and bruising.

Sertraline. Sertraline may displace warfarin from plasma proteins and may increase the prothrombin time. The drug interaction data on sertraline support a generally similar profile to that of fluoxetine, although sertraline does not interact as strongly with the CYP2D6 enzyme.

Paroxetine. Paroxetine has a higher risk for drug interactions than does either fluoxetine or sertraline because it is a more potent inhibitor of the

CYP2D6 enzyme. Cimetidine can increase the concentration of sertraline and paroxetine, and phenobarbital (Luminal) and phenytoin can decrease the concentration of paroxetine. Because of the potential for interference with the CYP2D6 enzyme, the coadministration of paroxetine with other antidepressants, phenothiazines, and antiarrhythmic drugs should be undertaken with caution. Paroxetine may increase the anticoagulant effect of warfarin. Coadministration of paroxetine and tramadol may precipitate serotonin syndrome in elderly persons.

Fluvoxamine. Among the SSRIs, fluvoxamine appears to present the most risk for drug–drug interactions. Fluvoxamine is metabolized by the enzyme CYP3A4, which may be inhibited by ketoconazole (Nizoral). Fluvoxamine may increase the half-life of alprazolam, triazolam (Halcion), and diazepam, and it should not be coadministered with these agents. Fluvoxamine may increase theophylline levels threefold and warfarin levels twofold, with significant clinical consequences; thus, the serum levels of the latter drugs should be closely monitored, and the doses adjusted accordingly. Fluvoxamine raises concentrations and may increase the activity of clozapine, carbamazepine, methadone (Dolophine, Methadose), propranolol (Inderal), and diltiazem (Cardizem). Fluvoxamine has no significant interactions with lorazepam (Ativan) or digoxin (Lanoxin).

Citalopram. Citalopram is not a potent inhibitor of any CYP enzymes. Concurrent administration of cimetidine increases concentrations of citalopram by about 40 percent. Citalopram does not significantly affect the metabolism of, nor is its metabolism significantly affected by, digoxin, lithium, warfarin, carbamazepine, or imipramine (Tofranil). Citalopram increases the plasma concentrations of metoprolol (Lopressor) twofold, but this usually does not affect BP or heart rate. Data on coadministration of citalopram and potent inhibitors of CYP3A4 or CYP2D6 are not available.

Escitalopram. Escitalopram is a moderate inhibitor of CYP2D6 and has been shown to raise desipramine (Norpramin) and metoprolol concentrations significantly.

Vilazodone. Vilazodone dose should be reduced to 20 mg when

coadministered with CYP3A4 potent inhibitors. Concomitant use with inducers of CYP3A4 can result in inadequate drug concentrations and may diminish effectiveness. The effect of CYP3A4 inducers on the systemic exposure of vilazodone has not been evaluated.

Laboratory Interferences

In general, the SSRIs do not interfere with any laboratory tests. There have been some reports of false-positive urine toxicology for benzodiazepines in patients taking sertraline; if this is suspected, additional confirmation can be sought via gas chromatography-mass spectrometry.

Dosage and Clinical Guidelines

Fluoxetine. Fluoxetine is available in 10- and 20-mg capsules, in a scored 10-mg tablet, as a 90-mg enteric-coated capsule for once-weekly administration, and as an oral concentrate (20 mg/5 mL). Fluoxetine is also marketed as Sarafem for PMDD. For depression, the initial dosage is usually 10 or 20 mg orally each day, usually given in the morning, because insomnia is a potential adverse effect of the drug. Fluoxetine should be taken with food to minimize possible nausea. The long half-lives of the drug and its metabolite contribute to 4 weeks to reach steady-state concentrations. Twenty milligrams is often as effective as higher doses for treating depression. The maximum dosage recommended by the manufacturer is 80 mg a day. To minimize the early side effects of anxiety and restlessness, some clinicians initiate fluoxetine use at 5 to 10 mg a day, either with the scored 10-mg tablet or by using the liquid preparation. Alternatively, because of the long half-life of fluoxetine, its use can be initiated with an every-other-day administration schedule. The dosage of fluoxetine (and other SSRIs) that is effective in other indications may differ from the dosage generally used for depression.

Sertraline. Sertraline is available in scored 25-, 50-, and 100-mg tablets. For the initial treatment of depression, sertraline use should be initiated with a dosage of 50 mg once daily. To limit the GI effects, some clinicians begin at 25 mg a day and increase to 50 mg a day after 3 weeks. Patients who do not respond after 1 to 3 weeks may benefit from dosage increases of 50 mg every week up to a maximum of 200 mg given once daily. Sertraline can be

administered in the morning or in the evening. Administration after eating may reduce the GI adverse effects. Sertraline oral concentrate (1 mL = 20 mg) has 12 percent alcohol content and must be diluted before use. When used to treat panic disorder, sertraline should be initiated at 25 mg to reduce the risk of provoking a panic attack.

Paroxetine. Immediate-release paroxetine is available in scored 20-mg tablets; in unscored 10-, 30-, and 40-mg tablets; and as an orange-flavored 10-mg/5-mL oral suspension. Paroxetine use for the treatment of depression is usually initiated at a dosage of 10 or 20 mg a day. An increase in the dosage should be considered when an adequate response is not seen in 1 to 3 weeks. At that point, the clinician can initiate upward dose titration in 10-mg increments at weekly intervals to a maximum of 50 mg a day. Persons who experience GI upset may benefit from taking the drug with food. Paroxetine can be taken initially as a single daily dose in the evening; higher dosages may be divided into two doses per day.

A delayed-release formulation of paroxetine, paroxetine CR, is available in 12.5-, 25-, and 37.5-mg tablets. The starting dosages of paroxetine CR are 25 mg/day for depression and 12.5 mg/day for panic disorder.

Paroxetine is the SSRI most likely to produce a discontinuation syndrome, because plasma concentrations decrease rapidly in the absence of continuous dosing. Paroxetine use should be tapered gradually to limit the symptoms of abrupt discontinuation, with dosage reductions every 2 to 3 weeks.

Fluvoxamine. Fluvoxamine is the only SSRI not approved by the FDA as an antidepressant. It is indicated for OCD. It is available in unscored 25-mg tablets and scored 50- and 100-mg tablets. The effective daily dosage range is 50 to 300 mg a day. A usual starting dosage is 50 mg once a day at bedtime for the first week, after which the dosage can be adjusted according to the adverse effects and clinical response. Dosages above 100 mg a day may be divided into twice-daily dosing. A temporary dosage reduction or slower upward titration may be necessary if nausea develops over the first 2 weeks of therapy. Although fluvoxamine can also be administered as a single evening dose to minimize its adverse effects, its short half-life may lead to interdose withdrawal. An extended-release formulation is available in 100- and 150-mg dose strengths. All fluvoxamine formulations should be

swallowed with food without chewing the tablet. Abrupt discontinuation of fluvoxamine may cause a discontinuation syndrome owing to its short half-life.

Citalopram. Citalopram is available in 20- and 40-mg scored tablets and as a liquid (10 mg/5 mL). The usual starting dosage is 20 mg a day for the first week, after which it usually is increased to 40 mg a day. For elderly persons or persons with hepatic impairment, 20 mg a day is recommended. Tablets should be taken once daily in either the morning or the evening with or without food.

Escitalopram. Escitalopram is available as 10- and 20-mg scored tablets, as well as an oral solution at a concentration of 5 mg/5 mL. The recommended dosage of escitalopram is 10 mg/day. In clinical trials, no additional benefit was noted when 20 mg/day was used.

Vilazodone. Vilazodone is available as 10-, 20-, and 40-mg tablets. The recommended therapeutic dose of vilazodone is 40 mg once daily. Treatment should be titrated, starting with an initial dose of 10 mg once daily for 7 days, followed by 20 mg once daily for an additional 7 days, and then an increase to 40 mg once daily. Vilazodone should be taken with food. If vilazodone is taken without food, inadequate drug concentrations may result, and the drug's effectiveness may be diminished. Vilazodone is not approved for use in children. The safety and efficacy of vilazodone in pediatric patients have not been studied. The patient's age should not influence the dose, nor does mild or moderate hepatic impairment. Vilazodone has not been studied in patients with severe hepatic impairment. No dose adjustment is recommended in patients with mild, moderate, or severe renal impairment.

Pregnancy and Breastfeeding. Except for paroxetine, the SSRIs are safe to take during pregnancy when deemed necessary for the treatment of the mother. There are no controlled human data regarding vilazodone use during pregnancy, nor are there human data regarding drug concentrations in breast milk. Transient QTc prolongation has been noted in newborns whose mother was being treated with an SSRI during pregnancy.

Loss of Efficacy. Some patients report a diminished response or total loss of response to SSRIs with recurrence of depressive symptoms while remaining on a full dose of medication. The exact mechanism of this so-called “poop-out” is unknown, but the phenomenon is genuine. Potential remedies for the attenuation of the response to SSRIs include increasing or decreasing the dosage, tapering drug use, and then rechallenging with the same medication, switching to another SSRI or non-SSRI antidepressant, and augmenting with bupropion or another augmentation agent.

Vortioxetine (Trintellix). Vortioxetine works mainly as an inhibitor of serotonin (5-HT) reuptake, but it has a more complex pharmacologic profile than other SSRIs. It also acts as an agonist at 5-HT_{1A} receptors, a partial agonist at 5-HT_{1B} receptors, and an antagonist at 5-HT_3, 5-HT_{1D}, and 5-HT_7 receptors. The contribution of each of these activities to the drug’s antidepressant effect has not been established, but it is the only compound with this combination of pharmacodynamic actions.

Side effects seen during the trials include, but are not limited to, nausea, constipation, and vomiting.

The recommended starting dose is 10 mg administered orally once daily without regard to meals. The dose should then be increased to 20 mg/day, as tolerated. A dose of 5 mg/day should be considered for patients who do not tolerate higher doses.

The maximum recommended dose of vortioxetine is 10 mg/day in known CYP2D6 poor metabolizers. Reduction of the dose of vortioxetine by one-half is suggested when patients are receiving a CYP2D6 potent inhibitor (e.g., bupropion, fluoxetine, paroxetine, or quinidine) concomitantly. The dose should be increased to the original dose of vortioxetine in patients who stop taking CYP inducers (e.g., rifampin, carbamazepine, or phenytoin). This adjustment is especially essential when a potent CYP inducer is coadministered for greater than 14 days. The maximum recommended dose should not exceed three times the original dose. The dose of vortioxetine should be reduced to the original level within 14 days when the inducer is discontinued.

Although vortioxetine can be abruptly discontinued, in placebo-controlled trials, patients experienced transient adverse reactions such as headache and

muscle tension following abrupt discontinuation of vortioxetine 15 or 20 mg/day. It is recommended that the dose be decreased to 10 mg/day for 1 week before full discontinuation of vortioxetine 15 or 20 mg/day.

Vortioxetine is available in 5-, 10-, 15-, and 20-mg tablets.

SELECTIVE SEROTONIN–NOREPINEPHRINE REUPTAKE INHIBITORS

There are currently four SNRIs approved for use in the United States: venlafaxine (Effexor and Effexor XR), desvenlafaxine succinate (DVS; Pristiq), duloxetine (Cymbalta), and levomilnacipran (Fetzima). A fifth SNRI, milnacipran (Savella), available in other countries as an antidepressant, has U.S. FDA approval in the United States as a treatment for fibromyalgia. The term SNRI reflects the belief that the therapeutic effects of these medications are mediated by concomitant blockade of neuronal serotonin (5-HT) and norepinephrine uptake transporters. The SNRIs are also sometimes referred to as dual reuptake inhibitors, a broader functional class of antidepressant medications that include TCAs such as clomipramine (Anafranil) and, to a lesser extent, imipramine (Tofranil) and amitriptyline (Elavil). What distinguishes the SNRIs from TCAs is their relative lack of affinity for other receptors, especially muscarinic, histaminergic, and the families of α- and β-adrenergic receptors. This distinction is an important one because the SNRIs have a more favorable tolerability profile than the older dual reuptake inhibitors.

Venlafaxine and Desvenlafaxine

Therapeutic Indications. Venlafaxine is approved for the treatment of four disorders: MDD, GAD, social anxiety disorder, and panic disorder. MDD is currently the only FDA-approved indication for DVS.

DEPRESSION. The FDA does not recognize any class of antidepressants as being more effective than any other. This does not mean that differences do not exist, but no study to date has sufficiently demonstrated such superiority. It has been argued that direct modulation of serotonin and norepinephrine may convey more significant antidepressant effects than are exerted by

medications that selectively enhance only noradrenergic or serotoninergic neurotransmission. This more significant therapeutic benefit could result from an acceleration of postsynaptic adaptation to increased neuronal signaling, simultaneous activation of two pathways for intracellular signal transduction, additive effects on the activity of relevant genes such as brain-derived neurotrophic factor, or, quite simply, broader coverage of depressive symptoms. Clinical evidence supporting this hypothesis first emerged in a pair of studies conducted by the Danish University Antidepressant Group, which found an advantage for the dual reuptake inhibitor clomipramine compared with SSRIs citalopram (Celexa) and paroxetine (Paxil). Another report, which compared the results of a group of patients prospectively treated with the combination of desipramine (Norpramin) and fluoxetine (Prozac) with a historical comparison group treated with desipramine alone, provided additional support. A meta-analysis of 25 inpatient studies comparing the efficacy of TCAs and SSRIs yielded the most substantial evidence. Specifically, although the TCAs were found to have a modest overall advantage, superiority versus SSRIs was almost entirely explained by the studies that used the TCAs that are considered to be dual reuptake inhibitors—clomipramine, amitriptyline, and imipramine. Meta-analyses of head-to-head studies suggest that venlafaxine has the potential to induce higher rates of remission in depressed patients than do the SSRIs. This difference in the venlafaxine advantage is about 6 percent. DVS has not been extensively compared with other classes of antidepressants concerning efficacy.

GENERALIZED ANXIETY DISORDER. The extended-release formulation of venlafaxine is approved for the treatment of GAD. In clinical trials lasting 6 months, dosages of 75 to 225 mg a day were effective in treating insomnia, poor concentration, restlessness, irritability, and excessive muscle tension related to GAD.

SOCIAL ANXIETY DISORDER. The extended-release formulation of venlafaxine is approved for the treatment of social anxiety disorder. Its efficacy was established in 12-week studies.

OTHER INDICATIONS. Case reports and uncontrolled studies have indicated that

venlafaxine may be beneficial in the treatment of obsessive-compulsive disorder, panic disorder, agoraphobia, social phobia, ADHD, and patients with a dual diagnosis of depression and cocaine dependence. It has also been used in chronic pain syndromes with good effect.

Precautions and Adverse Reactions. Venlafaxine has a safety and tolerability profile similar to that of the more widely prescribed SSRI class. Nausea is the most frequently reported treatment-emergent adverse effect associated with venlafaxine and DVS therapy. Initiating therapy at lower dosages may also attenuate nausea. When extremely problematic, treatment-induced nausea can be controlled by prescribing a selective 5-HT_3 antagonist or mirtazapine (Remeron).

Venlafaxine and DVS therapy is associated with sexual side effects, predominantly decreased libido and a delay to orgasm or ejaculation. The incidence of these side effects may exceed 30 to 40 percent when there is a direct, detailed assessment of sexual function.

Other common side effects include headache, insomnia, somnolence, dry mouth, dizziness, constipation, asthenia, sweating, and nervousness. Although several side effects are suggestive of anticholinergic effects, these drugs have no affinity for muscarinic or nicotinic receptors. Thus, noradrenergic agonism is likely to be the culprit.

Higher-dose venlafaxine therapy is associated with an increased risk of sustained elevations of BP. Experience with the instant-release (IR) formulation in studies of depressed patients indicated that sustained hypertension was dose-related, increasing from 3 to 7 percent at doses of 100 to 300 mg/day and 13 percent at doses greater than 300 mg/day. In this dataset, venlafaxine therapy did not adversely affect BP control of patients taking antihypertensives and lowered the mean values of patients with elevated BP readings before therapy. In controlled studies of the extended-release formulation, venlafaxine therapy resulted in only approximately 1 percent greater risk of high BP when compared with the placebo. Arbitrarily capping the upper dose of venlafaxine used in these studies, thus, significantly attenuated concerns about elevated BP. When higher doses of the extended-release formulation are used, however, monitoring of BP is recommended.

Venlafaxine and DVS are commonly associated with discontinuation

syndrome. This syndrome is characterized by the appearance of a constellation of adverse effects during a rapid taper or abrupt cessation, including dizziness, dry mouth, insomnia, nausea, nervousness, sweating, anorexia, diarrhea, somnolence, and sensory disturbances. It is recommended that, whenever possible, a slow taper schedule should be used when longer-term treatment must be stopped. On occasion, substituting a few doses of the sustained-release formulation of fluoxetine may help to bridge this transition.

There were no overdose fatalities in premarketing trials of venlafaxine, although electrocardiographic changes (e.g., prolongation of QT interval, bundle branch block, QRS interval prolongation), tachycardia, bradycardia, hypotension, hypertension, coma, serotonin syndrome, and seizures were reported. Fatal overdoses have been documented subsequently, typically involving venlafaxine ingestion in combination with other drugs, alcohol, or both.

Information concerning the use of venlafaxine and DVS by pregnant and nursing women is not available at this time. Venlafaxine and DVS are excreted in breast milk. Clinicians should carefully weigh the risks and benefits of venlafaxine use by pregnant and nursing women.

Drug Interactions. Venlafaxine is metabolized in the liver primarily by the CYP2D6 isoenzyme. Because the parent drug and principal metabolite are essentially equipotent, medications that inhibit this isoenzyme usually do not adversely affect therapy. Venlafaxine is itself a relatively weak inhibitor of CYP2D6, although it can increase levels of substrates, such as desipramine or risperidone (Risperdal). In vitro and in vivo studies have shown venlafaxine to cause little or no inhibition of CYP1A2, CYP2C9, CYP2C19, and CYP3A4.

Venlafaxine is contraindicated in patients taking MAOIs because of the risk of a pharmacodynamic interaction (i.e., serotonin syndrome). An MAOI should not be started for at least 7 days after stopping venlafaxine. Few data are available regarding the combination of venlafaxine with atypical neuroleptics, benzodiazepines, lithium (Eskalith), and anticonvulsants; therefore, clinical judgment should be exercised when combining medications.

Laboratory Interferences. Data are not currently available on laboratory

interferences with venlafaxine. There have been some reports of patients taking venlafaxine who have false-positive results on liquid chromatography testing for tramadol.

Dosage and Administration. Venlafaxine is available in 25-, 37.5-, 50-, 75-, and 100-mg tablets and 37.5-, 75-, and 150-mg extended-release capsules. The tablets and the extended-release capsules are equally potent, and persons stabilized with one can switch to an equivalent dosage of the other. Because the immediate-release tablets are rarely used due to their tendency to cause nausea and the need for multiple-daily doses, the dosage recommendations that follow refer to the use of the extended-release capsules.

In depressed persons, venlafaxine demonstrates a dose–response curve. The initial therapeutic dosage is 75 mg a day, given once a day. However, most persons are started at a dosage of 37.5 mg for 4 to 7 days to minimize adverse effects, particularly nausea. A convenient starter kit for the drug contains a 1-week supply of both the 37.5- and 75-mg strengths. If a rapid titration is preferred, the dosage can be raised to 150 mg/day after day 4. As a rule, the dosage can be raised in increments of 75 mg a day every 4 or more days. Although the recommended upper dosage of the extended-release preparation (venlafaxine XR) is 225 mg/day, it is approved by the FDA for use at dosages up to 375 mg a day. The dosage of venlafaxine should be halved in persons with significantly diminished hepatic or renal function. If discontinued, venlafaxine use should be gradually tapered over 2 to 4 weeks to avoid withdrawal symptoms.

There are minor differences in the doses used for major depression, GAD, and social anxiety disorder. In the treatment of these disorders, for example, a dose–response effect has not been found. Also, lower mean dosages are typically used, with most patients taking 75 to 150 mg/day.

DVS is available as 50- and 100-mg extended-release tablets. The therapeutic dose for most patients is 50 mg a day. Although some patients may need higher doses, in clinical trials, no more significant therapeutic benefit was noted when the dose was increased. At higher doses, adverse events and discontinuation rates were increased.

Duloxetine

Pharmacologic Actions. Duloxetine is formulated as a delayed-release capsule to reduce the risk of severe nausea associated with the drug. It is well absorbed, but there is a 2-hour delay before absorption begins. Peak plasma concentrations occur 6 hours after ingestion. Food delays the time to achieve maximum concentrations from 6 to 10 hours and reduces the extent of absorption by about 10 percent. Duloxetine has an elimination half-life of about 12 hours (range, 8 to 17 hours). Steady-state plasma concentrations occur after 3 days. Duloxetine's elimination is mainly through the isozymes CYP2D6 and CYP1A2. Duloxetine undergoes extensive hepatic metabolism to numerous metabolites. About 70 percent of the drug appears in the urine as metabolites, and about 20 percent is excreted in the feces. Duloxetine is 90 percent protein-bound.

Therapeutic Indications

DEPRESSION. In contrast to venlafaxine, a small number of studies have compared duloxetine with the SSRIs. Although these studies are suggestive of some advantage in efficacy, their findings are limited by the use of fixed, low starting doses of paroxetine and fluoxetine, but dosages of duloxetine in some studies were as high as 120 mg/day. Any inferences on whether duloxetine is superior to the SSRIs in any aspect of treatment for depression thus await more evidence from adequately designed trials.

NEUROPATHIC PAIN ASSOCIATED WITH DIABETES AND STRESS URINARY INCONTINENCE. Duloxetine is the first drug to be approved by the FDA as a treatment for neuropathic pain associated with diabetes. Its pain effects have not been compared with other agents, such as venlafaxine and the TCAs. Duloxetine is currently awaiting approval as a treatment for stress urinary incontinence, the inability to voluntarily control bladder voiding, which is the most frequent type of incontinence in women. The action of duloxetine in the treatment of stress urinary incontinence is associated with its effects in the sacral spinal cord, which in turn increases the activity of the striated urethral sphincter. Duloxetine is marketed under the name Yentreve for this indication which has not been approved for this indication in the United States but is available in the European Union and other countries.

PRECAUTIONS AND ADVERSE REACTIONS. The most common adverse reactions are nausea, dry mouth, dizziness, constipation, fatigue, decreased appetite, anorexia, somnolence, and increased sweating. Nausea was the most common side effect that led to treatment discontinuation in clinical trials. The true incidence of sexual dysfunction is unknown; the long-term effects on body weight are also unknown. In clinical trials, treatment with duloxetine was associated with mean increases in BP averaging 2 mm Hg systolic and 0.5 mm Hg diastolic versus placebo. No studies have compared the BP effects of venlafaxine and duloxetine at equivalent therapeutic doses.

Close monitoring is suggested when using duloxetine in patients who have or are at risk for diabetes. Duloxetine has been shown to increase blood sugar and hemoglobin A1C levels during long-term treatment.

Patients with substantial alcohol use should not be treated with duloxetine because of possible hepatic effects. It also should not be prescribed for patients with hepatic insufficiency and end-stage renal disease or patients with uncontrolled narrow-angle glaucoma.

Abrupt discontinuation of duloxetine should be avoided because it may produce a discontinuation syndrome similar to that of venlafaxine. A gradual dose reduction is recommended.

Clinicians should avoid the use of duloxetine by pregnant and nursing women unless the potential benefits justify the potential risks.

Drug Interactions. Duloxetine is a moderate inhibitor of CYP450 enzymes.

Laboratory Interferences. Data are not currently available on laboratory interferences with duloxetine.

Dosage and Administration. Duloxetine is available in 20-, 30-, and 60-mg tablets. The recommended therapeutic and maximum dosage is 60 mg/day. The 20- and 30-mg doses are useful for either initial therapy or for twice-daily use as strategies to reduce side effects. In clinical trials, dosages of up to 120 mg/day were studied, but no consistent advantage in efficacy was noted at doses higher than 60 mg/day. Duloxetine thus does not appear to demonstrate a dosage–response curve. However, there were difficulties in tolerability with single doses above 60 mg. Accordingly, when dosages of 80

and 120 mg/day were used, they were administered as 40 or 60 mg twice daily. It remains to be seen to what extent dosages above 60 mg/day will be necessary and whether this will require divided doses to make the drug tolerable.

Milnacipran and Levomilnacipran

Milnacipran is only FDA approved for the treatment of fibromyalgia. Although some countries have approved milnacipran for general use as an antidepressant, efficacy is not as well established. Compared with venlafaxine, milnacipran is approximately five times more potent for inhibition of norepinephrine uptake than for 5-HT reuptake inhibition. Milnacipran has a half-life of approximately 8 hours and shows linear pharmacokinetics between doses of 50 and 250 mg/day. Milnacipran has no active metabolites, and it is directly metabolized in the liver. The kidneys primarily excrete milnacipran.

Milnacipran is available as 12.5-, 25-, 50-, and 100-mg tablets. The standard recommended milnacipran dose is as follows: day 1, 12.5 mg once daily; days 2 and 3, 12.5 mg twice daily; days 4 to 7, 25 mg twice daily; and day 7 and beyond, 50 mg twice daily.

Levomilnacipran was approved in 2013 by the FDA as a treatment for MDD in adults. Levomilnacipran is an active enantiomer of the racemic drug milnacipran. In vitro studies have shown that it has greater potency for norepinephrine reuptake inhibition than for serotonin reuptake inhibition and does not directly affect the uptake of dopamine or other neurotransmitters. It is taken once daily as a sustained-release formulation. In clinical trials, doses of 40, 80, or 120 mg improved symptoms compared with placebo.

The most common adverse reactions in the placebo-controlled trials were nausea, constipation, hyperhidrosis, increased heart rate, erectile dysfunction, tachycardia, vomiting, and palpitations. Rates of adverse events were generally consistent across the 40- to 120-mg dose range. The only dose-related adverse events were urinary hesitation and erectile dysfunction.

BUPROPION

Bupropion (Wellbutrin, Wellbutrin SR, Wellbutrin XL) is an antidepressant drug that inhibits the reuptake of norepinephrine and, possibly, dopamine.

Most significantly, it does not act on the serotonin system like SSRI antidepressants. This novel action results in a side effect profile characterized by a low risk of sexual dysfunction and sedation and with modest weight loss during acute and long-term treatment. No withdrawal syndrome has been linked to discontinuation of bupropion. Although increasingly used as first-line monotherapy, a significant percentage of bupropion use occurs as add-on therapy to other antidepressants, usually SSRIs. Bupropion was marketed under the name Zyban for use in smoking cessation regimens. However, the manufacturer has discontinued that brand in the United States.

Pharmacologic Actions

Three formulations of bupropion are available: immediate-release (taken three times daily), sustained-release (taken twice daily), and extended-release (taken once daily). The different versions of the drug contain the same active ingredient but differ in their pharmacokinetics and dosing. There have been reports of inconsistencies in bioequivalence between various branded and generic versions of bupropion. If a patient experiences changes in adverse effects or efficacy, the clinician should inquire whether there has been a change of formulation.

Immediate-release bupropion is well absorbed from the GI tract. Peak plasma concentrations of bupropion are usually reached within 2 hours of oral administration, and peak levels of the sustained-release version are seen after 3 hours. The mean half-life of the compound is 12 hours, ranging from 8 to 40 hours. Peak levels of extended-release bupropion occur 5 hours after ingestion. This pharmacokinetic profile provides a longer time to maximum plasma concentration (t_{max}) but comparable peak and trough plasma concentrations. The 24-hour exposure occurring after administration of the extended-release version of 300 mg once daily is equivalent to that provided by sustained release of 150 mg twice daily. Clinically, this permits the drug to be taken once a day in the morning. Plasma levels are also reduced in the evening, making it less likely for some patients to experience treatment-related insomnia.

The mechanism of action for the antidepressant effects of bupropion is presumed to involve the inhibition of dopamine and norepinephrine reuptake. Bupropion binds to the dopamine transporter in the brain. The effects of

bupropion on smoking cessation may be related to its effects on dopamine reward pathways or inhibition of nicotinic acetylcholine receptors.

Therapeutic Indications

Depression. Although overshadowed by the SSRIs as first-line treatment for major depression, the therapeutic efficacy of bupropion in depression is well established in both outpatient and inpatient settings. Observed rates of response and remission are comparable to those seen with the SSRIs. Bupropion has been found to prevent seasonal major depressive episodes in patients with a history of seasonal patterns or affective disorder.

Smoking Cessation. Bupropion is indicated for use in combination with behavioral modification programs for smoking cessation. It is intended to be used in patients who are highly motivated and who receive some form of structured behavioral support. Bupropion is most effective when combined with nicotine substitutes (Nico-Derm, Nicotrol).

Bipolar Disorders. Bupropion is less likely than TCAs to precipitate mania in persons with bipolar I disorder and less likely than other antidepressants to exacerbate or induce rapid-cycling bipolar II disorder; however, the evidence about the use of bupropion in the treatment of patients with bipolar disorder is limited.

Attention-Deficit/Hyperactivity Disorder. Bupropion is used as a second-line agent, after the sympathomimetics, for treatment of ADHD. It has not been compared with proven ADHD medications such as methylphenidate (Ritalin) or atomoxetine (Strattera) for childhood and adult ADHD. Bupropion is an appropriate choice for persons with comorbid ADHD and depression or persons with comorbid ADHD, conduct disorder, or substance use disorder. It may also be considered for use in patients who develop tics when treated with psychostimulants.

Cocaine Detoxification. Bupropion may be associated with a euphoric feeling; thus, it may be contraindicated in persons with histories of substance abuse. However, because of its dopaminergic effects, bupropion has been explored as a treatment for cocaine cravings. Results have been inconclusive,

with some patients showing a reduction in drug craving and others finding their cravings increased.

Hypoactive Sexual Desire Disorder. Bupropion is often added to drugs such as SSRIs to counteract sexual side effects and may be helpful as a treatment for nondepressed individuals with a disorder of sexual desire. Bupropion may improve sexual arousal, orgasm completion, and sexual satisfaction.

Precautions and Adverse Reactions

Headache, insomnia, dry mouth, tremor, and nausea are the most common side effects. Restlessness, agitation, and irritability may also occur. Patients with severe anxiety or panic disorder should not be prescribed bupropion. Most likely, because of its potentiating effects on dopaminergic neurotransmission, bupropion can cause psychotic symptoms, including hallucinations, delusions, and catatonia, as well as delirium. Most notable about bupropion is the absence of significant drug-induced orthostatic hypotension, weight gain, daytime drowsiness, and anticholinergic effects. Some persons, however, may experience dry mouth or constipation and weight loss. Hypertension may occur in some patients, but bupropion causes no other significant cardiovascular or clinical laboratory changes. Bupropion exerts indirect sympathomimetic activity, producing positive inotropic effects in the human myocardium, an effect that may reflect catecholamine release. Some patients experience cognitive impairment, most notably word-finding difficulties.

Concern about seizure has deterred some physicians from prescribing bupropion. The risk of seizure is dose-dependent. Studies show that at dosages of 300 mg a day or less of sustained-
release bupropion, the incidence of seizures is 0.05 percent, which is no worse than the incidence of seizures with other antidepressants. The risk of seizures increases to about 0.1 percent with dosages of 400 mg a day.

Changes in electroencephalographic (EEG) waveforms have been reported to be associated with bupropion use. About 20 percent of individuals treated with bupropion exhibit spike waves, sharp waves, and focal slowing. The likelihood of women having sharp waves is higher than for men. The

presence of these waveforms in individuals taking a medication known to lower the seizure threshold may be a risk factor for developing seizures. Other risk factors for seizures include a history of seizures, use of alcohol, recent benzodiazepine withdrawal, organic brain disease, head trauma, or pretreatment epileptiform discharges on EEG.

The use of bupropion by pregnant women is not associated with a specific risk of an increased rate of congenital disabilities. Bupropion is secreted in breast milk, so the use of bupropion in nursing women should be based on the clinical circumstances of the patient and the judgment of the clinician.

Few deaths have been reported after overdoses of bupropion. Poor outcomes are associated with cases of massive doses and mixed-drug overdoses. Seizures occur in about one-third of all overdoses and are dose-dependent, with those having seizures ingesting a significantly higher median dose. Fatalities can involve uncontrollable seizures, sinus bradycardia, and cardiac arrest. Symptoms of poisoning most often involve seizures, sinus tachycardia, hypertension, GI symptoms, hallucinations, and agitation. All seizures are typically brief and self-limited. In general, however, bupropion is safer in overdose cases than are other antidepressants except for perhaps SSRIs.

Drug Interactions

Given the fact that bupropion is frequently combined with SSRIs or venlafaxine, potential interactions are significant. Bupropion has been found to affect the pharmacokinetics of venlafaxine. One study noted a significant increase in venlafaxine levels and a consequent decrease in its main metabolite O-desmethylvenlafaxine during combined treatment with sustained-release bupropion. Bupropion hydroxylation is weakly inhibited by venlafaxine. No significant changes in plasma levels of the SSRIs paroxetine and fluoxetine have been reported. However, few case reports indicate that the combination of bupropion and fluoxetine (Prozac) may be associated with panic, delirium, or seizures. Bupropion, in combination with lithium (Eskalith), may rarely cause CNS toxicity, including seizures.

Because of the possibility of inducing a hypertensive crisis, bupropion should not be used concurrently with MAOIs. At least 14 days should pass after the discontinuation of an MAOI before initiating treatment with

bupropion. In some cases, the addition of bupropion may permit persons taking antiparkinsonian medications to lower the doses of their dopaminergic drugs. However, delirium, psychotic symptoms, and dyskinetic movements may occur with the coadministration of bupropion and dopaminergic agents, such as antiparkinsonian meds. Sinus bradycardia may occur when bupropion is combined with metoprolol.

Carbamazepine (Tegretol) may decrease plasma concentrations of bupropion, and bupropion may increase plasma concentrations of valproic acid (Depakene).

In vitro biotransformation studies of bupropion have found that the formation of a major active metabolite, hydroxybupropion, is mediated by CYP2B6. Bupropion has a significant inhibitory effect on CYP2D6.

Laboratory Interferences

A report has appeared indicating that bupropion may give a false-positive result on urinary amphetamine screens. No other reports have appeared of laboratory interferences associated with bupropion treatment. Clinically nonsignificant changes in the electrocardiogram (premature beats and nonspecific ST-T changes) and decreases in the WBC count (by about 10 percent) have been reported in a small number of persons.

Dosage and Clinical Guidelines

Immediate-release bupropion is available in 75-, 100-, and 150-mg tablets. Sustained-release bupropion is available in 100-, 150-, 200-, and 300-mg tablets. Extended-release bupropion comes in 150- and 300-mg strengths.

There have been problems associated with one of the extended-release generic versions called Budeprion XL 300-mg tablets, which was found not to be therapeutically equivalent to Wellbutrin XL 300 mg and was removed from the market.

The initiation of immediate-release bupropion in the average adult person should be 75 mg orally twice a day. On the fourth day of treatment, the dosage can be increased to 100 mg three times a day. Because 300 mg is the recommended dose, the person should be maintained on this dose for several weeks before increasing it further. The maximum dosage, 450 mg a day, should be given as 150 mg three times a day. Because of the risk of seizures,

increases in dose should never exceed 100 mg in 3 days; a single dose of immediate-release bupropion should never exceed 150 mg, and the total daily dosage should not exceed 450 mg. The maximum of 400 mg of the sustained-release version should be used as a twice-a-day regimen of either 200 mg twice daily or 300 mg in the morning and 100 mg in the afternoon. A starting dosage of the sustained-release version, 100 mg once a day, can be increased to 100 mg twice a day after 4 days. Then 150 mg twice a day may be used. A single dose of sustained-release bupropion should never exceed 300 mg. The maximum dosage is 200 mg twice a day of the immediate-release or extended-release formulations. An advantage of the extended-release preparation is that the patient can take a single daily dose of 450 mg after proper titration.

For smoking cessation, the patient should start taking 150 mg a day of sustained-release bupropion 10 to 14 days before quitting smoking. On the fourth day, the dosage should be increased to 150 mg twice daily. Treatment generally lasts 7 to 12 weeks.

MIRTAZAPINE

Mirtazapine (Remeron) is unique among drugs used to treat major depression in that it increases both norepinephrine and serotonin through a mechanism other than reuptake blockade (as in the case of tricyclic agents or SSRIs) or monoamine oxidase inhibition (as in the case of phenelzine or tranylcypromine). Mirtazapine is also more likely to reduce rather than cause nausea and diarrhea, the result of its effects on serotonin 5-HT_3 receptors. Characteristic side effects include increased appetite and sedation.

Pharmacologic Actions

Mirtazapine is administered orally and is rapidly and completely absorbed. It has a half-life of about 30 hours. Peak concentration is achieved within 2 hours of ingestion, and steady state is reached after 6 days. Plasma clearance may be slowed up to 30 percent in persons with impaired hepatic function, up to 50 percent in those with impaired renal function, up to 40 percent slower in older men, and up to 10 percent slower in older women.

The mechanism of action of mirtazapine is the antagonism of central

presynaptic α_2-adrenergic receptors and blockade of postsynaptic serotonin 5-HT_2 and 5-HT_3 receptors. The α_2-adrenergic receptor antagonism causes the increased firing of norepinephrine and serotonin neurons. The potent antagonist of serotonin 5-HT_2 and 5-HT_3 receptors serves to decrease anxiety, relieve insomnia, and stimulate the appetite. Mirtazapine is a potent antagonist of histamine H_1 receptors and is a moderately potent antagonist at α_1-adrenergic and muscarinic-cholinergic receptors.

Therapeutic Indications

Mirtazapine is effective for the treatment of depression. It is highly sedating, making it a reasonable choice for use in depressed patients with severe or long-standing insomnia. Some patients find the residual daytime sedation associated with the initiation of treatment to be quite pronounced. However, the more extreme sedating properties of the drug generally lessen over the first week of treatment. Combined with the tendency to sometimes cause a ravenous appetite, mirtazapine is well suited for depressed patients with melancholic features such as insomnia, weight loss, and agitation. Elderly depressed patients, in particular, are good candidates for mirtazapine; young adults are more likely to object to this side effect profile.

Mirtazapine's blockade of 5-HT_3 receptors, a mechanism associated with medications used to combat the severe GI side effects of cancer chemotherapy agents, has led to the use of the drug in a similar role. In this population, sedation and stimulation of appetite clearly could be seen as being beneficial instead of unwelcome side effects.

Mirtazapine is often combined with SSRIs or venlafaxine to augment antidepressant response or counteract serotonergic side effects of those drugs, particularly nausea, agitation, and insomnia. Mirtazapine has no significant pharmacokinetic interactions with other antidepressants.

Precautions and Adverse Reactions

Somnolence, the most common adverse effect of mirtazapine, occurs in more than 50 percent of persons (Table 21-18). Persons starting mirtazapine should thus exercise caution when driving or operating dangerous machinery and even when getting out of bed at night. This adverse effect is why mirtazapine

is almost always given before sleep. Mirtazapine potentiates the sedative effects of other CNS depressants, so potentially sedating prescription or over-the-counter drugs and alcohol should be avoided during the use of mirtazapine. Mirtazapine also causes dizziness in 7 percent of persons. It does not appear to increase the risk of seizures. Mania or hypomania occurred in clinical trials at a rate similar to that of other antidepressant drugs.

Mirtazapine increases appetite in about one-third of patients. Mirtazapine may also increase serum cholesterol concentration to 20 percent or more above the upper limit of normal in 15 percent of persons and increase triglycerides to 500 mg/dL or more in 6 percent of persons. Elevations of alanine transaminase levels to more than three times the upper limit of normal were seen in 2 percent of mirtazapine-treated persons as opposed to 0.3 percent of placebo control subjects.

In limited premarketing experience, the absolute neutrophil count dropped to 500/mm^3 or less within 2 months of the onset of use in 0.3 percent of persons, some of whom developed symptomatic infections. This hematologic condition was reversible in all cases and was more likely to occur when other risk factors for neutropenia were present. Increases in the frequency of neutropenia have not, however, been reported during the extensive postmarketing period. Persons who develop fever, chills, sore throat, mucous membrane ulceration, or other signs of infection should nevertheless be evaluated medically. If a low WBC count is found, mirtazapine should be immediately discontinued, and the infectious disease status should be followed closely.



Table 21-18
Adverse Reactions Reported with Mirtazapine

Event	Patients (%)
Somnolence	54
Dry mouth	25
Increased appetite	17
Constipation	13

Weight gain	12
Dizziness	7
Myalgias	5
Disturbing dreams	4

A small number of persons experience orthostatic hypotension while taking mirtazapine. Although no data exist regarding the effects on fetal development, mirtazapine should be used with caution during pregnancy.

Mirtazapine use by pregnant women has not been studied, but because the drug may be excreted in breast milk, it should not be taken by nursing mothers. Because of the risk of agranulocytosis associated with mirtazapine use, persons should be attuned to signs of infection. Because of the sedating effects of mirtazapine, persons should determine the degree to which they are affected before engaging in driving or other potentially dangerous activities.

Drug Interactions

Mirtazapine can potentiate the sedation of alcohol and benzodiazepines. Mirtazapine should not be used within 14 days of use of an MAOI.

Laboratory Interferences

No laboratory interferences have yet been described for mirtazapine.

Dosage and Administration

Mirtazapine is available in 15-, 30-, and 45-mg scored tablets. Mirtazapine is also available in 15-, 30-, and 45-mg orally disintegrating tablets for persons who have difficulty swallowing pills. If persons fail to respond to the initial dose of 15 mg of mirtazapine before sleep, the dose may be increased in 15-mg increments every 5 days to a maximum of 45 mg before sleep. Lower dosages may be necessary for elderly persons or persons with renal or hepatic insufficiency.

NEFAZODONE AND TRAZODONE

Nefazodone (Serzone) and Trazodone (Desyrel) are mechanistically and

structurally related drugs approved as treatments for depression. Nefazodone (Serzone) is an analog of trazodone. When nefazodone was introduced in 1995, there were expectations that it would become widely used because it did not cause the sexual side effects and sleep disruption associated with the selective SSRIs. Although it was devoid of these side effects, it was nevertheless found to produce problematic sedation, nausea, dizziness, and visual disturbances. Consequently, nefazodone was never extensively adopted in clinical practice. This fact, as well as reports of rare cases of sometimes fatal hepatotoxicity, led the original manufacturer to discontinue the production of branded nefazodone in 2004. Generic nefazodone remains available in the United States.

Trazodone received FDA approval in 1981 as a treatment for MDD. Its novel triazolopyridine chemical structure distinguished it from the TCAs, and clinical trials suggested improved safety and tolerability compared with TCAs. There were high expectations that it would replace the older drugs as a mainstay of treatment for depression. However, the extreme sedation associated with trazodone, even at subtherapeutic doses, limited the clinical effectiveness of the drug. However, its soporific properties made trazodone a favorite alternative to standard hypnotics as a sleep-inducing agent. Unlike conventional sleeping pills, trazodone is not a controlled substance.

Nefazodone

Pharmacologic Actions. Nefazodone is rapidly and completely absorbed but is then extensively metabolized so that the bioavailability of active compounds is about 20 percent of the oral dose. Its half-life is 2 to 4 hours. Steady-state concentrations of nefazodone and its principal active metabolite, hydroxynefazodone, are achieved within 4 to 5 days. Metabolism of nefazodone in elderly persons, especially women, is about half of that seen in younger persons, so lowered doses are recommended for elderly persons. An important metabolite of nefazodone is meta-chlorophenylpiperazine (mCPP), which has some serotonergic effects and may cause migraine, anxiety, and weight loss.

Although nefazodone is an inhibitor of serotonin uptake and, more weakly, of norepinephrine reuptake, its antagonism of serotonin 5-HT_A receptors is thought to produce its antianxiety and antidepressant effects. Nefazodone is

also a mild antagonist of the α_1-adrenergic receptors, which predisposes some persons to orthostatic hypotension but is not sufficiently potent to produce priapism.

Therapeutic Indications. Nefazodone is effective for the treatment of major depression. The usual effective dosage is 300 to 600 mg a day. In direct comparison with SSRIs, nefazodone is less likely to cause inhibition of orgasm or decreased sexual desire. Nefazodone is also effective for the treatment of panic disorder and panic with comorbid depression or depressive symptoms, GAD, and PMDD and management of chronic pain. It is not useful for the treatment of OCD. Nefazodone increases rapid eye movement (REM) sleep and increases sleep continuity. Nefazodone is also of use in patients with PTSD and chronic fatigue syndrome. It may also be effective in patients who have been treatment-resistant to other antidepressant drugs.

Precautions and Adverse Reactions. The most common reasons for discontinuing nefazodone use are sedation, nausea, dizziness, insomnia, weakness, and agitation (Table 21-19). Many patients report no specific side effects but describe a vague sense of feeling medicated. Nefazodone also causes visual trails, in which patients see an afterimage when looking at moving objects or when moving their heads quickly.

A significant safety concern with the use of nefazodone is a severe elevation of hepatic enzymes, and, in some instances, liver failure. Accordingly, serial hepatic function tests need to be done when patients are treated with nefazodone. Hepatic effects can be seen early in treatment and are more likely to develop when nefazodone is combined with other drugs metabolized in the liver.

Some patients taking nefazodone may experience a decrease in BP that can cause episodes of postural hypotension. Nefazodone should therefore be used with caution by persons with underlying cardiac conditions or history of stroke or heart attack, dehydration, or hypovolemia or by persons being treated with antihypertensive medications. Patients switched from SSRIs to nefazodone may experience an increase in side effects, possibly because nefazodone does not protect against SSRI withdrawal symptoms. One of its metabolites, mCPP, may intensify these discontinuation symptoms. Patients have survived nefazodone overdoses above 10 g, but deaths have been

reported when it has been combined with alcohol. Nausea, vomiting, and somnolence are the most common signs of toxicity.

Table 21-19
Adverse Reactions Reported with Nefazodone (300–600 mg a day)

Reaction	Patients (%)
Headache	36
Dry mouth	25
Somnolence	25
Nausea	22
Dizziness	17
Constipation	14
Insomnia	11
Weakness	11
Lightheadedness	10
Blurred vision	9
Dyspepsia	9
Infection	8
Confusion	7
Scotomata	7

There are few studies or clinical reports of nefazodone during pregnancy. Nefazodone should, therefore, be used during pregnancy only if the potential benefit to the mother outweighs the potential risks to the fetus. It is not known whether nefazodone is excreted in human breast milk. Therefore, it should be used with caution by lactating mothers. The nefazodone dosage should be lowered in persons with severe hepatic disease, but no adjustment

is necessary for persons with renal disease.

Drug Interactions and Laboratory Interferences. Nefazodone should not be given concomitantly with MAOIs. Also, nefazodone has particular drug–drug interactions with the triazolobenzodiazepines triazolam (Halcion) and alprazolam (Xanax) because of the inhibition of CYP3A4 by nefazodone. Potentially elevated levels of each of these drugs can develop after administration of nefazodone, but the levels of nefazodone are generally not affected. The dose of triazolam should be lowered by 75 percent, and the dose of alprazolam should be lowered by 50 percent, when given concomitantly with nefazodone.

Nefazodone may slow the metabolism of digoxin; therefore, digoxin levels should be monitored carefully in persons taking both medications. Nefazodone also slows the metabolism of haloperidol (Haldol) so that the dosage of haloperidol should be reduced in persons taking both medications. The addition of nefazodone may also exacerbate the adverse effects of lithium carbonate (Eskalith).

There are no known laboratory interferences associated with nefazodone.

Dosage and Clinical Guidelines. Nefazodone is available in 50-, 200-, and 250-mg unscored tablets and 100- and 150-mg scored tablets. The recommended starting dosage of nefazodone is 100 mg twice a day, but 50 mg twice a day may be better tolerated, especially by elderly persons. The dosage should be slowly raised in increments of 100 to 200 mg a day at intervals of no less than 1 week per increase to limit the development of adverse effects. The optimal dosage is 300 to 600 mg daily in two divided doses. However, some studies report that nefazodone is effective when taken once a day, especially at bedtime. Geriatric persons should receive dosages about two-thirds of the usual nongeriatric dosages, with a maximum of 400 mg a day. Similar to other antidepressants, the clinical benefit of nefazodone usually appears after 2 to 4 weeks of treatment. Patients with the premenstrual syndrome are treated with a flexible dosage that averages about 250 mg a day.

Trazodone

Pharmacologic Actions. Trazodone is readily absorbed from the GI tract and reaches peak plasma levels in about 1 hour. It has a half-life of 5 to 9 hours. Trazodone is metabolized in the liver, and 75 percent of its metabolites are excreted in the urine.

Trazodone is a weak inhibitor of serotonin reuptake and a potent antagonist of serotonin 5-HT_{2A} and 5-HT_{2C} receptors. The active metabolite of trazodone is mCPP, which is an agonist at 5-HT_{2C} receptors and has a half-life of 14 hours. mCPP has been associated with migraine, anxiety, and weight loss. The adverse effects of trazodone are partially mediated by α_1-adrenergic receptor antagonism.

Therapeutic Indications

DEPRESSIVE DISORDERS. The main indication for the use of trazodone is MDD. There is a dose–response relationship, with dosages of 250 to 600 mg a day being necessary for trazodone to have therapeutic benefit. Trazodone increases total sleep time, decreases the number and the duration of nighttime awakenings, and decreases the amount of REM sleep. Unlike tricyclic drugs, trazodone does not decrease stage 4 sleep. Trazodone is thus useful for depressed persons with anxiety and insomnia.

INSOMNIA. Trazodone is a first-line agent for the treatment of insomnia because of its marked sedative qualities and favorable effects on sleep architecture (see above) combined with its lack of anticholinergic effects. Trazodone is effective for insomnia caused by depression or the use of drugs. When used as a hypnotic, the usual initial dosage is 25 to 100 mg at bedtime.

ERECTILE DISORDER. Trazodone is associated with an increased risk of priapism. Trazodone can potentiate erections resulting from sexual stimulation. It has thus been used to prolong erectile time and turgidity in some men with erectile disorder. The dosage for this indication is 150 to 200 mg a day. Trazodone-triggered priapism (an erection lasting more than 3 hours with pain) is a medical emergency. The use of trazodone for the treatment of male erectile dysfunction has diminished considerably since the introduction of phosphodiesterase (PDE)-5 agents (see Chapter 16).

OTHER INDICATIONS. Trazodone may be useful in low dosages (50 mg a day) for controlling severe agitation in children with developmental disabilities and elderly persons with dementia. At dosages above 250 mg a day, trazodone reduces the tension and apprehension associated with GAD. It has been used to treat depression in patients with schizophrenia. Trazodone may have a beneficial effect on insomnia and nightmares in persons with PTSD.

Precautions and Adverse Reactions. The most common adverse effects associated with trazodone are sedation, orthostatic hypotension, dizziness, headache, and nausea. Some persons experience dry mouth or gastric irritation. The drug is not associated with anticholinergic adverse effects, such as urinary retention, weight gain, and constipation. A few case reports have noted an association between trazodone and arrhythmias in persons with preexisting premature ventricular contractions or mitral valve prolapse. Neutropenia, usually not of clinical significance, may develop, which should be considered if persons have a fever or sore throat.

Trazodone may cause significant orthostatic hypotension 4 to 6 hours after a dose is taken, especially if taken concurrently with antihypertensive agents or if a large dose is taken without food. Administration of trazodone with food slows absorption and reduces the peak plasma concentration, thus reducing the risk of orthostatic hypotension.

Because suicide attempts often involve the ingestion of sleeping pills, it is essential to be familiar with the symptoms and treatment of trazodone overdose. Patients have survived trazodone overdoses of more than 9 g. Symptoms of overdose include lethargy, vomiting, drowsiness, headache, orthostasis, dizziness, dyspnea, tinnitus, myalgias, tachycardia, incontinence, shivering, and coma. Treatment consists of emesis or lavage and supportive care. Forced diuresis may enhance elimination. Treat hypotension and sedation as appropriate.

Trazodone causes priapism, prolonged erection in the absence of sexual stimuli, in one of every 10,000 men. Trazodone-induced priapism usually appears in the first 4 weeks of treatment but may occur as late as 18 months into treatment. It can appear at any dose. In such cases, trazodone use should be discontinued, and another antidepressant should be used. Painful erections or erections lasting more than 1 hour are warning signs that warrant immediate discontinuation of the drug and medical evaluation. The first step

in the emergency management of priapism is an intracavernosal injection of an α_1-adrenergic agonist pressor agent, such as metaraminol (Aramine) or epinephrine. In about one-third of reported cases, surgical intervention was required. In some cases, permanent impairment of erectile function or impotence resulted.

The use of trazodone is contraindicated in pregnant and nursing women. Trazodone should be used with caution in persons with hepatic and renal diseases.

Drug Interactions. Trazodone potentiates the CNS depressant effects of other centrally acting drugs and alcohol. Concurrent use of trazodone and antihypertensives may cause hypotension. No cases of hypertensive crisis have been reported when trazodone has been used to treat MAOI-associated insomnia. Trazodone can increase levels of digoxin and phenytoin. Trazodone should be used with caution in combination with warfarin. Drugs that inhibit CYP3A4 can increase levels of trazodone's primary metabolite, mCPP, leading to an increase in side effects.

Laboratory Interferences. No known laboratory interferences are associated with the administration of trazodone.

Dosage and Clinical Guidelines. Trazodone is available in 50-, 100-, 150-, and 300-mg tablets. Once-a-day dosing is as effective as divided dosing and reduces daytime sedation. The usual starting dose is 50 mg before sleep. The dosage can be increased in increments of 50 mg every 3 days if sedation or orthostatic hypotension does not become a problem. The therapeutic range for trazodone is 200 to 600 mg a day in divided doses. Some reports indicate that dosages of 400 to 600 mg a day are required for maximal therapeutic effects; other reports indicate that 250 to 400 mg a day is sufficient. The dosage may be titrated up to 300 mg a day; then, the person can be evaluated for the need for further dosage increases based on the presence or the absence of signs of clinical improvement.

TRICYCLICS AND TETRACYCLICS

The observation in 1957 that imipramine (Tofranil) had antidepressant effects

led to the development of a new class of antidepressant compounds, the tricyclics (TCAs). In turn, the finding that imipramine blocked reuptake of norepinephrine led to research into the role of catecholamines in depression. After the introduction of imipramine, several other antidepressant compounds were developed that shared a basic tricyclic structure and had relatively similar effects. Later, other heterocyclic compounds were also marketed that were somewhat similar in structure and that had relatively comparable secondary properties. At one time, amitriptyline (Elavil, Endep) and imipramine were the two most commonly prescribed antidepressants in the United States. However, because of their anticholinergic and antihistaminic side effects, they are rarely used for depression. Nortriptyline (Aventyl, Pamelor) and desipramine (Norpramin, Pertofrane) became preferred. Nortriptyline has the least effect on orthostatic hypotension, and desipramine is the least anticholinergic. Although introduced as antidepressants, the therapeutic indications for these agents have grown to include panic disorder, GAD, PTSD, OCD, and pain syndromes. The introduction of newer antidepressant agents with more selective actions on neurotransmitters or with unique mechanisms of action has sharply reduced the prescribing of TCAs and tetracyclics. The improved safety profiles of the newer drugs, especially when taken in overdose, also contributed to the decline in the use of older drugs. Nevertheless, the TCAs and tetracyclics remain unsurpassed in terms of their antidepressant efficacy. Table 21-20 lists TCA and tetracyclic drugs and their available preparations.

Pharmacologic Actions

The absorption of most TCAs is complete after oral administration, and there is significant metabolism from the first-pass effect. Peak plasma concentrations occur within 2 to 8 hours, and the half-lives of the TCAs vary from 10 to 70 hours; nortriptyline, maprotiline (Ludiomil), and particularly protriptyline (Vivactil) can have longer half-lives. The long half-lives allow all the compounds to be given once daily; 5 to 7 days is needed to reach steady-state plasma concentrations. Imipramine pamoate (Tofranil) is a depot form of the drug for intramuscular (IM) administration; indications for the use of this preparation are limited.

The TCAs undergo hepatic metabolism by the CYP450 enzyme system.

Clinically relevant drug interactions may result from competition for enzyme CYP2D6 among TCAs and quinidine, cimetidine (Tagamet), fluoxetine (Prozac), sertraline (Zoloft), paroxetine (Paxil), phenothiazines, carbamazepine (Tegretol), and the type IC antiarrhythmics propafenone (Rythmol) and flecainide (Tambocor). Concomitant administration of TCAs and these inhibitors may slow down the metabolism and raise the plasma concentrations of TCAs. Additionally, genetic variations in the activity of CYP2D6 may account for up to a 40-fold difference in plasma TCA concentrations in different persons. The dosage of the TCA may need to be adjusted to correct changes in the rate of hepatic TCA metabolism.



Table 21-20
Tricyclic and Tetracyclic Drug Preparations

Drug	Tablets (mg)	Capsules (mg)	Parenteral (mg/mL)	Solution
Imipramine (Tofranil)	10, 25, and 50	75, 100, 125, and 150	12.5	—
Desipramine (Norpramin, Pertofrane)	10, 25, 50, 75, 100, and 150	—	—	—
Trimipramine (Surmontil)	—	25, 50, and 100	—	—
Amitriptyline (Elavil)	10, 25, 50, 75, 100, and 150	—	10	—
Nortriptyline (Aventyl, Pamelor)	—	10, 25, 50, and 75	—	10 mg/5 mL
Protriptyline (Vivactil)	5 and 10	—	—	—
Amoxapine (Asendin)	25, 50, 100, and 150	—	—	—
Doxepin (Sinequan)	—	10, 25, 50, 75, 100, and 150	—	10 mg/mL
Maprotiline (Ludiomil)	25, 50, and 75	—	—	—
Clomipramine (Anafranil)	—	25, 50, and 75	—	—

The TCAs block the transporter site for norepinephrine and serotonin, thus increasing synaptic concentrations of these neurotransmitters. Each drug differs in its affinity for each of these transporters, with clomipramine (Anafranil) being the most serotonin selective and desipramine the most norepinephrine selective of the TCAs. Secondary effects of the TCAs include antagonism at the muscarinic acetylcholine, histamine H_1, and α_1- and α_2-adrenergic receptors. The potency of these effects on other receptors largely determines the side effect profile of each drug. Amoxapine, nortriptyline, desipramine, and maprotiline have the least anticholinergic activity; doxepin has the most antihistaminergic activity. Although they are more likely to cause constipation, sedation, dry mouth, or lightheadedness than the SSRIs, the TCAs are less prone to cause sexual dysfunction, significant long-term

weight gain, and sleep disturbances than the SSRIs. The half-lives and plasma clearance for most TCAs are very similar.

Therapeutic Indications

The indications are similar to those for the SSRIs, which have widely replaced the TCAs. However, the TCAs represent a reasonable alternative for persons who cannot tolerate the adverse effects of the SSRIs.

Major Depressive Disorder. The treatment of a major depressive episode and the prophylactic treatment of MDD are the principal indications for using TCAs. Although the TCAs are effective in the treatment of depression in persons with bipolar I disorder, they are more likely to induce mania, hypomania, or cycling than the newer antidepressants, most notably the SSRIs and bupropion. It is thus not advised that TCAs be routinely used to treat depression associated with bipolar I or bipolar II disorder.

Melancholic features, prior major depressive episodes, and a family history of depressive disorders increase the likelihood of therapeutic response. All of the available TCAs are equally effective in the treatment of depressive disorders. In the case of a person, however, one tricyclic or tetracyclic may be useful, and another one may be ineffective. The treatment of a major depressive episode with psychotic features almost always requires the coadministration of an antipsychotic drug and an antidepressant.

Although it is used worldwide as an antidepressant, clomipramine is only approved in the United States for the treatment of OCD.

Panic Disorder with Agoraphobia. Imipramine is the TCA most studied for panic disorder with agoraphobia, but other TCAs are also useful when taken at the usual antidepressant dosages. Because of the potential initial anxiogenic effects of the TCAs, starting dosages should be small, and the dosage should be titrated upward slowly. Small doses of benzodiazepines may be used initially to deal with this side effect.

Generalized Anxiety Disorder. The FDA approves the use of doxepin for the treatment of anxiety disorders. Some research data show that imipramine may also be useful. Although rarely used anymore, a chlordiazepoxide–amitriptyline combination (Limbitrol) is available for mixed anxiety and

depressive disorders.

Obsessive-Compulsive Disorder. Patients with OCD appear to respond specifically to clomipramine, as well as the SSRIs. Some improvement is usually seen in 2 to 4 weeks, but a further reduction in symptoms may continue for the first 4 to 5 months of treatment. None of the other TCAs appears to be nearly as effective as clomipramine for the treatment of this disorder. Clomipramine may also be a drug of choice for depressed persons with marked obsessive features.

Pain. The TCAs are widely used to treat chronic neuropathic pain and in the prophylaxis of migraine headache. Amitriptyline is the TCA most often used in this role. During the treatment of pain, doses are generally lower than those used in depression; for example, 75 mg of amitriptyline may be sufficient. These effects also appear more rapidly.

Other Disorders. Childhood enuresis is often treated with imipramine. Peptic ulcer disease can be treated with doxepin, which has marked antihistaminergic effects. Other indications for the TCAs are narcolepsy, nightmare disorder, and PTSD. The drugs are sometimes used for the treatment of children and adolescents with ADHD, sleepwalking disorder, separation anxiety disorder, and sleep terror disorder. Clomipramine has also been used to treat premature ejaculation, movement disorders, and compulsive behavior in children with autistic disorders; however, because the TCAs have caused sudden death in several children and adolescents, they should not be used in children.

Precautions and Adverse Reactions

The TCAs are associated with a wide range of problematic side effects and can be lethal when taken in overdose.

Psychiatric Effects. The TCAs can induce a switch to mania or hypomania in susceptible individuals. The TCAs may also exacerbate psychotic disorders in susceptible persons. At high plasma concentrations (levels above 300 ng/mL), the anticholinergic effects of the TCAs can cause confusion or delirium. Patients with dementia are particularly vulnerable to

this development.

Anticholinergic Effects. Anticholinergic effects often limit the tolerable dosage to relatively low ranges. Some persons may develop a tolerance for the anticholinergic effects with continued treatment. Anticholinergic effects include dry mouth, constipation, blurred vision, delirium, and urinary retention. Sugarless gum, candy, or fluoride lozenges can alleviate dry mouth. Bethanechol (Urecholine), 25 to 50 mg three or four times a day, may reduce urinary hesitancy and may be helpful in erectile dysfunction when the drug is taken 30 minutes before sexual intercourse. Anticholinergic drugs can also aggravate narrow-angle glaucoma, and the precipitation of glaucoma requires emergency treatment with a miotic agent. The TCAs should be avoided in persons with narrow-angle glaucoma, and an SSRI should be substituted. Severe anticholinergic effects can lead to CNS anticholinergic syndrome. This syndrome includes symptoms of confusion and delirium, especially if the TCAs are administered with DRAs or anticholinergic drugs. IM or IV physostigmine (Antilirium, Eserine) is used to diagnose and treat anticholinergic delirium.

Cardiac Effects. When administered in their usual therapeutic dosages, the TCAs may cause tachycardia, flattened T waves, prolonged QT intervals, and depressed ST segments in the electrocardiographic (ECG) recording. Imipramine has a quinidine-like effect at therapeutic plasma concentrations and may reduce the number of premature ventricular contractions. Because the drugs prolong conduction time, their use in persons with preexisting conduction defects is contraindicated. In persons with a history of any type of heart disease, the TCAs should be used only after SSRIs or other newer antidepressants have been found ineffective, and if used, they should be introduced at low dosages, with gradual increases in dosage and monitoring of cardiac functions. All of the TCAs can cause tachycardia, which may persist for months and is one of the most common reasons for drug discontinuation, especially in younger persons. At high plasma concentrations, as seen in overdoses, the drugs become arrhythmogenic.

Other Autonomic Effects. Orthostatic hypotension is the most common cardiovascular autonomic adverse effect, and the most common reason TCAs

are discontinued. It can result in falls and injuries in affected persons. Nortriptyline may be the drug least likely to cause this problem. Orthostatic hypotension is treated with avoidance of caffeine, intake of at least 2 L of fluid per day, and addition of salt to the diet unless the person is being treated for hypertension. In persons taking antihypertensive agents, reduction of the dosage may reduce the risk of orthostatic hypotension. Other possible autonomic effects are profuse sweating, palpitations, and increased BP. Although some persons respond to fludrocortisone (Florinef), 0.02 to 0.05 mg twice a day, the drug has potential toxicities. It is preferable to substitute an SSRI than to use a mineralocorticoid such as fludrocortisone. The TCAs' use should be discontinued several days before elective surgery because of the occurrence of hypertensive episodes during surgery in persons receiving TCAs.

Sedation. Sedation is a common effect of the TCAs and may be welcomed if sleeplessness has been a problem. The sedative effect of the TCAs is a result of anticholinergic and antihistaminergic activities. Amitriptyline, trimipramine, and doxepin are the most sedating agents; imipramine, amoxapine, nortriptyline, and maprotiline are less sedating; and desipramine and protriptyline are the least sedating agents.

Neurologic Effects. A fine, rapid tremor may occur. Myoclonic twitches and tremors of the tongue and the upper extremities are common. Rare effects include speech blockage, paresthesia, peroneal palsies, and ataxia.

Amoxapine is unique in causing parkinsonian symptoms, akathisia, and even dyskinesia because of the dopaminergic blocking activity of one of its metabolites. Amoxapine may also cause neuroleptic malignant syndrome in rare cases. Maprotiline may cause seizures when the dosage is increased too quickly or is kept at high levels for too long. Clomipramine and amoxapine may lower the seizure threshold more than other drugs in the class. As a class, however, the TCAs have a relatively low risk for inducing seizures except in persons who are at risk for seizures (e.g., persons with epilepsy and those with brain lesions). Although such persons can still use the TCAs, the initial dosages should be lower than usual, and subsequent dosage increases should be gradual.

Allergic and Hematologic Effects. Exanthematous rashes are seen in 4 to 5 percent of all persons treated with maprotiline. Jaundice is rare. Agranulocytosis, leukocytosis, leukopenia, and eosinophilia are rare complications of TCA treatment. However, a person who has a sore throat or a fever during the first few months of TCA treatment should have a CBC done immediately.

Hepatic Effects. Mild and self-limited increases in serum transaminase concentrations may occur and should be monitored. The TCAs can also produce a fulminant acute hepatitis in 0.1 to 1 percent of persons. This side effect can be life-threatening, and the antidepressant should be discontinued.

Other Adverse Effects. Modest weight gain is typical. Amoxapine exerts a DRA effect and may cause hyperprolactinemia, impotence, galactorrhea, anorgasmia, and ejaculatory disturbances. Other TCAs have also been associated with gynecomastia and amenorrhea. The syndrome of inappropriate secretion of antidiuretic hormone has also been reported with TCAs. Other effects include nausea, vomiting, and hepatitis.

TERATOGENICITY AND PREGNANCY-RELATED RISKS. A definitive link between the tricyclic compounds and tetracyclic compounds and teratogenic effects has not been established, but isolated reports of morphogenesis have been reported. TCAs cross the placenta, and neonatal drug withdrawal can occur. This syndrome includes tachypnea, cyanosis, irritability, and poor sucking reflex. If possible, tricyclic and tetracyclic medications should be discontinued 1 week before delivery. Recently, norepinephrine and serotonin transporters have been identified in the placenta. They appear to play an essential role in the clearance of these amines in the fetus. The understanding of the effects of reuptake inhibitors on these transporters during pregnancy is limited. However, one study compared intelligence and language development in 80 children exposed to TCAs during pregnancy with 84 children exposed to other nonteratogenic agents and found no deleterious effects of the TCAs. The TCAs are excreted in breast milk at concentrations similar to plasma. The actual quantity delivered, however, is small, so drug levels in the infant are usually undetectable or very low. Because the risk of relapse is a serious concern in patients with recurrent depression, and these

risks may be increased during pregnancy or the postpartum period, the risks and benefits of continuing or withdrawing treatment need to be discussed with the patient and weighed carefully.

Precautions. The TCAs may cause a withdrawal syndrome in newborns consisting of tachypnea, cyanosis, irritability, and poor sucking reflex. The drugs do pass into breast milk but at concentrations that are usually undetectable in the infant's plasma. We should use these drugs cautiously in patients with hepatic or renal diseases. The TCAs should not be administered during a course of ECT, primarily because of the risk of serious adverse cardiac effects.

Drug Interactions

Monoamine Oxidase Inhibitors. The TCAs should not be taken within 14 days of administration of an MAOI.

Antihypertensives. The TCAs block the therapeutic effects of antihypertensive medication. The TCAs may block the antihypertensive effects of the β-adrenergic receptor antagonists (e.g., propranolol [Inderal] and clonidine [Catapres]). The coadministration of a TCA and α-methyldopa (Aldomet) may cause behavioral agitation.

Antiarrhythmic Drugs. The antiarrhythmic properties of TCAs can be additive to those of quinidine, an effect that is further exacerbated by the inhibition of TCA metabolism by quinidine.

Dopamine Receptor Antagonists. Concurrent administration of TCAs and DRAs increases the plasma concentrations of both drugs. Desipramine plasma concentrations may increase twofold during concurrent administration with perphenazine (Trilafon). The DRAs also add to the anticholinergic and sedative effects of the TCAs. Concomitant use of SDAs also increases those effects.

Central Nervous System Depressants. Opioids, alcohol, anxiolytics, hypnotics, and over-the-counter cold medications have additive effects by causing CNS depression when coadministered with TCAs. Persons should be

advised to avoid driving or using dangerous equipment if sedated by TCAs.

Sympathomimetics. Tricyclic drug use with sympathomimetic drugs may cause serious cardiovascular effects.

Oral Contraceptives. Birth control pills may decrease TCA plasma concentrations through the induction of hepatic enzymes.

Other Drug Interactions. Nicotine may reduce TCA concentrations. Plasma concentrations may also be lowered by ascorbic acid, ammonium chloride, barbiturates, cigarette smoking, carbamazepine, chloral hydrate, lithium (Eskalith), and primidone (Mysoline). TCA plasma concentrations may be increased by concurrent use of acetazolamide (Diamox), sodium bicarbonate, acetylsalicylic acid, cimetidine, thiazide diuretics, fluoxetine, paroxetine, and fluvoxamine (Luvox). Plasma concentrations of the TCAs may rise three- to fourfold when administered concurrently with fluoxetine, fluvoxamine, and paroxetine.

Laboratory Interferences

The tricyclic compounds are present at low concentrations and are not likely to interfere with other laboratory assays. They may interfere with the determination of conventional neuroleptic blood concentrations because of their structural similarity and the low concentrations of some neuroleptics.

Dosage and Clinical Guidelines

Persons who intend to take TCAs should undergo routine physical and laboratory examinations, including a CBC, a WBC count with differential, and serum electrolytes with liver function tests. An ECG should be obtained for all persons, especially women older than 40 years of age and men older than 30 years of age. The TCAs are contraindicated in persons with a QT_C greater than 450 milliseconds. The initial dose should be small and should be raised gradually. Because of the availability of highly effective alternatives to TCAs, a newer agent should be used if there is any medical condition that may interact adversely with the TCAs.

Elderly persons and children are more sensitive to TCA adverse effects

than are young adults. In children, the ECG should be regularly monitored during the use of a TCA.

The available preparations of TCAs are presented in Table 21-20. The dosages and therapeutic blood levels for the TCAs vary among the drugs (Table 21-21). Except for protriptyline, all of the TCAs should be started at 25 mg a day and increased as tolerated. Divided doses at first reduce the severity of the adverse effects, although most of the dosage should be given at night to help induce sleep if a sedating drug such as amitriptyline is used. Eventually, the entire daily dose can be given at bedtime. A common clinical mistake is to stop increasing the dosage when the person is tolerating the drug but taking less than the maximum therapeutic dose and does not show clinical improvement. The clinician should routinely assess the person's pulse and orthostatic changes in BP while the dosage is being increased.

Nortriptyline use should be started at 25 mg a day. Most patients need only 75 mg a day to achieve a blood level of 100 mg/nL. However, the dosage may be raised to 150 mg a day if needed. Amoxapine use should be started at 150 mg a day and raised to 400 mg a day. Protriptyline use should be started at 15 mg a day and raised to 60 mg a day. Maprotiline has been associated with an increased incidence of seizures. This risk is most significant if the dosage is raised too quickly or is maintained at too high a level. Maprotiline use should be started at 25 mg a day and increased over 4 weeks to 225 mg a day. It should be kept at that level for only 6 weeks and then be reduced to 175 to 200 mg a day.

Persons with chronic pain may be particularly sensitive to adverse effects when TCA use is started. Therefore, treatment should begin with low dosages that are raised in small increments. However, persons with chronic pain may experience relief on long-term low-dosage therapy, such as amitriptyline or nortriptyline at 10 to 75 mg a day.



Table 21-21
General Information for the Tricyclic and Tetracyclic Antidepressants

Generic Name	Trade Name	Usual Adult Dosage Range (mg/day)	Therapeutic Plasma Concentrations (µg/mL)

Imipramine	Tofranil	150–300	150–300[a]
Desipramine	Norpramin, Pertofrane	150–300	150–300[a]
Trimipramine	Surmontil	150–300	150–300
Amitriptyline	Elavil, Endep	150–300	100–250[b]
Nortriptyline	Pamelor, Aventyl	50–150	50–150[a] (maximum)
Protriptyline	Vivactil	15–60	75–250
Amoxapine	Asendin	150–400	200–500[c]
Doxepin	Adapin, Sinequan	150–300	100–250[a]
Maprotiline	Ludiomil	150–230	150–300[a]
Clomipramine	Anafranil	130–250	50 to 250[c]

[a]Exact range may vary among laboratories.

[b]Includes parent compound and desmethyl metabolite.

[c]Estimated from limited studies.

The TCAs should be avoided in children except as a last resort. Dosing guidelines in children for imipramine include initiation at 1.5 mg/kg a day. The dosage can be titrated to no more than 5 mg/kg a day. In enuresis, the dosage is usually 50 to 100 mg a day, taken at bedtime. Clomipramine use can be initiated at 50 mg a day and increased to no more than 3 mg/kg per day or 200 mg a day.

When TCA treatment is discontinued, the dosage should first be decreased to three-fourths the maximal dosage for a month. At that time, if no symptoms are present, drug use can be tapered by 25 mg (5 mg for protriptyline) every 4 to 7 days. Slow tapering avoids a cholinergic rebound syndrome consisting of nausea, upset stomach, sweating, headache, neck pain, and vomiting. This syndrome can be treated by reinstituting a small dosage of the drug and tapering more slowly than before. Several case reports note the appearance of rebound mania or hypomania after the abrupt

discontinuation of TCA use.

Plasma Concentrations and Therapeutic Drug Monitoring. Clinical determinations of plasma concentrations should be conducted after 5 to 7 days on the same dosage of medication and 8 to 12 hours after the last dose. Because of variations in absorption and metabolism, there may be a 30- to 50-fold difference in the plasma concentrations in persons given the same dosage of a TCA. Nortriptyline is unique in its association with a therapeutic window—that is, plasma concentrations below 50 ng/mL or above 150 ng/mL may reduce its efficacy.

Plasma concentrations may be useful in confirming compliance, assessing reasons for drug failures, and documenting effective plasma concentrations for future treatment. Clinicians should always treat the person and not the plasma concentration. Some persons have adequate clinical responses with seemingly subtherapeutic plasma concentrations, and other persons only respond at supratherapeutic plasma concentrations without experiencing adverse effects. The latter situation, however, should alert the clinician to monitor the person's condition with, for example, serial ECG recordings.

Overdose Attempts. Overdose attempts with TCAs are severe and can often be fatal. Prescriptions for these drugs should be nonrefillable and for no longer than 1 week at a time for patients at risk for suicide. Amoxapine may be more likely than the other TCAs to result in death when taken in overdose. The newer antidepressants are safer in overdose.

Symptoms of overdose include agitation, delirium, convulsions, hyperactive deep tendon reflexes, bowel and bladder paralysis, dysregulation of BP and temperature, and mydriasis. The patient then progresses to coma and perhaps respiratory depression. Cardiac arrhythmias may not respond to treatment. Because of the long half-lives of TCAs, the patients are at risk of cardiac arrhythmias for 3 to 4 days after the overdose, so they should be monitored in intensive care medical settings.

MONOAMINE OXIDASE INHIBITORS

Introduced in the late 1950s, MAOIs were the first class of approved antidepressant drugs. The first of these drugs, isoniazid, was intended to be

used as a treatment for tuberculosis, but its antidepressant properties were discovered by chance when some treated patients experienced elevation of mood during treatment. Despite their effectiveness, prescription of MAOIs as first-line agents has always been limited by concern about the development of potentially lethal hypertension and the consequent need for a restrictive diet. The use of MAOIs declined further after the introduction of the SSRIs and other new agents. They are now mainly relegated to use in treatment-resistant cases. Thus, the second-line status of MAOIs has less to do with considerations of efficacy than with safety concerns. The currently available MAOIs include phenelzine (Nardil), isocarboxazid (Marplan), tranylcypromine (Parnate), rasagiline (Azilect), moclobemide (Manerix), and selegiline (Eldepryl).

Two subsequent advances in the field of antidepressant MAOIs involve the introduction of a selective reversible inhibitor of MAO_A (RIMA), moclobemide (Manerix), in the early 1990s into most countries except the United States, and in 2005, the introduction of a transdermal delivery form of selegiline (Emsam) in the United States that is used for the treatment of parkinsonism. Other RIMA agents, including brofaromine (Consonar) and befloxatone, have not been submitted for registration despite favorable outcomes in clinical trials.

Pharmacologic Actions

Phenelzine, tranylcypromine, and isocarboxazid are readily absorbed after oral administration and reach peak plasma concentrations within 2 hours. Whereas their plasma half-lives are in the range of 2 to 3 hours, their tissue half-lives are considerably longer. Because they irreversibly inactivate MAOs, the therapeutic effect of a single dose of irreversible MAOIs may persist for as long as 2 weeks. The RIMA moclobemide is rapidly absorbed and has a half-life of 0.5 to 3.5 hours. Because it is a reversible inhibitor, moclobemide has a much briefer clinical effect after a single dose than do irreversible MAOIs.

The MAO enzymes are found on the outer membranes of mitochondria, where they degrade cytoplasmic and extraneuronal monoamine neurotransmitters such as norepinephrine, serotonin, dopamine, epinephrine, and tyramine. MAOIs act in the CNS, the sympathetic nervous system, the

liver, and the GI tract. There are two types of MAOs, MAO_A and MAO_B. MAO_A primarily metabolizes norepinephrine, serotonin, and epinephrine; both MAO_A and MAO_B metabolize dopamine and tyramine.

The structures of phenelzine and tranylcypromine are similar to those of amphetamine and have similar pharmacologic effects in that they increase the release of dopamine and norepinephrine with attendant stimulant effects on the brain.

Therapeutic Indications

MAOIs are used for the treatment of depression. Some research indicates that phenelzine is more effective than TCAs in depressed patients with mood reactivity, extreme sensitivity to interpersonal loss or rejection, prominent anergia, hyperphagia, and hypersomnia—a constellation of symptoms conceptualized as atypical depression. Evidence also suggests that MAOIs are more effective than TCAs as a treatment for bipolar depression.

Patients with panic disorder and social phobia respond well to MAOIs. MAOIs have also been used to treat bulimia nervosa, PTSD, anginal pain, atypical facial pain, migraine, ADHD, idiopathic orthostatic hypotension, and depression associated with traumatic brain injury.

Precautions and Adverse Reactions

The most frequent adverse effects of MAOIs are orthostatic hypotension, insomnia, weight gain, edema, and sexual dysfunction. Orthostatic hypotension can lead to dizziness and falls. Thus, cautious upward tapering of the dosage should be used to determine the maximum tolerable dosage. Treatment for orthostatic hypotension includes avoidance of caffeine; intake of 2 L of fluid per day; addition of dietary salt or adjustment of antihypertensive drugs (if applicable); support stockings; and in severe cases, treatment with fludrocortisone (Florinef), a mineralocorticoid, 0.1 to 0.2 mg a day. Orthostatic hypotension associated with tranylcypromine use can usually be relieved by dividing the daily dosage.

Insomnia can be treated by dividing the dose, not giving the medication after dinner, and using a benzodiazepine hypnotic if necessary. Weight gain, edema, and sexual dysfunction often do not respond to any treatment and may warrant switching to another agent. When switching from one MAOI to

another, the clinician should taper and stop the use of the first drug for 10 to 14 days before beginning the use of the second drug.

Paresthesias, myoclonus, and muscle pains are occasionally seen in persons treated with MAOIs. Paresthesias may be secondary to MAOI-induced pyridoxine deficiency, which may respond to supplementation with pyridoxine, 50 to 150 mg orally each day. Occasionally, persons complain of feeling drunk or confused, perhaps indicating that the dosage should be reduced and then increased gradually. Reports that the hydrazine MAOIs are associated with hepatotoxic effects are relatively uncommon. MAOIs are less cardiotoxic and less epileptogenic than are the tricyclic and tetracyclic drugs.

The most common adverse effects of the RIMA moclobemide are dizziness, nausea, and insomnia or sleep disturbance. RIMAs cause fewer GI adverse effects than do SSRIs. Moclobemide does not have adverse anticholinergic or cardiovascular effects, and it has not been reported to interfere with sexual function.

MAOIs should be used with caution by persons with renal disease, cardiovascular disease, or hyperthyroidism. MAOIs may alter the dosage of a hypoglycemic agent required by persons with diabetes. MAOIs have been significantly associated with the induction of mania in persons in the depressed phase of bipolar I disorder and triggering a psychotic decompensation in persons with schizophrenia. MAOIs are contraindicated during pregnancy, although data on their teratogenic risk are minimal. Nursing women should not take MAOIs because the drugs can pass into breast milk.

Tyramine-Induced Hypertensive Crisis. The most worrisome side effect of MAOIs is the tyramine-induced hypertensive crisis. The amino acid tyramine is usually transformed via GI metabolism. However, MAOIs inactivate GI metabolism of dietary tyramine, thus allowing intact tyramine to enter the circulation. A hypertensive crisis may subsequently occur as a result of a powerful pressor effect of the amino acid. Tyramine-containing foods should be avoided for 2 weeks after the last dose of an irreversible MAOI to allow the resynthesis of adequate concentrations of MAO enzymes.

Accordingly, foods rich in tyramine (Table 21-22) or other sympathomimetic amines, such as ephedrine, pseudoephedrine (Sudafed), or dextromethorphan (Trocal), should be avoided by persons who are taking

irreversible MAOIs. Patients should be advised to continue the dietary restrictions for 2 weeks after they stop MAOI treatment to allow the body to resynthesize the enzyme. Bee stings may cause a hypertensive crisis. In addition to severe hypertension, other symptoms may include headache, stiff neck, diaphoresis, nausea, and vomiting. A patient with these symptoms should seek immediate medical treatment.



Table 21-22
Tyramine-Rich Foods to Be Avoided in Planning Monoamine Oxidase Inhibitor Diets

High Tyramine Content[a] (≥2 mg of tyramine a serving)
- Cheese: English Stilton, blue cheese, white (3 yr old), extra old, old cheddar, Danish blue, mozzarella, cheese snack spreads
- Fish, cured meats, sausage; pâtés and organs, salami, mortadella, air-dried sausage
- Alcoholic beverages[b]: Liqueurs and concentrated after-dinner drinks
- Marmite (concentrated yeast extract)
- Sauerkraut (Krakus)

Moderate Tyramine Content[a] (0.5–1.99 mg of tyramine a serving)
- Cheese: Swiss Gruyere, muenster, feta, parmesan, gorgonzola, blue cheese dressing, Black Diamond
- Fish, cured meats, sausage, pâtés and organs: Chicken liver (5 days old); bologna; aged sausage, smoked meat; salmon mousse
- Alcoholic beverages: Beer and ale (12 oz per bottle)—Amstel, Export Draft, Blue Light, Guinness Extra Stout, Old Vienna, Canadian, Miller Light, Export, Heineken, Blue Wines (per 4-oz glass)—Rioja (red wine)

Low Tyramine Content[a] (0.01 to >0.49 mg of tyramine a serving)
- Cheese: Brie, Camembert, Cambozola with or without rind
- Fish, cured meat, sausage, organs, and pâtés; pickled herring; smoked fish; kielbasa sausage; chicken liver; liverwurst (<2 days old)
- Alcoholic beverages: Red wines, sherry, scotch[c]

Others: Banana or avocado (ripe or not), banana peel

[a]Any food left out to age or spoil can spontaneously develop tyramine through fermentation.

[b]Alcohol can produce profound orthostasis interacting with monoamine oxidase inhibitors (MAOIs) but cannot produce direct hypotensive reactions.

[c]White wines, gin, and vodka have no tyramine content.

Table by Jonathan M. Himmelhoch, M.D.

An MAOI-induced hypertensive crisis should be treated with α-adrenergic antagonists—for example, phentolamine (Regitine) or chlorpromazine (Thorazine). These drugs lower BP within 5 minutes. IV furosemide (Lasix) can be used to reduce fluid load, and a β-adrenergic receptor antagonist can control tachycardia. A sublingual 10-mg dose of nifedipine (Procardia) can be given and repeated after 20 minutes. Persons with thyrotoxicosis or pheochromocytoma should not use MAOIs.

The risk of tyramine-induced hypertensive crises is relatively low for persons who are taking RIMAs, such as moclobemide and befloxatone. These drugs have relatively little inhibitory activity for MAO_B, and because they are reversible, regular activity of existing MAO_A returns within 16 to 48 hours of the last dose of a RIMA. Therefore, the dietary restrictions are less stringent for RIMAs, applying only to foods containing high concentrations of tyramine, which need to be avoided for 3 days after the last dose of a RIMA. A reasonable dietary recommendation for persons taking RIMAs is to avoid eating tyramine-containing foods 1 hour before and 2 hours after taking a RIMA.

Spontaneous, nontyramine-induced hypertensive crisis is a rare occurrence, usually shortly after the first exposure of an MAOI. Persons experiencing such a crisis should avoid MAOIs altogether.

Withdrawal. Abrupt cessation of regular doses of MAOIs may cause a self-limited discontinuation syndrome consisting of arousal, mood disturbances, and somatic symptoms. Dosages should be gradually tapered over several weeks to avoid these symptoms.

Overdose. It often takes 1 to 6 hours after an overdose for the toxic symptoms to occur. MAOI overdose is characterized by agitation that can progress to coma with hyperthermia, hypertension, tachypnea, tachycardia,

dilated pupils, and hyperactive deep tendon reflexes. Involuntary movements may be present, particularly in the face and the jaw. Acidification of the urine markedly hastens the excretion of MAOIs, and dialysis can be of some use. Phentolamine or chlorpromazine may be useful if hypertension is a problem. Moclobemide alone in overdosage causes relatively mild and reversible symptoms.

Drug Interactions

The significant drug–drug interactions involving MAOIs are listed in Table 21-23. Most antidepressants, as well as precursor agents, should be avoided. Persons should be instructed to tell any other physicians or dentists who are treating them that they are taking an MAOI. MAOIs may potentiate the action of CNS depressants, including alcohol and barbiturates. MAOIs should not be coadministered with serotonergic drugs, such as SSRIs and clomipramine (Anafranil), because this combination can trigger serotonin syndrome. The use of lithium or tryptophan with an irreversible MAOI may also induce serotonin syndrome. Initial symptoms of serotonin syndrome can include tremor, hypertonicity, myoclonus, and autonomic signs, which can then progress to hallucinosis, hyperthermia, and even death. Fatal reactions had occurred when MAOIs were combined with meperidine (Demerol) or fentanyl (Sublimaze).



Table 21-23
Drugs to Be Avoided during Monoamine Oxidase Inhibitor Treatment (Partial Listing)

Never Use

- Antiasthmatics
- Antihypertensives (methyldopa, guanethidine, reserpine)
- Buspirone
- Levodopa
- Opioids (especially meperidine, dextromethorphan, propoxyphene, tramadol; morphine or codeine may be less dangerous)
- Cold, allergy, or sinus medications containing dextromethorphan

or sympathomimetics
SSRIs, clomipramine, venlafaxine, sibutramine
Sympathomimetics (amphetamines, cocaine, methylphenidate, dopamine, epinephrine, norepinephrine, isoproterenol, ephedrine, pseudoephedrine, phenylpropanolamine)
L-Tryptophan
Use Carefully
Anticholinergics
Antihistamines
Disulfiram
Bromocriptine
Hydralazine
Sedative-hypnotics
Terpin hydrate with codeine
Tricyclics and tetracyclics (avoid clomipramine)

SSRI, selective serotonin reuptake inhibitor.

When switching from an irreversible MAOI to any other type of antidepressant drug, persons should wait at least 14 days after the last dose of the MAOI before beginning the use of the next drug to allow replenishment of the body's MAOs. When switching from an antidepressant to an irreversible MAOI, persons should wait 10 to 14 days (or 5 weeks for fluoxetine [Prozac]) before starting the use of the MAOI to avoid drug–drug interactions. In contrast, MAO activity recovers completely 24 to 48 hours after the last dose of a RIMA.

The effects of the MAOIs on hepatic enzymes are poorly studied. Tranylcypromine inhibits CYP2C19. Moclobemide inhibits CYP2D6, CYP2C19, and CYP1A2 and is a substrate for 2C19.

Cimetidine (Tagamet) and fluoxetine significantly reduce the elimination of moclobemide. Modest doses of fluoxetine and moclobemide administered concurrently may be well tolerated, with no significant pharmacodynamic or pharmacokinetic interactions.

Laboratory Interferences

MAOIs may lower blood glucose concentrations. MAOIs artificially raise

urinary metanephrine concentrations and may cause a false-positive test result for pheochromocytoma or neuroblastoma. MAOIs have been reported to be associated with a minimal false elevation in thyroid function test results.

Dosage and Clinical Guidelines

There is no definitive rationale for choosing one irreversible MAOI over another. Table 21-24 lists MAOI preparations and typical dosages. Phenelzine use should begin with a test dose of 15 mg on the first day. The dosage can be increased to 15 mg three times daily during the first week and increased by 15 mg a day each week after that until the dosage of 90 mg a day, in divided doses, is reached by the end of the fourth week. Tranylcypromine and isocarboxazid use should begin with a test dosage of 10 mg and may be increased to 10 mg three times daily by the end of the first week. Many clinicians and researchers have recommended upper limits of 50 mg a day for isocarboxazid and 40 mg a day for tranylcypromine. Administration of tranylcypromine in multiple small daily doses may reduce its hypotensive effects.

Table 21-24
Typical Dosage Forms and Recommended Dosages for Currently Available Monoamine Oxidase Inhibitors

Drug	Usual Dose (mg/day)	Maximum Dose (mg/day)	Dosage (Oral) Formulation
Isocarboxazid (Marplan)	20–40	60	10-mg tablets
Phenelzine (Nardil)	30–60	90	15-mg tablets
Tranylcypromine (Parnate)	20–60	60	10-mg tablets
Rasagiline	0.5–1.0	1.0	0.5- or 1.0-mg tablets
Selegiline	10	30	5-mg tablets

(Eldepryl)			
Moclobemide (Manerix)	300–600	600	100- or 150-mg tablets

Even though coadministration of MAOIs with TCAs, SSRIs, or lithium is generally contraindicated, these combinations have been used successfully and safely to treat patients with refractory depression. However, they should be used with extreme caution.

Hepatic transaminase serum concentrations should be monitored periodically because of the potential for hepatotoxicity, especially with phenelzine and isocarboxazid. Elderly persons may be more sensitive to MAOI adverse effects than are younger adults. MAO activity increases with age, so MAOI dosages for elderly persons are the same as those required for younger adults. We know little about using MAOIs in children.

Studies have suggested that transdermal selegiline has antidepressant properties. Although selegiline is a type B inhibitor at low doses, it becomes less selective as the dose is increased.

THYROID HORMONES

Thyroid hormones—levothyroxine (Synthroid, Levothroid, Levoxine) and liothyronine (Cytomel)—are used in psychiatry either alone or as an augmentation to treat persons with depression or rapid-cycling bipolar I disorder. They can convert an antidepressant-nonresponsive person into an antidepressant-responsive person. Thyroid hormones are also used as replacement therapy for persons treated with lithium (Eskalith) who have developed a hypothyroid state. Successful use of thyroid hormone as an intervention for treatment-resistant patients was first reported in the early 1970s. Study results since then have been mixed; however, most show that patients taking triiodothyronine (T_3) are twice as likely to respond to antidepressant treatment versus placebo. These studies have found that augmentation with T_3 is significant with TCAs and SSRIs. Nevertheless, many endocrinologists object to the use of thyroid hormones as antidepressant augmentation agents, citing such risks as osteoporosis and cardiac arrhythmias.

Pharmacologic Actions

Thyroid hormones are administered orally, and their absorption from the GI tract is variable. Absorption is increased if the drug is administered on an empty stomach. Thyroxine (T_4) crosses the blood–brain barrier and diffuses into neurons, where it is converted into T_3, which is the physiologically active form. The half-life of T_4 is 6 to 7 days, and that of T_3 is 1 to 2 days.

The mechanism of action for thyroid hormone effects on antidepressant efficacy is unknown. Thyroid hormone binds to intracellular receptors that regulate the transcription of a wide range of genes, including several receptors for neurotransmitters.

Therapeutic Indications

The primary indication for thyroid hormones in psychiatry is as an adjuvant to antidepressants. There is no clear correlation between the laboratory measures of thyroid function and the response to thyroid hormone supplementation of antidepressants. If a patient has not responded to a 6-week course of antidepressants at appropriate dosages, adjuvant therapy with either lithium or a thyroid hormone is an alternative. Most clinicians use adjuvant lithium before trying a thyroid hormone. Several controlled trials have indicated that liothyronine use converts about 50 percent of antidepressant nonresponders to responders.

The dosage of liothyronine is 25 or 50 µg a day added to the patient's antidepressant regimen. Liothyronine has been used primarily as an adjuvant for tricyclic drugs; however, evidence suggests that liothyronine augments the effects of all of the antidepressant drugs.

Thyroid hormones have not been shown to cause particular problems in pediatric or geriatric patients; however, the hormones should be used with caution in elderly persons who may have occult heart disease.

Precautions and Adverse Reactions

At the dosages usually used for augmentation—25 to 50 µg a day—adverse effects occur infrequently. The most common adverse effects associated with thyroid hormones are transient headache, weight loss, palpitations, nervousness, diarrhea, abdominal cramps, sweating, tachycardia, increased

BP, tremors, and insomnia. Osteoporosis may also occur with long-term treatment, but this has not been found in studies involving liothyronine augmentation. Overdoses of thyroid hormones can lead to cardiac failure and death.

Patients with cardiac disease, angina, or hypertension should not take thyroid hormones. The hormones are contraindicated in thyrotoxicosis and uncorrected adrenal insufficiency and persons with acute myocardial infarctions. Thyroid hormones can be administered safely to pregnant women, provided that laboratory thyroid indexes are monitored. Thyroid hormones are minimally excreted in breast milk and have not been shown to cause problems in nursing babies.

Drug Interactions

Thyroid hormones can potentiate the effects of warfarin (Coumadin) and other anticoagulants by increasing the catabolism of clotting factors. They may increase the insulin requirement for diabetic persons and the digitalis requirement for persons with cardiac disease. We should not coadminister with sympathomimetics, ketamine (Ketalar), or maprotiline (Ludiomil) because of the risk of cardiac decompensation. Administration of SSRIs, tricyclic and tetracyclic drugs, lithium, or carbamazepine (Tegretol) can mildly lower serum T_4 and raise serum thyrotropin concentrations in euthyroid persons or persons taking thyroid replacements. This interaction warrants close serum monitoring and may require an increase in the dosage or initiation of thyroid hormone supplementation.

Laboratory Interferences

Levothyroxine has not been reported to interfere with any laboratory test other than thyroid function indexes. Liothyronine, however, suppresses the release of endogenous T_4, thereby lowering the result of any thyroid function test that depends on the measure of T_4.

Thyroid Function Tests

Several thyroid function tests are available, including tests for T_4 by competitive protein binding (T_4 [D]) and by radioimmunoassay (T_4 RIA)

involving a specific antigen–antibody reaction. More than 90 percent of T_4 is bound to serum protein and is responsible for thyroid-stimulating hormone (TSH) secretion and cellular metabolism. Other thyroid measures include the free T_4 index (FT_4I), T_3 uptake, and total serum T_3 measured by radioimmunoassay (T_3 RIA). Those tests are used to rule out hypothyroidism, which can be associated with symptoms of depression. In some studies, up to 10 percent of patients complaining of depression and associated fatigue had an incipient hypothyroid disease. Lithium can cause hypothyroidism and, more rarely, hyperthyroidism. Neonatal hypothyroidism results in intellectual disability and is preventable if the diagnosis is made at birth.

Thyrotropin-Releasing Hormone Stimulation Test. The thyrotropin-releasing hormone (TRH) stimulation test is indicated for patients who have marginally abnormal thyroid test results with suspected subclinical hypothyroidism, which may account for clinical depression. It is also used in patients with possible lithium-induced hypothyroidism. The procedure entails an intravenous injection of 500 mg of protirelin (TRH), which produces a sharp increase in serum TSH levels are measured at 15, 30, 60, and 90 minutes. An increase in serum TSH of 5 to 25 mIU/mL above the baseline is normal. An increase of less than 7 mIU/mL is considered a blunted response, which may correlate with a diagnosis of depression. Eight percent of all patients with depression have some thyroid illness.

Dosage and Clinical Guidelines

Liothyronine is available in 5-, 25-, and 50-µg tablets. Levothyroxine is available in 12.5-, 25-, 50-, 75-, 88-, 100-, 112-, 125-, 150-, 175-, 200-, and 300-µg tablets; it is also available in a 200 and 500 µg parenteral form. The dosage of liothyronine is 25 or 50 µg a day added to the person's antidepressant regimen. Liothyronine has been used as an adjuvant for all of the available antidepressant drugs. An adequate trial of liothyronine supplementation should last 2 to 3 weeks. If liothyronine supplementation is successful, it should be continued for 2 months and then tapered off at a rate of 12.5 µg a day every 3 to 7 days.

NOVEL AGENTS

Several agents with novel mechanisms are currently approved for use. They are usually given in specialized centers, and clinicians should only administer after special training to manage their significant safety profile.

NMDA Receptor Antagonists

Ketamine, an anesthetic agent, is used in specialized centers for depression. It has a unique action, as after a single low dose, much lower than the usual anesthetic dose, it can produce rapid relief from depression. This effect often occurs within 4 hours of IV administration. The effects persist for several weeks, but then without additional intervention, the depression often relapses. The drug has several significant side effects, including its general anesthetic effect as well as its psychotomimetic effect. Most experts consider it safe at lower doses; however, the clinical team should be ready to provide medical treatment should the drug have a profound effect on a patient's breathing or cause significant bradycardia.

Ketamine's enantiomer, esketamine (Ketanest), was approved by the FDA for the treatment of depression in adults in 2019. It has the advantage over ketamine, in that in addition to an injectable form, it is available as a nasal spray. Esketamine's mechanism of action is similar to ketamine's, although not identical; for example, both are noncompetitive NMDA receptor antagonist and inhibit dopamine transporters. However, esketamine does not act on the sigma receptors, which may account for part of ketamine's antidepressant effect. This lack of effect on sigma receptors is thought to reduce the psychotomimetic properties of the drug. It must be provided in a clinical setting that has the staff and facilities to observe the patient for at least 2 hours following administration. The main side effects are sedation, difficulties with vision and speaking, and cognitive effects, including confusion, dissociation, and potentially delirium. It may also cause anxiety and increased BP in some patients.

$GABA_A$ Allosteric Modulators

Brexanolone, also called allopregnanolone, was approved in the United States in 2019 under the priority review process as a breakthrough therapy for the treatment of postpartum depression. It is a neurosteroid which positively modulates the $GABA_A$ receptor. It is also a negative allosteric modulator of

the nicotinic acetylcholine receptor and may have action on serotonin HT_3 as well. The mechanism of its antidepressant effect is not clear, as other positive allosteric modulators of $GABA_A$, such as benzodiazepines, do not have a significant antidepressant effect. However, like ketamine, it provides a rapid response in patients who respond to the drug, usually within 2 to 3 days. We know little about the long term efficacy of the drug.

The drug is marketed under the brand name Zulresso and is a Schedule IV controlled substance. It is only available through a national registry and has several other drawbacks limiting its widespread acceptance. One is its long administration time, as the drug must be given by continuous IV infusion over 60 hours, during which the dose is slowly increased. This extended hospitalization is to prevent excess sedation or syncope.

Perhaps the most significant drawback of the medication is its cost. Currently, the drug costs about $34,000 for a single dose, as well as the price of remaining in a medical facility for 60 hours, and is not currently covered by most insurance companies.

21.3 Mood Stabilizers

LITHIUM

The effectiveness of lithium for mania and the prophylactic treatment of manic–depressive disorder was established in the early 1950s as a result of research done by John F.J. Cade, an Australian psychiatrist. Concerns about toxicity limited initial acceptance of lithium use in the United States, but its use increased gradually in the late 1960s. It was not until 1970 that the FDA approved its labeling for the treatment of mania. The only other approved FDA indication came in 1974 when it was accepted as maintenance therapy in patients with a history of mania. For several decades, lithium was the only drug approved for both acute and maintenance treatment. It is also used as an adjunctive medication in the treatment of MDD.

Lithium (Li), a monovalent ion, is a member of the group IA alkali metals on the periodic table, a group that also includes sodium, potassium, rubidium, cesium, and francium. Lithium exists in nature as both ^{6}Li (7.42 percent) and

^{7}Li (92.58 percent). The latter isotope allows the imaging of lithium by magnetic resonance spectroscopy. Some 300 mg of lithium is contained in 1,597 mg of lithium carbonate (Li_2CO_3). Most lithium used in the United States is obtained from dry lake mining in Chile and Argentina.

Pharmacologic Actions

Lithium is rapidly and completely absorbed after oral administration, with peak serum concentrations occurring in 1 to 1.5 hours with standard preparations and in 4 to 4.5 hours with slow-release and controlled-release preparations. Lithium does not bind to plasma proteins, is not metabolized, and is excreted through the kidneys. The plasma half-life is initially 1.3 days and is 2.4 days after administration for more than 1 year. The blood–brain barrier permits only the slow passage of lithium, which is why a single overdose does not necessarily cause toxicity and why long-term lithium intoxication is slow to resolve. The elimination half-life of lithium is 18 to 24 hours in young adults but is shorter in children and more prolonged in elderly persons. Renal clearance of lithium is decreased with renal insufficiency. Equilibrium is reached after 5 to 7 days of regular intake. Obesity is associated with higher rates of lithium clearance. The excretion of lithium is complex during pregnancy; excretion increases during pregnancy but decreases after delivery. Lithium is excreted in breast milk and insignificant amounts in the feces and sweat. Thyroid and renal concentrations of lithium are higher than serum levels.

An explanation for the mood-stabilizing effects of lithium remains elusive. Theories include alterations of ion transport and effects on neurotransmitters and neuropeptides, signal transduction pathways, and second messenger systems.

Therapeutic Indications

Bipolar I Disorder

MANIC EPISODES. Lithium controls acute mania and prevents relapse in about 80 percent of persons with bipolar I disorder and in a somewhat smaller percentage of persons with mixed (mania and depression) episodes, rapid-cycling bipolar disorder, or mood changes in encephalopathy. Lithium has a

relatively slow onset of action when used and exerts its antimanic effects over 1 to 3 weeks. Thus, a benzodiazepine, DRA, SDA, or valproic acid is usually administered for the first few weeks. Patients with mixed or dysphoric mania, rapid cycling, comorbid substance abuse, or organicity respond less well to lithium than those with classic mania.

BIPOLAR DEPRESSION. Lithium is effective in the treatment of depression associated with bipolar I disorder, as well as in the role of add-on therapy for patients with severe MDD. Augmentation of lithium therapy with valproic acid (Depakene) or carbamazepine (Tegretol) is usually well tolerated, with little risk of precipitation of mania.

When a depressive episode occurs in a person taking maintenance lithium, the differential diagnosis should include lithium-induced hypothyroidism, substance abuse, and lack of compliance with lithium therapy. One approach is to increase the lithium concentration (up to 1 to 1.2 mEq/L). Alternately we may augment with valproate or carbamazepine. It may also help to add supplemental thyroid hormone (e.g., 25 µg a day of liothyronine [Cytomel]), even in the presence of normal findings on thyroid function tests. Antidepressants or ECT are sometimes added judiciously. After the acute depressive episode resolves, other therapies should be tapered in favor of lithium monotherapy, if clinically tolerated.

MAINTENANCE. Maintenance treatment with lithium markedly decreases the frequency, severity, and duration of manic and depressive episodes in persons with bipolar I disorder. Lithium provides relatively more effective prophylaxis for mania than for depression, and supplemental antidepressant strategies may be necessary either intermittently or continuously. Lithium maintenance is almost always indicated after the first episode of bipolar I disorder, depression or mania, and should be considered after the first episode for adolescents or for persons who have a family history of bipolar I disorder. Others who benefit from lithium maintenance are those who have poor support systems, had no precipitating factors for the first episode, have a high suicide risk, had a sudden onset of the first episode, or had a first episode of mania. Clinical studies have shown that lithium reduces the incidence of suicide in bipolar I disorder patients six- or sevenfold. Lithium is also an effective treatment for persons with severe cyclothymic disorder.

It is wise to initiate maintenance therapy after the first manic episode, based on several observations. First, each episode of mania increases the risk of subsequent episodes. Second, among people responsive to lithium, relapses are 28 times more likely after lithium use is discontinued. Third, case reports describe persons who initially responded to lithium, discontinued taking it, and then had a relapse but no longer responded to lithium in subsequent episodes. Continued maintenance treatment with lithium is often associated with increased efficacy and reduced mortality. Therefore, an episode of depression or mania that occurs after a relatively short time on lithium maintenance does not necessarily represent treatment failure. However, lithium treatment alone may begin to lose its effectiveness after several years of successful use. If this occurs, then supplemental treatment with carbamazepine or valproate may be useful.

Maintenance lithium dosages can often be adjusted to achieve plasma concentration somewhat lower than that needed for the treatment of acute mania. If lithium use is to be discontinued, then the dosage should be slowly tapered. Abrupt discontinuation of lithium therapy is associated with an increased risk of recurrence of manic and depressive episodes.

Major Depressive Disorder. Lithium is effective in the long-term treatment of major depression, but it is not more effective than antidepressant drugs. The most common role for lithium in MDD is as an adjuvant to antidepressant use in persons who have failed to respond to the antidepressants alone. About 50 to 60 percent of antidepressant nonresponders do respond when lithium, 300 mg three times daily, is added to the antidepressant regimen. In some cases, a response may be seen within days, but most often, several weeks are required to see the efficacy of the regimen. Lithium alone may effectively treat depressed persons who have bipolar I disorder but have not yet had their first manic episode. Lithium has been reported to be effective in persons with MDD whose disorder has marked cyclicity.

Schizoaffective Disorder and Schizophrenia. Persons with prominent mood symptoms—either bipolar type or depressive type—with schizoaffective disorder are more likely to respond to lithium than those with predominant psychotic symptoms. Although SDAs and DRAs are the

treatments of choice for persons with schizoaffective disorder, lithium is a useful augmentation agent. This approach is particularly true for persons whose symptoms are resistant to treatment with SDAs and DRAs. Lithium augmentation of an SDA or DRA medication may be an effective treatment for persons with schizoaffective disorder even in the absence of a prominent mood disorder component. Some persons with schizophrenia who cannot take antipsychotic drugs may benefit from lithium treatment alone.

Other Indications. Over the years, reports have appeared about the use of lithium to treat a wide range of other psychiatric and nonpsychiatric conditions (Tables 21-25 and 21-26). The effectiveness and safety of lithium for most of these disorders have not been confirmed. Lithium has an anti-aggressive activity that is separate from its effects on mood. Aggressive outbursts in persons with schizophrenia, violent prison inmates, and children with conduct disorder and aggression, or self-mutilation in persons with intellectual disability, can sometimes be controlled with lithium.



Table 21-25
Psychiatric Uses of Lithium

Well established (FDA approved)
- Bipolar disorder
  - Manic episode
  - Maintenance therapy

Reasonably well established
- Bipolar I disorder, depressive episode
- Bipolar II disorder
- Rapid-cycling bipolar I disorder
- Cyclothymic disorder
- Major depressive disorder
  - Acute depression (as an augmenting agent)
  - Maintenance therapy
- Schizoaffective disorder

Evidence of benefit in particular groups
- Schizophrenia

Aggression (episodic), explosive behavior, and self-mutilation
Conduct disorder in children and adolescents
Intellectual disability
Cognitive disorders
Aggressive prisoners

Anecdotal, controversial, unresolved, or doubtful

Alcohol and other substance-related disorders
Cocaine abuse
Substance-induced mood disorder with manic features
Obsessive-compulsive disorder
Phobias
Posttraumatic stress disorder
ADHD
Eating disorders
  Anorexia nervosa
  Bulimia nervosa
Impulse-control disorders
Kleine–Levin syndrome
Mental disorders caused by a general medical condition (e.g., mood disorder caused by a general medical condition with manic features)
Periodic catatonia
Periodic hypersomnia
Personality disorders (e.g., antisocial, borderline, emotionally unstable, schizotypal)
Premenstrual dysphoric disorder
Sexual disorders
  Transvestism
  Exhibitionism
  Pathologic hypersexuality

FDA, Food and Drug Administration; ADHD, attention-deficit/hyperactivity disorder.



Table 21-26
Nonpsychiatric Uses of Lithium[a]

- Neurologic
  - Epilepsy
  - Headache (chronic cluster, hypnic, migraine, particularly cyclic)
  - Ménière disease (not supported by controlled studies)
  - Movement disorders
  - Huntington disease
  - L-Dopa–induced hyperkinesias
  - On–off phenomenon in Parkinson disease (controlled study found decreased akinesia but development of dyskinesia in a few cases)
  - Spasmodic torticollis
  - Tardive dyskinesia (not supported by controlled studies, and pseudoparkinsonism has been reported)
  - Tourette disorder
  - Pain (facial pain syndrome, painful shoulder syndrome, fibromyalgia)
  - Periodic paralysis (hypokalemic and hypermagnesic but not hyperkalemic)
- Hematologic
  - Aplastic anemia
  - Cancer—chemotherapy-induced, radiotherapy-induced
  - Neutropenia (one study found increased risk of sudden death in patients with preexisting cardiovascular disorder)
  - Drug-induced neutropenia (e.g., from carbamazepine, antipsychotics, immunosuppressives, and zidovudine)
  - Felty syndrome
  - Leukemia
- Endocrine
  - Thyroid cancer as an adjunct to radioactive iodine
  - Thyrotoxicosis
  - Syndrome of inappropriate antidiuretic hormone secretion
- Cardiovascular
  - Antiarrhythmic agent (animal data only)
- Dermatologic
  - Genital herpes (controlled studies support topical and oral use)

Eczematoid dermatitis
Seborrheic dermatitis (controlled study supports)
Gastrointestinal
Cyclic vomiting
Gastric ulcers
Pancreatic cholera
Ulcerative colitis
Respiratory
Asthma (controlled study did not support)
Cystic fibrosis
Other
Bovine spastic paresis

[a]All the uses listed here do not have Food and Drug Administration(FDA)–approved labeling. There are conflicting reports about many of these uses—some have negative findings in controlled studies, and a few involve reports of possible adverse effects.

L-Dopa, levodopa.

Precautions and Adverse Effects

More than 80 percent of patients taking lithium experience side effects. It is crucial to minimize the risk of adverse events through monitoring of lithium blood levels and to use appropriate pharmacologic interventions to counteract unwanted effects when they occur. The most common adverse effects are summarized in Table 21-27. Patient education can play an essential role in reducing the incidence and severity of side effects. Patients taking lithium should be advised that changes in the body's water and salt content can affect the amount of lithium excreted, resulting in either increases or decreases in lithium concentrations. Excessive sodium intake (e.g., a dramatic dietary change) lowers lithium concentrations.



Table 21-27
Adverse Effects of Lithium

Neurologic
Benign, nontoxic: dysphoria, lack of spontaneity, slowed reaction time, memory difficulties
Tremor: postural, occasional extrapyramidal

	Toxic: coarse tremor, dysarthria, ataxia, neuromuscular irritability, seizures, coma, death
	Miscellaneous: peripheral neuropathy, benign intracranial hypertension, myasthenia gravis–like syndrome, altered creativity, lowered seizure threshold
Endocrine	
	Thyroid: goiter, hypothyroidism, exophthalmos, hyperthyroidism (rare)
	Parathyroid: hyperparathyroidism, adenoma
Cardiovascular	
	Benign T-wave changes, sinus node dysfunction
Renal	
	Concentrating defect, morphologic changes, polyuria (nephrogenic diabetes insipidus), reduced GFR, nephrotic syndrome, renal tubular acidosis
Dermatologic	
	Acne, hair loss, psoriasis, rash
Gastrointestinal	
	Appetite loss, nausea, vomiting, diarrhea
Miscellaneous	
	Altered carbohydrate metabolism, weight gain, fluid retention

GFR, glomerular filtration rate.

Conversely, too little sodium (e.g., fad diets) can lead to potentially toxic concentrations of lithium. Decreases in body fluid (e.g., excessive perspiration) can lead to dehydration and lithium intoxication. Patients should report whenever another clinician prescribes medications, because many commonly used agents can affect lithium concentrations.

Cardiac Effects. Lithium can cause diffuse slowing, widening of the frequency spectrum, and potentiation and disorganization of background rhythm on electrocardiography (ECG). Bradycardia and cardiac arrhythmias may occur, especially in people with cardiovascular disease. Lithium infrequently reveals Brugada syndrome, an inherited, life-threatening heart problem that some people may have without knowing it. It can cause a severe

abnormal heartbeat and other symptoms (such as severe dizziness, fainting, shortness of breath) that need medical attention right away. Before starting lithium treatment, clinicians should ask about known heart conditions, unexplained fainting, and family history of problems or sudden unexplained death before age 45.

Gastrointestinal Effects. GI symptoms—which include nausea, decreased appetite, vomiting, and diarrhea—can be diminished by dividing the dosage, administering the lithium with food, or switching to another lithium preparation. The lithium preparation least likely to cause diarrhea is lithium citrate. Some lithium preparations contain lactose, which can cause diarrhea in lactose-intolerant persons. Persons taking slow-release formulations of lithium who experience diarrhea caused by unabsorbed medication in the lower part of the GI tract may experience less diarrhea than with standard-release preparations. Diarrhea may also respond to antidiarrheal preparations such as loperamide (Imodium, Kaopectate), bismuth subsalicylate (Pepto-Bismol), or diphenoxylate with atropine (Lomotil).

Weight Gain. Weight gain results from a poorly understood effect of lithium on carbohydrate metabolism. Weight gain can also result from lithium-induced hypothyroidism, lithium-induced edema, or excessive consumption of soft drinks and juices to quench lithium-induced thirst.

Neurologic Effects

TREMOR. A lithium-induced postural tremor may occur that is usually 8 to 12 Hz and is most notable in outstretched hands, especially in the fingers, and during tasks involving fine manipulations. The tremor can be reduced by dividing the daily dosage, using a sustained-release formulation, reducing caffeine intake, reassessing the concomitant use of other medicines, and treating comorbid anxiety. β-Adrenergic receptor antagonists, such as propranolol, 30 to 120 mg a day in divided doses, and primidone (Mysoline), 50 to 250 mg a day, are usually effective in reducing the tremor. In persons with hypokalemia, potassium supplementation may improve the tremor. When a person taking lithium has a severe tremor, the possibility of lithium toxicity should be suspected and evaluated.

COGNITIVE EFFECTS. Lithium use has been associated with dysphoria, lack of spontaneity, slowed reaction times, and impaired memory. We should note these symptoms carefully because they are a frequent cause of noncompliance. The differential diagnosis for such symptoms should include depressive disorders, hypothyroidism, hypercalcemia, other illnesses, and other drugs. Some, but not all, persons have reported that fatigue and mild cognitive impairment decrease with time.

OTHER NEUROLOGIC EFFECTS. Uncommon neurologic adverse effects include symptoms of mild parkinsonism, ataxia, and dysarthria, although the last two symptoms may also be attributable to lithium intoxication. Lithium is rarely associated with the development of peripheral neuropathy, benign intracranial hypertension (pseudotumor cerebri), findings resembling myasthenia gravis, and increased risk of seizures.

Renal Effect. The most common adverse renal effect of lithium is polyuria with secondary polydipsia. The symptom is particularly a problem in 25 to 35 percent of persons taking lithium who may have a urine output of more than 3 L a day (reference range: 1 to 2 L a day). The polyuria primarily results from lithium antagonism to the effects of antidiuretic hormone, which thus causes diuresis. When polyuria is a significant problem, the person's renal function should be evaluated and followed up with 24-hour urine collections for creatinine clearance determinations. Treatment consists of fluid replacement, the use of the lowest effective dosage of lithium, and single daily dosing of lithium. Treatment can also involve the use of a thiazide or potassium-sparing diuretic. For example, amiloride (Midamor), spironolactone (Aldactone), triamterene (Dyrenium), or amiloride–hydrochlorothiazide (Moduretic) may help. If treatment with a diuretic is initiated, the lithium dosage should be halved, and the diuretic should not be started for 5 days, because the diuretic is likely to increase lithium retention.

The most severe renal adverse effects, which are rare and associated with continuous lithium administration for 10 years or more, involve the appearance of nonspecific interstitial fibrosis, associated with gradual decreases in glomerular filtration rate and increases in serum creatinine concentrations, and rarely with renal failure. Lithium is occasionally

associated with nephrotic syndrome and features of distal renal tubular acidosis. Another pathologic finding in patients with lithium nephropathy is the presence of microcysts. Magnetic resonance imaging (MRI) can be used to demonstrate renal microcysts secondary to chronic lithium nephropathy and therefore avoid renal biopsy. It is prudent for persons taking lithium to check their serum creatinine concentration, urine chemistries, and 24-hour urine volume at 6-month intervals. If creatinine levels do rise, then more frequent monitoring and MRI might be considered.

Thyroid Effects. Lithium causes a generally benign and often transient diminution in the concentrations of circulating thyroid hormones. Reports have attributed goiter (5 percent of persons), benign reversible exophthalmos, hyperthyroidism, and hypothyroidism (7 to 10 percent of persons) to lithium treatment. Lithium-induced hypothyroidism is more common in women (14 percent) than in men (4.5 percent). Women are at the highest risk during the first 2 years of treatment. Persons taking lithium to treat bipolar disorder are twice as likely to develop hypothyroidism if they develop rapid cycling. About 50 percent of persons receiving long-term lithium treatment have laboratory abnormalities, such as an abnormal TRH response, and about 30 percent have elevated concentrations of TSH. If symptoms of hypothyroidism are present, replacement with levothyroxine (Synthroid) is indicated. Even in the absence of hypothyroid symptoms, some clinicians treat persons with significantly elevated TSH concentrations with levothyroxine. In lithium-treated persons, TSH concentrations should be measured every 6 to 12 months. Lithium-induced hypothyroidism should be considered when evaluating depressive episodes that emerge during lithium therapy.

Cardiac Effects. The cardiac effects of lithium resemble those of hypokalemia on ECG. The displacement of intracellular potassium by the lithium ion causes this. The most common changes on the ECG are T-wave flattening or inversion. The changes are benign and disappear after lithium is excreted from the body.

Lithium depresses the pacemaking activity of the sinus node, sometimes resulting in sinus dysrhythmias, heart block, and episodes of syncope. Lithium treatment, therefore, is contraindicated in persons with sick sinus syndrome. In rare cases, ventricular arrhythmias and congestive heart failure

have been associated with lithium therapy. Lithium cardiotoxicity is prevalent in persons on a low-salt diet, those taking certain diuretics or ACEIs, and those with fluid–electrolyte imbalances or any renal insufficiency.

Dermatologic Effects. Dermatologic effects may be dose-dependent. They include acneiform, follicular, and maculopapular eruptions; pretibial ulcerations; and worsening of psoriasis. Occasionally, aggravated psoriasis or acneiform eruptions may force the discontinuation of lithium treatment. Alopecia has also been reported. Persons with many of those conditions respond favorably to changing to another lithium preparation and the usual dermatologic measures. Lithium concentrations should be monitored if tetracycline is used for the treatment of acne because it can increase the retention of lithium.

Lithium Toxicity and Overdoses. The early signs and symptoms of lithium toxicity include neurologic symptoms, such as coarse tremor, dysarthria, and ataxia; GI symptoms; cardiovascular changes; and renal dysfunction. The later signs and symptoms include impaired consciousness, muscular fasciculations, myoclonus, seizures, and coma. Signs and symptoms of lithium toxicity are outlined in Table 21-28. Risk factors include exceeding the recommended dosage, renal impairment, low-sodium diet, drug interaction, and dehydration. Elderly persons are more vulnerable to the effects of increased serum lithium concentrations. The greater the degree and duration of elevated lithium concentrations, the worse are the symptoms of lithium toxicity.



Table 21-28
Signs and Symptoms of Lithium Toxicity

1. Mild to moderate intoxication (lithium level, 1.5–2.0 mEq/L)	
GI	Vomiting Abdominal pain Dryness of mouth
Neurologic	Ataxia Dizziness

	Slurred speech
	Nystagmus
	Lethargy or excitement
	Muscle weakness
2. Moderate to severe intoxication (lithium level: 2.0–2.5 mEq/L)	
GI	Anorexia
	Persistent nausea and vomiting
Neurologic	Blurred vision
	Muscle fasciculations
	Clonic limb movements
	Hyperactive deep tendon reflexes
	Choreoathetoid movements
	Convulsions
	Delirium
	Syncope
	Electroencephalographic changes
	Stupor
	Coma
	Circulatory failure (lowered BP, cardiac arrhythmias, and conduction abnormalities)
3. Severe lithium intoxication (lithium level >2.5 mEq/L)	
	Generalized convulsions
	Oliguria and renal failure
	Death

Lithium toxicity is a medical emergency, potentially causing permanent neuronal damage and death. In cases of toxicity (Table 21-29), lithium should be stopped and dehydration treated. Unabsorbed lithium can be removed from the GI tract by ingestion of sodium polystyrene sulfonate (Kayexalate) or polyethylene glycol solution (GoLYTELY), but not activated charcoal. Ingestion of a single large dose may create clumps of medication in the stomach, which can be removed by gastric lavage with a wide-bore tube. The

value of forced diuresis is still debated. In severe cases, hemodialysis rapidly removes excessive amounts of serum lithium. Postdialysis serum lithium concentrations may increase as lithium is redistributed from tissues to the blood, so repeat dialysis may be needed. Neurologic improvement may lag behind the clearance of serum lithium by several days because lithium crosses the blood–brain barrier slowly.



Table 21-29
Management of Lithium Toxicity

1. Contact personal physician or go to a hospital emergency department
2. Lithium should be discontinued
3. Vital signs and a neurologic examination with complete formal mental status examination
4. Lithium level, serum electrolytes, renal function tests, and ECG
5. Emesis, gastric lavage, and absorption with Kayexalate or GoLYTELY
6. For any patient with a serum lithium level greater than 4.0 mEq/L, hemodialysis

ECG, electrocardiography.

Adolescents. The serum lithium concentrations for adolescents are similar to those for adults. Weight gain and acne associated with lithium use can be particularly troublesome to adolescents.

Elderly Persons. Lithium is a safe and effective drug for elderly persons. However, the treatment of elderly persons taking lithium may be complicated by the presence of other medical illnesses, decreased renal function, special diets that affect lithium clearance, and generally increased sensitivity to lithium. Elderly persons should initially be given low dosages, their dosages should be switched less frequently than those of younger persons, and a longer time must be allowed for renal excretion to equilibrate with absorption before lithium can be assumed to have reached its steady-state concentrations.

Pregnant Women. Lithium should not be administered to pregnant women in the first trimester because of the risk of congenital disabilities. The most common malformations involve the cardiovascular system, most commonly Ebstein anomaly of the tricuspid valves. The risk of Ebstein malformation in lithium-exposed fetuses is 1 in 1,000, which is 20 times the risk in the general population. The possibility of fetal cardiac anomalies can be evaluated with fetal echocardiography. The teratogenic risk of lithium (4 to 12 percent) is higher than that for the general population (2 to 3 percent) but appears to be lower than that associated with the use of valproate or carbamazepine. A woman who continues to take lithium during pregnancy should use the lowest effective dosage. The maternal lithium concentration must be monitored closely during pregnancy, and especially after pregnancy, because of the significant decrease in renal lithium excretion as the renal function returns to normal in the first few days after delivery. Adequate hydration can reduce the risk of lithium toxicity during labor. Lithium prophylaxis is recommended for all women with bipolar disorder as they enter the postpartum period. Lithium is excreted into breast milk and should be taken by a nursing mother only after careful evaluation of potential risks and benefits. Signs of lithium toxicity in infants include lethargy, cyanosis, abnormal reflexes, and sometimes hepatomegaly.

Miscellaneous Effects. Lithium should be used with caution in diabetic persons, who should monitor their blood glucose concentrations carefully to avoid diabetic ketoacidosis. Benign, reversible leukocytosis is commonly associated with lithium treatment. Dehydrated, debilitated, and medically ill persons are most susceptible to adverse effects and toxicity.

Drug Interactions

Lithium drug interactions are summarized in Table 21-30.

Lithium is commonly used in conjunction with DRAs. This combination is typically sufficient and safe. However, coadministration of higher dosages of a DRA and lithium may result in a synergistic increase in the symptoms of lithium-induced neurologic side effects and neuroleptic extrapyramidal symptoms. In rare instances, encephalopathy has been reported with this combination.

The coadministration of lithium and carbamazepine, lamotrigine, valproate, and clonazepam may increase lithium concentrations and aggravate lithium-induced neurologic adverse effects. Treatment with the combination should be initiated at slightly lower dosages than usual, and the dosages should be increased gradually. Changes from one to another treatment for mania should be made carefully, with as little temporal overlap between the drugs as possible.

Most diuretics (e.g., thiazide and potassium-sparing) can increase lithium concentrations; when treatment with such a diuretic is stopped, the clinician may need to increase the person's daily lithium dosage. Osmotic and loop diuretics, carbonic anhydrase inhibitors, and xanthines (including caffeine) may reduce lithium concentrations to below therapeutic concentrations. Whereas ACEIs may cause an increase in lithium concentrations, the AT_1 angiotensin II receptor inhibitors losartan (Cozaar) and irbesartan (Avapro) do not alter lithium concentrations. A wide range of NSAIDs can decrease lithium clearance, thereby increasing lithium concentrations. These drugs include indomethacin (Indocin), phenylbutazone (Azolid), diclofenac (Voltaren), ketoprofen (Orudis), oxyphenbutazone (Oxalid), ibuprofen (Motrin, Advil), piroxicam (Feldene), and naproxen (Naprosyn). Aspirin and sulindac (Clinoril) do not affect lithium concentrations.

The coadministration of lithium and quetiapine (Seroquel) may cause somnolence but is otherwise well tolerated. The coadministration of lithium and ziprasidone (Geodon) may modestly increase the incidence of tremors. The coadministration of lithium and calcium channel inhibitors should be avoided because of potentially fatal neurotoxicity.

A person taking lithium who is about to undergo ECT should discontinue taking lithium 2 days before beginning ECT to reduce the risk of delirium.

Laboratory Interferences

Lithium does not interfere with any laboratory tests, but lithium-induced alterations include an increased WBC count, decreased serum thyroxine, and increased serum calcium. Blood collected in a lithium–heparin anticoagulant tube will produce falsely elevated lithium concentrations.

Dosage and Clinical Guidelines

Initial Medical Workup. All patients should have a routine laboratory workup and physical examination before being started on lithium. The laboratory tests should include serum creatinine concentration (or a 24-hour urine creatinine if the clinician has any reason to be concerned about renal function), electrolytes, thyroid function (TSH, T_3 [triiodothyronine], and T_4 [thyroxine]), a CBC, ECG, and a pregnancy test in women of childbearing age.

Dosage Recommendations. Lithium formulations include immediate-release 150-, 300-, and 600-mg lithium carbonate capsules (Eskalith and generic), 300 mg lithium carbonate tablets (Lithotabs), 450-mg controlled-release lithium carbonate capsules (Eskalith CR and Lithonate), and 8 mEq/5 mL of lithium citrate syrup.

The starting dosage for most adults is 300 mg of the regular-release formulation three times daily. The starting dosage for elderly persons or persons with renal impairment should be 300 mg once or twice daily. After stabilization, dosages between 900 and 1,200 mg a day usually produce a therapeutic plasma concentration of 0.6 to 1 mEq/L, and a daily dose of 1,200 to 1,800 mg usually produces a therapeutic concentration of 0.8 to 1.2 mEq/L. Maintenance dosing can be given either in two or three divided doses of the regular-release formulation or in a single dosage of the sustained-release formulation equivalent to the combined daily dosage of the regular-release formulation. The use of divided doses reduces gastric upset and avoids single high-peak lithium concentrations. Discontinuation of lithium should be gradual to minimize the risk of early recurrence of mania and to permit recognition of early signs of recurrence.



Table 21-30
Drug Interactions with Lithium

Drug Class	Reaction
Antipsychotics	Case reports of encephalopathy, worsening of extrapyramidal adverse effects, and neuroleptic malignant syndrome; inconsistent reports of altered red blood cell and plasma

	concentrations of lithium, antipsychotic drug, or both
Antidepressants	Occasional reports of a serotonin-like syndrome with potent serotonin reuptake inhibitors
Anticonvulsants	No significant pharmacokinetic interactions with carbamazepine or valproate; reports of neurotoxicity with carbamazepine; combinations helpful for treatment resistance
NSAIDs	May reduce renal lithium clearance and increase serum concentration; toxicity reported (exception is aspirin)
Diuretics	
Thiazides	Well-documented reduced renal lithium clearance and increased serum concentration; toxicity reported
Potassium sparing	Limited data; may increase lithium concentration
Loop	Lithium clearance unchanged (some case reports of increased lithium concentration)
Osmotic (mannitol, urea)	Increase renal lithium clearance and decrease lithium concentration
Xanthine (aminophylline, caffeine, theophylline)	Increase renal lithium clearance and decrease lithium concentration
Carbonic anhydrase inhibitors (acetazolamide)	Increase renal lithium clearance
ACEIs	Reports of reduced lithium clearance, increased

	concentrations, and toxicity
Calcium channel inhibitors	Case reports of neurotoxicity; no consistent pharmacokinetic interactions
Miscellaneous	
Succinylcholine, pancuronium	Reports of prolonged neuromuscular blockade
Metronidazole	Increased lithium concentration
Methyldopa	Few reports of neurotoxicity
Sodium bicarbonate	Increased renal lithium clearance
Iodides	Additive antithyroid effects
Propranolol	Used for lithium tremor; possible slight increase in lithium concentration

NSAID, nonsteroidal anti-inflammatory drug; ACEI, angiotensin-converting enzyme inhibitor.

Laboratory Monitoring. The periodic measurement of serum lithium concentration is an essential aspect of patient care, but it should always be combined with sound clinical judgment. A laboratory report listing the therapeutic range as 0.5 to 1.2 mEq/L may lull a clinician into disregarding early signs of lithium intoxication in patients whose levels are around 1.2 mEq/L. Clinical toxicity, especially in elderly persons, has been well documented at or slightly above the upper limit of the therapeutic range.

Regular monitoring of serum lithium concentrations is essential. Lithium levels should be obtained every 2 to 6 months except when there are signs of toxicity, during dosage adjustments, and in persons suspected to be noncompliant with the prescribed dosages. Under these circumstances, levels may be done weekly. Baseline ECG studies are essential and should be repeated annually.

When obtaining blood for lithium levels, patients should be at steady-state lithium dosing (usually after 5 days of constant dosing), preferably using a twice-daily or thrice-daily dosing regimen, and the blood sample must be drawn 12 hours (±30 minutes) after a given dose. Lithium concentrations 12

hours postdose in persons treated with sustained-release preparations are generally about 30 percent higher than the corresponding concentrations obtained from those taking the regular-release preparations. Because available data are based on a sample population following a multiple-dosage regimen, regular-release formulations given at least twice daily should be used for the initial determination of the appropriate dosages. Factors that may cause fluctuations in lithium measurements include dietary sodium intake, mood state, activity level, body position, and use of an improper blood sample tube.

Laboratory values that do not seem to correspond to clinical status may result from the collection of blood in a tube with a lithium–heparin anticoagulant (which can give results falsely elevated by as much as 1 mEq/L) or aging of the lithium ion-selective electrode (which can cause inaccuracies of up to 0.5 mEq/L). After the daily dose has been set, it is reasonable to change to the sustained-release formulation given once daily.

Effective serum concentrations for mania are 1.0 to 1.2 mEq/L, a level associated with 1,800 mg a day. The recommended range for maintenance treatment is 0.4 to 0.8 mEq/L, which is usually achieved with a daily dose of 900 to 1,200 mg. A small number of persons will not achieve therapeutic benefit with a lithium concentration of 1.2 mEq/L, yet will have no signs of toxicity. For such persons, titration of the lithium dosage to achieve a concentration above 1.2 mEq/L may be warranted. Some patients can be maintained at concentrations below 0.4 mEq/L. There may be considerable variation from patient to patient, so it is best to follow the maxim, “treat the patient, not the laboratory results.” The only way to establish an optimal dose for a patient may be through trial and error.

Package inserts (US) for lithium products list effective serum concentrations for mania between 1.0 and 1.2 mEq/L (usually achieved with 1,800 mg of lithium carbonate daily) and for long-term maintenance between 0.6 and 1.2 mEq/L (usually achieved with 900 to 1,200 mg of lithium carbonate daily). The dose–blood level relationship may vary considerably from patient to patient. The likelihood of achieving a response at levels above 1.2 mEq/L is usually significantly outweighed by the increased risk of toxicity, although rarely a patient may both require and tolerate a higher-than-usual blood concentration.

What constitutes the lower end of the therapeutic range remains a matter of

debate. A prospective 3-year study found patients who maintained a concentration between 0.4 and 0.6 mEq/L (mean 0.54) were 2.6 times more likely to relapse than those who maintained between 0.8 and 1.0 mEq/L (mean 0.83). However, higher blood concentrations produced more adverse effects and were less well tolerated.

If there is no response after 2 weeks at a concentration that is beginning to cause adverse effects, then the person should taper off lithium over 1 to 2 weeks, and other mood-stabilizing drugs should be tried.

Patient Education. Lithium has a narrow therapeutic index, and many factors can upset the balance between lithium concentrations that are well tolerated and therapeutic, and those that produce side effects or toxicity. It is thus imperative that persons taking lithium be educated about signs and symptoms of toxicity, factors that affect lithium levels, how and when to obtain laboratory testing, and the importance of regular communication with the prescribing physician. Lithium concentrations can be disrupted by common factors such as excessive sweating from ambient heat or exercise or the use of widely prescribed agents such as ACEIs or NSAIDs. Patients may stop taking their lithium because they are feeling well or because they are experiencing side effects. They should be advised against discontinuing or modifying their lithium regimen. Table 21-31 lists some essential instructions for patients.

VALPROATE

Valproate (Depakene, Depakote), or valproic acid, is approved for the treatment of manic episodes associated with bipolar I disorder and is one of the most widely prescribed mood stabilizers in psychiatry. It has a rapid onset of action and is well-tolerated, and numerous studies suggest that it reduces the frequency and intensity of recurrent manic episodes over extended periods.

Chemistry

Valproate is a simple-chain branch carboxylic acid. It is called valproic acid because it is rapidly converted to the acid form in the stomach. Multiple formulations of valproic acid are marketed. These include valproic acid

(Depakene), divalproex sodium (Depakote), an enteric-coated delayed-release 1:1 mixture of valproic acid and sodium valproate available in tablet and sprinkle formulation (can be opened and spread on food); and sodium valproate injection (Depacon). Extended-release preparation is also available. Each of these is therapeutically equivalent because, at physiologic pH, valproic acid dissociates into valproate ion.



Table 21-31
Instructions to Patients Taking Lithium

Lithium can be remarkably effective in treating your disorder. If not used appropriately and not monitored closely, it can be ineffective and potentially harmful. It is important to keep the following instructions in mind.

Dosing

Take lithium exactly as directed by your doctor—never take more or less than the prescribed dose.

Do not stop taking without speaking to your doctor.

If you miss a dose, take it as soon as possible. If it is within 4 hr of the next dose, skip the missed dose (about 6 hr in the case of extended-release or slow-release preparations). Never double up doses.

Blood Tests

Comply with the schedule of recommended regular blood tests.

Despite their inconvenience and discomfort, your lithium blood levels, thyroid function, and kidney status need to be monitored as long as you take lithium.

When going to have lithium levels checked, you should have taken your last lithium dose 12 hr earlier.

Use of Other Medications

Do not start any prescription or over-the-counter medications without telling your doctor.

Even drugs such as ibuprofen (Advil, Motrin) and naproxen (Aleve) can significantly increase lithium levels.

Diet and Fluid Intake

Avoid sudden changes in your diet or fluid intake. If you do go on a diet, your doctor may need to increase the frequency of blood tests.

Caffeine and alcohol act as diuretics and can lower your lithium concentrations.

During treatment with lithium, it is recommended that you drink about 2 or 3 quarts of fluid daily and use normal amounts of salt.

Inform your doctor if you start or stop a low-salt diet.

Recognizing Potential Problems

If you engage in vigorous exercise or have an illness that causes sweating, vomiting, or diarrhea, consult your doctor because these might affect lithium levels.

Nausea, constipation, shakiness, increased thirst, frequency of urination, weight gain, or swelling of the extremities should be reported to your doctor.

Blurred vision, confusion, loss of appetite, diarrhea, vomiting, muscle weakness, lethargy, shakiness, slurred speech, dizziness, loss of balance, inability to urinate, or seizures could indicate severe toxicity and should prompt immediate medical attention.

Pharmacologic Actions

Regardless of how it is formulated, valproate is rapidly and completely absorbed 1 to 2 hours after oral administration, with peak concentrations occurring 4 to 5 hours after oral administration. The plasma half-life of valproate is 10 to 16 hours. Valproate is highly protein-bound. Protein binding becomes saturated at higher dosages, and concentrations of therapeutically effective free valproate increase at serum concentrations above 50 to 100 µg/mL. The unbound portion of valproate is considered to be pharmacologically active and can cross the blood–brain barrier. The extended-release preparation produces lower peak concentrations and higher minimum concentrations and can be given once a day. Valproate is metabolized primarily by hepatic glucuronidation and mitochondrial β

oxidation.

The biochemical basis of valproate's therapeutic effects remains poorly understood. Postulated mechanisms include enhancement of GABA activity, modulation of voltage-sensitive sodium channels, and action on extrahypothalamic neuropeptides.

Therapeutic Indications

Valproate is currently approved as monotherapy or adjunctive therapy of complex partial seizures, monotherapy and adjunctive therapy of simple and complex absence seizures, and adjunctive therapy for patients with multiple seizures that include absence seizures. Divalproex has additional indications for prophylaxis of migraine.

Bipolar I Disorder

ACUTE MANIA. About two-thirds of persons with acute mania respond to valproate. The majority of patients with mania usually respond within 1 to 4 days after achieving valproate serum concentrations above 50 µg/mL. The antimanic response is generally associated with levels greater than 50 µg/mL, in a range of 50 to 125 µg/mL. This serum concentration may be gradually achieved within 1 week of initiation of dosing, but rapid oral loading strategies achieve therapeutic serum concentrations in 1 day and can control manic symptoms within 5 days. The short-term antimanic effects of valproate can be augmented with the addition of lithium, carbamazepine (Tegretol), SDAs, or DRAs. Numerous studies have suggested that the irritable manic subtype responds significantly better to sodium valproate than lithium or placebo. Because of its more favorable profile of cognitive, dermatologic, thyroid, and renal adverse effects, valproate is preferred to lithium for the treatment of acute mania in children and elderly persons.

ACUTE BIPOLAR DEPRESSION. Valproate possesses some activity as a short-term treatment of depressive episodes in bipolar I disorder, but this effect is far less pronounced than for the treatment of manic episodes. Among depressive symptoms, valproate is more useful for the treatment of agitation than dysphoria. In clinical practice, valproate is most often used as add-on therapy to an antidepressant to prevent the development of mania or rapid

cycling.

PROPHYLAXIS. Studies suggest that valproate is useful in the prophylactic treatment of bipolar I disorder, resulting in fewer, less severe, and shorter manic episodes. In direct comparison, valproate is at least as effective as lithium and is better tolerated than lithium. It may be particularly useful in persons with rapid-cycling and ultrarapid-cycling bipolar disorders, dysphoric or mixed mania, and mania caused by a general medical condition as well as in persons who have comorbid substance use disorders or panic attacks and in persons who have not had complete favorable responses to lithium treatment.

Schizophrenia and Schizoaffective Disorder. Valproate may accelerate response to antipsychotic therapy in patients with schizophrenia or schizoaffective disorder. Valproate alone is generally less effective in schizoaffective disorder than in bipolar I disorder. Valproate alone is ineffective for the treatment of psychotic symptoms and is typically used in combination with other drugs in patients with these symptoms.

Other Mental Disorders. Valproate has been studied for possible efficacy in a broad range of psychiatric disorders. These include alcohol withdrawal and relapse prevention, panic disorder, PTSD, impulse control disorder, borderline personality disorder, and behavioral agitation and dementia. Evidence supporting use in these cases is weak, and any observed therapeutic effects may be related to the treatment of a comorbid bipolar disorder.

Precautions and Adverse Reactions

Although valproate treatment is generally well tolerated and safe, it carries quite a few black box warnings and other warnings (Table 21-32). The two most serious adverse effects of valproate treatment affect the pancreas and liver. Risk factors for potentially fatal hepatotoxicity include young age (younger than 3 years); concurrent use of phenobarbital; and the presence of neurologic disorders, especially inborn errors of metabolism. The rate of fatal hepatotoxicity in persons who have been treated with valproate alone is 0.85 per 100,000 persons; no persons older than the age of 10 years have been reported to have died from hepatotoxicity. Therefore, the risk of this adverse

reaction in adult psychiatric patients is low.

Table 21-32
Black Box Warnings and Other Warnings for Valproate

More Serious Side Effect	Management Considerations
Hepatotoxicity	Rare, idiosyncratic event Estimated risk, 1:118,000 (adults) Greatest risk profile (polypharmacy, younger than 2 yr of age, intellectual disability): 1:800
Pancreatitis	Rare, similar pattern to hepatotoxicity Incidence in clinical trial data is 2 in 2,416 (0.0008%) Postmarketing surveillance shows no increased incidence Relapse with rechallenge Asymptomatic amylase not predictive
Hyperammonemia	Rare; more common in combination with carbamazepine (Tegretol) Associated with coarse tremor and may respond to L-carnitine administration
Associated with urea cycle disorders	Discontinue valproate and protein intake Assess underlying urea cycle disorder Divalproex is contraindicated in patients with urea cycle disorders
Teratogenicity	Neural tube defect: 1–4% with valproate Preconceptual education and folate–vitamin B complex supplementation for all young women of childbearing potential
Somnolence in elderly persons	Slower titration than conventional doses Regular monitoring of fluid and nutritional intake

Thrombocytopenia	Decrease dose if clinically symptomatic (i.e., bruising, bleeding gums) Thrombocytopenia more likely with valproate levels ≥110 µg/mL (women) and ≥135 µg/mL (men)

Nevertheless, if symptoms of lethargy, malaise, anorexia, nausea and vomiting, edema, and abdominal pain occur in a person treated with valproate, the clinician must consider the possibility of severe hepatotoxicity. A modest increase in liver function test results does not correlate with the development of severe hepatotoxicity. Rare cases of pancreatitis have been reported; they occur most often in the first 6 months of treatment, and the condition occasionally results in death. The pancreatic function can be assessed and followed by serum amylase concentrations. Other potentially severe consequences of treatment include hyperammonemia-induced encephalopathy and thrombocytopenia. Thrombocytopenia and platelet dysfunction occur most commonly at high dosages and result in the prolongation of bleeding times.

There are multiple concerns regarding the use of valproate during pregnancy. Women who require valproate therapy should, therefore, inform their physicians if they intend to become pregnant. First-trimester use of valproate has been associated with a 3 to 5 percent risk of neural tube defects, as well as an increased risk of other malformations affecting the heart and other organ systems. Multiple reports have also indicated that in utero exposure to valproate may negatively affect cognitive development in children of mothers who take valproate during pregnancy. They have lower IQ scores at age 6 years compared with those exposed to other antiepileptic drugs. Fetal valproate exposure has dose-dependent associations with reduced cognitive abilities across a range of domains at 6 years of age. Valproate exposure may also increase the risk of autistic spectrum disorder.

Valproate is also associated with teratogenicity, most notably neural tube defects (e.g., spina bifida). The risk is about 1 to 4 percent of all women who take valproate during the first trimester of pregnancy. The risk of valproate-induced neural tube defects can be reduced with daily folic acid supplements (1 to 4 mg a day). All women with childbearing potential who take the drug

should be given folic acid supplements. Infants breastfed by mothers taking valproate develop serum valproate concentrations 1 to 10 percent of maternal serum concentrations, but no data suggest that this poses a risk to the infant. Valproate is not contraindicated in nursing mothers. We should avoid this drug in patients with hepatic diseases. Valproate may be especially problematic for adolescents and young women. Cases of polycystic ovarian disease have been reported in women using valproate. Even when the full syndromal criteria for this syndrome are not met, many of these women develop menstrual irregularities, hair loss, and hirsutism. These effects are thought to result from a metabolic syndrome that is driven by insulin resistance and hyperinsulinemia.

The common adverse effects associated with valproate (Table 21-33) are those affecting the GI system, such as nausea, vomiting, dyspepsia, and diarrhea. The GI effects are generally most common in the first month of treatment, significantly if the dosage is increased rapidly. Unbuffered valproic acid (Depakene) is more likely to cause GI symptoms than the enteric-coated "sprinkle" or the delayed-release divalproex sodium formulations. Other common adverse effects involve the nervous system, such as sedation, ataxia, dysarthria, and tremor. Valproate-induced tremor may respond well to treatment with β-adrenergic receptor antagonists or gabapentin. Treatment of the other neurologic adverse effects usually requires lowering the valproate dosage.

Weight gain is a common adverse effect, especially in long-term treatment, and can best be treated by strict limitation of caloric intake. Hair loss may occur in 5 to 10 percent of all persons treated, and rare cases of complete loss of body hair have been reported. Some clinicians have recommended treatment of valproate-associated hair loss with vitamin supplements that contain zinc and selenium. About 5 to 40 percent of persons experience a persistent but clinically insignificant elevation in liver transaminases up to three times the upper limit of normal, which is usually asymptomatic and resolves after discontinuation of the drug. High dosages of valproate (above 1,000 mg a day) may rarely produce mild to moderate hyponatremia, most likely because of some degree of the syndrome of secretion of inappropriate antidiuretic hormone, which is reversible upon lowering of the dosage. Overdoses of valproate can lead to coma and death.

Table 21-33
Adverse Effects of Valproate

Common
GI irritation
Nausea
Sedation
Tremor
Weight gain
Hair loss
Uncommon
Vomiting
Diarrhea
Ataxia
Dysarthria
Persistent elevation of hepatic transaminases
Rare
Fatal hepatotoxicity (primarily in pediatric patients)
Reversible thrombocytopenia
Platelet dysfunction
Coagulation disturbances
Edema
Hemorrhagic pancreatitis
Agranulocytosis
Encephalopathy and coma
Respiratory muscle weakness and respiratory failure

GI, gastrointestinal.

Drug Interactions

Valproate is commonly prescribed as part of a regimen involving other psychotropic agents. The only consistent drug interaction with lithium, if both drugs are maintained in their respective therapeutic ranges, is the exacerbation of drug-induced tremors, which can usually be treated with β-receptor antagonists. The combination of valproate and DRAs may result in increased sedation, as can be seen when valproate is added to any CNS

depressant (e.g., alcohol), and increased severity of extrapyramidal symptoms, which usually responds to treatment with antiparkinsonian drugs. Valproate can usually be safely combined with carbamazepine or SDAs. Perhaps the most worrisome interaction of valproate and a psychotropic drug involves lamotrigine. Since the approval of lamotrigine for the treatment of bipolar disorder, it is more likely that patients will be treated with both agents. Valproate more than doubles lamotrigine concentrations, increasing the risk of a severe rash (Stevens–Johnson syndrome and toxic epidermal necrolysis).

The plasma concentrations of carbamazepine, diazepam (Valium), amitriptyline (Elavil), nortriptyline (Pamelor), and phenobarbital (Luminal) may also be increased when these drugs are coadministered with valproate. Also, the plasma concentrations of phenytoin (Dilantin) and desipramine (Norpramin) may be decreased when they are combined with valproate. The plasma concentrations of valproate may be decreased when the drug is coadministered with carbamazepine and may be increased when coadministered with guanfacine (Tenex), amitriptyline, or fluoxetine (Prozac). Valproate can be displaced from plasma proteins by carbamazepine, diazepam, and aspirin. Persons who are treated with anticoagulants (e.g., aspirin and warfarin [Coumadin]) should also be monitored when valproate use is initiated to assess the development of any undesired augmentation of the anticoagulation effects. Interactions of valproate with other drugs are listed in Table 21-34.



Table 21-34
Interactions of Valproate with Other Drugs

Drug	Interactions Reported with Valproate
Lithium	Increased tremor
Antipsychotics	Increased sedation; increased extrapyramidal effects; delirium and stupor (single report)
Clozapine	Increased sedation; confusional syndrome (single report)
Carbamazepine	Acute psychosis (single report); ataxia, nausea,

	lethargy (single report); may decrease valproate serum concentrations
Antidepressants	Amitriptyline and fluoxetine may increase valproate serum concentrations
Diazepam	Serum concentration increased by valproate
Clonazepam	Absence status (rare; reported only in patients with preexisting epilepsy)
Phenytoin	Serum concentration decreased by valproate
Phenobarbital	Serum concentration increased by valproate; increased sedation
Other CNS depressants	Increased sedation
Anticoagulants	Possible potentiation of effect

CNS, central nervous system.

Laboratory Interferences

Valproate may cause a laboratory increase of serum-free fatty acids. Valproate metabolites may produce a false-positive test result for urinary ketones as well as falsely abnormal thyroid function test results.



Table 21-35
Recommended Laboratory Tests during Valproate Therapy

Before Treatment
Standard chemistry screen with special attention to liver function tests
CBC, including WBC and platelet count
During Treatment
Liver function tests at 1 mo; then every 6–24 mo if no abnormalities are found.
Complete blood count with platelet count at 1 mo; then every 6–24 mo if findings are normal.

If Liver Function Test Results Become Abnormal

Mild transaminase elevation (less than three times normal): monitoring every 1–2 wk; if stable and patient is responding to valproate, results are monitored monthly to every 3 mo.

Pronounced transaminase elevation (more than three times normal): dosage reduction or discontinuation of valproate; increase dose or rechallenge if transaminases normalize and if the patient is a valproate responder.

CBC, complete blood count; WBC, white blood cell.

Dosage and Clinical Guidelines

When starting valproate therapy, a baseline hepatic panel, CBC and platelet counts, and pregnancy testing should be ordered. Additional testing should include amylase and coagulation studies if baseline pancreatic disease or coagulopathy is suspected. In addition to baseline laboratory tests, hepatic transaminase concentrations should be obtained 1 month after initiation of therapy and every 6 to 24 months after that. However, because even frequent monitoring may not predict severe organ toxicity, it is more prudent to reinforce the need for prompt evaluation of any illnesses when reviewing the instructions with patients. Asymptomatic elevation of transaminase concentrations up to three times the upper limit of normal are typical and do not require any change in dosage. Table 21-35 lists the recommended laboratory tests for valproate treatment.

Valproate is available in several formulations (Table 21-36). For the treatment of acute mania, an oral loading strategy of initiation with 20 to 30 mg/kg a day can be used to accelerate the control of symptoms. This strategy is usually well tolerated but can cause excessive sedation and tremor in elderly persons. Agitated behavior can be rapidly stabilized with IV infusion of valproate. If acute mania is absent, it is best to initiate drug treatment gradually to minimize the common adverse effects of nausea, vomiting, and sedation. The dose on the first day should be 250 mg administered with a meal. The dosage can be raised to 250 mg orally three times daily for over 3 to 6 days. The plasma concentrations can be assessed in the morning before the first daily dose is administered. Therapeutic plasma concentrations for the control of seizures range between 50 and 100 µg/mL. It is reasonable to use

the same range for the treatment of mental disorders; most of the controlled studies have used 50 to 125 µg/mL. Most persons attain therapeutic plasma concentrations on a dosage between 1,200 and 1,500 mg a day in divided doses. After a person's symptoms are well controlled, the full daily dose can be taken all at once before sleep.



Table 21-36
Valproate Preparations Available in the United States

Generic Name	Trade Name, Form (Doses)	Time to Peak
Valproate sodium injection	Depacon injection (100-mg valproic acid/mL)	1 hr
Valproic acid	Depakene, syrup (250 mg/5 mL)	1–2 hr
	Depakene, capsules (250 mg)	1–2 hr
Divalproex sodium	Depakote, delayed-released tablets (125, 250, 500 mg)	3–8 hr
Divalproex sodium-coated particles in	Depakote, sprinkle capsules (125 mg)	Compared with divalproex tablets, divalproex sprinkle has earlier onset and slower absorption, with slightly lower peak plasma concentration

capsules		
Divalproex sodium extended-release	Depakote ER (250, 500 mg)	4–17 hr

LAMOTRIGINE

Lamotrigine (Lamictal) was developed as a result of screening folate antagonists as anticonvulsants. Lamotrigine proved effective in several animal models of epilepsy, was developed as an antiepileptic drug, and was marketed for the adjunctive treatment of partial seizures in the United States in 1995. Initial, postmarketing, open, clinical experience suggested efficacy in a variety of neurologic and psychiatric conditions, coupled with good tolerability (aside from the risk of rash). Later, double-blind, placebo-controlled studies revealed that lamotrigine was useful for some, but not all, of the neurologic and psychiatric conditions reported in open studies. Therefore, lamotrigine appeared effective as a maintenance treatment for bipolar disorder and was approved for the maintenance treatment of bipolar I disorder in 2003. Lamotrigine also appeared to have potential utility in acute bipolar depression. However, the magnitude of the effect was too modest to yield consistently superior performance compared with placebo, and hence lamotrigine did not receive approval for the treatment of acute bipolar depression.

Similarly, limited data suggested that lamotrigine had potential utility in rapid-cycling bipolar disorder. Lamotrigine did not appear to be as effective as the primary intervention in acute mania. Thus, lamotrigine has emerged as an agent that appears to “stabilize mood from below” in the sense that it may maximally impact the depressive component of bipolar disorders.

Pharmacologic Actions

Lamotrigine is wholly absorbed, has a bioavailability of 98 percent, and has a steady-state plasma half-life of 25 hours. However, the rate of metabolism of lamotrigine varies over a sixfold range, depending on which other drugs are administered concomitantly. Dosing is escalated slowly to twice-a-day

maintenance dosing. Food does not affect its absorption, and it is 55 percent protein-bound in the plasma; 94 percent of lamotrigine and its inactive metabolites are excreted in the urine. Among the better-delineated biochemical actions of lamotrigine is the blockade of voltage-sensitive sodium channels, which in turn modulates the release of glutamate and aspartate and has a slight effect on calcium channels. Lamotrigine modestly increases plasma serotonin concentrations, possibly through inhibition of serotonin reuptake, and is a weak inhibitor of serotonin 5-HT_3 receptors.

Therapeutic Indications

Bipolar Disorder. Lamotrigine is indicated in the treatment of bipolar disorder and may prolong the time between episodes of depression and mania. It is more effective in lengthening the intervals between depressive episodes than manic episodes. It is also useful as a treatment for rapid-cycling bipolar disorder.

Other Indications. There have been reports of therapeutic benefit in the treatment of borderline personality disorder and the treatment for various pain syndromes.

Precautions and Adverse Reactions

Lamotrigine is remarkably well tolerated. The absence of sedation, weight gain, and other metabolic effects are noteworthy. The most common adverse effects—dizziness, ataxia, somnolence, headache, diplopia, blurred vision, and nausea—are typically mild. Anecdotal reports of cognitive impairment and joint or back pain are standard.

The appearance of a rash, which is common and occasionally very severe, is a source of concern. About 8 percent of patients started on lamotrigine develop a benign maculopapular rash during the first 4 months of treatment, and the drug should be discontinued if a rash develops. Even though these rashes are benign, there is concern that, in some cases, they may represent early manifestations of Stevens–Johnson syndrome or toxic epidermal necrolysis. Nevertheless, even if lamotrigine is discontinued immediately upon the development of a rash or other signs of hypersensitivity reaction, such as fever and lymphadenopathy, this may not prevent subsequent

development of a life-threatening rash or permanent disfiguration.

Estimates of the rate of severe rash vary, depending on the source of the data. In some studies, the incidence of severe rashes was 0.08 percent in adult patients receiving lamotrigine as initial monotherapy and 0.13 percent in adult patients receiving lamotrigine as adjunctive therapy. German registry data, based on clinical practice, suggest that the risk of rash may be as low as 1 in 5,000 patients. The appearance of any rash necessitates immediate discontinuation of drug administration.

It is known that the likelihood of a rash increases if the recommended starting dose and speed of dose increase exceed what is recommended. Concomitant administration of valproic acid also increases risk and should be avoided if possible. If valproate is used, a more conservative dosing regimen is followed. Children and adolescents younger than age 16 years appear to be more susceptible to rash with lamotrigine. If patients miss more than 4 consecutive days of lamotrigine treatment, they need to restart therapy at the initial starting dose and titrate upward as if they had not already been on the medication.

Laboratory Testing

There is no proven correlation between lamotrigine blood concentrations and either antiseizure effects or efficacy in bipolar disorders. Laboratory tests are not useful in predicting the occurrence of adverse events.

Drug Interactions

Lamotrigine has significant, well-characterized drug interactions involving other anticonvulsants. The most potentially serious lamotrigine drug interaction involves concurrent use of valproic acid, which doubles serum lamotrigine concentrations. Lamotrigine decreases the plasma concentration of valproic acid by 25 percent. Sertraline (Zoloft) also increases plasma lamotrigine concentrations, but to a lesser extent than does valproic acid. Lamotrigine concentrations are decreased by 40 to 50 percent with concomitant administration of carbamazepine, phenytoin, or phenobarbital. Combinations of lamotrigine and other anticonvulsants have complex effects on the time of peak plasma concentration and the plasma half-life of lamotrigine.

Laboratory Interferences

Lamotrigine does not interfere with any laboratory tests.

Dosage and Administration

In the clinical trials leading to the approval of lamotrigine as a treatment for bipolar disorder, no consistent increase in efficacy was associated with doses above 200 mg/day. Most patients should take between 100 and 200 mg a day. In epilepsy, the drug is administered twice daily, but in bipolar disorder, the total dose can be taken once a day, either in the morning or night, depending on whether the patient finds the drug activating or sedating.

Lamotrigine is available as unscored 25-, 100-, 150-, and 200-mg tablets. The primary determinant of lamotrigine dosing is the minimization of the risk of rash. Lamotrigine should not be taken by anyone younger than 16 years of age. Because valproic acid markedly slows the elimination of lamotrigine, concomitant administration of these two drugs necessitates a much slower titration (Table 21-37). People with renal insufficiency should aim for a lower maintenance dosage. The appearance of any rash necessitates immediate discontinuation of lamotrigine administration. Lamotrigine should usually be discontinued gradually over 2 weeks unless a rash emerges, in which case it should be discontinued over 1 to 2 days.

Lamotrigine orally disintegrating tablets (Lamictal ODT) are available for patients who have difficulty swallowing. It is the only antiepileptic treatment that is available in an orally disintegrating formulation. It is available in 25, 50, 100, and 200 mg strengths and matches the dose of lamotrigine tablets. Chewable dispersible tablets of 2, 5, and 25 mg are also available.

CARBAMAZEPINE AND OXCARBAZEPINE

Carbamazepine (Tegretol) possesses some structural similarity to the TCA imipramine (Tofranil). It was approved for use in the United States for the treatment of trigeminal neuralgia in 1968 and temporal lobe epilepsy (complex partial seizures) in 1974. Interestingly, carbamazepine was first synthesized as a potential antidepressant, but because of its atypical profile in several animal models, it was initially developed for use in pain and seizure disorders. It is now recognized in most guidelines as a second-line mood

stabilizer useful in the treatment and prevention of both phases of bipolar affective disorder. A long-acting sustained release formulation (Equetro) was approved by the U.S. FDA for the treatment of acute mania in 2002.

Table 21-37
Lamotrigine Dosing (mg/day)

Treatment	Weeks 1–2	Weeks 3–4	Weeks 4–5
Lamotrigine monotherapy	25	50	100–200 (200 maximum)
Lamotrigine + carbamazepine	50	100	200–400 (400 maximum)
Lamotrigine + valproate	25 every other day	25	50–100 (100 maximum)

An analog of carbamazepine, oxcarbazepine (Trileptal), was marketed as an antiseizure medication in the United States in 2000. Previously it was used as a treatment for pediatric epilepsy in Europe since 1990. Because of its similarity to carbamazepine, many clinicians began to use it as a treatment for patients with bipolar disorder. Despite some reports that oxcarbazepine has mood-stabilizing properties, this has not been confirmed in large, placebo-controlled trials.

Carbamazepine

Pharmacologic Actions. The absorption of carbamazepine is slow and unpredictable. Food enhances absorption. Peak plasma concentrations are reached 2 to 8 hours after a single dose, and steady-state levels are reached after 2 to 4 days on a steady dosage. It is 70 to 80 percent protein-bound. The half-life of carbamazepine ranges from 18 to 54 hours, with an average of 26 hours. However, with chronic administration, the half-life of carbamazepine decreases to an average of 12 hours. This results from the induction of hepatic CYP450 enzymes by carbamazepine, specifically the autoinduction of carbamazepine metabolism. The induction of hepatic enzymes reaches its

maximum level after about 3 to 5 weeks of therapy.

The pharmacokinetics of carbamazepine are different for two long-acting preparations of carbamazepine, each of which uses slightly different technology. One formulation, Tegretol XR, requires food to ensure standard GI transit time. The other preparation, Carbatrol, relies on a combination of intermediate, extended-release, and very slow release beads, making it suitable for bedtime administration.

Carbamazepine is metabolized in the liver, and the 10,11-epoxide metabolite is active as an anticonvulsant. Its activity in the treatment of bipolar disorders is unknown. Long-term use of carbamazepine is associated with an increased ratio of the epoxide to the parent molecule.

The anticonvulsant effects of carbamazepine are thought to be mediated mainly by binding to voltage-dependent sodium channels in the inactive state and prolonging their inactivation. This action secondarily reduces voltage-dependent calcium channel activation and, thus, synaptic transmission. Additional effects include reduction of currents through *N*-methyl-D-aspartate (NMDA) glutamate-receptor channels, competitive antagonism of adenosine a_1-receptors, and potentiation of CNS catecholamine neurotransmission. Whether any or all of these mechanisms also result in mood stabilization is not known.

Therapeutic Indications

Bipolar Disorder

ACUTE MANIA. The acute antimanic effects of carbamazepine are typically evident within the first several days of treatment. About 50 to 70 percent of all persons respond within 2 to 3 weeks of initiation. Studies suggest that carbamazepine may be especially useful in persons who are not responsive to lithium (Eskalith), such as persons with dysphoric mania, rapid cycling, or an adverse family history of mood disorders. The antimanic effects of carbamazepine can be, and often are, augmented by concomitant administration of lithium, valproic acid (Depakene), thyroid hormones, DRAs, or SDAs. Some persons may respond to carbamazepine but not lithium or valproic acid and vice versa.

PROPHYLAXIS. Carbamazepine is effective in preventing relapses, particularly among patients with bipolar II disorder and schizoaffective disorder, and dysphoric mania.

ACUTE DEPRESSION. A subgroup of treatment-refractory patients with acute depression responds well to carbamazepine. Patients with more severe episodic and less chronic depression seem to be better responders to carbamazepine. Nevertheless, carbamazepine remains an alternative drug for depressed persons who have not responded to conventional treatments, including ECT.

OTHER DISORDERS. Carbamazepine helps to control symptoms associated with acute alcohol withdrawal, although benzodiazepines are more effective in this population. Carbamazepine has been suggested as a treatment for the recurrent paroxysmal component of PTSD. Uncontrolled studies suggest that carbamazepine is effective in controlling impulsive, aggressive behavior in nonpsychotic persons of all ages, including children and elderly persons. Carbamazepine is also effective in controlling nonacute agitation and aggressive behavior in patients with schizophrenia and schizoaffective disorder. Persons with prominent positive symptoms (e.g., hallucinations) may be likely to respond, as are persons who display impulsive aggressive outbursts.

Precautions and Adverse Reactions. Carbamazepine is relatively well tolerated. Mild GI (nausea, vomiting, gastric distress, constipation, diarrhea, and anorexia) and CNS (ataxia, drowsiness) side effects are the most common. The severity of these adverse effects is reduced if the dosage of carbamazepine is increased slowly and kept at the minimal effective plasma concentration. In contrast to lithium and valproate (other drugs used to manage bipolar disorder), carbamazepine does not appear to cause weight gain. Because of the phenomena of autoinduction, with consequent reductions in carbamazepine concentrations, side effect tolerability may improve over time. Most of the adverse effects of carbamazepine are correlated with plasma concentrations above 9 μg/mL. The rarest but most serious adverse effects of carbamazepine are blood dyscrasias, hepatitis, and severe skin reactions (Table 21-38).

BLOOD DYSCRASIAS. The drug's hematologic effects are not dose-related. Severe blood dyscrasias (aplastic anemia, agranulocytosis) occur in about 1 in 125,000 persons treated with carbamazepine. There does not appear to be a correlation between the degree of benign WBC suppression (leukopenia), which is seen in 1 to 2 percent of persons, and the emergence of life-threatening blood dyscrasias. Persons should be warned that the emergence of such symptoms as fever, sore throat, rash, petechiae, bruising, and easy bleeding can potentially herald a serious dyscrasia, and they should seek medical evaluation immediately. Routine hematologic monitoring in carbamazepine-treated persons is recommended at 3, 6, 9, and 12 months. If there is no significant evidence of bone marrow suppression by that time, many experts will reduce the interval of monitoring. However, even assiduous monitoring may fail to detect severe blood dyscrasias before they cause symptoms.



Table 21-38
Adverse Events Associated with Carbamazepine

Dosage-Related Adverse Effects	Idiosyncratic Adverse Effects
Double or blurred vision	Agranulocytosis
Vertigo	Stevens–Johnson syndrome
Gastrointestinal disturbances	Aplastic anemia
Task performance impairment	Hepatic failure
Hematologic effects	Rash Pancreatitis

HEPATITIS. Within the first few weeks of therapy, carbamazepine can cause both hepatitis associated with increases in liver enzymes, particularly transaminases, and cholestasis associated with elevated bilirubin and alkaline phosphatase. Mild transaminase elevations warrant observation only, but persistent elevations more than three times the upper limit of normal indicate the need to discontinue the drug. Hepatitis can recur if the drug is reintroduced to the person and can result in death.

DERMATOLOGIC EFFECTS. About 10 to 15 percent of those treated with carbamazepine develop a benign maculopapular rash within the first 3 weeks of treatment. Stopping the medication usually leads to resolution of the rash. Some patients may experience life-threatening dermatologic syndromes, including exfoliative dermatitis, erythema multiforme, Stevens–Johnson syndrome, and toxic epidermal necrolysis. The possible emergence of these serious dermatologic problems causes most clinicians to discontinue carbamazepine use in people who develop any rash. The risk of drug rash is about equal between valproic acid and carbamazepine in the first 2 months of use but is subsequently much higher for carbamazepine. If carbamazepine seems to be the only effective drug for a person who has a benign rash with carbamazepine treatment, a retrial of the drug can be undertaken. Many patients can be rechallenged without reemergence of the rash. Pretreatment with prednisone (Deltasone; 40 mg a day) may suppress the rash, although other symptoms of an allergic reaction (e.g., fever and pneumonitis) may develop even with steroid pretreatment.

RENAL EFFECTS. Carbamazepine is occasionally used to treat diabetes insipidus not associated with lithium use. This activity results from direct or indirect effects at the vasopressin receptor. It may also lead to the development of hyponatremia and water intoxication in some patients, mainly elderly persons, or when used in high doses.

OTHER ADVERSE EFFECTS. Carbamazepine decreases cardiac conduction (although less than the tricyclic drugs do) and can thus exacerbate preexisting cardiac disease. Carbamazepine should be used with caution in persons with glaucoma, prostatic hypertrophy, diabetes, or a history of alcohol abuse. Carbamazepine occasionally activates vasopressin receptor function, which results in a condition resembling the syndrome of secretion of inappropriate antidiuretic hormone, characterized by hyponatremia and, rarely, water intoxication. This effect is the opposite of the renal effects of lithium (i.e., nephrogenic diabetes insipidus). Augmentation of lithium with carbamazepine does not reverse the lithium effect, however. The emergence of confusion, severe weakness, or headache in a person taking carbamazepine should prompt the measurement of serum electrolytes.

Carbamazepine use rarely elicits an immune hypersensitivity response

consisting of fever, rash, eosinophilia, and possibly fatal myocarditis.

Cleft palate, fingernail hypoplasia, microcephaly, and spina bifida in infants may be associated with the maternal use of carbamazepine during pregnancy. Pregnant women should not use carbamazepine unless necessary. All women with childbearing potential should take 1 to 4 mg of folic acid daily, even if they are not trying to conceive. Carbamazepine is secreted in breast milk.

Drug Interactions. Carbamazepine decreases serum concentrations of numerous drugs as a result of the prominent induction of hepatic CYP3A4 (Table 21-39). Monitoring for a decrease in clinical effects is frequently indicated. Carbamazepine can decrease the blood concentrations of oral contraceptives, resulting in breakthrough bleeding and uncertain prophylaxis against pregnancy. Carbamazepine should not be administered with MAOIs, which should be discontinued at least 2 weeks before initiating treatment with carbamazepine. Grapefruit juice inhibits the hepatic metabolism of carbamazepine. When carbamazepine and valproate are used in combination, the dosage of carbamazepine should be decreased because valproate displaces carbamazepine binding on proteins, and the dosage of valproate may need to be increased.



Table 21-39
Carbamazepine: Drug Interactions

Effect of Carbamazepine on Plasma Concentrations of Concomitant Agents	Agents that May Affect Carbamazepine Plasma Concentrations
Carbamazepine may decrease drug plasma concentration of:	*Agents that may increase carbamazepine plasma concentration:*
Acetaminophen	Allopurinol
Alprazolam	Cimetidine
Amitriptyline	Clarithromycin
Bupropion	Danazol
Clomipramine	Diltiazem
Clonazepam	Erythromycin
Clozapine	

- Cyclosporine
- Desipramine
- Dicumarol
- Doxepin
- Doxycycline
- Ethosuximide
- Felbamate
- Fentanyl
- Fluphenazine
- Haloperidol
- Hormonal contraceptives
- Imipramine
- Lamotrigine
- Methadone
- Methsuximide
- Methylprednisolone
- Nimodipine
- Pancuronium
- Phensuximide
- Phenytoin
- Primidone
- Theophylline
- Valproate
- Warfarin

Carbamazepine may increase drug plasma concentrations of

- Clomipramine
- Phenytoin
- Primidone
- Fluoxetine
- Fluvoxamine
- Gemfibrozil
- Itraconazole
- Ketoconazole
- Isoniazid[a]
- Itraconazole
- Lamotrigine
- Loratadine
- Macrolides
- Nefazodone
- Nicotinamide
- Propoxyphene
- Terfenadine
- Troleandomycin
- Valproate[a]
- Verapamil
- Viloxazine

Drugs that may decrease carbamazepine plasma concentrations

- Carbamazepine (autoinduction)
- Cisplatin
- Doxorubicin HCl
- Felbamate
- Phenobarbital
- Phenytoin
- Primidone
- Rifampin[b]
- Theophylline
- Valproate

[a]Increased concentrations of the active 10,11-epoxide.

[b]Decreased concentrations of carbamazepine and increased concentrations of the 10,11-epoxide.

Table by Carlos A. Zarate, Jr., M.D. and Mauricio Tohen, M.D.

Laboratory Interferences. Circulating levels of thyroxine and triiodothyronine are associated with a decrease in TSH and may be associated with treatment. Carbamazepine is also associated with an increase in total serum cholesterol, primarily by increasing high-density lipoproteins. The thyroid and cholesterol effects are not clinically significant. Carbamazepine may interfere with the dexamethasone (Decadron) suppression test and may also cause false-positive pregnancy test results.

Dosing and Administration. The target dose for antimanic activity is 1,200 mg a day, although this varies considerably. Immediate-release carbamazepine needs to be taken three or four times a day, which leads to lapses in compliance. Extended-release formulations are thus preferred because they can be taken once or twice a day. One form of extended-release carbamazepine, Carbatrol, comes as 100-, 200-, and 300-mg capsules. Another form, Equetro, is identical to Carbatrol and is marketed as a treatment for bipolar disorder. These capsules contain tiny beads with three different types of coatings, so they dissolve at different times. Capsules should not be crushed or chewed. However, the contents can be sprinkled over food without affecting the extended-release qualities. This formulation can be taken either with or without meals. The entire daily dose can be given at bedtime. The rate of absorption is faster when it is given with a high-fat meal. Another extended-release form of carbamazepine, Tegretol XR, uses a different drug-delivery system than Carbatrol. It is available in 100-, 200-, and 300-mg tablets.

Preexisting hematologic, hepatic, and cardiac diseases can be relative contraindications for carbamazepine treatment. Persons with hepatic disease require only one-third to one-half the usual dosage; the clinician should be cautious about raising the dosage in such persons and should do so only slowly and gradually. The laboratory examination should include a CBC with platelet count, liver function tests, serum electrolytes, and an electrocardiogram in persons older than 40 years of age or with preexisting cardiac disease. An electroencephalogram is not necessary before the initiation of treatment, but it may be helpful in some cases for the documentation of objective changes correlated with clinical improvement. Table 21-40 presents a brief user's guide to carbamazepine in bipolar disorder.

Routine Laboratory Monitoring. Serum levels for antimanic efficacy have not been established. The anticonvulsant blood concentration range for carbamazepine is 4 to 12 µg/mL, and this range should be reached before determining that carbamazepine is not effective in the treatment of a mood disorder. A clinically insignificant suppression of the WBC count commonly occurs during carbamazepine treatment. This benign decrease can be reversed by adding lithium, which enhances colony-stimulating factor. Potential serious hematologic effects of carbamazepine, such as pancytopenia, agranulocytosis, and aplastic anemia, occur in about 1 in 125,000 patients. Complete laboratory blood assessments may be performed every 2 weeks for the first 2 months of treatment and quarterly after that, but the FDA has revised the package insert for carbamazepine to suggest that blood monitoring be performed at the discretion of the physician. Patients should be informed that fever, sore throat, rash, petechiae, bruising, or unusual bleeding may indicate a hematologic problem and should prompt immediate notification of a physician. This approach is probably more useful than is frequent blood monitoring during long-term treatment. It has also been suggested that liver and renal function tests be conducted quarterly, although the benefit of conducting tests this frequently has been questioned. It seems reasonable, however, to assess hematologic status, along with liver and renal functions, whenever a routine examination of the person is being conducted. A monitoring protocol is listed in Table 21-41.



Table 21-40
Carbamazepine in Bipolar Illness: A Brief User's Guide

1. Start with low (200 mg) bedtime dose in depression or euthymia; higher doses (600–800 mg/day in divided doses) in manic inpatients.
2. All bedtime dosing is reasonable with carbamazepine extended-release preparation.
3. Titrate slowly to the individual's response or side effects threshold.
4. Hepatic enzyme CYP450 (3A4) induction and autoinduction occur in 2–3 wk; slightly higher doses may be needed or tolerated at that time.

5. Warn regarding benign rash, which occurs in 5–10% of those taking the drug; progression to rare, severe rash is unpredictable, so the drug should be discontinued if any rash develops.
6. Benign white blood cell count decreases occur regularly (usually inconsequential).
7. Rarely, agranulocytosis and aplastic anemia may develop (several per million new exposures); warn regarding appearance of fever, sore throat, petechiae, and bleeding gums and to check with physician to obtain an immediate complete blood cell count.
8. Use adequate birth control methods, including higher dosage forms of estrogen (as carbamazepine lowers estrogen levels).
9. Avoid carbamazepine in pregnancy (spina bifida occurs in 0.5%; other severe adverse outcomes occur in about 8%).
10. Some people will respond well to carbamazepine and not other mood stabilizers (lithium) or anticonvulsants (valproic acid).
11. Combination treatment often required to maintain remission and prevent loss of effect via tolerance.
12. Major drug interactions associated with increases in carbamazepine and potential toxicity from 3A4 enzyme inhibition include calcium channel blockers (isradipine and verapamil); erythromycin and related macrolide antibiotics; and valproate.

Carbamazepine treatment should be discontinued, and a consult with a hematologist should be obtained, if the following laboratory values are found: total WBC count below 3,000/mm^3, erythrocytes below 4.0 × 10^6/mm^3, neutrophils below 1,500/mm^3, hematocrit less than 32 percent, hemoglobin less than 11 g/100 mL, platelet count below 100,000/mm^3, reticulocyte count below 0.3 percent, and a serum iron concentration below 150 mg/100 mL.



Table 21-41
Laboratory Monitoring of Carbamazepine for Adult Psychiatric Disorders

	Baseline	Weekly to Stability	Monthly for 6 Months	6–12 Months
CBC	+	+	+	+
Bilirubin	+		+	+
Alanine aminotransferase	+		+	+
Aspartate aminotransferase	+		+	+
Alkaline phosphatase	+		+	+
Carbamazepine level	+	+		+

CBC, complete blood count.

Oxcarbazepine

Although structurally related to carbamazepine, the usefulness of oxcarbazepine as a treatment for mania has not been established in controlled trials.

Pharmacokinetics. Absorption is rapid and unaffected by food. Peak concentrations occur after about 45 minutes. The elimination half-life of the parent compound is 2 hours, which remains stable over long-term treatment. The monohydroxide has a half-life of 9 hours. Most of the drug's anticonvulsant activity is presumed to result from this monohydroxy derivative.

Side Effects. The most common side effects are sedation and nausea. Less frequent side effects are cognitive impairment, ataxia, diplopia, nystagmus, dizziness, and tremor. In contrast to carbamazepine, oxcarbazepine does not have an increased risk of severe blood dyscrasias, so hematologic monitoring is not necessary. The frequency of benign rash is lower than observed with carbamazepine, and severe rashes are extremely rare. However, about 25 to 30 percent of patients who develop an allergic rash while taking carbamazepine also develop a rash with oxcarbazepine. Oxcarbazepine is

more likely to cause hyponatremia than carbamazepine. Approximately 3 to 5 percent of patients taking oxcarbazepine develop this side effect. It is advisable to obtain serum sodium concentrations early in the course of treatment because hyponatremia may be clinically silent. In severe cases, confusion and seizure may occur.

Dosing and Administration. Oxcarbazepine dosing for bipolar disorder has not been established. It is available in 150-, 300-, and 600-mg tablets. The dose range may vary from 150 to 2,400 mg/day, given in divided doses twice a day. In clinical trials for mania, the doses typically used were from 900 to 1,200 mg/day with a starting dose of 150 or 300 mg at night.

Drug Interactions. Drugs such as phenobarbital and alcohol, which induce CYP34A, increase clearance and reduce oxcarbazepine concentrations. Oxcarbazepine induces CYP3A4/5 and inhibits CYP2C19, which may affect the metabolism of drugs that use that pathway. Women taking oral contraceptives should be told to consult with their gynecologists, because oxcarbazepine may reduce concentrations of their contraceptive and thus decrease its efficacy.

OTHER ANTICONVULSANTS

The anticonvulsants described in this section were developed for the treatment of epilepsy but were also found to have beneficial effects in psychiatric disorders. Also, these agents are used as skeletal muscle relaxants and in neurogenic pain. These drugs have a variety of mechanisms, including increasing GABAergic function or decreasing glutamatergic function. This section includes seven anticonvulsants: gabapentin (Neurontin), levetiracetam (Keppra), pregabalin (Lyrica), tiagabine (Gabitril), topiramate (Topamax), and zonisamide (Zonegran), as well as one of the first used anticonvulsants, phenytoin (Dilantin). Carbamazepine (Tegretol), valproate (Depakene, Depakote), lamotrigine (Lamictal), and oxcarbazepine (Trileptal) are discussed in separate sections.

In 2008, the FDA issued a warning that these drugs may increase the risk of suicidal ideation or act in some persons compared with placebo; however, the relative risk for suicidality was higher in patients with epilepsy compared with those with psychiatric disorders. However, some published data contradict this warning. These studies suggest that anticonvulsants may have a protective effect on suicidal thoughts in bipolar disorder. Considering the inherently increased risk of suicide in persons with bipolar disorder, clinicians should be aware of these warnings.

Gabapentin

Gabapentin was first introduced as an antiepileptic drug and was found to have sedative effects that were useful in some psychiatric disorders, especially insomnia. It was also found to be beneficial in reducing neuropathic pain, including postherpetic neuralgia. It is used in anxiety disorders (social phobia and panic disorder), but not as the primary intervention in mania or treatment-resistant mood disorders.

Pharmacologic Actions. Gabapentin circulates in the blood primarily unbound and is not appreciably metabolized in humans. It is eliminated unchanged by renal excretion and can be removed by hemodialysis. Food only moderately affects the rate and extent of absorption. Clearance is decreased in elderly persons, requiring dosage adjustments. Gabapentin

appears to increase cerebral GABA and may inhibit glutamate synthesis as well. It increases human whole blood serotonin concentrations and modulates calcium channels to reduce monoamine release. It has antiseizure as well as antispastic activity and antinociceptive effects in pain.

Therapeutic Indications. In neurology, gabapentin is used for the treatment of both general and partial seizures. It is useful in reducing the pain of postherpetic neuralgia and other pain syndromes associated with diabetic neuropathy, neuropathic cancer pain, fibromyalgia, meralgia paresthetica, amputation, and headache. It is useful in some cases of chronic pruritus.

In psychiatry, gabapentin is used as a hypnotic agent because of its sedating effects. It has anxiolytic properties and benefits patients with social anxiety and panic disorder. It may decrease the craving for alcohol in some patients and improve mood as well; hence, it may have some use in depressed patients. Some bipolar patients have benefited when gabapentin is used adjunctively with mood stabilizers.

Precautions and Adverse Reactions. Adverse effects are mild, with the most common being daytime somnolence, ataxia, and fatigue, which are usually dose-related. Overdose (over 45 g) has been associated with diplopia, slurred screech, lethargy, and diarrhea, but all patients recovered. The drug was classified as pregnancy category C in the former FDA classification system and is excreted in breast milk, so it is best to avoid it in pregnant women and nursing mothers.

Drug Interactions. Gabapentin bioavailability may decrease as much as 20 percent when administered with antacids. In general, there are no drug interactions. Chronic use does not interfere with lithium administration.

Laboratory Interferences. Gabapentin does not interfere with any laboratory tests, although spontaneous reports of false-positive drug toxicology screenings for amphetamines, barbiturates, benzodiazepines, and marijuana have been reported.

Dosages and Clinical Guidelines. Gabapentin is well tolerated, and the dosage can be increased to the maintenance range within a few days. A

general approach is to start with 300 mg on day 1, increase to 600 mg on day 2, 900 mg on day 3, and subsequently increase up to 1,800 mg/day in divided doses as needed to relieve symptoms. Final total daily doses tend to be between 1,200 and 2,400 mg/day, but occasionally results may be achieved with dosages as low as 200 to 300 mg/day, especially in elderly persons. Sedation is usually the limiting factor in determining the dosage. Some patients have taken dosages as high as 4,800 mg/day.

Gabapentin is available as 100-, 300-, and 400-mg capsules and as 600- and 800-mg tablets. A 250-mg/5-mL oral solution is also available. Although abrupt discontinuation of gabapentin does not cause withdrawal effects, the use of all anticonvulsant drugs should be gradually tapered.

Topiramate

Topiramate (Topamax) was developed as an anticonvulsant and was found useful in a variety of psychiatric and neurologic conditions, including migraine prevention, treatment of obesity, bulimia, binge eating, and alcohol dependence.

Pharmacologic Actions. Topiramate has GABAergic effects and increases cerebral GABA in humans. It has 80 percent oral bioavailability and is not significantly altered by food. It is 15 percent protein-bound, and about 70 percent of the drug is eliminated by renal excretion. With renal insufficiency, topiramate clearance decreases by about 50 percent, so the dosage needs to be decreased. It has a half-life of around 24 hours.

Therapeutic Indications. Topiramate is used mainly as an antiepileptic medication and has been found superior to placebo as monotherapy in patients with seizure disorders. It is also used in the prevention of migraine, smoking cessation, pain syndromes (e.g., low back pain), PTSD, and essential tremor. The drug has been associated with weight loss, and that fact has been used to counteract the weight gain caused by many psychotropic drugs. It has also been used in general obesity and the treatment of bulimia and binge-eating disorder. Self-mutilating behavior may be decreased in borderline personality disorder. It is of little or no benefit in the treatment of psychotic disorders. In one study, the combination of topiramate with bupropion (Wellbutrin) showed some efficacy in bipolar depression, but

double-blind, placebo-controlled trials failed to demonstrate topiramate monotherapy efficacy in acute mania in adults.

Precautions and Adverse Reactions. The most common adverse effects of topiramate include paresthesias, weight loss, somnolence, anorexia, dizziness, and memory problems. Sometimes disturbances in the sense of taste occur. In many cases, the adverse effects are mild to moderate and can be attenuated by decreasing the dose. No deaths have been reported during an overdose. The drug affects acid–base balance (low serum bicarbonate), which can be associated with cardiac arrhythmias and the formation of renal calculi in about 1.5 percent of cases. Patients taking the drug should be encouraged to drink plenty of fluids. It is not known if the drug passes through the placenta or is present in breast milk, and pregnant women or nursing mothers should avoid it.

Drug Interactions. Topiramate has few drug interactions with other anticonvulsant drugs. Topiramate may increase phenytoin concentrations up to 25 percent and valproic acid 11 percent; it does not affect the concentration of carbamazepine, phenobarbital (Luminal), or primidone. Topiramate concentrations are decreased by 40 to 48 percent with concomitant administration of carbamazepine or phenytoin. We should not combine topiramate with other carbonic anhydrase inhibitors, as it increases the risk of nephrolithiasis or heat-related problems (oligohidrosis and hyperthermia). These include acetazolamide (Diamox) or dichlorphenamide (Daranide).

Laboratory Interferences. Topiramate does not interfere with any laboratory tests.

Dosages and Clinical Guidelines. Topiramate is available as unscored 25-, 100-, and 200-mg tablets. Topiramate dosage is titrated gradually over 8 weeks to a maximum of 200 mg twice a day to reduce the risk of adverse cognitive and sedative effects. Off-label topiramate is typically used adjunctively, starting with 25 mg at bedtime and increasing weekly by 25 mg as necessary and tolerated. Final doses in efforts to promote weight loss are often between 75 and 150 mg/day at bedtime. Doses higher than 400 mg are not associated with increased efficacy. All of the doses can be given at

bedtime to take advantage of the sedative effects. Persons with renal insufficiency should reduce doses by half.

Tiagabine

Tiagabine was introduced as a treatment for epilepsy in 1997 and was found to have efficacy in some psychiatric conditions, including acute mania. However, safety concerns (see later), along with a lack of controlled data, have limited the use of tiagabine in disorders other than epilepsy.

Pharmacologic Actions. Tiagabine is well absorbed with a bioavailability of about 90 percent and is extensively (96 percent) bound to plasma proteins. Tiagabine is a cytochrome P450 CYP3A substrate and is extensively transformed into inactive 5-oxo-tiagabine and glucuronide metabolites, with only 2 percent being excreted unchanged in the urine. The remainder is excreted as metabolites in the feces (65 percent) and the urine (25 percent). Tiagabine blocks the uptake of the inhibitory amino acid neurotransmitter GABA into neurons and glia, enhancing the inhibitory action of GABA at both $GABA_A$ and $GABA_B$ receptors, putatively yielding anticonvulsant and antinociceptive effects, respectively. It has mild blocking effects on histamine 1 (H_1), serotonin type 1B ($5\text{-}HT_{1B}$), benzodiazepine, and chloride channel receptors.

Therapeutic Indications. Tiagabine is rarely used for psychiatric disorders, and then it is used only for GAD and insomnia. Its main indication is in generalized epilepsy.

Precautions and Adverse Reactions. Tiagabine may cause withdrawal seizures, cognitive or neuropsychiatric problems (impaired concentration, speech or language problems, somnolence, and fatigue), status epilepticus, and sudden unexpected death in epilepsy (SUDEP). Acute oral overdoses of tiagabine have been associated with seizures, status epilepticus, coma, ataxia, confusion, somnolence, drowsiness, impaired speech, agitation, lethargy, myoclonus, stupor, tremors, disorientation, vomiting, hostility, temporary paralysis, and respiratory depression. Deaths have been reported in polydrug overdoses involving tiagabine. Cases of severe rash may occur, including

Stevens–Johnson syndrome.

Tiagabine was classified as pregnancy category C in the former FDA classification system because fetal loss and teratogenicity have been demonstrated in animals. It is not known if the drug is excreted in breast milk. Pregnant women and nursing mothers should not be given the drug.

Laboratory Tests. Tiagabine does not interfere with any laboratory tests.

Dosage and Administration. Tiagabine should not be rapidly loaded or rapidly initiated because of the risk of severe adverse effects. In adults and adolescents 12 years of age or older with epilepsy who are also taking enzyme inducers, tiagabine should be initiated at 4 mg/day and increased weekly by 4 mg/day during the first month. The dose should then be increased weekly by 4 to 8 mg/day for weeks 5 and 6, yielding 24 to 32 mg/day administered in two to four divided doses by week 6. In adults (but not adolescents), tiagabine doses may be further increased weekly by 4 to 8 mg/day to as high as 56 mg/day. Plasma concentrations in patients with epilepsy commonly range between 20 and 100 ng/mL but do not appear to be systematically related to antiseizure effects and thus are not routinely monitored.

Levetiracetam

Initially developed as a nootropic (memory enhancing) drug, levetiracetam proved to be a potent anticonvulsant and marketed as a treatment for partial seizures. It has been used to treat acute mania and anxiety and to augment antidepressant drug therapy.

Pharmacologic Actions. The CNS effects are not well understood, but it appears to enhance GABA inhibition indirectly. It is wholly and rapidly absorbed, and peak concentrations are reached in 1 hour. Food delays the rate of absorption and decreases the amount of absorption. Levetiracetam is not significantly plasma protein-bound and is not metabolized through the hepatic CYP system. Its metabolism involves the hydrolysis of the acetamide group. Serum concentrations are not correlated with therapeutic effects.

Therapeutic Indications. The primary indication is for the treatment of

convulsive disorders, including partial-onset seizures, myoclonic seizures, and idiopathic generalized epilepsy. In psychiatry, levetiracetam has been used off label to treat acute mania, as an add-on treatment for major depression, and as an anxiolytic agent.

Precautions and Adverse Reactions. The most common side effects of levetiracetam include drowsiness, dizziness, ataxia, diplopia, memory impairment, apathy, and paresthesias. Some patients develop behavioral disturbances during treatment, and hallucinations may occur. Suicidal patients may become agitated. It should not be used in pregnant or lactating women.

Drug Interactions. There are few if any interactions with other drugs, including other anticonvulsants. There is no interaction with lithium.

Laboratory Interferences. No laboratory interferences have been reported.

Dosages and Clinical Guidelines. The drug is available as 250-, 500-, 750-, and 1,000-mg tablets; 500-mg extended-release tablets; a 100-mg/mL oral solution; and a 100-mg/mL intravenous solution. In epilepsy, the typical adult daily dose is 1,000 mg.

Because of its renal clearance, dosages should be reduced in patients with impaired renal function.

Zonisamide

Zonisamide, another anticonvulsant, was also found to be useful in bipolar disorder, obesity, and binge-eating disorder.

Pharmacologic Actions. Zonisamide blocks sodium channels and may weakly potentiate dopamine and serotonin activity. It also inhibits carbonic anhydrase. Some evidence suggests that it may block calcium channels. The hepatic CYP3A system metabolizes zonisamide, so enzyme-inducing agents such as carbamazepine, alcohol, and phenobarbital increase the clearance and reduce the availability of the drug. Zonisamide does not affect the metabolism of other drugs. It has a long half-life of 60 hours, so it is

efficiently dosed once daily, preferably at nighttime.

Therapeutic Indications. Its primary use is in the treatment of generalized seizure disorders and refractory partial seizures. In psychiatry, controlled studies found it to be of use in obesity and binge-eating disorder. Uncontrolled trials have found it useful in bipolar disorder, particularly mania; however, further studies are warranted for this indication.

Precautions and Adverse Reactions. Zonisamide is a sulfonamide and thus may cause a fatal rash and blood dyscrasias, although these events are rare. About 4 percent of patients develop kidney stones. The most common side effects are drowsiness, cognitive impairment, insomnia, ataxia, nystagmus, paresthesia, speech abnormalities, constipation, diarrhea, nausea, and dry mouth. Weight loss is also a common side effect, which has been exploited as a therapy for patients who have gained weight during treatment with psychotropics or, as mentioned above, have ongoing difficulty controlling their eating. Zonisamide should not be used in pregnant women or breastfeeding mothers.

Drug Interactions. Zonisamide does not inhibit CYP isoenzymes and does not instigate drug interactions. It is important not to combine carbonic anhydrase inhibitors with zonisamide because of an increased risk of nephrolithiasis related to increased blood levels of urea.

Laboratory Interferences. Zonisamide can elevate hepatic alkaline phosphatase and increase blood urea nitrogen and creatinine.

Dosages and Clinical Guidelines. Zonisamide is available in 100- and 200-mg capsules. In epilepsy, the dosage range is 100 to 400 mg/day, with side effects becoming more pronounced at doses above 300 mg. Because of its long half-life, zonisamide can be given once a day.

Pregabalin

Pregabalin is pharmacologically similar to gabapentin. It is believed to work by inhibiting the release of excess excitatory neurotransmitters. It increases neuronal GABA levels, its binding affinity is six times greater than that of

gabapentin, and it has a longer half-life.

Pharmacologic Actions. Pregabalin exhibits linear pharmacokinetics. It is rapidly absorbed in proportion to its dose. The time to maximal plasma concentration is about 1 hour, and to steady state is within 24 to 48 hours. Pregabalin demonstrates high bioavailability, and it has a mean elimination half-life of about 6.5 hours. Food does not affect absorption. Pregabalin does not bind to plasma proteins and is excreted virtually unchanged (<2 percent metabolism) by the kidneys. It is not subject to hepatic metabolism and does not induce or inhibit liver enzymes. Dose reduction may be necessary for patients with creatinine clearance (CLcr) less than 60 mL/min. Daily doses should be further reduced by approximately 50 percent for each additional 50 percent decrease in CLcr. Pregabalin is highly cleared by hemodialysis so that additional doses may be needed for patients on chronic hemodialysis treatment after each hemodialysis treatment.

Therapeutic Indications. Pregabalin is approved for the management of diabetic peripheral neuropathy and postherpetic neuralgia and adjunctive treatment of partial-onset seizures. It is of benefit to some patients with GAD. In studies, no consistent dose–response relationship was found, although 300 mg of pregabalin per day was more effective than 150 or 450 mg. Some patients with panic disorder or social anxiety disorder may benefit from pregabalin, but little evidence supports its routine use in treating persons with these disorders. It was most recently approved for the treatment of fibromyalgia.

Precautions and Adverse Reactions. The most common adverse events associated with pregabalin use are dizziness, somnolence, blurred vision, peripheral edema, amnesia or loss of memory, and tremors. Pregabalin potentiates the sedating effects of alcohol, antihistamines, benzodiazepines, and other CNS depressants. It remains to be seen if pregabalin is associated with benzodiazepine-type withdrawal symptoms. There are scant data about its use in pregnant women or nursing mothers, and it is best avoided in these patients.

Drug Interactions. Given the absence of hepatic metabolism, pregabalin

lacks metabolic drug interactions.

Laboratory Interferences. There are no effects on laboratory tests.

Dosage and Clinical Guidelines. The recommended dose for postherpetic neuralgia is 50 or 100 mg orally three times a day. The recommended dose for diabetic peripheral neuropathy is 100 to 200 mg orally three times a day. Patients with fibromyalgia may require up to 450 to 600 mg/day given in divided doses. Pregabalin is available as 25-, 50-, 75-, 100-, 150-, 200-, 225-, and 300-mg capsules.

Phenytoin

Phenytoin sodium (Dilantin) is an antiepileptic drug and is related to the barbiturates in chemical structure. It is indicated for the control of generalized tonic–clonic (grand mal) and complex partial (psychomotor, temporal lobe) seizures and prevention and treatment of seizures occurring during or after neurosurgery. Studies have shown comparable efficacy of phenytoin to other anticonvulsants in bipolar disorder, but clinicians should take into account the danger of gingival hyperplasia, leukopenia, or anemia and the danger of toxicity caused by nonlinear pharmacokinetics.

Pharmacologic Action. Similar to other anticonvulsants, phenytoin causes blockade of voltage-activated sodium channels and hence is efficacious as an antimanic agent. The plasma half-life after oral administration averages 22 hours, with a range of 7 to 42 hours. Steady-state therapeutic levels are achieved at least 7 to 10 days (5 to 7 half-lives) after initiation of therapy, with recommended doses of 300 mg/day. Serum level should be obtained at least 5 to 7 half-lives after treatment initiation. Phenytoin is excreted in the bile, which is then reabsorbed from the intestinal tract and excreted in the urine. Urinary excretion of phenytoin occurs partly with glomerular filtration and by tubular secretion. Small incremental doses of phenytoin may increase the half-life and produce very substantial increases in serum levels. Patients should adhere strictly to the prescribed dosage, and serial monitoring of phenytoin levels is recommended.

Therapeutic Indications. Apart from its indication in generalized tonic–

clonic (grand mal) and complex partial (psychomotor, temporal lobe) seizures, phenytoin is also used for the treatment of acute mania in bipolar disorder.

Precautions and Adverse Reactions. The most common adverse reactions reported with phenytoin therapy are usually dose-related and include nystagmus, ataxia, slurred speech, decreased coordination, and mental confusion. Other side effects include dizziness, insomnia, transient nervousness, motor twitching, and headaches. There have been rare reports of phenytoin induced dyskinesias, similar to those induced by phenothiazine and other neuroleptic drugs. More severe side effects include thrombocytopenia, leukopenia, agranulocytosis, and pancytopenia, with or without bone marrow suppression.

Several reports have suggested the development of lymphadenopathy (local or generalized), including benign lymph node hyperplasia, pseudolymphoma, lymphoma, and Hodgkin's disease. Prenatal exposure to phenytoin may increase the risks for congenital malformations, and a potentially life-threatening bleeding disorder related to decreased levels of vitamin K–dependent clotting factors may occur in newborns exposed to phenytoin in utero. Hyperglycemia has been reported with phenytoin use; also, the agent may increase the serum glucose level in patients with diabetes.

Drug Interactions. Acute alcohol intake, amiodarone, chlordiazepoxide, cimetidine, diazepam, disulfiram, estrogens, fluoxetine, H_2-antagonists, isoniazid, methylphenidate, phenothiazines, salicylates, and trazodone may increase phenytoin serum levels. Drugs that may lower phenytoin levels include carbamazepine, chronic alcohol abuse, and reserpine.

Laboratory Interferences. Phenytoin may decrease serum concentrations of thyroxine. It may cause increased serum levels of glucose, alkaline phosphatase, and γ-glutamyl transpeptidase.

Dosage and Clinical Guidelines. Patients may be started on one 100-mg extended oral capsule three times daily, and the dosage then adjusted to suit individual requirements. Patients may then be switched to once-a-day dosing, which is more convenient. In this case, extended-release capsules may be

used. Serial monitoring of phenytoin levels is recommended, and the normal range is usually 10 to 20 µg/mL.

CALCIUM CHANNEL BLOCKERS

The intracellular calcium ion regulates the activity of multiple neurotransmitters such as serotonin and dopamine, and that action may account for its role as a treatment in mood disorders. Calcium channel inhibitors are used in psychiatry as antimanic agents for persons who are refractory to or cannot tolerate treatment with first-line mood-stabilizing agents such as lithium (Eskalith), carbamazepine (Tegretol), and sodium valproate (Depakote). Calcium channel inhibitors include nifedipine (Procardia, Adalat), nimodipine (Nimotop), isradipine (DynaCirc), amlodipine (Norvasc, Lotrel), nicardipine (Cardene), nisoldipine (Sular), nitrendipine, and verapamil (Calan). They are used for control of mania and ultradian bipolar disorder (mood cycling in less than 24 hours).

The results of a large genetic study have rekindled interest in the potential clinical uses of calcium channel blockers (CCBs). Two genome-wide findings implicated genes encoding L-type voltage-gated calcium channel subunits as susceptibility genes for bipolar disorder, schizophrenia, MDD, ADHD, and autism.

Pharmacologic Actions

The calcium channel inhibitors are nearly completely absorbed after oral use, with significant first-pass hepatic metabolism. Considerable intraindividual and interindividual variations are seen in the plasma concentrations of the drugs after a single dose. Peak plasma levels of most of these agents are achieved within 30 minutes. Amlodipine does not reach peak plasma levels for about 6 hours. The half-life of verapamil after the first dose is 2 to 8 hours; the half-life increases to 5 to 12 hours after the first few days of therapy. The half-lives of the other CCBs range from 1 to 2 hours for nimodipine and isradipine to 30 to 50 hours for amlodipine (Table 21-42).

Table 21-42
Half-Lives, Dosages, and Effectiveness of Selected Calcium Channel Inhibitors in Psychiatric Disorders

	Verapamil (Calan, Isoptin)	Nimodipine (Nimotop)	Isradipine (DynaCirc)	Amlodipine (Norvasc)
Half-Life	Short (5–12 hr)	Short (1–2 hr)	Short (1–2 hr)	Long (30–50 hr)
Starting Dosage	30 mg TID	30 mg TID	2.5 mg BID	5 mg HS
Peak Daily Dosage	480 mg	240–450 mg	20 mg	10–15 mg
Antimanic	++	++	++	[a]
Antidepressant	±	+	+	[a]
Antiultradian[b]	±	++	(++)	[a]

BID, twice a day; HS, bedtime; TID, three times a day.
[a]No systematic studies, only case reports.
[b]Rapid-cycling bipolar disorder.
Adapted from Robert M. Post, M.D.

The primary mechanism of action of CCBs in bipolar illness is not known. The calcium channel inhibitors discussed in this section inhibit the influx of calcium into neurons through L-type (long-acting) voltage-dependent calcium channels.

Therapeutic Indications

Bipolar Disorder. Nimodipine and verapamil have been demonstrated to be effective as maintenance therapy in persons with bipolar illness. Patients who respond to lithium appear to also respond to treatment with verapamil. Nimodipine may be useful for ultradian cycling and recurrent brief depression. The clinician should begin treatment with a short-acting drug such as nimodipine or isradipine, beginning with a low dosage and increasing the dosage every 4 to 5 days until clinical response is seen or adverse effects appear. When symptoms are controlled, a longer-acting drug, such as amlodipine, can be substituted as maintenance therapy. Failure to respond to verapamil does not exclude a favorable response to one of the other drugs. Verapamil has been shown to prevent antidepressant-induced mania. The CCBs can be combined with other agents, such as carbamazepine, in patients who are partial responders to monotherapy.

Depression. None of the CCBs is effective as a treatment for depression and may prevent response to antidepressants.

Other Psychiatric Indications. Nifedipine is used to treat hypertensive crises associated with the use of MAOIs. Isradipine may reduce the subjective response to methamphetamine. Calcium channel inhibitors may be beneficial in Tourette disorder, Huntington disease, panic disorder,

intermittent explosive disorder, and tardive dyskinesia.

Other Medical Uses. These drugs have been used to treat medical conditions such as angina, hypertension, migraine headaches, Raynaud phenomenon, esophageal spasm, premature labor, and headache. Verapamil has antiarrhythmic activity and has been used to treat superventricular arrhythmias.

Precautions and Adverse Reactions

The most common adverse effects associated with calcium channel inhibitors are those attributable to vasodilation: dizziness, headache, tachycardia, nausea, dysesthesias, and peripheral edema. Verapamil and diltiazem (Cardizem), in particular, can cause hypotension, bradycardia, and atrioventricular (AV) heart block, which necessitate close monitoring and sometimes discontinuation of the drugs. In all patients with cardiovascular disease, the drugs should be used with caution. Other common adverse effects include constipation, fatigue, rash, coughing, and wheezing. Adverse effects noted with diltiazem include hyperactivity, akathisia, and parkinsonism; with verapamil, delirium, hyperprolactinemia, and galactorrhea; with nimodipine, subjective sense of chest tightness and skin flushing; and with nifedipine, depression. The drugs have not been evaluated for safety in pregnant women and are best avoided. Because the drugs are secreted in breast milk, nursing mothers should also avoid the drugs.

Drug Interactions

All CCBs have the potential for drug–drug interactions. The types and risks of these interactions vary by the compound. Verapamil raises serum levels of carbamazepine, digoxin, and other CYP3A4 substrates. Verapamil and diltiazem but not nifedipine have been reported to precipitate carbamazepine-induced neurotoxicity. Calcium channel inhibitors should not be used by persons taking β-adrenergic receptor antagonists, hypotensives (e.g., diuretics, vasodilators, and ACEIs), or antiarrhythmic drugs (e.g., quinidine and digoxin) without consultation with an internist or cardiologist. Cimetidine (Tagamet) has been reported to increase plasma concentrations of nifedipine and diltiazem. Some patients who are treated with lithium and

calcium channel inhibitors concurrently may be at increased risk for the signs and symptoms of neurotoxicity, and deaths have occurred.

Laboratory Interferences

No known laboratory interferences are associated with the use of calcium channel inhibitors.

Dosage and Clinical Guidelines

Verapamil is available in 40-, 80-, and 120-mg tablets; 120-, 180-, and 240-mg sustained-release tablets; and 100-, 120-, 180-, 200-, 240-, 300-, and 360-mg sustained-release capsules. The starting dosage is 40 mg orally three times a day and can be increased in increments every 4 to 5 days up to 80 to 120 mg three times a day. The patient's BP, pulse, and electrocardiogram (in patients older than 40 years old or with a history of cardiac illness) should be routinely monitored.

Nifedipine is available in 10- and 20-mg capsules and 30-, 60-, and 90-mg extended-release tablets. The administration should be started at 10 mg orally three or four times a day and can be increased up to a maximum dosage of 120 mg a day.

Nimodipine is available in 30-mg capsules. It has been used at 60 mg every 4 hours for ultra–rapid-cycling bipolar disorder and sometimes briefly at up to 360 mg/day.

Isradipine is available in 2.5- and 5-mg capsules, with a maximum of 20 mg/day. An extended-release formulation of isradipine has been discontinued.

Amlodipine is available in 2.5-, 5-, and 10-mg tablets. The administration should start at 5 mg once at night and can be increased to a maximum dosage of 10 to 15 mg a day.

Diltiazem is available in 30-, 60-, 90-, and 120-mg tablets; 60-, 90-, 120-, 180-, 240-, 300-, and 360-mg extended-release capsules; and 60-, 90-, 120-, 180-, 240-, 300-, and 360-mg extended-release tablets. The administration should start with 30 mg orally four times a day and can be increased up to a maximum of 360 mg a day.

Elderly persons are more sensitive to calcium channel inhibitors than are younger adults. No specific information is available regarding the use of the

agents for children.

21.4 Anxiolytics

BENZODIAZEPINES AND DRUGS ACTING ON GABA RECEPTORS

The first benzodiazepine to be introduced was chlordiazepoxide (Librium), in 1959. In 1963, diazepam (Valium) became available. Over the next three decades, superior safety and tolerability helped the benzodiazepines replace the older antianxiety and hypnotic medications, such as the barbiturates and meprobamate (Miltown). Dozens of benzodiazepines and drugs acting on benzodiazepine receptors have been synthesized and marketed worldwide. Many of these agents are not available in the United States. Also, some benzodiazepines have been discontinued because of a lack of use. Table 21-43 lists agents currently available in the United States.

The benzodiazepines derive their name from their molecular structure. They share a common effect on receptors that have been termed benzodiazepine receptors, which in turn modulate GABA activity. Nonbenzodiazepine agonists, such as zolpidem (Ambien), zaleplon (Sonata), and eszopiclone (Lunesta), are similar to these drugs as they bind at a location close to the benzodiazepine receptor. However, we discuss these with the medications for insomnia. Flumazenil (Romazicon), a benzodiazepine receptor antagonist used to reverse benzodiazepine-induced sedation and in emergency care of benzodiazepine overdosage, is also covered here.

Because benzodiazepines have a rapid anxiolytic sedative effect, they are most commonly used for acute treatment of insomnia, anxiety, agitation, or anxiety associated with any psychiatric disorder. Also, the benzodiazepines are used as anesthetics, anticonvulsants, and muscle relaxants and as the preferred treatment for catatonia. Because of the risk of psychological and physical dependence associated with long-term use of benzodiazepines, ongoing assessment should be made as to the continued clinical need for these drugs in treating patients. In most patients, given the nature of their

disorders, it is often best if benzodiazepine agents are used in conjunction with psychotherapy and when alternative agents have been tried and proven ineffective or poorly tolerated. In many forms of chronic anxiety disorders, antidepressant drugs such as SSRIs and SNRIs are now used as primary treatments, with benzodiazepines used as adjuncts. Benzodiazepine abuse is rare, usually found in patients who abuse multiple prescription and recreational drugs.

Table 21-43
Preparations and Doses of Medications Acting on the Benzodiazepine Receptor Available in the United States

Medication	Brand Name	Dose Equivalent	Usual Adult Dose (mg)	How Supplied
Diazepam	Valium	5	2.5–40.0	2-, 5-, and 10-mg tablets 15-mg slow-release tablets
Clonazepam	Klonopin	0.25	0.5–4.0	0.5-, 1.0-, and 2.0-mg tablets
Alprazolam	Xanax	0.5	0.5–6.0	0.25-, 0.5-, 1.0-, and 2.0-mg tablets 1.5-mg sustained-release tablet
Lorazepam	Ativan	1	0.5–6.0	0.5-, 1.0-, and 2.0-mg tablets 4 mg/mL parenteral
Oxazepam	Serax	15	15–120	7.5-, 10.0-, 15.0-, and 30.0-mg capsules 15-mg tablets
Chlordiazepoxide	Librium	25	10–100	5-, 10-, and 25-mg capsules and tablets
Clorazepate	Tranxene	7.5	15–60	3.75-, 7.50-, and 15.00-mg tablets 11.25- and 22.50-mg slow-release tablets
Midazolam	Versed	0.25	1–50	5 mg/mL parenteral 1-, 2-, 5-, and 10-mL vials
Flurazepam	Dalmane	15	15–30	15- and 30-mg capsules
Temazepam	Restoril	15	7.5–30.0	7.5-, 15.0-, and 30.0-mg capsules
Triazolam	Halcion	0.125	0.125–0.250	0.125- and 0.250-mg tablets
Estazolam	ProSom	1	1–2	1- and 2-mg tablets
Quazepam	Doral	5	7.5–15.0	7.5- and 15.0-mg tablets
Zolpidem	Ambien Ambien CR	10 5	5–10 6.25–12.5	5- and 10-mg tablets 6.25- and 12.5-mg tablets
Zaleplon	Sonata	10	5–20	5- and 10-mg capsules
Eszopiclone	Lunesta	1	1–3	1-, 2- and 3-mg tablets
Flumazenil	Romazicon	0.05	0.2–0.5/min	0.1 mg/mL 5- and 10-mL vials

Pharmacologic Actions

All benzodiazepines except clorazepate (Tranxene) are completely absorbed after oral administration and reach peak serum levels within 30 minutes to 2 hours. Metabolism of clorazepate in the stomach converts it to desmethyldiazepam, which is then completely absorbed.

The absorption, the attainment of peak concentrations, and the onset of action are quickest for diazepam (Valium), lorazepam (Ativan), alprazolam (Xanax), triazolam (Halcion), and estazolam (ProSom). The rapid onset of effects is vital to persons who take a single dose of a benzodiazepine to calm an episodic burst of anxiety or to fall asleep rapidly. Several benzodiazepines are effective after intravenous (IV) injection, but only lorazepam and midazolam (Versed) have rapid and reliable absorption after intramuscular (IM) administration.

Diazepam, chlordiazepoxide, clonazepam (Klonopin), clorazepate, flurazepam (Dalmane), and quazepam (Doral) have plasma half-lives of 30 hours to more than 100 hours and are technically described as long-acting benzodiazepines. The plasma half-lives of these compounds can be as high as 200 hours in persons whose metabolism is genetically slow. Because the attainment of steady-state plasma concentrations of the drugs can take up to 2 weeks, persons may experience symptoms and signs of toxicity after only 7 to 10 days of treatment with a dosage that seemed initially to be in the therapeutic range.

Clinically, half-life alone does not necessarily determine the duration of therapeutic action for most benzodiazepines. The fact that all benzodiazepines are lipid-soluble to varying degrees means that benzodiazepines and their active metabolites bind to plasma proteins. The extent of this binding is proportional to their lipid solubility. The amount of protein binding varies from 70 to 99 percent. Distribution, onset, and termination of action after a single dose are thus determined mainly by benzodiazepine lipid solubility, not elimination half-life. Preparations with high lipid solubility, such as diazepam and alprazolam, are absorbed rapidly from the GI tract and distribute rapidly to the brain by passive diffusion along a concentration gradient, resulting in a rapid onset of action. However, as the concentration of the medication increases in the brain and decreases in the bloodstream, the concentration gradient reverses itself, and these medications leave the brain rapidly, resulting in fast cessation of drug effect. Drugs with longer elimination half-lives, such as diazepam, may remain in the bloodstream for a substantially more extended time than their actual pharmacologic action at benzodiazepine receptors because the concentration in the brain decreases rapidly below the level necessary for a noticeable effect.

In contrast, lorazepam, which has a shorter elimination half-life than diazepam but is less lipid-soluble, has a slower onset of action after a single dose because the drug is absorbed and enters the brain more slowly. However, the duration of action after a single dose is longer because it takes longer for lorazepam to leave the brain and for brain levels to decrease below the concentration that produces an effect. In chronic dosing, some of these differences are not as apparent because brain levels are in equilibrium with higher and more consistent steady-state blood levels, but additional doses still produce a more rapid but briefer action with diazepam than with lorazepam. Benzodiazepines are distributed widely in adipose tissue. As a result, medications may persist in the body after discontinuation longer than would be predicted from their elimination half-lives. Also, the dynamic half-life (i.e., duration of action on the receptor) may be longer than the elimination half-life.

The advantages of long–half-life drugs over short–half-life drugs include less frequent dosing, less variation in plasma concentration, and less severe withdrawal phenomena. The disadvantages include drug accumulation, increased risk of daytime psychomotor impairment, and increased daytime sedation.

The half-lives of lorazepam, oxazepam (Serax), temazepam (Restoril), and estazolam are between 8 and 30 hours. Alprazolam has a half-life of 10 to 15 hours, and triazolam has the shortest half-life (2 to 3 hours) of all the orally administered benzodiazepines. Rebound insomnia and anterograde amnesia are thought to be more of a problem with the short–half-life drugs than with the long–half-life drugs.

Because the administration of medications more frequently than the elimination half-life leads to drug accumulation, medications such as diazepam and flurazepam accumulate with daily dosing, eventually resulting in increased daytime sedation.

Some benzodiazepines (e.g., oxazepam) are conjugated directly by glucuronidation and are excreted. Most benzodiazepines are oxidized first by CYP3A4 and CYP2C19, often to active metabolites. These metabolites may then be hydroxylated to another active metabolite. For example, diazepam is oxidized to desmethyldiazepam, which, in turn, is hydroxylated to produce oxazepam. These products undergo glucuronidation to inactive metabolites. Several benzodiazepines (e.g., diazepam, chlordiazepoxide) have the same

active metabolite (desmethyldiazepam), which has an elimination half-life of more than 120 hours. Flurazepam (Dalmane), a lipid-soluble benzodiazepine used as a hypnotic that has a short elimination half-life, has an active metabolite (desalkylflurazepam) with a half-life greater than 100 hours. This is another reason that the duration of action of a benzodiazepine may not correspond to the half-life of the parent drug.

Therapeutic Indications

Insomnia. Because insomnia may be a symptom of a physical or psychiatric disorder, hypnotics should not be used for more than 7 to 10 consecutive days without a thorough investigation of the cause of insomnia. However, many patients have long-standing sleep difficulties and benefit significantly from the long-term use of hypnotic agents. Temazepam, flurazepam, and triazolam are benzodiazepines with a sole indication for insomnia.

Flurazepam, temazepam, quazepam, estazolam, and triazolam are the benzodiazepines approved for use as hypnotics. The benzodiazepine hypnotics differ principally in their half-lives; flurazepam has the most prolonged half-life, and triazolam has the shortest. Flurazepam may be associated with minor cognitive impairment on the day after its administration, and triazolam may be associated with mild rebound anxiety and anterograde amnesia. Quazepam may be associated with daytime impairment when used for a long time. Temazepam or estazolam may be a reasonable compromise for most adults. Estazolam produces rapid onset of sleep and a hypnotic effect for 6 to 8 hours.

Anxiety Disorders

GENERALIZED ANXIETY DISORDER. Benzodiazepines are highly effective for the relief of anxiety associated with GAD. Most persons should be treated for a predetermined, specific, and relatively brief period. However, because GAD is a chronic disorder with a high rate of recurrence, some persons with GAD may warrant long-term maintenance treatment with benzodiazepines.

PANIC DISORDER. Alprazolam and clonazepam, both high-potency benzodiazepines, are commonly used medications for panic disorder with or without agoraphobia. Although the SSRIs are also indicated for the treatment

of panic disorder, benzodiazepines have the advantage of working quickly and not causing significant sexual dysfunction and weight gain. However, the SSRIs are still often preferred because they target common comorbid conditions, such as depression or OCD. Benzodiazepines and SSRIs can be initiated together to treat acute panic symptoms; use of the benzodiazepine can be tapered after 3 to 4 weeks once the therapeutic benefits of the SSRI have emerged.

SOCIAL ANXIETY DISORDER. Clonazepam is an effective treatment for social anxiety disorder. Also, several other benzodiazepines (e.g., diazepam) have been used as adjunctive medications for the treatment of social anxiety disorder.

OTHER ANXIETY DISORDERS. Benzodiazepines are used adjunctively for treatment of adjustment disorder with anxiety, pathologic anxiety associated with life events (e.g., after an accident), OCD, and PTSD.

ANXIETY ASSOCIATED WITH DEPRESSION. Depressed patients often experience significant anxiety, and antidepressant drugs may cause initial exacerbation of these symptoms. Accordingly, benzodiazepines are indicated for the treatment of anxiety associated with depression.

Bipolar I and II Disorders

Clonazepam, lorazepam, and alprazolam are useful in the management of acute manic episodes and as an adjuvant to maintenance therapy instead of antipsychotics. As an adjuvant to lithium (Eskalith) or lamotrigine (Lamictal), clonazepam may result in an increased time between cycles and fewer depressive episodes. Benzodiazepines may help patients with bipolar disorder sleep better.

Catatonia. Lorazepam, sometimes in low doses (less than 5 mg/day) and sometimes in very high doses (12 mg/day or more), is regularly used to treat acute catatonia, which is more frequently associated with bipolar disorder than with schizophrenia. Other benzodiazepines have also been said to be helpful. However, there are no valid controlled trials of benzodiazepines in catatonia. Chronic catatonia does not respond as well to benzodiazepines.

The definitive treatment for catatonia is ECT.

Akathisia. The first-line drug for akathisia is most commonly a β-adrenergic receptor antagonist. However, benzodiazepines are also useful in treating some patients with akathisia.

Other Psychiatric Indications. Chlordiazepoxide (Librium) and clorazepate (Tranxene) are used to manage the symptoms of alcohol withdrawal. The benzodiazepines (especially IM lorazepam) are used to manage substance-induced and psychotic agitation in the emergency department. Benzodiazepines have been used instead of amobarbital (Amytal) for drug-assisted interviewing.

Flumazenil for Benzodiazepine Overdosage. Flumazenil is used to reverse the adverse psychomotor, amnestic, and sedative effects of benzodiazepine receptor agonists. Flumazenil is administered IV and has a half-life of 7 to 15 minutes. The most common adverse effects of flumazenil are nausea, vomiting, dizziness, agitation, emotional lability, cutaneous vasodilation, injection-site pain, fatigue, impaired vision, and headache. The most common serious adverse effect associated with the use of flumazenil is the precipitation of seizures, which is most likely to occur in persons with seizure disorders, those who are physically dependent on benzodiazepines, and those who have ingested large quantities of benzodiazepines. Flumazenil alone may impair memory retrieval.

In mixed-drug overdosage, the toxic effects (e.g., seizures and cardiac arrhythmias) of other drugs (e.g., TCAs) may emerge with the reversal of the benzodiazepine effects of flumazenil. For example, seizures caused by an overdosage of TCAs may have been partially treated in a person who had also taken an overdosage of benzodiazepines. With flumazenil treatment, the tricyclic-induced seizures or cardiac arrhythmias may appear and result in a fatal outcome. Flumazenil does not reverse the effects of ethanol, barbiturates, or opioids.

For the initial management of a known or suspected benzodiazepine overdosage, the recommended initial dosage of flumazenil is 0.2 mg (2 mL) administered IV over 30 seconds. If the desired consciousness is not obtained after 30 seconds, a further dose of 0.3 mg (3 mL) can be administered over 30

seconds. Further doses of 0.5 mg (5 mL) can be administered over 30 seconds at 1-minute intervals up to a cumulative dose of 3.0 mg. The clinician should not rush the administration of flumazenil. A secure airway and IV access should be established before the administration of the drug. Persons should be awakened gradually.

Most persons with a benzodiazepine overdosage respond to a cumulative dose of 1 to 3 mg of flumazenil; doses above 3 mg of flumazenil do not reliably produce additional effects. If a person has not responded 5 minutes after receiving a cumulative dose of 5 mg of flumazenil, the primary cause of sedation is probably not benzodiazepine receptor agonists, and additional flumazenil is unlikely to help.

Sedation can return in 1 to 3 percent of persons treated with flumazenil. It can be prevented or treated by giving repeated dosages of flumazenil at 20-minute intervals. For repeat treatment, no more than 1 mg (given as 0.5 mg a minute) should be given at any one time, and no more than 3 mg should be given in any 1 hour.

Precautions and Adverse Reactions

The most common adverse effect of benzodiazepines is drowsiness, which occurs in about 10 percent of all persons. Because of this adverse effect, persons should be advised to be careful while driving or using dangerous machinery when taking drugs. Drowsiness can be present during the day after the use of a benzodiazepine for insomnia the previous night, the so-called residual daytime sedation. Some persons also experience ataxia (fewer than 2 percent) and dizziness (less than 1 percent). These symptoms can result in falls and hip fractures, especially in elderly persons. The most severe adverse effects of the benzodiazepines occur when other sedative substances, such as alcohol, are taken concurrently. These combinations can result in marked drowsiness, disinhibition, or even respiratory depression. Infrequently, benzodiazepine receptor agonists cause mild cognitive deficits that may impair job performance. Persons taking benzodiazepine receptor agonists should be advised to exercise additional caution when driving or operating dangerous machinery.

High-potency benzodiazepines, especially triazolam, can cause anterograde amnesia. A paradoxical increase in aggression has been reported in persons

with preexisting brain damage. Allergic reactions to the drugs are rare, but a few studies report maculopapular rashes and generalized itching. The symptoms of benzodiazepine intoxication include confusion, slurred speech, ataxia, drowsiness, dyspnea, and hyporeflexia.

Triazolam has received significant attention in the media because of an alleged association with aggression. Therefore, the manufacturer recommends that the drug be used for no more than 10 days for the treatment of insomnia and that physicians carefully evaluate the emergence of any abnormal thinking or behavioral changes in persons treated with triazolam, giving appropriate consideration to all potential causes. Triazolam was banned in Great Britain in 1991.

Persons with hepatic disease and elderly persons are particularly likely to have adverse effects and toxicity from the benzodiazepines, including hepatic coma, especially when the drugs are administered repeatedly or in high dosages. Benzodiazepines can produce clinically significant impairment of respiration in persons with chronic obstructive pulmonary disease and sleep apnea. Alprazolam may exert a direct appetite stimulant effect and may cause weight gain. The benzodiazepines should be used with caution by persons with a history of substance abuse, cognitive disorders, renal disease, hepatic disease, porphyria, CNS depression, or myasthenia gravis.

Some data indicate that benzodiazepines are teratogenic; therefore, their use during pregnancy is not advised. Moreover, the use of benzodiazepines in the third trimester can precipitate a withdrawal syndrome in newborns. The drugs are secreted in the breast milk in sufficient concentrations to affect newborns. Benzodiazepines may cause dyspnea, bradycardia, and drowsiness in nursing babies.

Tolerance, Dependence, and Withdrawal. When benzodiazepines are used for short periods (1 to 2 weeks) in moderate dosages, they usually cause no significant tolerance, dependence, or withdrawal effects. The short-acting benzodiazepines (e.g., triazolam) may be an exception to this rule because some persons have reported increased anxiety the day after taking a single dose of the drug and then stopping its use. Some persons also report a tolerance for the anxiolytic effects of benzodiazepines and require increased doses to maintain the clinical remission of symptoms.

The appearance of a withdrawal syndrome, also called discontinuation

syndrome, depends on how long a person takes the drug, what dose, how quickly they taper it, and the half-life of the compound. Benzodiazepine withdrawal syndrome consists of anxiety, nervousness, diaphoresis, restlessness, irritability, fatigue, lightheadedness, tremor, insomnia, and weakness (Table 21-44). Abrupt discontinuation of benzodiazepines, particularly those with short half-lives, is associated with severe withdrawal symptoms, which may include depression, paranoia, delirium, and seizures. These severe symptoms are more likely to occur if flumazenil is used for the rapid reversal of the benzodiazepine receptor agonist effects. Some features of the syndrome may occur in as many as 90 percent of persons treated with the drugs. The development of severe withdrawal syndrome is seen only in persons who have taken high dosages for long periods. The syndrome may be delayed for 1 or 2 weeks in persons who had been taking benzodiazepines with long half-lives. Alprazolam seems to be significantly associated with an immediate and severe withdrawal syndrome and should be tapered gradually.

When the medication is to be discontinued, the drug must be tapered slowly (25 percent a week); otherwise, recurrence or rebound of symptoms is likely. Monitoring of any withdrawal symptoms (possibly with a standardized rating scale) and psychological support of the person are helpful in the successful accomplishment of benzodiazepine discontinuation. Concurrent use of carbamazepine (Tegretol) during benzodiazepine discontinuation has been reported to permit a more rapid and better-tolerated withdrawal than does a gradual taper alone. The dosage range of carbamazepine used to facilitate withdrawal is 400 to 500 mg a day. Some clinicians report particular difficulty in tapering and discontinuing alprazolam, especially in persons who have been receiving high dosages for long periods. There have been reports of successful discontinuation of alprazolam by switching to clonazepam, which is then gradually withdrawn.



Table 21-44
Signs and Symptoms of Benzodiazepine Withdrawal

Anxiety	Tremor
Irritability	Depersonalization
Insomnia	Hyperesthesia

Hyperacusis	Myoclonus
Nausea	Delirium
Difficulty concentrating	Seizures

Drug Interactions

The most common and potentially severe benzodiazepine receptor agonist interaction is excessive sedation and respiratory depression occurring when benzodiazepines are administered concomitantly with other CNS depressants, such as alcohol, barbiturates, tricyclic and tetracyclic drugs, DRAs, opioids, and antihistamines. Ataxia and dysarthria may be likely to occur when lithium, antipsychotics, and clonazepam are combined. The combination of benzodiazepines and clozapine (Clozaril) has been reported to cause delirium and should be avoided. Cimetidine (Tagamet), disulfiram (Antabuse), isoniazid, estrogen, and oral contraceptives increase the plasma concentration of diazepam, chlordiazepoxide, clorazepate, and flurazepam. However, antacids may reduce the GI absorption of benzodiazepines. The plasma concentrations of triazolam and alprazolam are increased to potentially toxic concentrations by nefazodone (Serzone) and fluvoxamine (Luvox).

The manufacturer of nefazodone recommends that the dosage of triazolam be lowered by 75 percent, and the dosage of alprazolam be lowered by 50 percent, when given concomitantly with nefazodone. Over-the-counter preparations of the kava plant, advertised as a “natural tranquilizer,” can potentiate the action of benzodiazepine receptor agonists through synergistic overactivation of GABA receptors. Carbamazepine can lower the plasma concentration of alprazolam. Antacids and food may decrease the plasma concentrations of benzodiazepines, and smoking may increase the metabolism of benzodiazepines. The benzodiazepines may increase the plasma concentrations of phenytoin and digoxin (Lanoxin). The SSRIs may prolong and exacerbate the severity of zolpidem-induced hallucinations. Deaths have been reported when parental lorazepam is given with parental olanzapine.

Dosage and Clinical Guidelines

The clinical decision to treat an anxious person with a benzodiazepine should be carefully considered. Medical causes of anxiety (e.g., thyroid dysfunction, caffeinism, and prescription medications) should be ruled out. Benzodiazepine use should be started at a low dosage, and the person should be instructed regarding the drug's sedative properties and abuse potential. An estimated length of therapy should be decided at the beginning of therapy, and the need for continued therapy should be reevaluated at least monthly because of the problems associated with long-term use. However, certain persons with anxiety disorders are unresponsive to treatments other than benzodiazepines in long-term use.

Benzodiazepines are available in a wide range of formulations. Clonazepam is available in a wafer formulation that facilitates its use in patients who have trouble swallowing pills. Alprazolam is available in an extended-release form, which reduces the frequency of dosing. Some benzodiazepines are more potent than others in that one compound requires a relatively smaller dosage than another compound to achieve the same effect. For example, clonazepam requires 0.25 mg to achieve the same effect as 5 mg of diazepam; thus, clonazepam is considered a high-potency benzodiazepine. Conversely, oxazepam has an approximate dosage equivalence of 15 mg and is a low-potency drug.

Table 21-43 lists preparations and doses of medications discussed in this chapter.

BUSPIRONE

Buspirone hydrochloride (BuSpar) is classified as an azapirone and is chemically distinct from other psychotropic agents. It acts on two types of receptors, serotonin (5-HT) and dopamine (D). It has a high affinity for the 5-HT_{1A} serotonin receptor, acting as an agonist or partial agonist, and moderate affinity for the D_2 dopamine receptor, acting as both an agonist and an antagonist. The approved indication for this psychotropic drug is for the treatment of GAD. It was initially believed to be a better alternative to the benzodiazepine drug group because buspirone does not possess anticonvulsant and muscle relaxant effects. Reports continue to appear that some patients benefit from the addition of buspirone to their antidepressant regimen. Its use in this role is more common than its use as an anxiolytic.

Interestingly, the antidepressant drug vilazodone (Viibryd) inhibits 5-HT reuptake and acts as a 5-HT_{1A} receptor partial agonist.

Pharmacologic Actions

Buspirone is well absorbed from the GI tract, but absorption is delayed by food ingestion. Peak plasma levels are achieved 40 to 90 minutes after oral administration. At doses of 10 to 40 mg, single-dose linear pharmacokinetics are observed. Nonlinear pharmacokinetics are observed after multiple doses. Because of its short half-life (2 to 11 hours), buspirone is dosed three times daily. An active metabolite of buspirone, 1-pyrimidinylpiperazine (1-PP), is about 20 percent less potent than buspirone but is up to 30 percent more concentrated in the brain than the parent compound. The elimination half-life of 1-PP is 6 hours.

Buspirone does not affect the GABA-associated chloride ion channel or the serotonin reuptake transporter, targets of other drugs that are effective in GAD. Buspirone also has activity at 5-HT_2 and dopamine type 2 (D_2) receptors, although the significance of the effects at these receptors is unknown. At D_2 receptors, it has properties of both an agonist and an antagonist.

Therapeutic Indications

Generalized Anxiety Disorder. Buspirone is a narrow-spectrum antianxiety agent with demonstrated efficacy only in the treatment of GAD. In contrast to the SSRIs or venlafaxine, buspirone is not effective in the treatment of panic disorder, OCD, or social phobia. Buspirone, however, has an advantage over these agents in that it does not typically cause sexual dysfunction or weight gain.

Some evidence suggests that, compared with benzodiazepines, buspirone is generally more useful for symptoms of anger and hostility, equally effective for psychic symptoms of anxiety, and less effective for somatic symptoms of anxiety. The full benefit of buspirone is evident only at dosages above 30 mg a day. Compared with the benzodiazepines, buspirone has a delayed onset of action and lacks any euphoric effect. Unlike benzodiazepines, buspirone has no immediate effects, and patients should be told that a full clinical response

may take 2 to 4 weeks. If an immediate response is needed, patients can be started on a benzodiazepine and then withdrawn from the drug after buspirone's effects begin. Sometimes the sedative effects of benzodiazepines, which are not found with buspirone, are desirable; however, these sedative effects may cause impaired motor performance and cognitive deficits.

Other Disorders. Many other clinical uses of buspirone have been reported, but most have not been confirmed in controlled trials. Evidence of the efficacy of high-dosage buspirone (30 to 90 mg a day) for depressive disorders is mixed. Buspirone appears to have weak antidepressant activity, which has led to its use as an augmenting agent in patients who have failed standard antidepressant therapy. In a large study, buspirone augmentation of SSRIs worked as well as other commonly used strategies. Buspirone is sometimes used to augment SSRIs in the treatment of OCD. There are reports that buspirone may be beneficial against the increased arousal and flashbacks associated with PTSD.

Because buspirone does not act on the GABA–chloride ion channel complex, the drug is not recommended for the treatment of withdrawal from benzodiazepines, alcohol, or sedative-hypnotic drugs, except as treatment of comorbid anxiety symptoms.

Scattered trials suggest that buspirone reduces aggression and anxiety in persons with organic brain disease or traumatic brain injury. It is also used for SSRI-induced bruxism and sexual dysfunction, nicotine craving, and ADHD.

Precautions and Adverse Reactions

Buspirone does not cause weight gain, sexual dysfunction, discontinuation symptoms, or significant sleep disturbance. It does not produce sedation or cognitive and psychomotor impairment. The most common adverse effects of buspirone are headache, nausea, dizziness, and (rarely) insomnia. No sedation is associated with buspirone. Some persons may report a minor feeling of restlessness, although that symptom may reflect an incompletely treated anxiety disorder. No deaths have been reported from overdoses of buspirone, and the median lethal dose is estimated to be 160 to 550 times the recommended daily dose. Buspirone should be used with caution by persons

with hepatic and renal impairment, pregnant women, and nursing mothers. Buspirone can be used safely by the elderly.

Drug Interactions

The coadministration of buspirone and haloperidol (Haldol) results in increased blood concentrations of haloperidol. Buspirone should not be used with MAOIs to avoid hypertensive episodes, and a 2-week washout period should pass between the discontinuation of MAOI use and the initiation of treatment with buspirone. Drugs or foods that inhibit CYP3A4, for example, erythromycin, itraconazole (Sporanox), nefazodone (Serzone), and grapefruit juice, increase buspirone plasma concentrations.

Laboratory Interferences

Single doses of buspirone can cause transient elevations in growth hormone, prolactin, and cortisol concentrations, although the effects are not clinically significant.

Dosage and Clinical Guidelines

Buspirone is available in single-scored 5- and 10-mg tablets and triple-scored 15- and 30-mg tablets; treatment is usually initiated with either 5 mg orally three times daily or 7.5 mg orally twice daily. The dosage can be raised 5 mg every 2 to 4 days to the usual dosage range of 15 to 60 mg a day.

Buspirone should not be used in patients with past hypersensitivity to buspirone, in cases of diabetes-associated metabolic acidosis, or patients with severely compromised liver or renal function.

Switching from a Benzodiazepine to Buspirone. Buspirone is not cross-tolerant with benzodiazepines, barbiturates, or alcohol. A common clinical problem, therefore, is how to initiate buspirone therapy in a person who is currently taking benzodiazepines. There are two alternatives. First, the clinician can start buspirone treatment gradually while the benzodiazepine is being withdrawn. Second, the clinician can start buspirone treatment and bring the person up to a therapeutic dosage for 2 to 3 weeks while the person is still receiving the regular dosage of the benzodiazepine and then slowly taper the benzodiazepine dosage. Patients who have received

benzodiazepines in the past, especially in recent months, may find that buspirone is not as effective as the benzodiazepines in the treatment of their anxiety. This difference might be explained by the absence of the immediate mildly euphoric and sedative effects of benzodiazepines. The coadministration of buspirone and benzodiazepines may be useful in the treatment of persons with anxiety disorders who have not responded to treatment with either drug alone.

β-ADRENERGIC RECEPTOR ANTAGONISTS

Because of the innervations of many, if not most, peripheral organs and vasculature by the sympathetic division of the autonomic nervous system, their functions are ultimately controlled, in part, by one of the two major classes of adrenergic receptors: α-receptors (see Table 33-2) and β-receptors. These receptors are further subdivided based on their action and location, and they are located both peripherally and in the CNS. Shortly after being introduced for cardiac indications, propranolol (Inderal) was reported to be useful for agitation, and its use in psychiatry spread rapidly. The five most commonly used β-receptor antagonists in psychiatry are propranolol, nadolol (Corgard), metoprolol (Lopressor, Toprol), pindolol (Visken), and atenolol (Tenormin) (Table 21-45).

Pharmacologic Actions

The β-receptor antagonists differ concerning lipophilicities, metabolic routes, β-receptor selectivity, and half-lives. The absorption of the β-receptor antagonists from the GI tract is variable. The agents that are most soluble in lipids (i.e., are lipophilic) are likely to cross the blood–brain barrier and enter the brain; those agents that are least lipophilic are less likely to enter the brain. When CNS effects are desired, a lipophilic drug may be preferred; when only peripheral effects are desired, a less lipophilic drug may be indicated.



Table 21-45
β-Adrenergic Drugs Used in Psychiatry

Drug	Pregnancy Category (former FDA classification system)	Protein Binding (%)	Lipophilic	ISA	Metabolism	Receptor Selectivity	Half-Life (hr)	Usual Starting Dosage (mg)	Usual Maximal Dosage (mg)
Atenolol (Tenormin)	D	6–16	No		Renal	$\beta_1 > \beta_2$	6–9	50 OD	50–100 OD
Metoprolol (Lopressor)	C	5–10	Yes		Hepatic	$\beta_1 > \beta_2$	3–4	50 bid	75–150 bid
Nadolol (Corgard)	C	30	No		Renal	$\beta_1 = \beta_2$	14–24	40 OD	80–240 OD
Propranolol (Inderal)	C	>90	Yes		Hepatic	$\beta_1 = \beta_2$	3–6	10–20 bid/tid	80–140 tid
Pindolol (Visken)	B	40	Yes	Minimal	Hepatic	$\beta_1 > \beta_2$	3–4	5 tid/qid	60 bid/tid

Note: In 2015, the FDA changed the pregnancy classification system from a letter-based system to a descriptive system. We refer here to the letter-based system in order to simplify the summary of data into a table.
ISA, intrinsic sympathomimetic activity; OD, once daily; bid, twice a day; tid, three times a day; qid, four times a day.

Whereas propranolol, nadolol, pindolol, and labetalol (Normodyne, Trandate) have nearly equal potency at both the β_1- and β_2-receptors, metoprolol and atenolol have a greater affinity for the β_1-receptor than for the β_2-receptor. Relative β_1-selectivity confers few pulmonary and vascular effects of these drugs, although they must be used with caution in persons with asthma because the drugs retain some activity at the β_2-receptors.

Pindolol has sympathomimetic effects in addition to its β-antagonist effects, which has allowed its use for augmentation of antidepressant drugs. Pindolol, propranolol, and nadolol possess some antagonist activity at the serotonin 5-HT_{1A} receptors.

Therapeutic Indications

Anxiety Disorders. Propranolol is useful for the treatment of social anxiety disorder, primarily of the performance type (e.g., disabling anxiety before a musical performance). Data are also available for its use in the treatment of panic disorder, PTSD, and GAD. In social anxiety disorder, the standard treatment approach is to take 10 to 40 mg of propranolol 20 to 30 minutes before the anxiety-provoking situation. The β-receptor antagonists are less useful for the treatment of panic disorder than are benzodiazepines or SSRIs.

Lithium-Induced Postural Tremor. The β-receptor antagonists are beneficial for lithium-induced postural tremor and other medication-induced postural tremors—for example, those induced by TCAs and valproate

(Depakene). The initial approach to this movement disorder includes lowering the dose of lithium (Eskalith), eliminating aggravating factors, such as caffeine, and administering lithium at bedtime. If these interventions are inadequate, however, propranolol in the range of 20 to 160 mg a day taken two or three times daily is generally sufficient for the treatment of lithium-induced postural tremor.

Neuroleptic-Induced Acute Akathisia. Many studies have shown that β-receptor antagonists can be effective in the treatment of neuroleptic-induced acute akathisia. They are generally more effective for this indication than are anticholinergics and benzodiazepines. The β-receptor antagonists are not effective in the treatment of such neuroleptic-induced movement disorders as acute dystonia and parkinsonism.

Aggression and Violent Behavior. The β-receptor antagonists may be useful in reducing the number of aggressive and violent outbursts in persons with impulse disorders, schizophrenia, and aggression associated with brain injuries such as trauma, tumors, anoxic injury, encephalitis, alcohol use disorders, and degenerative disorders (e.g., Huntington disease).

Alcohol Withdrawal. Propranolol is reported to be useful as an adjuvant to benzodiazepines but not as a sole agent in the treatment of alcohol withdrawal. The following dose schedule is suggested: no propranolol for a pulse rate below 50 beats/min; 50-mg propranolol for a pulse rate between 50 and 79 beats/min; and 100-mg propranolol for a pulse rate of 80 beats/min or above.

Antidepressant Augmentation. Pindolol has been used to augment and hasten the antidepressant effects of SSRIs, tricyclic drugs, and ECT. Small studies have shown that pindolol administered at the onset of antidepressant therapy may shorten the usual 2- to 4-week latency of antidepressant response by several days. Because the β-receptor antagonists may induce depression in some persons, augmentation strategies with these drugs need to be further clarified in controlled trials.

Other Disorders. Several case reports and controlled studies have reported

data indicating that β-receptor antagonists may be of modest benefit for persons with schizophrenia and manic symptoms. They have also been used in some cases of stuttering (see Table 21-46).

Precautions and Adverse Reactions

The β-receptor antagonists are contraindicated for use in people with asthma, insulin-dependent diabetes, congestive heart failure, significant vascular disease, persistent angina, and hyperthyroidism. The contraindication in diabetic persons is because the drugs antagonize the normal physiologic response to hypoglycemia. The β-receptor antagonists can worsen AV conduction defects and lead to complete AV heart block and death. If the clinician decides that the risk-to-benefit ratio warrants a trial of a β-receptor antagonist in a person with one of these coexisting medical conditions, a β_1-selective agent should be the first choice, and the patient should be monitored. All currently available β-receptor antagonists are excreted in breast milk and should be administered with caution to nursing women.

Table 21-46
Psychiatric Uses for β-Adrenergic Receptor Antagonists

Definitely effective
Performance anxiety
Lithium-induced tremor
Neuroleptic-induced akathisia
Probably effective
Adjunctive therapy for alcohol withdrawal and other substance-related disorders
Adjunctive therapy for aggressive or violent behavior
Possibly effective
Antipsychotic augmentation
Antidepressant augmentation

Table 21-47
Adverse Effects and Toxicity of β-Adrenergic Receptor

Antagonists

- Cardiovascular
  - Hypotension
  - Bradycardia
  - Congestive heart failure (in patients with compromised myocardial function)
- Respiratory
  - Asthma (less risk with β_1-selective drugs)
- Metabolic
  - Worsened hypoglycemia in diabetic patients on insulin or oral agents
- Gastrointestinal
  - Nausea
  - Diarrhea
  - Abdominal pain
- Sexual function
  - Impotence
- Neuropsychiatric
  - Lassitude
  - Fatigue
  - Dysphoria
  - Insomnia
  - Vivid nightmares
  - Depression (rare)
  - Psychosis (rare)
- Other (rare)
  - Raynaud phenomenon
  - Peyronie disease
- Withdrawal syndrome
  - Rebound worsening of preexisting angina pectoris when β-adrenergic receptor antagonists are discontinued

The most common adverse effects of β-receptor antagonists are hypotension and bradycardia. In persons at risk for these adverse effects, a test dosage of 20 mg a day of propranolol can be given to assess the reaction to the drug. Depression has been associated with lipophilic β-receptor antagonists, such as propranolol, but it is probably rare. Nausea, vomiting, diarrhea, and constipation can also be caused by treatment with these agents. The β-receptor antagonists may blunt cognition in some people. Other more severe CNS adverse effects (e.g., agitation, confusion, and hallucinations) are rare. Table 21-47 lists the possible adverse effects of β-receptor antagonists.

Drug Interactions

Antipsychotics, anticonvulsants, theophylline (Theo-Dur, Slo-bid), and levothyroxine (Synthroid) have increased plasma concentrations when given with propranolol. Other β-receptor antagonists may have similar effects. The β-receptor antagonists that are eliminated by the kidneys may have similar effects on drugs that are also eliminated by the renal route. Barbiturates, phenytoin (Dilantin), and cigarette smoking increase the elimination of β-receptor antagonists that are metabolized by the liver. Several reports have associated hypertensive crises and bradycardia with the coadministration of β-receptor antagonists and MAOIs. Depressed myocardial contractility and AV nodal conduction can occur from concomitant administration of a β-receptor antagonist and calcium channel inhibitors.

Laboratory Interferences

The β-receptor antagonists do not interfere with standard laboratory tests.

Dosage and Clinical Guidelines

Propranolol is available in 10-, 20-, 40-, 60-, 80-, and 90-mg tablets; 4-, 8-, and 80-mg/mL solutions; and 60-, 80-, 120-, and 160-mg sustained-release capsules. Nadolol is available in 20-, 40-, 80-, 120-, and 160-mg tablets. Pindolol is available in 5- and 10-mg tablets. Metoprolol is available in 50- and 100-mg tablets and 50-, 100-, and 200-mg sustained-release tablets. Atenolol is available in 25-, 50-, and 100-mg tablets. Acebutolol is available in 200- and 400-mg capsules.

For the treatment of chronic disorders, propranolol administration is

usually initiated at 10 mg by mouth three times a day or 20 mg by mouth twice daily. The dosage can be raised by 20 to 30 mg a day until a therapeutic effect emerges. The dosage should be leveled off at the appropriate range for the disorder under treatment. The treatment of aggressive behavior sometimes requires dosages up to 80 mg a day, and therapeutic effects may not be seen until the person has been receiving the maximal dosage for 4 to 8 weeks. For the treatment of social phobia, primarily the performance type, the patient should take 10 to 40 mg of propranolol 20 to 30 minutes before the performance.

Pulse and BP readings should be taken regularly, and the drug should be withheld if the pulse rate is below 50 beats/min or the systolic BP is below 90 mm Hg. The drug should be temporarily discontinued if it produces severe dizziness, ataxia, or wheezing. Treatment with β-receptor antagonists should never be discontinued abruptly. Propranolol should be tapered by 60 mg a day until a dosage of 60 mg a day is reached. After that, the drug should be tapered by 10 to 20 mg a day every 3 or 4 days.

The clinical guidelines for the other drugs listed in this chapter are similar to propranolol, taking into consideration the different doses used. For example, if propranolol is prescribed initially at the lowest available dose (e.g., 10 mg), then metoprolol should be prescribed at its lowest available dose (e.g., 50 mg).



Table 21-48
Barbiturate Dosages (Adult)

Drug	Trade Name	Available Preparations	Hypnotic Dose Range	Anticonvulsant Dose Range
Amobarbital	Amytal	200 mg	50–300 mg	65–500 mg IV
Aprobarbital	Alurate	40 mg/5 mL elixir	40–120 mg	Not established
Butabarbital	Butisol	15-, 30-, and 50-mg tablets 30 mg/5 mL elixir	45–120 mg	Not established
Mephobarbital	Mebaral	32, 50, and 100 mg tablets	100–200 mg	200–600 mg
Methohexital	Brevital	500 mg/50 mL	1 mg/kg for electroconvulsive therapy	Not established
Pentobarbital	Nembutal	50- and 100-mg capsules 50 mg/mL injection or elixir 30, 60, 120, and 200 mg suppository	100–200 mg	100 mg IV, each minute up to 500 mg
Phenobarbital	Luminal	Tablets range from 15–100 mg 20 mg/5 mL elixir 30–130 mg/mL injection	30–150 mg	100–300 mg IV, up to 600 mg/day
Secobarbital	Seconal	100-mg capsule, 50 mg/mL injection	100 mg	5.5 mg/kg IV

IV, intravenous.

BARBITURATES AND SIMILARLY ACTING DRUGS

The first barbiturate to be used in medicine was barbital (Veronal), which was introduced in 1903. It was followed by phenobarbital (Luminal), amobarbital (Amytal), pentobarbital (Nembutal), secobarbital (Seconal), and thiopental (Pentothal). Many others have been synthesized, but only a handful have been used clinically (Table 21-48). Many problems are associated with these drugs, including high abuse and addiction potential, a narrow therapeutic range with low therapeutic index, and unfavorable side effects. The use of barbiturates and similar compounds such as meprobamate (Miltown) has practically been eliminated by the benzodiazepines and hypnotics, such as zolpidem (Ambien), eszopiclone (Lunesta), and zaleplon (Sonata), which have a lower abuse potential and a higher therapeutic index than the barbiturates. Nevertheless, barbiturates still have a role in the treatment of certain mental and convulsive disorders.

Pharmacologic Actions

The barbiturates are well absorbed after oral administration. The binding of barbiturates to plasma proteins is high, but lipid solubility varies. The individual barbiturates are metabolized by the liver and excreted by the kidneys. The half-lives of specific barbiturates range from 1 to 120 hours. The barbiturates may also induce hepatic enzymes (cytochrome P450, CYP), thereby reducing the levels of both the barbiturate and any other concurrently administered drugs metabolized by the liver. The mechanism of action of barbiturates involves the GABA receptor–benzodiazepine receptor–chloride ion channel complex.

Therapeutic Indications

Electroconvulsive Therapy. Methohexital (Brevital) is commonly used as an anesthetic agent for ECT. It has lower cardiac risks than other barbiturate anesthetics. Used intravenously (IV), methohexital produces rapid unconsciousness, and because of its rapid redistribution, it has a brief duration of action (5 to 7 minutes). Typical dosing for ECT is 0.7 to 1.2 mg/kg. Methohexital can also be used to abort prolonged seizures in ECT or to limit postictal agitation.

Seizures. Phenobarbital (Solfoton, Luminal), the most commonly used barbiturate for treatment of seizures, has indications for the treatment of generalized tonic–clonic and simple partial seizures. Parenteral barbiturates are used in the emergency management of seizures independent of cause. Intravenous phenobarbital should be administered slowly at 10 to 20 mg/kg for status epilepticus.

Narcoanalysis. Amobarbital (Amytal) has been used historically as a diagnostic aid in several clinical conditions, including conversion reactions, catatonia, hysterical stupor, and unexplained muteness, and to differentiate stupor of depression, schizophrenia, and structural brain lesions.

The *Amytal interview* is performed by placing the patient in a reclining position and administering amobarbital IV at 50 mg a minute. Infusion is continued until lateral nystagmus is sustained or drowsiness is noted, usually at 75 to 150 mg. After this, 25 to 50 mg can be administered every 5 minutes to maintain narcosis. The patient should be allowed to rest for 15 to 30 minutes after the interview before attempting to walk.

Because of the risk of laryngospasm with IV amobarbital, diazepam has become the drug of choice for narcoanalysis.

Sleep. The barbiturates reduce sleep latency and the number of awakenings during sleep, although tolerance to these effects generally develops within 2 weeks. Discontinuation of barbiturates often leads to rebound increases in electroencephalographic measures of sleep and a worsening of insomnia.

Withdrawal from Sedative-Hypnotics

Barbiturates are sometimes used to determine the extent of tolerance to barbiturates or other hypnotics to guide detoxification. After intoxication has been resolved, a test dose of pentobarbital (200 mg) is given orally. One hour later, the patient is examined. Tolerance and dose requirements are determined by the degree to which the patient is affected. If the patient is not sedated, another 100 mg of pentobarbital can be administered every 2 hours, up to three times (maximum, 500 mg over 6 hours). The amount needed for mild intoxication corresponds to the approximate daily dose of barbiturate used. Phenobarbital (30 mg) may then be substituted for each 100 mg of pentobarbital. This daily dose requirement can be administered in divided

doses and gradually tapered by 10 percent a day, with adjustments made according to withdrawal signs.

Precautions and Adverse Reactions

Some adverse effects of barbiturates are similar to those of benzodiazepines, including paradoxical dysphoria, hyperactivity, and cognitive disorganization. Rare adverse effects associated with barbiturate use include the development of Stevens–Johnson syndrome, megaloblastic anemia, and neutropenia.

Before the advent of benzodiazepines, the widespread use of barbiturates as hypnotics and anxiolytics made them the most common cause of acute porphyria reactions. Severe attacks of porphyria have decreased, mainly because barbiturates are now seldom used and are contraindicated in patients with the disease.

A significant difference between the barbiturates and the benzodiazepines is the low therapeutic index of the barbiturates. An overdose of barbiturates can easily prove fatal. In addition to narrow therapeutic indexes, the barbiturates are associated with a significant risk of abuse potential and the development of tolerance and dependence. Barbiturate intoxication is manifested by confusion, drowsiness, irritability, hyporeflexia or areflexia, ataxia, and nystagmus. The symptoms of barbiturate withdrawal are similar to but more marked than those of benzodiazepine withdrawal.

Ten times the daily dose or 1 g of most barbiturates causes severe toxicity; 2 to 10 g generally proves fatal. Manifestations of barbiturate intoxication may include delirium, confusion, excitement, headache, and CNS and respiratory depression, ranging from somnolence to coma. Other adverse reactions include Cheyne–Stokes respiration, shock, miosis, oliguria, tachycardia, hypotension, hypothermia, irritability, hyporeflexia or areflexia, ataxia, and nystagmus. Treatment of overdose includes induction of emesis or lavage, activated charcoal, and saline cathartics; supportive treatment, including maintaining airway and respiration and treating shock as needed; maintaining vital signs and fluid balance; alkalinizing the urine, which increases excretion; forced diuresis if renal function is typical, or hemodialysis in severe cases.

Because of some evidence of teratogenicity, barbiturates should not be

used by pregnant women or women who are breastfeeding. Barbiturates should be used with caution by patients with a history of substance abuse, depression, diabetes, hepatic impairment, renal disease, severe anemia, pain, hyperthyroidism, or hypoadrenalism. Barbiturates are also contraindicated in patients with acute intermittent porphyria, impaired respiratory drive, or limited respiratory reserve.

Drug Interactions

The primary area for concern about drug interactions is the potentially dangerous effects of respiratory depression. Barbiturates should be used with great caution with other prescribed CNS drugs (including antipsychotic and antidepressant drugs) and nonprescribed CNS agents (e.g., alcohol). Also, caution must be exercised when prescribing barbiturates to patients who are taking other drugs that are metabolized in the liver, especially cardiac drugs and anticonvulsants. Because individual patients have a wide range of sensitivities to barbiturate-induced enzyme induction, it is not possible to predict the degree to which the metabolism of concurrently administered medications may be affected. Drugs that have their metabolism enhanced by barbiturate administration include opioids, antiarrhythmic agents, antibiotics, anticoagulants, anticonvulsants, antidepressants, β-adrenergic receptor antagonists, DRAs, contraceptives, and immunosuppressants.

Laboratory Interferences

No known laboratory interferences are associated with the administration of barbiturates.

Dose and Clinical Guidelines

Barbiturates and other drugs described later begin to act within 1 to 2 hours of administration. The doses of barbiturates vary, and treatment should begin with low doses that are increased to achieve a clinical effect. Children and older people are more sensitive to the effects of barbiturates than are young adults. The most commonly used barbiturates are available in a variety of dose forms. Barbiturates with half-lives in the 15- to 40-hour range are preferable, because long-acting drugs tend to accumulate in the body. Clinicians should instruct patients clearly about the adverse effects and the

potential for dependence associated with barbiturates.

Although determining plasma concentrations of barbiturates is rarely necessary for psychiatry, monitoring of phenobarbital concentrations is standard practice when the drug is used as an anticonvulsant. The therapeutic blood concentrations for phenobarbital in this indication range from 15 to 40 mg/L, although some patients may experience significant adverse effects in that range.

Barbiturates are contained in combination products with which the clinician should be familiar.

Other Similarly Acting Drugs

Several agents that act similarly to the barbiturates have been used in the treatment of anxiety and insomnia. Three such available drugs are paraldehyde (Paral), meprobamate, and chloral hydrate (Noctec). These drugs are rarely used because of their abuse potential and potential toxic effects and some of them are now banned in other countries such as the European Union and Canada.

Paraldehyde. Paraldehyde is a cyclic ether and was first used in 1882 as a hypnotic. It has also been used to treat epilepsy, alcohol withdrawal symptoms, and delirium tremens. Because of its low therapeutic index, it has been supplanted by the benzodiazepines and other anticonvulsants.

PHARMACOLOGIC ACTIONS. Paraldehyde is rapidly absorbed from the GI tract and intramuscular (IM) injections. It is primarily metabolized to acetaldehyde by the liver, and the lungs expire the unmetabolized drug. Reported half-lives range from 3.4 to 9.8 hours. The onset of action is 15 to 30 minutes.

THERAPEUTIC INDICATIONS. Paraldehyde is not indicated as an anxiolytic or a hypnotic and has little place in current psychopharmacology.

PRECAUTIONS AND ADVERSE REACTIONS. Paraldehyde frequently causes foul breath because of the expired unmetabolized drug. It can inflame pulmonary capillaries and cause coughing. It can also cause local thrombophlebitis with IV use. Patients may experience nausea and vomiting with oral use. Overdose leads to metabolic acidosis and decreased renal output. There is a risk of

abuse among drug addicts.

DRUG INTERACTIONS. Disulfiram (Antabuse) inhibits acetaldehyde dehydrogenase and reduces the metabolism of paraldehyde, leading to a possible toxic concentration of paraldehyde. Paraldehyde has additive sedating effects in combination with other CNS depressants such as alcohol or benzodiazepines.

LABORATORY INTERFERENCES. Paraldehyde can interfere with the metyrapone, phentolamine, and urinary 17-hydroxycorticosteroid tests.

DOSING AND CLINICAL GUIDELINES. Paraldehyde is available in 30-mL vials for oral, IV, or rectal use. For seizures in adults, up to 12 mL (diluted to a 10 percent solution) can be administered by gastric tube every 4 hours. For children, the oral dose is 0.3 mg/kg.

Meprobamate. Meprobamate, a carbamate, was introduced shortly before the benzodiazepines, specifically to treat anxiety. It is also used for muscle relaxant effects.

PHARMACOLOGIC ACTIONS. Meprobamate is rapidly absorbed from the GI tract and IM injections. It is metabolized primarily by the liver, and a small portion is excreted unchanged in the urine. The plasma half-life is approximately 10 hours.

THERAPEUTIC INDICATIONS. Meprobamate is indicated for the short-term treatment of anxiety disorders. It has also been used as a hypnotic and is prescribed as a muscle relaxant.

PRECAUTIONS AND ADVERSE REACTIONS. Meprobamate can cause CNS depression and death in overdose and carries the risk of abuse by patients with drug or alcohol use disorders. Abrupt cessation after long-term use can lead to withdrawal syndrome, including seizures and hallucinations. Meprobamate can exacerbate acute intermittent porphyria. Other rare side effects include hypersensitivity reactions, wheezing, hives, paradoxical excitement, and leukopenia. It should not be used in patients with hepatic

compromise.

DRUG INTERACTIONS. Meprobamate has additive sedating effects in combination with other CNS depressants, such as alcohol, barbiturates, or benzodiazepines.

LABORATORY INTERFERENCES. Meprobamate can interfere with the metyrapone, phentolamine, and urinary 17-hydroxycorticosteroid tests.

DOSING AND CLINICAL GUIDELINES. Meprobamate is available in 200-, 400-, and 600-mg tablets; 200- and 400-mg extended-release capsules; and various combinations, for example, aspirin 325 mg and 200 mg of meprobamate (Equagesic) for oral use. For adults, the usual dose is 400 to 800 mg twice daily. Elderly patients and children aged 6 to 12 years require half the adult dose.

Chloral Hydrate. Chloral hydrate is a hypnotic agent rarely used in psychiatry because numerous safer options, such as benzodiazepines, are available.

PHARMACOLOGIC ACTIONS. Chloral hydrate is well absorbed from the GI tract. The parent compound is metabolized within minutes by the liver to the active metabolite trichloroethanol, which has a half-life of 8 to 11 hours. A dose of chloral hydrate induces sleep in about 30 to 60 minutes and maintains sleep for 4 to 8 hours. It probably potentiates GABAergic neurotransmission, which suppresses neuronal excitability.

THERAPEUTIC INDICATIONS. The FDA lists chloral hydrate as an unapproved drug that is still prescribed by clinicians, however it has no FDA-approved indication. The primary use in clinical practice for chloral hydrate is to induce sleep. It should be used for no more than 2 or 3 days because longer-term treatment is associated with an increased incidence and severity of adverse effects. Tolerance develops to the hypnotic effects of chloral hydrate after 2 weeks of treatment. The benzodiazepines are superior to chloral hydrate for all psychiatric uses.

PRECAUTIONS AND ADVERSE REACTIONS. Chloral hydrate has adverse effects on the CNS, GI system, and skin. High doses (>4 g) may be associated with stupor, confusion, ataxia, falls, or coma. The GI effects include nonspecific irritation, nausea, vomiting, flatulence, and an unpleasant taste. With long-term use and overdose, gastritis and gastric ulceration can develop. Along with tolerance, dependence on chloral hydrate can occur. The symptoms are similar to those of alcohol dependence. With a lethal dose between 5,000 and 10,000 mg, chloral hydrate is a particularly poor choice for potentially suicidal persons.

DRUG INTERACTIONS. Because of metabolic interference, chloral hydrate should be strictly avoided with alcohol, a notorious concoction known as a *Mickey Finn*. Chloral hydrate may displace warfarin (Coumadin) from plasma proteins and enhance anticoagulant activity; this combination should be avoided.

LABORATORY INTERFERENCES. Chloral hydrate administration can lead to false-positive results for urine glucose determinations that use cupric sulfate (e.g., Clinitest) but not in tests that use glucose oxidase (e.g., Clinistix and Tes-Tape). Chloral hydrate can also interfere with the determination of urinary catecholamines and 17-hydroxycorticosteroids.

DOSING AND CLINICAL GUIDELINES. Chloral hydrate is available in 500-mg capsules; 500-mg/5-mL solution; and 324-, 500-, and 648-mg rectal suppositories. The standard dose of chloral hydrate is 500 to 2,000 mg at bedtime. Because the drug is a GI irritant, it should be administered with excess water, milk, other liquids, or antacids to decrease gastric irritation.

Propofol. Propofol (Diprivan) is a $GABA_A$ agonist that is used as an anesthetic. It induces the presynaptic release of GABA and dopamine (the latter possibly through action on $GABA_B$ receptors) and is a partial agonist at dopamine D_2 and NMDA receptors. Because it is very lipid-soluble, it crosses the blood–brain barrier readily and induces anesthesia in less than 1 minute. Rapid redistribution from the CNS results in an offset of action within 3 to 8 minutes after the infusion is discontinued. It is well tolerated

when used for conscious sedation, but it has a potential for acute adverse effects, including respiratory depression, apnea, and bradyarrhythmias, and the prolonged infusion can cause acidosis and mitochondrial myopathies. The carrier used for the infusion is a soybean emulsion that can be a culture medium for various organisms. Also, the carrier can impair macrophage function and cause hematologic and lipid abnormalities and anaphylactic reactions.

Etomidate. Etomidate is a carboxylated imidazole that acts at the β_2 and β_3 subunits of the $GABA_A$ receptor. It has a rapid onset (1 minute) and short duration (less than 5 minutes) of action. The propylene glycol vehicle has been linked to hyperosmolar metabolic acidosis. It has both proconvulsant and anticonvulsant properties, and it inhibits cortisol release, with possible adverse consequences after long-term use. It has been used in anesthesia as it is unlikely to affect blood pressure, however it has largely been replaced by newer medications.

21.5 Drugs Used to Treat Sleep Disorders

NONBENZODIAZEPINE GABA AGONISTS

Several of the benzodiazepines are commonly used for sleep, such as flurazepam, temazepam, quazepam, estazolam, and triazolam. Temazepam, flurazepam, and triazolam are benzodiazepines used for insomnia; these are discussed with the other benzodiazepines under the anxiolytic section. Nonbenzodiazepine agonists, such as zolpidem (Ambien), zaleplon (Sonata), and eszopiclone (Lunesta)—the so-called “Z drugs”—are similar to these drugs as they bind at a location close to the benzodiazepine receptor

Pharmacologic Action

Zaleplon, zolpidem, and eszopiclone are structurally distinct and vary in their binding to the GABA receptor subunits. Benzodiazepines activate all three specific GABA–benzodiazepine (GABA–BZ) binding sites of the $GABA_A$-receptor, which opens chloride channels and reduces the rate of neuronal and

muscle firing. Zolpidem, zaleplon, and eszopiclone have selectivity for specific subunits of the GABA receptor. This specificity may account for their selective sedative effects and relative lack of muscle relaxant and anticonvulsant effects.

Zolpidem, zaleplon, and eszopiclone are rapidly and well absorbed after oral administration, although absorption can be delayed by as long as 1 hour if they are taken with food. Zolpidem reaches peak plasma concentrations in 1.6 hours and has a half-life of 2.6 hours. Zaleplon reaches peak plasma concentrations in 1 hour and has a half-life of 1 hour. Eszopiclone reaches peak plasma concentrations in 1 hour. If taken immediately after a high-fat or heavy meal, the peak is delayed by approximately 1 hour, reducing the effects of eszopiclone on sleep onset. The terminal-phase elimination half-life is approximately 6 hours in healthy adults. Eszopiclone is weakly bound to plasma protein (52 to 59 percent).

The rapid metabolism and lack of active metabolites of zolpidem, zaleplon, and eszopiclone avoid the accumulation of plasma concentrations compared with the long-term use of benzodiazepines such as flurazepam or diazepam.

Flumazenil for Benzodiazepine Overdosage. Similar to the benzodiazepines, flumazenil can reverse the adverse psychomotor, amnestic, and sedative effects of zolpidem and zaleplon.

Therapeutic Indications

Insomnia. Zolpidem, zaleplon, and eszopiclone are also indicated only for insomnia. Although these “Z drugs” are not usually associated with rebound insomnia after the discontinuation of their use for short periods, some patients experience increased sleep difficulties the first few nights after discontinuing their use. The use of zolpidem, zaleplon, and eszopiclone for periods more extended than 1 month is not associated with the delayed emergence of adverse effects. No development of tolerance to any parameter of sleep measurement was observed over 6 months in clinical trials of eszopiclone.

γ-Hydroxybutyrate (GHB, Xyrem), which is approved for the treatment of narcolepsy and improves slow-wave sleep, is also an agonist at the $GABA_A$ receptor, where it binds to specific GHB receptors. GHB has the capacity

both to reduce drug craving and to induce dependence, abuse, and absence seizures as a result of complex actions on tegmental dopaminergic systems.

Parkinson Disease. A small number of persons with idiopathic Parkinson disease respond to long-term use of zolpidem with reduced bradykinesia and rigidity. Zolpidem dosages of 10 mg four times daily may be tolerated without sedation for several years.

Precautions and Adverse Reactions

Zolpidem and zaleplon are generally well tolerated. At zolpidem dosages of 10 mg/day and zaleplon dosages above 10 mg/day, a small number of persons will experience dizziness, drowsiness, dyspepsia, or diarrhea. Zolpidem and zaleplon are secreted in breast milk and are therefore contraindicated for use by nursing mothers. The dosage of zolpidem and zaleplon should be reduced in elderly persons and persons with hepatic impairment.

Zolpidem (Ambien) has been associated with automatic behavior and amnesia. In rare cases, zolpidem may cause hallucinations and behavioral changes. The coadministration of zolpidem and SSRIs may extend the duration of hallucinations in susceptible patients. Eszopiclone exhibits a dose–response relationship in elderly adults for the side effects of pain, dry mouth, and unpleasant taste.

Zolpidem and zaleplon can produce a mild withdrawal syndrome lasting 1 day after prolonged use at higher therapeutic dosages. Rarely, a person taking zolpidem has self-titrated up the daily dosage to 30 to 40 mg a day. Abrupt discontinuation of such a high dosage of zolpidem may cause withdrawal symptoms for 4 or more days. Tolerance does not develop to the sedative effects of zolpidem and zaleplon.

Drug Interactions

These agents are most dangerous when used with other CNS depressants, such as alcohol, barbituates, opioids, and similar drugs. Cimetidine increases the plasma concentrations of zaleplon.

Rifampin (Rifadin), phenytoin (Dilantin), carbamazepine, and phenobarbital (Solfoton, Luminal) significantly increase the metabolism of zaleplon.

The CYP3A4 and CYP2E1 enzymes are involved in the metabolism of eszopiclone. Eszopiclone did not show any inhibitory potential on CYP450 1A2, 2A6, 2C9, 2C19, 2D6, 2E1, and 3A4 in cryopreserved human hepatocytes.

Laboratory Interferences

No known laboratory interferences are associated with the use of zolpidem and zaleplon.

Dosage and Clinical Guidelines

Zaleplon is available in 5- and 10-mg capsules. A single 10-mg dose is the usual adult dose. The dose can be increased to a maximum of 20 mg as tolerated. A single dose of zaleplon can be expected to provide 4 hours of sleep with minimal residual impairment. For persons older than age 65 years or persons with hepatic impairment, an initial dose of 5 mg is advised.

Eszopiclone is available in 1-, 2-, and 3-mg tablets. The starting dose should not exceed 1 mg in patients with severe hepatic impairment or those taking potent CYP3A4 inhibitors. The recommended dosing to improve sleep onset or maintenance is 2 or 3 mg for adult patients (ages 18 to 64 years) and 2 mg for older adult patients (ages 65 years and older). The 1-mg dose is for sleep onset in older adult patients whose primary complaint is difficulty falling asleep.

Zolpidem is available in 5- and 10-mg capsules, and 6.25- and 12.5-mg capsules for the controlled-release capsules. There is also a sublingual tablet and an oropharyngeal spray. The controlled-release form is usually for problems with sleep maintenance. The sublingual and oropharyngeal forms are for difficulties with taking the pill, or in particular circumstances. For example, the sublingual can be used for middle of the night awakening. With the oral capsule, the starting dose is 5 mg for females and 5 to 10 mg for males before bedtime. A single 10-mg dose is the usual adult dose (6.25- or 12.5-mg of the controlled release). The capsule should only be taken at the beginning of the night, as the drug can impair a person if they attempt activity within 7 to 8 hours of using the drug. For persons older than age 65 years or persons who are medically frail, 5 mg is the recommended dose.

MELATONIN AGONISTS: RAMELTEON AND MELATONIN

There are two melatonin receptor agonists commercially available in the United States: (1) melatonin, a dietary supplement available in various preparations in health food stores, and not under FDA regulations; (2) and ramelteon (Rozerem), an FDA-approved drug for the treatment of insomnia characterized by difficulties with sleep onset. Both exogenous melatonin and ramelteon are thought to exert their effects by interaction with central melatonin receptors.

Ramelteon

Ramelteon (Rozerem) is a melatonin receptor agonist used to treat sleep-onset insomnia. Unlike the benzodiazepines, ramelteon has no appreciable affinity for the GABA receptor complex.

Pharmacologic Actions. Ramelteon essentially mimics melatonin's sleep-promoting properties and has a high affinity for melatonin MT_1 and MT_2 receptors in the brain. These receptors are thought to be critical in the regulation of the body's sleep–wake cycle.

Ramelteon is rapidly absorbed and eliminated over a dose range of 4 to 64 mg. Maximum plasma concentration (C_{max}) is reached approximately 45 minutes after administration, and the elimination half-life is 1 to 2.6 hours. The total absorption of ramelteon is at least 84 percent, but extensive first-pass metabolism results in a bioavailability of approximately 2 percent. Ramelteon is metabolized primarily through the CYP1A2 pathway and eliminated principally in urine. Repeated once-daily dosing does not appear to result in accumulation, likely because of the compound's short half-life.

Therapeutic Indications. The FDA-approved ramelteon for the treatment of insomnia characterized by difficulty with sleep onset. Potential off-label use is centered on the application in circadian rhythm disorders, predominantly jet lag, delayed sleep phase syndrome, and shift work sleep disorder.

Clinical trials and animal studies have failed to demonstrate evidence of rebound insomnia or withdrawal effects.

Precautions and Adverse Events. Headache is the most common side effect of ramelteon. Other adverse effects may include somnolence, fatigue, dizziness, worsening insomnia, depression, nausea, and diarrhea. We should not use the drug in patients with severe hepatic impairment. It is also not recommended in patients with severe sleep apnea or severe chronic obstructive pulmonary disease. Prolactin levels may be increased in women. The drug should be used with caution, if at all, in nursing mothers and pregnant women.

Ramelteon has been found sometimes to decrease blood cortisol and testosterone and to increase prolactin. Female patients should be monitored for the cessation of menses and galactorrhea, decreased libido, and fertility problems. The safety and effectiveness of ramelteon in children has not been established.

Drug Interactions. CYP1A2 is the major isozyme involved in the hepatic metabolism of ramelteon. Accordingly, fluvoxamine (Luvox) and other CYP1A2 inhibitors may increase the side effects of ramelteon.

Ramelteon should be administered with caution in patients taking CYP1A2 inhibitors, potent CYP3A4 inhibitors such as ketoconazole, and strong CYP2C inhibitors such as fluconazole (Diflucan). No clinically meaningful interactions were found when ramelteon was coadministered with omeprazole, theophylline, dextromethorphan, midazolam, digoxin, and warfarin.

Dosing and Clinical Guidelines. The usual dose of ramelteon is 8 mg within 30 minutes of going to bed. It should not be taken with or immediately after high-fat meals.

Melatonin

Melatonin (*N*-acetyl-5-methoxytryptamine) is a hormone produced mainly at night in the pineal gland. Ingested melatonin can reach and bind to melatonin-binding sites in the brains of mammals and produce somnolence when used at high doses. Melatonin is available as a dietary supplement and is not a medication. Few well-controlled clinical trials have been conducted to determine its effectiveness in treating such conditions as insomnia, jet lag, and sleep disturbances related to shift work.

Pharmacologic Actions. Melatonin's secretion is stimulated by the dark and inhibited by the light. It is naturally synthesized from the amino acid tryptophan, which is converted to serotonin and finally converted to melatonin. The suprachiasmatic nuclei (SCN) of the hypothalamus have melatonin receptors, and melatonin may have a direct action on SCN to influence circadian rhythms, which are relevant for jet lag and sleep disturbances. In addition to the pineal gland, melatonin is also produced in the retina and GI tract.

Melatonin has a very short half-life of 0.5 to 6 minutes. Plasma concentrations are a function of the dose administered and the endogenous rhythm. Approximately 90 percent of melatonin is cleared through the first-pass metabolism by way of the CYP1A1 and CYP1A2 pathways. Elimination occurs principally in urine.

Exogenous melatonin interacts with the melatonin receptors that suppress neuronal firing and promote sleep. There does not appear to be a dose–response relationship between exogenous melatonin administrations and sleep effects.

Therapeutic Indications. Melatonin is approved for insomnia in the European Union and several other countries, however, it is not approved by the FDA for any medical use. It is sold over the counter in the United States and Canada but requires a prescription in some other countries such as the United Kingdom. Individuals have used exogenous melatonin to address sleep difficulties (insomnia, circadian rhythm disorders), cancer (breast, prostate, colorectal), seizures, depression, anxiety, and seasonal affective disorder. Some studies suggest that exogenous melatonin may have some antioxidant effects and antiaging properties.

Precautions and Adverse Reactions. Adverse events associated with melatonin include fatigue, dizziness, headache, irritability, and somnolence. Disorientation, confusion, sleepwalking, vivid dreams, and nightmares have also been observed, often with effects resolving after melatonin administration was suspended.

Melatonin may reduce fertility in both men and women. In men, exogenous melatonin reduces sperm motility, and long-term administration has been shown to inhibit testicular aromatase levels. In women, exogenous

melatonin may inhibit ovarian function, and for that reason, it has been evaluated as a contraceptive, but with inconclusive results.

Drug Interactions. As a dietary supplement preparation, exogenous melatonin is not regulated by the FDA and has not been subjected to the same type of drug interaction studies that were performed for ramelteon. Caution is suggested in coadministering melatonin with blood thinners (e.g., warfarin [Coumadin], aspirin, and heparin), antiseizure medications, and medications that lower BP.

Laboratory Interference. Melatonin is not known to interfere with any commonly used clinical laboratory tests.

Dosage and Administration. Over-the-counter melatonin is available in the following formulations: 1-, 2.5-, 3-, and 5-mg capsules; 1-mg/4-mL liquid; 0.5- and 3-mg lozenges; 2.5-mg sublingual tablets; and 1-, 2-, and 3-mg timed-release tablets.

Standard recommendations are to take the desired melatonin dose at bedtime, but some evidence from clinical trials suggests that dosing up to 2 hours before habitual bedtime may produce a more significant improvement in sleep onset.

Agomelatine (Valdoxan). Agomelatine is structurally related to melatonin and is used in Europe as a treatment for MDD. It acts as an agonist at melatonin (MT_1 and MT_2) receptors. It also acts as a serotonin antagonist. Analysis of agomelatine clinical trial data raised serious questions about the efficacy and safety of the drug. The drug is not being marketed in the United States.

PRAZOSIN

Prazosin (Minipress) is a quinazoline derivative and one of a new chemical class of antihypertensives. It is an α_1-adrenergic receptor antagonist as opposed to the drugs mentioned above, which are α_2-blockers.

Pharmacologic Actions

The exact mechanism of the hypotensive action of prazosin, and how it suppresses nightmares, is unknown. Prazosin causes a decrease in total peripheral resistance that is related to its action as an α_1-adrenergic receptor antagonist. BP is lowered in both the supine and standing positions. This effect is most pronounced on the diastolic BP. After oral administration, human plasma concentrations reach a peak at about 3 hours with a plasma half-life of 2 to 3 hours. The drug is highly bound to plasma protein. Tolerance has not been observed to develop with long-term therapy.

Therapeutic Action

Prazosin is used in psychiatry to suppress nightmares, particularly those associated with PTSD.

Precautions and Adverse Reactions

During clinical trials and subsequent marketing experience, the most frequent reactions were dizziness (10.3 percent); headache (7.8 percent); drowsiness (7.6 percent); lack of energy (6.9 percent); weakness (6.5 percent); palpitations (5.3 percent); and nausea (4.9 percent). In most instances, side effects disappeared with continued therapy or have been tolerated with no decrease in the dose of the drug. Prazosin should not be used in nursing mothers or during pregnancy.

Drug Interactions

No adverse drug interactions have been reported.

Laboratory Interferences

No laboratory interferences have been reported.

Dosage and Clinical Guidelines

The drug is supplied in 1-, 2-, and 5-mg capsules and a nasal spray. The therapeutic dosages most commonly used have ranged from 6 to 15 mg daily given in divided doses. Doses higher than 20 mg do not increase efficacy. When adding a diuretic or other antihypertensive agent, the dose should be reduced to 1 or 2 mg three times a day, and retitration then carried out.

Concomitant use with a PDE-5 inhibitor can result in additive BP-lowering effects and symptomatic hypotension; therefore, PDE-5 inhibitor therapy should be initiated at the lowest dose in patients taking prazosin.

21.6 Stimulants

STIMULANT DRUGS AND ATOMOXETINE

Stimulant drugs increase motivation, mood, energy, and wakefulness. They are also called sympathomimetics because they mimic the physiologic effects of the neurotransmitter epinephrine. Several chemical classes are included in this group.

Currently, these drugs are most commonly used to treat symptoms of poor concentration and hyperactivity in children and adults with ADHD. Paradoxically, many patients with ADHD find that these drugs can have a calming effect. Sympathomimetics are also approved for use in increasing alertness in narcolepsy.

Amphetamines were the first stimulants to be synthesized. They were created in the late 19th century and were used by Bavarian soldiers in the mid-1880s to maintain wakefulness, alertness, energy, and confidence in combat. They have been used similarly in most wars since then. They were not widely used clinically until the 1930s when they were marketed as Benzedrine inhalers for relief of nasal congestion. When their psychostimulant effects were noted, these drugs were used to treat sleepiness associated with narcolepsy. They have been classified as controlled drugs because of their rapid onset, immediate behavioral effects, and propensity to develop tolerance, which leads to the risk of abuse and dependence in vulnerable individuals. State and federal agencies regulate their manufacture, distribution, and use. In 2005, pemoline was withdrawn from the market because of significant risks of treatment-emergent hepatotoxicity.

Sympathomimetics have been widely used in persons with ADHD and narcolepsy because no equally effective agents have been available. They have also been found useful in treating certain cognitive disorders that result in secondary depression or profound apathy (e.g., AIDS, multiple sclerosis,

poststroke depression, dementia, closed head injury) as well as in the augmentation of antidepressant medications in specific treatment-resistant depressions.

Atomoxetine is included in this section because it is used to treat ADHD, even though it is not a psychostimulant.

Pharmacologic Actions

All of these drugs are well absorbed from the GI tract. Amphetamine (Adderall) and dextroamphetamine (Dexedrine, Dextrostat) reach peak plasma concentrations in 2 to 3 hours and have a half-life of about 6 hours, thereby necessitating once- or twice-daily dosing. Methylphenidate is available in immediate-release (Ritalin), sustained-release (Ritalin SR), and extended-release (Concerta) formulations. Immediate-release methylphenidate reaches peak plasma concentrations in 1 to 2 hours and has a short half-life of 2 to 3 hours, thereby necessitating multiple-daily dosing. The sustained-release formulation reaches peak plasma concentrations in 4 to 5 hours and doubles the effective half-life of methylphenidate. The extended-release formulation reaches peak plasma concentrations in 6 to 8 hours and is designed to be effective for 12 hours in once-daily dosing. Dexmethylphenidate (Focalin) reaches peak plasma concentration in about 3 hours and is prescribed twice daily.

Lisdexamfetamine dimesylate, also known as L-lysine D-amphetamine (Vyvanse), is an amphetamine prodrug. In this formulation, dextroamphetamine is coupled with the amino acid L-lysine. Lisdexamfetamine becomes active upon cleavage of the lysine portion of the molecule by enzymes in the red blood cells. This results in the gradual release of dextroamphetamine into the bloodstream. Apart from having an extended duration of action, this formulation reduces its abuse potential. It is the only prodrug of its kind. Lisdexamfetamine is indicated for the treatment of ADHD in children 6 years and up and in adults as an integral part of a total treatment program that may include other measures (i.e., psychological, educational, social). The safety and efficacy of lisdexamfetamine dimesylate in patients 3 to 5 years old has not been established. In contrast to Adderall, which contains approximately 75 percent dextroamphetamine and 25 percent levoamphetamine, lisdexamfetamine is a single, dextro-enantiomer

amphetamine molecule. In most cases, this makes the drug better tolerated, but some patients experience a more significant benefit from the mixed isomer preparation.

Methylphenidate, dextroamphetamine, and amphetamine are indirectly acting sympathomimetics, with the primary effect causing the release of catecholamines from presynaptic neurons. Their clinical effectiveness is associated with increased release of both dopamine and norepinephrine. Dextroamphetamine and methylphenidate are also weak inhibitors of catecholamine reuptake and inhibitors of monoamine oxidase.

In recent years, there has been an explosion of new brand name stimulants. Most of these are variations of the generic methylphenidate or mixed amphetamine salts that provide additional options to customize treatment in terms of dosing and duration of effect.

For modafinil, the specific mechanism of action is unknown. Narcolepsy–cataplexy results from a deficiency of hypocretin, a hypothalamic neuropeptide. Hypocretin-producing neurons are activated after modafinil administration. Modafinil does not appear to work through a dopaminergic mechanism. It does have α_1-adrenergic agonist properties, which may account for its alerting effects, because the wakefulness induced by modafinil can be attenuated by prazosin, an α_1-adrenergic antagonist. Some evidence suggests that modafinil has some norepinephrine reuptake blocking effects. Armodafinil (Nuvigil) is the R-enantiomer of modafinil. Both drugs have similar clinical effects and side effects.

Therapeutic Indications

Attention-Deficit/Hyperactivity Disorder (ADHD). Sympathomimetics are the first-line drugs for the treatment of ADHD in children and are effective about 75 percent of the time. Methylphenidate and dextroamphetamine are equally effective and work within 15 to 30 minutes. Sympathomimetic drugs decrease hyperactivity, increase attentiveness, and reduce impulsivity. They may also reduce comorbid oppositional behaviors associated with ADHD. Many persons take these drugs throughout their schooling and beyond. In responsive persons, the use of a sympathomimetic may be a critical determinant of scholastic success.

Sympathomimetics improve the core ADHD symptoms of hyperactivity,

impulsivity, and inattentiveness and permit improved social interactions with teachers, family, other adults, and peers. The success of long-term treatment of ADHD with sympathomimetics, which are efficacious for most of the various constellations of ADHD symptoms present from childhood to adulthood, supports a model in which ADHD results from a genetically determined neurochemical imbalance that requires lifelong pharmacologic management.

Methylphenidate is the most commonly used initial agent at a dosage of 5 to 10 mg every 3 to 4 hours. Dosages may be increased to a maximum of 20 mg four times daily or 1 mg/kg a day. Use of the 20-mg sustained-release formulation to achieve 6 hours of benefit and eliminate the need for dosing at school is supported by many experts, although other authorities believe it is less effective than the immediate-release formulation. Dextroamphetamine is about twice as potent as methylphenidate on a per milligram basis and provides 6 to 8 hours of benefit. Some 70 percent of nonresponders to one sympathomimetic may benefit from another. All of the sympathomimetic drugs should be tried before switching to drugs of a different class. The previous dictum that sympathomimetics worsen tics and, therefore, should be avoided by persons with comorbid ADHD and tic disorders has been questioned. Small dosages of sympathomimetics do not appear to cause an increase in the frequency and severity of tics. Alternatives to sympathomimetics for ADHD include bupropion (Wellbutrin), venlafaxine (Effexor), guanfacine (Tenex), clonidine (Catapres), and tricyclic drugs. Further studies are needed to determine whether modafinil improves the symptoms of ADHD.

Short-term use of sympathomimetics induces a euphoric feeling; however, tolerance develops for both the euphoric feeling and the sympathomimetic activity.

Narcolepsy and Hypersomnolence. Narcolepsy consists of sudden sleep attacks (*narcolepsy*), sudden loss of postural tone (*cataplexy*), loss of voluntary motor control going into (hypnagogic) or coming out of (hypnopompic) sleep (*sleep paralysis*), and hypnagogic or hypnopompic *hallucinations*. Sympathomimetics reduce narcoleptic sleep attacks and improve wakefulness in other types of hypersomnolent states. Modafinil is approved as an antisomnolence agent for the treatment of narcolepsy, for

people who cannot adjust to night shift work, and for those who do not sleep well because of obstructive sleep apnea.

Other sympathomimetics are also used to maintain wakefulness and accuracy of motor performance in persons subject to sleep deprivation, such as pilots and military personnel. Persons with narcolepsy, unlike persons with ADHD, may develop a tolerance for the therapeutic effects of sympathomimetics.

In direct comparison with amphetamine-like drugs, modafinil is equally useful at maintaining wakefulness, with a lower risk of excessive activation.

Depressive Disorders. Sympathomimetics may be used for treatment-resistant depressive disorders, usually as augmentation of standard antidepressant drug therapy. Possible indications for the use of sympathomimetics as monotherapy include depression in elderly persons, who are at increased risk for adverse effects from standard antidepressant drugs; depression in medically ill persons, especially persons with AIDS; obtundation caused by chronic use of opioids; and clinical situations in which a rapid response is necessary but for which ECT is contraindicated. Depressed patients with abulia and anergia may also benefit.

Dextroamphetamine may be useful in differentiating pseudodementia of depression from dementia. A depressed person generally responds to a 5-mg dose with increased alertness and improved cognition. Sympathomimetics are thought to provide only short-term benefit (2 to 4 weeks) for depression because most persons rapidly develop a tolerance for the antidepressant effects of the drugs. However, some clinicians report that long-term treatment with sympathomimetics can benefit some persons.

Encephalopathy Caused by Brain Injury. Sympathomimetics increase alertness, cognition, motivation, and motor performance in persons with neurologic deficits caused by strokes, trauma, tumors, or chronic infections. Treatment with sympathomimetics may permit earlier and more robust participation in rehabilitative programs. Poststroke lethargy and apathy may respond to long-term use of sympathomimetics.

Obesity. Sympathomimetics are used in the treatment of obesity because of their anorexia-inducing effects. Because tolerance develops for the anorectic

effects and because of the drugs' high abuse potential, their use for this indication is limited. Of the sympathomimetic drugs, phentermine (Adipex-P, Fastin) is the most widely used for appetite suppression. Phentermine was the second half of "fen-phen," an off-label combination of fenfluramine and phentermine, widely used to promote weight loss. Fenfluramine and dexfenfluramine were withdrawn from the market because of their adverse effect, including cardiac valvular insufficiency, primary pulmonary hypertension, and irreversible loss of cerebral serotoninergic nerve fibers. The toxicity of fenfluramine is attributed to the fact that it stimulates the release of massive amounts of serotonin from nerve endings, a mechanism of action not shared by phentermine. The use of phentermine alone has not been reported to cause the same adverse effects like those caused by fenfluramine or dexfenfluramine.

Careful limitations of caloric intake and judicious exercise are at the core of any successful weight loss program. Sympathomimetic drugs facilitate the loss of, at most, an additional fraction of a pound per week. Sympathomimetic drugs are effective appetite suppressants only for the first few weeks of use; then, the anorexigenic effects tend to decrease.

Fatigue. Between 70 and 90 percent of individuals with multiple sclerosis experience fatigue. Modafinil, armodafinil, amphetamines, methylphenidate, and the dopamine receptor agonist amantadine (Symmetrel) are sometimes useful in combating this symptom. Other causes of fatigue, such as chronic fatigue syndrome, respond to stimulants in many cases.

Precautions and Adverse Reactions

The most common adverse effects associated with amphetamine-like drugs are stomach pain, anxiety, irritability, insomnia, tachycardia, cardiac arrhythmias, and dysphoria. Sympathomimetics cause a decreased appetite, although tolerance usually develops for this effect. The treatment of common adverse effects in children with ADHD is usually straightforward (Table 21-49). The drugs can also cause increases in heart rate and BP and may cause palpitations. Less common adverse effects include the possible induction of movement disorders, such as tics, Tourette disorder–like symptoms, and dyskinesias, all of which are often self-limited over 7 to 10 days. If a person

taking a sympathomimetic develops one of these movement disorders, a correlation between the dose of the medication and the severity of the disorder must be firmly established before adjustments are made in the medication dosage. In severe cases, augmentation with risperidone (Risperdal), clonidine (Catapres), or guanfacine (Tenex) is necessary. Methylphenidate may worsen tics in one-third of persons; these persons fall into two groups: those whose methylphenidate-induced tics resolve immediately upon the metabolism of the dosage, and a smaller group in whom methylphenidate appears to trigger tics that persist for several months but eventually resolve spontaneously.

Table 21-49
Management of Common Stimulant-Induced Adverse Effects in Attention-Deficit/Hyperactivity Disorder

Adverse Effect	Management
Anorexia, nausea, weight loss	• Administer stimulant with meals. • Use caloric-enhanced supplements. Discourage forcing meals.
Insomnia, nightmares	• Administer stimulants earlier in day. • Change to short-acting preparations. • Discontinue afternoon or evening dosing. • Consider adjunctive treatment (e.g., antihistamines, clonidine, antidepressants).
Dizziness	• Monitor BP. • Encourage fluid intake. • Change to long-acting form.
Rebound phenomena	• Overlap stimulant dosing. • Change to long-acting preparation or combine long- and short-acting preparations. • Consider adjunctive or alternative treatment (e.g., clonidine, antidepressants).
Irritability	• Assess timing of phenomena (during peak or

	withdrawal phase). • Evaluate comorbid symptoms. • Reduce dose. • Consider adjunctive or alternative treatment (e.g., lithium, antidepressants, anticonvulsants).
Dysphoria, moodiness, agitation	• Consider comorbid diagnosis (e.g., mood disorder). • Reduce dosage or change to long-acting preparation. • Consider adjunctive or alternative treatment (e.g., lithium, anticonvulsants, antidepressants).

BP, blood pressure.

From Wilens TE, Blederman J. The stimulants. In: Shaffer D, ed. *The Psychiatric Clinics of North America: Pediatric Psychopharmacology*. Philadelphia, PA: Saunders; 1992, with permission.

Longitudinal studies do not indicate that sympathomimetics cause growth suppression. Sympathomimetics may exacerbate glaucoma, hypertension, cardiovascular disorders, hyperthyroidism, anxiety disorders, psychotic disorders, and seizure disorders.

High dosages of sympathomimetics can cause dry mouth, pupillary dilation, bruxism, formication, excessive ebullience, restlessness, emotional lability, and occasionally seizures. Long-term use of high dosages can cause a delusional disorder that resembles paranoid schizophrenia. Seizures can be treated with benzodiazepines, cardiac effects with β-adrenergic receptor antagonists, fever with cooling blankets, and delirium with DRAs. Overdosages of sympathomimetics result in hypertension, tachycardia, hyperthermia, toxic psychosis, delirium, hyperpyrexia, convulsions, coma, chest pain, arrhythmia, heart block, hypertension or hypotension, shock, and nausea. Toxic effects of amphetamines can be seen at 30 mg, but idiosyncratic toxicity can occur at doses as low as 2 mg. Conversely, survival has been reported up to 500 mg.

The most limiting adverse effect of sympathomimetics is their association with psychological and physical dependence. At the doses used for the treatment of ADHD, the development of psychological dependence virtually

never occurs. The more considerable concern is the presence of adolescent or adult cohabitants who might confiscate the supply of sympathomimetics for abuse or sale.

The use of sympathomimetics should be avoided during pregnancy, especially during the first trimester. Dextroamphetamine and methylphenidate pass into the breast milk, and it is not known whether modafinil or armodafinil do.

Drug Interactions

Sympathomimetics decrease the metabolism of several drugs, thus increasing their plasma levels. These drugs include tricyclic or tetracyclic antidepressants, warfarin (Coumadin), primidone (Mysoline), phenobarbital (Luminal), phenytoin (Dilantin), or phenylbutazone (Butazolidin). Sympathomimetics decrease the therapeutic efficacy of many antihypertensive drugs, especially guanethidine (Esimil, Ismelin). The sympathomimetics should be used with extreme caution with MAOIs.

Laboratory Interferences

Dextroamphetamine may elevate plasma corticosteroid levels and interfere with some assay methods for urinary corticosteroids.

Dosage and Administration

Many psychiatrists believe that governmental authorities have overly regulated amphetamine use. Amphetamines are listed as schedule II drugs by the Drug Enforcement Agency. Some states keep a registry of patients who receive amphetamines. Such mandates worry both patients and physicians about breaches in confidentiality, and physicians are concerned that official agencies may misinterpret their prescribing practices. Consequently, some physicians may withhold prescription of sympathomimetics, even from persons who may benefit from the medications.

The dosage ranges and the available preparations for sympathomimetics are presented in Table 21-50. Vyvanse dosing is a particular case because many patients are switched to this formulation after being treated with other stimulants. A conversion table is shown in Table 21-51. It is available in 20-, 30-, 40-, 50-, 60-, and 70-mg capsules. Dosage should be individualized

according to the therapeutic needs and response of the patient. Lisdexamfetamine (Vyvanse) should be administered at the lowest effective dosage. In patients who are either starting treatment for the first time or switching from another medication, 30 mg once daily in the morning is the recommended dose. Dosages may go up or down in 10 or 20 mg increments in intervals of approximately 1 week. Patients should avoid taking the drug in the afternoon, as this may cause insomnia. The drug may be taken with or without food.

Dextroamphetamine, methylphenidate, amphetamine, benzphetamine, and methamphetamine are schedule II drugs and, in some states, require triplicate prescriptions. Phendimetrazine (Adipost, Bontril) and phenmetrazine (Prelude) are schedule III drugs, and modafinil, armodafinil, phentermine, diethylpropion (Tenuate), and mazindol (Mazanor, Sanorex) are schedule IV drugs.

Pretreatment evaluation should include an evaluation of the patient's cardiac function, with particular attention to the presence of hypertension or tachyarrhythmias. The clinician should also examine the patient for the presence of movement disorders, such as tics and dyskinesia, because the administration of sympathomimetics can exacerbate these conditions. If tics are present, many experts will not prescribe sympathomimetics but will instead choose clonidine or antidepressants. However, recent data indicate that sympathomimetics may cause only a mild increase in motor tics and may suppress vocal tics. Liver function and renal function should be assessed, and dosages of sympathomimetics should be reduced for persons with impaired metabolism.



Table 21-50
Selected Sympathomimetics Commonly Used in Psychiatry

Generic Name	Trade Name	Preparations	Initial Daily Dose	Usual Daily Dose for ADHD[a]	Usual Daily Dose for Disorders Associated with Excessive Daytime Somnolence	Maximum Daily Dose
Amphetamine–dextroamphetamine	Adderall	5-, 10-, 20-, and 30-mg tablets	5–10 mg	20–30 mg	5–60 mg	Children: 40 mg Adults: 60 mg
Armodafinil	Nuvigil	50-, 150-, and 250-mg tablets	50–150 mg	150–250 mg	250 mg	
Atomoxetine	Strattera	10-, 18-, 25-, 40-, and 60-mg tablets	20 mg	40–80 mg	Not used	Children: 80 mg Adults: 100 mg
Dexmethylphenidate	Focalin	2.5-, 5-, and 10-mg capsules	5 mg	5–20 mg	Not used	20 mg
Dextroamphetamine	Dexedrine, Dextrostat	5-, 10-, and 15-mg ER capsules; 5- and 10-mg tablets	5–10 mg	20–30 mg	5–60 mg	Children: 40 mg Adults: 60 mg
Lisdexamfetamine	Vyvanse	20-, 30-, 40-, 50-, 60-, and 70-mg capsules	20–30 mg			70 mg
Methamphetamine	Desoxyn	5-mg tablets; 5-, 10-, and 15-mg ER tablets	5–10 mg	20–25 mg	Not generally used	45 mg
Methylphenidate	Ritalin, Methidate, Methylin, Attenade	5-, 10-, and 20-mg tablets; 10- and 20-mg SR tablets	5–10 mg	5–60 mg	20–30 mg	Children: 80 mg Adults: 90 mg
	Concerta	18- and 36-mg ER tablets	18 mg	18–54 mg	Not yet established	54 mg
Modafinil*	Provigil	100- and 200-mg tablets	100 mg	Not used	400 mg	400 mg

*Obstructive sleep apnea, narcolepsy, and shift work disorder
[a]For children 6 years of age and older.
ER, extended release; SR, sustained release.



Table 21-51 Lisdexamfetamine (Vyvanse) Dosage Equivalency Conversions

Vyvanse and Adderall XR	
Vyvanse	Adderall XR
20 mg	5 mg
30 mg	10 mg
40 mg	15 mg
50 mg	20 mg
60 mg	25 mg
70 mg	30 mg
Vyvanse, Adderall IR, and Dexedrine	

Vyvanse	Adderall IR	Dexedrine
70 mg	30 mg	22.5 mg
50 mg	20 mg	15 mg
30 mg	10 mg	7.5 mg

XR, extended release; IR, immediate-release.

Persons with ADHD can take immediate-release methylphenidate at 8 AM, 12 noon, and 4 PM. Dextroamphetamine, Adderall, sustained-release methylphenidate, or 18 mg of extended-release methylphenidate may be taken once at 8 AM. The starting dose of methylphenidate ranges from 2.5 mg of regular to 20 mg of the sustained-release formulation. If this is inadequate, it may be increased to a maximum dose of 80 mg in children and 90 mg daily in adults. The dosage of dextroamphetamine is 2.5 to 40 mg a day up to 0.5 mg/kg a day.

The starting dosage of modafinil is 200 mg in the morning in medically healthy individuals and 100 mg in the morning in persons with hepatic impairment. Some persons take a second 100- or 200-mg dose in the afternoon. The maximum recommended daily dosage is 400 mg, although dosages of 600 to 1,200 mg a day have been used safely. Adverse effects become prominent at dosages greater than 400 mg a day. Compared with amphetamine-like drugs, modafinil promotes wakefulness but produces less attentiveness and less irritability. Some persons with excessive daytime sleepiness extend the activity of the morning modafinil dose with an afternoon dose of methylphenidate. Armodafinil is virtually identical to modafinil but is dosed differently, the dosing range being 50 to 250 mg daily.

ATOMOXETINE (STRATTERA)

Atomoxetine is the first nonstimulant drug to be approved by the FDA as a treatment of ADHD in children, adolescents, and adults. It is included in this chapter because it shares this indication with the stimulants described earlier.

Pharmacologic Actions

Atomoxetine is believed to produce a therapeutic effect through selective

inhibition of the presynaptic norepinephrine transporter. It is well absorbed after oral administration and is minimally affected by food. High-fat meals may decrease the rate but not the extent of absorption. Maximum plasma concentrations are reached after approximately 1 to 2 hours. At therapeutic concentrations, 98 percent of atomoxetine in plasma is bound to protein, mainly albumin. Atomoxetine has a half-life of approximately 5 hours and is metabolized principally by the CYP2D6 pathway. Poor metabolizers of this compound reach a fivefold higher area under the curve and fivefold higher peak plasma concentration than normal or extensive metabolizers. This variation is critical to consider in patients receiving medications that inhibit the CYP2D6 enzyme. For example, the antidepressant-like pharmacology of atomoxetine has led to its use as an add-on to SSRIs or other antidepressants. Drugs such as fluoxetine (Prozac), paroxetine (Paxil), and bupropion (Wellbutrin) are CYP2D6 inhibitors and may raise atomoxetine levels.

Therapeutic Indications

Atomoxetine is used for the treatment of ADHD. It should be considered for use in patients who find stimulants too activating or who experience other intolerable side effects. Because atomoxetine has no abuse potential, it is a reasonable choice in the treatment of patients with both ADHD and substance abuse, patients who complain of ADHD symptoms but are suspected of seeking stimulant drugs, and patients who are in recovery.

Atomoxetine may enhance cognition when used to treat patients with schizophrenia. It may also be used as an alternative or add-on to antidepressants in patients who fail to respond to standard therapies.

Precautions and Adverse Reactions

Common side effects of atomoxetine include abdominal discomfort, decreased appetite with resulting weight loss, sexual dysfunction, dizziness, vertigo, irritability, and mood swings. Minor increases in BP and heart rate have also been observed. There have been cases of severe liver injury in a small number of patients taking atomoxetine. The drug should be discontinued in patients with jaundice (yellowing of the skin or whites of the eyes, itching) or laboratory evidence of liver injury. Atomoxetine should not be taken at the same time as, or within 2 weeks of taking, an MAOI or by

patients with narrow-angle glaucoma.

The effects of overdose greater than twice the maximum recommended daily dose in humans are unknown. No specific information is available on the treatment of overdose with atomoxetine.

Dosage and Clinical Guidelines

Atomoxetine is available as 10-, 18-, 25-, 40-, and 60-mg capsules. In children and adolescents who weigh up to 70 kg, atomoxetine should be initiated at a total daily dose of approximately 0.5 mg/kg and increased after a minimum of 3 days to a target total daily dose of approximately 1.2 mg/kg, administered either as a single daily dose in the morning or as evenly divided doses in the morning and late afternoon or early evening. The total daily dose in smaller children and adolescents should not exceed 1.4 mg/kg or 100 mg, whichever is less. Dosing of children and adolescents who weigh more than 70 kg and adults should start at a total daily dose of 40 mg and then be increased after a minimum of 3 days to a target total daily dose of approximately 80 mg. The doses can be administered either as a single daily dose in the morning or as evenly divided doses in the morning and late afternoon or early evening. After 2 to 4 additional weeks, the dose may be increased to a maximum of 100 mg in patients who have not achieved an optimal response. The maximum recommended total daily dose in children and adolescents over 70 kg and adults is 100 mg.

▲ 21.7 Drugs Used to Treat Substance Use Disorders

OPIOID RECEPTOR AGONISTS

Opioid receptor agonists are a structurally diverse group of compounds that are used for pain management. These drugs are also called narcotics. Although highly effective as analgesics, they often cause dependence and are frequently diverted for recreational use. Commonly used opioid agonists for pain relief include morphine, hydromorphone (Dilaudid), codeine, meperidine (Demerol), oxycodone (OxyContin), buprenorphine (Buprenex),

hydrocodone (Robidone), tramadol (Ultram), and fentanyl (Durogesic). Heroin is used as a street drug. Methadone is used both for pain management and for the treatment of opiate addiction. This chapter focuses on the μ-opioid receptor agonists that are most likely to be used in the treatment of psychiatric disorders instead of pain management.

It is now recognized that the pharmacology of the opioid system is complex. There are multiple types of opioid receptors, with μ- and κ-opioid receptors representing functionally opposing endogenous systems (Table 21-52). All of the compounds above, which represent the most extensively used narcotic analgesics, are agonists at μ-opioid receptors. However, analgesic effects also result from agonist effects on the κ-opioid receptor. Buprenorphine has mixed receptor effects, being primarily a μ-opioid receptor agonist as well as a κ-opioid antagonist.

There is growing interest in the use of some drugs that act on opioid receptors as alternative treatments for a subpopulation of patients with refractory depression, as well as treatment for cutting behavior in patients with borderline personality disorder.

Ongoing, regular use of opioids produces dependence and tolerance and may lead to maladaptive use, functional impairment, and withdrawal symptoms. This well-known fact tempers consideration of such off-label use. The prevalence of opioid use, abuse, and dependence, particularly with prescription opioids, has risen in recent years.



Table 21-52
μ- and κ-Opiate Receptors

Receptor	Agonist Effects	Antagonist Effects
Mu (μ)	Analgesia Euphoria Antidepressant Anxiety	Anxiety Hostility
Kappa (κ)	Analgesia Dysphoria Depression	Antidepressant

Before using opioid receptor agonists with patients who have failed on multiple conventional therapeutic agents, screen carefully for a history of drug use disorders, document the rationale for off-label use, establish treatment ground rules, obtain written consent, consult with primary care physician, and monitor closely. Avoid replacing “lost” prescriptions and providing early prescription renewals.

Pharmacologic Actions

Methadone and buprenorphine are absorbed rapidly from the GI tract. Hepatic first-pass metabolism significantly affects the bioavailability of each of the drugs but in markedly different ways. For methadone, hepatic enzymes reduce the bioavailability of an oral dosage by about half, an effect that is easily managed with dosage adjustments.

For buprenorphine, first-pass intestinal and hepatic metabolism eliminates oral bioavailability almost wholly. When used in opioid detoxification, buprenorphine is given sublingually in either a liquid or a tablet formulation.

The peak plasma concentrations of oral methadone are reached within 2 to 6 hours, and the plasma half-life initially is 4 to 6 hours in opioid-naive persons and 24 to 36 hours after steady dosing of any type of opioid. Methadone is highly protein-bound and equilibrates widely throughout the body, which ensures little post dosage variation in steady-state plasma concentrations.

Elimination of a sublingual dosage of buprenorphine occurs in two phases: an initial phase with a half-life of 3 to 5 hours and a terminal phase with a half-life of more than 24 hours. Buprenorphine dissociates from its receptor binding site slowly, which permits an every-other-day dosing schedule.

Methadone acts as a pure agonist at μ-opioid receptors and has negligible agonist or antagonist activity at κ- or δ-opioid receptors. Buprenorphine is a partial agonist at μ-receptors, a potent antagonist at κ-receptors, and neither an agonist nor an antagonist at δ-receptors.

Therapeutic Indications

Methadone. Methadone is used for short-term detoxification (7 to 30 days),

long-term detoxification (up to 180 days), and maintenance (treatment beyond 180 days) of opioid-dependent individuals. For these purposes, it is only available through designated clinics called methadone maintenance treatment programs (MMTPs) and in hospitals and prisons. Methadone is a schedule II drug, which means that specific federal laws and regulations tightly govern its administration.

Enrollment in a methadone program reduces the risk of death by 70 percent; reduces illicit use of opioids and other substances of abuse; reduces criminal activity; reduces the risk of infectious diseases of all types, most importantly HIV and hepatitis B and C infection; and in pregnant women, reduces the risk of fetal and neonatal morbidity and mortality. The use of methadone maintenance frequently requires lifelong treatment.

Some opioid-dependence treatment programs use a stepwise detoxification protocol in which a person addicted to heroin switches first to the potent agonist methadone; then to the weaker agonist buprenorphine; and finally to maintenance on an opioid receptor antagonist, such as naltrexone (ReVia). This approach minimizes the appearance of opioid withdrawal effects, which, if they occur, are mitigated with clonidine (Catapres). However, compliance with opioid receptor antagonist treatment is low outside of settings using intensive cognitive-behavioral techniques. In contrast, noncompliance with methadone maintenance precipitates opioid withdrawal symptoms, which serve to reinforce the use of methadone and make cognitive-behavioral therapy less than essential. Thus, some well-motivated, socially integrated former heroin addicts can use methadone for years without participation in a psychosocial support program.

Data pooled from many reports indicate that methadone is more effective when taken at dosages above 60 mg a day. The analgesic effects of methadone are sometimes used in the management of chronic pain when less addictive agents are ineffective.

PREGNANCY. Methadone maintenance, combined with effective psychosocial services and regular obstetric monitoring, significantly improves obstetric and neonatal outcomes for women addicted to heroin. Enrollment of a heroin-addicted pregnant woman in such a maintenance program reduces the risk of malnutrition, infection, preterm labor, spontaneous abortion, preeclampsia, eclampsia, abruptio placenta, and septic thrombophlebitis.

The dosage of methadone during pregnancy should be the lowest effective dosage, and no withdrawal to abstinence should be attempted during pregnancy. Methadone is metabolized more rapidly in the third trimester, which may necessitate higher dosages. The daily dose can be administered in two divided doses during the third trimester to avoid potentially sedating postdose peak plasma concentrations. Methadone treatment has no known teratogenic effects.

NEONATAL METHADONE WITHDRAWAL SYMPTOMS. Withdrawal symptoms in newborns frequently include tremor, a high-pitched cry, increased muscle tone and activity, poor sleep and eating, mottling, yawning, perspiration, and skin excoriation. Convulsions that require aggressive anticonvulsant therapy may also occur. Withdrawal symptoms may be delayed in onset and prolonged in neonates because of their immature hepatic metabolism. Women taking methadone are sometimes counseled to initiate breastfeeding as a means of gently weaning their infants from methadone dependence, but they should not breastfeed their babies while still taking methadone.

Buprenorphine. The analgesic effects of buprenorphine are sometimes used in the management of chronic pain when less addictive agents are ineffective. Because buprenorphine is a partial agonist rather than a full agonist at the μ-receptor and is a weak antagonist at the κ-receptor, this agent produces a milder withdrawal syndrome and has a wider margin of safety than the full μ-agonist compounds generally used in treatment. Buprenorphine has a ceiling effect beyond which dose increases prolong the duration of action of the drug without further increasing the agonist effects. Because of this, buprenorphine has a high clinical safety profile, with limited respiratory depression, therefore decreasing the likelihood of lethal overdose. Buprenorphine does have the capacity to cause typical side effects associated with opioids, including sedation, nausea and vomiting, constipation, dizziness, headache, and sweating. A relevant pharmacokinetic consideration when using buprenorphine is the fact that it requires hepatic conversion to become analgesic (*N*-dealkylation catalyzed by CYP3A4). This may explain why some patients do not benefit from buprenorphine. Genetics, grapefruit juice, and many medications (including fluoxetine and fluvoxamine) can reduce a person’s ability to metabolize buprenorphine into its bioactive form.

Buprenorphine has been combined with the narcotic antagonist naloxone for sublingual administration (Suboxone). This combination reduces the likelihood of abusing buprenorphine via the IV route. Because the sublingual route poorly absorbs naloxone, when the combination drug is taken sublingually, there is no effect of the naloxone on the efficacy of buprenorphine. If an opioid-dependent individual injects the combination medication, the naloxone precipitates a withdrawal reaction, therefore reducing the likelihood of illicit injection use of the sublingual preparation.

Inducting and stabilizing a patient on buprenorphine is analogous to inducting and stabilizing a patient on methadone except that, as a partial agonist, buprenorphine has the potential to cause precipitated withdrawal in patients who have recently taken full agonist opioids. Thus, a patient must avoid short-acting opioids for 12 to 24 hours before starting buprenorphine. Similarly, they should abstain from longer-acting opioids such as methadone for 24 to 48 hours or longer. The physician must assess the patient clinically and determine that the patient is in mild to moderate opioid withdrawal with objectively observable withdrawal signs before initiating buprenorphine.

In most instances, a relatively low dose of buprenorphine (2 to 4 mg) can then be administered, with additional doses given in 1 to 2 hours if withdrawal signs persist. The goal for the first 24 hours is to suppress withdrawal signs and symptoms, and the total 24-hour dose to do so can range from 2 to 16 mg on the first day. In subsequent days, the dose can be adjusted upward or downward to resolve withdrawal fully and, as with methadone, to achieve an absence of craving, adequate tolerance to prevent reinforcement from the use of other opioids, and ultimately abstinence from other opioids while minimizing side effects. Dose-ranging studies have demonstrated that dosages of 6 to 16 mg/day are associated with improved treatment outcomes compared with lower doses of buprenorphine (1 to 4 mg). Sometimes patients seem to need dosages higher than 16 mg/day, although there is no evidence for any benefit of dosages beyond 32 mg/day. For the treatment of opioid dependence, a dose of approximately 4 mg of sublingual buprenorphine is the equivalent of a daily dose of 40 mg of oral methadone. It has also been demonstrated that daily, alternate-day or three-times-per-week administration has equivalent effects in suppressing the symptoms of opioid withdrawal for dependent individuals. The combination tablet is recommended for most clinical purposes, including induction and

maintenance. The buprenorphine mono (Subutex) should be used only for pregnant patients (although recent studies are investigating whether the combination of buprenorphine/naloxone is safe in pregnancy) or for patients who have a documented anaphylactic reaction to naloxone.

Newer forms of buprenorphine delivery, including a transdermal skin patch, a long-acting depot intramuscular injection that provides therapeutic plasma levels for several weeks (Sublocade), and subcutaneous buprenorphine implants that may provide therapeutic plasma levels for 6 months (Probuphine), are also FDA approved, although the latter treatment requires special training to administer.

Tramadol. There are multiple reports of tramadol's antidepressant effects, both as monotherapy and augmentation agent in treatment-resistant depression. Clinical and experimental data suggest that tramadol has an inherent antidepressant-like activity. Tramadol has complex pharmacology. It is a weak μ-opioid receptor agonist, a 5-HT releasing agent, a DA-releasing agent, a 5-HT_{2C} receptor antagonist, a norepinephrine reuptake inhibitor, an NMDA receptor antagonist, a nicotinic acetylcholine receptor antagonist, a $TRPV_1$ receptor agonist, and an M_1 and M_3 muscarinic acetylcholine receptor antagonist. Consistent with the evidence of its antidepressant effects is the fact that tramadol has a close structural similarity to the antidepressant venlafaxine.

Both venlafaxine and tramadol inhibit norepinephrine/serotonin reuptake and inhibit the reserpine-induced syndrome completely. Both compounds also have an analgesic effect on chronic pain. Venlafaxine may have an opioid component, and naloxone reverses the antipain effect of venlafaxine. Nonopioid activity is demonstrated by the fact that the μ-opioid receptor antagonist naloxone does not fully antagonize its analgesic effect. Indicative of their structural similarities, venlafaxine may cause false-positive results on liquid chromatography tests to detect urinary tramadol levels.

Another relevant property of tramadol is its relatively long half-life, which reduces the potential for misuse. Its habituating effects are found to be much less than other opiate agonists, but abuse, withdrawal, and dependence are risks. Tramadol requires metabolism to become analgesic: individuals who are CYP2D6 "poor metabolizers" or use drugs that are CYP2D6 inhibitors

reduce the efficacy of tramadol (the same is true of codeine).

Precautions and Adverse Reactions

The most common adverse effects of opioid receptor agonists are lightheadedness, dizziness, sedation, nausea, constipation, vomiting, perspiration, weight gain, decreased libido, inhibition of orgasm, and insomnia or sleep irregularities. Opioid receptor agonists are capable of inducing tolerance as well as producing physiologic and psychological dependence. Other CNS adverse effects include depression, sedation, euphoria, dysphoria, agitation, and seizures. Delirium has been reported in rare cases. Occasional non-CNS adverse effects include peripheral edema, urinary retention, rash, arthralgia, dry mouth, anorexia, biliary tract spasm, bradycardia, hypotension, hypoventilation, syncope, antidiuretic hormone-like activity, pruritus, urticaria, and visual disturbances. Menstrual irregularities are frequent in women, especially in the first 6 months of use. Various abnormal endocrine laboratory indexes of little clinical significance may also be seen.

Most persons develop tolerance to the pharmacologic adverse effects of opioid agonists during long-term maintenance, and relatively few adverse effects are experienced after the induction period.

Overdosage. The acute effects of opioid receptor agonist overdosage include sedation, hypotension, bradycardia, hypothermia, respiratory suppression, miosis, and decreased GI motility. Severe effects include coma, cardiac arrest, shock, and death. The risk of overdosage is most significant in the induction stage of treatment and in persons with slow drug metabolism caused by preexisting hepatic insufficiency. Deaths have been caused during the first week of induction by methadone dosages of only 50 to 60 mg a day.

The risk of overdosage with buprenorphine appears to be lower than with methadone. However, deaths have been caused by the use of buprenorphine in combination with benzodiazepines.

Withdrawal Symptoms. Abrupt cessation of methadone use triggers withdrawal symptoms within 3 to 4 days, which usually reaches peak intensity on the sixth day. Withdrawal symptoms include weakness, anxiety, anorexia, insomnia, gastric distress, headache, sweating, and hot and cold

flashes. The withdrawal symptoms usually resolve after 2 weeks. However, a protracted methadone abstinence syndrome is possible that may include restlessness and insomnia.

The withdrawal symptoms associated with buprenorphine are similar to but less marked than those caused by methadone. In particular, buprenorphine is sometimes used to ease the transition from methadone to opioid receptor antagonists or abstinence because of the relatively mild withdrawal reaction associated with discontinuation of buprenorphine.

Drug–Drug Interactions

Opioid receptor agonists can potentiate the CNS-depressant effects of alcohol, barbiturates, benzodiazepines, other opioids, low-potency DRAs, tricyclic and tetracyclic drugs, and MAOIs. Carbamazepine (Tegretol), phenytoin (Dilantin), barbiturates, rifampin (Rimactane, Rifadin), and heavy long-term consumption of alcohol may induce hepatic enzymes, which may lower the plasma concentration of methadone or buprenorphine and thereby precipitate withdrawal symptoms. In contrast, however, hepatic enzyme induction may increase the plasma concentration of active levomethadyl metabolites and cause toxicity.

Acute opioid withdrawal symptoms may be precipitated in persons on methadone maintenance therapy who take pure opioid receptor antagonists such as naltrexone, nalmefene (Revex, Selincro), and naloxone (Narcan); partial agonists such as buprenorphine; or mixed agonist–antagonists such as pentazocine (Talwin). These symptoms may be mitigated by the use of clonidine, a benzodiazepine, or both.

Competitive inhibition of methadone or buprenorphine metabolism after short-term use of alcohol or administration of cimetidine (Tagamet), erythromycin, ketoconazole (Nizoral), fluoxetine (Prozac), fluvoxamine (Luvox), loratadine (Claritin), quinidine (Quinidex), and alprazolam (Xanax) may lead to higher plasma concentrations or a prolonged duration of action of methadone or buprenorphine. Medications that alkalinize the urine may reduce methadone excretion.

Methadone maintenance may also increase plasma concentrations of desipramine (Norpramin, Pertofrane) and fluvoxamine. The use of methadone may increase zidovudine (Retrovir) concentrations, which

increases the possibility of zidovudine toxicity at otherwise standard dosages. Moreover, in vitro human liver microsome studies demonstrate competitive inhibition of methadone demethylation by several protease inhibitors, including ritonavir (Norvir), indinavir (Crixivan), and saquinavir (Invirase). The clinical relevance of this finding is unknown.

Fatal drug–drug interactions with the MAOIs are associated with the use of the opioids fentanyl (Sublimaze) and meperidine (Demerol) but not with the use of methadone, levomethadyl, or buprenorphine.

Tramadol may interact with drugs that inhibit serotonin reuptake. Such combinations can trigger seizures and serotonin syndrome. These events may also develop during tramadol monotherapy, either at routine or excessive doses. The risk of interactions is increased when tramadol is combined with virtually all classes of antidepressants and with drugs that lower the seizure threshold, especially the antidepressant bupropion.

Laboratory Interferences

Methadone and buprenorphine can be tested separately in urine toxicology to distinguish them from other opioids. No known laboratory interferences are associated with the use of methadone or buprenorphine.

Dosage and Clinical Guidelines

Methadone. Methadone is supplied in 5-, 10-, and 40-mg dispersible scored tablets; 40-mg scored wafers; 5-mg/5-mL, 10-mg/5-mL, and 10-mg/mL solutions; and a 10-mg/mL parenteral form. In maintenance programs, methadone is usually dissolved in water or juice, and dose administration is directly observed to ensure compliance. For induction of opioid detoxification, an initial methadone dose of 15 to 20 mg will usually suppress craving and withdrawal symptoms. However, some individuals may require up to 40 mg a day in single or divided doses. Higher dosages should be avoided during induction of treatment to reduce the risk of acute toxicity from overdosage.

Over several weeks, the dosage should be raised to at least 70 mg a day. The maximum dosage is usually 120 mg a day, and higher dosages require prior approval from regulatory agencies. Dosages above 60 mg a day are associated with much more complete abstinence from the use of illicit opioids

than are dosages less than 60 mg a day.

The duration of treatment should not be predetermined but should be based on response to treatment and assessment of psychosocial factors. All studies of methadone maintenance programs endorse long-term treatment (i.e., several years) as more effective than short-term programs (i.e., less than 1 year) for the prevention of relapse into opioid abuse. In actual practice, however, a minority of programs are permitted by policy or approved by insurers to provide even 6 months of continuous maintenance treatment. Moreover, some programs encourage withdrawal from methadone in less than 6 months after induction. This practice is quite ill-conceived, because more than 80 percent of persons who terminate methadone maintenance treatment eventually return to illicit drug use within 2 years. In programs that offer both maintenance and withdrawal treatments, the overwhelming majority of participants enroll in the maintenance treatment.

Buprenorphine. Buprenorphine is supplied as a 0.3-mg/mL solution in 1-mL ampules. Sublingual tablet formulations of buprenorphine containing buprenorphine only or buprenorphine combined with naloxone in a 4:1 ratio are used for opioid maintenance treatment. Physicians must be trained and certified to carry out this therapy in their private offices. There are several approved training programs in the United States. As noted above, an extended release and subcutaneous extended formulation are also available but require some special skill and training.

Tramadol. No controlled trials exist that establish the appropriate dosing schedule for tramadol when used for conditions other than pain. Tramadol is available in many formulations. These range from capsules (regular and extended-release) to tablets (regular, extended-release, chewable tablets) that can be taken sublingually, suppositories, and injectable ampules. It also comes as tablets and capsules containing acetaminophen or aspirin. Doses in case reports of treatment for depression or OCD range from 50 to 200 mg a day and involve short-term use. The long-term use of tramadol in the treatment of psychiatric disorders has not been studied.

OPIOID RECEPTOR ANTAGONISTS: NALTREXONE, NALMEFENE, AND NALOXONE

Naltrexone (Revia, Depade) and naloxone (Narcan) are competitive opioid antagonists. They bind to opioid receptors without causing their activation. Because these drugs induce opioid withdrawal effects in people using full opioid agonists, these drugs are classified as opioid antagonists.

Naltrexone is the most widely used of these drugs. It has a relatively long half-life, is orally effective, is not associated with dysphoria, and is administered once daily. Naloxone, which predated naltrexone to treat narcotic overdose, became less widely used for preventing relapse to opiate use in detoxified opiate addicts. Since its introduction, naltrexone has been tried for the treatment of a wide range of psychiatric disorders, including, among others, eating disorders, autism, self-injurious behavior, cocaine dependence, gambling, and alcoholism. Naltrexone was approved for the treatment of alcohol dependence in 1994. Several generic formulations are also available. An extended-release, once-a-month injectable suspension (Vivitrol) was also approved in 2006. Nalmefene (Revex) is indicated for the complete or partial reversal of opioid drug effects and in the management of known or suspected opioid overdose. An oral formulation of nalmefene is available in some countries but not in the United States. Nalmefene (Revex, Selincro) is an opioid receptor antagonist that is sometimes used in the management of alcohol dependence.

Pharmacologic Actions

The GI tract rapidly absorbs oral opioid receptor antagonists. However, because of first-pass hepatic metabolism, only 60 percent of a dose of naltrexone and 40 to 50 percent of a dose of nalmefene reach the systemic circulation unchanged. Peak concentrations of naltrexone and its active metabolite, 6-β-naltrexol, are achieved within 1 hour of ingestion. The half-life of naltrexone is 1 to 3 hours, and the half-life of 6-β-naltrexol is 13 hours. Peak concentrations of nalmefene are achieved in about 1 to 2 hours, and the half-life is 8 to 10 hours. Clinically, a single dose of naltrexone effectively blocks the rewarding effects of opioids for 72 hours. Traces of 6-β-naltrexol may linger for up to 125 hours after a single dose.

Naltrexone and nalmefene are competitive antagonists of opioid receptors. Understanding the pharmacology of opioid receptors can explain the difference in adverse effects caused by naltrexone and nalmefene. Opioid

receptors in the body are typed pharmacologically as μ, κ, or δ. Whereas activation of the κ- and δ-receptors is thought to reinforce opioid and alcohol consumption centrally, activation of μ-receptors is more closely associated with central and peripheral antiemetic effects. Because naltrexone is a relatively weak antagonist of κ- and δ-receptors and a potent μ-receptor antagonist, dosages of naltrexone that effectively reduce opioid and alcohol consumption also strongly block μ-receptors and, therefore, may cause nausea. Nalmefene, in contrast, is an equally potent antagonist of all three opioid receptor types, and dosages of nalmefene that effectively reduce opioid and alcohol consumption have no significantly increased effect on μ-receptors. Thus, nalmefene is associated clinically with few GI adverse effects.

Naloxone has the highest affinity for the μ-receptor but is a competitive antagonist at the μ-, κ-, and δ-receptors.

Whereas the effects of opioid receptor antagonists on opioid use are easily understood in terms of competitive inhibition of opioid receptors, the effects of opioid receptor antagonists on alcohol dependence are less straightforward. That likely relates to the fact that the desire for and the effects of alcohol consumption appear to be regulated by several neurotransmitter systems, both opioid and nonopioid.

Therapeutic Indications

The combination of a cognitive-behavioral program plus the use of opioid receptor antagonists is more successful than either the cognitive-behavioral program or the use of opioid receptor antagonists alone. Naloxone is used as a screening test to ensure that the patient is opioid-free before the induction of therapy with naltrexone (see “Naloxone Challenge Test” in Table 21-53).



Table 21-53
Naloxone (Narcan) Challenge Test

The naloxone challenge test should not be performed in a patient showing clinical signs or symptoms of opioid withdrawal or in a patient whose urine contains opioids. The naloxone challenge test may be administered by either the intravenous (IV) or the

subcutaneous route.

IV challenge: After appropriate screening of the patient, 0.8 mg of naloxone should be drawn into a sterile syringe. If the IV route of administration is selected, 0.2 mg of naloxone should be injected, and while the needle is still in the patient's vein, the patient should be observed for 30 s for evidence of withdrawal signs or symptoms. If there is no evidence of withdrawal, the remaining 0.6 mg of naloxone should be injected and the patient observed for an additional 20 min for signs and symptoms of withdrawal.

Subcutaneous challenge: If the subcutaneous route is selected, 0.8 mg should be administered subcutaneously and the patient observed for signs and symptoms of withdrawal for 20 min.

Conditions and technique for observation of patient: During the appropriate period of observation, the patient's vital signs should be monitored, and the patient should be monitored for signs of withdrawal. It is also important to question the patient carefully. The signs and symptoms of opioid withdrawal include, but are not limited to, the following:

- **Withdrawal signs:** Stuffiness or running nose, tearing, yawning, sweating, tremor, vomiting, or piloerection
- **Withdrawal symptoms:** Feeling of temperature change, joint or bone and muscle pain, abdominal cramps, and formication (feeling of bugs crawling under skin)

Interpretation of the challenge: Warning—the elicitation of the enumerated signs or symptoms indicates a potential risk for the subject, and naltrexone should not be administered. If no signs or symptoms of withdrawal are observed, elicited, or reported, naltrexone may be administered. If there is any doubt in the observer's mind that the patient is not in an opioid-free state or is in continuing withdrawal, naltrexone should be withheld for 24 hr and the challenge repeated.

Opioid Dependence. Patients in detoxification programs are usually weaned from potent opioid agonists such as heroin over days to weeks, during which emergent adrenergic withdrawal effects are treated as needed

with clonidine (Catapres). A serial protocol is sometimes used in which potent agonists are gradually replaced by weaker agonists followed by mixed agonist–antagonists and then finally by pure antagonists. For example, an abuser of the potent agonist heroin would switch first to the weaker agonist methadone (Dolophine), then to the partial agonist buprenorphine (Buprenex) or levomethadyl acetate (ORLAAM)—commonly called LAAM—and finally, after a 7- to 10-day washout period, to a pure antagonist, such as naltrexone or nalmefene. However, even with gradual detoxification, some persons continue to experience mild adverse effects or opioid withdrawal symptoms for the first several weeks of treatment with naltrexone.

As the opioid receptor agonist potency diminishes, so do the adverse consequences of discontinuing the drug. Thus, because there are no pharmacologic barriers to discontinuation of pure opioid receptor antagonists, the social environment and frequent cognitive-behavioral intervention become significant factors supporting continued opioid abstinence. Because of poorly tolerated adverse symptoms, most persons not simultaneously enrolled in a cognitive-behavioral program stop taking opioid receptor antagonists within 3 months. Compliance with the administration of an opioid receptor antagonist regimen can also be increased with participation in a well-conceived voucher program.

Issues of medication compliance should be a central focus of treatment. If a person with a history of opioid addiction stops taking a pure opioid receptor antagonist, the person’s risk of relapse into opioid abuse is exceedingly high, because the reintroduction of a potent opioid agonist would yield a very rewarding subjective “high.” In contrast, compliant persons do not develop tolerance to the therapeutic benefits of naltrexone even if it is administered continuously for 1 year or longer. Individuals may undergo several relapses and remissions before achieving long-term abstinence.

Persons taking opioid receptor antagonists should also be warned that sufficiently high dosages of opioid agonists can overcome the receptor antagonism of naltrexone or nalmefene, which may lead to hazardous and unpredictable levels of receptor activation (see “Precautions and Adverse Reactions”).

Rapid Detoxification. The recommended period of opioid abstinence before starting opioid receptor antagonists is 7 to 10 days. However, to avoid

this, rapid detoxification protocols have been developed. Continuous administration of adjunct clonidine—to reduce the adrenergic withdrawal symptoms—and adjunct benzodiazepines, such as oxazepam (Serax)—to reduce muscle spasms and insomnia—can permit the use of oral opioid receptor antagonists on the first day of opioid cessation. Detoxification can thus be completed within 48 to 72 hours, at which point opioid receptor antagonist maintenance is initiated. Moderately severe withdrawal symptoms may be experienced on the first day, but they taper off rapidly after that.

Because of the potential hypotensive effects of clonidine, the BP of persons undergoing rapid detoxification must be closely monitored for the first 8 hours. Outpatient rapid detoxification settings must therefore be adequately prepared to administer emergency care.

The main advantage of rapid detoxification is that the transition from opioid abuse to maintenance treatment occurs over just 2 or 3 days. The completion of detoxification in as little time as possible minimizes the risk that the person will relapse into opioid abuse during the detoxification protocol.

Alcohol Use Disorder. Opioid receptor antagonists are also used as adjuncts to cognitive-behavioral programs for the treatment of alcohol use disorder. Opioid receptor antagonists reduce alcohol craving and alcohol consumption, and they ameliorate the severity of relapses. Concomitant opioid receptor antagonist treatment added to an effective cognitive-behavioral program may halve the risk of relapse into heavy alcohol consumption.

The newer agent, nalmefene, has several potential pharmacologic and clinical advantages over its predecessor, naltrexone, for the treatment of alcohol use disorder. Whereas naltrexone may cause reversible transaminase elevations in persons who take dosages of 300 mg a day (which is six times the recommended dosage for treatment of alcohol and opioid dependence [50 mg a day]), nalmefene has not been associated with any hepatotoxicity. Clinically effective dosages of naltrexone are discontinued by 10 to 15 percent of persons because of adverse effects, most commonly nausea. In contrast, discontinuation of nalmefene because of an adverse event is rare at the clinically effective dosage of 20 mg a day and in the range of 10 percent at excessive dosages—that is, 80 mg a day. Because of its pharmacokinetic

profile, a given dosage of nalmefene may also produce a more sustained opioid antagonist effect than does naltrexone.

The efficacy of opioid receptor antagonists in reducing alcohol craving may be augmented with an SSRI, although data from large trials are needed to assess this potential synergistic effect more fully.

Precautions and Adverse Reactions

Because opioid receptor antagonists are used for sustaining a drug-free state after opioid detoxification, great care must be taken to ensure that an adequate washout period elapses. This detox requires at least 5 days for a short-acting opioid such as heroin and at least 10 days for longer-acting opioids such as methadone—after the last dose of opioids and before the first dose of an opioid receptor antagonist is taken. Self-report and urine toxicology screens should determine the opioid-free state. If any question persists of whether opioids are in the body despite a negative urine screen result, then a *naloxone challenge test* should be performed. Naloxone challenge is used because its opioid antagonism lasts less than 1 hour, but those of naltrexone and nalmefene may persist for more than 24 hours. Thus, any withdrawal effects elicited by naloxone will be relatively short-lived (see "Dosage and Clinical Guidelines"). Symptoms of acute opioid withdrawal include drug craving, a feeling of temperature change, musculoskeletal pain, and GI distress. Signs of opioid withdrawal include confusion, drowsiness, vomiting, and diarrhea. Naltrexone and nalmefene should not be taken if naloxone infusion causes any signs of opioid withdrawal except as part of a supervised rapid detoxification protocol.

A set of adverse effects resembling a vestigial withdrawal syndrome tends to affect up to 10 percent of persons who take opioid receptor antagonists. Up to 15 percent of persons taking naltrexone may experience abdominal pain, cramps, nausea, and vomiting, which may be limited by transiently halving the dosage or altering the time of administration. Adverse CNS effects of naltrexone, experienced by up to 10 percent of persons, include headache, low energy, insomnia, anxiety, and nervousness. Joint and muscle pains may occur in up to 10 percent of persons taking naltrexone, as may rash.

Naltrexone may cause dosage-related hepatic toxicity at dosages well above 50 mg a day; 20 percent of persons taking 300 mg a day of naltrexone

may experience serum aminotransferase concentrations 3 to 19 times the upper limit of normal. The hepatocellular injury of naltrexone appears to be a dose-related toxic effect rather than an idiosyncratic reaction. At the lowest dosages of naltrexone required for significant opioid antagonism, hepatocellular injury is not typically observed. However, naltrexone dosages as low as 50 mg a day may be hepatotoxic in persons with underlying liver disease, such as persons with cirrhosis of the liver caused by chronic alcohol abuse. Serum aminotransferase concentrations should be monitored monthly for the first 6 months of naltrexone therapy and after that based on clinical suspicion. Hepatic enzyme concentrations usually return to normal after discontinuation of naltrexone therapy.

If analgesia is required while a dose of an opioid receptor antagonist is pharmacologically active, opioid agonists should be avoided in favor of benzodiazepines or other nonopioid analgesics. Persons taking opioid receptor antagonists should be instructed that low dosages of opioids will not affect them. However, larger dosages could overcome the receptor blockade and suddenly produce symptoms of profound opioid overdosage, with sedation possibly progressing to coma or death. Use of opioid receptor antagonists is contraindicated in persons who are taking opioid agonists, small amounts of which may be present in over-the-counter antiemetic and antitussive preparations; in persons with acute hepatitis or hepatic failure; and in persons who are hypersensitive to the drugs.

Because naltrexone is transported across the placenta, opioid receptor antagonists should only be taken by pregnant women if a compelling need outweighs the potential risks to the fetus. It is not known whether opioid receptor antagonists are distributed into breast milk.

Opioid receptor antagonists are relatively safe drugs, and the ingestion of high doses of opioid receptor antagonists should be treated with supportive measures combined with efforts to decrease GI absorption.

Because buprenorphine has a high affinity and slow displacement from the opioid receptors, nalmefene may not wholly reverse buprenorphine-induced respiratory depression.

Drug Interactions

Many drug interactions involving opioid receptor antagonists have been

discussed earlier, including those with opioid agonists associated with drug abuse as well as those involving antiemetics and antitussives. Because of its extensive hepatic metabolism, naltrexone may affect or be affected by other drugs that influence hepatic enzyme levels. However, the clinical importance of these potential interactions is not known.

One potentially hepatotoxic drug that has been used in some cases with opioid receptor antagonists is disulfiram (Antabuse). Although no adverse effects were observed, frequent laboratory monitoring is indicated when such combination therapy is contemplated. Opioid receptor antagonists have been reported to potentiate the sedation associated with the use of thioridazine (Mellaril), an interaction that probably applies equally to all low-potency DRAs.

Intravenous nalmefene has been administered after benzodiazepines, inhalational anesthetics, muscle relaxants, and muscle relaxant antagonists were administered in conjunction with general anesthetics without any adverse reactions. Care should be taken when using flumazenil (Romazicon) and nalmefene together because both of these agents have been shown to induce seizures in preclinical studies.

Laboratory Interferences

The potential for false-positive urine for opiates using less specific urine screens such as enzyme-multiplied immunoassay technique (EMIT) may exist, given that naltrexone and nalmefene are derivatives of oxymorphone. Naltrexone does not interefere with thin-layer, gas-liquid, and high-pressure liquid chromatographic methods used for the detection of opiates in the urine.

Dosage and Clinical Guidelines

We should take several steps to ensure that the person is opioid-free. These steps are meant to avoid the possibility of precipitating an acute opioid withdrawal syndrome. Within a supervised detoxification setting, at least 5 days should elapse after the last dose of short-acting opioids, such as heroin, hydromorphone (Dilaudid), meperidine (Demerol), or morphine, and at least 10 days should elapse after the last dose of longer-acting opioids, such as methadone, before opioid antagonists are initiated. Some rapid detoxification protocols use shorter abstinence periods, however these require special

expertise to do safely. Urine toxicologic screens should demonstrate no opioid metabolites, and detoxification should be complete. However, an individual may have a negative urine opioid screen result, yet still be physically dependent on opioids and thus susceptible to antagonist-induced withdrawal effects. Therefore, after the urine screen result is negative, a naloxone challenge test is recommended unless an adequate period of opioid abstinence can be reliably confirmed by observers (Table 21-53).

The initial dosage of naltrexone for the treatment of opioid or alcohol dependence is 50 mg a day, which should be achieved through a gradual introduction, even when the naloxone challenge test result is negative. Various authorities begin with 5, 10, 12.5, or 25 mg and titrate up to the 50-mg dosage over a period ranging from 1 hour to 2 weeks while continually monitoring for evidence of opioid withdrawal. When a daily dose of 50 mg is well tolerated, it may be averaged over a week by giving 100 mg on alternate days or 150 mg every third day. Such schedules may increase compliance. The corresponding therapeutic dosage of nalmefene is 20 mg a day, divided into two equal doses. Gradual titration of nalmefene to this daily dose is probably a wise strategy, although clinical data on dosage strategies for nalmefene are not yet available.

It is recommended that family members directly observe the ingestion of each dose. This strategy helps to maximize compliance. Random urine tests for opioid receptor antagonists and their metabolites, as well as for ethanol or opioid metabolites, should also be taken. Opioid receptor antagonists should be continued until the person is no longer considered psychologically at risk for relapse into opioid or alcohol abuse. This treatment generally requires at least 6 months but may take longer, mainly if there are external stresses.

Nalmefene is available as a sterile solution for intravenous, intramuscular, and subcutaneous administration in two concentrations, containing 100 µg or 1.0 mg of nalmefene free base per milliliter. The 100 µg/mL concentration contains 110.8 µg of nalmefene hydrochloride, and the 1.0 mg/mL concentration contains 1.108 mg of nalmefene hydrochloride per milliliter. Both concentrations contain 9.0 mg of sodium chloride per milliliter, and the pH is adjusted to 3.9 with hydrochloric acid. Pharmacodynamic studies have shown that nalmefene has a longer duration of action than naloxone at fully reversing opiate activity.

Rapid Detoxification. Rapid detoxification has been standardized using naltrexone, although nalmefene would be expected to be equally effective with fewer adverse effects. In rapid detoxification protocols, the addicted person stops opioid use abruptly and begins the first opioid-free day by taking clonidine 0.2 mg orally every 2 hours for nine doses, to a maximum dose of 1.8 mg, during which time the BP is monitored every 30 to 60 minutes for the first 8 hours. Naltrexone 12.5 mg is administered 1 to 3 hours after the first dose of clonidine. To reduce muscle cramps and later insomnia, a short-acting benzodiazepine, such as oxazepam 30 to 60 mg, is administered simultaneously with the first dose of clonidine, and half of the initial dose is readministered every 4 to 6 hours as needed. The maximum daily dosage of oxazepam should not exceed 180 mg. The person undergoing rapid detoxification should be accompanied home by a reliable escort. On the second day, similar doses of clonidine and the benzodiazepine are administered but with a single dose of naltrexone 25 mg taken in the morning. Relatively asymptomatic persons may return home after 3 to 4 hours. Administration of the daily maintenance dose of 50 mg of naltrexone is begun on the third day, and the dosages of clonidine and the benzodiazepine are gradually tapered off over 5 to 10 days.

DISULFIRAM AND ACAMPROSATE

Disulfiram (Antabuse) and acamprosate (Campral) are drugs used to treat alcohol use disorder. Disulfiram has suffered from a reputation as a dangerous medication only suitable for highly motivated and strictly supervised drinkers because of the severe physical reactions the drug causes after drinking. Experience has shown, however, that at recommended doses, it is an acceptable and safe medication for dependent drinkers seeking to sustain abstinence. The properties that constitute disulfiram's main therapeutic effect (i.e., the ability to produce unpleasant symptoms after alcohol intake, also known as disulfiram–alcohol reaction) have created that perception of dangerousness.

In the most severe cases, when disulfiram is combined with alcohol, severe clinical conditions can occur. These include respiratory depression, cardiovascular collapse, acute heart failure, convulsions, loss of consciousness, and death in rare cases. These potential complications, as well

as the development of alternative antialcohol medications, have been the limiting factors for the broader use of disulfiram. Unlike disulfiram, acamprosate, the other drug discussed in this section, does not produce aversive side effects. Acamprosate is now prescribed more commonly than disulfiram in outpatient settings, but disulfiram is prescribed more often in inpatient settings because it helps facilitate initial abstinence.

Other drugs that are useful in reducing alcohol consumption include naltrexone (ReVia, Trexan), nalmefene (Revex), topiramate (Topamax), and gabapentin (Neurontin). These agents are discussed in their respective sections.

Disulfiram

Pharmacologic Actions. Disulfiram is almost completely absorbed from the GI tract after oral administration. Its half-life is estimated to be 60 to 120 hours. Therefore, it may take 1 or 2 weeks to eliminate disulfiram from the body after taking the last dose.

The metabolism of ethanol proceeds through oxidation via alcohol dehydrogenase to the formation of acetaldehyde, which is further metabolized to acetyl-coenzyme A (acetyl-CoA) by aldehyde dehydrogenase. Disulfiram is an aldehyde dehydrogenase inhibitor that interferes with the metabolism of alcohol by producing a marked increase in blood acetaldehyde concentration. The accumulation of acetaldehyde (to a level up to 10 times higher than occurs in the normal metabolism of alcohol) produces a wide array of unpleasant reactions, called the *disulfiram–alcohol reaction,* characterized by nausea, throbbing headache, vomiting, hypertension, flushing, sweating, thirst, dyspnea, tachycardia, chest pain, vertigo, and blurred vision. The reaction occurs almost immediately after the ingestion of one alcoholic drink and may last from 30 minutes to 2 hours.

BLOOD CONCENTRATIONS IN RELATION TO ACTION. Plasma concentrations of disulfiram vary among individuals because of several factors, most notably age and hepatic function. In general, the severity of disulfiram–alcohol reaction is proportional to the amount of the ingested disulfiram and alcohol. Nevertheless, disulfiram plasma levels are rarely obtained in clinical practice. The positive correlation between plasma concentrations of alcohol and the

intensity of the reaction is described as follows: in sensitive individuals, as little as a 5 to 10 mg/100 mL increase of the plasma alcohol level may produce mild symptoms; fully developed symptoms occur at alcohol levels of 50 mg/100 mL, and levels as high as 125 to 150 mg/100 mL result in loss of consciousness and coma.

Therapeutic Indications. The primary indication for disulfiram use is as an aversive conditioning treatment for alcohol use disorder. Either the fear of having a disulfiram–alcohol reaction or the memory of having had one is meant to condition the person not to use alcohol. Usually, describing the severity and the unpleasantness of the disulfiram–alcohol reaction graphically enough discourages the person from imbibing alcohol. Disulfiram treatment should be combined with such treatments as psychotherapy, group therapy, and support groups such as Alcoholics Anonymous (AA). Treatment with disulfiram requires careful monitoring because a person can decide not to take the medication.

Precautions and Adverse Reactions

WITH ALCOHOL CONSUMPTION. The intensity of the disulfiram–alcohol reaction varies with each person. In extreme cases, it is marked by respiratory depression, cardiovascular collapse, myocardial infarction, convulsions, and death. Therefore, disulfiram is contraindicated for persons with significant pulmonary or cardiovascular disease. Also, disulfiram should be used with caution, if at all, by persons with nephritis, brain damage, hypothyroidism, diabetes, hepatic disease, seizures, polydrug dependence, or an abnormal electroencephalogram. Most fatal reactions occur in persons who take more than 500 mg a day of disulfiram and who consume more than 3 oz of alcohol. The treatment of a severe disulfiram–alcohol reaction is primarily supportive to prevent shock. The use of oxygen, intravenous vitamin C, ephedrine, and antihistamines have been reported to aid in recovery.

WITHOUT ALCOHOL CONSUMPTION. The adverse effects of disulfiram in the absence of alcohol consumption include fatigue, dermatitis, impotence, optic neuritis, a variety of mental changes, and hepatic damage. A metabolite of disulfiram inhibits dopamine-

β-hydroxylase, the enzyme that metabolizes dopamine into norepinephrine and epinephrine, and thus may exacerbate psychosis in persons with psychotic disorders. Catatonic reactions may also occur.

Drug Interactions. Disulfiram increases the blood concentration of diazepam (Valium), paraldehyde, phenytoin (Dilantin), caffeine, tetrahydrocannabinol (the active ingredient in marijuana), barbiturates, anticoagulants, isoniazid (Nydrazid), and tricyclic drugs. Disulfiram should not be administered concomitantly with paraldehyde because paraldehyde is metabolized to acetaldehyde in the liver.

Laboratory Interferences. In rare instances, disulfiram has been reported to interfere with the incorporation of iodine-131 into protein-bound iodine. Disulfiram may reduce urinary concentrations of homovanillic acid, the major metabolite of dopamine, because it inhibits dopamine hydroxylase.

Dosage and Clinical Guidelines. Disulfiram is supplied in 250- and 500-mg tablets. The usual initial dosage is 500 mg a day, taken by mouth for the first 1 or 2 weeks, followed by a maintenance dosage of 250 mg a day. The dosage should not exceed 500 mg a day. The maintenance dosage range is 125 to 500 mg a day.

Persons taking disulfiram must be instructed that the ingestion of even the smallest amount of alcohol will bring on a disulfiram–alcohol reaction, with all of its unpleasant effects. Also, persons should be warned against ingesting any alcohol-containing preparations, such as cough drops, tonics of any kind, and alcohol-containing foods and sauces. Some reactions have occurred in patients who used alcohol-based lotions, toilet water, colognes, or perfumes and inhaled the fumes; therefore, precautions must be explicit and should include any topically applied preparations containing alcohol, such as perfume.

Disulfiram should not be administered until the person has abstained from alcohol for at least 12 hours. Persons should be warned that the disulfiram–alcohol reaction may occur as long as 1 or 2 weeks after the last dose of disulfiram. Persons taking disulfiram should carry identification cards describing the disulfiram–alcohol reaction and listing the name and telephone number of the physician to be called.

Acamprosate

Pharmacologic Actions. Acamprosate's mechanism of action is not fully understood, but it is thought to antagonize neuronal overactivity related to the actions of the excitatory neurotransmitter glutamate. In part, this may result from antagonism of NMDA receptors.

Indications. Acamprosate is used for treating alcohol-dependent individuals seeking to continue to remain alcohol-free after they have stopped drinking. Its efficacy in promoting abstinence has not been demonstrated in persons who have not undergone detoxification and who have not achieved alcohol abstinence before beginning treatment.

Precautions and Adverse Effects. Side effects are mostly seen early in treatment and are usually mild and transient. The most common side effects are headache, diarrhea, flatulence, abdominal pain, paresthesias, and various skin reactions. No adverse events occur after abrupt withdrawal of acamprosate, even after long-term use. There is no evidence of addiction to the drug. Patients with severe renal impairment (creatinine clearance of less than 30 mL/min) should not be given acamprosate.

Drug Interactions. Alcohol and acamprosate do not affect each other's pharmacokinetics. The same is true when adding disulfiram or diazepam to acamprosate. Coadministration of naltrexone with acamprosate produces an increase in concentrations of acamprosate. No adjustment of dosage is recommended in such patients. The pharmacokinetics of naltrexone and its major metabolite 6-β-naltrexol were unaffected after coadministration with acamprosate. During clinical trials, patients taking acamprosate concomitantly with antidepressants more commonly reported both weight gain and weight loss compared with patients taking either medication alone.

Laboratory Interferences. Acamprosate has not been shown to interfere with commonly performed laboratory tests.

Dosage and Clinical Guidelines. It is important to remember that acamprosate should not be used to treat alcohol withdrawal symptoms. It should only be started after the individual has been successfully weaned off

alcohol. Patients should show a commitment to remaining abstinent, and treatment should be part of a comprehensive management program that includes counseling or support group attendance.

Each tablet contains acamprosate calcium 333 mg, which is equivalent to 300 mg of acamprosate. The dose of acamprosate is different for different patients. The recommended dosage is two 333-mg tablets (each dose should total 666 mg) taken three times daily. Although dosing may be done without regard to meals, dosing with meals was used during clinical trials and is suggested as an aid to compliance in patients who regularly eat three meals daily. A lower dose may be useful in some patients. A missed dose should be taken as soon as possible. However, if it is almost time for the next dose, the missed dose should be skipped, and then the regular dosing schedule should be resumed. Doses should not be doubled up. For patients with moderate renal impairment (creatinine clearance of 30 to 50 mL/min), a starting dosage of one 333-mg tablet taken three times daily is recommended. People with severe renal insufficiency should not take acamprosate.

CLONIDINE AND GUANFACINE

Clonidine and guanfacine are marketed as hypertension treatments. Guanfacine is more selective and less potent than clonidine, but clonidine is the most widely used α_2-agonist.

Pharmacologic Actions

Clonidine and guanfacine are presynaptic α_2-receptor agonists. They inhibit sympathetic outflow and cause vasodilation of blood vessels. Clonidine and guanfacine are well absorbed from the GI tract and reach peak plasma levels 1 to 3 hours after oral administration. The half-life of clonidine is 6 to 20 hours, and that of guanfacine is 10 to 30 hours.

The agonist effects of clonidine and guanfacine on presynaptic α_2-adrenergic receptors in the sympathetic nuclei of the brain result in a decrease in the amount of norepinephrine released from the presynaptic nerve terminals. This decrease serves to reset the body's sympathetic tone at a lower level and decrease arousal. Table 21-54 provides a summary of α_2-adrenergic receptor agonists used in psychiatry.

Therapeutic Indications

There is considerably more experience in clinical psychiatry with clonidine than with guanfacine. There is recent interest in the use of guanfacine for the same indications that respond to clonidine due to guanfacine's longer half-life and relative lack of sedative effects.

Withdrawal from Opioids, Alcohol, Benzodiazepines, or Nicotine. Clonidine and guanfacine are effective in reducing the autonomic symptoms of rapid opioid withdrawal (e.g., hypertension, tachycardia, dilated pupils, sweating, lacrimation, and rhinorrhea) but not the associated subjective sensations. Clonidine administration (0.1 to 0.2 mg two to four times a day) is initiated before detoxification and is then tapered off over 1 to 2 weeks (Table 21-55).

Clonidine and guanfacine can reduce symptoms of alcohol and benzodiazepine withdrawal, including anxiety, diarrhea, and tachycardia. They can reduce craving, anxiety, and irritability symptoms of nicotine withdrawal. The transdermal patch formulation of clonidine is associated with better long-term compliance for purposes of detoxification than is the tablet formulation.

Tourette Disorder. Clonidine and guanfacine are effective drugs for the treatment of Tourette disorder. Most clinicians begin treatment for Tourette disorder with the standard DRAs haloperidol (Haldol) and pimozide (Orap) and the SDAs risperidone (Risperdal) and olanzapine (Zyprexa). However, if concerned about the adverse effects of these drugs, the clinician may begin treatment with clonidine or guanfacine. The starting dose of clonidine for children is 0.05 mg a day; it can be increased to 0.3 mg a day in divided doses. Clonidine may take long periods (e.g., 4 to 6 months) to positively effect the symptoms of the disorder. The response rate has been reported to be up to 70 percent.



Table 21-54
α_2-Adrenergic Receptor Agonists Used in Psychiatry[a]

Drug	Preparations	Usual Child Starting Dosage	Usual Child Dosage Range	Usual Adult Starting Dosage	Usual Adult Dosage
Clonidine tablets (Catapres)	0.1, 0.2, 0.3 mg	0.05 mg/day	Up to 0.3 mg/day tablets in divided doses	0.1–0.2 mg, two to four times a day (0.2–0.8 mg/day)	0.3–1.2 mg/day, two to three times a day (1.2 mg/day maximal dosage)
Clonidine transdermal system (Catapres-TTS)	0.1, 0.2, 0.3 mg/day	0.05 mg/day	Up to 0.3 mg/day patch every 5 days (0.5 mg/day every 5 days maximal dosage)	0.1 mg/day every 7 days	0.1 mg/day patch per week (0.6 mg/day every 7 days)
Guanfacine (Tenex)	1- and 2-mg tablets	1 mg/day at bedtime	1–2 mg/day at bedtime (3 mg/day maximal dosage)	1 mg/day at bedtime	1–2 mg at bedtime (3 mg/day maximal dosage)

*Dosages for medical indications, such as hypertension, vary.

Other Tic Disorders. Clonidine and guanfacine reduce the frequency and severity of tics in persons with tic disorder with or without comorbid ADHD.

Hyperactivity and Aggression in Children. Clonidine and guanfacine can be useful alternatives for the treatment of ADHD. They are used in place of sympathomimetics and antidepressants, which may produce a paradoxical worsening of hyperactivity in some children with intellectual disability, aggression, or features on the spectrum of autism. Clonidine and guanfacine can improve mood, reduce activity level, and improve social adaptation. Some impaired children may respond favorably to clonidine, but others may become sedated. The starting dose is 0.05 mg a day; it can be raised to 0.3 mg a day in divided doses. The efficacy of clonidine and guanfacine for control of hyperactivity and aggression often diminishes over several months of use.



Table 21-55
Oral Clonidine Protocols for Opioid Detoxification

Clonidine 0.1–0.2 mg PO four times a day; hold for systolic BP <90 mm Hg or bradycardia; stabilize for 2–3 days, then taper over 5–10 days

OR

Clonidine 0.1–0.2 mg PO q4–6h as needed for withdrawal signs or symptoms; stabilize for 2–3 days, then taper over 5–10 days

OR

Test dose with clonidine 0.1–0.2 mg PO or SL (for patients weighing

over 200 lb); check BP after 1 hr. If diastolic BP >70 mm Hg and no symptoms of hypotension, begin treatment as follows:	
Weight (lb)	Number of Clonidine Patches
<110	1 patch
110–160	2 patches
160–200	2 patches
>200	2 patches
OR	
Test dose of oral clonidine 0.1 mg; check BP after 1 hr (if systolic BP <90 mm Hg, do not give patch)	
Place two TTS-2 clonidine patches (or three patches if patient weighs >150 lb) on hairless area of upper body; then	
For first 23 hr after patch application, give oral clonidine 0.2 mg q6h; then	
For next 24 hr, give oral clonidine 0.1 mg q6h	
Change patches weekly	
After 2 wk of two patches, switch to one patch (or two patches if patient weighs >150 lb)	
After 1 wk of one patch, discontinue patches	

BP, blood pressure; PO, oral; q, every; SL, sublingual; TTS, through the skin.

From American Society of Addiction Medicine. Detoxification: principle and protocols. In: *The Principles Update Series: Topics in Addiction Medicine*, Section 11. American Society of Addiction; 1997, with permission.

Clonidine and guanfacine can be combined with methylphenidate (Ritalin) or dextroamphetamine (Dexedrine) to treat hyperactivity and inattentiveness, respectively. A small number of cases have been reported of the sudden death of children taking clonidine together with methylphenidate; however, it has not been conclusively demonstrated that these medications contributed to these deaths. The clinician should explain to the family that the efficacy and

safety of this combination have not been investigated in controlled trials. Periodic cardiovascular assessments, including vital signs and electrocardiograms, are warranted if this combination is used.

Posttraumatic Stress Disorder. Acute exacerbations of PTSD may be associated with hyperadrenergic symptoms such as hyperarousal, exaggerated startle response, insomnia, vivid nightmares, tachycardia, agitation, hypertension, and perspiration. Preliminary reports suggested that these symptoms may respond to the use of clonidine or, especially for an overnight benefit, to the use of guanfacine. More recent studies have failed to demonstrate that guanfacine produces an improvement in PTSD symptoms.

Other Disorders. Other potential indications for clonidine include other anxiety disorders (panic disorder, phobias, obsessive-compulsive disorder, and GAD) and mania, in which it may be synergistic with lithium (Eskalith) or carbamazepine (Tegretol). There are anecdotal accounts suggesting the efficacy of clonidine in schizophrenia and tardive dyskinesia. A clonidine patch can reduce the hypersalivation and dysphagia caused by clozapine. Low-dose use has been reported effective in hallucinogen-persisting perceptive disorders.

Precautions and Adverse Reactions

The most common adverse effects associated with clonidine are dry mouth and eyes, fatigue, sedation, dizziness, nausea, hypotension, and constipation, which result in discontinuation of therapy by about 10 percent of all persons taking the drug. Some persons also experience sexual dysfunction. Tolerance may develop to these adverse effects. A similar but milder adverse profile is seen with guanfacine, especially in doses of 3 mg or more per day. Adults should not take clonidine and guanfacine if their BP is below 90/60 mm Hg or if they have cardiac arrhythmias, especially bradycardia. Development of bradycardia warrants gradual, tapered discontinuation of the drug. Clonidine, in particular, is associated with sedation, and tolerance does not usually develop to this adverse effect. Uncommon CNS adverse effects of clonidine include insomnia, anxiety, and depression; rare CNS adverse effects include vivid dreams, nightmares, and hallucinations. Fluid retention associated with clonidine treatment can be treated with diuretics.

The transdermal patch formulation of clonidine may cause local skin irritation, which can be minimized by rotating the sites of application.

Overdose. Persons who take an overdose of clonidine may present with coma and constricted pupils, symptoms similar to those of an opioid overdose. Other symptoms of overdose are decreased BP, pulse, and respiratory rate. Guanfacine overdose produces a milder version of these symptoms. Clonidine and guanfacine should be avoided during pregnancy and by nursing mothers. Elderly persons are more sensitive to the drug than are younger adults. Children are susceptible to the same adverse effects as are adults.

Withdrawal. Abrupt discontinuation of clonidine can cause anxiety, restlessness, perspiration, tremor, abdominal pain, palpitations, headache, and a dramatic increase in BP. These symptoms may appear about 20 hours after the last dose of clonidine, and these may also be seen if one or two doses are skipped. A similar set of symptoms occasionally occurs 2 to 4 days after discontinuation of guanfacine, but the usual course is a gradual return to baseline BP over 2 to 4 days. Because of the possibility of discontinuation symptoms, doses of clonidine and guanfacine should be tapered slowly.

Drug Interactions

Clonidine and guanfacine cause sedation, especially early in therapy, and when administered with other centrally active depressants, such as barbiturates, alcohol, and benzodiazepines, the potential for additive sedative effects should be considered. Dose reduction may be required in patients receiving agents that interfere with AV node and sinus node conduction such as β-blockers, CCBs, and digitalis. This combination increases the risk of AV block and bradycardia. Clonidine should not be given with TCAs, which can inhibit the hypotensive effects of clonidine.

Laboratory Interferences

No known laboratory interferences are associated with the use of clonidine or guanfacine.

Dosage and Clinical Guidelines

Clonidine is available in 0.1-, 0.2-, and 0.3-mg tablets. The usual starting dosage is 0.1 mg orally twice a day; the dosage can be raised by 0.1 mg a day to an appropriate level (up to 1.2 mg/day). Clonidine must always be tapered when it is discontinued to avoid rebound hypertension, which may occur about 20 hours after the last clonidine dose. A weekly transdermal formulation of clonidine is available at doses of 0.1, 0.2, and 0.3 mg/day. The usual starting dosage is the 0.1-mg-a-day patch, which is changed each week for adults and every 5 days for children; the dose can be increased, as needed, every 1 to 2 weeks. The transition from the oral to the transdermal formulations should be accomplished gradually by overlapping them for 3 to 4 days.

Guanfacine is available in 1- and 2-mg tablets. The usual starting dosage is 1 mg before sleep, and this can be increased to 2 mg before sleep after 3 to 4 weeks, if necessary. Regardless of the indication for which clonidine or guanfacine is being used, the drug should be withheld if a person becomes hypotensive (BP below 90/60 mm Hg).

An extended-release preparation of guanfacine (Intuniv) is also available. Extended-release guanfacine should be dosed once daily. Tablets should not be crushed, chewed, or broken before swallowing because this will increase the rate of guanfacine release. It should not be administered with high-fat meals due to increased exposure. The extended-release formulation should not be substituted for immediate-release guanfacine tablets on a milligram-per-milligram basis because of differing pharmacokinetic profiles. If switching from immediate-release guanfacine, discontinue that treatment, and titrate with extended-release guanfacine according to the following recommended schedule:

1. ;Begin at a dose of 1 mg/day, and adjust in increments of no more than 1 mg/wk for both monotherapy and adjunctive therapy to a psychostimulant.
2. For both monotherapy and adjunctive therapy (with a stimulant), maintain the dose within the range of 1 to 4 mg once daily, depending on clinical response and tolerability. In clinical trials, patients were randomized or dose optimized to doses of 1, 2, 3, or 4 mg and received extended-release guanfacine once daily in the morning in monotherapy trials and once daily in the morning or the evening in the adjunctive

therapy trial.

3. In monotherapy trials, clinically relevant improvements were observed beginning at doses in the range 0.05 to 0.08 mg/kg once daily. Efficacy increased with increasing weight-adjusted dose (mg/kg). If well tolerated, doses up to 0.12 mg/kg once daily may provide additional benefit. Doses above 4 mg/day have not been systematically studied in controlled clinical trials.
4. In the adjunctive trial, the majority of subjects reached optimal doses in the 0.05 to 0.12 mg/kg/day range.

In clinical trials, there were dose-related and exposure-related risks for several clinically significant adverse reactions (e.g., hypotension, bradycardia, sedative events). Thus, consideration should be given to dosing an extended-release preparation of guanfacine on a milligram-per-kilogram basis in order to balance the exposure-related potential benefits and risks of treatment.

▲ 21.8 Drugs Used for Cognitive Enhancement

CHOLINESTERASE INHIBITORS

Donepezil (Aricept), rivastigmine (Exelon), and galantamine (Reminyl) are cholinesterase inhibitors used to treat mild to moderate cognitive impairment in dementia of the Alzheimer type. They reduce the inactivation of the neurotransmitter acetylcholine and, thus, potentiate cholinergic neurotransmission, which in turn produces a modest improvement in memory and goal-directed thought. Memantine (Namenda) is not a cholinesterase inhibitor, producing its effects through blockade of NMDA receptors. Unlike the cholinesterase inhibitors, which are indicated for the mild to moderate stages of Alzheimer disease, memantine is indicated for the moderate to severe stages of the disease. Tacrine (Cognex), the first cholinesterase inhibitor to be introduced, is no longer used because of its multiple-daily dosing regimens, its potential for hepatotoxicity, and the consequent need for

frequent laboratory monitoring. Clinicians often combine a cholinesterase inhibitor with memantine, as this combination may provide a beneficial response compared with only cholinesterase inhibitor pharmacotherapy.

Pharmacologic Actions

Donepezil is entirely absorbed from the GI tract. Peak plasma concentrations are reached about 3 to 4 hours after oral dosing. The half-life of donepezil is 70 hours in elderly persons, and it is taken only once daily. Steady-state levels are achieved within about 2 weeks. The presence of stable alcoholic cirrhosis reduces the clearance of donepezil by 20 percent. Rivastigmine (Exelon) is rapidly and completely absorbed from the GI tract and reaches peak plasma concentrations in 1 hour, but this is delayed by up to 90 minutes if rivastigmine is taken with food. The half-life of rivastigmine is 1 hour, but because it remains bound to cholinesterases, a single dose is therapeutically active for 10 hours, and it is taken twice daily. Galantamine (Reminyl) is an alkaloid similar to codeine and is extracted from daffodils of the plant *Galanthus nivalis*. It is readily absorbed, with maximum concentrations reached after 30 minutes to 2 hours. Food decreases the maximum concentration by 25 percent. The elimination half-life of galantamine is approximately 6 hours.

Tacrine (Cognex) is absorbed rapidly from the GI tract. Peak plasma concentrations are reached about 90 minutes after oral dosing. The half-life of tacrine is about 2 to 4 hours, thereby necessitating four-times-daily dosing.

Cholinesterase inhibitors are reversible, nonacylating inhibitors of either acetylcholinesterase or butyrylcholinesterase, the enzymes that catabolize choline-based esthers in the CNS. The enzyme inhibition increases synaptic concentrations of acetylcholine, especially in the hippocampus and cerebral cortex. Unlike tacrine, which is nonselective for all forms of acetylcholinesterase, donepezil appears to be selectively active within the CNS and has little activity in the periphery. Donepezil’s favorable side effect profile appears to correlate with its lack of inhibition of cholinesterases in the GI tract. Rivastigmine appears to have somewhat more peripheral activity than donepezil and is thus more likely to cause GI adverse effects than is donepezil.

Therapeutic Indications

Cholinesterase inhibitors are effective for the treatment of mild to moderate cognitive impairment in dementia of the Alzheimer type. In long-term use, they slow the progression of memory loss and diminish apathy, depression, hallucinations, anxiety, euphoria, and purposeless motor behaviors. Functional autonomy is less well preserved. Some persons note an immediate improvement in memory, mood, psychotic symptoms, and interpersonal skills. Others note little initial benefit but can retain their cognitive and adaptive faculties at a relatively stable level for many months. A practical benefit of cholinesterase inhibitor use is a delay or reduction of the need for nursing home placement.

Donepezil and rivastigmine may be beneficial for patients with Parkinson disease and Lewy body disease and treatment of cognitive deficits caused by traumatic brain injury. Donepezil is under study for the treatment of mild cognitive impairment that is less severe than that caused by Alzheimer disease. People with vascular dementia may respond to acetylcholinesterase inhibitors. Occasionally, cholinesterase inhibitors elicit an idiosyncratic catastrophic reaction, with signs of grief and agitation, which is self-limited after the drug is discontinued. The use of cholinesterase inhibitors to improve cognition by nondemented individuals should be discouraged.

Precautions and Adverse Reactions

Donepezil. Donepezil is generally well tolerated at recommended dosages. Fewer than 3 percent of those taking donepezil experience nausea, diarrhea, and vomiting. These mild symptoms are more frequent with a 10-mg dose than with a 5-mg dose, and when present, they tend to resolve after 3 weeks of continued use. Donepezil may cause weight loss. Donepezil treatment has been infrequently associated with bradyarrhythmias, especially in those with underlying cardiac disease. A small number of persons experience syncope.

Rivastigmine. Rivastigmine is generally well tolerated but recommended dosages may need to be scaled back in the initial period of treatment to limit GI and CNS adverse effects. These mild symptoms are more common at dosages above 6 mg a day, and when present, they tend to resolve after the dosage is lowered. The most common adverse effects associated with

rivastigmine are nausea, vomiting, dizziness, headache, diarrhea, abdominal pain, anorexia, fatigue, and somnolence. Rivastigmine may cause weight loss, but it does not appear to cause hepatic, renal, hematologic, or electrolyte abnormalities.

Galantamine. The most common side effects of galantamine are dizziness, headache, nausea, vomiting, diarrhea, and anorexia. These side effects tend to be mild and transient.

Tacrine. Tacrine is the least used of the cholinesterase inhibitors but requires more discussion than the others because it is cumbersome to titrate and use, and it poses the risk of potentially significant elevations in hepatic transaminase levels. These increases occur in 25 to 30 percent of persons. Aside from elevated transaminase levels, the most common specific adverse effects associated with tacrine treatment are nausea, vomiting, myalgia, anorexia, and rash, but only nausea, vomiting, and anorexia have been found to have a clear relation to the dosage. Transaminase elevations characteristically develop during the first 6 to 12 weeks of treatment, and cholinergically mediated events are dosage related.

HEPATOTOXICITY. Tacrine is associated with increases in the plasma activities of alanine aminotransferase (ALT) and aspartate aminotransferase (AST). The ALT measurement is the more sensitive indicator of the hepatic effects of tacrine. About 95 percent of patients who develop elevated ALT serum levels do so in the first 18 weeks of treatment. The average length of time for elevated ALT concentrations to return to normal after stopping tacrine treatment is 4 weeks.

For routine monitoring of hepatic enzymes, AST and ALT activities should be measured weekly for the first 18 weeks, every month for the second 4 months, and every 3 months after that. Weekly assessments of AST and ALT should be performed for at least 6 weeks after an increase in dosage. Patients with mildly elevated ALT activity should be monitored weekly and not be rechallenged with tacrine until the ALT activity returns to the normal range. For any patient with elevated ALT activity and jaundice, tacrine treatment should be stopped, and the patient should not be given the drug again.

Table 21-56
Incidence of Major Adverse Side Effects with Cholinesterase Inhibitors (%)

Drug	Dose (mg/day)	Nausea	Vomiting	Diarrhea	Dizziness	Muscle Cramps	Insomnia
Donepezil	5	4	3	9	15	9	7
Donepezil	10	17	10	17	13	12	8
Rivastigmine	1–4	14	7	10	15	NR	NR
Rivastigmine	6–12	48	27	17	24	NR	NR
Galantamine	8	5.7	3.6	5	NR	NR	NR
Galantamine	16	13.3	6.1	12.2	NR	NR	NR
Galantamine	24	16.5	9.9	5.5	NR	NR	NR

NR, not reported from clinical trial data; incidence less than 5%.

Table 21-56 summarizes the incidence of significant adverse side effects associated with each of the cholinesterase inhibitors.

Drug Interactions. All cholinesterase inhibitors should be used cautiously with drugs that also possess cholinomimetic activity. Such drugs include succinylcholine (Anectine) and bethanechol (Urecholine). The coadministration of cholinesterase inhibitors and drugs that have cholinergic antagonist activity (e.g., tricyclic drugs) is probably counterproductive. Paroxetine is the most anticholinergic of the SSRIs, and we avoid it in patients taking cholinesterase inhibitors.

Donepezil undergoes extensive metabolism via both CYP2D6 and 3A4 isozymes. The metabolism of donepezil may be increased by phenytoin (Dilantin), carbamazepine (Tegretol), dexamethasone (Decadron), rifampin (Rifadin), and phenobarbital (Solfoton). Commonly used agents such as paroxetine, ketoconazole (Nizoral), and erythromycin can significantly increase donepezil concentrations. Donepezil is highly protein-bound, but it does not displace other protein-bound drugs, such as furosemide (Lasix), digoxin (Lanoxin), or warfarin (Coumadin). Rivastigmine circulates mostly unbound to serum proteins and has no significant drug interactions.

Similar to donepezil, galantamine is metabolized by both CYP2D6 and 3A4 isozymes and thus may interact with drugs that inhibit these pathways. Paroxetine and ketoconazole should be used with great caution.

Laboratory Interferences. No laboratory interferences have been associated with the use of cholinesterase inhibitors.

Dosage and Clinical Guidelines. Before initiation of cholinesterase inhibitor therapy, potentially treatable causes of dementia should be ruled out, and the diagnosis of dementia of the Alzheimer type established.

Donepezil is available in 5- and 10-mg tablets. Treatment should be initiated at 5 mg each night. If well tolerated and of some discernible benefit after 4 weeks, the dosage should be increased to a maintenance dosage of 10-mg each night. Donepezil absorption is unaffected by meals.

Rivastigmine is available in 1.5-, 3-, 4.5-, and 6-mg capsules. The recommended initial dosage is 1.5 mg twice daily for a minimum of 2 weeks, after which increases of 1.5 mg a day can be made at intervals of at least 2 weeks to a target dosage of 6 mg a day, taken in two equal dosages. If tolerated, the dosage may be further titrated upward to a maximum of 6 mg twice daily. The administration of rivastigmine with food can reduce the risk of adverse GI events.

Galantamine is available in 4-, 8-, and 16-mg tablets. The suggested dose range is 16 to 32 mg/day given twice a day. The higher dose is better tolerated than lower doses. The initial dosage is 8 mg/day, and after a minimum of 4 weeks, the dose can be raised. All subsequent dosage increases should occur at 4-week intervals and should be based on tolerability.

Tacrine is available in 10-, 20-, 30-, and 40-mg capsules. Before the initiation of tacrine treatment, a complete physical and laboratory examination should be conducted, with particular attention to liver function tests and baseline hematologic indexes. Treatment should be initiated at 10 mg four times a day and then raised by increments of 10 mg a dose every 6 weeks up to 160 mg a day; the person’s tolerance of each dosage is indicated by the absence of unacceptable side effects and lack of elevation of ALT activity. Tacrine should be given four times daily—ideally 1 hour before meals because the absorption of tacrine is reduced by about 25 percent when it is taken during the first 2 hours after meals. If tacrine is used, the specific guidelines for tacrine-induced ALT listed above should be followed.

MEMANTINE

Pharmacologic Actions

Memantine is well absorbed after oral administration. It reaches peak

concentrations in about 3 to 7 hours. Food does not affect the absorption of memantine. Memantine has linear pharmacokinetics over the therapeutic dosage range and has a terminal elimination half-life of about 60 to 80 hours. Plasma protein binding is 45 percent.

Memantine undergoes little metabolism, with the majority (57 to 82 percent) of an administered dose excreted unchanged in the urine; the remainder is converted primarily to three polar metabolites: the *N*-gludantan conjugate, 6-hydroxy memantine, and 1-nitroso-deaminated memantine. These metabolites possess minimal NMDA receptor antagonist activity. Memantine is a low- to moderate-affinity NMDA receptor antagonist. It is thought that the overexcitation of NMDA receptors by the neurotransmitter glutamate may play a role in Alzheimer disease because glutamate plays an integral role in the neural pathways associated with learning and memory. Excess glutamate overstimulates NMDA receptors to allow too much calcium into nerve cells, leading to the eventual cell death observed in Alzheimer disease. Memantine may protect cells against excess glutamate by partially blocking NMDA receptors associated with the abnormal transmission of glutamate while allowing for physiologic transmission associated with normal cell functioning.

Therapeutic Indications

Memantine is the only approved therapy in the United States for moderate to severe Alzheimer disease.

Precautions and Adverse Reactions

Memantine is safe and well tolerated. The most common adverse effects are dizziness, headache, constipation, and confusion. The use of memantine in patients with severe renal impairment is not recommended. In a documented case of an overdose with up to 400 mg of memantine, the patient experienced restlessness, psychosis, visual hallucinations, somnolence, stupor, and loss of consciousness. The patient recovered without permanent sequelae.

Drug Interactions

In vitro studies conducted with marker substrates of CYP450 enzymes (CYP1A2, 2A6, 2C9, 2D6, 2E1, and 3A4) showed minimal inhibition of

these enzymes by memantine. No pharmacokinetic interactions with drugs metabolized by these enzymes are expected.

Because memantine is eliminated in part by tubular secretion, coadministration of drugs that use the same renal cationic system, including hydrochlorothiazide triamterene (Dyrenium), cimetidine (Tagamet), ranitidine (Zantac), quinidine, and nicotine, could potentially result in altered plasma levels of both agents. Coadministration of memantine with a combination of hydrochlorothiazide and triamterene did not affect the bioavailability of either memantine or triamterene, and the bioavailability of hydrochlorothiazide decreased by 20 percent.

Urine pH is altered by diet, drugs (e.g., carbonic anhydrase inhibitors, topiramate [Topamax], sodium bicarbonate), and the clinical state of the patient (e.g., renal tubular acidosis or severe infections of the urinary tract). The clearance of memantine is reduced by about 80 percent under alkaline urine conditions at pH 8. Therefore, alterations of urine pH toward the alkaline condition may lead to an accumulation of the drug with a possible increase in adverse effects. Hence, memantine should be used with caution under these conditions.

Laboratory Interferences

No laboratory interferences have been associated with the use of memantine.

Dosage and Clinical Guidelines

Memantine is available in 5- and 10-mg tablets, with a recommended starting dose of 5 mg daily. The recommended target dose is 20 mg/day. The drug is administered twice daily in separate doses with 5-mg increment increases weekly depending on tolerability.

Patients with mild to moderate disease receiving memantine in combination with a cholinesterase inhibitor have not been found to experience significantly greater benefit in cognition or overall function than those who receive a cholinesterase inhibitor alone.

21.9 Drugs Used to Treat Sexual

Disorders

PHOSPHODIESTERASE-5 INHIBITORS

Phosphodiesterase (PDE)-5 inhibitors, such as sildenafil (Viagra), which was developed in 1998, revolutionized the treatment of erectile disorder. Two congeners have since come on the market—vardenafil (Levitra) and tadalafil (Cialis). All have a similar method of action and have changed people's expectations of sexual functioning. Although indicated only for the treatment of male erectile dysfunction, there is anecdotal evidence of these drugs being effective in women. They are also being misused as recreational drugs to enhance sexual performance. These drugs have been used by more than 20 million men around the world.

The development of sildenafil provided important information about the physiology of erection. Sexual stimulation causes the release of the neurotransmitter nitric oxide (NO), which increases the synthesis of cyclic guanosine monophosphate (cGMP), causing smooth muscle relaxation in the corpus cavernosum that allows blood to flow into the penis and results in turgidity and tumescence. The concentration of cGMP is regulated by the enzyme PDE-5, which, when inhibited, allows cGMP to increase and enhance erectile function. Because sexual stimulation is required to cause the release of NO, PDE-5 inhibitors do not affect in the absence of such stimulation, an important point to understand when providing information to patients about their use. The congeners vardenafil and tadalafil work in the same way, by inhibiting PDE-5, thus allowing an increase in cGMP and enhancing the vasodilatory effects of NO. For this reason, these drugs are sometimes referred to as NO enhancers.

Pharmacologic Actions

All three substances are relatively rapidly absorbed from the GI tract, with maximum plasma concentrations reached in 30 to 120 minutes (median, 60 minutes) in the fasting state. Because it is lipophilic, concomitant ingestion of a high-fat meal delays the rate of absorption by up to 60 minutes and reduces the peak concentration by one-quarter. These drugs are principally metabolized by the CYP3A4 system, which may lead to clinically significant

drug–drug interactions, not all of which have been documented. Excretion of 80 percent of the dose is via feces, and another 13 percent is eliminated in the urine. Elimination is reduced in persons older than age 65 years, which results in plasma concentrations 40 percent higher than in persons ages 18 to 45 years. Elimination is also reduced in the presence of severe renal or hepatic insufficiency.

The mean half-lives of sildenafil and vardenafil are 3 to 4 hours, and that of tadalafil is about 18 hours. Tadalafil can be detected in the bloodstream 5 days after ingestion, and because of its long half-life, it has been marketed as effective for up to 36 hours—the so-called weekend pill. The onset of sildenafil occurs about 30 minutes after ingestion on an empty stomach; tadalafil and vardenafil act somewhat more quickly.

Clinicians need to be aware of the crucial clinical observation that these drugs do not by themselves create an erection. Instead, the mental state of sexual arousal brought on by erotic stimulation must first lead to activity in the penile nerves, which then release NO into the cavernosum, triggering the erectile cascade, the resulting erection being prolonged by the NO enhancers. Thus, a full advantage may be taken of a sexually exciting stimulus, but the drug is not a substitute for foreplay and emotional arousal.

Therapeutic Indications

Erectile dysfunctions have traditionally been classified as organic, psychogenic, or mixed. Over the past 20 years, the prevailing view of the cause of erectile dysfunction has shifted away from psychological causes toward organic causes. The latter include diabetes mellitus, hypertension, hypercholesterolemia, cigarette smoking, peripheral vascular disease, pelvic or spinal cord injury, pelvic or abdominal surgery (especially prostate surgery), multiple sclerosis, peripheral neuropathy, and Parkinson disease. Erectile dysfunction is often induced by alcohol, nicotine, and other substances of abuse and prescription drugs.

These drugs are effective regardless of the baseline severity of erectile dysfunction, race, or age. Among those responding to sildenafil are men with coronary artery disease, hypertension, other cardiac diseases, peripheral vascular disease, diabetes mellitus, depression, coronary artery bypass graft surgery, radical prostatectomy, transurethral resection of the prostate, spina

bifida, and spinal cord injury, as well as persons taking antidepressants, antipsychotics, antihypertensives, and diuretics. However, the response rate is variable.

Sildenafil has been reported to reverse SSRI-induced anorgasmia in men. There are anecdotal reports of sildenafil having a therapeutic effect on sexual inhibition in women as well.

Precautions and Adverse Reactions

A significant potential adverse effect associated with the use of these drugs is myocardial infarction (MI). The U.S. FDA distinguished the risk of MI caused directly by these drugs from that caused by underlying conditions such as hypertension, atherosclerotic heart disease, diabetes mellitus, and other atherogenic conditions. The FDA concluded that when used according to the approved labeling, the drugs do not by themselves confer an increased risk of death. However, there is increased oxygen demand and stress placed on the cardiac muscle by sexual intercourse. Thus, coronary perfusion may be severely compromised, and cardiac failure may occur as a result. For that reason, any person with a history of MI, stroke, renal failure, hypertension, or diabetes mellitus and any person older than the age of 70 years should discuss plans to use these drugs with an internist or a cardiologist. The cardiac evaluation should specifically address exercise tolerance and the use of nitrates.

The use of PDE-5 inhibitors is contraindicated in persons who are taking organic nitrates in any form. Also, amyl nitrate (poppers), a popular substance of abuse used by gay men to enhance the intensity of orgasm, should not be used with any of the erection-enhancing drugs. The combination of organic nitrates and PDE inhibitors can cause a precipitous lowering of BP and can reduce coronary perfusion to the point of causing MI and death.

Adverse effects are dose-dependent, occurring at higher rates with higher dosages. The most common adverse effects are headache, flushing, and stomach pain. Other less common adverse effects include nasal congestion, urinary tract infection, abnormal vision (colored tinge [usually blue], increased sensitivity to light, or blurred vision), diarrhea, dizziness, and rash. No cases of priapism were reported in premarketing trials. Supportive

management is indicated in cases of overdosage. Tadalafil has been associated with back and muscle pain in about 10 percent of patients.

Recently, there have been 50 reports and 14 verified cases of a severe condition in men taking sildenafil called nonarteritic anterior ischemic optic neuropathy. This condition is an eye ailment that causes restriction of blood flow to the optic nerve and can result in permanent vision loss. The first symptoms appear within 24 hours after the use of sildenafil and include blurred vision and some degree of vision loss. The incidence of this effect is infrequent—1 in 1 million. In the reported cases, many patients had preexisting eye problems that may have increased their risk, and many had a history of heart disease and diabetes, which may indicate vulnerability in these men to endothelial damage.

In 2010, possible hearing loss was reported based on 29 incidents of the problem since the introduction of these drugs. Hearing loss usually occurs within hours or days of using the drug. In some cases, the loss is both unilateral and temporary.

No data are available on the effects on human fetal growth and development or testicular morphologic or functional changes. However, because these drugs are not considered an essential treatment, they should not be used during pregnancy.

Treatment of Priapism

Phenylephrine (Neo-Synephrine) is the drug of choice and first-line treatment of priapism, because the drug has almost pure α-agonist effects and minimal β activity. In short-term priapism (less than 6 hours), especially for drug-induced priapism, intracavernosal injection of phenylephrine can be used to cause detumescence. A mixture of 1 ampule of phenylephrine (1 mL/1,000 µg) should be diluted with an additional 9 mL of normal saline. 0.3 to 0.5 mL should be injected into the corpora cavernosa, using a 29-gauge needle with 10 to 15 minutes between injections. Vital signs should be monitored, and compression should be applied to the area of injection to help prevent hematoma formation.

Phenylephrine can also be used orally, 10 to 20 mg every 4 hours as needed, but it may not be as effective or act as rapidly as the injectable route.

Drug Interactions

The primary route of PDE-5 metabolism is through CYP3A4, and the minor route is through CYP2C9. Inducers or inhibitors of these enzymes will therefore affect the plasma concentration and half-life of sildenafil. For example, 800 mg of cimetidine (Tagamet), a nonspecific CYP inhibitor, increases plasma sildenafil concentrations by 56 percent, and erythromycin (E-mycin) increases plasma sildenafil concentrations by 182 percent. Other, more potent inhibitors of CYP3A4 include ketoconazole (Nizoral), itraconazole (Sporanox), and mibefradil (Posicor). In contrast, rifampicin, a CYP3A4 inducer, decreases plasma concentrations of sildenafil.

Laboratory Interferences

No laboratory interferences have been described.

Dosage and Clinical Guidelines

Sildenafil is available as 25-, 50-, and 100-mg tablets. The recommended dose of sildenafil is 50 mg taken by mouth 1 hour before intercourse. However, sildenafil may take effect within 30 minutes. The duration of the effect is usually 4 hours, but in healthy young men, the effect may persist for 8 to 12 hours. Based on effectiveness and adverse effects, the dose should be titrated between 25 and 100 mg. Sildenafil is recommended for use no more than once a day. The dosing guidelines for use by women as off-label use are the same as those for men.

Increased plasma concentrations of sildenafil may occur in persons older than 65 years of age and those with cirrhosis or severe renal impairment or using CYP3A4 inhibitors. A starting dose of 25 mg should be used in these circumstances.

An investigational nasal spray formulation of sildenafil has been developed that acts within 5 to 15 minutes of administration. This formulation is highly water-soluble, and it is rapidly absorbed directly into the bloodstream. Such a formulation would permit more ease of use.

Vardenafil is supplied in 2.5-, 5-, 10-, and 20-mg tablets. The initial dose is usually 10 mg taken with or without food about 1 hour before sexual activity. The dose can be increased to a maximum of 20 mg or decreased to 5 mg based on efficacy and side effects. The maximum dosing frequency is once

per day. As with sildenafil, dosages may have to be adjusted in patients with hepatic impairment or patients using specific CYP3A4 inhibitors. A 10 mg orally disintegrating form of vardenafil (Staxyn) is available. It is placed on the tongue approximately 60 minutes before sexual activity and should not be used more than once a day.

Tadalafil is available in 2.5-, 5-, or 20-mg tablets for oral administration. The recommended dose of tadalafil is 10 mg before sexual activity, which may be increased to 20 mg or decreased to 5 mg depending on efficacy and side effects. Once-a-day use of the 2.5- or 5-mg pill is acceptable for most patients. Similar cautions apply as mentioned earlier in patients with hepatic impairment and those taking concomitant potent inhibitors of CYP3A4. As with other PDE-5 inhibitors, concomitant use of nitrates in any form is contraindicated.

YOHIMBINE

Yohimbine (Yocon) is an α_2-adrenergic receptor antagonist that is used as a treatment for both idiopathic and medication-induced erectile disorder. Currently, sildenafil (Viagra) and its congeners and alprostadil (Impulse, Edex) are considered more efficacious for this indication than yohimbine. Yohimbine is derived from an alkaloid found in *Rubiaceae* and related trees and the *Rauwolfia serpentina* plant.

Pharmacologic Actions

Yohimbine is erratically absorbed after oral administration, with bioavailability ranging from 7 to 87 percent. There is extensive hepatic first-pass metabolism. Yohimbine affects the sympathomimetic autonomic nervous system by increasing plasma concentrations of norepinephrine. The half-life of yohimbine is 0.5 to 2 hours. Clinically, yohimbine produces increased parasympathetic (cholinergic) tone.

Therapeutic Indications

Yohimbine has been used to treat erectile dysfunction. Penile erection has been linked to cholinergic activity and to the α_2-adrenergic blockade, which theoretically results in an increased penile inflow of blood, decreased penile

outflow of blood, or both. Yohimbine is reported to help counteract the loss of sexual desire and the orgasmic inhibition caused by some serotonergic antidepressants (e.g., SSRIs). It has not been found useful in women for these indications.

Precautions

The side effects of yohimbine include anxiety, elevated BP and heart rate, increased psychomotor activity, irritability, tremor, headache, skin flushing, dizziness, urinary frequency, nausea, vomiting, and sweating. Patients with panic disorder show heightened sensitivity to yohimbine and experience increased anxiety, increased BP, and increased plasma 3-methoxy-4-hydroxyphenylglycol (MHPG).

Yohimbine should be used with caution in female patients and should not be used in patients with renal disease, cardiac disease, glaucoma, or a history of gastric or duodenal ulcers.

Drug Interactions

Yohimbine blocks the effects of clonidine, guanfacine, and other α_2-receptor agonists.

Laboratory Interferences

No known laboratory interferences are associated with yohimbine use.

Dosage and Clinical Guidelines

Yohimbine is available in 5.4-mg tablets. The dosage of yohimbine in the treatment of erectile disorder is approximately 16 mg a day, given in doses that range from 2.7 to 5.4 mg three times a day. In the event of significant adverse effects, the dose should first be reduced and then gradually increased again. Yohimbine should be used judiciously in psychiatric patients because it may worsen their mental status. Because yohimbine has no consistent effect on erectile dysfunction, its use remains controversial. Phosphodiesterase-5 (PDE-5) inhibitors are the preferred medication for this disorder.

21.10 Drugs Used to Treat the Side Effects of Psychotropic Drugs

ANTICHOLINERGIC AGENTS

Anticholinergic drugs block the actions of atropine. In the clinical practice of psychiatry, the anticholinergic drugs are primarily used to treat medication-induced movement disorders, particularly neuroleptic-induced parkinsonism, neuroleptic-induced acute dystonia, and medication-induced postural tremor.

Anticholinergics

Pharmacologic Actions. All anticholinergic drugs are well absorbed from the GI tract after oral administration, and all are sufficiently lipophilic to enter the CNS. Trihexyphenidyl (Artane) and benztropine (Cogentin) reach peak plasma concentrations in 2 to 3 hours after oral administration, and their duration of action is 1 to 12 hours. Benztropine is absorbed equally rapidly by intramuscular (IM) and intravenous (IV) administration; IM administration is preferred because of its low risk for adverse effects.

All six anticholinergic drugs listed in this section (Table 21-57) block muscarinic acetylcholine receptors, and benztropine has some antihistaminergic effects. None of the available anticholinergic drugs has any effect on the nicotinic acetylcholine receptors. Of these drugs, trihexyphenidyl is the most stimulating agent, perhaps acting through dopaminergic neurons, and benztropine is the least stimulating and thus is least associated with abuse potential.



Table 21-57
Anticholinergic Drugs

Generic Name	Brand Name	Tablet Size	Injectable	Usual Daily Oral Dosage	Short-Term Intramuscular or Intravenous Dosage
Benztropine	Cogentin	0.5, 1, 2 mg	1 mg/mL	1–4 mg one to three times	1–2 mg
Biperiden	Akineton	2 mg	5 mg/mL	2 mg one to three times	2 mg
Ethopropazine	Parsidol	10, 50 mg	—	50–100 mg one to three times	—
Orphenadrine	Norflex, Disipal	100 mg	30 mg/mL	50–100 mg three times	60 mg IV given over 5 min
Procyclidine	Kemadrin	5 mg	—	2.5–5 mg three times	—
Trihexyphenidyl	Artane, Trihexane, Trihexy-5	2, 5 mg elixir 2 mg/5 mL	—	2–5 mg two to four times	—

IV, intravenous.

Therapeutic Indications. The primary indication for the use of anticholinergics in psychiatric practice is for the treatment of *neuroleptic-induced parkinsonism*, characterized by tremor, rigidity, cogwheeling, bradykinesia, sialorrhea, stooped posture, and festination. All of the available anticholinergics are equally effective in the treatment of parkinsonian symptoms. Neuroleptic-induced parkinsonism is most common in elderly persons and is most frequently seen with high-potency DRAs, for example, haloperidol (Haldol). The onset of symptoms usually occurs after 2 or 3 weeks of treatment. The incidence of neuroleptic-induced parkinsonism is lower with the newer antipsychotic drugs of the SDA class.

Another indication for the use of anticholinergics is for the treatment of *neuroleptic-induced acute dystonia*, which is most common in young men. The syndrome often occurs early in the course of treatment; it is commonly associated with high-potency DRAs (e.g., haloperidol); and most commonly affects the muscles of the neck, tongue, face, and back. Anticholinergic drugs are useful both in the short-term treatment of dystonias and in prophylaxis against neuroleptic-induced acute dystonias.

Akathisia is characterized by a subjective and objective sense of restlessness, anxiety, and agitation. Although a trial of anticholinergics for the treatment of neuroleptic-induced acute akathisia is reasonable, these drugs are not generally considered as effective as the β-adrenergic receptor antagonists, the benzodiazepines, and clonidine (Catapres).

Precautions and Adverse Reactions. The adverse effects of the anticholinergic drugs result from the blockade of muscarinic acetylcholine receptors. Anticholinergic drugs should be used cautiously, if at all, by persons with prostatic hypertrophy, urinary retention, and narrow-angle glaucoma. The anticholinergics are occasionally used as drugs of abuse

because of their mild mood-elevating properties, most notably, trihexyphenidyl.

The most severe adverse effect associated with anticholinergic toxicity is anticholinergic intoxication. The intoxication includes symptoms of delirium, coma, seizures, agitation, hallucinations, severe hypotension, supraventricular tachycardia, and peripheral manifestations (flushing, mydriasis, dry skin, hyperthermia, and decreased bowel sounds). Treatment should begin with the immediate discontinuation of all anticholinergic drugs. We can both diagnose and treat anticholinergic intoxication with physostigmine (Antilirium, Eserine), an inhibitor of anticholinesterase, 1 to 2 mg IV (1 mg every 2 minutes) or IM every 30 or 60 minutes. Because physostigmine can lead to severe hypotension and bronchial constriction, it should be used only in severe cases, and only when emergency cardiac monitoring and life-support services are available.

Drug Interactions. The most common drug–drug interactions with the anticholinergics occur when they are coadministered with psychotropics that also have high anticholinergic activity, such as DRAs, tricyclic and tetracyclic drugs, and MAOIs. Many other prescription drugs and over-the-counter cold preparations also induce significant anticholinergic activity. The coadministration of those drugs can result in a life-threatening anticholinergic intoxication syndrome. Also, anticholinergic drugs can delay gastric emptying, thereby decreasing the absorption of drugs that are broken down in the stomach and usually absorbed in the duodenum (e.g., levodopa [Larodopa] and DRAs).

Laboratory Interferences. No known laboratory interferences have been associated with anticholinergics.

Dosage and Clinical Guidelines. The six anticholinergic drugs discussed in this chapter are available in a range of preparations (see Table 21-57).

NEUROLEPTIC-INDUCED PARKINSONISM. For the treatment of neuroleptic-induced parkinsonism, the equivalent of 1 to 3 mg of benztropine should be given one to two times daily. The anticholinergic drug should be administered for 4 to 8 weeks, and then it should be discontinued to assess

whether the person still requires the drug. Anticholinergic drugs should be tapered over 1 to 2 weeks.

Treatment with anticholinergics as prophylaxis against the development of neuroleptic-induced parkinsonism is usually not indicated because the onset of its symptoms is usually sufficiently mild and gradual to allow the clinician to initiate treatment only after it is indicated. In young men, prophylaxis may be indicated, however, mainly if a high-potency DRA is being used. The clinician should attempt to discontinue the antiparkinsonian agent in 4 to 6 weeks to assess whether its continued use is necessary.

NEUROLEPTIC-INDUCED ACUTE DYSTONIA. For the short-term treatment and prophylaxis of neuroleptic-induced acute dystonia, 1 to 2 mg of benztropine or its equivalent in another drug should be given IM. The dose can be repeated in 20 to 30 minutes, as needed. If the person still does not improve in another 20 to 30 minutes, a benzodiazepine (e.g., 1 mg IM or IV lorazepam [Ativan]) should be given. Laryngeal dystonia is a medical emergency and should be treated with benztropine, up to 4 mg in 10 minutes, followed by 1 to 2 mg of lorazepam, administered slowly by the IV route.

Prophylaxis against dystonias is indicated in persons who have had one episode or in persons at high risk (young men taking high-potency DRAs). Prophylactic treatment is given for 4 to 8 weeks and then gradually tapered over 1 to 2 weeks to allow assessment of its continued need. The prophylactic use of anticholinergics in persons requiring antipsychotic drugs has mostly become a moot issue because of the availability of SDAs, which are relatively free of parkinsonian effects.

AKATHISIA. As mentioned, anticholinergics are not the drugs of choice for this syndrome. The β-adrenergic receptor antagonists and perhaps the benzodiazepines and clonidine are preferable drugs to try initially.

ANTIHISTAMINES

Antihistamines are frequently used in the treatment of a variety of psychiatric disorders because of their sedative and anticholinergic activities. Certain antihistamines (antagonists of histamine H_1 receptors) are used to treat neuroleptic-induced parkinsonism and neuroleptic-induced acute dystonia

and as hypnotics and anxiolytics. Diphenhydramine (Benadryl) is used to treat neuroleptic-induced parkinsonism and neuroleptic-induced acute dystonia and sometimes as a hypnotic. Hydroxyzine hydrochloride (Atarax) and hydroxyzine pamoate (Vistaril) are used as anxiolytics. Promethazine (Phenergan) is used for its sedative and anxiolytic effects. Cyproheptadine (Periactin) has been used for the treatment of anorexia nervosa and inhibited male and female orgasms caused by serotonergic agents. The antihistamines most commonly used in psychiatry are listed in Table 21-58. Second-generation, "nonsedating" H_1 blockers, such as fexofenadine (Allegra), loratadine (Claritin), and cetirizine (Zyrtec) are less commonly used in psychiatric practice. The H_2-receptor antagonists, such as cimetidine, work primarily on gastric mucosa, inhibiting gastric secretion.

Table 21-59 lists antihistaminic drugs not used in psychiatry, but that may have psychiatric adverse effects or drug–drug interactions.

Pharmacologic Actions

The H_1 antagonists used in psychiatry are well absorbed from the GI tract. The antiparkinsonian effects of intramuscular (IM) diphenhydramine have their onset in 15 to 30 minutes and the sedative effects of diphenhydramine peak in 1 to 3 hours. The sedative effects of hydroxyzine and promethazine begin after 20 to 60 minutes and last for 4 to 6 hours. Because all three drugs are metabolized in the liver, persons with hepatic disease, such as cirrhosis, may attain high plasma concentrations with long-term administration. Cyproheptadine is well absorbed after oral administration, and its metabolites are excreted in the urine.



Table 21-58
Histamine Antagonists Commonly Used in Psychiatry

Generic Name	Trade Name	Duration of Action (hr)
Diphenhydramine	Benadryl	4–6
Hydroxyzine	Atarax, Vistaril	6–24
Promethazine	Phenergan	4–6

Cyproheptadine	Periactin	4–6

Table 21-59
Other Histamine Antagonists often Prescribed

Class	Generic Name	Trade Name
Second-generation histamine 1 receptor antagonists	Cetirizine	Zyrtec
	Loratadine	Claritin
	Fexofenadine	Allegra
Histamine 2 receptor antagonists	Nizatidine	Axid
	Famotidine	Pepcid
	Ranitidine	Zantac
	Cimetidine	Tagamet

Activation of H_1 receptors stimulates wakefulness; therefore, receptor antagonism causes sedation. All four agents also possess some antimuscarinic cholinergic activity. Cyproheptadine is unique among the drugs because it has both potent antihistamine and serotonin 5-HT_2-receptor antagonist properties.

Therapeutic Indications

Antihistamines are useful as a treatment for neuroleptic-induced parkinsonism, neuroleptic-induced acute dystonia, and neuroleptic-induced akathisia. They are an alternative to anticholinergics and amantadine for these purposes. The antihistamines are relatively safe hypnotics, but they are not superior to the benzodiazepines, which have been much better studied in terms of efficacy and safety. The antihistamines have not been proven effective for long-term anxiolytic therapy; therefore, the benzodiazepines, buspirone (BuSpar), or SSRIs are preferable for such treatment. Cyproheptadine is sometimes used to treat impaired orgasms, especially

delayed orgasm resulting from treatment with serotonergic drugs.

Because it promotes weight gain, cyproheptadine may be of some use in the treatment of eating disorders, such as anorexia nervosa. Cyproheptadine can reduce recurrent nightmares with posttraumatic themes. The antiserotonergic activity of cyproheptadine may counteract the serotonin syndrome caused by concomitant use of multiple serotonin-activating drugs, such as SSRIs and MAOIs.

Precautions and Adverse Reactions

Antihistamines are commonly associated with sedation, dizziness, and hypotension, all of which can be severe in elderly persons, who are also likely to experience the anticholinergic effects of those drugs. Paradoxical excitement and agitation is an adverse effect seen in a small number of persons. Poor motor coordination can result in accidents; therefore, persons should be warned about driving and operating dangerous machinery. Other common adverse effects include epigastric distress, nausea, vomiting, diarrhea, and constipation. Because of mild anticholinergic activity, some people experience dry mouth, urinary retention, blurred vision, and constipation.

For this reason, also, antihistamines should be used only at very low doses, if at all, by persons with narrow-angle glaucoma or obstructive GI, prostate, or bladder conditions. Either cyproheptadine or diphenhydramine may induce a central anticholinergic syndrome with psychosis. The use of cyproheptadine in some persons has been associated with weight gain, which may contribute to its reported efficacy in some persons with anorexia nervosa.

In addition to the above adverse effects, antihistamines have some potential for abuse. The coadministration of antihistamines and opioids can increase the euphoria experienced by persons with substance use disorders. Overdoses of antihistamines can be fatal. Antihistamines are excreted in breast milk, so nursing mothers should avoid their use. Because of some potential for teratogenicity, pregnant women should avoid the use of antihistamines.

Drug Interactions

The sedative property of antihistamines can be additive with other CNS

depressants, such as alcohol, other sedative-hypnotic drugs, and many psychotropic drugs, including tricyclic drugs and DRAs. Anticholinergic activity can be cumulative when using other anticholinergic drugs and may sometimes result in severe anticholinergic symptoms or intoxication.

Laboratory Interferences

H_1 antagonists may eliminate the wheal and induration that form the basis of allergy skin tests. Promethazine may interfere with pregnancy tests and may increase blood glucose concentrations. Diphenhydramine may yield a false-positive urine test result for phencyclidine (PCP). Hydroxyzine use can falsely elevate the results of specific tests for urinary 17-hydroxycorticosteroids.

Dosage and Clinical Guidelines

The antihistamines are available in a variety of preparations (Table 21-60). IM injections should be deep because superficial administration can cause local irritation.



Table 21-60
Dosage and Administration of Common Histamine Antagonists

Medication	Route	Preparation	Common Dosage
Diphenhydramine (Benadryl)	PO	Capsules and tablets: 25 mg, 50 mg	Adults: 25–50 mg three to four times per day
		Liquid: 12.5 mg/5.0 mL	Children: 5 mg/kg three to four times per day, not to exceed 300 mg/day
	Deep IM or IV	Solution: 10 or 50 mg/mL	Same as oral

Hydroxyzine Hydrochloride (Atarax)	PO	Tablets: 10, 25, 50, and 100 mg	Adults: 50–100 mg three to four times daily
		Syrup: 10 mg/5 mL	Children younger than 6 yr of age: 2 mg/kg/day in divided doses
			Children older than 6 yr of age: 12.5–25.0 mg three to four times daily
	IM	Solution: 25 or 50 mg/mL	Same as oral
Hydroxyzine Pamoate (Vistaril)	PO	Suspension: 25 mg/mL	Same as dosages for hydroxyzine hydrochloride
		Capsules: 25, 50, and 100 mg	
Promethazine (Phenergan)	PO	Tablets: 15.2, 25.0, and 50.0 mg	Adults: 50–100 mg three to four times daily for sedation
		Syrup: 3.25 mg/5 mL	Children: 12.5–25.0 mg at night for sedation
	Rectal	Suppositories: 12.5, 25.0, and 50.0 mg	
	IM	Solution: 25 and 50 mg/mL	

Cyproheptadine (Periactin)	PO	Tablets: 4 mg Syrup: 2 mg/5 mL	Adults: 4–20 mg/day Children 2–7 yr of age: 2 mg two to three times daily (maximum, 12 mg/day) Children 7–14 yr of age: 4 mg two to three times daily (maximum of 16 mg/day)

IM, intramuscular; IV, intravenous; PO, oral.

Intravenous (IV) administration of 25 to 50 mg of diphenhydramine is an effective treatment for neuroleptic-induced acute dystonia, which may immediately disappear. Treatment with 25 mg three times a day—up to 50 mg four times a day, if necessary—can be used to treat neuroleptic-induced parkinsonism, akinesia, and buccal movements. Diphenhydramine can be used as a hypnotic at a 50-mg dose for mild transient insomnia. Doses of 100 mg are not superior to doses of 50 mg, but they produce more anticholinergic effects than doses of 50 mg.

Hydroxyzine is most commonly used as a short-term anxiolytic. Hydroxyzine should not be given IV because it is irritating to the blood vessels. Dosages of 50 to 100 mg given orally four times a day for long-term treatment or 50 to 100 mg IM every 4 to 6 hours for short-term treatment are usually effective.

SSRI-induced anorgasmia may sometimes be reversed with 4 to 16 mg a day of cyproheptadine taken by mouth 1 or 2 hours before anticipated sexual activity. Several case reports and small studies have also reported that cyproheptadine may be of some use in the treatment of eating disorders, such as anorexia nervosa. Cyproheptadine is available in 4-mg tablets and a 2-mg/5-mL solution. Children and elderly patients are more sensitive to the effects of antihistamines than are young adults.

DOPAMINE RECEPTOR AGONISTS AND PRECURSORS

Dopamine agonists activate dopamine receptors in the absence of endogenous dopamine and have been widely used to treat idiopathic Parkinson disease, hyperprolactinemia, and certain pituitary tumors (prolactinoma). Because dopamine stimulates the heart and increases blood flow to the liver, kidneys, and other organs, low levels of dopamine are associated with low BP and low cardiac input. Dopamine agonist drugs are also administered to treat shock and congestive heart failure.

Their use in psychiatry has been limited to treat such adverse effects of antipsychotic drugs as parkinsonism, extrapyramidal symptoms, akinesia, focal perioral tremors, hyperprolactinemia, galactorrhea, and neuroleptic malignant syndrome. The drugs in this class most commonly prescribed are bromocriptine (Parlodel), levodopa (also called L-Dopa; Larodopa), carbidopa-levodopa (Sinemet), and amantadine (Symmetrel). Amantadine is used primarily for the treatment of medication-induced movement disorders, such as neuroleptic-induced parkinsonism. It is also used as an antiviral agent for the prophylaxis and treatment of influenza A infection and Cotard syndrome, a rare neuropsychiatric disorder in which a person holds a delusional belief that he or she is dead. There are also a few reports of amantadine's role in augmenting antidepressant medications in patients with treatment-resistant depression.

New dopamine receptor agonists include ropinirole (Requip), pramipexole (Mirapex), apomorphine (Apokyn), and pergolide (Permax). Of these drugs, pramipexole is the most widely prescribed in psychiatry as an augmenter of antidepressants. In 2007, pergolide was removed from the market because of the risk of severe damage to patients' heart valves. In 2012, the U.S. FDA notified health care professionals about a possible increased risk of heart failure with pramipexole. This warning was based on studies that suggested a potential risk of heart failure; however, further review is required because of study limitations.

Pharmacologic Actions

L-Dopa is rapidly absorbed after oral administration, and peak plasma levels are reached after 30 to 120 minutes. The half-life of L-Dopa is 90 minutes.

Absorption of L-Dopa can be significantly reduced by changes in gastric pH and by ingestion with meals. Bromocriptine and ropinirole are rapidly absorbed but undergo first-pass metabolism such that only about 30 to 55 percent of the dose is bioavailable. Peak concentrations are achieved 1.5 to 3 hours after oral administration. The half-life of ropinirole is 6 hours. Pramipexole is rapidly absorbed with little first-pass metabolism and reaches peak concentrations in 2 hours. Its half-life is 8 hours. Oral forms of apomorphine have been studied, but this form is not available in the United States. Subcutaneous apomorphine injection results in rapid and controlled systemic delivery, with linear pharmacokinetics over a dose ranging from 2 to 8 mg.

After L-Dopa enters the dopaminergic neurons of the CNS, it is converted into the neurotransmitter dopamine. Apomorphine, bromocriptine, ropinirole, and pramipexole act directly on dopamine receptors. L-Dopa, pramipexole, and ropinirole bind about 20 times more selectively to dopamine D_3 than D_2 receptors; the corresponding ratio for bromocriptine is less than 2 to 1. Apomorphine binds selectively to D_1 and D_2 receptors, with little affinity for D_3 and D_4 receptors. L-Dopa, pramipexole, and ropinirole have no significant activity at nondopaminergic receptors. However, bromocriptine binds to serotonin 5-HT_1 and 5-HT_2 and α_1-, α_2-, and β-adrenergic receptors.

Therapeutic Indications

Medication-Induced Movement Disorders. In present-day clinical psychiatry, dopamine receptor agonists are used for the treatment of medication-induced parkinsonism, extrapyramidal symptoms, akinesia, and focal perioral tremors. Their use has diminished sharply, however, because the incidence of medication-induced movement disorders is much lower with the use of the newer, atypical antipsychotics (SDAs). Dopamine receptor agonists are effective in treating idiopathic restless legs syndrome and may also be helpful when this is a medication side effect. Ropinirole can treat restless legs syndrome.

For the treatment of medication-induced movement disorders, most clinicians rely on anticholinergics, amantadine, and antihistamines because they are equally effective and have few adverse effects. Bromocriptine

remains in use in the treatment of neuroleptic malignant syndrome; however, the incidence of this disorder is diminishing with the decreased use of DRAs.

Dopamine receptor agonists are also used to counteract the hyperprolactinemic effects of DRAs, which result in the side effects of amenorrhea and galactorrhea.

Mood Disorders. Bromocriptine has long been used to enhance response to antidepressant drugs in refractory patients. Ropinirole has been reported to be useful as an augmentation of antidepressant therapy and as a treatment for medication-resistant bipolar II depression. Ropinirole may also be helpful in the treatment of antidepressant-induced sexual dysfunction. Pramipexole is often used in the augmentation of antidepressants in treatment-resistant depression. Some studies have found pramipexole to be superior to sertraline (Zoloft) in the treatment of depression in Parkinson disease, as well as in reducing anhedonia in Parkinson patients.

Sexual Dysfunction. Dopamine receptor agonists improve erectile dysfunction in some patients. However, they are rarely used because they frequently cause adverse effects at therapeutic dosages. Phosphodiesterase-5 inhibitor agents are better tolerated and more effective (see Chapter 16).

Precautions and Adverse Reactions

Adverse effects are common with dopamine receptor agonists, thus limiting the usefulness of these drugs. Adverse effects are dosage-dependent and include nausea, vomiting, orthostatic hypotension, headache, dizziness, and cardiac arrhythmias. To reduce the risk of orthostatic hypotension, the initial dosage of all dopamine receptor agonists should be relatively low, with incremental increases at intervals of at least 1 week. These drugs should be used with caution in persons with hypertension, cardiovascular disease, and hepatic disease. After long-term use, persons, particularly elderly persons, may experience choreiform and dystonic movements and psychiatric disturbances—including hallucinations, delusions, confusion, depression, and mania—and other behavioral changes.

Long-term use of bromocriptine can produce retroperitoneal and pulmonary fibrosis, pleural effusions, and pleural thickening.

In general, ropinirole and pramipexole have a similar but milder adverse

effect profile than L-Dopa and bromocriptine. Pramipexole and ropinirole may cause irresistible sleep attacks that occur suddenly without warning and have caused motor vehicle accidents.

The most common adverse effects of apomorphine are yawning, dizziness, nausea, vomiting, drowsiness, bradycardia, syncope, and perspiration. Hallucinations have also been reported. Apomorphine's sedative effects are exacerbated with concurrent use of alcohol or other CNS depressants.

Dopamine receptor agonists are contraindicated during pregnancy, especially for nursing mothers, because they inhibit lactation.

Drug Interactions

DRAs are capable of reversing the effects of dopamine receptor agonists, but this is not usually clinically significant. The concurrent use of tricyclic drugs and dopamine receptor agonists has been reported to cause symptoms of neurotoxicity, such as rigidity, agitation, and tremor. They may also potentiate the hypotensive effects of diuretics and other antihypertensive medications. Dopamine receptor agonists should not be used in conjunction with MAOIs, including selegiline (Eldepryl), and MAOIs should be discontinued at least 2 weeks before the initiation of dopamine receptor agonist therapy.

Benzodiazepines, phenytoin (Dilantin), and pyridoxine may interfere with the therapeutic effects of dopamine receptor agonists. We should not combine ergot alkaloids and bromocriptine because they may cause hypertension and myocardial infarction. Progestins, estrogens, and oral contraceptives may interfere with the effects of bromocriptine and may raise plasma concentrations of ropinirole. Ciprofloxacin (Cipro) can raise plasma concentrations of ropinirole, and cimetidine (Tagamet) can raise plasma concentrations of pramipexole.

Laboratory Interferences

L-Dopa administration has been associated with false reports of elevated serum and urinary uric acid concentrations, urinary glucose test results, urinary ketone test results, and urinary catecholamine concentrations. No laboratory interferences have been associated with the administration of the other dopamine receptor agonists.

Dosage and Clinical Guidelines

Table 21-61 lists the various dopamine receptor agonists and their formulations. For the treatment of antipsychotic-induced parkinsonism, the clinician should start with a 100-mg dose of levodopa three times a day, which may be increased until the person is functionally improved. The maximum dosage of L-Dopa is 2,000 mg a day, but most persons respond to dosages below 1,000 mg/day. The dosage of the carbidopa component of the L-Dopa-carbidopa formulation should total at least 75 mg a day.

The dosage of bromocriptine for mental disorders is uncertain, although it seems prudent to begin with low dosages (1.25 mg twice daily) and to increase the dosage gradually. Bromocriptine is usually taken with meals to help reduce the likelihood of nausea.

The starting dosage of pramipexole is 0.125 mg three times daily, which is increased to 0.25 mg three times daily in the second week and is increased by 0.25 mg/dose each week until therapeutic benefit or adverse effects emerge. Persons with idiopathic Parkinson disease usually experience benefits at total daily doses of 1.5 mg, and the maximum daily dose is 4.5 mg.

For ropinirole, the starting dosage is 0.25 mg three times daily and is increased by 0.25 mg/dose each week to a total daily dose of 3 mg, then by 0.5 mg/dose each week to a total daily dose of 9 mg, and then by 1 mg/dose each week to a maximum dosage of 24 mg a day until therapeutic benefit or adverse effects emerge. The average daily dose for persons with idiopathic Parkinson disease is about 16 mg.

The recommended subcutaneous dose of apomorphine in Parkinson disease is 0.2 to 0.6 mL subcutaneously during acute hypomobility episodes delivered via metered injector pen. Apomorphine can be administered three times daily, with a maximum dose of 0.6 mL five times daily.



Table 21-61
Available Preparations of Dopamine Receptor Agonists and Carbidopa

Generic Name	Trade Name	Preparations
Amantadine	Symmetrel	100-mg capsule, 50-mg/5-mL

Bromocriptine	Parlodel	syrup (teaspoon) 2.5-, 5-mg tablets
Carbidopa	Lodosyn	25 mg[a]
Levodopa (L-Dopa)	Larodopa	100-, 250-, 500-mg tablets
Levodopa-carbidopa (co-careldopa)	Sinemet, Atamet	100/10-mg, 100/25-mg, 250/25-mg tablets; 100/25-, 200/50-mg extended-release tablets
Pramipexole	Mirapex	0.125-, 0.375-, 0.75-, 1.5-, 3-, 4-mg extended-release tablets
Ropinirole	Requip	0.25-, 0.5-, 1-, 2-, 5-mg tablets

[a]Drug only available directly through the manufacturer.

Amantadine

Amantadine is an antiviral drug used for the prophylaxis and treatment of influenza. It was found to have antiparkinsonian properties and is now used to treat that disorder as well as akinesias and other extrapyramidal signs, including focal perioral tremors (rabbit syndrome).

Pharmacologic Actions. Amantadine is well absorbed from the GI tract after oral administration, reaches peak plasma concentrations in approximately 2 to 3 hours, has a half-life of about 12 to 18 hours, and attains steady-state concentrations after approximately 4 to 5 days of therapy. Amantadine is excreted unmetabolized in the urine. Amantadine plasma concentrations can be twice as high in elderly persons as in younger adults. Patients with renal failure accumulate amantadine in their bodies.

Amantadine augments dopaminergic neurotransmission in the CNS; however, the precise mechanism is unknown. The mechanism may involve dopamine release from presynaptic vesicles, blocking the reuptake of dopamine into presynaptic nerve terminals, or an agonist effect on postsynaptic dopamine receptors.

Therapeutic Indications. The primary indication for amantadine use in

psychiatry is to treat extrapyramidal signs and symptoms, such as parkinsonism, akinesia, and rabbit syndrome (focal perioral tremor of the choreoathetoid type) caused by the administration of DRA or SDA drugs. Amantadine is as effective as the anticholinergics (e.g., benztropine [Cogentin]) for these indications and results in an improvement in approximately half of all persons who take it. Amantadine, however, is not generally considered as effective as the anticholinergics for the treatment of acute dystonic reactions and is not effective in treating tardive dyskinesia and akathisia.

Amantadine is a reasonable compromise for persons with extrapyramidal symptoms who would be sensitive to additional anticholinergic effects, particularly those taking a low-potency DRA or the elderly. Elderly persons are susceptible to anticholinergic adverse effects, both in the CNS, such as anticholinergic delirium, and in the peripheral nervous system, such as urinary retention. Amantadine is associated with less memory impairment than are the anticholinergics.

Amantadine has been reported to be of benefit in treating some SSRI-associated side effects, such as lethargy, fatigue, anorgasmia, and ejaculatory inhibition.

Amantadine is used in general medical practice for the treatment of parkinsonism of all causes, including idiopathic parkinsonism.

Precautions and Adverse Effects. The most common CNS effects of amantadine are mild dizziness, insomnia, and impaired concentration (dosage related), which occur in 5 to 10 percent of all persons. Irritability, depression, anxiety, dysarthria, and ataxia occur in 1 to 5 percent of all persons. More severe CNS adverse effects, including seizures and psychotic symptoms, have been reported. Nausea is the most common peripheral adverse effect of amantadine. Headache, loss of appetite, and blotchy spots on the skin have also been reported.

Livedo reticularis of the legs (a purple discoloration of the skin caused by dilation of blood vessels) has been reported in up to 5 percent of persons who take the drug for longer than 1 month. It usually diminishes with the elevation of the legs and resolves in almost all cases when drug use is terminated.

Amantadine is relatively contraindicated in persons with renal disease or a

seizure disorder. Amantadine should be used with caution in persons with edema or cardiovascular disease. Some evidence indicates that amantadine is teratogenic and, therefore, should not be taken by pregnant women. Because amantadine is excreted in breast milk, breastfeeding women should not take the drug.

Suicide attempts with amantadine overdosages are life-threatening. Symptoms can include toxic psychoses (confusion, hallucinations, aggressiveness) and cardiopulmonary arrest. Emergency treatment beginning with gastric lavage is indicated.

Drug Interactions. Coadministration of amantadine with phenelzine (Nardil) or other MAOIs can result in a significant increase in resting BP. The coadministration of amantadine with CNS stimulants can result in insomnia, irritability, nervousness, and possibly seizures or irregular heartbeat. Amantadine should not be coadministered with anticholinergics because unwanted side effects—such as confusion, hallucinations, nightmares, dry mouth, and blurred vision—may be exacerbated.

Dosage and Clinical Guidelines. Amantadine is available in 100-mg capsules and as a 50-mg/5-mL syrup. The usual starting dosage of amantadine is 100 mg given twice a day orally, although the dosage can be cautiously increased up to 200 mg given twice a day orally if indicated. Amantadine should be used in persons with renal impairment *only* in consultation with the physician treating the renal condition. If amantadine is successful in the treatment of the drug-induced extrapyramidal symptoms, it should be continued for 4 to 6 weeks and then discontinued to see whether the person has become tolerant to the neurologic adverse effects of the antipsychotic medication. Amantadine should be tapered over 1 to 2 weeks after a decision has been made to discontinue the drug. Persons taking amantadine should not drink alcoholic beverages.

DRUGS TO TREAT PSYCHOTROPIC-INDUCED WEIGHT GAIN

Weight management is a critical element of psychotropic drug treatment because obesity is common among persons with mental disorders. Thus,

medical conditions such as hypertension, diabetes mellitus, and hyperlipidemia need to be taken into account when selecting medications. With few exceptions, most psychotropic drugs used to manage mood disorders, anxiety disorders, and psychosis are associated with a significant risk of weight gain as a side effect. Many patients may refuse or discontinue treatment if weight gain occurs, even if the drug is effective in treating their symptoms. For this and other reasons, clinicians need to be well informed about treatment strategies for mitigating drug-induced weight gain and obesity in general.

The standard recommendation for weight loss regimens consists of attempting to manage body weight through consistent dietary modifications and regular physical activity. This behavioral approach may be difficult for patients struggling with psychiatric symptoms, because their mental disorder can compromise their ability to be disciplined in this effort. Also, the physiologic effects of some psychotropic drugs on the regulation of satiety and body metabolism are difficult, if not impossible, to overcome through diet and exercise alone. For these reasons, it may be necessary to use prescription medications to facilitate weight loss.

In this section, drugs used to manage obesity are categorized in two ways: (1) drugs approved by the U.S. FDA as diet pills; and (2) drugs with primary indications other than weight loss but produce weight loss as a side effect.

Drugs with U.S. Food and Drug Administration Approval for Weight Loss

All of the drugs approved by the FDA as weight loss agents are specifically indicated as an adjunct to a reduced-calorie diet and increased physical activity for chronic weight management in adult patients with an initial body mass index (BMI) of 30 kg/m^2 or greater (obese) or 27 kg/m^2 or greater (overweight) in the presence of at least one weight-related comorbidity such as hypertension, type 2 diabetes mellitus, or dyslipidemia.

Phentermine. Phentermine hydrochloride (Adipex-P) is a sympathomimetic amine with pharmacologic activity similar to the amphetamines. It is indicated as a short-term adjunct in a regimen of weight reduction, but in fact, many patients use the drug for extended periods. As

with all sympathomimetics, contraindications include advanced arteriosclerosis, cardiovascular disease, moderate to severe hypertension, hyperthyroidism, known hypersensitivity or idiosyncrasy to the sympathomimetic amines, agitated states, and glaucoma.

The drug should be prescribed with caution to patients with a history of drug abuse. Hypertensive crises may result if phentermine is used during or within 14 days following the administration of MAOIs. Insulin requirements in diabetes mellitus may be altered in association with the use of phentermine hydrochloride and the concomitant dietary regimen. Phentermine hydrochloride may decrease the hypotensive effect of guanethidine. Phentermine was pregnancy Category X in the former FDA classification system and thus contraindicated during pregnancy. Studies have not been performed with phentermine hydrochloride to determine the potential for carcinogenesis, mutagenesis, or impairment of fertility.

Phentermine should be taken on an empty stomach once daily, before breakfast. Tablets may be broken or cut in half but should not be crushed. It should be dosed early in the day to avoid disrupting normal sleep patterns. If taking more than one dose a day, the last dose should be taken approximately 4 to 6 hours before going to bed. The recommended dose of phentermine may be different for different patients. Adults under age 60 taking phentermine using 15- to 37.5-mg capsules should take them once per day before breakfast or 1 to 2 hours after breakfast. Those using 15- to 37.5-mg tablets should take them once per day before breakfast or 1 to 2 hours after breakfast. Instead of taking it once a day, some patients may take 15 to 37.5 mg in divided doses a half hour before meals. An oral resin formulation is available in 15- and 30-mg capsules, which should be taken once per day before breakfast.

Phentermine/Topiramate Extended Release (Qsymia).

This drug is a combination of phentermine and topiramate (Topamax). The FDA approved the phentermine/topiramate combination in 2012 as an extended-release formulation. Both active agents in this formulation are associated with weight loss through separate mechanisms.

Adverse events associated with the use of this drug may include, but are not limited to, paresthesia, dizziness, dysgeusia, insomnia, constipation, dry mouth, kidney stones, metabolic acidosis, and secondary angle-closure glaucoma. The use of this drug is associated with a fivefold increased risk of

infants with cleft palate and was classified as pregnancy Category X in the former FDA classification system. It can only be prescribed by clinicians who have been certified in the use of this drug.

It is available as a tablet and should be administered once daily in the morning with or without food. Avoid dosing with the drug in the evening due to the possibility of insomnia. The recommended dose is as follows: Start treatment with 3.75 mg/
23 mg (phentermine/topiramate extended-release) daily for 14 days; after 14 days, increase the dose to 7.5 mg/46 mg once daily. Evaluate for weight loss after 12 weeks of treatment at the 7.5 mg/
46 mg dose. If at least 3 percent of baseline body weight has not been lost on 7.5 mg/46 mg, discontinue the drug or escalate the dose. To escalate the dose, increase to 11.25 mg/69 mg daily for 14 days, followed by dosing 15 mg/92 mg daily. Evaluate weight loss following dose escalation to 15 mg/92 mg after an additional 12 weeks of treatment. If at least 5 percent of baseline body weight has not been lost on 15 mg/92 mg, discontinue the medication gradually.

Phendimetrazine (Bontril PDM, Adipost, Phendiet, Statobex). Phendimetrazine is a sympathomimetic amine that is closely related to the amphetamines. It is classified by the Drug Enforcement Agency (DEA) as a Schedule III controlled substance.

The overall prescribing of this agent is limited. The most commonly used formulation is the 105-mg extended-release capsule, which approximates the action of three 35-mg immediate-release doses taken at 4-hour intervals. The average half-life of elimination, when studied under controlled conditions, is about 3.7 hours for both the extended-release and immediate-release forms. The absorption half-life of the drug from the immediate-release 35-mg phendimetrazine tablets is appreciably more rapid than the absorption rate of the drug from the extended-release formulation. The primary route of elimination is via the kidneys, where most of the drug and metabolites are excreted.

Phendimetrazine contraindications are similar to those of phentermine. They include a history of cardiovascular disease (e.g., coronary artery disease, stroke, arrhythmias, congestive heart failure, uncontrolled hypertension, pulmonary hypertension); use during or within 14 days

following the administration of MAOIs; hyperthyroidism; glaucoma; agitated states; a history of drug abuse; pregnancy; nursing; use in combination with other anorectic agents or CNS stimulants; and known hypersensitivity or idiosyncratic reactions to sympathomimetics. Given the lack of systematic research, phendimetrazine should not be used in combination with over-the-counter preparations and herbal products that claim to promote weight loss.

Phendimetrazine tartrate was considered pregnancy Category X in the former FDA classification system and is contraindicated during pregnancy because weight loss offers no potential benefit to a pregnant woman and may result in fetal harm. Studies with phendimetrazine tartrate sustained release have not been performed to evaluate carcinogenic potential, mutagenic potential, or effects on fertility.

Interactions may occur with MAOIs, alcohol, insulin, and oral hypoglycemic agents. Phendimetrazine may decrease the hypotensive effect of adrenergic neuron blocking drugs. The effectiveness and safety of phendimetrazine in pediatric patients have not been established. It is not recommended in patients less than 17 years of age.

Adverse reactions reported with phendimetrazine include sweating, flushing, tremor, insomnia, agitation, dizziness, headache, psychosis, and blurred vision. Elevated BP, palpitations, and tachycardia are frequent. GI side effects include dry mouth, nausea, stomach pain, diarrhea, and constipation. Genitourinary side effects include frequency, dysuria, and changes in libido.

Phendimetrazine tartrate is related chemically and pharmacologically to the amphetamines. Amphetamines and related stimulant drugs have been extensively abused, and the possibility of abuse of phendimetrazine should be kept in mind when evaluating the desirability of including a drug as part of a weight reduction program.

Acute overdose with phendimetrazine may manifest itself by restlessness, confusion, belligerence, hallucinations, and panic states. Fatigue and depression usually follow central stimulation. Cardiovascular effects include tachycardia, arrhythmias, hypertension or hypotension, and circulatory collapse. GI symptoms include nausea, vomiting, diarrhea, and abdominal cramps. Poisoning may result in convulsions, coma, and death. The management of acute overdose is mainly symptomatic. It includes lavage and sedation with a barbiturate. If hypertension is marked, the use of a nitrate or

rapid-acting α-receptor–blocking agent should be considered.

Diethylpropion (Tenuate). Diethylpropion preceded its analog, the antidepressant drug bupropion (Wellbutrin). Diethylpropion comes in two formulations: a 25-mg tablet and a 75-mg extended-release tablet (Tenuate Dospan). It is usually taken three times a day, 1 hour before meals (regular tablets), or once a day in midmorning (extended-release tablets). The extended-release tablets should be swallowed whole, never crushed, chewed, or cut. The maximum daily dose is 75 mg.

Side effects include dry mouth, unpleasant taste, restlessness, anxiety, dizziness, depression, tremors, upset stomach, vomiting, and increased urination. Side effects that warrant medical attention include tachycardia, palpitations, blurred vision, skin rash, itching, difficulty breathing, chest pain, fainting, swelling of the ankles or feet, fever, sore throat, chills, and painful urination. Diethylpropion was classified as pregnancy Category B in the former FDA classification system and has low abuse potential. It is listed as a Schedule IV drug by the DEA.

Orlistat (Xenical, Alli). Orlistat interferes with the absorption of dietary fats, causing reduced caloric intake. It works by inhibiting gastric and pancreatic lipases, the enzymes that break down triglycerides in the intestine. When lipase activity is blocked, triglycerides from the diet are not hydrolyzed into absorbable free fatty acids and are excreted undigested instead. Only trace amounts of orlistat are absorbed systemically; it is almost entirely eliminated through the feces.

The effectiveness of orlistat in promoting weight loss is definite, though modest. When used as part of a weight loss program, between 30 and 50 percent of patients can expect a 5 percent or more significant decrease in body mass. About 20 percent achieve at least a 10 percent decrease in body mass. After orlistat is stopped, up to a third of people gain the weight they lose.

Among the benefits of orlistat treatment is a decrease in BP and a reduced risk of developing type 2 diabetes.

The most common subjective side effects of orlistat are GI related and include steatorrhea, flatulence, fecal incontinence, and frequent or urgent bowel movements. Patients should avoid foods with high-fat content to

minimize these effects. Also, they should eat a reduced-calorie diet. Because of this side effect, orlistat can be used with high-fat content diets to treat constipation that results from treatment with some psychotropic drugs, such as the TCAs. Side effects are most severe when beginning therapy and may decrease in frequency with time. Hepatic and renal injuries are potentially severe side effects of orlistat use. In 2010, new safety information about rare cases of severe liver injury was added to the product label of orlistat. The rate of acute kidney injury is more common among orlistat users than nonusers. It should be used with caution in patients with impaired liver function and renal function, as well as those with an obstructed bile duct and pancreatic disease. Orlistat is contraindicated in malabsorption syndromes, hypersensitivity to orlistat, reduced gallbladder function, and in pregnancy and breastfeeding. Orlistat was rated pregnancy Category X in the former FDA classification system.

The use of orlistat inhibits the absorption of fat-soluble vitamins and other fat-soluble nutrients. Multivitamin supplements that contain vitamins A, D, E, K, as well as β-carotene, should be taken once a day, preferably at bedtime.

Orlistat can reduce plasma levels of the immunosuppressant cyclosporine (Sandimmune), so the two drugs should, therefore, not be administered concomitantly. Orlistat can also impair the absorption of the antiarrhythmic amiodarone (Nexterone).

At the standard prescription dose of 120 mg three times daily before meals, orlistat prevents approximately 30 percent of dietary fat from being absorbed. Higher doses have not been shown to produce more pronounced effects.

An over-the-counter formulation of orlistat (Alli) is available as 60-mg capsules—half the dosage of prescription orlistat.

Drugs without U.S. Food and Drug Administration Approval for Weight Loss

Topiramate. Topiramate and zonisamide (Zonegran) are discussed more fully among the mood stabilizers earlier in this chapter, but are mentioned here because both agents can have a substantial effect on weight loss.

Topiramate is approved as an antiepileptic drug and for prevention of migraine headaches in adults. The degree of weight loss associated with

topiramate may be comparable to the weight loss that other FDA-approved antiobesity drugs induce. Small studies and extensive anecdotal reports indicate that topiramate can help to offset weight gain associated with SSRIs and SGA drugs. Its impact on body weight may be due to its effects on both appetite suppression and satiety enhancement. These may be the result of a combination of pharmacologic effects including augmenting GABA activity, modulation of voltage-gated ion channels, inhibition of excitatory glutamate receptors, or inhibition of carbonic anhydrase.

The duration and dosage of treatment affect the weight loss benefits of topiramate. Weight loss is higher when the drug is prescribed at doses of 100 to 200 mg/day for more than a month compared with less than a month. In a large study, it was shown that compared to those who took a placebo, topiramate-treated patients were seven times more likely to lose more than 10 percent of their body weight. In clinical practice, many patients experience weight loss at a starting dose of 25 mg/day.

The most common side effects of topiramate are paresthesias, typically around the mouth, impaired taste (taste perversion), and psychomotor disturbances, including slowed cognition and reduced physical movements. Concentration and memory impairment, often characterized by word finding and name recall problems, is often reported. Some patients may experience emotional lability and mood changes. Medical side effects include increased risk of kidney stones and acute-angle closure glaucoma. Patients should report any change in visual acuity. Those with a history of kidney stones should be instructed to drink adequate amounts of fluid.

Topiramate is available as 25-, 50- 100-, and 200-mg tablets and as 15-, 25-, and 50-mg capsules.

Zonisamide. Zonisamide is a sulfonamide-related drug, similar in many ways to topiramate. Its exact mechanism of action is not known. Like topiramate, it can cause cognitive problems, but the incidence is lower than that with topiramate.

Zonisamide was assigned to pregnancy Category C in the former FDA classification system. Animal studies have revealed evidence of teratogenicity. Fetal abnormalities or embryo-fetal deaths have been reported in animal tests at zonisamide dosage and maternal plasma levels similar to, or lower than, human therapeutic levels. Therefore, the use of this drug in

human pregnancy may expose the fetus to significant risk.

The most common side effects include drowsiness, loss of appetite, dizziness, headache, nausea, and agitation or irritability. Zonisamide has also been associated with hypohidrosis. There is a 2 to 4 percent risk of kidney stones. Other drugs known to provoke stones, such as topiramate or acetazolamide (Diamox), should not be combined with zonisamide. Severe, but rare, adverse drug reactions include Stevens–Johnson syndrome, toxic epidermal necrolysis, and metabolic acidosis.

Typical dosing for weight loss has not been established. Generally, zonisamide is started at 100 mg at night for 2 weeks, and increased by 100 mg daily every 2 weeks to a target dose of 200 to 600 mg/day in one or two daily doses.

Metformin (Glucophage). Metformin is a medication for type 2 diabetes mellitus. Its actions include reduction of hepatic glucose production, reduced intestinal glucose absorption, increased insulin sensitivity, and improved peripheral glucose uptake and regulation. It does not increase insulin secretion.

When used as an adjunct to SGAs, it has consistently been shown to reduce body weight and waist circumference. Metformin probably has the best evidence of therapeutic benefit for the treatment of antipsychotic drug-induced metabolic syndrome. In several studies, metformin has been shown to attenuate or reverse some of the weight gain induced by antipsychotics. The degree of effect on body weight compares favorably with the effect of other treatment options that are approved for weight reduction. The weight loss effect of adjunctive metformin appears to be stronger in drug-naive patients treated with SGA medications. This effect is most evident for those being treated with clozapine (Clozaril) and olanzapine (Zyprexa). Based on the existing evidence, if weight gain occurs after SGA initiation, despite lifestyle intervention, metformin should be considered.

Common side effects include nausea, vomiting, abdominal pain, and loss of appetite. GI side effects can be mitigated by dividing the dose, taking the drug after meals, or using delayed-release formulations.

One serious treatment risk is that of lactic acidosis. This side effect is more common in those with reduced renal function than other patients. Although very rare (approximately 9 in 100,000 persons/yr), it has a 50 percent

mortality rate. Alcohol use, along with metformin, can increase the risk of acidosis. Renal function monitoring and alcohol avoidance are essential.

The weight loss effects of metformin are also evident in chronically ill patients with schizophrenia. Long-term use of metformin appears to be safe and effective.

There is no established dose range for metformin when used as an adjunct for weight loss. In most reports, the usual dose ranged from 500 to 2,000 mg/day. The maximum dose used in treating diabetes is 850 mg three times daily. Patients usually start with a low dose to see how the drug affects them.

Metformin is available in 500-, 850-, and 1,000-mg tablets, all now generic. Metformin SR (slow release) or XR (extended-release) is available in 500- and 750-mg strengths. These formulations are intended to reduce GI side effects and to increase patient compliance by reducing the pill burden.

Amphetamine. Amphetamine is a psychostimulant approved for the treatment of ADHD and narcolepsy. It has the effect of reducing appetite and has been used off label for that purpose for many years. Some of the drugs discussed above have amphetamine-like properties, which account for their effectiveness. Amphetamines and other psychostimulants are discussed earlier in the stimulant section of this chapter.

▲ 21.11 Nutritional Supplements and Related

NUTRITIONAL SUPPLEMENTS AND MEDICAL FOODS

Thousands of herbal and dietary supplements are being marketed today. Some are purported to have psychoactive properties. A number have even shown promise in the treatment of specific psychiatric symptoms. Although certain compounds may be beneficial, in many cases, the quantity and quality of data have been insufficient to make definitive conclusions. Nevertheless, some patients prefer to use these substances in place of, or in conjunction with, standard pharmaceutical treatments. If electing to use herbal drugs or nutritional supplements, bear in mind that their use may come at the expense

of proven interventions and that adverse effects are possible. Though more research is needed, information published to date is still of clinical interest in diagnosing and treating patients who may be taking dietary supplements.

Additionally, herbal and nonherbal supplements may augment or antagonize the actions of prescription and nonprescription drugs. Thus, clinicians need to remain informed on the latest research involving these substances. Because of the paucity of clinical trials, the clinician must be extraordinarily alert to the possibility of adverse effects as a result of drug–drug interactions, mainly if psychotropic agents are prescribed, because many phytomedicines have ingredients that produce physiologic changes in the body.

Nutritional Supplements

In the United States, the term *nutritional supplement* is used interchangeably with the term *dietary supplement*. The Dietary Supplement Health and Education Act (DHSEA) of 1994 defined nutritional supplements as items taken by mouth that contain a "dietary ingredient" meant to supplement the diet. These ingredients may include vitamins, minerals, herbs, botanicals, amino acids, and substances such as enzymes, tissues, glandulars, and metabolites. By law, such products must be labeled as supplements and may not be marketed as conventional food.

The DSHEA places dietary supplements in a select category, and therefore the regulations governing them are laxer than those for prescription and over-the-counter drugs. Unlike pharmaceutical drugs, nutritional supplements do not need the approval of the U.S. FDA, and the FDA does not evaluate their effectiveness. Because the FDA does not regulate dietary supplements, the contents and quality on store shelves vary dramatically. Contamination, mislabeling, and misidentification of herbs and supplements are problems. Table 21-62 provides a list of dietary supplements used in psychiatry.

Medical Foods

In recent years the FDA has introduced a new category of nutritional supplement called *medical foods*. According to the FDA, medical food, as defined in the Orphan Drug Act, is "a food which is formulated to be consumed or administered enterally under the supervision of a physician and

which is intended for the specific dietary management of a disease or condition for which distinctive nutritional requirements, based on recognized scientific principles, are established by medical evaluation."

A clear distinction can be made between the regulatory classifications of medical foods and dietary supplements. Medical foods must be shown, by medical evaluation, to meet the distinctive nutritional needs of a specific population of patients with a specific disease being targeted. Dietary supplements, on the other hand, are intended for normal, healthy adults and may not require proof of the efficacy of the finished product. Medical foods are distinguished from the broader category of foods for special dietary use and from foods that make health claims by the requirement that medical foods are to be used under medical supervision.

Medical foods do not have to undergo premarket approval by the FDA. However, medical food firms must comply with other requirements, such as acceptable manufacturing practices and registration of food facilities. Medical foods do have some additional regulations that dietary supplements do not because medical foods are intended to treat illnesses. For example, a compliance program requires annual inspections of all medical food manufacturers.

In summary, to be considered a medical food, a product must, at a minimum, meet the following criteria: (1) The product must be a food for oral or tube feeding; (2) the product must be labeled for the dietary management of a specific medical disorder, disease, or condition for which there are distinctive nutritional requirements; and (3) the product must be intended to be used under medical supervision. The most common medical foods with psychoactive claims are listed in Table 21-63.



Table 21-62
Dietary Supplements Used in Psychiatry

Name	Ingredients/What Is It?	Uses	Adverse Effects	Interactions	Dosage	Comments
Docosahexaenoic acid (DHA)	Omega-3 polyunsaturated fatty acid	ADD, dyslexia, cognitive impairment, dementia	Anticoagulant properties, mild GI distress	Warfarin	Varies with indication	Stop using prior to surgery
Choline	Choline	Fetal brain development, manic conditions, cognitive disorders, tardive dyskinesia, cancers	Restrict in patients with primary genetic trimethyluria, sweating, hypotension, depression	Methotrexate, works with B_6, B_{12}, and folic acid in metabolism of homocysteine	300–1,200 mg, doses >3 g associated with fishy body odor	Needed for structure and function of all cells
L-α-Glyceryl-phosphorylcholine (α-GPC)	Derived from soy lecithin	To increase growth hormone secretion, cognitive disorders	None known	None known	500 mg–1 g daily	Remains poorly understood
Phosphatidylcholine	Phospholipid that is part of cell membranes	Manic conditions, Alzheimer disease, and cognitive disorders, tardive dyskinesia	Diarrhea, steatorrhea in those with malabsorption, avoid with antiphospholipid antibody syndrome	None known	3–9 g/day in divided doses	Soybeans, sunflower, and rapeseed are major sources
Phosphatidylserine	Phospholipid isolated from soya and egg yolks	Cognitive impairment including Alzheimer disease, may reverse memory problems	Avoid with antiphospholipid antibody syndrome, GI side effects	None known	For soya-derived variety, 100 mg tid	Type derived from bovine brain carries hypothetical risk of bovine spongiform encephalopathy
Zinc	Metallic element	Immune impairment, wound healing, cognitive disorders, prevention of neural tube defects	GI distress, high doses can cause copper deficiency, immunosuppression	Bisphosphonates, quinolones, tetracycline, penicillamine, copper, cysteine-containing foods, caffeine, iron	Typical dose 15 mg/day, adverse effects >30 mg	Claims that zinc can prevent and treat the common cold are supported in some studies but not in others; more research needed
Acetyl-L-carnitine	Acetyl ester of L-carnitine	Neuroprotection, Alzheimer disease, Down syndrome, strokes, antiaging, depression in geriatric patients	Mild GI distress, seizures, increased agitation in some with Alzheimer disease	Nucleoside analogs, valproic acid, and pivalic acid–containing antibiotics	500 mg–2 g daily in divided doses	Found in small amounts in milk and meat
Huperzine A	Plant alkaloid derived from Chinese club moss	Alzheimer disease, age-related memory loss, inflammatory disorders	Seizures, arrhythmias, asthma, irritable bowel disease	Acetylcholinesterase inhibitors and cholinergic drugs	60 µg–200 µg/day	*Huperzia serrata* has been used in Chinese folk medicine for the treatment of fevers and inflammation
NADH (nicotinamide adenine dinucleotide)	Dinucleotide located in mitochondria and cytosol of cells	Parkinson disease, Alzheimer disease, chronic fatigue, CV disease	GI distress	None known	5 mg/day or 5 mg bid	Precursor of NADH is nicotinic acid
S-Adenosyl-L-methionine (SAMe)	Metabolite of essential amino acid L-methionine	Mood elevation, osteoarthritis	Hypomania, hyperactive muscle movement, caution in patients with cancer	None known	200–1,600 mg daily in divided doses	Several trials demonstrate some efficacy in the treatment of depression
5-Hydroxytryptophan (5-HTP)	Immediate precursor of serotonin	Depression, obesity, insomnia, fibromyalgia, headaches	Possible risk of serotonin syndrome in those with carcinoid tumors or taking MAOIs	SSRIs, MAOIs, methyldopa, St. John's wort, phenoxybenzamine, 5-HT antagonists, 5-HT receptor agonists	100 mg–2 g daily, safer with carbidopa	5-HTP along with carbidopa is used in Europe for the treatment of depression
Phenylalanine	Essential amino acid	Depression, analgesia, vitiligo	Contraindicated in patients with PKU, may exacerbate tardive dyskinesia or hypertension	MAOIs and neuroleptic drugs	Comes in 2 forms: 500 mg–1.5 g daily for DL-phenylalanine, 375 mg–2.25 g for D-phenylalanine	Found in vegetables, juices, yogurt, and miso
Myoinositol	Major nutritionally active form of inositol	Depression, panic attacks, OCD	Caution in patients with bipolar disorder, GI distress	Possible additive effects with SSRIs and 5-HT receptor agonists (sumatriptan)	12 g in divided doses for depression and panic attacks	Studies have *not* shown effectiveness in treating Alzheimer disease, autism, or schizophrenia
Vinpocetine	Semisynthetic derivative of vincamine (plant derivative)	Cerebral ischemic stroke, dementias	GI distress, dizziness, insomnia, dry mouth, tachycardia, hypotension, flushing	Warfarin	5–10 mg daily with food, no more than 20 mg/day	Used in Europe, Mexico, and Japan as pharmaceutical agent for treatment of cerebrovascular and cognitive disorders
Vitamin E family	Essential fat-soluble vitamin, family made of tocopherols and tocotrienols	Immune-enhancing, antioxidant, some cancers, protection in CV disease, neurologic disorders, diabetes, premenstrual syndrome	May increase bleeding in those with propensity to bleed, possible increased risk of hemorrhagic stroke, thrombophlebitis	Warfarin, antiplatelet drugs, neomycin, may be additive with statins	Depends on form: tocotrienols, 200–300 mg daily with food; tocopherols, 200 mg/day	Stop members of vitamin E family 1 mo prior to surgical procedures
Glycine	Amino acid	Schizophrenia, alleviating spasticity, and seizures	Avoid in those who are anuric or have hepatic failure	Additive with antispasmodics	1 g/day in divided doses for supplement; 40–90 g/day for schizophrenia	
Melatonin	Hormone of pineal gland	Insomnia, sleep disturbances, jet lag, cancer	May inhibit ovulation in 1-g doses, seizures, grogginess, depression, headache, amnesia	Aspirin, NSAIDs, β-blockers, INH, sedating drugs, corticosteroids, valerian, kava kava, 5-HTP, alcohol	0.3–3 mg hs for short periods of time	Melatonin sets the timing of circadian rhythms and regulates seasonal responses
Fish oil	Lipids found in fish	Bipolar disorder, lowering triglycerides, hypertension, decrease blood clotting	Caution in hemophiliacs, mild GI upset, "fishy"-smelling excretions	Coumadin, aspirin, NSAIDs, garlic, ginkgo	Varies depending on form and indication—usually about 3–5 g daily	Stop prior to any surgical procedure

ADD, attention-deficit disorder; CV, cardiovascular; OCD, obsessive-compulsive disorder; GI, gastrointestinal; hs, at night; MAOIs; monamine oxidase inhibitors; PKU, phenylketonuria; SSRIs, serotonin reuptake inhibitors; NSAIDs, nonsteroidal anti-inflammatory drugs; INH, isoniazid; 5-HTP, 5-hydroxytryptophan; tid, three times a day; bid, twice a day.
Table by Mercedes Blackstone, M.D.

Table 21-63

Some Common Medical Foods

Medical Food	Indication	Mechanism of Action
Caprylic-triglyceride (Axona)	Alzheimer disease	Increases plasma concentration of ketones as an alternative energy source in the brain; metabolized in the liver
L-methylfolate (Deplin)	Depression	Regulates synthesis of serotonin, norepinephrine, and dopamine; adjunctive to selective serotonin reuptake inhibitors (SSRIs); 15 mg/day
S-adenosyl-L-methionine (SAMe)	Depression	Naturally occurring molecule involved in synthesis of hormones and neurotransmitters including serotonin and norepinephrine
L-Tryptophan	Sleep disturbance Depression	Essential amino acid; precursor of serotonin; reduces sleep latency; usual dose 4–5 g/day
Omega-3 fatty acid	Depression Cognition	Eicosapentaenoic (EPA) and docosahexaenoic (DHA) acids; direct effect on lipid metabolism; used for augmentation of antidepressant drugs
Theramine (Sentra)	Sleep disturbances Cognitive enhancer	Cholinergic modulator; increases acetylcholine and glutamate
N-Acetylcysteine	Depression Obsessive-	Amino acid that attenuates glutamatergic

	compulsive disorder	neurotransmission; used to augment SSRIs
L-Tyrosine	Depression	Amino acid precursor to biogenic amines epinephrine and norepinephrine
Glycine	Depression	Amino acid that activates *N*-methyl-D-aspartate (NMDA) receptors; may facilitate excitatory transmission in the brain
Citicoline	Alzheimer disease Ischemic brain injury	Choline donor involved in synthesis of brain phosopholipids and acetylcholine; 300–1,000 mg/day; may improve memory
Acetyl L-carnitine (Alcar)	Alzheimer disease Memory loss	Antioxidant that may prevent oxidative damage in the brain

PHYTOMEDICINES

The term *phytomedicines* (from the Greek *phyto*, meaning "plant") refers to herb and plant preparations that are used or have been used for centuries for the treatment of a variety of medical conditions. Phytomedicines are categorized as dietary supplements, not drug products, and are therefore exempt from the regulations that govern prescription and over-the-counter medications. Manufacturers of phytomedicines are not required to provide the FDA with safety information before marketing a product or give the FDA postmarketing safety reports. Thousands of herbal drugs are being marketed today; the most common with psychoactive properties are listed in Table 21-64. Ingredients, when identified, are listed, as are indications, adverse events, dosages, and comments, particularly on interactions with commonly prescribed drugs used in psychiatry. For example, St. John's wort (*wort* is an old English word meaning "root or herb" and a familiar term for home

brewers), which is used to treat depression, decreases the effectiveness of certain psychotropic drugs such as amitriptyline (Elavil), alprazolam (Xanax), paroxetine (Paxil), and sertraline (Zoloft), among others. Kava kava, which is used to treat anxiety states, has been associated with liver toxicity.

Adverse Effects

Adverse effects are possible, and toxic interactions with other drugs may occur with all phytomedicines, dietary supplements, and medicinal foods. Adulteration is possible, especially with phytomedicines. There are few or no consistent standard preparations available for most herbs. The FDA does not test medical foods; however, strict voluntary compliance is required. Safety profiles and knowledge of adverse effects of most of these substances have not been studied rigorously, however. Because of the paucity of clinical trials, all of these agents should be avoided during pregnancy; some herbs may act as abortifacients, for example. Because most of these substances or their metabolites are secreted in breast milk, they are contraindicated during lactation.

Clinicians should always attempt to obtain a history of herbal use or the use of medical foods or nutritional supplements during the psychiatric evaluation.

It is important to be nonjudgmental in dealing with patients who use these substances. Many do so for various reasons: (1) as part of their cultural tradition, (2) because they mistrust physicians or are dissatisfied with conventional medicine, or (3) because they experience relief of symptoms with the particular substance. Because patients will be more cooperative with traditional psychiatric treatments if they are allowed to continue using their preparations, psychiatrists should keep an open mind and not attribute all effects to suggestion. If psychotropic agents are prescribed, the clinician must be extraordinarily alert to the possibility of adverse effects as a result of drug–drug interactions, because many of these compounds have ingredients that produce physiologic changes in the body.



Table 21-64
Phytomedicinals with Psychoactive Effects

Name	Ingredients	Use	Adverse Effects[a]	Interactions	Dosage[a]	Comments
Arctic weed, golden root	MAOI and β endorphin	Anxiolytic, mood enhancer, antidepressant	No side effect yet documented in trials		100 mg bid to 200 mg tid	Use caution with drugs that mimic MAOIs
Areca, areca nut, betel nut, *L. Areca catechu*	Arecoline, guvacoline	For alteration of consciousness to reduce pain and elevate mood	Parasympathomimetic overload: increased salivation, tremors, bradycardia, spasms, GI disturbances, ulcers of the mouth	Avoid with parasympathomimetic drugs; atropine-like compounds reduce effect	Undetermined; 8–10 g is toxic dose for humans	Used by chewing the nut; used in the past as a chewing balm for gum disease and as a vermifuge; long-term use may result in malignant tumors of the oral cavity
Ashwaganda	Also called Indian Winter Cherry or Indian Ginseng, native to India Flavonoids	Anticoxidant, may decrease anxiety levels Improved libido in men and women May lower levels of the stress hormone cortisol	Drowsiness and sleepiness	None	Dosage is 1 tablet twice daily before meals with a gradual increase to 4 tablets per day	None
Belladonna, *L. Atropa belladonna*, deadly nightshade	Atropine, scopolamine, flavonoids[b]	Anxiolytic	Tachycardia, arrhythmias, xerostomia, mydriasis, difficulties with micturition and constipation	Synergistic with anticholinergic drugs; avoid with TCAs, amantadine, and quinidine	0.05–0.10 mg a day; maximum single dose is 0.20 mg	Has a strong smell, tastes sharp and bitter, and is poisonous
Biota, *Platycladus orientalis*	Plant derivative	Used as a sedative. Other uses are to treat heart palpitations, panic, night sweats, and constipation. May be useful in ADHD	No known adverse effects	None	No clear established doses exist	None
Bitter orange flower, *citrus aurantium*	Flavonoids, limonene	Sedative, anxiolytic, hypnotic	Photosensitization	Undetermined	Tincture, 2–3 g/day; drug, 4–6 g/day; extract, 1–2 g/day	Contradictory evidence; some refer to it as a gastric stimulant
Black cohosh, *L. Cimicifuga racemosa*	Triterpenes, isoferulic acid	For PMS, menopausal symptoms, dysmenorrhea	Weight gain, GI disturbances	Possible adverse interaction with male or female hormones	1–2 g/day; over 5 g can cause vomiting, headache, dizziness, cardiovascular collapse	Estrogen-like effects questionable because root may act as an estrogen-receptor blocker
Black haw, cramp bark, *L. Viburnum prunifolium*	Scopoletin, flavonoids, caffeic acids, triterpenes	Sedative, antispasmodic action on uterus; for dysmenorrhea	Undetermined	Anticoagulant-enhanced effects	1–3 g/day	Insufficient data
California poppy, *L. Eschscholtzia californica*	Isoquinoline alkaloids, cyanogenic glycosides	Sedative, hypnotic, anxiolytic; for depression	Lethargy	Combination of California poppy, valerian, St. John's wort, and passion flowers can result in agitation	2 g/day	Clinical or experimental documentation of effects is unavailable
Casein	Casein peptides	Used as antistress agent; May improve sleep	Usually consumed though milk products. May interact with antihypertensive medicine and lower the blood pressure. May cause drowsiness and should be avoided when taking alcohol or benzodiazepines	None	One to two tablets once or twice daily	
Catnip, *L. Nepeta cataria*	Valeric acid	Sedative, antispasmodic; for migraine	Headache, malaise, nausea, hallucinogenic effects	Undetermined	Undetermined	Delirium produced in children
Chamomile, *L. Matricaria chamomilla*	Flavonoids	Sedative, anxiolytic	Allergic reaction	Undetermined	2–4 g/day	May be GABAergic
Coastal water hyssop		Anxiolytic, sedative, epilepsy, asthma	Mild GI discomfort	May stimulate	300–450 mg qid	Insufficient data
Cordyceps sinensis	A genus of fungi that includes about 400 described species, found primarily in the high altitudes of the Tibetan plateau in China; Antioxidant	Has been used for weakness, fatigue, to improve sexual drive in the elderly	GI discomfort, dry mouth, and nausea	None	Dosage in ranges of 3–6 g daily	None
Corydalis, *L. Corydalis cava*	Isoquinoline alkaloids	Sedative, antidepressant; for mild depression	Hallucination, lethargy	Undetermined	Undetermined	Clonic spasms and muscular tremor with overdose
Cyclamen, *L. Cyclamen europaeum*	Triterpene	Anxiolytic; for menstrual complaints	Small doses (e.g., 300 mg) can lead to nausea, vomiting, and diarrhea	Undetermined	Undetermined	High doses can lead to respiratory collapse
Echinacea, *L. Echinacea purpurea*	Flavonoids, polysaccharides, caffeic acid derivatives, alkamides	Stimulates immune system; for lethargy, malaise, respiratory infections and lower UTIs	Allergic reaction, fever, nausea, vomiting	Undetermined	1–3 g/day	Use in HIV and AIDS patients is controversial; may not be effective in coryza
Ephedra, ma-huang *L. Ephedra sinica*	Ephedrine, pseudoephedrine	Stimulant; for lethargy, malaise, diseases of respiratory tract	Sympathomimetic overload: arrhythmias, increased BP, headache, irritability, nausea, vomiting	Synergistic with sympathomimetics, serotonergic agents; avoid with MAOIs	1–2 g/day	Tachyphylaxis and dependence can occur (taken off market)
Ginkgo, *L. Ginkgo biloba*	Flavonoids, ginkgolide A, B	Symptomatic relief of delirium, dementia; improves concentration and memory deficits; possible antidote to SSRI-induced sexual dysfunction	Allergic skin reactions, GI upset, muscle spasms, headache	Anticoagulant: use with caution because of its inhibitory effect on PAF; increased bleeding possible	120–240 mg/day	Studies indicate improved cognition in persons with Alzheimer disease after 4–5 wk of use, possibly because of increased blood flow

Ginseng, *L. Panax ginseng*	Triterpenes, ginsenosides	Stimulant; for fatigue, elevation of mood, immune system	Insomnia, hypertonia, and edema (called ginseng abuse syndrome)	Not to be used with sedatives, hypnotic agents, MAOIs, antidiabetic agents, or steroids	1–2 g/day	Several varieties exist; Korean (most highly valued), Chinese, Japanese, American (*Panax quinquefolius*)
Heather, *L. Calluna vulgaris*	Flavonoids, triterpenes	Anxiolytic, hypnotic	Undetermined	Undetermined	Undetermined	Efficacy for claimed uses is not documented
Holy Basil formula, *Ocimum tenuiflorum*	*Ocimum tenuiflorum*, an aromatic plant native to the tropics, part of the *Lamiaceae* family Flavonoids	Used to combat stress, also used for common colds, headaches, stomach disorders, inflammation, heart disease	No data exist regarding the long-term effects. May prolong clotting time, increase the risk of bleeding during surgery, and lower blood sugar	None	Dosage depends on the formulation type, recommended dose is 2 softgel capsules taken with 8 oz water daily	None
Hops, *L. Humulus lupulus*	Humulone, lupulone, flavonoids	Sedative, anxiolytic, hypnotic; for mood disturbances, restlessness	Contraindicated in patients with estrogen-dependent tumors (breast, uterine, cervical)	Hyperthermia effects with phenothiazine antipsychotics and with CNS depressants	0.5 g/day	May decrease plasma levels of drugs metabolized by CYP450 system
Horehound, *L. Ballota nigra*	Diterpenes, tannins	Sedative	Arrhythmias, diarrhea, hypoglycemia, possible spontaneous abortions	May enhance serotonergic drug effects, may augment hypoglycemic effects of drugs	1–4 g/day	May cause abortion
Jambolan, *L. Syzygium cumini*	Oleic acid, myristic acid, palmitic and linoleic acids, tannins	Anxiolytic, antidepressant	Undetermined	Undetermined	1–2 g/day	In folk medicine, a single dose is 30 seeds (1.9 g) of powder
Kanna, *Sceletium tortuosum*	Alkaloid, mesembrine	Anxiolytic, mood enhancer, empathogen, COPD treatment	Sedation, vivid dreams, headache	Potentiates cannabis, PDE inhibitor	50–100 mg	Insufficient data
Kava kava, *L. Piperis methysticum*	Kava lactones, kava pyrone	Sedative, hypnotic, antispasmodic	Lethargy, impaired cognition, dermatitis with long-term usage, liver toxicity	Synergistic with anxiolytics, alcohol; avoid with levodopa and dopaminergic agents	600–800 mg/day	May be GABAergic; contraindicated in patients with endogenous depression; may increase the danger of suicide
Kratom, *Mitragyna speciosa*	Alkaloid	Stimulant and depressant	Priapism, testicular enlargement, withdrawal, depression, fatigue, insomnia	Structurally similar to yohimbine	Undetermined	Chewed, extracted into water, tar formulations
Lavender, *L. Lavandula angustifolia*	Hydroxycoumarin, tannins, caffeic acid	Sedative, hypnotic	Headache, nausea, confusion	Synergistic with other sedatives	3–5 g/day	May cause death in overdose
Lemon balm, sweet Mary, *L. Melissa officinalis*	Flavonoids, caffeic acid, triterpenes	Hypnotic, anxiolytic, sedative	Undetermined	Potentiates CNS depressant; adverse reaction with thyroid hormone	8–10 g/day	Insufficient data
L-methylfolate	Folate is a B vitamin found in some foods, needed to form healthy cells, especially red blood cells. L-methylfolate and levomefolate are names for the active form of folic acid	Adjunctive L is used for major depression, not an antidepressant when used alone. Folate and L- methylfolate are also used to treat folic acid deficiency in pregnancy, to prevent spinal cord birth defects	GI side effects reported	None	15 mg once a day by mouth with or without food	Considered a "medical food" by the FDA and only available by prescription. Safe to take during pregnancy when used as directed
Mistletoe, *L. Viscum album*	Flavonoids, triterpenes, lectins, polypeptides	Anxiolytic; for mental and physical exhaustion	Berries said to have emetic and laxative effects	Contraindicated in patients with chronic infections (e.g., tuberculosis)	10 per day	Berries have caused death in children
Mugwort, *L. Artemisia vulgaris*	Sesquiterpene lactones, flavonoids	Sedative, antidepressant, anxiolytic	Anaphylaxis, contact dermatitis, may cause hallucinations	Potentiates anticoagulants	5–15 g/day	May stimulate uterine contractions, can induce abortion
N-acetylcysteine (NAC)	Amino acid	Used as an antidote for acetaminophen overdose, treatment of trichotillomania	Rash, cramps, and angioedema may occur	Activated charcoal, ampicillin, carbamazepine, cloxacillin, oxacillin, nitroglycerin, and penicillin G	1,200–2,400 mg/day	Acts as an antioxidant and a glutamate modulating agent. When used as an antidote for acetaminophen overdose, the dose is 20–40 times higher than those used in OCD trials. It has not been shown to be effective in treating schizophrenia
Nux vomica, *L. Strychnos nux vomica*, poison nut	Indole alkaloids: strychnine and brucine, polysaccharides	Antidepressant; for migraine, menopausal symptoms	Convulsions, liver damage, death; severely toxic because of strychnine	Undetermined	0.02–0.05 g/day	Symptoms of poisoning can occur after ingestion of one bean; lethal dose is 1–2 g
Oats, *L. Avena sativa*	Flavonoids, oligo and polysaccharides	Anxiolytic, hypnotic; for stress, insomnia, opium, and tobacco withdrawal	Bowel obstruction or other bowel dysmotility syndromes, flatulence	Undetermined	3 g/day	Oats have sometimes been contaminated with aflatoxin, a fungal toxin linked with some cancers

Omega-3 fatty acid	Comes in three forms, eicosapentaenoic acid (EPA), docosahexaenoic acid (DHA), and alpha-linolenic acid (LNA)	Used as a supplement in the treatment of heart disease, high cholesterol, high blood pressure. May also be helpful in treatment of depression, bipolar disorder, schizophrenia, and ADHD. May reduce the risk of ulcers when used in conjunction with NSAID pain relievers	Can cause gas, bloating, belching, and diarrhea	May increase effectiveness of blood thinners, may increase fasting blood sugar levels when used with diabetes medications such as insulin and metformin	Doses vary from 1 to 4 g/day	Can be contaminated with mercury and PCBs
Passion flower, *L. Passiflora incarnata*	Flavonoids, cyanogenic glycosides	Anxiolytic, sedative, hypnotic	Cognitive impairment	Undetermined	4–8 g/day	Overdose causes depression
Phosphatidylserine and Phosphatidylcholine	Phospholipids	Used for Alzheimer disease, age-related decline in mental function, improving thinking skills in young people, ADHD, depression, preventing exercise-induced stress, and improving athletic performance	Insomnia and stomach upset	None	100 mg three times daily	None
Polygala	Polygala is a genus of about 500 species of flowering plants belonging to the family *Polygalaceae*, commonly known as milkwort or snakeroot	Used for insomnia, forgetfulness, mental confusion, palpitation, seizures, anxiety, and listlessness	Contraindicated in patients who have ulcers or gastritis, should not be used long term	None	Dosage of polygala is 1.5–3 g of dried root, 1.5–3 g of a fluid extract, or 2.5–7.5 g of a tincture. A polygala tea can also be made, with a maximum of three cups per day	None
Rehmannia	Iridoid glycosides	Stimulates the release of cortisol. Used in lupus, rheumatoid arthritis (RA), fibromyalgia, and multiple sclerosis. May improve asthma and urticaria. Used to treat menopause, hair loss, and impotence	Loose bowel movements, bloating, nausea, and abdominal cramps	None	Exact dosage unknown	None
Rhodiola rosea	Potentiator, monoterpene alcohols, flavonoids					
S-adenosyl-L-methionine (SAMe)	S-adenosyl-L-methionine	Used for arthritis and fibromyalgia, may be effective as an augmentation strategy for SSRI in depression	GI symptoms, anxiety, nightmares, insomnia, and worsening of Parkinson symptoms	Use with SSRIs or SNRIs may result in serotonin syndrome. Interacts with levodopa, meperidine, pentazocine, and tramadol	400–1,600 mg/day	A naturally occurring molecule made from the amino acid methionine and ATP, serves as a methyl donor in human cellular metabolism
Scarlet Pimpernel, *L. Anagallis arvensis*	Flavonoids, triterpenes, cucurbitacins, caffeic acids	Antidepressant	Overdose or long-term doses may lead to gastroenteritis and nephritis	Undetermined	1.8 g of powder four times a day	Flowers are poisonous
Skullcap, *L. Scutellaria latwiflora*	Flavonoid, monoterpenes	Amiolytic, sedative, hypnotic	Cognitive impairment, hepatotoxicity	Disulfiram-like reaction may occur if used with alcohol	1–2 g/day	Little information exists to support the use of this herb in humans
St. John's wort, *L. Hypericum perforatum*	Hypericin, flavonoids, xanthones	Antidepressant, sedative, anxiolytic	Headaches, photosensitivity (may be severe), constipation	Report of manic reaction when used with sertraline (Zoloft); do not combine with SSRIs or MAOIs: possible serotonin syndrome; do not use with alcohol, opioids	100–950 mg/day	Mixed results on effectiveness for depression; may act as MAOI or SSRI; 4- to 6-wk trial for mild depressive moods; if no apparent improvement, another therapy should be tried
Strawberry leaf, *L. Fragaria vesca*	Flavonoids, tannins	Anxiolytic	Contraindicated with strawberry allergy	Undetermined	1 g/day	Little information exists to support the use of this herb in humans
Tarragon, *L. Artemisia dracunculus*	Flavonoids, hydroxycoumarins	Hypnotic, appetite stimulant	Undetermined	Undetermined	Undetermined	Little information exists to support the use of this herb in humans
Valerian, *L. Valeriana officinalis*	Valepotriates, valerenic acid, caffeic acid	Sedative, muscle relaxant, hypnotic	Cognitive and motor impairment, GI upset, hepatotoxicity; long-term use: contact allergy, headache, restlessness, insomnia, mydriasis, cardiac dysfunction	Avoid concomitant use with alcohol or CNS depressants	1–2 g/day	May be chemically unstable
Wild lettuce, *Lactuca Virosa*	Flavonoids, coumarins, lactones	Sedative, anesthetic, galactagogue	Tachycardia, tachypnea, visual disturbance, diaphoresis		Undetermined	Bitter taste, added to salad or drinks, active compound closely resembles opium
Winter cherry, *withania somnifera*	Alkaloids, steroidal lactones	Sedative, treatment for arthritis, possible anticarcinogenic	Thyrotoxicosis, unfavorable effects on heart and adrenal gland		Undetermined	Smoke inhaled

ADHD, attention-deficit/hyperactivity disorder; AIDS, acquired immunodeficiency syndrome; ATP, adenosine triphosphate; bid, twice a day; BP, blood pressure; CNS, central nervous system; COPD, chronic obstructive pulmonary disease; FDA, U.S. Food and Drug Administration; GABA, γ-aminobutyric acid; GI, gastrointestinal; MAOI, monoamine oxidase inhibitor; PAF, platelet-activating factor; PCB, polychlorinated biphenyl; PDE, phosphodiesterase; PMS, premenstrual syndrome; NSAID, nonsteroidal anti-inflammatory drug; OCD, obsessive-compulsive disorder; qid, four times a day; SNRI, serotonin and norepinephrine reuptake inhibitor; SSRI, selective serotonin reuptake inhibitor; TCA, tricyclic antidepressant; tid, three times a day; UTI, urinary tract infection.

[a]There are no reliable, consistent, or valid data exist on dosages or adverse effects of most phytomedicinals.

[b]Flavonoids are common to many herbs. They are plant byproducts that act as antioxidants (i.e., agents that prevent the deterioration of material such as deoxyribonucleic acid [DNA] via oxidation).

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Introduction

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22 Other Somatic Therapies

ELECTROCONVULSIVE THERAPY

Convulsive therapies for major psychiatric illnesses predate the modern therapeutic era, with the use of camphor reported as early as the 16th century and the existence of several accounts of camphor convulsive therapies from the late 1700s to the mid-1800s.

Unaware of the history of camphor convulsive therapy, the Hungarian neuropsychiatrist Ladislas von Meduna observed that the brains of people with epilepsy had more glial cells than other brains. In contrast, the brains of patients with schizophrenia had fewer glial cells, and von Meduna hypothesized that there might be a biologic antagonism between convulsions and schizophrenia. Following animal experimentation, camphor was (again) selected as the appropriate agent to use for the therapeutic induction of seizures. In 1934, the first catatonic psychotic patient was successfully treated using intramuscular injections of camphor in oil to produce therapeutic seizures. Later, other chemicals were used, including pentylenetetrazol. Insulin coma was also used in the 1940s and 1950s, which also sometimes induced seizures. Lucio Bini and Ugo Cerletti were interested in the use of electricity to induce seizures, and, after a series of animal experiments and observation of the use of electricity commercially, they were able to safely apply current across the heads of animals for this purpose. In 1938, the first electroconvulsive treatment was administered to a delusional and incoherent patient, who improved with one treatment and remitted after 11 treatments. Electrical induction of convulsive therapy could be made more reliable and shorter-acting than chemically induced convulsive therapies, and, by the early 1940s, it had replaced them. In 1940, the first use of electroconvulsive

therapy (ECT) occurred in the United States.

Explorations of nondominant electrode placement and alternative, more efficient waveforms were then undertaken to reduce the retrograde memory problems that persisted for some patients after the initial recovery period post-ECT. The practice of ECT also benefited from the introduction of controlled trials methodology, which demonstrated its safety and efficacy, and from refinements made in diagnostic systems and the process of informed consent. In the 1980s and 1990s, efforts to ensure uniformly high standards of practice were underway with the publication of recommendations for treatment delivery, education, and training by professional organizations in the United States, England, Scandinavia, and Canada, among others.

With the widespread use of pharmacologic agents as first-line treatments for major psychiatric disorders, ECT is now more commonly used for patients with resistance to those treatments, except in the case of life-threatening illness due to inanition, severe suicidal symptoms, or catatonia. Although subconvulsive stimulation was ineffective and a seizure is necessary, it is now known that there is a dose–response relationship with right unilateral ECT and bilateral ECT is likely to be ineffective with ultrabrief pulse widths. Work continues to explore the underlying mechanisms and biologic characteristics of effective ECT treatments, with interest in having the treatment focus on appropriate neural networks with a more efficient stimulus as a method of reducing cognitive side effects. With the growing understanding that depression is a chronic disease for many patients, more emphasis has been placed on continuation and maintenance treatments following an acute course of ECT. Utilization of ECT has diminished since the middle of the 20th century, but because ECT remains the most effective treatment for major depression and is a rapidly effective treatment for life-threatening psychiatric conditions, ECT, unlike its contemporaneous somatic therapies, such as insulin coma, remains in the active treatment portfolio of modern therapeutics. Its use has shifted from public to private institutions, and it is estimated that approximately 100,000 patients have received ECT annually over the past few decades in the United States (Table 22-1).

The Nobel Laureate Paul Greengard has suggested that the term *electrocortical therapy* might be used to replace the current term

electroconvulsive therapy (ECT). Greengard has acknowledged that if the mechanism of action of ECT, as yet unknown, turns out to be subcortical, then the term might have limited use. Until that time, however, the authors of this text think Greengard's suggestion deserves consideration. It would help diminish the fear associated with the word convulsion and help destigmatize a very effective treatment method.

Electrophysiology in Electroconvulsive Therapy

Neurons maintain a resting potential across the plasma membrane and may propagate an action potential, which is a transient reversal of the membrane potential. Regular brain activity is desynchronized; that is, neurons fire action potentials asynchronously. A convulsion, or seizure, occurs when a large percentage of neurons fire in unison. Such rhythmical changes in the extracellular potential entrain neighboring neurons, propagate the seizure activity across the cortex and into deeper structures, and eventually engulf the entire brain in high-voltage synchronous neuronal firing. Cellular mechanisms work to contain the seizure activity and to maintain cellular homeostasis, and the seizure eventually ends. In epilepsy, any of possibly several hundred genetic defects can alter the balance in favor of unrestrained activity. In ECT, seizures are triggered in normal neurons by application through the scalp of pulses of current, under conditions that are carefully controlled to create a seizure of a particular duration over the entire brain.

Ohm's law can describe the qualities of the electricity used in ECT: $E = IR$, or $I = E/R$, in which E is voltage, I is current, and R is resistance. The intensity or dose of electricity in ECT is measured in terms of charge (milliampere-seconds or millicoulombs) or energy (watt-seconds or joules). Resistance is synonymous with impedance, and, in the case of ECT, both the electrode's contact with the body and the nature of the bodily tissues are the significant determinants of resistance. The skull has a high impedance; the brain has a low impedance. Because scalp tissues are much better conductors of electricity than bone, only about 20 percent of the applied charge enters the skull to excite neurons. The ECT machines that are now widely used can be adjusted to administer the electricity under conditions of constant current, voltage, or energy.

Table 22-1
Milestones in the History of Convulsive Therapy

1500s	Paracelsus induces seizures by administering camphor (by mouth) to treat psychiatric illness.
1785	First published report of the use of seizure induction to treat mania, again using camphor.
1934	Ladislas von Meduna begins the modern era of convulsive therapy using intramuscular injection of camphor for catatonic schizophrenia. Camphor is soon replaced with pentylenetetrazol.
1938	Ugo Cerletti and Lucio Bini conduct the first electrical induction of a series of seizures in a catatonic patient and produce a successful treatment response
1940s	ECT is introduced to the United States.
	Curare developed for use as a muscle relaxant at ECT.
1951	Introduction of succinylcholine.
1958	First controlled study of unilateral ECT.
1960	Attenuation of seizure expression with an anticonvulsant agent (lidocaine) reduces the efficacy of ECT. Subconvulsive treatment produces only weak clinical responses; the hypothesis that seizure activity is necessary and sufficient for efficacy is upheld.
1960s	Randomized clinical trials of the efficacy of ECT versus medications in the treatment of depression yield response rates that are significantly higher with ECT.
	Comparisons of neuroleptics and ECT show that neuroleptic medication is superior for acute treatment, although ECT may be more effective in the long term.
1970	The most common electrode positioning for right unilateral

	ECT developed.
1976	A constant current, brief pulse ECT device, the prototype for modern devices, is developed.
1978	The American Psychiatric Association publishes the first Task Force Report on ECT with the aim of establishing standards for consent and the technical and clinical aspects of the conduct of ECT.
Late 1970s–early 1980s	Randomized, controlled trials demonstrate that ECT is more effective than sham treatment for major depression.
1985	The National Institutes of Health and National Institute of Mental Health Consensus Conference on ECT endorse a role for the use of ECT and advocate research and national standards of practice.
1987	The belief that the seizure in itself is sufficient for clinical response is challenged by H. A. Sackheim and collaborators, who report that the combination of dosage just above seizure threshold and right unilateral electrode placement, while producing a seizure of sufficient duration, is ineffective.
1988	Randomized, controlled clinical trials of ECT versus lithium demonstrate them to be equally effective in mania.
2000	In controlled trials, the dose–response relationship for right unilateral ECT is validated; high-dose right unilateral and bilateral ECT show equal response rates in major depression, but right unilateral electrode placement is associated with fewer adverse cognitive effects.
	Convulsive treatment is induced with magnetic stimulation by S. H. Lisanby and colleagues.

2001	The largest modern controlled trial of relapse prevention post-ECT with continuation pharmacotherapy demonstrates a significantly better outcome for combined treatment with a tricyclic antidepressant (nortriptyline) plus lithium compared with nortriptyline alone or placebo during the first 6 months post-ECT.

ECT, electroconvulsive therapy.

Mechanism of Action

The induction of a bilateral generalized seizure is necessary for both the beneficial and adverse effects of ECT. Although a seizure superficially seems as though it is an all-or-none event, some data indicate that not all generalized seizures involve all the neurons in deep brain structures (e.g., the basal ganglia and the thalamus); recruitment of these deep neurons may be necessary for full therapeutic benefit. After the generalized seizure, the electroencephalogram (EEG) shows about 60 to 90 seconds of postictal suppression. This period is followed by high-voltage delta and theta waves, with a return to preseizure activity in about 30 minutes. During a series of ECT treatments, the interictal EEG is generally slower and of greater amplitude than usual, but the EEG returns to pretreatment appearance 1 month to 1 year after the end of the course of treatment.

Positron emission tomography (PET) studies of both cerebral blood flow and glucose use have shown that during seizures, cerebral blood flow, use of glucose and oxygen, and permeability of the blood–brain barrier increase. After the seizure, blood flow and glucose metabolism are decreased, perhaps most markedly in the frontal lobes. Some research indicates that the degree of decrease in cerebral metabolism is correlated with therapeutic response.

Seizure foci in idiopathic epilepsy are hypometabolic during interictal periods; ECT itself acts as an anticonvulsant because its administration is associated with an increase in the seizure threshold as treatment progresses. Recent data suggest that for 1 to 2 months following a session of ECT, EEGs record a massive increase in slow-wave activity located over the prefrontal cortex in patients who responded well to the ECT. High-intensity, bilateral stimulation produced the best response; low-intensity, unilateral stimulation,

the weakest. These data are of unclear significance, however, because the specific EEG correlate disappears 2 months after ECT, whereas the clinical benefit persists.

ECT affects the cellular mechanisms of memory and mood regulation and raises the seizure threshold. The latter effect may be blocked by the opioid antagonist naloxone.

Neurochemical research into the mechanisms of action of ECT has focused on changes in neurotransmitter receptors and, recently, changes in second-messenger systems. Virtually every neurotransmitter system is affected by ECT, but a series of ECT sessions results in the downregulation of postsynaptic β-adrenergic receptors, the same receptor change observed with virtually all antidepressant treatments. The effects of ECT on serotonergic neurons remain controversial. Various research studies have reported an increase in postsynaptic serotonin receptors, no change in serotonin receptors, and a change in the presynaptic regulation of serotonin release. ECT has also been reported to effect changes in the muscarinic, cholinergic, and dopaminergic neuronal systems. In second-messenger systems, ECT has been reported to affect the coupling of G-proteins to receptors, the activity of adenylyl cyclase and phospholipase C, and the regulation of calcium entry into neurons.

Recently, there has been increased interest in structural changes in the brain associated with psychiatric syndromes and response to treatment. This interest has been particularly so for microscopic changes associated with electroconvulsive stimulation, as well as antidepressants and other medications. In animals, mostly rodents, synaptic plasticity in the hippocampus, including mossy fiber sprouting, alterations in cytoskeletal structure, increased connectivity in perforant pathways, promotion of neurogenesis, and suppression of apoptosis have been observed. We also see many of these structural events, although less often, with antidepressant medications such as fluoxetine. These reports have also galvanized controversy over various aspects of the technical validity of the observations. It is unknown whether such changes occur clinically and, if they do, what significance to efficacy and cognitive side effects might be discovered.

Indications

Major Depressive Disorder. The most common indication for ECT is major depressive disorder, for which ECT is the fastest and most effective available therapy. ECT should be considered for use in patients who have failed medication trials, have not tolerated medications, have severe or psychotic symptoms, are acutely suicidal or homicidal, or have marked symptoms of agitation or stupor. Controlled studies have shown that up to 70 percent of patients who fail to respond to antidepressant medications may respond positively to ECT. Table 22-2 presents the indications for the use of ECT.

ECT is effective for depression in both major depressive disorder and bipolar I disorder. Delusional or psychotic depression has long been considered exceptionally responsive to ECT, but recent studies have indicated that major depressive episodes with psychotic features are no more responsive to ECT than nonpsychotic depressive disorders. Nevertheless, because major depressive episodes with psychotic features respond poorly to antidepressant pharmacotherapy alone, ECT should be considered much more often as the first-line treatment for patients with the disorder. Major depressive disorder with melancholic features (e.g., markedly severe symptoms, psychomotor retardation, early morning awakening, diurnal variation, decreased appetite and weight, and agitation) is considered likely to respond to ECT. ECT is particularly indicated for persons who are severely depressed, who have psychotic symptoms, who show suicidal intent, or who refuse to eat. Depressed patients less likely to respond to ECT include those with somatic symptom disorder. Elderly patients tend to respond to ECT more slowly than do young patients. ECT is a treatment for major depressive episode and does not provide prophylaxis unless it is administered on a long-term maintenance basis.



Table 22-2
Indications for the Use of Electroconvulsive Therapy

Major diagnostic indications
Major depressive disorder
Mania, including mixed episodes
Schizophrenia with acute exacerbation or catatonic subtype

Schizoaffective disorder

Other diagnostic indications

Parkinson disease

Neuroleptic malignant disorder

Clinical indications

Primary use

Rapid definitive response required on medical or psychiatric grounds

Risks of alternative treatments outweigh benefits

Past history of poor response to medications or good response to ECT

Patient preference

Secondary use

Failure to respond to pharmacotherapy in the current episode

Intolerance of pharmacotherapy in the current episode

Rapid definitive response necessitated by deterioration of the patient's condition

ECT, electroconvulsive therapy.

Manic Episodes. ECT is at least equal to lithium for the treatment of acute manic episodes. The pharmacologic treatment of manic episodes, however, is so effective in the short term and for prophylaxis that the use of ECT to treat manic episodes is generally limited to situations with specific contraindications to all available pharmacologic approaches. The relative rapidity of the ECT response indicates its usefulness for patients whose manic behavior has produced dangerous levels of exhaustion. ECT should not be used for a patient who is receiving lithium because lithium can lower the seizure threshold and may be more likely to predispose the patient to delirium.

Schizophrenia. Although an effective treatment for the symptoms of acute schizophrenia, ECT is not for those of chronic schizophrenia. Patients with schizophrenia who have marked positive symptoms, catatonia, or affective symptoms are considered most likely to respond to ECT. In such patients, the efficacy of ECT is about equal to that of antipsychotics, but improvement

may occur faster.

Other Indications. Small studies have found ECT effective in the treatment of catatonia, a symptom associated with mood disorders, schizophrenia, and medical and neurologic disorders. ECT is also reportedly useful to treat episodic psychoses, atypical psychoses, obsessive-compulsive disorder (OCD), and delirium and such medical conditions as a neuroleptic malignant syndrome, hypopituitarism, intractable seizure disorders, and the on–off phenomenon of Parkinson disease. ECT may also be the treatment of choice for depressed, suicidal pregnant women who cannot take medication. It also may be the treatment of choice for geriatric and medically ill patients who cannot take antidepressant drugs safely. It may also be appropriate for severely depressed and suicidal children and adolescents who may be less likely to respond to antidepressant drugs compared with adults. ECT is not effective in somatic symptom disorder (unless accompanied by depression), personality disorders, or anxiety disorders.

Clinical Guidelines

Patients and their families are often apprehensive about ECT; therefore, clinicians must explain both beneficial and adverse effects and alternative treatment approaches. The informed-consent process should be documented in the patients' medical records and should include a discussion of the disorder, its natural course, and the option of receiving no treatment. Printed literature and videotapes about ECT may be useful in attempting to obtain genuinely informed consent. The use of involuntary ECT is rare today and should be reserved for patients who urgently need treatment and who have a legally appointed guardian who has agreed to its use. Clinicians must know local, state, and federal laws about the use of ECT.

Pretreatment Evaluation. Pretreatment evaluation should include standard physical, neurologic, and preanesthesia examinations and complete medical history. Laboratory evaluations should include blood and urine chemistries, a chest x-ray, and an electrocardiogram (ECG). A dental examination to assess the state of patients' dentition is advisable for elderly patients and patients who have had inadequate dental care. An x-ray of the spine is needed if other evidence of a spinal disorder is seen. Computed

tomography (CT) or magnetic resonance imaging (MRI) should be performed if a clinician suspects the presence of a seizure disorder or a space-occupying lesion. Practitioners of ECT no longer consider even a space-occupying lesion to be an absolute contraindication to ECT, but with such patients, the procedure should be performed only by experts.

CONCOMITANT MEDICATIONS. Patients' ongoing medications should be assessed for possible interactions with the induction of a seizure, for effects (both positive and negative) on the seizure threshold, and drug interactions with the medications used during ECT. Most antidepressants except bupropion, monoamine oxidase inhibitors, and antipsychotics are generally acceptable. Benzodiazepines should be withdrawn because of their anticonvulsant activity; lithium should be withdrawn because it can result in increased postictal delirium and can prolong seizure activity; clozapine and bupropion should be withdrawn because they are associated with the development of late-appearing seizures. Lidocaine should not be administered during ECT because it markedly increases the seizure threshold; theophylline is contraindicated because it increases the duration of seizures. Reserpine is also contraindicated because it is associated with the further compromise of the respiratory and cardiovascular systems during ECT.

Premedications, Anesthetics, and Muscle Relaxants. Patients should not be given anything orally for 6 hours before treatment. Just before the procedure, the patient's mouth should be checked for dentures and other foreign objects, and an intravenous (IV) line should be established. A bite block is inserted in the mouth just before the treatment is administered to protect the patient's teeth and tongue during the seizure. Except for the brief interval of electrical stimulation, 100 percent oxygen is administered at a rate of 5 L a minute during the procedure until spontaneous respiration returns. Emergency equipment for establishing an airway should be immediately available in case it is needed.

MUSCARINIC ANTICHOLINERGIC DRUGS. Muscarinic anticholinergic drugs are administered before ECT to minimize oral and respiratory secretions and to block bradycardias and asystoles unless the resting heart rate is above 90 beats a minute. Some ECT centers have stopped the routine use of

anticholinergics as premedications, although their use is still indicated for patients taking β-adrenergic receptor antagonists and those with ventricular ectopic beats. The most commonly used drug is atropine, which can be administered 0.3 to 0.6 mg intramuscularly (IM) or subcutaneously (SC) 30 to 60 minutes before the anesthetic or 0.4 to 1.0 mg IV 2 or 3 minutes before the anesthetic. An option is to use glycopyrrolate (0.2 to 0.4 mg IM, IV, or SC), which is less likely to cross the blood–brain barrier and less likely to cause cognitive dysfunction and nausea, although it is thought to have less cardiovascular protective activity than does atropine.

ANESTHESIA. The administration of ECT requires general anesthesia and oxygenation. The depth of anesthesia should be as light as possible, not only to minimize adverse effects but also to avoid elevating the seizure threshold associated with many anesthetics. Methohexital (0.75 to 1.0 mg/kg IV bolus) is the most commonly used anesthetic because of its shorter duration of action and lower association with postictal arrhythmias than thiopental (usual dose 2 to 3 mg/kg IV). However, this difference in cardiac effects is not universally accepted. Four other anesthetic alternatives are etomidate, ketamine, alfentanil, and propofol. Etomidate (0.15 to 0.3 mg/kg IV) is sometimes used because it does not increase the seizure threshold; this effect is particularly useful for elderly patients because the seizure threshold increases with age. Ketamine (6 to 10 mg/kg IM) is sometimes used because it does not increase the seizure threshold, although its use is limited by the association of psychotic symptoms with emergence from anesthesia with this drug. Alfentanil (2 to 9 mg/kg IV) is sometimes coadministered with barbiturates to allow the use of low doses of the barbiturate anesthetics and, thus, reduce the seizure threshold less than usual, although its use can be associated with an increased incidence of nausea. Propofol (0.5 to 3.5 mg/kg IV) has anticonvulsant properties.

MUSCLE RELAXANTS. After the onset of the anesthetic effect, usually within a minute, a muscle relaxant is administered to minimize the risk of bone fractures and other injuries resulting from motor activity during the seizure. The goal is to produce profound relaxation of the muscles, not necessarily to paralyze them, unless the patient has a history of osteoporosis or spinal injury or has a pacemaker and, therefore, is at risk for injury related to motor

activity during the seizure. Succinylcholine, an ultrafast-acting depolarizing blocking agent, has gained virtually universal acceptance for the purpose. Succinylcholine is usually administered in a dose of 0.5 to 1 mg/kg as an IV bolus or drip. Because succinylcholine is a depolarizing agent, its action is marked by the presence of muscle fasciculations, which move in a rostrocaudal progression. The disappearance of these movements in the feet or the absence of muscle contractions after peripheral nerve stimulation indicates maximal muscle relaxation. In some patients, tubocurarine (3 mg IV) is administered to prevent myoclonus and increases in potassium and muscle enzymes; these reactions can be a problem in patients with musculoskeletal or cardiac disease. A blood pressure cuff may be inflated at the ankle to apply pressure over the systolic pressure before infusion of the muscle relaxant to allow observation of relatively innocuous seizure activity in the foot muscles to monitor the duration of the convulsion.

If a patient has a known history of pseudocholinesterase deficiency, atracurium (0.5 to 1 mg/kg IV) or curare can be used instead of succinylcholine. In such a patient, the metabolism of succinylcholine is disrupted, and prolonged apnea may necessitate emergency airway management. In general, however, because of the short half-life of succinylcholine, the duration of apnea after its administration is generally shorter than the delay in regaining consciousness caused by the anesthetic and the postictal state.

ELECTRODE PLACEMENT. Historically, most practitioners have used bifrontotemporal electrode placement because of its reliability in producing efficacy and its ease of use. This electrode placement is also associated with more short- and long-term adverse cognitive effects and is more likely to produce delirium, which may require interrupting a course of ECT and perhaps even terminating it before optimal therapeutic effects have been obtained. Hence, when bifrontotemporal ECT is used, attention should be paid to restricting the dose to a moderately suprathreshold level to attenuate adverse cognitive effects as much as possible. It should be emphasized that the combination of ultrabrief pulse and bifrontotemporal electrode placement has not been demonstrated to be effective. Treatment with bilateral electrode placements, particularly a bifrontal configuration, is more likely to manifest EEG seizure without a motor seizure, and EEG monitoring can be

particularly useful in detecting its occurrence.

Newer electrode placements include bifrontal configuration and asymmetrical placements. There are limitations to these strategies, imposed by the fact that the high impedance of the skull and scalp causes the spreading of the electrical stimulus and restricts possibilities for localization of the stimulus. Bifrontal electrode placement, located sufficiently lateral to minimize interference with impedance relations, has been investigated. There have been several demonstrations that bifrontal electrode placements are equally effective as bifrontotemporal and adequately dosed right unilateral electrode configurations. Evidence of advantages in sparing of cognitive effects is quite preliminary, and adequately powered investigations with more extensive and sensitive cognitive batteries are needed. The seizure threshold is likely to be relatively higher with bifrontal ECT.

The relatively better cognitive side effect profile of right unilateral ECT (also called the d'Elia placement) should encourage wider use now that the efficacy of this electrode placement can be ensured with adequate dosing strategies. In contrast to bilateral ECT, a dose closer to 500 percent above the seizure threshold is more likely to ensure efficacy. ECT devices in the United States are restricted to an output in the range of 504 to 576 mCi. Approximately 90 percent of patients have seizure thresholds that can accommodate optimal dosing with brief pulse right unilateral ECT, and the combination of right unilateral electrode placement with ultrabrief pulse width extends the range of US devices, so that most patients can be treated within these constraints. Individuals with an exceptionally high seizure threshold may require bilateral electrode placements to remain within the device restrictions. Maximizing interelectrode distance by using the d'Elia placement may also be optimal. Many other right unilateral placements have been described, but there is little work to support their use (Fig. 22-1).

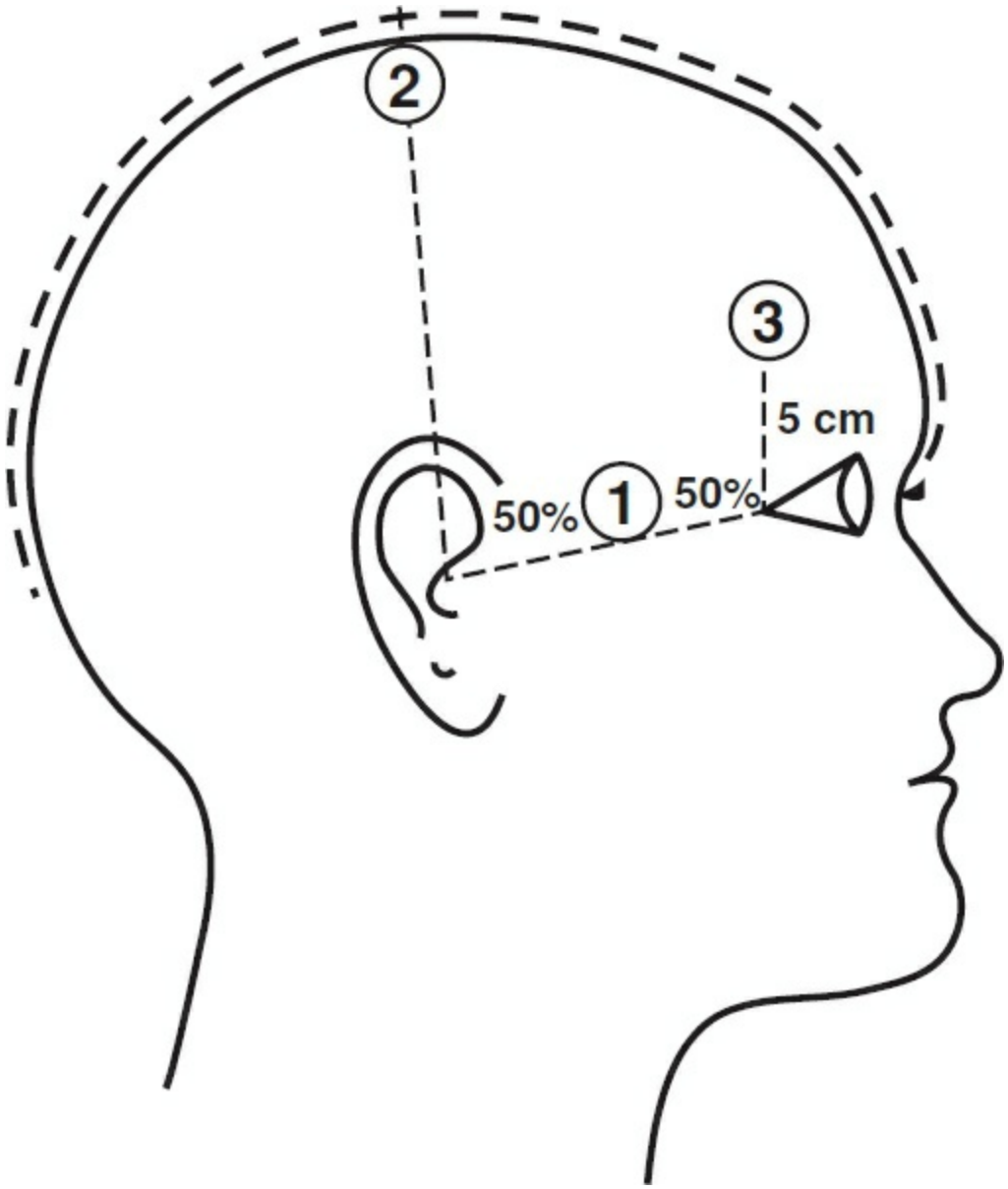


FIGURE 22-1
Electrode placements. Position 1 represents the frontotemporal position, used for both electrodes, one on each side of the head, in conducting bilateral electroconvulsive therapy (ECT). For right unilateral ECT, one electrode is in the right frontotemporal position, and the other is just to the right of the vertex at position 2. (Courtesy of American Psychiatric Association, with permission.)

There has been some concern that left-handed patients may require different electrode placement than right-handed patients, especially if a unilateral placement is desired. Even when handedness is lateralized to the left, the anatomic localization of language function in 70 percent of left-handed individuals is the same as in those who are right-handed.

Furthermore, there is evidence for independent lateralization of affect, with the right hemisphere involved in sustaining depressed mood regardless of handedness. Because of limited indications that affective function and efficacy of ECT are associated with handedness, handedness is not generally used to guide the choice of electrode placement.

Electrical Stimulus. The electrical stimulus must be sufficiently strong to reach the seizure threshold (the level of intensity needed to produce a seizure). The electrical stimulus is given in cycles, and each cycle contains a positive and a negative wave. Old machines use a sine wave; however, this type of machine is now considered obsolete because of the inefficiency of that wave shape. When a sine wave is delivered, the electrical stimulus in the sine wave before the seizure threshold is reached and after the seizure is activated is unnecessary and excessive. Modern ECT machines use a brief pulse waveform that administers the electrical stimulus, usually in 1 to 2 milliseconds, at a rate of 30 to 100 pulses a second. Machines that use an ultrabrief pulse (0.5 milliseconds) are not as effective as brief pulse machines.

Establishing a patient's seizure threshold is not straightforward. A 40 times variability in seizure thresholds occurs among patients. Also, during ECT treatment, a patient's seizure threshold may increase 25 to 200 percent. The seizure threshold is also higher in men than in women and higher in older than in younger adults. A common technique is to initiate treatment at an electrical stimulus that is thought to be below the seizure threshold for a particular patient and then to increase this intensity by 100 percent for unilateral placement and by 50 percent for bilateral placement until the seizure threshold is reached. A debate in the literature concerns whether a minimally suprathreshold dose, a moderately suprathreshold dose (one and a half times the threshold), or a high suprathreshold dose (three times the threshold) is preferable. The debate about stimulus intensity resembles the debate about electrode placement. Essentially, the data support the conclusion that doses of three times the threshold are the most rapidly effective and that minimal suprathreshold doses are associated with the fewest and least severe cognitive adverse effects.

Induced Seizures. A brief muscular contraction, usually strongest in a

patient's jaw and facial muscles, is seen concurrently with the flow of stimulus current, regardless of whether a seizure occurs. The first behavioral sign of the seizure is often a plantar extension, which lasts 10 to 20 seconds and marks the tonic phase. This phase is followed by rhythmic (i.e., clonic) contractions that decrease in frequency and finally disappear. The tonic phase is marked by high-frequency, sharp EEG activity on which a higher-frequency muscle artifact may be superimposed. During the clonic phase, bursts of polyspike activity coincide with the muscular contractions but usually persist for at least a few seconds after the clonic movements stop.

MONITORING SEIZURES. A physician must have an objective measure that a bilateral generalized seizure has occurred after the stimulation. The physician should be able to observe either some evidence of tonic–clonic movements or electrophysiologic evidence of seizure activity from the EEG or electromyogram (EMG). Seizures with unilateral ECT are asymmetrical, with higher ictal EEG amplitudes over the stimulated hemisphere than over the nonstimulated hemisphere. Occasionally, unilateral seizures are induced; for this reason, at least a single pair of EEG electrodes should be placed over the contralateral hemisphere when using unilateral ECT. For a seizure to be effective in the course of ECT, it should last at least 25 seconds.

FAILURE TO INDUCE SEIZURES. If a particular stimulus fails to cause a seizure of sufficient duration, up to four attempts at seizure induction can be tried during a course of treatment. The onset of seizure activity is sometimes delayed as long as 20 to 40 seconds after the stimulus administration. If a stimulus fails to result in a seizure, the contact between the electrodes and the skin should be checked, and the intensity of the stimulus should be increased by 25 to 100 percent. The clinician can also change the anesthetic agent to minimize increases in the seizure threshold caused by the anesthetic. Additional procedures to lower the seizure threshold include hyperventilation and administration of 500 to 2,000 mg IV of caffeine sodium benzoate 5 to 10 minutes before the stimulus.

PROLONGED AND TARDIVE SEIZURES. Prolonged seizures (seizures lasting more than 180 seconds) and status epilepticus can be terminated either with additional doses of the barbiturate anesthetic agent or with IV diazepam (5 to

10 mg). Management of such complications should be accompanied by intubation because the oral airway is insufficient to maintain adequate ventilation over an extended apneic period. Tardive seizures—that is, additional seizures appearing sometime after the ECT treatment—may develop in patients with preexisting seizure disorders. Rarely, ECT precipitates the development of an epileptic disorder in patients. Such situations should be clinically managed as if they were pure epileptic disorders.

Number and Spacing of Treatments. ECT treatments are usually administered two to three times a week; twice-weekly treatments are associated with less memory impairment than thrice-weekly treatments. In general, the course of treatment of major depressive disorder can take 6 to 12 treatments (although up to 20 sessions are possible); the treatment of manic episodes can take 8 to 20 treatments; the treatment of schizophrenia can take more than 15 treatments, and the treatment of catatonia and delirium can take as few as 1 to 4 treatments. Treatment should continue until the patient achieves what is considered the maximal therapeutic response. Further treatment does not yield any therapeutic benefit but increases the severity and duration of the adverse effects. The point of maximal improvement is usually thought to occur when a patient fails to continue to improve after two consecutive treatments. If a patient is not improving after 6 to 10 sessions, bilateral placement and high-density treatment (three times the seizure threshold) should be attempted before ECT is abandoned.

MULTIPLE-MONITORED ELECTROCONVULSIVE THERAPY. Multiple-monitored ECT (MMECT) involves giving multiple ECT stimuli during a single session, most commonly two bilateral stimuli within 2 minutes. This approach may be warranted in severely ill patients and those at exceptionally high risk from the anesthetic procedures. MMECT is associated with the most frequent occurrences of serious cognitive adverse effects.

Maintenance Treatment. A short-term course of ECT induces remission in symptoms but does not, of itself, prevent a relapse. Post-ECT maintenance treatment should always be considered. Maintenance therapy is generally pharmacologic, but maintenance ECT treatments (weekly, biweekly, or

monthly) have been reported to be effective relapse prevention treatments, although data from extensive studies are lacking. Indications for maintenance ECT treatments can include rapid relapse after initial ECT, severe symptoms, psychotic symptoms, and the inability to tolerate medications. If ECT was used because a patient was unresponsive to a specific medication, then, following ECT, the patient should be given a trial of different medications.

Failure of Electroconvulsive Therapy Trial. Patients who fail to improve after a trial of ECT should have another medication trial, even if that agent previously failed. Although the data are primarily anecdotal, many reports indicate that patients who had previously failed to improve while taking an antidepressant drug do improve while taking the same drug after receiving a course of ECT treatments, even if the ECT seemed to be a therapeutic failure. Nonetheless, with the increased availability of drugs that act at diverse receptor sites, it is less often necessary to return to a drug that has failed than it was formerly.

Adverse Effects

Contraindications. ECT has no absolute contraindications, only situations in which a patient is at increased risk and has an increased need for close monitoring. Pregnancy is not a contraindication for ECT, and fetal monitoring is generally considered unnecessary unless the pregnancy is high risk or complicated. Patients with space-occupying central nervous system lesions are at increased risk for edema and brain herniation after ECT. If the lesion is small, however, pretreatment with dexamethasone is given, and hypertension is controlled during the seizure, and the risk of serious complications minimized for these patients. Patients who have increased intracerebral pressure or are at risk for cerebral bleeding (e.g., those with cerebrovascular diseases and aneurysms) are at risk during ECT because of the increased cerebral blood flow during the seizure. This risk can be lessened, although not eliminated, by control of the patient’s blood pressure during the treatment. Patients with recent myocardial infarctions are another high-risk group, although the risk is greatly diminished 2 weeks after the myocardial infarction and is even further reduced 3 months after the infarction. Patients with hypertension should be stabilized on their

antihypertensive medications before ECT is administered. Propranolol and sublingual nitroglycerin can also be used to protect such patients during treatment.

Mortality. The mortality rate with ECT is about 0.002 percent per treatment and 0.01 percent for each patient. These numbers compare favorably with the risks associated with general anesthesia and childbirth. ECT death is usually from cardiovascular complications and is most likely to occur in patients whose cardiac status is already compromised.

Central Nervous System Effects. Common adverse effects associated with ECT are headache, confusion, and delirium shortly after the seizure while the patient is coming out of anesthesia. Marked confusion may occur in up to 10 percent of patients within 30 minutes of the seizure and can be treated with barbiturates and benzodiazepines. Delirium is usually most pronounced after the first few treatments and in patients who receive bilateral ECT or who have coexisting neurologic disorders. The delirium characteristically clears within days or a few weeks at the longest.

MEMORY. The most significant concern about ECT is the association between ECT and memory loss. About 75 percent of all patients given ECT say that memory impairment is the worst adverse effect. Although memory impairment during a course of treatment is almost the rule, follow-up data indicate that almost all patients are back to their cognitive baselines after 6 months. Some patients, however, complain of persistent memory difficulties. For example, a patient may not remember the events leading up to the hospitalization and ECT, and such autobiographical memories may never be recalled. The degree of cognitive impairment during treatment and the time it takes to return to baseline are related, in part, to the amount of electrical stimulation used during treatment. Memory impairment is most often reported by patients who have experienced little improvement with ECT. Despite the memory impairment, which usually resolves, no evidence indicates brain damage caused by ECT. This subject has been the focus of several brain-imaging studies, using a variety of modalities; virtually all concluded that permanent brain damage is not an adverse effect of ECT. Neurologists and epileptologists generally agree that seizures that last less

than 30 minutes do not cause permanent neuronal damage.

Other Adverse Effects of Electroconvulsive Therapy. Fractures often accompanied treatments in the early days of ECT. With routine use of muscle relaxants, fractures of long bones or vertebrae should not occur. Some patients, however, may break teeth or experience back pain because of contractions during the procedure. Muscle soreness can occur in some individuals, but it often results from the effects of muscle depolarization by succinylcholine and is most likely to be particularly troublesome after the first session in a series. This soreness can be treated with mild analgesics, including nonsteroidal anti-inflammatory drugs (NSAIDs). A significant minority of patients experience nausea, vomiting, and headaches following an ECT treatment. Nausea and vomiting can be prevented by treatment with antiemetics at the time of ECT (e.g., metoclopramide, 10 mg IV, or prochlorperazine, 10 mg IV; ondansetron is an acceptable alternative if adverse effects preclude the use of dopamine receptor antagonists).

ECT can be associated with headaches, although this effect is usually readily manageable. Headaches often respond to NSAIDs given in the ECT recovery period. In patients with severe headaches, pretreatment with ketorolac (30 to 60 mg IV), an NSAID approved for brief parenteral use, can be helpful. Acetaminophen, tramadol, and more potent analgesia provided by opioids can be used individually or in various combinations to manage more intractable headache. ECT can induce migrainous headache and related symptoms; sumatriptan (6 mg SC or 25 mg orally) may be a useful addition to the agents described above. Ergot compounds can exacerbate cardiovascular changes observed during ECT and probably should not be a component of ECT pretreatment.

Investigations in Electrical Brain Stimulation Treatment

There is interest in continued refinements of ECT techniques. Common themes in these approaches are focusing the treatment spatially to optimize dosing in brain areas associated with putative neural networks involved in depression and other psychopathologies that are indications for ECT, diminishing dosing in areas associated with adverse cognitive effects, and improving the efficiency of a noninvasive electrical stimulus in direction and

amplitude, even to a subconvulsive level. This research is parallel to investigations in magnetic stimulation (e.g., repetitive transcranial magnetic stimulation [TMS]) and the renaissance of invasive electrical techniques (e.g., vagal nerve stimulation and deep brain stimulation [DBS]).

OTHER BRAIN STIMULATION METHODS

Brain stimulation in psychiatric practice and research uses electrical currents or magnetic fields to alter neuronal firing. There is a growing list of tools capable of eliciting such neuromodulation, each with a different spectrum of action. These tools either apply electrical or magnetic fields transcranially or involve the surgical implantation of electrodes to deliver electrical currents to a cranial nerve or the brain directly. The transcranial techniques include cranial electrical stimulation (CES), ECT, transcranial direct current stimulation (tDCS, also called direct current polarization), TMS, and magnetic seizure therapy (MST). The surgical techniques include cortical brain stimulation (CBS), DBS, and vagus nerve stimulation (VNS).

In 1985, nearly 50 years after the first use of ECT, Anthony Barker and colleagues published on the first use of pulsed magnetic fields to stimulate the brain with a procedure called *transcranial magnetic stimulation.* TMS was initially used in neurology for studies of nerve conduction, but it quickly caught the attention of psychiatrists eager to explore other, less invasive alternatives to ECT. This nonconvulsive stimulation method through TMS is under active study, with some promising results in the treatment of various psychiatric disorders, including depression, anxiety, and schizophrenia. A convulsive treatment derived from the application of more powerful magnetic stimulation has been under investigation in nonhuman primates and human studies in both the United States and Europe. The first MST procedure was performed in an animal in 1998 and a human in 2000. MST is under development as a more focal means of inducing seizures in an attempt to retain the thus-far-unparalleled efficacy of ECT with fewer cognitive side effects.

Two more additions to brain stimulation methods, DBS and VNS, were introduced about a decade following the first trials of TMS. Both were first approved by the U.S. Food and Drug Administration (FDA) in 1997 in the realm of treating sequelae of neurologic syndromes. DBS was initially

approved for the treatment of essential tremor and Parkinson tremors, whereas VNS was approved for the treatment of epilepsy. Five years later, in 2002, indications for DBS were expanded to include treatment of all symptoms of Parkinson disease, including tremor, slowness, and stiffness, as well as involuntary movements induced by medications. TMS, DBS, and VNS originated in the field of neurology. Psychiatrists quickly saw the potential for those tools in the treatment of psychiatric conditions, however, and as a result of clinical trials in depression, VNS subsequently received FDA approval for the long-term adjunctive treatment of chronic or recurrent depression in adults. Also, human studies are underway to validate the efficacy of DBS in the treatment of depression and OCD.

Therapeutic Neuromodulation: Treating Psychiatric Disorders Through Brain Stimulation

Mechanism of Action

ELECTRICAL STIMULATION—COMMON PATHWAY. The brain stimulation modalities just reviewed generate either electrical or magnetic pulses. However, both of these share a common final pathway—they affect the neurons electrically. That electrical effect may either be through the direct application of electricity or the indirect induction of electricity via magnetic stimulation. The direct forms of electrical stimulation are exemplified in either *transcranial* delivery, as with ECT, CES, and tDCS, or *intracerebral* delivery, as in the case of DBS or direct cortical stimulation (epidural or subdural). The indirect forms of electrical stimulation include TMS and MST, which induce electrical fields in the brain through the application of alternating magnetic fields. Of note, both the epidural and intracerebral modalities are more focal than the transcranial application of electricity because electrodes are placed directly in the neuronal tissue, bypassing the impedance of the scalp and skull. The relatively more contemporary magnetic stimulation methods (TMS and MST) also bypass the impedance of the scalp and skull and are thus likewise more focal. However, magnetic stimulation is, in fact, an example of an indirect method of electrical brain stimulation, in that the changing magnetic fields from these devices induce electricity in the brain, the latter acting as a conductor, according to the principle first

described by Michael Faraday in a law that bears his name and later incorporated into James Clerk Maxwell's equations, which unify all of electromagnetism. The magnetic modalities achieve their enhanced focality noninvasively, in contrast to the intracerebral and epidural methods, and are thus at the center of intensive research in that they offer the promise of an unparalleled degree of spatial specificity without the need for surgery.

All but one of the brain stimulation modalities described here act by stimulating neurons. The one exception is tDCS, which does not stimulate but instead polarizes. In this sense, the "S" of tDCS is a misnomer. It is more accurate to conceptualize tDCS as exerting a polarizing effect that may alter the likelihood of neuronal firing.

The action of the subconvulsive modalities of stimulation relies on the effects of the repeated stimulation of the targeted neural circuitry. However, in the case of the convulsive modalities (ECT and MST), the action depends on the seizure induced by the stimulation and the effects of repeated seizure induction on brain processes. This is not to say that the form of stimulation that triggers the seizure does not affect the outcome. Indeed, it is well replicated that electrode placement and electrical stimulus parameters have a profound effect on the efficacy and side effects of ECT. Whether the same will be valid for MST is under active investigation.

ACUTE VERSUS PROLONGED EFFECTS. Brain stimulation can have immediate or lasting effects. A single electrical pulse delivered at sufficient intensity can induce depolarization, trigger an action potential, release neurotransmitters at the synapse, and result in transsynaptic propagation with subsequent activation of a functional circuit. For example, brain stimulation applied to the hand area of the primary motor cortex may activate the corticospinal tract and induce a muscle twitch in the contralateral hand. Such stimulation can result acutely in the induction of either a positive effect, as in the case of a muscle twitch or visualization of phosphenes, or a disruptive effect, as in the case of visual masking.

Repetitive pulses delivered at fixed frequencies can exert even more powerful effects. Epstein and colleagues described in 1999 how repetitive TMS (rTMS) applied to the language-dominant hemisphere induced an arrest of speech. After termination of the stimulation, speech returned to normal.

Some more invasive brain stimulation modalities, such as DBS or VNS,

are programmed to operate chronically, thus extending the acute action for as long as the stimulation is turned on. In the case of DBS, the pulses are typically given continuously at a high frequency, whereas in the case of VNS, the pulses are given in trains lasting up to 30 seconds and typically repeated every 5 minutes. The less invasive modalities, such as rTMS, tDCS, CES, and even ECT, presumably require the induction of some form of neuroplasticity for their effect to become lasting.

Transcranial Magnetic Stimulation

Definition. TMS is the application of a rapidly changing magnetic field to the superficial layers of the cerebral cortex, which locally induces small electric currents, also referred to as *eddy* currents. This induction was initially discovered by Faraday through his experiments in 1831 and later quantified in Maxwell's equations of electromagnetism. Thus, TMS may be referred to as electrical stimulation without an electrode, in that it uses magnetic fields to induce electrical pulses indirectly. TMS devices deliver intense magnetic pulses via a coil that is held on the scalp. Because magnetic fields are unaffected by the electrical impedance of the scalp and skull, this method of stimulation enables the focal stimulation of smaller areas of the brain than is possible with other noninvasive devices that use either alternating (ECT, CES) or direct (tDCS) electrical current for primary stimulation. TMS is an example of noninvasive stimulation of focal regions of the brain and, as such, can be used for research or therapeutically without the need for anesthesia.

Mechanisms of Action. At sufficient intensity, electrical currents will stimulate neuronal depolarization, which can result in an action potential. For example, when the TMS coil is positioned over the hand area of the cerebral cortex's motor strip, the changing magnetic field generated by the repetitive pulses induces local currents immediately below the site of stimulation that cause the neurons in area M1 to fire. In turn, this action potential propagates through the polysynaptic corticospinal tract and results in a twitch in the contralateral hand muscle. In essence, TMS uses magnetic fields to indirectly induce focal electrical currents in the brain, thereby triggering the firing of functional neuronal circuits that can lead to observable behavioral effects. This effect can be easily demonstrated by single TMS pulses that can be used

to map the homunculus by merely moving the TMS coil across the cortical representation of neighboring muscle groups and simultaneously to study the excitability of the corticospinal system.

Single TMS pulses can exert other effects when moved to different cortical areas. When positioned over the primary visual cortex (V1), scotomas, or "blind spots," are often elicited. This illustrates that TMS can transiently disrupt functions.

Activation of motor neurons resulting in a muscle twitch and disruption of visual perception with a single-pulse TMS represent examples of the acute effects of TMS-induced neuronal depolarization, as shown in Table 22-3. The effects of single TMS pulses are believed to be immediate and short-lived. The muscle twitch, as induced by TMS to area M1, is nearly instantaneous, with the hand movement occurring approximately 20 milliseconds after the TMS pulse is applied. The visual masking likewise operates on a similar time scale measured in milliseconds. TMS can, however, exert longer-lasting effects when the pulses are repeated at regular intervals in the process of rTMS. It also occurs when they are paired with other forms of stimulation in which TMS pulses are coupled with electrical stimulation of a peripheral nerve (as in paired associative stimulation [PAS]) or when TMS is paired with audiovisual stimuli, as in the example of classical conditioning of the brain response to TMS. The mechanisms underlying these lasting effects of TMS have been described by various researchers and are thought to be related to neuroplasticity and alterations in synaptic efficacy.

Treatment of psychiatric disorders with rTMS has been informed by attempts to focally alter pathologic cortical excitability, believed to be linked to a specific illness. Reduced activity in the left dorsolateral prefrontal cortex has been implicated in several studies as a physiologic correlate of affective disorders. Numerous studies have applied high frequencies of rTMS, which have been reported to increase excitability to the left dorsolateral prefrontal cortex (DLPFC) in an attempt to normalize activity in this region. In a related approach, some investigators who implicated abnormal interhemispheric balance in activation between the right and left DLPFC applied low-frequency rTMS, which has been reported to be inhibitory, to the right DLPFC in an attempt to normalize this balance.

Table 22-3
Acute and Prolonged Mechanisms of Action

Acute Effects
Phasic activation of neural circuits
Observable motor responses (e.g., twitch)
Temporary disruption (e.g., speech arrest) or facilitation of ongoing processing (e.g., speeds reaction time)
Prolonged Effects
Neuroplasticity
• Change in synaptic efficacy, akin to long-term potentiation or depression
• Alterations in neurotrophic factors
• Modulation of cortical excitability
• Modulation of functional connectivity

Side Effects, Interactions with Medications, and Other Risks. Administration of TMS is a noninvasive, relatively benign procedure when applied by a knowledgeable professional to a subject who has been appropriately evaluated. However, it is not entirely without risk. The most severe known risk of TMS is an unintended seizure. Several factors may contribute to seizure risk. Primarily, these include the form of TMS, with single-pulse stimulation less likely to result in a seizure than rTMS, and, in an equally important manner, the dose, which is the combination of treatment parameters including frequency, power, train duration, and intertrain interval. Also, subject factors can be significant, such as the presence of a neurologic disorder (epilepsy or a focal brain lesion) or the use of seizure-lowering medications.

Single-pulse TMS is generally considered to be of minimal risk when administered to appropriately screened adults without seizure risk factors. On the other hand, rTMS can induce seizures in individuals without predisposing conditions when given at sufficiently high doses.

Patient Selection. Patients who have failed a trial of one or more antidepressant medications or have untoward side effects to medications may

be good candidates for TMS. However, given the lower effect size of TMS, for urgent or severely refractory cases, ECT would remain the gold standard treatment.

Future Directions and Controllable Pulse Shape TMS. TMS and other forms of magnetic stimulation hold a tremendous promise in psychiatric treatment due to their focality and noninvasiveness. However, much research is needed to replicate preliminary findings, improve optimal dosing, establish the patient characteristics that predict response, and examine the influence of concomitant medications on the TMS effect. Posttreatment relapse prevention is one of many areas that have to be adequately explored. Other vigorously pursued directions are attempts to develop stimulation coils that will allow deeper brain penetration and work on pulse shapes that may be more physiologically optimal for human stimulation.

Transcranial Direct Current Stimulation

Definition. tDCS is a noninvasive form of treatment that uses very weak (1 to 3 mA) direct electrical current applied to the scalp. Because direct current (DC) polarizes rather than stimulates with discrete pulses, its action does not appear directly to result in action potential firing in cortical neurons. It is also this DC form of electrical stimulation that distinguishes it from devices that use alternating currents (AC) as found in CES, ECT, VNS, and DBS, which produce discrete pulse stimulation. Also, because tDCS works via polarization and does not affect action potential firing in cortical neurons, the term *transcranial direct current polarization* is favored by some modern investigators, and both terms appear interchangeably in the literature today. The small device is very portable and usually operated by readily available DC batteries.

Side Effects. There are no known severe adverse effects of tDCS. It is well tolerated, with reported common side effects in the literature listing mostly minimal tingling at the site of stimulation, with a few reported cases of skin irritation.

Mechanism of Action. tDCS is believed to act via the alteration of

neuronal membrane polarization, but little is known about the actual mechanism of action of tDCS. Polarization may affect the firing and conductance of neurons by either lowering or raising the threshold of activation. Because tDCS involves the application of low currents to the scalp via cathodal and anodal electrodes, depending on the direction of current flow, polarization can either inhibit (cathodal) or facilitate (anodal) function.

Clinical Studies. Preliminary research suggests that tDCS may enhance certain brain functions independent of mood; however, tDCS technology and its use in psychiatry are in the early stages of exploration. Research is focusing on its potential effectiveness in facilitating recovery from stroke and certain forms of dementia.

Future Directions. Most of the current tDCS devices use large, saline solution–soaked electrodes. Future device development will most likely investigate electrode shape and contact material to optimize the intended clinical effects and further improve ease of use. However, fundamental questions of efficacy, indications, and dose–response relationships, as well as predictors of response, will need to be explored first.

Cranial Electrical Stimulation

Definition. CES, like tDCS, uses a weak (1- to 4-mA) current. However, with CES, the current is alternating. It is traditionally applied via saline-soaked, felt-covered electrodes clipped onto the earlobes. Other placement strategies are also being investigated.

Mechanism of Action. The exact mechanism of action has not been elicited, and there is no agreement among researchers on the predominant mode of action. Previous hypotheses proposed that the stimulation with the alternating microcurrent affects the thalamic and hypothalamic brain tissue and facilitates the release of neurotransmitters. Claims have been made that through interaction with cell membranes, the stimulation produces changes in signal transduction associated with classic second-messenger pathways, including calcium channels and cyclic adenosine monophosphate (AMP). There are summary reports that CES causes increases in plasma serotonin,

norepinephrine, dopamine, and monoamine oxidase type B (MAO_B) in blood platelets and cerebrospinal fluid (CSF), as well as the release of 5-hydroxyindoleacetic acid, dehydroepiandrosterone (DHEA), and enkephalins and reduction of cortisol and tryptophan. However, most of these reports have not been validated through modern research.

Side Effects. It is believed that the CES stimulation is not harmful, primarily due to its low-voltage power supply (9-V battery) and lack of any reported adverse event by the FDA. Local skin effects, as well as a general feeling of dizziness, have been reported, however, and device manufacturers do not advise the use of the device during pregnancy, in those with low blood pressure, or in people who have arrhythmias or pacemakers.

Clinical Studies. In a meta-analysis by the Harvard School of Public Health, 18 human clinical trials were examined that used CES to treat depression, anxiety, drug addiction, insomnia, headache, and pain. The overall pooled result showed CES to be better than sham treatment for anxiety at a statistically significant level.

Current Status in Treatment Algorithms, Patient Selection, and Dosing. The use of CES has not been studied sufficiently in the United States, and it does not have a specific place in any algorithm of standard US psychiatric practice.

Future Directions. As with tDCS, fundamental issues of indications, patient selection, dose–response relationship, and efficacy are under active research and remain to be optimized.

Magnetic Seizure Therapy

Definition. MST is a novel form of convulsive treatment that is under development in several research institutions in the United States and Europe. The treatment uses an alternating magnetic field that crosses the scalp and the calvarium bone unaffected by their high electrical impedance, to in turn induce a more-localized electrical current in the targeted regions of the cerebral cortex than is possible with ECT. The aim is to produce a seizure

whose focus and patterns of spread may be controlled.

MST is a convulsive treatment that is similar to ECT. It is performed under general anesthesia. It requires approximately the same preparation and infrastructure as ECT. However, MST is given using a modified TMS device, one that can administer higher output than the conventional TMS devices and thus relies on magnetic stimulation, unlike electric stimulation in ECT. The MST procedure is performed under general anesthesia with a muscle relaxant. MST is under investigation and is not FDA approved.

Mechanism of Action. Induction of a seizure is hypothesized to be the underlying event responsible for the likely multiple specific mechanisms of action of MST treatment. As in ECT, these are not fully understood. However, due to its focality, MST appears to represent a tool better suited than ECT to study the mechanisms of action of convulsive therapy through its potential of inducing seizures initiated in different regions of the brain.

Side Effects. Adverse effects from MST, like those of ECT, are primarily connected to the risks associated with anesthesia and generalized seizure. Also, the MST magnetic coil produces a clicking noise that may potentially affect hearing. Earplugs should be worn by both the patient and members of the treating team to mitigate that risk and prevent any cumulative damage. Studies suggest that MST results in less retrograde and anterograde amnesia than ECT, although this result should be replicated in a larger trial.

Current Status in Treatment Algorithms. No clinical algorithms exist for MST, given that it is still an investigational protocol, and treatments outside of research are not FDA approved. Assuming the hypothesis that MST can approach the efficacy of ECT (but with fewer side effects) is correct, this magnetically induced convulsive treatment will play an essential role before referral for ECT.

Future Directions. MST is a novel treatment in the early phases of clinical testing. Clinical treatment variables, including dosing, optimal coil placement, patient selection, and mechanisms of action, are the topics of ongoing and future studies.

Vagus Nerve Stimulation

Definition. VNS is the direct, intermittent electrical stimulation of the left cervical vagus nerve via an implanted pulse generator, usually in the left chest wall. The electrode is wrapped around the left vagus nerve in the neck and is connected to the generator subcutaneously.

Mechanisms of Action. The majority of the fibers contained in the left vagus are afferents. It is estimated that as many as 80 percent of these fibers are up-going afferents. Thus chronic stimulation of these nerve fibers predominantly changes activity in the brainstem nuclei such as the nucleus of the tractus solitarius and other neighboring nuclei (e.g., raphe) that alter serotonergic activity in cortical and limbic structures. Also, persistent stimulation of the vagal afferents is anticonvulsant, an effect that appears to depend on the norepinephrine-producing locus ceruleus.

Side Effects and Contraindications. To date, reasonably comprehensive literature confirms that VNS is generally well tolerated. The adverse events that are most frequently reported are voice alteration, dyspnea, and neck pain. Besides of the risk of perioperative infection, the surgical implantation carries a small risk of vocal cord paralysis, bradycardia, or asystole.

Current Status in Treatment Algorithms. The FDA indicated VNS for the long-term adjunctive treatment of chronic or recurrent depression in patients 18 years or older experiencing a major depressive episode in the setting of unipolar or bipolar disease who have not had an adequate response to four or more adequate antidepressant treatments. Consultation with another clinician experienced with treatment-resistant depression and VNS is recommended.

VNS treatment success rates are lower than those with ECT. Its onset of action is also comparatively slow—typically, an approximately 30 percent response rate is observed after 1 year. VNS may be worth considering, therefore, when patients have failed to respond to less invasive treatments, ECT was ineffective, or post-ECT relapse cannot be prevented with less invasive means. VNS might be helpful with longer-term relapse prevention,

but results of controlled trials would be useful to guide practice.

Patient Selection. VNS is approved as a long-term adjunctive treatment for chronic or recurrent depressive episodes in adults with a major depressive episode who have not had a satisfactory response to four or more adequate antidepressant trials. The efficacy of VNS in other disorders is unknown.

ECT can be safely used in patients with an implant as long as the VNS generator is turned off during the convulsive treatment. This precaution is needed because of the anticonvulsant effects of VNS. It remains to be studied whether VNS could be useful in relapse prevention post-ECT.

Dosing. The optimal dosing for psychiatric applications of VNS is still mostly an area of investigation. The published studies do not identify optimal dosing parameters like time on, time off, frequency, current, or pulse width. However, the epilepsy literature suggests that there is a threshold current for efficacy. Given current knowledge of VNS dosing, electrical current is typically increased up to greater than 1 mA, and clinical benefit is assessed over several months. Because the adverse effects of VNS are known to be dose-dependent, treatment parameters are often chosen to mitigate specific side effects. For example, lowering the pulse width reduces neck pain, allowing patients to tolerate higher currents.

Future Directions. More research is required to establish dose–response relationships for VNS. Future studies may explore optimal medication strategies to augment responses, test the potential role of VNS for long-term relapse prevention (e.g., after ECT), and study its mechanism of action.

Implanted Cortical Stimulation

Definition. CBS is a novel neurosurgical approach in which electrodes are implanted over the surface of the cortex to provide electrical brain stimulation in a targeted superficial region. This approach is being studied for the treatment of conditions like stroke, tinnitus, and treatment-resistant depression.

NEUROSURGICAL TREATMENTS AND DEEP BRAIN

STIMULATION

After a long and checkered history, neurosurgical treatments for psychiatric illness have reemerged as a focus of great interest. Many still associate psychiatric neurosurgery with the bygone era of crude freehand "psychosurgery" when prefrontal lobotomy saw broad and indiscriminate use. Those primitive operations, which predated modern psychopharmacology, yielded modest reductions in symptoms but were accompanied by unacceptable adverse effects. Over nearly five decades, techniques, and significantly, procedures and practices have evolved tremendously. First, ablative lesions are now accurately, precisely, and reproducibly placed in specific brain targets stereotactically guided by MRI and specialized software. Alternative methods include radiosurgery, which allows stereotactic lesion placement without craniotomy. DBS, while requiring craniotomy to implant stimulating electrodes in specific brain targets, is intentionally nonablative and allows flexible and reversible modulation of brain function. Second, strict criteria for patient selection are observed, and the process of determining appropriate candidacy has been formalized.

Currently, surgical intervention is predominantly reserved for patients with severe, incapacitating major depression or OCD who have failed an exhaustive array of standard treatments. Surgery is not approved unless a multidisciplinary committee reaches consensus regarding its appropriateness for a given candidate, and the patient renders informed consent. Although a large body of clinical data has already been collected that indicates the effectiveness and safety of modern neurosurgical interventions, major centers providing these treatments continue to gather information prospectively, and controlled trials are underway or planned. With these advances in neurosurgical techniques and better-established selection criteria and long-term follow-up procedures, available data suggest that psychiatric neurosurgery yields a substantial improvement in symptoms and functioning in approximately 40 to 70 percent of cases, with morbidity and mortality drastically lower than for earlier procedures.

Although lesion procedures have been influenced by theories implicating corticolimbic systems in disordered behavior, they were initially mainly developed empirically. Although psychiatric neurosurgery is sometimes

criticized for this reason, as for any clinical therapy, the relevant issues are safety and efficacy, not a correction of pathophysiologic processes that are not yet fully understood. However, psychiatric neurosurgery is now developing in a scientific context where the translation of data between clinical results to cross-species anatomical, neuroimaging, and physiologic studies of neural networks involved a promise to illuminate mechanisms of therapeutic action.

History

Trephination performed by ancient civilizations probably represents the earliest form of surgical intervention for psychopathology. In 1891, the first formal report of neurosurgical treatment in psychiatry was published, describing bilateral cortical excisions in demented and depressed patients, which yielded mixed results. After four decades in which little progress was made, in 1935, John Fulton and Charles Jacobsen presented their research on primate behavior following frontal cortical ablation. They observed that lobectomized chimpanzees showed a reduction in “experimental neurosis” and were less fearful while retaining an ability to perform complex tasks. António Egaz Moniz, a renowned Portuguese neurologist, pioneered prefrontal leukotomy in collaboration with his neurosurgical colleague, Almeida Lima. First, by using absolute alcohol injections and subsequently by mechanical means with a leucotome, Moniz and Lima performed “psychosurgery” on 20 severely ill institutionalized patients; 14 were said to have exhibited worthwhile improvement. In an era of overflowing asylums and few effective treatments for chronic debilitating psychiatric illness, this mode of therapy was initially enthusiastically embraced, and Moniz won the 1949 Nobel Prize in Physiology or Medicine for this contribution.

From the mid-1930s until the emergence of the phenothiazines in the mid-1950s, these techniques proliferated globally. Walter Freeman, a neuropsychiatrist, was perhaps the most zealous promoter of psychosurgery in the United States. Pioneering a series of freehand procedures to achieve prefrontal lobotomy (i.e., severing the white matter connections between the prefrontal cortex and the rest of the brain), Freeman, together with neurosurgeon James Watts, reported on their first 200 cases by 1942. Although the benefits of the surgery were highlighted, others acknowledged a

significant complication rate, including frontal lobe syndrome, seizures, and even deaths. At its peak, lobotomy was being performed on approximately 5,000 patients per year in the United States alone. A review of the results of 10,365 prefrontal lobotomies performed from 1942 to 1954 in Britain concluded that, while 70 percent showed an improvement, adverse effects included 6 percent mortality, seizures in 1 percent, and disinhibition syndromes in 1.5 percent. There were widespread reports of blunted personality and socially inappropriate behavior. In the late 1940s and early 1950s, recognition of these risks prompted attempts to develop modified stereotactic surgical procedures that might yield better results. For example, Ernest Spiegel and Henry Wycis, who began stereotactic neurosurgery in humans, reported in the 1940s that dorsomedial thalamotomies improved obsessive-compulsive symptoms. However, with the introduction of chlorpromazine in 1954, medical management of psychiatric illness became possible. Following this, psychiatric neurosurgery was all but abandoned in favor of nonsurgical therapies.

Patient Selection: Indications and Contraindications

Although limited reports have suggested efficacy across a broad range of psychiatric conditions and research is expanding rapidly, as of this writing, the best-established indications for psychiatric neurosurgery remain major depression and OCD. In evaluating candidates, several factors are considered:

1. *Primary diagnosis:* The patient must meet clinical criteria for the diagnostic indication, and this disorder should be a primary cause of the patient's debility and suffering.
2. *Severity:* The patient must have a chronic, severe, and debilitating illness; the duration of the primary illness must exceed 1 year and typically exceeds 5 years. Severity is gauged on standardized instruments (e.g., patients with OCD typically have Yale-Brown Obsessive-Compulsive Scale scores of 25 to 30; patients with major depression typically have Beck Depression Inventory scores of 30 or higher), while a low level of functioning should indicate debility (e.g., a Global Assessment of Functioning score of 50 or less) and a low quality of life.
3. *Adequacy of previous treatment:* Patients must have already undergone

an exhaustive array of other available established treatments, which are documented in detail.

4. *Psychiatric comorbidity:* Appropriate treatment must have been rendered for any comorbid psychiatric disorder; the presence of psychoactive substance use or severe personality disorders are considered strong relative contraindications.
5. *Medical comorbidity and surgical fitness:* Structural brain lesions or significant central nervous system injuries are strong contraindications. Medical conditions that increase neurosurgical risks (e.g., cardiopulmonary disease) and age greater than 65 years are relative contraindications for lesion procedures, while for DBS, the relative age restriction might be older. A history of past seizures is a risk factor for perioperative seizures after lesion procedures and must be weighed in the overall risk–benefit assessment (again, data are currently less clear in this regard for psychiatric DBS).
6. *Access to postoperative care:* The psychiatric neurosurgery procedures themselves represent the beginning of a new episode of care. Patients must have access to adequate postoperative treatment, including a psychiatrist (typically the referring physician) who will accept responsibility for managing the case after discharge. Arrangements for postoperative care (e.g., intensive behavior therapy) should be confirmed ahead of time. Notably, after lesion procedures, care can generally be delivered in standard treatment settings without the need for highly specialized psychiatric neurosurgery teams. For DBS, access to such teams is essential over the long term. Once implanted, patients require clinical monitoring and device adjustment, which can be intensive and time-consuming, especially early in treatment. Device monitoring and replacements may need to occur on a relatively urgent basis. The continuing costs incurred can be substantial, and adequacy of third-party reimbursement needs to be ensured in advance to the extent possible. After either lesion procedures or DBS, family or significant others may be needed to support and accompany patients to follow-up care, similar to the level of support that is usually necessary during the intensive evaluation process.
7. *Informed consent:* Under no circumstances should psychiatric neurosurgery be performed on patients against their will. The patient must be able and willing to render informed consent. Formal consent

monitoring may be used to ensure that the consent process is adequate. In rare instances, these procedures are performed with the assent of the patient and formal consent from a legal guardian. In this context, age less than 18 years also represents a relative contraindication.

Postoperative Care

Immediate postoperative care includes the standard medical and surgical considerations following any stereotactic neurosurgical procedure. Special attention is paid to signs or symptoms of potential surgical complications, including infection, hemorrhage, seizures, or altered mental status. A postoperative MRI should be obtained to document the placement and extent of lesions. Intensive postoperative psychiatric treatment is recommended since the efficacy of the surgery may rely on some synergy between the neurosurgical intervention itself and enhanced response to pharmacologic or behavioral therapies. Although dosages of psychotropic medications may be reduced during the immediate perioperative period, the medication regimen should be readjusted as tolerated postoperatively. Moreover, in the case of OCD, intensive behavior therapy should be initiated as soon as possible, preferably within the first month postoperatively.

For DBS, electrode implantation is usually followed by a several-week delay to enable resolution of local edema and stabilization of other factors that might influence the response to stimulation. Then, systematic outpatient adjustment of stimulation parameters is performed before initial settings are determined. This adjustment is often a time-consuming process, lasting hours over one or more days. Ongoing protocols for DBS entail frequent follow-up, especially during the approximate 6 months after implantation, to enable optimization of stimulation parameters, monitoring of the patient, and coordination of other pharmacologic and behavioral therapies.

Lesion Procedures

Although numerous approaches have been tried, four lesion procedures have evolved as the safest and most effective for treating psychiatric disorders. All four entail bilateral lesions and are performed using modern stereotactic methods.

Subcaudate Tractotomy. Subcaudate tractotomy was introduced by Geoffrey Knight in Great Britain in 1964 as one of the first attempts to limit adverse effects by restricting lesion size. By targeting the substantia innominata (just inferior to the head of the caudate nucleus), the goal was to interrupt white matter tracts connecting the orbitofrontal cortex and subcortical structures. The surgery involved the placement of radioactive yttrium-90 seeds at the desired centroid, yielding lesion volumes of approximately 2 cc on each side. Indications for subcaudate tractotomy are major depression, OCD, and other severe anxiety disorders.

Anterior Cingulotomy. Anterior cingulotomy remains the most commonly employed neurosurgical treatment for psychiatric disease in North America. The surgery is conducted under local anesthesia, and two or three approximately 1-cc lesions are made on each side by thermocoagulation through bilateral burr holes. The target is within the anterior cingulate cortex (Brodmann areas 24 and 32), at the margin of the white matter bundle known as the cingulum. Initially, the placement of lesions was determined by ventriculography; however, since 1991, anterior cingulotomy has been conducted via MRI guidance. Approximately 40 percent of patients return several months following the first operation for a second procedure to extend the first set of lesions. The indications for anterior cingulotomy include major depression and OCD.

Limbic Leukotomy. Desmond Kelly and colleagues introduced limbic leukotomy in England in 1973. The procedure combines the targets of subcaudate tractotomy and anterior cingulotomy. The lesions have typically been made via thermocoagulation or with a cryoprobe. Historically, the precise placement of the lesions was guided by intraoperative stimulation; pronounced autonomic responses were believed to designate the optimal lesion site. The indications for limbic leukotomy include major depression, OCD, and other severe anxiety disorders. More recently, there is also some evidence that this procedure might be beneficial for repetitive self-injurious behaviors or in the context of severe tic disorders.

Anterior Capsulotomy. Anterior capsulotomy or its newer variant, Gamma Knife (Elekta, Stockholm) capsulotomy, are used in Scandinavia, the

United States, Belgium, Brazil, and elsewhere. The procedure places lesions within the anterior limb of the internal capsule, which impinges on the adjacent ventral striatum, thereby interrupting fibers of passage between the prefrontal cortex and subcortical nuclei, including the dorsomedial thalamus. Although the original anterior capsulotomy procedure is performed using thermocoagulation via burr holes in the skull, for the past 15 years, capsulotomy has used the Gamma Knife as an alternative. This radiosurgical instrument makes craniotomy unnecessary. Typically, gamma capsulotomy lesions are smaller than those induced by thermocapsulotomy, remaining within the ventral portion of the anterior capsule. Hence, the term *gamma ventral capsulotomy* is coming into use to describe this procedure.

In contrast to thermocapsulotomy, gamma ventral capsulotomy may be performed as an outpatient procedure, and at most, the patient might require one night of hospitalization. The relative advantages and disadvantages of this radiosurgical approach are the focus of ongoing research. Some data suggest, unsurprisingly, that the rates of neuropsychiatric adverse effects may be considerably lower for gamma ventral capsulotomy than for earlier procedures in which much larger tissue volumes were lesioned. Indications for anterior capsulotomy include major depression, OCD, and other severe anxiety disorders.

Deep Brain Stimulation

DBS for psychiatric illness is not a new idea, although the devices, surgical techniques, and theoretical models of relevant neurocircuitry have all advanced. The procedure involves the placement of small-diameter brain “leads” (e.g., approximately 1.3 mm) with multiple electrode contacts into subcortical nuclei or specific white matter tracts. The surgeon drills burr holes in skull bone under local anesthesia and then places the leads, guided by multimodal imaging and precise stereotactic landmarking. Usually, this is done bilaterally. The patient is typically sedated but awake during surgery. Later, the “pacemaker” (also known as an implantable neurostimulator or pulse generator) is implanted subdermally (e.g., in the upper chest wall) and connects via extension wires tunneled under the skin to the brain leads. The goals of DBS are to achieve improved efficacy and more favorable adverse effect profiles in comparison with ablation. Because various combinations of

electrodes can be activated, at adjustable polarity, intensity, and frequency, DBS allows more flexible modulation of brain function, referred to as *neuromodulation.* Thus, parameters can be optimized for individual patients, but the process, typically performed by a specially trained psychiatrist in the outpatient setting, can be quite time-consuming and requires attentive long-term follow-up. In cases where no beneficial settings can be identified despite extensive efforts, the electrodes can be inactivated, and devices may be removed. In that event, devices are usually only partly explanted, with the brain electrodes left in place given the small risk of hemorrhage upon removal. The relative advantages and disadvantages of DBS are the focus of very active research.

Treatment Outcome

For all four contemporary ablative procedures, outcomes cannot be fairly assessed for a considerable period postoperatively, which could extend from 6 months to 2 years. In the first two or three decades of this work, clinical reports usually employed measures of global improvements, such as the Pippard Postoperative Rating Scale, which rates outcomes as follows: (1) symptom-free, (2) much improved, (3) slightly improved, (4) unchanged, and (5) worse. Most studies have operationalized significant improvement as categories 1 and 2. Also, many of the reports employ a measure of symptom severity that is specific to the indication for the procedure (e.g., the Yale-Brown Obsessive-Compulsive Scale for OCD and the Beck Depression Inventory for major depression). Most studies focus on one or another of the procedures and are best reviewed according to the surgical approach.

Outcome with Subcaudate Tractotomy. Significant improvement was seen in 68 percent of patients with major depression, 50 percent of patients with OCD, and 62.5 percent of patients with other anxiety disorders. Patients with schizophrenia, substance abuse, or personality disorders did poorly. Short-term side effects include transient headache and confusion or somnolence, which typically resolve in less than 1 week. Patients are usually ambulatory by the third postoperative day. Transient disinhibition syndromes were common. In 1994, a large-scale review of 1,300 cases was conducted and concluded that the procedure enables 40 to 60 percent of patients to lead

normal or near-normal lives, with a reduction in suicide rate to 1 percent versus 15 percent in a similarly affected control group with major affective disorders.

Outcome with Anterior Cingulotomy. Significant improvement occurred in 62 percent of patients with affective disorders, 56 percent with OCD, and 79 percent with other anxiety disorders. Among patients with unipolar depression, 60 percent responded favorably; among patients with bipolar disorder, 40 percent responded favorably; and among patients with OCD, 27 percent were classified as responders with another 27 percent categorized as possible responders. Short-term side effects include headache, nausea, or difficulty with urination; however, these typically resolve within a few days. Patients are usually ambulatory within 12 hours following the operation and discharged on the third to fifth postoperative day. Over the past 10 years, the practice of treating patients who experience perioperative seizures with chronic anticonvulsant therapy has been discontinued, and no cases of new-onset recurrent seizures have been seen. Although patients have occasionally (5 percent or less) noted transient problems with memory, an independent analysis of 34 patients was performed. It demonstrated no significant intellectual or behavioral impairments attributable to anterior cingulotomy; a subsequent study of 57 patients likewise found no evidence for lasting neurologic or behavioral adverse effects.

Outcome with Limbic Leukotomy. Significant improvement occurred in 89 percent of patients with OCD, 78 percent with major depression, and 66 percent with other anxiety conditions. Short-term side effects include headache, lethargy or apathy, confusion, and lack of sphincter control, which may last from a few days to a few weeks. In particular, it is common for postoperative confusion to last at least several days, and patients are often not discharged in less than 1 week. There were no seizures and no deaths; however, one patient suffered severe memory loss due to improper lesion placement, and enduring lethargy was present in 12 percent of cases.

Outcome with Anterior Capsulotomy

THERMOCAPSULOTOMY. A favorable response occurred in 50 percent of those

with OCD and 48 percent of those with major depression. Short-term side effects can include transient headache or incontinence. Postoperative confusion often lasts for up to 1 week. Recovery from gamma capsulotomy is swifter and characterized by less discomfort and virtually no confusion, but side effects from radiation exposure, principally cerebral edema, may be delayed for up to 8 to 12 months. For the open capsulotomy, patients are typically ambulatory in a matter of hours to days following the operation, although the duration of confusion may influence the length of hospital stay. Weight gain has been noted to be a common enduring side effect with a mean increase in the mass of 10 percent.

GAMMA VENTRAL CAPSULOTOMY. Gamma capsulotomy was generally well tolerated and effective for patients with otherwise intractable OCD. Adverse events included cerebral edema and headache, small asymptomatic caudate infarctions, and possible exacerbation of preexisting bipolar mania. A therapeutic response, defined conservatively, was seen in 60 percent of over 50 patients receiving a gamma capsulotomy procedure, in which pairs of lesions in the ventral capsule are made bilaterally, impinging on the ventral striatum. The therapeutic benefit was achieved over 1 to 2 years and was essentially stable by 3 years. Adverse effects of gamma ventral capsulotomy include significant radiation-induced edema, appearing months after the procedure, apparently due to differential sensitivity to radiation that remains poorly understood. Long-term follow-up will be necessary to clarify the risks and benefits of gamma ventral capsulotomy. The same applies to any neurosurgery, including lesion procedures and DBS.

Outcomes with DBS

OBSESSIVE-COMPULSIVE DISORDER. Long-term outcomes of open stimulation in 26 patients showed clinically significant symptom reductions and functional improvement in about two-thirds of patients overall. Conservatively defined responses (35 percent or more significant reductions on the Yale-Brown Obsessive-Compulsive Scale) were seen in one-third of patients in the initial group, irrespective of the study center, while the response rate was over 70 percent in the second and third patient cohorts treated. The development of psychiatric DBS is following the path of

stimulation for movement disorders, where several targets have been pursued with therapeutic benefit. As in movement disorders, overlapping or converging effects of DBS at different anatomical sites on the neurocircuitry involved are likely and are a focus of active research. The same reasoning applies to DBS for depression.

MAJOR DEPRESSION. A body of functional neuroimaging research implicates the subgenual cingulate cortex as a node in circuits involved in the normal experience of sadness, symptoms of depressive illness, and responses to depression treatments. Chronic DBS for up to 6 months was associated with sustained remission of depression in four of the six patients studied. Another line of research on DBS for depression was prompted by the OCD research discussed above and also by the reported antidepressant effects of anterior capsulotomy on which the VC/VS (ventral capsule/ventral striatum) stimulation target was initially based. The OCD patients, who had very high rates of comorbid depression, characteristically responded to stimulation onset with mood enhancement and reductions in nonspecific as well as OCD-related anxiety. Such effects were accompanied, or even preceded by, improvements in social interaction and daily functioning. Worsening in these same clinical domains was noted in some patients with the cessation of VC/VS stimulation. Moreover, DBS-induced changes in mood and nonspecific anxiety often seemed to precede reductions in core OCD symptoms.

Outcome Across Contemporary Neurosurgical Procedures. Although the field is developing rapidly, the conclusion reached is that 40 to 70 percent of carefully selected psychiatric patients should meaningfully benefit from contemporary neurosurgical treatment. Twenty-five percent or more might be expected to show outstanding improvement. Responses to ablative procedures have appeared marginally superior for major depression than for OCD generally. The adverse effect profiles of this group of procedures are influenced by lesion size, the surgical approach, and whether radiosurgical methods (in which the tempo of lesion development is very slow vs. thermocoagulation) are used. However, adverse effects are greatly minimized in comparison with the procedures of the past. Although minor short-term side effects may be expected after some modern ablative

procedures, severe or enduring adverse consequences are relatively rare. These can include seizures in about 1 to 5 percent of cases. Although frontal syndromes, confusion, or subtle cognitive deficits can still be seen, overall cognitive function, as indicated by the standard intelligence quotient, is generally enhanced, a finding that has been attributed to the overriding beneficial effects of symptomatic improvement. Psychiatric neurosurgery likely reduces mortality, as evidenced by data on comparative suicide rates. Nonetheless, patients who undergo and fail to benefit from these procedures are at exceptionally high risk for completed suicide. Therefore, as with any therapy, the potential risks and benefits of psychiatric neurosurgery must be weighed against the potential risks and benefits of undergoing this brand of treatment.

The advent of DBS in psychiatry has created tremendous interest and considerable research activity. This therapy is intentionally nonablative, can be optimized for individual patients, is reversible, and is based on devices that are (to varying degrees) removable. DBS may therefore be accepted by patients who would not choose to undergo lesion procedures (although the reverse is also true). With all of its advantages, DBS requires that patients be treated by highly specialized teams willing and able to provide long-term care. The logistics and expenses involved can represent significant barriers.

In contrast, psychiatric care can be delivered in standard treatment settings after lesion procedures. However, although the relative risks of enduring adverse effects after psychiatric DBS remain to be established, at this stage, ablative methods appear to carry a more significant risk. Because rates of adverse outcomes are low when modern lesion procedures are performed at highly experienced centers, there may be a particularly strong rationale for referral of appropriate patients to such expert centers.

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23 Psychotherapy

This chapter provides an overview of common psychotherapeutic approaches, covering a broad spectrum of therapy modalities.

PSYCHOANALYSIS AND PSYCHOANALYTIC PSYCHOTHERAPY

As broadly practiced today, psychoanalytic treatment encompasses a wide range of uncovering strategies used in varying degrees and blends. Despite the inevitable blurring of boundaries in actual application, we describe the original modality of classic psychoanalysis and major modes of psychoanalytic psychotherapy (expressive and supportive) separately here (Table 23-1). Analytical practice, in all its complexity, resides on a continuum. The individual technique is always a matter of emphasis, as the therapist titrates the treatment according to the needs and capacities of the patient at every moment.

Psychoanalysis is virtually synonymous with the renowned name of its founding father, Sigmund Freud. It is also referred to as “classic” or “orthodox” psychoanalysis to distinguish it from more recent variations known as *psychoanalytic psychotherapy*.

Psychoanalysis is based on the theory of sexual repression and traces the unfulfilled infantile libidinal wishes in the individual’s unconscious memories. It remains unsurpassed as a method to discover the meaning and motivation of behavior, especially the unconscious elements that inform thoughts and feelings.

Psychoanalysis

Psychoanalytic Process. The psychoanalytic process involves bringing to the surface repressed memories and feelings through a scrupulous unraveling of hidden meanings of verbalized material and of the unwitting ways in which the patient wards off underlying conflicts through defensive forgetting and repetition of the past.

The overall process of analysis is one in which unconscious neurotic conflicts are recovered from memory and verbally expressed, reexperienced in the transference, reconstructed by the analyst, and, ultimately, resolved through understanding. Freud referred to these processes as *recollection, repetition,* and *working through,* which make up the totality of remembering, reliving, and gaining insight. *Recollection* entails the extension of memory back to early childhood events, a time in the distant past when the core of neurosis developed. The actual reconstruction of these events comes through reminiscence, associations, and autobiographical linking of developmental events. *Repetition* involves more than mere mental recall; it is an emotional replay of former interactions with significant individuals in the patient's life. The replay occurs within the unique context of the analyst as a projected parent, a fantasized object from the patient's past with whom the latter unwittingly reproduces forgotten, unresolved feelings and experiences from childhood. Finally, *working through* is both an affective and cognitive integration of previously repressed memories that are brought into consciousness and through which the patient is gradually set free (cured of neurosis). The analytical course includes three major stages (Table 23-2).

Indications and Contraindications. In general, all of the so-called *psychoneuroses* are suitable for psychoanalysis. These include anxiety disorders, obsessional thinking, compulsive behavior, conversion disorder, sexual dysfunction, depressive states, and many other nonpsychotic conditions, such as personality disorders. Significant suffering must be present so that patients are motivated to make the sacrifices of time and financial resources required for psychoanalysis. Patients who enter analysis must have a genuine wish to understand themselves, not a desperate hunger for symptomatic relief. They must be able to withstand frustration, anxiety, and other strong affects that emerge in the analysis without fleeing or acting out their feelings in a self-destructive manner. They must also have a reasonable, mature superego that allows them to be honest with the analyst.

Intelligence must be at least average, and above all, they must be psychologically minded in the sense that they can think abstractly and symbolically about the unconscious meanings of their behavior.

Many contraindications for psychoanalysis are the flip side of the indications. The absence of suffering, poor impulse control, inability to tolerate frustration and anxiety, and low motivation to understand are all contraindications. The presence of extreme dishonesty or antisocial personality disorder contraindicates analytic treatment. Concrete thinking or the absence of psychological mindedness is another contraindication. Some patients who might ordinarily be psychologically minded are not suitable for analysis because they are in the midst of a major upheaval or life crisis, such as a job loss or a divorce. Serious physical illness can also interfere with a person's ability to invest in a long-term treatment process. Patients of low intelligence generally do not understand the procedure or cooperate in the process. An age older than 40 years was once considered a contraindication, but today analysts recognize that patients are malleable and analyzable in their 60s or 70s. One final contraindication is a close relationship with the analyst. Analysts should avoid analyzing friends, relatives, or persons with whom they have other involvements.

Patient Requisites. Table 23-3 lists the most important patient requisites for psychoanalysis.



Table 23-1
Scope of Psychoanalytic Practice: A Clinical Continuum

Features	"Classic" Analytic	Expressive Mode (Exploratory)	Support Mode
Frequency	Regular 4–5 times/wk; "50-minute hour"	Regular 1–3 times/wk; 1/2 to full hour	Flexible or less needed full ho
Duration	Long term; usually 3–5+ yr; maybe considerably	Short or long term; several sessions to months or	Short or interm long te

	longer	years	single to lifet
Setting	Patient primarily on couch with analyst out of view	Patient and therapist face to face; occasional use of couch	Patient a therap to-face contra
Modus operandi	Systematic analysis of all positive and negative transference and resistance; primary focus on analyst and intrasession events; transference neurosis facilitated; regression encouraged	Partial analysis of dynamics and defenses; focus on current interpersonal events and transference to others outside of sessions; analysis of negative transference; positive transference left unexplored unless it impedes progress; limited regression encouraged	Formatic therap allianc real ob relatio analys transfe contra with ra excep focus consci extern regres discou
Analyst/therapist role	Absolute neutrality; frustration of patient; reflector/mirror role	Modified neutrality; implicit gratification of patient and greater activity of therapist	Neutralit suspe limited gratific directi disclos
Mutative change agents	Insight predominates within relatively	Insight within more empathic environment;	Auxiliary surrog as a te

	deprived environment	identification with benevolent object	substit holdin enviro insight degree
Patient population	Neuroses; mild character psychopathology	Neuroses; mild to moderate character psychopathology, especially narcissistic and borderline disorders	Severe d disord latent manife psych acute physic
Patient requisites	High motivation, psychological mindedness, good previous object relationships, ability to maintain transference neuroses, good frustration tolerance	High moderate motivation and psychological-mindedness, ability to form therapeutic alliance, some frustration tolerance	Some de motiva ability therap allianc
Basic goals	Structural reorganization of personality; resolution of unconscious conflicts; insight into intrapsychic events; symptom relief	Partial reorganization of personality and defenses; resolution of preconscious and conscious derivatives of conflicts; insight	Reintegr self ar to cop stabiliz restora preexi equilib streng of defe

	an indirect result	into current interpersonal events; improved object relations; symptom relief a goal or prelude to further exploration	better adjust accep pathol sympt and enviro restruc primar
Major techniques	Free association method predominates; full dynamic interpretation (including confrontation, clarification, and working through), with emphasis on genetic reconstruction	Limited free association; confrontation, clarification, and partial interpretation predominate, with emphasis on here-and-now interpretation and limited genetic interpretation	Free ass metho contra sugge (advic predor abreac useful confro clarific and interpr the he now a secon geneti interpr contra
Adjunct treatment	Primarily avoided; if applied, all negative and positive meanings and implications thoroughly analyzed more	May be necessary, e.g., psychotropic drugs as temporary measure; if applied, negative implications are	Often ne e.g., psych drugs, rehabi therap hospit if appl

	recently, medication adjunct	explored and diffused	positiv implica empha

This division is not categorical; all practice resides on a clinical continuum.

Ms. M, a 29-year-old unmarried woman who worked in a low-level capacity for a magazine, presented for consultation with the chief complaints of considerable sadness and distress over her parent's reaction when they found that she had had a homosexual relationship. She also realized that she had been working far below her potential. She had never sought any treatment before. She was clearly intelligent, sensitive, self-reflective, and insightful. When she learned about the possibility of psychoanalysis as a treatment option, she worried that meant she was "sicker." Ms. M, however, began reading Freud, realized that analysis was actually recommended for those who are higher functioning, and became intrigued by the idea. She agreed to come 4 days a week for 50-minute sessions.

She was the oldest of three children and the only girl. Ms. M's father, a successful professional, was described as very demanding and intrusive; someone who never thought anything was good enough. He had always expected his children to do the "extra credit" assignments as part of their regular work. Ms. M, however, was very proud of her father's accomplishments. She spoke of her mother in conflicting terms as well: she was a homemaker, weak, and sometimes acquiescent to the powerful father but also an individual in her own right who was involved in community volunteer work and could be a powerful public speaker.

Just before beginning her analysis, Ms. M had had her wallet stolen. In her first analytic session, she spoke of losing all of her identification cards, and to her, it seemed as if she were starting analysis "with a completely new identity." Initially, she was somewhat hesitant to use the couch because she wanted to see her analyst's reactions, but Ms. M quickly appreciated that she could associate more easily without seeing the analyst.

As her analysis proceeded, through dreams and free associations, Ms. M became quite focused on the analyst. She became extremely curious

about the analyst's life. What emerged from her associations to seeing the analyst's appointment book on the desk was that she felt "slotted in." Whenever Ms. M saw other patients, she felt the office was "like an assembly line." Further associations led to her feeling slotted in by her parents as they ran from one activity to another. Her resistance manifested itself in Ms. M, often coming as much or more than 15 minutes late to her sessions. Her associations led to her admitting that she did not want her analyst to think that she was "too eager." Ms. M was able to see that she needed to devalue her analyst and her importance to Ms. M as a defense against an overwhelmingly positive and even erotic transference toward her.

For example, Ms. M wanted to improve her appearance so that the therapist, whom she called a "role model," would find her more attractive. Her negative transference, however, was never far from the surface, and she denigrated the analyst by wondering if the analyst were a "clotheshorse" who was financing her wardrobe with the patient's payments.

Her conflicts about her sexual orientation were a central preoccupation in the course of her analysis, particularly because her father was homophobic. Early on, Ms. M felt awkward and uncomfortable when she went to a lesbian bar, and when asked if she qualified for the "lesbian discount," she said she did not. At one point, she began seeing several men, including a male psychologist. The analyst made the transference interpretation, which Ms. M accepted, that a date with this man seemed as if it were a date with the analyst, and sleeping with him would be equivalent to sleeping with the analyst. Ms. M was also able to see that her transient choice of dating a male therapist was a defensive compromise. Although her homosexual object choice was multidetermined, Ms. M came to appreciate, through her work in analysis, that at least a part of her conflicts about homosexuality stemmed from her relationship with her father. It was a means of securing his attention as well as infuriating him.

Over 4 years, Ms. M performed considerably better at work and earned a promotion to a job commensurate with her potential. She was also able to deal better with both her parents and particularly her father, regarding her sexual orientation. She became much more comfortable with her

"new identity" and became involved in a relationship with a professional woman. At the end of therapy, Ms. M and this woman were committed to each other and considering adopting a child. (Courtesy of T. Byram Karasu, M.D. and S. R. Karasu, M.D.)

Goals. In developmental terms, psychoanalysis aims at the gradual removal of amnesias rooted in early childhood based on the assumption that filling all gaps in memory will lead to cessation of the morbid condition because the patient no longer needs to repeat or remain fixated to the past. The patient should be better able to relinquish former regressive patterns and to develop new, more adaptive ones, particularly as he or she learns the reasons for his or her behavior. A related goal of psychoanalysis is for the patient to achieve some measure of self-understanding or insight.



Table 23-2
Stages of Psychoanalysis

Stage one: Patient becomes familiar with the methods, routines, and requirements of analysis, and a realistic therapeutic alliance is formed between patient and analyst. Basic rules are established; the patient describes his or her problems; there is some review of history, and the patient gains initial relief through catharsis and a sense of security before delving more deeply into the source of the illness. The patient is primarily motivated by the wish to get well.

Stage two: Transference neurosis emerges that substitutes for the actual neurosis of the patient and in which the wish for health comes into direct conflict with the simultaneous wish to receive emotional gratification from the analyst. There is a gradual surfacing of unconscious conflicts; an increased irrational attachment to the analyst, with regressive and dependent concomitants of that bond; a developmental return to earlier forms of relating (sometimes compared with that of mother and infant); and a repetition of childhood patterns and recall of traumatic memories through transfer to the analyst of unresolved libidinal wishes.

Stage three: The termination phase is marked by the dissolution of the analytical bond as the patient prepares for leave-taking. The irrational attachment to the analyst in the transference neurosis has subsided because it has been worked through, and more rational aspects of the psyche preside, providing greater mastery and more mature adaptation to the patient's problems. Termination is not a hard-and-fast event, and the patient invariably has to continue to work through any problems outside of the therapy situation without the analyst or may need intermittent assistance after analysis has technically terminated.

Courtesy of T. Byram Karasu, M.D.



Table 23-3
Patient Prerequisites for Psychoanalysis

High motivation. The patient needs a strong motivation to persevere, in light of the rigors of intense and lengthy treatment. The desire for health and self-understanding must surpass the neurotic need for unhappiness. The patient must be willing to face issues of time and money and to endure the pain and frustration associated with sacrificing rapid relief in favor of future cure and with foregoing the secondary gains of illness.

Ability to form a relationship. The capacity to form and maintain, as well as to detach from, a trusting object relationship is essential. The patient also has to withstand a frustrating and regressive transference without decompensating or becoming excessively attached. Patients with a history of impaired or transient interpersonal relations who cannot establish a viable connection to another human make poor candidates for psychoanalysis.

Psychological mindedness and capacity for insight. As an introspective process, psychoanalysis requires curiosity about oneself and the capacity for self-scrutiny. Those who are unable to articulate and comprehend their inner thoughts and feelings cannot negotiate with the fundamental analytical coin words and their meanings. The inability to examine one's own motivations and

behaviors precludes benefits from the analytical method.

Ego strength. Ego strength is the integrative capacity to oscillate appropriately between two antithetical types of ego functioning: On the one hand, the patient must be able to reflect temporarily, to relinquish reality for fantasy, and to be dependent and passive. On the other hand, the patient has to be able to accept analytical rules, to integrate interpretations, to defer important decisions, to shift perspectives to become an observer of his or her intrapsychic processes, and to function in a sustained interpersonal relationship as a responsible adult.

Courtesy of T. Byram Karasu, M.D.

Psychoanalytic goals often seem formidable (e.g., a total personality change), involving the radical reorganization of old developmental patterns based on earlier affects and the entrenched defenses built up against them. Goals may also be elusive, framed as they are in theoretical intrapsychic terms (e.g., greater ego strength) or conceptually ambiguous ones (resolution of the transference neurosis). Criteria for successful psychoanalysis may be largely intangible and subjective, and it is best to view them as conceptual endpoints of treatment that the therapist must translate into more realistic and practical terms.

In practice, the goals of psychoanalysis for any patient naturally vary, as do the many manifestations of neuroses. The form that the neurosis takes—unsatisfactory sexual or object relationships, inability to enjoy life, underachievement, and fear of work or academic success, or excessive anxiety, guilt, or depressive ideation—determines the focus of attention and the general direction of treatment, as well as the specific goals. Such goals may change at any time during analysis, especially as many years of treatment may be involved.

Major Approach and Techniques. Structurally, *psychoanalysis* usually refers to individual (dyadic) treatment that is frequent (four or five times per week) and long term (several years). All three features take their precedent from Freud himself.

The dyadic arrangement is a direct function of the Freudian theory of neurosis as an intrapsychic phenomenon, which takes place within the person

as instinctual impulses continually seek discharge. Because it is crucial to resolve dynamic conflicts internally if structural personality reorganization is to take place, the individual's memory and perceptions of the repressed past are pivotal.

Freud initially saw patients 6 days a week for 1 hour each day, a routine now reduced to four or five sessions per week of the classic 50-minute hour, which leaves time for the analyst to take notes and organize relevant thoughts before the next patient. Therapists should avoid long intervals between sessions so that the momentum gained in uncovering conflictual material is not lost, and confronted defenses do not have time to restrengthen.

Freud's belief that successful psychoanalysis always takes a long time because profound changes in the mind occur slowly still holds. The process is similar to the fluid sense of time that is characteristic of our unconscious processes. Moreover, because psychoanalysis involves a detailed recapitulation of present and past events, any compromise in time presents the risk of losing pace with the patient's mental life.

PSYCHOANALYTIC SETTING. As with other forms of psychotherapy, psychoanalysis takes place in a professional setting, apart from the realities of everyday life, in which the therapist offers the patient a temporary sanctuary in which to ease psychic pain and reveal intimate thoughts to an accepting expert. The design of the psychoanalytic environment promotes relaxation and regression. The setting is usually spartan and sensorially neutral and minimizes external stimuli.

Use of the Couch. The couch has several clinical advantages that are both real and symbolic: (1) the reclining position is relaxing because it is associated with sleep and so eases the patient's conscious control of thoughts; (2) it minimizes the intrusive influence of the analyst, thus curbing unnecessary cues; (3) it permits the analyst to make observations of the patient without interruption; and (4) it holds a symbolic value for both parties, a tangible reminder of the Freudian legacy that gives credibility to the analyst's professional identity, allegiance, and expertise. The reclining position of the patient with analyst nearby can also generate threat and discomfort, however, as it recalls anxieties derived from the earlier parent–child configuration that it physically resembles. It may also have personal meanings—for some, a

portent of dangerous impulses or submission to an authority figure; for others, relief from confrontation by the analyst (e.g., fear of the use of the couch and over-eagerness to lie down may reflect resistance and, thus, need to be analyzed). Although the use of the couch is requisite to the analytical technique, it is not applied automatically; the therapist introduces it gradually and may suspend its use whenever additional regression is unnecessary or countertherapeutic.

Fundamental Rule. The fundamental rule of free association requires patients to tell the analyst everything that comes into their heads—however disagreeable, unimportant, or nonsensical—and to let themselves go as they would in a conversation that leads from "cabbages to kings." It differs decidedly from ordinary conversation—instead of connecting personal remarks with a rational thread, the therapist asks the patient to reveal those very thoughts and events that are objectionable precisely because of being averse to doing so.

This directive represents an ideal because free association does not arise freely but is guided and inhibited by a variety of conscious and unconscious forces. The analyst must not only encourage free association through the physical setting and a nonjudgmental attitude toward the patient's verbalizations but also examine those very instances when the flow of associations is diminished or comes to a halt—they are as important analytically as the content of the associations. The analyst should also be alert to how individual patients use or misuse the fundamental rule.

Aside from its primary purpose of eliciting recall of deeply hidden early memories, the fundamental rule reflects the analytical priority placed on verbalization, which translates the patient's thoughts into words, so the patient does not channel them physically or behaviorally. As a direct concomitant of the fundamental rule, which prohibits action in favor of verbal expression, patients should postpone making significant alterations in their lives, such as marrying or changing careers, until they discuss and analyze them within the context of treatment.

Principle of Evenly Suspended Attention. As a reciprocal corollary to the rule that patients communicate everything that occurs to them without criticism or selection, the principle of evenly suspended attention requires the analyst to

suspend judgment and to give impartial attention to every detail equally. The method consists simply of making no effort to concentrate on anything specific while maintaining a neutral, quiet attentiveness to all that the patient says.

Analyst as Mirror. A second principle is the recommendation that the analyst is impenetrable to the patient and, like a mirror, only reflect what the patient shows. Analysts should be neutral blank screens and not bring their personalities into treatment. They are not to bring their values or attitudes into the discussion or to share personal reactions or mutual conflicts with their patients, although they may sometimes feel the temptation to do so. The bringing in of reality and external influences can interrupt or bias the patient's unconscious projections. Neutrality also allows the analyst to accept without censure all forbidden or objectionable responses.

Rule of Abstinence. The fundamental rule of abstinence does not mean corporal or sexual abstinence but refers to the frustration of emotional needs and wishes that the patient may have toward the analyst or part of the transference. It allows the patient's longings to persist and serve as driving forces for analytical work and motivation to change. Freud advised that the analyst proceeds through the analytical treatment in a state of renunciation. The analyst must deny the patient who is longing for love the satisfaction he or she craves.

LIMITATIONS. At present, the predominant treatment constraints are often economic, relating to the high cost in time and money, both for patients and in the training of future practitioners. In addition, because clinical requirements emphasize such requisites as psychological mindedness, verbal and cognitive ability, and stable life situation, psychoanalysis may be unduly restricted to a diagnostically, socioeconomically, or intellectually advantaged patient population. Other intrinsic issues pertain to the use and misuse of its stringent rules, whereby overemphasis on technique may interfere with an authentic human encounter between analyst and patient, and to the major long-term risk of interminability, in which protracted treatment may become a substitute for life. The reification of the classic analytical tradition may interfere with a more open and flexible application of its tenets to meet

changing needs. It may also obstruct a comprehensive view of patient care that includes a greater appreciation of other treatment modalities in conjunction with or as an alternative to psychoanalysis.

Ms. A, a 25-year-old articulate and introspective medical student began analysis complaining of mild, chronic anxiety, dysphoria, and a sense of inadequacy, despite above-average intelligence and performance. She also expressed difficulty in long-term relationships with her male peers.

Ms. A began the initial phase of analysis with enthusiastic self-disclosure, frequent reports of dreams and fantasies, and over idealization of the analyst; she tried to please him by being a compliant, good patient, just as she had been a good daughter to her father (a professor of medicine) by going to medical school.

Over the next several months, Ms. A gradually developed a strong attachment to the analyst and settled into a phase of excessive preoccupation with him. Simultaneously, however, she began dating an older psychiatrist and proceeded to complain about the analyst's coldness and unresponsiveness, even considering dropping out of the analysis because he did not meet her demands.

In the course of analysis, through dreams and associations, Ms. A recalled early memories of her ongoing competition with her mother for her father's attention and realized that failing to obtain his exclusive love, she had tried to become like him. She was also able to see how her increasing interest in becoming a psychiatrist (rather than following her original plan to be a pediatrician), as well as her recent choice of a man to date, were recapitulations of the past vis-à-vis the analyst. As this repeated pattern was recognized, the patient began to relinquish her intense erotic and dependent tie to the analyst, viewing him more realistically and beginning to appreciate how his quiet presence reminded her of her mother. She also became less disturbed by the similarities she shared with her mother and was able to disengage from her father more comfortably. By the fifth year of analysis, she was happily married to a former classmate, was pregnant, and was a pediatric chief resident. Her anxiety had attenuated and become situation-specific (i.e., she was concerned about motherhood and the termination of analysis). (Courtesy

of T. Byram Karasu, M.D.)

Psychoanalytic Psychotherapy

Psychoanalytic psychotherapy, based on fundamental dynamic formulations and techniques that derive from psychoanalysis, is designed to broaden its scope. Psychoanalytic psychotherapy, in its narrowest sense, is the use of insight-oriented methods only. As generically applied today to an ever-larger clinical spectrum, it incorporates a blend of uncovering and suppressive measures.

The strategies of psychoanalytic psychotherapy currently range from expressive (insight-oriented, uncovering, evocative, or interpretive) techniques to supportive (relationship-oriented, suggestive, suppressive, or repressive) techniques. Although those two methods may seem antithetical, their precise definitions and the distinctions between them are by no means absolute.

The duration of psychoanalytic psychotherapy is generally shorter and more variable than in psychoanalysis. Treatment may be brief, even with an initially agreed-upon or fixed time limit, or may extend to a less definite number of months or years. Brief treatment is chiefly used for selected problems or highly focused conflict, whereas more prolonged treatment may be applied for more chronic conditions or for intermittent episodes that require ongoing attention to deal with pervasive conflict or recurrent decompensation. Unlike psychoanalysis, psychoanalytic psychotherapy rarely uses the couch; instead, the patient and therapist sit face to face. This posture helps to prevent regression because it encourages the patient to look at the therapist as a real person from whom to receive direct cues, even though transference and fantasy will continue. The couch is unnecessary because the free-association method is rarely used, except when the therapist wishes to gain access to fantasy material or dreams to enlighten a particular issue.

Expressive Psychotherapy

INDICATIONS AND CONTRAINDICATIONS. Diagnostically, psychoanalytic psychotherapy in its expressive mode is suited to a range of psychopathology with mild to moderate ego weakening, including neurotic conflicts, symptom

complexes, reactive conditions, and the whole realm of nonpsychotic character disorders, including those disorders of the self that are among the more transient and less profound on the severity-of-illness spectrum, such as narcissistic personality disorders. It is also one of the treatments recommended for patients with borderline personality disorders (BPDs), although special variations may be required to deal with the associated turbulent personality characteristics, primitive defense mechanisms, tendencies toward regressive episodes, and irrational attachments to the analyst.

Ms. B, an intelligent and verbal 34-year-old divorced woman, presented with complaints of being unappreciated at work. Always angry and irritable, she considered quitting her job and even leaving the city. Her social life was also being negatively affected; her boyfriend had threatened to leave her because of her extremely hostile, clinging behavior (the same reason her ex-husband had given when he left her 9 years earlier after only 16 months of marriage). Her past included promiscuity and experimentation with various drugs, and, currently, she indulged in heavy drinking on weekends and occasionally smoked marijuana. She had held many jobs and had lived in various cities.

The eldest of three children of a middle-class family, she came from an unhappy and unstable home: her brother had been in and out of psychiatric hospitals; her sister had left home at the age of 16 after becoming pregnant and being forced to marry; and her overly controlling parents had subjected their children to psychological (and occasionally physical) abuse, alternating between heated arguments and passionate reconciliations.

Initially, Ms. B attempted to contain her rage in treatment, but it frequently surfaced and alternated with child-like helplessness; she interrogated the psychiatrist regarding his credentials, ridiculed psychodynamic concepts, constantly challenged statements, and would demand practical advice but then denigrate or fail to follow the guidance given. The psychiatrist remained unprovoked by her aggression and explored with her the need to engage him negatively. Her response was to question and test his continued concern.

When her boyfriend left her, she attempted suicide (she cut her wrists superficially), was briefly hospitalized, and, on discharge, was placed on a selective serotonin reuptake inhibitor (SSRI) for 6 months for her minor, but protracted, depression. The psychiatrist maintained their regular frequency of sessions despite her greater demands. Although she was puzzled by the steadiness of his interest, she gradually felt safe enough to express her vulnerabilities. As they explored her lack of full commitment to work, friends, and therapy, she began to understand the meaning of her anger in terms of the early abusive relationship with her parents and her tendency to bring it into contemporary relationships. With the psychiatrist's encouragement, she also began to seek work and make small strides in relationship-oriented efforts. By the end of her second year of treatment, she decided to remain in the city, to stay at her place of employment, and to continue therapy. She needed to experience and practice her somewhat fragile new self, which included greater intimacy in relationships, additional mastery of work skills, and a more cohesive sense of self. (Courtesy of T. Byram Karasu, M.D.)

The persons best suited for the expressive psychotherapy approach have reasonably well-integrated egos and the capacity to both sustain and detach from a bond of dependency and trust. They are, to some degree, psychologically minded and self-motivated, and they are generally able, at least temporarily, to tolerate doses of frustration without decompensating. They must also have the ability to manage the rearousal of painful feelings outside the therapy hour without additional contact. Patients must have some capacity for introspection and impulse control, and they should be able to recognize the cognitive distinction between fantasy and reality.

GOALS. The overall goals of expressive psychotherapy are to increase the patient's self-awareness and to improve object relations through the exploration of current interpersonal events and perceptions. In contrast to psychoanalysis, expressive psychotherapy modifies major structural changes in ego function and defenses in light of patient limitations. The aim is to achieve a more limited and, thus, select and focused understanding of one's problems. Rather than uncovering deeply hidden and past motives and tracing

them back to their origins in infancy, the major thrust is to deal with preconscious or conscious derivatives of conflicts as they became manifest in present interactions. Although the therapy seeks insight, it is less extensive; instead of delving to a genetic level, greater emphasis is on clarifying recent dynamic patterns and maladaptive behaviors in the present.

MAJOR APPROACH AND TECHNIQUES. The primary modus operandi involves the establishment of a therapeutic alliance and early recognition and interpretation of negative transference. Only limited or controlled regression is encouraged, and positive transference manifestations are generally left unexplored unless they are impeding therapeutic progress; even here, the emphasis is on shedding light on current dynamic patterns and defenses.

LIMITATIONS. A general limitation of expressive psychotherapy, as of psychoanalysis, is the problem of emotional integration of cognitive awareness. The primary danger for patients who are at the more disorganized end of the diagnostic spectrum, however, may have less to do with the over intellectualization that is sometimes seen in neurotic patients than with the threat of decompensation from or acting out of, deep or frequent interpretations that the patient is unable to integrate properly.

Some therapists fail to accept the limitations of a modified insight-oriented approach and so apply it inappropriately to modulate the techniques and goals of psychoanalysis. Overemphasis on dreams and fantasies, zealous efforts to use the couch, indiscriminate deep interpretations, and continual focus on the analysis of transference may have less to do with the patient's needs than with those of a therapist who is unwilling or unable to be flexible.

Ms. S was an attractive 30-year-old unmarried woman working as a secretary when she presented for consultation. Her chief complaints at the time were feeling "only anger and tension" and an inability to apply herself to studying voice, "which is one of the most important things to me."

In obtaining a history, the therapist noted that Ms. S had never completed anything: she had dropped out of college, never pursued a music degree, and switched from job to job, and even city to city. What

initially seemed like a woman with diverse interests (e.g., jobs as a research assistant; freelance copyeditor; part-time radio announcer; manager of data entry for a software company; and, most recently, secretary) actually reflected a woman with a chaotic lifestyle and serious difficulties committing to anyone or anything. Although obviously intelligent, Ms. S presented with unrealistic expectations regarding her consultation. For example, after the first consultative session, Ms. S said she felt good afterward but felt there were "no revelations yet." Because of Ms. S's inability to commit and her somewhat disorganized life, the therapist recommended a course of psychotherapy, beginning twice a week, rather than something more intense like psychoanalysis. The therapist also realized throughout the consultation that Ms. S would have difficulty with free association without getting disorganized. The therapist also thought that Ms. S might regress unproductively on the couch without visual contact with the therapist.

Ms. S was the second oldest of four children—two brothers and a younger sister, with whom she was most competitive and who seemed the mother's favorite. She described her mother as a successful professor who was demanding and critical as if she had a "raised eyebrow" in disapproval. For example, much to her mother's chagrin, Ms. S had once wanted a sandwich "with everything on it." Ms. S was also disappointed when she received one piece of luggage rather than a complete set for a Christmas gift. She was able to accept the therapist's interpretation that she felt "part of a set" by being one of four siblings. Ms. S initially idealized her father, who was active in the family church, but eventually saw him as disappointing and rejecting.

Ms. S's ideal therapist would be "flexible," by which she meant a therapist who might do hypnosis one session, psychotherapy the next, and, maybe, analysis another session. In fact, within the first week of beginning therapy, Ms. S had simultaneously consulted a hypnotherapist, which she mentioned in passing only weeks later, for her neck pain and tension. Although she did not pursue hypnosis, she did maintain a chiropractor for most of her therapy, also something she mentioned many months after beginning therapy. She did speak of wanting to be "on best behavior" and "follow the rules." Her tremendous sense of entitlement, however, was evident: she expected "cut-rate prices" on everything from

haircuts and car repairs to doctors' visits. Her initial fee was a much reduced one, which she paid late and begrudgingly.

Although she had therapy only twice a week, Ms. S developed intense feelings for her therapist. Mostly she experienced rage when she saw evidence of the therapist's other patients, such as footprints on the waiting room floor after a snowstorm or a coat hanger turned around. She expressed the wish to keep some of her things, like bobby pins and hairspray, in the therapist's bathroom. She vacillated between feelings she wanted to move in and feelings that the therapist did not exist. For example, before she took a plane flight, she wondered who would tell her therapist if something happened to her. She had never given the therapist's name to anyone, nor did she have her name in her weekly appointment book. The therapist interpreted that she had a simultaneous wish to devalue her and not to share her with anyone else. Associations to a dream with an image of a string of baroque pearls led to thoughts that these pearls—irregular and imperfect—defective and even lopsided, represented how she viewed herself.

Over the next few years, Ms. S was able to commit to regularly coming to therapy, although the course was somewhat tumultuous, with many threats of quitting and much withholding of information. At one point, she even tried to provoke the therapist by seeking a consultation with another therapist in order to "tattle" on her, just as she had tattled on her siblings. Her therapist remained unprovoked and continued to provide a safe environment for Ms. S to explore her ambivalence to the therapist and the therapeutic situation. The therapist was also able to contain Ms. S's tendency to regress, particularly with separations, by providing her with the therapist's telephone number.

She had actually entered therapy with an unconscious wish to become a world-famous singer who would win her mother's approval and praise. Her narcissism and sense of entitlement made it difficult for her to give up on that fantasy despite repeated evidence that she did not have sufficient talent. She was finally able to settle on a compromise: she began to work diligently and closely as a research assistant to her mother, who was writing a book, and as Ms. S became more focused and organized over time, she even thought she might write a book about the church. (Courtesy of T. Byram Karasu, M.D. and S. R. Karasu, M.D.)

Supportive Psychotherapy. Supportive psychotherapy aims at the creation of a therapeutic relationship as a temporary buttress or bridge for the deficient patient. It has roots in virtually every therapy that recognizes the ameliorative effects of emotional support and a stable, caring atmosphere in the management of patients. As a nonspecific attitude toward mental illness, it predates scientific psychiatry, with foundations in 18th-century moral treatment, whereby for the first time, patients received understanding and kindness in a humane, interpersonal treatment environment free from mechanical restraints.

Supportive psychotherapy has been the primary form used in the general practice of medicine and rehabilitation, frequently to augment other measures, such as prescriptions of medication to suppress symptoms, rest to remove the patient from excessive stimulation, or hospitalization to provide a structured therapeutic environment, protection, and control of the patient. It can serve as primary or ancillary treatment. The global perspective of supportive psychotherapy (often part of a combined treatment approach) places major etiologic emphasis on external rather than intrapsychic events, particularly on stressful environmental and interpersonal influences on a severely damaged self.

INDICATIONS AND CONTRAINDICATIONS. Supportive psychotherapy is generally indicated for those patients for whom classic psychoanalysis or insight-oriented psychoanalytic psychotherapy is typically contraindicated—those who have poor ego strength and whose potential for decompensation is high. Amenable patients fall into the following major areas: (1) individuals in acute crisis or a temporary state of disorganization and inability to cope (including those who might otherwise be well functioning) whose intolerable life circumstances have produced extreme anxiety or sudden turmoil (e.g., individuals going through grief reactions, illness, divorce, job loss, or who were victims of crime, abuse, natural disaster, or accident); (2) patients with chronic severe pathology with fragile or deficient ego functioning (e.g., those with latent psychosis, impulse disorder, or severe character disturbance); (3) patients whose cognitive deficits and physical symptoms make them particularly vulnerable and, thus, unsuitable for an insight-oriented approach

(e.g., certain psychosomatic or medically ill persons); and (4) individuals who are psychologically unmotivated, although not necessarily characterologically resistant to a depth approach (e.g., patients who come to treatment in response to family or agency pressure and are interested only in immediate relief or those who need assistance in very specific problem areas of social adjustment as a possible prelude to more exploratory work).

Mr. C, a 50-year-old married man with two sons, the owner of a small construction company, was referred by his internist after recovery from bypass surgery because of frequent, unfounded physical complaints. He was taking minor tranquilizers in increasing doses, not adhering to his daily regimen, avoiding sexual contact with his wife, and had dropped out of group therapy for postsurgical patients after one session.

He came to his first appointment 20 minutes late, after having "forgotten" two previous appointments. He was extremely anxious, often lost in his train of thought, and was semi-delusional about his wife and sons, suggesting that they might want to have him locked up. He briefly told his life history, which included his coming from a strict and hard-working but caring middle-class family and the death of his mother when he was only 11 years old. He had joined his father's business (taking over after his father's death 2 years earlier), with both of his sons as associates. Describing himself as successful in work and marriage, he claimed that "the only test I ever failed was the stress test."

Mr. C explained his lack of adherence to dietary restrictions as a lack of will and his constant contact with the internist as his having real physical problems not yet diagnosed; he rejected the idea of addiction to tranquilizers, insisting that he could quit any time. He had no fantasy life, remembered no dreams, made it clear that he had entered treatment on his internist's instruction only, and started each session by stating that he had nothing to discuss.

After suggesting that Mr. C was coming to sessions just to pass the "sanity test" and that there was no reason to have him locked up, the psychiatrist encouraged the patient to join him in figuring out the real reasons for his anxiety. Initial sessions were devoted to discussing the patient's medical condition and providing factual information about heart

and bypass surgery. The therapist likened the patient's condition to that of an older house getting new plumbing, trying to allay his unrealistic fears of impending death. As Mr. C's anxiety declined, he became less defensive and more psychologically accessible. As the therapist began to explore his difficulty in accepting help, Mr. C was able to talk about his inability to admit problems (i.e., weaknesses). The therapist's explicit recognition of the patient's strength in admitting his weaknesses encouraged the patient to reveal more about himself—how he had welcomed his father's death and his belief that perhaps his illness was punishment. The psychiatrist also encouraged him to speak about his unrealistic guilt and, at the same time, helped him recognize his suspicion of his sons as the reflection of his own wishes concerning his father and his lack of commitment to his medical regimen as a wish to die to expiate guilt. After steady urging by the therapist, Mr. C returned to work. He agreed to meet monthly with the psychiatrist and to taper off his use of tranquilizers. He even agreed that he might see the psychiatrist for "deep analysis" in the future because his wife now jokingly complained of his obsessive dieting, his uncompromising exercise regimens, and his regularly scheduled sexual activities. (Courtesy of T. Byram Karasu, M.D.)

Because support forms a tacit part of every therapeutic modality, it is rarely contraindicated as such. The typical attitude regards better-functioning patients as unsuitable not because a supportive approach will harm them, but because they will not be sufficiently benefited by it. In aiming to maximize the patient's potential for further growth and change, supportive therapy is relatively restricted and superficial and, thus, is not recommended as the treatment of choice if the patient is available for, and capable of, a more in-depth approach.

GOALS. The general aim of supportive treatment is the amelioration or relief of symptoms through behavioral or environmental restructuring within the existing psychic framework. Supportive therapy helps the patient to adapt better to problems and to live more comfortably with his or her psychopathology. To restore the disorganized, fragile, or decompensated

patient to a state of relative equilibrium, the therapist's primary goal is to suppress or control symptomatology and to stabilize the patient in a protective and reassuring benign atmosphere that militates against overwhelming external and internal pressures. The ultimate goal is to maximize the integrative or adaptive capacities so that the patient increases the ability to cope while decreasing vulnerability by reinforcing assets and strengthening defenses.

MAJOR APPROACH AND TECHNIQUES. Supportive therapy uses several methods, either singly or in combination, including warm, friendly, strong leadership; partial gratification of dependency needs; support in the ultimate development of legitimate independence; help in developing pleasurable activities (e.g., hobbies); adequate rest and diversion; removal of excessive strain, when possible; hospitalization, when indicated; medication to alleviate symptoms; and guidance and advice in dealing with current issues. This therapy uses techniques to help patients feel secure, accepted, protected, encouraged, safe, and not anxious.

LIMITATIONS. To the extent that much supportive therapy is spent on practical, everyday realities and on dealing with the external environment of the patient, it may be viewed as more mundane and superficial than depth approaches. Because those patients attend therapy sessions intermittently and less frequently, the interpersonal commitment may not be as compelling on the part of either the patient or the therapist. Greater severity of illness (and possible psychoses) also makes such treatment potentially more erratic, demanding, and frustrating. The need for the therapist to deal with other family members, caretakers, or agencies (auxiliary treatment, hospitalization) can become an additional complication because the therapist comes to serve as an ombudsman to negotiate with the outside world of the patient and with other professional peers. Finally, the supportive therapist must be able to accept personal limitations and the patient's limited psychological resources and to tolerate the often unrewarded efforts until the patient makes small gains.

Mr. W was a 42-year-old widowed businessman who was referred by his

internist because of the sudden death of his wife, who had had an intracranial hemorrhage, about 2 months earlier. Mr. W had two children, a boy and a girl, ages 10 and 8 years, respectively.

Mr. W had never been to a psychiatrist before, and when he arrived, he admitted he was not sure what a psychiatrist could do for him. He just had to get over his wife's death. He was not sure how talking about anything could help. He had been married for 15 years. He admitted to having difficulty sleeping, particularly awakening in the middle of the night with considerable anxiety about the future. One of his relatives had given him some of her clonazepam for his anxiety, which helped tremendously, but he feared getting dependent on it. He was also drinking more than he thought he should. He was most concerned about raising his children alone and felt somewhat overwhelmed by the responsibility. He was beginning to appreciate just how wonderful a mother his wife had been and now saw how critical he had been of her for spending so much time with the children. "It really does take a lot of effort," he said.

Mr. W did admit to feelings of guilt. For one thing, he admitted to some sense that he could now start over. He had been somewhat restless in the marriage recently before his wife's death and had been unfaithful for a brief period early in the marriage. He also felt some guilt that had he been awake the night of his wife's hemorrhage, and maybe he could have saved his wife. In reality, there was nothing he could have done.

Mr. W agreed to come for a few sessions to talk about his wife. At this point, only 2 months after her death, he seemed to have an uncomplicated mourning reaction. Although he talked easily in session, he was clearly worried that he might like "being here too much." The therapist chose not to interpret his dependency conflicts. Mr. W seemed to have good coping skills and used humor as a high-functioning defense. For example, in giving a eulogy for his wife (who had been a very popular member of their congregation), he looked around at the enormous crowd of people at the church service and said he had never seen so many people attending church before, adding, "Sorry, Reverend."

After about four sessions, Mr. W said that he felt better and no longer saw the need for further sessions. He was sleeping better and had stopped drinking excessively. The therapist suggested that he might want to continue to talk more about his guilt and his life as he went forward

without his wife. The therapist was also reassuring that there seemed to be nothing else Mr. W could have done to save his wife. He also encouraged the patient to begin dating when he felt ready, something that Mr. W's in-laws were not encouraging. For now, however, Mr. W was not interested in any further therapy. He was appreciative of the therapist and felt that talking about his wife's death had been helpful. The therapist accepted his wish to discontinue their sessions but encouraged Mr. W to keep in touch to let him know how he was doing. (Courtesy of T. Byram Karasu, M.D. and S. R. Karasu, M.D.)

CORRECTIVE EMOTIONAL EXPERIENCE. The relationship between therapist and patient allows a therapist to display behavior different from the destructive or unproductive behavior of a patient's parent. At times, such experiences seem to neutralize or reverse some effects of the parents' mistakes. If the patient had overly authoritarian parents, the therapist's friendly, flexible, nonjudgmental, nonauthoritarian—but at times firm and limit setting—attitude allows the patient to adjust to, be led by, and identify with a new parent figure. Franz Alexander described this process as a corrective emotional experience. It draws on elements of both psychoanalysis and psychoanalytic psychotherapy.

BRIEF PSYCHODYNAMIC PSYCHOTHERAPY

Brief psychodynamic psychotherapy has gained widespread popularity, partly because of the great pressure on health care professionals to contain treatment costs. It is also easier to evaluate treatment efficacy by comparing groups of persons who have had short-term therapy for mental illness with control groups than it is to measure the results of long-term psychotherapy. Thus, short-term therapies have been the subject of much research, especially on outcome measures, which have found them to be effective. Other short-term methods include interpersonal therapy and cognitive-behavioral therapy (CBT, discussed later in this chapter).

Brief psychodynamic psychotherapy is a time-limited treatment (10 to 12 sessions) based on psychoanalysis and psychodynamic theory. It is useful in helping persons with depression, anxiety, and posttraumatic stress disorder

(PTSD), among others. There are several methods, each having its own treatment technique and specific criteria for selecting patients; however, they are more similar than different.

In 1946, Franz Alexander and Thomas French identified the basic characteristics of brief psychodynamic psychotherapy. They described a therapeutic experience designed to put patients at ease, to manipulate the transference, and to use trial interpretations flexibly. Alexander and French conceived psychotherapy as a corrective emotional experience capable of repairing traumatic events of the past and convincing patients that new ways of thinking, feeling, and behaving are possible. At about the same time, Eric Lindemann established a consultation service at Massachusetts General Hospital in Boston for persons experiencing a crisis. He developed new treatment methods to deal with these situations and eventually applied these techniques to persons who were not in crisis, but who were experiencing various kinds of emotional distress. Since then, the field has been influenced by many workers such as David Malan in England, Peter Sifneos in the United States, and Habib Davanloo in Canada.

Types

Brief Focal Psychotherapy (Tavistock–Malan). Brief focal psychotherapy was originally developed in the 1950s by the Balint team at the Tavistock Clinic in London. Malan, a member of the team, reported the results of the therapy. Malan's selection criteria for treatment included eliminating absolute contraindications, rejecting patients for whom certain dangers seemed inevitable, clearly assessing patients' psychopathology, and determining patients' capacities to consider problems in emotional terms, face disturbing material, respond to interpretations, and endure the stress of the treatment. Malan found that high motivation invariably correlated with a successful outcome. Contraindications to treatment were serious suicide attempts, substance dependence, chronic alcohol abuse, incapacitating chronic obsessional symptoms, incapacitating chronic phobic symptoms, and gross destructive or self-destructive acting out.

REQUIREMENTS AND TECHNIQUES. In Malan's routine, therapists should identify the transference early and interpret it and the negative transference.

They should then link the transferences to patients' relationships with their parents. Both patients and therapists should be willing to become deeply involved and to bear the ensuing tension. Therapists should formulate a circumscribed focus and set a termination date in advance, and patients should work through grief and anger about termination. An experienced therapist should allow about 20 sessions as an average length for the therapy; a trainee should allow about 30 sessions. Malan himself did not exceed 40 interviews with his patients.

Time-Limited Psychotherapy (Boston University–Mann). A psychotherapeutic model of exactly 12 interviews focusing on a specified central issue was developed at Boston University by James Mann and his colleagues in the early 1970s. In contrast with Malan's emphasis on clear-cut selection and rejection criteria, Mann was not as explicit about the appropriate candidates for time-limited psychotherapy. Mann considered the major emphases of his theory to be determining a patient's central conflict reasonably correctly and exploring young persons' maturational crises with many psychological and somatic complaints. Mann's exceptions, similar to his rejection criteria, include persons with major depressive disorder that interferes with the treatment agreement, those with acute psychotic states, and desperate patients who need, but cannot tolerate, object relations.

REQUIREMENTS AND TECHNIQUES. Mann's technical requirements included strict limitation to 12 sessions, positive transference predominating early, specification and strict adherence to a central issue involving transference, positive identification, making separation a maturational event for patients, absolute prospect of termination to avoid development of dependence, clarification of present and past experiences and resistances, active therapists who support and encourage patients, and education of patients through direct information, reeducation, and manipulation. The conflicts likely to be encountered included independence versus dependence, activity versus passivity, unresolved or delayed grief, and adequate versus inadequate self-esteem.

Short-Term Dynamic Psychotherapy (McGill University–Davanloo). As conducted by Davanloo at McGill University, short-term

dynamic psychotherapy encompasses nearly all varieties of brief psychotherapy and crisis intervention. Davanloo's series classified patients treated as those whose psychological conflicts are predominantly oedipal, those whose conflicts are not oedipal, and those whose conflicts have more than one focus. Davanloo also devised a specific psychotherapeutic technique for patients with severe, long-standing neurotic problems, specifically those with incapacitating obsessive-compulsive disorders (OCDs) and phobias.

Davanloo's selection criteria emphasize evaluating those ego functions of primary importance to psychotherapeutic work: the establishment of a psychotherapeutic focus; the psychodynamic formulation of the patient's psychological problems; the ability to interact emotionally with evaluators; a history of give-and-take relationships with a significant person in the patient's life; the patient's ability to experience and tolerate anxiety, guilt, and depression; the patient's motivations for change, psychological mindedness, and an ability to respond to interpretation and to link evaluators with persons in the present and past. Both Malan and Davanloo emphasized a patient's responses to interpretation as an essential selection and prognostic criterion.

REQUIREMENTS AND TECHNIQUES. The highlights of Davanloo's psychotherapeutic approach are flexibility (therapists should adapt the technique to the patient's needs), control, the patient's regressive tendencies, active intervention to avoid having the patient develop overdependence on a therapist, and the patient's intellectual insight and emotional experiences in the transference. These emotional experiences become corrective as a result of the interpretation.

Ana, a divorced 60-year-old woman, sought psychiatric help following a severe depressive episode lasting several months. This episode, which was one of many in her life, was especially severe in terms of loss of energy, interest, motivation, and the intensity of her sadness and wish to die. Only her profound religious convictions protected her from acting on these wishes. Ana had lost much weight, had trouble sleeping, experienced many nightmares, and had difficulty with concentration. Pervasive feelings of hatred for her mother plagued Ana; her mother was

very old, ill, and dependent on Ana, who was unable to forgive her for abandoning her in an orphanage when she was 5 or 6 years of age.

After an extensive assessment, the dynamic formulation of Ana's problem was:

1. *Life problems:* Recurrent depressive episodes plagued by feelings of guilt and self-reproach; problems with men involving choosing partners who are commonly cold, distant, or otherwise unavailable; involuntary and painful emotional distance from her children, friends, and other close relationships; and unproductive and unrewarding work life, despite considerable intellectual gifts.
2. *Dynamics:* Ambivalent relationship with her mother, whom she blames for most of the tragedies of her life; guilt and need for punishment concerning her unrelenting hatred for her mother; and pathologic grief reaction for the loss of an idealized and more optimal relationship with her mother, the one she remembers she had before her orphanage placement. From this focus, there flows a melancholic conviction of the inevitable failure of human relationships.
3. *Pathogenic foci:* Grief and inability to mourn the loss of her mother after going to live in the orphanage, with attendant rage and guilt; pathologic grief for the loss of her father, who, because of severe alcoholism, abandoned the family first, a move that caused the mother to place her children in an orphanage in order to be able to work and ultimately recover their care. Unconsciously, she blamed her mother for the family catastrophe, thus "protecting" an idealized view of her father, to whom she was profoundly attached.

For Ana, the initial phase of treatment focused on the clarification and the experience of her destructive impulses toward her mother, which, as they were worked through, made possible the appearance of a modicum of empathy with her mother's painful life situation around the time she placed Ana and her sisters in the orphanage. Next, the therapy focused on Ana's father. She experienced deep feelings of idealization, disappointment, anger, and grief with increasing clarity and intensity, frequently via displaced feelings in the transference and after overcoming considerable resistance. The last phase of treatment permitted the

development of realistic feelings of empathy and appreciation for her mother, now without anger or emotional distancing, and the reawakening within Ana of feelings of joy and hope, as well as professional ambition. (Courtesy of M. Trujillo, M.D.)

Short-Term Anxiety-Provoking Psychotherapy (Harvard University–Sifneos). Sifneos developed short-term anxiety-provoking psychotherapy at the Massachusetts General Hospital in Boston during the 1950s. He used the following criteria for selection: a circumscribed chief complaint (implying a patient's ability to select one of a variety of problems to be given top priority and the patient's desire to resolve the problem in treatment), one meaningful or give-and-take relationship during early childhood, the ability to interact flexibly with an evaluator and to express feelings appropriately, above-average psychological sophistication (implying not only above-average intelligence but also an ability to respond to interpretations), a specific psychodynamic formulation (usually a set of psychological conflicts underlying a patient's difficulties and centering on an oedipal focus), a contract between therapist and patient to work on the specified focus and the formulation of minimal expectations of outcome, and good to excellent motivation for change, not just for symptom relief.

Chris, a 31-year-old single man, sought help for a moderate depressive episode precipitated by the loss of his relationship with his girlfriend, Joanna. She had broken off the relationship after approximately 1 year, tired of Chris's erratic work ethic and emotional instability and discouraged by his fear of commitment to the future of their relationship. This cycle of infatuation, increasing fear of commitment, and relationship loss had become a pattern in Chris's interpersonal life. His work life experienced similar problems. He frequently lost jobs because of serious conflict and threatening confrontations with his superiors. As conflicts arose at both work and home, Chris typically suffered increasing anxiety and episodic panic attacks. After the loss of each relationship, Chris usually confronted moderate depressive feelings, at times accompanied by suicidal ideation.

After an assessment, the dynamic hologram for Chris was:

1. *Life problems:* Recurrent episodes of anxiety and depression; work problems; unstable interpersonal relationships; conflict with authority figures; antagonism toward, and emotional distance from, his father, brother, and male friends; and fears of heterosexual intimacy and commitment.
2. *Dynamic forces:* Ongoing hostility and envy toward males, authority figures, and successful people, and compulsive and possessive seeking of female love objects with a serious inability to consider, fulfill, or tolerate their independent needs.
3. *Genetic pathogenic foci:* Unconscious loss of maternal objects precipitated by birth of a brother when Chris was age 2 years; uncontrolled grief for that loss with a compulsive drive to experience child-like possession of love objects; and compulsive hostility toward others perceived as rivals.

The therapist's active inquiry yielded additional confirmation of the persistence of repressed sexual feelings toward his mother and the presence of hostile feelings toward all rivals for his mother's affection. A memory suffused with very visceral feelings emerged in this phase as a result of the therapist's active inquiry. In this memory, Chris saw himself in his mother's arms in a dark room. He remembered the intense pleasure of contact with his mother's warm skin, the texture of her clothes, and the smell of her perfume vividly. While narrating this memory to the therapist, Chris became so absorbed in the experience that he blushed intensely. He also described the painful termination of this moment of pleasure by his father's sudden and disruptive opening of the door and the flood of light that disturbed his pleasurable absorption. This sequence gave way to experiencing grief at the loss of the intense and exclusive bond with his mother after his brother's birth and a reexperiencing of anger, impotence, and loneliness. These feelings were all too familiar in his present life when his romantic attachments would be threatened or lost. The affective link between this childhood experience and his intimacy problems in the present became very obvious to Chris, and the acceptance of this link enhanced his capacity to work through this essential component of his pathology. A parallel conflict appeared in the

transference as the patient resented the “intrusion” of the inquiring therapist into the zealously guarded privacy of this primal fantasy of material possession. (Courtesy of M. Trujillo, M.D.)

REQUIREMENTS AND TECHNIQUES. Treatment includes four major phases: patient–therapist encounter, early therapy, height of treatment, and evidence of change and termination. Therapists use the following techniques during the four phases.

Patient–Therapist Encounter. A therapist establishes a working alliance by using the patient’s quick rapport with, and positive feelings for, the therapist that appear in this phase. The judicious use of open-ended and forced-choice questions enables the therapist to outline and concentrate on a therapeutic focus. The therapist specifies the minimal expectations of outcome to be achieved by the therapy.

Early Therapy. Transference involves clarifying feelings for the therapist as soon as they appear, a technique that establishes a true therapeutic alliance.

Height of the Treatment. Height of treatment emphasizes active concentration on the oedipal conflicts that have been chosen as the therapeutic focus; repeated use of anxiety-provoking questions and confrontations; avoidance of pregenital characterologic issues, which the patient uses defensively to avoid dealing with the therapist’s anxiety-provoking techniques; avoidance at all costs of a transference neurosis; repetitive demonstration of the patient’s neurotic ways or maladaptive patterns of behavior; concentration on the anxiety-laden material, even before the defense mechanisms have been clarified; repeated demonstrations of parent-transference links by the use of properly timed interpretations based on material given by the patient; establishment of a corrective emotional experience; encouragement and support of the patient, who becomes anxious while struggling to understand the conflicts; new learning and problem-solving patterns; and repeated presentations and recapitulations of the patient’s psychodynamics until the defense mechanisms used in dealing with oedipal conflicts are understood.

Evidence of Change and Termination of Psychotherapy. The final phase of

therapy emphasizes the tangible demonstration of change in the patient's behavior outside therapy, evidence that the patient is using adaptive patterns of behavior, and initiation of talk about terminating the treatment.

Overview and Results

The shared techniques of all the brief psychotherapies described above outdistance their differences. They share the therapeutic alliance or dynamic interaction between therapist and patient, the use of transference, the active interpretation of a therapeutic focus or central issue, the repetitive links between parental and transference issues, and the early termination of therapy.

The outcomes of these brief treatments have undergone extensive investigation. Contrary to prevailing ideas that the therapeutic factors in psychotherapy are nonspecific, controlled studies and other assessment methods (e.g., interviews with unbiased evaluators, patients' self-evaluations) point to the importance of the specific techniques used. The capacity for genuine recovery in certain patients is far greater than was thought. A certain type of patient receiving brief psychotherapy can benefit greatly from a practical working through of his or her nuclear conflict in the transference. Such patients can be recognized in advance through a process of dynamic interaction because they are responsive, motivated, and able to face disturbing feelings and because they have a circumscribed focus formulated for them. The more radical the technique in terms of transference, depth of interpretation, and the link to childhood, the more radical the therapeutic effects will be. For some disturbed patients, a carefully chosen partial focus can be therapeutically effective.

GROUP PSYCHOTHERAPY, COMBINED INDIVIDUAL AND GROUP PSYCHOTHERAPY, AND PSYCHODRAMA

Group psychotherapy is a modality that employs a professionally trained leader who selects, composes, organizes, and leads a collection of members to work together toward the maximal attainment of the goals for each individual in the group and for the group itself. Certain properties present in groups, such as mutual support, can be harnessed in the service of providing relief from psychological suffering and supply peer support to counter

isolation experienced by many who seek psychiatric help. Similarly, homogeneously composed small groups are ideal settings for the dissemination of accurate information about a condition shared by group members. Medical illness, substance use disorders, and chronic and persistent severe psychiatric conditions, including schizophrenia and major affective disorders, are cases in point.

A widely accepted psychiatric treatment modality, group psychotherapy uses therapeutic forces within the group, constructive interactions among members, and interventions of a trained leader to change the maladaptive behaviors, thoughts, and feelings of emotionally distressed individuals. Group therapy applies to inpatient and outpatient settings, institutional work, partial hospitalization units, halfway houses, community settings, and private practice. Group psychotherapy is also widely used by those who are not mental health professionals in the adjuvant treatment of physical disorders. Application of group psychotherapy principles has also found success in the business and education fields in the form of training, sensitivity, and role-playing.

Group psychotherapy is a treatment in which carefully selected persons who are emotionally ill meet in a group guided by a trained therapist and help one another effect personality change. By using a variety of technical maneuvers and theoretical constructs, the leader directs group members' interactions to bring about changes.

Classification

Group therapy, at present, has many approaches. Many individually derived forms of psychotherapy have their group counterparts. A representative sample of major theoretical orientations toward group therapy can be broadly subsumed under the psychodynamic, interpersonal, and cognitive-behavioral schools of thought. Table 23-4 illustrates some of the similarities and differences among representative group formats.

Patient Selection

To determine a patient's suitability for group psychotherapy, a therapist needs to gather a great deal of information in a screening interview. The psychiatrist should take a psychiatric history and perform a mental status

examination to obtain certain dynamic, behavioral, and diagnostic information.

Robert entered therapy seeking to understand why he was unable to maintain close or lasting relationships. A handsome and successful businessman, he had made a painful and courageous transition away from self-centered, dysfunctional parents early in his life. Although Robert made good initial impressions in his jobs, he was always puzzled and disappointed when his superiors gradually lost interest in him, and his colleagues avoided him. In one-on-one therapy, he was charming and entertaining, but perceived narcissistic slights caused him to become angry and attack. His therapist recommended group psychotherapy when the patient's transference feelings remained intense, and therapy was at a seeming impasse. Initially, Robert charmed the group and strove to be the center of attention. Visibly annoyed whenever he felt the group leader was paying more attention to other members, Robert was especially critical and hostile toward older people in the group and displayed little empathy for others. After repeated and forceful confrontations from the group about his antagonistic behavior, he gradually realized that he was repeating childhood patterns in his family of desperately seeking the attention of unloving parents and then entering violent rages when they lost interest. (Courtesy of Normund Wong, M.D.)

Diagnosis. The diagnosis of patients' disorders is essential in determining the best therapeutic approach and in evaluating patients' motivations for treatment, capacities for change, and personality structure strengths and weaknesses. Few contraindications exist for group therapy. Antisocial patients generally do poorly in a heterogeneous group setting because they cannot adhere to group standards, but if the group is composed of other antisocial patients, they may respond better to peers than to perceived authority figures. Depressed patients profit from group therapy after they have established a trusting relationship with the therapist. Patients who are actively suicidal or severely depressed should not be treated solely in a group setting. Patients who are manic are disruptive but, once under pharmacologic control, do well in the group setting. Therapists should exclude patients who

are delusional and who may incorporate the group into their delusional system, as well as patients who pose a physical threat to other members because of uncontrollable aggressive outbursts.



Table 23-4
Comparison of Major Group Therapy Orientations

	Supportive	Psychodynamic	Self-Help	Cognitive/Behavioral
	Group therapy	Group therapy	Groups	Group therapy
Frequency	1–5 times/wk	1–2 times/wk	7 days/wk	1–3 times/wk
Individual screening	Usually	Always	None	Always
Group size	8–15 members	5–9 members	No size limits	5–10 members
Goals	Better adaptation for daily living	Reconstruction of personality dynamics	Social support	Relief of specific symptoms
Indications	Crisis situations; severe emotional	Neuroses; mild personality disorders	Shared life experiences	Phobias; anxiety disorders
Group composition	Homogeneous for level of psychopathology	Balance of similarities and differences	Homogeneous	Homogeneous for similar symptoms
Group focus	"Here and now"; family, vocational, environmental factors	Past and present; intragroup and extragroup dynamics	Education; emotional "sharing"	Training in methods that control symptoms
Use of confrontation	No	Yes	No	Rarely
Therapist activity	Actively structures and leads the group	Active around interpretation	No formal group leader	Very active in teaching skills to patient
Extragroup contacts	Encouraged	Prohibited or discouraged	Encouraged; formal peer sponsorship	Discouraged
Transference	Not analyzed	Utilized extensively	Not analyzed	Not relevant to the group work
Therapeutic factors	Cohesion; universality; reality testing instillation of hope imparting information	Cohesion; catharsis; family replay; interpersonal learning	Cohesion; universality; education; peer support	Cohesion; universality; education; reinforcement

Preparation

Patients prepared by a therapist for a group experience tend to continue in treatment longer and report less initial anxiety than those who are not prepared. The preparation consists of having a therapist explain the procedure in as much detail as possible and answer the patient's questions before the first session.

Size. Group therapy has been successful with as few as 3 members and as many as 15, but most therapists consider 8 to 10 members the optimal size. Interaction may be insufficient with fewer members unless they are especially verbal, and with more than 10 members, the interaction may be too great for the members or the therapist to follow.

Frequency and Length of Sessions. Most group psychotherapists conduct group sessions once a week. Maintaining continuity in sessions is essential. When there are alternate sessions, the group meets twice a week, once with and once without the therapist. Group sessions generally last anywhere from 1 to 2 hours, but the time limit should be constant.

Marathon groups were most popular in the 1970s but are much less common today. In time-extended therapy (marathon group therapy), the group meets continuously for 12 to 72 hours. Enforced interactional proximity and, during the most prolonged time-extended sessions, sleep deprivation break down certain ego defenses, release affective processes, and theoretically promote open communication. Time-extended sessions, however, can be dangerous for patients with weak ego structures, such as persons with schizophrenia or BPD.

Homogeneous versus Heterogeneous Groups. Most therapists believe that groups should be as heterogeneous as possible to ensure maximal interaction. Members with different diagnostic categories and varied behavioral patterns, from all races, social levels, and educational backgrounds, and of varying ages and both sexes, should be brought together. Patients between the ages of 20 and 65 years can be included effectively in the same group. Age differences help in developing parent–child and brother–sister models, and patients have the opportunity to relive and rectify interpersonal difficulties that may have appeared insurmountable.

Both children and adolescents benefit the most in groups comprising mostly persons in their age groups. Some adolescent patients are capable of assimilating the material of an adult group, regardless of content, but they may benefit significantly from a constructive peer experience that they might otherwise not have.

Open versus Closed Groups. Closed groups have a set number and composition of patients. If members leave, no new members are accepted. In open groups, membership is more fluid, and new members join whenever old members leave.

Mechanisms

Group Formation. Each patient approaches group therapy differently, and, in this sense, groups are microcosms. Patients use typical adaptive abilities, defense mechanisms, and ways of relating, and when these tactics are ultimately reflected back to them by the group, they learn to be introspective about their personality functioning. A process inherent in group formation requires that patients suspend their previous ways of coping. In entering the group, they allow their executive ego functions—reality testing, adaptation to and mastery of the environment, and perception—to be assumed, to some degree, by the collective assessment provided by the total membership, including the leader.

Therapeutic Factors. Table 23-5 outlines 20 significant therapeutic factors that account for change in group psychotherapy.



Table 23-5
Therapeutic Factors

Factor	Definition
Abreaction	A process by which repressed material, particularly a painful experience or conflict, is brought back to consciousness. During the process, the person not only recalls, but also relives the material, which is accompanied by the appropriate emotional response; insight usually results from the experience.
Acceptance	The feeling of being accepted by other members of the group; differences are tolerated, and there is an absence of censure.
Altruism	The act of one member's being of help to another; putting another person's need before one's own and learning that there is value in giving to others. The term was originated by Auguste Comte (1798–1857), and Freud believed it was a major factor in establishing group cohesion and

	community feeling.
Catharsis	The expression of ideas, thoughts, and suppressed material that is accompanied by an emotional response that produces a state of relief in the patient.
Cohesion	The sense that the group is working together toward a common goal; also referred to as a sense of we-ness; believed to be the most important factor related to positive therapeutic effects.
Consensual validation	Confirmation of reality by comparing one's own conceptualizations with those of other group members; interpersonal distortions are thereby corrected. The term was introduced by Harry Stack Sullivan. Trigant Burrow had used the phrase consensual observation to refer to the same phenomenon.
Contagion	The process in which the expression of emotion by one member stimulates the awareness of a similar emotion in another member.
Corrective familial experience	The group recreates the family of origin for some members who can work through conflicts psychologically through group interaction (e.g., sibling rivalry, anger toward parents).
Empathy	A capacity of a group member to put himself or herself into the psychological frame of reference of another group member and, thereby, understand his or her thinking, feeling, or behavior.
Identification	An unconscious defense mechanism in which the person incorporates the characteristics and qualities of another person or object into his or her ego system.

Imitation	The conscious emulation or modeling of one's behavior after that of another (also called role modeling); also known as "spectator therapy," as one patient learns from another.
Insight	Conscious awareness and understanding of one's own psychodynamics and symptoms of maladaptive behavior. Most therapists distinguish two types: (1) intellectual insight—knowledge and awareness without any changes in maladaptive behavior and (2) emotional insight—awareness and understanding leading to positive changes in personality and behavior.
Inspiration	The process of imparting a sense of optimism to group members; the ability to recognize that one has the capacity to overcome problems; also known as installation of hope.
Interaction	The free and open exchange of ideas and feelings among group members; effective interaction is emotionally charged.
Interpretation	The process during which the group leader formulates the meaning or significance of a patient's resistance, defenses, and symbols; the result is that the patient develops a cognitive framework within which to understand his or her behavior.
Learning	Patients acquire knowledge about new areas, such as social skills and sexual behavior; they receive advice, obtain guidance, and attempt to influence and are influenced by other group members.
Reality testing	Ability of the person to evaluate objectively the world outside the self; includes the capacity to perceive oneself and other group members accurately.

Transference	Projection of feelings, thoughts, and wishes onto the therapist, who has come to represent an object from the patient's past. Such reactions, although perhaps appropriate for the condition prevailing in the patient's earlier life, are inappropriate and anachronistic when applied to the therapist in the present. Patients in group therapy may also direct such feelings toward one another, a process called multiple transferences.
Universalization	The awareness of the patient that he or she is not alone in having problems; others share similar complaints or difficulties in learning; the patient is not unique.
Ventilation	The expression of suppressed feelings, ideas, or events to other group members; the sharing of personal secrets that ameliorates a sense of sin or guilt (also referred to as self-disclosure).

Role of the Therapist

Although opinions differ about how active or passive a group therapist should be, the consensus is that the therapist's role is primarily facilitative. Ideally, the group members themselves are the primary source of cure and change. The climate produced by the therapist's personality is a potent agent of change. The therapist is more than an expert applying techniques; he or she exerts a personal influence that taps such variables as empathy, warmth, and respect.

Inpatient Group Psychotherapy

Group therapy is an integral part of hospitalized patients' therapeutic experiences. Group organization can take many forms on a ward. In a community meeting, an entire inpatient unit meets with all the staff members (e.g., psychiatrists, psychologists, and nurses). In team meetings, 15 to 20

patients and staff members meet; a regular or small group comprising 8 to 10 patients may meet with 1 or 2 therapists, as in traditional group therapy. Although the goals of each group vary, they all have common purposes: to increase patients' awareness of themselves through their interactions with the other group members, who provide feedback about their behavior; to provide patients with improved interpersonal and social skills; to help the members adapt to an inpatient setting; and, to improve communication between patients and staff. In addition, one type of group meeting is attended only by inpatient hospital staff in order to improve communication among the staff members and to provide mutual support and encouragement in their day-to-day work with patients. Community meetings and team meetings are more helpful for dealing with patient treatment problems than they are for providing insight-oriented therapy, which is the province of the small-group therapy meeting.

Group Composition. Two critical factors of inpatient groups common to all short-term therapies are the heterogeneity of the members and the rapid turnover of patients. Outside the hospital, therapists have large caseloads from which to select patients for group therapy. On the ward, therapists have a limited number of patients to choose from and have restrictions in terms of those patients who are both willing to participate and suitable for a small-group experience. In certain settings, group participation may be mandatory (e.g., in substance use disorder units), but mandatory attendance does not usually apply in a general psychiatry unit. Most group experiences are more productive when the patients themselves choose to enter them.

More sessions are preferable to fewer. During patients' hospital stays, groups may meet daily to allow interactional continuity and the carryover of themes from one session to the next. A new group member can be brought up to date quickly, either by the therapist in an orientation meeting or by another group member. A newly admitted patient has often learned many details about the small-group program from another patient before actually attending the first session. The less frequently the group sessions are held, the greater the need for a therapist to structure the group and be active in it.

Inpatient versus Outpatient Groups. Although the therapeutic factors that account for change in small inpatient groups are similar to those in the

outpatient settings, there are qualitative differences. For example, the relatively high turnover of patients in inpatient groups complicates the process of cohesion. However, the fact that all the group members are together in the hospital aids cohesion, as do the therapists' efforts to foster the process. Sharing of information, universalization, and catharsis are the main therapeutic factors at work in inpatient groups. Although insight more likely occurs in outpatient groups because of their long-term nature, some patients can obtain a new understanding of their psychological makeup within the confines of a single group session. A unique quality of inpatient groups is the patients' extra-group contacts, which are extensive because they live together on the same ward. Verbalizing their thoughts and feelings about such contacts in the therapy sessions encourages interpersonal learning. In addition, conflicts between patients or between patients and staff members can be anticipated and resolved.

Twelve former psychiatric inpatients who attended the monthly medication clinic would meet for 1 hour before their individual appointments with the psychiatrist to review their current social situation and medications. All received treatment from the same ward doctor and had known one another while on the inpatient service. The psychiatrist who performed the medication reviews also served as the group leader. Periodically, a staff member who was also familiar with the patients assisted with the group. Coffee was available, and the patients often brought pastries from home. The patients socialized with one another during the hour and frequently exchanged helpful ideas and tips about job opportunities. Those without cars shared rides with other members. The group was open-ended and well attended. Most of the patients were single and had a long history of psychotic illness. For most, this meeting was their only opportunity to socialize and be among peers. Frequently, when a member entered the hospital again, many in the group would visit their colleague on the ward. (Courtesy of Normund Wong, M.D.)

Self-Help Groups

Leaderless or self-help group experiences, usually homogeneously composed

of participants with a common issue, represent an additional use of the group setting as a vehicle for change. Although these groups are not strictly considered a form of psychotherapy, they do share several parameters and can have therapeutic benefits for participants. Perhaps the best-known examples are the Twelve Step groups of Alcoholics Anonymous (AA), Al-Anon, Adult Children of Alcoholics (ACoA), and other substance use disorder centered groups. In addition, many self-help groups develop around health and illness, loss, or shared life experience. Cancer support groups, groups for widows or widowers, parents without partners, and bipolar support groups are but a few well-known self-help groups.

Combined Individual and Group Psychotherapy

In combined individual and group psychotherapy, patients see a therapist individually and also take part in group sessions. The therapist for the group and individual sessions is usually the same person. Groups can vary in size from 3 to 15 members, but the most helpful size is 8 to 10. Patients must attend all group sessions. Attendance at individual sessions is also essential, and the therapeutic process should examine failure to attend either group or individual sessions.

Combined therapy is a particular treatment modality, not a system by which to augment individual therapy by an occasional group session or a group therapy in which a participant meets alone with a therapist from time to time. Instead, it is an ongoing plan in which meaningful integration of the group experience with the individual sessions yields reciprocal feedback to help form an integrated therapeutic experience. Although the one-to-one doctor–patient relationship makes an in-depth examination of the transference reaction possible for some patients, it may not provide other patients with the corrective emotional experiences necessary for therapeutic change. The group gives patients a variety of persons with whom they can have transferential reactions. In the microcosm of the group, patients can relive and work through familial and other important influences.

Techniques. The combined therapy format may include different techniques based on varying theoretical frameworks. Some clinicians increase the frequency of individual sessions to encourage the emergence of

the transference neurosis. In the behavioral model, individual sessions occur regularly, but they tend to be less frequent than in other approaches. Whether patients use a couch or a chair during individual sessions depends on a therapist's orientation. Other techniques include alternate meetings or "after-sessions" without the therapist present. A combined therapy approach called *structured interactional group psychotherapy* has a different group member as the focus of each weekly group session who is discussed in depth by the other members.

Results. Most workers in the field believe that combined therapy has the advantages of both dyadic and group settings, without sacrificing the qualities of either. Generally, the dropout rate in combined therapy is lower than that in group therapy alone. In many cases, combined therapy appears to bring problems to the surface and to resolve them more quickly than might be possible with either method alone.

Psychodrama

Psychodrama is a method of group psychotherapy originated by the Viennese-born psychiatrist Jacob Moreno that explores personality makeup, interpersonal relationships, conflicts, and emotional problems through special dramatic methods. Therapeutic dramatization of emotional problems includes the protagonist or patient, the person who acts out problems with the help of auxiliary egos, persons who enact varying aspects of the patient, and the director, psychodramatist, or therapist, the person who guides those in the drama toward the acquisition of insight.

Roles

DIRECTOR. The director is the leader or therapist and so must be an active participant. He or she has a catalytic function by encouraging the members of the group to be spontaneous. The director must also be available to meet the group's needs without superimposing his or her values. Of all the group psychotherapies, psychodrama requires the most participation from the therapist.

PROTAGONIST. The protagonist is the patient in conflict. The patient chooses

the situation to portray in the dramatic scene, or the therapist chooses it if the patient so desires.

AUXILIARY EGO. An auxiliary ego is another group member who represents something or someone in the protagonist's experience. The auxiliary egos help account for the great range of therapeutic effects available in psychodrama.

GROUP. The members of the psychodrama and the audience make up the group. Some are participants, and others are observers, but all benefit from the experience to the extent that they can identify with the ongoing events. The concept of spontaneity in psychodrama refers to the ability of each group member, especially the protagonist, to experience the thoughts and feelings of the moment and to communicate emotion as authentically as possible.

Techniques. The psychodrama can focus on any particular area of functioning (a dream, a family, or a community situation), a symbolic role, an unconscious attitude, or an imagined future situation. The group can also act out such symptoms as delusions and hallucinations. Techniques to advance the therapeutic process and to increase productivity and creativity include the soliloquy (a recital of overt and hidden thoughts and feelings), role reversal (the exchange of the patient's role for the role of a significant person), the double (an auxiliary ego acting as the patient), the multiple double (several egos acting as the patient did on varying occasions), and the mirror technique (an ego imitating the patient and speaking for him or her). Other techniques include the use of hypnosis and psychoactive drugs to modify the acting behavior in various ways.

Ethical and Legal Issues

Confidentiality. Except where the law requires disclosure, the group therapist legally and ethically gives information about the group members to others only after obtaining appropriate patient consent. The therapist is obligated to take appropriate steps to be responsible to society, as well as to patients, when patients pose a danger to themselves or others. The guidelines for ethics of the American Group Psychotherapy Association state that

therapists must obtain specific permission to confer with the referring therapist or with the individual therapist when the patient is in conjoint therapy.

Although the group members, as well as the therapist, should protect the identity of the members and maintain confidentiality, the group members are not legally bound to do so. During the preparation of patients for group psychotherapy, therapists should routinely instruct the prospective members to keep all material discussed in the group confidential. Theoretically, in a legal case, one member of a group can be asked to testify against another, but such a situation is likely to be rare.

A therapist must exercise clinical judgment and caution in placing a patient in a group if he or she thinks that the burdens of maintaining secrets will be too great for some potential members or if a prospective group patient harbors a secret of such magnitude or notoriety that membership in a group would not be wise.

Violence and Aggression. Although reports of violence and aggression are rare, the potential exists that a group member may physically attack another patient or a therapist. The attack may occur within the group or outside the group. Careful selection of group members can diminish the likelihood of such an event. Patients with a demonstrated history of assaultive behavior and psychotic patients who pose a potential for violence should not participate in a group. In institutional settings, in which group therapy is common, sufficient safeguards must be in place to discourage any physical danger to others—for example, guards or attendants can act as observers.

Sexual Behavior. For therapists, sexual intercourse with a patient or a former patient is unethical; in many states, such behavior is considered a criminal act. The issue is complicated in group psychotherapy, however, because members may engage in sexual activities with one another. The issues of pregnancy, rape, and sexually transmitted infections by group members are open questions. If an injury results from sexual activity by group members, the therapist could be held accountable for not preventing such behavior. The therapist should advise prospective group members that each patient is responsible for reporting any sexual contact between members. The therapist cannot anticipate every group sexual encounter or

prevent sexual relationships from developing, but he or she is obligated to provide patients with guidelines of acceptable behavior. The therapist should identify sexual, vulnerable, or exploitive patients in the selection and preparation of patients for the group. Explicitly inform sociopathic patients who sexually exploit others that such behavior is not acceptable in the group and should be verbalized rather than acted out. The therapist must not encourage or tacitly allow sexual activity in the conduct of the group. Patients with AIDS are encouraged to reveal that they harbor the virus.

FAMILY THERAPY AND COUPLES THERAPY

Family Therapy

The family is the foundation of most societies. The study of families in different cultures has been a subject of fascination and scientific interest from viewpoints as diverse as sociology, group dynamics, anthropology, ethnicity, race, evolutionary biology, and, of course, the mental health field. The confluence of information gleaned from family studies has set the backdrop against which the contemporary practice of family therapy has evolved.

Family therapy is any psychotherapeutic endeavor that explicitly focuses on altering the interactions between or among family members and seeks to improve the functioning of the family as a unit. Both family therapy and couple therapy aim at some change in relational functioning. In most cases, they also aim at some other change, typically in the functioning of specific individuals in the family. Family therapy meant to heal a rift between parents and their adult children is an example of the use of family therapy centered on relationship goals. Family therapy aimed at increasing the family's coping with schizophrenia and at reducing the family's expressed emotion is an example of family therapy aimed at individual goals (in this case, the functioning of the person with schizophrenia), as well as family goals. In the early years of family therapy, experts viewed change in the family system as sufficient to produce individual change. More recent treatments aimed at change in individuals, as well as in the family system, tend to supplement the interventions that focus on interpersonal relationships with specific strategies that focus on individual behavior.

Indications. The presence of a relational difficulty is a clear indication for family and couple therapy. Couples and family therapies are the only treatments with evidence to support their use for such problems as marital maladjustment, and other methods, such as individual therapy, often have deleterious effects in these situations. Couples and family therapy also have a clear and important role in the treatment of numerous specific psychiatric disorders, often as a component within a multimethod treatment.

Of course, as with any therapy, the indications for family and couple therapy are broad and vary from case to case. Family therapy is a therapeutic collage of ideas regarding the underpinnings of family and individual stability and change, psychopathology, and problems in living, as well as relational ethics. Family therapy might better be called *systemically sensitive therapy*, and, in this sense, reflects a basic worldview as much as a clinical treatment methodology. For therapists thus inclined, then, all clinical problems involve salient interactional components; thus, some kind of family (or other functionally significant other's) involvement in therapy is always indicated, even in treatment that emphasizes individual problems.

An impressive array now exists of common clinical disorders and problems, including child, adolescent, and adult disorders, for which research has demonstrated family or couple treatment methods to be effective. In a few instances, couple and family interventions are probably even the treatment of choice, and for several disorders, the research argues for family intervention to be an essential part of treatment.

Techniques

INITIAL CONSULTATION. Family therapy is familiar enough to the general public for families with a high level of conflict to request it specifically. When the initial complaint is about an individual family member, however, pretreatment work may be needed. Underlying resistance to a family approach typically includes fears by parents that the therapist will blame them for their child's difficulties, that the entire family will be pronounced sick, that a spouse will object, and that open discussion of one child's misbehavior will have a negative influence on siblings. Refusal by an adolescent or young adult patient to participate in family therapy is frequently a disguised collusion with the fears of one or both parents.

INTERVIEW TECHNIQUE. The unique quality of a family interview springs from two important facts. A family comes to treatment with its history and dynamics firmly in place. To a family therapist, the established nature of the group, more than the symptoms, constitutes the clinical problem. Family members usually live together and, at some level, depend on one another for their physical and emotional well-being. Whatever transpires in the therapy session is known to all. Central principles of technique also derive from these facts. For example, the therapist must carefully channel the catharsis of anger by one family member toward another. The person who is the object of the anger will react to the attack, and the anger may escalate into violence and fracture relationships, with one member or more withdrawing from therapy. For another example, free association is inappropriate in family therapy because it can encourage one person to dominate a session. Thus, therapists must always control and direct the family interview.

FREQUENCY AND LENGTH OF TREATMENT. Unless an emergency arises, sessions occur no more than once a week. Each session, however, may require as much as 2 hours. Extended sessions can include an intermission to give the therapist time to organize the material and plan a response. A flexible schedule is necessary when geography or personal circumstances make it physically difficult for the family to get together. The length of treatment depends both on the nature of the problem and on the therapeutic model. Therapists who use problem-solving models exclusively may accomplish their goals in a few sessions, whereas therapists using growth-oriented models may work with a family for years and may schedule sessions at long intervals.

Schools of Family Therapy. Many schools of family therapy exist, none of which is superior to the others. The particular model used depends on the training received, the context in which therapy occurs, and the personality of the therapist. Table 23-6 outlines the major schools of family therapy, the influential practitioners of each method, and the focus of each technique.



Table 23-6
Schools of Family and Couple Therapy

Therapeutic Approach (Examples)	Core Concepts	Typical Goals	Common Strategies and Techniques	Comments
Behavioral/cognitive-behavioral (James Alexander, Neil Jacobson, Donald Baucom, Gerald Patterson)	Functional analysis Social learning theory Communication and problem solving Attributional style	Resolution of presenting problem Enhanced communication and problem-solving skills Balance between change and acceptance	Communication and problem-solving training Acceptance training/ reattribution techniques Parent management, emphasis on behavior/ consequences Therapist as educator	Functional analysis, not treatment techniques, is defining characteristic Arguably the most empirically supported of all family and couple methods
Bowen family systems (Murray Bowen, Philip Guerin, Michael Kerr, Monica McGoldrick)	Differentiation of self Triangulation Emotional cutoffs Family emotional system Sibling position	Increased differentiation Detriangulation Cutoffs resolved Improved ability to manage anxiety	Use of genogram Therapist as "coach" Education about multigenerational family processes	Often conducted with only a single patient Influence continues but has waned since Bowen's death
Experiential–humanistic (Susan Johnson and Leslie Greenberg, Virginia Satir, Carl Whitaker)	Communication styles (e.g., placater–blamer) Psychotherapy of the absurd Attachment theory	Fostering creativity (comfort with "craziness") Increased family cohesion Personal growth, self-esteem Tolerance for conflict	Resolving the battle for structure and the battle for initiative Frequent use of cotherapists Family sculpture Use of self Family reconstruction	Since Whitaker's death, symbolic–experiential approach receding in visibility Emotionally focused therapy one of few nonbehavioral or psychodynamic couple methods increasing in influence
Integrative (Alan Gurman, William Pinsof, Douglas Snyder, Froma Walsh)	Importance of and relationships among multiple levels of experience Theoretical and technical eclecticism	Improved communication and problem solving Enhanced interactional insight Presenting problems	Combinations of cognitive-behavioral and psychodynamic methods, in general systems context	Along with behavioral and psychoeducational methods, the most sensitive to research findings Attempts to match interventions to problem and family
Mental Research Institute brief therapy (Don Jackson, John Weakland, Paul Watzlawick)	Distinction between "difficulties" and "problems" Communication processes Symptom as communication First- and second-order change	Resolution of presenting problem Second-order change	Reframing Therapist maneuverability	Historically overlaps strategic, but with important differences (e.g., "function" of symptoms)
Feminist (Virginia Goldner, Rachel Hare-Mustin, Betty Carter)	Raising gender consciousness Underlying effect of gender	Gender equality Recognizing gender inequities	Raising consciousness Identifying and challenging culturally constructed narratives	Not so much a way of intervening as a set of issues to address; has had a pervasive effect on family and couple therapy
Narrative (David Epston, Michael White, Harlene Anderson)	Constructivism "Languaging" "Stories" socially created Problem-saturated descriptions	Resolution of presenting problem Enhancing undiscovered parts of self	Externalization of problems Focusing on "unique outcomes" Creation of new meaning via "restorying" Therapeutic letters	Currently very popular but rarely acknowledges overlaps with cognitive and behavioral theory
Psychodynamic–psychoanalytic (Nathan Ackerman, Henry Dicks, Clifford Sager, Jill Scharff and David Scharff)	Object relations Projective identification Splitting Scapegoating	Improved insight Genetic (historical) awareness Enhanced empathy De-emphasize presenting problem Disentangle interlocking pathologies	Transference, resistance, and countertransference analysis Creation of holding environment Interpretation Emphasis on therapeutic alliance	Probably influences family therapists' clinical work more than is usually acknowledged

Psychoeducational (Carol Anderson, Ian Falloon, William McFarlane, David Miklowitz)	Biopsychosocial theory Stress-diathesis Expressed emotion	Enhanced family coping skills Improved communication and problem-solving skills Relapse prevention	Survival skills workshops Family management Family information and education Concurrent use of psychopharmacology	Without doubt, the most effective family-based approach for families with a member with major psychiatric disorder
Solution focused (Steve de Shazer and Insoo Berg, William O'Hanlon, Michele Weiner-Davis)	Focus on solutions, not problems Unimportance of problem etiology Disbelief in resistance	Resolution of presenting problem Creation of solutions	Scaling questions Miracle questions Exception questions Client empowerment Coping questions Inquire about changes made since deciding to take action ("pre-session changes")	Grows out of basic Mental Research Institute philosophy, emphasizing solutions more than problems. Has made extraordinary claims of effectiveness
Strategic (Milton Erickson, Jay Haley, Cloe Madanes, James Keim)	Power and control Family life cycle Symptom-maintaining sequences Function of problems	Resolve presenting problem Disruption of problem-maintaining sequences	Persuasion Paradoxical injections Insight downplayed Pretend and ordeal techniques	Unfortunate de-emphasis on biology, psychiatric diagnosis, and personality theory
Structural (Harry Aponte, Salvador Minuchin, Jose Szapocznik)	Boundaries Hierarchies Coalitions and alliances Complementarity Engagement–enmeshment	Increased flexibility Adaptability to developmental change Balance between connectedness and differentiation Subsystem functioning	Unbalancing Enactment Joining	Arguably the most influential family therapy approach
Transgenerational (Ivan Boszormenyi-Nagy, James Framo, Laura Roberto-Forman)	Invisible loyalties Ledgers and debts Family mourning Personal authority	More universal than family-specific goals Increased trust Repair ruptured relationships	Multidirected partiality Family-of-origin sessions Use of cotherapy	Almost all multi-/transgenerational methods have significant psychodynamic underpinnings

Modifications of Techniques

FAMILY GROUP THERAPY. Family group therapy combines several families into a single group. Families share mutual problems and compare their interactions with those of the other families in the group. Treatment of families impacted by schizophrenia has been effective in multiple family groups. Parents of disturbed children may also meet together to share their situations.

SOCIAL NETWORK THERAPY. In social network therapy, the social community or network of a disturbed patient meets in group sessions with the patient. The network includes those with whom the patient comes into contact in daily life, not only the immediate family but also relatives, friends, tradespersons, teachers, and coworkers.

PARADOXICAL THERAPY. With the paradoxical therapy approach, which evolved from the work of Gregory Bateson, a therapist suggests that the patient intentionally engage in the unwanted behavior (called the paradoxical

injunction) and, for example, avoid a phobic object or perform a compulsive ritual. Although paradoxical therapy and the use of paradoxical injunctions seem to be counterintuitive, the therapy can create new insights for some patients. It is used in individual therapy as well as in family therapy.

REFRAMING. Reframing, also known as *positive connotation,* is a relabeling of all negatively expressed feelings or behavior as positive. When the therapist attempts to get family members to view behavior from a new frame of reference, "This child is impossible," becomes "This child is desperately trying to distract and protect you from what he or she perceives as an unhappy marriage." Reframing is an important process that allows family members to view themselves in new ways that can produce change.

Goals. Family therapy has several goals: to resolve or reduce pathogenic conflict and anxiety within the matrix of interpersonal relationships; to enhance the perception and fulfillment by family members of one another's emotional needs; to promote appropriate role relationships between the sexes and generations; to strengthen the capacity of individual members and the family as a whole to cope with destructive forces inside and outside the surrounding environment; to influence family identity and values to orient members toward health and growth. The therapy ultimately aims to integrate families into the large systems of society, extended family, and community groups and social systems, such as schools, medical facilities, and social, recreational, and welfare agencies.

Couples (Marital) Therapy

Couples or marital therapy is a form of psychotherapy designed to psychologically modify the interaction of two persons in conflict with each other over one parameter or a variety of parameters—social, emotional, sexual, or economic. In couples therapy, a trained person establishes a therapeutic contract with a patient-couple and, through definite types of communication, attempts to alleviate the disturbance, to reverse or change maladaptive patterns of behavior, and to encourage personality growth and development.

Marriage counseling may be considered more limited in scope than marriage therapy: the focus is only surrounding a particular familial conflict,

and the counseling is primarily task-oriented, geared to solving a specific problem, such as child-rearing. Marriage therapy, by contrast, emphasizes restructuring a couple's interaction and sometimes explores the psychodynamics of each partner. Both therapy and counseling emphasize helping marital partners cope effectively with their problems. Most important is the definition of appropriate and realistic goals, which may involve extensive reconstruction of the union or problem-solving approaches or a combination of both.

Types of Therapies

INDIVIDUAL THERAPY. In individual therapy, the partners may consult different therapists, who do not necessarily communicate with each other and indeed may not even know each other. The goal of treatment is to strengthen each partner's adaptive capacities. At times, only one of the partners is in treatment, and, in such cases, it is often helpful for the person who is not in treatment to visit the therapist. Not only can the visiting partner provide data about the patient that the therapist might otherwise overlook, but the therapist can identify and deal with overt or covert anxiety in the visiting partner resulting from change in the patient, correct irrational beliefs about treatment events, and examine conscious or unconscious attempts by the partner to sabotage the patient's treatment.

INDIVIDUAL COUPLES THERAPY. In individual couples therapy, each partner is in therapy, which is either concurrent, with the same therapist, or collaborative, with each partner seeing a different therapist.

CONJOINT THERAPY. In conjoint therapy, the most common treatment method in couples therapy, either one or two therapists treat the partners in joint sessions. Cotherapy with therapists of both sexes prevents a particular patient from feeling ganged up on when confronted by two members of the opposite sex.

FOUR-WAY SESSION. In a four-way session, each partner is seen by a different therapist, with regular joint sessions in which all four persons participate. A variation of the four-way session is the roundtable interview, developed by

William Masters and Virginia Johnson for the rapid treatment of sexually dysfunctional couples. Two patients and two opposite-sex therapists meet regularly.

GROUP PSYCHOTHERAPY. Group therapy for couples allows a variety of group dynamics to affect the participants. Groups usually consist of three to four couples and one or two therapists. The couples identify with one another and recognize that others have similar problems; each gains support and empathy from fellow group members of the same or opposite sex. They explore sexual attitudes and have an opportunity to gain new information from their peer groups, and each receives specific feedback about his or her behavior, either negative or positive, which may have more meaning and be better assimilated coming from a neutral, nonspouse member, for example, than from the spouse or the therapist.

During the middle phase of a couples group comprising four couples, the theme of whether to have children arose. One couple had just come from a visit to the gynecologist, who informed them that they were running out of time because of the wife's age. The woman in the couple did not want to have children, but her husband did. His complaint about the marriage was that his wife never was demonstrative in showing her loving feelings for him. He felt her to be detached, distant, and sexually inhibited.

The prevailing sentiment among the other couples who had children was that children only added additional stress to an already stressed relationship. One other couple, however, voiced their different view by describing how their children had enriched their lives.

As the talk about going forward and getting pregnant progressed, the group leader noted the nonverbal communication between the ambivalent couple. Whenever the tone of the group leaned toward having children, the wife would reach out and grasp the hand of her husband tenderly. This action invariably had the effect of stopping him from pursuing the topic for fear of the withdrawal of her affection. All this occurred without words. Once identified, this repetitive nonverbal pattern was available for examination in the group, and the supportive elements provided by other members and the leader encouraged a frank, direct, and open

conversation between the partners, who eventually chose to go forward and attempt to have a child. (Courtesy of H. I. Spitz, M.D. and S. Spitz, ACSW.)

COMBINED THERAPY. Combined therapy refers to all or any of the preceding techniques used concurrently or in combination. Thus, a particular patient-couple may begin treatment with one or both partners in individual psychotherapy, continue in conjoint therapy with the partner, and terminate therapy after a course of treatment in a married couples' group. The rationale for combined therapy is that there is no evidence to support that a single approach to marital problems is superior to another. Familiarity with a variety of approaches thus allows therapists flexibility that provides maximal benefit for couples in distress.

Indications. Whatever the specific therapeutic technique, initiation of couples therapy is appropriate when individual therapy has failed to resolve the relationship difficulties, when the onset of distress in one or both partners is clearly a relational problem, and when a couple in conflict requests couples therapy. Problems in communication between partners are a prime indication for couples therapy. In such instances, one spouse may be intimidated by the other, may become anxious when attempting to tell the other about thoughts or feelings, or may project unconscious expectations onto the other. The therapy works toward enabling each partner to see the other realistically.

Conflicts in one or several areas, such as the partners' sexual life, are also indications for treatment. Similarly, difficulty in establishing satisfactory social, economic, parental, or emotional roles implies that a couple needs help. Clinicians should evaluate all aspects of the marital relationship before attempting to treat only one problem, which could be a symptom of a pervasive marital disorder.

Contraindications. Contraindications for couples therapy include patients with severe forms of psychosis, particularly patients with paranoid elements and those in whom the marriage's homeostatic mechanism is a protection against psychosis, marriages in which one or both partners really want to divorce, marriages in which one spouse refuses to participate because of

anxiety or fear, and when there is a threat of domestic violence.

Goals. Nathan Ackerman defined the aims of couples therapy as follows: the goals of therapy for partner relational problems are to alleviate emotional distress and disability and to promote the levels of well-being of both partners together and of each as an individual. Ideally, therapists move toward these goals by strengthening the shared resources for problem-solving, by encouraging the substitution of adequate controls and defenses for pathogenic ones, by enhancing both the immunity against the disintegrative effects of emotional upset and the complementarity of the relationship, and by promoting the growth of the relationship and of each partner.

Part of a therapist's task is to persuade each partner in the relationship to take responsibility for understanding the psychodynamic makeup of personality. The emphasis is on each person's accountability for the effects of behavior on his or her own life, the life of the partner, and the lives of others in the environment. The result is often a deep understanding of the problems that created marital discord.

Couples therapy does not ensure the maintenance of any marriage or relationship. Indeed, in certain instances, it may show the partners that they are in a nonviable union and should consider dissolution. In these cases, couples may continue to meet with therapists to work through the difficulties of separating and obtaining a divorce, a process referred to as *divorce therapy*.

DIALECTICAL BEHAVIOR THERAPY

Dialectical behavior therapy (DBT) is the psychosocial treatment that has received the most empirical support for patients with BPD. Put simply, the overarching goal of DBT is to help create a life worth living for patients who often suffer tremendously from chronic and pervasive problems across many areas of their lives. DBT is a type of psychotherapy initially developed for chronically self-injurious patients with BPD and parasuicidal behavior. In recent years, its use has extended to other forms of mental illness. The method is eclectic, drawing on concepts derived from supportive, cognitive, and behavioral therapies. Some elements trace back to Franz Alexander's view of therapy as a corrective emotional experience and other elements from

certain Eastern philosophical schools (e.g., Zen).

Weekly visits aim to improve interpersonal skills and decrease self-destructive behavior using techniques involving advice, metaphor, storytelling, and confrontation, among others. Patients with BPD especially are helped to deal with the ambivalent feelings that are characteristic of the disorder. Marsha Linehan, Ph.D., developed the treatment method based on her theory that such patients cannot identify emotional experiences and cannot tolerate frustration or rejection. As with other behavioral approaches, DBT assumes all behavior (including thoughts and feelings) is learned and that patients with BPD behave in ways that reinforce or even reward their behavior, regardless of how maladaptive it is.

Functions of DBT

As described by its originator, there are five essential "functions" in treatment: (1) to enhance and expand the patient's repertoire of skillful behavioral patterns; (2) to improve patient motivation to change by reducing reinforcement of maladaptive behavior, including dysfunctional cognition and emotion; (3) to ensure that new behavioral patterns generalize from the therapeutic to the natural environment; (4) to structure the environment so that effective behaviors, rather than dysfunctional behaviors, are reinforced; and (5) to enhance the motivation and capabilities of the therapist so that the treatment rendered is effective. Figure 23-1 illustrates how DBT breaks the cycle of problem behavior being used to avoid emotional distress.

The four modes of treatment in DBT are as follows: (1) group skills training, (2) individual therapy, (3) phone consultations, and (4) consultation team. Descriptions of these modes of treatment are below. Other ancillary treatments used are pharmacotherapy and hospitalization when needed.

Group Skills Training. In group format, patients learn specific behavioral, emotional, cognitive, and interpersonal skills. Unlike traditional group therapy, observations about others in the group are discouraged. Instead, DBT takes a didactic approach, using specific exercises from a skills training manual, many of which emphasize control of emotional dysregulation and impulsive behavior.

Individual Therapy. Sessions in DBT occur weekly, generally for 50 to 60

minutes, in which review of skills learned during group training takes place, along with examination of life events from the previous week. Therapists note episodes of pathologic behavioral patterns that could have been corrected if learned skills had been put into effect. Patients are encouraged to record their thoughts, feelings, and behaviors on diary cards, which therapists analyze in the session.

Telephone Consultation. Therapists are available for phone consultation 24 hours per day. Patients are encouraged to call when they feel themselves heading toward some crisis that might lead to injurious behavior to themselves or others. Calls are intended to be brief and usually last about 10 minutes.

Consultation Team. Therapists meet in weekly meetings to review their work with their patients. By doing so, they provide support for one another and maintain motivation in their work. The meetings enable them to compare the techniques used and to validate those that are most effective (Table 23-7).

Results

Several studies evaluating the effect of DBT for patients with BPD found that such therapy was positive. Patients had a low dropout rate from treatment, the incidence of parasuicidal behaviors declined, self-report of angry affect decreased, and social adjustment and work performance improved. The method is now being applied to other disorders, including substance use disorders, eating disorders, schizophrenia, and PTSD.



Table 23-7
Consultation Team Agreements in Dialectical Behavior Therapy

1. Meet weekly for 1–2 hr
2. Discuss cases according to the treatment hierarchy (i.e., self-injurious/life-threatening behavior, treatment-interfering behavior, and quality-of-life–interfering behavior)
3. Accept a dialectical philosophy

4. Consult with the patient on how to interact with other therapists, and do not tell other therapists how to interact with the patient

5. Do not expect consistency of therapists with one another (even across the same patient)

6. Allow all therapists to observe their own limits without fear of judgmental reactions from other consultation group members

7. Search for nonpejorative, empathic interpretation of patient's behavior

8. Acknowledge that all therapists are fallible

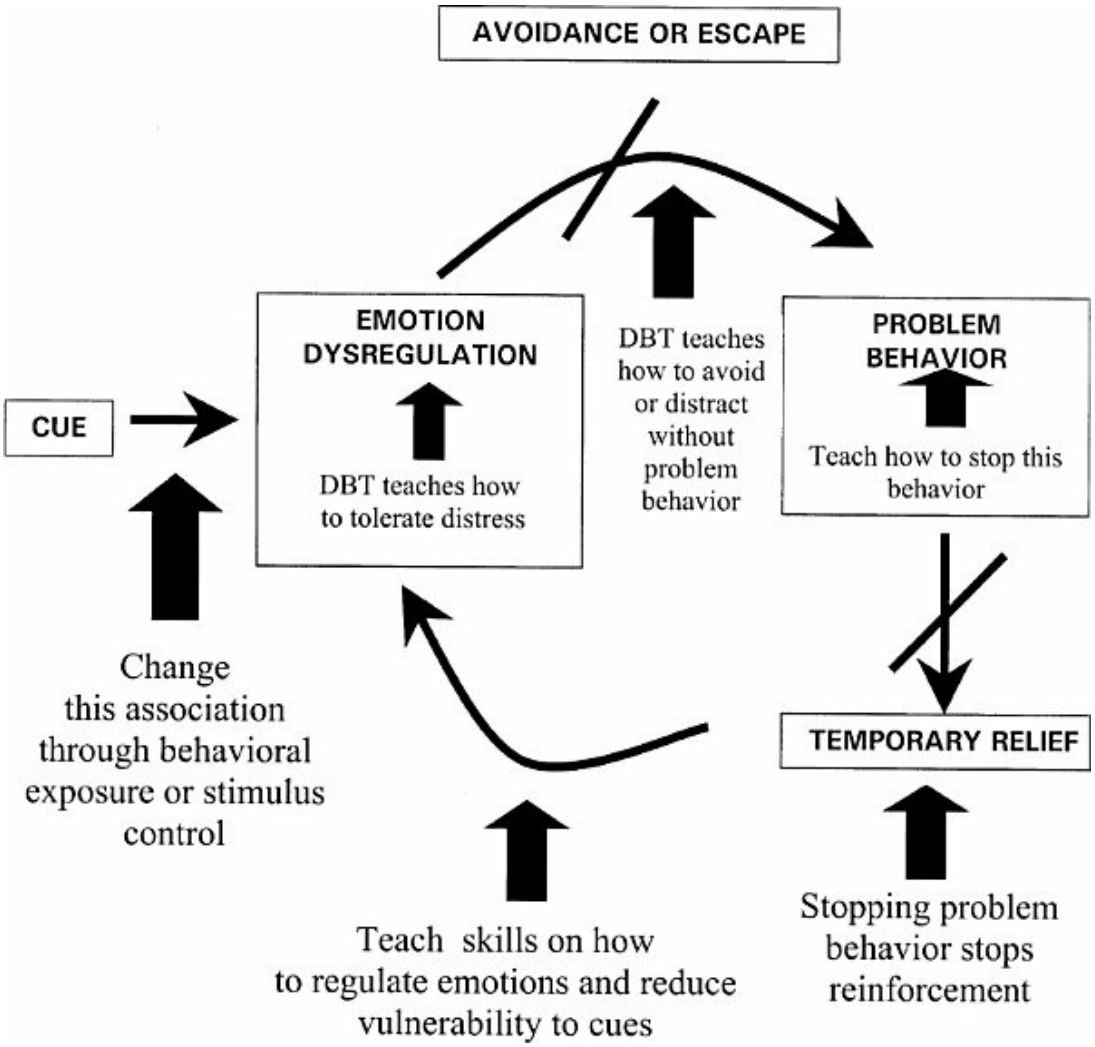


FIGURE 23-1
How dialectical behavior therapy (DBT) works.

BIOFEEDBACK

Biofeedback involves the recording and display of small changes in the physiologic levels of the feedback parameter. The display can be visual, such as a big meter or a bar of lights, or auditory. Therapists instruct patients to change the levels of the parameter, using the feedback from the display as a guide. The basis of biofeedback is the idea that the autonomic nervous system can come under voluntary control through operant conditioning. Biofeedback can be used by itself or in combination with relaxation. For example, patients with urinary incontinence use biofeedback alone to regain control over the pelvic musculature. The rehabilitation of neurologic disorders also uses biofeedback. The relaxation that patients learn to facilitate may augment the benefits of biofeedback.

Theory

Neal Miller demonstrated the medical potential of biofeedback by showing that the normally involuntary autonomic nervous system is susceptible to operant conditioning by the use of appropriate feedback. Utilizing instruments, patients acquire information about the status of involuntary biologic functions, such as skin temperature and electrical conductivity, muscle tension, blood pressure, heart rate, and brain wave activity. Patients then learn to regulate one or more of these physiologic states that affect symptoms. For example, a person can learn to raise the temperature of his or her hands to reduce the frequency of migraines, palpitations, or angina pectoris. Presumably, patients lower the sympathetic activation and voluntarily self-regulate arterial smooth muscle vasoconstrictive tendencies.

Methods

Instrumentation. The feedback instrument used depends on the patient and the specific problem. The most effective instruments are the electromyogram (EMG), which measures the electrical potentials of muscle fibers; the electroencephalogram (EEG), which measures alpha waves that occur in relaxed states; the galvanic skin response (GSR) gauge, which shows decreased skin conductivity during a relaxed state; and the thermistor, which measures skin temperature (which drops during tension because of peripheral

vasoconstriction). Patients are attached to one of the instruments that measure a physiologic function and translate the measurement into an audible or visual signal that patients use to gauge their responses. For example, in the treatment of bruxism, an EMG is attached to the masseter muscle. The EMG emits a high tone when the muscle contracts and a low tone when at rest. Patients can learn to alter the tone to indicate relaxation. Patients receive feedback about the masseter muscle, the tone reinforces the learning, and the condition ameliorates—all of these events interacting synergistically.

Many less-specific clinical applications (e.g., treating insomnia, dysmenorrhea, and speech problems; improving athletic performance; treating volitional disorders; achieving altered states of consciousness; managing stress; and supplementing psychotherapy for treating anxiety associated with somatic symptom and related disorders) use a model in which frontalis muscle EMG biofeedback is combined with thermal biofeedback and verbal instructions in progressive relaxation.

Relaxation Therapy. Muscle relaxation is used as a component of treatment programs (e.g., systematic desensitization) or as a treatment in its own right (relaxation therapy). Relaxation involves (1) immobility of the body, (2) control over the focus of attention, (3) low muscle tone, and (4) cultivation of a specific frame of mind, described as contemplative, nonjudgmental, detached, or mindful.

Edmund Jacobson developed progressive relaxation in 1929. Jacobson observed that when an individual lies “relaxed,” in the ordinary sense, the following clinical signs reveal the presence of residual tension: respiration is slightly irregular in time or force; the pulse rate, although often normal, is in some instances moderately increased as compared with later tests; voluntary or local reflex activities are evident in such slight marks as wrinkling of the forehead, frowning, movements of the eyeballs including frequent or rapid winking, restless shifting of the head, a limb, or even a finger; and finally, the mind continues to be active, and once started, worry or oppressive emotion will persist. It is incredible that a faint degree of tension can be responsible for all of this.

Learning relaxation, therefore, involves cultivating a muscle sense. To develop the muscle sense further, patients learn to isolate and contract specific muscles or muscle groups, one at a time. For example, patients flex

the forearm while the therapist holds it back to observe tenseness in the biceps muscle. (Jacobson used the word “tenseness” rather than “tension” to emphasize the patient’s role in tensing the muscles.) Once the patient reports this sensation, Jacobson would say, “This is your doing! What we wish is the reverse of this—simply not doing.” The therapist repeatedly reminds the patient that relaxation involves no effort. In fact, “making an effort is being tense and therefore is not to relax.” As the session progresses, the therapist instructs the patient to let go further and further, even past the point when the body part seems perfectly relaxed.

Patients would work in this fashion with different muscle groups, often over more than 50 sessions. For example, an entire session might involve relaxing the biceps muscle. Another feature of Jacobson’s method was to give instructions tersely so they would not interfere with a patient’s focus on muscle sensations; suggestions commonly used today (e.g., “*Your arm is becoming limp*”) were avoided. Patients were also frequently left alone, while the therapist attended to other patients.

In psychiatry, relaxation therapy is primarily a component of multifaceted broad-spectrum programs. Its use in desensitization was mentioned previously. Relaxing breathing exercises are often helpful for patients with panic disorder, especially when considered to be related to hyperventilation. In the treatment of patients with anxiety disorders, relaxation can serve as an occasion-setting stimulus (i.e., as a context of safety in which to try other specific interventions confidently). Table 23-8 describes some common relaxation techniques.



Table 23-8
Example Relaxation Techniques

Technique	Approach
Autogenic training	The patient is instructed to concentrate on feelings of warmth, heaviness, and relaxation throughout their body.
Biofeedback-assisted relaxation	The patient uses real-time biofeedback data to concentrate on certain physiologic functions, such as muscle

	tension.
Deep breathing/breathing exercises	The patient focuses on taking slow, deep, even breaths.
Guided imagery	The patient is taught to focus on pleasant images as a replacement for negative ones.
Progressive relaxation/progressive muscle relaxation	The patient is instructed to tighten and relax various muscle groups.
Self-hypnosis	The patient is taught to produce a relaxation response when prompted by a suggestion (usually a phrase or a nonverbal cue).

Adapted from National Center for Complementary and Integrative Health. Relaxation technique for health. https://www.nccih.nih.gov/health/relaxation-techniques-for-health

Later Adaptation of Progressive Muscular Relaxation. Joseph Wolpe chose progressive relaxation as a response incompatible with anxiety when designing his systematic desensitization treatment (discussed below). For this purpose, Jacobson's original method was too lengthy to be practical. Wolpe abbreviated the program to 20 minutes during the first six sessions (devoting the remainder of these sessions to other things, such as behavioral analysis). In a later modification of progressive relaxation, patients completed work with all the principal muscle groups in one session. Once patients have mastered this procedure (typically after three sessions), they learn to combine these groups into larger groups. Finally, patients practice relaxation by recall (i.e., without tensing the muscles).

Autogenic Training. Autogenic training is a method of self-suggestion that originated in Germany. It involves the patients directing their attention to specific bodily areas and hearing themselves think certain phrases reflecting a relaxed state. In the original German version, patients progressed through six themes over many sessions. Table 23-9 contains a list of these six areas,

along with representative autogenic phrases. Autogenic relaxation is an American modification of autogenic training, in which one session covers all six areas.

Applied Tension. Applied tension is a technique that is the opposite of relaxation; applied tension can counteract the fainting response. The treatment extends over four sessions. In the first session, patients learn to tense the muscles of the arms, legs, and torso for 10 to 15 seconds (as if they were bodybuilders). The tension is maintained long enough for a sensation of warmth to develop in the face. The patients then release the tension but do not progress to a state of relaxation. The maneuver is repeated five times at half-minute intervals. Feedback of the patient's blood pressure during the muscle contraction can augment this method; increased blood pressure suggests the achievement of appropriate muscle tension. The patients continue to practice the technique five times a day. An adverse effect of treatment that sometimes develops is headache. In this case, the patient reduces the intensity of the muscle contraction and the frequency of treatment.



Table 23-9
Sample Autogenic Phrases

Theme	Examples of Self-Statements
Heaviness	"My left arm is heavy."
Warmth	"My left arm is warm."
Cardiac regulation	"My heartbeat is calm and regular."
Breathing adjustment	"It breathes me."
Solar plexus	"My solar plexus is warm."
Forehead	"My forehead is cool."

Patients with blood and injury phobia show a unique, biphasic response when exposed to a phobic stimulus. The first phase is associated with increased heart rate and blood pressure. In the second phase, however, blood

pressure suddenly falls, and the patient faints. To treat the problem, the therapist shows the patient a series of slides that are provocative (e.g., mutilated bodies). The patient learns to identify early warning signs of fainting, such as queasiness, cold sweats, or dizziness, and to apply the learned muscle tension response quickly, contingent on these warning signs. Patients can also perform applied tension while donating blood or watching a surgical operation. The technique of isometric tension raises blood pressure, which prevents fainting.



Table 23-10
Steps in Applied Relaxation

Technique	Instructions
Progressive relaxation	Session 1: hands, arms, face, neck, and shoulders. Session 2: back, chest, stomach, breathing, hips, legs, and feet.
Release-only relaxation	As with progressive relaxation, except that the tension phase is omitted; when release-only relaxation is mastered, the patient can relax within 5–7 min.
Cue-controlled relaxation	A stimulus—the word *relax*—is presented just before exhalation; patients focus on their breathing while already in a relaxed state; the therapist says the word *inhale* just before each inhalation and the word *relax* just before each exhalation; after approximately five cycles, the patient mentally says these words (optionally dropping the *inhale*).
Differential relaxation	Patients can remain relaxed and move at the same time by differentially keeping muscles unrelated to the movement in a relaxed state; after achieving a relaxed state, patients lift an arm or a leg or look around in the room, while keeping movements and tension in other body parts at a minimum; patients

	also perform differential relaxation in other settings, including sitting in different chairs, sitting at a desk while writing, talking on the phone, and walking.
Rapid relaxation	Patients relax by taking one to three breaths with slow exhalations, thinking the word *relax* before each exhalation and scanning their bodies for areas of tension; with this practice, relaxation is shortened to 20–30 s; patients are instructed to relax in this manner 15–20 times per day at certain predetermined events in their natural environment (e.g., when they look at a watch or make a telephone call. As a reminder, colored dots might be taped on the watch or phone. After some time, the dots are changed to a different color to keep their reminding power fresh).
Application training	Patients relax just before entering the target situation; they stay in the situation for 10–15 min, using their relaxation skills as a coping technique; patients may initially be accompanied by the therapist; alternatively, if the patient's problem is panic attacks or generalized anxiety, imagery or physical exercise is used to induce fearful sensations, which then are used for application training.

Applied Relaxation. Applied relaxation involves eliciting a relaxation response in the stressful situation itself. The previous discussion showed that this is not advisable right away because of the possible ironic effects of relaxation. Therefore, patients should first practice relaxation in nonstressful circumstances. The method developed by Lars-Göran Öst and coworkers in Sweden has been proven efficacious for panic disorder and generalized anxiety disorder. Establishing the relaxation response in the patient's natural environment consists of seven phases of one to two sessions each: progressive relaxation, release-only relaxation, cue-controlled relaxation, differential relaxation, rapid relaxation, application training, and

maintenance. Table 23-10 provides details.

Results

Biofeedback, progressive relaxation, and applied tension are effective treatment methods for a broad range of disorders. They form one basis of behavioral medicine in which the patient changes (or learns how to change) behavior that contributes to illness. They form a basis on which many complementary and alternative medical procedures are effective (e.g., yoga and Reiki) in which relaxation is a critical component. Relaxation also informs more mainstream treatments, such as hypnosis.

COGNITIVE THERAPY

Cognitive therapy, developed by Aaron T. Beck, MD, is based on the cognitive model, which has been widely studied and has received substantial empirical support. The cognitive model posits that it is not situations that directly impact one's reaction (emotional, behavioral, and physiologic). Rather, it is one's automatic *perception* of the situation that is more closely associated with the reaction. A depressed individual, for example, who is reading this chapter (situation), might think, "I can't concentrate. There is something seriously wrong with me. I'll never get better" ("automatic thoughts"). As a result, his sadness and hopelessness increase (emotional reaction); he stops reading altogether (behavioral reaction); and he feels a weight in his chest (physiologic reaction). On the other hand, another individual might think, "I don't want to read this. I don't want to know anything about this topic, but I have to." She then feels annoyed, her body becomes somewhat tense, and she skims the pages instead of reading the article carefully. One important component of cognitive therapy involves teaching patients to identify the thoughts that lead to distress or dysfunction, to evaluate their thinking, and to effectively respond to these cognitions. Figure 23-2 depicts the cognitive model.

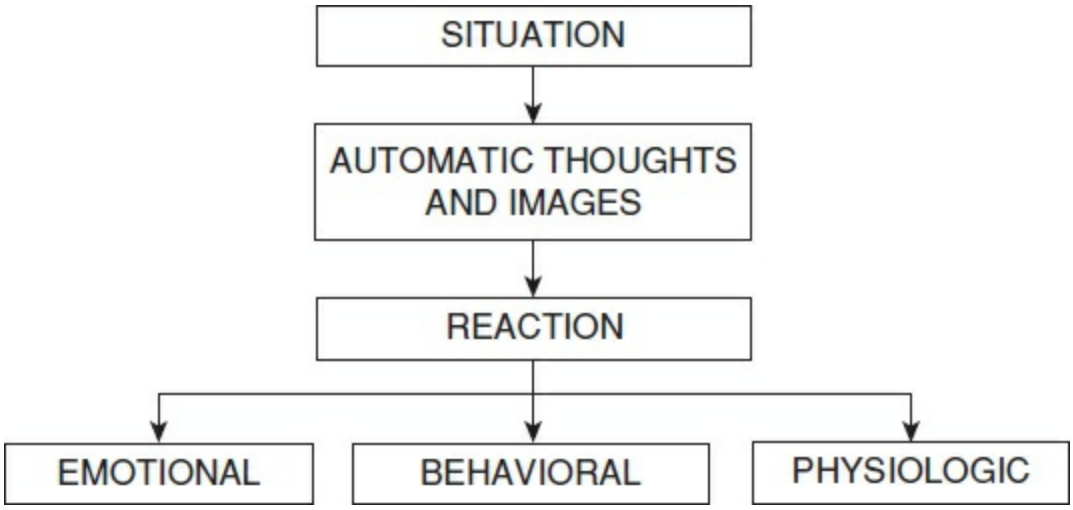


FIGURE 23-2
Cognitive model.

A strong therapeutic alliance is essential for effective treatment, and the relationship is collaborative. Both patients and clinicians are active in solving problems, evaluating cognitions, and modifying behavior. Cognitive therapy is educative; clinicians teach patients how to use the techniques at home that they have employed in session. The clinician's stated goal is to help patients become their own therapists. They emphasize that the way patients get better is to apply what they have learned in session to make small changes in their thinking and behavior every day.

Originally designed as a short-term treatment for depression, cognitive therapy has evolved over the years. Patients with straight-forward cases of anxiety and depression, who were relatively psychologically healthy before the onset of their disorder, often benefit from a short course of treatment, perhaps 6 to 12 sessions. At the other end of the spectrum are patients whose disorders are chronic or complex, or those for whom standard treatment needs to be varied due to personality disorders. They may need many more sessions over a significantly longer period of time. Those with recurrent episodes or with serious mental illnesses may require intermittent treatment through their lifetime.

Cognitive Theory of Depression

According to the cognitive theory of depression, cognitive dysfunctions are the core of depression, and affective and physical changes and other

associated features of depression are consequences of cognitive dysfunctions. For example, apathy and low energy result from a person's expectation of failure in all areas. Similarly, paralysis of will stems from a person's pessimism and feelings of hopelessness. From a cognitive perspective, depression can be explained by the cognitive triad, which explains that negative thoughts are about the self, the world, and the future.

The goal of therapy is to alleviate depression and to prevent its recurrence by helping patients to identify and test negative cognitions, to develop alternative and more flexible schemas, and to rehearse both new cognitive and behavioral responses. Changing the way a person thinks can alleviate the psychiatric disorder.

Strategies and Techniques

The goal of cognitive therapy is to facilitate remission and prevent or reduce relapse by teaching skills patients can use for their lifetime, within a highly supportive, collaborative relationship. Both patients and clinicians are active participants, jointly solving current problems, evaluating patients' cognitions in specific situations, and working toward needed behavior change. As a result, patients' symptoms decrease and their mood and day-to-day functioning improves.

As with other psychotherapies, therapists' attributes are important to successful therapy. Therapists must exude warmth, understand the life experience of each patient, and be genuine and honest with themselves and with their patients. They must be able to relate skillfully and interactively with their patients. Cognitive therapists set the agenda at the beginning of each session, assign homework between sessions, and teach new skills. The therapist and patient collaborate actively. There are several techniques utilized in cognitive therapy; three common components are described below.

Didactic Aspects. The therapy's didactic aspects include explaining to patients the cognitive triad, schemas, and faulty logic. Therapists must tell patients that they will formulate hypotheses together and test them throughout the treatment. Cognitive therapy requires a full explanation of the relation between symptoms and thinking, affect, and behavior, as well as the rationale for all aspects of treatment. This explanation contrasts with

psychoanalytically oriented therapies, which require little explanation.

Cognitive Techniques. The therapy's cognitive approach includes four processes: eliciting automatic thoughts, testing automatic thoughts, identifying maladaptive underlying assumptions, and testing the validity of maladaptive assumptions.

ELICITING AUTOMATIC THOUGHTS. Automatic thoughts, also called *cognitive distortions,* are cognitions that intervene between external events and a person's emotional reaction to the event. For example, the belief that "people will laugh at me when they see how badly I bowl" is an automatic thought that occurs to someone who receives an invitation to go bowling and responds negatively. Another example is the thought, "She doesn't like me," when someone passes in the hall without saying, "Hello." Examples of cognitive distortions are illustrated in Figure 23-3.

TESTING AUTOMATIC THOUGHTS. Acting as a teacher, a therapist helps a patient test the validity of automatic thoughts. The goal is to encourage the patient to reject inaccurate or exaggerated automatic thoughts after careful examination. Patients often blame themselves when things that are outside their control go awry. The therapist reviews the entire situation with the patient and helps reassign the blame or cause of the unpleasant events. Generating alternative explanations for events is another way of undermining inaccurate and distorted automatic thoughts.

IDENTIFYING MALADAPTIVE ASSUMPTIONS. As the patient and therapist continue to identify automatic thoughts, patterns usually become apparent. The patterns represent rules or maladaptive general assumptions that guide a patient's life. Samples of such rules are, "In order to be happy, I must be perfect," and, "If anyone doesn't like me, I'm not lovable." Such rules inevitably lead to disappointments and failure and, ultimately, to psychiatric symptoms such as depression. See Figure 23-4 for a sample automatic thought record.

All-or-nothing thinking *(also called black-and-white, polarized, or dichotomous thinking)*: You view a situation in only two categories instead of on a continuum. Example: "If I'm not a total success, I'm a failure."

Catastrophizing *(also called fortune-telling)*: You predict the future negatively without considering other, more likely outcomes. Example: "I'll be so upset, I won't be able to function at all."

Disqualifying or discounting the positive: You unreasonably tell yourself that positive experiences, deeds, or qualities do not count. Example: "I did that project well, but that doesn't mean I'm competent; I just got lucky."

Emotional reasoning: You think something must be true because you "feel" (actually believe) it so strongly, ignoring or discounting evidence to the contrary. Example: "I know I do a lot of things okay at work, but I still feel like I'm a failure."

Labeling: You put a fixed, global label on yourself or others without considering that the evidence might more reasonably lead to a less disastrous conclusion. Example: "I'm a loser. He's no good."

Magnification/minimization: When you evaluate yourself, another person, or a situation, you unreasonably magnify the negative and/or minimize the positive. Example: "Getting a mediocre evaluation proves how inadequate I am. Getting high marks doesn't mean I'm smart."

Mental filter *(also called selective abstraction)*: You pay undue attention to one negative detail instead of seeing the whole picture. Example: "Because I got one low rating on my evaluation [which also contained several high ratings] it means I'm doing a lousy job."

Mind reading: You believe you know what others are thinking, failing to consider other, more likely possibilities. Example: "He's thinking that I don't know the first thing about this project."

Overgeneralization: You make a sweeping negative conclusion that goes far beyond the current situation. Example: "[Because I felt uncomfortable at the meeting] I don't have what it takes to make friends."

Personalization: You believe others are behaving negatively because of you, without considering more plausible explanations for their behavior. Example: "The repairman was curt to me because I did something wrong."

"Should" and "must" statements *(also called imperatives)*: You have a precise, fixed idea of how you or others should behave, and you overestimate how bad it is that these expectations are not met. Example: "It's terrible that I made a mistake. I should always do my best."

Tunnel vision: You only see the negative aspects of a situation. Example: "My son's teacher can't do anything right. He's critical and insensitive and lousy at teaching."

FIGURE 23-3
Cognitive distortions. (From Beck JS. *Cognitive Behavior Therapy: Basics and Beyond*. 2nd ed; 2011. Reprinted with permission of Guilford Press.)

TESTING THE VALIDITY OF MALADAPTIVE ASSUMPTIONS. Testing the accuracy of maladaptive assumptions is similar to testing the validity of automatic thoughts. In a particularly effective test, therapists ask patients to defend the validity of their assumptions. For example, patients may state that they should always work up to their potential, and a therapist may ask, "*Why is that so important to you?*" Table 23-11 gives examples of some interventions designed to elicit, identify, test, and correct the cognitive distortions that lead to depressive and other painful affects.

A woman presented for therapy with anger control problems. Before therapy, she sent a slew of hostile voicemail and e-mail messages to a colleague, alienated her neighbors with her noise complaints, and received a request to leave her bowling league after two physical altercations with members of other teams. A careful review of the patient's thoughts and beliefs surrounding these situations revealed a common denominator of a sense of *mistrust* and *entitlement.* In each situation, she believed that others had gone out of their way to mistreat her. Furthermore, she had an exaggerated sense of self-importance represented by beliefs such as, "Nobody has the right to treat me that way," "I shouldn't have to deal with these people and their stupidity," and "I have to show them they can't ever push me around." To this patient, her anger was justified, as she was trying to defend herself from the misbehavior of others. However, to the outside observer, the patient was a "loose cannon" who took offense at the drop of a hat and whose behavior was outrageous and indefensible. In therapy, the patient at first was not open to viewing her anger problem in the manner just described. However, as she learned to recognize the activation of her schemas of *mistrust* and *entitlement,* she became more willing to consider ways in which she could modify her viewpoints and behaviors. This positive change was facilitated by the therapist's empathic responses to the patient's more credible stories of mistreatment she had received from her family, whose abusive behavior gave her the message that she should never trust anyone, and that she should never put up with being mistreated again. (Courtesy of C. F. Newman, Ph.D. and A. T. Beck, M.D.)

Behavioral Techniques. Behavioral and cognitive techniques go hand in hand; behavioral techniques test and change maladaptive and inaccurate cognitions. The overall purposes of such techniques are to help patients understand the inaccuracy of their cognitive assumptions and learn new strategies and ways of dealing with issues.

AUTOMATIC THOUGHT RECORD

Directions: When you notice your mood getting worse, ask yourself, **"What's going through my mind right now?"** and as soon as possible jot down the thought or mental image in the Automatic Thoughts column.

DATE/ TIME	SITUATION	AUTOMATIC THOUGHT(S)	EMOTION(S)	ALTERNATIVE RESPONSE	OUTCOME
	1. What event, daydream, or recollection led to the unpleasant emotion? 2. What (if any) distressing physical sensations did you have?	1. What thought(s) and/or image(s) went through your mind? 2. How much did you believe each one at the time?	1. What emotion(s) (sad, anxious, angry, etc.) did you feel at the time? 2. How intense (0-100%) was the emotion?	1. (optional) What cognitive distortion did you make? (e.g., all-or-nothing thinking, mind-reading, catastrophizing) 2. Use questions at bottom to compose a response to the automatic thought(s). 3. How much do you believe each response?	1. How much do you now believe each automatic thought? 2. What emotion(s) do you feel now? How intense (0-100%) is the emotion? 3. What will or did you do?
Friday 7:30 PM	I called Sally to go out, as we talked about. I got her answering machine. Felt a sinking sensation.	1) They have all gone out and forgotten about me, because I'm not important to them anymore. (90% believable) 2) I'm left out again. (90% believable) 3) I'm going to have to spend another Friday night alone. (100% believable) 4) I just don't fit in anywhere in this world. (70% believable)	1) Angry (60% intensity) 2) Lonely (95% intensity) 3) Depressed (95% intensity)	I'm engaging in arbitrary inference, overgeneralization, personalization, and catastrophization. 1) It could all be an innocent misunderstanding. (40% believable) 2) I have spent a lot of time with Sally and the others and I know they like me. (60% believable) 3) Being at home alone is not the end of the world. (50% believable)	1) 30% 2) 10% 3) 50% 4) 0% Angry (5%) Lonely (40%) Depressed (20%) Calm (70%) I will call back in an hour if I don't hear from Sally.

Questions to help compose an alternative response: (1) What is the evidence that the automatic thought is true? Not true? (2) Is there an alternative explanation? (3) What's the worst that could happen? Could I live through it? What's the best that could happen? What's the most realistic outcome? (4) What's the effect of my believing the automatic thought? What could be the effect of changing my thinking? (5) What should I do about it? (6) If _____ (friend's name) was in the situation and had this thought, what would I tell him/her?

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FIGURE 23-4
Sample automatic thought record.

Among the behavioral techniques in cognitive therapy are scheduling activities, mastery and pleasure, graded task assignments, cognitive rehearsal, self-reliance training, role-playing, and diversion techniques. One of the first tasks in therapy is to schedule activities on an hourly basis. Patients keep records of the activities and review them with the therapist. In addition to scheduling activities, patients rate the amount of mastery and pleasure their activities bring them. Patients often express surprise that they have much more mastery of activities and enjoy them more than they had thought.



Table 23-11
Cognitive Errors Derived from Assumptions

Cognitive Error	Assumption	Intervention
Overgeneralizing	If it's true in one case, it applies to any case that is even	Exposure of faulty logic. Establish criteria of which cases are similar

	slightly similar.	to what degree.
Selective abstraction	The only events that matter are failures, deprivation, etc. Should measure self by errors, weaknesses, etc.	Use log to identify successes patient forgot.
Excessive responsibility (assuming personal causality)	I am responsible for all bad things, failures, etc.	Disattribution technique.
Assuming temporal causality (predicting without sufficient evidence)	If it has been true in the past, it's always going to be true.	Expose faulty logic. Specify factors that could influence outcome other than past events.
Self-references	I am the center of everyone's attention—especially my bad performances. I am the cause of misfortunes.	Establish criteria to determine when patient is the focus of attention and also the probable facts that cause bad experiences.
Catastrophizing	Always think of the worst. It's almost likely to happen to you.	Calculate real probabilities. Focus on evidence that the worst did not happen.
Dichotomous thinking	Everything is either one extreme or another (black or	Demonstrate that events may be evaluated on a continuum.

white, good or bad).

From Beck AT, Rush AJ, Shaw BF, Emery G. *Cognitive Therapy of Depression*. New York: Guilford; 1979:48, with permission.

To simplify the situation and allow mini-accomplishments, therapists often break tasks into subtasks, as in graded task assignments, to show patients that they can succeed. In cognitive rehearsal, patients imagine and rehearse the various steps in meeting and mastering a challenge.

Patients (especially inpatients) are encouraged to become self-reliant by doing such simple things as making their own beds, doing their own shopping, and preparing their own meals. This process is called self-reliance training. Role-playing is a particularly powerful and useful technique to elicit automatic thoughts and to learn new behaviors. Diversion techniques are effective in helping patients get through difficult times and include physical activity, social contact, work, play, and visual imagery.

Imagery or thought stoppage can treat impulsive or obsessive behavior. For instance, patients imagine a stop sign with a police officer nearby or another image that evokes inhibition at the same time that they recognize an impulse or obsession that is alien to the ego. Similarly, obesity can be treated by having patients visualize themselves as thin, athletic, trim, and well-muscled, and then training them to evoke this image whenever they have an urge to eat. Hypnosis or autogenic training can enhance such imagery. In a technique called guided imagery, therapists encourage patients to have fantasies that can be interpreted as wish fulfillments or attempts to master disturbing affects or impulses.

Indications

Cognitive therapy may be used to (1) eliminate or moderate the symptoms of a psychiatric disorder or psychological distress as an independent treatment or in combination with medication; (2) modify underlying beliefs and schemas that contribute to the development and maintenance of disorders; (3) address psychosocial problems (e.g., marital discord, job stress, caregiver burnout) that may have preceded, coincided with, or been caused by the disorder; (4) reduce the likelihood of relapse or recurrence of a disorder after symptoms have resolved; (5) increase adherence to recommended medical treatment (e.g., medication, weight loss, diabetes management); or moderate

the impact of a medical condition on level of functioning (e.g., chronic pain, cancer).

BEHAVIOR THERAPY

The term *behavior* in *behavior therapy* refers to a person's observable actions and responses. Behavior therapy involves changing the behavior of patients to reduce dysfunction and to improve quality of life. Behavior therapy includes a methodology, referred to as *behavior analysis*, for the strategic selection of behaviors to change, and a technology to bring about behavior change, such as modifying antecedents or consequences or giving instructions. Behavior therapy has not only influenced mental health care but, under the rubric of behavioral medicine, it has also made inroads into other medical specialties.

Behavior therapy represents clinical applications of the principles developed in learning theory. Behavioral psychology, or behaviorism, arose in the early 20th century in reaction to the method of introspection that dominated psychology at the time. John B. Watson, the father of behaviorism, had initially studied animal psychology. This background made it a small conceptual leap to argue that psychology should concern itself only with publicly observable phenomena (i.e., overt behavior). According to behavioristic thinking, because mental content is not publicly observable, it cannot be subjected to rigorous scientific inquiry. Consequently, behaviorists developed a focus on overt behaviors and their environmental influences.

Today, different behavioral schools continue to share a focus on verifiable behavior. Behavioral views differ from cognitive views in holding that physical, rather than mental, events control behavior. According to behaviorism, mental phenomena or speculations about them are of little or no scientific interest.

History

As early as the 1920s, scattered reports about the application of learning principles to the treatment of behavioral disorders began to appear, but they had little effect on the mainstream of psychiatry and clinical psychology. Not until the 1960s did behavior therapy emerge as a systematic and comprehensive approach to psychiatric (behavioral) disorders; at that time, it

arose independently on three continents. Joseph Wolpe and his colleagues in Johannesburg, South Africa, used Pavlovian techniques to produce and eliminate experimental neuroses in cats. From this research, Wolpe developed systematic desensitization, the prototype of many current behavioral procedures for the treatment of maladaptive anxiety produced by identifiable stimuli in the environment. At about the same time, a group at the Institute of Psychiatry of the University of London, particularly Hans Jurgen Eysenck and M. B. Shapiro, stressed the importance of an empirical, experimental approach to understanding and treating individual patients, using controlled, single-case experimental paradigms and modern learning theory. The third origin of behavior therapy was work inspired by the research of Harvard psychologist B. F. Skinner. Skinner's students began to apply his operant-conditioning technology, developed in animal-conditioning laboratories, to human beings in clinical settings.

Systematic Desensitization

The foundation of systematic desensitization is the behavioral principle of counterconditioning, whereby a person overcomes maladaptive anxiety elicited by a situation or an object by approaching the feared situation gradually, in a psychophysiological state that inhibits anxiety. In systematic desensitization, patients attain a state of complete relaxation and are then exposed to the stimulus that elicits the anxiety response. The relaxed state inhibits the negative reaction of anxiety, a process called *reciprocal inhibition.* Rather than using actual situations or objects that evoke fear, patients and therapists prepare a graded list or hierarchy of anxiety-provoking scenes associated with a patient's fears. Treatment systematically pairs the learned relaxation state and the anxiety-provoking scenes. Thus, systematic desensitization consists of three steps: relaxation training, hierarchy construction, and desensitization of the stimulus.

Relaxation Training. Relaxation produces physiologic effects opposite to those of anxiety: slow heart rate, increased peripheral blood flow, and neuromuscular stability. A variety of relaxation methods exist. Some, such as yoga and Zen, have been known for centuries. Most methods use the so-called progressive relaxation, discussed above. Patients relax major muscle

groups in a fixed order, beginning with the small muscle groups of the feet and working cephalad or vice versa. Some clinicians use hypnosis to facilitate relaxation or use tape-recorded exercises to allow patients to practice relaxation on their own. Mental imagery is a relaxation method in which patients imagine themselves in a place associated with pleasant, relaxed memories. Such images allow patients to enter a relaxed state or experience (as Herbert Benson termed it) the *relaxation response*.

The physiologic changes that take place during relaxation are the opposite of those induced by the adrenergic stress responses that are part of many emotions. Muscle tension, respiration rate, heart rate, blood pressure, and skin conductance decrease. Finger temperature and blood flow to the finger usually increase. Relaxation increases respiratory heart rate variability, an index of parasympathetic tone.

Hierarchy Construction. When constructing a hierarchy, clinicians determine all the conditions that elicit anxiety, and then patients create a hierarchy list of 10 to 12 scenes in order of increasing anxiety. For example, an acrophobic hierarchy may begin with a patient imagining standing near a window on the second floor and end with being on the roof of a 20-story building, leaning on a guard rail and looking straight down.

Desensitization of the Stimulus. In the final step, called *desensitization,* patients proceed systematically through the list from the least to the most anxiety-provoking scene while in a deeply relaxed state. Their responses to the stimuli determine the rate at which patients progress through the list. When patients can vividly imagine the most anxiety-provoking scene of the hierarchy with equanimity, they experience little anxiety in the corresponding real-life situation.

ADJUNCTIVE USE OF DRUGS. Clinicians have used various drugs to hasten relaxation, but drugs should be used cautiously and only by clinicians trained and experienced in potential adverse effects. Either the ultra-rapidly acting barbiturate sodium methohexital or diazepam is given intravenously in subanesthetic doses. By implementing the procedural details carefully, almost all patients find the procedure pleasant, with few unpleasant side effects. The advantages of pharmacologic desensitization are a shortening of preliminary

training in relaxation, the ability for nearly all patients to relax adequately, and a more rapid progression of treatment than without the drugs.

INDICATIONS. Systematic desensitization works best in cases of a clearly identifiable anxiety-provoking stimulus. Phobias, obsessions, compulsions, and certain sexual disorders have been treated successfully with this technique.

Therapeutic-Graded Exposure

Therapeutic-graded exposure is similar to systematic desensitization, except that relaxation training is not involved, and treatment usually occurs in a real-life context. The individual must be brought in contact with (i.e., be exposed to) the warning stimulus to learn firsthand that no dangerous consequences will ensue. Exposure is graded according to a hierarchy. Patients afraid of cats, for example, might progress from looking at a picture of a cat to holding one.

Flooding

Flooding (sometimes called *implosion*) is similar to graded exposure in that it involves exposing the patient to the feared object in vivo; however, there is no hierarchy. The basis of flooding is the premise that escaping from an anxiety-provoking experience reinforces the anxiety through conditioning. Thus, clinicians can extinguish the anxiety and prevent the conditioned avoidance behavior by not allowing patients to escape the situation. Clinicians encourage patients to confront feared situations directly, without a gradual buildup, as in systematic desensitization or graded exposure. No relaxation exercises are used, as in systematic desensitization. Patients experience fear, which gradually subsides after a time. The success of the procedure depends on having patients remain in the fear-generating situation until they are calm and feel a sense of mastery. Prematurely withdrawing from the situation or prematurely terminating the fantasized scene is equivalent to an escape, which then reinforces both the conditioned anxiety and the avoidance behavior and produces the opposite of the desired effect. In a variant, called *imaginal flooding,* the feared object or situation is confronted only in the imagination, not in real life. Many patients refuse flooding

because of the psychological discomfort involved. Contraindications include situations when intense anxiety would be hazardous to a patient (e.g., those with heart disease or fragile psychological adaptation). The technique works best with specific phobias. The case study below presents an example of in vivo flooding.

The patient was a 33-year-old woman with social fears of eating in public. In particular, she was afraid of being observed by others when chewing and swallowing, particularly at dinner parties. The therapist arranged a contrived situation in which the patient came to the session with a prepared meal and drink. She entered a conference room in which five persons in professional attire were already seated at a table. The therapist instructed the patient to eat her meal in front of these individuals. Between bites, she was instructed to look at them often, and they had been coached to avoid staring contests. She was not to distract herself from her anxiety symptoms. She was to eat her meal slowly, paying attention to the behavior of the observers and to her anxiety symptoms (e.g., dry mouth or difficulty swallowing). No conversation between the patient and the observers was permitted. The observers would look at her and observe her chewing and swallowing behaviors, at times writing comments in a notebook. Occasionally, observers would communicate by whispering to each other, exchanging written notes, or giving knowing glances and smiles.

The only other communication occurred between the patient and therapist, and this was limited to the patient providing her subjective units of distress rating. The session lasted 90 minutes. Note: this situation may seem quite traumatizing. Because the exposure session is long and continues until ratings decline, the patient becomes desensitized. (Courtesy of Rolf G. Jacob, M.D. and William H. Pelham, M.D.)

In most cases, flooding (exposure) is accompanied by response prevention. Response prevention involves discontinuing all avoidance and other anxiety-reducing behaviors. In order for exposure to produce extinction, avoidance behaviors that typically reduce fear need to be prevented (e.g., looking at the floor when talking can reduce social anxiety; during exposure, therefore, to

incorporate response prevention, the patient looks directly at people and not at the floor). Over time, while the patient remains in the fear-producing situation, the anxious emotional response will decrease. *Exposure and response prevention* (ERP) is actually the treatment of choice for obsessive compulsive disorder. In ERP, patients are exposed to feared stimuli and obsessions while rituals, which typically serve to reduce anxiety, are prevented.

Participant Modeling

In participant modeling, patients learn a new behavior by imitation, primarily by observation, without having to perform the behavior until they feel ready. Just as a person can acquire irrational fears by learning, a person may unlearn them by observing a fearless model confront the feared object. The technique is effective with phobic children who observe other children of their own age and sex who approach the feared object or situation. With adults, a therapist may describe the feared activity in a calm manner that a patient can identify. Or, the therapist may act out the process of mastering the feared activity with a patient. Sometimes a hierarchy of activities is established, and the patient tackles the least anxiety-provoking activity first. The participant-modeling technique is successful in treating agoraphobia by having a therapist accompany a patient into the feared situation. In a variant of the procedure, called *behavior rehearsal,* the patient acts out real-life problems under a therapist's observation or direction.

The following is a self-report by a patient with a contamination phobia, who is afraid to touch objects for fear of being infected or contaminated. She describes her reactions.

[The therapist] started touching everything very slowly. I was told to follow behind and touch everything she touched. It was like we were spreading the contamination. She touched doorknobs, light switches, walls, pictures, and woodwork. She opened drawers in each bedroom and touched the contents. She opened closets and touched clothes hanging on the rods. She touched the towels and sheets in the linen closet. She went through the children's rooms, touching dolls, stuffed animals, models, Star Wars figures, Transformers, and books.

[The therapist] kept talking to me quietly and calmly all the time we went along. I had been anxious when we started, but as we continued, my anxiety level decreased. At one point, when I had begun to think the worst was over, she pointed to the attic door and said we were going inside. I said, "No, that's where the mice were." She told me I didn't want to have a place in my home that was off-limits. I agreed but became very anxious. It was very hard for me to go inside. I began touching the boxes too, but I was very upset. Then, she put her hands down on the floor and wanted me to do the same. I said, "I can't. I just can't." [The therapist] said, "Yes, you can."

[The therapist] spent several hours with me that day. Before she left, she made a list of things for me to do by myself. Twice a day, I was to go through the house, touching everything the way she had done with me. I was to invite my friend who had a pet to come and visit, as well as friends of my children who had pets. (Courtesy of Rolf G. Jacobs, M.D. and William H. Pelham, M.D.)

Exposure to Stimuli Presented in Virtual Reality

Advances in computer technology have made it possible to present environmental cues in virtual reality for exposure treatment. Beneficial effects have been reported with virtual reality exposure of patients with height phobia, fear of flying, spider phobia, and claustrophobia. Much experimental work is ongoing. One model uses an avatar of the patient walking through a crowded supermarket filled with other avatars (including one of the therapist) as a way of conquering agoraphobia.

Assertiveness Training

Assertiveness is assertive behavior that enables a person to act in his or her own best interest, to stand up for herself or himself without undue anxiety, to express honest feelings comfortably, and to exercise personal rights without denying the rights of others.

Two types of situations frequently call for assertive behaviors: (1) setting limits on pushy friends or relatives and (2) commercial situations, such as countering a sales pitch or being persistent when returning defective

merchandise. Early assertiveness training programs tended to define specific behaviors as assertive or nonassertive. For example, individuals were encouraged to assert themselves if somebody got in front of them in a supermarket checkout line. Context has received increasing attention; that is, what would be assertive behavior in this situation depends on circumstances.

Social Skills Training

The negative symptoms in patients with schizophrenia constitute behavioral deficits that go beyond difficulties with assertiveness. These patients have inadequate expressive behaviors and inappropriate stimulus control of their social behaviors (i.e., they do not pick up social cues). Similarly, patients with depression often experience a lack of social reinforcement because of a lack of social skills, and studies have found social skills training to be efficacious for depression. Patients with social phobia similarly often have not acquired adolescents' social skills. In fact, their social defensive behaviors (e.g., avoiding eye contact, making brief statements, and minimizing self-disclosure) increase the probability of the rejection that they fear.

Social skills training programs for patients with schizophrenia cover skills in the following areas: conversation, conflict management, assertiveness, community living, friendship and dating, work and vocation, and medication management. Each of these skills has several components. For example, assertiveness skills include making requests, refusing requests, making complaints, responding to complaints, expressing unpleasant feelings, asking for information, making apologies, expressing fear, and refusing alcohol and street drugs. Each component involves specific steps. For example, conflict management includes skills in negotiating, compromising, tactful disagreeing, responding to untrue accusations, and leaving overly stressful situations. A situation in which a patient would use conflict management skills could be when the patient and a friend decide to go to a movie, and their choice of movie differs.

Negotiating and compromising, for example, involves the following steps:

1. Explain one's viewpoint briefly.
2. Listen to the other person's viewpoint.
3. Repeat the other person's viewpoint.

4. Suggest a compromise.

At his initial appointment, Phillip described severe symptoms of obsessive-compulsive disorder (OCD). He was 23 years old and living at home because he was no longer able to work or go to school. His days were consumed with behaviors related to checking, repeating, and hoarding. Phillip was unable to throw away anything—he saved junk mail, used tissues and napkins, old papers and magazines, and any kind of receipt for fear that he might lose something important. Phillip spent many hours checking his trash, his car, and his home to be sure that he had not thrown away anything important. He also checked everything he wrote (e.g., checks, school exams and papers, letters and e-mails) to be sure that he had not made a mistake, and he read and reread books, magazines, and articles to be sure he understood the written material adequately. Phillip constantly worried that he had done something wrong and would disappoint his parents. He was also depressed because he was unable to function well in life, and he had tremendous social anxiety that had plagued him for many years, making it difficult to make and keep friends.

By the end of Phillip's second session, his therapist was beginning to get a good idea of the general nature and severity of his symptoms and some of the maintaining factors. However, to plan the treatment in more detail and to get a better idea of how the symptoms occurred during his daily life, she asked Phillip to keep daily records over the next week using a form that she had prepared for him. The form had a place for recording the amount of time he spent in rituals each morning, afternoon, and evening, as well as another place to record more details about at least one episode of rituals each day (e.g., what was happening before, during, and after the rituals; see Table 23-12).

Phillip's therapist determined that his difficulties with obsessions, rituals, depression, and social fears reflected a core fear of negative evaluation. Phillip was overly concerned with making mistakes, being imperfect, and disappointing others. Even as a child, Phillip was worried about not doing well enough, and he had difficulty making friends for fear that others would not like him. His highly anxious parents provided

much adulation when Phillip did things well (e.g., learned to ride a bike, got good grades in school), and they spent a great deal of time instructing him on improving his performance when an activity or grade was not perfect. As Phillip took on more responsibility at school and with part-time work, he became more concerned about doing things right. He learned that going back and checking his work relieved his anxiety. He also learned that saving his papers for future checking reassured him that he would be able to fix any unrecognized mistakes at a later time. His parents helped him to reduce his anxiety when he was uncertain about his work by reassuring him that he was doing okay. As Phillip progressed from elementary school to middle school to high school, his workload and anxiety gradually increased, but he was able to manage things with some moderate checking and saving. When he began attending college, however, the workload increased extensively, and he found himself doing even more checking and hoarding to reduce his fears about making mistakes. Phillip began to feel that these behaviors were getting out of control, but he could not stop them. He had to check and recheck to be sure that he was not making mistakes. The cycle of anxiety to ritual to reduced anxiety was so powerfully reinforcing that he could not stop. He needed help to break this cycle and to address his persistent fear of negative evaluation.

Phillip's therapist decided to begin treatment with a course of exposure and response prevention (ERP) to get his obsessions and rituals under control and begin to address his core fear of making mistakes and being evaluated negatively. Given that Phillip's depression had grown from the disability associated with his OCD, the therapist expected that a successful course of ERP might also help to reduce his depressive symptoms. ERP for Phillip began with a home visit, where the therapist helped him to complete common daily activities with adherence to his RP plan, which included the following:

- No more checking: After eating, leave the table immediately without inspecting your plate and the surrounding areas (including under the table and chair) for lost items. Leave the restroom immediately after using it, without checking the toilet, trash, and sink for lost items. When leaving the car, do not check seats, floors, and windows.

Write everything (papers, checks, etc.) only once; no checking to be sure that letters and words are correct.

- No more repeating: No more rereading books. No staring repeatedly at items to ensure that nothing is lost.
- No more saving. Throw tissues away immediately after using them. Discard trash and junk mail immediately. Do not look into the trash can for lost items.

Phillip's parents also were asked to stop reassuring him and to discontinue doing rituals for him. This session was very difficult for Phillip and his family, but they understood the logic of ERP, and they were willing to try anything.

For the next 3 weeks, Phillip and his therapist met three times a week to conduct in vivo exposure sessions that helped him to face his core fears. For many of these sessions, the therapist asked Phillip to bring hoarded items from home and to discard all unnecessary items during the therapy session. At first, this created tremendous anxiety, but over time, Phillip was able to throw things away with less fear of losing something important. He also developed the ability to conduct self-directed exposure at home. Other exposure sessions involved writing letters and mailing them without checking, reading passages from magazines and books only once, and sorting through junk mail to make quick decisions about what to save or discard. As Phillip was able to take on more responsibility for home-based exposure, session frequency decreased to two times per week, and then to once per week. After 3 months of treatment, Phillip's scores on the YBOCS (Yale-Brown Obsessive-Compulsive Scale) and BDI (Beck Depression Inventory) had decreased to 20 and 19, respectively, demonstrating significant improvement in obsessive-compulsive symptoms and depression. His SPAI (Social Phobia and Anxiety Inventory) score, however, remained relatively unchanged, suggesting that he was still experiencing significant social anxiety.

Next, while Phillip worked on maintaining the gains he had made following ERP, he and his therapist conducted some role-plays to evaluate his social skills. It was apparent that Phillip had extreme difficulty with initiating and maintaining conversations. His eye contact also was quite poor in social interactions. Thus, the therapist devised a

plan for teaching and practicing new skills, which also involved additional exposure to Phillip's core fears as the therapist asked him to resume contact with old friends and identify activities where he could meet new people. He practiced new behaviors first in session with his therapist and then developed a hierarchy of feared social situations in which he could practice his new behaviors. These practice exercises also involved a form of exposure, as the therapist asked Phillip to make social contact, which produced fears of negative evaluation. After another 3 months of treatment focused on social skills training (and associated exposure), Phillip's scores on the YBOCS and BDI had decreased further (YBOCS = 15; BDI = 13), and his SPAI score had decreased to 100. Phillip had gone back to school to take one class, he was spending small amounts of time with old friends, and he was volunteering a few hours each week at his church. (Courtesy of M. A. Stanley, Ph.D. and D. C. Beidel, Ph.D.)



Table 23-12
Daily Monitoring of Rituals

Each day, record the amount of time spent doing rituals in the morning, afternoon, and evening.							
	Tuesday	Wednesday	Thursday	Friday	Saturday	Sunday	Monday
Morning	2 hr	1.5 hr					
Afternoon	3 hr	2 hr					
Evening	1.5 hr	3 hr					

Once a day, record the following details about an episode of rituals:						
Day	Time	Situation	Feelings	Thoughts (Obsessions)	Type of Ritual	Feelings After Rituals
Saturday	8 AM	Finished breakfast	Afraid Scared Worried	Shouldn't have thrown away my napkin Might have left something under my plate What if I lost something important?	Checking through trash Looking under plate Staring to see if I lost something	Better For now, I think I have not lost anything
Sunday	2 PM	At the store; signed a check	Worried Anxious	Did I sign my name correctly? Did I write the correct amount? What if I give them the check and it is wrong?	Staring at the check Tracing the lines I wrote Standing there	Anxious because I couldn't finish checking

Courtesy of M. A. Stanley, Ph.D. and D. C. Beidel, Ph.D.

Aversion Therapy

When a noxious stimulus (punishment) is presented immediately after a specific behavioral response, theoretically, the response is eventually inhibited and extinguished. Aversion therapy can use many types of noxious

stimuli: electric shocks, substances that induce vomiting, corporal punishment, and social disapproval. The therapist pairs the negative stimulus with the behavior, resulting in suppression of the behavior. The unwanted behavior may disappear after a series of such sequences. Aversion therapy has been used for alcohol abuse, paraphilias, and other behaviors with impulsive or compulsive qualities, but this therapy is controversial for many reasons. For example, punishment does not always lead to the expected decreased response and can sometimes be positively reinforcing.

Eye Movement Desensitization and Reprocessing

Saccadic eye movements are rapid oscillations of the eyes that occur when a person tracks an object that is moved back and forth across the line of vision. Several studies have demonstrated that inducing saccades while a person is imagining or thinking about an anxiety-producing event can yield a positive thought or image that results in decreased anxiety. Eye movement desensitization and reprocessing therapy is effective in the treatment of PTSDs and phobias.

Positive Reinforcement

When a generally rewarding event, such as food or praise, follows a behavioral response, the response tends to be strengthened and to occur more frequently than before the reward. This principle can be applied to a variety of situations. On inpatient hospital wards, patients with mental disorders receive a reward for performing a desired behavior, such as tokens that they can use to purchase luxury items or certain privileges. The process, known as *token economy*, has successfully altered behavior.

Behavioral Activation

Based on behavioral theory, depression persists because: (a) reinforcement for nondepressed (healthy) behavior is low or nonexistent, (b) depressed behavior produces positive (e.g., sympathy from others) or negative reinforcement (decrease in responsibilities or avoidance of undesirable situations), (c) exposure to aversive or unpleasant life experiences is significant, or (d) some combination of these factors. Severely depressed mood leads to behavioral withdrawal, which in turn results in loss of

reinforcement from pleasurable events or activities.

Behavioral activation emphasizes structured attempts to increase overt behaviors that bring patients into contact with reinforcing environmental contingencies and corresponding improvements in thoughts, mood, and quality of life. Patients begin by engaging in a self-monitoring (or daily diary) exercise to examine already occurring daily activities to provide a baseline measurement and to provide some potential activities to target during treatment. For each behavior, they indicate their levels of reward, difficulty, and importance. Emphasis then shifts to identifying patients' values and goals within life areas that include family, social, and intimate relationships, education, employment/career, hobbies/recreation, volunteer work/charity, physical/health issues, spirituality, and anxiety-eliciting situations. An activity hierarchy is constructed, where activities are rated from "easiest" to "most difficult" to accomplish. Patients progressively move through the hierarchy, moving from easier behaviors to the more difficult.

Charles was a 70-year-old retired business executive. Throughout his life, his work consumed him. Although he married and had a family, his job was his primary focus. He went to the office early and came home late. He enjoyed what he did—it was stimulating and made him feel important and useful. But as he got older, his performance was not what it used to be, and he decided it was time to retire. However, his mood was pretty low when he no longer had a job. He did not have the energy to get more involved in his church or to develop other hobbies, so he sat around all day, without any social contacts. His wife and best friend encouraged him to talk to someone. The therapist suggested that they try behavioral activation. Charles was somewhat skeptical, as it seemed too simple, but he needed to do something. The therapist spent some time with Charles talking about the kinds of activities that used to make him feel good and some of the things he used to enjoy. They then put together a list of things he might be able to do—even if he did not feel much like it—just to see what would happen. The list included looking for volunteer work where he could use his job skills, spending more time with his wife in some of the activities they once had enjoyed (e.g., watching movies, taking walks), and rejuvenating an old hobby from his college days—

fishing. Charles initially agreed to do some easy activities—go to one movie a week, take one walk a week, and contact his church activity leader about possible volunteer activities. He was surprised to find that even these "baby steps" helped him feel better. He had the chance to talk with other people and began to see that, even in retirement, he could find useful and fun things to do. (Courtesy of M. A. Stanley, Ph.D. and D. C. Beidel, Ph.D.)

Results

Behavior therapy has a wide range of applications and is useful for a wide variety of psychiatric disorders (e.g., anxiety, depression, schizophrenia, and bipolar disorder), psychological problems (e.g., marital discord, school refusal, and stress), and medical problems (e.g., adjustment to chronic illness, medication adherence, and healthy behaviors). Behavioral treatments have also been administered effectively with individual patients, couples, families, and groups, and they are useful in a range of treatment settings, including inpatient, outpatient, day treatment, and community settings.

HYPNOSIS

The concept of hypnosis conjures up myriad perceptions among clinicians and the lay public. Even the term *hypnosis* can be misleading, coming as it does from the Greek root *hypnos* (meaning "sleep"). In reality, hypnosis is not sleeping. It is more likely a complex process that requires alert focus and receptive attention. Hypnosis is a powerful means of directing innate capabilities of imagination, imagery, and attention. Many also believe the myth that the clinician projects the hypnotic trance onto the patient or has the power to influence the patient. In reality, it is the patient who has the hypnotic gift, and the clinician's role is to assess the patient's capacity to capitalize on this asset and to help the patient discover and use it effectively. Patient motivation, personality style, and biologic predisposition may contribute to the manifestation of this talent.

The hypnotic trance enhances focal attention and imagination and simultaneously decreases peripheral awareness. A hypnotist may induce this trance through formalized induction procedures, but it can also occur

spontaneously. The capacity to be hypnotized and, relatedly, the occurrence of spontaneous trance states, is a trait that varies among individuals but is relatively stable throughout a person's life cycle.

History

Descriptions of trance states, ecstatic states, and spontaneous dissociative states abound in the Eastern and Western religious, literary, and philosophical traditions. Anton Franz Anton Mesmer (1734–1815) first formally described hypnosis as a therapeutic modality in the 18th century and believed it to be the result of magnetic energy or an invisible fluid that the therapist channels into the patient to correct imbalances, restoring health. James Braid (1795–1860), an English physician and surgeon, used eye fixation and closure to induce trance states. Later, Jean Martin Charcot (1825–1893) theorized the hypnotic state to be a neurophysiologic phenomenon that was a sign of mental illness. Contemporaneously, Hippolyte Bernheim (1840–1919) believed it to be a function of the normal brain.

Early in his career, Sigmund Freud (1856–1939) used hypnosis as part of his psychoanalysis and noticed that patients in a trance could relive traumatic events, a process called *abreaction.* Later, Freud switched from hypnosis to free association because he wanted to minimize the transference that sometimes accompanies the trance state. Importantly, the switch did not eliminate the occurrence of spontaneous trance during the analysis.

World War I produced many shell-shocked soldiers, and Ernst Simmel (1882–1947), a German psychoanalyst, developed a technique for accessing repressed material that he named *hypnoanalysis.* During World War II, hypnosis played a prominent role in the treatment of pain, combat fatigue, and neurosis. Formal recognition of hypnosis as a therapeutic modality did not occur, however, until the 1950s. The British Medical Society recommended its teaching in medical schools in 1955, and the American Medical Association and American Psychiatric Association officially stated its safety and efficacy in 1958.

Definition

The current understanding of hypnosis is that it is a normal activity of a normal mind through which attention is more focused, critical judgment is

partially suspended, and peripheral awareness is diminished. The trance state, being a function of the subject's mind, cannot be forcibly projected by an outside person. The hypnotist, however, may aid in the achievement of the state and use its uncritical, intense focus to facilitate the acceptance of new thoughts and feelings, thereby accelerating therapeutic change. For the subject, hypnosis typically involves a sense of involuntariness, and movements seem automatic.

Trait of Hypnotizability

A person's degree of hypnotizability is a trait that is relatively stable throughout the life cycle and is measurable. The process of hypnosis takes the hypnotizability trait and transforms it into the hypnotized state. Experiencing the hypnotic concentration state requires a convergence of three essential components: absorption, dissociation, and suggestibility.

Absorption is an ability to reduce peripheral awareness that results in greater focal attention. A metaphorical description is that of a psychological zoom lens that increases attention to the given thought or emotion to the increasing exclusion of all contexts, even including orientation to time and space.

Dissociation is the separating out from consciousness elements of the patient's identity, perception, memory, or motor response as the hypnotic experience deepens. The result is that components of self-awareness, time, perception, and physical activity can occur without being known to the patient's consciousness and so may seem involuntary.

Suggestibility is the tendency of the hypnotized patient to accept signals and information with a relative suspension of normal critical judgment; it is controversial whether a person can completely suspend critical judgment. This trait will vary from an almost compulsive response to input in the highly hypnotizable to a sense of automaticity in the less hypnotizable individual.

Quantification of Hypnotizability. Quantifying a patient's degree of hypnotizability is useful in a clinical setting because it predicts the effectiveness of hypnosis as a therapeutic modality. Quantification also provides useful information about the way patients relate to themselves and the social environment. Highly hypnotizable patients have an increased

incidence of spontaneous trance-like states and so may be unduly influenced by ideas and emotions that are not being appropriately self-critiqued.

Induction

Many different induction protocols follow the same basic principles and pattern but may be better suited to patients with different levels of hypnotizability.

Doctor: Take a long, deep breath—inhale and exhale; now close your eyes and relax. Pay particular attention to the muscles in and about your eyes—relax them to the point that they just won't work. Are you trying to do that? Good. If you really have them relaxed, right at this very moment, no matter how hard you try, they just won't open. Test them. The harder you try, the faster they stick together, just as if they were glued together. That's fine!

Now you can open your eyes; that's good. When I tell you to and not before, open and close your eyes once more, and, when you close them this time, you will be ten times as relaxed as you are right now. Go ahead, open and close, and feel that surge of relaxation go through your whole body, from the top of your head to the tip of your toes. Very good!

Now once again, open and close your eyes, and this time, when you close them, you will double the relaxation that you currently have. Fine.

If you have followed my suggestions, right at this very moment, when I lift your hand and let it drop into your lap, it will drop like a wet cloth, heavy and limp. That's very, very good.

You now have good physical relaxation, but medical hypnosis consists of two phases: physical, which you currently have, and mental, which I will now show you how to achieve.

When I ask you to, and not before, I want you to start counting backward from 100. I know you can count; that is not what we're after. I just want you to relax mentally. As you say each number, pause momentarily until you feel a wave of relaxation cover your whole body, from the top of your head to the tip of your toes. When you feel this wave of relaxation, then say the next number, and each time you

say a number, you will double the relaxation you had before you said the number. If you do this properly, an interesting thing will happen—as you say the numbers and relax, the succeeding numbers will start to disappear and vanish from your mind. Command your mind to dispel these numbers. Now, aloud and slowly, start counting backward from 100.

Patient: One hundred.

Doctor: Very good.

Patient: Ninety-nine.

Doctor: Make them start to disappear now.

Patient: Ninety-eight.

Doctor: Now they're fading away, and after the next number, they'll all be gone. Make them disappear. Let the numbers go.

Patient: Ninety-seven.

Doctor: And now they're all gone. Are they gone? Fine. If there are any numbers still lurking in your mind, when I lift your hand and drop it, they will all disappear.

(Courtesy of William Holt, M.D.)

Indications

A patient's degree of hypnotizability and the technique of hypnosis are clinically useful in diagnosis and treatment, respectively. The existence of spontaneous, trance-like states in everyday life and the potential of individuals to uncritically accept emotions and information in these states make a person's degree of hypnotizability a factor in the way the world is viewed and processed.

Therapeutically, hypnosis's effectiveness in facilitating acceptance of new thoughts and feelings makes it useful in treating habitual problems and also with symptom management. Smoking, overeating, phobias, anxiety, conversion symptoms, and chronic pain are all indications for hypnosis. They can often be treated in a single session, in which a patient learns to perform self-hypnosis. Hypnosis can also aid in psychotherapy, notably for PTSD, and may also be useful for memory retrieval.

A 32-year-old man presented to the emergency department with a severe headache. He was a chronic migraine sufferer and was unable to control the pain on this occasion with his propranolol. The emergency department recognized that he had a high hypnotic capacity. The physician suggested imagery of an icepack resting on his forehead. Initially, they placed some real ice on his forehead to help. The patient was able to control his pain completely with this imagery. He did not require narcotics, as he had on previous visits. On follow-up several weeks later, the patient reported being able to use this strategy to control, as well as prevent, migraine attacks, and he no longer had to rely on frequent emergency department visits for pain relief. (Courtesy of A. D. Axelrad, M.D., D. Brown, Ph.D., and H. J. Wain, Ph.D.)

A 22-year-old male patient presented to the emergency room with bilateral blindness. An evaluation by ophthalmology revealed that the blindness was psychogenic. The emergency physician referred the patient to psychiatry. After an initial evaluation and the development of a therapeutic alliance, the psychiatrist used hypnosis to take the patient to a safe place and then back to the time immediately before the blindness. After two sessions, the patient described seeing his wife in an adulterous relationship. At that moment, the patient vocalized a desire to harm his wife and her suitor. Immediately after this vocalization, he became amnesic for the event and blind. On describing this under hypnosis, the therapist suggested that, when he became alert, “He would only remember what he felt comfortable remembering.” After the patient became alert, he had no idea what had occurred, and each day after initiation of the hypnotic intervention, the patient’s anger was reframed. When the patient felt comfortable, he then confronted his wife. The patient became aware that the amnesia was being used to prevent him from acting out. The use of a psychodynamic, cognitive reframing approach with a hypnotic milieu helped this patient to gain control and understanding of his symptoms. The therapist referred the patient and his wife for marital counseling. (Courtesy of A. D. Axelrad, M.D., D. Brown, Ph.D., and H. J. Wain, Ph.D.)

A 29-year-old woman presented for evaluation and treatment of ongoing facial pain that was not responding to traditional methods of intervention. The neurologic evaluation showed no objective physical correlations. Initially, hypnotic intervention controlled the pain, but it returned 24 hours later. Her self-hypnotic technique ceased to be effective. The therapist decided to explore more completely the meaning of the pain. Using age regression under hypnosis, the patient regressed to a time before the pain. She related that her brother had been injured by a car while he was running in the street. The patient was babysitting at the time, and her father was so angry that he hit her. Recently her friend's dog ran away, and she felt responsible. As she began to recognize her need to punish herself because of her guilt over what occurred, she was able to understand her feelings and reframe her thoughts more productively. An "affect bridge" was also used, and the patient went back to a previous time when she felt guilty and received punishment. She then described her feelings of being hit by her alcoholic, abusive father. She continued to gain insight and mastery over the past and was able to ablate her pain. (Courtesy of A. D. Axelrad, M.D., D. Brown, Ph.D., and H. J. Wain, Ph.D.)

A 42-year-old married mother of three children was kidnapped and locked in a large packing trunk. After she freed herself and broke out, her abductors stabbed her multiple times, tied her up, put her back in the trunk, and threw her down a cliff. She eventually managed to break out and crawl to safety. Eventually, a passerby called 911, and paramedics transported her to a hospital. She reported that others saw her lying on the road and appeared frightened to approach her.

Following medical stabilization, she was discharged from the hospital and found herself developing nightmares, reexperiencing avoidance, and having hyperarousal symptoms. Her internist referred her for treatment. The psychiatrist initiated sertraline. She was found to be a mid- to high-range hypnotic subject. She learned to go to a safe place and to use a

split-screen technique. The therapist also gave her permission to describe her nightmares, reexperiences, and overwhelming anxieties and fears that she faced while being captive, as well as her feelings of abandonment while lying on the road. The therapist reinforced her ingenuity in breaking out of the trunk. She responded to a reframing of the feeling of blame for her capture while under hypnosis. She learned to calm herself and to reframe her negative feelings about her helplessness. Hypnotic age regression helped her master her experiences and facilitate their becoming like a bad movie. Initially, her startle response was a signal for her to go to her comfort zone. Age progression helped her to rehearse the future. The treatment used the milieu of hypnosis along with exposure, cognitive reframing, psychodynamic approaches, and pharmacology. (Courtesy of A. D. Axelrad, M.D., D. Brown, Ph.D., and H. J. Wain, Ph.D.)

Contraindications

No intrinsic dangers to the hypnotic process exist. Because of the increased dependence that the hypnotized patient has toward the therapist, a strong transference may occur, however, in which the patient exhibits feelings for the therapist that are inappropriate in regard to their relationship. Strong attachments may occur, and these must be respected and properly interpreted. Negative emotions may also surface in the patient, especially in those who are emotionally fragile or who have poor reality testing. To minimize the likelihood of this negative transference, the therapist must use caution when choosing patients who have problems with basic trust, such as those who are paranoid or who require high levels of control. The hypnotized patient also has a reduced ability to evaluate hypnotic suggestions critically and, thus, the hypnotist must have a strong ethical value system. Controversy exists about whether patients can perform acts during a trance state that they would otherwise find repugnant or that run contrary to their moral system.

INTERPERSONAL THERAPY

Interpersonal psychotherapy (ITP) is a time-limited treatment for major depressive disorder developed by Gerald L. Klerman and Myrna Weissman

in the 1970s. The initial formulation of ITP was an attempt to represent the current practice of psychotherapy for depression. It assumes that the development and maintenance of some psychiatric illnesses occur in a social and interpersonal context and that interpersonal relations between the patient and significant others influence the onset, response to treatment, and outcomes of these illnesses. The overall goal of ITP is to reduce or eliminate psychiatric symptoms by improving the quality of the patient's current interpersonal relations and social functioning.

The typical course of ITP lasts 12 to 20 sessions over a 4- to 5-month period. ITP moves through three defined phases: (1) the initial phase identifies the problem area that will be the target for treatment; (2) the intermediate phase works on the target problem area(s); and (3) the termination phase focuses on consolidating gains made during treatment and preparing the patients for future work on their own (Table 23-13).

Techniques

Individual Interpersonal Psychotherapy

INITIAL PHASE. Sessions one through five typically constitute the initial phase of ITP. After assessing the patient's current psychiatric symptoms and obtaining a history of these symptoms, the therapist gives the patient a formal diagnosis. The therapist and patient then discuss the diagnosis, as well as what to expect from treatment. Assignment of the sick role during this phase serves the dual function of granting the patient both the permission to recover and the responsibility to recover. The therapist explains the rationale of ITP, underscoring that therapy will focus on identifying and altering dysfunctional interpersonal patterns related to psychiatric symptomatology. To determine the precise focus of treatment, the therapist conducts an interpersonal inventory with the patient and develops an interpersonal formulation based on this. In the interpersonal formulation, the therapist links the patient's psychiatric symptomatology to one of the four interpersonal problem areas—grief, interpersonal deficits, interpersonal role disputes, or role transitions. The patient's concurrence with the therapist's identification of the problem area and agreement to work on this area are essential before beginning the intermediate treatment phase.

Table 23-13
Phases of Time-Limited Interpersonal Psychotherapy

Initial Phase: Sessions 1–5

Give the syndrome a name; provide information about the prevalence and characteristics of the disorder

Describe the rationale and nature of the therapy

Conduct the interpersonal inventory to identify the current interpersonal problem area(s) associated with the onset or maintenance of the psychiatric symptoms

Review significant relationships, past and present

Identify interpersonal precipitants of episodes of psychiatric symptoms

Select and reach consensus about the interpersonal problem area(s) and treatment plan with patient

Intermediate Phase: Sessions 6–15

Implement strategies specific to the identified problem area(s)

Encourage and review work on goals specific to the problem area

Illuminate connections between symptoms and interpersonal events during the week

Work with the patient to identify and manage negative or painful affects associated with his or her interpersonal problem area

Termination Phase: Sessions 16–20

Discuss termination explicitly

Educate the patient about the end of treatment as a potential time of grieving; encourage the patient to identify associated emotions

Review progress to foster feelings of accomplishment and competence

Outline goals for remaining work; identify areas and warning signs of anticipated future difficulty

Formulate specific plans for continued work after termination of treatment

INTERMEDIATE PHASE. The intermediate phase—typically sessions 6 to 15—constitutes the "work" of the therapy. An essential task throughout the intermediate phase is to strengthen the connections the patient makes between the changes in his or her interpersonal life and the changes in psychiatric symptoms. During the intermediate phase, the therapist implements the treatment strategies specific to the identified problem area, as specified in Table 23-14.

TERMINATION PHASE. In the termination phase (usually, sessions 16 through 20), the therapist discusses termination explicitly with the patient and assists him or her in understanding that the end of treatment is a potential time of grief. During this phase, the therapist encourages patients to describe specific changes in their psychiatric symptoms, especially as they relate to improvements in the identified problem area(s). The therapist also assists the patient in evaluating and consolidating gains, detailing plans for maintaining improvements in the identified interpersonal problem area(s), and outlining remaining work for the patient to continue on his or her own. The therapist also encourages patients to identify early warning signs of symptom recurrence and to identify plans of action.



Table 23-14
Interpersonal Problem Areas: Description, Goals, and Strategies

Problem Area	Description	Goals	Strategies
Grief	Complicated bereavement after death of a loved one	Facilitate the mourning process Help the patient to establish interest in new activities and relationships to substitute for what has been	Reconstruct t patient's relationship with the deceased Explore associated feelings (negative and positive

		lost	Consider way of becomin reinvolved with others
Interpersonal deficits	A history of social impoverishment, inadequate or unsustaining interpersonal relationships	Reduce patient's social isolation Enhance quality of any existing relationships Encourage the formation of new relationships	Review past significant relationship including negative ar positive aspects Explore repetitive patterns in relationship Note problematic interperson patterns in the session and relate them to similar patterns in the patient's life
Interpersonal role disputes	Conflicts with a significant other —a partner, other family member, coworker, or close friend	Identify the nature of the dispute Explore options to resolve dispute Modify	Determine the stage of the dispute: renegotiatic (calm down participants to facilitate

		expectations and faulty communication to bring about a satisfactory resolution If modification is unworkable, encourage patient to reassess the expectations for the relationship and to generate options to either resolve it or dissolve it and mourn its loss	resolution); impasse (increase disharmony to reopen negotiation) dissolution (assist mourning and adaptation) Understand how nonreciproc role expectation relate to the dispute Identify available resources t bring about change in th relationship
Role transitions	Economic or family change—the beginning or end of a relationship or career, a move, promotion, retirement, graduation, diagnosis of a medical illness	Mourn and accept the loss of the old role Recognize the positive and negative aspects of the new role and assets and liabilities of the old role	Review positiv and negativ aspects of old and new roles Explore feelings about what lost Encourage developme

	Restore self-esteem by developing a sense of mastery regarding the demands of the new role	of social support system and new skills called for in new role

Reprinted with permission from Wilfley D, Stein R, Welch R. Interpersonal psychotherapy. In: Treasure J, van Furth E, Schmidt U, eds. *Handbook of Eating Disorders*. 2nd ed. Hoboken, NJ: Wiley; 2003:253.

Ms. G is a 51-year-old woman who presented for treatment of binge eating disorder. She is college-educated, has her own business, and is a divorced mother of one adult son in his early 20s. Before treatment, she had a body mass index (BMI) of 42, and her binge eating occurred approximately 10 to 15 days per month over the past 8 years. Along with her current diagnosis of binge eating disorder, Ms. G struggled with recurrent major depression.

During the initial phase, Ms. G and her therapist began to review her history and the interpersonal events that were associated with her binge eating. Ms. G shared that she began overeating and gaining weight at age 14. When she was 18 years of age, she moved to a foreign country with her parents. Soon after the move, Ms. G's father left her and her mother to return to the United States. Ms. G was enraged at her father for leaving them and still gets very tearful and angry when discussing the separation. She and her mother decided to stay abroad because she had started university, and her mother was working. Both had developed strong social ties and felt comfortable in their new home. During this time, Ms. G continued to gain weight and started dieting. Shortly after graduating from university, Ms. G met and married a foreign national and, at the age of 28, delivered their only son. Two years later, she and her husband went through a very bitter divorce. Although Ms. G described this as a terrible time in her life, she maintained close ties with her friends and her mother. During this time, she began to diet and reached her lowest adult weight.

At the age of 35, when her mother died of a heart condition, Ms. G had her first episode of major depression, which was treated and resolved with antidepressants and a brief course of psychotherapy. Although she had previous cycles of weight loss and weight regain, she did not evidence any sign of eating disturbance at this point. She continued to maintain close social ties and enjoyed her close relationship with her son. When Ms. G was in her early 40s, an economic downturn in her adopted country forced her to return to the United States. Having lost all of her savings, she struggled financially while she looked for work. During this time, she started binge eating and gaining weight. Within 1 year of this move, Ms. G's son decided to return to live with his father (who was very wealthy). Ms. G felt angry and betrayed. Yet, when her son would visit, she would assume a subservient role with him, because she was afraid of losing his affection. He, in turn, became quite demanding and critical of her. Before seeking treatment, her heightened feelings of isolation and loneliness were leading to increased binge eating, depression, and weight gain.

By session three of the initial phase, Ms. G's therapist began to consider which problem area would be the focus of the remainder of treatment. Ms. G had a history of important relationship losses and subsequent grief—the loss of her father, her husband, her mother, and, most recently, her son. However, none of these losses led to the development of binge eating problems (although her dieting clearly linked to her feelings of anger after the divorce from her husband, and her depression intimately related to her mother's death). Ms. G's anger at her son for returning to live with the enemy was clearly a role dispute, yet her binge eating had begun 2 years before his departure (although it clearly worsened after he left). Because neither of these problem areas directly linked to the onset of the eating disorder, Ms. G's therapist decided that the focus of treatment would be to assist her in managing her role transition. Her move back to the United States, with the subsequent loss of her support and friendship networks, was clearly associated with the onset and continued maintenance of her binge eating. During session four of the initial phase, Ms. G's therapist shared her formulation of the problem area with her: "From what you have described, your binge eating really began after you returned to the United States. After that transition,

you felt more isolated and alone than you have ever been. It seems that binge eating was a way for you to manage that transition and the subsequent feelings of isolation and loneliness. Your transition has also negatively impacted your relationship with your son. Even though you are a very social person and enjoy the company of others, you have yet to develop the kind of support that you had before you moved. Although you have struggled with some very significant issues throughout your life—your father leaving, the pain of the divorce, and the death of your mother—your friends and support systems sustained you. If we work together to help you find and develop more intimate and supportive relationships here, I believe you will be much less likely to turn to food and binge eating as a source of support or comfort."

Ms. G agreed with the formulation and worked with her therapist to establish some treatment goals to help her resolve the problem area. First, the therapist encouraged her to become more aware of her feelings (notably isolation and loneliness) when she was binge eating and of how binge eating seemed to be the way she managed those feelings. A second goal was for her to take steps to increase her social contacts and develop more friendships. The third goal, which was a secondary problem area, centered on helping Ms. G resolve the role dispute with her son. Specifically, the therapist developed a goal with her to help her establish a clearer parental role with her son.

During the intermediate phase, the therapist helped Ms. G to grieve the loss of her previous role and the extensive support that she once had. Ms. G and her therapist worked to identify several sources of support and friendships of which she had not been aware. Soon after, Ms. G reported significant progress in initiating and establishing relationships with others. This change appeared to help give her confidence in her new roles. In fact, she began to receive a few social invitations. She became more attuned to the ways that she would rely on food, especially when she felt lonely or felt that she was not receiving enough time from others. The connection between the lack of supportive contacts and binge eating was becoming very clear to her in these intermediate sessions. During this phase, the therapist also assisted her in setting appropriate limits in her relationship with her adult son and in recognizing his adult-like responses in return. By the termination phase, Ms. G reported that she no

longer felt so lonely and isolated and that her binge eating had all but disappeared. She remarked how the quality of her relationship with her son had changed dramatically. He was more supportive and respectful, visited more frequently, and stayed with her for more extended periods. In the final sessions, she talked about her need to let go of the past and move on with her life as it is now, assuming her new roles more fully. She worked closely with her therapist to develop a plan to maintain the gains that she made in treatment and used the final session to review the important work that she accomplished. (Courtesy of D. E. Wilfley, Ph.D. and R. W. Guynn, M.D.)

Interpersonal Psychotherapy Delivered in a Group Format. ITP delivered in a group format has many potential benefits in comparison with individual treatment. For example, a group format based on members' diagnostic similarity (e.g., depression, social phobia, eating disorders) can help alleviate patients' concerns that they are the only one with a particular psychiatric disorder, while offering a social environment for patients who are isolated, withdrawn, or disconnected from others. Given the number and different types of interpersonal interactions in a group setting, the interpersonal skills that develop may be more readily transferable to the patient's outside social life than are the relationship patterns that develop in a one-on-one setting. Moreover, a group modality has therapeutic features not present in individual psychotherapy (e.g., interpersonal learning). The group format also facilitates the identification of problems common to many patients and provides a cost-effective alternative to individual treatment. Table 23-15 links the phases of ITP to the stages of group development.

TIMELINE AND STRUCTURE OF TREATMENT. The typical course of group ITP lasts 20 sessions over 5 months. The recommended group size ranges from six to nine members, with one or two group leaders, depending on resources and training needs. Three individual meetings (pregroup, midgroup, and postgroup), sequenced to correspond with critical time points in the three phases of ITP, in combination with other techniques, maintain the exclusive and strategic focus on individual patients' interpersonal problem areas—the hallmark of ITP.

Table 23-15
Linking the Phases of Interpersonal Psychotherapy to the Stages of Group Development

Phases/Tasks	Group Stages	Members' Work	Therapist Interventions
Initial: Sessions 1–5; identify problem areas	Engagement: Sessions 1–2	Members look for structure as they grapple with the anxiety of being in a group and sharing their problems	Establish a structure that encourages appropriate self-disclosure; facilitate norms for effective communication
	Differentiations: Sessions 3–5	Members work to manage negative feelings over interpersonal differences as they emerge in the group	Help members to understand their reactions in the context of interpersonal differences in their outside social lives
Middle: Sessions 6–15; work on goals	Work: Sessions 6–15	Members work out differences and strive toward common goals	Facilitate connections among members as they share their work with each other; encourage practice of newly acquired

			interpersonal skills in and outside of the group
Final: Sessions 15–20; consolidate treatment	Terminations: Sessions 16–20	Members struggle with how to manage the impending loss of connection with other group members	Help members to consolidate their work and to plan continued work; assist members in grieving the loss of the group

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PREGROUP MEETING. The pretreatment meeting is crucial for facilitating a patient's individualized work in the first phase of group ITP. The focus of the 2-hour pretreatment meeting is to identify interpersonal problem areas, establish an explicit treatment contract to work on problem areas, and prepare patients for group treatment. After identifying a patient's interpersonal problem(s) (i.e., interpersonal deficits, role disputes, role transitions, or grief), the therapist works collaboratively with the patient to formulate concrete prescriptions for change, in addition to the specific steps the patient will take to improve social relationships and patterns of relating. The therapist expresses these goals of treatment in language that is as specific and personally meaningful to the patient as possible. Before the start of the group, each group member receives a written summary of his or her goals and instructions that these goals will guide his or her work in the group.

Another important element of the pregroup meeting involves adequately preparing patients for group treatment. That is, the therapist encourages patients to think of the group as an "interpersonal laboratory" in which they can experiment with new approaches to handle challenging interpersonal situations. In this regard, patients hear about the important interpersonal skills

that they will learn while participating in a group (e.g., interpersonal confrontation, honest communication, expression of feelings) and receive encouragement to learn from others as they see changes occur. The therapist stresses to patients the importance of keeping their work in the group focused on changing their current interpersonal situations or intensifying important existing relationships and not using the group as a substitute social network.

INITIAL PHASE. The first five sessions of the group treatment comprise the initial phase in group ITP. During this phase, the therapist works to cultivate positive group norms and group cohesion, while emphasizing the commonality of symptoms among members and how the group context will address these. During this phase, group members review their goals with the group and begin to make some initial changes in their respective interpersonal problem areas. As members begin to experiment with the changes outlined in their goals, the therapist works collaboratively with each group member to refine and make any alterations in the target areas before the beginning of the intermediate phase.

INTERMEDIATE PHASE. During the intermediate "work" phase of group ITP (sessions 6 through 15), the therapist works to facilitate connections among members as they share the work on their goals with one another. In contrast to other interactive group approaches, the group interpersonal psychotherapist is much less likely to focus on intragroup processes and relationships unless they are specific to the work on a member's interpersonal problem area (e.g., interpersonal deficits). The therapist, however, consistently and continuously encourages group members to practice newly acquired interpersonal skills both inside and, most importantly, outside the group. As is the case with individual ITP, an essential task throughout the intermediate phase is to strengthen the connections the group members make between difficulties in their interpersonal lives and their psychiatric problems.

MIDTREATMENT MEETING. The midtreatment meeting is held midway (usually between sessions 10 and 11) through the intermediate phase. This meeting provides an opportunity to conduct a detailed review of each group member's progress on his or her individual problems and to refine interpersonal goals. The therapist recontracts with group members during this meeting to outline

and emphasize the work that remains, both inside and outside of the group, before treatment concludes.

TERMINATION PHASE. In the termination phase (sessions 16 through 20), the therapist discusses termination explicitly with the group members and begins to help them recognize that the end of treatment is a time of possible grief and loss. The therapist helps members recognize their own progress and the progress made by other group members. During this phase, group members describe the specific changes in their psychiatric symptoms, especially as they relate to improvements in the identified problem area(s) and relationships. Although it is common for group members to want to keep meeting on their own or to have frequent reunions, the therapist encourages group members to use this phase of the group to formally say goodbye to one another and the therapist. The therapist also uses this time to encourage members to detail their plans for maintaining improvements in their identified interpersonal problem area(s) and to outline their remaining work.

POSTTREATMENT MEETING. The posttreatment meeting occurs within 1 week after the final group session. The therapist uses this final individual meeting to develop an individualized plan for each group member's continued work on his or her interpersonal goals. The therapist reviews the group experience and the changes the patient has made in his or her interpersonal problem area and significant relationships.

NARRATIVE PSYCHOTHERAPY

More than anything else psychiatrists do, they listen to stories. These stories so saturate the clinical encounter that it would be impossible to imagine a clinical encounter without them. In the very first meeting between psychiatrist and patient, the psychiatrist begins with an open-ended invitation to a story: "*What brings you here?*" or "*What seems to be the problem?*" Patients respond to these questions by telling psychiatrists about their lives, their troubles, when the troubles began, what seems to have caused them, how they create difficulty, and what kinds of problem-solving they have tried. Such stories may be rudimentary, they may be only partially worked out, and they may even be baffled and confused. The patient may also be perplexed enough to answer, "I don't know why I came," or "I'm not really sure what's wrong; my family sent me." Nonetheless, the patient's response to the psychiatrist's initial questions always involves a story.

Narrative psychotherapy emerges out of this increased interest in clinical stories. The two main tributaries that lead to narrative psychotherapy come from the two different sides of psychiatry: narrative medicine and narrative psychotherapy. Narrative psychiatrists are psychiatrists who combine the wisdom of these two domains. Following the lead of narrative medicine, narrative psychiatrists recognize that psychiatric patients, like medical patients, come to clinics with intense stories to tell. Contemporary narrative medicine has developed from 30 years of work in bioethics and medical humanities devoted to humanizing the clinical encounter through a better understanding of patient stories. The term *narrative medicine* comes from Rita Charon, an internist and literary scholar, who used it to describe an approach to medicine that uses narrative approaches to augment scientific understandings of illness. Narrative medicine brings together insights from human-centered medical models, such as George Engel's biopsychosocial model and Eric Cassel's person-centered model, with research and insights from phenomenology, the humanities, and interpretive social sciences.

Narrative medicine uses these resources to understand the illness experience better, "to recognize, absorb, interpret, and be moved by the stories of illness." As Charon argued, when clinicians possess narrative competency, they can enter the clinical setting with a nuanced capacity for

“attentive listening…, adopting alien perspectives, following the narrative thread of the story of another, being curious about other people’s motives and experiences, and tolerating the uncertainty of stories.” She further argued that doctors “*need* rigorous and disciplined training” in narrative reading and writing not just for their own sake (helping them to deal with the strains and traumas of clinical work), but also “*for the sake of their practice.*” Without such narrative competency, clinicians lack the ability to understand their client’s experience of illness fully. For Charon and others in narrative medicine, narrative study is not a mere adornment to a doctor’s medical training; it is a crucial and basic science that physicians should master for medical practice.

A major task of narrative medicine, and therefore narrative psychotherapy, is to be a good listener and to connect empathically with the patient’s story. A narrative psychiatrist, like a narrative physician, seeks to understand the patient first and foremost. This understanding brings patient and clinician together into a shared experience of the patient’s world. This narrative understanding is much more than a causal explanation of problem A or problem B that the patient might have. It does not simply abstract from the person’s situation a categorical label that groups problems under a well-known abstract grid. Instead, narrative understanding tunes in to the uniqueness of the individual and the unrepeatability of their experience and difficulties. Narrative understanding, in short, is a deep appreciation of the person as a whole—what it feels like for this person, in this particular context, going through these particular problems.

In addition to following the lead of narrative medicine colleagues, narrative psychiatrists also follow the lead of contemporary colleagues in narrative psychotherapy. The history of narrative psychotherapy goes back to Sigmund Freud’s early work at the inception of psychoanalysis. At that time, Freud lamented how his case histories sounded more like narrative fiction than hard science.

Contemporary narrative psychotherapy’s motivation for returning to the role of narrative comes partly from the broader turn to narrative in humanities, psychology, and social science and partly from the history of psychotherapy since Freud. The past century of psychotherapy has been a century of strife, with one faction after another splitting off from psychoanalysis. Leading alternatives to psychoanalysis included behavioral,

humanistic, family, cognitive, feminist, and interpersonal, just to name some. These splits are characterized by further splits within splits, fragmenting the field of psychotherapy to the point of more than 400 approaches to psychotherapy. Narrative approaches emerge at this particular moment as part of an important trend away from further fragmentation and toward psychotherapy reintegration. Narrative approaches are invaluable for psychotherapy integration because they provide a metatheoretical orientation from which to understand and practice psychotherapy.

Metaphor

Metaphor performs this function by allowing us to understand and experience one thing in terms of something else. The metaphor selects, accentuates, and backgrounds aspects of two systems of ideas so that they come to be seen as similar: “Men are seen to be more like wolves after the wolf metaphor is used, and wolves seem to be more human.”

In narrative psychotherapy, understanding metaphor connects to broader work in continental linguistic philosophy, and that work, as a whole, attempts to shift standard ideas about truth and objectivity. It sidesteps the binary concepts of relativism (anything goes) and realism (there is only one correct or true way to describe the world). Rather than using the binary distinction between true and false, narrative psychotherapy instead uses a postmodern language of semiotic realism and pluridimensional consequences.

Plot

Plot works like metaphor in that it also orders experiences and provides form for narratives. Plot, or the process of emplotment, adds to metaphor two key dimensions: (1) it brings together what would otherwise be separate and heterogeneous elements, and (2) it organizes understanding and experience or time, or what could be called temporal perception.

The critical function of plot for narrative is that plot creates a narrative synthesis between multiple individual events and brings them together into a single story, making an intelligible connection between them. Remarkably, plot can create a synthesis between events and elements that are surprisingly incongruous or heterogeneous—events that do not seem to fit together.

Plot also configures these multiple elements into a temporal order. This

temporal order is of two sorts. First, each plot comprises a discrete series of incidents, of theoretically infinite *nows.* Second, each plot takes these infinite nows, proceeding one after another in succession, and organizes them into a humanly manageable experience.

Character

In narrative theory, the concept of character connects directly to contemporary controversy surrounding the related and, some may argue, more basic concept of identity. The controversy around identity may be understood as a tension between essentialist and nonessentialist approaches. Essentialist notions of identity tell us that each person has a fixed personality, perhaps biologically stamped, that authentically belongs to that person and is at the core of that person's being. This "true self" or "core self" may be distorted or covered over, but it is nonetheless there for the discovery if individuals apply themselves patiently and persistently to the task. Nonessentialist critiques, however, have deconstructed this ideal of identity and its notion of an integral, originary, and unified self. One of the most productive ways to navigate the tension between essentialist and nonessentialist understandings of identity is to draw a comparison between identity (in life) and character (in fiction). Rather than adopting a linear logic that understands identity as a more fundamental concept to character, this approach uses a circular logic to argue that people understand themselves in the same way they understand characters.

Narrative approaches to identity allow people to navigate the tension between essentialist and nonessentialist identities because narrative identity allows for a kind of continuity over time, a relative stability of self, without implying a substantial or essentialist core to this stability. People's interpretations of themselves use the cultural stories surrounding them to tell a story of self that escapes the two poles of random change and absolute identity. In this way, a narrative identity is also a cultural identification. A person's identification may seem original, but he or she narrates them with the resources of history, language, and culture.

Narrative Psychiatry

With this brief introduction to narrative medicine, narrative psychotherapy,

and narrative theory, it is possible further to draw out the meaning of narrative for psychiatry.

Fortunately, one of the most helpful aspects of narrative theory for psychiatry is that it provides an overarching, or metatheoretical, rationale for understanding how these many psychotherapies work. From a narrative perspective, all therapies involve a process of storytelling and story retelling. No matter which style of psychotherapy one uses, the process of therapy involves an initial presentation of problems that the client is unable to resolve. The client and therapist work together to bring additional perspectives to these problems, allowing the client to understand them in a new way. These additional perspectives vary greatly depending on the style of psychotherapy used. It matters, in other words, whether the therapy is psychodynamic, cognitive, humanistic, feminist, spiritual, or expressive. From the vantage point of narrative theory, however, what these different approaches all have in common is that they rework, or "reauthor," the patient's initial story into a new story. This new story allows new degrees of flexibility for understanding the past and provides new strategies for moving into the future.

Future Directions

Recent work in narrative medicine, narrative psychotherapy, and narrative theory has opened the door for the development of narrative psychiatry. This development provides a critical corrective to contemporary psychiatric practice that helps self-correct from an overreliance on biologic models without falling into a morass of antipsychiatry and without throwing away the bioscience baby with the bathwater. When psychiatrists take a narrative turn, they do not throw out their other skills and knowledge. The shift to narrative is, as much as anything else, an attitude shift and an opening out to additional sources of information. It starts by bringing to the foreground that the clinical encounter is a human encounter, and it follows by opening out to colleagues in the humanities, interpretive social sciences, and the arts to help to understand this human encounter better.

Most of all, narrative psychotherapy joins with other contemporary efforts in psychiatry—such as the recovery movement—to make clinical encounters much more client-focused and collaborative. Narrative psychotherapy, at its

core, recognizes that there are many ways to tell the story of one's life. The choice among these different options is a key way in which people create their identity. These choices should not be reduced to expert choices or scientific choices because they are always also personal and ethical choices. In the end, they are choices about what kind of life one wants to live.

Furthermore, clinicians must come to understand the value of biography, autobiography, and literature for developing a repertoire of narrative frames and options. In the end, narrative competency in psychiatry means a tremendous familiarity with the many possible stories of psychic pain and psychic difference. The more stories clinicians know, the more likely they are to help their clients to find a narrative frame that works for them.

For patients and potential service users, a narrative understanding means that there is a range of possible therapists and healing solutions that might be helpful. An approach that is right for one person may not be right for another. There must be a fit between the person and the approach, and people should feel empowered to take their intuitions and feelings seriously. If the person getting help does not feel this fit, he or she is likely right. There may well be another approach that would work better with the person's proclivities. Like everything else, however, judgment is critical. Therapeutic experiences of all kinds can be frustrating, slow, and uncertain. How, for example, does one know when an approach misses his or her needs and when it is something that will take time, patience, and perseverance to be helpful? From a narrative perspective, there can be no gold standard or simple answers. Only judgment, wisdom, and trial and error can decide.

COMBINED PSYCHOTHERAPY AND PHARMACOTHERAPY

The use of psychotropic drugs, in combination with psychotherapy, has become widespread. It has become the standard of care for many patients seen by psychiatrists. In this therapeutic approach, the use of pharmacologic agents augments psychotherapy. It should not be a system in which the therapist meets with the patient only on an occasional or irregular basis to monitor the effects of medication, to make notations on a rating scale to assess progress, or to assess side effects. Instead, it should be a system in which both therapies are integrated and synergistic. In many cases, studies

demonstrate that the results of combined therapy are superior to either type of therapy used alone. The term *pharmacotherapy-oriented psychotherapy* is used by some practitioners to refer to the combined approach. The methods of psychotherapy used can vary immensely, and all can be combined with pharmacotherapy when indicated.

Indications for Combined Therapy

A major indication for using medication when conducting psychotherapy, particularly for those patients with major mental disorders such as schizophrenia or bipolar disorder, is that psychotropics reduce anxiety and hostility. This improves the patient's capacity to communicate and to participate in the psychotherapeutic process. Another indication for combined therapy is to relieve distress when the signs and symptoms of the patient's disorder are so prominent that they require more rapid amelioration than psychotherapy alone may be able to offer. In addition, each technique may facilitate the other; psychotherapy may enable the patient to accept a much-needed pharmacologic agent, and the psychoactive drug may allow the patient to overcome resistance to entering or continuing psychotherapy (Table 23-16).

The reduction of symptoms, especially anxiety, does not decrease the patient's motivation for psychoanalysis or other insight-oriented psychotherapy. In practice, drug-induced symptom reduction improves communication and motivation. All therapies have a cognitive base, and anxiety generally interferes with the patient's ability to gain a cognitive understanding of the illness. Drugs that decrease anxiety facilitate cognitive understanding. They can improve attention, concentration, memory, and learning in patients who suffer from anxiety disorders.

Number of Treating Clinicians

Any number of clinicians can be involved in the treatment of a psychiatric disorder. In *one-person therapy*, the psychiatrist provides individual psychotherapy and medication treatment. Multiperson therapy is a form of treatment in which one clinician conducts psychotherapy while the other clinician prescribes medications. Other therapists may oversee marriage or family therapy or group therapy. The terms *cotherapy* or *triangular therapy*

also describe permutations of multiperson therapy.



Table 23-16
Benefits of Combined Therapy

Improved medication adherence
Better monitoring of clinical status
Decreased number and length of hospitalizations
Decreased risk of relapse
Improved social and occupational functioning

Communication among Therapists

Whenever more than one clinician is involved in treatment, there should be regular exchanges of information. Some patients split the transference between the two; the patient may view one therapist as giving and nurturing and may view the other as withholding and aloof. Similarly, countertransference issues, such as one therapist's identifying with the patient's idealized or devalued image of the other therapist, can interfere with therapy. Therapists must address those issues with the patient, and the cotherapists must be compatible and respectful of each other's orientation so that the therapy program can succeed.

A therapist may have some concerns about the quality of the psychopharmacology or a need to reevaluate the existing regimen. For example, a patient may not be doing well on medication, experiencing significant side effects, or showing a lack of sufficient improvement. Some patients may also be taking many different medications. When and if it is deemed in the patient's interest to question the medication regimen or the prescriber's skill, these misgivings should not be shared with the patient without first conferring with the prescribing physician.

If the therapist or pharmacologist, after a good-faith effort to understand the methods and course of treatment, still has misgivings about treatment, he or she should inform his or her counterpart that a second opinion would be useful. The therapist can then suggest to the patient that he or she consider a second opinion without raising undue alarm. Communication between treating clinicians should take place as frequently as needed. No standard

exists for how frequent that should be.

Orientations of Treating Clinicians

The orientation of the treating psychiatrist or other clinician can influence the therapeutic process during combination treatment. Clinicians invariably bring a theoretical bias to the treatment setting. Some, for example, are oriented, by preference and training, to practice a specific form of psychotherapy, such as psychoanalysis, CBT, or group therapy. These clinicians view psychotherapy as the primary treatment modality, with pharmacologic agents being an adjunct to treatment. Conversely, psychopharmacologically oriented psychiatrists view psychotherapy as augmenting the use of medication. Although disagreement may arise on which approach represents the most active ingredient in clinical response, the optimal use of both modalities should complement each other.

In addition to having extensive training in one or more psychoanalytic or psychotherapeutic techniques, the psychiatrist who practices pharmacotherapy-oriented psychotherapy must have a comprehensive knowledge of psychopharmacology. That knowledge must include a thorough understanding of the indications for the use of each drug, the contraindications, the pharmacokinetics and pharmacodynamics, the drug–drug interactions (with all pharmacologic agents, not only the psychoactive agents), and the adverse effects of medications. The psychiatrist must be able both to identify adverse effects and to treat them.

Nonpsychiatric physicians often use psychoactive agents inaccurately (too small or too large a dose for too short or too long a course), because they lack the requisite psychopharmacological knowledge, training, and experience. Psychotherapists who work with primary care physicians instead of psychiatrists should understand the limitations in depth of knowledge that these practitioners have and should seek a consultation with a psychiatrist if a patient is not responding to, or tolerating, medication. In some situations, the same clinician should deliver both psychotherapy and pharmacotherapy; however, this is often not possible for a variety of reasons, including therapist availability, time limitations, and economic restraints, among others.

Therapist Attitudes. Psychiatrists trained primarily as psychotherapists

may prescribe medication more reluctantly than those more oriented toward biologic psychiatry. Conversely, those who view medication as the preferred intervention for most psychiatric disorders may be reluctant to refer patients for psychotherapy. Therapists who are pessimistic about the value of psychotherapy or who misjudge the patient's motivation may prescribe medications because of their own beliefs; others may withhold medication if they overvalue psychotherapy or undervalue pharmacologic treatments. When a patient is in psychotherapy with someone other than the clinician prescribing medication, it is essential to recognize treatment bias and to avoid contentious turf battles that put the patient in the middle of such conflict.

Linkage Phenomenon. At some point, patients may view the improvement achieved in therapy as the result of a conscious or unconscious linkage between the psychopharmacological agent and the therapist. In fact, after being weaned from medication, patients often carry a pill with them for reassurance. In that sense, the pill acts as a transitional object between the patient and the therapist. Some patients with anxiety disorders, for example, may carry a single benzodiazepine tablet, which they take when they think they are about to have an anxiety attack. Then, the patient may report that the medication aborted the attack—before the medication could even have entered the bloodstream. In other cases, the patient never takes the pill because the patient knows that the pill is available and gains reassurance from that fact. The linkage phenomenon is usually not seen unless the patient is in a positive transference to the therapist. Indeed, the therapist may use this phenomenon to his or her advantage by suggesting that the patient carry medication to use as needed. Eventually, the behavior has to be analyzed, and patients frequently attribute magical properties to the therapist and then transfer those properties to the medication. Some clinicians believe this effect results from conditioning. After repeated trials, the sight of the medicine can decrease anxiety. The positive transference may also cause *transference cure* or *flight into health*, in which the patient feels better in an unconscious attempt to meet the presumed expectations of the prescribing physician. Therapists should consider this phenomenon if the patient reports rapid improvement well before a particular medication may reach its therapeutic level.

Rachel, a 25-year-old woman, presented with depressive symptoms and abdominal pain. After an extensive psychiatric and medical evaluation, she received a diagnosis of major depression of moderate severity and irritable bowel disorder. She began a course of CBT targeting her negative attributional style and low self-esteem, and she learned relaxation and distraction techniques for her pain. After a 12-week trial, she experienced only partial remission of her symptoms, and the psychiatrist offered her an antidepressant, citalopram at 20 mg per day. Her depressive symptoms remitted within 1 month, and she was able to function better at work but socially remained hesitant to engage with her peers. Her abdominal pain persisted, and she began to exhibit a pattern of disordered eating, severely restricting her intake to 500 calories per day due to the "pain." She experienced a 15-lb weight loss over the next several months. An intensive behavioral plan to target eating was begun, as well as continued probing of her negative cognitions relating to eating, pain, and newly emerging concerns that she would regain the weight too quickly and would become "fat." She did not meet the criteria for anorexia nervosa, although her cognitive distortions about her body image were significant. These new concerns resulted in a relapse of her depressive symptoms, including suicidal ideation, and the psychiatrist increased her citalopram to 40 mg/day. She reported severe akathisia on this dose and refused to take any more medication, including an antidepressant of another class. Rachel did agree to intensify her therapy to twice weekly, and this allowed her to explore some of the conflicts, feelings, and thoughts that fostered her treatment-refractory illness. The psychiatrist used a combination of psychotherapy and hypnosis for this work. Over the next 6 months, Rachel revealed a history of sexual abuse as a child, and this made her feel that she did not "deserve" to live or to eat and that the pain served to "punish" her for being bad. She also admitted that she resisted the medication "psychologically" because she felt that she did not deserve to get well. Her newly found insight, as well as the coping skills she developed in therapy, resulted in a reduction of her depressive symptoms, marked improvement in her eating habits with normalization of her weight, and decreased abdominal pain. She maintained these gains over the next year, including the normalization of

her daily functioning, a promotion at work, and the ability to tolerate the intimacy of a boyfriend. (Courtesy of E. M. Szigethy, M.D., Ph.D. and E. S. Friedman, M.D.)

Adherence and Patient Education

Adherence. Adherence is the degree to which a patient carries out the recommendations of the treating physician. A positive doctor–patient relationship fosters adherence, and the patient's refusal to take medication may provide insight into a negative transferential situation. In some cases, the patient acts out hostilities by nonadherence, rather than by becoming aware of, and ventilating, such negative feelings toward the doctor. Medication nonadherence may provide the psychiatrist with the first clue that a negative transference is present in an otherwise adherent patient who had appeared to be agreeable and cooperative.

Education. Patients should know the target signs and symptoms that the drug is supposed to reduce, the length of time they will be taking the drug, the expected and unexpected adverse effects, and the treatment plan to be followed if the current drug is unsuccessful. Although some psychiatric disorders interfere with patients' abilities to comprehend that information, the psychiatrist should relay as much of the information as possible. The clear presentation of such material is often less frightening than are patients' fantasies about drug treatment. The psychiatrist should tell patients when they may expect to begin receiving benefits from the drug. That information is most critical when the patient has a mood disorder and may not observe any therapeutic effects for 3 to 4 weeks.

Some patients' ambivalent attitudes toward drugs often reflect the confusion about drug treatment that exists in the field of psychiatry. Patients often believe that taking a psychotherapeutic medication means they are not in control of their lives, or they may become addicted to the drug and have to take it forever. Psychiatrists should explain the difference between drugs of abuse that affect the normal brain and psychiatric medications that treat emotional disorders. They should also point out to patients that antipsychotics, antidepressants, and antimanic drugs are not addictive in the

way in which, for example, heroin is addictive. The psychiatrist's straightforward and honest explanation of how long the patient should take the drug helps the patient adjust to the idea of chronic maintenance medication if that is the treatment plan. In some cases, the psychiatrist may appropriately give the patient increasing responsibility for adjusting the medications as the treatment progresses. Doing so often helps the patient feel less controlled by the drug and supports a collaborative role with the therapist.

Attribution Theory

Attribution theory is concerned with how persons perceive the causes of behavior. According to attribution theory, persons are likely to attribute changes in their own behavior to external events but are likely to attribute another's behavior to internal dispositions, such as that person's personality traits. Research on drug effects by attribution theorists has shown that, when patients take medication and their behaviors change, they attribute it to the drug and not to any changes that occur within themselves. Accordingly, it may be unwise to describe a drug as extremely strong or effective, because if it does have the desired effect, the patient may believe that is the only reason he or she got better. If the drug does not work, the patient may assume his or her condition is incurable. Therapists do best by presenting the use of drugs and psychotherapy as complementary or adjunctive, as neither standing alone and both being needed for improvements or cure to occur.

Mental Disorders

Depressive Disorders. Some patients and clinicians fear that medication covers over the depression and impedes psychotherapy. Instead, clinicians should view medication as a facilitator in overcoming the anergia that can inhibit the communication process between doctor and patient. The psychiatrist should explain to the patient that depression interferes with interpersonal activity in a variety of ways. For instance, depression produces withdrawal and irritability, which alienate significant others who may otherwise gratify the strong dependency needs that make up much of depressive psychodynamics.

When the patient is no longer taking medication, the psychiatrist should be alert for signs and symptoms of a recurrent major depressive episode. The psychiatrist may need to reinstitute medication. Before doing so, however, carefully review any stress, especially rejections, that could have precipitated recurrent major depressive disorder. A new episode of depression may occur because the patient is in a stage of negative transference, and the psychiatrist must try to elicit negative feelings. In many cases, the ventilation of angry feelings toward the therapist without an angry response can serve as a corrective emotional experience and can forestall a major depressive episode necessitating medication. Depressed patients are generally maintained on their medication for 6 months or longer after clinical improvement. The cessation of pharmacotherapy before that time is likely to result in a relapse.

Combined treatment is superior to either therapy used alone in the treatment of major depression. It is associated with improved social and occupational functioning and enhanced quality of life compared with either therapy alone.

Bipolar I Disorder. Patients taking lithium or other treatments for bipolar I disorder are usually medicated for an indefinite period to prevent episodes of mania or depression. Most psychotherapists insist that patients with bipolar I disorder take medication before starting any insight-oriented therapy. Without such premedication, most patients with bipolar I disorder are unable to make the necessary therapeutic alliance. When those patients are depressed, their abulia seriously disrupts their flow of thoughts, and the sessions are less productive. When they are manic, their flow of associations can be rapid, and their pressured speech can flood the therapist with material and prevent him or her from making appropriate interpretations or assimilating the material into the patient's disrupted cognitive framework.

Combined therapy for bipolar disorder increases adherence, decreases relapse, and reduces the need for hospitalization.

Substance Use Disorders. Patients with substance use disorders present the most difficult challenge in combined therapy. They are often impulsive, and, although they may promise not to use a substance, they may do so repeatedly. In addition, they frequently withhold information from the psychiatrist about episodes of substance use. For that reason, some

psychiatrists do not prescribe any medication to such patients, especially not those substances with high abuse potential, such as benzodiazepines, barbiturates, and amphetamines. Drugs with no abuse potential, such as antidepressants, have an important role in treating the anxiety or depression that frequently accompanies substance-related disorders. The psychiatrist conducting psychotherapy with such patients should have no reservations about sending the patient to a laboratory for random urine toxicologic tests.

Anxiety, Obsessive-Compulsive, and Trauma-Related Disorders. These disorders encompass OCD, PTSD, generalized anxiety disorder, phobic disorders, and panic disorder with or without agoraphobia. Many drugs are effective in managing distressing signs and symptoms. As the medication controls patients' symptoms, patients are reassured and develop confidence that the disorder will not incapacitate them. That effect is particularly strong in panic disorder, which is often associated with anticipatory anxiety about the attack. Depression can also complicate the symptom picture in patients with anxiety disorders and requires pharmacologic and psychotherapeutic intervention. Studies have shown that patients with anxiety disorders who receive ongoing psychotherapy are less likely to experience relapse compared with patients who receive medication alone.

Schizophrenia and Other Psychotic Disorders. Included in the group of schizophrenia and other psychotic disorders are schizophrenia, delusional disorder, schizoaffective disorder, schizophreniform disorder, and brief psychotic disorder. Drug treatment for those disorders is always indicated, and hospitalization is often necessary for diagnostic purposes, to stabilize medication, to prevent danger to self or others, and to establish a psychosocial treatment program that may include individual psychotherapy. In attempting individual psychotherapy, the therapist must establish a treatment relationship and a therapeutic alliance with the patient. The patient with schizophrenia defends against closeness and trust and often becomes suspicious, anxious, hostile, or regressed in therapy. Before the advent of psychotropics, many psychiatrists were fearful for their safety when working with such patients. Indeed, many assaults occurred.

Individual psychotherapy for schizophrenia is labor-intensive, expensive,

and not often attempted. The recognition that combined psychotherapy and pharmacotherapy have a greater chance of success than either type of therapy alone may reverse that situation. The psychiatrist who conducts such combined therapy must be especially empathic and must be able to tolerate the bizarre manifestations of the illness. The patient with schizophrenia is exquisitely sensitive to rejection, and the therapist should never start individual psychotherapy unless he or she is willing to make a total commitment to the process.

Other Issues

Evidence suggests that therapy can induce physical changes in the nervous system. Eric Kandel has provided elegant proof, winning the Nobel Prize for demonstrating that environmental stimuli produce lasting changes in the synaptic architecture of living organisms. Imaging studies have shown that patients who exhibit clinical improvement from psychotherapy show changes in brain metabolism that are similar to those seen in patients successfully treated with medications.

Still, some patients do well on only one form of treatment. Even with identical diagnoses, not all patients respond to the same treatment regimens. Success may be as dependent on the knowledge and quality of the clinician as on the potential benefit of a particular drug.

A real dilemma when combining treatment is the additional direct costs of two treatments. Although successful treatment results in reduced costs to society, the cost of treatment is usually narrowly defined by the patient as out-of-pocket expenses and by insurance and managed care companies as payments to the physician or hospital. Restrictions placed on the frequency and cost of visits to mental health professionals by managed care organizations, however, encourage the use of medication rather than psychotherapy.

MENTALIZATION-BASED TREATMENT

Mentalizing is the social cognitive ability to understand actions by other people and oneself in terms of mental states, including thoughts, feelings, wishes, and desires; it is a very human capability that underpins everyday interactions. In nontechnical language, it is attentiveness to thinking and

feeling in oneself and others. It is beyond question that mental states influence behavior. Beliefs, wishes, feelings, and thoughts, whether within or outside people's awareness, always influence what people do. Mentalizing involves a whole spectrum of capacities: critically, this includes the ability to experience one's *own* behavior as coherently organized by mental states, and to differentiate oneself psychologically from others. Individuals with a personality disorder, who lose cognitive and emotional coherence, particularly at moments of interpersonal (relational) stress, have reduced mentalizing capacities. Mentalization-based treatment (MBT) is a psychotherapy developed for BPD that focuses specifically on the mentalizing vulnerabilities of the patient in the context of an understanding of the attachment process, which is the developmental context in which individuals initially acquire mentalizing.

Techniques

MBT is a group and individual treatment. It is anticipated that, at times, the patient will experience strong affect while focusing on identified problems in treatment sessions, and their mentalizing will be limited or failing, and/or the patient's understanding of the way mental states link to behavior is inadequate.

The clinician addresses this by a structured process (the sessional intervention trajectory) of: (1) empathy and validation about problem areas; (2) clarification, exploration, and where necessary, challenge; (3) following a structured process to gently expand mentalizing and encourage the patient to identify the mental states previously outside their awareness.

The process is primarily in the here and now of the session, but increasingly, as the patient's mentalizing improves, it comes to concern core attachment relationships, including how these are activated with the clinician and key figures in the patient's life and how they influence mentalizing itself. Gradually, improvements in mentalizing serve to enable the patient to address their distorted representations of personal and social relationships.

The assessment and introductory process in MBT facilitates the alliance between patient and clinician (see Table 23-17) and introduces the patient to the treatment frame. An MBT-Introductory group of 10 to 12 sessions assists in the development of the formulation and facilitates the alliance. This

psychoeducational intervention covers all areas of mentalizing, attachment processes, personality disorder, emotion management, and the treatment itself. This preparatory work aims to ensure patients know what they are facing in trying to address their problems and are fully aware of the method and focus of treatment.

MBT is collaborative. Nothing can occur without joint discussion, taking into account the mental experiences and ideas of both patient and clinician. The process of mentalizing requires an authentic desire to understand the mental processes of oneself and others. This applies as much to the clinician as to the patient. The therapeutic process has to become a shared endeavor aimed at extending the influence of explicit, reflective, cognitive, internally focused mentalizing. Initial goals, on the road to improved mentalizing, are jointly developed and given focus. The goals cannot solely be those of the patient, although the patient's aims take priority unless they are antithetical to the whole process of treatment. The sharing of responsibility for the therapeutic process is at the core of the effectiveness of the treatment approach in the pursuit of improved mental state understanding.

Assessment involves delineation of the patient's mentalizing vulnerabilities and mentalizing profile, identification of nonmentalizing cycles, and a shared formulation, which includes specific detail of attachment patterns and areas of vulnerability to emotional dysregulation. This has to be understood by the patient and is for both patient and clinician. The formulation identifies common relational fears, for example, abandonment, which stimulate the patient's attachment system and result in the use of maladaptive attachment strategies in interpersonal interactions. In brief, the pattern of the patient's relationships informs an understanding of the relationship in treatment, and the relationship in treatment is used to reappraise the relationships in life outside treatment. Finally, it is important that the patient and clinician consider establishing a goal of improving social function. This goal will include work, social activity, voluntary work, education, and other constructive life-affirming activity.



Table 23-17
Alliance Building in MBT

Identification of patient's mentalizing vulnerabilities in an

understandable form
Formulation of problems—agreed between patient and clinician
Identification of patient's risk profile and crisis management strategies
Agreement of short- and long-term goals

Indications

MBT is effective in treatment for severe BPD. Patients in the early studies of the intervention had made serious suicide attempts, been admitted to a psychiatric hospital for risk, and/or had acted on thoughts of self-harm. Trials involved both men and women, and patients showed high levels of comorbidity. Analysis of the data suggested that patients with comorbidity for a number of personality disorders, including antisocial personality disorder, did preferentially better in MBT than comparison treatment. At a clinical level, patients with marked interpersonal problems who have a personality disorder rooted in mentalizing vulnerability and attachment problems may benefit from MBT.

Contraindications

MBT is a generic treatment constructed to optimize access both by creating a low-demand treatment protocol and by facilitating access to training by a range of professionals. Individuals who have problems with mentalizing rooted in nonattachment contexts—for example, those with psychosis—may not benefit from MBT; if they do, the mechanism of change is likely to be different from that in BPD. Individuals with relatively simple problems, such as phobias or uncomplicated depression, may do better with more direct approaches, such as CBT. Even within a population of patients with BPD, patients with more complex presentations (more than one personality disorder diagnosis) are more likely to require MBT than those with a single BPD diagnosis, who may do as well in structured clinical management.

PSYCHOEDUCATIONAL INTERVENTIONS

Psychoeducation is the process of helping individuals to understand a psychiatric illness affecting them or their loved ones. Through this process,

patients and/or family members learn about common signs and symptoms of the disease, treatment options, and prognosis for recovery. Psychoeducation moves beyond merely providing information and includes teaching individuals problem-solving skills, coping strategies, support resources, and tools to improve communication. Essentially, psychoeducational interventions provide strategies to support patients and families as they adjust to living with and managing a psychiatric illness.

Psychoeducation can be delivered individually or in a group setting. Interventions can be brief and incorporated into other treatment modalities (e.g., medication management, inpatient hospitalization, outpatient psychotherapy) or they can be structured and take place across several weeks or months. Common areas addressed by psychoeducation include treatment adherence, identifying signs of relapse, lifestyle factors relevant to the illness, and management of crises.

Individuals and families dealing with serious mental illnesses, such as schizophrenia and bipolar disorder, often benefit from psychoeducation. However, psychoeducation may be useful for a variety of disorders, including, for example, major depression, anxiety, autism, dementia, personality disorders, eating disorders, and substance use disorders.

Effectiveness of Psychoeducational Interventions

Psychoeducational interventions have been shown to improve outcomes for patients and families, particularly for those dealing with serious or chronic mental illness. A Cochrane review demonstrated that individuals with schizophrenia who participated in psychoeducation had improved medication adherence and lower relapse rates as compared to those without psychoeducational intervention. Individuals with bipolar disorder who participate in psychoeducation are also reported to have improved treatment adherence and reduced relapse, particularly reduced manic/hypomanic relapse. In addition, psychoeducational interventions for depression have been associated with better illness prognosis and reduced psychosocial burden for family members.

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24

Psychiatric Rehabilitation and Other Interventions

PSYCHIATRIC REHABILITATION

Psychiatric rehabilitation denotes a wide range of interventions designed to help people with disabilities caused by mental illness improve their functioning and quality of life by enabling them to acquire the skills and supports needed to be successful in usual adult roles and the environments of their choice. Normative adult roles include living independently, attending school, working in competitive jobs, relating to family, having friends, and having intimate relationships. Psychiatric rehabilitation emphasizes independence rather than reliance on professionals, community integration rather than isolation in segregated settings for persons with disabilities, and patient preferences rather than professional goals.

Vocational Rehabilitation

In the late 1980s, the emphasis of vocational rehabilitation began to shift to place-and-train models, termed supported employment, which were borrowed from the field of developmental disabilities. According to the federal definition, supported employment involves "competitive work in integrated settings…consistent with the strengths, resources, priorities, concerns, abilities, capabilities, interests, and informed choice of the individuals, for individuals with the most significant disabilities for whom competitive employment has not traditionally occurred; or for whom competitive employment has been interrupted or intermittent as a result of a significant disability." In contrast to other vocational approaches, supported employment

programs (a) did not screen people for work readiness but helped anyone who says they wanted to work; (b) did not provide intermediate work experiences, such as prevocational work units, transitional employment, or sheltered workshops; (c) actively facilitated job acquisition, often accompanying clients on interviews; and (d) provided ongoing support once the person is employed.

Antonio is a 45-year-old man who has been a client of a mental health agency for more than 10 years. He attended the rehabilitative day treatment program until it was converted to a supported employment program. His case manager encouraged him to think about the possibility of working part-time. Antonio told his case manager that he could not work because of his schizophrenia and because he was helping to raise his two kids and needed to be home at 3 PM when they returned from school every day. The case manager explained to Antonio that getting a job does not necessarily mean working 40 hours a week and that lots of people in the agency's supported employment program were working in part-time jobs, even jobs that only require a few hours a week.

Antonio agreed to meet one of the employment specialists to discuss the possibility of work. Over the next couple of weeks, the employment specialist met with Antonio several times, read his clinical record, and talked with his case manager and psychiatrist. The employment specialist learned that Antonio loved to drive his car. He also learned that Antonio had attendance problems in past jobs because he felt unappreciated. The employment specialist found Antonio to be a sociable and likable person.

Antonio told the employment specialist that he was willing to do any job. He did not have one specific job in mind. After discussing options with Antonio and with the team, the employment specialist suggested a job at Meals on Wheels as a driver for the lunch delivery. Antonio was hired and loved it right from the start. Absenteeism was never a problem because he liked driving around and knew that people were counting on him for their meals. The hours were perfect (10 AM to 2 PM), so Antonio could be at home when his kids returned from school. He became good friends with the other workers. He told his case manager that it was wonderful to be bringing home a paycheck again. Moreover, best of all,

he said, was that his kids saw him going to work just like their friends' dads. (Courtesy of Robert E. Drake, M.D., Ph.D. and Alan S. Bellack, Ph.D.)

Social Skills Rehabilitation

Social dysfunction is a defining characteristic of schizophrenia. People with the illness have difficulty fulfilling social roles, such as worker, spouse, and friend, and they have difficulty meeting their needs when social interaction is required (e.g., negotiating with merchants, requesting assistance to solve problems). Social dysfunction is semi-independent of symptomatology and plays an essential role in the course and outcome of the illness. As shown in Table 24-1, social competence involves three component skills: (1) social perception or receiving skills; (2) social cognition, or processing skills; and (3) behavioral response, or expressive skills. Social perception is the ability to read or decode social inputs accurately, including accurate detection of affect cues, such as facial expressions and nuances of voice, gesture, and body posture, as well as verbal content and contextual information. Social cognition involves an effective analysis of the social stimulus, integration of current information with historical information, and planning an adequate response. *Social problem-solving* is another term to describe this domain.

Methods. The primary modality of social skills training is the role-play of simulated conversations. The trainer first provides instructions on how to perform the skill and then models the behavior to demonstrate its performance. After identifying a relevant social situation in which the skill might be appropriate, the patient engages in role-playing with the trainer. The trainer next provides feedback and positive reinforcement, which are followed by suggestions for how the response can be improved. The sequence of role-playing followed by feedback and reinforcement repeats until the patient can perform the response adequately. Training typically occurs in small groups (six to eight patients), in which case patients each practice role-playing for three to four trials and provide feedback and reinforcement to one another. The therapist tailors the teaching to the individual—for example, a highly impaired group member might simply practice saying "no" to a simple request, whereas a less cognitively impaired

peer might learn to negotiate and compromise.



Table 24-1
Components of Social Skill

Expressive behaviors
Speech content
Paralinguistic features
Voice volume
Speech rate
Pitch
Intonation
Nonverbal behaviors
Eye contact (gaze)
Posture
Facial expression
Proxemics
Kinesics
Receptive skills (social perception)
Attention to and interpretation of relevant cues
Emotion recognition
Processing skills
Analysis of the situation demands
Incorporation of relevant contextual information
Social problem solving
Interactive behaviors
Response timing
Use of social reinforcers
Turn taking
Situational factors
Social “intelligence” (knowledge of social mores and the demands of the specific situation)

Richard was a single white man first diagnosed with schizophrenia at age 22 when he was in college. He was hospitalized briefly but was unable to return to school and moved back home with his parents. He attended a day treatment program intermittently over the next 6 years before he received a referral for help with getting a job and dating.

Richard had missed out on a critical period of adult development and had never learned dating skills or the social skills needed to get or maintain a job. He was appropriately groomed and did not present himself as a patient, but he seemed quite uncomfortable in social interactions. He scarcely made eye contact, staring at the floor when he spoke, and did not initiate conversation, responding to questions with brief answers.

Richard joined a social skills training group for 3 months with six other patients. The focus of the group was employment skills. Patients were taught critical social skills for getting and maintaining a job, such as how to participate in job interviews; how to approach a supervisor to understand how to do a job or for help with work-related problems; how and when to make requests or explain problems, such as getting to work late because of traffic or needing to leave early to go to a doctor's appointment; and socializing with coworkers. Simultaneously, Richard enrolled in a supported employment program and worked with a case manager to find a job as a computer support person. He found a 24-hour-per-week job at a small company and continued to attend the skills group, using the sessions to work on interpersonal issues at work, including engaging in casual conversation with coworkers and dealing with unreasonable requests from people.

When the vocational skills group ended, Richard joined a dating group with seven other male and female patients who had similar interests. This group focused on finding someone to date, dating etiquette, asking someone out (or being asked out), appropriate conversation for dates, sexual interactions, and safe sex practices. In addition to role play and discussion, the group shared ideas on how to meet people and what to do on dates.

Richard responded well to treatment. He had maintained the computer job at follow-up, 6 months after he concluded the dating skills group. His

case manager also reported that he had a girlfriend, a woman whom he had met at his church group. He had also expressed an interest in enrolling in college classes at night. He was still living at home with his parents, but, for the first time, was seriously considering what he would need to do to move out. (Courtesy of Robert E. Drake, M.D., Ph.D. and Alan S. Bellack, Ph.D.)

Goals. In a treatment setting, there are four major goals of social skills training: (1) improved social skills in specific situations, (2) moderate generalization of acquired skills to similar situations, (3) acquisition or relearning of social and conversational skills, and (4) decreased social anxiety. Learning, however, is tedious or almost nonexistent when patients are floridly ill with positive symptoms and high levels of distractibility.

Some findings limit the applicability of social skills training. It is more challenging to teach complex conversational skills than to teach briefer, more discrete verbal and nonverbal responses in social situations. Because complex behaviors are more critical for generating social support in the community, therapy includes methods to improve the learning and durability of conversational skills. These training methods, focusing on training in social skills and information-processing skills, are discussed below.

Training in Social Perception Skills. There is a need for strategies for training patients in affect and social cue recognition. Patients with chronic psychotic disorders, such as schizophrenia, often have difficulty perceiving and interpreting the subtle affective and cognitive cues that are critical elements of communication. Social perception abilities are considered the first step in effective interpersonal problem solving; difficulties in this area are likely to lead to a cascade of deficits in social behavior. Training skills in social perception address these deficits and help provide a foundation for developing more specific social and coping skills.

Despite attending several social gatherings, Matt felt apart from the rest of the group. He reported that these events seemed like “a jumble of sights and sounds.” His therapist, recognizing Matt’s difficulty with social perception, gave him a series of questions designed to help him

organize and give meaning to the social stimuli he encountered. For example, when Matt was confused about a conversation someone was having with him, he would ask himself, “What is this person’s short-term goal? At what level of disclosure should I be? Should I be talking now or listening?” Identifying the rules and goals of a particular social interaction provided a template for Matt to recognize and react to a greater variety of social cues, thus enhancing his behavioral repertoire. (Courtesy of Robert Paul Liberman, M.D., Alex Kopelowicz, M.D., and Thomas E. Smith, M.D.)

INFORMATION-PROCESSING MODEL OF TRAINING. Methods of training that follow a cognitive perspective teach patients to use a set of generative rules that can be adapted for use in various situations. For example, a six-step problem-solving strategy has developed as an outline for helping patients overcome interpersonal dilemmas: (1) adopt a problem-solving attitude, (2) identify the problem, (3) brainstorm alternative solutions, (4) evaluate solutions and pick one to implement, (5) plan the implementation and carry it out, and (6) evaluate the efficacy of the effort and, if ineffective, choose another alternative. Although the step-wise, structured, linear process of problem-solving occurs intuitively, without conscious awareness in normal persons, it can be a useful interpersonal crutch to help cognitively impaired mental patients cope with the information needed to fill their social and personal needs.

Milieu Therapy

The locus of milieu is a living, learning, or working environment. The defining characteristics of treatment are the use of a team to provide treatment and the time the patient spends in the environment. Recent adaptations of milieu therapy include 24-hour-a-day programs situated in community locales frequented by patients, which provide in vivo support, case management, and training in living skills.

Most milieu therapy programs emphasize group and social interaction; peer pressure mediates rules and expectations for normalization of adaptation. When clinical staff views patients as responsible human beings, the patient role becomes blurred. Milieu therapy stresses a patient’s rights to goals and to

have freedom of movement and informal relationship with staff; it also emphasizes interdisciplinary participation and goal-oriented, clear communication.

Token Economy. The use of tokens, points, or credits as secondary or generalized reinforcers normalizes a mental hospital or day hospital environment with a program mimicking society's use of money to meet instrumental needs. Token economies establish the rules and culture of a hospital inpatient unit or partial hospitalization program, offering coherence and consistency to the interdisciplinary team as it struggles to promote therapeutic progress with challenging patients. These programs are challenging to establish, however, and their widespread dissemination has suffered because of the organizational prerequisites and the additional resources and rewards needed to create a truly positively reinforcing environment.

Cognitive Rehabilitation

Increased recognition of the prevalence and importance of neurocognitive deficits over the past decade has stimulated a growing interest in remediation strategies. Much of the work in this area has focused on psychopharmacological approaches, especially on the new-generation antipsychotics. New-generation medications appear to have a positive effect on neurocognitive test performance. However, the effect size for any of the medications is small to medium, and little evidence indicates that these medications have a clinically meaningful impact on neurocognitive functioning in the community. As a result, a parallel interest has arisen in the potential for *rehabilitation* or *cognitive remediation.* This body of work is distinct from cognitive-behavioral therapy and cognitive therapy, which focus on reducing psychotic symptoms.

Ethical Issues

The ethics of conducting rehabilitation strategies are generally the same as for conducting other psychotherapies. Two issues come up regularly, however: avoiding infantilization and maintaining confidentiality. The first issue concerns the risk of viewing the patient as unable to make adult

choices, such as whether to participate in rehabilitation, where to live, whether or not to work, and whether or not to use drugs and alcohol. Although it may be more of a value than an ethical standard, psychiatric rehabilitation assumes that the practitioner and the patient are in a partnership to facilitate recovery and improve quality of life. The basic model involves collaboration and shared decision-making and does not portray the practitioner as an authority or parental figure. When patients make what appear to be bad choices, the practitioner must consider the patient's right to choose and whether the choice is dangerous versus merely not the choice the practitioner would make. If the choice is potentially harmful, a collaborative process of considering alternatives is more likely to produce good quality choices than an authoritative, admonitory approach.

Failure to consider the patient as a partner also leads to violations of confidentiality. Practitioners sometimes assume that they are the primary arbiters of what information to share with parents, other clinicians, and other agencies. In fact, in most circumstances that do not involve the safety of patients or others, the patient should be the arbiter of what information to share with whom. For example, in supported employment, the patients always determine whether to disclose information about their illnesses to employers.

GENETIC COUNSELING

Medical geneticists and specially trained and qualified genetic counselors have traditionally provided genetic counseling to patients in need of such help. Many psychiatrists, however, are also well placed to provide genetic education and counseling because they often have knowledge of their patient's needs and family histories and have ongoing therapeutic relationships. The ideal approach for providing psychiatric genetic counseling is through a multidisciplinary team approach, with collaboration between genetics and mental health professionals. Genetic professionals are available for consultation with psychiatry regarding advanced risk assessment, collection and construction of complicated family medical history (FMH), and the availability and limitations of genomic testing. Precision medicine is evolving to include the use of panels of genetic variants and genome sequencing to assess the origins of conditions in families where multiple

individuals are affected, or psychiatric symptoms accompany signs of rare disorders.

Definitions

Genetic counseling is the process of helping people to understand and adapt to the medical, psychological, and familial implications of genetic contributions to disease. According to the National Society of Genetic Counseling, it integrates three factors: (1) interpretation of family and medical histories to assess the chance of disease occurrence or recurrence; (2) education about inheritance, testing, management, prevention, resources, and research; and (3) counseling to promote informed choices and adaptation to the risk or condition. The process aims to minimize distress and facilitate adaptation, to increase one's feeling of personal control, and to facilitate informed decision-making and life planning.

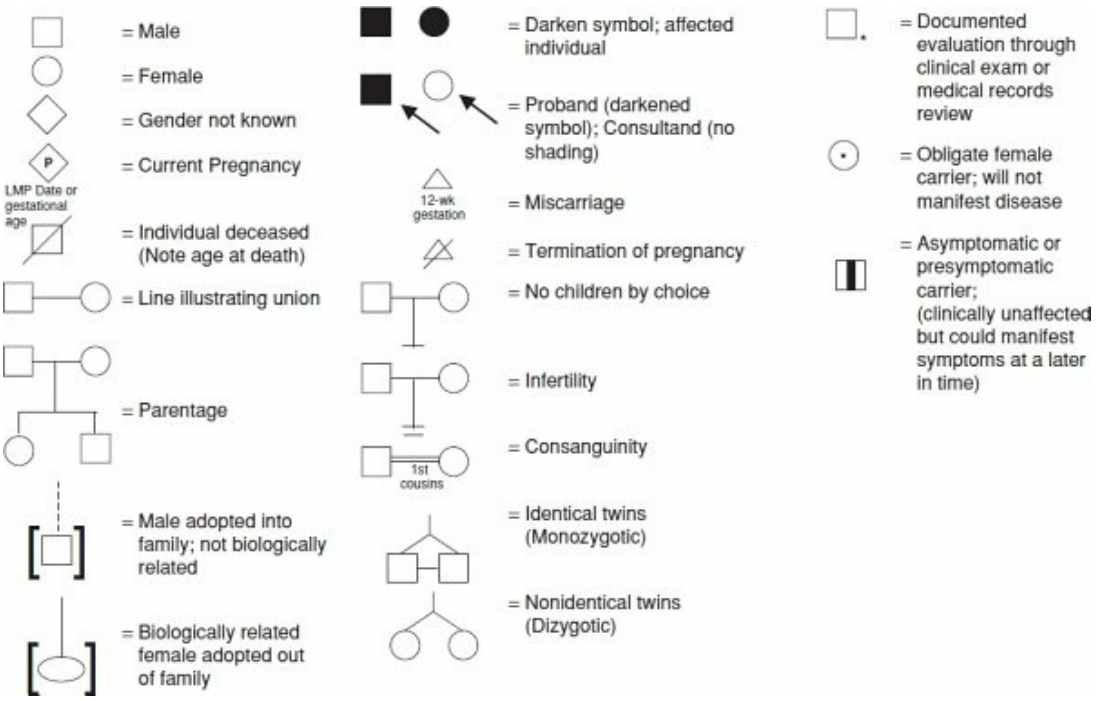


FIGURE 24-1
Standard symbols used in pedigree construction. LMP, last menstrual period.

Although the term *genetic counseling* suggests a primary focus on the genetic contributions to disease, it is not limited to that domain. Genetic counseling entails risk assessment, education, and counseling for common,

complex diseases and necessarily considers both *genetic* and *environmental* components of the presenting illness. The growing awareness of the complex interplay between genes and the environment dictates a comprehensive approach to care.

To facilitate comprehension and application of the material in this section, Figure 24-1 demonstrates standard symbols used in pedigree construction, and Figure 24-2 illustrates a complex FMH presented in the form of a pedigree.

Genetics and Psychiatric Disorders

It is widely accepted that psychiatric illnesses tend to run in families (Table 24-2). Disorders can recur in families for several reasons, including the functioning of the genome (rarely, due to one genomic variation with a large effect on susceptibility, or more commonly due to multiple variations that each have a small effect on susceptibility) or shared environmental exposures. Thus, genetic counseling for a complex disorder entails discussion around *both* genetic and environmental risk factors. For example, consider a case in which a father with obsessive-compulsive disorder (OCD) has concerns about symptoms developing in his children. Genetic counseling may focus on the following: (1) the family history, indicative of both genetic and shared environmental risk in the family; (2) risk factors for the children, stemming from those genetic and environmental factors; (3) early symptom identification and early intervention (aiming to modify the environment and thus affect genetically mediated disease progression); (4) a discussion of the benefits and limitations of environmental modifications aimed at reducing the risk or minimizing emerging symptoms.

Table 24-2
Examples of Psychiatric Disorders Recognized as Having a Genetic Component to Their Etiology

Psychotic disorders: Schizophrenia, schizoaffective disorder
Mood disorders: Bipolar disorder, recurrent unipolar depression
Personality disorders: Antisocial personality disorder, schizotypal disorder

Anxiety disorders: Generalized anxiety disorder, obsessive-compulsive disorder, panic disorder, phobia
Substance-related disorders: Substance dependence and abuse
Eating disorders: Anorexia nervosa, bulimia
Childhood disorders: Attention-deficit/hyperactivity disorder, autism, chronic tic disorders, including Tourette syndrome
Memory disorders: Alzheimer disease

The concept of complex disease (which has replaced *multifactorial* as an etiologic category for common disorders) not only takes into account the interplay of the genome and the environment but also includes the broader biologic and physiologic mechanisms of evolution and development that operate with societies and cultures. The onset of most cases of psychiatric disease, like the onset of other common disorders, is thought to arise when an individual has several risk-increasing genomic variants and has experienced one or more environmental triggers; development, growth, and maturation influence the overall risk. The modern perception that psychiatric symptomatology causes disadvantageous “disease” is related to the effects of the symptoms on the ability of the individual to interact successfully with his or her social and cultural environment.

The first evidence that genetic factors were operating in psychiatric disorders came from the analysis of families with multiple affected members, twin studies, and adoption studies. Family studies indicated familial etiology, although the relative contributions of genes and environmental factors were unclear. Twin and adoption studies provided insights into the relative contributions of etiologic risk factors, and for all major psychiatric disorders, consistently indicated the importance of genes and environment. Family, twin, and adoption studies also provided insight into the genetic heterogeneity and phenotypic variability of mental disorders.

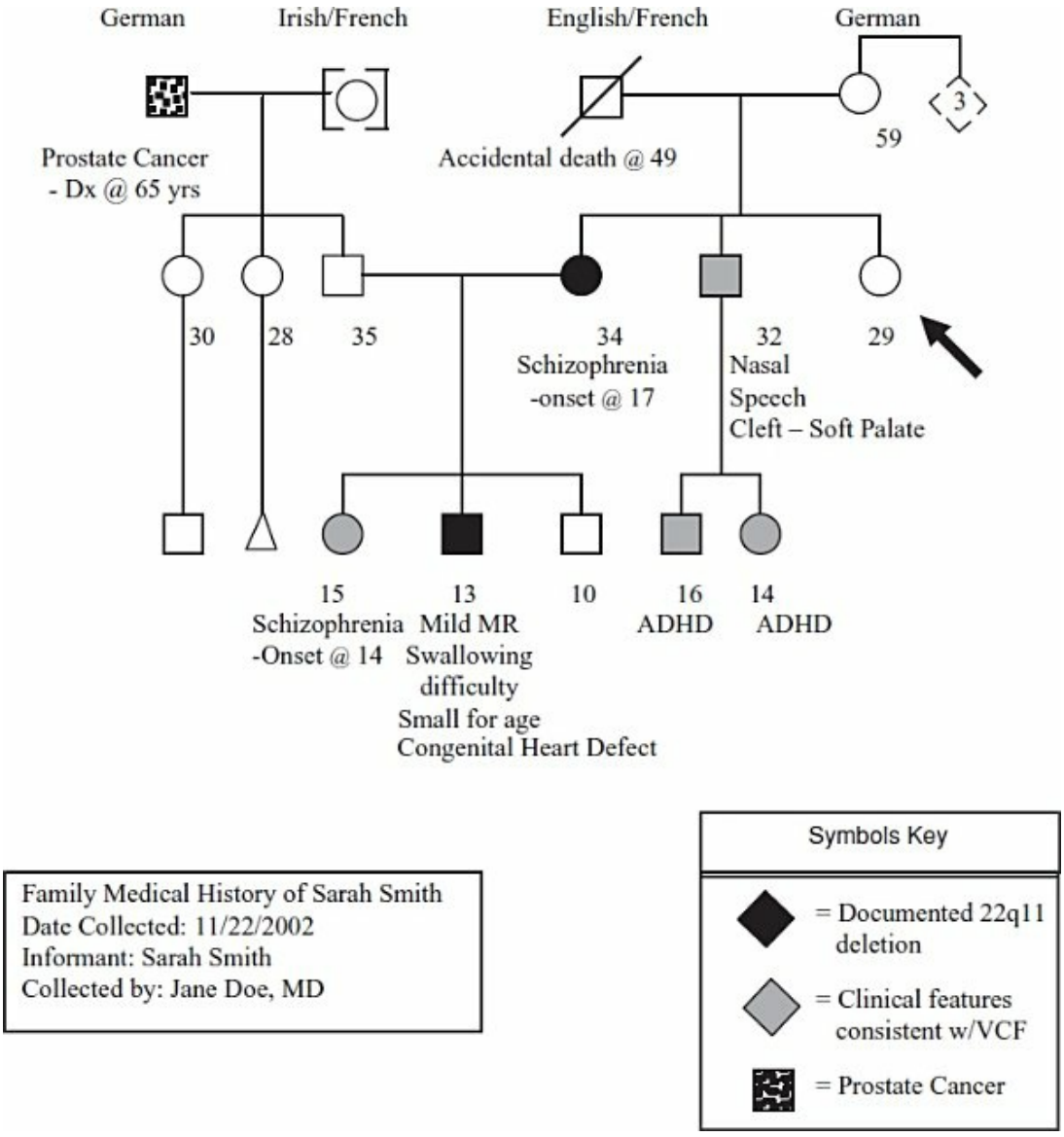


FIGURE 24-2
Pedigree of a family with velocardiofacial (VCF) syndrome. ADHD, attention-deficit/hyperactivity disorder; Dx, diagnosed; MR, mental retardation. (From Sadock BJ, Sadock VA, Ruiz P, eds. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia: Lippincott Williams & Wilkins; 2009, with permission.)

Priorities of Psychiatric Genetic Counseling

Requests for genetic counseling often arise from the patient's or relatives' questions about the disorder that is present in the family. One can conceptualize the provision of information on the potential genomic origins of the condition and the options for genome testing or assessment of the family as shared decision-making. Patient preference dictates risk assessment

and genome testing. It is important to weigh how much the patient prefers to pursue and for what purpose against the estimated likelihood that useful information will come from testing. Table 24-3 outlines the essential components of genetic counseling.

Obtaining the Data Needed for Risk Assessment. The FMH is used to assess the likely etiology of the disorder and estimate risks for recurrence in the family. Confirmation or clarification of the diagnosis or diagnoses in the family is essential to the provision of valid information.



Table 24-3
Steps and Process of Genetic Counseling

Solicit and clarify the client's presenting questions and goals for genetic counseling.
Explore client's existing explanation for cause of illness.
Collect and review medical and family medical history.
Identify support systems.
Verify diagnoses in proband and other affected family members when possible.
Address issues and concerns identified through the genetic counseling process.
Assess the client's emotional and intellectual capacity before proceeding to determine the approach to the provision of education and counseling.
Provide information at cognitively appropriate levels.
Note that the processing of emotional reactions is intertwined with the provision of information.
Assess the personal meaning of the information and the client's willingness to negotiate various risks and burdens.
When applicable, assist the client in arriving at a decision by discussing available options; discuss benefits and limitations of each alternative.
Assist the client with adapting to the risk status in the family.
Assist in formulating a plan that the client is able to carry out.

Provide follow-up counseling and support. Continue to assess the client's and family's understanding of the information and the effect of the information, risks, or decisions.

Topics that are specific to psychiatric disorders and are essential to include in the FMH are: (1) any psychiatric diagnosis, whether childhood, adult, postpartum, or geriatric; (2) age at symptom onset and diagnosis; (3) subjective assessment of illness severity and treatment response; (4) potentially significant environmental exposures (e.g., obstetrical complications, cannabis use, head injury); (5) symptomatic, undiagnosed individuals or undiagnosed individuals treated with psychiatric medication; (6) developmental history (e.g., has the individual achieved typical milestones for age, such as independent living and employment for adults?); (7) social history; (8) substance abuse; (9) suicide and self-harm; (10) congenital disabilities, intellectual disability or learning disabilities, and unusual medical conditions; (11) the age and sex of at-risk family members.

As mental health providers will be aware, challenges to collecting the psychiatric FMH may include guilt and stigma that limit knowledge of diagnoses among family members, the frequency of undiagnosed psychiatric illness, and the dynamic nature of emerging psychiatric symptoms that may result in diagnostic change over time.

Risk Communication. Different clients seek different types of risk counseling, and they should be queried about their preferences before risk communication begins. Some may seek broad information about the underpinnings of risk for psychiatric disorders, while others prefer a qualitative discussion of the risk magnitude. These individuals may not welcome or benefit from receiving numerical risk information. For others who prefer a detailed numerical estimation of risk for recurrence, risks based on FMH should be presented as a range to reflect the ambiguity of the estimates. In some cases, it may be possible to suggest that one end of the range is more appropriate for a specific client than midrange.

Psychosocial Counseling and Support

Genetic counseling integrates psychological and social issues that result from

living with a psychiatric condition in the family with FMH and increasingly will also include genomic risk information. Historically, genetic counseling has been nondirective, which is deemed particularly appropriate within the context of reproductive decisions and presymptomatic testing for untreatable disorders, such as Huntington disease. However, nondirectiveness is challenging to practice and has been replaced with a person-centered approach, as improved treatments and risk behaviors realize health benefits. The following case exemplifies the personal experiences and meaning of psychiatric disorders.

A couple in their mid-30s with a 10-year history of infertility had been trying to adopt a child for several years. Recently, the adoption agency they were working with told them of a baby who was being placed for adoption because the biologic mother was affected with bipolar disorder and did not feel that she could provide adequate care for the baby. The FMH collected on the newborn baby did not identify others in his family with psychiatric disorders. The recurrence risk for bipolar disorder to the newborn was, therefore, estimated to be between 5 and 20 percent, with additional risks for other psychiatric disorders. The couple individually reacted quite disparately to the estimated risks. In attempting to help them clarify the factors contributing to their feelings regarding the risks, the husband shared his experience with a childhood neighbor who had “some kind of mental illness” and detailed the “torment and agony” that the child brought to the family. Retorting, the woman shared the fact that her coworker also had bipolar disorder and did “just fine” at work with the help of medication. She, therefore, did not feel that the risks for psychiatric disorders were of concern. The psychiatrist facilitated the couple’s discussion of the spectrum and meaning of mental illness, along with recurrence risks in the context of a genetic education and counseling session. Although the couple did not agree during that meeting over the potential for adopting the child, they did feel that the information and sharing of experiences and perspectives about psychiatric disorders were beneficial. They agreed to return in 1 week after further considering the issues to decide on the adoption. (Courtesy of Holly L. Peay, M.S. and Donald W. Hadley, M.S.)

Challenges Posed by Presymptomatic and Susceptibility Genetic Testing

As genomic technologies continue to improve and knowledge about variants that can increase vulnerability for psychiatric illness grows, we will more commonly identify individuals at a higher risk for psychiatric illness. Appropriate applications of such developments need careful consideration from both economic and ethical perspectives but may have critical clinical uses.

Psychiatrists will be on the front line for receiving requests for genetic counseling and testing because of their established relationship between patients and families with psychiatric disorders. The clinical identification of genomic risks will most likely occur before most individuals have access to preventative options. The option of knowing risks without preventative options raises concerns regarding the impact of such knowledge on the individual's mood, anxiety, distress, self-image, reproductive decisions, career decisions, family relationships, insurability, employment, and, potentially, other areas.

Across many disease indications, the majority of persons undergoing genomic testing and receiving their identified risk status do not experience long-term adverse effects. However, those receiving results indicating increased disease risk may experience elevated levels of disease-related anxiety and depression soon after receiving results, and a small number have prolonged difficulty in adjusting to their results. Supportive counseling should be available to facilitate adjustment.

The pursuit of genomic counseling and testing by an individual has clear and profound implications for other family members. Helping the client understand the potential reactions and effect that his or her decision to pursue genomic testing could have on close family members is an important consideration. The health professional can play a role in assisting the client in considering ways to share the information with selected significant others. Similarly, the effect of such information on the spouse or significant other requires consideration.

Ethical, Legal, and Social Considerations

Although there are many benefits of genomics knowledge, scientists and the lay public have worried, from the beginning of genetic research, about the potential for harmful consequences of knowing one's genetic makeup. How will family members, neighbors, communities, employers, health care providers, and insurance companies react to such information? Might health care providers request that those at increased genomic risk take psychiatric medications prophylactically? Might employers limit job opportunities for those with increased risk for psychiatric illness, keeping them in positions with lower levels of stress? Will persons at increased risk be pressured not to have children? Might providing such risk information induce significant enough anxiety/stress to trigger the onset of illness? Furthermore, how will families and the public be educated about the availability of susceptibility testing for psychiatric illness?

Some individuals and families may experience significant levels of stigma associated with the identification of a genetic disorder, a situation already familiar to individuals and families with psychiatric illness. The added knowledge of a hereditary component may heighten stigmatization. Conversely, having an identified, biologic basis may supplant current public perceptions that mental illness is somehow a personal or family failure in moral, spiritual, or attitudinal perspectives.

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25

Consultation to Other Disciplines

Psychiatrists are often asked to consult with other doctors and health professionals to help with the psychiatric issues in their patients. Consultation-Liaison Psychiatry is devoted to this role. However, several other subspecialties have large consultative roles. These include geriatric psychiatrists, who may work in multidisciplinary settings with diseases that cross all disciplines (such as the dementias) or forensic psychiatrists who consult with other doctors, lawyers, and the courts. Much of the forensic psychiatrists' expertise concerns legal and ethical issues, and we discuss these in Chapters 27 and 28. Psychiatrists also work in the emergency department to help assess and treat psychiatric emergencies. In some larger hospitals, Emergency Psychiatry exists as a separate division with its own location. However, in most hospitals, psychiatrists consult in the general emergency setting. Some child psychiatrists may work in a consultative setting as well, particularly emergency settings. In this chapter, we cover the consultative role of consultation-liaison psychiatrists, geriatric psychiatrists, and both general and child psychiatrists working in the emergency setting.

25.1 Consultation-Liaison Psychiatry

Consultation-Liaison Psychiatry and the field of Psychosomatic Medicine have been a specific area of concern within psychiatry for more than 50 years. The term *psychosomatic* is derived from the Greek words *psyche* (soul) and *soma* (body). The term refers to how the mind affects the body. Unfortunately, it has come to be used, at least by the lay public, to describe an individual with medical complaints that have no physical cause and are

“all in your head.” For this and related reasons, in 2018, the Academy of Psychosomatic Medicine changed its name to the Academy of Consultation-Liaison Psychiatry. Before that, the DSM deleted the nosologic term psychophysiological (or psychosomatic) disorders and replaced it with psychological factors affecting physical conditions.

Nonetheless, the term continues to be used by researchers and is in the title of significant journals in the field (e.g., *Psychosomatic Medicine*, and the *Journal of Psychosomatic Research*). It is also used by some major national organizations in the field (the American Psychosomatic Society) as well as international organizations (e.g., the European Association for Consultation-Liaison Psychiatry and Psychosomatics). In 2003 the American Board of Medical Specialties and the American Board of Psychiatry and Neurology approved the specialty of Psychosomatic Medicine and in 2017 they changed the name to Consultation-Liaison psychiatry.

HISTORY

As Edward Shorter discusses in detail in his summary of the history of psychosomatic illness, ways of presenting illness vary over history because patients unconsciously select symptoms that are thought to represent actual somatic illnesses. Psychosomatic presentations have thus varied throughout recent history. Before 1800, physicians did not conduct clinical evaluations and could not distinguish somatic from psychogenic illness. As a result, the diagnoses of hysteria and hypochondriasis could easily be made in the presence of genuine medical illnesses and did not suggest any specific disease.

Sigmund Freud helped to bring psyche and soma together. He demonstrated the importance of emotions in producing mental disturbances and somatic disorders. His early psychoanalytic formulations detailed the role of psychic determinism in somatic conversion reactions. Using Freud’s insight, several workers in the early decades of the 20th century tried to expand the understanding of the interrelationship of psyche and soma. Karl Abraham proposed the influence on adult organ tissue of various unresolved pregenital impulses in 1927. Sándor Ferenczi described the application of the idea of conversion reaction to organs under the control of the autonomic nervous systems in 1926, and Georg Groddeck suggested the attaching of

symbolic meaning to fever and hemorrhage in 1929.

The field of consultation-liaison psychiatry has existed for many decades, and George W Henry described the practice in the United States in the 1920s. Divisions of consultation-liaison psychiatry began springing up in various academic departments, as did fellowship programs. The American Board of Psychiatry and Neurology offered the first examination for the subspecialty in 2005.

In Europe, particularly in Germany, pioneers such as Thure von Uexküll saw psychosomatic medicine more broadly, and not merely a subspecialty of psychiatry. The field was understood as central to the concept of all medical diseases. Diseases were conceived as being multifactorial, with both organic and psychological etiologies. It is notable that in Europe, most departments of psychosomatic medicine are independent of departments of psychiatry.

CURRENT TRENDS

The practice of Consultation-Liaison psychiatry has evolved considerably since its early clinical origins and has come to focus on psychiatric illnesses that occur in the setting of physical health care. In large part, this evolution has occurred as a result of the increased complexity of medicine, the increased understanding of the relationship of medical illness to psychiatric illness, and the greater appreciation of mind and body as one. Clinical care is now delivered in a variety of health care settings and utilizes an ever-expanding set of diagnostic tools, as well as many effective somatic and psychotherapeutic interventions. Research in the area has progressed to include a greater understanding of the relationship between chronic medical conditions and psychiatric disorders. It has examined the pathophysiologic relationships, the epidemiology of comorbid medical and psychiatric disorders, and the role that specific interventions play in physiologic, clinical, and economic outcomes (Table 25-1).



Table 25-1
Summary of Clinical Problems in Psychosomatic Medicine

Type of Clinical Problem	Example
Psychiatric symptoms	Delirium, dementia

secondary to medical conditions	
Psychiatric symptoms as a reaction to medical conditions or treatments	Anxiety related to chemotherapy, depression related to limb amputation
Psychiatric complications of medical conditions and treatments	Depression secondary to interferon treatment
Psychological factors precipitating medical symptoms	Somatic Symptom and related disorders
Medical complications of psychiatric conditions or treatments	Neuroleptic malignant syndrome, acute withdrawal from alcohol or another substance
Co-occurring medical and psychiatric conditions	Recurrence of depressive disorder in the setting of cancer treatment (conditions occur independently); schizophrenia in a patient with end-stage renal disease
Psychiatric/psychosocial assessment	Capacity evaluation; evaluation prior to organ transplantation

Psychiatric morbidity is prevalent in patients with medical conditions, with a prevalence ranging from 20 to 67 percent, depending on the illness and setting. Patients in the general hospital have the highest rate of psychiatric disorders when compared with community samples or patients in ambulatory primary care. For example, compared with community samples, depressive disorders in the general hospital are more than twice as prevalent, and substance abuse is two to three times as common. Delirium occurs in 18 percent of patients. Similarly, increased rates are seen in primary and long-term care.

Psychiatric morbidity has severe effects for medically ill patients and is often a risk factor for their medical conditions. It is well established that

depression is both a risk factor and a poor prognostic indicator in coronary artery disease. Psychiatric illness worsens cardiac morbidity and mortality in patients with a history of myocardial infarction, diminishes glycemic control in patients with diabetes, and decreases return to functioning in patients experiencing a stroke. Depressive and anxiety disorders compound the disability associated with stroke. In the context of neurodegenerative diseases such as Parkinson's or Alzheimer's, depression, psychosis, and behavioral disturbances are significant predictors of functional decline, institutionalization, and caregiver burden. Hospitalized patients with delirium are significantly less likely to improve in function compared with patients without delirium. Delirium is associated with worse outcomes after surgery, even after controlling for the severity of medical illness.

Also, depression and other mental disorders significantly impact the quality of life and the ability of patients to adhere to treatment regimens (e.g., in patients with diabetes mellitus or cancer). Psychiatric disorders are linked to nonadherence with antiretroviral therapy, adversely affecting the survival of human immunodeficiency virus (HIV)-infected patients. Psychiatric disorders are also linked to nonadherence with safe sex guidelines and with the use of sterile needles in HIV-infected injection drug users, thus having significant public health implications.

EVALUATION PROCESS IN PSYCHOSOMATIC MEDICINE

Psychiatric assessment in the medical setting includes a standard psychiatric assessment as well as a particular focus on the medical history and context of physical health care. In addition to obtaining a complete psychiatric history, including history, family history, developmental history, and a review of systems, the medical history and current treatment should be reviewed and documented. A full mental status examination, including a cognitive examination, should be completed, and components of a neurologic and physical examination may be indicated depending on the nature of the presenting problem.

Another essential objective of the psychiatric evaluation is to gain an understanding of the patient's experience of his or her illness. In many cases, this becomes the central focus for both the psychiatric assessment and

interventions. It is often helpful to develop an understanding of the patient's developmental and personal history as well as critical dynamic conflicts, which in turn may help to make the patient's experience with illness more comprehensible. Such an evaluation can include the use of the concepts of stress, personality traits, coping strategies, and defense mechanisms. Observations and hypotheses that are developed can help to guide a patient's psychotherapy aimed at diminishing distress and may also be helpful for the primary medical team in their interactions with the patient.

Finally, a full report synthesizing the information should be completed and include specific recommendations for additional evaluations and intervention. Ideally, the report should be accompanied by a discussion with the referring physician.

TREATMENTS USED IN PSYCHOSOMATIC MEDICINE

A host of interventions have been successfully utilized in psychosomatic medicine. Specific consideration must be given to medical illness and treatments when making recommendations for psychiatric medications. Psychotherapy also plays a vital role in psychosomatic medicine and may vary in its structure and outcomes as compared with therapy that occurs in mental health practice.

Psychopharmacologic recommendations need to consider several essential factors. In addition to targeting a patient's active symptoms, considering the history of illness and treatments, and weighing the particular side-effect profile of a particular medication, several other factors must be considered that relate to the patient's medical illness and treatment. It is critical to evaluate potential drug–drug interactions and contraindications to the use of potential psychiatric agents. Because the majority of psychiatric medications used are metabolized in the liver, awareness of liver function is essential. General appreciation of side effects, such as weight gain, risk of developing diabetes, and cardiovascular risk, must be considered in the choice of medications. Also, it is critical to incorporate knowledge of recent data that outlines the effectiveness and specific risks involved for patients with co-occurring psychiatric and physical disorders. For example, a greater understanding of the side effects of antipsychotic medications has raised concerns about the use of these medications in patients with dementia.

The use of psychosocial interventions also requires adaptation when used in this population. The methods and the goals of psychosocial interventions used in the medically ill are often determined by the consideration of disease onset, etiology, course, prognosis, treatment, and understanding of the nature of the presenting psychiatric symptoms in addition to an understanding of the patient's existing coping skills and social support networks. However, there are ample data that psychosocial interventions are effective in addressing a series of identified problems and that such interventions, in many cases, are associated with a variety of positive clinical outcomes.

Common C-L Problems

Suicide Attempt or Threat. Suicide rates are higher in persons with medical illness than in those without medical or surgical problems. High-risk factors for suicide are men over 45 years of age, no social support, alcohol dependence, previous suicide attempt, and incapacitating or catastrophic medical illness, especially if accompanied by severe pain. If suicide risk is present, the patient should be transferred to a psychiatric unit or started on 24-hour nursing care.

Depression. Suicidal risk must be assessed in every depressed patient. Depression without suicidal ideation is not uncommon in hospitalized patients, and treatment with antidepressant medication can be started if necessary. A careful assessment of drug–drug interactions must be made before prescribing, which should be undertaken in collaboration with the patient's primary physician.

Agitation. Agitation is often related to the presence of a cognitive disorder or associated with withdrawal from drugs (e.g., opioids, alcohol, sedative-hypnotics). Antipsychotic medications may be beneficial for excessive agitation. Physical restraints should be used with great caution and only as a last resort as sedation is preferable and safer. The patient should be examined for command hallucinations or paranoid ideation to which they are responding in an agitated manner. Toxic reactions to medications that cause agitation should always be ruled out.

Hallucinations. A common cause of hallucinations in the general hospital setting is delirium tremens, which usually begin 3 to 4 days after hospitalization. Patients in intensive care units (ICUs) who experience sensory isolation may respond with hallucinatory activity. Conditions such as brief psychotic disorder, schizophrenia, and neurocognitive disorders are associated with hallucinations, and they respond rapidly to antipsychotic medication. Fornication, in which the patient believes that bugs are crawling over the skin, is often associated with cocaine use.

Sleep–Wake Disorders. A common cause of insomnia in hospitalized patients is pain, which, when treated, solves the sleep problem. Early morning awakening is associated with depression, and difficulty falling asleep is associated with anxiety. Depending on the cause, antianxiety or antidepressant agents may be prescribed. Early substance withdrawal as a cause of insomnia should be considered in the differential diagnosis.

Confusion. Delirium is the most common cause of confusion or disorientation among hospitalized patients in general hospitals. The causes are myriad and relate to metabolic status, neurologic findings, substance abuse, and mental illness, among many others. Small doses of antipsychotics may be used when significant agitation occurs in conjunction with the confused state; however, sedatives, such as benzodiazepines, can worsen the condition and cause sundowner syndrome (ataxia, disorientation). If sensory deprivation is a contributing factor, the environment can be modified so that the patient has sensory cues (e.g., radio, clock, no curtains around the bed). Table 25-2 lists the probable causes of confusional states that require urgent attention.

Noncompliance or Refusal to Consent to Procedure. The relationship between the patient and their doctor may be a critical underlying factor when patients are noncompliant or refuse to consent to a procedure. A negative transference toward the physician is a common cause of noncompliance. Patients who fear medication or a procedure may respond to education and reassurance. Patients whose refusal to give consent is related to impaired judgment can be declared incompetent, but only by a judge. The leading cause of impaired judgment in hospitalized patients is cognitive

disorders.

No Physiologic Basis for Symptoms. The C-L psychiatrist is often called when the physician cannot find evidence of medical or surgical disease to account for the patient's symptoms. In these instances, several psychiatric conditions must be considered, including conversion disorder, somatization disorder, factitious disorders, and malingering. Glove and stocking anesthesia with autonomic nervous system symptoms is seen in conversion disorder; multiple bodily complaints are present in somatization disorder; the wish to be in the hospital occurs in factitious disorder, and apparent secondary gain is observed in patients who are malingering (e.g., compensation cases).

C-L Psychiatry in Special Situations

Intensive Care Units. All ICUs deal with patients who experience anxiety, depression, and delirium. ICUs also impose extraordinarily high stress on staff and patients, which is related to the intensity of the problems. Patients and staff members alike frequently observe cardiac arrests, deaths, and medical disasters, which leave them all autonomically aroused and psychologically defensive. ICU nurses and their patients experience exceptionally high levels of anxiety and depression. As a result, nurse burnout and high turnover rates are frequent.

The problem of stress among ICU staff receives much attention, especially in the nursing literature. Much less attention is given to the house staff, especially those on the surgical services. All persons in ICUs must be able to deal directly with their feelings about their extraordinary experiences and challenging emotional and physical circumstances. Regular support groups in which persons can discuss their feelings are recommended for the ICU staff and the house staff. Such support groups protect staff members from the otherwise predictable psychiatric morbidity that some may experience and also protect their patients from the loss of concentration, decreased energy, and psychomotor-retarded communications that some staff members otherwise exhibit.

Hemodialysis Units. Hemodialysis units present a paradigm of complicated modern medical treatment settings. Patients are coping with

lifelong, debilitating, and limiting disease; they are dependent on a multiplex group of caretakers for access to a machine controlling their well-being. Dialysis is scheduled three times a week and takes 4 to 6 hours; thus, it disrupts patients' previous living routines.



Table 25-2
Some Clues to Causes of Delirium Demanding Urgent Attention

Metabolic Disorders

1. Hypoglycemia: history of diabetes or alcoholism; reduced level of consciousness, shaky, sweaty, perhaps combative
2. Hyperglycemia: history of diabetes; complaints of increased thirst, urination, or flulike symptoms
3. Hyponatremia: underlying illness like lung cancer, recent stroke, chronic pulmonary infections, heart failure, cirrhosis, diuretic use
4. Hypernatremia: dehydration from inadequate fluid intake or excessive fluid loss without replacement
5. Hypercalcemia: underlying disorder such as cancer metastatic to bone, sarcoidosis, lung and renal cell cancer, multiple myeloma, and/or prolonged immobilization
6. Hypoxia: inadequate oxygen supplied to the brain because of poor pulmonary or cardiac function or carbon monoxide poisoning
7. Hypercarbia: history of chronic lung disease characterized by carbon dioxide retention; may use oxygen at home
8. Hepatic encephalopathy: history of chronic liver disease or alcoholism; probably jaundiced; ascites
9. Uremia: history of kidney disease, enlarged prostate, recent inability to pass urine
10. Thiamine deficiency (Wernicke encephalopathy): variable degrees of ophthalmoplegia, ataxia, and mental disturbance; history of nutritional deficiency secondary to alcoholism, particularly of thiamine; since remaining thiamine in the body is rapidly used when the patient is given intravenous glucose, any

patient with alcoholism should immediately receive intramuscular thiamine before glucose infusion to prevent precipitating this encephalopathy; untreated, the disorder rapidly progresses to a permanent memory disorder (Korsakoff syndrome) and, in some advanced cases, death

11. Hypothyroidism: history of progressive fatigue, constipation, sensitivity to cold, weight gain, coarsening of hair and skin, mental slowing; examination shows abnormally low temperature and enlarged heart and slow pulse; may be precipitated by the effects of lithium on thyroid function
12. Hyperthyroidism: patient may be either hyperactive or apathetic; history may reveal rapid weight loss, diarrhea, heat intolerance, and emotional instability; examination shows goiter, silky fine hair, warm moist skin, proptosis and wide-eyed stare, fine tremor, rapid or irregular pulse; in elderly patients, muscle weakness and heart failure may be most apparent

Systemic Illness

1. Decreased cardiac output from various causes, such as congestive heart failure, arrhythmia, pulmonary embolus, and myocardial infarction; acute myocardial infarction presents with confusion as the major symptom in 13% of elderly patients; aged patients do not complain of typical pain; often they complain of indigestion; vital signs may be abnormal, and patient may look ill (ashen coloring, weak, nauseated, sweaty) and be confused

2. Pneumonia: recent history of a cold, becoming bedridden and aspirating; fever may not be apparent, but tachycardia or hypotension are evident on vital signs

3. Urinary tract infection: especially in patients with indwelling urinary catheters, prostatic hypertrophy, diabetes, neurogenic bladder

4. Anemia: especially with acute blood loss (injury, intestinal bleeding), chronic illness, occult gastrointestinal malignancy

5. Acute surgical emergencies: infarction of the bowel, appendicitis,

and volvulus are common and often present only with confusion and no other complaints

6. Hypertension: sustained or rapid increase in blood pressure may cause encephalopathy; often has history of elevated blood pressure; may occur in patient on MAO inhibitor antidepressants who has eaten food containing tyramine

7. Vasculitides: e.g., systemic lupus erythematosus; confusion arises from cerebral involvement or treatment with steroids

8. Any febrile illness and infection can cause confusion in the aged

Central Nervous System Disorders

1. Subdural or epidural hematoma: may or may not have history of head trauma; fluctuating mental status often present; may have no focal neurologic signs

2. Seizure: unwitnessed seizure may be suggested if patient was found on floor with evidence of incontinence or vomiting; history of seizure disorder or alcoholism

3. Stroke: history of transient ischemic attacks or strokes; may have no signs except confusion

4. Infection: meningitis (bacterial, fungal, or tuberculous), viral encephalitis

5. Tumor, primary or metastatic: with a growing mass, raised intracranial pressure may cause local compression of vital structures or herniation of the brain; in the elderly, brain atrophy allows for greater space inside the skull so that symptoms may not appear until the mass is quite large

6. Normal pressure hydrocephalus: presents with triad of gait disturbance, incontinence, dementia; surgery may be curative

Drugs and Medication

1. Almost all drugs are capable of causing confusion in the elderly; the most commonly implicated drugs include those with strong

anticholinergic effects (antidepressants, antipsychotics, and antiparkinsonian drugs, and many over-the-counter preparations), sedative-hypnotics (barbiturates, benzodiazepines), cardiac medications (digoxin, propranolol, lidocaine, quinidine), antihypertensives, anticonvulsants, cimetidine, nonnarcotic and narcotic analgesics, and corticosteroids

2. Alcohol: intoxication and withdrawal syndromes occur as in young patients, but poor health in the elderly may put geriatric patients at greater risk

3. Drug abuse: far less common in elderly persons, but chronic intoxication with bromides, prescription tranquilizers occurs

From Minden SL. Elderly psychiatric emergency patients. In: Bassuk EL, Birk AW, eds. *Emergency Psychiatry*. New York: Plenum; 1984:360, with permission.

In this context, patients first and foremost fight the disease. Invariably, however, they also must come to terms with a level of dependence on others probably not experienced since childhood. Predictably, patients entering dialysis struggle for their independence; regress to childhood states; show denial by acting out against doctor's orders (by breaking their diet or by missing sessions); show anger directed against staff members; bargain and plead; or become infantilized and obsequious; however, most often they are accepting and courageous. The determinants of patients' responses to entering dialysis include personality styles and previous experiences with this or another chronic illness. Patients who have time to react and adapt to their disease are less challenged by the need to adapt to new circumstances. Those with recent renal failure and machine dependence may have more difficulty.

Although little has been written about social factors, the effects of culture in reaction to dialysis and the management of the dialysis unit are known to be necessary. Units are run with a firm hand, which is consistent in dealing with patients; clear contingencies are in place for behavioral failures; and adequate psychological support is available for staff members, which tend to produce the best results.

Complications of dialysis treatment can include psychiatric problems, such as depression, and suicide is not rare. Sexual problems can be neurogenic,

psychogenic, or related to gonadal dysfunction and testicular atrophy. Dialysis dementia is a rare condition that includes loss of memory, disorientation, dystonias, and seizures. The disorder occurs in patients who have been receiving dialysis treatment for many years. Although not entirely understood, the leading cause is probably aluminum toxicity, and reducing dialysate aluminum levels and minimizing aluminum intake decreases the incidence of the disease. Chelating agents, such as deferoxamine, may also help if given early. However, it should be used sparingly because of its serious adverse effects.

The psychological treatment of dialysis patients falls into two areas. First, careful preparation before dialysis, including the work of adaptation to chronic illness, is essential, especially in dealing with denial and unrealistic expectations. Before dialysis, all patients should have a psychosocial evaluation. Second, once in a dialysis program, patients need periodic specific inquiries about adaptation that do not encourage dependence or the sick role. Staff members should be sensitive to the likelihood of depression and sexual problems. Group sessions function well for support, and patient self-help groups restore a useful social network, self-esteem, and self-mastery. When needed, antidepressants can be used for dialysis patients. Psychiatric care is most effective when brief and problem-oriented.

The use of home dialysis units has improved attitudes toward treatment. Patients treated at home can integrate the treatment into their daily lives more efficiently, and they feel more autonomous and less dependent on others for their care than do those who are treated in the hospital.

Surgical Units. Some surgeons believe that patients who expect to die during surgery are more likely to die. This belief now seems less superstitious than it once did. Some patients who are scheduled for surgery and show evident depression or anxiety but deny it have a higher risk for morbidity and mortality than those who can express it. Even better results occur in those with a positive attitude toward impending surgery. The factors that contribute to an improved outcome for surgery are informed consent and education so that patients know what they can expect to feel, where they will be (e.g., it is useful to show patients the recovery room), what loss of function to expect, what tubes and gadgets will be in place, and how to cope with the anticipated pain. If patients will not be able to talk or see after surgery, it is helpful to

explain before surgery what they can do to compensate for these losses. If postoperative states such as confusion, delirium, and pain can be predicted, they should be discussed with patients in advance, so they do not experience them as unwarranted or as signs of danger. Constructive family support members can help both before and after surgery.

Transplantation Issues. Transplantation programs have expanded over the past decade, and C-L psychiatrists play an essential role in helping patients and families deal with the many psychosocial issues involved: (1) which and when patients on a waiting list will receive organs, (2) anxiety about the procedure, (3) fear of death, (4) organ rejection, and (5) adaptation to life after successful transplantation. After the transplant, patients require complicated aftercare, and achieving compliance with medication may be difficult without supportive psychotherapy. This challenge is particularly relevant to patients who have received liver transplants as a result of hepatitis C brought on by promiscuous sexual behavior and patients with a history of substance use disorder who used contaminated needles.

Group therapy with patients who have had similar transplantation procedures benefits members who can support one another and share information and feelings about particular stressors related to their disease. Groups may be conducted or supervised by the psychiatrist. Psychiatrists must be especially concerned about psychiatric complications. Within 1 year of transplant, almost 20 percent of patients experience major depression or an adjustment disorder with depressed mood. In such cases, evaluation for suicidal ideation and risk is essential. In addition to depression, another 10 percent of patients experience signs of posttraumatic stress disorder, with nightmares and anxiety attacks related to the procedure. Other issues concern whether or not the transplanted organ came from a cadaver or from a living donor who may or may not be related to the patient. Pretransplant consulting sessions with potential organ donors help them to deal with fears about surgery and concerns about who will receive their donated organ. Sometimes, both the recipient and donor may be counseled together, as with a family member donating a kidney. Peer support groups with both donors and recipients have also been used to facilitate coping with transplantation issues.

PSYCHO-ONCOLOGY

Psycho-oncology seeks to study both the impact of cancer on psychological functioning and the role that psychological and behavioral variables may play in cancer risk and survival. A hallmark of psycho-oncology research has been intervention studies that attempt to influence the course of illness in patients with cancer. A landmark study by David Spiegel found that women with metastatic breast cancer who received weekly group psychotherapy survived an average of 18 months longer than control patients randomly assigned to routine care. In another study, patients with malignant melanoma who received structured group intervention exhibited a statistically significant lower recurrence of cancer and a lower mortality rate than patients who did not receive such therapy. Patients with malignant melanoma who received the group intervention also exhibited significantly more large granular lymphocytes and natural killer (NK) cells as well as indications of increased NK cell activity, suggesting an increased immune response. Another study used a group behavioral intervention (relaxation, guided imagery, and biofeedback training) for patients with breast cancer, who demonstrated higher NK cell activity and lymphocyte mitogen responses than the controls.

Because new treatment protocols, in many cases, have transformed cancer from an incurable to frequently chronic and often curable disease, the psychiatric aspects of cancer—the reactions to both the diagnosis and the treatment—are increasingly important. At least half of the persons who contract cancer in the United States each year are alive 5 years later. Currently, an estimated 3 million cancer survivors have no evidence of the disease.

About half of all cancer patients have mental disorders. The largest groups are those with adjustment disorder (68 percent), and major depressive disorder (13 percent) and delirium (8 percent) are the next most common diagnoses. Most of these disorders are thought to be reactive to the knowledge of having cancer.

When persons learn that they have cancer, their psychological reactions include fear of death, disfigurement, and disability; fear of abandonment and loss of independence; fear of disruption in relationships, role functioning, and financial standings; and denial, anxiety, anger, and guilt. Although suicidal thoughts and wishes are frequent in persons with cancer, the actual incidence of suicide is only slightly higher than that in the general population.

Psychiatrists should make a careful assessment of psychiatric and medical

issues in every patient. Particular attention should be given to family factors, in particular, preexisting intrafamily conflicts, family abandonment, and family exhaustion.

25.2 Geriatric Psychiatry

PSYCHIATRIC PROBLEMS OF OLDER PERSONS

Despite the ubiquity of loss in old age, the prevalence of major depressive disorder and dysthymia is less than in younger age groups. Several explanations for this phenomenon have been proposed: rarity of late-onset depression, higher mortality among persons with depression, and a general decrease in disorders caused by emotional upheavals or substance abuse in older persons. Depression may, however, be underrecognized in the elderly as it sometimes presents differently than in younger persons. Depression in old persons is often accompanied by physical symptoms or cognitive changes that may mimic dementia.

The incidence of suicide among older persons is high (40 per 100,000 population) and is highest for older white men. The suicide of older persons is perceived differently by surviving friends and family members based on gender: men are thought to have been physically ill, and women are thought to have been mentally ill.

The relation between good mental and good physical health is apparent in older persons. Adverse effects on the course of chronic medical illness are correlated with emotional problems.

MENTAL DISORDERS OF OLD AGE

The National Institute of Mental Health's Epidemiologic Catchment Area (ECA) program found that the most common mental disorders of old age are depressive disorders, cognitive disorders, phobias, and alcohol use disorders. Older adults also have a high risk of suicide and drug-induced psychiatric symptoms. Many mental disorders of old age can be prevented, ameliorated, or even reversed. Of particular importance are the reversible causes of delirium and dementia; if not diagnosed accurately and treated in a timely

fashion, however, these conditions can progress to an irreversible state. Table 25-3 lists the general cognitive domains assessed in neuropsychological evaluation, with the tests used to measure that skill and a description of the specific behaviors measured by each test. The tests listed in the table constitute a comprehensive test battery generally appropriate for use with a geriatric population. The use of a comprehensive battery is preferable for confident determination of the presence and type of dementia or other cognitive disorder in elderly persons; in some circumstances, however, administering a several-hour battery is not possible.



Table 25-3
Cognitive Domains

Gross Cognitive Functioning
Mini-Mental State Examination: *orientation, repetition, following commands, naming, constructional skill, written expression, memory, mental flexibility, and calculations*
Intelligence
Wechsler Adult Intelligence Scale-Revised (WAIS-R) or Wechsler Intelligence Scale-III (WAIS-III): *verbal and nonverbal intelligence*
Basic Attention
WAIS-R or WAIS-III Digit Span: *repetition of digits forward and backward*
Information-Processing Speed
WAIS-R or WAIS-III Digit Symbol: *rapid graphomotor tracking*
Trailmaking Part A: *rapid graphomotor tracking*
Stroop A and B: *rapid word reading and color naming*
Motor Dexterity
Finger tapping: *right and left index finger dexterity*
Language
Boston Naming Test: *word retrieval*
WAIS-R or WAIS-III Vocabulary: *vocabulary range*
Visual Perceptual/Spatial

WAIS-R or WAIS-III Picture Completion: *visual perception*
WAIS-R or WAIS-III Block Design: *constructional ability*
Rey–Osterrieth Complex Figure Test: *paper-and-pencil copy of complex design*
Beery Developmental Test of Visual Motor Integration: *paper-and-pencil copy of simple-to-complex designs*

Learning and Memory

An 8- to 10-item word list learning task: *learning and recall of rote verbal information*
Wechsler Memory Scale-Revised (WMS-R) or Wechsler Memory Scale-III (WMS-III)
Logical Memory subtest: *immediate and delayed recall of paragraph information*
Visual Reproduction subtest: *immediate and delayed recall of visual designs*
Rey–Osterrieth Complex Figure Test and Recognition Trial: *3-minute immediate recall and 30-minute delayed recall of a complex design*

Executive Functions

Trailmaking Part B: *rapid alternation between tasks*
Stroop C: *inhibition of an overlearned response*
Wisconsin Card Sorting Test: *categorization and mental flexibility*
Verbal fluency (FAS and category): *rapid word generation*
Design fluency: *rapid generation of novel designs*

Courtesy of Kyle Brauer Boone, Ph.D.

Several psychosocial risk factors also predispose older persons to mental disorders. These risk factors include loss of social roles, loss of autonomy, the deaths of friends and relatives, declining health, increased isolation, financial constraints, and decreased cognitive functioning.

Many drugs can cause psychiatric symptoms in older adults. These symptoms can result from age-related alterations in drug absorption, a prescribed dosage that is too large, not following instructions and taking too large a dose, sensitivity to the medication, and conflicting regimens presented by several physicians. Drugs can cause almost the entire spectrum of mental

disorders.

Dementing Disorders

Only arthritis is a more common cause of disability among adults age 65 and older than dementia, a generally progressive and irreversible impairment of the intellect, the prevalence of which increases with age. About 5 percent of persons in the United States older than age 65 have severe dementia, and 15 percent have mild dementia. Of persons older than age 80, about 20 percent have severe dementia. Known risk factors for dementia are age, family history, and female sex.

In contrast to intellectual disability, the intellectual impairment of dementia develops over time—that is, previously achieved mental functions are lost gradually. The characteristic changes of dementia involve cognition, memory, language, and visuospatial functions, but behavioral disturbances are common as well and include agitation, restlessness, wandering, rage, violence, shouting, social and sexual disinhibition, impulsiveness, sleep disturbances, and delusions. Delusions and hallucinations occur during the dementias in nearly 75 percent of patients.

Cognition is impaired by many conditions, including brain injuries, cerebral tumors, acquired immune deficiency syndrome (AIDS), alcohol, medications, infections, chronic pulmonary diseases, and inflammatory diseases. Although dementias associated with advanced age typically are caused by primary degenerative central nervous system (CNS) disease and vascular disease, many factors contribute to cognitive impairment; in older persons, mixed causes of dementia are common.

About 10 to 15 percent of all patients who exhibit symptoms of dementia have potentially treatable conditions. The treatable conditions include systemic disorders, such as heart disease, renal disease, and congestive heart failure; endocrine disorders, such as hypothyroidism; vitamin deficiency; medication misuse; and primary mental disorders, most notably depressive disorders.

Depending on the site of the cerebral lesion, dementias are classified as cortical and subcortical. Subcortical dementia occurs in Huntington disease, Parkinson disease, normal pressure hydrocephalus, vascular dementia, and Wilson disease. The subcortical dementias are associated with movement

disorders, gait apraxia, psychomotor retardation, apathy, and akinetic mutism, which can be confused with catatonia. Table 25-4 lists some potentially reversible conditions that may resemble dementia. The cortical dementias occur in dementia of the Alzheimer type, Creutzfeldt–Jakob disease (CJD), and Pick disease, which frequently manifests aphasia, agnosia, and apraxia. In clinical practice, the two types of dementias overlap, and, in most cases, an accurate diagnosis can be made only by autopsy. Human prion diseases result from coding mutations in the prion protein gene (*PRNP*) and may be inherited, acquired, or sporadic. They include familial CJD, Gerstmann–Sträussler–Scheinker syndrome, and fatal familial insomnia. These are inherited as autosomal-dominant mutations. The acquired diseases include kuru and iatrogenic CJD. Kuru was an epidemic prion disease of the Fore people of Papua, New Guinea, caused by cannibalistic funeral rituals, which peaked in incidence in the 1950s. Iatrogenic disease is rare and is caused, for example, by the use of contaminated dura mater and corneal grafts and treatment with human cadaveric pituitary-derived growth hormone and gonadotropin. Sporadic CJD accounts for 85 percent of the human prion diseases and occurs worldwide, with a uniform distribution and an incidence of about 1 in 1 million per annum, with a mean age at onset of 65 years. It is exceedingly rare in individuals under 30 years of age.

Depressive Disorders

Depressive symptoms are present in about 15 percent of all older adult community residents and nursing home patients. Age itself is not a risk factor for the development of depression, but being widowed and having a chronic medical illness are associated with vulnerability to depressive disorders. High rates of recurrence characterize late-onset depression.



Table 25-4
Some Potentially Reversible Conditions that May Resemble Dementia

Substance
Anticholinergic agents
Antihypertensives

- Antipsychotics
- Corticosteroids
- Digitalis
- Narcotics
- Nonsteroidal anti-inflammatory agents
- Phenytoin
- Polypharmacotherapy
- Sedative hypnotics

Psychiatric Disorders

- Anxiety
- Depression
- Mania
- Delusional disorders

Metabolic and Endocrine Disorders

- Addison disease
- Cushing syndrome
- Hepatic failure
- Hypercarbia (chronic obstructive pulmonary disease)
- Hypernatremia
- Hyperparathyroidism
- Hyperthyroidism
- Hypoglycemia
- Hyponatremia
- Hypothyroidism
- Renal failure
- Volume depletion

Miscellaneous Conditions

- Fecal impaction
- Hospitalization
- Impaired hearing or vision

Courtesy of Gary W. Small, M.D.

The common signs and symptoms of depressive disorders include reduced energy and concentration, sleep problems (especially early morning awakening and multiple awakenings), decreased appetite, weight loss, and

somatic complaints. The presenting symptoms may be different in older depressed patients from those seen in younger adults because of an increased emphasis on somatic complaints in older persons. Older persons are particularly vulnerable to major depressive episodes with melancholic features, characterized by depression, hypochondriasis, low self-esteem, feelings of worthlessness, and self-accusatory trends (especially about sex and sinfulness) with paranoid and suicidal ideation. A geriatric depression scale is given in Table 25-5.

Cognitive impairment in depressed geriatric patients is referred to as depression-related cognitive dysfunction (sometimes misleadingly called pseudodementia), which can be confused easily with nonreversible dementias. In common dementias such as that caused by Alzheimer disease, intellectual performance usually is global, and impairment is consistently poor; in depression-related cognitive dysfunction, deficits in attention and concentration are variable. Compared with patients who have neurodegenerative dementias, patients with depression-related cognitive dysfunction are less likely to have language impairment and to confabulate; when uncertain, they are more likely to say "I don't know"; and their memory difficulties are more limited to free recall than to recognition on cued recall tests. Depression-related cognitive dysfunction occurs in about 15 percent of depressed older patients, and 25 to 50 percent of patients with neurodegenerative dementias are depressed.



Table 25-5
Geriatric Depression Scale (Short Version)

Answers indicating depression are boldfaced. Each answer counts one point; scores greater than 5 indicate probable depression.

1. Are you basically satisfied with your life?	Yes/**No**
2. Have you dropped many of your activities and interests?	**Yes**/No
3. Do you feel that your life is empty?	**Yes**/No
4. Do you often get bored?	**Yes**/No
5. Are you in good spirits most of the time?	Yes/**No**

6. Are you afraid that something bad is going to happen to you?	**Yes**/No
7. Do you feel happy most of the time?	Yes/**No**
8. Do you often feel helpless?	**Yes**/No
9. Do you prefer to stay at home, rather than going out and doing new things?	**Yes**/No
10. Do you feel you have more problems with memory than most?	**Yes**/No
11. Do you think it is wonderful to be alive now?	Yes/**No**
12. Do you feel pretty worthless the way you are now?	**Yes**/No
13. Do you feel full of energy?	Yes/**No**
14. Do you feel that your situation is hopeless?	**Yes**/No
15. Do you think that most people are better off than you are?	**Yes**/No

Special Instructions. The scale can be used as a self-rating or observer-rated metric. It has also been used as an observer-rated scale in mildly demented subjects.

From Yesavage JA. Geriatric depression scale. *Psychopharmacol Bull*. 1988;24:709, with permission.

Schizophrenia

Schizophrenia usually begins in late adolescence or young adulthood and persists throughout life. Although the first episodes diagnosed after age 65 are rare, a late-onset type beginning after age 45 has been described. Women are more likely to have a late onset of schizophrenia than men. Another difference between early-onset and late-onset schizophrenia is the greater prevalence of paranoid schizophrenia in the late-onset type. About 20 percent of persons with schizophrenia show no active symptoms by age 65; 80

percent show varying degrees of impairment. Psychopathology becomes less marked as patient's age.

The residual type of schizophrenia occurs in about 30 percent of persons with schizophrenia. It primarily presents with negative symptoms, including emotional blunting, social withdrawal, eccentric behavior, and illogical thinking. Delusions and hallucinations are uncommon. Because most persons with residual schizophrenia cannot care for themselves, long-term hospitalization is required.

Older persons with schizophrenic symptoms respond reasonably well to antipsychotic drugs. Medication must be administered judiciously, and lower-than-usual dosages often are useful for older adults.

Delusional Disorder

The age of onset of delusional disorder usually is between ages 40 and 55, but it can occur at any time during the geriatric period. Delusions can take many forms; the most common are persecutory—patients believe that they are being spied on, followed, poisoned, or harassed in some way. Persons with delusional disorder may become violent toward their supposed persecutors. Some persons lock themselves in their rooms and live reclusive lives. Somatic delusions, in which persons believe they have a fatal illness, also can occur in older persons. In one study of persons older than 65 years of age, pervasive persecutory ideation was present in 4 percent of persons sampled.

Among those who are vulnerable, delusional disorder can occur under physical or psychological stress. The death of a spouse can precipitate it, loss of a job, retirement, social isolation, adverse financial circumstances, debilitating medical illness or surgery, visual impairment, and deafness. Delusions also can accompany other disorders—such as dementia of the Alzheimer type, alcohol use disorders, schizophrenia, depressive disorders, and bipolar I disorder—which need to be ruled out. Delusional syndromes also can result from prescribed medications or be early signs of a brain tumor. The prognosis is fair to good in most cases; the best results are achieved through a combination of psychotherapy and pharmacotherapy.

Persecutory delusions characterize a late-onset delusional disorder called *paraphrenia*. It develops over several years and is not associated with

dementia. Some workers believe that the disorder is a variant of schizophrenia that first becomes manifest after age 60. Patients with a family history of schizophrenia show an increased rate of paraphrenia.

Anxiety Disorders

Anxiety disorders usually begin in early or middle adulthood, but some appear for the first time after age 60. The initial onset of panic disorder in older persons is rare but can occur. The ECA study determined that the 1-month prevalence of anxiety disorders in persons age 65 and older is 5.5 percent. By far, the most common disorders are phobias (4 to 8 percent). The rate for panic disorder is 1 percent.

The signs and symptoms of phobia in older adults are less severe than those that occur in younger persons, but the effects are equally, if not more, debilitating for older patients. Existential theories help explain anxiety when no specifically identifiable stimulus exists for a chronically anxious feeling. Older persons must come to grips with death. The person may deal with the thought of death with a sense of despair and anxiety, rather than with equanimity and Erikson's "sense of integrity." The fragility of the autonomic nervous system in older persons may account for the development of anxiety after a significant stressor. Because of concurrent physical disability, older persons react more severely to PTSD than younger persons.

Obsessive-Compulsive Disorders

Obsessions and compulsions may appear for the first time in older adults, although older adults with obsessive-compulsive disorder (OCD) usually had demonstrated evidence of the disorder (e.g., being orderly, perfectionistic, punctual, parsimonious) when they were younger. When symptomatic, patients become excessive in their desire for orderliness, rituals, and sameness. They may become generally inflexible and rigid and have compulsions to check things again and again. OCD (in contrast to obsessive-compulsive personality disorder) is characterized by ego-dystonic rituals and obsessions and may begin late in life.

Somatic Symptom Disorders

Disorders characterized by physical symptoms resembling medical diseases

are relevant to geriatric psychiatry because somatic complaints are common among older adults. More than 80 percent of persons over 65 years of age have at least one chronic disease—usually arthritis or cardiovascular problems. After age 75, 20 percent have diabetes, and an average of four diagnosable chronic illnesses that require medical attention.

Hypochondriasis is common in persons over 60 years of age, although the peak incidence is in those 40 to 50 years of age. The disorder usually is chronic, and the prognosis guarded. Repeated physical examinations help reassure patients that they do not have a fatal illness, but invasive and high-risk diagnostic procedures should be avoided unless medically indicated.

Telling patients that their symptoms are imaginary is counterproductive and usually engenders resentment. Clinicians should acknowledge that the complaint and pain are real to the patient, and a psychological or psychopharmacological approach to the problem is indicated.

Alcohol and Other Substance Use Disorder

Older adults with alcohol dependence usually give a history of excessive drinking that began in young or middle adulthood. They usually are medically ill, primarily with liver disease, and are either divorced, widowed, or are men who never married. Many have arrest records and are numbered among homeless persons. A large number have a chronic dementing illness, such as Wernicke encephalopathy or Korsakoff syndrome. Of nursing home patients, 20 percent have alcohol dependence.

Overall, alcohol and other substance use disorders account for 10 percent of all emotional problems in older persons, and dependence on such substances as hypnotics, anxiolytics, and narcotics is more common in old age than is generally recognized. Substance-seeking behavior characterized by crime, manipulativeness, and antisocial behavior is rarer in older than in younger adults. Older patients may abuse anxiolytics to allay chronic anxiety or to try to help sleep. The maintenance of chronically ill cancer patients with narcotics prescribed by a physician produces dependence, but the need to provide pain relief takes precedence over the possibility of narcotic dependence and is justified.

The clinical presentation of older patients with alcohol and other substance use disorders varies and includes confusion, poor personal hygiene,

depression, malnutrition, and the effects of exposure and falls. The sudden onset of delirium in older persons hospitalized for medical illness is most often caused by alcohol withdrawal. Alcohol abuse also should be considered in older adults with chronic gastrointestinal (GI) problems.

Older persons may misuse over-the-counter substances, or nicotine and caffeine. Over-the-counter analgesics are used by 35 percent of older persons, and 30 percent use laxatives. Unexplained GI, psychological, and metabolic problems should alert clinicians to over-the-counter substance abuse.

Sleep Disorders

Advanced age is the single most important factor associated with the increased prevalence of sleep disorders. Sleep-related phenomena reported more frequently by older than by younger adults are sleeping problems, daytime sleepiness, daytime napping, and the use of hypnotic drugs. Clinically, older persons experience higher rates of breathing-related sleep disorder and medication-induced movement disorders than younger adults.

In addition to altered regulatory and physiologic systems, the causes of sleep disturbances in older persons include primary sleep disorders, other mental disorders, general medical disorders, and social and environmental factors. Among the primary sleep disorders, dyssomnias are the most frequent, mainly primary insomnia, nocturnal myoclonus, restless legs syndrome, and sleep apnea. Of the parasomnias, rapid eye movement (REM) sleep behavior disorder occurs almost exclusively among older men. The conditions that commonly interfere with sleep in older adults also include pain, nocturia, dyspnea, and heartburn. The lack of a daily structure and social or vocational responsibilities contributes to poor sleep.

As a result of the decreased length of their daily sleep–wake cycle, older persons without daily routines, especially patients in nursing homes, may experience an advanced sleep phase, in which they go to sleep early and awaken during the night.

Even modest amounts of alcohol can interfere with the quality of sleep and can cause sleep fragmentation and early morning awakening. Alcohol can also precipitate or aggravate obstructive sleep apnea. Many older persons use alcohol, hypnotics, and other CNS depressants to help them fall asleep, but data show that these persons experience more early morning awakening than

trouble falling asleep. When prescribing sedative-hypnotic drugs for older persons, clinicians must monitor the patients for unwanted cognitive, behavioral, and psychomotor effects, including memory impairment (anterograde amnesia), residual sedation, rebound insomnia, daytime withdrawal, and unsteady gait.

Changes in sleep structure among persons over 65 years of age involve both REM sleep and nonrapid eye movement (NREM) sleep. The REM changes include the redistribution of REM sleep throughout the night, more REM episodes, shorter REM episodes, and less total REM sleep. The NREM changes include the decreased amplitude of delta waves, a lower percentage of stages 3 and 4 sleep, and a higher percentage of stages 1 and 2 sleep. Also, older persons experience increased awakening after sleep onset.

Altered timing and consolidation of the sleep account for much of the deterioration in the quality of sleep. This is due, in part, to decreasing melatonin and alternated responses to the adenosine system with age. With advanced age, persons have a lower amplitude of circadian rhythms, a 12-hour sleep-propensity rhythm, and shorter circadian cycles.

SUICIDE RISK

Elderly persons have a higher risk of suicide than any other population. The suicide rate for white men over the age of 65 is five times higher than that of the general population. One-third of elderly persons report loneliness as the principal reason for considering suicide. Approximately 10 percent of elderly individuals with suicidal ideation report financial problems, poor medical health, or depression as reasons for suicidal thoughts. Suicide victims differ demographically from individuals who attempt suicide. About 60 percent of those who commit suicide are men; 75 percent of those who attempt suicide are women. Suicide victims, as a rule, use guns or hang themselves, whereas 70 percent of suicide attempters take a drug overdose, and 20 percent cut or slash themselves. Psychological autopsy studies suggest that most elderly persons who commit suicide have had a psychiatric disorder, most commonly depression. Psychiatric disorders of suicide victims, however, often do not receive medical or psychiatric attention. More elderly suicide victims are widowed, and fewer are single, separated, or divorced than is true of younger adults. Violent methods of suicide are more common in the elderly, and

alcohol use and psychiatric histories appear to be less frequent. The most common precipitants of suicide in older individuals are physical illness and loss, whereas problems with employment, finances, and family relationships are more frequent precipitants in younger adults. Many elderly persons who commit suicide communicate their suicidal thoughts to family or friends before the act of suicide.

Older patients with significant medical illnesses or a recent loss should be evaluated for depressive symptomatology and suicidal ideation or plans. Thoughts and fantasies about the meaning of suicide and life after death may reveal information that the patient cannot share directly. There should be no reluctance to question patients about suicide because no evidence indicates that such questions increase the likelihood of suicidal behavior.

OTHER CONDITIONS OF OLD AGE

Vertigo

Feelings of vertigo or dizziness, a common complaint of older adults, cause many older adults to become inactive because they fear falling. The causes of vertigo vary and include anemia, hypotension, cardiac arrhythmia, cerebrovascular disease, basilar artery insufficiency, middle ear disease, acoustic neuroma, benign postural vertigo, and Ménière disease. Most cases of vertigo have a vital psychological component, and clinicians should ascertain any secondary gain from the symptom. The overuse of anxiolytics can cause dizziness and daytime somnolence. Treatment with meclizine, 25 to 100 mg daily, has been successful in many patients with vertigo.

Syncope

The sudden loss of consciousness associated with syncope results from a reduction of cerebral blood flow and brain hypoxia. A thorough medical workup is required to rule out potential causes. Causes of syncope are listed in Table 25-6.

Hearing Loss

About 30 percent of persons over age 65 have significant hearing loss (presbycusis). After age 75, that figure rises to 50 percent. The causes vary.

Clinicians should be sensitive to hearing loss in patients who complain they can hear but cannot understand what is being said or who asks that questions be repeated. Most elderly persons with hearing loss can be treated with hearing aids.

Elder Abuse

An estimated 10 percent of persons above 65 years of age are abused. The American Medical Association defines elder abuse as "an act or omission which results in harm or threatened harm to the health or welfare of an elderly person." Mistreatment includes abuse and neglect—physically, psychologically, financially, and materially. Sexual abuse does occur. Acts of omission include withholding food, medicine, clothing, and other necessities.



Table 25-6
Causes of Syncope

Cardiac Disorders
- Anatomical/valvular
  - Aortic stenosis
  - Mitral prolapse and regurgitation
  - Hypertrophic cardiomyopathy
  - Myxoma
- Electrical
  - Tachyarrhythmia
  - Bradyarrhythmia
  - Heart block
  - Sick sinus syndrome
- Functional
  - Ischemia and infarct

Situational Hypotension
- Dehydration (diarrhea, fasting)
- Orthostatic hypotension
- Postprandial hypotension
- Micturition, defecation, coughing, swallowing

Abnormal Cardiovascular Reflexes
- Carotid sinus syndrome
- Vasovagal syncope

Drugs
- Vasodilators
- Calcium channel blockers
- Diuretics
- β-Blockers

Central Nervous System Abnormalities
- Cerebrovascular insufficiency
- Seizures

Metabolic Abnormalities
- Hypoxemia
- Hypoglycemia or hyperglycemia
- Anemia

Pulmonary Disorders
- Chronic obstructive pulmonary disease
- Pneumonia
- Pulmonary embolus

Family conflicts and other problems often underlie elder abuse. The victims tend to be very old and frail. They often live with their assailants, who may be financially dependent on the victims. Both the victim and the perpetrator tend to deny or minimize the presence of abuse. Interventions include providing legal services, housing, and medical, psychiatric, and social services.

SPOUSAL BEREAVEMENT

Demographic data suggest that 51 percent of women and 14 percent of men over the age of 65 will be widowed at least once. Spousal loss is among the most stressful of all life experiences. As a group, older adults appear to have a more favorable outcome than expected following the death of a spouse. Depressive symptoms peak within the first few months after death but decline

significantly within a year. A relationship exists between spousal loss and subsequent mortality. Elderly survivors of spouses who committed suicide are especially vulnerable, as are those with psychiatric illness.

PSYCHOPHARMACOLOGICAL TREATMENT OF GERIATRIC DISORDERS

Specific guidelines should be followed regarding the use of all drugs in older adults. A pretreatment medical evaluation is essential, including an electrocardiogram (ECG). It is especially useful to have the patient or a family member bring in all currently used medications because multiple drug use could be contributing to the symptoms.

Most psychotropic drugs should be given in equally divided doses three or four times over 24 hours. Older patients may not be able to tolerate a sudden rise in drug blood levels resulting from one sizeable daily dose. Any changes in blood pressure and pulse rate and other side effects should be watched. For patients with insomnia, however, giving the significant portion of an antipsychotic or antidepressant at bedtime takes advantage of its sedating and soporific effects. Liquid preparations are useful for older patients who cannot or will not swallow tablets. Clinicians should frequently reassess all patients to determine the need for maintenance medication, changes in dosage, and development of adverse effects. If a patient is taking psychotropic drugs at the time of the evaluation, the clinician should discontinue these medications, if possible, and, after a washout period, reevaluate the patient during a drug-free baseline state.

Adults over 65 years of age use the most significant number of medications of any age group; 25 percent of all prescriptions are written for them. Adverse drug reactions caused by medications result in the hospitalization of nearly 250,000 persons in the United States each year. Psychiatric drugs are among the most commonly prescribed, along with cardiovascular and diuretic medications; 40 percent of all hypnotics dispensed in the United States each year are to those older than 75 years of age, and 70 percent of older persons use over-the-counter medications, compared with only 10 percent of young adults.

Principles

The primary goals of the pharmacologic treatment of older persons are to improve the quality of life, maintain persons in the community, and delay or avoid their placement in nursing homes. The individualization of dosage is the basic tenet of geriatric psychopharmacology.

Alterations in drug dosages are required because of the physiologic changes that occur as we age. Renal disease is associated with a decreased renal clearance of drugs; liver disease results in a decreased ability to metabolize drugs; cardiovascular disease and reduced cardiac output can affect both renal and hepatic drug clearance. GI disease and decreased gastric acid secretion also influence drug absorption. As a person ages, the ratio of lean to fat body mass also changes. With normal aging, lean body mass decreases, and body fat increases. Changes in the ratio of lean to fat body mass that accompany aging affect the distribution of drugs. Many lipid-soluble psychotropic drugs are distributed more widely in fat than in lean tissue, so a drug's action can be unexpectedly prolonged in older persons. Similarly, changes in end-organ or receptor-site sensitivity must be taken into account. In older persons, the increased risk of orthostatic hypotension from psychotropic drugs is related to reduced functioning of blood pressure–regulating mechanisms.

As a general rule, the lowest possible dose should be used to achieve the desired therapeutic response. Clinicians must know the pharmacodynamics, pharmacokinetics, and biotransformation of each drug prescribed and the effects of the interaction of the drug with other drugs that a patient is taking. An adage in geriatric medicine regarding the use of drugs is: Start low, go slow.

PSYCHOTHERAPY FOR GERIATRIC PATIENTS

The standard psychotherapeutic interventions—such as insight-oriented psychotherapy, supportive psychotherapy, cognitive therapy, group therapy, and family therapy—should be available to geriatric patients. According to Sigmund Freud, persons older than 50 years are not suited for psychoanalysis because their mental processes lack elasticity. In the view of many who followed Freud, however, psychoanalysis is possible after age 50. Advanced age certainly limits the plasticity of the personality, but as Otto Fenichel stated, "It does so in varying degrees and at very different ages so that no

general rule can be given." Insight-oriented psychotherapy may help remove a specific symptom, even in older persons. It is of most benefit when patients have possibilities for libidinal and narcissistic gratification, but it is contraindicated if it would bring only the insight that life has been a failure and that the patient has no opportunity to make up for it.

Common age-related issues in therapy involve the need to adapt to recurrent and diverse losses (e.g., the deaths of friends and loved ones), the need to assume new roles (e.g., the adjustment to retirement and the disengagement from previously defined roles), and the need to accept mortality. Psychotherapy helps older persons to deal with these issues and the emotional problems surrounding them and to understand their behavior and the effects of their behavior on others. In addition to improving interpersonal relationships, psychotherapy increases self-esteem and self-confidence, decreases feelings of helplessness and anger, and improves the quality of life.

Psychotherapy helps relieve tensions of biologic and cultural origins and helps older persons work and play within the limits of their functional status and as determined by their past training, activities, and self-concept in society. In patients with impaired cognition, psychotherapy can produce remarkable gains in both physical and mental symptoms. In one study conducted in a nursing home, 43 percent of the patients receiving psychotherapy showed less urinary incontinence, improved gait, greater mental alertness, improved memory, and better hearing than before psychotherapy.

Therapists must be more active, supportive, and flexible in conducting therapy with older than with younger adults, and they must be prepared to act decisively at the first sign of an incapacity that requires the active involvement of another physician, such as an internist or that requires consulting with, or enlisting the aid of, a family member.

Older persons usually seek therapy for a therapist's unqualified and unlimited support, reassurance, and approval. Patients often expect a therapist to be all-powerful, all-knowing, and able to effect a magical cure. Most patients eventually recognize that the therapist is human and that they are engaged in a collaborative effort. In some cases, however, the therapist may have to assume the idealized role, especially when the patient is unable or unwilling to test reality effectively. With the help of the therapist, the patient deals with problems that had been avoided previously. As the therapist offers

direct encouragement, reassurance, and advice, the patient's self-confidence increases as conflicts are resolved.

▲ 25.3 Psychiatric Emergencies

A psychiatric emergency is any disturbance in thoughts, feelings, or actions for which immediate therapeutic intervention is necessary. For a variety of reasons—such as the growing incidence of violence, the increased appreciation of the role of medical disease in altered mental status, and the epidemic of alcoholism and other substance use disorders—the number of emergency psychiatry patients is on the rise. The widening scope of emergency psychiatry goes beyond general psychiatric practice to include such specialized problems as the abuse of substances, children, and spouses; violence in the form of suicide, homicide, and rape; and such social issues as homelessness, aging, and competence. The emergency psychiatrist must be up to date on medicolegal issues and managed care. This section provides an overview of psychiatric emergencies in general and in adults in particular. Psychiatric emergencies in children is covered later in this chapter.

TREATMENT SETTINGS

Nonpsychiatrists do most emergency psychiatric evaluations in a general medical emergency room setting, but specialized psychiatric services are increasingly favored. Regardless of the type of setting, an atmosphere of safety and security must prevail. An adequate number of staff members—including psychiatrists, nurses, aides, and social workers—must be present at all times. Additional personnel to help out in times of overcrowding should be available. Specific responsibilities, such as the use of restraints, should be clearly defined and practiced by the entire emergency team. Clear communication and lines of authority are essential. The organization of the staff into multidisciplinary teams is desirable.

Children and young adolescents are best served in a pediatric setting. Unless there is a risk of behavioral problems or of their leaving the hospital against advice, they need not be sent to the adult psychiatric emergency service.

Immediate access to the medical emergency department and appropriate diagnostic services is necessary because one-third of medical conditions present with psychiatric manifestations. The full spectrum of psychopharmacological options should be available to the psychiatrist.

Violence in the emergency service cannot be condoned or tolerated. The code of conduct expected of staff members and patients must be posted and understood from the time of the patient's arrival in the emergency room. Security is best managed as a clinical issue by the clinical staff, not by law enforcement personnel. Whenever possible, agitated and threatening patients should be sequestered from the nonagitated. Seclusion and restraint rooms should be located close to the nursing station for close observation.

The entire staff must understand that patients in physical and emotional distress are fragile and that various expectations and fantasies, often unrealistic, influence their responses to treatment. For example, a man with impaired reality testing who is brought in by the police against his will may not understand that the clinician is interested in helping him. Other patients, influenced by previous unsatisfactory treatment experiences, may be hostile. Some patients believe that psychiatrists can read minds or are only interested in admitting patients to lock them away. Such people see little point in openly discussing their problems. Many people have an inaccurate understanding of their rights as patients. All clinical interventions must take those expectations and attitudes into account to minimize the possibility of misunderstanding and consequent problems.

EPIDEMIOLOGY

Psychiatric emergency rooms are used equally by men and women and more by single than by married persons. About 20 percent of these patients are suicidal, and about 10 percent are violent. The most common diagnoses are depressive disorders, manic episodes, schizophrenia, and alcohol dependence. About 40 percent of all patients seen in psychiatric emergency rooms require hospitalization. Most visits occur during the night hours, but usage difference is not based on the day of the week or the month of the year. Contrary to popular belief, studies have not found that the use of psychiatric emergency rooms increases during the full moon or the Christmas season.

EVALUATION

The primary goal of an emergency psychiatric evaluation is the timely assessment of the patient in crisis. To that end, the physician must make an initial diagnosis, identify the precipitating factors and immediate needs, and begin treatment or refer the patient to the most appropriate treatment setting. Given the unpredictable nature of emergency medicine work, with many patients presenting both physical and emotional complaints, and because of the limited space and the competition for ancillary services, a pragmatic approach to the patient is required. Sometimes, moving the patient out of the emergency department into the most appropriate diagnostic or treatment setting is best for the patient. Medical emergencies are generally better managed elsewhere in the system. Keeping the number of emergency patients in one place to a minimum reduces the chance of agitation and violence.

The standard psychiatric interview—consisting of a history, a mental status examination, and, when appropriate and depending on the rules of the emergency department, a full physical examination and ancillary tests—is the cornerstone of the emergency psychiatry evaluation. The emergency department psychiatrist, however, must be ready to introduce modifications as needed. For example, the emergency psychiatrist may have to structure the interview with a rambling manic patient, medicate or restrain an agitated patient, or forgo the usual rules of confidentiality to assess an adolescent's risk of suicide. In general, any strategy introduced in the emergency department to accomplish the goal of assessing the patient is considered consistent with good clinical practice as long as the rationale for the strategy is documented in the medical record.

What constitutes a psychiatric emergency is highly subjective. The emergency department has increasingly come to serve as an admitting area, a holding area, a detoxification center, and a private medical office. Such medical conditions as head traumas, acute intoxications, withdrawal states, and AIDS encephalopathies may present with acute psychiatric manifestations. The emergency psychiatrists must rapidly assess and distinguish the genuine emergency psychiatric patients from those who are less acutely ill and from nonpsychiatric emergencies. A triage system using psychiatrists, nurses, and psychiatric social workers is an efficient and effective way to identify emergency, urgent, and nonurgent patients, who can

then be prioritized for care (Fig. 25-1).

In one model, every patient who comes to the emergency department is assessed by a triage nurse on arrival to ascertain the patient's chief complaint, clinical condition, and vital signs. The psychiatrist then briefly meets with the patient and other significant people involved in the case—family members, emergency medical service technicians, and police—to assign the patient to one of the three categories—emergency, urgent, and nonurgent—or to refer the patient to an appropriate treatment setting, such as the medical emergency medicine service. Having a senior clinician perform that task ensures rapid identification of the most urgent and troublesome cases, an appropriate allocation of resources, and an answer to the most common question heard in the emergency department: "When am I going to see a doctor?"

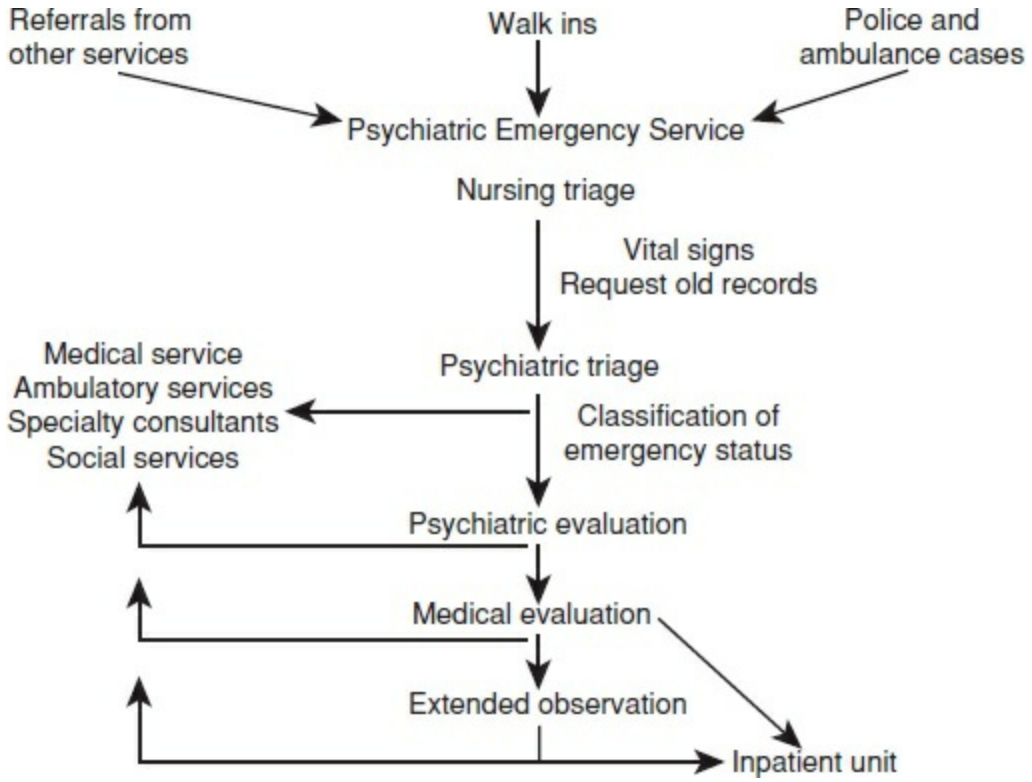


FIGURE 25-1
Evaluation and treatment of psychiatric emergencies.

The psychiatrist then assigns clinical responsibility for each patient to the appropriate personnel. As the evaluation often stretches over more than one shift, a careful procedure to transfer responsibility and to pass along information from shift to shift must be built into the system by using visual,

oral, and written communications. A request for old records should be made automatically for every patient who is evaluated in the emergency department. Each emergency should be judged on its own merits. However, information from previous records and workers in the field and family members can be of crucial importance in assessing patients, especially patients who are psychotic, frightened, or otherwise unable or unwilling to cooperate in giving a good history.

The psychiatrist should have access to multilingual staff and other translation services. The use of the patient's friends or family members as translators is not desirable because of the possibility of unconscious or deliberate denial or distortion of the clinical picture stemming from their involvement with the patient.

An initial assessment of the patient's total biopsychosocial needs is optimal, but the patient's emergency status, other patients, waiting to be seen, and the constraints of the emergency setting often make such a full assessment a moot point. At a minimum, the emergency evaluation should address the following five questions before any disposition is decided on: (1) Is it safe for the patient to be in the emergency department? (2) Is the problem physiologic, functional, or a combination? (3) Is the patient psychotic? (4) Is the patient suicidal or homicidal? (5) To what degree is the patient capable of self-care? Table 25-7 provides a general strategy for evaluating patients.

Patient Safety

Physicians should consider the question of the patient's safety before evaluating every patient. The answer must address the issues of the emergency department physical layout, staffing patterns and communication, and patient population. Psychiatrists must then take stock of themselves: Are they in the proper frame of mind to conduct an evaluation? Do any issues in the case spark countertransference reactions? The self-assessment should go on throughout the evaluation. The physical and emotional safety of the patient takes priority over all other considerations. If verbal interventions fail or are contraindicated, the use of medication or restraints must be considered and, if necessary, ordered. Careful attention to the possible outbreak of agitation or disruptive behavior beyond acceptable limits is often the best

insurance against untoward occurrences.



Table 25-7
General Strategy in Evaluating the Patient

I. Self-protection

A. Know as much as possible about the patients before meeting them.

B. Leave physical restraint procedures to those who are trained.

C. Be alert to risks of impending violence.

D. Attend to the safety of the physical surroundings (e.g., door access, room objects).

E. Have others present during the assessment if needed.

F. Have others in the vicinity.

G. Attend to developing an alliance with the patient (e.g., do not confront or threaten patients with persecutory psychoses).

II. Prevent harm

A. Prevent self-injury and suicide. Use whatever methods are necessary to prevent patients from hurting themselves during the evaluation.

B. Prevent violence toward others. During the evaluation, briefly assess the patient for the risk of violence. If the risk is deemed significant, consider the following options:

1. Inform the patient that violence is not acceptable.

2. Approach the patient in a nonthreatening manner.

3. Reassure, calm, or assist the patient’s reality testing.

4. Offer medication.

5. Inform the patient that restraint or seclusion will be used if necessary.

6. Have teams ready to restrain the patient.

7. When patients are restrained, always closely observe them, and frequently check their vital signs. Isolate restrained patients from surrounding agitating stimuli. Immediately plan a further approach—medication, reassurance, medical evaluation.

III. Rule out nonpsychiatric etiologies.

IV. Rule out impending psychosis.

Medical or Psychiatric?

The most important question for the emergency psychiatrist to address is whether the problem is medical, psychiatric, or both. Medical conditions—such as diabetes mellitus, thyroid disease, acute intoxications, withdrawal states, AIDS, and head traumas—can present with prominent mental status changes that mimic common psychiatric illnesses. Such conditions may be life-threatening if not treated promptly. Generally, the treatment of medical illness is more definitive, and the prognosis is better than for a functional psychiatric disorder. The psychiatrist must consider all casual possibilities.

Once patients are labeled psychiatric, their complaints may not be taken seriously by nonmental health professionals. However, such patients' conditions may deteriorate. Because of such factors as deinstitutionalization, homelessness, and chronic alcoholism, the mentally ill are at significant risk of tuberculosis, vitamin deficiencies, and other easily overlooked but easily treated conditions. Symptoms such as paranoia, internal preoccupation, and acute psychosis can make a routine medical diagnosis exceedingly tricky. Each patient must be assessed for the possibility that a nonpsychiatric medical illness is combined with an underlying psychiatric illness. A young man who comes to the emergency department intoxicated or in alcohol withdrawal two or three times a month may one day come with a subdural hematoma as a result of a fall. Table 25-8 lists feature that point to a medical cause of a mental disorder.



Table 25-8
Features that Point to a Medical Cause of a Mental

Disorder
Acute onset (within hours or minutes, with prevailing symptoms)
First episode
Older age
Current medical illness or injury
Significant substance abuse
Nonauditory disturbances of perception
Neurologic symptoms—loss of consciousness, seizures, head injury, change in headache pattern, change in vision
Classic mental status signs—diminished alertness, disorientation, memory impairment, impairment in concentration and attention, dyscalculia, concreteness
Other mental status signs—speech, movement, or gait disorders
Constructional apraxia—difficulties in drawing clock, cube, intersecting pentagons, Bender–Gestalt test

SPECIFIC INTERVIEW SITUATIONS

Psychosis

Whether the patient is psychotic refers less to diagnosis and more to the severity of symptoms and life disruption. The patient's degree of withdrawal from objective reality, level of affectivity, intellectual functioning, and degree of regression are other important parameters. Impairment in any of those areas may lead to difficulties in conducting an evaluation. Agitated, assaultive behavior or failure to comply with treatment recommendations may also result. A paranoid, hypervigilant patient may misperceive a staff member's offer of help as an attack and may lash out in self-defense. Command auditory hallucinations may cause a patient to deny symptoms and to throw prescriptions in the garbage immediately after leaving the emergency department. The psychiatrist should be alert to the complications that can arise with patients whose reality testing is impaired, and the psychiatrist should modify the approach accordingly.

All communication with patients must be straightforward. All clinical interventions should be briefly explained using language the patient can

understand. Psychiatrists should not assume that the patient trusts or believes them or even wants their help. Clinicians must be prepared to structure or to terminate an interview to limit the potential for agitation and regression.

Depression and Potentially Suicidal Patients

The clinician should always ask about suicidal ideas as part of every mental status examination, especially if the patient is depressed. The patient may not realize that such symptoms as waking during the night and increased somatic complaints are related to depressive disorders. The patient should be asked directly, "*Are you or have you ever been suicidal?*" "*Do you want to die?*" "*Do you feel so bad that you might hurt yourself?*" Eight of 10 persons who eventually kill themselves give warnings of their intent. If the patient admits to a plan of action, that is a particularly dangerous sign. If a patient who has been threatening suicide becomes quiet and less agitated than before, that may be an ominous sign. The clinician should be primarily concerned with the factors listed in Table 25-9.



Table 25-9
History, Signs, and Symptoms of Suicidal Risk

1. Previous attempt or fantasized suicide
2. Anxiety, depression, exhaustion
3. Availability of means of suicide
4. Concern for effect of suicide on family members
5. Verbalized suicidal ideation
6. Preparation of a will, resignation after agitated depression
7. Proximal life crisis, such as mourning or impending surgery
8. Family history of suicide
9. Pervasive pessimism or hopelessness

A suicide note, a family history of suicide, or previous suicidal behavior on the part of the patient increases the risk of suicide. Evidence of impulsivity or pervasive pessimism about the future also places the patient at risk. If the physician decides that the patient is in imminent risk for suicidal behavior, the patient must be hospitalized or otherwise protected. A difficult situation

arises when the risk does not seem to be immediate, but the potential for suicide is present as long as the patient remains depressed. If the psychiatrist decides not to hospitalize the patient immediately, the doctor should insist that the patient promise to call whenever the suicidal pressure mounts.

Violent Patients

Patients may be violent for many reasons, and the interview with a violent patient must attempt to ascertain the underlying cause of violent behavior because cause determines intervention. The differential diagnosis of violent behavior includes psychoactive substance-induced organic mental disorder, antisocial personality disorder, catatonia, medical infections, cerebral neoplasms, decompensating obsessive-compulsive personality disorder, dissociative disorders, impulse control disorders, sexual disorders, alcohol idiosyncratic intoxication, delusional disorder, paranoid personality disorder, schizophrenia, temporal lobe epilepsy, bipolar disorder, and uncontrollable violence secondary to interpersonal stress. The psychiatric interview must include questions that attempt to sort out the differential for violent behavior and questions directed toward the prediction of violence.

The best predictors of violent behavior are (1) excessive alcohol intake; (2) a history of violent acts, with arrests or criminal activity; and (3) a history of childhood abuse. Table 25-10 lists some of the most significant factors in assessing and predicting violence.

Rape and Sexual Abuse

Rape is the forceful coercion of an unwilling victim to engage in a sexual act, usually sexual intercourse, although anal intercourse and fellatio can also be acts of rape. As with other acts of violence, rape is a psychiatric emergency that requires immediate, appropriate intervention. Rape victims may suffer sequelae that persist for a lifetime. Rape is a life-threatening experience, and the victim has almost always been threatened with physical harm, often with a weapon. In addition to rape, other forms of sexual abuse include genital manipulation with foreign objects, infliction of pain, and forced sexual activity.

Most rapists are male, and most victims are female. Male rape does occur, however, often in institutions where men are detained (e.g., prisons). Women

between the ages of 16 and 24 years are in the highest risk category, but female victims as young as 15 months and as old as 82 years have been raped. More than a third of all rapes are committed by rapists known to the victim, 7 percent by close relatives. About one-quarter of rapes involve more than one rapist (gang rape).



Table 25-10
Assessing and Predicting Violent Behavior

1. Signs of impending violence
    a. Very recent acts of violence, including property violence
    b. Verbal or physical threats (menacing)
    c. Carrying weapons or other objects that may be used as weapons (e.g., forks, ashtrays)
    d. Progressive psychomotor agitation
    e. Alcohol or drug intoxication
    f. Persecutory ideation in a psychotic patient
    g. Command violent auditory hallucinations—some but not all patients are at high risk
    h. Mental disorders due to a medical condition, global or with frontal lobe findings; less commonly with temporal lobe findings (controversial)
    i. Patients with catatonic excitement
    j. Certain patients with mania
    k. Certain patients with agitated depression
    l. Personality disorder patients prone to rage, violence, or impulse dyscontrol
2. Assess the risk of violence
    a. Consider violent ideation, wish, intention, plan, availability of means, implementation of plan, wish for help.
    b. Consider demographics—sex (male), age (15–24), socioeconomic status (low), social supports (few).
    c. Consider past history: violence, nonviolent antisocial acts, impulse dyscontrol (e.g., gambling, substance abuse, suicide or self-injury, psychosis).
    d. Consider overt stressors (e.g., marital conflict, real or

symbolic loss).

Typical reactions in both rape and sexual abuse victims include shame, humiliation, anxiety, confusion, and outrage. Many victims wonder whether they are partly responsible and somehow invited the assault. Victim behavior is less critical in precipitating a rape than it is in precipitating a homicide or a robbery. Rape and sexual abuse victims are often confused after the assault. Clinicians should be reassuring, supportive, and nonjudgmental. Inform the patient about the availability of medical and legal services and about rape crisis centers that provide multidisciplinary services.

If possible, a female clinician should evaluate the patient because the victim may find it easier to talk with a woman than with a man. The evaluation should take place in private. When rape or sexual abuse has not been acknowledged openly, it is usually because many victims hesitate to discuss the assault and thus avoid the topic. If the patient appears to be anxious when questioned about sexual history and avoids the discussion, it is vital to validate the patient's avoidance. Recognize that the rape victim has undergone unanticipated, life-threatening stress. It is legally and therapeutically essential to take a detailed and complete history of the attack.

With the patient's written consent, collect evidence, such as semen and pubic hair, that may be used to identify the rapist. Take photographs of the evidence, if possible. The medical record may be used as evidence in criminal proceedings; therefore, meticulous objective documentation of all aspects of the evaluation is essential.

TREATMENT OF EMERGENCIES

Psychotherapy

In an emergency psychiatric intervention, all attempts are made to help patients' self-esteem. Empathy is critical to healing in a psychiatric emergency. The acquired knowledge of how biogenetic, situational, developmental, and existential forces converge at one point to create a psychiatric emergency is tantamount to the maturation of skill in emergency psychiatry. Adjustment disorder in all age groups may result in tantrum-like outbursts of rage. These outbursts are particularly common in marital

quarrels, and police are often summoned by neighbors distressed by the sounds of a violent altercation. Such family quarrels should be approached with caution because they may be complicated by alcohol use and the presence of dangerous weapons. The warring couple frequently turns their combined fury on an unwary outsider. Wounded self-esteem is a significant issue, and clinicians must avoid patronizing or contemptuous attitudes and try to communicate an attitude of respect and an authentic peacemaking concern.

In family violence, psychiatrists should note the unique vulnerability of selected close relatives. A wife or husband may have a curious masochistic attachment to the spouse and can provoke violence by taunting and otherwise undermining a partner's self-esteem. Such relationships often end in the murder of the provoking partner and sometimes in the suicide of the other partner—the dynamics behind most so-called suicide pacts. As with many suicidal patients, many violent patients require hospitalization and usually accept the offer of inpatient care with a sense of relief.

More than one psychotherapist or type of psychotherapy is frequently used in emergency therapy. For example, a 28-year-old man, depressed and suicidal after a colostomy for intractable colitis, whose wife was threatening to leave him because of his irritability and their constant altercations, may be referred to a psychiatrist for supportive psychotherapy and antidepressant medication, to a marital therapist with his wife to improve their marital functioning, and to a colostomy support group to learn ways of coping with a colostomy. Emergency psychiatric clinicians are pragmatic; they use every necessary mode of therapeutic intervention available to resolve the crisis and facilitate value exploration and growth, with less concern than usual about diluting a therapeutic relationship. Emergency therapy emphasizes how various psychiatric modalities act synergistically to enhance recovery.

No single approach is appropriate for all persons in similar situations. What should a doctor say to a patient and family experiencing an emergency, such as a suicide attempt or a schizophrenic break? For some, a genetic rationale helps; the information that an illness has a vital biologic component relieves some persons. For others, however, this approach underlines a lack of control and increases depression and anxiety. All feel helpless because neither the family nor the patient can alter the behavior to minimize the likelihood of recurrence. Some persons may benefit from an explanation of family or individual dynamics. Others only want someone to listen to them;

in time, they reach their understanding.

In an emergency, as in any other psychiatric situation, when a clinician does not know what to say, the best approach is to listen. Persons in crisis reveal how much they need support, denial, ventilation, and words to conceptualize the meaning of their crisis and to discover paths to resolution.

Pharmacotherapy

The primary indications for the use of psychotropic medication in an emergency room include violent or assaultive behavior, massive anxiety or panic, and extrapyramidal reactions, such as dystonia and akathisia as adverse effects of psychiatric drugs. Laryngospasm is a rare form of dystonia, and psychiatrists should be prepared to maintain an open airway with intubation if necessary.

Persons who are delusional or in a state of catatonic excitement require tranquilization. Episodic outbursts of violence may respond to antipsychotics (haloperidol, second-generation antipsychotics), benzodiazepines and antihistamines. If history suggests a seizure disorder, use clinical studies to confirm the diagnosis and evaluation to ascertain the cause. If the findings are positive, anticonvulsant therapy is initiated, or appropriate surgery is provided (e.g., in the case of a cerebral mass). Conservative measures may suffice for intoxication from drugs of abuse. Sometimes, drugs such as haloperidol (5 to 10 mg every half-hour to an hour) are needed until a patient is stabilized. Benzodiazepines may be used instead of, or in addition to, antipsychotics (to reduce the antipsychotic dosage). When a recreational drug has strong anticholinergic properties, benzodiazepines are more appropriate than antipsychotics.

Violent, struggling patients are subdued most effectively with an appropriate sedative or antipsychotic. Diazepam (5 to 10 mg) or lorazepam (2 to 4 mg) may be given slowly intravenously (IV) over 2 minutes. Clinicians must give IV medication with great care to avoid respiratory arrest. Patients who require IM medication can be sedated with haloperidol (5 to 10 mg IM). If the furor is caused by alcohol or is part of a postseizure psychomotor disturbance, the sleep produced by a relatively small amount of an IV medication may go on for hours. On awakening, patients are often entirely alert and rational and typically have complete amnesia about the violent

episode.

If the disturbance is part of an ongoing psychotic process and returns as soon as the IV medication wears off, continuous medication may be given. It is sometimes better to use small IM or oral doses at half-hour to 1-hour intervals (e.g., haloperidol, 2 to 5 mg, or diazepam, 20 mg) until the patient is controlled than to use large dosages initially, which can result in an overmedicated patient. As the disturbing behavior is brought under control, successively smaller and less frequent doses should be used. During the preliminary treatment, a patient's blood pressure and other vital signs should be monitored.

Restraints

Restraints are used when patients are so dangerous to themselves or others that they pose a severe threat that cannot be controlled in any other way. Patients may be restrained temporarily to receive medication or for long periods if we cannot use medications. Usually, patients in restraints quiet down after a time. On a psychodynamic level, such patients may even welcome the control of their impulses provided by restraints. Table 25-11 provides a summary of the use of restraints.

Disposition

In some cases, the usual option of admitting or discharging the patient is not considered optimal. Suspected toxic psychoses, brief decompensations in a patient with a personality disorder, and adjustment reactions to traumatic events, for example, may be best managed in an extended-observation setting. Allowing the patient more time in a secure environment can result in sufficient improvement or clarification of the issues to make traditional inpatient treatment unnecessary. It can also spare the patient the trauma and stigma of psychiatric admission and can free up bed space for needier patients. Crisis intervention for victims of rape and other traumas can also be done in an extended-observation setting.

When the decision is to admit the patient to the hospital, it is preferable to do so voluntarily. Allowing patients that option gives them a sense of control over their lives and of participation in the treatment decisions. Patients who meet involuntary admission criteria based on dangerousness to themselves or

others cannot leave the hospital without further review and can always be converted to involuntary status if warranted.

Table 25-11
Use of Restraints

Recommended principles for physical restraint use

- Use only after nonphysical approaches fail
- Maintain the privacy and dignity of the patient throughout the intervention
- The limitation on movement should be as nonrestrictive as safely possible
- Medical personnel should be properly trained for in how to use and monitor this intervention
- The ED should set appropriate and consistent guidelines for restraint, including
  - what treatments should be given
  - what monitoring is necessary
  - how the patient will be evaluated for ending the restraint
- The patient should be explicitly and continuously observed by the treatment team
- The use of physical restraint should be consistent with all standards and rules

Adapted from: American College of Emergency Physicians (ACEP). Use of patient restraints. Policy statement. *Ann Emerg Med*. 2014;64(5):574.

Because the initial evaluation is often inconclusive, definitive treatment is best deferred until the patient can be further assessed in the inpatient unit or the outpatient department. When the diagnosis is clear, however, and the patient's response to previous treatment is known, nothing is gained by delay. For example, a patient with chronic schizophrenia that has decompensated after discontinuing the usual regimen of antipsychotic medication is best served by a prompt resumption of treatment.

Even if patients feel comfortable coming to the emergency department in times of need, the emergency psychiatrist should always direct or redirect

them to the most appropriate treatment setting. Patients in the psychopharmacology clinic who have missed their regular appointments should be given only enough medication to sustain them until they can be seen in the clinic. Feedback to others treating them should be a matter of course.

The emergency department is often the gateway to the department of psychiatry or the general hospital. First impressions carry a great deal of weight. The kind of attention and concern showed to patients on arrival in the emergency room strongly affects how they will respond to staff members and treatment recommendations and even their treatment compliance long after they have left the emergency room.

Documentation

In the interests of good care, respect for patients' rights, cost control, and medicolegal concerns, documentation has become a central focus for the emergency physician. The medical record should convey a concise picture of the patient, highlighting all pertinent positive and negative findings. Gaps in information and their reason should be mentioned. The names and the telephone numbers of interested parties should be noted. A provisional diagnosis or differential diagnosis must be made. An initial treatment plan or recommendations should follow from the findings of the patient's history, mental status examination and other diagnostic tests, and the medical evaluation. The writing must be legible. The emergency physician has unusual latitude under the law to perform an adequate initial assessment; however, all interventions and decisions must be thought out, discussed, and documented in the patient's record.

Specific Psychiatric Emergencies

Table 25-12 outlines common psychiatric emergencies in alphabetical order. Readers are referred to the index and relevant chapters of this textbook for a thorough discussion of each disorder.

PSYCHIATRIC EMERGENCIES IN CHILDREN

Few children or adolescents seek psychiatric intervention on their own, even

during a crisis; thus, most of their emergency evaluations are initiated by parents, relatives, teachers, therapists, physicians, and child protective service workers. Some referrals are for the evaluation of life-threatening situations for the child or others, such as suicidal behavior, physical abuse, and violent or homicidal behavior. Other urgent but non–life-threatening referrals pertain to children and adolescents with exacerbations of clear-cut serious psychiatric disorders, such as mania, depression, florid psychosis, and school referral. Less diagnostically obvious situations occur when children and adolescents present with a history of a wide range of disruptive, aberrant behaviors and are accompanied by an overwhelmed, anxious, and distraught adult who perceives the child's actions as an emergency, despite the absence of life-threatening behavior of an evident psychiatric disorder. In those cases, the spectrum of contributing factors is not immediately apparent, and the emergency psychiatrist must assess the entire family or system involved with the child. Familial stressors and parental discord can contribute to the evolution of a crisis for a child. For example, immediate evaluations are sometimes legitimately indicated for a child caught in the crossfire of feuding parents or a seemingly irreconcilable conflict between a set of parents and a school, therapist, or protective service worker regarding the needs of the child (Table 25-13).



Table 25-12
Common Psychiatric Emergencies

Syndrome	Emergency Manifestations	Treatment Issues
Abuse of child or adult	Signs of physical trauma	Management of medica problems; psychiatric evaluation; report to authorities
Acquired immune deficiency syndrome (AIDS)	Changes in behavior secondary to medical causes; changes in	Management of neurologic illness; management of psychological concomitants;

	behavior secondary to fear and anxiety; suicidal behavior	reinforcement of social support
Adolescent crises	Suicidal attempts and ideation; substance abuse, truancy, trouble with law, pregnancy, running away; eating disorders; psychosis	Evaluation of suicidal potential, extent of substance abuse, family dynamics; crisis-oriented family and individual therapy; hospitalization if necessary; consultation with appropriate extrafamilial authorities
Agoraphobia	Panic; depression	Alprazolam, 0.25–2 mg antidepressant medication
Agranulocytosis (clozapine induced)	High fever, pharyngitis, oral and perianal ulcerations	Discontinue medication immediately; administer granulocyte colony-stimulating factor
Akathisia	Agitation, restlessness, muscle discomfort; dysphoria	Reduce antipsychotic dosage; propranolol (30–120 mg a day); benzodiazepines; diphenhydramine orally or IV; benztropine IM

Alcohol-Related Emergencies

Alcohol delirium	Confusion, disorientation, fluctuating consciousness and perception, autonomic hyperactivity; may be fatal	A benzodiazepine; haloperidol for psychotic symptoms may be added if necessary
Alcohol intoxication	Disinhibited behavior, sedation at high doses	With time and protectiv environment, symptoms abate
Alcohol-persisting amnestic disorder	Confusion, loss of memory even for all personal identification data	Hospitalization; hypnosis; rule out organic cause
Alcohol-persisting dementia	Confusion, agitation, impulsivity	Rule out other causes for dementia; no effective treatment; hospitalization if necessary
Alcohol psychotic disorder with hallucinations	Vivid auditory (at times visual) hallucinations with affect appropriate to content (often fearful); clear sensorium	Haloperidol for psychotic symptoms
Alcohol seizures	Grand mal seizures; rarely status epilepticus	Diazepam, phenytoin; prevent by using chlordiazepoxide during detoxification
Alcohol withdrawal	Irritability, nausea,	Fluid and electrolytes

	vomiting, insomnia, malaise, autonomic hyperactivity, shakiness	maintained; sedation with benzodiazepine restraints; monitoring of vital signs; 100 mg thiamine IM
Idiosyncratic alcohol intoxication	Marked aggressive or assaultive behavior	Generally no treatment required other than protective environment
Korsakoff syndrome	Alcohol stigmata, amnesia, confabulation	No effective treatment; institutionalization often needed
Wernicke encephalopathy	Oculomotor disturbances, cerebellar ataxia; mental confusion	Thiamine, 100 mg IV o IM, with $MgSO_4$ give before glucose loading
Amphetamine (or related substance) intoxication	Delusions, paranoia; violence; depression (from withdrawal); anxiety, delirium	Antipsychotics; restraints; hospitalization if necessary; no need for gradual withdrawal; antidepressants may be necessary
Anorexia nervosa	Loss of 25% of body weight of the norm for age and sex	Hospitalization; electrocardiogram (ECG), fluid and electrolytes; neuroendocrine evaluation
Anticholinergic	Psychotic	Discontinue drug, IV

intoxication	symptoms, dry skin and mouth, hyperpyrexia, mydriasis, tachycardia, restlessness, visual hallucinations	physostigmine, 0.5–2 mg, for severe agitation or fever, benzodiazepines; antipsychotics contraindicated
Anticonvulsant intoxication	Psychosis; delirium	Dosage of anticonvulsant is reduced
Benzodiazepine intoxication	Sedation, somnolence, and ataxia	Supportive measures; flumazenil, 7.5–45 m a day, titrated as needed, should be used only by skilled personnel with resuscitative equipment available
Bereavement	Guilt feelings, irritability; insomnia; somatic complaints	Must be differentiated from major depressiv disorder; antidepressants not indicated; benzodiazepines for sleep; encourageme of ventilation
Borderline personality disorder	Suicidal ideation and gestures; homicidal ideations and gestures; substance abuse; micropsychotic	Suicidal and homicidal evaluation (if great, hospitalization); sma dosages of antipsychotics; clear follow-up plan

	episodes; burns, cut marks on body	
Brief psychotic disorder	Emotional turmoil, extreme lability; acutely impaired reality testing after obvious psychosocial stress	Hospitalization often necessary; low dosage of antipsychotics may b necessary but often resolves spontaneously
Bromide intoxication	Delirium; mania; depression; psychosis	Serum levels obtained (>50 mg a day); bromide intake discontinued; large quantities of sodium chloride IV or orally; agitation, paraldehyd or antipsychotic is used
Caffeine intoxication	Severe anxiety, resembling panic disorder; mania; delirium; agitated depression; sleep disturbance	Cessation of caffeine-containing substances; benzodiazepines
Cannabis intoxication	Delusions; panic; dysphoria; cognitive impairment	Benzodiazepines and antipsychotics as needed; evaluation o suicidal or homicidal risk; symptoms usually abate with time and reassuranc
Catatonia	Marked	Rapid tranquilization

	psychomotor disturbance (either excitement or stupor); exhaustion; can be fatal	with benzodiazepine or antipsychotics; monitor vital signs
Cimetidine psychotic disorder	Delirium; delusions	Reduce dosage or discontinue drug
Clonidine withdrawal	Irritability; psychosis; violence; seizures	Symptoms abate with time, but antipsychotics may b necessary; gradual lowering of dosage
Cocaine intoxication and withdrawal	Paranoia and violence; severe anxiety; manic state; delirium: schizophreniform psychosis; tachycardia, hypertension, myocardial infarction, cerebrovascular disease; depression and suicidal ideation	Antipsychotics and benzodiazepines; antidepressants or electroconvulsive therapy (ECT) for withdrawal depressio if persistent; hospitalization
Delirium	Fluctuating sensorium; suicidal and homicidal risk; cognitive clouding; visual, tactile, and	Evaluate all potential contributing factors and treat each accordingly; reassurance, structure, clues to orientation;

	auditory hallucinations; paranoia	benzodiazepines and low-dosage, high-potency antipsychotics must be used with extreme care because of their potential to act paradoxically and increase agitation
Delusional disorder	Most often brought into emergency department involuntarily; threats directed toward others	Antipsychotics if patient will comply (IM if necessary); intensive family intervention; hospitalization if necessary
Dementia	Unable to care for self; violent outbursts; psychosis; depression and suicidal ideation; confusion	Small dosages of high-potency antipsychotics; clues to orientation; organic evaluation, including medication use; family intervention
Depressive disorders	Suicidal ideation and attempts; self-neglect; substance abuse	Assessment of danger to self; hospitalization if necessary, nonpsychiatric cause of depression must be evaluated
L-Dopa intoxication	Mania; depression; schizophreniform disorder, may induce rapid cycling in patients with bipolar I	Lower dosage or discontinue drug

	disorder	
Dystonia, acute	Intense involuntary spasm of muscles of neck, tongue, face, jaw, eyes, or trunk	Decrease dosage of antipsychotic; benztropine or diphenhydramine IM
Group hysteria	Groups of people exhibit extremes of grief or other disruptive behavior	Group is dispersed with help of other health care workers; ventilation, crisis-oriented therapy; if necessary, small dosages of benzodiazepines
Hallucinogen-induced psychotic disorder with hallucinations	Symptom picture is result of interaction of type of substance, dose taken, duration of action, user's premorbid personality, setting; panic; agitation; atropine psychosis	Serum and urine screens; rule out underlying medical o mental disorder; benzodiazepines (2–20 mg) orally; reassurance and orientation; rapid tranquilization; often responds spontaneously
Homicidal and assaultive behavior	Marked agitation with verbal threats	Seclusion, restraints, medication
Hypertensive crisis	Life-threatening hypertensive reaction secondary to ingestion of tyramine-	Antihypertensive medications as per emergency medicine team make sure symptoms are not secondary to

	containing foods in combination with monoamine oxidase inhibitors (MAOIs); headache, stiff neck, sweating, nausea, vomiting	hypotension (side effect of MAOIs alone)
Hyperthermia	Extreme excitement or catatonic stupor or both; extremely elevated temperature; violent hyperagitation	Hydrate and cool; may be drug reaction, so discontinue any drug rule out infection
Hyperventilation	Anxiety, terror, clouded consciousness; giddiness, faintness; blurring vision	Shift alkalosis by havin patient breathe into paper bag; patient education; antianxiet agents
Hypothermia	Confusion; lethargy; combativeness; low body temperature and shivering; paradoxical feeling of warmth	IV fluids and rewarming cardiac status must be carefully monitored; avoidanc of alcohol
Incest and sexual abuse of child	Suicidal behavior; adolescent crises; substance abuse	Corroboration of charg protection of victim; contact social services; medical an psychiatric evaluatio

		crisis intervention
Insomnia	Depression and irritability; early morning agitation; frightening dreams; fatigue	Hypnotics only in short term; treat any underlying mental disorder; rules of sleep hygiene
Intermittent explosive disorder	Brief outbursts of violence; periodic episodes of suicide attempts	Benzodiazepines or antipsychotics for short term; long-term evaluation with computed tomography (CT) scan, sleep-deprived electroencephalogra (EEG), glucose tolerance curve
Jaundice	Uncommon complication of low-potency phenothiazine use (e.g., chlorpromazine)	Change drug to low dosage of a low-potency agent in a different class
Leukopenia and agranulocytosis	Side effects within the first 2 mo of treatment with antipsychotics	Patient should call immediately for sore throat, fever, etc., an obtain immediate blood count; discontinue drug; hospitalize if necessary
Lithium toxicity	Vomiting; abdominal pain; profuse diarrhea; severe	Lavage with wide-bore tube; osmotic diuresis; medical

	tremor, ataxia; coma; seizures; confusion; dysarthria; focal neurologic signs	consultation; may require intensive car unit treatment
Major depressive episode with psychotic features	Major depressive episode symptoms with delusions; agitation, severe guilt; ideas of reference; suicide and homicide risk	Antipsychotics plus antidepressants; evaluation of suicide and homicide risk; hospitalization and ECT if necessary
Manic episode	Violent, impulsive behavior; indiscriminate sexual or spending behavior; psychosis; substance abuse	Hospitalization; restraints if necessary; rapid tranquilization with antipsychotics; restoration of lithium levels
Marital crises	Precipitant may be discovery of an extramarital affair, onset of serious illness, announcement of intent to divorce, or problems with children or work; one or both members of the couple may be in therapy or may be	Each should be questioned alone regarding extramarita affairs, consultations with lawyers regarding divorce, and willingness to work in crisis-oriente or long-term therapy to resolve the problem; sexual, financial, and psychiatric treatment

	psychiatrically ill; one spouse may be seeking hospitalization for the other	histories from both, psychiatric evaluatio at the time of presentation; may be precipitated by onset of untreated mood disorder or affective symptoms caused by medical illness or insidious-onset dementia; referral for management of the illness reduces immediate stress and enhances the healthier spouse's coping capacity; children may give insights available onl to someone intimatel involved in the social system
Migraine	Throbbing, unilateral headache	Sumatriptan 6 mg IM
Mitral valve prolapse	Associated with panic disorder; dyspnea and palpitations; fear and anxiety	Echocardiogram; alprazolam or propranolol
Neuroleptic malignant syndrome	Hyperthermia; muscle rigidity; autonomic instability; parkinsonian	Discontinue antipsychotic; IV dantrolene; bromocriptine orally; hydration and cooling

	symptoms; catatonic stupor; neurologic signs; 10% to 30% fatality; elevated creatine phosphokinase (CPK)	monitor CPK levels
Nitrous oxide toxicity	Euphoria and light-headedness	Symptoms abate without treatment within hours of use
Nutmeg intoxication	Agitation; hallucinations; severe headaches; numbness in extremities	Symptoms abate within hours of use without treatment
Opioid intoxication and withdrawal	Intoxication can lead to coma and death; withdrawal is not life-threatening	IV naloxone, narcotic antagonist; urine and serum screens; psychiatric and medical illnesses (e.g., AIDS) may complicate picture
Panic disorder	Panic, terror; acute onset	Must differentiate from other anxiety-producing disorders, both medical and psychiatric; ECG to rule out mitral valve prolapse; short-acting benzodiazepines for symptomatic management; long-

		term management may include an antidepressant
Parkinsonism	Stiffness, tremor, bradykinesia, flattened affect, shuffling gait, salivation, secondary to antipsychotic medication	Oral antiparkinsonian drug for 4 wk to 3 mo decrease dosage of the antipsychotic
Perioral (rabbit) tremor	Perioral tumor (rabbit-like facial grimacing) usually appearing after long-term therapy with antipsychotics	Decrease dosage or change to a medication in another class
Phencyclidine (or phencyclidine-like intoxication)	Psychosis; can lead to death; acute danger to self and others	Serum and urine assay benzodiazepines may interfere with excretion; antipsychotics may worsen symptoms because of anticholinergic side effects; medical monitoring and hospitalization for severe intoxication
Phenelzine-induced psychotic disorder	Psychosis and mania in predisposed people	Reduce dosage or discontinue drug

Phenylpropanolamine toxicity	Psychosis; paranoia; insomnia; restlessness; nervousness; headache	Symptoms abate with dosage reduction or discontinuation (foun in over-the-counter diet aids and oral an nasal decongestants
Phobias	Panic, anxiety; fear	Treatment same as for panic disorder
Photosensitivity	Easy sunburning secondary to use of antipsychotic medication	Patient should avoid strong sunlight and use high-level sunscreens
Pigmentary retinopathy	Reported with dosages of thioridazine of 800 mg a day or above	Remain below 800 mg day of thioridazine
Postpartum psychosis	Childbirth can precipitate schizophrenia, depression, reactive psychoses, mania, and depression; affective symptoms are most common; suicide risk is reduced during pregnancy but increased in the postpartum period	Danger to self and others (including infant) must be evaluated and prope precautions taken; medical illness presenting with behavioral aberration is included in the differential diagnosis and must be sought and treated; care must be paid to the effects on father, infant, grandparents, and other children

Posttraumatic stress disorder	Panic, terror; suicidal ideation; flashbacks	Reassurance; encouragement of return to responsibilities; avoi hospitalization if possible to prevent chronic invalidism; monitor suicidal ideation
Priapism (trazodone induced)	Persistent penile erection accompanied by severe pain	Intracorporeal epinephrine; mechanical or surgical drainage
Propranolol toxicity	Profound depression; confusional states	Reduce dosage or discontinue drug; monitor suicidality
Rape	Not all sexual violations are reported; silent rape reaction is characterized by loss of appetite, sleep disturbance, anxiety, and, sometimes, agoraphobia; long periods of silence, mounting anxiety, stuttering, blocking, and physical symptoms during the interview when the sexual	Rape is a major psychiatric emergency; victim may have enduring patterns of sexual dysfunction; crisis-oriented therapy, social support, ventilation, reinforcement of healthy traits, and encouragement to return to the previous level of functioning a rapidly as possible; legal counsel; thorough medical examination and test

		history is taken; fear of violence and death and of contracting a sexually transmitted disease or being pregnant	to identify the assailant (e.g., obtaining samples of pubic hairs with a pubic hair comb, vaginal smear to identify blood antigens in semen); i a woman, methoxyprogesteron or diethylstilbestrol orally for 5 days to prevent pregnancy; i menstruation does n commence within 1 wk of cessation of th estrogen, all alternatives to pregnancy, including abortion, should be offered; if the victim has contracted a venereal disease, appropriate antibiotics; witnesse written permission is required for the physician to examine photograph, collect specimens, and release information t the authorities; obtai consent, record the history in the patient' own words, obtain required tests, recor

		the results of the examination, save al clothing, defer diagnosis, and provide protection against disease, psychic trauma, and pregnancy; men's ar women's responses to rape affectively ar reported similarly, although men are more hesitant to talk about homosexual assault for fear they will be assumed to have consented
Reserpine intoxication	Major depressive episodes; suicidal ideation; nightmares	Evaluation of suicidal ideation; lower dosage or change drug; antidepressant of ECT may be indicated
Schizoaffective disorder	Severe depression; manic symptoms; paranoia	Evaluation of dangerousness to se or others; rapid tranquilization if necessary; treatmen of depression (antidepressants alone can enhance schizophrenic symptoms); use of antimanic agents

Schizophrenia	Extreme self-neglect; severe paranoia; suicidal ideation or assaultiveness; extreme psychotic symptoms	Evaluation of suicidal and homicidal potential; identificatio of any illness other than schizophrenia; rapid tranquilization
Schizophrenia in exacerbation	Withdrawn; agitation; suicidal and homicidal risk	Suicide and homicide evaluation; screen fo medical illness; restraints and rapid tranquilization if necessary; hospitalization if necessary; reevaluation of medication regimen
Sedative, hypnotic, or anxiolytic intoxication and withdrawal	Alterations in mood, behavior, thought —delirium; derealization and depersonalization; untreated, can be fatal; seizures	Naloxone to differentiat from opioid intoxication; slow withdrawal with benzodiazepine; hospitalization
Seizure disorder	Confusion; anxiety; derealization and depersonalization; feelings of impending doom; gustatory or olfactory hallucinations; fugue-like state	Immediate EEG; admission and sleep deprived and 24-hr EEG; rule out pseudoseizures; anticonvulsants
Substance withdrawal	Abdominal pain;	Symptoms of

	insomnia, drowsiness; delirium; seizures; symptoms of tardive dyskinesia may emerge; eruption of manic or schizophrenic symptoms	psychotropic drug withdrawal disappea with time or disappea with reinstitution of the substance; symptoms of antidepressant withdrawal can be successfully treated with anticholinergic agents, such as atropine; gradual withdrawal of psychotropic substances over 2–4 wk generally obviate development of symptoms
Sudden death associated with antipsychotic medication	Seizures; asphyxiation; cardiovascular causes; postural hypotension; laryngeal–pharyngeal dystonia; suppression of gag reflex	Specific medical treatments
Sudden death of psychogenic origin	Myocardial infarction after sudden psychic stress; voodoo and hexes; hopelessness, especially	Specific medical treatments; folk healers

	associated with serious physical illness	
Suicide	Suicidal ideation; hopelessness	Hospitalization, antidepressants
Sympathomimetic withdrawal	Paranoia; confusional states; depression	Most symptoms abate without treatment; antipsychotics; antidepressants if necessary
Tardive dyskinesia	Dyskinesia of mouth, tongue, face, neck, and trunk; choreoathetoid movements of extremities; usually but not always appearing after long-term treatment with antipsychotics, especially after a reduction in dosage; incidence highest in the elderly and brain damaged; symptoms are intensified by antiparkinsonian drugs and masked but not cured by	No effective treatment reported; may be prevented by prescribing the least amount of drug possible for as little time as is clinically feasible and using drug-free holidays fo patients who need to continue taking the drug; decrease or discontinue drug at first sign of dyskineti movements

	increased dosages of antipsychotic	
Thyrotoxicosis	Tachycardia; gastrointestinal dysfunction; hyperthermia; panic, anxiety, agitation; mania; dementia; psychosis	Thyroid function test (T_3, T_4, thyroid-stimulating hormone [TSH]); medical consultation
Toluene abuse	Anxiety; confusion; cognitive impairment	Neurologic damage is nonprogressive and reversible if toluene use in discontinued early
Vitamin B_{12} deficiency	Confusion; mood and behavior changes; ataxia	Treatment with vitamin B_{12}
Volatile nitrates	Alternations of mood and behavior; light-headedness; pulsating headache	Symptoms abate with cessation of use

An emergency setting is often the site of an initial evaluation of chronic problem behavior. For example, an identified problem—such as severe tantrums, violence, and destructive behavior in a child—may have been present for months or even years. Nevertheless, the initial contact with the mental health system in the emergency room or private office may be the first opportunity for the child or adolescent to disclose underlying stressors, such as physical or sexual abuse.

Given the integral relation of severe family dysfunction to childhood behavioral disturbance, the emergency psychiatrist must assess familial discord and psychiatric disorder in family members during an urgent evaluation. One way to make the assessment is to interview the child and the individual family members, both alone and together, and to obtain a history from informants outside the family whenever possible. Noncustodial parents, therapists, and teachers may add valuable information regarding the child's daily functioning. Many families, especially those with mental illness and severe dysfunction, may have little or no inclination to seek psychiatric help on a nonurgent basis; therefore, the emergency evaluation becomes the only way to engage them in an extensive psychiatric treatment program.



Table 25-13
Familial Risk Factors

Physical and sexual abuse
Recent family crisis: loss of a parent, divorce, loss of job, family move
Severe family dysfunction, including parental mental illness

Life-Threatening Emergencies

Suicidal Behavior

ASSESSMENT. Suicidal behavior is the most common reason for an emergency evaluation in adolescents. Despite the minimal risk for a complete suicide in a child less than 12 years of age, suicidal ideation or behavior in a child of any age must be carefully evaluated, with particular attention to the psychiatric status of the child and the ability of the family or the guardians to provide the appropriate supervision. The assessment must determine the circumstances of the suicidal ideation or behavior, its lethality, and the persistence of the suicidal intention. An evaluation of the family's sensitivity, supportiveness, and competence must be done to assess their ability to monitor the child's suicidal potential. Ultimately, during an emergency evaluation, the psychiatrist must decide whether the child may return home to

a safe environment and receive outpatient follow-up care or whether hospitalization is necessary. A psychiatric history, a mental status examination, and an assessment of family functioning help establish the general level of risk.

MANAGEMENT. When self-injurious behavior has occurred, the adolescent likely requires hospitalization in a pediatric unit for treatment of the injury or the observation of medical sequelae after toxic ingestion. If the adolescent is medically clear, the psychiatrist must decide whether the adolescent needs psychiatric admission. If the patient persists in suicidal ideation and shows signs of psychosis, severe depression (including hopelessness), or marked ambivalence about suicide, psychiatric admission is indicated. An adolescent who is taking drugs or alcohol should not be released until an assessment can be done when the patient is not intoxicated. Patients with high-risk profiles—such as late-adolescent males, especially those with substance abuse and aggressive behavior disorders, and those who have severe depression or who have made prior suicide attempts, particularly with lethal weapons—warrant hospitalization. Young children who made suicide attempts, even when the attempt had a low lethality, need psychiatric admission if the family is so chaotic, dysfunctional, and incompetent that follow-up treatment is unlikely.

Violent Behavior and Tantrums

ASSESSMENT. The first task in an emergency evaluation of a violent child or adolescent is to make sure that both the child and the staff members are physically protected so that nobody gets hurt. If the child appears to be calming down in the emergency area, the clinician may indicate to the child that it would be helpful if the child recounted what happened and may ask whether the child feels in sufficient control to do so. If the child agrees and the clinician judges the child to be in reasonable control, the clinician may approach the child with the appropriate backup close at hand. If not, the clinician may either give the child several minutes to calm down before reassessing the situation or, with an adolescent, suggest that medication may help the adolescent relax.

If the adolescent is combative, physical restraint may be necessary before anything else is attempted. Some rageful children and adolescents brought to

an emergency setting by overwhelmed families can regain control of themselves without the use of physical or pharmacologic restraint. Children and adolescents are most likely to calm down if approached calmly in a nonthreatening manner and given a chance to tell their side of the story to a nonjudgmental adult. At this time, the psychiatrist should look for any underlying psychiatric disorder that may be mediating the aggression. The psychiatrist should speak to family members and others who have been witnessing the episode to understand the context in which it occurred and the extent to which the child has been out of control.

MANAGEMENT. Prepubertal children, in the absence of major psychiatric illness, rarely require medication to keep them safe, because they are generally small enough to be physically restrained if they begin to hurt themselves or others. It is not immediately necessary to administer medication to a child or an adolescent who was in a rage but is in a calm state when examined. Adolescents and older children who are assaultive, extremely agitated, or overtly self-injurious and who may be difficult to subdue physically may require medication before a dialogue can take place.

Children who have a history of repeated, self-limited, severe tantrums may not require admission to a hospital if they can calm down during the evaluation. However, the pattern, no doubt, will reoccur unless ongoing outpatient treatment for the child and the family is arranged. For adolescents who continue to pose a danger to themselves or others during the evaluation period, admission to a hospital is necessary.

Fire Setting

ASSESSMENT. A sense of emergency and panic often surrounds the parents of a child who has set a fire. Parents or teachers often request an emergency evaluation, even for a very young child who has accidentally lit a fire. Many children, during normal development, become interested in fire, but in most cases, a school-aged child who has set a fire has done so accidentally while playing with matches and seeks help to put it out. When a child has a strong interest in playing with matches, the level of supervision by family members must be clarified so that no further accidental fires occur. The clinician must distinguish between a child who accidentally or even impulsively sets a

single fire and a child who engages in repeated fire setting with premeditation and subsequently leaves the fire without making any attempt to extinguish it. In repeated fire setting, the risk is more significant than in a single occurrence, and the psychiatrist must determine whether underlying psychopathology exists in the child or the family members. The psychiatrist should also evaluate family interactions because any factors that interfere with adequate supervision and communication—such as high levels of marital discord and harsh, punitive parenting styles—can impede appropriate intervention.

Fire setting is one of a triad of symptoms—enuresis, cruelty to animals, and fire setting—that were believed, some years ago, to be typical of children with conduct disorders; however, no evidence indicates that the three symptoms are genuinely linked, although conduct disorder is the most frequent psychiatric disorder that occurs with pathologic fire setting.

MANAGEMENT. The critical component of management and treatment for firesetters is to prevent further incidents while treating any underlying psychopathology. In general, fire setting alone is not an indication for hospitalization, unless a continued direct threat exists that the patient will set another fire. The parents of children with a pattern of fire setting must be emphatically counseled that the child must not be left alone at home and should never be left to take care of younger siblings without direct adult supervision. Children who exhibit a pattern of concurrent aggressive behaviors and other forms of destructive behavior are likely to have a poor outcome. Outpatient treatment should be arranged for children who repeatedly set fires. Behavioral techniques that involve both the child and the family help decrease the risk for further fire setting, as is positive reinforcement for alternate behaviors.

Child Abuse: Physical and Sexual

Assessment. Physical and sexual abuse occurs in girls and boys of all ages, in all ethnic groups, and at all socioeconomic levels. The abuses vary widely for severity and duration, but any form of continued abuse constitutes an emergency for a child. No single psychiatric syndrome is a sine qua non of physical or sexual abuse, but fear, guilt, anxiety, depression, and ambivalence

regarding disclosure commonly surround the child who has been abused.

Young children who are being sexually abused may exhibit precocious sexual behavior with peers and present detailed sexual knowledge that reflects exposure beyond their developmental level. Often, children who endure sexual or physical abuse display sadistic and aggressive behaviors themselves. Abused children are likely to be threatened with severe and frightening consequences by the perpetrator if they reveal the situation to anyone. Frequently, an abused child who is victimized by a family member is placed in the irreconcilable position of having to choose between enduring the abuse or risk destroying the family, not being believed or being abandoned.

In cases of suspected abuse, the child and other family members must be interviewed individually to give each member a chance to speak privately. If possible, the clinician should observe the child with each parent individually to get a sense of the spontaneity, warmth, fear, anxiety, or other prominent features of the relationships. One observation is generally not sufficient to make a final judgment about the family relationship. However, abused children almost always have mixed emotions toward abusive parents.

Physical indicators of sexual abuse in children include sexually transmitted diseases (e.g., gonorrhea); pain, irritation, and itching of the genitalia and the urinary tract; and discomfort while sitting and walking. In many instances of suspected sexual abuse, however, physical evidence is not present. Thus, a careful history is essential. The physician should speak directly about the issues without leading the child in any direction, because already frightened children may be easily influenced to endorse what they think the examiner wants to hear. Furthermore, children who have been abused often retract all or part of what has been disclosed during an interview.

The use of anatomically correct dolls in the assessment of sexual abuse can help the child identify body parts and show what has happened, but no conclusive evidence supports sexual play with dolls as a means of validating abuse.

Neglect: Failure to Thrive

Assessment. In child neglect, a child’s physical, mental, or emotional condition has been impaired because of the inability of a parent or caretaker

to provide adequate food, shelter, education, or supervision. Similar to abuse, any form of continued neglect is an emergency for the child. Parents who neglect their children range widely and may include parents who are very young and ignorant about the emotional and concrete needs of a child, parents with depression and significant passivity, substance-abusing parents, and parents with a variety of incapacitating mental illnesses.

In its extreme form, neglect can contribute to failure to thrive—that is, an infant, usually under 1 year of age, becomes malnourished in the absence of an organic cause. Failure to thrive typically occurs under circumstances in which adequate nourishment is available, but a disturbance within the relationship between the caretaker and the child results in a child who does not eat sufficiently to grow and develop. A negative pattern may exist between the mother and the child in which the child refuses feedings, and the mother feels rejected and eventually withdraws. She may then avoid offering food as frequently as the infant needs it. Observation of the mother and the child together may reveal a nonspontaneous, tense interaction, with withdrawal on both sides, resulting in a seeming apathy in the mother. Both the mother and the child may seem depressed.

A rare form of failure to thrive in children who are at least several years old and are not necessarily malnourished is the syndrome of psychosocial dwarfism. In that syndrome, marked growth retardation and delayed epiphyseal malnutrition accompany a disturbed relationship between the parent and the child, along with bizarre social and eating behaviors in the child. Those behaviors sometimes include eating from garbage cans, drinking toilet water, binging and vomiting, and diminished outward response to pain. Half of the children with the syndrome have decreased growth hormone. Once the children are removed from the troubled environment and placed in another setting, such as a psychiatric hospital with appropriate supervision and guidance regarding meals, the endocrine abnormalities normalize, and the children begin to grow at a more rapid rate.

Management. In cases of child neglect, as with physical and sexual abuse, the most crucial decision to be made during the initial evaluation is whether the child is safe in the home environment. Whenever neglect is suspected, it must be reported to the local child protective service agency. In mild cases, the decision to refer the family for outpatient services, as opposed to

hospitalizing the child, depends on the clinician's conviction that the family is cooperative and willing to be educated and to enter into treatment and that the child is not in danger. Before a neglected child is released from an emergency setting, a follow-up appointment must be made.

Education for the family must begin during the evaluation; the family must be told, in a nonthreatening manner, that failure to thrive can become life-threatening, that the entire family needs to monitor the child's progress, and that they will receive some help in overcoming the many possible obstacles interfering with the child's emotional and physical well-being.

Anorexia Nervosa

Anorexia nervosa occurs in females about 10 times as often as in males. It is characterized by the refusal to maintain body weight, leading to weight at least 15 percent below the expected weight, by a distorted body image, by a persistent fear of becoming fat, and by the absence of at least three menstrual cycles. The disorder usually begins after puberty, but it has occurred in children of 9 to 10 years of age, in whom expected weight gain does not occur, rather than a loss of 15 percent of body weight. The disorder reaches medical emergency proportions when the weight loss approaches 30 percent of body weight or when metabolic disturbances become severe. Hospitalization then becomes necessary to control the ongoing process of starvation, potential dehydration, and the medical complications of starvation, including electrolyte imbalances, cardiac arrhythmias, and hormonal changes.

Acquired Immune Deficiency Syndrome

Assessment. AIDS, which is caused by the human immunodeficiency virus (HIV), occurs in neonate through perinatal transmission from an infected mother, in children and adolescents secondary to sexual abuse by an infected person, and adolescents through intravenous drug abuse with an infected person or intravenous drug abuse with infected needles and sexual activities with infected partners. Child and adolescent hemophiliac patients may contract AIDS through tainted blood transfusions.

Children and adolescents may present for emergency evaluations at the urging of a family member of a peer; in some cases, they take the initiative

themselves when they are faced with anxiety or panic about high-risk behavior. Early screening of high-risk persons may lead to the treatment of asymptomatic infected patients with such drugs as azidothymidine (AZT) and possibly other new medications that may slow the course of the disease. During the assessment of the risks for HIV infection, the doctor can start to educate the patient and family so that an adolescent who is not infected but exhibits high-risk behavior can be counseled about that behavior and safe-sex practices.

In children, the brain is often a primary site for HIV infection; encephalitis, decreased brain development, and such neuropsychiatric symptoms as impairment in memory, concentration, and attention span may be present before the diagnosis is made. The virus can be present in the cerebrospinal fluid before it shows up in the bloodstream. Changes in cognitive function, frontal lobe disinhibition, social withdrawal, slowed information processing, and apathy constitute some common symptoms of the AIDS dementia complex. Mood disorders, personality changes, and frank psychosis can also occur in patients infected with HIV.

Munchausen Syndrome by Proxy

Assessment. Munchausen syndrome by proxy, essentially, is a form of child abuse in which a parent, usually the mother, or a caretaker repeatedly fabricates or inflicts injury or illness in a child for whom the medical intervention is then sought, often in an emergency setting. Although it is a rare scenario, mothers who inflict injury often have some prior knowledge of medicine, leading to sophisticated symptoms; the mothers sometimes engage in inappropriate camaraderie with the medical staff regarding the treatment of the child. Careful observation may reveal that the mothers often do not exhibit appropriate signs of distress on hearing the details of the child's medical symptoms. Prototypically, such mothers tend to present themselves as highly accomplished professionals in ways that seem inflated or blatantly untrue.

The illnesses appearing in the child can involve any organ system, but specific symptoms are commonly presented: bleeding from one or many sites, including the GI tract, the genitourinary system, and the respiratory system; seizures; and CNS depression. At times, the illness is simulated,

rather than inflicted.

Urgent Non–Life-Threatening Situations

School Refusal

ASSESSMENT. Refusal to go to school may occur in a young child who is first entering school or in an older child or adolescent who is making a transition into a new grade or school, or it may emerge in a vulnerable child without an apparent external stressor. In any case, school refusal requires immediate intervention, because the longer the dysfunctional pattern continues, the more difficult it is to interrupt.

School refusal is generally associated with separation anxiety, in which the child's distress is related to the consequences of being separated from the parent, so the child resists going to school. School refusal can also occur in children with school phobia, in which the school is the target of fear and distress. In either case, a severe disruption of the child's life occurs. Although mild separation anxiety is universal, particularly among very young children who are first facing the school, treatment is required when a child actually cannot attend school. Severe psychopathology, including anxiety and depressive disorders, is often present when school refusal occurs for the first time in an adolescent. Children with separation anxiety disorder typically present extreme worries that catastrophic events will befall their mothers, attachment, or themselves as a result of the separation. Children with separation anxiety disorder may also exhibit many other fears and symptoms of depression, including such somatic complaints such as headaches, stomachaches, and nausea. Severe tantrums and desperate pleas may ensue when preoccupation that a parent will be harmed during the separation is frequently verbalized; in adolescents, the stated reasons for refusing to go to school are often physical complaints.

As part of an urgent assessment, the psychiatrist must ascertain the duration of the patient's absence from school and must assess the parents' ability to participate in a treatment plan that will undoubtedly involve firm parental guidelines to ensure the child's return to school. The parents of a child with separation anxiety disorder often exhibit excessive separation anxiety or other anxiety disorders themselves, thereby compounding the

child's problem. When the parents are unable to participate in a treatment program from home, hospitalization should be considered.

MANAGEMENT. When school refusal caused by separation anxiety is identified during an emergency evaluation, the underlying disorder can be explained to the family, and an intervention can be started immediately. In severe cases, however, a multidimensional, long-term family-oriented treatment plan is necessary. Whenever possible, a separation-anxious child should be brought back to school the next school day, despite the distress, and a contact person within the school (counselor, guidance counselor, or teacher) should be involved to help the child stay in school while praising the child for tolerating the school situation.

When school refusal has been going on for months or years or when the family members are unable to cooperate, a treatment program to move the child back to school from the hospital should be considered. When the child's anxiety is not diminished by behavioral methods alone, antidepressants are helpful. Medication is generally prescribed not at the initial evaluation, but after a behavioral intervention has been tried.

Other Childhood Disturbances

Posttraumatic Stress Disorder. Children who have been subjected to a severe catastrophic or traumatic event may present for a prompt evaluation because they have extreme fears of the specific trauma occurring again or sudden discomfort with familiar places, people, or situations that previously did not evoke anxiety. Within weeks of a traumatic event, a child may recreate the event in play, in stories, and in dreams that directly replay the terrifying situation. A sense of reliving the experience may occur, including hallucinations and flashback (dissociative) experiences, and intrusive memories of the event come and go. Many traumatized children, over time, go on to reproduce parts of the event through their victimization behaviors toward others, without being aware that those behaviors reflect their own traumatic experiences.

Dissociative Disorders. Dissociative states—including the extreme form, multiple personality disorder—are believed most likely to occur in children

who have been subjected to severe and repetitive physical, sexual, or emotional abuse. Children with dissociative symptoms may be referred for evaluation because family members or teachers observe that the children sometimes seem to be spaced out or distracted or act like different persons. Dissociative states are occasionally identified during the evaluation of violent and aggressive behavior, particularly in patients who honestly do not remember chunks of their behavior.

When a child who dissociates is violent or self-destructive or endangers others, hospitalization is necessary. A variety of psychotherapy methods have been used in the complex treatment of children with dissociative disorders, including play techniques and, in some cases, hypnosis.

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26 Level of Care

Before the 1960s in the United States, most psychiatric care for serious mental illnesses took place in hospitals. The rest was individual therapy with a psychiatrist or psychologist or other highly trained professionals; this treatment was reserved mainly for patients considered “neurotic” and having the means to pay for hourly sessions that would often occur several days a week.

Several forces changed this. The community mental health movement was motivated by social justice, occurring at a time defined by the civil rights movement. Experts argued that the housing of seriously mentally ill patients in a public hospital was unjust. The treatment there emphasized long-term custodial care rather than rehabilitation, and few patients improved. Visionaries such as Walter Barton, then medical director for the American Psychiatric Association, argued that the federal government must lead the way to develop alternatives to the current system of institutionalization. The result was The Mental Retardation Facilities and Community Mental Health Centers Construction Act of 1963, championed by President Kennedy, who proposed a “wholly new emphasis and approach to care for the mentally ill.”

The act and the amendments that followed it under President Johnson initiated the federal government’s role in mental health care, something previously left to the states. It established a grant program for states to establish local mental health centers, under the overview of the National Institutes of Health. The purpose was to build a robust network of mental health centers for community-based care as an alternative to long-term institutionalization.

Only about half of the proposed centers were built, and funding for these sites has been chronically inadequate. Thus, the movement was more

successful at deinstitutionalizing mental health care than it has been at offering quality alternatives. Part of the reason for this is another crucial force in the US health care system, the rise of managed care. Managed care emphasizes seeking the lowest level of care possible and the use of lower-cost professionals as a way of saving money.

The result is that we now have a myriad of treatment settings, services, and options. Most rely on grants and third-party payers, although some require considerable patient expenditures. The details and terminology differ by institution and region. However, they roughly fall into several categories. In this chapter, we briefly describe these options as well as the requirements and goals for each treatment. Our purpose is to help the reader navigate these options with an emphasis on choosing the treatment that is best for the patient. Although analogous services exist for children and adolescents, this chapter focuses on adult levels of care.

Table 26-1 summarizes the main levels of care that we should consider for most psychiatric patients.

INPATIENT ADMISSION

Inpatient admission units are usually specializing in intensive psychiatric care. The patients may voluntarily agree to hospitalization, or clinicians may hospitalize them against their will in a manner consistent with state law. The units are usually locked and equipped to restrain or seclude patients when required. Nurses and other staff are present around the clock. Physicians are present on the unit at least 5 days a week, and a covering physician is always available.

The hospitalization allows an intense treatment, with regular observation throughout a day and robust treatments, including medical adjustments, group, and individual psychotherapies, case management, and milieu treatments. Previously the most common form of psychiatric care, inpatient hospitalization is now reserved for patients who are unsafe in or unlikely to benefit from other treatments.

The patient’s risk of illness or severity is usually high. Many patients are hospitalized because they are at high risk of imminently harming themselves or others. Alternatively, their disorder may be so impairing that they cannot function adequately outside the hospital or attend to even basic activities of

daily living (ADLs).

Usually, the treatments required are ones that clinicians cannot safely or practically give in other settings. These include treatments that require daily observation for serious side effects. They may be given with such frequency, such as some intense psychotherapies, that other settings are not practical. In the latter case, the clinicians have often already tried other settings and have decided that the patient is likely to deteriorate should they continue the attempts to treat the patient in another setting.

In some cases, the practicality of alternative treatments arises from a patient's poor insight or inability to cooperate with treatment. Such patients may require involuntary hospitalization, and an inpatient unit is the only setting that would be appropriate to hold them against their will until the treatment allows them an ability to make their ordinary appropriate judgments. In the most severe cases, staff may have to restrain a violent or self-harming patient physically.

Except for some specialized care facilities, the goals of treatment are short term. Treatments aim to manage the crisis and stabilize the patient to the point that treatment can continue in a less restrictive setting for the patient. Crisis interventions, intense and frequent therapy, and medications are usual treatments. Longer-term treatments may begin in this setting, but usually to continue the treatments following the hospitalization.

The length of stay varies widely. A 2010 Centers for Disease Control and Prevention (CDC) survey reported the average length of stay for patients with schizophrenia as about 7.2 days, and for major depressive disorder, about 10.6 days. This number varies significantly by regional practice. Internationally, it varies by health care system, economic resources, and many other factors. Figure 26-1 gives some example comparisons of the average lengths of stay in 2010 for various countries.

The goal of treatment is to return the patient to a safe and functional state so that the team can then transition the patient to a lower level of care. This goal includes reducing the patient's threat to themselves or others, decreasing the symptoms of the disorder, and improving the patient's functional level.



Table 26-1
Main Levels of Care

	Inpatient Hospitalization	Partial Hospital	Intensive Outpatient Treatment	Outpatient	Residential Care
Risk	High: risk of imminent danger to self/others	Moderate but not imminent risk	No imminent risk	Low risk of harm	Moderate risk
Disease severity	Severe symptoms	Moderate	Moderate	Mild to moderate	Moderately severe
Functioning, activities of daily living (ADLs)	Poor, cannot attend to basic ADLs without assistance	Moderate dysfunction	Mild to moderate dysfunction	Adequate to attend to ADLs independently or with some assistance	Severe
Treatments	Require around-the-clock administration, regular observation for side effects or adjustment, or are given at a frequency making other settings impractical	Fully staffed unit offering a range of services similar to inpatient, however, limited to the working day and given daily, usually 5 days a week	May vary from several times a week to weekly or monthly, depending on the level of severity and treatment. Includes both psychotherapies and medications	May vary from several times a week to weekly or monthly, depending on the level of severity and treatment. Includes both psychotherapies and medications	Psychosocial treatments and medication management, although not necessarily at the level of an inpatient unit
Patient cooperation	Voluntary or involuntary	Voluntary	Voluntary	Voluntary	Voluntary or court-ordered
Need for restraints	May need restraining for safety purposes to prevent violence or injury	None	None	None	None
Goals of treatment	Reduce safety concern, reduce symptoms, restore functioning	Improvements in symptoms and functioning to transition the patient to a less restrictive treatment	Achieve and maintain the usual premorbid level of functioning	Achieve and maintain the usual premorbid level of functioning	Improvement in symptoms and functioning to a level where the patient can live in a less restrictive setting

Discharge planning usually begins on day one of the hospitalization. It helps to identify early on the likely next step in treatment, should it be back home, to a transitional setting or, less often, to long-term custodial care.

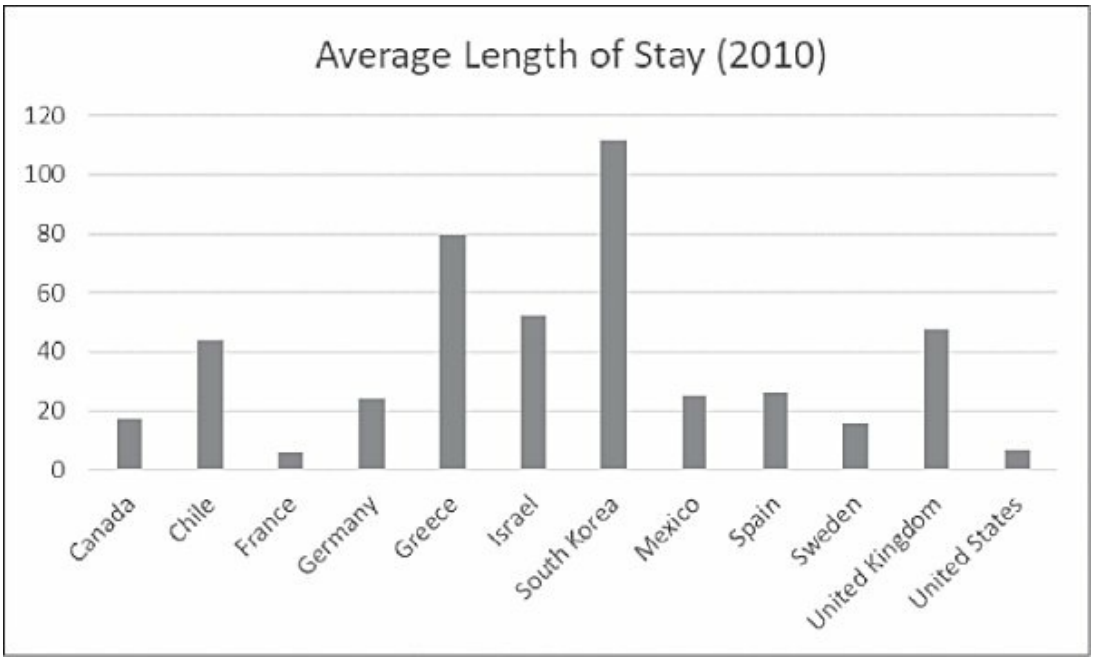


FIGURE 26-1

Average lengths of stay (LOS) for mental health disorders for selected countries in 2010. (Source: Organisation for Economic Co-Operation and Development [OECD.Stat].)

PARTIAL HOSPITAL TREATMENT

Partial hospitals are facilities that provide a near hospital level of care, the exception being that the care takes place for about 6 to 8 hours a day, after which the patients can go home. The service is fully staffed in a manner typical to a psychiatric hospital unit, however only during the day. The units usually offer a full range of treatment from various psychotherapies and psychosocial options to medication management. The patient usually attends daily, for 5 days a week.

Appropriate patients for a partial hospital program include patients who may be at significant risk for harm to themselves or others. However, the treatment team assesses that the patient is sufficiently stable such that the reinforcement of the intensive treatment is sufficient to prevent such harm. Thus, the patient is not an imminent risk or is not too impulsive to be trusted to be independent during the evening.

The patients are usually of moderate severity in symptoms. Similarly, they often have moderate dysfunction such that their self-care may be substandard. They often have ongoing trouble maintaining normal relationships.

The program usually offers a variety of psychotherapy treatments, often in a group setting. They have the benefit of milieu therapy in the unit environment. Case management is also available. Psychiatrists are present and help manage the patient's medication and overall treatment plan.

The average length of treatment is from 1 to a few weeks, although in some cases it may be longer. The treatment should continue until the patient is sufficiently stable for a less intensive mode of treatment.

INTENSIVE OUTPATIENT TREATMENT

Intensive outpatient treatment (IOP) is a treatment in the outpatient setting that involves 3 to 4 hours of psychosocial treatment for between 1 and 4 days of treatment, with an average of 6 to 12 hours of treatment per week.

This care is usually appropriate when the level of danger is sufficiently stable so as not to be at imminent risk of harming themselves or others.

Patients usually have moderate symptoms and mild to moderate functioning. For example, their self-care may be mildly below what we would expect for them. However, they should be sufficiently functional to be able to live independently or with assistance from a caregiver.

Although sufficiently stable to live independently, the psychiatrist should be sure that the patient's home environment is sufficiently safe and without illness triggers to allow for improvement with treatment.

The treatments emphasize a group format but often include some individual psychotherapy treatments or medication management. Psychotherapies may include cognitive behavioral therapy, interpersonal therapy, or other therapies in a group format.

The length of treatment varies widely and depends on the illness, but usually lasts 1 to several weeks. The goal is to address the patient's symptoms sufficiently that a less intensive model of care is feasible.

OUTPATIENT CARE

Outpatient care is pharmacologic or psychosocial management, provided by a health care professional, usually in an office setting.

This care is usually appropriate when the danger to the patient is considered reasonably low risk. Patients need not be free of all thoughts of harm to themselves or others; however, this risk should be low, and the clinician should have a reasonable belief that the patient can warn them should their situation worsen.

Also, the patient's disorder should be likely to improve with outpatient care. This situation, fortunately, includes most psychiatric disorders. The modes of treatment may include psychotherapy or medication or a combination of both.

Patients are voluntary and require some level of motivation for their treatment, sufficient so that they can show up regularly for their appointments. At times of symptomatic worsening, family members or other caretakers may assist with providing some external motivation until the patient can resume on their own.

The length of treatment varies widely depending on the disorder. The frequency of treatment also varies. In the case of an acute illness, such as an episode of major depressive disorder, a manic episode, or a worsening of

anxiety, the treatment may occur weekly or more frequently. Generally, the frequency of treatment will decrease as the patient improves, but as many disorders are chronic, patients often require ongoing medication or therapy to maintain remission. For patients who have been stable for an extended time, twice a year follow-ups may be reasonable. However, the clinician should work out with the patient a plan for increasing the follow-ups when necessary, as it less likely the clinician will catch a pending recurrence of the illness early.

RESIDENTIAL CARE

Residential care is a 24-hour facility that offers a level of treatment less intensive than traditional inpatient treatment. It generally does not have the level of security typical of an inpatient unit. Although medical and nursing staff are available, the units do not have the same level of staffing, and the treatments are not as frequent as an inpatient unit.

Patients may be at some risk of harming themselves or others, but not at imminent risk and thus not requiring the level of observation or restraint available in an inpatient unit. Their symptoms are moderately severe, as they can have severe functional limitations such that they cannot safely live independently. The patients are usually voluntary, although they may be court-ordered for treatment; however, in those cases, they are willfully participating. The units are usually not locked.

The facilities offer a variety of treatments, often in a group setting. The units differ from long-term rehabilitation units, such as nursing homes, in that the goals are improvement in symptoms and functioning and return to a lower level of care.

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C Other Issues Relevant to Psychiatry



27 Ethics and Professionalism

Ethical guidelines and knowledge of ethical principles help psychiatrists avoid *ethical conflicts* (which are tensions between what one wants to do and what is ethically right to do) and think through *ethical dilemmas* (conflicts between ethical perspectives or values).

Ethics deal with the relations between people in different groups and often entail balancing rights. *Professional ethics* refers to the appropriate way to act when in a professional role. Professional ethics derive from a combination of morality, social norms, and the parameters of the relationship people have agreed to have.

PROFESSIONAL CODES

Most professional organizations and many business groups have codes of ethics that reflect a consensus about the general standards of appropriate professional conduct. The American Medical Association's (AMA's) *Principles of Medical Ethics* and the American Psychiatric Association's (APA's) *Principles of Medical Ethics with Annotations Especially Applicable to Psychiatry* articulate ideal standards of practice and professional virtues of practitioners. These codes include exhortations to use skillful and scientific techniques; to self-regulate misconduct within the profession; and to respect the rights and needs of patients, families, colleagues, and society.

BASIC ETHICAL PRINCIPLES

Four ethical principles that psychiatrists ought to weigh in their work are respect for autonomy, beneficence, nonmaleficence, and justice. At times, the principles conflict, and we must make decisions concerning how to balance

them.

Respect for Autonomy

Autonomy requires that a person act intentionally after being given sufficient information and time to understand the benefits, risks, and costs of all reasonable options. It may mean honoring an individual's right not to hear every detail and even choosing someone else (e.g., family or doctor) to decide the best course of treatment.

Psychiatrists need to provide patients with a rational understanding of their disorder and options for treatment. Patients need conceptual understanding; the psychiatrist should not merely state isolated facts. Patients also need time to think and to talk with friends and family about their decision. Finally, if a patient is not in a state of mind to make decisions for himself or herself, the psychiatrist should consider mechanisms for alternative decision-making, such as guardianship, conservators, and health care proxy.

A young adult experienced a schizophrenic episode in which his religious fervor turned into psychotic delusions. After being involuntarily hospitalized because he became suicidal, he insistently refused medication, claiming that his physicians were trying to poison him. His psychiatrist decided to respect his refusal of medication as long as he could control his suicidal tendencies. As his mental suffering became more intense, in 1 week, the patient changed his mind about medication and agreed to try it. The therapeutic relationship with his psychiatrist deepened, and the patient left the hospital willing to continue with both antipsychotic medication and psychotherapy. Although not all cases work out so well, this one illustrates the benefits of negotiation about treatment even when hospitalization is involuntary.

Beneficence

The requirement for psychiatrists to act with beneficence derives from their fiduciary relationship with patients and the profession's belief that they also have an obligation to the society. As a result of the role obligation of trust, psychiatrists must heed their patients' interests, even to the neglect of their

own.

The expression of the principle is called *paternalism*, the use of the psychiatrist's judgment about the best course of action for the patient or research subject. *Weak paternalism* is acting beneficently when the patient's impaired faculties prevent an autonomous choice. *Strong paternalism* is acting beneficently despite the patient's intact autonomy.

There are times when it is appropriate to permit beneficence to overrule patient autonomy. For example, when the patient faces substantial harm or risk of harm, we would choose the paternalistic act to ensure the optimal combination of maximal harm reduction, low added risk, and minimal necessary infringement on patient autonomy.

Nonmaleficence

To adhere to the principle of nonmaleficence (*primum non nocere* or *first, do no harm*), psychiatrists must be careful in their decisions and actions and must ensure that they have had adequate training for what they do. They also need to be open to seeking second opinions and consultations. They need to avoid creating risks for patients by an action or inaction.

Justice

The concept of justice concerns the issues of reward and punishment and the equitable distribution of social benefits. Relevant issues include whether resources should be distributed equally to those in greatest need, whether they should go to where they can have the most significant impact on the well-being of each individual served, or to where they will ultimately have the highest impact on society.

SPECIFIC ISSUES

From a practical point of view, several specific issues most frequently involve psychiatrists. These include (1) sexual boundary violations, (2) nonsexual boundary violations, (3) violations of confidentiality, (4) mistreatment of the patient (incompetence, double agentry), and (5) illegal activities (insurance, billing, insider stock trading).

Sexual Boundary Violations

For a psychiatrist to engage a patient in a sexual relationship is unethical. Furthermore, legal sanctions against such behavior make the ethical question moot. Psychiatrists who violate this ethical principle are subject to various criminal law statutes. Rape charges have been used against such psychiatrists; sexual assault and battery charges also have been used to convict psychiatrists.

Also, patients who have been victimized sexually by psychiatrists and other physicians have won damages in malpractice suits. Insurance carriers for the APA and the AMA no longer insure against patient–therapist sexual relations, and the carriers exclude liability for any such sexual activity.

The issue of whether sexual relations between an ex-patient and a therapist violate an ethical principle, however, remains controversial. Proponents of the view that "Once a patient, always a patient," insist that any involvement with an ex-patient—even if it leads to marriage—is unethical. They maintain that a transferential reaction that always exists between the patient and the therapist prevents a rational decision about their emotional or sexual union. Others insist that, if a transferential reaction still exists, the therapy is incomplete and that as autonomous human beings, ex-patients should not be subjected to paternalistic moralizing by physicians. Accordingly, they believe that no sanctions should prohibit emotional or sexual involvement by ex-patients and their psychiatrists. Some psychiatrists maintain that a reasonable time should elapse before such a liaison. The length of the "reasonable" period remains controversial: some have suggested 2 years. Other psychiatrists maintain that any period of prohibited involvement with an ex-patient is an unnecessary restriction. The majority of psychiatrists, as well as the APA, believe that it is always unethical, and their code of ethics states: "Sexual activity with a current or *former* patient is unethical."

Although not spelled out in *The Principles*, sexual activity with a patient's family member is also unethical. This principle is most important when the psychiatrist is treating a child or adolescent. Most training programs in child and adolescent psychiatry emphasize that the parents are patients too and that the ethical and legal proscriptions apply to parents (or parent surrogates) as well as to the child. Nevertheless, some psychiatrists misunderstand this concept. Sexual activity between a doctor and a patient's family member is also unethical.

As an example of an egregious violation, the *Medical Board of California*

Action Report (July 2006) reported on a psychiatrist who had a 7-year affair with a patient who had schizophrenia. The doctor not only had sex with the patient but also had her procure prostitutes with whom he and the patient had group sex. He paid for their services by providing them with prescriptions for controlled substances and went so far as to bill Medi-Cal for these encounters as group therapy. The state revoked the physician's license, and he was also criminally convicted of fraud.

Nonsexual Boundary Violations

The *doctor–patient relationship* is the relationship between a doctor and a patient to provide and obtain treatment. That relationship has boundaries both around it and within it. Either person may cross the boundary.

Not all boundary-crossings are boundary violations. For example, a patient may say to a doctor, "I left my money at home, can you lend me cash for the parking garage?" The patient has invited the doctor to cross the doctor–patient boundary and set up a lender–borrower relationship as well. Depending on the doctor's theoretical orientation, the clinical situation with the patient, and other factors, the doctor may elect to cross the boundary. Whether the boundary-crossing is also a boundary violation is debatable. A *boundary violation* is a boundary-crossing that is exploitative. It gratifies the doctor's needs at the expense of the patient. The doctor is responsible for preserving the boundary and for ensuring that they hold boundary-crossings to a minimum and that exploitation does not occur.

A supervisor told a psychiatry resident never to accept gifts from patients. In the course of treating a young girl with schizophrenia, the patient offered her a modest Christmas gift (a cotton scarf), which she refused to accept. The resident explained, gently, that this was against the "rules of the hospital." The next day the patient attempted suicide. She experienced the resident's refusal to accept the gift as a profound rejection, which she could not tolerate. The case illustrates the need to understand the dynamics of gift-giving and the transferential meaning to the patients of rejecting (or accepting) the gift.

There is a story (possibly apocryphal) of how Freud, who was an inveterate cigar smoker, was offered a box of difficult-to-find Havana

cigars by a patient during his analysis. Freud accepted the cigars and then proceeded to ask his patient to explore his motivations in offering the gift. Freud's reasons for accepting the cigars are more evident than the patient's unconscious motivation for giving them, about which no information is available.

Harm to the patient is not a component of a boundary violation. For example, using information supplied by the patient (e.g., a stock tip) is an unethical boundary violation, although no apparent harm may come to the patient. For purposes of discussion, nonsexual boundary violations come in several arbitrary (overlapping and not mutually exclusive) categories.

Business. Almost any business relationship with a former patient is problematic, and almost any business relationship with a current patient is unethical. Naturally, the circumstance and location may play a significant role in this admonition. In a rural area or a small community, a doctor might be treating the only pharmacist (or plumber or couch upholsterer) in town; then, when doing business with the pharmacist-patient, the doctor tries to keep boundaries in check. Ethical psychiatrists try to avoid doing business with a patient or a patient's family member or asking a patient to hire one of their family members. Ethical psychiatrists avoid investing in a patient's business and collaborating with a patient in a business deal.

Ideologic Issues. Ideologic issues can cloud judgment and may lead to ethical lapses. Any clinical decision should reflect what is best for the patient; the psychiatrist's ideology should not be a factor in the decision. A psychiatrist who is consulted by a patient with an illness should tell the patient what forms of treatment are available to treat the illness and allow the patient to decide on a course of treatment. Naturally, psychiatrists should recommend the treatment that they feel is in the best interest of the patient, but ultimately, the patient should be free to choose.

Social. We should consider the particular locale and circumstances in any discussion of the behavior of an ethical psychiatrist in social situations. The overarching principle is that the boundaries of the psychiatrist–patient relationship should be respected. Furthermore, if options exist, they should be

exercised in favor of the patient. Problems often arise in treatment situations when friendships develop between the psychiatrist and the patient. Objectivity is compromised, therapeutic neutrality is impaired, and factors outside the consciousness of either party may play a destructive role. During treatment, the psychiatrist should avoid such a friendship. Similarly, psychiatrists should not treat their social friends for the same set of reasons. Obviously, in an emergency, a person does what a person must.

Financial. For psychiatrists who practice in the private sector, dealing with the patient about money is a part of treatment. Issues surrounding setting the fee, collecting the fee, and other financial matters are grist for the mill. Even so, they must observe ethical concerns. *The Principles* advises the doctor on such matters as charging for missed appointments and other contractual problems. Financial issues are a frequent source of ethics complaints against doctors. The doctor must recognize the power that these issues have in the therapeutic relationship. Because the psychotherapeutic relationship is so much like a social relationship—the office looks like a living room; the doctor wears regular clothes; some patients might, without recognizing it, assume that a friendship exists that forgives payment of a fee. When they receive the bill, their feelings, even though they are unconscious, are ruffled. We cannot sufficiently emphasize the point that psychiatrists dispense their services in a contractual context. Early in their careers, psychiatrists are often reluctant to discuss fees openly out of a sense of embarrassment over discussing money or a sense of protecting the patient.

How an ethical psychiatrist handles the situation when a patient temporarily or permanently runs out of money is essential. Many options are available—some more problematic than others. The psychiatrist can undoubtedly lower the fee, but caution is needed because a fee lowered to the point of inadequate compensation for the treatment may evoke countertransference resentment. The number of patients receiving a reduced fee is a similar consideration. Running up a bill can also be a problem. Is there an expectation of eventually being paid? Ultimately, the psychiatrist may have to alter the frequency of sessions. Any psychiatrist who sees private patients will eventually face these problems.

Confidentiality

Confidentiality refers to the therapist's responsibility not to release information learned in the course of treatment to third parties. *Privilege* refers to the patient's right to prevent disclosure of information from treatment in judicial hearings. Psychiatrists must maintain confidentiality because it is an essential ingredient of psychiatric care; it is a prerequisite for patients to be willing to speak freely to therapists. Violating confidentiality by gossiping embarrasses people and violates nonmaleficence. Violation of confidentiality also breaks the promise that a psychiatrist has explicitly or implicitly made to keep material confidential.

Confidentiality must also give way to the responsibility to protect others when a patient makes a credible threat to harm someone. The situation becomes complicated when the risk is not to a particular individual, such as when a doctor is impaired, or someone's mental state adversely affects his or her performance of a dangerous job, such as police work, firefighting, or use of dangerous machinery. Erosion has also arisen from the demands of an insurance company for detailed information. Psychiatrists should tell patients if they are going to release their information, under what circumstances, and when they do. They are not required to warn a patient when they report the abuse of a child or a threat to others.

There are various settings in which we may use patient data ethically. The general rule for doing so is to disclose only that information that is truly necessary. In teaching, research, and supervision, we should avoid using personal information that might allow others to identify a patient unless it is necessary. In ward rounds and case conferences, in which one presents patient material, we should remind those present that what they hear should not be repeated.

Confidentiality endures after death, with the ethical obligation to withhold information unless the next of kin provides consent. A subpoena is not an automatic license to release the entire record. A psychiatrist can petition the judge for an in-camera (private) review to define what precise information they must disclose.

Ethics in Managed Care

Psychiatrists have specific responsibilities toward patients treated in managed care settings, including the responsibilities to disclose all treatment options,

exercise appeal rights, continue emergency treatment, and cooperate reasonably with utilization reviewers.

Responsibility to Disclose. Psychiatrists have a continuing responsibility to the patient to obtain informed consent for treatments or procedures. The psychiatrist should disclose all the treatment options, even those not covered under the terms of a managed care plan. Most states have enacted legislation making gag rules illegal that limit information about treatment provided to patients under managed care.

Responsibility to Appeal. The AMA Council on Ethical and Judicial Affairs states that physicians have an ethical obligation to advocate for any care that they believe will materially benefit their patients, regardless of any allocation guidelines or gatekeeper directives.

Responsibility to Treat. Physicians are liable for failure to treat their patients within the defined standard of care. The treating physician has the sole responsibility to determine what is medically necessary. Psychiatrists must be careful not to discharge suicidal or violent patients prematurely merely because a managed care company does not approve continued coverage of benefits.

Responsibility to Cooperate with Utilization Review. The psychiatrist should cooperate with utilization reviewers' requests for information on proper authorization from the patient. When the company denies benefits, it is crucial to understand and follow grievance procedures carefully, return telephone calls from review agencies, and provide documented justification for continued treatment.

With the advent of managed care and the need to send periodic progress reports and documentation of signs and symptoms to third-party reviewers to pay for treatment, some psychiatrists may diminish or exaggerate symptomatology. The following case report and discussion illustrate the ethical difficulties psychiatrists face in dealing with managed care.

Mrs. P admitted herself to the hospital because she was afraid she might kill herself. She was experiencing a major depressive episode, but she

improved markedly during the first weeks in Dr. A's ward. Although Dr. A believed that Mrs. P was no longer suicidal, he thought she would benefit significantly from continued hospitalization. Because he knew that Mrs. P could not afford to pay for hospitalization and that the insurance company would only pay if the patient were suicidally depressed, he decided not to document Mrs. P's improvement. He noted in the chart that "the patient continues to have a risk of suicide."

Does Dr. A engage in deception? Yes, he intentionally misleads by what he writes and what he omits writing in the chart. Although what he writes is true in some literal sense, his statement is misleading in the context of treatment. Mrs. P is not suicidally depressed in the way that she was.

What Dr. A omits from the chart is also deceptive. Whether a particular omission is deceptive depends, in part, on the roles and expectations of the people involved. Not telling a colleague that one dislikes his tie is not a deception. It is merely polite unless the role or relationship involves the expectation that one offers a candid opinion. Dr. A's case is different. His professional role is to document the patient's course, and the expectation is that he will note any significant improvement. Thus, his failure to document Mrs. P's progress accurately is a kind of deception.

The second and more difficult question is whether deception is justified in this instance. The answer depends on the reasons for and against the deception and the alternatives available. The reasons for this deception are apparent. Dr. A's aim and primary obligation are to help the patient. He believes that Mrs. P would benefit significantly from continued hospitalization that she cannot afford. He may also believe that it is unfair for the insurance company to refuse to pay for the inpatient treatment of nonsuicidal depression and that his deception rectifies that unfair practice.

Important reasons also exist against this deception. The first concerns honesty and social trust. It is a good thing if people can rely on what others say and write. Without some honesty and trust, many social exchanges and practices would be impossible. Deception, even for beneficent purposes, has real potential to damage social trust. A risk exists that deception may damage people's trust in the profession of

psychiatry and even patients' trust in their psychiatrists. Damage to trust may, in turn, compromise treatment.

The second reason concerns future medical treatment. If Mrs. P seeks medical treatment in the future, the physicians who attend her will read the misleading notes. If they believe that the notes are an accurate account of the previous treatment, they may suggest an inappropriate treatment for the present problem. Even if they doubt the record's accuracy, the psychiatrist has deprived their colleagues of an accurate history and report. In either case, the prior deception can hinder treatment.

The third reason concerns obligations and coverage policies. Dr. A seems to ignore the obligation that he has to the population that is covered by the insurance policy. He shifts a burden onto this population by forcing the insurance company to pay for treatment that it did not agree to cover. Perhaps the insurance company should pay for inpatient treatment in cases such as Mrs. P's; perhaps its policies are unreasonable and unfair. However, Dr. A's deception does not challenge the insurance company and pressure it to change its policy, nor does his deception encourage patients and their families to contest the company's policies. The use of deception simply circumvents, in an ad hoc way, a policy that should be challenged and discussed.

Dr. A also seems to ignore his obligation to future patients. By introducing an inaccuracy into the chart, he compromises the value of medical records research. His deception works, in a small way, to deprive future patients of the benefit of the research that relies on medical records.

One should also consider alternatives to deception. One alternative is to tailor the chart. Another alternative is to describe Mrs. P's response accurately and to discharge her to outpatient care. However, a third alternative exists. Dr. A can accurately document the patient's course and can recommend continued hospitalization. He can petition the insurance company for coverage. If the insurance company decides not to approve further inpatient care for the patient, Dr. A can appeal that decision. This alternative is more time consuming, and nothing guarantees that it will succeed, but it avoids all the problems associated with the use of deception.

Impaired Physicians

A physician may become impaired as the result of psychiatric or medical disorders or the use of mind-altering and habit-forming substances (e.g., alcohol and drugs). Many illnesses can interfere with the cognitive and motor skills required to provide competent medical care. Although the legal responsibility to report an impaired physician varies, depending on the state, the ethical responsibility remains universal. We should report an incapacitated physician to an appropriate authority, and the reporting physician is required to follow the specific hospital, state, and legal procedures. A physician who treats an impaired physician should not be required to monitor the impaired physician's progress or fitness to return to work. An independent physician or group of physicians who have no conflicts of interest should perform the monitoring.

Most states have a division that investigates professional misconduct, usually overseen by their board of health. The definitions of professional misconduct differ by state but invariably include practicing while impaired. Professional misconduct complaints derive mainly from the public in addition to insurance companies, law enforcement agencies, and doctors, among others. Many states also have a physician health committee, often independent of the disciplinary board, which helps to ensure that impaired physicians receive appropriate treatment and monitoring.

Physicians in Training

It is unethical to delegate authority for patient care to anyone who is not appropriately qualified and experienced, such as a medical student or a resident, without adequate supervision from an attending physician. Residents are physicians in training and, as such, must provide a good deal of patient care. Within a healthy, ethical teaching environment, residents and medical students may care for many ill patients. However, this care should be supervised and supported by highly trained and experienced physicians. Patients have the right to know the level of training of their care providers and should be informed about the resident's or medical student's level of training. Residents and medical students should know and acknowledge their

limitations and should ask for supervision from experienced colleagues as necessary.

Physician Charter on Professionalism

In 2001, a movement to clarify the concept of "professionalism" was begun by the American Board of Internal Medicine. A set of principles called the *Physician Charter on Professionalism* was developed, which describes what it means for physicians to perform at their highest and most ethical level. Table 27-1 lists the principles and commitments of professional behaviors in the *Physician Charter on Professionalism* to which all physicians (including psychiatrists) should adhere.

Table 27-2 considers some common ethical questions that psychiatrists may face.



Table 27-1
Physician Charter on Professionalism

Fundamental Principles

- **Primacy of patient welfare**. Altruism contributes to the trust central to doctor–patient relationships. Market forces, societal pressures, and administrative exigencies must not compromise this principle.
- **Patient autonomy**. Physicians must be honest with patients and empower them to make informed decisions about treatment.
- **Social justice**. Physicians should work actively to eliminate discrimination in health care, whether based on race, gender, socioeconomic status, ethnicity, religion, or any other social category.

A Set of Commitments

- **Professional competence**. Physicians must be committed to lifelong learning. The profession as a whole must strive to see that all of its members are competent.
- **Honesty with patients**. Physicians must ensure that patients are completely and honestly informed before consenting to a treatment; they must be empowered to decide about the course

of therapy. Physicians should also acknowledge that medical errors that injure patients sometimes occur. If a patient is injured through error, he or she should be informed promptly because failure to do so seriously compromises patient and societal trust.

- **Patient confidentiality**. Fulfilling the commitment to confidentiality is more pressing now than ever before given the widespread use of electronic information systems for compiling patient data.
- **Maintaining appropriate relations with patients**. Physicians should never exploit patients for any sexual advantage, personal financial gain, or other private purpose.
- **Improving quality of care**. This commitment entails both maintaining clinical competence and working collaboratively with other professionals to reduce medical error, increase patient safety, minimize overuse of health care resources, and optimize the outcomes of care.
- **Improving access to care**. Physicians must individually and collectively strive to reduce barriers to equitable health care.
- **Just distribution of finite resources**. Physicians should be committed to working with other physicians, hospitals, and payers to develop guidelines for cost-effective care. The physician's professional responsibility for appropriate allocation of resources requires scrupulous avoidance of superfluous tests and procedures.
- **Scientific knowledge**. Physicians have a duty to uphold scientific standards, to promote research, and to create new knowledge and ensure its appropriate use.
- **Maintaining trust by managing conflicts of interest**. Physicians have an obligation to recognize, disclose to the general public, and address conflicts of interest. Relationships between industry and opinion leaders should be disclosed.
- **Professional responsibilities**. Physicians are expected to participate in the process of self-regulation, including remediation and discipline of members who have failed to meet professional standards.

Table 27-2
Ethical Questions and Answers

Topic	Question	Answer
Abandonment	How can psychiatrists avoid being charged with patient abandonment on retirement?	Retiring psychiatrists are not abandoning patients if they provide their patients with sufficient notice and make every reasonable effort to find follow-up care for the patients.
	Is it ethical to provide only outpatient care to a seriously ill patient who may require hospitalization?	This could constitute abandonment unless the outpatient practitioner or agency arranges for their patients to receive inpatient care from another provider.
Bequests	A dying patient bequeaths his or her estate to his or her treating psychiatrist. Is this ethical?	No. Accepting the bequest seems improper and exploitational of the therapeutic relationship. However, it may be ethical to accept a token bequest from a deceased patient who named his or her psychiatrist in the will without that psychiatrist's knowledge.
Competency	Is it ethical for psychiatrists to perform vaginal examinations? Hospital physical examinations?	Psychiatrists may provide nonpsychiatric medical procedures if they are competent to do so and if the procedures do not preclude effective psychiatric

		treatment by distorting the transference. Pelvic examinations carry a high risk of distorting the transference and would be better performed by another clinician.
	Can ethics committees review issues of physician competency?	Yes. Incompetency is an ethical issue.
Confidentiality	Must confidentiality be maintained after the death of a patient?	Yes. Ethically, confidences survive a patient's death. Exceptions include protecting others from imminent harm or proper legal compulsions.
	Is it ethical to release information about a patient to an insurance company?	Yes, if the information provided is limited to that which is needed to process the insurance claim.
	Can a videotaped segment of a therapy session be used at a workshop for professionals?	Yes, if informed, uncoerced consent has been obtained, anonymity is maintained, the audience is advised that editing makes this an incomplete session, and the patient knows the purpose of the videotape.
	Should a physician report mere	No. A physician must make several assessments before

	suspicion of child abuse in a state requiring reporting of child abuse?	deciding whether to report suspected abuse. One must consider whether abuse is ongoing, whether abuse is responsive to treatment, and whether reporting will cause potential harm. Check specific statutes. Make safety for potential victims the top priority.
Conflict of interest	Is there a potential ethical conflict if a psychiatrist has both psychotherapeutic and administrative duties in dealing with students or trainees?	Yes. You must define your role in advance to the trainees or students. Administrative opinions should be obtained from a psychiatrist who is not involved in a treatment relationship with the trainee or student.
Diagnosis without examination	Is it ethical to offer a diagnosis based only on review of records to determine, for insurance purposes, if suicide was the result of illness?	Yes.
	Is it ethical for a supervising psychiatrist to sign a diagnosis on an insurance form for services	Yes, if the psychiatrist ensures that proper care is given and the insurance form clearly indicates the role of supervisor and supervisee.

	provided by a supervisee when the psychiatrist has not examined the patient?	
Exploitation (also see Bequests)	What constitutes exploitation of the therapeutic relationship?	Exploitation occurs when the psychiatrist uses the therapeutic relationship for personal gain. This includes adopting or hiring a patient as well as sexual or financial relationships.
Fee splitting	What is fee splitting?	Fee splitting occurs when one physician pays another for a patient referral. This would also apply to lawyers giving a forensic psychiatrist referrals in exchange for a percentage of the fee. Fee splitting may occur in an office setting if the psychiatrist takes a percentage of his or her office mates' fees for supervision or expenses. Costs for such items or services must be arranged separately. Otherwise, it would appear that the office owner could benefit from referring patients to a colleague in the office. Fee splitting is illegal.
Informed consent	Is it ethical to refuse to divulge	No. It is the patient's decision, not the therapist's.

	information about a patient who has agreed to give this information to those requesting it?	
	Is informed consent needed when presenting or writing about case material?	Not if the patient is aware of the supervisory or teaching process and confidentiality is preserved.
Moonlighting	Can psychiatric residents ethically “moonlight”?	They can if their duties are not beyond their ability, if they are properly supervised, and if the moonlighting does not interfere with their residency training.
Reporting	Should psychiatrists expose or report unethical behavior of a colleague or colleagues? Can a spouse bring an ethical complaint?	Psychiatrists are obligated to report colleagues’ unethical behavior. A spouse with knowledge of unethical behavior can bring an ethical complaint as well.
Research	How can ethical research be performed with subjects who cannot give informed consent?	Consent can be given by a legal guardian or via a living will. Incompetent persons have the right to withdraw from the research project at any time.
Retirement	See Abandonment.	

Supervision	What are the ethical requirements when a psychiatrist supervises other mental health professionals?	The psychiatrist must spend sufficient time to ensure that proper care is given and that the supervisee is not providing services that are outside the scope of his or her training. It is ethical to charge a fee for supervision.
Taping and recording	Can videotapes of patient interviews be used for training purposes on a national level (e.g., workshops, board examination preparation)?	Appropriate and explicit informed consent must be obtained. The purpose and scope of exposure of the tape must be emphasized in addition to the resulting loss of confidentiality.

Table by Eugene Rubin, M.D. Data derived from the American Medical Association's Principles of Medical Ethics.

Military Psychiatry

Psychiatrists in the military face unique ethical problems because confidentiality does not exist under the military code of conduct. Psychiatrists, when treating military patients, are obligated to warn them regarding these limits. The reasoning is related to concepts of military necessity; however, many experts have pointed out the problems with this rule, as many soldiers may avoid needed care rather than risk exposure. Given the concerns about the rates of suicide in the military, there is growing pressure to revise this standard. Anecdotally many military psychiatrists choose to use their judgment as to what they are required to reveal to a patient's superiors.

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28 Forensic and Legal Issues

"*Forensic*" is from Latin and means "of the forum." In Rome, criminal cases were presented in a public forum, hence the modern connotation of the term is relating to courts of law. At various times, psychiatry and the law intersect. Forensic psychiatry is the branch of psychiatry that examines this intersection. This chapter considers some of the legal aspects of psychiatry.

MEDICAL MALPRACTICE

Medical malpractice is a tort or civil wrong. It is a wrong resulting from a physician's negligence. Simply put, negligence means doing something that a physician with a duty to care for the patient should not do, or not doing something they should, as defined by current medical practice. Usually, expert witnesses help to establish the standard of care. Also, journal articles; professional textbooks, such as the *Kaplan and Sadock's Comprehensive Textbook of Psychiatry;* professional practice guidelines; and ethical practices promulgated by professional organizations all help to establish the standard of care.

To prove malpractice, the plaintiff (e.g., patient, or family) must establish by a preponderance of evidence that (1) a doctor–patient relationship existed that created a *duty* of care, (2) a *deviation* from the standard of care occurred, (3) the patient was *damaged*, and (4) the deviation *directly* caused the damage.

These elements of a malpractice claim are sometimes called the *4 Ds* (duty, deviation, damage, and direct causation).

Each of the four elements of a malpractice claim must be present, or there can be no finding of liability. For example, a psychiatrist whose negligence is

the direct cause of harm to an individual (physical, psychological, or both) is not liable for malpractice if no doctor–patient relationship existed to create a duty of care. Psychiatrists are not likely to be successfully sued if they advise on a radio program, particularly if they begin with the caveat that there is no doctor–patient relationship intended from the general advice-giving. No malpractice claim will be sustained against a psychiatrist if a patient's worsening condition is unrelated to negligent care. Not every adverse outcome is the result of negligence. Psychiatrists cannot guarantee correct diagnoses and treatments. When the psychiatrist provides due care, they may make mistakes without necessarily incurring liability. Most psychiatric cases are complicated. Psychiatrists make judgment calls when selecting a particular treatment course among the many options that may exist. In hindsight, the decision may prove wrong but not be a deviation in the standard of care.

In addition to negligence suits, patients can sue psychiatrists for the intentional torts of assault, battery, false imprisonment, defamation, fraud or misrepresentation, invasion of privacy, and intentional infliction of emotional distress. In an intentional tort, wrongdoers intend harm to another person or realize, or should have realized, that such harm is likely to result from their actions. For example, telling a patient that sex with the therapist is therapeutic perpetrates a fraud. Most malpractice policies do not provide coverage for intentional torts.

Negligent Prescription Practices

Negligent prescription practices usually include exceeding recommended dosages and then failing to adjust the medication level to therapeutic levels, unreasonable mixing of drugs, prescribing medications that are not indicated, prescribing too many drugs at one time, and failing to disclose medication effects. Elderly patients frequently take a variety of drugs prescribed by different physicians. We should only prescribe multiple psychotropic medications with special care because of possible harmful interactions and adverse effects. We should also carefully document our reasons for prescribing multiple medications.

Psychiatrists who prescribe medications must explain the diagnosis, risks, and benefits of the drug within reason and as circumstances permit (Table 28-

1). Obtaining informed consent can be problematic if a psychiatric patient has diminished cognitive capacity because of mental illness or chronic brain impairment. In these cases, a substitute health care decision maker may need to provide consent.

We should obtain informed consent each time we change a medication and introduce a new drug. If a medication injures a patient and the doctor did not inform them of this risk, there may be sufficient grounds for a malpractice action.

The question is often asked: How frequently should patients be seen for medication follow-up? The answer is that we should see patients according to their clinical needs. There is no stock answer about the frequency of visits. The longer the time interval between visits, however, the higher the likelihood of adverse drug reactions and clinical developments. Patients taking medications should probably not go beyond 6 months for follow-up visits. Managed care policies that do not reimburse for frequent follow-up appointments can result in a psychiatrist prescribing large amounts of medications. The psychiatrist is duty-bound to provide appropriate treatment to the patient, quite apart from managed care or other payment policies.

There are several other areas of negligence involving medication that may result in malpractice actions. A psychiatrist may fail to treat adverse effects that they have, or should have, recognized. They may also fail to monitor a patient's compliance with prescription limits or fail to prescribe the appropriate amount of medication. Prescribing addictive drugs to a vulnerable patient creates a risk, as can failing to seek appropriate consultation when indicated. Finally, stopping a medication inappropriately can be a cause of a malpractice action.

Split Treatment

In split treatment, the psychiatrist provides medication, and a nonmedical therapist conducts the psychotherapy. The following vignette illustrates a possible complication.



Table 28-1
Informed Consent: Reasonable Information to Be Disclosed

Although there exists no consistently accepted standard for information disclosure for any given medical or psychiatric situation, as a rule of thumb, five areas of information are generally provided:

1. Diagnosis—description of the condition or problem
2. Treatment—nature and purpose of proposed treatment
3. Consequences—risks and benefits of the proposed treatment
4. Alternatives—viable alternatives to the proposed treatment, including risks and benefits
5. Prognosis—projected outcome with and without treatment

Table by RI Simon, M.D.

A psychiatrist provided medications for a depressed 43-year-old woman. A master's level counselor saw the patient for outpatient psychotherapy. The psychiatrist saw the patient for 20 minutes during the initial evaluation and prescribed a tricyclic drug, and the patient was prescribed sufficient drugs for follow-up in 3 months. The psychiatrist initially diagnosed recurrent major depression. The patient denied suicidal ideation but had markedly diminished appetite and sleep. The patient had a long history of recurrent depression with suicide attempts. There were no further discussions between the psychiatrist and the counselor, who saw the patient once a week for 30 minutes in psychotherapy. Within 3 weeks, after a failed romantic relationship, the patient stopped taking her antidepressant medication, started to drink heavily, and committed suicide with an overdose of alcohol and antidepressant drugs. The patient's family sued the counselor and psychiatrist for negligent diagnosis and treatment.

Psychiatrists must do an adequate evaluation, obtain prior medical records, and understand that no such thing as a partial patient exists. Split treatments are potential malpractice traps because patients can "fall between the cracks" of fragmented care. The psychiatrist retains full responsibility for the patient's care in a split-treatment situation. This responsibility does not preempt the responsibility of the other mental health professionals involved

in the patient's treatment. Section V, annotation 3 of the *Principles of Medical Ethics with Annotations Especially Applicable to Psychiatry*, states: "When the psychiatrist assumes a collaborative or supervisory role with another mental health worker, he/she must expend sufficient time to assure that proper care is given."

In managed care or other settings, a marginalized role of merely prescribing medication apart from a working doctor–patient relationship does not meet generally accepted standards of good clinical care. The psychiatrist must be more than just a medication technician. Fragmented care in which the psychiatrist only dispenses medication while remaining uninformed about the patient's overall clinical status constitutes substandard treatment that may lead to a malpractice action. At a minimum, such a practice diminishes the efficacy of the drug treatment itself or may even lead to the patient's failure to take the prescribed medication.

Split-treatment situations require that the psychiatrist remain fully informed of the patient's clinical status as well as the nature and quality of treatment the patient is receiving from the nonmedical therapist. In a collaborative relationship, the psychiatrist and therapist share the responsibility for the patient's care according to their qualifications and limitations. The responsibilities of each discipline do not diminish those of the other disciplines. The treatment team should explain the separate responsibilities of each discipline to the patient. The psychiatrist and the nonmedical therapist must periodically evaluate the patient's clinical condition and requirements to determine whether the collaboration should continue. On termination of the collaborative relationship, both parties treating the patient should inform the patient either separately or jointly. In split treatments, if the patient sues the nonmedical therapist, the collaborating psychiatrist will likely be sued also and vice versa.

Psychiatrists who prescribe medications in a split-treatment arrangement should be able to hospitalize a patient if it becomes necessary. If the psychiatrist does not have admitting privileges, they should make prearrangements with other psychiatrists who can hospitalize patients if emergencies arise. Many community mental health organizations, group psychiatry practices, and managed care companies rely on split treatment, and this is a potential malpractice minefield.

PRIVILEGE AND CONFIDENTIALITY

Privilege

Privilege is the right to maintain secrecy or confidentiality in the face of a subpoena. Privileged communications are statements made by certain persons within a relationship—such as husband–wife, priest–penitent, or doctor–patient—that the law protects from forced disclosure on the witness stand. In the case of the doctor–patient relationship, the right of privilege belongs to the patient, not to the physician so that the patient can waive the right.

Psychiatrists, who are licensed to practice medicine, may claim medical privilege, but privilege has some qualifications. For example, this privilege does not exist in military courts, regardless of whether the physician is in the military. Also, the state where the trial takes place is irrelevant even if the state recognizes such privilege as military courts are not under the jurisdiction of the states.

In 1996, the United States Supreme Court recognized a psychotherapist–patient privilege in *Jaffee v. Redmon.* Emphasizing the critical public and private interests served by the psychotherapist–patient privilege, the court wrote:

Because we agree with the judgment of the state legislatures and the Advisory Committee that a psychotherapist-patient privilege will serve a "public good transcending the normal predominant principle utilizing all rational means for ascertaining truth"… we hold that confidential communications between a licensed psychotherapist and her patients in the course of diagnosis or treatment are protected from compelled disclosure under Rule 501 of the Federal Rules of Evidence.

Confidentiality

A long-held premise of medical ethics binds physicians to hold secret all information given by patients. This professional obligation is called *confidentiality.* Confidentiality applies to certain populations and not to others; a group that is within the circle of confidentiality shares information without receiving specific permission from a patient. Such groups include, in addition to the physician, other staff members treating the patient, clinical

supervisors, and consultants.

A subpoena can force a psychiatrist to breach confidentiality, and courts must be able to compel witnesses to testify for the law to function adequately. A subpoena ("under penalty") is an order to appear as a witness in court or at a deposition. Physicians usually are served with a *subpoena duces tecum*, which requires that they also produce their relevant records and documents. Although the power to issue subpoenas belongs to a judge, they are routinely issued at the request of an attorney representing a party to an action.

In bona fide emergencies, we can release information in as limited a way as feasible to carry out necessary interventions. We should try, time allowing, to obtain the patient's permission, and we should always debrief the patient after an emergency as to what information we released.

As a rule, clinical information may be shared with the patient's permission —preferably written permission, although oral permission suffices with proper documentation. Each release is good for only one piece of information, and the psychiatrist should obtain permission for each subsequent release, even to the same party. Permission overcomes only the legal barrier, not the clinical one; the release is permission, not an obligation. If a clinician believes that the information may be destructive, they should discuss this with the patient, and they can refuse the release, with some exceptions.

Third-Party Payers and Supervision. Increased insurance coverage for health care is precipitating a concern about confidentiality and the conceptual model of psychiatric practice. Today, insurance covers about 70 percent of all health care bills. An insurance carrier must be able to obtain information with which it can assess the administration and costs of various programs to provide coverage.

Quality control of care necessitates that confidentiality is not absolute; it also requires a review of individual patients and therapists. The therapist in training must breach a patient's confidence by discussing the case with a supervisor. Institutionalized patients who have been ordered by a court to get treatment must have their individualized treatment programs submitted to a mental health board.

Discussions about Patients. In general, psychiatrists have multiple

loyalties: to patients, to society, and the profession. Through their writings, teaching, and seminars, they can share their acquired knowledge and experience and provide information that may be valuable to other professionals and the public. It is not easy to write or talk about a psychiatric patient without breaching the confidentiality of the relationship. Unlike physical ailments, which we can usually discuss without anyone recognizing the patient, a psychiatric history usually entails a discussion of distinguishing characteristics. Psychiatrists have an obligation not to disclose identifiable patient information (and, perhaps, any descriptive patient information) without appropriate informed consent. Failure to obtain informed consent could result in a claim based on breach of privacy, defamation, or both.

The Internet and Social Media. Psychiatrists and other mental health professionals must be aware of the legal implications of discussing patients over the Internet. Internet communications about patients are not confidential, are subject to hacking, and are open to legal subpoenas. Some psychiatrists have blogged about patients thinking they were sufficiently disguised only to find that others do recognize them, including the involved patient. Some professional organizations have electronic mailing lists in which they ask advice about patients from their colleagues or make referrals and, in so doing, provide detailed information about the patient, which others can trace. Similarly, using social media to communicate about patients is equally risky.

Child Abuse. In many states, all physicians must take a course on child abuse for medical licensure. All states now legally require that psychiatrists (along with others), when suspecting child abuse, must immediately report this to an appropriate agency. In this situation, confidentiality is decisively limited by legal statute because potential or actual harm to vulnerable children outweighs the value of confidentiality in a psychiatric setting. Although many complex psychodynamic nuances accompany the required reporting of suspected child abuse, such reports generally are considered ethically justified.

HIGH-RISK CLINICAL SITUATIONS

Tardive Dyskinesia

It is estimated that at least 10 to 20 percent of patients and perhaps as high as 50 percent of patients treated with first-generation antipsychotics for more than 1 year exhibit some tardive dyskinesia. These figures are even higher for elderly patients. Despite the possibility for many tardive dyskinesia–related suits, relatively few psychiatrists have been sued. Also, patients who develop tardive dyskinesia may not have the physical energy and psychological motivation to pursue litigation. Allegations of negligence involving tardive dyskinesia occur when there is a failure to evaluate a patient properly, a failure to obtain informed consent, a negligent diagnosis of a patient's condition, and a failure to monitor.

Suicidal Patients

Psychiatrists may face legal action when their patients commit suicide. This is particularly true when psychiatric inpatients kill themselves. Psychiatrists are assumed to have more control over inpatients, making the suicide preventable.

The evaluation of suicide risk is one of the most complex, dauntingly difficult clinical tasks in psychiatry. Suicide is a rare event. In our current state of knowledge, clinicians cannot accurately predict when or if a patient will commit suicide. No professional standards exist for predicting who will or will not commit suicide. Professional standards do exist for assessing suicide risk, but at best, only the degree of suicide risk can be judged clinically after a comprehensive psychiatric assessment.

A review of the case law on suicide reveals that certain affirmative precautions should be taken with a suspected or confirmed suicidal patient. For example, failing to perform a reasonable assessment of a suicidal patient's risk for suicide or implement an appropriate precautionary plan will likely render a practitioner liable. The law tends to assume that suicide is preventable if it is foreseeable. Courts carefully scrutinize suicide cases to determine if a patient's suicide was foreseeable. *Foreseeability* is a deliberately vague legal term that has no comparable clinical counterpart, a common sense rather than a scientific construct. It does not (and should not) imply that clinicians can predict suicide. Foreseeability should not be confused with preventability, however. In hindsight, many suicides seem preventable that were not foreseeable.

Violent Patients

Psychiatrists who treat violent or potentially violent patients may be sued for failure to control aggressive outpatients and for the discharge of violent inpatients. Psychiatrists could be sued for failing to protect society from the violent acts of their patients if it was reasonable for the psychiatrist to have known about the patient's violent tendencies and if the psychiatrist could have done something that could have safeguarded the public. In the landmark case *Tarasoff v. Regents of the University of California*, the California Supreme Court ruled that mental health professionals have a duty to protect identifiable, endangered third parties from imminent threats of serious harm made by their outpatients. Since then, courts and state legislatures have increasingly held psychiatrists to a fictional standard of having to predict future behavior (dangerousness) of their potentially violent patients. Research has consistently demonstrated that psychiatrists cannot predict future violence with any dependable accuracy.

The duty to protect patients and endangered third parties is primarily a professional and moral obligation and, only secondarily, a legal duty. Most psychiatrists acted to protect both their patients and others threatened by violence long before *Tarasoff*.

If a patient threatens harm to another person, most states require that the psychiatrist perform some intervention that might prevent the harm from occurring. In states with duty-to-warn statutes, the states define the options available to psychiatrists and psychotherapists. In states offering no such guidance, health care providers should use their clinical judgment and act to protect endangered third persons. Typically, a variety of options to warn and protect are clinically and legally available, including voluntary hospitalization, involuntary hospitalization (when meeting civil commitment requirements), warning the intended victim of the threat, notifying the police, adjusting medication, and seeing the patient more frequently. Warning others of danger, by itself, is usually insufficient. Psychiatrists should consider the *Tarasoff* duty to be a national standard of care, even if they practice in states that do not have a duty to warn and protect.

Tarasoff I. The 1976 case *Tarasoff v. Regents of the University of California* (now known as *Tarasoff I*) raised the issue of the legal duty to warn. In this

case, Prosenjit Poddar, a student and a voluntary outpatient at the mental health clinic of the University of California, told his therapist that he intended to kill a student readily identified as Tatiana Tarasoff. Realizing the seriousness of the intention, the therapist, with the concurrence of a colleague, decided to commit Poddar for observation under a 72-hour emergency psychiatric detention provision of the California commitment law. The therapist notified the campus police, both orally and in writing that Poddar was dangerous and should be committed.

Concerned about the breach of confidentiality, the therapist's supervisor vetoed the recommendation and ordered all records relating to Poddar's treatment destroyed. At the same time, the campus police temporarily detained Poddar but released him on his assurance that he would "stay away from that girl." Poddar stopped going to the clinic when he learned from the police about his therapist's recommendation to commit him. Two months later, he carried out his previously announced threat to kill Tatiana. The young woman's parents thereupon sued the university for negligence.

As a consequence, the California Supreme Court, which deliberated the case for an unprecedented time of about 14 months, ruled that a physician or a psychotherapist who has reason to believe that a patient may injure or kill someone should warn the potential victim.

The discharge of the duty imposed on the therapist to warn intended victims against danger may take one or more forms, depending on the case. Therefore, stated the court, it may call for the therapist to notify the intended victim or others likely to notify the victim of the danger, to notify the police, or to take whatever other steps are reasonably necessary under the circumstances.

The *Tarasoff I* ruling does not require therapists to report a patient's fantasies; instead, it requires them to report an intended homicide. The therapist must exercise sound judgment to decide when this is the case.

Tarasoff II. In 1982, the California Supreme Court issued a second ruling in the case of *Tarasoff v. Regents of the University of California* (now known as *Tarasoff II*), which broadened its earlier ruling, extending the duty to warn, to include the duty to protect.

The *Tarasoff II* ruling has stimulated intense debates in the medicolegal field. Lawyers, judges, and expert witnesses argue the definition of

protection, the nature of the relationship between the therapist and the patient, and the balance between public safety and individual privacy.

Clinicians argue that the duty to protect hinders treatment because a patient may not trust a doctor if confidentiality is not maintained. Furthermore, because it is not easy to determine whether a patient is sufficiently dangerous to justify long-term incarceration, unnecessary involuntary hospitalization may occur because of a therapist's defensive practices.

As a result of such debates in the medicolegal field, since 1976, the state courts have not made a uniform interpretation of the *Tarasoff II* ruling (the duty to protect). In general, clinicians should note whether a specific identifiable victim seems to be in imminent and probable danger from the threat of an action contemplated by a mentally ill patient. The imminent harm should also be potentially serious or severe. Usually, the patient must be a danger to another person and not to property; the therapist should take clinically reasonable action.

HOSPITALIZATION

All states provide for some form of involuntary hospitalization. Such action usually is taken when psychiatric patients present a danger to themselves or others in their environment to the extent that their urgent need for treatment in a closed institution is evident. Individual states allow involuntary hospitalization when patients are unable to care for themselves adequately.

The doctrine of *parens patriae* allows the state to intervene and to act as a surrogate parent for those who are unable to care for themselves or who may harm themselves. In English common law, *parens patriae* ("father of his country") dates to the time of King Edward I and originally referred to a monarch's duty to protect the people. The US. common law transformed the doctrine into a paternalism in which the state acts for persons who are mentally ill and for minors.

The statutes governing hospitalization of persons who are mentally ill generally have been designated commitment laws, but psychiatrists have long considered the term to be undesirable. *Commitment* legally means a warrant for imprisonment. The American Bar Association and the American Psychiatric Association have recommended that state governments replace the term *commitment* with the less offensive and more accurate term

hospitalization, which most states have adopted. Although this change does not correct past mistakes, the emphasis on hospitalization is in keeping with psychiatrists' interest in treatment rather than punishment.

Procedures of Admission

Four procedures of admission to psychiatric facilities have been endorsed by the American Bar Association to safeguard civil liberties and to make sure that no one can capriciously force a person into a mental hospital. Although each of the 50 states has the power to enact its laws on psychiatric hospitalization, the procedures outlined here are gaining much acceptance.

Informal Admission. Informal admission operates on the general hospital model, in which the procedure for a psychiatric admission is the same as that for other patients. Under such circumstances, the ordinary doctor–patient relationship applies, with the patient free to enter and to leave, even against medical advice.

Voluntary Admission. In cases of voluntary admission, patients apply in writing for admission to a psychiatric hospital. They may come to the hospital on the advice of a personal physician, or they may seek help on their own. In either case, we would admit patients if an examination reveals the need for hospital treatment. The patient is free to leave, even against medical advice.

Temporary Admission. We may use a temporary admission for patients who are so senile or so confused that they require hospitalization and are not able to make decisions on their own and for patients who are so acutely disturbed that they must be admitted immediately to a psychiatric hospital on an emergency basis. Under the procedure, a hospital would admit a person on the written recommendation of one physician. After admitting the patient, a psychiatrist or the hospital staff must confirm the need for hospitalization. The procedure is temporary because we cannot hold these patients against their will for more than a set time, usually 15 days.

Involuntary Admission. Involuntary admission involves the question of whether patients are suicidal and, thus, a danger to themselves or homicidal

and thus a danger to others. Because these persons lack insight into their illness, a relative or friend may make the application for admission to a hospital. After making the application, at least two physicians must examine the patient, and if both physicians confirm the need for hospitalization, the hospital can admit the patient.

Involuntary hospitalization involves an established procedure for written notification of the next of kin. Furthermore, the patients have access at any time to legal counsel, who can bring the case before a judge. If the judge does not think that there is an indication for hospitalization, that judge can order the patient's release.

Involuntary admission allows us to hospitalize a patient for a set time, usually defined by the state and often in the range of 60 days. After this time, if the patient is to remain hospitalized, the case must be reviewed periodically by a board consisting of psychiatrists, nonpsychiatric physicians, lawyers, and other citizens not connected with the institution.

Involuntarily hospitalized persons who believe that they should be released have the right to file a petition for a writ of habeas corpus. Under the law, those who believe that they have been illegally deprived of liberty can proclaim a writ of habeas corpus. The legal procedure asks a court to decide whether a patient was hospitalized without due process of law. The courts must hear this case immediately, regardless of how the motion is filed. Hospitals are obligated to submit the petitions to the court immediately.

RIGHT TO TREATMENT

Among the rights of patients, the right to the standard quality of care is fundamental. This right has been litigated in highly publicized cases under the slogan of "right to treatment."

In 1966, Judge David Bazelon, speaking for the District of Columbia Court of Appeals in *Rouse v. Cameron,* noted that the purpose of involuntary hospitalization is treatment and concluded that the absence of treatment draws into question the constitutionality of the confinement. Treatment in exchange for liberty is the logic of the ruling. In this case, the patient was discharged on a writ of habeas corpus, the basic legal remedy to ensure liberty. Judge Bazelon further held that if alternative treatments that infringe less on personal liberty are available, involuntary hospitalization cannot take

place.

Alabama Federal Court judge Frank Johnson was more venturesome in the decree he rendered in 1971 in *Wyatt v. Stickney*. The *Wyatt* case was a class-action proceeding brought under newly developed rules that sought not release but treatment. Judge Johnson ruled that persons civilly committed to a mental institution have a constitutional right to receive such individual treatment as will give them a reasonable opportunity to be cured or to have their mental condition improved. Judge Johnson set out minimum requirements for staffing, specified physical facilities, and nutritional standards and required individualized treatment plans.

The new codes, more detailed than the old ones, include the right to be free from excessive or unnecessary medication; the right to privacy and dignity; the right to the least restrictive environment; the unrestricted right to be visited by attorneys, clergy, and private physicians; and the right to refuse lobotomies, electroconvulsive treatments, and other procedures without fully informed consent. Patients can be required to perform therapeutic tasks but not hospital chores unless they volunteer for them and receive the federal minimum wage as payment. This requirement is an attempt to eliminate the practice of peonage, in which state hospitals forced psychiatric patients to work at menial tasks, without payment, for the benefit of the state.

In several states today, medication or electroconvulsive therapy cannot be forcibly administered to a patient without first obtaining court approval, which may take as long as 10 days.

RIGHT TO REFUSE TREATMENT

The right to refuse treatment is a legal doctrine that holds that, except in emergencies, we cannot force patients to accept treatment against their will. An “emergency” is a condition in clinical practice that requires immediate intervention to prevent death or severe harm to the patient or another person or to prevent deterioration of the patient’s clinical state.

In the 1976 case of *O’Connor v. Donaldson*, the Supreme Court of the United States ruled that harmless mentally ill patients cannot be confined against their will without treatment if they can survive outside. According to the Court, a finding of mental illness alone cannot justify a state’s confining persons in a hospital against their will. Instead, involuntarily confined

patients must be considered dangerous to themselves or others or possibly so unable to care for themselves that they cannot survive outside. As a result of the 1979 case of *Rennie v. Klein*, patients have the right to refuse treatment and to use an appeal process. As a result of the 1981 case of *Roger v. Oken*, patients have an absolute right to refuse treatment, but a guardian may authorize treatment.

Many rightly question psychiatrists' ability to predict dangerousness. Also, many psychiatrists worry about the risk entailed by involuntarily hospitalizing a patient if the patient can then sue them for monetary damages for infringing on the patient's civil rights.

CIVIL RIGHTS OF PATIENTS

Because of several clinical, public, and legal movements, criteria for the civil rights of persons who are mentally ill, apart from their rights as patients, have been both established and affirmed.

Least Restrictive Alternative

The principle of *least restrictive alternative* holds that patients have the right to receive the least restrictive means of treatment for the requisite clinical effect. Therefore, if it is possible to treat a patient as an outpatient, we should not pursue commitment. Similarly, if it is safe to treat a patient in an open ward, we should not seclude them.

Although reasonably straightforward on first reading, the difficulty arises when clinicians attempt to apply the concept to choose among involuntary medication, seclusion, and restraint as the intervention of choice. Distinguishing among these interventions based on restrictiveness proves to be a purely subjective exercise fraught with personal bias. Moreover, each of these three interventions is both more and less restrictive than each of the other two. Nevertheless, we should make an effort to think in terms of restrictiveness when deciding how to treat patients.

Visitation Rights

Patients have the right to receive visitors and to do so at reasonable hours (customary hospital visiting hours). Hospitals have to allow for the fact that

at certain times, a patient's clinical condition may not permit visits. We should document when this is the case, as we should not suspend such rights without good reason.

Specific categories of visitors are not limited to the regular visiting hours; these include a patient's attorney, private physician, and members of the clergy—all of whom, broadly speaking, have unrestricted access to the patient, including the right to privacy in their discussions. Even here, a bona fide emergency may delay such visits. Again, the patient's needs come first. Under similar reasoning, we should curtail specific noxious visits (e.g., a patient's relative bringing drugs into the ward).

Communication Rights

Patients should generally have free and open communication with the outside world by telephone or mail, but this right varies regionally to some degree. Some jurisdictions charge the hospital administration with responsibility for monitoring the communications of patients. In some areas, hospitals must make available reasonable supplies of paper, envelopes, and stamps for patient's use.

Specific circumstances affect communication rights. A patient hospitalized with an associated criminal charge of making harassing or threatening phone calls should not have unrestricted access to the telephone, and similar considerations apply to mail. As a rule, however, patients should be allowed private telephone calls, and we should not invade their privacy by listening in or opening their mail.

Private Rights

Patients have several rights to privacy. In addition to confidentiality, they are allowed private bathroom and shower space, secure storage space for clothing and other belongings, and adequate floor space per person. They also have the right to wear their clothes and to carry their own money.

Economic Rights

Apart from special considerations related to incompetence, psychiatric patients generally are permitted to manage their financial affairs. One feature of this fiscal right is the requirement that we pay patients if they work in the

institution (e.g., gardening or preparing food). This right often creates tension between the valid therapeutic need for activity, including jobs, and exploitative labor. A consequence of this tension is that some hospitals have had to eliminate valuable occupational, vocational, and rehabilitative programs because of the failure of legislatures to supply the funding to pay wages to patients who participate in these programs.

Table 28-2
Indications and Contraindications for Seclusion and Restraint

Indications
Prevent clear, imminent harm to the patient or others
Prevent significant disruption to treatment program or physical surroundings
Assist in treatment as part of ongoing behavior therapy
Decrease sensory overstimulation[a]
Patient's voluntary reasonable request
Contraindications
Extremely unstable medical and psychiatric conditions[b]
Delirious or demented patients who are unable to tolerate decreased stimulation[b]
Overtly suicidal patients[b]
Patients with severe drug reactions or overdoses or who require close monitoring of drug dosages[b]
For punishment or convenience of staff

[a]Seclusion only.

[b]Unless close supervision and direct observation are provided.

Data from *The Psychiatric Uses of Seclusion and Restraint (Task Force Report No. 22)*. Washington, DC: American Psychiatric Association.

SECLUSION AND RESTRAINT

Seclusion and restraint raise complex legal issues. Seclusion and restraint have both indications and contraindications (Table 28-2). Seclusion and

restraint have become increasingly regulated over the past decade.

Legal challenges to the use of restraints and seclusion have been brought on behalf of institutionalized persons with psychiatric illnesses or cognitive disabilities. Typically, these lawsuits do not stand alone but are part of a challenge to a wide range of alleged abuses.

In general, courts hold, or consent decrees provide that hospitals can only implement restraints and seclusion when a patient creates a risk of harm to self or others, and no less restrictive alternative is available. Table 28-3 lists additional restrictions.



Table 28-3
Restrictions for Seclusion and Restraint

Restraints and seclusion can be implemented only when a patient creates a risk of harm to self or others and no less restrictive alternative is available.
Restraint and seclusion can only be implemented by a written order from an appropriate medical official.
Orders are to be confined to specific, time-limited periods.
A patient's condition must be regularly reviewed and documented.
Any extension of an original order must be reviewed and reauthorized.

INFORMED CONSENT

Lawyers representing an injured claimant now invariably add to a claim of negligent performance of procedures (malpractice) an informed consent claim as another possible area of liability. Ironically, this is one claim under which there is no requirement for expert testimony. The usual claim of medical malpractice requires the litigant to produce an expert to establish that the defendant physician departed from accepted medical practice. However, in a case in which the physician did not obtain informed consent, the fact that the treatment was technically well performed, was in accord with the generally accepted standard of care, and effected a complete cure is immaterial. As a practical matter, however, unless the treatment had adverse consequences, a complainant will not get far with a jury for actions based solely on an

allegation that the doctor performed treatment without consent.

In the case of minors, the parent or guardian must give consent to medical treatment. By statute, most states, however, list specific diseases and conditions that a minor can consent to have treated—including venereal disease, pregnancy, substance dependence, alcohol abuse, and contagious diseases. In an emergency, a physician can treat a minor without parental consent. The trend is to adopt the so-called mature minor rule, which allows minors to consent to treatment under ordinary circumstances. As a result of the Supreme Court's 1967 *Gault* decision, all juveniles must now be represented by counsel, must be able to confront witnesses, and must be given proper notice of any charges. Emancipated minors have the rights of an adult when they can show that they are living as adults with control over their own lives.

Consent Form

The essential elements of a consent form should include a fair explanation of the procedures and their purposes. The form should identify those procedures that are experimental. Also, the form should describe any attendant discomforts and risks than can be expected, as well as the benefits that are reasonable to expect. The form should also disclose appropriate alternative procedures that may be advantageous to the patient with an offer to answer any inquiries concerning the procedures. Finally, the form should explain that the patient is free to withdraw patient consent and to discontinue participation in the project or activity at any time without prejudice.

Some theorists suggest that a standardized discussion that covers the issues noted above can substitute for a form. In such a case, it would be essential to document the discussion.

CHILD CUSTODY

The action of a court in a child custody dispute centers on the child's best interests. The maxim reflects the idea that a natural parent does not have an inherent right to be named a custodial parent, but the presumption, although a bit eroded, remains in favor of the mother in the case of young children. As a rule, the courts presume that the welfare of a child of tender years generally is best served by maternal custody when the mother is a good and fit parent.

Giving the mother custody may also serve her best interest, as many parents may not otherwise ever resolve the pain of losing a child, but her best interest is not to be equated ipso facto with the best interest of the child. Care and protection proceedings are the court's interventions in the welfare of a child when the parents are unable to care for the child.

More fathers are asserting custodial claims. In about 5 percent of all cases, fathers are named custodians. With more women going to work outside the home, and more men assuming the primary role in child care, the traditional rationale for maternal custody has less force today than it did in the past.

Currently, every state has a statute allowing a court, usually a juvenile court, to assume jurisdiction over a neglected or abused child and to remove the child from parental custody. It usually orders that a welfare or probation department supervise the care and custody of the child.

TESTAMENTARY AND CONTRACTUAL CAPACITY AND COMPETENCE

Courts may ask psychiatrists to evaluate patients' testamentary capacities or their competence to make a will. Three psychological abilities are necessary to prove this competence. Patients must know the nature and the extent of their bounty (property), the fact that they are making a bequest, and the identities of their natural beneficiaries (spouse, children, and other relatives).

When probating a will, one of the heirs or another person may challenge its validity. Judgment in such cases must rely on reconstruction, using data from documents and from expert psychiatric testimony, of the testator's mental state at the time they wrote the will. When someone cannot, or chooses not, to write a will, all states provide for the distribution of property to the heirs. If there are no heirs, the estate goes to the public treasury.

Witnesses at the signing of a will, which might include a psychiatrist, may attest that the testator was rational at the time of executing the will. In unusual cases, a lawyer may videotape the signing to safeguard the will from attack. Ideally, persons who are thinking of making a will and believe that there may be questions about their testamentary competence should hire a forensic psychiatrist to perform a dispassionate examination antemortem to validate and record their capacity.

An incompetence proceeding and the appointment of a guardian may be

necessary when a family member is spending the family's assets, and the property is in danger of dissipation, as in the case of patients who are elderly, have cognitive disabilities, are dependent on alcohol, or have psychosis. An issue is whether such persons are capable of managing their affairs. A guardian appointed to take control of the property of one deemed incompetent, however, cannot make a will for the ward (the incompetent person).

We determine competence based on a person's ability to make a sound judgment—to weigh, to reason, and to make reasonable decisions. Competence is task-specific, not general; the capacity to weigh decision-making factors (competence) often is best demonstrated by a person's ability to ask pertinent and knowledgeable questions after explaining the risks and benefits. Although physicians (especially psychiatrists) often give opinions on competence, only a judge's ruling converts the opinion into a finding; a patient is not competent or incompetent until the court so rules. The diagnosis of a mental disorder is not, in itself, sufficient to warrant a finding of incompetence. Instead, the mental disorder must cause impairment in judgment for the specific issues involved. After they have been declared incompetent, persons lose specific rights: they cannot make contracts, marry, start a divorce action, drive a vehicle, handle their property, or practice their professions. A formal courtroom proceeding decides incompetence, and the court usually appoints a guardian who will best serve a patient's interests. Another hearing is necessary to declare a patient competent. Admission to a mental hospital, even involuntarily, does not automatically mean that a person is incompetent.

Competence also is essential in contracts because a contract is an agreement between parties to do a specific act. A contract is invalid if, when it was signed, one of the parties was unable to comprehend the nature and effect of his or her act. The marriage contract is subject to the same standard and thus can be voided if either party did not understand the nature, duties, obligations, and other characteristics entailed at the time of the marriage. In general, however, the courts are unwilling to declare a marriage void based on incompetence.

Whether competence is related to wills, contracts, or the making or breaking of marriages, the fundamental concern is a person's state of awareness and capacity to comprehend the significance of the particular

commitment made.

Durable Power of Attorney

A modern development that permits persons to make provisions for their anticipated loss of decision-making capacity is called a *durable power of attorney.* The document permits the advance selection of a substitute decision maker who can act without the necessity of court proceedings when the signatory becomes incompetent through illness or progressive dementia.

CRIMINAL LAW

Competence to Stand Trial

The Supreme Court of the United States stated that the prohibition against trying someone who is mentally incompetent is fundamental to the US system of justice. Accordingly, the court, in *Dusky v. the United States,* approved a test of competence that seeks to ascertain whether a criminal defendant "has sufficient present ability to consult with his lawyer with a reasonable degree of rational understanding—and whether he has a rational as well as factual understanding of the proceedings against him."

Competence to Be Executed

One of the new areas of competence to emerge in the interface between psychiatry and the law is the question of a person's competence to be executed. The requirement for competence in this area is believed to rest on three general principles. First, a person's awareness of what is happening is supposed to heighten the retributive element of the punishment. Punishment is meaningless unless the person is aware of it and knows the punishment's purpose. Second, a competent person who is about to be executed is believed to be in the best position to make whatever peace is appropriate with his or her religious beliefs, including confession and absolution. Third, a competent person who is about to be executed preserves, until the end, the possibility (admittedly slight) of recalling a forgotten detail of the events or the crime that exonerate the person.

The need to preserve competence was supported recently in the Supreme Court case of *Ford v. Wainwright.* However, no matter the outcome of legal

struggles with this question, most medical bodies have gravitated toward the position that it is unethical for any clinician to participate, no matter how remotely, in state-mandated executions; a physician's duty to preserve life transcends all other competing requirements. Major medical societies, such as the American Medical Association (AMA), believe that doctors should not participate in the death penalty. A psychiatrist who agrees to examine a patient slated for execution may find the person incompetent based on a mental disorder and may recommend a treatment plan, which, if implemented, would ensure the person's fitness to be executed. Although room exists for a difference of opinion regarding whether or not a psychiatrist should become involved, the authors of this text agree with the AMA.

Criminal Responsibility

According to criminal law, committing a socially harmful act is not the sole criterion of whether a crime has been committed. Instead, the objectionable act must have two components: voluntary conduct (*actus reus*) and evil intent (*mens rea*). An evil intent cannot exist when an offender's mental status is so deficient, so abnormal, or so diseased to have deprived the offender of the capacity for rational intent. The law can be invoked only when an illegal intent is implemented. Neither behavior, however harmful, nor the intent to harm is, in itself, a ground for criminal action.

M'Naghten Rule. The precedent for determining legal responsibility was established in 1843 in the British courts. The so-called M'Naghten rule, which, until recently, has determined criminal responsibility in most of the United States, holds that persons are not guilty by reason of insanity if they labored under a mental disease such that they were unaware of the nature, the quality, and the consequences of their acts or if they were incapable of realizing that their acts were wrong. Moreover, to absolve persons from punishment, a delusion used as evidence must be one that, if true, would be an adequate defense. If the delusional idea does not justify the crime, such persons are presumably held responsible, guilty, and punishable. The M'Naghten rule is known commonly as the right–wrong test.

The M'Naghten rule derives from the famous M'Naghten case of 1843. When Daniel M'Naghten murdered Edward Drummond, the private secretary

of Robert Peel, M'Naghten had been suffering from delusions of persecution for several years, had complained to many persons about his "persecutors," and finally had decided to correct the situation by murdering Robert Peel. When Drummond came out of Peel's home, M'Naghten shot Drummond, mistaking him for Peel. The jury, as instructed under the prevailing law, found M'Naghten not guilty by reason of insanity. In response to questions about what guidelines could be used to determine whether a person could plead insanity as a defense against criminal responsibility, the English chief judge wrote:

1. To establish a defense on the ground of insanity, it must be clearly proved that, at the time of committing the act, the party accused was laboring under such a defect of reason, from disease of the mind, as not to know the nature and quality of the act he was doing, or if he did know it, he did not know he was doing what was wrong.
2. Where a person labors under partial delusions only and is not in other respects insane and as a result commits an offense, he must be considered in the same situation regarding responsibility as if the facts with respect to which the delusion exists were real.

According to the M'Naghten rule, the question is not whether the accused knows the difference between right and wrong in general, instead, it is whether the defendant understood the nature and the quality of their act and whether they knew the difference between right and wrong—specifically whether the defendant knew the act was wrong or perhaps thought the act was correct, a delusion causing the defendant to act in legitimate self-defense.

Jeffery Dahmer killed 17 young men and boys between June 1978 and July 1991. Most of his victims were either gay or bisexual. He would meet and select his prey at gay bars or bathhouses and then lure them by offering them money for posing for photographs or simply to enjoy some beer and videos. Then he would drug them, strangle them, masturbate on the body or have sex with the corpse, dismember the body, and dispose of it. Sometimes he would keep the skull or other body parts as souvenirs.

On July 13, 1992, Dahmer changed his plea to guilty by means of insanity. The jury, however, decided that Dahmer's meticulous murder

plans and the systematic way that he disposed of his victims' bodies was evidence that Dahmer could control his behavior. All of the testimony bolstered the notion that, as with most serial killers, Dahmer knew what he was doing and knew right from wrong. Finally, the jury did not accept the defense that Dahmer experienced a mental illness to the degree that it had disabled his thinking or behavioral controls. Dahmer was sentenced to 15 consecutive life terms or a total of 957 years in prison. An inmate killed him on November 28, 1994.

Irresistible Impulse. In 1922, a committee of jurists in England reexamined the M'Naghten rule. The committee suggested broadening the concept of insanity in criminal cases to include the irresistible impulse test, which rules that a person charged with a criminal offense is not responsible for an act if the act was committed under an impulse that the person was unable to resist because of mental disease. The courts interpret this concept such that it has been called the *policeman-at-the-elbow law.* In other words, the court grants an impulse to be irresistible only when it can be determined that the accused would have committed the act even if a policeman had been at the accused person's elbow. To most psychiatrists, this interpretation is unsatisfactory because it covers only a small, select group of those who are mentally ill.

Durham Rule. In the case of *Durham v. the United States,* Judge Bazelon handed down a decision in 1954 in the District of Columbia Court of Appeals. The decision resulted in the product rule of criminal responsibility, namely that an accused person is not criminally responsible if his or her unlawful act was the product of mental disease or mental defect. In the Durham case, Judge Bazelon expressly stated that the purpose of the rule was to get good and complete psychiatric testimony. He sought to release the criminal law from the theoretical straitjacket of the M'Naghten rule, but judges and juries in cases using the *Durham* rule became mired in confusion over the terms *product, disease,* and *defect.* In 1972, some 18 years after the rule's adoption, the Court of Appeals for the District of Columbia, in *United States v. Brawner* discarded the rule. The court—all nine members, including Judge Bazelon—decided in a 143-page opinion to throw out its *Durham* rule

and to adopt in its place the test recommended in 1962 by the American Law Institute in its model penal code, which is the law in the federal courts today.

Model Penal Code. In its model penal code, the American Law Institute recommended the following test of criminal responsibility: Persons are not responsible for criminal conduct if, at the time of such conduct, as a result of mental disease or defect, they lacked substantial capacity either to appreciate the criminality (wrongfulness) of their conduct or to conform their conduct to the requirement of the law. The term *mental disease or defect* does not include an abnormality manifested only by repeated criminal or otherwise antisocial conduct.

Subsection 1 of the American Law Institute rule contains five operative concepts: mental disease or defect, lack of substantial capacity, appreciation, wrongfulness, and conformity of conduct to the requirements of law. The rule's second subsection, stating that repeated criminal or antisocial conduct is not, of itself, to be taken as mental disease or defect, aims to keep the sociopath or psychopath within the scope of criminal responsibility.

Guilty but Mentally Ill. Some states have established an alternative verdict of guilty but mentally ill. Under guilty but mentally ill statutes, this alternative verdict is available to the jury if the defendant pleads not guilty by reason of insanity. Under an insanity plea, four outcomes are possible: not guilty, not guilty by reason of insanity, guilty but mentally ill, and guilty.

The problem with finding someone guilty but mentally ill is that it is an alternative verdict without a difference. It is the same as finding the defendant just plain guilty. The court must still impose a sentence on the convicted person. Although the convicted person supposedly receives psychiatric treatment, if necessary, this treatment provision is available to all prisoners.

OTHER AREAS OF FORENSIC PSYCHIATRY

Emotional Damage and Distress

A rapidly rising trend in recent years is to sue for psychological and emotional damage, both secondary to physical injury or as a consequence of witnessing a stressful act and from the suffering endured under the stress of

such circumstances as concentration camp experiences. The German government heard many of these claims from persons detained in Nazi camps during World War II. In the United States, the courts have moved from a conservative to a liberal position in awarding damages for such claims. Psychiatric examinations and testimony are sought in these cases, often by both the plaintiffs and the defendants.

Recovered Memories

Patients alleging recovered memories of abuse have sued parents and other alleged perpetrators. In several instances, the alleged victimizers have sued therapists who, they claim, negligently induced false memories of sexual abuse. In an about-face, some patients have recanted and joined forces with others (usually their parents) to sue therapists.

Courts have handed down multimillion-dollar judgments against mental health practitioners. A fundamental allegation in these cases is that the therapist abandoned a position of neutrality to suggest, persuade, coerce, and implant false memories of childhood sexual abuse. The guiding principle of clinical risk management in recovered memory cases is the maintenance of therapist neutrality and the establishment of sound treatment boundaries. Table 28-4 lists the risk management principles that should be considered when evaluating or treating a patient who recovers memories of abuse in psychotherapy.

Table 28-4
Risk Management Principles for Cases of Recovered Memories of Abuse in Psychotherapy

1. Maintain therapist neutrality: Do not suggest abuse.
2. Stay clinically focused: Provide adequate evaluation and treatment for patients presenting problems and symptoms.
3. Carefully document the memory-recovery process.
4. Manage personal bias and countertransference.
5. Avoid mixing treater and expert witness roles.
6. Closely monitor supervisory and collaborative therapy relationships.

7. Clarify nontreatment roles with family members.
8. Avoid special techniques (e.g., hypnosis or sodium amobarbital [Amytal]) unless clearly indicated; obtain consultation first.
9. Stay within professional competence: Do not take cases that you cannot handle.
10. Distinguish between narrative truth and historical truth.
11. Obtain consultation in problematic cases.
12. Foster patient autonomy and self-determination: Do not suggest lawsuits.
13. In managed care settings, inform patients with recovered memories that more than brief therapy may be required.
14. When making public statements, distinguish personal opinions from scientifically established facts.
15. Stop and refer, if uncomfortable with a patient who is recovering memories of childhood abuse.
16. Do not be afraid to ask about abuse as part of a competent psychiatric evaluation.

Worker's Compensation

The stresses of employment can cause or accentuate mental illness. Patients are entitled to be compensated for their job-related disabilities or to receive disability retirement benefits. A psychiatrist is often called on to evaluate such situations.

Civil Liability

Psychiatrists who sexually exploit their patients are subject to civil and criminal actions in addition to ethical and professional licensure revocation proceedings. Malpractice is the most common legal action (Table 28-5).



Table 28-5
Sexual Exploitation: Legal and Ethical Consequences

Civil lawsuit
Negligence

Loss of consortium
Breach-of-contract action
Criminal sanctions (e.g., statutory, adultery, sexual assault, rape)
Civil action for intentional tort (e.g., battery, fraud)
License revocation
Ethical sanctions
Dismissal from professional organizations

Table by RI Simon, M.D.



Table 28-6
Patients' Rights under the Privacy Rule

Physicians must give the patient a written notice of his or her privacy rights; the privacy policies of the practice; and how patient information is used, kept, and disclosed. A written acknowledgment should be taken from the patient verifying that he or she has seen such notice.
Patients should be able to obtain copies of their medical records and to request revisions to those records within a stated amount of time (usually 30 days). Patients do not have the right to see psychotherapy notes.
Physicians must provide the patient with a history of most disclosures of their medical history on request. Some exceptions exist. The APA Committee on Confidentiality has developed a model document for this requirement.
Physicians must obtain authorization from the patient for disclosure of information other than for treatment, payment, and health care operations (these three are considered to be routine uses, for which consent is not required). The APA Committee on Confidentiality has developed a model document for this requirement.
Patients may request another means of communication of their protected information (e.g., request that the physician contact them at a specific phone number or address).
Physicians cannot generally limit treatment to obtaining patient

authorization for disclosure of their information for nonroutine uses. Patients have the right to complain about Privacy Rule violations to the physician, their health plan, or to the Secretary of HHS.

APA, American Psychiatric Association; HHS, Department of Health and Human Services.

Health Insurance Portability and Accountability Act

The Health Insurance Portability and Accountability Act (HIPAA) was passed in 1996 to address the medical delivery system's mounting complexity and its rising dependence on electronic communication. The act orders that the federal Department of Health and Human Services (HHS) develop rules protecting the transmission and confidentiality of patient information, and all units under HIPAA must comply with such rules.

The Privacy Rule, administered by the Office for Civil Rights (OCR) at HHS, protects the confidentiality of patient information (Table 28.6).

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29

End-of-Life Issues and Palliative Care

29.1 Death, Dying, and Bereavement

DEATH AND DYING

Definitions

The terms *death* and *dying* require definition: Whereas *death* may be considered the absolute cessation of vital functions, *dying* is the process of losing these functions. Dying may also be seen as a developmental concomitant of living, a part of the birth-to-death continuum. Living may entail numerous mini-deaths—the end of growth and its potential, health-compromising illnesses, multiple losses, decreasing vitality and growing dependency with aging, and dying. Dying, and the individual's awareness of it, imbues humans with values, passions, wishes, and the impetus to make the most of the time.

Two terms that have been used with increased frequency in recent years refer to the quality of living as death comes near. A *good death* is one that is free from avoidable distress and suffering for patients, families, and caregivers and is reasonably consistent with clinical, cultural, and ethical standards. A *bad death*, in contrast, is characterized by needless suffering, a dishonoring of the patient or family's wishes or values, and a sense among participants or observers that norms of decency have been offended.

Uniform Determination of Death Act. The President's Commission for the Study of Ethical Problems in Medicine and Biomedical and Behavioral

Research published its definition of death in 1981. Working with the American Bar Association, the American Medical Association (AMA), and the National Conference of Commissioners on Uniform State Laws, the commission established that one who has sustained either (1) irretrievable cessation of circulatory and respiratory functions or (2) irretrievable cessation of all functions of the entire brain, including the brainstem, is dead. The determination of death must be per accepted medical standards.

Generally accepted criteria for determining brain death require a series of neurologic and other assessments. For children, particular guidelines apply. They generally specify two assessments separated by an interval of at least 48 hours for those between the ages of 1 week and 2 months, 24 hours for those between the ages of 2 months and 1 year, and 12 hours for older children; additional confirmatory tests may also be advisable under some circumstances. Brain death criteria usually are not applied to infants younger than 7 days. Table 29-1 lists the clinical criteria for brain death in adults and children.

Legal Aspects of Death

According to law, physicians must sign the death certificate, which attests to the cause of death (e.g., congestive heart failure or pneumonia). They must also attribute the death to natural, accidental, suicidal, homicidal, or unknown causes. A medical examiner, coroner, or pathologist must examine anyone who dies unattended by a physician and perform an autopsy to determine the cause of death. In some cases, a psychological autopsy is performed: A person's sociocultural and psychological background is examined retrospectively by interviewing friends, relatives, and doctors to determine whether mental illness, such as a depressive disorder, was present. For example, a determination can be made that a person died because he or she was pushed (murder) or because he or she jumped (suicide) from a high building. Each situation has clear medical and legal implications.

Stages of Death and Dying

Elisabeth Kübler-Ross, a psychiatrist and thanatologist, made a comprehensive and useful organization of reactions to impending death. A dying patient seldom follows a regular series of responses that can be

identified; no established sequence applies to all patients. Nevertheless, the following five stages proposed by Kübler-Ross are widely encountered.

Stage 1: Shock and Denial. On being told that they are dying, persons initially react with shock. They may appear dazed at first and then may refuse to believe the diagnosis; they may deny that anything is wrong. Some persons never pass beyond this stage and may go from doctor to doctor until they find one who supports their position. The degree to which denial is adaptive or maladaptive appears to depend on whether a patient continues to obtain treatment even while denying the prognosis. In such cases, physicians must communicate to patients and their families, respectfully and directly, necessary information about the illness, its prognosis, and the options for treatment. For effective communication, physicians must allow for patients' emotional responses and reassure them that they will not be abandoned.

Stage 2: Anger. Persons become frustrated, irritable, and angry at being ill. They commonly ask, "Why me?" They may become angry at God, their fate, a friend, or a family member; they may even blame themselves. They may displace their anger onto the hospital staff members and the doctor, whom they blame for the illness. Patients in the stage of anger are challenging to treat. Doctors who have difficulty understanding that anger is a predictable reaction and is a displacement may withdraw from patients or transfer them to other doctors' care.



Table 29-1
Clinical Criteria for Brain Death in Adults and Children

Coma
Absence of motor responses
Absence of pupillary responses to light and pupils at midposition with respect to dilatation (4–6 mm)
Absence of corneal reflexes
Absence of caloric responses
Absence of gag reflex
Absence of coughing in response to tracheal suctioning

Absence of sucking and rooting reflexes
Absence of respiratory drive at a $Paco_2$ that is 60 or 20 mm Hg above normal baseline values
Interval between two evaluations, according to patient's age
Term to 2 mo old, 48 hr
>2 mo to 1 yr old, 24 hr
>1 yr to <18 yr old, 12 hr
≥18 yr old, interval optional
Confirmatory tests
Term to 2 mo old, two confirmatory tests
>2 mo to 1 yr old, one confirmatory test
>1 yr to <18 yr old, optional
≥18 yr old, optional

$PaCO_2$, partial pressure of arterial carbon dioxide.

Reprinted from Wijdicks EF. The diagnosis of brain death. *N Engl J Med*. 2001;344:1215–1221, with permission.

Physicians treating angry patients must realize that the anger being expressed cannot be taken personally. An empathic, nondefensive response can help defuse patients' anger and can help them refocus on their deep feelings (e.g., grief, fear, loneliness) that underlie the anger. Physicians should also recognize that anger may represent patients' desire for control in a situation in which they feel completely out of control.

Stage 3: Bargaining. Patients may attempt to negotiate with physicians, friends, or even God; in return for a cure, they promise to fulfill one or many pledges, such as giving to charity and attending church regularly. Some patients believe that if they are good (compliant, nonquestioning, cheerful), the doctor will make them better. The treatment of such patients involves making it clear that they will be taken care of to the best of the doctor's abilities and that everything that can be done will be done, regardless of any action or behavior on the patients' part. Patients must also be encouraged to participate as partners in their treatment and to understand that being a good patient means being as honest and straightforward as possible.

Stage 4: Depression. In the fourth stage, patients show clinical signs of depression—withdrawal, psychomotor retardation, sleep disturbances, hopelessness, and, possibly, suicidal ideation. The depression may be a reaction to the effects of the illness on their lives (e.g., loss of a job, economic hardship, helplessness, hopelessness, and isolation from friends and family) or it may be in anticipation of the loss of life that will eventually occur. A major depressive disorder with vegetative signs and suicidal ideation may require treatment with antidepressant medication or electroconvulsive therapy (ECT). All persons feel some sadness at the prospect of their death, and normal sadness does not require biologic intervention. However, major depressive disorder and active suicidal ideation can be alleviated and should not be accepted as normal reactions to impending death. A person who suffers from major depressive disorder may be unable to sustain hope, which can enhance the dignity and quality of life and even prolong longevity. Studies have shown that some terminally ill patients can delay their death until after a loved one's significant event, such as the graduation of a grandson from college.

Stage 5: Acceptance. In the stage of acceptance, patients realize that death is inevitable, and they accept the universality of the experience. Their feelings can range from a neutral to a euphoric mood. Under ideal circumstances, patients resolve their feelings about the inevitability of death and can talk about facing the unknown. Those with strong religious beliefs and a conviction of life after death sometimes find comfort in the ecclesiastical maxim, "Fear not death; remember those who have gone before you and those who will come after."

Near-Death Experiences

Near-death descriptions are often strikingly similar. They often involve an out-of-body experience of viewing one's body and overhearing conversations, feelings of peace, hearing a distant noise, entering a dark tunnel, leaving the body behind, meeting dead loved ones, witnessing beings of light, returning to life to complete unfinished business, and deep sadness on leaving this new dimension. This pattern of sensations and perceptions is usually described as peaceful and loving; it feels real to participants, who

distinguish it from dreams or hallucinations. These experiences provoke sweeping lifestyle changes, such as fewer material concerns, a heightened sense of purpose, a belief in God, joy of life, compassion, less fear of death, an enhanced approach to life, and intense feelings of love. In a similar vein, hospice nurses have described experiences among terminally ill patients of visions that may include a sense of the presence of departed loved ones, of spiritual beings, of bright light, or of being in a particular place, often described with a sense of warmth and love. Although such "visions" do not readily lend themselves to scientific investigation and thus are not legitimized, patients may benefit from discussing them with clinicians. A term to describe this experience is *unio mystica*, which refers to an oceanic feeling of mystic unity with infinite power.

Life Cycle Considerations about Death and Dying

The clinical diversity of death-related attitudes and behaviors between children and adults has its roots in developmental factors and age-dependent differences in causes of death. As opposed to adults, who usually die from chronic illness, children are apt to die from sudden, unexpected causes. Almost half of the children who die between the ages of 1 and 14 years and nearly 75 percent of those who died in late adolescence and early adulthood die from accidents, homicides, and suicides. With their characteristics of violence, suddenness, and mutilation, such unnatural causes of death are extraordinary stressors for grieving survivors. Bereaved parents and siblings of dead young children and teenagers often feel victimized and traumatized by their losses; their grief reactions resemble posttraumatic stress disorder (PTSD). Devastating family disruptions can occur, and surviving siblings risk having their emotional needs put on the back burner, ignored, or completely unnoticed.

Children. Children's attitudes toward death mirror their attitudes toward life. Although they share with adolescents, adults, and elderly adults similar fears, anxieties, beliefs, and attitudes about dying, some of their interpretations and reactions are age specific. None welcome it without ambivalence, and all temper their acceptance with healthy doses of denial and avoidance. Dying children are often aware of their condition and want to

discuss it. They often have more sophisticated views about dying than their medically well counterparts, engendered by their failing health, separation from parents, subjection to painful procedures, and the deaths of hospital chums.

At the preschool, preoperational stage of cognitive development, death is seen as a temporary absence, incomplete and reversible, like departure or sleep. Separation from the primary caretaker(s) is the main fear of preschool-age children. This fear surfaces as an increase in nightmares, more aggressive play, or concern about the deaths of others rather than direct discourse. Terminally ill children may assume responsibility for their death, feeling guilty for dying. Preschool children may be unable to relate the treatment to the illness, instead of viewing treatment as punishment and family separation as rejection. They need reassurance that they are loved, have done nothing wrong, are not responsible for their illness, and will not be abandoned.

School-age children manifest concrete-operational thinking and recognize death as a final reality. They, however, view death as something that happens to older people, not to them. Between the ages of 6 and 12 years, children have active fantasies of violence and aggression, often dominated by themes of death and killing. School-age children ask questions about serious illness and death if encouraged to do so; however, if they receive cues that the subject is taboo, they may withdraw and participate less fully in their care. Facilitating open discussion and updating children with relevant information, including prognostic changes, can be very helpful. Also, children may need help coping with peers and school demands. Teachers should be informed and updated. Classmates may need education and assistance to help them understand the situation and respond appropriately.

Adolescents. Capable of formal cognitive operations, adolescents understand that death is inevitable and final but may not accept that their death is possible. The significant fears of dying teenagers parallel those of all teenagers. These include losing control, being imperfect, and being different. Concerns about body image, hair loss, or loss of bodily control can generate resistance to continuing treatment. Alternating emotions of despair, rage, grief, bitterness, numbness, terror, and joy are typical. The potential for withdrawal and isolation is high because teenagers may equate parental support with loss of independence or may deny their fears of abandonment by

actually repulsing friendly gestures. Teenagers must be included in all decision-making processes surrounding their deaths. Many are capable of great courage, grace, and dignity in facing death.

Adults. Some of the most often expressed fears of adult patients entering hospice care, listed in the approximate order of frequency, include fears of (1) separation from loved ones, homes, and jobs; (2) becoming a burden to others; (3) losing control; (4) what will happen to dependents; (5) pain or other worsening symptoms; (6) being unable to complete life tasks or responsibilities; (7) dying; (8) being dead; (9) the fears of others (reflected fears); (10) the fate of the body; and (11) the afterlife. Problems in communication arise out of trepidation, making it essential for those involved in health care to provide environments of trust and safety in which people can begin to talk about uncertainties, anxieties, and concerns.

Late-age adults often accept that their time has come. Their main fears include long, painful, and disfiguring deaths; prolonged vegetative states; isolation; and loss of control or dignity. Elderly patients may talk or joke openly about dying and sometimes welcome it. In their 70s and beyond, they rarely harbor illusions of indestructibility—most have already had several close calls: Their parents have died, and they have gone to funerals of friends and relatives. Although they may not be happy to die, they can be reconciled to it.

According to Erik Erikson, the eighth and final stage in the life cycle brings a sense of either integrity or despair. As elderly adults enter the last phase of their lives, they reflect on their pasts. When they have taken care of their affairs, have been relatively successful, and have adapted to the triumphs and disappointments of life, they can look back with satisfaction and only a few regrets. The integrity of the self allows people to accept inevitable disease and death without fear of succumbing helplessly. If elderly individuals look back on life as a series of missed opportunities or personal misfortunes, however, they feel a sense of bitter despair, a preoccupation with what might have been if only this or that had happened. Then death is fearsome because it symbolizes emptiness and failure.

Management

Caring for a dying patient is highly individual. Caretakers need to deal with death honestly, tolerate wide ranges of affects, connect with suffering patients and bereaved loved ones, and resolve routine issues as they arise. Although each therapeutic relationship between a patient and health provider has a uniqueness derived from the patient's and health provider's gender, constitution, life experience, age, stage of life, resources, faith, culture, and other considerations, major themes confront all health providers caring for dying patients.

BEREAVEMENT, GRIEF, AND MOURNING

Bereavement, *grief*, and *mourning* are terms that apply to the psychological reactions of those who survive a significant loss. Grief is the subjective feeling precipitated by the death of a loved one. The term is used synonymously with mourning, although, in the strictest sense, mourning is the process by which grief is resolved; it is the societal expression of post bereavement behavior and practices. Bereavement means the state of being deprived of someone by death and refers to being in the state of mourning. Regardless of the fine points that differentiate these terms, the experiences of grief and bereavement have sufficient similarities to warrant a syndrome that has signs, symptoms, a demonstrable course, and an expected resolution.

Normal Bereavement Reactions

The first response to loss, *protest*, is followed by a more extended period of *searching* behavior. As hope to reestablish the attachment bond diminishes, searching behaviors give way to *despair* and *detachment* before bereaved individuals eventually *reorganize* themselves around the recognition that the lost person will not return. Although the bereaved ultimately learn to accept the reality of death, they also find psychological and symbolic ways of keeping the memory of the deceased person very much alive. Grief work allows the survivor to redefine his or her relationship to the deceased person and to form new but enduring ties.

Duration of Grief. Most societies mandate modes of bereavement and time for grieving. In contemporary America, bereaved individuals are expected to return to work or school in a few weeks, to establish equilibrium within a few

months, and to be capable of pursuing new relationships within 6 months to 1 year. Ample evidence suggests that the bereavement process does not end within a prescribed interval; certain aspects persist indefinitely for many otherwise high-functioning, typical individuals.

The most lasting manifestation of grief, especially after spousal bereavement, is loneliness. Often present for years after the death of a spouse, loneliness may, for some, be a daily reminder of the loss. Other common manifestations of protracted grief occur intermittently. For example, a man who has lost his wife may experience elements of acute grief every time he hears her name or sees her picture on the nightstand. Usually, these reactions become increasingly short-lived over time, dissipating within minutes and become tinged with positive and pleasant affects. Such bittersweet memories may last a lifetime. Thus, most grief does not fully resolve or permanently disappear; instead, grief becomes circumscribed and submerged only to reemerge in response to specific triggers.

Anticipatory Grief

In *anticipatory grief*, grief reactions are brought on by the slow dying process of a loved one through injury, illness, or high-risk activity. Although anticipatory grief may soften the blow of the eventual death, it can also lead to premature separation and withdrawal while not necessarily mitigating later bereavement. At times, the intensification of intimacy during this period may heighten the actual sense of loss even though it prepares the survivor in other ways.

Anniversary Reactions. When the trigger for an acute grief reaction is a special occasion, such as a holiday or birthday, the rekindled grief is called an *anniversary reaction.* It is not unusual for anniversary reactions to occur each year on the same day the person died or, in some cases, when the bereaved individual becomes the same age the deceased person was at the time of death. Although these anniversary reactions tend to become relatively mild and brief over time, they can be experienced as the reliving of one’s original grief and prevail for hours or days.

Mourning

From earliest history, every culture records its own beliefs, customs, and behaviors related to bereavement. Specific patterns include rituals for mourning (e.g., wakes or Shiva), for disposing of the body, for invocation of religious ceremonies, and periodic official remembrances. The funeral is the prevailing public display of bereavement in contemporary North America. The funeral and burial service acknowledge the real and final nature of the death, countering denial; they also garner support for the bereaved, encouraging tribute to the dead, uniting families, and facilitating community expressions of sorrow. If cremation replaces burial, ceremonies associated with the dissemination of the ashes perform similar functions. Visits, prayers, and other ceremonies allow for continuing support, coming to terms with reality, remembering, emotional expression, and concluding unfinished business with the deceased. Several cultural and religious rituals provide purpose and meaning, protect the survivors from isolation and vulnerability, and set limits on grieving. Subsequent holidays, birthdays, and anniversaries serve to remind the living of the dead and may elicit grief as real and fresh as the original experience; over time, these anniversary grievings become attenuated but often remain in some form.

Bereavement

Because bereavement often evokes depressive symptoms, it may be necessary to demarcate normal grief reactions from major depressive disorder (Table 29-2). DSM-5 took a significant step from DSM-IV in that it removed the bereavement exclusion from a major depressive disorder so that we can diagnose individuals experiencing a recent loss as having major depression if we believe that is the case. To acknowledge that some persons may have a syndrome of disabling grief that is not depression, DSM-5 proposed criteria for persistent complex bereavement, which they included as a condition for further study. ICD-10 considers this an adjustment disorder and includes the grief reaction as a category of that disorder. Table 29-3 summarizes the criteria for each approach.

Complicated Bereavement

Complicated bereavement has a confusing array of terms to describe it —*abnormal, atypical, distorted, morbid, traumatic*, and *unresolved*, to name

a few types. Three patterns of complicated, dysfunctional grief syndromes have been identified—chronic, hypertrophic, and delayed grief. These are not diagnostic categories within DSM-5 but are descriptive syndromes that, if present, may be prodromal of a major depressive disorder.

Table 29-2
Differentiating the Depressive Symptoms Associated with Bereavement from Major Depression

Bereavement	Major Depressive Disorder
Symptoms may meet syndromal criteria for major depressive episode, but the survivor rarely has morbid feelings of guilt and worthlessness, suicidal ideation, or psychomotor retardation	Any symptoms as defined by DSM-5
Considers self-bereaved	May consider self-weak, defective, or bad
Dysphoria often triggered by thoughts or reminders of the deceased	Dysphoria is often autonomous and independent of thoughts or reminders of the deceased
Onset is within the first 2 mo of bereavement	Onset at any time
Duration of depressive symptoms is less than 2 mo	Depression often becomes chronic, intermittent, or episodic
Functional impairment is transient and mild	Clinically significant distress or impairment

No family or personal history of major depression	Family or personal history of major depression

Table 29-3
Approaches to Diagnosing Bereavement

	DSM-5	ICD-10
Diagnostic name	Persistent Complex Bereavement Disorder (list under “conditions for further study”)	Adjustment Disorder, Grief Reaction
Duration	Experiences symptoms from each group of symptoms below, more days than not, for 12 mo (adults) or 6 mo (children)	Occurs while adapting to a period of life change
Symptoms	Since the death of someone close to them, experiencing following symptoms, out of proportion or inconsistent with sociocultural norms (Symptom group B) • Longing for the deceased • Sorry/pain about the death • Preoccupied with the lost person • Preoccupied with the circumstances surrounding the death	Distress/Emotional disturbance • Depression • Anxiety • Worry • Difficulty coping • Difficulty planning • Difficulty carrying out daily routines • Disturbance of conduct (adolescents)

	(Symptom group C) • Difficulty accepting the death • Cannot believe or is numb to the loss • Difficulty thinking about positive memories • Bitter or angry about the loss • Self-blame • Avoids things that remind them of the lost person • Wants to die to be with the person • Cannot trust others • Feels alone/detached • Feels empty or unable to function without the lost person • Trouble seeing their purpose independent of the lost person • Trouble/reluctance pursuing other interests	
Required number of symptoms	Symptom group B: 1 symptom Symptom group C: 6 symptoms	Any combination of the above
Psychosocial consequences of symptoms	Distress of functional impairment	Usually interferes with functioning

Symptoms specifiers	With traumatic bereavement: the death was violent (homicide, suicide), and the bereaved is preoccupied with the suffering that preceded death

Chronic Grief. The most common type of complicated grief is chronic grief, often highlighted by bitterness and idealization of the dead person. Chronic grief is most likely to occur when the relationship between the bereaved and the deceased had been extremely close, ambivalent, or dependent; or when social supports are lacking, and friends and relatives are not available to share the sorrow over the extended time needed for most mourners.

Hypertrophic Grief. Most often seen after a sudden and unexpected death, bereavement reactions are extraordinarily intense in hypertrophic grief. Customary coping strategies are ineffectual to mitigate anxiety, and withdrawal is frequent. When one family member is experiencing a hypertrophic grief reaction, disruption of family stability can occur. Hypertrophic grief frequently takes on a long-term course, albeit one attenuated over time.

Delayed Grief. Absent or inhibited grief when one usually expects to find overt signs and symptoms of acute mourning is referred to as delayed grief. This pattern is marked by prolonged denial; anger and guilt may complicate its course.

Traumatic Bereavement. Traumatic bereavement refers to grief, which is both chronic and hypertrophic. This syndrome is characterized by recurrent, intense pangs of grief with persistent yearning, pining, and longing for the deceased; recurrent intrusive images of the death; and a distressing admixture of avoidance and preoccupation with reminders of the loss. Positive memories are often blocked or excessively sad, or they are experienced in prolonged states of reverie that interfere with daily activities. A history of

psychiatric illness appears to be frequent in this condition, as is a very close, identity-defining relationship with the deceased person.

Medical or Psychiatric Illnesses Associated with Bereavement. Medical complications include exacerbations of existing diseases and vulnerability to new ones; fear for one's health and more trips to the doctor; and an increased mortality rate, especially in men. The highest relative mortality risk is found immediately after bereavement, particularly from ischemic heart disease. The greatest effect of bereavement on mortality is for men younger than 65 years. Higher mortality rates in bereaved men than in bereaved women are due to increase in the relative risk of death by suicide, accident, cardiovascular disease, and some infectious diseases. In widows, the relative risk of death from cirrhosis and suicide may increase. In both sexes, bereavement appears to exacerbate health-compromising behaviors, such as increased alcohol consumption, smoking, and the use of over-the-counter medications.

Psychiatric complications of bereavement include an increased risk for major depressive disorder, prolonged anxiety, panic, and posttraumatic stress–like syndrome; increased alcohol, drug, and cigarette consumption; and an increased risk of suicide. Because of their psychosocial, emotional, and cognitive immaturity, bereaved children may be especially vulnerable to psychopathology.

Bereavement and Depression. Although symptoms overlap, grief can be distinguished from a full depressive episode. Most bereaved individuals experience intense sadness, but only a few meet DSM-5 criteria for a major depressive episode. Grief is a complex experience in which positive emotions take their place beside the negative ones. Grief is fluid and changing, an evolving state in which emotional intensity gradually lessens, and positive, comforting aspects of the lost relationship come to the fore. Pangs of grief are stimulus bound, related to internal and external reminders of the deceased person. This condition differs from depression, which is more pervasive and characterized by significant difficulties experiencing self-validating, positive feelings. Grief is a fluctuating state with individual variability, in which cognitive and behavioral adjustments are progressively made until the bereaved individual can hold the deceased person in a comfortable place in

memory, and a satisfying life can be resumed. By contrast, a major depressive episode consists of a recognizable and stable cluster of debilitating symptoms accompanied by a protracted, enduring low mood. Major depressive episode tends to be persistent and associated with poor work and social functioning, pathologic psychoneuroimmunological function, and other neurobiologic changes unless treated.

Bereavement and Posttraumatic Stress Disorder. Unnatural and violent deaths, such as homicide, suicide, or death in the context of terrorism, are much more likely to precipitate PTSD in surviving loved ones than are natural deaths. In such circumstances, themes of violence, victimization, and volition (i.e., the choice of death over life, as in the case of suicide) are intermixed with other aspects of grief, and traumatic distress marked by fear, horror, vulnerability, and disintegration of cognitive assumptions ensues. Disbelief, despair, anxiety symptoms, preoccupation with the deceased person, and the circumstances of the death, withdrawal, hyperarousal, and dysphoria are more intense and more prolonged than they are under nontraumatic circumstances, and an increased risk may exist for other complications. Although treatment studies in survivors of sudden death are few and far between, most experts agree that initial attention should be focused on traumatic distress, that a role is seen for both pharmacotherapy and psychotherapy, and that self-help support groups can be enormously beneficial.

Biologic Perspectives

Grief is both a physiologic and emotional response. During acute grief (as with other stressful events), persons may experience disruption of biologic rhythms. Grief is also accompanied by impaired immune functioning, including decreased lymphocyte proliferation and impaired functioning of natural killer cells. Whether the immune changes are clinically significant has not been established, but the mortality rate for widows and widowers following the death of a spouse is higher than that in the general population. Widowers appear to be at risk longer than widows.

Phenomenology of Grief. Bereavement reactions include intense feeling states; invoke a variety of coping strategies; and lead to alterations in

interpersonal relationships, biopsychosocial functioning, self-esteem, and world view that can last indefinitely. Manifestations of grief reflect the individual's personality, previous life experiences, and past psychological history; the significance of the loss; the nature of the bereaved person's relationship with the deceased person; the existing social network; intercurrent life events; health; and other resources. Despite individual variations in the bereavement process, investigators have proposed grieving process models, which include at least three partially overlapping phases or states: (1) initial shock, disbelief, and denial; (2) an intermediate period of acute discomfort and social withdrawal; and (3) a culminating period of restitution and reorganization. As with Kübler-Ross' stages of dying, the grieving stages do not prescribe a correct course of grief; instead, they are general guidelines describing an overlapping and fluid process that varies with the survivors (Table 29-4).



Table 29-4
Phases of Grief

Shock and denial (min, days, wk)
Disbelief and numbness
Searching behaviors: pining, yearning, and protest
Acute anguish (wk, mo)
Waves of somatic distress
Withdrawal
Preoccupation
Anger
Guilt
Lost patterns of conduct
Restless and agitated
Aimless and amotivational
Identification with the bereaved
Resolution (mo, yr)
Have grieved
Return to work
Resume old roles

Acquire new roles
Reexperience pleasure
Seek companionship and love of others

LIFE CYCLE PERSPECTIVES ABOUT BEREAVEMENT

Bereavement during Childhood and Adolescence

Approximately 4 percent of North American children lose one or both parents by the age of 15 years; sibling death is the second most commonly experienced bereavement. Grief reactions are colored by developmental levels and concepts of death and may not resemble adult reactions. Children may display minimal grief at the time of death and experience the full effect of the loss later. Grieving children may not withdraw and dwell on the person who died, but instead, may throw themselves into activities. Indifference, anger, or misbehavior may be displayed rather than the sadness; behaviors can be erratic and labile. Strong feelings of anger and fears of abandonment or death may show up in the behavior of grieving children. Children often play death games as a way of working out their feelings and anxieties. These games are familiar to the children and provide safe opportunities to express their feelings. Although they may seem to show grief only occasionally and briefly, in reality, a child's grief often lasts longer than that of an adult.

Mourning in children may need to be addressed again and again as the child ages. Children will think about the loss repeatedly, especially during critical times in their lives, such as going to camp, graduating from school, getting married, or giving birth to their children. A child's grief can be influenced by his or her age, personality, developmental stage, earlier experiences with death, and relationship with the deceased person. The surroundings, cause of death, and family members' ability to communicate with one another and to continue as a family after death can also affect grief. The child's ongoing need for care, his or her opportunity to share feelings and memories, the parent's ability to cope with stress, and the child's steady relationships with other adults are other factors that may influence grief. Even older children frequently feel abandoned or rejected when a parent dies and may show hostility toward the deceased or the surviving parent, now perceived as one who might also "abandon" them. They may feel responsible

because of earlier misbehavior or because they said or wished that that person would die at some time.

Children younger than 2 years may show loss of speech or diffuse distress. Children younger than 5 years are apt to respond with eating, sleeping, and bowel and bladder dysfunctions. Intense feelings of sadness, fear, and anxiety can occur, but these feelings are not persistent and tend to alternate between longer lasting normal states. School-age children may become phobic or hypochondriacal, withdrawn, or pseudomature, and school performance and peer relations often suffer. Adolescents, as with adults, run the gamut in expressing bereavement, ranging from behavioral problems, somatic symptoms, and erratic moods to stoicism. Whereas adolescent boys losing a parent may become delinquent, girls may turn to a sexual pattern for comfort and reassurance. Behavioral disturbances and depression are familiar at all ages. Rates of depressive episodes in bereaved children and adolescents are as high as in bereaved adults.

Bereaved children must be treated in a manner that considers their levels of emotional and cognitive maturity. They need to be told that death is real and irreversible and that they are blameless. Feelings and concerns should be expressed, and questions should be invited and answered with simplicity, candor, and clarity. Children, as with adults, need rituals to commemorate their loved ones; attendance at the funeral and participation in mourning may be beneficial first steps.

Bereavement during Adulthood

No consensus exists on which type of loss is associated with the most severe reactions. Although the death of a spouse is often ranked as the most stressful life event, some have argued that losing a child is even more profound. The death of a child is a particular sorrow, a lifelong loss for surviving mothers, fathers, brothers, sisters, grandparents, and other family members. A child’s death is a life-altering experience. The deaths of parents and siblings in adult life have not achieved much systematic study, but they are generally considered relatively mild compared with the loss of a spouse or child.

Grief appears most intense for the mother in late perinatal losses (stillbirths or neonatal deaths rather than miscarriages) and often is reexperienced during subsequent pregnancies. Sudden infant death syndrome is particularly

problematic in that death is sudden and unexpected. Parents may experience extra guilt or blame each other, often resulting in subsequent marital difficulties.

The surviving family members, friends, or lovers of individuals who have died from acquired immunodeficiency syndrome (AIDS) are uniquely challenged. The illness carries with it the stigmata of the illness itself and the gay community in general; it carries with it caretakers' fear of contracting the illness, and it is most prevalent in people who are in the prime of life. Asymptomatic infection may permit the infected person and those close to him or her time to adapt to the diagnosis. When a person who is human immunodeficiency virus (HIV) positive begins to manifest symptoms of opportunistic infection or associated cancer; however, the illness again becomes a threat. Coping with emotional reality is arduous and complicated. Often caretakers, as well as HIV-positive patients, wish for death, which can evoke feelings of guilt. For bereaved lovers, their own HIV status, multiple losses, and other concurrent stressors can complicate recovery. Gay men who have lost lovers to AIDS may be more depressed, consider suicide more often, and be more vulnerable to illicit drug use than are other bereaved individuals.

Elderly adults face more losses than individuals at other phases of the life cycle, and intense loneliness may be a lasting memorial to those who have died. For highly impaired elders who lose a spouse they depended on for daily functions or who was their sole source of companionship, bereavement reactions are profound.

Grief Therapy

Persons in normal grief seldom seek psychiatric help because they accept their reactions and behaviors as appropriate. Accordingly, a bereaved person should not routinely see a psychiatrist or psychologist unless a markedly divergent reaction to the loss is noted. For example, under usual circumstances, a bereaved person does not make a suicide attempt; if someone seriously contemplates suicide, psychiatric intervention is indicated.

When professional assistance is sought, it usually involves a request for sleeping medication from a family physician. A mild sedative to induce sleep may be useful in some situations, but antidepressant medication or

antianxiety agents are rarely indicated in normal grief. Bereaved persons may have to go through the mourning process, however painful it is, for a successful resolution to occur. Narcotizing patients with drugs interferes with the normal process that ultimately can lead to a desired outcome.

Because grief reactions can develop into a depressive disorder or pathologic mourning, specific counseling sessions for bereaved individuals are often valuable. Grief therapy is an increasingly important skill. In regularly scheduled sessions, grieving persons are encouraged to talk about feelings of loss and about the person who has died. Many bereaved persons have difficulty recognizing and expressing angry or ambivalent feelings toward a deceased person, and they must be reassured that these feelings are normal.

Grief therapy need not be conducted only on a one-to-one basis; group counseling is also useful. Self-help groups also have great value in some instances. About 30 percent of widows and widowers report that they become isolated from friends, withdraw from social life, and thus experience feelings of isolation and loneliness. Self-help groups offer companionship, social contacts, and emotional support; they eventually enable their members to reenter society in a meaningful way. Bereavement care and grief therapy have been most effective with widows and widowers. The necessity for this therapy stems, in part, from the contraction of the family unit; extended family members are no longer available to provide the needed emotional support and guidance during the mourning period.

29.2 Palliative Care

Psychological symptoms are nearly universal at the end of life. Psychiatric syndromes occur with an increased but definable frequency and have different age and gender distribution. For example, anxiety and depression are as common in men as in women. Psychiatric classification remains an essential framework on which to base clinical observations, but it was not designed with dying patients in mind. Hence, for such patients, it is pragmatically useful to think of a few syndromes for which this exists. The most common ones are anxiety states, depressive states, and confusional

states. These frequently coexist and overlap. Rarely, specific phobias of needles, enclosures, and the like may interfere with comfort and should be addressed, adapting the usual treatments to the patient's medical status. Occasionally, an emotional crisis or exacerbation of symptoms can be identified as an adjustment disorder, but it is occurring against the backdrop of other severe symptoms, so technically, it does not meet diagnostic criteria. However, this should not prevent the consultant from identifying the precipitating factor and defusing the response in the usual ways. Major psychotic disorders become submerged by the increasing symptomatology of the active dying process and only require specific attention when the patient is not actively dying and when the psychotic symptoms are separated from and superimposed on the symptoms of the illness.

PREVALENCE

Much of the research on prevalence has been done in cancer and AIDS and shows a marked increase in psychiatric conditions near the end of life. Severe depressive symptoms were found in from 25 to 77 percent in a sample of hospitalized cancer patients, although stricter criteria find 15 percent with major depression and another 15 percent having severe depressive symptoms. The prevalence of delirium rises from a range from 25 to 40 percent to as high as 85 percent with increasingly advanced disease. The association of psychiatric symptoms with pain was demonstrated in one of several consistent studies in which 39 percent of patients with a psychiatric diagnosis reported significant pain compared with 19 percent for those without a psychiatric diagnosis. In a sample of inpatients with AIDS, 65 percent had an organic mental disorder, and 27 percent had major depression. The financial cost of psychiatric disorders can be inferred from a study in which patients with a psychiatric diagnosis remained in the hospital 60 days longer than those without a psychiatric diagnosis.

Most commonly, terminally ill patients demonstrate an intertwining of anxiety and depression. These are difficult to tease apart, and the term *negative affect* has been suggested to define them as a symptom complex. It is hard not to feel grief at what is being lost and fear about the ultimate unknown. Individuals whose deep faith in an afterlife animates their spirit are an exception, and, even among them, many describe coexisting regret at the

loss of their temporal life and its furnishings.

GENERAL TREATMENT PRINCIPLES

Because the improved quality of life is the primary goal, pharmacologic treatment and any other measures that bring symptom relief should be instituted rapidly while an integrated plan for psychological and family interventions is being designed and set into motion. Psychiatric syndromes in this group are often secondary to medical conditions; hence, an etiologic diagnosis often yields useful clues to prevention and improved management. It should be sought simultaneously as long as the search is in line with treatment goals.

ANXIETY IN PATIENTS WITH ADVANCED DISEASE

Anxiety can be the presenting symptom for almost all medical disorders and can occur as a side effect from many medications. However, in patients with advanced disease, it usually presents with somatic symptoms, such as restlessness, hyperactivity, tachycardia, gastrointestinal (GI) distress, nausea, insomnia, shortness of breath, numbness, or tremor. It lowers the threshold for pain, worsens functional impairment, and increases the distress experienced in all comorbid conditions. It often interferes with the patient's ability to cooperate with other treatments or to relate optimally with loved ones. Patients refer to fear, worry, apprehension, or ruminations more often than to anxiety per se.

Mr. S, a 50-year-old physical therapist with newly diagnosed advanced lung cancer, was noted by his family to be anxious to the point of having panic symptoms when his wife would leave his bedside to attend to chores. He would start hyperventilating, would feel short of breath, would become restless and unable to concentrate on anything, and would be overwhelmed with morbid ruminations about his future. He was upset and felt guilty at having become overly dependent on his wife. He was taught relaxation and breathing exercises and was treated with clonazepam, which brought about a marked resolution of his anxiety. Mr. S felt more relaxed, less anxious, and more resilient and became able to

withstand periods of solitude without difficulty. (Courtesy of Marguerite Lederberg, M.D.)

DEPRESSION IN PATIENTS WITH ADVANCED DISEASE

Depressive symptoms are also common in advanced disease and are associated with the same existential factors found with anxiety. Studies have found a prevalence ranging from 9 percent, using the strictest criteria, to 58 percent, with less demanding ones. Risk factors include a previous history or family history. Having a way of differentiating somatic effects of disease from the neurovegetative criteria of major depression is especially daunting in terminally ill patients.

Table 29-5 describes the Endicott Substitution Criteria, which have been found to perform as well as the DSM criteria for depression and go further in that they also reflect the clinical observation which classically described that depressive thoughts and feelings are not universal in terminally ill patients and, when present, reflect depression just as they do in physically healthy patients.



Table 29-5
Endicott Substitution Criteria for Depression

Physical-Somatic Symptoms	Psychological Symptom Substitute
Changes in appetite or weight, or both	Tearfulness, depressed appearance
Sleep disturbance	Social withdrawal, decreased talkativeness
Fatigue, loss of energy	Brooding, self-pity, pessimism
Memory and concentration deficits, indecisiveness	Lack of reactivity

Although there are not many studies on the treatment of depression in terminally ill individuals, the available studies and a large body of clinical

experience with medically ill patients show that the pharmacologic treatment of depression can be useful even when definable medical causes exist and even in the last days of life.

A patient with diabetes and end-stage renal disease who had been on dialysis for 2 years was diagnosed with depression with marked insomnia and was started on 15 mg by mouth every day of mirtazapine, which helped her sleep immediately and showed an antidepressant effect within 3 weeks. Four weeks later, she was admitted to the hospital in congestive heart failure for what was to be her terminal admission. She did not wish to give up the antidepressant but now felt oversedated and wished that she could be more alert during the day to interact with her loved ones. She was started on methylphenidate 5 mg by mouth twice daily, after which she was more alert, engaged, and communicative with her family. On her death, her family was grateful to have been able to connect with her until the end.

CONFUSIONAL STATES IN PATIENTS WITH ADVANCED DISEASE

The prevalence of delirium rises to 85 percent in patients with advanced disease. If one includes the last hours of life, it nears 100 percent unless the patient lapses rapidly into a coma or dies of an acute event, such as pulmonary embolus. Acute events are always unexpected and traumatize the survivors even though they knew full well that the patient was near the end of life. Some patients slip into an irreversible coma, leaving families in a bedside vigil, which may give them a period of adaptation before the final instant of death. However, for some 75 to 85 percent of patients, death is associated with a period of delirium.

Patients frequently experience some disorientation, impaired memory, concentration, and altered arousal as they become increasingly ill. These symptoms may remain mild or may be the harbingers of full delirium. Clinicians should be aware that mild and early signs of delirium are often mistaken for depression, anxiety, and poor coping.

A psychiatric consultation was sought to evaluate depression in a 56-year-old practicing attorney with pancreatic cancer. His moderately severe back pain was being treated with morphine. The patient was noted by the inpatient staff to be more withdrawn, disengaged, and quiet, making poor eye contact and sleeping most of the day.

On examination, the psychiatric consultant found Mr. K to have disturbed arousal and to be mildly confused and disoriented. His speech was slow and his thought process disorganized. Mr. K admitted to intermittently experiencing visual hallucinations that he had been too embarrassed to report earlier to the nursing staff.

Mr. K was diagnosed with a hypoactive delirium secondary to opioid medications. He was treated with a low dose of olanzapine, 2.5 mg at bedtime. Mr. K's sensorium improved dramatically. He became alert, fully oriented, and better related, without any perceptual disturbances or thought disorder. This treatment was accomplished without needing to decrease his much-needed pain medications.

DISEASE-SPECIFIC CONSIDERATIONS

Different illnesses bring special issues. For example, a patient on dialysis can opt for death three times a week and is more vulnerable to acting on feelings of depression, anger, hopelessness, and reactions to family neglect or conflict. Patients with cancer must acknowledge possible death while hoping for a cure or remission. Treatment decisions in the face of a disease that is less and less treatable become increasingly difficult and can cause acute anxiety. Stem cell transplant patients experience high levels of anxiety and depression because they are getting a last chance with high stakes and high risks. The availability of organ transplants has created a large population of patients who wait, knowing that they may die while waiting for a cure that could be just around the corner. Neurodegenerative disorders are associated with increased physical disability and dependence. When there is associated loss of cognitive faculties, the problems may be behavioral. Otherwise, depression is frequent, although not an inevitable problem. Many patients state that they will not tolerate complete immobility and dependence, yet when the time comes, they go on life supports and stay on them. It has been

shown in several settings that as patients become increasingly ill, they accept an increasingly limited quality of life as worth living and opt for more onerous, less promising treatments if they offer even a small chance of help.

PATIENT–FAMILY UNIT

The intensity of family relationships becomes even more significant during the terminal period. The response of family members can be a conspiracy of silence. Nothing is sadder than a bedside where family members are tense and silent because they want to protect the patient by not talking about dying, and the patient is tense and silent because he or she is protecting the family by not talking about things that will upset them. Instead of closeness, expressions of gratitude, apologies, reminiscences, and farewells, there is a distance, and the patient is dying alone even though others physically surround the patient.

The psychiatrist can use family sessions to open patient–family dialogues. He or she can identify discrepant views of the illness, can deal with conflicts regarding treatment, and can explore concerns regarding an absent member, all of which undermine patient support and medical management. A major crisis, such as the imminent loss of a member, destabilizes the family structure, creating an opportunity for the psychiatrist to promote adaptive change and reconstitution. *Family-centered grief therapy,* which initially includes the patient and continues after the patient's death, provides a natural setting for such interventions.

When an older adolescent dropped out of college to take over her dying mother's duties, the family subtly discouraged her return to school after her mother's death. Psychiatric intervention allowed the patient to play a part in reorganizing family roles in a way that made it more possible for her daughter to continue her studies.

DECISION POINTS, ADVANCE DIRECTIVES, PROXIES, AND SURROGATES

This section reviews the transitions and decisions that characterize the end of

life.

Transition to Palliative Care

The transition to palliative care is not always clear. As soon as a diagnosis of an incurable disease is made, a cure is no longer the goal of care. However, if death is distant or even if some life extension can still be obtained, the patient and family focus on this positive goal. The physician is under no illusion about the future but has the delicate task of promoting short-term gains without obliterating the awareness of what lies ahead. Only when the nearness of death is acknowledged can thoughtful decisions be made about palliative care.

Where to Die?

Unattended Deaths. Many traumatic or unexpected deaths become known from a catastrophic phone call. The traumatized family then needs help to absorb the loss, to cope with the circumstances, and to come to some kind of closure. It is not realized how much the experience of living through the illness, death, and the funeral rites is crucial to the normal resolution of the mourning process.

An unexpected death, even an understandable one, such as a heart attack, leaves an aching sense of unfinished business over and above the expected grief. If the person was the victim of a crime, obsessive thoughts might be difficult to repress, and grief may turn to unquenchable anger with profound psychological disruption. It is also difficult, if not impossible, for survivors to make peace with suicide.

Emergency departments, police departments, and religious and community institutions should be equipped with a list of referral resources to help survivors of traumatic deaths. Psychiatric input can include program development and consultation to a wide range of professionals and individuals. Families must be helped to construct their ritual whereby they acknowledge to themselves and others the finality of their loss and perhaps create a place, if not a grave, around which to center their memorializing.

Attended Deaths. Patients can die in an acute hospital, in nursing homes, in hospices, and at home, with or without hospice support.

Most patients still die in acute care hospitals, having received active care until shortly before they die. This may occur because death is sudden or because the family or patient needed to be in a place where “everything is being done.” Fortunately, a growing number of hospitals have palliative care teams that provide appropriate care in the acute hospital setting.

Many patients die in nursing homes without the benefit of special care. This unfortunate situation could be remedied by bringing formal hospice care into nursing homes, but funding sources and turf issues need to be settled before it can become routine practice.

Inpatient hospice care was the first model of care to be developed and is warmly remembered by grateful families whose multiple human needs and those of their loved ones had been met in ways that they had not been trained to expect. As hospice gained acceptance, the insufficiency of inpatient beds encouraged the development of home hospice. Its existence has, in turn, encouraged more families to elect to keep dying patients at home.

In a home hospice program, the patient is evaluated and accepted in the usual manner but stays at home. The patient and family receive extensive instruction about what to expect. They are helped to obtain necessary materials, taught how to use them, and helped to obtain as much home help as they need. All the while, they receive medical supervision, nursing care, and emotional support with 24-hour phone availability and routine daily contacts.

Without this kind of help, a good death at home can be challenging to achieve. With it, patients feel safe from the abandonment that is so commonly feared, and families feel safe from the terror of an unmanageable event. The families of these patients work hard, but they are more likely to feel competent and in control. They experience more of a sense of achievement and less of the gnawing sense of inadequacy that is otherwise common.

CARING FOR THE DYING PATIENT

Marguerite S. Lederberg, M.D., at Memorial Sloan Kettering Cancer Center in New York makes the following observation:

A dying human being whose physical, social, emotional, and spiritual needs

are being effectively attended to seldom demands to be helped to commit suicide, and the family members—given proper help and support—derive a deep sense of peace from having helped their loved one to die feeling loved and secure.

One of the most critical tasks for a physician caring for a dying patient is to determine when the time for curative care has ceased. It is only then that palliative care can begin. Some physicians are so upset by death that they are reluctant to use palliative methods; instead, they continue to treat the patient knowing that efforts are futile. Alternatively, they resort to using so-called heroic methods that do not prevent death, and that may produce needless suffering. Ideally, physicians should strive to extend the patient's life and decrease suffering; at the same time, they must accept death as a defining characteristic of life. Some physicians, however, develop dysfunctional attitudes about death, which they reinforce throughout their lives in their experiences and training. It has been postulated that doctors are more frightened of death than members of other professional groups and that many enter the study of medicine so they may gain control of their mortality using the defense mechanism of intellectualization. Risk factors that can interfere with a physician's ability to care optimally for dying patients are listed in Table 29-6. These factors range from overidentifying with the patient to being fearful of death, as mentioned.

Physicians able to deal with death and dying can communicate effectively in several areas: diagnosis and prognosis, the nature of the terminal illness, advance directives about life-sustaining treatment, hospice care, legal and ethical issues, grief and bereavement, and psychiatric care. Also, palliative care physicians must be skilled in pain management, especially in the use of potent opioids—the gold standard of drugs used for pain relief. In 1991, the American Board of Pain Medicine was established to ensure that physicians treating patients in pain were both qualified to do so and were kept up to date on the latest advances in the field.



Table 29-6
Risk Factors for the Development of Aversive Reactions in Physicians

The physician:

- Identifies with the patient: looks, profession, age, character, and so on.
- Identifies the patient with someone in his or her own life.
- Is currently dealing with a sick family member.
- Is recently bereaved or dealing with unresolved loss or grief issues.
- Feels professionally insecure.
- Is fearful of death and disability.
- Is unconsciously reflecting feelings felt or expressed by the patient or family.
- Cannot tolerate high and protracted levels of ambiguity or uncertainty.
- Carries a psychiatric diagnosis, such as depression or substance abuse.

Adapted from Meier DE, Back AL, Morrison RS The inner life of physicians and care of the seriously ill. *JAMA*. 2001;286:3007–3014, with permission.

COMMUNICATION

After a diagnosis and prognosis have been made, physicians need to talk to the patient and the patient's family. Formerly, doctors subscribed to a conspiracy of silence, believing that their patient's chances for recovery would improve if they knew less, because news of impending death might bring despair. The standard of care is to be honest and open toward a patient. The question is no longer whether to tell a patient, but when and how. In 1972, the American Hospital Association drafted the Patient's Bill of Rights, declaring that patients have the "right to obtain complete, current information regarding diagnosis, treatment, and prognosis in terms the patient can be reasonably expected to understand."

Breaking Bad News

When breaking news of impending death to the patient, as when relating any bad news, diplomacy and compassion should be the guiding principles. Often, the bad news is not entirely related during one meeting but instead is absorbed gradually over a series of separate conversations. Preparations,

including scheduling sufficient time for the visit, researching pertinent information such as test results and facts about the case, and even arranging furniture can help make the patient feel comfortable.

If possible, these conversations should take place in a private, suitable space with the patient on equal terms with the physician (i.e., the patient dressed and the physician seated). If it is possible and desired by the patient, the patient's spouse or partner should be present. The treating physician should explain the current situation to the patient in clear, simple language even when speaking to highly educated patients. Information may need to be repeated, or additional meetings may be necessary to communicate all of the information. A gentle, sensible approach will help modulate the patient's denial and acceptance. At no time should physicians take their patient's angry comments personally, and they should never criticize the patient's response to the bad news.

Physicians can signal their availability for honest communication by encouraging and answering questions from patients. Estimates on how long a patient has to live are usually inaccurate and thus should not be given, or given with that caveat. Also, physicians should make it clear to their patients that they are willing to see them through until death occurs. Ultimately, physicians must choose how much information to give and when based on each patient's needs and capacities.

The same general approaches apply as physicians seek to comfort members of the patient's family. Helping family members deal with feelings about the patient's illness can be just as important as comforting the patient because family members are often the primary source of emotional support for patients.

Telling the Truth

Tactful honesty is the doctor's most valuable aid. Honesty, however, need not preclude hope or guarded optimism. It is essential to be aware that if 85 percent of patients with a particular disease die in 5 years, 15 percent are still alive after that time. When deciding when and what to tell a patient about their prognosis, we should consider our commitment to the principle of "do no harm." In general, most patients want to know the truth about their condition. Various studies of patients with malignancies show that 80 to 90

percent want to know their diagnosis.

Doctors, however, should ask patients how much they want to know because some persons do not want to know all the facts about their illness. Such patients, if told the truth, deny that they ever were told, and they cannot participate in end-of-life decisions, such as the use of life-sustaining equipment. The patients who openly request that they not be given “bad news” are often those who most fear death. Physicians should deal with these fears directly, but if the patient still cannot bear to hear the truth, someone closely related to the patient must be informed.

Informed Consent

In the United States, informed consent is legally required for both conventional and experimental treatment. Patients must be given sufficient information about their diagnosis, prognosis, and treatment options to make knowledgeable decisions. This information includes a discussion of potential risks and benefits, available alternative treatments, and the results of not receiving treatment. This approach may come at some psychological cost; severe anxiety and occasional psychiatric decompensation can occur when patients feel overburdened by demands to make decisions. Nevertheless, patients respond best to doctors who explain the various options in detail. Physicians must be prepared to deal with difficult questions posed by patients. Some of them are listed in Table 29-7.

End-of-life discussions are challenging, mainly because they can influence how patients make informed choices.

TERMINAL CARE DECISIONS

Modern society is poorly equipped to cope with the life-and-death decisions spawned by technology. When it first emerged, cardiopulmonary resuscitation was enthusiastically supported by the medical profession. It was endowed with magical power and eventually became a ritualized rite rather than an optional medical treatment. That practice played into the therapeutic activism characteristic of many physicians. By the end of the 20th century, however, a countermovement began. First, the right to refuse treatment was established, mostly because of synergy between the consumer movement and the bioethics movement, with its emphasis on patient autonomy. Next, the

legality of DNR orders and the moral equivalence of stopping and not starting treatment were established. The medical profession was less enthusiastic than the public about these changes, perhaps because practitioners know too well the emotional ambiguities that surround death and must repeatedly experience them.



Table 29-7
Some Difficult Questions from Patients

"Why me?"
"Why didn't you catch this earlier? Did you make a mistake?"
"How long do I have?"
"What would you do in my shoes?"
"Should I try experimental therapy?"
"Should I go to a 'medical mecca' for treatment or a second opinion?"
"If my suffering gets really bad, will you help me die?"
"Will you work with me all the way through to my death, no matter what?"

From Quill TF. Perspectives on care at the close of life. Initiating end-of-life discussions with seriously ill patients: Addressing the "elephant in the room". *JAMA*. 2000;284:2502–2507, with permission.



Table 29-8
Persistent Vegetative State

No evidence of awareness of self or environment; no interaction with others
No meaningful response to stimuli
No receptive or expressive language
Return of sleep–wake cycles, arousal, even smiling, frowning, yawning
Preserved brainstem or hypothalamic autonomic functions to permit survival
Bowel and bladder incontinence
Variably preserved cranial nerve and spinal reflexes

Brain Death and Persistent Vegetative State

In an attempt to deal with these ambiguities, the concept of brain death emerged. Brain death is associated with the loss of higher brain functions (e.g., cognition) and all brain stem functions (e.g., pupillary and reflex eye movement), respiration, and gag and corneal reflexes. Determination of brain death is a generally accepted criterion for death. Some clinicians advocate an absence of brain waves on electroencephalography (EEG) to confirm the diagnosis.

The American Academy of Neurology defined the persistent vegetative state as a condition in which no awareness exists of self or environment associated with severe neurologic damage (Table 29-8). The medical treatment provides no benefits to patients in a persistent vegetative state, and after the diagnosis is established, DNR and do-not-intubate (DNI) orders can be followed and life-sustaining methods (e.g., feeding tubes, ventilators) can be removed.

In 1976, the case of Karen Ann Quinlan made international headlines when her parents sought the assistance of a judge to discontinue the use of a ventilator in their daughter, who was in a persistent vegetative state. Ms. Quinlan's physician had refused her parents' request to remove the ventilator because they said they feared that they might be held civilly or even criminally liable for her death. The New Jersey Supreme Court ruled that competent persons have a right to refuse life-sustaining treatment and that this right should not be lost when a person becomes incompetent. Because the court believed that the physicians were unwilling to withdraw the ventilator because of the fear of legal liability, not precepts of medical ethics, it devised a mechanism to grant the physicians prospective legal immunity for taking this action. The New Jersey Supreme Court ruled that after a prognosis, confirmed by a hospital ethics committee, that "no reasonable possibility of a patient returning to a cognitive, sapient state," exists, life-sustaining treatment can be removed, and no one involved, including the physicians, can be held civilly or criminally responsible for the death.

The publicity surrounding the Quinlan case motivated two independent developments: It encouraged states to enact "living will" legislation that

provided legal immunity to physicians who honored patients' written "advance directives," specifying how they would want to be treated if they ever became incompetent. It also encouraged hospitals to establish ethics committees that could attempt to resolve similar treatment disputes without going to court. (Courtesy of Annas GJ. "Culture of Life" politics at the bedside—the case of Terri Schiavo. *N Eng J Med*. 2005;352:1710–1715.)

Advance Directives

Advance directives are wishes and choices about medical intervention when the patient's condition is considered terminal. Advance directives, which are legally binding in all 50 states, include three types: living will, health care proxy, and DNR and DNI orders.

Living Will. In a living will, a mentally competent patient gives specific instructions that doctors must follow when the patient cannot communicate with them because of illness. These instructions may include rejection of feeding tubes, artificial airways, and any other measures to prolong life.

Health Care Proxy. Also known as a *durable power of attorney*, the health care proxy gives another person the power to make medical decisions if the patient cannot do so. That person, also known as the surrogate, is empowered to make all decisions about terminal care based on what he or she thinks the patient would want.

Do-Not-Resuscitate and Do-Not-Intubate Orders. These orders prohibit doctors from attempting to resuscitate (DNR) or intubate (DNI) the patient who is in extremis. DNR and DNI orders are made by the patient who is competent to do so. They can be made part of the living will or expressed by the health care proxy. Figure 29-1 is an infographic that includes a checklist of important items to consider when making an advanced directive.

The Uniform Rights of the Terminally Ill Act, drafted by the National Conference of Commissioners on Uniform State Laws was approved and recommended for enactment in all states. This act authorizes an adult to control the decisions regarding the administration of life-sustaining treatment

by executing a declaration instructing a physician to withhold or to withdraw life-sustaining treatment if the person is in a terminal condition and cannot participate in medical treatment decisions. In 1991, the Federal Patient Self-Determination Act became law in the United States. It required that all health care facilities (1) provide each patient admitted to a hospital with written information about the right to refuse treatment, (2) ask about advance directives, and (3) keep written records of whether the patient has an advance directive or has designated a health care proxy.

Today, patients who have left no advance directives or who are legally incompetent to do so have access to hospital ethics committees that hold active legal and ethical debates about these issues. These ethics committees are also of help to doctors, who can gain both legal and moral support when recommending that no further treatment occur. It is much easier for all parties, however, if the patient has advance directives or a proxy. Ideally, physicians should initiate discussions with patients about advance directives and proxies early even while the patient is healthy. The patient should be reminded that these early formulations can be modified but that even having preliminary advance directives will ensure that treating physicians observe the patient's wishes in the event of an emergency.

CARING FOR THE FAMILY

Family members play an essential role as caregivers to terminally ill patients and have needs of their own that often go unrecognized. Their responsibilities can be overwhelming, especially if only one family member is available or if family members themselves are infirm or elderly. Table 29-9 lists some family caregiving tasks. Many of these tasks require long hours of work or supervision that can lead to physical and emotional fatigue. One study of caregivers reports that 25 to 30 percent lost their jobs, and more than half moved to lower-paying jobs to accommodate the need for flexibility. The highest stress level was found in families who cared for a terminally ill patient at home, mainly when the death occurred in the home and realized in retrospect that they would have preferred an environment in which death occurs in the presence of skilled caretakers.

GETTING YOUR AFFAIRS IN ORDER:
Advance Care Planning

Making healthcare decisions for yourself or someone who is no longer able to do so can be overwhelming. That's why it's important to get a clear idea about preferences and arrangements while you can make decisions and participate in legal and financial planning together.

Use this checklist to ensure healthcare and financial arrangements are in place before serious illness or a healthcare crisis.

- ☑ **Start discussions** early with your loved one while everyone can still help make decisions.



- ☑ **Create documents** that communicate **healthcare**, **financial management**, and **end of life** wishes for yourself and the people you care for, with legal advice as needed.



- ☑ **Review plans regularly**, and update documents as circumstances change.



- ☑ **Put important papers in one place.** Make sure a trusted family member or friend knows the location and any instructions.



- ☑ **Make copies** of **healthcare directives** to be placed in all medical files, including information on every doctor seen.



- ☑ **Give permission** in advance for a doctor or lawyer to talk directly with a caregiver as needed.



- ☑ **Reduce anxiety** about funeral and burial arrangements by planning ahead.



Visit https://www.nia.nih.gov/health/caregiving/advance-care-planning to learn more about advance care planning.

NIH National Institute on Aging

FIGURE 29-1
Advance Care planning checklist. (Source: National Institute on Aging, https://www.nia.nih.gov/health/infographics/getting-your-affairs-order-advance-care-planning.)

Table 29-9
Tasks of Family Members of Dying Individuals

1. Administering medications
2. Dealing with adverse effects of medications
3. Providing help with, or actually performing, activities of daily living
4. Changing wound dressing
5. Managing ambulatory infusion pumps or other equipment
6. Providing symptom management (e.g., for pain, nausea and vomiting, shortness of breath, seizures, and terminal agitation)
7. Notifying the nurse or doctor when they are needed
8. Shopping for needed items and picking up prescriptions
9. Providing a presence and companionship
10. Attending to spiritual and religious needs
11. Carrying out advance directives
12. Managing financial matters

Dying at Home

Depending on the patient's wishes and the specific disease, the choice to die at home is one that should be explored. Although it is more burdensome on a family than dying in a hospital or hospice, death at home can be a welcome alternative for the patient and family seeking to spend quality time together. A home care team can assess a home for its suitability and suggest ways to facilitate activities of daily living, including modifications to furniture; hospital bed leasing; and installation of assistive devices, such as handrails and commodes. The family's care can be supplemented with house calls by

physicians, nurses, therapists, and chaplains. In any case, the family must know what their responsibilities are and must be well prepared to care for the patient. Recently, hospice home care was approved by Medicare and is being more widely used.

Family therapy sessions allow family members to explore feelings about death and dying. They serve as a forum in which anticipatory grief and mourning can take place. The ability to share feelings can be cathartic, especially if guilt is involved. Family members often have to deal with feelings of guilt about past interactions with the dying patient. Family sessions also help to achieve consensus about the patient's advance directives. If family members disagree about the patient's wishes, the medical staff may be unable to act. In such cases, legal action may be needed to resolve family disputes about what course of action to pursue.

PALLIATIVE CARE

Palliative care is the most critical part of end-of-life care. It refers to providing relief from the suffering caused by pain or other symptoms of a terminal disease. Although this is most commonly associated with analgesic drug administration, many other medical interventions and surgical procedures fall under the umbrella of palliative care because they can make the patient more comfortable. Monitors and their alarms, peripheral and central lines, phlebotomy, measurement of signs, and even supplemental oxygen are usually discontinued to allow the patient to die peacefully. Relocating the patient to a quiet, private room (as opposed to an intensive care unit) and allowing family members to be present is another critical palliative care modality.

The shift from active, curative treatment to palliative care is sometimes the first tangible sign that the patient will die, a transition that is emotionally difficult for everyone concerned about the patient to accept. The discontinuation of machines and measurements, which, up until this point, have been an integral part of the hospital experience, can be extremely disconcerting to the patient, family members, and even other physicians. Indeed, if these parties are not active in planning this transition, it can easily seem that persons have given up on the patient.

Because of this difficulty, palliative care is sometimes avoided altogether

(i.e., curative treatment is continued until the patient dies). This approach is likely to cause problems if it is adopted merely to avoid the reality of impending death. A well-negotiated transition to palliative care often decreases anxiety after the patient and family go through an appropriate anticipatory grief reaction. Furthermore, a positive emotional outcome is much more likely if the physician and staff project a conviction that palliative care will be an active, involved process, without a hint of withdrawal or abandonment. When this does not occur or when the family cannot tolerate the transition, the ensuing stress frequently results in a need for psychiatric consultation.

A 36-year-old physician with end-stage leukemia was seen in psychiatric consultation because he reported seeing the “angel of death” at the foot of his hospital bed. He described the experience as frightening and inexplicable. The consultant asked the patient, “Are you afraid that you are going to die?” That was the first time anyone had mentioned death or dying in any context to the patient. He welcomed the opportunity to talk openly about his fears to the medical staff and his family and eventually died a peaceful death.

Psychiatric consultation is indicated for patients who become severely anxious, suicidal, depressed, or overtly psychotic. In each instance, appropriate psychiatric medication can be prescribed to provide relief. Patients who are suicidal do not always have to be transferred to a psychiatric service. An attendant or nurse can be assigned to the patient on a 24-hour basis (one-on-one coverage). In such instances, the relationship that develops between the observer and the patient may have therapeutic overtones, especially with patients whose depression is related to a sense of abandonment. Patients who are terminal and who are at high risk for suicide are usually in pain. When pain is relieved, suicidal ideation is likely to diminish. A careful evaluation of suicide potential is required for all patients. A premorbid history of past suicide attempts is a high-risk factor for suicide in terminally ill patients. In patients who become psychotic, impaired cognitive function secondary to metastatic lesions to the brain must always be considered. Such patients respond to antipsychotic medications, and

psychotherapy may also be of use.

PAIN MANAGEMENT

Types of Pain

Dying patients are subject to several different kinds of pain, as summarized in Table 29-10. The distinctions are important because they call for different treatment strategies; whereas somatic and visceral pain is responsive to opiates, neuropathic and sympathetically maintained pain may require adjuvant medications in addition to opiates. Most patients with advanced cancer, for example, have more than one kind of pain and require complex treatment regimens.

Treatment of Pain

It cannot be overemphasized that pain management should be aggressive, and treatment should be multimodal. A good pain regimen may require several drugs or the same drug used in different ways and administered via different routes. For example, intravenous morphine can be supplemented by self-administered oral “rescue” doses, or intravenous bolus doses can supplement a continuous epidural drip. Transdermal patches may provide baseline concentrations in patients for whom intravenous or oral intake is difficult. Patient-controlled analgesia systems for intravenous opiate administration result in better pain relief with lower amounts dispensed than in staff-administered dosing.



Table 29-10
Types of Pain

Nociceptive Pain	
Somatic pain	Usually, but not always constant, aching, gnawing, and well localized (e.g., bone metastases)
Visceral pain	Usually, but not always constant, deep, squeezing, poorly localized, with possible cutaneous referral (e.g., pleural effusion leading to deep chest pain,

	diaphragmatic irritation referred to the shoulder)
Neuropathic pain	Burning dysesthetic pain with shock-like paroxysms associated with direct damage to peripheral receptors, afferent fibers, or the central nervous system, leading to loss of central inhibitory modulation and spontaneous firing (e.g., phantom limb pain; can involve sympathetic somatic afferents)
Psychogenic pain	Variable characteristics secondary to psychological factors in the absence of medical factors; rare as a pure phenomenon in patients with cancer but often an additional factor in the presence of organic pain

Courtesy of Marguerite S. Lederberg, M.D. and Jimmie C. Holland, M.D.

Opioids commonly cause delirium and hallucinations. A frequent mechanism of psychotoxicity is the accumulation of drugs or metabolites whose durations of analgesia are shorter than their plasma half-lives (morphine, levorphanol, and methadone). The use of drugs such as hydromorphone, which have half-lives closer to their analgesic duration, can relieve the problem without loss of pain control. Cross-tolerance is incomplete between opiates; hence, several should be tried in any patient with the dosage lowered when switching drugs. Table 29-11 lists opioid analgesics.

The benefits of maintenance analgesia administration in terminally ill patients compared with as-needed administration cannot be overemphasized. Maintenance dosing improves pain control, increases drug efficiency, and relieves patient anxiety, but as-needed orders allow pain to increase while waiting for the drug to be given. Moreover, as-needed analgesia administration perversely sets up the patient for staff complaints about drug-seeking behavior. Even when maintenance treatment is used, extra doses of medication should be available for breakthrough pain, and repeated use of these medications should signal the need to raise the maintenance dose. Depending on their previous experiences with opioid analgesics and their weight, it is not unusual for some patients to require 2 g or more of morphine

per day for relief of symptoms.

Knowing doses of different drugs and different routes of administration is essential to avoid accidental undermedication. For example, when changing a patient from intramuscular to oral morphine use, the intramuscular dose must be multiplied by 6 to avoid causing the patient pain and provoking drug-seeking behavior. Many adjuvant drugs used for pain are psychotropic with which psychiatrists are familiar, but in some cases, their analgesic effect is separate from their primary psychotropic effect. Commonly used adjuvants include antidepressants, mood stabilizers (e.g., gabapentin) phenothiazines, butyrophenones, antihistamines, amphetamines, and steroids. They are particularly crucial in neuropathic and sympathetically maintained pain, for which they can be the mainstay of treatment.



Table 29-11
Opioid Analgesics for Management of Pain

Drug and Equianalgesic Dose Relative Potency	Dose (mg IM or Oral)	Plasma Half-Life (hr)[a]	Starting Oral Dose[b] (mg)	Available Commercial Preparations
Morphine	10 IM 60 oral	3–4	30–60	Oral: tablet, liquid, slow-release tablet Rectal: 5–30 mg Injectable: SC, IM, IV, epidural, intrathecal
Hydromorphone	1.5 IM 7.5 oral	2–3	2–18	Oral: tablets: 1, 2, 4 mg Injectable: SC, IM, IV 2 mg/mL, 3 mg/mL, and 10 mg/mL
Methadone	10 IM 20 oral	12–24	5–10	Oral: tablets, liquid Injectable: SC, IM, IV
Levorphanol	2 IM 4 oral	12–16	2–4	Oral: tablets Injectable: SC, IM, IV
Oxymorphone	1	2–3	NA	Rectal: 10 mg Injectable: SC, IM, IV
Heroin	5 IM 60 oral	3–4	NA	NA
Meperidine	75 IM 300 oral	3–4 (normeperidine 12–16)	75	Oral: tablets Injectable: SC, IM, IV
Codeine	130 oral 200 oral	3–4	60	Oral: tablets in combination with acetylsalicylic acid, acetaminophen, liquid
Oxycodone[c]	15 oral 30 oral	—	5	Oral: tablets, liquid, oral formulation in combination with acetaminophen (tablet and liquid) and aspirin (tablet)

IM, intramuscular; IV, intravenous; NA, not applicable; SC, subcutaneous.

[a]The time of peak analgesia in nontolerant patients ranges from ½ to 1 hour and the duration from 4 to 6 hours. The peak analgesic effect is delayed, and the duration is prolonged after oral administration.

[b]Recommended starting IM doses; the optimal dose for each patient is determined by titration, and the maximal dose is limited by adverse effects.

[c]A long-acting sustained-release form of oxycodone (Oxycontin) has been abused by drug addicts, and its use has been criticized because of this; however, it is a very useful preparation available in 10-, 20-, 40-, and 160-mg doses that need to be taken once every 12 hours. It is used as a maintenance therapy for severe persistent pain.

Adapted from Foley K. Management of cancer pain. In: DeVita VT, Hellman S, Rosenberg SA, eds. *Cancer: Principles and Practice of Oncology*. 4th ed. Philadelphia, PA: JB Lippincott; 1993:936, with permission.

Other developments in pain management include more intrusive procedures, such as nerve blocks or the use of continuous epidural infusions.

Additionally, radiation therapy, chemotherapy, and even surgical resection can be considered as pain management modalities in palliative care. Short courses of radiotherapy or chemotherapy can be used to shrink tumors or manage metastatic lesions that cause pain or impairment. In patients with end-stage Hodgkin disease, for example, systemic chemotherapy can improve the patient's quality of life by decreasing tumor burden. Surgical resection of invasive tumors, most notably breast carcinomas, can be useful for the same reason.

PALLIATION OF OTHER SYMPTOMS

Symptom management is a high priority in palliative care. Patients are often more concerned about the day-to-day distress of their symptoms than they are about their impending death, which may not be as real to them. Table 29-12 lists common end-of-life symptoms. A comprehensive approach to palliation involves attending to these end-of-life symptoms as well as pain. Sources of distress include psychiatric symptoms, such as anxiety and physical symptoms. Foremost among physical symptoms are those involving the GI system, including diarrhea, constipation, anorexia, nausea, vomiting, and bowel obstruction. Other significant symptoms include insomnia, confusion, mouth sores, dyspnea, cough, pruritus, decubitus ulcers, and urinary frequency or incontinence. Caretakers should follow these symptoms carefully and establish appropriate early and aggressive care for these symptoms before they become burdensome.

An effective treatment for nausea and vomiting associated with chemotherapy is the use of Δ-tetrahydrocannabinol (THC), the active ingredient of marijuana. The oral synthetic cannabinoid, dronabinol (Marinol), is used in 1- to 2-mg doses every 8 hours. The use of marijuana cigarettes to deliver THC is believed to be more effective than pills. Proponents say that its absorption is faster and antiemetic properties are more potent via the pulmonary system. Repeated attempts to legalize marijuana cigarettes for medical use have met with only limited success in this country.

A 47-year-old man with incurable lung cancer who had been treated unsuccessfully with chemotherapy and radiotherapy had been suffering

from intractable dyspnea for 1 week. His family, nursing, and other staff were increasingly upset by his difficulty breathing and his pleas for relief. The attending physician refused to prescribe anything more substantial than codeine. The palliative care team at the hospital intervened at the family's request. Relief was obtained with the use of 5 to 10 mg of an intravenous bolus of morphine every 15 minutes. When the patient became comfortable, a continuous drip of intravenous morphine was instituted, complemented by subcutaneous morphine as needed.



Table 29-12
Common End-of-Life Symptoms/Signs

Symptom/Sign	Comments	Management/Care
Cachexia	All terminal disease states are associated with cachexia secondary to anorexia and dehydration	Feeding tubes useful in some cases; small sips of water of help
Delusions	Common in terminal state	Antipsychotic medication useful
Delirium/Confusion	Occurs in nearly 90% of all terminal patients but is reversible in over 50%	Can be reversed if cause is found and treatable; may respond to antipsychotic and/or pain medication
Depression or anxiety	Psychological factors, e.g., fear of death, abandonment and/or physiologic factors, e.g., pain, hypoxia	Antianxiety and antidepressant medication of use; opioids have strong antianxiety effects
Dysphagia	Seen in neurologic	Attention to oral care,

	disease, e.g., multiple sclerosis, amyotrophic lateral sclerosis	e.g., ice chips, lip balm; adjust to upright position when feeding
Dyspnea or cough	Associated with severe anxiety; fear of suffocation in extreme case; common in lung cancer patients	Opioids, supplemental oxygen, bronchodilators of use
Fatigue	Most common occurrence in terminal illness	Psychostimulants can be used for relief
Incontinence	Associated with radiation induced fistulas	Keep patient clean and dry; use indwelling or condom catheter if necessary
Nausea or vomiting	Side effect of radiation and chemotherapy	Antiemetics, e.g., metoclopramide, prochlorperazine; marijuana of use
Loss of skin integrity	Decubiti most common on weight-bearing areas	Turn body frequently; elbow and hip pads; inflating mattresses
Pain	Pain medications can be administered orally, sublingually, by injection or infusion, or via skin patch	Opioids are the gold standard

Data from National Coalition on Health Care (NCHC) and the Institute for Healthcare Improvement (IHI). Promises to Keep: Changing the Way We Provide Care at the End of Life, release, October 12, 2000.

The AMA supports the position that patients with a terminal condition require substantial doses of opioids regularly and should not be denied drugs for fear of producing physical dependence.

HOSPICE CARE

In 1967, the founding of St. Christopher's Hospice in England by Cicely Saunders launched the modern hospice movement. Several factors in the 1960s propelled the development of hospices, including concerns about inadequately trained physicians, inept terminal care, gross inequities in health care, and neglect of elderly adults. Life expectancy had increased, and heart disease and cancer were becoming more common. Saunders emphasized an interdisciplinary approach to symptom control, care of the patient and family as a unit, the use of volunteers, continuity of care (including home care), and follow-up with family members after a patient's death. The first hospice in the United States, Connecticut Hospice, opened in 1974. By 2000, more than 3,000 hospices were open in the United States. Round-the-clock pain control with opioids is an essential component of hospice management.

In 1983, Medicare began reimbursing hospice care. Medicare hospice guidelines emphasize home care, with benefits provided for a broad spectrum of physician, nursing, psychosocial, and spiritual services at home or, if necessary, in a hospital or nursing home. The patient is eligible for these services if the physician certifies them as having 6 months or less to live. By electing hospice care, patients agree to receive palliative rather than curative treatment. Many hospice programs are hospital-based, sometimes in separate units and sometimes in the form of hospice beds interspersed throughout the facility. Other program models include free-standing hospices and programs, hospital-affiliated hospices, nursing home hospice care, and home care programs. Nursing homes are the site of death for many elderly patients with incurable chronic illness, yet dying nursing home residents have limited access to palliative and hospice care. Families generally express satisfaction with their involvement in hospice care. Savings with hospice care vary, but home care programs generally cost less than conventional institutional care, particularly in the final months of life. Hospice patients are less likely to receive diagnostic studies or such intensive therapy as surgery or chemotherapy; however, a new trend is to allow treatment programs to

continue while the patient remains in the hospice. Hospice care is a proven, viable alternative for patients who elect a palliative approach to terminal care. Also, hospice goals of dignified, comfortable death for terminally ill patients and care for patients and families together have been increasingly adopted in mainstream medicine.

NEONATAL AND INFANT END-OF-LIFE CARE

Advances in reproductive medicine have increased the number of infants born prematurely as well as the number of multiple births. These advances have increased the need for life-sustaining methods of care and have made decisions about when to use palliative care more complicated. Some bioethicists believe that withholding life-sustaining interventions is appropriate under certain circumstances; others maintain that life-sustaining methods should not be used at all. An extensive study of attitudes among neonatologists about end-of-life decisions found no consensus about if and when to terminate life.

For newborns, the decision to forego life-sustaining procedures is usually limited to those children whose death is imminent. Even if their future quality of life is determined to be bleak, most physicians believe that some life is better than no life at all. Physicians who support withholding intensive care consider the following quality-of-life issues: (1) extent of bodily damage (e.g., severe neurologic impairment), (2) the burden that a disabled child will place on the family, and (3) the ability of the child to derive some pleasure from existence (e.g., having an awareness of being alive and being able to form relationships).

The American Academy of Pediatrics permits nontreatment decisions for newborns when the infant is irreversibly comatose or when treatment would be futile and only prolong the process of dying. These standards do not permit the parents to have any input in the decision-making process. In a well-publicized case in England in 2000, doctors decided to surgically separate conjoined twins, even though one would die from the procedure. This procedure occurred despite the objections of the parents, who believed that nature should take its course. Neonatal end-of-life decisions remain in a state of limbo. No clear-cut criteria exist about which patients should receive intensive care and which should receive palliative care.

CHILD END-OF-LIFE CARE

After accidents, cancer is the second most common cause of death in children. Although many childhood cancers are treatable, palliative care is necessary for children with cancers that are not. Children require more support than adults in coping with death. On average, a child does not view death as permanent until the age of about 10 years. Before then, death is viewed as sleep or separation. Therefore, children should be told what they can understand. If they are capable, they should be involved in the decision-making process about treatment plans. Assurances that patients are pain-free and physically comfortable are just as crucial for children as they are for adults.

A unique aspect of end-of-life care in children involves addressing their fear of being separated from their parents. It is helpful to have parents participate in end-of-life care tasks within their capacities. Family sessions with the child in attendance allow feelings to emerge and questions to be answered.

SPIRITUAL ISSUES

There is increasing awareness of the importance of this area to patients, families, and many staff members as well. Several studies have shown that religious beliefs are often associated with mature and active coping methods, and the field of psychological and spiritual interfaces in terminally ill patients is spawning a whole new area of psychological research within the traditional medical establishment. The psychiatric consultant should inquire about faith, its meaning, associated religious practices, and impact on the coping response. It can be a source of strength or guilt at all stages of the disease, ranging from the earliest “What did I do to cause this?” through “Will God give me only what I can carry?” to the poignant life review of the late stage. It is often a primary factor in the reactions to suicidality and attitudes toward terminal care decisions. Mental health professionals should deal with these areas in an unselfconscious and noncondescending manner and work to help patients fully integrate this aspect of their personality into their current crisis. The professional should also work in harmony with the patient’s spiritual guide if one is available. Sometimes an experienced chaplain working with

the appropriate patient can achieve positive results more directly than any psychotherapy. The following case exemplifies how creative pastoral care can relieve suffering.

A young woman was admitted to Calvary Hospital in a terminal state. She was experiencing a severe depression, which she attributed to not being able to see her oldest daughter receive her first communion. Arrangements were made for a ceremonial communion for her daughter to take place at Calvary Hospital. After the ceremony, the patient's mood improved markedly as one of her fears was alleviated, and a religious need was satisfied. As her mood improved, she was able to address other unresolved issues and had quality visits with her children in her remaining days. (From O'Neil MT. Pastoral care. In: Cimino JE, Brescia MJ, eds. *Calvary Hospital Model for Palliative Care in Advanced Cancer*. Bronx, NY: Palliative Care Institute; 1998, with permission.)

ALTERNATIVE AND COMPLEMENTARY MEDICINE

Many patients, when they are told they are terminally ill, seek alternative treatments, ranging from innocuous programs aimed at enhancing general health to more aggressive, harmful, or fraudulent regimens. Although most patients combine the alternative and the traditional, a substantial number favor complementary medicine as the only treatment for their disease.

Complementary methods to cure terminal illness, especially cancer, emphasize a holistic approach, involving purification of the body, detoxification through internal cleansing, and attention to nutritional and emotional well-being. Despite their widespread appeal, not one of these methods has been demonstrated to cure cancer or prolong life, yet all have strong followings bolstered by anecdotal accounts of their efficacy. The popular metabolic therapy attributes cancer and other potentially fatal illnesses to toxins and waste materials accumulating in the body; treatment is based on reversing this process by diet, vitamins, minerals, enzymes, and colonic irrigations. Another approach includes macrobiotic diets or megavitamins to enhance the body's capacity to destroy malignancy. In 1987, the National Research Council recommended minimizing carcinogenic

substances and fat in the diet and increasing whole-grain, fruit, and vegetable consumption as preventive guidelines. Psychological approaches cite maladaptive personality and coping styles as contributors to fatal diseases; treatment consists of shaping a positive attitude. Spiritual approaches aim at achieving harmony between the patient and nature. Some groups use spirituality as a way to ward off illness, which is sometimes seen as an external evil to be exorcised. Immunotherapies have gained popularity in recent years; cancer is attributed to a defective immune system, and restoration of immunocompetency is seen as the cure. Many patients find increased strength to endure the suffering of terminal illness with the help of alternative medicine even though the course of the disease may not be affected. (For a further discussion of alternative medicine, see Chapter 21.)

▲ 29.3 Euthanasia and Physician-Assisted Suicide

EUTHANASIA

From the Greek term for a good death, *euthanasia* means compassionately allowing, hastening, or causing the death of another. Generally, someone resorts to euthanasia to relieve suffering, maintain dignity, and shorten the course of dying when death is inevitable. Euthanasia can be *voluntary* if the patient has requested it or *involuntary* if the decision is made against the patient's wishes or without the patient's consent. Euthanasia can be *passive*—simply withholding heroic lifesaving measures—or *active*—deliberately taking a person's life. Euthanasia assumes that the physician intends to aid and abet the patient's wish to die.

Arguments for euthanasia revolve around patient autonomy and dignified death. One of the most dramatic ways patients can exercise their right to self-determination is by asking that life-sustaining treatment to be withdrawn. If the patient is mentally competent, physicians must respect such wishes. Proponents of active, voluntary euthanasia argue that the same rights should be extended to patients who are not on life-sustaining treatment but also choose to have their physicians help them die.

Opponents of euthanasia also provide strong ethical and medical justification for their position. First, active euthanasia, even if the patient voluntarily requests it, is a form of killing and should never be sanctioned. Second, many patients who request aid in dying may be suffering from depression, which, when treated, will change the patient's mind about wanting to die.

Most medical, religious, and legal groups in the United States are against euthanasia. Both the American Psychiatric Association (APA) and the AMA condemn active euthanasia as illegal and contrary to medical ethics; however, few individuals have been convicted of euthanasia. Most physicians and medical groups in other parts of the world also oppose legalizing euthanasia. In the United Kingdom, for example, the British Medical Association believes that euthanasia is "alien to the traditional ethos and moral focus of medicine" and, if legalized, "would irrevocably change the context of health care for everyone, but especially for the most vulnerable."

The World Medical Association issued the following declaration on euthanasia in October 2019:

The WMA is firmly opposed to euthanasia and physician-assisted suicide… no physician should be forced to participate in euthanasia or assisted suicide, nor should any physician be obliged to make referral decisions to this end.

Separately, the physician who respects the basic right of the patient to decline medical treatment does not act unethically in forgoing or withholding unwanted care, even if respecting such a wish results in the death of the patient. (*Adopted by the 70**th** WMA General Assembly, Tbilisi, Georgia, October 2019*, https://www.wma.net/policies-post/declaration-on-euthanasia-and-physician-assisted-suicide/)

Similarly, the New York State Committee on Bioethical Issues issued a statement declaring its opposition to euthanasia. The committee stated that physicians are obligated to relieve pain and suffering for the dying patient, even at the risk of hastening a patient's death. Physicians, however, should not perform active euthanasia or participate in assisted suicide. The Committee believed that support, comfort, respect for patient autonomy, good communication, and adequate pain control would dramatically decrease

the demand for euthanasia and assisted suicide. They argued that the societal risks of involving physicians in medical interventions to cause a patient's death were too significant to condone active euthanasia or physician-assisted suicide. In response to shifting public opinion and lobbying groups with different views, state laws that banned physician-assisted death in Washington State and New York were sent to the United States Supreme Court, challenging the constitutionality of these prohibitions. In June 1997, the Court unanimously held that terminally ill patients do not have the right to physician aid in dying. The ruling, however, left room for continuing debate and future policy initiatives at the state level.

PHYSICIAN-ASSISTED SUICIDE

In the United States, most of the debate centers on physician-assisted suicide rather than on euthanasia. Some have argued that physician-assisted suicide is a humane alternative to active euthanasia in that the patient maintains more autonomy, remains the actual agent of death, and may be less likely to be coerced. Others believe that the distinctions are capricious in that the intent in both cases is to bring about a patient's death. Indeed, it may be difficult to justify providing a lethal dose of medication to a terminally ill patient (physician-assisted suicide) while ignoring the desperate pleas of another patient who may be even more ill and distressed but who cannot complete the action because of problems with swallowing, dexterity, or strength.

Several degrees are seen to which a physician may assist the suicidal patient to end his or her life. Physician-assisted suicide can involve providing information on ways of committing suicide, supplying a prescription for a lethal dose of medication or a means of inhaling a lethal amount of carbon monoxide, or perhaps even providing a suicide device that the patient can operate.

The controversy over physician-suicide came to national attention surrounding the activities of retired pathologist Jack Kevorkian, who, in 1989, provided his suicide machine to a 54-year-old woman with probable Alzheimer disease. After the woman killed herself with his device, Kevorkian was charged with first-degree murder. The charges were later dismissed because Michigan had no law against physician-assisted suicide. Since that first case, Kevorkian assisted in several more suicides, often for persons he

met on only a few occasions and frequently for persons who did not have a terminal illness. Claiming to have helped more than 130 people take their lives, Kevorkian was sent to prison in 1999, was released in 2006, and died in 2011. His attorneys and followers applauded his courage in easing pain and suffering; his detractors countered that he was a serial mercy killer. Opponents of Kevorkian's methods charged that, without safeguards, consultations, and thorough psychiatric evaluations, patients might search out suicide not because of terminal illness or intractable pain but because of untreated depressive disorders. They argued that suicide rarely occurs in the absence of psychiatric illness. Finding more effective treatments for pain and depression, rather than inventing more sophisticated devices to help desperate patients kill themselves, defines compassionate and effective physician care.

In 1994, Oregon passed a ballot initiative legalizing physician-assisted suicide (Death with Dignity Act), making Oregon the first state in the United States to permit assisted suicides (Table 29-13). An assessment of the first 4 years revealed the following: Patients dying from physician-assisted suicide represent approximately eight of 10,000 deaths. The most common underlying illnesses were cancer, amyotrophic lateral sclerosis, and chronic lower respiratory disease. The three most common end-of-life concerns were loss of autonomy (85 percent), a decreasing ability to participate in activities that made life enjoyable (77 percent), and losing control of bodily functions (63 percent). Eighty percent of the patients were enrolled in hospice programs, and 91 percent died at home. The prescribing physician was present in 52 percent of the cases.

In 2001, Attorney General John Ashcroft attempted to prosecute Oregon doctors who helped terminally ill patients die, claiming that doctor-assisted suicide is not a legitimate medical purpose. The case was brought to the Supreme Court, which in 2006 supported the Oregon law and said the "authority claimed by the attorney general is both beyond his expertise and incongruous with the statutory purposes and design." Since 2001, three other states—Washington (2008), Montana (2009), and Vermont (2011)—have passed laws similar to the one in Oregon.



Table 29-13
Oregon's Assisted Suicide Law

- The patient must be terminally ill and expected to die within 6 mo; mentally competent; fully informed about his or her diagnosis, prognosis, risks, and alternatives, such as comfort care; and be making a voluntary choice.
- A second doctor must agree that the patient is terminally ill, acting on his or her own free will, fully informed, and capable of making health care decisions.
- If either doctor thinks that the patient is suffering from any form of mental illness that could affect his or her judgment, they must refer the patient for counseling.
- The patient must make one written request and two spoken requests.
- The doctor must ask the patient to tell the next of kin, but the patient may decide not to do so.
- The patient is free to change his or her mind at any time.
- There is a 15-day waiting period between the patient making the request and the doctor writing the prescription.
- All information must be written down in the medical records.
- Only people who normally live in Oregon may use the Act.
- Mercy killing, lethal injection, and active euthanasia are not permitted.
- Pharmacists must be told of the prescribed medication's ultimate use.
- Physicians, pharmacists, and health care systems are under no obligation to participate in the Death with Dignity Act.

Despite the abhorrence that many physicians and medical ethicists express regarding physician-assisted suicide, poll after poll shows that as many as two-thirds of Americans favor the legalization of physician-assisted suicide in certain circumstances, and evidence even indicates that the formerly uniform opposition to physician-assisted suicide within the medical community has eroded. Consistent with their positions on active euthanasia, the AMA, APA, and American Bar Association, however, continue to oppose physician-assisted suicide. Recently, the American College of Physicians-American Society of Internal Medicine (ACP-ASIM) expressed its commitment to improving care for patients at the end of life while

recommending against the legalization of physician-assisted suicide. The ACP-ASIM believes physician-assisted suicide raises grave ethical concerns, undermines the physician–patient relationship, and the trust necessary to sustain it alters the medical profession's role in society, and endangers the values American society places on life, especially on the lives of disabled, incompetent, and vulnerable individuals.

The American Association of Suicidology in its 1996 *Report of the Committee on Physician-Assisted Suicide and Euthanasia* concluded that involuntary euthanasia could never be condoned; the report also stated, however, that "intolerable, prolonged suffering of persons in extremis should never be insisted upon, against their wishes, in single-minded efforts to preserve life at all cost." This position acknowledges that patients can die as a result of treatment given to them for the explicit purpose of relieving suffering. However, death associated with palliative care differs significantly from physician-assisted suicide in that death is not the goal of treatment and is not intentional.

How to Deal with Requests for Suicide

To help guide clinicians facing requests for physician-assisted suicide, the AMA's Institute for Ethics proposed an eight-step clinical protocol. Table 29-14 lists these steps.



Table 29-14
The AMA's Recommended Approach to Requests of Physician-Assisted Suicide

1. Evaluation of the patient for depression or other psychiatric conditions that could cause disordered thought
2. Evaluation of the patient's "decision-making competence"
3. Discussion with the patient about his or her goals for care
4. Evaluation and response to the patient's "physical, mental, social, and spiritual suffering"
5. Discussion with the patient about the full range of treatment and

care options

6. Consultation by the attending physician with other professional colleagues

7. Assurance that care plans chosen by the patient are being followed, including removal of unwanted treatment and the provision of adequate pain and symptom relief

8. Discussion with the patient explaining why physician-assisted suicide is to be avoided and why it is not compatible with the principled nature of the care protocol

Psychiatrists view suicide as an irrational act that is the product of mental illness, usually depression. In almost every case in which a patient asks to be put to death, a triad exists of depression associated with an incurable medical condition that causes intolerable pain. In these instances, every effort should be made to provide antidepressants or psychostimulants for depression and opioids for pain. Psychotherapy, spiritual counseling, or both may also be needed. Also, family therapy to help with the stress of dealing with a dying patient may be necessary. Family therapy is also useful because some patients may ask to be put to death because they do not wish to be a burden to their families; others may feel coerced by their families into believing that they are, or will be, a burden and may choose death as a result. Currently, no professional codes countenance euthanasia or assisted suicide in the United States. Therefore, psychiatrists must stand on the side of responsible rescue and treatment.

A distinction also is needed between major depression and suffering. Psychiatrists have not sufficiently studied the nature of suffering. It remains the province of theologians and philosophers. Suffering is a complex mix of spiritual, emotional, and physical factors that transcends pain and other symptoms of a terminal illness. Physicians are more skilled at dealing with depression than with suffering. Anatole Broyard, who chronicled his death in his book *Intoxicated by My Illness*, wrote the following:

> I see no reason or need for my doctor to love me nor would I expect him to suffer with me. I wouldn’t demand a lot of my doctor’s time; I just

wish he would brood on my situation for perhaps 5 minutes, that he would give me his whole mind just once, be bonded with me for a brief space, survey my soul as well as my flesh, to get at my illness, for each man is ill in his own way.

FUTURE DIRECTIONS

Advances in technology bring more complex medical, legal, moral, and ethical controversies regarding life, death, euthanasia, and physician-assisted suicide. Some forms of euthanasia have found a place in modern medicine, and expansion of the boundaries of patients' rights and their ability to choose the way they live and die is inevitable. Both patients and physicians need to be better educated about depression, pain management, palliative care, and quality of life. Medical schools and residency training programs need to give the topics of death, dying, and palliative care the attention they deserve. Society must ensure that economics, ageism, and racism do not get in the way of adequate and humane management of patients with chronic terminal illnesses. Finally, a national health care policy must provide adequate insurance coverage, home care, and hospice services to all appropriate patients. If these mandates are followed, the argument for physician assistance in dying will lose much of its impact.

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30 Community Psychiatry

INTRODUCTION

The subject area of public psychiatry embodies a fundamental core of experience and tradition. In the context of the reexamination of American health care initiated by the effect of managed care and the health care reform, the experience of public psychiatry is poised to serve as the foundation for a transformation of behavioral health care.

The term *public* can refer to psychiatric programs, treatments, or institutions paid for by public funds or as objects of public policy, whether paid for or not. The traditional concept of public psychiatry has been expanded to include medical and psychosocial initiatives directed for the public good, whether funded by public or private funds and directed in particular to those who are economically disadvantaged.

The care and treatment offered under public psychiatry are delivered in a variegated mosaic of inpatient and community-based services that are more or less integrated into a coherent network sponsored by public agencies. Funding for public psychiatric services tends to be provided by federally legislated appropriations that are passed through the state, county, and municipal government agencies (such as departments of mental health; substance use treatment services; children, youth, and family services; and public health, social services, education, adult corrections, and juvenile justice agencies). Ultimately, most public psychiatry and community services are provided by not-for-profit community mental health, substance use treatment, child guidance, or health care organizations that are either funded by or subcontracted to government agencies. Thus, the very existence of public and community psychiatric services and the policies and resources

determining how they are delivered are dependent on legislative mandates and fiscal appropriations from all levels of public government.

CONTEMPORARY PUBLIC AND COMMUNITY PSYCHIATRY

There are five themes around which the discussion of contemporary public and community psychiatry is structured: public health, public agencies, evidence-based psychiatry, roles for psychiatrists, and delivery systems.

Public Health

Public health is not merely publicly funded health care but is instead a specific discipline and tradition. It is a complicated field that historically has been defined negatively by the dominance of *personal health*, that is, the health care delivery systems that take care of individual patients. Until the advent of managed care in the 1990s, U.S. health care was an industry mainly organized in terms of individual doctor-entrepreneurs. Each jurisdiction uniquely defined and organized its public health programs, and the particular pattern of personal health practices had a substantial influence—hence the wide variation in public health programs. Nonetheless, as a discipline and tradition, public health's mission is to assure the conditions in which people can be healthy. Public health consists of organized community efforts aimed at the prevention of disease and the promotion of health. It involves many disciplines but rests on the scientific core of epidemiology. It provides an essential template for contemporary public and community psychiatry.

The Surgeon General's report on mental health underscored the necessity of a public health approach to care and rehabilitation for people experiencing mental illness that is "broader in focus than medical models that concentrate on diagnosis and treatment." Although diagnosis and treatment are core areas of expertise for all psychiatrists, the Surgeon General recommended that even when psychiatric diagnosis or treatment is the primary focus for practitioners, researchers, or educators, they should ground their professional activities in a vision and knowledge base that is "*population-based*… encompass[ing] a focus on epidemiologic surveillance, health promotion, disease prevention, and access to services." These fundamental public health functions define the public health perspective in public and community psychiatry. In the

discussion that follows, these four public health components are framed in current terms of health care reform to define a coherent "public health" strategy for public and community psychiatry.

Health Promotion

Psychiatric professionals can contribute substantially to the promotion of public health by working with primary health care professionals and educators to identify and provide front-line treatment and referrals to adults and children who have undetected psychiatric symptoms, subthreshold syndromes, and psychiatric disorders. Collaborative care models bring psychiatric professionals into primary care and school settings as consultants for medical, nursing, and education professionals. These professionals work in those settings, as well as for educators and direct treatment providers to at-risk or psychiatrically impaired individuals.

Furthermore, people with identified mental illness or addictive disorders can benefit from enhanced physical and mental health care, as well as from the alleviation or management of psychiatric symptoms. Achieving or regaining physical or mental health (i.e., recovery) depends not only on genetic and biologic factors but also on a person's or family's access to social and psychological resources and integration into supportive social networks. Illness management (also known as *disease management* or *chronic illness care management*) is a framework that has been adapted from medicine to guide mental health professionals in delivering services that go beyond traditional diagnostic treatment to promote the health and recovery of people with mental illness or addiction. *Illness management* has been defined as "professional-based interventions designed to help people collaborate with professionals in the treatment of their mental illness, reduce their susceptibility to relapses, and cope more effectively with their symptoms… [to] improve self-efficacy and self-esteem and to foster skills that help people pursue their personal goals." Several approaches to illness management have been scientifically and clinically evaluated and have been found to enhance standard psychiatric treatment.

Prevention. Psychiatric disorders most often follow a course over time that begins with an often lengthy period in which prodromal or subthreshold

symptoms or functional problems precede the full onset of a disorder with marked impairment. Intervention with adults, adolescents, or children who are not clinically impaired but who are at high risk or who are manifesting preclinical symptoms or functional problems is an approach to prevention that is cost-effective because it targets a relatively small group of individuals promptly. Examples of high-risk patients include those with a family history of psychiatric or addictive disorders or exposure to extreme stressors, such as violence, neglect, or antisocial behavior. Also, prodromal symptoms and behaviors that would argue for early intervention include periodic or pervasive dysphoria, problems with separation from caregivers, or involvement with deviant peers.

Application of the traditional public health concepts of *primary, secondary,* and *tertiary psychiatry* has been confusing in psychiatry. *Primary prevention* involves addressing the root causes of illness with healthy individuals, to prevent illness before it occurs. *Secondary prevention* involves the early identification and early treatment of individuals with acute or subclinical disorders or *high-risk* persons to reduce morbidity. *Tertiary prevention* attempts to reduce the effects of a disorder on an individual through rehabilitation and chronic illness care management. The Institute of Medicine, to help clarify different aspects of prevention, developed a classification system with three different categories. *Universal* interventions are those intended for the general public, such as immunizations or media campaigns, providing information about illnesses, early warning signs, and resources for health promotion and timely treatment. *Selective* interventions focus on individuals at higher-than-average risk (e.g., persons with prodromal symptoms or a family history of psychiatric disorders) to reduce morbidity by enhancing resilience and preventing the onset of illness. *Indicated interventions* target individuals who are experiencing impairment as a result of illness as early as possible in the course of the illness, to reduce the burden of the illness on the individual, family, community, and treatment system. Psychiatric services most often take the form of indicated interventions, but the smaller number of psychiatric practitioners and researchers who conduct and evaluate selective or universal interventions in public and community settings is making a substantial contribution to the broader health of society.

Prevention interventions are effective with adults with a variety of risk factors or preclinical problems. For example, women who have been raped

are less likely to develop posttraumatic stress disorder (PTSD) if they receive a five-session cognitive-behavioral treatment than if their recovery is left to chance. People with subthreshold symptoms of depression who are identified early by their primary care providers are less likely to develop the full syndrome if their medical treatment includes education about depression and learning skills for coping actively with depressive symptoms or stressors. Prevention with adults must be judiciously designed to address the specific factors that place a person at risk for illness or that enhance the person's ability to cope effectively. For example, brief supportive meetings with people who have experienced a traumatic stressor (e.g., a mass disaster or life-threatening accident) tend to have little benefit and may inadvertently intensify posttraumatic stress. In contrast, a focused cognitive-behavioral approach to teaching skills for coping with traumatic memories and stress symptoms is effective in preventing posttraumatic stress and depressive disorders with adult and child disaster or accident survivors.

Several prevention programs have been developed and evaluated to address physical and mental health risks in childhood and adolescence, incorporating several elements that influence intervention effectiveness. Their broad and systematic implementation, however, has been halting. Some states have made an effort to implement *school-based* interventions involving teachers and the peer group. These tend to be more effective than programs exclusively relying on intervening with parents or children alone. Such interventions in middle childhood have been successful in influencing peer group norms regarding alcohol and substance use, violence and bullying, and depression, thus achieving the dual outcome of reducing immediate initiation of alcohol and substance use and increasing the long-term support within the peer group for sustained abstinence into adolescence. Thus, systems-based *multimodal* interventions simultaneously targeting and developing enhanced relationships among the child, peer group, school personnel, parents, and the wider community tend to be most effective as universal or selected approaches to early prevention of what otherwise may become lifelong behavioral, legal, academic, and addictive problems.

Access to Effective Mental Health Care. Access is a significant problem for most people with severe mental illness or addictions. In the United States, the National Comorbidity Survey (NCS) and the National

Comorbidity Survey-Replication (NCS-R) found that fewer than 40 percent of people with severe psychiatric disorders had received any mental health treatment in the previous year, and fewer than one in six (15 percent) had received minimally adequate mental health services. Young adults, African Americans, people residing in specific geographical areas, people with psychotic disorders, and patients treated by medical but not mental health providers were at the highest risk for inadequate psychiatric treatment. Although income was not a predictor of inadequate treatment, many of the people who did not receive adequate mental health services were likely to be uninsured or underinsured. Such people often cannot access mental health care other than through a primary care provider or clinic or in the public mental health system.

Even when mental illness is identified, people with socioeconomic adversities often do not, or cannot, get adequate mental health services in their communities. For example, although it is estimated that more than 200,000 incarcerated adults in the United States have psychiatric disorders, few were detected or received treatment until they reached jail or prison. Federal and state correctional systems have instituted mental health screening and treatment programs to address psychiatric disorders as a health problem for incarcerated adults and to manage the problematic behavior that can occur in controlled settings as a result of mental illness. On returning to the community, the vast majority of prisoners with psychiatric disorders cease to receive more than minimal mental health services: A recent study found that fewer than one in six (16 percent—strikingly similar to the NCS finding) received steady mental health services, and only one in 20 with addictions received steady substance abuse recovery services. Thus, access to mental health and addiction treatment services is far better *in prison* than in the community! This disparity has caused great concern because of the possibility that correctional facilities may be a de facto system of care for low-income people (often of minority backgrounds) with serious mental illness.

Evidence of severe and pervasive barriers to accessing mental health care can be found in several social crises. Impoverished children and adults increasingly are deferring physical and mental health care until illnesses become chronic and severe and then often do not know or cannot gain entry into any setting for services except public hospital emergency departments.

Children with severe psychiatric or behavioral disorders are being held in emergency department facilities for days and even weeks because the staff cannot locate any available treatment facilities with an appropriate level of care, especially if they are underinsured. People who cannot afford private services thus face daunting obstacles when seeking appropriate mental health care as a result of a severe underfunding of practitioners and programs. The public policy dilemmas that are driven by the ever-increasing cost of health care bear directly on the field of public psychiatry. It also tragically affects the lives of tens or hundreds of thousands of people who do not receive adequate care.

Psychiatry and Public Agencies

Psychiatry's relationship to public sector agencies has, with some specific exceptions, been one of detachment because of the dominance of the private practice models before the advent of managed care and also mainly because of the nature and structure of U.S. social welfare programs. This separation is a divide that often begins during academic preparation, in that many psychiatry training programs eschew meaningful rotations in public sector settings. In contrast to most industrial nations in which comprehensive social welfare reforms have been initiated, U.S. social welfare programs grew incrementally and categorically, that is, one category of service at a time. Large-scale initiatives, such as President Johnson's War on Poverty, have been implemented in piecemeal fashion by fragmented federal, state, and local government bureaucracies and have been vastly reduced through subsequent initiatives, such as the recent changes in federal regulations to "end welfare as we know it." Various social service agencies have been created through a process, and set of alliances called the *iron triangle.* Advocates form an organization to champion a particular cause. Example causes may include blindness, developmental disabilities, primary care, or mental health, among others. The advocates then find key legislative sponsors to advance the cause through legislation and appropriation. This creates a bureaucracy and bureaucrats that join the alliance. Through successive legislative sessions, the alliance of advocates, lawmakers, and bureaucrats builds a more durable and more reliable categorical agency, for example, adults and families with dependent children or people with

biologically based severe mental illness. From the 1930s through the end of the twentieth century, U.S. social welfare agencies were created on this pattern; the system has resulted in generous funding but a hopeless fragmentation of services at local service delivery levels in which each agency depends on separate funding silos that also deliver conflicting rules and regulations.

It has been said that "mental health is not a place." Psychiatrists have a role at the receiving end of every categorical silo, as the clients in each agency experience mental illness. Services for children with severe emotional, mental, or behavioral disturbances are a dramatic case. Five categorical agencies—child welfare, education, primary health, substance abuse, and juvenile justice—all have a responsibility to care for these children (and, indirectly, for their caregivers, including parents and families). Especially for the children with the most severe disabilities, the protective service worker taking a child into court, the special education teacher working day to day with the child, the juvenile probation officer, the substance abuse counselor, and the child's pediatrician all need the consultation and support of a skilled psychiatrist. The situation is similar for adults with severe mental illness, for whom case managers, vocational rehabilitation counselors, basic needs benefits specialists, social workers, psychotherapists, substance abuse counselors, parole or probation officers, legal conservators, visiting nurses, peer specialists, and physicians all may be mandated to deliver services.

Workforce Influences on Contemporary Practice

Shortages in the ranks of psychiatry have been well documented historically, and some project this to continue. However, as the field of psychiatry connects with its science and evidence base (e.g., neuroscience, genetics, pharmacology, outcomes science) while retaining its strength in the meaningful connection with patients and their families, some would argue that the reversal of the shortage trend has begun. The National Institutes of Health (NIH) sponsored Decade of the Brain (the 1990s), and Decade of the Mind (the 2000s) contributed to this change. Medical students are increasingly stimulated to enter the field by recognizing the importance of psychiatry to health care in general, realizing that this is still a field in which the practitioner (and by extension, the interdisciplinary team) is an instrument

of change through integration of contemporary knowledge, skillful learned interactions with patients, and a willingness to commit to providing value. The national aggregate numbers, however, cannot adequately capture the extreme variations that exist geographically; rural and frontier sites are dramatically more likely to experience shortages than more urban environments.

Because subsets of the public sector populations have special needs, responses to their needs have to be correspondingly tailored:

- Children and family needs are expressed differently than those for adults and require different responses.
- Mental illnesses have culturally driven ways of presenting, and interventions must be culturally competent.
- Issues of language (including American Sign Language with deaf and hard of hearing individuals) can complicate diagnosis and treatment.
- The rapidly expanding elderly population will pose significant new challenges for effective treatment.

Just these few examples suggest the sophistication with which education and training must be engineered to meet the existing and changing demands of people in need of services.

Contemporary Evidence Base for Effective Public and Community Psychiatry

Since the 1980s, several structured interventions have been developed to address the gap between what historically has been taught in most psychiatry training programs—typically, an office-, clinic-, or hospital-based approach focusing on diagnoses and pharmacotherapy supplemented by psychotherapy—and the competencies required to deliver or to support the delivery of the full array of psychiatric rehabilitation services. Public and community approaches to psychiatric rehabilitation involve not only pharmacotherapy but also an array of complementary services. These services must be coordinated to assist people with severe mental illness or behavioral disorders. The services much also help patient's families and support systems. Patient's and their loved ones need help with managing symptoms, accessing resources, and using these resources effectively. The goal of this

support is to help patients gain the highest degree of autonomy in the least restrictive setting possible. The competencies required to effect these interventions go beyond the scope of psychiatry, thus requiring that psychiatrists effectively collaborate with other rehabilitation and mental health specialists. Although psychiatrists rarely implement the educational psychotherapy and other interventions involved, psychiatrists should be aware of them and understand how to reinforce them. Hence, familiarity with the manuals that describe how to implement these interventions with fidelity is strongly advised—and increasingly incorporated into psychiatry training.

Evidence-Based Manualized Interventions for Child Psychiatric Rehabilitation. Children with severe emotional disturbances and adolescents with severe behavioral disorders traditionally have been removed from their families and placed in restrictive psychiatric or juvenile justice settings (e.g., psychiatric inpatient wards, residential group homes, and juvenile detention centers). These placements separate the child from the natural family, school, peer group, and community environments, which may provide some benefit by reducing the child's exposure to addiction, conflict, violence, or deviant behavior. However, the placements also deprive the child and family of the opportunity to build better relationships with one another and with other children, families, teachers, and community groups. The second key ingredient to effective public and community child psychiatry comprises interventions that complement pharmacotherapy. Since the 1970s, several rehabilitative approaches for children with severe disorders (and their families) have been developed, tested, and disseminated in replicable manuals and training programs.

Implementation of evidence-based practices is not always easy. It rarely occurs even in clinics and practice groups that are well trained and motivated to use evidence-based treatment models, given that most current systems of payment and service delivery were not designed to accommodate or support the use of science-based interventions such as those highlighted here. Also, evidence-based dissemination and implementation strategies must be deliberately designed to support the initial adoption and sustained use of evidence-based mental health treatment models. For example, a recent study of the implementation of one of the most widely disseminated evidence-based mental health interventions for children and families, multisystemic

therapy (MST), demonstrated that initial training yielded a very positive early rate of adoption of the program with reasonable fidelity to the model. However, only teams that were provided with regular ongoing supervision and support (for supervisors as well as therapists) from expert consultants were able to sustain the initial successes.

Roles of Psychiatrists in Public and Community Psychiatry Multidisciplinary Teams

Psychiatrists in the public sector rarely work in isolation. Most often, psychiatrists work within a team of professionals. The team includes many disciplines, including psychology, social work, nursing, occupational therapy, rehabilitation or addiction counseling, social services, housing, and employment specialists. It also includes nondegreed direct care workers, such as bachelor's degree-level counselors or case managers, high school graduate indigenous outreach workers, peer-support specialists, and family advocates. Each of these members brings unique skills and experience to address the needs of people with severe and persistent mental illnesses. The team, rather than any single provider, assumes responsibility for the ongoing care of each patient across the many levels of services and often for many years. Its success is based on effective communication. Every communication should be explicitly focused on the *client's* (i.e., patient and family) *stated goals*, as well as on the team's technical and logistical issues, to maximize the client's motivation to participate actively and productively in all services by ensuring that the services indeed are patient-centered and collaborative.

Psychiatrists play three primary roles in multidisciplinary psychiatric rehabilitation teams: Conducting psychiatric evaluations, providing pharmacotherapy, and serving as the team's medical director (and, at times, as the administrative supervisor or team leader).

Psychiatric Evaluation and Diagnosis. As with any practice setting, the sine qua non in the public and community setting is a thorough evaluation. This involves a complete history and examination leading to an accurate diagnosis. The goal of evaluation and diagnosis is to develop the most clinically useful individualized approach to treatment and rehabilitation for each patient. Psychiatric evaluations in public and community settings must

include a careful review of the individual's psychosocial strengths and resources. The focus on strengths and resources often is lost or obscured when systemic factors (e.g., eligibility regulations for government funded services or benefits) emphasize disability. It can also limit the individual's or family's access to services or benefits (e.g., welfare-to-work regulations that place a time or other eligibility restrictions on types of temporary aid, such as food stamps or vouchers for household supplies or housing).

Pharmacotherapy. The psychiatrist's most visible role usually is providing pharmacotherapy. The most difficult challenge in public sector mental health settings often is not the technical formulation of an effective medication regimen but instead the arranging of a plan of care so that the patient reliably follows the prescribed regimen (i.e., *adherence* or *compliance*). A recent review of interventions designed to enhance adherence to psychotropic medication regimens by patients with schizophrenia found that education alone often was the only strategy used. This is true despite evidence showing that the optimal strategy includes such approaches as concrete problem solving, motivational techniques, and practical assistant (reminders or cues for example). Although the psychiatrist must address technical, medical issues around diagnosis and treatment, effective pharmacotherapy largely depends on providing practical assistance. This approach includes helping patients manage the stresses and problems that might affect adherence. Such practical assistance might be direct, as in encounters with patients, or indirect, by working closely with nonpsychiatric mental health professionals and nondegreed care workers.

Team Leadership. The psychiatrist also often plays a leadership role as the project or program medical director—with the attendant responsibilities of monitoring the medical safety and well-being of all patients and establishing or supporting management and clinical procedures that support the quality of care and a cohesive treatment team. As the team leader or only as a team member, the psychiatrist serves as a role model for compassionate and professional behavior in relation not only to patients but also to all other team members. Formal or informal leadership is especially important when psychiatry residents and medical students work on a team as a training experience. The team psychiatrist serves not only as a mentor and a role

model for the core aspects of psychiatric practice but also to demonstrate the values and skills necessary to integrate psychiatry within the framework of multimodal community-based longitudinal psychosocial interventions.

Psychiatrists who work in these settings and desire a leadership role must go beyond their prescription pads to acquire more expertise. This expertise requires gaining the knowledge, attitudes, and skills relevant to the contemporary practice guidelines for psychiatric rehabilitation. It also includes respecting and supporting other team members and earning their respect and support in return. Also, the psychiatrist must cope with large caseloads, and collaborate with agencies and programs to ensure continuity, consistency, and coordination of care. If psychiatrists can incorporate these attitudes and skills, they can expect reciprocity and cohesion from other team members, and clients will benefit. Surveys of psychiatrists in the public sector indicate that those who embrace the rehabilitation perspective by expanding their roles to be consultants and teachers for patients, families, trainees, and colleagues from other mental health disciplines are more satisfied professionally than those who focus on diagnosis and treatment alone.

New Paradigms. Virtually all health care should be integrated and coordinated to deliver practical, rational, and cost-efficient care. The broader, contemporary definition of care must include public health. Linking public health models with acute care management, which, in turn, is coupled with chronic illness care management, rehabilitation, or recovery models, represents the continuum for an integrated system.

However, this continuum is insufficient without four other components: (1) the application of new research to improve care and the application of bench-to-bedside models to translational research, (2) meaningful involvement of patients and families in a shared decision-making paradigm, (3) integration with primary and specialty care systems, and (4) the development of the clinical system as a system of advocacy. Training preprofessionals in such a model is one approach to introduce meaningful and sustained change to health care and its delivery and is already operational in some systems.

New Models for Service Delivery and Treatment

Organized Systems of Care. The focus of reform has moved increasingly to coherent, efficient, and accountable delivery systems. The basic idea of the *system of care* for children with severe emotional disturbances can be described in two ways—in terms of interagency structures and clinical processes. Five categorical agencies in a given community form a *strong consortium* that targets a specific population of children (and their families) to whose needs none of the agencies can adequately respond on their own. The consortium agencies commit themselves to treat the child and family with a *common plan of care* and to find ways to pool their resources to do so appropriately. This care often requires that in the central bureaucracies at the state level, a parallel interagency commitment be in place that supports the local system of care initiative, resolves any regulatory conflicts that may arise, and gives permission to innovative aspects of the delivery system. The collaborative interagency structure provides the resources and flexibility for effective clinical work.

From the viewpoint of *clinical processes,* the system of care provides a full enactment of the traditional clinical practice that meets the requirements of medical quality assurance. What is different is that it is carried out in-home and community settings, and it involves participants from different disciplines—child welfare, special education, and juvenile justice—in one standardized treatment methodology. The center of the system of care is the *child and family team* that is made up of the child and family, clinicians and agency representatives involved in the case, and significant supportive individuals identified by the child and family. Current issues are reviewed—strengths, problems, and needs—diagnostic issues are considered, clinical goals are articulated, and appropriate treatment strategies are identified. The expected outcome of each intervention is specified, and progress toward it is systematically charted. The system of care requires a full array of flexible services. The essential *starters* include clinical diagnostic services, care coordination or case management, crisis intervention services, and a flexible essential supportive service—child care specialists who can be assigned to support the child and family in any situation. Particularly effective systems of care have relied on substantial community organization and collaboration or innovative funding models, or both, that blend or braid funding streams to focus on clear assignment of responsibility for a particular child and family,

as well as adequate resources.

The basic model for community support for adults was called *assertive community treatment* (ACT). ACT was implemented with multidisciplinary teams with psychiatrists, nurses, psychologists, social workers, psychiatric aides, and paraprofessionals. The ACT team would assume the care of a designated number of adult patients with severe and persistent mental illness and would be available around the clock, 7 days a week. The team would help find housing, manage money, organize household routines, find social contacts, find work, and support the individual's adjustment to workplace settings. Concurrently, medications would be managed and help provided to facilitate an individual adjusting to community living. At the heart of the program was the necessary clinical process that developed and maintained an individualized treatment plan, which was adjusted continuously to the changing needs of each client. The ACT model has been modified in various ways as it has been implemented in different states. Innovative funding models have been developed in several states by using bundled rates and case rates, which make it easier to implement and to sustain than traditional fee-for-service payment systems.

In contemporary managed care terms, ACT teams are *disease and disability management models* that assign accountability to provide systems that may not assume risk in managing community support of persons with disabling conditions. The National Alliance on Mental Illness (NAMI) has developed program models and protocols for the Program for Assertive Community Treatment (PACT) to encourage public agencies to contract for ACT services.

Effective Treatment Models. The previous discussions have outlined the various treatment models that have been introduced since the 1980s to establish an evidence base for the effectiveness of specific interventions and approaches to care. The attention to *evidence-based* treatment models has been a response to the call for quality and accountability for service outcomes demanded from health care services in general by purchasers and policymakers and part of an effort to cope with the difficulties of evaluating service delivery models or systems of care.

The movement to evidence-based services is necessary to break down components of the service delivery system to determine the relative

effectiveness of specific service interventions using the evaluative tools and methods that are available. Eventually, the case will be made to address the broader policy questions concerning the value of coherent and rationally organized service delivery in mitigating the effects of mental disability on the development of the child and enabling the recovery process for adults with severe and persistent mental illness.

Table 30-1 summarizes some of the evidence-based treatments used in adult psychiatric rehabilitation, and Table 30-2 summarizes treatment for children.



Table 30-1
Evidence-Based Manualized Interventions for Adult Psychiatric Rehabilitation

Modality	Goals and Targeted Areas	Results of Evaluation Research
Social and independent living skills	Teaches skills for managing symptoms and preventing relapse (mental illness and addictions), improving social communication with providers, peers, and family, and choosing and engaging in activities that match the person's abilities, limitations, and	Social and independent living skills evaluated in randomized, clinical trials show positive outcome, with increased successful community living and reduced rehospitalization

	interests	
Assertive community treatment (ACT)	Provides ongoing care coordination and case management to increase ability to live independently in the community and prevent/manage psychiatric crises	Randomized effectiveness trials with adults and adolescents with serious and persistent mental illness and co-occurring addictive disorders demonstrate increased psychiatric, residential, and vocational stability
Cognitive-behavior therapy (CBT)	Teaches skills for recognizing and modifying thoughts and beliefs that increase or prolong distress, and strengthening thoughts and beliefs that are self-affirming and interpersonally assertive	Meta-analyses support CBT's efficacy in reducing the severity of anxiety disorders, somatoform disorders, bulimia, anger control problems, and generalized stress reactions, and offer modest support for CBT as an adjunctive treatment with addictive disorders and serious mental illness
Dialectical behavior therapy	Teaches skills for regulating intense emotions, tolerating extreme	Evaluated in randomized clinical trials with parasuicidal adult patients with borderline personality disorder, resulting in reduced behavioral and

	emotional distress, achieving interpersonal effectiveness, and developing mindfulness	psychiatric crises greater than that achieved by expert psychiatrists' treatment
	Originally developed for adults with borderline personality disorder, but has been adapted for use with a wide range of Axis I disorders in which extreme dysregulation and poor impulse control are problematic	Randomized clinical trials and observational studies show positive psychosocial outcomes with adults with severe and persistent mental illness and chronic addictions and with adolescents who are suicidal or self-harming, including those who have substance abuse or posttraumatic stress disorders (PTSDs)
Acceptance and commitment therapy	Teaches skills for identifying and managing (rather than avoiding and feeling controlled by) the symptoms of psychosis, affective, anxiety, eating, traumatic stress, and addictive disorders.	Randomized clinical trials show evidence of increased awareness but reduced distress and impairment due to psychiatric symptoms across a variety of psychiatric disorders in outpatient and inpatient treatment settings

	Acceptance of symptoms facilitated by explanation that psychotic symptoms are extreme versions of normal perception and cognitive phenomena (private experiences)	
Relapse prevention therapy	Teaches skills for identifying the warning signs of potential or actual resumption of substance use or psychiatric symptoms Uses cognitive-behavioral skills for proactively modifying patterns of thought and behavior that are likely to escalate into a full relapse	Randomized clinical trials show efficacy in delaying, reducing the severity of, and preventing substance use relapse with adults with chronic addictions and co-occurring or comorbid psychiatric disorders
Interpersonal psychotherapy	Teaches skills for interacting and communicating	Randomized clinical trials show evidence of reduced severity and risk of relapse

	effectively in close relationships, with a particular focus on overcoming four core interpersonal problems: unresolved grief, role conflict (e.g., dissatisfaction with parenting), role transition (e.g., marital separation), and interpersonal deficits (e.g., isolation and anger management)	of moderate to severe depression, PTSD, anxiety disorders, and eating disorders
Behavioral therapy	Provides systematic reinforcement contingencies to increase adaptive behavior and to extinguish maladaptive behavior	Randomized effectiveness trials show evidence of reduced alcohol, drug, and other addictive behaviors
Motivational enhancement therapy	Teaches skills for recognizing and aligning choices and behavior	Randomized effectiveness trials show evidence of reduced alcohol, drug, and other addictive behaviors

	with personal motivations	
Trauma-focused CBT for PTSD	Teaches skills for safely recalling rather than avoiding troubling memories of traumatic events and for recognizing and managing PTSD symptoms, such as intrusive memories or flashbacks, emotional numbing, dissociation, and distress (e.g., guilt, grief), and problems with sleep or anger, or concentration problems due to hyperarousal or hypervigilance	Randomized clinical trials of seven approaches with adults (prolonged exposure therapy, cognitive processing therapy, eye movement desensitization and reprocessing, narrative exposure therapy, seeking safety, skills training for affect/interpersonal regulation with modified prolonged exposure, and emotion-focused trauma therapy) show evidence of safety and efficacy for adults with severe PTSD, co-occurring addictive disorders, problems with anger, dissociation, grief, and guilt, including one study with adults with chronic mental illness
Self-regulation focused CBT for PTSD	Teaches emotion regulation skills for recognizing and modulating traumatic stress symptoms without requiring	Randomized effectiveness trials of three approaches with adults (trauma affect regulation: guide for education and therapy, patient-centered therapy, and skills training in

	reworking of traumatic memories	affect/interpersonal regulation) show evidence of safety and efficacy for adults with severe PTSD, co-occurring addictive disorders, problems with anger, dissociation, grief, and guilt, including in individuals with comorbid serious mental illness, and incarcerated women



Table 30-2
Evidence-Based Manualized Intervention for Child Psychiatric Rehabilitation

Modality	Goals and Targeted Areas	Results of Evaluation
Problem-solving skills training (PSST)	Teaches children skills and supports parents in helping children with self-monitoring, prosocial goal setting, developing peer positive environments, setting limits with friends, and developing problem-solving and communication skills	Randomized, clinical trials show evidence of improved problem-solving and interpersonal skills and modest reductions in oppositional and inattentive behavior
Cognitive-behavior therapy	Teaches skills for recognizing and modifying thoughts and beliefs that increase or prolong	Meta-analyses demonstrate efficacy in reducing children's anxiety and depressive

	distress, and strengthening thoughts and beliefs that are self-affirming and interpersonally prosocial and assertive	symptoms
Parent behavior/Teacher classroom management training	Parent management training focuses on classroom strategies to help teachers and parents manage problematic child behavior and promote social, emotional, and academic competence	Randomized effectiveness studies with two approaches (Incredible Years; Oregon PMT) demonstrate reduced antisocial, oppositional, impulsive behavior, extending to adolescence for children with severe early antisocial problems
Brief strategic family therapy (BSFT)	A short-term, problem-focused intervention with an emphasis on modifying maladaptive patterns of interactions. It focuses on ways to help families interact more effectively	In a randomized, clinical trial with inner-city families whose children had a variety of behavior problems and varying degrees of juvenile justice involvement, BSFT was superior to usual community care services

Functional family therapy (FFT)	FFT teaches skills for problem solving and behavior management and delivers to parents and children together	Randomized and quasi-experimental studies show reduced juvenile recidivism for youths receiving FFT
Multidimensional family therapy (MDFT)	MDFT identifies family interaction patterns and restructures them to increase trust, cooperation, and emotional support among all members of the family; MDFT has evolved from a primarily clinic-based intervention to delivering services in the home and community settings to enhance youth engagement and the immediacy of services; MDFT is developing a variant for traumatized youth incorporating trauma affect regulation: guide for education and therapy (TARGET)	Randomized clinical trials of MDFT across a variety of ethnocultural and socioeconomic subpopulations have shown that it is associated with reduced youth substance abuse and legal problems and enhanced family functioning and school involvement
Multisystemic therapy (MST)	MST involves up to 6 mo of individualized, community-based	In randomized clinical trials, MST has been shown to

	contact by an M.A. or Ph.D. level therapist, supported by a 24-hr backup team; MST addresses all relevant environments by empowering parents with the skills and resources needed to independently prevent and manage crises and maintain family rules; variants of MST are under development for sexual offending youth, youth in psychiatric crisis, and youth who have experienced childhood maltreatment	reduce juvenile delinquent recidivism and crime severity, psychiatric symptoms, out-of-home placements, and drug use while improving family functioning and school attendance; large-scale dissemination studies in the United States, Canada, and Scandinavia show that MST is superior to usual services on some, but not all, psychosocial outcomes
Intensive in-home child and adolescent psychiatric services (IICAPS)	IICAPS provides assistance in the home for families who have children with serious emotional disorders using a clinical team consisting of clinicians and counselors who are supervised by a child and adolescent psychiatrist	One randomized effectiveness trial and observational studies with a system of community agencies treating psychiatrically impaired, socioeconomically disadvantaged children

		demonstrate reduced psychiatric morbidity
Trauma-informed affect regulation therapies for posttraumatic stress disorder (PTSD)	Dyadic therapies for young children and their primary caregivers, and group and individual therapies for adolescents, teach children and caregivers skills for affect regulation to reduce PTSD-related emotional and behavioral problems	Randomized controlled trials with young traumatized children (child–parent psychotherapy, parent–child interaction therapy) and youth (trauma affect regulation: guide for education and therapy; trauma and grief components therapy for adolescents) demonstrate improvements in parent–child relationships and child/youth emotional, behavioral, and interpersonal functioning
Multidimensional treatment foster care (MTFC)	MTFC provides training and ongoing support, parent education, family therapy, and youth mentoring to	Three randomized controlled trials showed that MTFC reduced adolescents’

	enable foster parents to maintain a structured therapeutic environment for teaching children skills, setting limits, and modeling communication and problem-solving strategies	psychiatric rehospitalization, arrests and incarceration, running away, and drug use while improving vocational and educational outcome
Cognitive-behavioral intervention for trauma in the schools (CBITS)	CBITS provides 10-session groups in school mental health or counseling settings for preadolescents or adolescents with PTSD due to family or community violence; CBITS includes education about traumatic stress, skills for anxiety management, and desensitization of trauma memories	Two randomized, controlled trials with inner-city children (one with Spanish-speaking youth) showed CBITS to be associated with reductions in PTSD and depression but not in teacher-reported behavioral or emotional problems
Trauma-focused cognitive–behavioral therapy (TF-CBT)	TF-CBT guides traumatized children and their parents in recognizing and managing PTSD symptoms and in desensitizing trauma memories	Randomized, controlled trials with sexually and physically abused children and children who experienced traumatic losses, disasters, or violence showed

	reductions in PTSD and depression and improvements in social functioning, but no improvement in disruptive or oppositional behavior

Chronic Illness Care Model: Psychiatry in the Context of Primary Care. Finally, the care of persons with long-term mental health problems is included in innovative developments in the provision of primary health care for persons with chronic illnesses (http://www.improvingchroniccare.org). The Health Resources Services Administration (HRSA), the federal agency responsible for the community health centers or Federally Qualified Health Centers (FQHCs), has adopted the chronic illness care model in its training and technical assistance efforts for community health centers. The model grows out of the current concern for the quality of care and accountability for health care delivery systems for effective outcomes. Depression management is one of the four chronic health conditions that the HRSA has selected for its training collaboratives.

The model begins with the assumption that the health care delivery system is part of a community context and must be responsive in its interactions with the community. Four areas of focus are essential in implementing the model in the health care delivery system: self-management support, delivery system design, decision support, and clinical tracking system. *Self-management support* gives patients a central role in determining their care, fostering a sense of responsibility for their health. Patients collaborate with the primary care team to establish goals, create treatment plans, and solve problems along the way. *Delivery system design* requires a reorganization of how the health system operates so that up-to-date information about a given patient is centralized, and follow-up responsibility is assigned as a standard procedure, and so on. *Decision support* requires that treatment decisions are based on explicit, proven practice guidelines supported by at least one defining study.

Guidelines are discussed with patients and providers, and treatment team members are trained continuously in the latest proven methods. Finally, *clinical tracking systems* track individual patients and populations of patients with similar problems. These systems must be practical and operational—able to check an individual's treatment at any point to confirm that the treatment conforms to recommended guidelines. The real integration of psychiatric care into primary care is an important innovation to consider that would improve physical health care for individuals with psychiatric disorders and eliminate the separation imposed by placing individuals with psychiatric problems into community mental health centers.

THE ROLE OF PUBLIC AND COMMUNITY PSYCHIATRY IN TWENTY-FIRST CENTURY HEALTH CARE

Public psychiatry emerged historically as an attempt to provide humane care for people with severe mental disabilities who did not have the resources to be protected from the stigma and approbation that shielded so-called "eccentrics" and "black sheep" fortunate enough to have been born into the wealthier social strata. Community psychiatry evolved as an answer to concerns that public psychiatric treatment was a de facto form of marginalization and oppression, if not inhumane exile and imprisonment, of economically disadvantaged persons with mental illness. The best ideas from community and public psychiatry are sometimes lost amidst the competition for scarce resources. The spirit and skills of advocacy that sparked the development and continue to characterize the best practices of public and community psychiatry, as well as the dedication to bring demonstrably effective services to the people who are most in need but are least served, have never been more needed by the entire mental health field.

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31 Global and Cultural Issues in Psychiatry

Mental disorders are highly prevalent in all regions of the world and represent a significant source of disability and social burden worldwide. Treatments for all these disorders are available and are efficacious in both developed and developing countries. However, mental disorders are remarkably undertreated worldwide, especially in low-income countries. National mental health policies are lacking in several countries, especially low-income ones. Resources for mental health care are scarce and unequally distributed. World psychiatry focuses on these and other issues, such as the stigma attached to mental disorders, the relationships between mental and physical diseases, and the ethics of mental health care.

PREVALENCE AND BURDEN OF MENTAL DISORDERS WORLDWIDE

The World Health Organization (WHO) estimates that more than 25 percent of individuals worldwide develop one or more mental disorders during their lifetime. Among people seen by primary health care professionals, more than 20 percent have one or more current mental disorders. In a study carried out by the WHO at 14 sites in Africa, Asia, the Americas, and Europe, the average current prevalence of any mental disorder was 24 percent, without consistent differences between low- and high-income countries. The most common diagnoses were those of depression (average, 10.4 percent) and generalized anxiety disorder (average, 7.9 percent). Female rates were 1.89 times higher than male rates for depression, but male rates were higher for alcohol-related disorders so that there was no sex difference in the proportion

of people having at least one mental disorder. Physical ill-health and educational disadvantage were both significantly associated with a diagnosis of mental disorder.

To quantify the burden of the various diseases and injuries, the WHO, in collaboration with the Harvard School of Public Health and the World Bank, introduced the disability-adjusted life year (DALY). DALY for a given disease or injury is the sum of the years of life lost due to premature mortality plus the years lost due to disability for incident cases of that disease or injury in the general population. In the original estimates for the year 1990, mental and neurologic disorders accounted for 10.5 percent of the total DALYs lost due to all diseases and injuries. The estimate for the year 2000 was 12.3 percent, with two mental disorders (depression and alcohol use disorders) and suicide ranking in the top 20 causes of DALYs for all ages.

In the estimates for the year 2005, mental and neurologic disorders accounted for 13.5 percent of all DALYs in the world, and is the main contributor to burden among noncommunicable diseases (Table 31-1).

Table 31-2 summarizes the global burden of disease for mental and substance use disorders at various times in the last few decades.

By 2030, mental and neurologic disorders will likely account for 14.4 percent of all global DALYs and 25.4 percent of those due to noncommunicable diseases. Depression will rank number 2 in the percentage of total DALYs in that year (5.7 percent), following HIV/AIDS and preceding ischemic heart disease (Table 31-3). It will be number 1 in high-income countries (9.8 percent), number 2 in middle-income countries (6.7 percent), and number 3 in low-income countries (4.7 percent).

The WHO estimates that one of four families worldwide has at least one member with a mental disorder. The objective and subjective burden related to caring for people with severe mental disorders (in terms of disruption of family relationships; constraints in social, leisure, and work activities; financial difficulties; detrimental effects on physical health; feelings of loss, depression, and embarrassment in social situations; and the stress of coping with disturbing behaviors) is substantial and significantly higher than that related to caring for people with long-term physical diseases such as diabetes and heart, kidney, or lung diseases. Cross-cultural differences exist in some dimensions of family burden.

Suicide is among the 10 leading causes of death for all ages in most of the

countries for which information is available. In some countries (e.g., China), it is the leading cause of death for people between 15 and 34 years of age.

According to WHO estimates, about 849,000 people died from suicide worldwide in the year 2001. In that year, the number of suicide deaths overtook the number of deaths by violence (500,000) and war (230,000). In 2020, approximately 1.53 million people will die from suicide worldwide, based on current trends, and 10 to 20 times more people will attempt suicide. Reported suicide rates vary considerably across countries; for instance, in many Eastern and Central European countries, the annual suicide rates are between 48.0 and 79.3 per 100,000. In contrast, many Islamic and Latin American countries report rates of less than 4.0 per 100,000. More than 85 percent of suicides occur in low- or middle-income countries. However, this figure may represent an underestimate due to the low reliability of official statistics in those countries: for example, when epidemiologists used a verbal autopsy in South India, the observed rates for suicide exceeded official national estimates 10-fold. In the Asia-Pacific region, perhaps 300,000 cases of suicide per year occur by self-poisoning with pesticides.

Suicide rates are higher in men than in women (3.2:1 in 1950, 3.6:1 in 1995, and an estimated 3.9:1 in 2020). China is the only country where suicide rates in women are consistently higher than those in men, especially in rural areas. Over the past few decades, suicide rates are generally stable worldwide, but there is a rising trend among young men ages 15 to 19 years. A systematic review covering 15,629 cases in the general population worldwide estimated that 98 percent of those who committed suicide had a diagnosable mental disorder, with mood disorders accounting for 35.8 percent, substance-related disorders for 22.4 percent, personality disorders for 11.6 percent, and schizophrenia for 10.6 percent of cases.

TREATMENT GAP AND PROJECTED POPULATION-LEVEL TREATMENT EFFECTIVENESS WORLDWIDE

Clinical trials have convincingly proved the efficacy of pharmacologic and psychosocial treatments for mood, anxiety, psychotic, and substance-related disorders carried out in low- and middle-income, as well as in high-income countries. However, the treatment gap is substantial for all mental disorders worldwide, particularly in low-income countries.

Table 31-1
Disorder-Specific Global Sheehan Disability Scale Ratings for Commonly Occurring Mental and Chronic Physical Disorders in Developed and Developing WMH Countries

	Proportion Rated Severely Disabling			
	Developed		Developing	
	%	(SE)	%	(SE)
I. Physical Disorders				
Arthritis	23.3	(1.5)	22.8	(3.0)
Asthma	8.2	(1.4)	26.9	(5.4)
Back/Neck	34.6	(1.5)	22.7	(1.8)
Cancer	16.6	(3.2)	23.9	(10.3)
Chronic pain	40.9	(3.6)	24.8	(3.8)
Diabetes	13.6	(3.4)	23.7	(6.1)
Headaches	42.1	(1.9)	28.1	(2.1)
Heart disease	26.5	(3.9)	27.8	(5.2)
High blood pressure	5.3	(0.9)	23.8	(2.6)
Ulcer	15.3	(3.9)	18.3	(3.6)
II. Mental Disorders				
ADHD	37.6	(3.6)	24.3	(7.4)
Bipolar	68.3	(2.6)	52.1	(4.9)
Depression	65.8	(1.6)	52.0	(1.8)
GAD	56.3	(1.9)	42.0	(4.2)
IED	36.3	(2.8)	27.8	(3.6)
ODD	34.2	(6.0)	41.3	(10.3)
Panic disorder	48.4	(2.6)	38.8	(4.7)
PTSD	54.8	(2.8)	41.2	(7.3)
Social phobia	35.1	(1.4)	41.4	(3.6)
Specific phobia	18.6	(1.1)	16.2	(1.6)

35.8 percent, substance-related disorders for 22.4 percent, personality disorders for 11.6 percent, and schizophrenia for 10.6 percent of cases.

In the World Mental Health Surveys, failure and delays in treatment seeking were generally higher in low-income countries, older cohorts, men, and cases with earlier ages of onset. People with mood disorders had earlier treatment contact than other mental disorders, this is likely because many countries have targeted this disorder, with screening campaigns and public informational campaigns.

RESOURCES FOR MENTAL HEALTH CARE WORLDWIDE

According to the *Mental Health Atlas 2005,* only 62.1 percent of countries worldwide, accounting for 68.3 percent of the world population, have a mental health policy (i.e., a document of the government or ministry of health specifying the goals for improving the mental health situation of the country, the priorities among those goals, and the main directions for attaining them). Mental health policies are present in 58.8 percent of low-income and 70.5 percent of high-income countries. In Africa, only 50 percent of countries have a mental health policy. In Southeast Asia, only about half of countries have a mental health policy, and only about a quarter of the population are affected by these policies (Table 31-4).

Community care facilities exist in only 68.1 percent of the countries (51.7 percent of low-income and 93 percent of high-income countries). Only 60.9 percent of the countries report providing treatment facilities for severe mental disorders at the primary care level (55.2 percent of low-income and 79.5 percent of high-income countries). About one-fourth of low-income countries do not provide even rudimentary antidepressant medications in primary care settings. In many others, the supply does not cover all regions of the country or is very irregular. Because medicines are often not available in health care facilities, patients and families must pay for them out of pocket.

Whereas 61.5 percent of European countries spend more than 5 percent of their health budget on mental health care, 70 percent of countries in Africa and 50 percent of countries in Southeast Asia spend less than 1 percent. Out-of-pocket payment is the most important method of financing mental health care in 38.6 percent of countries in Africa and 30 percent of countries in Southeast Asia, but it is not the primary method of financing mental health

care in any European country (Table 31-5). All countries with out-of-pocket payment as the dominant method of financing mental health care belong to low-income or lower-middle-income categories, but almost all countries with social insurance as the dominant method of financing belong to high-income or upper-middle-income categories.

The median number of psychiatrists per 100,000 population ranges from 0.04 in Africa and 0.2 in Southeast Asia to 9.8 in Europe (Table 31-6). It is 0.1 in low-income countries compared with 9.2 in high-income countries. Two-thirds of low-income countries have less than one psychiatrist per 100,000 population. Chad, Eritrea, and Liberia (with populations of 9, 4.2, and 3.5 million, respectively) each have just one psychiatrist per 100,000 population. Afghanistan, Rwanda, and Togo (with populations of 25, 8.5, and 5 million respectively) each have just two psychiatrists per 100,000 population. Large-scale migration of psychiatrists from low- and middle-income to high-income countries, as part of the larger picture of migration of health professionals in general, has been consistently documented. India and some sub-Saharan African countries are the most important contributors to the mental health workforce in the United Kingdom, although the United Kingdom has 110 psychiatrists per million population. However, India has 2 per million and sub-Saharan Africa less than 1 per million. The median number of psychologists working in mental health care per 100,000 population ranges from 0.03 in Southeast Asia and 0.05 in Africa to 3.1 in Europe. Approximately 69 percent of low-income countries have less than one psychologist per 100,000 population. The median number of psychiatric nurses per 100,000 population ranges from 0.1 in Southeast Asia and 0.2 in Africa to 24.8 in Europe.



Table 31-2
Global Burden of Disease: Mental and Substance Use Disorders

	Mental Disorders	Substance Use Disorders	Before 1998	1998–2005	2006–2013	Total
All causes			100.0%	100.0%	100.0%	100.0%
Mental and substance use disorders			37.8%	58.5%	35.6%	67.6%
	Schizophrenia		17.0%	9.0%	3.7%	19.1%
	Alcohol use disorders		19.7%	28.7%	14.9%	31.4%
	Drug use disorders		20.7%	47.3%	26.1%	51.6%
		Opioid use disorders	12.8%	17.6%	2.7%	19.7%
		Cocaine use disorders	6.9%	31.9%	5.9%	34.6%
		Amphetamine use disorders	6.4%	23.9%	8.0%	27.7%
		Cannabis use disorders	16.0%	42.0%	20.7%	46.8%
		Other drug use disorders	—	—	—	—
	Depressive disorders		19.7%	23.9%	11.2%	33.0%
		Major depressive disorder	19.7%	23.9%	11.2%	33.0%
		Dysthymia	9.0%	13.8%	5.3%	18.6%
	Bipolar disorder		8.5%	16.0%	3.7%	18.6%
	Anxiety disorders		12.8%	21.8%	5.3%	26.1%
	Eating disorders		10.6%	12.2%	4.3%	14.9%
		Anorexia nervosa	10.1%	12.2%	4.3%	14.4%
		Bulimia nervosa	8.5%	11.7%	3.2%	14.9%
	Autistic spectrum disorders		5.3%	5.9%	3.7%	9.6%
		Autism	5.3%	5.3%	3.7%	9.6%
		Asperger syndrome	1.6%	4.8%	1.6%	5.3%
	Attention-deficit or hyperactivity disorder		10.6%	10.1%	4.8%	19.1%
	Conduct disorder		5.9%	6.4%	1.6%	11.2%
	Idiopathic intellectual disability		6.4%	3.2%	1.1%	7.4%
	Other mental and substance use disorders		0.5%	0.5%	0.0%	1.1%

Table 31-3
Leading Causes of Disability-Adjusted Life Year (DALY) Worldwide as Estimated for the Year 2030

Disease or Injury	Percent of Total
1. HIV/AIDS	12.1
2. Unipolar depressive disorders	5.7
3. Ischemic heart disease	4.7
4. Road traffic accidents	4.2
5. Perinatal conditions	4.0
6. Cerebrovascular disease	3.9

7. Chronic obstructive pulmonary disease	3.1
8. Lower respiratory infections	3.0
9. Hearing loss, adult onset	2.5
10. Cataracts	2.5

From Mathers CD, Loncar D. Projections of global mortality and burden of disease from 2002 to 2030. *PLoS Med*. 2006;3(11):e442.

Table 31-4
Presence of a Mental Health Policy in Countries of Each Region of the World Health Organization (WHO)

WHO Region	Percent of Countries	Percent of Population Coverage
Africa	50.0	69.4
Americas	72.7	64.2
Eastern Mediterranean	72.7	93.8
Europe	70.6	89.1
Southeast Asia	54.5	23.6
Western Pacific	48.1	93.8

From *World Health Organization: Mental Health Atlas 2005*. Geneva: World Health Organization; 2005.

Table 31-5
Countries in Which Out-of-Pocket Payment Is the Most Common Method of Financing Mental Health Care in Each Region of the World Health Organization (WHO)

WHO Region	Percent of Countries
Africa	38.6
Americas	12.9

Eastern Mediterranean	15.8
Europe	0
Southeast Asia	30.0
Western Pacific	18.5

From *World Health Organization: Mental Health Atlas 2005*. Geneva: World Health Organization; 2005.

From these figures, it is clear that resources for mental health care are grossly inadequate compared with the needs, and that inequalities across countries are substantial, especially between low- and high-income countries. Moreover, resources tend to be concentrated in urban areas, especially in low-income countries, leaving vast regions without any form of mental health care.

Even worse is the situation concerning child and adolescent mental health care. According to the WHO, only 7 percent of countries worldwide have a specific child and adolescent mental health policy. In less than one-third of all countries, is it possible to identify an institution or a governmental entity with overall responsibility for child mental health. School-based services are almost exclusively present in high-income countries; and even in Europe, only 17 percent of countries have a sufficient number of these services. There are no pediatric beds for mental health identified in low-income countries, but such beds exist in 50 percent of high-income countries. In all African countries outside South Africa, there are as little as 10 psychiatrists trained to work with children. In European countries, the number of child psychiatrists ranges from one per 5,300 to one per 51,800. In more than 70 percent of countries worldwide, there is no list of essential psychotropic medications for children. In 45 percent of countries worldwide, psychostimulants are either prohibited or unavailable for use in children with attention-deficit/hyperactivity disorder.

Table 31-6
Median Number of Mental Health Professionals per 100,000 Population in Each Region of the World Health Organization (WHO)

WHO Region	Psychiatrists	Psychiatric Nurses	Psychologists Working in Mental Health
Africa	0.04	0.20	0.05
Americas	2.00	2.60	2.80
Eastern Mediterranean	0.95	1.25	0.60
Europe	9.80	24.80	3.10
Southeast Asia	0.20	0.10	0.03
Western Pacific	0.32	0.50	0.03

From *World Health Organization: Mental Health Atlas 2005*. Geneva: World Health Organization; 2005.

PRINCIPLES FOR MENTAL HEALTH PROGRAM DEVELOPMENT AND BARRIERS TO CHANGE WORLDWIDE

According to the WHO, the development of mental health programs worldwide should adhere to the following principles: (1) providing treatment in primary care; (2) making psychotropic medications available; (3) giving care in the community; (4) educating the public; (5) involving communities, families, and consumers; (6) establishing national policies and legislation; (7) developing human resources; (8) linking with other relevant sectors; (9) monitoring community mental health; and (10) supporting more research.

The guiding principles proposed for the prevention of suicide worldwide include (1) reducing access to means of suicide (e.g., pesticides, firearms), (2) treating people with mental disorders, (3) improving media portrayal of suicide, (4) training primary health care personnel, (5) implementing school-based programs, and (6) developing hotlines and crisis centers.

The most significant barriers to the implementation of the preceding principles worldwide, according to the WHO, include the following: (1) some stakeholders may be resistant to the changes; (2) health authorities may not believe in the effectiveness of mental health interventions; (3) there may be

no consensus among the country's stakeholders about how to formulate or implement the new policy; (4) financial and human resources may be scarce; (5) other essential health priorities may compete with mental health care for funding; (6) primary care teams may feel overburdened by their workload and refuse to accept the introduction of the new policy; and (7) many mental health specialists may not want to work in community facilities or with primary care teams, preferring to remain in hospitals. The suggested solutions include (1) adopting an "all-winners approach" that ensures countries take into account the needs of all stakeholders, (2) developing pilot projects and evaluating their effect on health and consumer satisfaction, (3) asking for technical reports from international experts, (4) focusing the implementation of the mental health policy on a demonstration area and performing cost-effectiveness studies, (5) linking mental health programs to other health priorities, and (6) showing primary care practitioners that people with mental disorders are already a hidden part of their burden and that the burden will decrease if they identify and treat these disorders.

STIGMATIZING ATTITUDES TOWARD PEOPLE WITH MENTAL DISORDERS

Stigmatizing attitudes toward people with mental disorders are widespread in the general public and even among mental health professionals. Although stigma may be less severe in some Asian and African countries, a study carried out in India within the Stigma Programme of the World Psychiatric Association (WPA), in which the investigators interviewed 463 persons with schizophrenia and 651 family members in four cities, reported that two-thirds of the respondents had experienced discrimination. Women and people living in urban areas were the most stigmatized. Whereas men experienced more significant discrimination in the job area, women experienced more problems in the family and social areas.

Unlike people with physical disabilities, those with mental disorders are often perceived by the public to be in control of their disabilities and responsible for causing them. The view that "weakness," "laziness," or "lack of willpower" contributes to the development of mental disorders still exists in many countries. The stigmatization of people with mental disorders may result in public avoidance, systematic discrimination, and reduced help-

seeking behavior. In a survey carried out in 1996 in a probability sample of 1,444 adults in the United States, more than half of the respondents were unwilling to spend an evening socializing with, work next to, or have a family member marry a person with mental illness. Although most countries have some provision for disability benefits, people with mental illness are often explicitly excluded from such entitlements. Moreover, mental disorders are frequently not considered in social and private insurance schemes for health care. Shame is one of the main barriers to seeking help for mental disorders in both developed and developing countries.

We can subdivide the strategies for addressing the stigmatization of people with mental disorders into three groups: protest, education, and contact. There is some evidence that protest campaigns may be useful in reducing stigmatizing behaviors against people with mental disorders. Education may promote a better understanding of mental illness, and educated people may be less likely to endorse stigma and discrimination. There is an inverse relationship between having contact with a person with mental illness and endorsing stigmatizing behavior.

RELATIONSHIPS BETWEEN MENTAL AND PHYSICAL DISEASES

People with severe mental disorders have higher mortality from physical illnesses compared with the general population. In a follow-up study carried out in the United Kingdom, the standardized mortality ratio (SMR) for natural causes in people with schizophrenia was 2.32 (i.e., death was more than twice higher than in the general population). The SMR for causes "avoidable by appropriate treatment" was 4.68. The highest SMRs were those for endocrine, nervous, respiratory, circulatory, and gastrointestinal diseases. There is also an increased nonsuicide all-cause mortality for bipolar disorder (SMR 1.9 for men and 2.1 for women) and dementia (relative risk [RR] 2.63; 95 percent confidence interval [CI] 2.17 to 3.21). A meta-analysis of 15 population-based studies of the effect of a diagnosis of depression on subsequent all-cause mortality yielded a pooled odds ratio of 1.7 (CI 1.5 to 2.0). Evidence from low-income countries is limited, but a large population study conducted in Ethiopia found high mortality rates for major depression (SMR 3.55, 95 percent CI 1.97 to 6.39) and schizophrenia (almost 5 percent

per year).

People with mental disorders also have a higher prevalence of several physical diseases compared with the general population. In a study carried out in the United States, people with psychotic disorders were more likely than other people to develop diabetes, hypertension, heart disease, asthma, gastrointestinal disorders, skin infections, malignant neoplasms, and acute respiratory disorders. This high rate remained true even when only considering patients without a concomitant substance use disorder. In a study conducted in Nigeria, 55.2 percent of persons with schizophrenia-spectrum disorders referred for the first time to a psychiatric clinic had at least one physical disease, but in persons with neurotic disorders, the rate was 11.8 percent. A strong prospective association exists between depression and coronary heart disease outcomes, including fatal myocardial infarction; on the other hand, the incidence of depression is increased after myocardial infarction, especially in the first month after the event. Depression also increases the risk of type II diabetes. In South Asia, there are reports of an association between maternal perinatal depression and infant malnutrition and stunting at 6 months.

People with severe mental disorders are at increased risk of contracting HIV infection, although prevalence rates vary substantially worldwide. A large multicenter study conducted in sub-Saharan Africa, Asia, Latin America, Europe, and the United States reported a higher prevalence of the depressive disorder among symptomatic HIV-seropositive people than in asymptomatic HIV-seropositive cases and seronegative control participants. Evidence from both developed and developing countries shows that adherence to highly active antiretroviral therapy (HAART) is negatively affected by depression, cognitive impairment, and substance abuse.

In a study carried out in the United Kingdom, people with severe mental illness were significantly more likely to be obese (body mass index higher than 30) and morbidly obese (body mass index higher than 40) than the general population; the respective figures were 35.0 versus 19.4 percent and 3.7 versus 1.3 percent. When these figures were broken down by age and sex, 28.7 percent of men with severe mental illness between 18 and 44 years of age were obese compared with 13.6 percent in the general population, and 3.7 versus 0.4 percent were morbidly obese. Even more striking were the figures concerning women of the same age; 50.6 versus 16.6 percent and 7.4 versus

2.0 percent, respectively.

A meta-analysis of worldwide studies confirmed that there is a highly significant association between schizophrenia and current smoking. The weighted average odds ratio was 5.9; it was 7.2 in men and 3.3 in women. The association remained significant when controls with severe mental illness were used (odds ratio, 1.9). Heavy smoking and high nicotine dependence were also more frequent in people with schizophrenia than in the general population.

The quality of physical health care received by patients with severe mental illness is often worse than the general population. A study conducted in the United States found that adverse events during medical and surgical hospitalizations were significantly more frequent in patients with schizophrenia than in the other people, including infections due to medical care, postoperative respiratory failure, postoperative deep venous thrombosis or pulmonary embolism, and postoperative sepsis. All of these adverse events were associated with a significantly increased odds of admission to an intensive care unit and of death.

The decreased access of people with mental disorders to medical services relates to several factors concerning the health care system. There is a well-documented effect of lack of insurance and the cost of care. In a study carried out in the United States, people with mental disorders were twice as likely as those without mental disorders to have been denied insurance because of a preexisting condition (odds ratio, 2.18). Having a mental disorder conferred a higher risk of having delayed seeking care because of cost (odds ratio, 1.76) and of having been unable to obtain needed medical care (odds ratio, 2.30).

Even when a doctor sees people with mental illness, their physical diseases often remain undiagnosed. Primary care providers may misperceive the medical complaints of people with mental disorders as “psychosomatic” or be unskilled or feel uncomfortable in dealing with this population. Underlying stigmatization may be involved. Moreover, during hospitalizations in medical and surgical wards, health care professionals may not be experienced in dealing with the unique needs of patients with schizophrenia, may minimize or misinterpret their somatic symptoms, and may make inappropriate use of restraints or sedative drugs or fail to consider possible interactions of psychotropic drugs with other medications. On the other hand, many psychiatrists are unable or unwilling to perform physical and even neurologic

examinations or are not up to date on the management of even common physical diseases.

The first step is raising awareness of the problem among mental health care professionals, primary care providers, and patients with schizophrenia and their families. Education and training of mental health professionals and primary care providers is a further essential step. We should train mental health professionals to perform at least basic medical tasks. They should understand the importance of recognizing physical illness in people with severe mental disorders and become familiar with the most common reasons for underdiagnosis or misdiagnosis of physical illness in these patients. On the other hand, primary care providers should overcome their reluctance to treat people with severe mental illness and learn practical ways to interact and communicate with them; it is not only an issue of knowledge and skills but also most of all one of attitudes.

A well-identified professional should be responsible for physical health care in each person with a severe mental disorder. Mental health services should be able to provide at least a standard routine assessment of their patients to identify or at least suspect the presence of physical health problems. Guidelines about the management of patients receiving antipsychotic drugs should be known and applied by all mental health services. Patients should be involved as much as possible; for instance, mental health professionals should encourage patients to monitor and chart their weight. Mental health services should routinely provide dietary and exercise programs. Flexible smoking cessation programs, which have shown some degree of success, could be considered in some settings.

ETHICAL ISSUES IN MENTAL HEALTH CARE

The protection and promotion of the human rights of people with mental disorders are emerging as a priority worldwide. In 1991, the United Nations (UN) issued Resolution 46/119 for the Protection of Persons with Mental Illness and the Improvement of Mental Health Care. In that resolution, the UN codified the human rights of people with mental disorders, and their right to treatment. The 25 principles covered the following areas: definition of mental illness; protection of confidentiality; standards of care and treatment, including involuntary admission and consent to treatment; rights of persons

with mental disorders in mental health facilities; protection of minors; provision of resources for mental health facilities; the role of community and culture; review mechanisms providing for the protection of the rights of offenders with mental disorders; and procedural safeguards protecting rights of persons with mental disorders. The UN encouraged national governments to promote the principles of the resolution by appropriate legislative, juridical, administrative, educational, and other provisions. However, violations of the human rights of people with mental disorders still exist in many countries, and in a substantial number of low- and middle-income countries, patients in mental hospitals are physically restrained or secluded for long periods.

In Latin America, an influential document has been the Declaration of Caracas, adopted in 1990 by the Regional Conference on Restructuring Psychiatric Care in Latin America, which states that resources, care, and treatment for people with mental disorders should safeguard their dignity and human and civil rights and strive to maintain them in their communities. The Declaration also states that mental health legislation should safeguard the human rights of people with mental disorders and that countries should organize services so that they can enforce these rights.

In Africa, the Banjul Charter on Human and Peoples' Rights, a legally binding document supervised by the African Commission on Human and Peoples' Rights, in Article 5 addresses the right to respect for the dignity inherent in human beings and the prohibition of all forms of degradation, including cruel, inhuman, or degrading treatment.

According to the WHO, mental health legislation should cover the following issues: access to fundamental mental health care, least restrictive care, informed consent to treatment, voluntary and involuntary admission to treatment, competence issues, periodical review mechanism, confidentiality, rehabilitation, accreditation of professionals and facilities, and rights of families and caregivers. Specific legislation in the field of mental health is present in 74 percent of low-income countries versus 92.7 percent of high-income countries.

In 1996, the WPA released the Madrid Declaration, which contains the ethical principles by which all national psychiatric societies should abide. The declaration includes seven general guidelines focusing on the aims of psychiatry: (1) psychiatrists must serve patients by providing the best therapy

available, consistent with accepted scientific knowledge and ethical principles, and should devise therapeutic interventions that are the least restrictive to the freedom of the patient; (2) it is the duty of psychiatrists to keep abreast of scientific developments of their specialty and to convey updated knowledge to others; (3) the patient should be accepted as a partner by right in the therapeutic process, and the therapist–patient relationship must be based on mutual trust and respect, to allow the patient to make free and informed decisions; (4) treatment must always be in the best interest of the patient, and no treatment should be provided against the patient's will, unless withholding treatment would endanger life of the patient or those who surround him or her; (5) when psychiatrists are requested to assess a person, it is their duty to inform the person being assessed about the purpose of the intervention; (6) information contained in the therapeutic relationship should be kept in confidence and used exclusively for the purpose of improving the mental health of the patient; and (7) because psychiatric patients are particularly vulnerable research subjects, extra caution should be taken to safeguard their autonomy, as well as their mental and physical integrity.

INTERNATIONAL ORGANIZATIONS ACTIVE IN THE MENTAL HEALTH FIELD

Many international organizations are active in the mental health field. They include the WHO, which is the world's leading public health agency; some professional associations, among which the largest is the WPA, representing the psychiatric profession worldwide; and several organizations with a membership of users and families (such as the World Fellowship for Schizophrenia and Allied Disorders and the Global Alliance of Mental Illness Advocacy Networks [GAMIAN]) or both mental health professionals and users and families (such as the World Federation for Mental Health [WFMH]).

The WHO is a UN agency that is funded by 192 member states. It has six regional groupings: Africa, the Americas, Eastern Mediterranean, Europe, Southeast Asia, and Western Pacific. The Department of Mental Health and Substance Abuse is in Geneva, and it has advisors for mental health in each of its regional offices. Its main functions are direction and coordination of international health work and technical cooperation with countries. Among

the numerous recent WHO activities in the mental health field, of particular interest, are the release of the *World Health Report 2001* and the development of the report *Mental Health: New Understanding, New Hope*, which was devoted entirely to mental health. It provided a summary of the current and projected impact of mental disorders and the principles of mental health policy and service provision, as well as a set of recommendations for future action that countries can adapt to their various needs and resources. Project Atlas aims to collect information on mental health resources across the world. Global and regional analyses of those resources were first published in 2001 and updated in 2005. Reports from the project focus, for example, on resources for child and adolescent mental health and psychiatric education and training worldwide (the latter in collaboration with the WPA).

The WPA is an association of national psychiatric societies aimed at increasing the knowledge and skills necessary for work in the field of mental health and the care for the mentally ill. Its member societies number 134, and it spans 122 countries and represents more than 200,000 psychiatrists. The WPA organizes the World Congress of Psychiatry every 3 years. It also organizes international and regional congresses and meetings and thematic conferences. It has 65 scientific sections, which disseminate information and promoting collaborative work in specific domains of psychiatry. It has produced several educational programs, books, and consensus statements (including the Declaration of Madrid on ethical principles for psychiatric practice). It has an official journal, *World Psychiatry*, which is in English, Spanish, and Chinese and is indexed in PubMed and Current Contents and reaches more than 33,000 psychiatrists worldwide.

The WFMH is a multidisciplinary advocacy and education organization aimed at promoting the advancement of mental health awareness, prevention, advocacy, and best-practice recovery-focused interventions worldwide. Among its activities is the organization of the World Mental Health Day, observed every year on October 10, each time with a different theme.

FUTURE PERSPECTIVES

There are several statements from various groups and organizations about the priorities for future action in the mental health field at the international level. Of particular interest is a document produced by 39 leaders comprising the

Lancet Commission on global mental health and sustainable development, or the Global Mental Health Commission. This document identifies five main goals: (1) placing mental health on the public health priority agenda, (2) improving the organization of mental health services, (3) integrating the availability of mental health into general health care, (4) developing human resources for mental health, and (5) strengthening public mental health leadership.

Among the strategies proposed to place mental health on the public health priority agenda are the development and use of uniform and understandable messages for mental health advocacy and the education of decision-makers within governments and donor agencies on the evidence concerning the public health significance of mental disorders and the cost-effectiveness of mental health care. Strategies suggested to improve the organization of mental health services include, among others, the provision of incentive arrangements to overcome vested interests blocking change and the organization of international technical support to learn from countries that have experienced successful mental health reforms. The authors proposed that mental health professionals be appointed and trained specially to support and supervise primary health care staff, to help integrate mental and general health care. It is also crucial that we increase and diversify the professional and specialist workforce is increased and diversified and improve the quality of mental health training to ensure that it is practical and widely practiced in community or primary care settings.

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D Contributions from the Sciences and Social Sciences to Psychiatry

32

Normal Development and Aging

32.1 Infant, Child, and Adolescent Development

The transactional nature of development in infancy, childhood, and adolescence, consisting of a continuous interplay between biologic predisposition and environmental experiences, forms the basis of current conceptualizations of development. There is much evidence that observed developmental outcomes evolve from interactions between particular biologic substrates and specific environmental events. For example, the serotonin transporter gene sensitizes a child with early adverse experiences of abuse or neglect to increased risk for later development of a depressive disorder. Also, the degree of resilience and adaptation, that is, the ability to withstand adversity without damaging effects, is likely to be mediated by endogenous glucocorticoids, cytokines, and neurotrophins. Thus, allostasis, the process of achieving stability in the face of adverse environmental events, results from interactions between specific environmental challenges and particular genetic backgrounds that combine to result in a response. Adverse childhood experiences (ACEs) are likely to alter the trajectory of development in a given individual and that during early development, the brain is especially vulnerable to injury. Future studies may uncover windows of plasticity in older children and adolescents that affect vulnerability as well. Acquiring subtle social skills, for example, relates to changes in the adolescent brain. Adolescents' keen abilities, competencies, and interests in a host of technologic advances—including the Internet, social media sites, and

smartphones, to name a few—shed some light on their potential to adapt to new and challenging demands.

PRENATAL, INFANT, AND CHILD

The phases of development described in this section are defined as follows: prenatal is the time frame from conception to 8 weeks; the fetus, from 8 weeks to birth; infancy, from birth to 15 months; the toddler period, from 15 months to 2½ years; the preschool period, from 2½ years to 6 years; and the middle years, from 6 to 12 years.

PRENATAL

Historically, the analysis of human development began with birth. The influence of endogenous and exogenous in utero factors, however, now requires that developmental schemes consider intrauterine events. The infant is not a tabula rasa, a smooth slate upon which outside influences etch patterns. On the contrary, myriad factors within the womb have already influenced the newborn. For example, the studies of Stella Chess and Alexander Thomas (described later) have demonstrated a wide range of temperamental differences among newborns. Maternal stress, through the production of adrenal hormones, also influences the behavioral characteristics of newborns.

The time frame in which the development of the embryo and fetus occurs is known as the prenatal period. After implantation, the egg begins to divide and is known as an embryo. Growth and development occur at a rapid pace; by the end of 8 weeks, the shape is recognizably human, and the embryo has become a fetus. Figure 32-1 illustrates a sonogram of a 9-week and 15-week fetus in utero.

The fetus maintains an internal equilibrium that, with variable effects, interacts continuously with the intrauterine environment. In general, most disorders that occur are multifactorial—the result of a combination of effects, some of which can be cumulative. Damage at the fetal stage usually has a more global impact than damage after birth, because rapidly growing organs are the most vulnerable. Boys are more vulnerable to developmental damage than girls are; geneticists recognize that in humans and animals, female

fetuses show a propensity for greater biologic vigor than male fetuses, possibly because of the second X chromosome in the female.

Prenatal Life

Much biologic activity occurs in utero. A fetus is involved in a variety of behaviors that are necessary for adaptation outside the womb. For example, a fetus sucks on thumb and fingers; folds and unfolds its body, and eventually assumes a position in which its occiput is in an anterior vertex position, which is the position in which fetuses usually exit the uterus.

Behavior. Pregnant women are extraordinarily sensitive to fetal movements. They describe their unborn babies as active or passive, as kicking vigorously or rolling around, as quiet when the mothers are active, but as kicking as soon as the mothers try to rest.

Women usually detect fetal movements 16 to 20 weeks into the pregnancy; the fetus can be artificially set into total body motion by in utero stimulation of its ventral skin surfaces by the 14th week. The fetus may be able to hear by the 18th week, and it responds to loud noises with muscle contractions, movements, and an increased heart rate. A bright light flashed on the abdominal wall of the 20-week pregnant woman causes changes in fetal heart rate and position. The retinal structures begin to function at that time. Eyelids open at 7 months. Smell and taste develop then, and the fetus can respond to substances injected into the amniotic sac, such as contrast medium. Some reflexes present at birth exist in utero: the grasp reflex, which appears at 17 weeks; the Moro (startle) reflex, which appears at 25 weeks; and the sucking reflex, which appears at about 28 weeks.

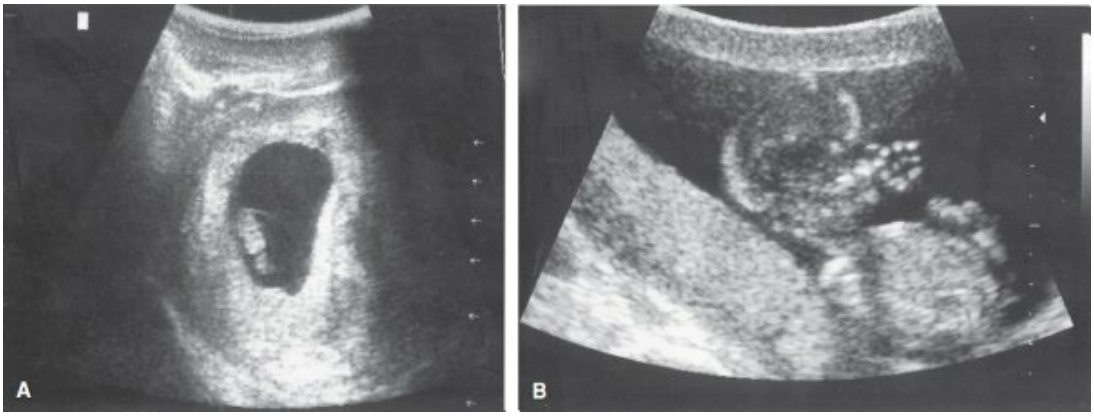


FIGURE 32-1
A: Sonogram of fetus at 9 weeks. **B:** Same fetus at 15 weeks. (Courtesy K.C. Attwell, M.D.)

Nervous System. The nervous system arises from the neural plate, which is a dorsal ectodermal thickening that appears on about day 16 of gestation. By the sixth week, part of the neural tube becomes the cerebral vesicle, which later becomes the cerebral hemispheres (Fig. 32-2).

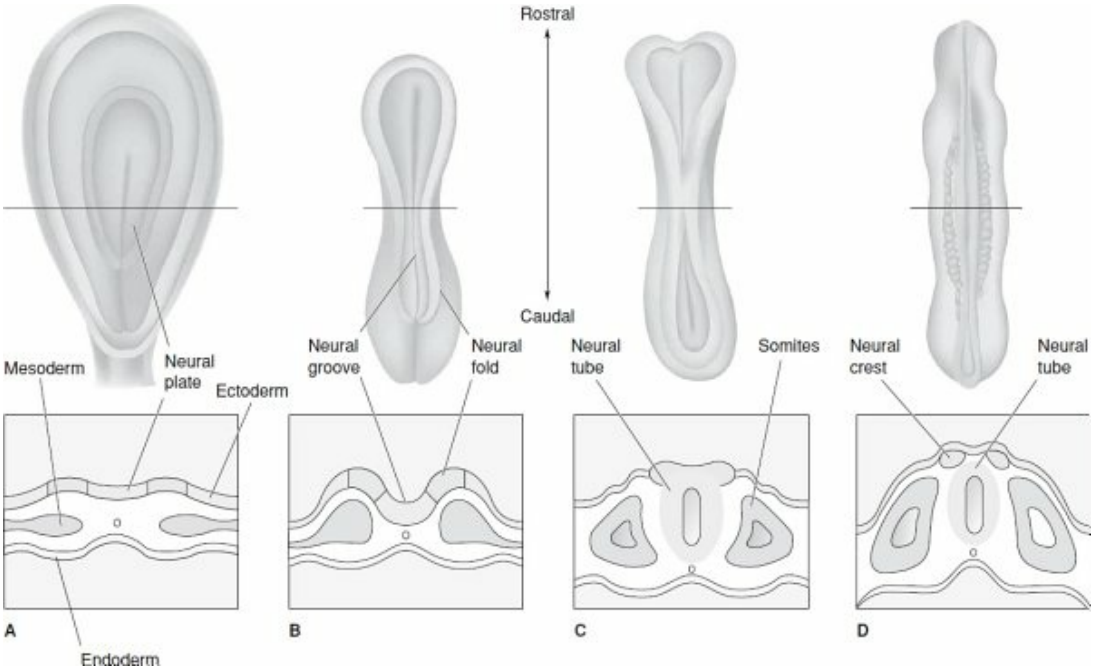


FIGURE 32-2
Formation of the neural tube and neural crest. These schematic illustrations follow the early development of the nervous system in the embryo. The drawings above are dorsal views of the embryo; those below are cross sections. **A:** The primitive embryonic central nervous system (CNS) begins as a thin sheet of ectoderm. **B:** The first important step in the development of the nervous system is the formation of the neural groove. **C:** The walls of the groove, called neural folds, come together and fuse, forming the neural tube. **D:** The bits of neural ectoderm that are pinched off when the tube rolls up are called the neural crest, from which the peripheral nervous system (PNS) will develop. The somites are mesoderm that will give rise to much of the skeletal system and the muscles. (Reprinted from Bear MF, Conners BW, Paradiso MA, eds. *Neuroscience: Exploring the Brain*. 2nd ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2001:179, with permission.)

The cerebral cortex begins to develop by the 10th week, but layers do not appear until the sixth month of pregnancy; the sensory cortex and the motor cortex form before the association cortex. Some brain function has been

detected in utero by fetal encephalographic responses to sound. The human brain weighs about 350 g at birth and 1,450 g at full adult development, a fourfold increase, mainly in the neocortex. This increase is almost entirely because of the growth in the number and branching of dendrites establishing new connections. After birth, the number of new neurons is negligible. Uterine contractions can contribute to fetal neural development by causing the developing neural network to receive and transmit sensory impulses.

Pruning. Pruning refers to the programmed elimination during the development of neurons, synapses, axons, and other brain structures from the original number, present at birth, to a lesser number. Thus, the developing brain contains structures and cellular elements that are absent in the older brain. The fetal brain generates more neurons than it will need for adult life. For example, in the visual cortex, neurons increase in number from birth to 3 years of age, at which point they diminish in number. The adult brain contains fewer neural connections than were present during the early and middle years of childhood. About twice as many synapses are present in certain parts of the cerebral cortex during early postnatal life than during adulthood.

Pruning occurs to rid the nervous system of cells that have served their function in the development of the brain. Some neurons, for example, exist to produce neurotrophic or growth factors and are programmed to die once they complete their task, a process called *apoptosis.*

These observations imply that the immature brain can be vulnerable in locations that lack sensitivity to injury later on. The developing white matter of the human brain before 32 weeks of gestation is especially sensitive to damage from hypoxic and ischemic injury and metabolic insults. Neurotransmitter receptors located on synaptic terminals are subject to injury from excessive stimulation by excitatory amino acids, (e.g., glutamate, aspartate), a process referred to as excitotoxicity. This process may be relevant to the etiology of disorders such as schizophrenia.

Maternal Stress

Maternal stress correlates with high levels of stress hormones (epinephrine, norepinephrine, and adrenocorticotropic hormone) in the fetal bloodstream,

which act directly on the fetal neuronal network to increase blood pressure, heart rate, and activity level. Mothers with high levels of anxiety are more likely to have babies who are hyperactive, irritable, and of low birth weight, and who have problems feeding and sleeping than are mothers with low anxiety levels. Fever in the mother causes the fetus's temperature to rise.

Genetic Disorders

In many cases, genetic counseling depends on prenatal diagnosis. The diagnostic techniques used include amniocentesis (transabdominal aspiration of fluid from the amniotic sac), ultrasound examinations, x-ray studies, fetoscopy (direct visualization of the fetus), fetal blood and skin sampling, chorionic villus sampling, and α-fetoprotein screening. In about 2 percent of women tested, the results are positive for some abnormality, including X-linked disorders, neural tube defects (detected by high levels of α-fetoprotein), chromosomal disorders (e.g., trisomy 21), and various inborn errors of metabolism (e.g., Tay–Sachs disease and the lipoidoses). Figure 32-3 illustrates hypertelorism of the eyes.

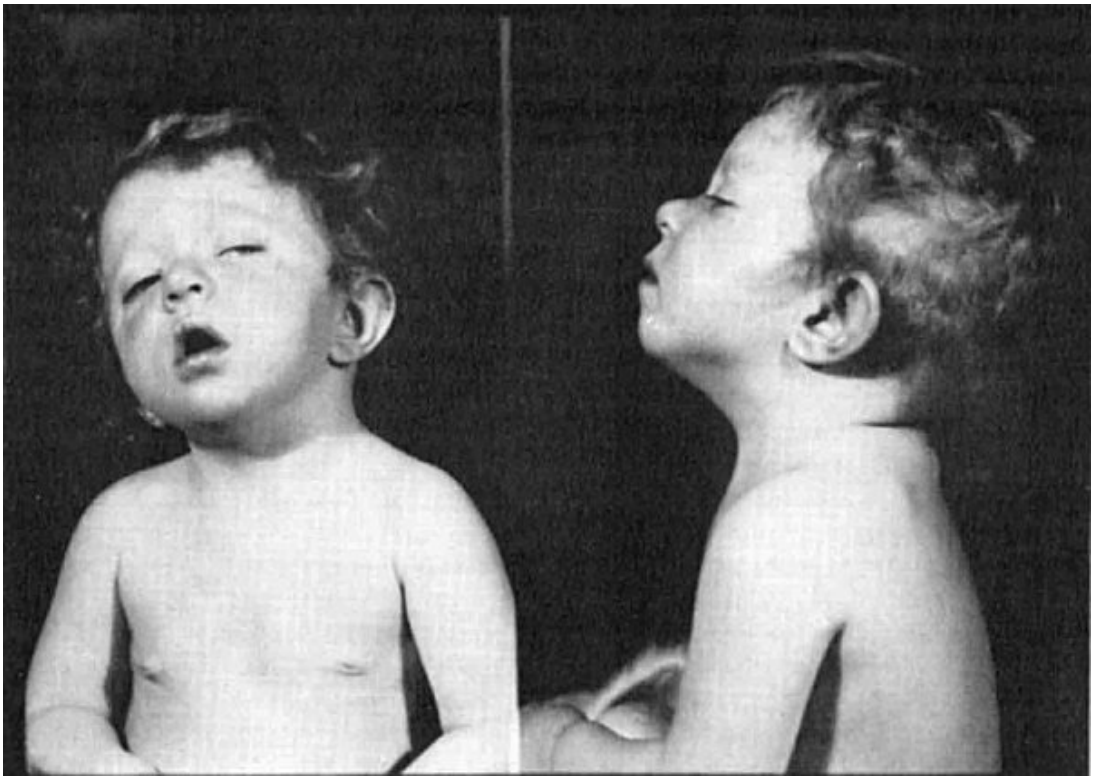


FIGURE 32-3
Hypertelorism. Note the wide distance between the eyes, flat nasal bridge, and

external strabismus. (Courtesy of Michael Malone, M.D. Children's Hospital, Washington, DC.)

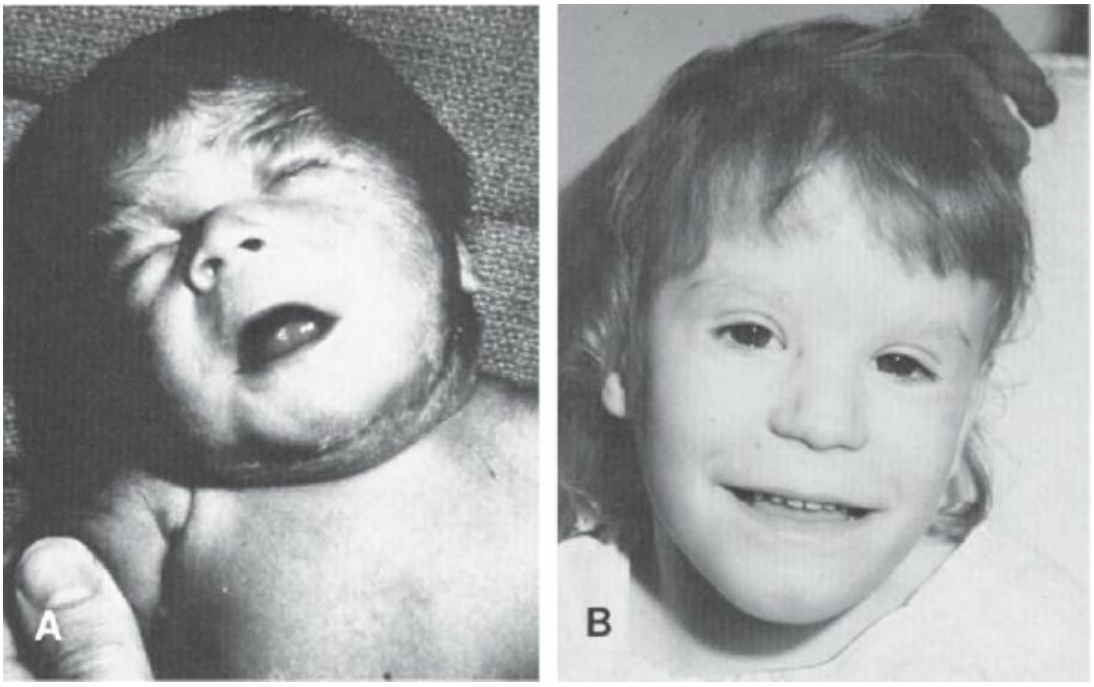


FIGURE 32-4
Photographs of children with "fetal-alcohol syndrome." **A:** Severe case. **B:** Slightly affected child. Note in both children the short palpebral fissures and hypoplasia of the maxilla. Usually, the defect includes other craniofacial abnormalities. Cardiovascular defects and limb deformities are also common symptoms of the fetal alcohol syndrome. (From Langman J. *Medical Embryology*. 7th ed. Philadelphia, PA: Williams & Wilkins; 1995:108, with permission.)

Some diagnostic tests carry a risk; for instance, about 5 percent of women who undergo fetoscopy miscarry. Amniocentesis, usually performed between the 14th and 16th weeks of pregnancy, causes fetal damage or miscarriage in less than 1 percent of women tested. Fully 98 percent of all prenatal tests in pregnant women reveal no abnormality in the fetus. Women older than 35 years of age and those with a family history of a congenital disability should have prenatal testing.

Parental reactions to congenital disabilities can include feelings of guilt, anxiety, or anger. The loss of the fantasized perfect child may cause depression. Some women may choose to terminate a pregnancy because of a known or suspected congenital malformation.

Maternal Drug Use

Alcohol. Alcohol use in pregnancy is a significant cause of severe physical and mental congenital disabilities in children. Each year, up to 40,000 babies are born with some degree of alcohol-related damage. The National Institute on Drug Abuse (NIDA) reports that 19 percent of pregnant women used alcohol during their pregnancy, the highest rate being among white women.

Fetal alcohol syndrome (Fig. 32-4) affects about one-third of all infants born to alcoholic women. Table 32-1 lists the characteristics of the disorder. The incidence of infants born with fetal alcohol syndrome is about 0.5 per 1,000 live births.

Some studies suggest that alcohol use during pregnancy may also contribute to attention-deficit/hyperactivity disorder (ADHD).

Smoking. Smoking during pregnancy may cause both premature births and below-average infant birth weight. Some reports have associated sudden infant death syndrome (SIDS) with mothers who smoke.



Table 32-1
Characteristics of Fetal Alcohol Syndrome

Growth retardation of prenatal origin (height, weight)
Facial Dysmorphism
Microcephaly (head circumference below the third percentile)
Hypertelorism (large distance between eyes)
Microphthalmia (small eyeballs)
Short palpebral fissures
Inner epicanthal folds
Midface hypoplasia (underdevelopment)
Smooth or short philtrum
Thin upper lip
Short, turned up nose
Cardiac Defects
Central nervous system (CNS) manifestations
Delayed development
Hyperactivity
Attention deficits
Learning disabilities

Intellectual deficits Seizures	

Other Substances. Chronic marijuana use is associated with low infant birth weight, prematurity, and withdrawal-like symptoms, including excessive crying, tremors, and hyperemesis (severe and chronic vomiting). Crack cocaine use by women during pregnancy may cause behavioral abnormalities such as increased irritability and crying and decreased desire for human contact. Infants born to mothers dependent on narcotics go through a withdrawal syndrome at birth.



Table 32-2
Causes of Human Malformations Observed During the First Year of Life

Suspected Cause	% of Total
Genetic	
Autosomal genetic disease	15–20
Cytogenic (chromosomal abnormalities)	5
Unknown	
Polygenic	
Multifactorial (genetic–environmental interactions)	
Spontaneous error of development	
Synergistic interactions of teratogens	
Environmental	
Maternal conditions: diabetes; endocrinopathies; nutritional deficiencies, starvation; drug and substance addictions	4
Maternal infections: rubella, toxoplasmosis, syphilis, herpes, cytomegalic inclusion disease, varicella,	3

Venezuelan equine encephalitis, parvovirus B19	
Mechanical problems (deformations): abnormal cord constrictions, disparity in uterine size and uterine contents	1–2
Chemicals, drugs, radiation, hyperthermia	<1
Preconception exposures (excluding mutagens and infectious agents)	<1

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Prenatal exposure to various prescribed medications can also result in abnormalities. Common drugs with teratogenic effects include antibiotics (tetracyclines), anticonvulsants (valproate, carbamazepine, phenytoin), progesterone-estrogens, lithium, and warfarin. Table 32-2 outlines the etiologies of malformations that may emerge during the first year of life.

INFANCY

The average newborn weighs about 3,400 g (7.5 lb). Small fetuses (birth weight below the 10th percentile for their gestational age) occur in about 7 percent of all pregnancies. From the 26th to the 28th week of gestation, the prematurely born fetus has a good chance of survival. Arnold Gesell described developmental landmarks that are widely used in both pediatrics and child psychiatry. These landmarks outline the sequence of children's motor, adaptive, and personal–social behavior from birth to 6 years (Table 32-3).

Premature infants are born before 34 weeks or weigh less than 2,500 g (5.5 lb). Such infants are at increased risk for learning disabilities, such as dyslexia, emotional and behavioral problems, mental retardation, and child abuse. With each 100 g increment of weight, beginning at about 1,000 g (2.2 lb), infants have a progressively better chance of survival. A 36-week-old fetus has less chance of survival than a 3,000 g (6.6 lb) fetus born close to term. Figure 32-5 gives examples of some differences between full-term and premature infants.



FIGURE 32-5
Premature baby at breast during a gavage feed. (Photo courtesy of Mike McCarter, Connecticut Children's Medical Center.)

Postmature infants are born 2 weeks or more beyond the expected date of birth. The incidence of postmaturity is high if based on menstrual history alone. The postmature baby typically has long nails, scanty lanugo, more scalp hair than usual, and increased alertness.

Developmental Milestones in Infants

Reflexes and Survival Systems at Birth. Primitive reflexes are present at birth. They include the rooting reflex (puckering of the lips in response to perioral stimulation), the grasp reflex, the plantar (Babinski) reflex, the knee reflex, the abdominal reflexes, the startle (Moro) reflex (Fig. 32-6), and the tonic neck reflex. In healthy children, the grasp reflex, the startle reflex, and the tonic neck reflex disappear by the fourth month. The Babinski reflex usually disappears by the 12th month.



FIGURE 32-6
The Moro reflex. The head is allowed to fall backward. The arms abduct and extend.

Survival systems—breathing, sucking, swallowing, and circulatory and temperature homeostasis—are relatively functional at birth, but the sensory organs are still developing. Further differentiation of neurophysiologic functions depends on an active process of stimulatory reinforcement from the external environment, such as persons touching and stroking the infant. The newborn infant is awake for only a short period each day; rapid eye movement (REM) and non-REM sleep are present at birth. Other spontaneous behaviors include crying, smiling, and penile erection in males. Infants 1 day old can detect the smell of their mother's milk, and those 3 days old distinguish their mother's voice.

Table 32-3
Landmarks of Normal Behavioral Development

Age	Motor and Sensory	Adaptive	Personal and

	Behavior	Behavior	Social Behavior
Birth to 4 wk	Hand-to-mouth reflex, grasping reflex Rooting reflex (puckering lips in response to perioral stimulation), Moro reflex (digital extension when startled), sucking reflex, Babinski reflex (toes spread when sole of foot is touched) Differentiates sounds (orients to human voice) and sweet and sour tastes Visual tracking Fixed focal distance of 8 in Makes alternating crawling movements Moves head laterally when placed in prone position	Anticipatory feeding-approach behavior at 4 days Responds to sound of rattle and bell Regards moving objects momentarily	Responsiveness to mother's face, eyes, and voice within first few hours of life Endogenous smile Independent play (until 2 yr) Quiets when picked up Impassive face
4 wk	Tonic neck reflex positions predominate Hands fisted Head sags but can hold head erect for a	Follows moving objects to the midline Shows no interest and drops object	Regards face and diminishes activity Responds to speech Smiles

	few seconds Visual fixation, stereoscopic vision (12 wk)	immediately	preferentially to mother
16 wk	Symmetrical postures predominate Holds head balanced Head lifted 90 degrees when prone on forearm Visual accommodation	Follows a slowly moving object well Arms activate on sight of dangling object	Spontaneous social smile (exogenous) Aware of strange situations
28 wk	Sits steadily, leaning forward on hands Bounces actively when placed in standing position	One-hand approach and grasp of toy Bangs and shakes rattle Transfers toys	Takes feet to mouth Pats mirror image Starts to imitate mother’s sounds and actions
40 wk	Sits alone with good coordination Creeps Pulls self to standing position Points with index finger	Matches two objects at midline Attempts to imitate scribble	Separation anxiety manifest when taken away from mother Responds to social play, such as pat-a-cake and peek-a-boo Feeds self cracker and holds own bottle
52 wk	Walks with one hand held Stands alone briefly	Seeks novelty	Cooperates in dressing
15 mo	Toddles		Points or vocalizes

	Creeps up stairs		wants Throws objects in play or refusal
18 mo	Coordinated walking, seldom falls Hurls ball Walks up stairs with one hand held	Builds a tower of three or four cubes Scribbles spontaneously and imitates a writing stroke	Feeds self in part, spills Pulls toy on string Carries or hugs a special toy, such as a doll Imitates some behavioral patterns with slight delay
2 yr	Runs well, no falling Kicks large ball Goes up and down stairs alone Fine motor skills increase	Builds a tower of six or seven cubes Aligns cubes, imitating train Imitates vertical and circular strokes Develops original behaviors	Pulls on simple garment Domestic mimicry Refers to self by name Says “no” to mother Separation anxiety begins to diminish Organized demonstrations of love and protest Parallel play (plays side by side but does not interact with other children)

3 yr	Rides tricycle Jumps from bottom steps Alternates feet going up stairs	Builds tower of 9 or 10 cubes Imitates a three-cube bridge Copies a circle and a cross	Puts on shoes Unbuttons buttons Feeds self well Understands taking turns
4 yr	Walks down stairs one step to a tread Stands on one foot for 5–8 s	Copies a cross Repeats four digits Counts three objects with correct pointing	Washes and dries own face Brushes teeth Associative or joint play (plays cooperatively with other children)
5 yr	Skips, using feet alternately Usually has complete sphincter control Fine coordination improves	Copies a square Draws a recognizable person with a head, a body, and limbs Counts 10 objects accurately	Dresses and undresses self Prints a few letters Plays competitive exercise games
6 yr	Rides two-wheel bicycle	Prints name Copies triangle	Ties shoelaces

Adapted from Arnold Gessell, M.D., Ph.D. and Stella Chess, M.D.



Table 32-4
Language Development

Age and Stage of Development	Mastery of Comprehension	Mastery of Expression
0–6 mo	Shows startle	Has vocalizations other

	response to loud or sudden sounds; Attempts to localize sounds, turning eyes or head; Appears to listen to speakers, may respond with smile; Recognizes warning, angry, and friendly voices; Responds to hearing own name	than crying; Has differential cries for hunger, pain; Makes vocalizations to show pleasure; Plays at making sounds Babbles (a repeated series of sounds)
7–11 mo Attending-to-Language	Shows listening selectivity (voluntary control over responses to sounds); Listens to music or singing with interest; Recognizes “no,” “hot,” own name; Looks at pictures being named for up to 1 min; Listens to speech without being distracted by other sounds	Responds to own name with vocalizations; Imitates the melody of utterances; Uses jargon (own language); Has gestures (shakes head for no); Has exclamation (“oh-oh”); Plays language games (pat-a-cake, peekaboo)
12–18 mo Single-Word	Shows gross discriminations between dissimilar sounds (bells vs. dog vs. horn vs.	Uses single words (mean age of first word is 11 mo; by age 18 mo, child is using up to 20 words); “Talks” to toys, self, or

	mother’s or father’s voice); Understands basic body parts, names of common objects; Acquires understanding of some new words each week; Can identify simple objects (baby, ball, etc.) from a group of objects or pictures; Understands up to 150 words by age 18 mo	others using long patterns of jargon and occasional words; Approximately 25% of utterances are intelligible; All vowels articulated correctly; Initial and final consonants often omitted
12–24 mo Two-Word Messages	Responds to simple directions (“Give me the ball”) Responds to action commands (“Come here,” “Sit down”) Understands pronouns (me, him, her, you) Begins to understand complex sentences (“When we go to the store, I’ll buy you some candy”)	Uses two-word utterances (“Mommy sock,” “all gone,” “ball here”); Imitates environmental sounds in play (“moo,” “mmm, mmm,” etc.); Refers to self by name, begins to use pronouns; Echoes two or more last words of sentences; Begins to use three-word telegraphic utterances (“all gone ball,” “me go now”); Utterances 26–50% intelligible; Uses language to ask for needs
24–36 mo	Understands small	Uses real sentences with

Grammar Formation	body parts (elbow, chin, eyebrow); Understands family name categories (grandma, baby) Understands size (little one, big one) Understands most adjectives Understands functions (why do we eat, why do we sleep)	grammatical function words (can, will, the, a); Usually announces intentions before acting “Conversations” with other children, usually just monologues Jargon and echolalia gradually drop from speech Increased vocabulary (up to 270 words at 2 yr, 895 words at 3 yr); Speech 50–80% intelligible P, b, m articulated correctly; Speech may show rhythmic disturbances
36–54 mo Grammar Development	Understands prepositions (under, behind, between) Understands many words (up to 3,500 at 3 yr, 5,500 at 4 yr) Understands cause and effect (What do you do when you’re hungry? cold?) Understands analogies (Food is to eat, milk is to ____)	Correct articulation of *n, w, ng, h, t, d, k, g*; Uses language to relate incidents from the past; Uses wide range of grammatical forms: plurals, past tense, negatives, questions; Plays with language: rhymes, exaggerates; Speech 90% intelligible, occasional errors in the ordering of sounds within words; Able to define words; Egocentric use of language rare; Can repeat a 12-syllable sentence correctly;

		Some grammatical errors still occur
55 mo on True Communication	Understands concepts of number, speed, time, space; Understands left and right; Understands abstract terms; Is able to categorize items into semantic classes	Uses language to tell stories, share ideas, and discuss alternatives; Increasing use of varied grammar; spontaneous self-correction of grammatical errors; Stabilizing of articulation f, v, s, z, l, r, th, and consonant clusters; Speech 100% intelligible

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Language and Cognitive Development. At birth, infants can make noises, such as crying, but they do not vocalize until about 8 weeks. At that time, guttural or babbling sounds occur spontaneously, especially in response to the mother. The persistence and further evolution of children's vocalizations depend on parental reinforcement. Language development occurs in well-delineated stages, as outlined in Table 32-4.

By the end of infancy (about 2 years), infants have transformed reflexes into voluntary actions that are the building blocks of cognition. They begin to interact with the environment, to experience feedback from their bodies, and to become intentional in their actions. By the end of the second year of life, children begin to use symbolic play and language.



Table 32-5
Piaget's Stages of Cognitive Development

Period of Development	Cognitive Spatial Stages and Associated Concepts	Cognitive Achievements
Gestational		Fetus can "learn"

		sounds and respond differentially to them after birth
Infancy: Birth–2 yr	**Sensorimotor**	Infants “think” with their eyes, ears, and senses
Birth–1 mo	Reflective; egocentric (newer research refutes this)	Newborns can learn to associate stroking with sucking
4–8 mo	Secondary circular: looks for objects partially hidden	Newborns can learn to suck to produce certain visual displays or music
8–12 mo	Secondary circulation coordinated: peek-a-boo, finds hidden objects	Can remember for 1-mo periods Can play with parent by looking for partially hidden objects
12–18 mo	Tertiary circular: explores properties and drops objects	Memory improves
18 mo–2 yr	Mental representation, make-believe play; memory of objects	Body parts used as objects Can stack one object within another Remembers hidden objects Drops objects over crib Knows animal sounds;

		names objects Knows body parts and familiar pictures Can understand causes not visible
Early Childhood: 2–7 yr	**Preoperational**	
2–7 yr	Egocentrism: “I want you to eat this too.” Animistic: “I’m afraid of the moon.” Lack of hierarchy: “Where do these blocks go?” Centration: “I want it now, not after dinner.” Irreversibility: “I don’t know how to go back to that room.”	Preschoolers use symbols Development of language and make-believe No sign of logic 3-yr-olds can count 2–3 objects; know colors and age 4-yr-olds can fantasize without concrete props
2–5 yr	Transductive reasoning: “We have to go this way because that’s the way Daddy goes.”	5–6-yr-olds get humor; understand good and bad; can do some chores 7–11-yr-olds have good memory; recall; can solve problems
Middle Childhood: 7–11 yr	**Concrete operational**	
	Hierarchical classification—arranges cars by types Reversibility—can play	Children begin to think logically Understand

	games backward and forward (e.g., checkers, triple kings) Conservation—lose two dimes and look for same Decentration—worry about small details, obsessive Spatial operations—likes models for directions Horizontal decalage—conservation of weight, logic Transitive inference—syllogisms; compare everything, brand names important	conservation of matter Frozen milk same amount as melted Can organize objects into hierarchies Children seem rational and organized
Adolescence: 11–19 yr	**Formal operational**	
11 yr onward	Hypothetical-deductive reasoning; adolescent quick thinking or excuses Imaginary audience—everyone is looking at them Personal fable—inflated opinion of themselves Propositional thinking—logic	Abstraction and reason Can think of all possibilities

Jean Piaget (1896–1980), a Swiss psychologist, observed the growing capacity of young children (including his own) to think and to reason. Table

32-5 outlines Piaget's stages of cognitive development.

Emotional and Social Development. By the age of 3 weeks, infants imitate the facial movements of adult caregivers. They open their mouths and thrust out their tongues in response to adults who do the same. By the third and fourth months of life, we can easily elicit these behaviors. These imitative behaviors are likely the precursors of the infant's emotional life. The smiling response occurs in two phases: the first phase is endogenous smiling, which occurs spontaneously within the first 2 months and is unrelated to external stimulation; the second phase is exogenous smiling, usually stimulated by the mother, and occurs by the 16th week.

The stages of emotional development parallel those of cognitive development. The caregiving person provides the primary stimulus for both aspects of mental growth. Human infants depend totally on adults for survival. Through warm and predictable interactions, an infant's social and emotional repertoire expands with the interplay of caregivers' social responses (Table 32-6).



Table 32-6
Emotional Development

Stages First Seen	Emotional Skills	Emotional Behavior
Gestational–Infancy: 0–2 yr		
0–2 mo onward	Love, evoked by touching	Social smile and joy shown
	Fear, evoked by loud noise	Responds to emotions of others
	Rage, evoked by body restrictions	All emotions there
	Brain pathways for emotion forming	

3–4 mo onward	Self-regulation of emotions starts; brain pathways of emotion growing	Laughter possible and more control over smiles; anger shown
7–12 mo	Self-regulation of emotion grows	Able to elicit more responsiveness
	Increased intensity of basic three	Denies to cope with stress
1–2 yr	Shame and pride appear; envy, embarrassment appear Displaces onto other children	Some indications of empathy starting; expressions of feeling: “I like you, Daddy” “I’m sorry” Likes attention and approval; enjoys play alone or next to peers
Early Childhood: 2–5 yr		
3–6 yr	Can understand causes of many emotions Can begin to find ways for regulating emotions and for expressing them Identifies with adult to cope	Empathy increases with understanding More response and less reaction; self-regulation: “Use your words to say that you are angry with him” Aggression becomes competition By age 5, shows sensitivity to criticism and cares about feelings of others

Middle Childhood: 5–11 yr		Ego rules until age 6
7–11 yr	Can react to the feelings of others More aware of other's feelings	Empathy becomes altruism: "I feel so bad about their fire, I'm going to give them some of my things" Superego dominates

In the first year, infants' moods are highly variable and intimately related to internal states such as hunger. Toward the second two-thirds of the first year, infants' moods grow increasingly related to external social cues; a parent can get even a hungry infant to smile. When the infant is internally comfortable, a sense of interest and pleasure in the world and its primary caregivers should prevail. Prolonged separation from the mother (or other primary caregivers) during the second 6 months of life can lead to depression that may persist into adulthood as part of an individual's character.

Temperamental Differences

Infants vary in their autonomic reactivity and temperament. Chess and Thomas identified nine behavioral dimensions in which reliable differences among infants can be observed (Table 32-7).

In studies of temperament, most temperamental dimensions of individual children remain stable through adulthood, but some do not persist. This is likely due to genetic and environmental effects on personality. A complex interplay exists among the initial characteristics of infants, the mode of parental interactions, and children's subsequent behavior. Observations of the stability and plasticity of certain temperamental traits support the importance of interactions between genetic endowment (nature) and environmental experience (nurture) in behavior.

Attachment

Bonding is the term used to describe the intense emotional and psychological

relationship a mother develops for her baby. Attachment is the relationship the baby develops with its caregivers. Infants in the first months after birth become attuned to social and interpersonal interaction. They show a rapidly increasing responsivity to the external environment and an ability to form a special relationship with significant primary caregivers—that is, to form an attachment. Table 32-8 lists the commonly observed attachment styles.

Harry Harlow. Harry Harlow studied social learning and the effects of social isolation on monkeys. Harlow placed newborn rhesus monkeys with two types of surrogate mothers—one a wire-mesh surrogate with a feeding bottle and the other a wire-mesh surrogate covered with terry cloth. The monkeys preferred the terry-cloth surrogates, which provided contact and comfort, to the feeding surrogate. (When hungry, the infant monkeys would go to the feeding bottle but then would quickly return to the terry-cloth surrogate.) When frightened, monkeys raised with terry-cloth surrogates showed intense clinging behavior and appeared to be comforted, whereas those raised with wire-mesh surrogates gained no comfort and appeared to be disorganized. The results of Harlow's experiments suggest that infant attachment is not merely the result of feeding.



Table 32-7
Temperament—Newborn to 6 Years

Dimension	Description
Activity level	Percent of time spent in activities
Distractibility	Degree to which stimuli are allowed to alter behavior
Adaptability	Ease moving into change
Attention span	Amount of time spent on attending
Intensity	Energy level
Threshold of responsiveness	Intensity required for response

Quality of mood	Amount positive compared to amount negative behavior
Rhythmicity	Regulation of functions
Approach/withdrawal	Response to new situations



Table 32-8
Types of Attachment

Secure Attachment	Children show fewer adjustment problems; however, these children have typically received more consistent and developmentally appropriate parenting for most of their life. The parents of securely attached children are likely better able to maintain these aspects of parenting through a divorce. Given that the family factors that lead to divorce also impact the children, there could be fewer securely attached children in divorcing families
Insecure/Avoidant Attachment	Children become anxious, clinging, and angry with the parent. These children typically come from families with adults who were also insecurely attached to their families and, thus, were unable to provide the kind of consistency, emotional responsiveness, and care that securely attached parents could offer. Such parents have a more difficult time with divorce, and are more likely to become rejecting.
Insecure/Ambivalent Attachment	Children generally are raised with disorganized, neglecting, and inattentive parenting. The parents are even less able to provide stability and psychological

strength for them after a divorce and, as a result, the children are even more likely to become clinging but inconsolable in their distress, as well as to act out, suffer mood swings, and become oversensitive to stress.

Both types of surrogate-reared monkeys were subsequently unable to adjust to life in a monkey colony and had extraordinary difficulty learning to mate. When impregnated, the female monkeys failed to mother their young.

John Bowlby. John Bowlby studied the attachment of infants to mothers and concluded that early separation of infants from their mothers had severe adverse effects on children's emotional and intellectual development. He described attachment behavior, which develops during the first year of life, as the maintenance of physical contact between the mother and child when the child is hungry, frightened, or in distress.

Mary Ainsworth. Mary Ainsworth expanded on Bowlby's observations and found that the interaction between mother and baby during the attachment period influences the baby's current and future behavior significantly. Many observers believe that patterns of infant attachment affect future adult emotional relationships. Patterns of attachment vary among babies; for example, some babies signal or cry less than others. Sensitive responsiveness to infant signals, such as cuddling the baby when it cries, causes infants to cry less in later months. Close bodily contact with the mother when the baby signals for her fosters self-reliance, rather than clinging dependence. Unresponsive mothers produce anxious babies.

Ainsworth also confirmed that attachment serves to reduce anxiety. What she called the secured base effect enables a child to move away from the attachment figure and explore the environment. Inanimate objects, such as a teddy bear or a blanket (called the transitional object by Donald Winnicott), also serve as a secure base, one that often accompanies children as they investigate the world. A growing body of literature derived from direct observation of mother–infant interactions and longitudinal studies has expanded on and refined Ainsworth's original descriptions. Maternal

sensitivity and responsiveness are the main determinants of secure attachment. However, when the attachment is insecure, the type of insecurity (avoidant, anxious, or ambivalent) is determined by infant temperament. Overall, male infants are less likely to have secure attachments and are more vulnerable to changes in maternal sensitivity than are female infants.

The birth of a second child decreases the attachment of the firstborn child. This decrease is most notable when the firstborn is 2 to 5 years of age as opposed to younger. Not surprisingly, the extent of the decrease also depends on the mother's sense of security, confidence, and mental health.

Social Deprivation Syndromes and Maternal Neglect. Investigators, especially René Spitz, have long documented the severe developmental retardation that accompanies maternal rejection and neglect. Infants in institutions characterized by low staff-to-infant ratios and frequent turnover of personnel tend to display marked developmental retardation, even with adequate physical care and freedom from infection. The same infants, placed in adequate foster or adoptive care, exhibit marked acceleration in development.

Fathers and Attachment. Babies become attached to fathers as well as to mothers, but the attachment is different. Generally, mothers hold babies for caregiving, and fathers hold babies for purposes of play. Given a choice of either parent after separation, infants usually go to the mother, but if the mother is unavailable, they turn to the father for comfort. Babies raised in extended families or with multiple caregivers can establish many attachments.

Stranger Anxiety. A developmentally expected fear of strangers first appears in infants at about 26 weeks of age and is more fully developed by 32 weeks. At the approach of a stranger, infants cry and cling to their mothers. Babies exposed to only one caregiver are more likely to have stranger anxiety than babies exposed to a variety of caregivers. Stranger anxiety likely results from a baby's growing ability to distinguish caregivers from all other persons.

Separation anxiety, which occurs between 10 and 18 months of age, is related to stranger anxiety but is not identical to it. Separation from the

person to whom the infant is attached precipitates separation anxiety. Stranger anxiety, however, occurs even when the infant is in the mother's arms. The infant learns to separate as it starts to crawl and move away from the mother, but the infant continually looks back and frequently returns to the mother for reassurance.

Margaret Mahler (1897–1985) proposed a theory to describe how young children acquire a sense of identity separate from that of their mothers'. Her observations of children and their mothers lead to her theory of separation-individuation. Table 32-9 lists Mahler's stages of separation-individuation.



Table 32-9
Stages of Separation-Individuation Proposed by Mahler

1. Normal autism (birth–2 mo)
   Periods of sleep outweigh periods of arousal in a state reminiscent of intrauterine life.
2. Symbiosis (2–5 mo)
   Developing perceptual abilities gradually enable infants to distinguish the inner from the outer world; mother–infant is perceived as a single fused entity.
3. Differentiation (5–10 mo)
   Progressive neurologic development and increased alertness draw infants' attention away from self to the outer world. Physical and psychological distinctiveness from the mother is gradually appreciated.
4. Practicing (10–18 mo)
   The ability to move autonomously increases children's exploration of the outer world.
5. Rapprochement (18–24 mo)
   As children slowly realize their helplessness and dependence, the need for independence alternates with the need for closeness. Children move away from their mothers and come back for reassurance.
6. Object constancy (2–5 yr)
   Children gradually comprehend and are reassured by the permanence of mother and other important people, even

when not in their presence.

Infant Care

Clinicians are now beginning to view infants as important actors in the family drama, ones who partly determine its course. Infants' behavior controls mothers' behavior, just as mothers' behavior modulates infants' behavior. A calm, smiling, predictable infant is a powerful reward for tender maternal care. A jittery, irregular, irritable infant tries a mother's patience. When a mother's capacity for giving is marginal, such infant traits may cause her to turn away from her child and thus complicate the child's already-troubled beginnings.

Parental Fit

Parental fit describes how well the mother or father relates to the newborn or developing infant; the idea takes into account the temperamental characteristics of both parent and child. Each newborn has innate psychophysiological characteristics, which are known collectively as temperament. Chess and Thomas identified a range of standard temperamental patterns, from the difficult child at one end of the spectrum to the easy child at the other end.

Difficult children, who make up 10 percent of all children, have a hyperalert physiologic makeup. They overreact to stimuli (cry easily at loud noises), sleep poorly, eat at unpredictable times, and are difficult to comfort. Easy children, who make up 40 percent of all children, are regular in eating, eliminating, and sleeping; they are flexible, can adapt to change and new stimuli with a minimum of distress, and are easily comforted when they cry. The other 50 percent of children are mixtures of these two types. The difficult child is harder to raise and places higher demands on the parent than the easy child. Chess and Thomas used the term *goodness of fit* to characterize the harmonious and consonant interaction between a mother and a child in their motivations, capacities, and styles of behavior. Poor fit is likely to lead to distorted development and maladaptive functioning. A difficult child must be recognized because parents of such infants often have feelings of inadequacy and believe that they are doing something wrong to account for the child's difficulty in sleeping and eating and their problems comforting the child.

Also, most difficult children have emotional disturbances later in life.

Good-Enough Mothering. Winnicott believed that infants begin life in a state of nonintegration, with unconnected and diffuse experiences, and that mothers provide the relationship that enables infants' incipient selves to emerge. Mothers supply a holding environment in which infants are contained and experienced. During the last trimester of pregnancy and for the first few months of a baby's life, the mother is in a state of primary maternal preoccupation, absorbed in fantasies about, and experiences with, her baby. The mother need not be perfect, but she must provide good-enough mothering. She plays a vital role in bringing the world to the child and offering empathic anticipation of the infant's needs. If the mother can resonate with the infant's needs, the baby can become attuned to its bodily functions and drives that are the basis for the gradually evolving sense of self.

TODDLER PERIOD

Accelerated motor and intellectual development occur in the second year. The ability to walk gives toddlers some control over their actions; this mobility enables children to determine when to approach and when to withdraw. The acquisition of speech profoundly extends their horizons. Typically, children learn to say "no" before they learn to say "yes." Toddlers' negativism is vital to the development of independence, but if it persists, oppositional behavior connotes a problem.

Learning language is a crucial task in the toddler period. Vocalizations become distinct, and toddlers can name a few objects and make needs known in one or two words. Near the end of the second year and into the third year, toddlers sometimes use short sentences. The pace of language development varies considerably from child to child, and although a small number of children are genuinely late developers, most child experts recommend a hearing test if the child is not making two-word sentences by age 2.

Developmental Milestones in Toddlers

Language and Cognitive Development. Toddlers begin to listen to explanations that can help them tolerate delay. They create new behaviors

from old ones (originality) and engage in symbolic activities, for instance, using words and playing with dolls when the dolls represent something, such as a feeding sequence. Toddlers have varied capacities for concentration and self-regulation.

Emotional and Social Development. In the second year, pleasure and displeasure become further differentiated. Social referencing is often apparent at this age; the child looks to parents and others for emotional cues about how to respond to novel events. Toddlers show exploratory excitement, assertive pleasure, and pleasure in discovery and in developing new behavior (e.g., new games), including teasing and surprising or fooling the parent (e.g., hiding). The toddler has capacities for an organized demonstration of love, as when the toddler runs up and hugs, smiles, and kisses the parent at the same time, and of protest when the toddler turns away, cries, bangs, bites, hits, yells, and kicks. Comfort with family and apprehension with strangers may increase. Anxiety appears to be related to disapproval and the loss of a loved caregiver and can be disorganizing.

Sexual Development. Sexual differentiation is evident from birth when parents start dressing and treating infants differently because of the expectations evoked by sex typing. Through imitation, reward, and coercion, children assume the behaviors that their cultures define as appropriate for their sexual roles. Children exhibit curiosity about anatomical sex. When their curiosity is recognized as healthy and elicits age-appropriate replies, children acquire a sense of the wonder of life and are comfortable with their roles. If the subject of sex is taboo and parents rebuff children’s questions, shame and discomfort may result.

Gender identity, the conviction of being male or female, begins to manifest at 18 months of age and is often fixed by 24 to 30 months. It was once widely believed that gender identity was primarily a function of social learning. John Money reported on children with ambiguous or damaged external genitalia who were raised as the sex opposite to their chromosomal sex. Long-term follow-up of those individuals suggests that the major part of gender identity is innate and that rearing may not affect the genetic diathesis.

Gender role describes the behavior that society deems appropriate for one sex or another, and it is not surprising that significant cultural differences

exist. There may be different expectations for boys and girls in what and with whom they play, their tone of voice, the expression of emotions, and how they dress. Nevertheless, some generalizations are possible. Boys are more likely than girls to engage in rough and tumble play. Mothers talk more to girls than to boys, and by the time the child is 2 years of age, fathers generally pay more attention to boys. Many educated, middle-class parents determined to raise nonsexist children are startled to see their children's determined preference for sex-stereotyped toys: girls want to play with dolls and boys with guns.

Toilet Training. The second year of life is a period of increasing social demands on children. Toilet training serves as a paradigm of the family's general training practices; that is, the parent who is overly severe in the area of toilet training is likely to be punitive and restrictive in other areas also. Control of daytime urination is usually complete by the age of 2½, and control of nighttime urination is usually complete by the age of 4 years. Since 1900, the pendulum has swung between extremes of permissiveness and control in toilet training. The trend in the United States has been toward delayed training, but in the last few years, this trend appears to be shifting back to early training.

Toddlers may have sleep difficulties related to fear of the dark, which may be helped with a nightlight. Most toddlers generally sleep about 12 hours a day, including a 2-hour nap. Parents must be aware that children of this age may need reassurance before going to bed and that the average 2-year-old takes about 30 minutes to fall asleep.

Parenting Challenges. In infancy, the primary responsibility for parents is to meet the infant's needs sensitively and consistently. The parental task in the toddler stage requires firmness about the boundaries of acceptable behavior and encouragement of the child's progressive emancipation. Parents must be careful not to be too authoritarian at this stage; children must be allowed to operate for themselves and to learn from their mistakes and must be protected and assisted when challenges are beyond their abilities.

During the toddler period, children are likely to struggle for the exclusive affection and attention of their parents. They may compete with siblings or one or another parent for the starring role in the family. Although children are

beginning to be able to share, they do so reluctantly. When the demands for exclusive possession are not resolved effectively, the result is likely to be jealous competitiveness in relationships with peers and lovers. The fantasies aroused by the struggle lead to fear of retaliation and displacement of fear onto external objects. In an equitable, loving family, a child elaborates on a moral system of ethical rights. Parents need to balance between punishment and permissiveness and set realistic limits on a toddler's behavior.

PRESCHOOL PERIOD

Marked physical and emotional growth occurs during the preschool period. Generally, between 2 and 3 years of age, children reach half their adult height. The 20 baby teeth are in place at the beginning of the stage, and by the end, they begin to fall out. Children are ready to enter school by the time the stage ends at age 5 or 6. They have mastered the tasks of primary socialization—to control their bowels and urine, to dress and feed themselves, and to control their tears and temper outbursts, at least most of the time.

The term preschool for the age group of 2½ to 6 years may be a misnomer; many children are already in school-like settings, such as preschool nurseries and daycare centers, where working mothers must often place their children. Preschool education can be valuable, but stressing academic advancement too far beyond a child's capabilities can be counterproductive.

Developmental Milestones in Preschoolers

Language and Cognitive Development. In the preschool period, children's use of language expands, and they use sentences. Individual words have regular and consistent meanings at the beginning of the period, and children begin to think symbolically. In general, however, their thinking is egocentric; they cannot place themselves in the position of another child and are incapable of empathy. Children think intuitively and prelogically and do not understand causal relations.

Emotional and Social Behavior. At the start of the preschool period, children can express such complex emotions as love, unhappiness, jealousy,

and envy, both preverbally and verbally. Their emotions are still easily influenced by somatic events, such as tiredness and hunger. Although they still think mostly egocentrically, children's capacity for cooperation and sharing is emerging. Various losses, such as of a loved person or approval, may cause anxiety. Although still potentially disorganizing, preschool children can tolerate anxiety better than when they were younger. Four-year-olds are learning to share and to have concern for others and may express feelings of tenderness.

By the end of the preschool period, children have many relatively stable emotions. The child balances expansiveness, curiosity, pride, and gleeful excitement related to the self and the family with coyness, shyness, fearfulness, jealousy, and envy. Shame and humiliation are evident. Capacities for empathy and love are developed but are fragile and quickly lost if competitive or jealous strivings intervene. Anxiety and fears are related to bodily injury and loss of respect, love, and emerging self-esteem. Guilt feelings are possible.

Children between the ages of 3 and 6 years are aware of their bodies and differences between the sexes. Their physical awareness extends beyond genitalia. Children may be preoccupied with illness or injury, and some call this the "the Band-Aid phase." Every injury must be examined and taken care of by a parent.

Children develop a division between what they want and what others tell them to do. The division increases until a gap grows between their set of expanded desires, their exuberance at unlimited growth, and their parents' restrictions; they gradually turn parental values into self-obedience, self-guidance, and self-punishment.

At the end of the preschool stage, the child's conscience is evolving. The development of a conscience sets the tone for the moral sense of "right and wrong." Until about 7 years of age, children typically experience rules as "absolute" and as existing for their own sake. They do not understand that more than one point of view on a moral issue may exist; a violation of the rules calls for absolute retribution—that is, children have the notion of immanent justice.

SIBLING RIVALRY. In the preschool period, children relate to others in new ways. The birth of a sibling (a common occurrence during this time) tests a

preschool child's capacity for further cooperation and sharing but may also evoke sibling rivalry, which is most likely to occur at this time. Sibling rivalry depends on child-rearing practice. Favoritism, for any reason, commonly aggravates such rivalry. Children who get special treatment because they are gifted, are defective in some way, or have a preferred gender are likely to receive angry feelings from their siblings. Experiences with siblings can influence growing children's relationships with peers and authority; for example, a problem may result if the needs of a new baby prevent the mother from attending to a firstborn child's needs. If not handled properly, the displacement of the firstborn can be a traumatic event.

PLAY. In the preschool years, children begin to distinguish reality from fantasy, and play reflects this growing awareness. Pretend games are popular and playfully help test real-life situations. Dramatic play in which children act out a role is typical. One-to-one play relationships advance to complicated patterns with rivalries, secrets, and two-against-one intrigues. Children's play behavior reflects their level of social development.

Between 2½ and 3 years, children commonly engage in parallel play, solitary play alongside another child with no interaction between them. By age 3, play is often associative, that is, playing with the same toys in pairs or small groups, but still with no real interaction among them. By age 4, children are usually able to share and engage in cooperative play. Real interactions and taking turns become possible.

Between 3 and 6 years of age, drawings may trace growth. A child's first drawing of a human being is a circular line with marks for the mouth, nose, and eyes; they add ears and hair later; arms and stick-like fingers appear next, and then legs appear. Last to appear is a torso in proportion to the rest of the body. Intelligent children can deal with details in their art. Drawings express creativity throughout a child's development: They are representational and formal in early childhood, make use of perspective in middle childhood, and become abstract and affect-laden in adolescence. Drawings also reflect children's body image concepts and sexual and aggressive impulses.

IMAGINARY COMPANIONS. Imaginary companions most often appear during preschool years, usually in children with above-average intelligence and usually in the form of persons. Imaginary companions may also be things,

such as anthropomorphized toys. Some studies suggest that up to 50 percent of children between the ages of 3 and 10 years have imaginary companions at one time or another. Their significance is not clear, but these figures are usually friendly, relieve loneliness, and reduce anxiety. In most instances, imaginary companions disappear by age 12, but they can occasionally persist into adulthood.

MIDDLE YEARS

The period between age 6 and puberty is the middle years. During this time, children enter elementary school. The formal demands for academic learning and accomplishment become significant determinants of further personality development.

Developmental Milestones in School-Age Children

Language and Cognitive Development. In the middle years, language expresses complex ideas with relations among several elements. Logical exploration tends to dominate fantasy, and children show an increased interest in rules and orderliness and an increased capacity for self-regulation. During this period, children's conceptual skills develop, and thinking becomes organized and logical. The ability to concentrate is well established by age 9 or 10, and by the end of the period, children begin to think in abstract terms. Improved gross motor coordination and muscle strength enable children to write fluently and draw artistically. They are also capable of complex motor tasks and activities, such as tennis, gymnastics, golf, baseball, and skateboarding.

Recent evidence has shown that changes in thinking and reasoning during the middle years result from maturational changes in the brain. Children are now capable of increased independence, learning, and socialization. Theorists consider moral development a gradual, stepwise process spanning childhood, adolescence, and young adulthood.

In the middle years, both girls and boys make new identifications with other adults, such as teachers and counselors. These identifications may so influence girls that their goals of wanting to marry and have babies, as their mothers did, may be combined with a desire for a career or may be postponed

or abandoned entirely.

Girls who cannot identify with their mothers or whose fathers are overly attached may become fixated at about a 6-year-old level; as a result, they may fear men or women or both or become seductively close to them. A similar situation can occur in boys who have been unable to identify successfully with fathers who were aloof, brutal, or absent. As a result, boys may enter this period with a variety of problems. They may be fearful of men, unsure of their sense of masculinity, or unwilling to leave their mothers (sometimes manifested by a school phobia); they may lack initiative and be unable to master school tasks, thus incurring academic problems.

The school-age period is a time when peer interaction assumes significant importance. Interest in relationships outside the family takes precedence over those within the family. Nevertheless, a special relationship exists with the same-sex parent, with whom children identify and who is now an ideal and a role model.

Empathy and concern for others begin to emerge early in the middle years; by the time children are 9 or 10, they have well-developed capacities for love, compassion, and sharing. They have a capacity for long-term, stable relationships with family, peers, and friends, including best friends. Emotions about sexual differences begin to emerge as either excitement or shyness with the opposite sex. School-age children prefer to interact with children of the same sex. Although the middle years have sometimes been referred to as a latency period—a moratorium on psychosexual exploration and play until the eruption of sexual impulses with puberty—it is now recognized that a considerable amount of sexual interest continues through these years. Sex play and curiosity are typical, especially among boys, but also among girls. Boys compare genitals and sometimes engage in group or mutual masturbation. Interest in anal humor and toilet jokes is also typical. Children this age often start using sexual and excretory words as expletives.

BEST FRIEND. Harry Stack Sullivan postulated that a buddy, or best friend, is an influential phenomenon during the school years. By about 10 years of age, children develop a close same-sex relationship, which Sullivan believed is necessary for further healthy psychological growth. Moreover, Sullivan believed that the absence of a close friend during the middle years of childhood is an early harbinger of schizophrenia.

SCHOOL REFUSAL. Some children refuse to go to school at this time, generally because of separation anxiety. A fearful mother may transmit her fear of separation to a child, or a child who has not resolved dependence needs panics at the idea of separation. School refusal is usually not an isolated problem; children with the problem typically avoid many other social situations.

Sex Role Development

Persons' sex roles are similar to their gender identity; persons see themselves as male or female. The sex-role also involves identification with culturally acceptable masculine or feminine ways of behaving, but changing expectations in society (particularly in the United States) of what constitutes masculine and feminine behavior can create ambiguity.

Parents react differently to their male and female children. Independence, physical play, and aggressiveness are encouraged in boys; dependence, verbalization, and physical intimacy are encouraged in girls. Nowadays, however, boys are encouraged to verbalize their feelings and to pursue interests traditionally associated with girls, whereas girls are encouraged to pursue careers traditionally dominated by men and to participate in competitive sports. As society grows more tolerant in its expectations of the sexes, roles become less rigid, and opportunities for boys and girls enlarge and broaden.

Biologically, boys are more physically aggressive than girls; and parental expectations, particularly the expectations of fathers, reinforce this trait. Differences also exist between boys and girls in the influence of persons outside the family. Girls tend to respond to the expectations and opinions of girls and teachers of either sex but to ignore boys. Boys, on the other hand, tend to respond to other boys but to ignore girls and teachers.

Dreams and Sleep

Children's dreams can have a profound effect on behavior. During the first year of life, when reality and fantasy are not yet fully differentiated, dreams may be experienced as if they were, or could be, real. At age 3, many children believe dreams are shared directly by more than one person, but most 4-year-olds understand that dreams are unique to each person. Children

view dreams either with pleasure or, as is most often reported, with fear. The dream content depends on the children's life experience, developmental stage, mechanisms used during dreaming, and sex.

Disturbing dreams are most common when children are 3, 6, and 10 years of age. Two-year-old children may dream about being bitten or chased; at the age of 4, they may have many animal dreams and also dream of persons who either protect or destroy. At age 5 or 6, dreams of being killed or injured, of flying and being in cars, and of ghosts become prominent; the role of conscience, moral values, and increasing conflicts are concerned with these themes. In early childhood, aggressive dreams rarely seem to occur; instead, dreamers are in danger, a state that perhaps reflects children's dependent position. By about the age of 5, children realize that their dreams are not real; before then, they believed them to be real events. By age 7, children know that they create their dreams themselves.

Between the ages of 3 and 6 years, children usually want to keep their bedroom door open or to have a nightlight, so that they can either maintain contact with their parents or view the room in a realistic, nonfearful way. At times, children resist going to sleep to avoid dreaming. Disorders associated with falling asleep, therefore, are often connected with dreaming. Children often create rituals to protect themselves in the withdrawal from the world of reality into the world of sleep. Parasomnias, such as sleepwalking, sleep talking, enuresis (bed-wetting), and night terrors, are common at this age. They usually occur during stage 4 sleep when dreaming is minimal, and they do not indicate emotional trouble or underlying psychopathology. Most children grow out of parasomnias by adolescence.

Periods of REM occur about 60 percent of the time during the first few weeks of life, a period when infants sleep two-thirds of the time. Premature babies sleep even longer than full-term babies, and a higher proportion of their sleep is REM sleep. The sleep-wake cycle of newborns is about 3 hours long. Adults spend about 20 percent of sleep dreaming. Even newborns have brain activity similar to that of the dreaming state.

Birth Spacing

For women in the United States, 10 percent of conceptions that lead to live births are considered unwanted, and 20 percent are wanted but considered ill-

timed.

Children born close together have higher rates of premature or underweight births and malnutrition; they develop more slowly and are at increased risk of contracting and dying from childhood infectious diseases. Studies have shown that a mother and child's health risks are reduced when the child is born 3 to 5 years after a previous birth. Compared with 24- to 29-month intervals, children born in 36- to 41-month intervals have a 28 percent reduction in stunting and a 29 percent reduction of low birth weight. Women who have children at 27- to 32-month intervals are 1.3 times more likely to avoid anemia, 1.7 times more likely to avoid third-trimester bleeding, and 2.5 times more likely to survive childbirth.

Birth Order

The effects of birth order vary. Firstborn children may be more highly valued or given more attention than subsequent children. Firstborn children appear to be more achievement-oriented and motivated to please their parents than subsequent children born to the same parents. Some studies show that people in certain competitive occupational areas, such as architecture, accounting, and engineering, tend to be firstborn children.

Second and third children have the advantage of their parents' previous experience. Younger children also learn from their older siblings. For example, they may show a more sophisticated use of pronouns at an earlier age than firstborns did. When spacing children too tightly, however, there may not be enough time for each child. The arrival of new children in the family affects not only the parents but also the siblings. Firstborn children may resent the birth of a new sibling, who threatens their sole claim on parental attention. In some cases, regressive behavior, such as enuresis or thumb-sucking, occurs.

Some researchers have suggested that there is a relationship between birth order and personality. However, extensive multicohort analyses have not supported this.

Children and Divorce

Many children live in homes in which divorce has occurred. Approximately 30 to 50 percent of all children in the United States live in homes in which

one parent (usually the mother) is the sole head of the household, and 61 percent of all children born in any given year can expect to live with only one parent before they reach the age of 18 years. A child's age at the time of the parents' divorce affects the child's reaction to the divorce. Immediately after a divorce, an increase in behavioral and emotional disorders appears in all age groups. Infants do not understand anything about separation or divorce; however, they do notice changes in their parents' responses to them and may experience changes in their eating or sleeping patterns; have bowel problems; and seem more fretful, fearful, or anxious. Children 3 to 6 years of age may not understand what is happening, and those who do understand often assume that they are somehow responsible for the divorce. Older children, especially adolescents, comprehend the situation and may believe that they could have prevented the divorce had they intervened in some way, but they are still hurt, angry, and critical of their parents' behavior.

Some children harbor the fantasy that their parents will reunite. Such children may show animosity toward a parent's real or potential new mate. Adaptation to the effects of divorce in children typically takes several years; however, up to about one-third of children from divorced homes may have lasting psychological trauma. Among boys, physical aggression is a common sign of distress. Adolescents tend to spend more time away from the parental home after the divorce. Children who adapt best to divorce are typically in a situation in which both parents make genuine efforts to spend time and relate to the child despite the child's potential anger about the divorce. The pre-divorce stability of the household is another influence, as are economic factors. To facilitate adaptation in children, a divorced couple who are amicable and avoid arguing with one another is most likely to succeed. Table 32-10 lists the potential psychological effects of divorce on children.



Table 32-10
Effects of Divorce on Children

- Children in homes with absent fathers are more likely to suffer from antisocial personality disorder, child conduct disorder, and attention-deficit hyperactivity disorder.
- The divorce rate of children of divorced parents doubles that of children from stable families.

- Children of divorce are far more likely to be delinquent, engage in premarital sex, and bear children out of wedlock during adolescence and young adulthood.
- Children from divorced homes function more poorly than children from continuously married parents across a variety of domains, including academic achievement, social relations, and conduct problems.
- Children from divorced homes have more psychological problems than those from homes disrupted by the death of a parent.
- Children from disrupted marriages experience greater risk of injury, asthma, headaches, and speech defects than children from intact families.
- Children of divorce tend to be impulsive, irritable, socially withdrawn, lonely, unhappy, anxious, and insecure.
- Children of divorce, especially boys, are more aggressive than children whose parents stayed married.
- Suicide rates for children of divorce are much higher than for children from intact families.
- Twenty to 25% have significant adjustment problems as teenagers

Data adapted from Americans for Divorce Reform, Arlington, Virginia. Table by Nitza Jones.



Table 32-11
Types of Step-Families

Neo-Traditional Families	• Resembles “traditional” families • Absent biologic parent is included at times. • Discipline, boundaries and limits, and expectations are discussed openly. • Family coalitions and “side-taking” are better avoided.
Romantic Families	• Expect to be a “traditional family” immediately • The absent biologic parent is expected to disappear and is often criticized. • Stepparent/stepchild difficulties are common.

	• Stress is unbearable. • Few open and frank discussions about problems
Matriarchal Families	• Run by a highly competent mom and her companion follows • Companion is a “buddy” to the children, not to the parent. • Birth of a step-sibling causes problems.

Stepparents. Although many possible scenarios that may occur after a divorce and remarriage, several potential scenarios have been outlined in Table 32-11. These include (1) Neo-traditional, (2) Romantic, and (3) Matriarchal. When remarriage occurs, children must learn to adapt to the stepparent and the “blended” family. Adaptation is often challenging, especially when a child feels that a stepparent is nonsupportive, resents the stepchild, or favors his or her natural children. Of step-families, 25 percent tend to dissolve within the first 2 years, whereas 75 percent grow to find a new balance in their blended family. A biologic child born to a new couple with a stepchild already in the home may receive more attention than the stepchild, leading to sibling rivalry. After 5 years, about 20 percent of adolescents in step-families suggest that they move out and try living with their other biologic parent.

Family Factors in Child Development

Family Stability. Parents and children living under the same roof in harmonious interaction is the expected cultural norm in Western society. Within this framework, childhood development presumably proceeds most expeditiously. Deviations from the norm, such as divorced- and single-parent families, are associated with a broad range of problems in children, including low self-esteem, increased risk of child abuse, and increased incidence of divorce when they eventually marry, and increased incidence of mental disorders, particularly depressive disorders and antisocial personality disorder as adults. Why some children from unstable homes are less affected than others (or even immune to these deleterious effects) is of great interest. Michael Rutter postulates that sex (boys are more affected than girls), age

(older children are less vulnerable than younger ones), and inborn personality characteristics influence vulnerability. For example, children who have a placid temperament are less likely to be victims of abuse within a family than are hyperactive children; they may be less affected by the emotional turmoil surrounding them.

Adverse Events. It is now well known that significant adverse events, especially in early childhood, such as sexual and physical abuse, neglect, or loss of a parent, interact with genetic background in a given child and influences the trajectory of development. For example, as mentioned earlier, early severe maltreatment such as sexual abuse increases the risk of multiple psychosocial difficulties and the emergence of many psychiatric disorders. Among young maltreated children, those with particular genetics, that is, who have the "short" variant of the serotonin transporter gene (short 5-HTTLPR polymorphism), may be more vulnerable to chronic depression in adulthood. This example of specific gene–environment interaction plays a vital role in a child's development as well as in the risk for future psychopathology. Current investigations are also seeking insight into what factors lead to resilience in youth exposed to adverse events, yet maintain allostasis, that is, stability in the face of stressful events. Hormones of the adrenal glands, thyroid, gonads, as well as metabolic hormones, play a role in the brain's ability to maintain stability upon exposure to stress, and the prefrontal cortex, hippocampus, and amygdala play critical roles in regulating emotionality, aggression, and resilience.

Daycare Centers. The role of daycare centers for children is under continuous investigation, and various studies have produced different results. One study found that children placed in daycare centers before the age of 5 are less assertive and less adequately toilet trained than home-reared children. Another study found that children in daycare to be more advanced in social and cognitive development than children who were not in daycare. The National Institute of Child Health and Human Development reported that 4½-year-olds who had spent more than 30 hours a week in child care were more demanding, more aggressive, and more noncompliant than those raised at home and showed higher cognitive skills, particularly in math and reading. These same children continued to score higher in math and reading skills but

had more poor work habits and social skills in third grade. The researchers were careful to note that this behavior was within the normal range, however.

All studies of daycare must take into account the quality of both the daycare center and the homes from which children come. For example, a child from a disadvantaged home may be better off at a daycare center than a child from an advantaged home. Similarly, a woman who wishes to leave home to work for financial or other reasons and cannot do so may resent remaining in the home in a child-rearing role, which may adversely affect the child.

Parenting Styles. How children are raised, vary considerably between and within cultures. Rutter has clustered this diversity into four general styles. Subsequent research has confirmed that certain styles tend to correlate with particular behavior in the children, although the outcomes are by no means absolute. The authoritarian style, characterized by strict, inflexible rules, can lead to low self-esteem, unhappiness, and social withdrawal. The indulgent-permissive style, which includes little or no limit setting coupled with unpredictable parental harshness, can lead to low self-reliance, poor impulse control, and aggression. The indulgent-neglectful style, one of noninvolvement in the child's life and rearing, puts the child at risk for low self-esteem, impaired self-control, and increased aggression. The authoritative-reciprocal style, marked by firm rules and shared decision-making in a warm, loving environment, is believed to be the style most likely to result in self-reliance, self-esteem, and a sense of social responsibility.

Development and Expression of Psychopathology

The expression of psychopathology in children can be related to both age and developmental level. Specific developmental disorders, particularly developmental language disorders, often are diagnosed in the preschool years. Delayed development of language is a common parental concern. Children who do not use words by 18 months or phrases by 2½ to 3 years may need assessment, particularly if they do not appear to understand typical verbal cues or much language at all. Mild mental retardation or specific learning problems often are not diagnosed until after the child begins elementary school. Disruptive behavior disorder will become apparent at that

time as the child begins to interact with peers. Similarly, attention-deficit disorders may only be apparent in school with its need for sustained attention. Other conditions, particularly schizophrenia and bipolar disorder, are rare in preschool and school-aged children.

ADOLESCENCE

Adolescence, marked by the physiologic signs and surging sexual hormones of puberty, is the period of maturation between childhood and adulthood. Adolescence is a transitional period in which peer relationships deepen, autonomy in decision-making grows, and they seek intellectual pursuits and social belonging. Adolescence is mostly a time of exploration and making choices, a gradual process of working toward an integrated concept of self. Adolescents are "works in progress," characterized by increasing ability for mastery over complex challenges of academic, interpersonal, and emotional tasks while searching for new interests, talents, and social identities. A body of growing literature of the specific mechanisms of brain development in adolescence has increased our understanding of broadening social skills in adolescents, in addition to the three expected developmental changes in adolescence: increased risk-taking, increased sexual behavior, and a move toward peer affiliation rather than a primary family attachment. The total cortical gray matter is at its peak at about age 11 years in girls and 13 years in boys, which enhances the ability to understand subtle social situations, control impulses, make long-range plans, and think ahead. White matter volume increases throughout childhood and adolescence, which may allow for increased "connectivity," thereby enhancing the abilities of adolescents to acquire new competencies, such as those needed to master today's technology.

What Is Normal Adolescence?

The concept of normality in adolescent development refers to the degree of psychological adaptation an adolescent achieves while navigating the hurdles and meeting the milestones characteristic of this period of growth. For up to approximately 75 percent of youth, adolescence is a period of successful adaptation to physical, cognitive, and emotional changes, mostly continuous with their previous functioning. Psychological maladjustment, self-loathing,

disturbance of conduct, substance abuse, affective disorders, and other impairing psychiatric disorders emerge in approximately 20 percent of the adolescent population.

Adolescent adjustment is continuous with previous psychological function; thus, psychologically disturbed children are at higher risk for psychiatric disorders during adolescence. Adolescents with psychiatric disorders are at increased risk for more significant conflicts with families and for feeling alienated from their families. Although up to 60 percent of adolescents endorse occasional distress or a psychiatric symptom, this group of adolescents functions well academically and with peers and describes themselves as generally satisfied with their lives.

The developmentalist Erik Erikson characterizes the normative task of adolescence as identity versus role confusion. The integration of past experiences with current changes takes place in what Erikson calls *ego identity*. Adolescents explore various aspects of their psychological selves by becoming fans of heroes or other well-known musical or political idols. Some adolescents appear consumed by their identification with a particular idol, whereas others are more moderate in their expression. Adolescents who feel accepted by a peer group and are involved in a variety of activities are less likely to become consumed by adoration of an idol. Socially isolated adolescents, feel socially rejected and become overly identified with an idol to the exclusion of all other activities, are at higher risk for serious emotional problems, and require psychiatric intervention.

Erickson uses the term moratorium to describe the interim period between the concrete thinking of childhood and a more evolved complex ethical development. Erikson defines identity crisis as a normative part of adolescence in which adolescents pursue alternative behaviors and styles and, then, successfully mold these different experiences into a stable identity. A failure to do so would result in identity diffusion, or role confusion, in which the adolescent lacks a cohesive or confident sense of identity. Adolescence is the time to bond with peers, experiment with new beliefs and styles, fall in love for the first time, and explore creative ideas for future endeavors.

Most adolescents go through this developmental process with optimism, develop good self-esteem, maintain good peer relationships, and sustain harmonious relationships with their families.

Stages of Adolescence

Early Adolescence. Early adolescence, from 12 to 14 years of age, is the period in which the most striking initial changes occur—physically, attitudinally, and behaviorally. Growth spurts often begin in these years for boys, whereas girls may have already had rapid growth for 1 to 2 years. At this stage, boys and girls begin to criticize usual family habits, insist on spending time with peers with less supervision, have a greater awareness of style and appearance, and may question previously accepted family values. A new awareness of sexuality may be displayed by increased modesty and embarrassment with their current physical development or may exhibit itself in an increased interest in the opposite sex.

Early adolescents engage in subtle or overt displays of their growing desire for autonomy, sometimes with challenging behaviors toward authority figures, including teachers and school administrators, and exhibit disdain for rules themselves. At this age, some adolescents begin to experiment with cigarettes, alcohol, and marijuana.

During early adolescence, there is a normal variation in when new defining behaviors are acquired. Overall, although many early adolescents make new friends and modify their public image, most maintain positive connections to family members, old friends, and their family's values. However, for some, early adolescence can a time of overwhelming turmoil, during which there is a dramatic rejection of family, friends, and lifestyle, resulting in a powerful alienation of the adolescent.

Jake, a 13-year-old adolescent, had just started the 8th grade. In the past, he has been a jovial, fun-loving, and cooperative student, but this year he found the school rules increasingly irritating and felt that his teachers were too strict. He had always been a good student while putting in a minimum of work. His older brother Sean, now in 11th grade, had established himself as a compliant, well-liked, and well-behaved student who always put maximal effort into school projects in the same school, so Jake was compared with his brother regularly by many teachers. Jake resented these comparisons because, unlike his brother, whom Jake felt was a "nerd," Jake was more rebellious, took more risks, and made

friends with more popular peers. To distinguish himself from his older brother in school and at home, Jake began to challenge the rules at school, stating that they were “stupid” and “meaningless.” Jake began to cut classes, to stay out late, and to experiment with alcohol and marijuana. He rejected his best friends from sixth and seventh grade and began to hang out with more daring peers. When Jake was at home, he was able to relate to his older brother Sean only when they played basketball and video games.

Jake’s grades began to deteriorate only slightly, but his parents noticed that on his report cards, his teachers rated his effort and behavior as unsatisfactory. During the second month of school, Jake’s parents received a phone call that Jake was going to be suspended due to possession of a small amount of marijuana on the school grounds during recess. During a subsequent meeting with the assistant principal and school counselor, Jake argued that the suspension was unfair because his grades were still good, and did not understand why his marijuana possession had triggered a suspension. When confronted with the fact that he had broken not only the school rules, but also violated the law, and that he was fortunate because the school did not involve the police, Jake became angry and continued to insist that they were mistreating him. He also blurted out that all of his teachers and his parents favored his older brother Sean and treated him like a second-class citizen. The school suspended Jake for 5 days but indicated that they would report the incident to the police unless Jake and his family initiated immediate counseling.

Jake begrudgingly began psychotherapy and entered into a weekly therapy group specializing in substance use for teens. Jake’s parents also sought therapy to work on becoming more unified in their parenting. Jake remained in psychotherapy for the next 1½ years, during which time his attitude and reasoning style changed and evolved considerably. At age 15, Jake was able to understand why his school had suspended him for possession of marijuana and came to appreciate their willingness to give him the chance to seek counseling, rather than be turned over to the police. Over time, Jake was able to admit the dangers of using drugs and took responsibility for his ill-advised behaviors. Alcohol and drug use continued to be a focus of his therapy and, by 15, Jake had virtually lost

interest in alcohol, and admitted to smoking marijuana rarely at parties. Jake became more open to making friends with a variety of peers, and he disclosed that he liked himself better now than when he was 13. He now treated his brother respectfully when alone or with friends, and he felt that his parents appreciated him for "who he was." (Courtesy of Caroly S. Pataki, M.D.)

Middle Adolescence. During the middle phase of adolescence (roughly between the ages of 14 and 16), adolescents' lifestyles may reflect their efforts to pursue their own stated goals of being independent. Their abilities to combine abstract reasoning with realistic decision-making and judgment are put to the test. In this phase, sexual behavior intensifies, making romantic relationships more complicated, and self-esteem becomes a pivotal influence on positive and negative risk-taking behaviors.

In this phase of development, adolescents tend to identify with a group of peers who become highly influential in their choices of activities, styles, music, idols, and role models. Adolescents' underestimation of the risks associated with a variety of recreational behaviors and their sense of "omnipotence," mixed with their drive to be autonomous, frequently cause some conflict with parental requests and expectations. Most teens can achieve the challenge of defining themselves as unique and different from their families while still connected with family members.

Jenna, a 16-year-old junior in high school, had just gotten her driver's license. She realized that she was lucky to have been given a brand new car at 16 because many of her friends did not yet have cars. Jenna was upset that her parents disapproved of her agreement to drive all of her friends to places that she did not even want to go. Jenna was an attractive and well-liked adolescent who had always been an "A" and "B" student, and never had school conflicts. She played the flute in the school's orchestra and was not involved in any team sports. Jenna started "going out" with a boy in her grade at school, Brett, who was also 16 years old, shortly after getting her license, and even though they did not know each other that well, Jenna felt that they had a close relationship. Since he did not yet have a car, she was the "identified driver" whenever they went out

or to parties. Jenna was glad about this because she did not like alcohol and was relieved that Brett would not be driving, given that he drank quite a bit at parties. Jenna got along reasonably well with her parents, who were considered very “easy-going” by her friends, and she felt that she and her parents had similar values and ideas.

Things were going well until Brett began to pressure her to go further in their sexual relationship. When Jenna told him that she was not ready, Brett hounded her more. When the subject of sex had come up with her parents “hypothetically” in the past, they had dismissed the subject, indicating that when it was the right time for her, Jenna would know. Jenna knew that she was not ready to have sex, although many of her classmates were sexually active. Jenna was not an impulsive person and liked to plan things carefully so that they would feel right to her. Jenna realized that she could not agree with Brett’s request, but she was confident that she could make him understand. One of Jenna’s friends suggested that Brett might break up with her as a result, but Jenna was willing to take that risk. Jenna carefully told Brett that she loved him, but she was not yet ready for sex. Jenna was slightly surprised that instead of pressuring her more or breaking up with her, Brett accepted her decision. He seemed a little relieved.

Jenna and Brett continued their relationship into their senior year of high school, and, toward the end of her senior year, Jenna desired to be sexually active with him. They decided to go to a community clinic known for its positive attitude toward adolescents, to learn about birth control methods, and pick one without including their parents. Jenna and Brett took the time to learn about a variety of birth control methods and chose to use condoms. When they left the clinic, Jenna and Brett felt closer than they had before and realized how they had both grown in their relationship. Jenna and Brett both felt that they were doing the right thing. (Courtesy of Caroly S. Pataki, M.D.)

Late Adolescence. Late adolescence (between the ages of 17 and 19) is a time when continued exploration of academic pursuits, musical and artistic tastes, athletic participation, and social bonds lead a teen toward a greater definition of self and a sense of belonging to specific groups or subcultures

within mainstream society. Well-adjusted adolescents can be comfortable with current choices of activities, tastes, hobbies, and friendships, yet remain aware that they will continue to refine their “identities” during young adulthood.

Joey was a second-semester freshman in college, living away from home, and had just turned 18 years of age. He reflected on the fact that he was no longer a “minor” and could make almost any decision for himself without involving his parents.

Joey felt liberated, but at the same time, he was confused and a little lost. Since 10th grade, Joey had planned to pursue a career in medicine like his father, so he had taken a heavy load of science courses in the first semester, all of which he had despised. This semester, however, Joey signed up for liberal arts classes. He did not mention this to his father. He then added classes that ranged from art history to architectural drafting to sociology, philosophy, and music. He had been influenced, he believed, by his roommate Tony, who was in the architecture program, and by his girlfriend, Lisa, who was majoring in studio art.

As the semester progressed, Joey found that his favorite course was the drafting class. Joey could not help but wonder whether he liked the drafting class so much because of how much he idolized Tony, or because he enjoyed the class. Joey talked this over with Lisa, who suggested that he chill out and not figure out the rest of his life right now. She recommended that he take at least two more semesters of varied classes, including those in the architecture curriculum, before making a final decision about a career. Joey realized that Lisa’s approach to college and life was so relaxed, the opposite of his approach, following his parents’ pressure to plan, make commitments early, and see them through, regardless of how it felt. Lisa’s approach left more room for reflecting on experiences and then making a choice, rather than jumping into what he was “supposed” to do. Joey took her advice and allowed himself another year to try out majors and then decide on a career. After experiencing courses in many varied subjects, Joey decided that he did truly enjoy architecture and was able to switch his focus from premed to architecture. (Courtesy of Caroly S. Pataki M.D.)

Components of Adolescence

Physical Development. Puberty is the process by which adolescents develop physical and sexual maturity, along with reproductive ability. The first signs of the pubertal process are an increased rate of growth in both height and weight. This process begins in girls by approximately 10 years of age. By the age of 11 or 12, many girls noticeably tower over their male classmates, who do not experience a growth spurt, on average, until they reach 13 years of age. By age 13, many girls have experienced menarche, and most have developed breasts and pubic hair.

Wide variation exists in the normal range of onset and timing of pubertal development and its components. A set sequence occurs, however, in the order in which pubertal development proceeds. Thus, secondary sexual characteristics in boys, such as increased length and width of the penis, for example, will occur after the release of androgens from developed enlarged testes.

Sexual maturity ratings (SMRs), also referred to as Tanner Stages, range from SMR 1 (prepuberty) to SMR 5 (adult). The SMR ratings include stages of genital maturity in boys and breast development in girls, as well as pubic hair development. Table 32-12 outlines SMRs for boys and girls.

The primary female sex characteristic is ovulation, the release of eggs from ovarian follicles, approximately once every 28 days. When adolescent girls reach SMR 3 to 4, ovarian follicles are producing enough estrogen to result in menarche, the onset of menstruation. When adolescent girls reach SMR 4 to 5, an ovarian follicle matures monthly, and ovulation occurs. Estrogen and progesterone promote sexual maturation, including further development of fallopian tubes and breasts.

For adolescent boys, the primary sex characteristic is the development of sperm by the testes. In boys, sperm development occurs in response to follicle-stimulating hormone acting on the seminiferous tubules within the testes. The pubertal process in boys consists of the growth of the testes stimulated by luteinizing hormone. An adolescent boy’s ability to ejaculate generally emerges within 1 year of reaching SMR 2. Secondary sexual characteristics in boys include thickening of the skin, broadening of the

shoulders, and the development of facial hair.

Cognitive Maturation. Cognitive maturation in adolescence encompasses a wide range of expanded abilities that fall within the global category of executive functions of the brain. These include the transition from concrete thinking to more abstract thinking, an increased ability to draw logical conclusions in scientific pursuits, with peer interactions, and in social situations; and new abilities for self-observation and self-regulation. Adolescents acquire increased awareness of their intellectual, artistic, and athletic gifts and talents, yet it often takes many more years into young adulthood to establish a practical application for these abilities.



Table 32-12
Sexual Maturity Ratings for Male and Female Adolescents

Sexual Maturity Rating	Girls	Boys
Stage 1	Preadolescent, papilla elevated No pubic hair	Penis, testes, scrotum preadolescent No pubic hair
Stage 2	Breast bud, small mound; areola diameter increased Sparse long pubic hair, mainly along labia	Penis size same, testes and scrotum enlarged, with scrotal skin reddened Sparse long pubic hair, mainly at the base of penis
Stage 3	Breast and areola larger; no separation of contours Pubic hair darker and coarser; spread over pubic area	Penis elongated, with increased size of testes and scrotum Pubic hair darker and coarser; spread over pubic area
Stage 4	Breast size increased	Penis increased in length

		and width
	Areola and papilla raised	Testes and scrotum larger
	Pubic hair coarse and thickened; covers less area than in adults, does not extend to thighs	Pubic hair coarse and thickened; covers less area than in adults, does not extend to thighs
Stage 5	Breasts resemble adult female breast; areola has recessed to breast contour Pubic hair increased in density; area extends to thighs	Penis, testes, scrotum appear mature Pubic hair increased in density; area extends to thighs

The central cognitive change that occurs gradually during adolescence is the shift from concrete thinking (concrete operational thinking, according to Jean Piaget) to the ability to think abstractly (formal operational thinking, in Piaget's terminology). This evolution occurs as an adaptation to stimuli that demand an adolescent to produce hypothetical responses, as well as in response to the adolescent's expanded abilities to provide generalizations from specific situations. The development of abstract thinking is not a sudden epiphany but, rather, a gradual process of expanding logical deductions beyond concrete experiences and achieving the capacity for idealistic and hypothetical thinking based on everyday life.

Adolescents often use an omnipotent belief system that reinforces their sense of immunity from danger, even when confronted with logical risks. Some degree of child-like magical thinking continues to coexist with more mature abstract thinking in many adolescents. Despite the persistence of magical thinking into adolescence, adolescent cognition departs from that of younger children insofar as the increased ability for self-observation and development of strategies to promote strengths and compensate for weaknesses.

One of the essential cognitive tasks in adolescence is to identify and gravitate toward those pursuits that seem to match the adolescent's cognitive strengths, in academic courses, and in thinking about future aspirations.

Piaget believed that cognitive adaptation in adolescence is profoundly influenced by social relationships and the dialogue between adolescents and peers, making social cognition an integral part of cognitive development in adolescence.

Socialization. Socialization in adolescence encompasses the ability to find acceptance in peer relationships, as well as the development of more mature social cognition. The skills to develop a sense of belonging to a peer group are of central importance to a sense of well-being. Being viewed as socially competent by peers is a critical component in building good self-esteem for most early adolescents. Peer influences are powerful and can foster positive social interactions, as well as apply pressure in less socially accepted behaviors or even high-risk behavior. Belonging to a peer group is, in general, a sign of adaptation and a developmentally appropriate step in separating from parents and turning the focus of loyalty toward friends. Children between the ages of 6 and 12 can engage in exchanges of ideas and opinions and acknowledge feelings of peers, but the relationships often wax and wane in a discontinuous way based on altercations and good times. Friendships deepen with repeated good times, but, for some school-aged children, a variety of peers are often interchangeable—that is, a child may seek a companion when that child has free time, rather than out of a desire to spend time with a specific friend. As adolescence ensues, friendships become more individualized, and adolescents may share personal secrets with a friend rather than a family member. The adolescent may achieve a comfort level with one or several early adolescent peers, and the group may "stick together," spending most of their free time together. In early adolescence, a blend of the above two social modes may emerge, small "cliques" arise, and, even within the cliques, competition and jealousies regarding which dyads are "preferred" or higher ranked within the clique may result in some discontinuities in the relationships. In later adolescence, the peer group solidifies, leading to increased stability in the friendships and a greater mutuality in the quality of the interactions.

Moral Development. Morality is a set of values and beliefs about codes of behavior that conform to those shared by others in society. Adolescents, as do younger children, tend to develop patterns of behaviors characteristic of their

family and educational environments and by imitation of specific peers and adults whom they admire. Moral development does not strictly follow chronologic age but, instead, is an outgrowth from cognitive development.

Piaget described moral development as a gradual process parallel to cognitive development, with expanded abilities in differentiating the best interests for society from those of individuals occurring during late adolescence. Preschool children simply follow the rules set forth by the parents; in the middle years, children accept rules but show an inability to allow for exceptions; and during adolescence, young persons recognize rules in terms of what is good for the society at large.

Lawrence Kohlberg integrated Piaget's concepts and described multiple stages of moral development within three significant levels of morality. The first level is preconventional morality, in which punishment and obedience to the parent are the determining factors. The second level is the morality of conventional role-conformity, in which children try to conform to gain approval and to maintain good relationships with others. The third and highest level is the morality of self-accepted moral principles, in which children voluntarily comply with rules based on a concept of ethical principles and make exceptions to rules in certain circumstances.

Although Kohlberg's and Piaget's notions of moral development focus on a unified theory of cognitive maturation for both sexes, Carol Gilligan emphasizes the social context of moral development leading to divergent patterns in moral development. Gilligan points out that, in women, compassion and the ethics of caring are dominant features of moral decision-making, whereas, for men, predominant features of moral judgments are related more to a perception of justice, rationality, and a sense of fairness. Although influential, many psychologists and scholars criticize Gilligan's work as primarily observational and lacking scientific rigor. Feminist scholars note that Gilligan rarely considers the role of societal expectations in influencing gender-based behavior.

Modern psychologists have incorporated insights from cognitive psychology and suggest that moral development goes beyond the rationality of Piaget and Kolberg. Psychologists such as Jonathan Haidt emphasize the role of intuition and preconceived beliefs based on the fundamental moral foundations found in a society. He suggests that the logical explanations one gives to justify one's actions are post hoc rationalizations meant to justify

one's emotional (i.e., irrational) reactions to the moral dilemmas we face in society.

Self-Esteem. Self-esteem is a measure of one's sense of self-worth based on perceived success and achievements, as well as a perception of how much one is valued by peers, family members, teachers, and society in general. The most important correlates of good self-esteem are one's perception of attractive physical appearance and high value to peers and family. Secondary features of self-esteem relate to academic achievement, athletic abilities, and unique talents. Adolescent self-esteem is mediated, to a significant degree, by the positive feedback received from a peer group and family members, and adolescents often seek out a peer group that offers acceptance, regardless of negative behaviors associated with that group. Adolescent girls have more of a problem maintaining self-esteem than do boys. Girls continued to rate themselves with generally lower self-esteem into adulthood.

Current Environmental Influences and Adolescence

Adolescent Sexual Behavior. Sexual experimentation in adolescents often begins with fantasy and masturbation in early adolescence, followed by noncoital genital touching with the opposite sex or, in some cases, same-sex partners, oral sex with partners, and initiation of sexual intercourse at a later point in development. By high school, most male adolescents report experience with masturbation, and more than half of adolescent girls report masturbation. The balance between healthy adolescent sexual experimentation and emotionally and physically safe sexual practices is one of the significant challenges for society.

Estimates vary, but about 50 percent of 9th- to 12th-grade students reported having had sexual intercourse. The median age at first intercourse is about 16 years for boys and 17 years for girls. Boys generally have more sexual partners than do girls, and boys are less likely than girls to seek emotional attachments with their sexual partners.

FACTORS INFLUENCING ADOLESCENT SEXUAL BEHAVIOR. Factors that affect sexual behavior in adolescents include personality traits, gender, cultural and religious background, racial factors, family attitudes, and sexual education

and prevention programs.

Personality factors are associated with sexual behavior, as well as sexual risk-taking. Higher levels of impulsivity are associated with a younger age at the first experience of sexual intercourse; a higher number of sexual partners; sexual intercourse without the use of contraception, including condoms; and a history of a sexually transmitted disease (STD, chlamydia).

Historically, male adolescents have initiated sexual intercourse at a younger age than female adolescents. The younger a teenage girl is when she has sex for the first time, the more likely she is to have had unwanted sexual activity. Close to four of ten girls who had first intercourse at 13 or 14 years of age report it was either not voluntary or unwanted. Three of four girls and over half of boys report that girls who have sex do so because their boyfriends want them to. In general, adolescents who initiate sexual intercourse at younger ages are also more likely to have a higher number of sexual partners.

The additive effects of more highly educated families, social and religious youth groups, and school-based educational programs can decrease high-risk sexual behavior among adolescents. Female empowerment strategies can play an essential role in decreasing unwanted sexual behavior and teenage pregnancies. The primary reason that teenage girls who have never had intercourse give for abstaining from sex is that having sex would be against their religious or moral values. Other reasons include the desire to avoid pregnancy, fear of contracting a STD, and not having met the appropriate partner.

CONTRACEPTIVES. Currently, 98 percent of teenagers 15 to 19 years are using at least one method of birth control. The two most common methods are condoms and birth control pills. STDs, despite the use of condoms, are still at high levels in teens—approximately one in four sexually active teens contracts an STD every year. Approximately half of all new human immunodeficiency virus (HIV) infections occur in people younger than age 25.

PREGNANCY. Each year 750,000 to 850,000 teenage girls younger than age 19 become pregnant. Of this number, 432,000 give birth, a 19 percent decline from 532,000 in 1991; the rest (418,000) obtain abortions. The largest decline

in teen pregnancy by race is for black women. Hispanic teen births have declined 20 percent, but continue to have the highest teen birth rates compared with other races.

Teenage pregnancy creates a plethora of health risks for both mother and child. Children born to teenage mothers have a higher chance of dying before the age of 5 years. Those who survive are more likely to perform poorly in school and are at higher risk of abuse and neglect. Teenage mothers are less likely to gain adequate weight during pregnancy, increasing the risk of premature births and low–birth-weight infants. Low–birth-weight babies are more likely to have organs that are not fully developed, resulting in bleeding in the brain, respiratory distress syndrome, and intestinal problems. Teenage mothers are also less likely to seek regular prenatal care and to take recommended daily multivitamins, and they are more likely to smoke, drink, or use drugs during pregnancy. Only one-third of teenage mothers obtain high school diplomas, and only 1.5 percent have a college degree by the age of 30.

The average adolescent mother who cannot care for her child has the child either placed in foster care or raised by the teenager's already overburdened parents or other relatives. Few teenage mothers marry the fathers of their children; the fathers, usually teenagers, cannot care for themselves, much less the mothers of their children. If the two do marry, they usually divorce. Many are more likely to end up on welfare.

ABORTION. Nearly four of ten teen pregnancies end in abortion. Almost all the girls are unwed mothers from low socioeconomic groups; their pregnancies result from sex with boys to whom they felt emotionally attached. Most (61 percent) teenagers elect to have abortions with their parents' consent, but laws of mandatory parental consent put two rights into a competition: a girl's claim to privacy and a parent's need to know. Most adults believe that teenagers should have parental permission for an abortion, but when parents refuse to give their consent, most states prohibit parents from vetoing the teenager's decision.

The abortion rate in many European countries tends to be far lower than that in the United States. The Control and Prevention (CDC) reports that the abortion rate in the United States for girls between the ages of 15 and 19 is about 30 per 1,000. In France, about 10.5 of every 1,000 girls under the age

of 20 had an abortion, according to World Health Organization statistics. The rate of abortion in Germany was 6.8; in Italy, 6.3; and in Spain, 4.5. Britain has a higher rate, 18.5. Family planning experts believe that more sex education and the availability of contraceptive devices help keep the number of abortions down. In Holland, where contraceptives are freely available in schools, the teenage pregnancy rate is among the lowest in the world.

Risk-Taking Behavior. Reasonable risk-taking is a necessary endeavor in adolescence, leading to confidence both in forming new relationships and in sports and social situations. High-risk behaviors among adolescents are associated with severe negative consequences, however, and can take many forms, including drug and alcohol use, unsafe sexual practices, self-injurious behaviors, and reckless driving.

Drug Use

ALCOHOL. About 30 percent of 12th graders report having five or more drinks in a row within 2 weeks. The average age when youths first try alcohol is 11 years for boys and 13 years for girls. The national average age at which Americans begin regular drinking is 15.9 years of age. People ages 18 to 25 show the highest prevalence of binge and heavy drinking. Drunk driving has declined since 2002. Alcohol dependence, along with other drugs, is associated with depression, anxiety, oppositional defiant disorder, antisocial personality disorder, and an increased rate of suicide.

NICOTINE. The number of young American smokers has declined since 1990. However, the rate of smoking among teenagers is still as high as or higher than that of adults. According to the American Cancer Society, on average more than one of five students have smoked cigarettes. Each day, more than 4,000 teenagers try their first cigarette, and another 2,000 become regular, daily smokers. Cigarette smokers are more likely to get into fights, carry weapons, attempt suicide, suffer from mental health problems such as depression, and engage in high-risk sexual behaviors. One of three will eventually die from smoking-related diseases. Cigarettes are the most common type of tobacco used among middle-school students, followed by cigars, smokeless tobacco, and pipes.

CANNABIS. Marijuana is the most popular illicit drug, with 14.6 million people using it (6.2 percent of the population), two-thirds being under the age of 18. Its use, however, is slowly declining. About 6 percent of 12th graders report daily use of marijuana.

One of the primary reasons for such prevalence of marijuana use among teenagers is because many find that marijuana is easier to get than alcohol or cigarettes. This belief has declined in recent years. Once teenagers are dependent on marijuana, they often tumble into truancy, crime, and depression.

COCAINE. About 13 percent of high school seniors use cocaine, exceeding the national average of 3.6 percent. Also, about 1 percent of 12th graders admit to using phencyclidine (PCP). Crystal methamphetamine (ice) has an annual prevalence in 12th graders of about 2 percent.

OPIOIDS. In recent years, the number of teens using prescription pain relievers for nonmedical reasons has increased. Prescription drug abuse by people ages 18 to 25 has increased by 15 percent. Drugs of specific concern are the pain relievers oxycodone and hydrocodone. Oxycodone has gained ground among high school students since its emergence in 2001, with 5 percent of 12th graders, 3.5 percent of 10th graders, and 1.7 percent of 8th graders reporting use. About 9 percent of 12th graders use hydrocodone, 6 percent of 10th graders, and 2.5 percent of 8th graders.

HEROIN. Heroin use is prevalent among adolescents, although less so than cocaine. The average age of use is 19, but almost 2 percent of 12th graders use it. The nasal route (snorting) is the most common method of use.

Violence. Although rates of violent crime have decreased throughout the United States in recent years, violent crimes by young offenders are on the increase. Homicides are the second leading cause of death among people aged 15 to 25. (Accidents are first; suicides are third.) Black male teenagers are far more likely to be murder victims than are boys from any other racial or ethnic group or girls of any race. The factor most strongly associated with violence among adolescent boys is growing up in a household without a father or father surrogate; this factor aside, race, socioeconomic status, and

education show no effect on the propensity toward violence.

BULLYING. *Bullying* is the use of one's strength or status to intimidate, injure, or humiliate another person of lesser strength or status. It can be physical, verbal, or social. Physical bullying involves physical injury or threat of injury to someone. Verbal bullying refers to teasing or insulting someone. Social bullying refers to the use of peer rejection or exclusion to humiliate or isolate a victim.

About 30 percent of 6th- through 10th-grade students experience moderate-to-frequent bullying, as a bully, a target, or both. Approximately 1.7 million children within this age group bully other children. Boys are more likely to be involved in bullying and violent behavior than girls. Girls tend to use verbal bullying rather than physical.

An estimated 160,000 students miss school each day because of fear of attack or intimidation from peers; some may drop out. Stresses of "victimization" can interfere with student's engagement and learning in school. Children who bully other children are at risk for engaging in more violent severe behaviors, such as frequent fighting and carrying a weapon.

Cyberbullying. During the last decade, electronic or internet bullying has become of great concern to adolescents. Cyberbullying is defined broadly to convey the use of electronic means to intimidate or harm someone intentionally. The reported prevalence of cyberbullying is variable, reports ranging from 1 to 62 percent of youth reporting that they were victims of cyber victimization. A study of about 700 Australian students, recruited at age 10 years, and followed until ages 14 to 15 years, found that 15 percent had engaged in cyberbullying, 21 percent had engaged in traditional bullying, and 7 percent had engaged in both. Another study of self-reported information collected from 399 teens in the 8th to 10th grades found that involvement in cyberbullying, either as a victim or a bully, specifically contributed to the prediction of depressive symptoms and suicidal ideation. This correlation between cyberbullying and depressive symptomatology is more reliable than the association of traditional bullying and affective disorder.

Gangs. Gang violence is a problem throughout the United States. There are

2,000 different youth gangs around the country, with more than 200,000 teens and young adults as members. Most members are between the ages of 12 and 24 years, with an average of 17 to 18 years. Gang membership is a brief phase for many teenagers; one-half to two-thirds leave the gang by the 1-year mark. Boys are more likely to join gangs than girls; however, female gang membership may be underrepresented. Female gang members are more likely to be found in small cities and rural areas and tend to be younger than male gang members. Female gang members are also involved in less delinquent or criminal activity than males and commit fewer violent crimes.

WEAPONS. Each day, on average, nearly 10 American children younger than the age of 18 years are killed in handgun suicides, homicides, and accidents. Many more are wounded. One in five youths in grades 9 to 12 carries a weapon: knife, gun, or club.

By law, anyone younger than 18 years cannot purchase a firearm. Two-thirds of students in grades 6 to 12 say that they can get a firearm within 24 hours, however. More than 22 million children live in a home with a firearm. In 40 percent of these homes, at least one gun is not locked, and 13 percent are unlocked and loaded. Two of three students involved in school shootings acquired their guns from their own home or that of a relative. At least 60 percent of suicide deaths in teens involve the use of a handgun.

SCHOOL VIOLENCE. According to the CDC, of all youth homicides in 2010, about 2 percent occurred in schools. Approximately 7 percent of teachers report they have been threatened with injury or physically attacked by a student from their school. Also, among students in grades 9 through 12, about 6 percent reported carrying a weapon on school property on one or more days in the 30 days before the survey.

Many factors can lead to violent acts in teenagers. Some inherited traits include impulsivity, learning difficulties, low IQ, or fearlessness. A correlation also exists between witnessing violent acts and involvement in violence. Children who witness violent acts are more aggressive and grow up more likely to become involved in violence—either as a victimizer or as a victim. Table 32-13 lists some of the early and imminent warning signs of school violence.

On April 20, 1999, two teenage boys, ages 17 and 18 years, went on a shooting rampage through Columbine High School of Littleton, Colorado. Armed with shotguns, a semiautomatic rifle, and a pistol, they laughed and shouted as they shot classmates and teachers at point-blank range while hurling homemade explosives. Fifteen were killed, including the 2 gunmen, who also injured 25 people.

The gunmen were members of the "trench coat mafia" at the high school, a clique of social misfits who stood out at the school for their gothic style of dress and nihilistic attitude. The two gunmen were obsessed with violent video games and intrigued with Nazi culture, even though one was part Jewish. They chose the date of the attack to coincide with Adolf Hitler's birthday.

On March 21, 2005, a 16-year-old boy went on a shooting rampage at Red Lake High School on the Red Lake Indian Reservation in far northern Minnesota. He began his shooting spree by killing his grandfather and the grandfather's companion. He then donned his grandfather's police-issue gun belt and bulletproof vest before heading to the school, where he killed a security guard, a teacher, five students, and then himself. About 15 others were injured.

The gunman had a troubled childhood; his father committed suicide in 1997, and his mother suffered head injuries in an auto accident. He expressed admiration for Adolf Hitler on a neo-Nazi website, using the handle "Todesengel," which is German for "Angel of Death." He had bouts of depression, suicide ideation and was taking fluoxetine. He was a member of a clique of about five students known as "The Darkers," who wore black clothes and chains, spiked or dyed their hair, and loved heavy-metal music. The gunman usually wore a long black trench coat, eyeliner, and combat boots. Peers described him as a quiet teenager.



Table 32-13
Warning Signs of School Violence

Early Warning Signs
Social withdrawal
Excessive feelings of isolation and being alone

- Excessive feelings of rejection
- Being a victim of violence
- Feelings of being picked on and persecuted
- Expression of violence in writings and drawings
- Uncontrolled anger
- Patterns of impulsive and chronic hitting, intimidating, and bullying behaviors
- History of discipline problems
- History of violent and aggressive behavior
- Intolerance for differences and prejudicial attitudes
- Drug and alcohol use
- Affiliation with gangs
- Inappropriate access to, possession of, and use of firearms
- Serious threats of violence

Imminent Warning Signs

- Serious physical fighting with peers or family members
- Severe destruction of property
- Severe rage for seemingly minor reasons
- Detailed threats of lethal violence
- Possession and/or use of firearms and other weapons
- Other self-injurious behaviors or threats of suicide

SEXUAL OFFENSE. Adolescents younger than age 18 years account for 20 percent of arrests for all sexual offenses (excluding prostitution), 20 to 30 percent of rape cases, 14 percent of aggravated sexual assault offenses, and 27 percent of child sexual homicides. These adolescent offenders account for the victimization of approximately one-half of boys and one-fourth of girls who are molested or sexually abused. Most instances have involved adolescent male perpetrators.



Table 32-14
Juvenile Sex Offender Subtypes

Juvenile Offenders Who Sexually Offend against Peers or Adults

- Predominantly assault females and strangers or casual

- acquaintances
- Sexual assaults occur in association with other types of criminal activity (e.g., burglary)
- Have histories of nonsexual criminal offenses, and appear more generally delinquent and conduct disordered
- Commit their offenses in public areas
- Display higher levels of aggression and violence in the commission of their sexual crimes
- More likely to use weapons and to cause injuries to their victims

Juvenile Offenders Who Sexually Offend against Children

- Most victims are male and are related to them, either siblings or other relatives
- Almost half of the offenders have had at least one male victim
- The sexual crimes tend to reflect a greater reliance on opportunity and guile than injurious force. This appears to be particularly true when their victim is related to them. These youths may “trick” the child into complying with the molestation, use bribes, or threaten the child with loss of the relationship
- Within the overall population of juveniles who sexually assault children are certain youths who display high levels of aggression and violence. Generally, these are youths who display more severe levels of personality and/or psychosexual disturbances, such as psychopathy, sexual sadism, and so on
- Suffer from deficits in self-esteem and social competency
- Many show evidence of depression

Characteristics Common to Both Groups

- High rates of learning disabilities and academic dysfunction (30–60%)
- The presence of other behavioral health problems, including substance abuse, and disorders of conduct (up to 80% have some diagnosable psychiatric disorder)
- Observed difficulties with impulse control and judgment

There appear to be two types of juvenile sex offenders: those who target children and those who offend against peers or adults. The age difference between the victim and the offender is the main difference between the two groups. Table 32-14 lists the differences and similarities between these two groups.

Etiologic factors of juvenile sex offending include maltreatment experiences, exposure to pornography, substance abuse, and exposure to aggressive role models. A significant number of offending adolescents have a childhood history of physical abuse (25 to 50 percent) or sexual abuse (10 to 80 percent). Half of the adolescent offenders lived with both parents and one other juvenile at the time of their offending. Evidence also suggests that most juvenile sex offenders are likely to become adult sex offenders. The most common psychosocial deficits of adolescent sexual offenders include low self-esteem, few social skills, minimal assertive skills, and poor academic performance. The most common psychiatric diagnoses are conduct disorder, substance abuse disorder, adjustment disorder, ADHD, specific phobia, and mood disorders. Male offenders are more often diagnosed with paraphilias and antisocial behavior, whereas female offenders are more likely to be diagnosed with mood disorders and engage in self-mutilation.

Prostitution. Teenagers constitute a large portion of all prostitutes, with estimates ranging up to 1 million teenagers involved in prostitution. The average age of a recruit is 13 years; however, some are young as 9 years of age. Most adolescent prostitutes are girls, but boys may become same-sex prostitutes. Most teenagers who enter a life of prostitution come from broken homes; however, a growing number of teenage prostitutes come from middle- to upper-middle-class homes. Many have been victims of rape or abuse as children. Most teenagers ran away from home and were taken in by pimps and substance abusers; the adolescents themselves then became substance abusers. Twenty-seven percent of teenage prostitution occurs in large cities, and incidents usually take place at an outside location, such as highways, roads, alleys, fields, woods, or parking lots. Teenage prostitutes are at high risk for acquired immunodeficiency syndrome (AIDS), and many (up to 70 percent in some studies) have HIV.

As many as 17,500 individuals are smuggled into the United States each year as “sex slaves.” They arrive under the pretenses of a better life and job

opportunities, but once they are in the United States, they are forced into prostitution, making little money while traffickers make thousands of dollars from their services. Many times they are raped and abused.

Tattoos and Body Piercing. Body piercing and tattoos have become more prevalent among adolescents since the 1980s. In the general population, approximately 10 to 13 percent of adolescents have tattoos. Of the more than 500 adolescents surveyed in a study, 13.2 percent report at least one tattoo, and 26.9 percent report at least one body piercing, other than in their ear lobe, at some point in their lives. Both tattoos and body piercing are more common in girls than in boys. Adolescents who endorsed possession of at least one tattoo or body piercing are more likely to endorse the use of gateway drugs (cigarettes, alcohol, marijuana), as well as experience with hard drugs (cocaine, crystal methamphetamine, and ecstasy).

32.2 Adulthood

INTRODUCTION

For most of the history of developmental psychology, the predominant theory held that development ended with childhood and adolescence. Adults were considered to be finished products in whom the ultimate developmental states had been reached. Beyond adolescence, the developmental point of view was relevant only insofar as success or failure to reach adult levels or to maintain them determined the maturity or immaturity of the adult personality.

In contradistinction were the long-recognized ideas that adult experiences, such as pregnancy, marriage, parenthood, and aging, had an obvious and significant impact on mental processes and experience in the adult years. This view of adulthood suggests that the patient, of any age, is still in the process of ongoing development, as opposed to merely having a past that influences mental processes and is the primary determinant of current behavior. Although the debate continues, the idea that development continues throughout life is increasingly accepted.

Development in adulthood, as in childhood, is always the result of the

interaction among body, mind, and environment, never exclusively the result of any one of the three variables. Most adults are forced to confront and adapt to similar circumstances: establishing an independent identity, forming a marriage or other partnership, raising children, building and maintaining careers, and accepting the disability and death of one's parents.

In modern Western societies, adulthood is the most prolonged phase of human life. Although the exact age of consent varies from person to person, adulthood can be divided into three main parts: young or early adulthood (ages 20 to 40), middle adulthood (ages 40 to 65), and late adulthood or old age.

YOUNG ADULTHOOD (20 TO 40 YEARS OF AGE)

Usually considered to begin at the end of adolescence (about age 20) and to end at age 40, early adulthood is characterized by peaking biologic development, the assumption of major social roles, and the evolution of an adult self and life structure. The successful passage into adulthood depends on the satisfactory resolution of childhood and adolescent crises.

During late adolescence, young persons generally leave home and begin to function independently. Sexual relationships become serious, and the quest for intimacy begins. The 20s are spent, for the most part, exploring options for occupation and marriage or alternative relationships and making commitments in various areas.

Early adulthood requires choosing new roles (e.g., husband, father) and establishing an identity congruent with those new roles. It involves asking and answering the questions "Who am I?" and "Where am I going?" The choices made during this time may be tentative; young adults may make several false starts.

Transition from Adolescence to Young Adulthood

The transition from adolescence to young adulthood is characterized by real and intrapsychic separation from the family of origin and the engagement of new, phase-specific tasks (Table 32-15). It involves many important events, such as graduating from high school, starting a job or entering college, and living independently. During these years, the individual resolves the issue of childhood dependency sufficiently to establish self-reliance and begins to

formulate new, young-adult goals that eventually result in the creation of new life structures that promote stability and continuity.

Developmental Tasks

Establishing a self that is separate from parents is a significant task of young adulthood. For most individuals, the emotional detachment from parents that takes place in adolescence and young adulthood is followed by a new inner definition of themselves as comfortably alone and competent, able to care for themselves in the real world. This shift away from the parents continues long after marriage, and parenthood results in the formation of new relationships that replace the progenitors as the most influential individuals in the young adult's life.

Psychological separation from the parents is followed by the synthesis of mental representations from the childhood past and the young-adult present. The psychological separation from parents in adolescence has been called the *second individuation*, and the continued elaboration of these themes in young adulthood has been called the *third individuation.* The continuous process of elaboration of self and differentiation from others that occurs in the developmental phases of young and middle age continues to influence our meaningful adult relationships.



Table 32-15
Development Tasks of Young Adulthood

To develop a young-adult sense of self and other: the third individuation
To develop adult friendships
To develop the capacity for intimacy; to become a spouse
To become a biologic and psychological parent
To develop a relationship of mutuality and equality with parents while facilitating their midlife development
To establish an adult work identity
To develop adult forms of play
To integrate new attitudes toward time

Table 32-16
Psychological Development Concepts

Concept	Definition	Example
Transition	The bridge between two successive stages	Late adolescence
Normative crisis	A period of rapid change or turmoil that strains a person's adaptive capacities	Midlife crisis
Stage	Period of consolidation of skills and capacities	Mature adulthood
Plateau	Period of developmental stability	Adulthood up to midlife
Rite of passage	Social ritual that facilitates a transition	Graduation; marriage

Adapted from Wolman T, Thompson T. Adult and later-life development. In: Stoudemire A, ed. *Human Behavior*. Philadelphia, PA: Lippincott-Raven; 1998, with permission.

Several different models have been proposed for understanding adult development. They are all theoretical and somewhat idealized. They all use metaphors to describe complex social, psychological, and interpersonal interactions. The models are heuristic: They provide a conceptual framework for thinking about important common experiences. They are descriptive rather than prescriptive; that is, they offer a useful way of looking at what many persons do, not a formula for what all persons should do. Some of the terms and concepts commonly used are explained in Table 32-16. These periods involve individuation, that is, leaving the family of origin and becoming one's own man or woman, passing through midlife, and preparing in middle adulthood for the transition into late adulthood.

Work Identity. The transition from learning and play to work may be gradual or abrupt. Socioeconomic group, gender, and race affect the pursuit and development of particular occupational choices. Blue-collar workers generally enter the workforce directly after high school; white-collar workers

and professionals usually enter the workforce after college or professional school. Depending on the choice of career and opportunity, work may become a source of ongoing frustration or activity that enhances self-esteem. Symptoms of job dissatisfaction are a high rate of job changes, absenteeism, mistakes at work, accident proneness, and even sabotage.

UNEMPLOYMENT. The effects of unemployment transcend those of loss of income; the psychological and physical tolls are enormous. The incidence of alcohol dependence, homicide, violence, suicide, and mental illness rises with unemployment. One's core identity, which is often tied to occupation and work, is seriously damaged when a job is lost, whether through firing, attrition, or early or sometimes even regular retirement.

A young adult female patient had much enjoyed her 5 years in college, and only reluctantly accepted a job with a large real estate firm. During college, she had limited interest in her appearance, and she began work in clothing borrowed from family and friends. She scoffed when her boss began to criticize her dress and gave her an advance to buy an upscale wardrobe, but she then began to enjoy the fine clothing and the respect engendered by her appearance and position. As her income began to rise, work became a source of pleasure and self-esteem and the way to acquire some of the trappings of adulthood. (Courtesy of Calvin Colarusso, M.D.)

Developing Adult Friendships. In late adolescence and young adulthood, before marriage and parenthood, friendships are often the primary source of emotional sustenance. Roommates, apartment-mates, sorority sisters, and fraternity brothers, as indicated by the names used to describe them, are substitutes for parents and siblings, temporary stand-ins until more permanent replacements are found.

Friendships mostly meet the emotional needs of closeness and confidentiality. All major developmental issues are discussed with friends, particularly those in similar circumstances. As marriages occur and children are born, the central emotional importance of friendships diminishes. Some friendships are abandoned at this point because the spouse objects to the friend, recognizing at some level that they are competitors. Gradually, there is

a movement toward a new form of friendship, couples friendships. They reflect the newly committed status but are more difficult to form and to maintain because four individuals must be compatible, not just two.

As children begin to move out of the family into the community, parents follow. Dance classes and Little League games provide the progenitors with a new focus and the opportunity to make friends with others who are at the same point developmentally and who are receptive to the formation of relationships that help explain and cushion the pressures of young adult life.

Sexuality and Marriage. The developmental shift from sexual experimentation to the desire for intimacy is experienced in young adulthood as intense loneliness, resulting from the awareness of an absence of committed love similar to that experienced in childhood with their parents. Brief sexual encounters in short-lived relationships no longer significantly boost self-esteem. Increasingly, the desire is for emotional involvement in a sexual context. The young adult who fails to develop the capacity for intimate relationships runs the risk of living in isolation and self-absorption in midlife.

For most individuals in Western culture, the experience of intimacy increases the desire for marriage. Most persons in the United States marry for the first time in their mid-to-late 20s. The median age of first marriage has been rising steadily since 1950 for both men and women, and the number of persons who never marry has been increasing. Today, approximately 50 percent of all adults ages 18 and older are not married, compared with only 28 percent in 1960. The proportion of 30- to 34-year-olds who never married has almost tripled, and the ratio of never-married 35- to 39-year-olds doubled.

INTERRACIAL MARRIAGE. Mixed-race marriages were banned in 19 states until a U.S. Supreme Court decision in 1967. In 1970, such marriages accounted for only 2 percent of all marriages. The trend has been steadily upward. Currently, interracial marriages account for about 1.5 million marriages in the United States.

Despite the trend toward more interracial marriages, they remain a small proportion of all marriages. Most persons are more likely to marry someone from the same racial and ethnic background. Marriages between Hispanic whites and non-Hispanic whites and between Asians and whites are more

common than those between blacks and whites.

SAME-SEX MARRIAGE. Same-sex marriage is recognized as legal by many states in the United States and by the U.S. Supreme Court, as well as in several countries around the world (e.g., France and Denmark). It differs from same-sex civil unions granted by states, which do not provide the same federal protections or benefits as marriage. No reliable estimates are available for the number of same-sex marriages in the United States; however, in 2013, it was estimated to be about 80,000. There is growing consensus in the United States and around the world that gay and lesbian persons should be allowed the same marital rights and privileges as straight adults. Same-sex marriage can be subject to more stress than heterosexual marriage because of continued prejudice toward such unions among certain conservative political or religious groups who oppose such unions.

MARITAL PROBLEMS. Although marriage tends to be regarded as a permanent tie, unsuccessful unions can be terminated, as indeed they are in most societies. Nevertheless, many marriages that do not end in separation or divorce are disturbed. In considering marital problems, clinicians are concerned with both the persons involved and with the marital unit itself. How any marriage works relates to the partner selected, the personality organization or disorganization of each, the interaction between them, and the original reasons for the union. Persons marry for a variety of reasons—emotional, social, economic, and political, among others. One person may look to the spouse to meet unfulfilled childhood needs for good parenting. Another may see the spouse as someone to be saved from an otherwise unhappy life. Irrational expectations between spouses increase the risk of marital problems.

MARRIAGE AND COUPLES THERAPY. When families consist of grandparents, parents, children, and other relatives living under the same roof, assistance for marital problems can sometimes be obtained from a member of the extended family with whom one or both partners have rapport. With the contraction of the extended family in recent times, however, this source of informal help is no longer as accessible as it once was. Similarly, religion once played a more important role than it does now in the maintenance of

family stability. Wise religious leaders are available to provide counseling. Still, they are not sought out to the extent they once were, which reflects the decline in religious influence among large segments of the population. Traditionally, a couple's extended family and their religion guided them during distressing times. In addition to these supports, social pressure helped to prevent the dissolution of marriages. As family, religious, and societal pressures have been relaxed, legal procedures for relatively easy separation and divorce have expanded. Concurrently, the need for formalized marriage counseling services has developed.

Marital therapy is a form of psychotherapy for married persons in conflict with each other. A trained person establishes a professional contract with the patient-couple and, through definite types of communication, attempts to alleviate the disturbance, to reverse or change maladaptive patterns of behavior, and to encourage personality growth and development.

In *marriage counseling*, only a particular conflict related to the immediate concerns of the family is discussed; marriage counseling is conducted much more superficially by persons with less psychotherapeutic training than is marital therapy. *Marital therapy* places greater emphasis on restructuring the interaction between the couple, including, at times, exploration of the psychodynamics of each partner. Both therapy and counseling emphasize helping marital partners cope effectively with their problems.

Parenthood. Parenthood intensifies the relationship between new parents. Through their physical and emotional union, the couple has produced a fragile, dependent being that needs them in the interlocking roles of father and mother. This recognition expands their internal images of each other to include thoughts and feelings emanating from the role of parent. As they live together as a family, the lovers' relationship with each other changes. They become parents relating to one another and their children.

Parent–child problems do arise, however. In addition to the economic burden of raising a child (estimated to be $250,000 for a middle-class family whose child goes to college), there are emotional costs. Children may reawaken conflicts that parents themselves had as children, or children may have chronic illnesses that challenge families' resources. In general, men have been more concerned with their work and occupational advancement than with child-rearing, and women have been more concerned about their

role as mothers than with advancement in their occupation. However, this emphasis is changing dramatically for both sexes. A small but growing number of couples are choosing to split a job (or work at two part-time jobs) and share child-rearing duties.

Parenting has been described as a continuing process of letting go. Children must be allowed to separate from parents and, in some cases, must be encouraged to do so. Letting go involves separation from children who are starting school. School phobias and school refusal syndromes that are accompanied by extreme separation anxiety may have to be handled. Often, a parent who cannot let go of a child accounts for this situation; some parents want their children to remain tightly bound to them emotionally. Family therapy that explores these dynamics may be needed to resolve such problems.

As children get older and enter adolescence, the process of establishing identity assumes great importance. Peer relationships become crucial to a child's development, and overprotective parents who keep a child from developing friendships or having the freedom to experiment with friends that the parents disapprove of can interfere with the child's passage through adolescence. Parents need not refrain from exerting influence over their children; guidance and involvement are crucial. But they must recognize that adolescents especially require parental approval. However, rebellious on the surface, adolescents are much more tractable than they appear, provided parents are not overbearing or generally punitive.

SINGLE-PARENT FAMILIES. More than 10 million single-parent families exist with one or more children under the age of 18; of these families, 20 percent are single-parent homes in which a woman is the sole head of the household. The increase in the number of single-parent families has risen by almost 200 percent since 1980.

ALTERNATIVE LIFESTYLE PARENTING. Single, partnered, and married gay men and women are choosing to raise children. In most cases, such children are obtained through adoption. Some, however, may be born to a lesbian through artificial insemination or obtained from a willing mother surrogate. The number of such family units is increasing. The data about the development of children in these homes indicate that they are at no greater risk for emotional

problems (or for a gay orientation) than children raised in conventional households.

ADOPTION. Since the turn of the century, adoption or foster placement has replaced institutional care as the preferred way to raise children who are neglected, unwanted, or abandoned. Many couples who are unable to conceive (and some couples who already have children) turn to adoption.

In addition to the full range of typical parent–child developmental issues, adoptive parents face particular problems. They must decide how and when to tell the child about the adoption. They must deal with the child's possible desire for information about his or her biologic parents. Adopted children are more likely to develop conduct disorders, problems with drug abuse, and antisocial personality traits. It is unclear whether these problems result from the process of adoption or whether parents who give up children for adoption are more likely to pass along a genetic predisposition for these behaviors.

With the widespread use of birth control and access to safe abortions, the number of infants available for adoption has declined steeply. Wealthy parents may prefer to arrange for private adoption rather than wait many uncertain years for institutional adoption. (In private adoptions, a biologic mother is paid for her legal and medical expenses but not for the baby. Baby selling is a felony in all states.) International adoptions (especially from Bosnia, Latin America, Eastern Europe, and China) have also become more common. Questionable regulation in these countries has raised concerns that some infants put up for adoption in developing countries may not be orphans but are being sold by impoverished mothers.

MIDDLE ADULTHOOD (40 TO 65 YEARS OF AGE)

Middle adulthood is the golden age of adulthood, similar to the latency years in childhood, but much longer. Physical health, emotional maturity, competence, and power in the work situation, and gratifying relationships with spouse, children, parents, friends, and colleagues all contribute to a normative sense of satisfaction and well-being. Concerning occupation, many persons begin to experience the gap between early aspirations and current achievements. They may wonder whether the lifestyle and the commitments they chose in early adulthood are worth continuing; they may feel that they

would like to live their remaining years in a different, more satisfying way, without knowing exactly how. As children grow up and leave home, parental roles change, and persons redefine their roles as husbands and wives.

Significant gender-specific changes occur in middle adulthood. Many women who no longer need to nurture young children can release their energy into independent pursuits that require assertiveness and a competitive spirit, traits that were traditionally considered masculine. Alternatively, men in middle adulthood may develop qualities that enable them to express their emotions and recognize their dependency needs, traits that were usually considered feminine. With the new balance of the masculine and the feminine, a person may now be able to relate more effectively to someone of the other sex than in the past.

Transition from Young to Middle Adulthood

The transition from young adulthood to middle adulthood is slow and gradual, with no sharp physical or psychological demarcation. The aging process picks up speed and becomes a powerful organizing influence on intrapsychic life, but the change is gradual, unlike during adolescence. Mental change is experienced similarly, slow and imperceptible, without a sense of disruption.

Development in young adulthood is embedded in close relationships. Intimacy, love, and commitment are related to the mastery of the relationships most immediate to personal experience. The transition from young adulthood to middle age includes widening concern for the more extensive social system and differentiation of one's own social, political, and historical system from others. Authors have described middle adulthood in terms of generativity, self-actualization, and wisdom.



Table 32-17
Features Salient to Middle Adulthood

Issues	Positive Features	Negative Features
Prime of life	Responsible use of power; maturity; productivity	Winner–loser view; competitiveness
Stock taking:	Possibility; alternatives;	Closure; fatalism

what to do with the rest of one's life	organization of commitments; redirection	
Fidelity and commitments	Commitment to self, others, career, society; filial maturity	Hypocrisy; self-deception
Growth-death (to grow is to die); juvenescence and rejuvenation fantasies	Naturality regarding body, time	Obscene or frenetic efforts (e.g., to be youthful); hostility and envy of youth and progeny; longing
Communication and socialization	Matters understood; continuity; picking up where left off; large social network; rootedness of relationships, places, and ideas	Repetitiveness; boredom; impatience; isolation; conservatism; confusion; rigidity

Adapted from Robert N. Butler, M.D.

Developmental Theorists

Robert Butler described several underlying themes in middle adulthood that appear to be present regardless of marital and family status, gender, or economic level (Table 32-17). These themes include aging (as changes in bodily functions are noticed in middle adulthood); taking stock of accomplishments and setting goals for the future; reassessing commitments to family, work, and marriage; dealing with parental illness and death, and attending to all the developmental tasks without losing the capacity to experience pleasure or to engage in playful activities.

Erik Erikson. Erikson described middle adulthood as characterized either by generativity or by stagnation. Erikson defined *generativity* as the process by which persons guide the oncoming generation or improve society. This

stage includes having and raising children, but wanting or having children does not ensure generativity. A childless person can be generative by (1) helping others, (2) being creative, and (3) contributing to society. Parents must be secure in their own identities to raise children successfully: They cannot be preoccupied with themselves and act as if they were, or wished to be, the child in the family.

To be *stagnant* means that a person stops developing. For Erikson, stagnation was anathema, and he referred to adults without any impulses to guide the new generation or to those who produce children but don't care for them as being "within a cocoon of self-concern and isolation." Such persons are in great danger. Because they are unable to negotiate the developmental tasks of middle adulthood, they are unprepared for the next stage of the life cycle, old age, which places more demands on the psychological and physical capacities than all the preceding stages.

George Vaillant. In his longitudinal study of 173 men who were interviewed at 5-year intervals after they graduated from Harvard, Vaillant found a strong correlation between physical and emotional health in middle age. Also, those with the most deficient psychological adjustment during college years had a high incidence of physical illness in middle age. No single factor in childhood accounted for adult mental health, but an overall sense of stability in the parental home predicted well-adjusted adulthood. A close sibling relationship during college years was correlated with emotional and physical well-being in middle age. In another study, Vaillant found that childhood and adult work habits were related and that adult mental health and good interpersonal relationships were associated with the capacity to work in childhood. Vaillant's studies are ongoing and represent the most extended continuous study of adulthood ever performed.

Calvin Colarusso and Robert Nemiroff. Based on their experience as clinicians and psychoanalysts, Calvin Colarusso and Robert Nemiroff propose a broad theoretical foundation for adult development by suggesting that the developmental process is the same in the adult as in the child because, like the child, the adult is always in the midst of an ongoing dynamic process, continually influenced by a constantly changing environment, body, and mind. Whereas child development focuses primarily

on the formation of psychic structure, adult development is concerned with the continuing evolution of existing psychic structure and with its use. Although the fundamental issues of childhood continue in altered form as central aspects of adult life, attempts to explain all adult behavior and pathology in terms of the experiences of childhood are considered reductionistic. The adult past must be taken into account in understanding adult behavior in the same way that the childhood past is considered. The aging body is understood to have a profound influence on psychological development in adulthood, as is the growing midlife recognition and acceptance of the finiteness of time and the inevitability of personal death.

Developing Midlife Friendships

Unlike friendships in latency and adolescence and, to some extent, in young adulthood, midlife friendships do not usually have a sense of urgency or the need for the frequent or nearly constant physical presence of the friend. Midlife individuals have neither the need to build a new psychic structure (as do latency-age children and adolescents) nor the pressing need to find new relationships (as do young adults). They may have many sources of gratification available through relationships with spouses, children, and colleagues.

As their firstborn sons progressed through high school, two women in their mid-40s became fast friends. In addition to raising money for the school activities in which their sons were involved, thus maintaining a close involvement with the boys, they spent many hours talking about the boys' activities, girlfriends, and college plans. Their husbands, who liked each other, became acquaintances, not friends. They directed their feelings about their sons into other relationships. After the boys left for college, the intensity of the friendship diminished, tending to peak again during vacation periods. (Courtesy of Calvin Colarusso, M.D.)

Because of their unique position in the life cycle, midlife adults are easily able to initiate and sustain friendships with individuals of different ages, as well as chronologic peers. In the face of a disrupted marriage or intimacy or

the pressure of other midlife developmental themes, friendships may quickly become vehicles for the direct expression of impulses.

Reappraising Relationships. Midlife is a time of serious reappraisal of marriage and committed relationships. In the process, individuals struggle with the question of whether to settle for what they have or to search for greater perfection with a new partner. For some, the conflict rages internally and is kept from others; others express it through actions that take the form of affairs, trial separations, and divorce.

Recent research on happy marriage indicates that these couples, despite internal and real conflict, have found or achieved a special *goodness-of-fit* between their individual needs, wishes, and expectations. In the eyes of these couples, marital success is based on the ongoing, successful engagement of several psychological tasks. Among the most important are providing a safe place for conflict and difference, holding a double vision of the other, and maintaining a satisfying sexual life.

The decision to leave a long-standing, committed relationship has great consequences, not only for the two individuals involved but also for their friends and loved ones. The effect on children, in particular, is especially profound, extending far beyond childhood. The impact on the abandoned spouse, parents, and close relatives may be nearly as severe.

Various forms of therapeutic intervention, such as marital counseling, individual psychotherapy, and psychoanalysis, can be extremely effective in helping uncertain individuals decide what to do or in helping those who leave deal with the consequences of their decision on the abandoned partner, children, and other loved ones. Problems relating to intimacy, love, and sex can occupy a prominent position in an outpatient practice.

The four case studies presented here by Calvin Colarusso, M.D., illustrate some of the issues described above.

A couple in their late 50s sought treatment to decide their marriage. Both had been unhappy for years and wanted to divorce, feeling that they had to act now while there was still time to begin new relationships that would fulfill them. Their concerns were for their children and grandchildren. How would they react? Would they respect their decision

to end a relationship of more than 30 years or attempt to stop the separation? As the work progressed, they decided that seeking happiness in the hopefully 20 or 30 years that they had left to live had to come ahead of the feelings of their loved ones. The fact that their decision was a mutual one facilitated their family's gradual acceptance of the divorce.

A 43-year-old patient, Mr. S, was continually preoccupied with his marriage during this 4-year psychoanalysis. Sexually inhibited during adolescence, he "married the only girl in the world who knew less about sex than I did." Both were virgins on their wedding night. As the marriage progressed, the couple gradually developed a "satisfactory" sex life, but the patient always wondered what he had missed. As his sexual inhibitions were explored, Mr. S's sense of having "missed out on a lot of opportunities" lead to visits to massage parlors and prostitutes. Eventually, such behavior ceased because of the recognition that his wife was a wonderful mother and loving wife and not the cause of the lack of sexual experience that he brought into the marriage. "I'll always feel that I missed out when I was young, but I've got so much going for me now, I'm not going to mess it up over something that I can't change."

A 38-year-old woman entered therapy after her husband discovered that she was having sexual relations with men in their early and middle 20s. She explained that she loved her husband, but he seemed to take her for granted. He no longer made her feel attractive and wanted. As the therapy progressed, it became clear that she felt that as long as she could attract younger men, she was still young and sexually desirable. Struggling with the early signs of physical aging, the realization that the young men were only using her to satisfy their own sexual needs was sobering and distressing. As she began to see that such behavior was self-destructive, she approached her husband about starting marital therapy.

Fifty-year-old Mrs. T left her "wonderful" husband because "I've missed

something. I just have to get out on my own." Married at 18 years of age, "after going from my parents' home to his home," she recognized that her rage at her husband for "not being all the other men I could have married, for closing off all the living I could have done" was irrational but uncontrollable. "I have to live on my own for a while to see if I can do it before it's too late." Fully intending to return to her husband, she continued exploring the infantile and adult issues that precipitated the separation, leaving the future of the marriage in doubt.

Sexuality

Whereas the young adult is preoccupied with developing the capacity for intimacy, the midlife individual is focused on maintaining intimacy in the face of deterring physical, psychological, and environmental pressures. In a long-standing relationship, these pressures include real and imaginary concerns about diminished sexual capability, emotional withdrawal because of preoccupation with developmental tasks, and the realistic pressures related to work and providing for dependent children and, sometimes, elderly parents as well. In relationships that begin in midlife, the maintenance of intimacy can be compromised by the absence of a common past, age and generational differences in interests and activities, and the difficulties involved in forming a stepfamily.

For sexual intimacy to continue, the participants must (1) accept the appearance of the partner's middle-aged body, (2) continue to find it sexually stimulating, and (3) accept the normative changes that occur in sexual functioning. For those who master these developmental issues, the partner's body remains sexually stimulating. Diminished sexual ability is compensated for by feelings of love and tenderness generated over the years by a satisfying relationship. Those who cannot accept the changes in the partner's body or their stop having sex, begin affairs or leave the relationship, usually in search of a younger partner.

Normative changes in midlife sexual functioning include diminished sexual drive and an increase in mechanical problems. Men have greater difficulty getting and sustaining erections and experience a longer refractory period after ejaculation. Because of diminished estrogen production, women experience a thinning of the vaginal mucosa, a decrease in secretions, and

fewer contractions at the time of orgasm. Women do not reach their sexual prime until their mid-30s; consequently, they have a greater capacity for orgasm in middle adulthood than in young adulthood. Women, however, are more vulnerable than men to narcissistic blows to their self-esteem as they lose their youthful appearance, which is overvalued in today's society. During middle adulthood, they may feel less sexually desirable than in early adulthood and, thus, feel less entitled to adequate sex life. An inability to deal with changes in body image prompts many women and men to undergo cosmetic surgery to maintain their youthful appearance.

The demands of raising children interfere with the privacy and emotional equilibrium required for intimacy, as do the pressures and responsibilities of work. Fatigue and diminished interest are common denominators in these circumstances. Patients with deeply rooted problems with sexuality or relationships may use aging, work, and relationships with children or elderly parents as a means of rationalizing their conflicts and refusing to analyze them.

Climacterium

Middle adulthood is the time of the male and female climacterium, the period in a life characterized by decreased biologic and physiologic functioning. For women, the menopausal period is considered the climacterium, and it may start anywhere from the 40s to the early 50s. Bernice Neugarten studied this period and found that more than 50 percent of women described menopause as an unpleasant experience. Still, a significant portion believed that their lives had not changed in any meaningful way, and many women experienced no adverse effects. Because they no longer had to worry about becoming pregnant, some women report feeling sexually freer after menopause than before its onset. Generally, the female climacterium has been stereotyped as a sudden or radical psychophysiological experience. Still, it is more often a gradual experience as estrogen secretion decreases with changes in the flow, timing, and eventual cessation of the menses. Vasomotor instability (hot flashes) can occur, and menopause can extend over several years. Some women experience anxiety and depression, but women who have a history of poor adaptation to stress are more predisposed to the menopausal syndrome.

For men, the climacterium has no clear demarcation; male hormones stay

reasonably constant through the 40s and 50s and then begin to decline. Nevertheless, men must adapt to a decline in biologic functioning and overall physical vigor. About age 50, a slight decrease in healthy sperm and seminal fluid occurs, not sufficient, however, to preclude insemination. Coincident with a decreased testosterone level may be fewer and less firm erections and decreased sexual activity generally. Some men experience a so-called midlife crisis during this period. The crisis can be mild or severe, characterized by a sudden drastic change in work or marital relationships, severe depression, increased use of alcohol or drugs, or a shift to an alternate lifestyle.

Midlife Transition and Crisis

The *midlife transition* has been defined as an intensive reappraisal of all aspects of life precipitated by the growing recognition that life is finite and approaching an end. It is characterized by mental turmoil, not action. For most people, the reappraisal results in decisions to keep most life structures, such as marriages and careers, which have been painstakingly built over time. When significant changes are made, they are thoughtful and considered, even when they include major shifts, such as divorce or a job change. The developmentally aware clinician recognizes that every patient in this age group is engaged in a midlife transition (whether the patient is talking about it or not) and facilitates the process by making it conscious and verbal.

A true *midlife crisis* is a major, revolutionary turning point in life, involving changes in commitments to career or spouse, or both, and accompanied by significant, ongoing emotional turmoil for the individual and others. It is an upheaval of substantial proportions. A flurry of impulsive actions follows a period of internal agitation, for example, leaving spouse and children, becoming involved with a new sexual partner, and quitting a job, all within days or weeks of each other. Although unrecognized warning signs may have existed, those who are left behind are often shocked by the suddenness and abruptness of the change.

Efforts by family members or therapists to get the individual to stop and to reconsider usually fall on deaf ears. The overwhelming need is to avoid anyone who counsels restraint and to ignore therapists who recommend examining motivations and feelings before making such major decisions. Usually, during the crisis, the therapist is left with the painful job of helping

those who have been left to deal with their shock and grief.

Empty-Nest Syndrome. Another phenomenon described in middle adulthood has been called the *empty-nest syndrome,* a depression that occurs in some men and women when their youngest child is about to leave home. Most parents, however, perceive the departure of the youngest child as a relief rather than stress. If no compensating activities have been developed, particularly by the mother, some parents become depressed. This is especially true of women whose predominant role in life has been mothering or of couples who decided to stay in an otherwise unhappy marriage "for the sake of the children."

Other Tasks of Middle Adulthood

As persons approach the age of 50, they clearly define what they want from work, family, and leisure. Men who have reached their highest level of advancement in work may experience disillusionment or frustration when they realize they can no longer anticipate new work challenges. For women who have invested themselves entirely in mothering, this period leaves them with no suitable identity after the children leave home. Sometimes, social rules become rigidly established; lack of freedom in lifestyle and a sense of entrapment can lead to depression and a loss of confidence. Also, unique financial burdens can occur in middle age, produced by pressures to care for aged parents at one end of the spectrum and children at the other end.

Daniel Levinson described a transitional period between the ages of 50 and 55, during which a developmental crisis may occur when persons feel incapable of changing an intolerable life structure. Although no single event characterizes the transition, the physiologic changes that begin to appear may have a dramatic effect on a person's sense of self. For example, a person may experience a decrease in cardiovascular efficiency that accompanies aging. Chronologic age and physical infirmity are not linear; however, those who exercise regularly, who do not smoke, and who eat and drink in moderation can maintain their physical health and emotional well-being.

Middle adulthood is when persons frequently feel overwhelmed by too many obligations and duties, but it is also a time of great satisfaction for most persons. They have developed a wide array of acquaintances, friendships, and

relationships, and the satisfaction they express about their network of friends predicts positive mental health. Some social ties, however, may be a source of stress when demands either cannot be met or assault a person's self-esteem. Middle-aged persons most generally possess power, leadership, wisdom, and understanding, and if their health and vitality remain intact, it is indeed the prime of life.

DIVORCE

Divorce is a significant crisis in life. Spouses often grow, develop, and change at different rates; one spouse may discover that the other is not the same as when they first married. In truth, both partners have changed and evolved, not necessarily in complementary directions. Frequently, one spouse blames a third person for alienation of affections and refuses to examine his or her role in the marital problems. Certain aspects of marital deterioration and divorce seem to be related to specific qualities of middle life—the need for change, weariness with acting responsibly, fear of facing up to oneself.

Types of Separation

Paul Bohannan, an anthropologist with expertise in marriage and divorce, described the types of separations that take place at the time of divorce.

Psychic Divorce. In psychic divorce, the love object is given up, and a grief reaction about the death of the relationship occurs. Sometimes a period of anticipatory mourning sets in before the divorce. Separating from a spouse forces a person to become autonomous, to change from a position of dependence. The separation may be difficult to achieve, mainly if both are used to being dependent on each other (as happens typically in marriage) or if one was so dependent as to be afraid or incapable of becoming independent. Most persons report such feelings as depression, ambivalence, and mood swings at the time of divorce. Studies indicate that recovery from divorce takes about 2 years; by then, the ex-spouse may be viewed neutrally, and each spouse accepts his or her new identity as a single person.

Legal Divorce. Legal divorce involves going through the courts so that each of the parties is remarriageable. Of divorced women and divorced men,

75 percent and 80 percent, respectively, remarry within 3 years of divorce. No-fault divorce, in which neither person is judged to be the guilty party, has become the most widely used legal mechanism for divorce.

Economic Divorce. Economic divorce involves significant concerns about the division of the couple's property between them and economic support for the wife. Many men who are ordered by the courts to pay alimony or child support flout the law and create a major social problem.

Community Divorce. The social network of the divorced couple changes markedly. A few relatives and friends are retained from the community, and new ones are added. The task of meeting new friends is often tricky for divorced persons, who may realize how dependent they were on their spouses for social exchanges.

Coparental Divorce. Coparental divorce is the separation of a parent from the child's other parent. Being a single parent differs from being a married parent.

Custody

The parental right doctrine is a legal concept that awards custody to the more fit natural parent and attempts to ensure that the best interests of the child are served. In the past, mothers were almost always awarded custody, but custody is now given to fathers in about 15 percent of cases. Custodial fathers are likely to be white, married, older, and better educated than custodial mothers. Women who are granted custody have a better chance of being awarded child support and of actually receiving payment than do men who are given custody. Nevertheless, women who accept payments still have lower incomes than men who receive compensation.

The types of custody include *joint custody,* in which a child spends equal time with each parent, an increasingly common practice; *split custody,* in which siblings are separated and each parent has custody of one or more of the children; and *single custody,* in which the children live solely with one parent and the other parent has rights of visitation that may be limited in some way by the court. Child support payments are more likely to be made when parents have joint custody or when the noncustodial parent is given

visitation rights.

Problems can surface in the parent–child relationship with the custodial or the noncustodial parent. The absence of the noncustodial parent in the home represents the reality of the divorce, and the custodial parent may become the target of the child's anger about the divorce. The parent under such stress may not be able to deal with the child's increased needs and emotional demands.

The noncustodial parent must cope with limits placed on time spent with the child. This parent loses the day-to-day gratification and the responsibilities involved with parenting. Emotional distress is common in parent and child. Joint custody offers a solution with some advantages, but it requires substantial maturity on the part of the parents and can present some problems. Parents must separate their child-rearing practices from their postdivorce resentments, and they must develop a spirit of cooperation about rearing the child. They must also be able to tolerate frequent communication with the ex-spouse.

Reasons for Divorce

Divorce tends to run in families, and rates are highest in couples who marry as teenagers or come from different socioeconomic backgrounds. Every marriage is psychologically unique, and so is each divorce. If a person's parents were divorced, he or she might choose to resolve a marital problem in the same way, through a divorce. Expectations of the spouse may be unrealistic: One partner may expect the other to act as an all-giving mother or a magically protective father. The parenting experience places the most significant strain on a marriage. In surveys of couples with and without children, those without children reported getting more pleasure from their spouse than those with children. Illness in the child creates the greatest strain of all and more than 50 percent of marriages in which a child has died through illness or accident end in divorce.

Other causes of marital distress are problems with sex and money. Both areas may be used as a means of control, and withholding sex or money is a means of expressing aggression. Also, less social pressure to remain married currently exists. As discussed above, the easing of divorce laws and the declining influence of religion and the extended family make divorce an

acceptable course of action today.

Intercourse Outside of Marriage. Adultery is defined as voluntary sexual intercourse between a married person and someone other than his or her spouse. For men, the first extramarital affair is often associated with the wife's pregnancy, when coitus may be interdicted. Most of these incidents are kept secret from the spouse and, if known, rarely account for divorce. Nevertheless, the infidelity can serve as the catalyst for basic dissatisfactions in the marriage to the surface, and these problems may then lead to its dissolution. Adultery may decline, as potentially fatal STDs such as acquired immune deficiency syndrome (AIDS) serve as sobering deterrents.

ADULT MATURITY

Success and happiness in adulthood are made possible by achieving a modicum of maturity—a mental state, not an age. The capacity for maturity, however, is a direct outgrowth of the engagement and mastery of the developmental tasks of young and middle adulthood. From a developmental perspective, maturity can be defined as a mental state found in healthy adults that is characterized by detailed knowledge of the parameters of human existence, a sophisticated level of self-awareness based on an honest appraisal of one's own experience within those basic parameters, and the ability to use this intellectual and emotional knowledge and insight caringly concerning one's self and others.

The achievement of maturity in midlife leads to the emergence of the capacity for wisdom. Those who possess wisdom have learned from the past and are fully engaged in life in the present. Just as important, they anticipate the future and make the necessary decisions to enhance prospects for health and happiness. In other words, a philosophy of life has been developed that includes understanding and acceptance of the person's place in the order of human existence. Unfortunately, the joys of midlife do not last forever. Old age lies ahead. Although the hope and statistical expectation is for many years of mental competence and independence, physical and mental decline, increased dependence, and, eventually, death must be anticipated. Late adulthood has its great pleasure, when there is a focus on continued mental and physical activity, a dominant preoccupation with the present and the

future, and involvement with and facilitation of the young. Then, death can be met with feelings of satisfaction and acceptance, the natural endpoint of human existence that follows a life lived and well loved.

32.3 Old Age

When we progress from youth to old age, the focus may change from pursuing wealth to maintaining health. In late adulthood, the aging body increasingly becomes a central concern, replacing the midlife preoccupations with career and relationships. This change is so because of diminution in function, altered physical appearance, and the increased incidence of physical illness. Despite these occurrences, the body in late adulthood can still be a source of considerable pleasure and can convey a sense of competence, particularly when giving attention to regular exercise, a healthy diet, adequate rest, and preventive maintenance medical care. The normal state in the aged is physical and mental health, not illness and debilitation. The developmental tasks of late adulthood that lead to mental health are listed in Table 32-18.



Table 32-18
Developmental Tasks of Late Adulthood

To maintain the body image and physical integrity
To conduct the life review
To maintain sexual interests and activities
To deal with the death of significant loved ones
To accept the implications of retirement
To accept the genetically programmed failure of organ systems
To divest oneself of the attachment to possessions
To accept changes in the relationship with grandchildren

Old age, or late adulthood, usually refers to the stage of the life cycle that begins at age 65. Gerontologists—those who study the aging process—divide older adults into two groups: young-old, ages 65 to 74, and old-old, ages 75 and beyond. Some use the term oldest-old to refer to those over 85. Older

adults can also be described as well-old (healthy persons) and sick-old (persons who have an infirmity that interferes with functioning and requires medical or psychiatric attention). The health needs of older adults have grown enormously as the population ages, and geriatric physicians and psychiatrists play significant roles in treating this population.

DEMOGRAPHICS

The number of individuals over age 65 is rapidly expanding. In 1900, for example, 4 percent of the US population was older than 65 years. By 2012 it was 13.7 percent, and by 2050, it is projected to be about 20 percent. That increase far exceeds the general population growth—tenfold compared with just over threefold between 1900 and 2000—and is projected to continue (e.g., 2.5 times vs. just over 1.5 times between 1990 and 2050) (Table 32-19).



Table 32-19
Aging Population of the United States: 1900–2050

Year	Median Age	Mean Age	Population, in Millions and as a Percentage of Total Population				
			All Ages	65 and Over		85 and Over	
			(N)	(N)	(%)	(N)	(%)
1900			76.0	3.1	4.1	0.1	0.1
1950			150.1	12.3	8.2	0.6	0.4
1990			248.7	31.1	12.5	3.0	1.2
2000	35.7	36.5	276.2	35.3	12.8	4.3	1.6
2010	37.2	37.8	300.4	40.1	13.3	6.0	2.0
2030	38.5	39.9	350.0	70.2	20.1	8.8	2.5
2050	38.1	40.3	392.0	80.1	20.4	18.9	4.8

Population: U.S. Bureau of the Census. Current Population Reports, Special Studies, P23-190, 65+ in the United States. Washington, DC: U.S. Government Printing Office; 1996.

Mean/Median Age, 2000–2050: Day JC. Population projections of the United States by age, sex, race, and Hispanic origin: 1995 to 2050. In: U.S. Bureau of the Census, Current Population Reports, P25–1130. Washington, DC: U.S. Government Printing Office; 1996.

The life expectancy for women at birth is projected to continue to exceed that for men by 7 years until 2050. By 2050, the composition of the US population by age and sex is estimated to differ markedly from that today. Such changes are bound to influence income and marital statistics, the percentage of elderly persons living alone or in long-term care facilities, and other aspects of the social network. A summary of demographic highlights of the aged is given in Table 32-20.

The accuracy of these projections, however, depends on the accuracy of other predications such as birth rates, immigration, and emigration—all of which are more difficult to gauge for the future than the remaining variables, death rates, or life expectancies. Projections concerning life expectancy, for example, can change substantially within a single decade.

BIOLOGY OF AGING

The aging process, or senescence (from the Latin *senescere*, "to grow old"), is characterized by a gradual decline in the functioning of all of the body's systems—cardiovascular, respiratory, genitourinary, endocrine, and immune, among others. However, the belief that old age is invariably associated with profound intellectual and physical infirmity is a myth. Many older persons retain their cognitive abilities and physical capacities to a remarkable degree.

An overview of the biologic changes that accompany old age is given in Table 32-21. The various decrements listed do not occur linearly in all systems. Not all organ systems deteriorate at the same rate, nor do they follow a similar pattern of decline for all persons. Each person is genetically endowed with one or more vulnerable systems, or a system may become vulnerable because of environmental stressors or intentional misuse, such as excessive ultraviolet exposure, smoking, or alcohol use. Moreover, not all organ systems deteriorate at the same time. Any one of several organ systems begins to deteriorate, and this deterioration then leads to illness or death.

Aging generally means the aging of cells. In the most commonly held theory, each cell has a genetically determined life span, during which it can replicate itself a limited number of times before it dies. Structural changes occur in cells with age. In the central nervous system, for example, age-related cell changes occur in neurons, which show signs of degeneration. In senility (characterized by severe memory loss and a loss of intellectual functioning), signs of degeneration are much more severe. An example is the neurofibrillary degeneration seen most commonly in dementia of the Alzheimer's type.

Structural changes and mutations in deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) are also found in aging cells; these have been attributed to genotypic programming, x-rays, chemicals, and food products, among others. Probably no single cause of aging exists, and all areas of the body are affected to some degree. Genetic factors have been implicated in disorders that commonly occur in older persons, such as hypertension, coronary artery disease, arteriosclerosis, and neoplastic disease. Family studies indicate inheritance factors for breast and stomach cancer, colon polyps, and certain mental disorders of old age. Huntington disease shows an autosomal dominant mode of inheritance with complete penetrance. The

average age of onset is between 35 and 40 years, but cases have occurred as late as 70 years.

Longevity

Longevity has been studied since the beginning of recorded history and has always been a topic of great interest. The research about longevity reveals that a family history of longevity is the best indicator of a long life; of persons who live past 80, half of their fathers also lived past 80. Nevertheless, many conditions leading to a shortened life can be prevented, ameliorated, or delayed with effective intervention. Heredity is but one factor—one beyond a person's control. Predictors of longevity that are within a person's control include regular medical checkups, minimal or no caffeine or alcohol consumption, work gratification, and a perceived sense of the self as being socially useful in an altruistic role, such as a spouse, teacher, mentor, parent, or grandparent. Healthy eating and adequate exercise are also associated with health and longevity.



Table 32-20
Demographic Highlights of the Aged

- The older population (65+) numbered 4.04 million in 2010, an increase of 5.4 million for 15.3 since 2000.
- The number of Americans aged 45–64—who will reach 65 over the next two decades—increased by 31% during this decade.
- Over one in every eight, or 13.1%, of the population is an older American.
- Persons reaching age 65 have an average life expectancy of an additional 18.8 yr (20 yr for females and 17.3 for males).
- Older women outnumber older men at 23 million older women to 17.5 million older men.
- In 2010, 20% of persons 65+ were minorities—8.4% were African American.[a] Persons of Hispanic origin (who may be of any race) represented 6.9% of the older population. About 3.5% were Asian or Pacific Islanders,[a] and less than 1% were American Indian or Native Alaskan.[a]

- In addition, 0.8% of persons 65+ identified themselves as being of two or more races.
- Older men were much more likely to be married than older women —72% of men vs. 42% of women. In 2010, 40% older women were widows.
- About 29% (11.3 million) of noninstitutionalized older persons live alone (8.1 million women, 3.2 million men).
- Almost half of older women (47%) age 75+ live alone.
- About 485,000 grandparents aged 65 or more had the primary responsibility for their grandchildren who lived with them.
- The population 65 and over has increased from 35 million in 2000 to 40 million in 2010 (a 15% increase) and is projected to increase up to 55 million in 2020 (a 36% increase for that decade).
- The 85+ population is projected to increase from 5.5 million in 2010 and then 6.6 million in 2020 (19%) for that decade.
- Minority populations have increased from 5.7 million in 2000 (16.3% of the elderly population) to 8.1 million in 2010 (20% of the elderly) and are projected to increase to 13.1 million in 2020 (24% of the elderly).
- The median income of older persons in 2010 was $25,704 for males and $15,072 for females. Median money income (after adjusting for inflation) of all households headed by older people fell 1.5% (not statistically significant) from 2009 to 2010. Households containing families headed by person 65+ reported a median income in 2010 of $45,763.
- The major sources of income as reported by older persons in 2009 were Social Security (reported by 87% of older persons), income from assets (reported by 53%), private pensions (reported by 28%), government employee pensions (reported by 14%), and earnings (reported by 26%).
- Social Security constituted 90% or more of the income received by 35% of beneficiaries in 2009 (22% married couples and 43% of nonmarried beneficiaries).
- Almost 3.5 million elderly persons (9.0%) were below the poverty level in 2010. This poverty rate is not statistically different from the

poverty rate in 2009 (8.9%). During 2011, the U.S. Census Bureau also released a new Supplemental Poverty Measure (SPM) that takes into account regional variations in the living costs, noncash benefits received, and nondiscretionary expenditures but does not replace the official poverty measure. The SPM shows a poverty level for older persons of 15.9%, an increase of over 75% over the official rate of 9% mainly due to medical out-of-pocket expenses.

- About 11% (3.7 million) of older Medicare enrollees received personal care from a paid or unpaid source in 1999.

[a]Principle source data for the profile are from the U.S. Census Bureau, the National Center for Health Statistics, and the Bureau of Labor Statistics. The profile incorporates the latest data (2010) available but not all items are updated on an annual basis.



Table 32-21
Biologic Changes Associated with Aging

Cellular Level

Change in cellular DNA (deoxyribonucleic acid) and RNA (ribonucleic acid) structures: intracellular organelle degeneration

Neuronal degeneration in central nervous system, primarily in superior temporal precentral and inferior temporal gyri; no loss in brainstem nuclei

Receptor sites and sensitivity altered

Decreased anabolism and catabolism of cellular transmitter substances

Intercellular collagen and elastin increase

Immune System

Impaired T-cell response to antigen

Increase in function of autoimmune bodies

Increased susceptibility to infection and neoplasia

Leukocytes unchanged, T lymphocytes reduced

Increased erythrocyte sedimentation (nonspecific)

Musculoskeletal

Decrease in height because of shortening of spinal column (2-in

- loss in both men and women from the second to the seventh decades)
- Reduction in lean muscle mass and muscle strength; deepening of thoracic cage
- Increase in body fat
- Elongation of nose and ears
- Loss of bone matrix, leading to osteoporosis
- Degeneration of joint surfaces may produce osteoarthritis
- Risk of hip fracture is 10–25% by age 90
- Continual closing of cranial sutures (parietomastoid suture does not attain complete closure until age 80)
- Men gain weight until about age 60, then lose; women gain weight until age 70, then lose

Integument

- Graying of hair results from decreased melanin production in hair follicles (by age 50, 50% of all persons male and female are at least 50% gray; pubic hair is last to turn gray)
- General wrinkling of skin
- Less active sweat glands
- Decrease in melanin
- Loss of subcutaneous fat
- Nail growth slowed

Genitourinary and Reproductive

- Decreased glomerular filtration rate and renal blood flow
- Decreased hardness of erection, diminished ejaculatory spurt
- Decreased vaginal lubrication
- Enlargement of prostate
- Incontinence

Special Senses

- Thickening of optic lens, reduced peripheral vision
- Inability to accommodate (presbyopia)
- High-frequency sound hearing loss (presbyacusis)—25% show loss by age 60, 65% by age 80

- Yellowing of optic lens
- Reduced acuity of taste, smell, and touch
- Decreased light-dark adaption

Neuropsychiatric

- Takes longer to learn new material, but complete learning still occurs
- Intelligence quotient (IQ) remains stable until age 80
- Verbal ability maintained with age
- Psychomotor speed declines

Memory

- Tasks requiring shifting attentions performed with difficulty
- Encoding ability diminishes (transfer of short- to long-term memory and vice versa)
- Recognition of right answer on multiple-choice tests remains intact
- Simple recall declines

Neurotransmitters

- Norepinephrine decreases in central nervous system
- Increased monoamine oxidase and serotonin in brain

Brain

- Decrease in gross brain weight, about 17% by age 80 in both sexes
- Widened sulci, smaller convolutions, gyral atrophy
- Ventricles enlarge
- Increased transport across blood–brain barrier
- Decreased cerebral blood flow and oxygenation

Cardiovascular

- Increase in size and weight of heart (contains lipofuscin pigment derived from lipids)
- Decreased elasticity of heart valves
- Increased collagen in blood vessels
- Increased susceptibility to arrhythmias
- Altered homeostasis of blood pressure

Cardiac output maintained in absence of coronary heart disease

Gastrointestinal (GI) System

At risk for atrophic gastritis, hiatal hernia, diverticulosis
Decreased blood flow to gut, liver
Diminished saliva flow
Altered absorption from GI tract (at risk for malabsorption syndrome and avitaminosis)
Constipation

Endocrine

Estrogen levels decrease in women
Adrenal androgen decreases
Testosterone production declines in men
Increase in follicle-stimulating hormone (FSH) and luteinizing hormone (LH) in postmenopausal women
Serum thyroxine (T_4) and thyroid-stimulating hormone (TSH) normal, triiodothyronine (T_3) reduced
Glucose tolerance test result decreases

Respiratory

Decreased vital capacity
Diminished cough reflex
Decreased bronchial epithelium ciliary action

Life Expectancy

In the United States, the average life expectancy has increased throughout the last century—from 48 years in 1900 to 77.4 years for men and 82.2 years for women in 2013. The projected life expectancy at birth, and age 65 is indicated in Table 32-22. Changes in morbidity and mortality have also occurred. Over the past 30 years, for example, a 60 percent decline has occurred in mortality from cerebrovascular disease and a 30 percent decline in mortality from coronary artery disease. In contrast, mortality from cancer, which rises steeply with age, has increased, especially cancer of the lung, colon, stomach, skin, and prostate.

The oldest-old, persons over 85 years of age, is the most rapidly growing

segment of the older population. Over the past 25 years, the population of all older persons increased by 100 percent, compared with 45 percent for the entire US population, but the increase for the 85 and older group exceeded 275 percent. It is expected that by 2050, the oldest-old will make up about 25 percent of the elderly population and 5 percent of the total population in the United States. Figure 32-7 gives projected percentages for the average annual growth rate of the elderly population to 2050.

Table 32-22
Projected Life Expectancy at Birth and Age 65, by Sex: 1990–2050 (in Years)

	At Birth			At Age 65		
Year	Men	Women	Difference	Men	Women	Difference
1990	72.1	79.0	6.9	15.0	19.4	4.4
2000	73.5	80.4	6.9	15.7	20.3	4.6
2010	74.4	81.3	6.9	16.2	21.0	4.8
2020	74.9	81.8	6.9	16.6	21.4	4.8
2030	75.4	82.3	6.9	17.0	21.8	4.8
2040	75.9	82.8	6.9	17.3	22.3	5.0
2050	76.4	83.3	6.9	17.7	22.7	5.0

Data from U.S. Bureau of the Census, Washington, DC.

The leading causes of death among older persons are heart disease, cancer, and stroke. Accidents are among the leading causes of death of persons over 65. Most fatal accidents are caused by falls, pedestrian incidents, and burns. Falls are most commonly the result of cardiac arrhythmias and hypotensive episodes.

Some gerontologists consider death in very old persons (over 85) to result from an aging syndrome characterized by diminished elastic-mechanical properties of the heart, arteries, lungs, and other organs. Death results from trivial tissue injuries that would not be fatal to a younger person; accordingly,

senescence is viewed as the cause of death.

Ethnicity and Race

The proportion of older persons in the black, Hispanic, and Asian populations is smaller than that in the white population, but it is increasing rapidly. By 2050, 20 percent of older persons will be nonwhite. The proportion of older persons who are Hispanic will increase from 4 percent to approximately 14 percent over the same period. According to the U.S. Census Bureau, Hispanic refers to persons "whose origins are Mexican, Puerto Rican, Cuban, Central or South American, and other Hispanic or Latino, regardless of race" (Fig. 32-8).

Sex Ratios

On average, women live longer than men and are more likely than men to live alone. The number of men per 100 women decreases sharply from ages 65 to 85 (Fig. 32-9).

Geographic Distribution

The most populous states have the largest number of older persons. California has the most (3.3 million), followed by New York, Pennsylvania, Texas, Michigan, Illinois, Florida, and Ohio, each with more than 1 million. States with high proportions of older persons include Pennsylvania, Florida, Nebraska, and North Dakota. Florida's high percentage of older adults related its popularity as a retirement locale. In other states, the reason is usually more young persons leaving.

Exercise, Diet, and Health

Diet and exercise play a role in preventing or ameliorating chronic diseases of older persons, such as arteriosclerosis and hypertension. Hyperlipidemia, which correlates with coronary artery disease, can be controlled by reducing body weight, decreasing the intake of saturated fat, and limiting the intake of cholesterol. Increasing the daily intake of dietary fiber can also help decrease serum lipoprotein levels. A daily intake of 1 oz (about 30 mL) of alcohol has been correlated with longevity and elevated high-density lipoproteins (HDLs). Studies have also clearly demonstrated that statin drugs that reduce

cholesterol have a dramatic effect on reducing cardiovascular disease in persons with diet-resistant or exercise-resistant hyperlipidemia.

Low salt intake (less than 3 g a day) is associated with a lowered risk of hypertension. Hypertensive geriatric patients can often correct their condition by moderate exercise and decreased salt intake without the addition of drugs.

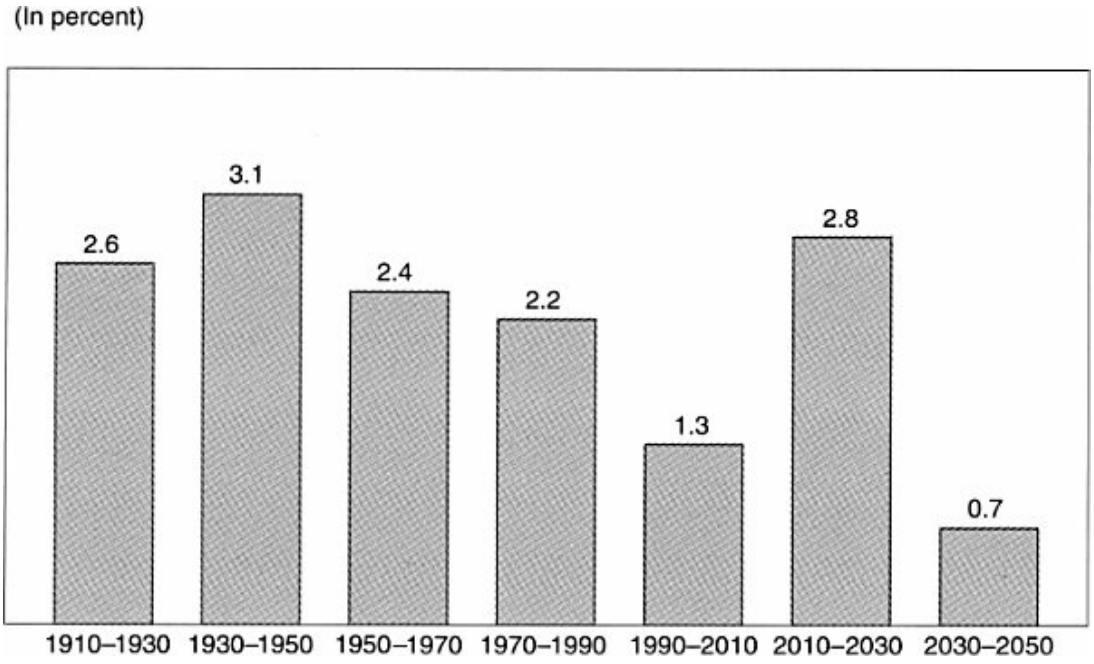


FIGURE 32-7
Average annual growth rate of the elderly population. (Data from U.S. Bureau of the Census.)

A regimen of moderate daily exercise (walking for 30 minutes a day) has been associated with a reduction in cardiovascular disease, decreased incidence of osteoporosis, improved respiratory function, the maintenance of ideal weight, and a general sense of well-being. Exercise has been shown to improve strength and function even among the very old. In many cases, a disease process has been reversed and even cured by diet and exercise, without additional medical or surgical intervention.

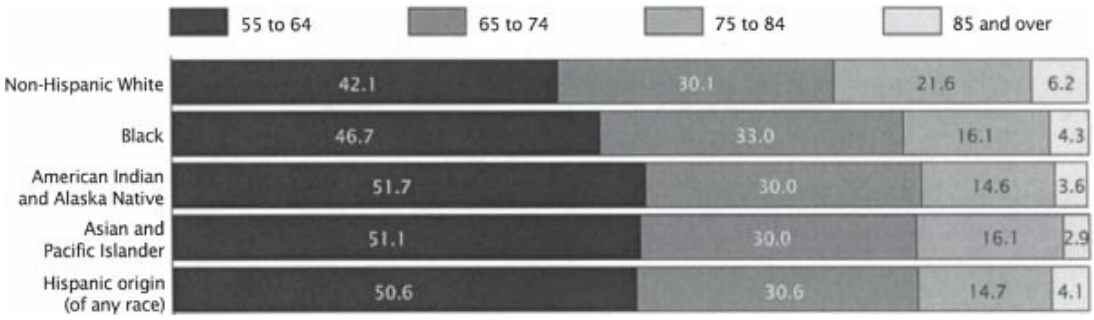


FIGURE 32-8
Percent distribution of people 55 years and over by race, Hispanic origin, and age: 2002. (Data from U.S. Bureau of the Census.)

Table 32-23 lists the biologic changes associated with diet and exercise. A comparison with Table 32-19 reveals that almost every biologic change associated with aging is positively affected by diet and exercise.

STAGE THEORIES OF PERSONALITY DEVELOPMENT

Early personality theorists proposed that development was completed by the end of childhood or adolescence. One of the first development theorists to propose that personality continues to develop and grow over the life span was Erik Erikson. Erikson believed that development proceeded through a series of psychosocial stages, each with its conflict that is resolved by the individual with greater or lesser success. Erikson termed the crisis of the last epoch of life integrity versus despair and believed that successful resolution of this crisis involved a process of life review and achieving a sense of peace and wisdom through coming to terms with how one's life was lived. For example, Erikson proposed that successful resolution of this crisis would be characterized by a sense of having lived one's life well, whereas a less successful resolution would be characterized by feeling that life was too short, that one did not choose wisely, and bitterness that one will not have a chance to live life over.

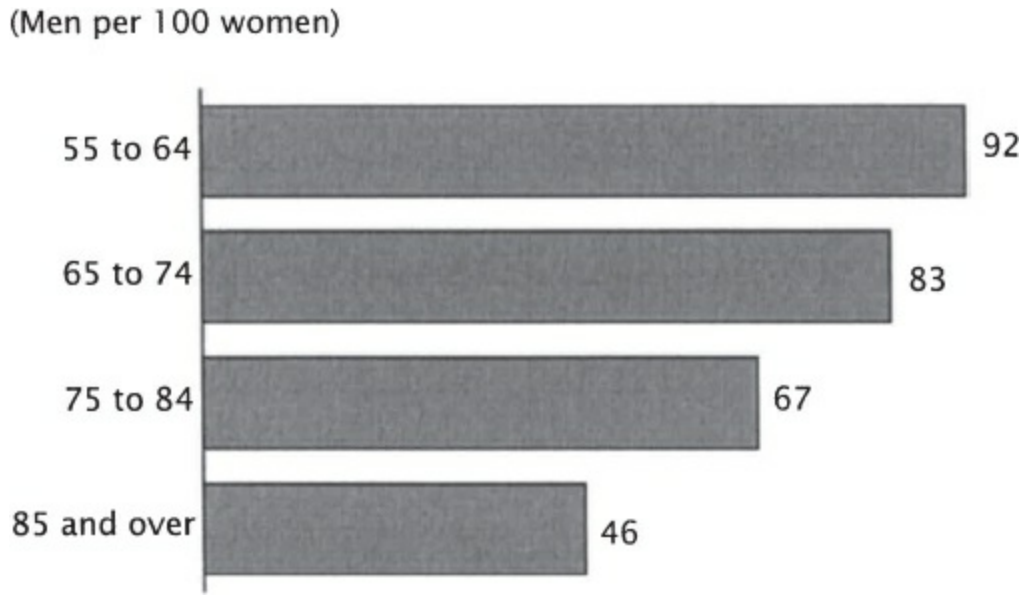


FIGURE 32-9
Sex ratio of people 55 years and over by age: 2002. (Data from U.S. Bureau of the Census.)

Table 32-23
Positive and Healthy Physiologic Effects of Exercise and Nutrition

Increases

- Strength of bones, ligaments, and muscles
- Muscle mass and body density
- Articular cartilage thickness
- Skeletal muscle ATP (adenosine triphosphatase), CRP (C-reactive protein), K^+ (potassium), and myoglobin
- Skeletal muscle oxidative enzyme content and mitochondria
- Skeletal muscle arterial collaterals and capillary density
- Heart volume and weight
- Blood volume and total circulating hemoglobin
- Cardiac stroke volume
- Myocardial contractility
- Maximal CO_2 (A-V)
- Maximal blood lactate concentration
- Maximal pulmonary ventilation

Maximal respiratory work
Maximal oxygen diffusing capacity
Maximal exercise capacity as measured by the maximal oxygen intake, exercise time, and distance
Serum high-density lipoprotein concentration
Anaerobic threshold
Plasma insulin concentration with submaximal exercise

Decreases

Heart rate at rest and during submaximal exercise
Blood lactate concentration during submaximal exercise
Pulmonary ventilation during submaximal work
Respiratory quotient during submaximal work
Serum triglyceride concentration
Body fatness
Serum low-density lipoprotein concentration
Systolic blood pressure
Core temperature threshold for initiation of sweating
Sweat sodium and chloride content
Plasma epinephrine and norepinephrine with submaximal exercise
Plasma glucagon and growth hormone concentrations with submaximal exercise
Relative hemoconcentration with submaximal exercise in the heat

Reprinted from Buskirk ER. In: White PL, Monderka T, eds. *Diet and Exercise: Synergism in Health Maintenance*. Chicago: American Medical Association; 1982:133, with permission.



Table 32-24
Old Age Developmental Theorists

Sigmund Freud	Increasing control of the ego and id with aging results in increased autonomy. Regression may permit primitive modes of functioning to reappear.
Erik Erikson	The central conflict in old age is between integrity, the sense of satisfaction people feel reflecting on a life lived productively, and despair, the sense that life

	has little purpose or meaning. Contentment in old age comes only with getting beyond narcissism and into intimacy and generativity.
Heinz Kohut	Old people must continually cope with narcissistic injury as they attempt to adapt to the biologic, psychological, and social losses associated with the aging process. The maintenance of self-esteem is a major task of old age.
Bernice Neugarten	The major conflict of old age relates to giving up the position of authority and evaluating achievements and former competence. It is a time of reconciliation with others and resolution of grief over the death of others and the approaching death of self.
Daniel Levinson	Ages 60–65 is a transition period (“the late adult transition”). People who are narcissistic and too heavily invested in body appearance are liable to become preoccupied with death. Creative mental activity is a normal and healthy substitute for reduced physical activity.

Several studies have attempted to validate aspects of Erikson’s theory. In one study, a sample of more than 400 men was studied prospectively, and the highest Eriksonian life stage each achieved was rated according to data gathered on the circumstances of his life. For example, if a man had achieved independence from his family of origin and was self-sufficient but was unable to develop an intimate relationship, the highest life stage achieved would be the identity stage, not the intimacy stage. This study found that Eriksonian stages are passed through in sequential order, although often not at the same age for every individual, and that the stages are surprisingly universal in populations that are ethnically and socioeconomically diverse.

A longitudinal study of approximately 500 subjects from two age cohorts found that the earlier age cohort scored significantly higher on integrity than the latter age cohort, and scores for both age cohorts on integrity had declined significantly by the final time of testing. These data suggest that the conflict

of integrity versus despair may have a more favorable outcome in earlier age cohorts than in later ones, raising the possibility that changing societal values have hurt the struggle for integrity. Another study found that wisdom, a construct related to integrity, bore a more substantial relation to life satisfaction in elderly adults than other variables, including finances, health, and living situation.

A survey of theories of development in old age is given in Table 32-24.

Personality Over the Life Span: Stability or Change?

Although Erikson and other stage theorists focused on unique developmental tasks and stages central to each phase of life, other theorists focused on defining core personality traits within the individual and determining their course over the life span. For example, do those who are gregarious or extroverted during early childhood and adolescence remain extroverted through midlife and old age? Several well-designed longitudinal studies that have followed individuals over periods ranging from 10 to 50 years have found strong evidence for stability in five essential personality traits: extroversion, neuroticism, agreeableness, openness to experience, and conscientiousness. Some studies found slight decreases in extroversion and slight increases in agreeableness as individuals move into the oldest-old category, which contrasts with early theories that proposed that personality rigidifies as individuals age.

Is the fact that personality appears to have considerable stability over time inconsistent with the basic tenets of stage theories? Perhaps not. It may be that although individuals are consistent over time in their basic personality structure, the themes and conflicts with which they struggle change considerably over the life span, from concerns about developing identity and a stable sense of self to finding a life partner, to issues related to life review, as hypothesized by the stage theories. Also, in developing theories about personality change, few studies have examined the impact of significant historical events on personality; thus, how these events may result in personality change have not been studied systematically.

PSYCHOSOCIAL ASPECTS OF AGING

Social Activity

Healthy older persons usually maintain a level of social activity that is only slightly changed from that of earlier years. For many, old age is a period of continued intellectual, emotional, and psychological growth. In some cases, however, physical illness or the death of friends and relatives may preclude continued social interaction. Moreover, as persons experience an increased sense of isolation, they may become vulnerable to depression. Growing evidence indicates that maintaining social activities is valuable for physical and emotional well-being. Contact with younger persons is also essential. Old persons can pass on cultural values and provide care services to the younger generation and thereby maintain a sense of usefulness that contributes to self-esteem.

Ageism

Ageism, a term coined by Robert Butler, refers to discrimination toward old persons and to the negative stereotypes about old age that are held by younger adults. Old persons may themselves resent and fear other old persons and discriminate against them. In Butler's scheme, persons often associate old age with loneliness, poor health, senility, and general weakness or infirmity. The experience of older persons, however, does not consistently support this attitude. For example, although 50 percent of young adults expect poor health to be a problem for those over 65 years old, 75 percent of persons 65 to 74 years of age describe their health as good. Two-thirds of persons 75 and older feel the same way. Health problems, when they do exist, more often involve chronic than acute conditions. More than four of five persons over the age of 65 have at least one chronic condition (Table 32-25).

Good health, however, is not the sole determinant of a good quality of life in old age. Surveys of old persons show that social contacts are at least as highly valued. The factors affecting good aging appear to be multidimensional. Aging "robustly" means considering aging in terms of productive involvement, affective status, functional status, and cognitive status. These four indicators are only minimally correlated. The most robustly aging individuals report more significant social contact, better health and vision, and fewer significant life events in the past 3 years than their less robustly aging counterparts. A linear, age-related decrease occurs in

robustness, but it can still be found among the oldest-old.

Table 32-25
Top Ten Chronic Conditions for People 65+, by Age and Race (Number Per 1,000 People)

	Age				Race (65+)		
Condition	65+	45–64	65–74	75+	White	Black	Black as of White
Arthritis	483.0	253.8	437.3	554.5	483.2	522.6	108
Hypertension	380.6	229.1	383.8	375.6	367.4	517.7	141
Hearing impairment	286.5	127.7	239.4	360.3	297.4	174.5	59
Heart disease	278.9	118.9	231.6	353.0	286.5	220.5	77
Cataracts	156.8	16.1	107.4	234.3	160.7	139.8	87
Deformity or orthopedic impairment	155.2	155.5	141.4	177.0	156.2	150.8	97
Chronic sinusitis	153.4	173.5	151.8	155.8	157.1	125.2	80
Diabetes	88.2	58.2	89.7	85.7	80.2	165.9	207
Visual impairment	81.9	45.1	69.3	101.7	81.1	77.0	95
Varicose veins	78.1	57.8	72.6	86.6	80.3	64.0	80

Data from National Center for Health Statistics, Washington, DC.

George Vaillant followed up a group of Harvard freshmen into old age and found the following about emotional health at age 65: Having been close to brothers and sisters during college correlated with emotional well-being; undergoing early traumatic life experiences, such as the death of a parent or parental divorce, did not correlate with poor adaptation in old age; being depressed at some point between ages 21 and 50 predicted emotional problems at age 65, and possessing the personality traits of pragmatism and dependability as a young adult was associated with a sense of well-being at age 65.

Transference

Several forms of transference, some of them unique to adulthood, are present in older adults. First is the well-recognized parental transference, in which the patient reacts to the therapist as a child to a parent. Peer or sibling transference, expressions of experiences from a variety of nonparental relationships, is also common. In this form of transference, the patient looks to the therapist to share experiences with siblings, spouses, friends, and associates. At first, therapists may be surprised by older patients' ability to ignore their age in creating such transferences.

In son or daughter transference, quite common in middle-aged individuals and the elderly, the therapist is cast in the role of the patient's child, grandchild, or son-in-law or daughter-in-law. The themes expressed in this form of transference are multiple and often center on defenses against dependency feelings, activity and dominance versus passivity and submission, and attempts to rework unsatisfying aspects of relationships with children before time runs out. Finally, sexual transferences in older individuals are frequent and intense, and the therapist needs to be able to accept them and manage his or her countertransference responses.

Countertransference

Older individuals are dealing with illness and signs of aging, the loss of spouses and friends, and the constant awareness of time limitation and the nearness of death. These are painful issues that are just beginning to come into focus for younger therapists who would prefer not to confront them with great intensity daily.

The second source of countertransference responses centers on the older patient's sexuality. The presence of a vivid fantasy life, masturbation, and intercourse is disconcerting in and of themselves if the therapist has not had much experience in working with individuals who are the same age as their parents and grandparents. Consider the experience presented in the case study of a 31-year-old female therapist who was treating a 62-year-old man.

> Early in the treatment process, Mr. E's sexual feelings emerged. His well-groomed appearance and adolescent-like nervousness caused the therapist discomfort. Her concern was how to engender respect and develop a therapeutic alliance with a patient who approached each session as a date, mainly because he was old enough to be her grandfather. At first shocked by his open expression of sexual interest in her, with the help of supervision and her therapy, she was able to recognize that she and the patient had similar conflicts to resolve, despite the 30-year age difference between them. She had hoped that Mr. E would be "all grown up," devoid of issues that she was grappling with also. She came to recognize that failure to help him understand the relation between his past and still vibrant sexuality would do the patient a great disservice. It would spring

from her a lack of understanding of late-life sexual development and her countertransference reaction to him based on her conflicted attitudes toward the sexuality of her parents and grandparents. (Courtesy of Calvin A. Colarusso, M.D.)

Socioeconomics

The economics of old age is of paramount importance to older persons themselves and society at large. The past 30 years have seen a dramatic decline in the proportion of the US elderly population who are poor, primarily as a result of the availability of Medicare, Social Security, and private pensions. In 1959, 35.2 percent of persons over 65 lived below the poverty line, but by 2012 this figure had declined to 9.1 percent. Persons over age 65 make up 12 percent of the population, but they include only 9 percent of those living at low socioeconomic levels. Women are more likely than men to be impoverished. Income sources vary for persons age 65 and older. Despite overall economic gains, many older persons are so preoccupied with money worries that their enjoyment of life is lessened. Obtaining proper medical care may be incredibly difficult when personal funds are not available or are insufficient.

Medicare (Title 18) provides both hospital and medical insurance for those over age 65. About 150 million medical bills are reimbursed under the Medicare program each year, but only about 40 percent of all medical expenses incurred by older persons are covered under Medicare. The rest is paid by private insurance, state insurance, or personal funds. Some services—such as outpatient psychiatric treatment, skilled nursing care, physical rehabilitation, and preventive physical examinations—are covered minimally or not at all.

In addition to Medicare, the Social Security program pays benefits to persons over age 65 (over age 67 in 2027) and pays benefits at reduced rates starting at age 62. To qualify for benefits, a person must have worked long enough to become insured: A worker must have worked for 10 years to be eligible for benefits. Benefits are also paid to widows, widowers, and dependent children if those receiving benefits or contributing to Social Security die (survivor benefits). Social Security is not a pension scheme but a pay-as-you-go income supplement to prevent mass destitution among older

persons. Benefits are paid by those currently working to those who are retired. Serious difficulties for Social Security are forecast for the next three decades when the number of baby boomers reaching old age will significantly exceed the number of younger workers paying into the plan.

Retirement

For many older persons, retirement is a time for the pursuit of leisure and freedom from the responsibility of previous working commitments. For others, it is a time of stress, mainly when retirement results in economic problems or a loss of self-esteem. Ideally, employment after age 65 should be a matter of choice. With the passage of the Age Discrimination in Employment Act of 1967 and its amendments, forced retirement at age 70 has been virtually eliminated in the private sector, and it is not legal in federal employment.

Most of those who retire voluntarily reenter the workforce within 2 years, for a variety of reasons, including adverse reactions to being retired, feelings of being unproductive, economic hardship, and loneliness. The amount of time spent in retirement has increased as the life span has nearly doubled since 1900. Currently, the number of years spent in retirement is almost equal to the number of years spent working.

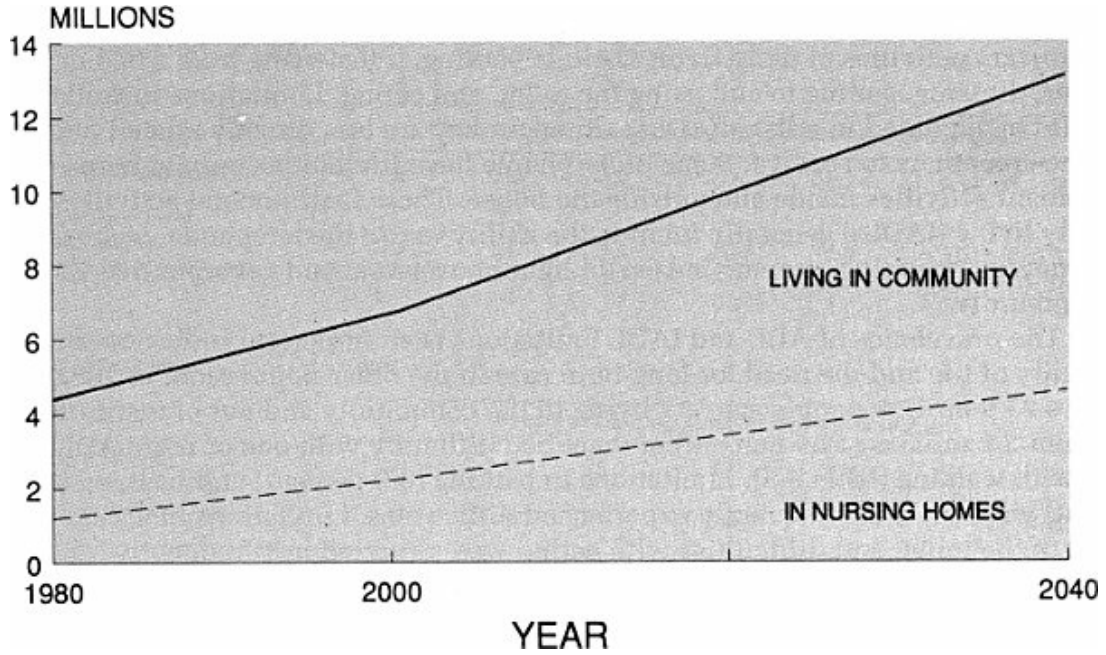


FIGURE 32-10

People age 65 or older in need of long-term care: 1980–2040. (Reprinted from Manton B, Saldo J. *Dynamics of Health Changes in the Oldest Old: New Perspectives and Evidence*. Milbank Q; 1985;63:12, with permission.)

Sexual Activity

The frequency of orgasm, from coitus or masturbation, decreases with age in men and women. The most critical factors in determining the level of sexual activity with age are the health and survival of the spouse, one's health, and the level of past sexual activity. Although some degree of declining sexual interest and function is inevitable with age, social and cultural factors appear to be more responsible for the sexual changes observed than for the psychological changes of aging per se. Although satisfying sexual activity is possible for reasonably healthy elderly, many do not actualize this potential. The widely held notion that the elderly are essentially asexual is often a self-fulfilling prophecy.

Long-Term Care

Many older persons who are infirm require institutional care. Although only 5 percent are institutionalized in nursing homes at any one time, about 35 percent of older persons require care in a long-term facility at some time during their lives (Fig. 32-10). Older nursing home residents are mainly widowed women, and about 50 percent are over age 85.

Medicare does not cover nursing home care costs; they range from $20,000 to $1 million a year. About 20,000 long-term nursing care institutions are available in the United States. This number is not enough to meet the need. Those older persons who do not require skilled nursing care can be managed in other types of health-related facilities, such as centers they attend during the daytime hours, but the need for care far exceeds the availability of such centers.

Besides institutions, the older person's children tend to care for them. The most common caregivers are daughters and daughters-in-law, wives, and other women (Fig. 32-11). More than 50 percent of these women caregivers also work, and about 40 percent also care for their children. In general, women end up as caregivers more often than men because of cultural and societal expectations. According to the American Association of Retired Persons, daughters with jobs spend an average of 12 hours a week providing

care and currently spend about $150 a month for travel, telephone calls, special foods, and medication for older persons.

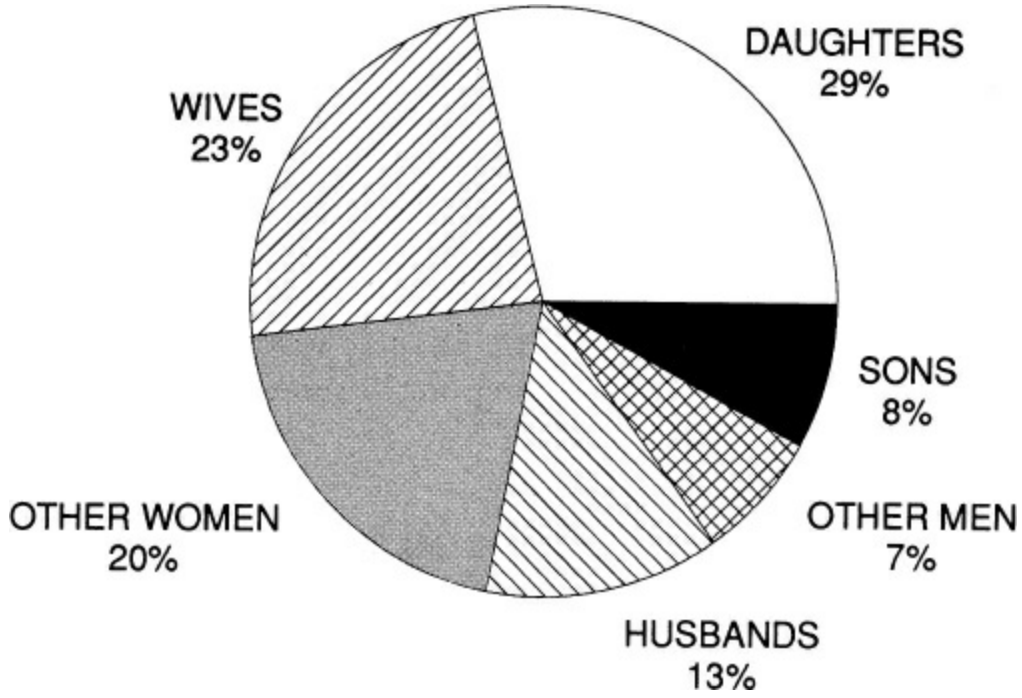


FIGURE 32-11
Caretakers and their relationship to the elderly care recipient. (Data from Select Committee on Aging, U.S. House of Representatives.)

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33 Contributions from the Neurosciences

The human brain is responsible for our cognitive processes, emotions, and behaviors—that is, everything we think, feel, and do. Although the early development and adult function of the brain are shaped by multiple factors (e.g., epigenetic, environmental, and psychosocial experiences), the brain is the final integrator of these influences. Despite the many advances in neural sciences, there has not been a genuine transformational advance in the treatment of mental disorders for more than half a century. The most apparent reason for the absence of more progress is the profound complexity of the human brain. A perhaps less apparent reason is the current practice of psychiatric diagnosis, which, for most clinicians, is based on syndrome-based classification systems.

The purpose of this chapter is to introduce the neural sciences sections, which describe the anatomy and function of the human brain, and then to discuss how an evolution of thinking toward a brain-based or biologically based diagnostic system for mental illness might facilitate our efforts to advance brain research, to develop better treatments, and to improve patient care.

In other fields of medicine, diagnosis is based on physical signs and symptoms, a medical history, and results of laboratory and radiologic tests. In psychiatry, a diagnosis is based primarily on the clinician's impression of the patient's interpretation of his or her thoughts and feelings. The patient's symptoms are then cross-referenced to a diagnostic or classification manual (e.g., *Diagnostic and Statistical Manual of Mental Disorders* [DSM-5], *International Statistical Classification of Diseases and Related Health Problems* [ICD]) containing hundreds of potential syndromes, and one or more diagnoses are applied to the particular patient. These standard

classification systems represent significant improvements in reliability over previous diagnostic systems, but there is little reason to believe that these diagnostic categories are valid, in the sense that they represent discrete, biologically distinct entities. Although a patient with no symptoms or complaints can be diagnosed as having diabetes, cancer, or hypertension based on blood tests, x-rays, or vital signs, a patient with no symptoms cannot be diagnosed with schizophrenia, for example, because there are no currently recognized objective, independent assessments.

The goals of clinicians and researchers are to reduce human suffering by increasing our understanding of diseases, developing new treatments to prevent or cure diseases, and caring for patients optimally. If the brain is the organ of focus for mental illnesses, then it may be time to be more ambitious in building the classification of patients with mental illnesses directly from our understanding of biology, rather than only from the assessment of a patient's symptoms.

THE HUMAN BRAIN

The following neural sciences sections each address a field of brain biology. Each of these fields could be relevant to the pathophysiology and treatment of mental illnesses. Although the complexity of the human brain is daunting compared with other organs of the body, progress can only be made if one approaches this complexity consistently, methodically, and bravely.

The neuronal and glial cells of the human brain are organized in a characteristic manner, which has been increasingly clarified through modern neuroanatomical techniques. Also, our knowledge of normal human brain development has become more robust in the last decade. The human brain evolved from the brains of lower animal species, allowing inferences to be made about the human brain from animal studies. Neurons communicate with one another through chemical and electrical neurotransmission. The major neurotransmitters are the monoamines, amino acids, and neuropeptides. Other chemical messengers include neurotrophic factors and an array of other molecules, such as nitric oxide. Electrical neurotransmission occurs through a wide range of ion channels. Chemical and electrical signals received by a neuron subsequently initiate various molecular pathways within other neurons that regulate the biology and function of individual neurons,

including the expression of individual genes and the production of proteins.

Also, to the central nervous system (CNS), the human body contains two other systems that have complex internal communicative networks: the endocrine system and the immune system. The recognition that these three systems communicate with each other has given birth to the fields of psychoneuroendocrinology and psychoneuroimmunology. Another property shared by the CNS, the endocrine system, and the immune system is the regular changes they undergo with time (e.g., daily, monthly), which is the basis of the field of chronobiology.

Psychiatry and the Human Brain

In the first half of the 20th century, our emphasis on psychodynamic and social psychiatry divorced psychiatric research from the study of the human brain. Since the 1950s, the appreciation of the effectiveness of medications in treating mental disorders and the mental effects of illicit drugs have reestablished a biologic view of mental illness, which had already been seeded by the introduction of electroconvulsive therapy (ECT) and James Papez's description of the limbic circuit in the 1930s. This biologic view has been reinforced further by the development of brain imaging techniques that have helped reveal how the brain performs in normal and abnormal conditions. During this period, countless discoveries have been made in basic neural science research using experimental techniques to assess the development, structure, biology, and function of the CNS of humans and animals.

Psychopharmacology. The effectiveness of drugs in the treatment of mental illness has been a significant feature of the last half-century of psychiatric practice. The first five editions of this textbook divided psychopharmacological treatment into four chapters on antipsychotic, antidepressant, antianxiety, and mood-stabilizing drugs. The prior division of psychiatric drugs into four classes is less valid now than it was in the past for the following reasons: (1) Many drugs of one class are used to treat disorders previously assigned to another class; (2) drugs from all four categories are used to treat disorders not previously treatable by drugs (e.g., eating disorders, panic disorders, and impulse control disorders); and (3) drugs such

as clonidine, propranolol, and verapamil can effectively treat a variety of psychiatric disorders and do not fit easily into the other categories of psychiatric drugs.

The primary motivation for this change was that the variety and application of the drug treatments no longer fit a division of disorders into psychosis, depression, anxiety, and mania. In other words, the clinical applications of biologically based treatments did not neatly align with our syndrome-based diagnostic system. An implication of this observation could be that drug response might be a better indicator of underlying biologic brain dysfunction than any particular group of symptoms. For example, although the DSM-5 distinguishes major depressive disorder from generalized anxiety disorder, most clinicians are aware that these are often overlapping symptoms and conditions in clinical practice. Moreover, the same drugs are used to treat both conditions.

The animal models that are used to identify new drug treatments may also have affected our ability to advance research and treatment. Many major classes of psychiatric drugs were discovered serendipitously. Specifically, the drugs were developed originally for nonpsychiatric indications, but observant clinicians and researchers noted that psychiatric symptoms improved in some patients, which led to a focused study of these drugs in psychiatric patients. The availability of these effective drugs, including monoaminergic antidepressants and antipsychotics, led to the development of animal models that could detect the effects of these drugs (e.g., tricyclic antidepressants [TCAs] increase the time mice spend trying to find a submerged platform in a “forced swim” test). These animal models were then used to screen new compounds in an attempt to identify drugs that were active in the same animal models. The potential risk of this overall strategy is that these animal models are merely a method for detecting a particular molecular mechanism of action (e.g., increasing serotonin concentrations), rather than a model for a true behavioral analog of human mental illness (e.g., behavioral despair in a depressed patient).

Endophenotypes. A possible diagnosis-related parallel to how this textbook separated the four classes of psychotropic drugs into approximately 30 different categories is the topic of *endophenotypes* in psychiatric patients. An endophenotype is an internal phenotype, which is a set of objective

characteristics of an individual that are not visible to the unaided eye. Because there are so many steps and variables that separate a particular set of genes from the final functioning of a whole human brain, it may be more tractable to consider intermediate assessments such as endophenotypes. This hypothesis is based on the assumption that the number of genes that are involved in an endophenotype might be fewer than the number of genes involved in causing what we would conceptualize as a disease. For psychiatry, an endophenotype is based on neuropsychological, cognitive, neurophysiologic, neuroanatomical, biochemical, and brain imaging data. Such an endophenotype, for example, might include specific cognitive impairments as just one of its objectively measured features. This endophenotype would not be limited to patients with a diagnosis of schizophrenia because it might also be found in some patients with depression or bipolar disorder.

The potential role of an endophenotype can be further clarified by stating what it is not. An endophenotype is not a symptom, and it is not a diagnostic marker. A classification based on the presence or absence of one or more endophenotypes would be based on objective biologic and neuropsychological measures with specific relationships to genes and brain function. A classification based on endophenotypes might also be a productive approach toward the development of more relevant animal models of mental illnesses, and thus the development of novel treatments.

Psychiatry and the Human Genome

Perhaps 70 to 80 percent of the 25,000 human genes are expressed in the brain, and because most genes code for more than one protein, there may be 100,000 different proteins in the brain. Perhaps 10,000 of these are known proteins with somewhat identified functions, and no more than 100 of these are the targets for existing psychotherapeutic drugs.

The study of families with the use of population genetic methods over the last 50 years has consistently supported a genetic, heritable component to mental disorders. More recent techniques in molecular biology have revealed that specific chromosomal regions and genes are associated with particular diagnoses. A potentially compelling application of these techniques has been to study transgenic models of behavior in animals. These transgenic models

can help us understand the effects of individual genes as well as discover completely novel molecular targets for drug development.

It may be a natural response to resist “simple” genetic explanations for human features. Nonetheless, research on humans generally has found that approximately 40 to 70 percent of aspects of cognition, temperament, and personality are attributable to genetic factors. Because these are the very domains that are affected in mentally ill patients, it would not be surprising to discover a similar level of genetic influence on mental illness, mostly if we were able to assess this impact at a more discrete level, such as with endophenotypes.

Individual Genes and Mental Disorders. Several types of data and observations suggest that any single gene is likely to have only a modest effect on the development of a mental disorder and that when a mental disorder is present in an individual, it represents the effects of multiple genes, speculatively on the order of five to ten genes. This hypothesis is also supported by our failure to find single genes with significant effects on mental illnesses. Some researchers, however, still consider it a possibility that genes with significant effects will be identified.

“Nature” and “Nurture” within the CNS. In 1977, George Engel, at the University of Rochester, published a paper that articulated the biopsychosocial model of disease, which stressed an integrated approach to human behavior and disease. The biologic system refers to the anatomical, structural, and molecular substrates of disease; the psychological system refers to the effects of psychodynamic factors; and the social system examines cultural, environmental, and familial influences. Engel postulated that each system affects and is affected by the others.

The observation that a significant percentage of identical twins are discordant for schizophrenia is one example of the type of data that supports the understanding that there are many significant interactions between the genome and the environment (i.e., the biologic basis of the biopsychosocial concept). Studies in animals have also demonstrated that many factors—including activity, stress, drug exposure, and environmental toxins—can regulate the expression of genes and the development and functioning of the brain.

Mental Disorders Reflect Abnormalities in Neuroanatomical Circuits and Synaptic Regulation. Although genes lead to the production of proteins, the actual functioning of the brain needs to be understood at the level of regulation of complex pathways of neurotransmission and intraneuronal signaling, and networks of neurons within and between brain regions. In other words, the downstream effects of abnormal genes are modifications in discrete attributes such as axonal projections, synaptic integrity, and specific steps in intraneuronal molecular signaling.

Why Not a Genetic-Based Diagnostic System? Some researchers have proposed moving psychiatry toward a completely genetic-based diagnostic system. This proposal, however, seems premature based on the complexity of the genetic factors presumably involved in psychiatric disorders, the current absence of sufficient data to make these genetic connections, and the importance of epigenetic and environmental influences on the final behavioral outcomes resulting from an individual's genetic information.

Lessons from Neurology

Clinical and research neurologists seem to have been able to think more than psychiatrists about their diseases of interest and their causes, perhaps because the symptoms are generally nonbehavioral. Neurologists have biologically grounded differential diagnoses and treatment choices. This clarity of approach has helped lead to significant advances in neurology in the last two decades, for example, clarification of the amyloid precursor protein abnormalities in some patients with Alzheimer disease, the presence of trinucleotide repeat mutations in Huntington disease and spinocerebellar ataxia, and the appreciation of alpha-synucleinopathies, such as Parkinson disease and Lewy body dementia.

The continued separation of psychiatry from neurology is, in itself, a potential impediment to good patient care and research. Many neurologic disorders have psychiatric symptoms (e.g., depression in patients following a stroke or with multiple sclerosis or Parkinson disease), and several of the most severe psychiatric disorders have been associated with neurologic

symptoms (e.g., movement disorders in schizophrenia). This overlap is not surprising given that the brain is the organ shared by psychiatric and neurologic diseases, and the division between these two disease areas is arbitrary. For example, patients with Huntington disease are at much greater risk for a wide range of psychiatric symptoms and syndromes, and thus many different DSM-5 diagnoses. Because we know that Huntington disease is an autosomal dominant genetic disorder, the observation that it can manifest with so many different DSM-5 diagnoses does not speak to a reliable biologic distinction among the existing DSM-5 categories.

Examples of Complex Human Behaviors

The goal to understand the human brain and its normal and abnormal functioning is genuinely one of the last frontiers for humans to explore. Trying to explain why a particular individual is the way he or she is, or what causes schizophrenia, for example, will remain too large a challenge for some decades. It is more approachable to consider more discrete aspects of human behavior.

It is not the role of textbooks to set policies or to write diagnostic manuals, but rather to share knowledge, generate ideas, and encourage innovation. The authors believe, however, that it is time to reap the insights of decades of neural science and clinical brain research and to build the classification of mental illnesses on fundamental principles of biology and medicine. Regardless of official diagnostic systems, however, clinicians and researchers should fully understand the biologic component of the biopsychosocial model, and not let research or patient care suffer because of a diagnostic system that is not founded on biologic principles.

FUNCTIONAL NEUROANATOMY

The sensory, behavioral, affective, and cognitive phenomena and attributes experienced by humans are mediated through the brain. It is the organ that perceives and affects the environment and integrates past and present. The brain is the organ of the mind that enables persons to sense, do, feel, and think.

Sensory systems create an internal representation of the external world by processing external stimuli into neuronal impulses. A separate map is formed

for each sensory modality. *Motor systems* enable persons to manipulate their environment and to influence the behavior of others through communication. In the brain, sensory input, representing the external world, is integrated with internal drivers, memories, and emotional stimuli in *association units,* which in turn drive the actions of motor units. Although psychiatry is concerned primarily with the brain's association function, an appreciation of information processing of the sensory and motor systems is essential for sorting logical thought from the distortions introduced by psychopathology.

Brain Organization

The human brain contains approximately 86 billion *neurons* (nerve cells) and approximately 85 billion *glial cells.* Neurons most classically consist of a *soma,* or cell body, which contains the nucleus; usually multiple *dendrites,* which are processes that extend from the cell body and receive signals from other neurons; and a single *axon,* which extends from the cell body and transmits signals to other neurons. Connections between neurons are made at *axon terminals;* there, the axons of one neuron generally contact the dendrite or cell body of another neuron. Neurotransmitter release occurs within axon terminals and is one of the central mechanisms for intraneuronal communications, and also for the effects of psychotropic drugs.

There are three types of glial cells, and although they have often been thought of as having only a supportive role for neuronal functioning, glia has been increasingly appreciated as potentially involved in brain functions that may contribute more directly to both normal and disease mental conditions. The most common type of glial cells are *astrocytes,* which have several functions, including the nutrition of neurons, deactivation of some neurotransmitters, and integration with the blood–brain barrier (BBB). The *oligodendrocytes* in the CNS and the *Schwann cells* in the peripheral nervous system wrap their processes around neuronal axons, resulting in *myelin sheaths* that facilitate the conduction of electrical signals. The third type of glial cells, the *microglia,* which are derived from macrophages, are involved in removing cellular debris following neuronal death.

The neurons and glial cells are arranged in regionally distinct patterns within the brain. Neurons and their processes form groupings in many different ways, and these patterns of organization, or architecture, can be

evaluated by several approaches. The pattern of distribution of nerve cell bodies, called *cytoarchitecture*, is revealed by aniline dyes called Nissl stains that stain ribonucleotides in the nuclei and the cytoplasm of neuronal cell bodies. The Nissl stains show the relative size and packing density of the neurons and, consequently, reveal the organization of the neurons into the different layers of the cerebral cortex.

Sensory Systems

The external world offers an infinite amount of potentially relevant information. In this overwhelming volume of sensory information in the environment, the sensory systems must both detect and discriminate stimuli; they winnow relevant information from the mass of confounding input by applying filtration at all levels. Sensory systems first transform external stimuli into neural impulses and then filter out irrelevant information to create an internal image of the environment, which serves as a basis for reasoned thought. Feature extraction is the quintessential role of sensory systems, which achieve this goal with their hierarchical organizations, first by transforming physical stimuli into neural activity in the primary sense organs and then by refining and narrowing the neural activity in a series of higher cortical processing areas. This neural processing eliminates irrelevant data from higher representations and reinforces crucial features. At the highest levels of sensory processing, neural images are transmitted to the association areas to be acted on in the light of emotions, memories, and drives.

Somatosensory System. The *somatosensory system,* an intricate array of parallel point-to-point connections from the body surface to the brain, was the first sensory system to be understood in anatomical detail. The six somatosensory modalities are light touch, pressure, pain, temperature, vibration, and proprioception (position sense). The organization of nerve bundles and synaptic connections in the somatosensory system encodes spatial relationships at all levels so that the organization is strictly *somatotopic* (Fig. 33-1).

Within a given patch of skin, various receptor nerve terminals act in concert to mediate distinct modalities. The mechanical properties of the skin’s mechanoreceptors and thermoreceptors generate neural impulses in

response to dynamic variations in the environment while they suppress static input. Nerve endings are either fast or slow responders; their depth in the skin also determines their sensitivity to sharp or blunt stimuli. Thus the representation of the external world is significantly refined at the level of the primary sensory organs.

The receptor organs generate coded neural impulses that travel proximally along the sensory nerve axons to the spinal cord. These far-flung routes are susceptible to varying systemic medical conditions and pressure palsies. Pain, tingling, and numbness are the typical presenting symptoms of peripheral neuropathies.

All somatosensory fibers project to and synapse in the thalamus. The thalamic neurons preserve the somatotopic representation by projecting fibers to the somatosensory cortex, located immediately posterior to the Sylvian fissure in the parietal lobe. Despite the considerable overlap, several bands of cortex roughly parallel to the Sylvian fissure are segregated by a somatosensory modality. Within each band is the sensory “homunculus,” the culmination of the careful somatotopic segregation of the sensory fibers at the lower levels. The clinical syndrome of *tactile agnosia* (*astereognosis*) is defined by the inability to recognize objects based on touch, although the primary somatosensory modalities—light touch, pressure, pain, temperature, vibration, and proprioception—are intact. This syndrome, localized at the border of the somatosensory and association areas in the posterior parietal lobe, appears to represent an isolated failure of only the highest order of feature extraction, with preservation of the more basic levels of the somatosensory pathway.

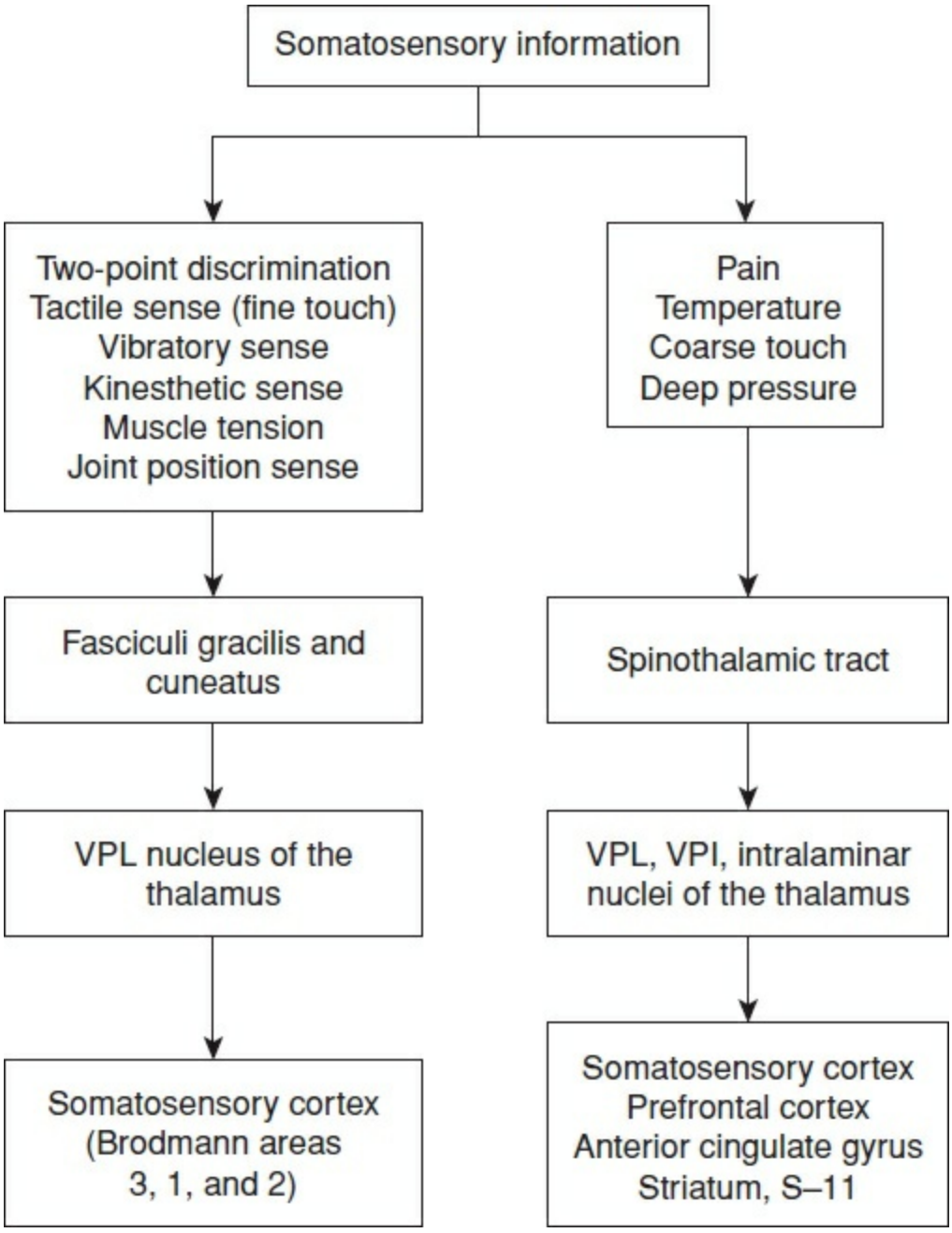


FIGURE 33-1
Pathway of somatosensory information processing. (Adapted from Patestas MA, Gartner LP. *A Textbook of Neuroanatomy*. Malden, MA: Blackwell; 2006:149.)

Reciprocal connections are a critical anatomical feature of crucial importance to conscious perception—as many fibers project down from the cortex to the thalamus as project up from the thalamus to the cortex. These reciprocal fibers play a critical role in filtering sensory input. In normal states, they facilitate the sharpening of internal representations, but in

pathologic states, they can generate false signals or inappropriately suppress sensation. Such cortical interference with sensory perception is thought to underlie many psychosomatic syndromes, such as the hemisensory loss that characterizes conversion disorder.

The prenatal development of the strict point-to-point pattern that characterizes the somatosensory system remains an area of active study. Patterns of sensory innervation result from a combination of axonal guidance by particular molecular cues and pruning of exuberant synaptogenesis based on an organism's experience. Leading hypotheses weigh contributions from a genetically determined molecular map—in which the arrangement of fiber projections is organized by fixed and diffusible chemical cues—against contributions from the modeling and remodeling of projections based on coordinated neural activity. Thumbnail calculations suggest that the 30,000 to 40,000 genes in human deoxyribonucleic acid (DNA) are far too few to completely encode the position of all the trillions of synapses in the brain. Genetically determined positional cues probably steer growing fibers toward the general target, and activity-dependent mechanisms fine-tune the pattern of projections. Recent data suggest that well-established adult thalamocortical sensory projections can be gradually remodeled as a result of a reorientation of coordinated sensory input or in response to the loss of part of the somatosensory cortex, for instance, in stroke.

DEVELOPMENT OF THE SOMATOSENSORY SYSTEM. A strict somatotopic representation exists at each level of the somatosensory system. During development, neurons extend axons to connect to distant brain regions; after arriving at the destination, a set of axons must therefore sort itself to preserve the somatotopic organization. A classic experimental paradigm for this developmental process is the representation of a mouse's whiskers in the somatosensory cortex. The murine somatosensory cortex contains a barrel field of cortical columns, each of which corresponds to one whisker. When mice are inbred to produce fewer whiskers, fewer somatosensory cortex barrels appear. Each barrel is expanded in area, and the entire barrel field covers the same area of the somatosensory cortex as it does in normal animals. This experiment demonstrates that specific higher cortical structures can form in response to peripheral input and that different input complexities determine different patterns of synaptic connectivity. Although the

mechanisms by which peripheral input molds cortical architecture are mostly unknown, animal model paradigms are beginning to yield clues. For example, in a mutant mouse that lacks monoamine oxidase A and, thus, has exceptionally high cortical levels of serotonin, barrels fail to form in the somatosensory cortex. This result indirectly implicates serotonin in the mechanism of barrel field development.

In adults, the classic mapping studies of Wilder Penfield suggested the existence of a homunculus, an immutable cortical representation of the body surface. More recent experimental evidence from primate studies and stroke patients, however, has promoted a more plastic conception than that of Penfield. Minor variations exist in the cortical pattern of ordinary individuals, yet dramatic shifts in the map can occur in response to the loss of cortex from stroke or injury. When a stroke ablates a significant fraction of the somatosensory homunculus, the homuncular representation begins to contract and shift proportionately to fill the remaining intact cortex.

Moreover, the cortical map can be rearranged solely in response to a change in the pattern of tactile stimulation of the fingers. The somatotopic representation of the proximal and distal segments of each finger typically forms a contiguous map, presumably because both segments contact surfaces simultaneously. However, under experimental conditions in which the distal segments of all fingers are simultaneously stimulated while contact of the distal and proximal parts of each finger is separated, the cortical map gradually shifts 90 degrees to reflect the new sensory experience. In the revised map, the cortical representation of the proximal segment of each finger is no longer contiguous with that of the distal segment.

These data support the notion that the internal representation of the external world, although static in gross structure, can be continuously modified at the level of synaptic connectivity to reflect relevant sensory experiences. The cortical representation also tends to shift to fit entirely into the available amount of cortex.

These results also support the notion that cortical representations of sensory input, or memories, may be holographic rather than spatially fixed: The pattern of activity, rather than the physical structure, may encode information. In sensory systems, this plasticity of cortical representation allows recovery from brain lesions; the phenomenon may also underlie learning.

Visual System. Visual images are transduced into neural activity within the retina and are processed through a series of brain cells, which respond to increasingly complex features, from the eye to the higher visual cortex. The neurobiologic basis of feature extraction is best understood in the finest detail in the visual system. Beginning with classic work in the 1960s, research in the visual pathway has produced two main paradigms for all sensory systems. The first paradigm, mentioned earlier for the somatosensory system, evaluates the contributions of genetics and experience—or nature and nurture—in the formation of the final synaptic arrangement. Transplantation experiments, resulting in an accurate point-to-point pattern of connectivity, even when the eye was surgically inverted, have suggested an innate, genetically determined mechanism of synaptic pattern formation. The crucial role of early visual experience in establishing the adult pattern of visual connections, on the other hand, crystallized the hypothesis of the activity-dependent formation of synaptic connectivity. The final adult pattern is the result of both factors.

The second central paradigm, most revealed in the visual system, is that of highly specialized brain cells that respond exclusively to particular stimuli. Recent work, for example, has identified cells in the inferior temporal cortex that respond only to faces viewed at a specific angle. An individual's response to a particular face requires the activity of large neural networks and may not be limited to a single neuron. Nevertheless, the cellular localization of specific feature extraction is of critical importance in defining the boundary between sensory and association systems, but only in the visual system has this significant question been posed experimentally.

In the primary visual cortex, columns of cells respond specifically to lines of a specific orientation. The cells of the primary visual cortex project to the secondary visual cortex. There the cells respond specifically to particular movements of lines and angles. In turn, these cells project to two association areas, where additional features are extracted and conscious awareness of images forms.

The inferior temporal lobe detects the shape, form, and color of the object—the *what* questions; the posterior parietal lobe tracks the location, motion, and distance—the *where* questions. The posterior parietal lobe contains distinct sets of neurons that signal the intention either to look into a specific

part of visual space or to reach for a particular object. In the inferior temporal cortices (ITCs), adjacent cortical columns respond to complex forms. Responses to facial features tend to occur in the left ITC, and responses to complex shapes tend to occur in the right ITC. The brain devotes specific cells to the recognition of facial expressions and the aspect and position of faces of others for the individual.

The crucial connections between the feature-specific cells and the association areas involved in memory and conscious thought remain to be delineated. Much elucidation of feature recognition is based on invasive animal studies. In humans, the clinical syndrome of *prosopagnosia* describes the inability to recognize faces in the presence of preserved recognition of other environmental objects. Based on the pathologic and radiologic examination of individual patients, prosopagnosia is thought to result from the disconnection of the left ITC from the visual association area in the left parietal lobe. Such lesional studies are useful in identifying necessary components of a mental pathway, but they may be inadequate to define the entire pathway. One noninvasive technique that is still being perfected and is beginning to reveal the full anatomical relation of the human visual system to conscious thought and memory is functional neuroimaging.

As is valid for language, there appears to be a hemispheric asymmetry for individual components of visuospatial orientation. Although both hemispheres cooperate in perceiving and drawing complex images, the right hemisphere, especially the parietal lobe, contributes the overall contour, perspective, and right-left orientation, and the left hemisphere adds internal detail, embellishment, and complexity. The brain can be fooled in optical illusions.

Neurologic conditions such as strokes and other focal lesions have permitted the definition of several disorders of visual perception. *Apperceptive visual agnosia* is the inability to identify and draw items using visual cues, with preservation of other sensory modalities. It represents a failure of transmission of information from the higher visual sensory pathway to the association areas and is caused by bilateral lesions in the visual association areas. *Associative visual agnosia* is the inability to name or use objects despite the ability to draw them. It is caused by bilateral medial occipitotemporal lesions and can occur along with other visual impairments. Color perception may be ablated in lesions of the dominant occipital lobe that

include the splenium of the corpus callosum. *Color agnosia* is the inability to recognize a color despite being able to match it. *Color anomia* is the inability to name a color despite being able to point to it. *Central achromatopsia* is a complete inability to perceive color.

Anton syndrome is a failure to acknowledge blindness, possibly owing to the interruption of fibers involved in self-assessment. It is seen with bilateral occipital lobe lesions. The most common causes are hypoxic injury, stroke, metabolic encephalopathy, migraine, herniation resulting from mass lesions, trauma, and leukodystrophy. *Balint syndrome* consists of a triad of optic ataxia (the inability to direct optically guided movements), *oculomotor apraxia* (inability to direct gaze rapidly), and *simultanagnosia* (inability to integrate a visual scene to perceive it as a whole). Balint syndrome is seen in bilateral parietooccipital lesions. *Gerstmann syndrome* includes agraphia, calculation difficulties (acalculia), right-left disorientation, and finger agnosia. It has been attributed to lesions of the dominant parietal lobe.

DEVELOPMENT OF THE VISUAL SYSTEM. In humans, the initial projections from both eyes intermingle in the cortex. During the development of visual connections in the early postnatal period, there is a window of time during which binocular visual input is required for the development of ocular dominance columns in the primary visual cortex. Ocular dominance columns are stripes of cortex that receive input from only one eye, separated by stripes innervated only by fibers from the other eye. Occlusion of one eye during this critical period eliminates the persistence of its fibers in the cortex and allows the fibers of the active eye to innervate the entire visual cortex. In contrast, when normal binocular vision is allowed during the critical development window, the usual dominance columns form, occluding one eye after the completion of innervation of the cortex, produces no subsequent alteration of the ocular dominance columns. This paradigm crystallizes the importance of early childhood experience on the formation of adult brain circuitry.

Auditory System. Sounds are instantaneous, incremental changes in ambient air pressure. The pressure changes cause the ear’s tympanic membrane to vibrate; the vibration is then transmitted to the ossicles (malleus, incus, and stapes) and thereby to the endolymph or fluid of the cochlear spiral. Vibrations of the endolymph move cilia on hair cells, which

generate neural impulses. The hair cells respond to sounds of different frequencies in a tonotopic manner within the cochlea, like an extended, spiral piano keyboard. Neural impulses from the hair cells travel in a tonotopic arrangement to the brain in the fibers of the cochlear nerve. They enter the brainstem cochlear nuclei, are relayed through the lateral lemniscus to the inferior colliculi, and then to the medial geniculate nucleus (MGN) of the thalamus. MGN neurons project to the primary auditory cortex in the posterior temporal lobe. Dichotic listening tests, in which different stimuli are presented to each ear simultaneously, demonstrate that most of the input from one ear activates the contralateral auditory cortex and that the left hemisphere tends to be dominant for auditory processing.

Sonic features are extracted through a combination of mechanical and neural filters. The representation of sound is roughly tonotopic in the primary auditory cortex, whereas *lexical processing* (i.e., the extraction of vowels, consonants, and words from the auditory input) occurs in higher language association areas, especially in the left temporal lobe. The syndrome of *word deafness,* characterized by intact hearing for voices but an inability to recognize speech, may reflect damage to the left parietal cortex. This syndrome is thought to result from the disconnection of the auditory cortex from Wernicke area. A rare, complementary syndrome, *auditory sound agnosia*, is defined as the inability to recognize nonverbal sounds, such as a horn or a cat's meow, in the presence of intact hearing and speech recognition. Researchers consider this syndrome to be the right hemisphere correlate of pure word deafness.

DEVELOPMENT OF THE AUDITORY SYSTEM. Some children are unable to process auditory input and therefore have impaired speech and comprehension of spoken language. Studies on some of these children have determined that they can discriminate speech if the consonants and vowels—the phonemes—are slowed two- to fivefold by a computer. Based on this observation, a tutorial computer program was designed that initially asked questions in a slowed voice and, as subjects answered questions correctly, gradually increased the rate of phoneme presentation to approximate typical rates of speech. Subjects gained some ability to discriminate routine speech over 2 to 6 weeks and appeared to retain these skills after the tutoring period was completed. This finding probably has therapeutic applicability to 5 to 8

percent of children with speech delay, but ongoing studies may expand the eligible group of students. This finding, moreover, suggests that neuronal circuits required for auditory processing can be recruited and be made more efficient long after language is usually learned, provided that the circuits are allowed to finish their task correctly, even if this requires slowing the rate of input. Circuits thus functioning with high fidelity can then be trained to speed their processing.

A recent report has extended the age at which language acquisition may be acquired for the first time.

> A boy who had intractable epilepsy of one hemisphere was mute because the uncontrolled seizure activity precluded the development of organized language functions. At the age of 9 years, he had the abnormal hemisphere removed to cure epilepsy. Although up to that point in his life he had not spoken, he initiated an accelerated acquisition of language milestones beginning at that age and ultimately gained language abilities only a few years delayed relative to his chronologic age.

Researchers cannot place an absolute upper limit on the age at which language abilities can be learned, although acquisition at ages beyond the usual childhood period is usually incomplete. Anecdotal reports document the acquisition of reading skills after the age of 80 years.

Olfaction. Odorants, or volatile chemical cues, enter the nose, are solubilized in the nasal mucus, and bind to odorant receptors displayed on the surface of the sensory neurons of the olfactory epithelium. Each neuron in the epithelium displays a unique odorant receptor, and cells displaying a given receptor are arranged randomly within the olfactory epithelium. Humans possess several hundred distinct receptor molecules that bind the massive variety of environmental odorants; researchers estimate that humans can discriminate 10,000 different odors. Odorant binding generates neural impulses, which travel along the axons of the sensory nerves through the cribriform plate to the olfactory bulb. Within the bulb, all axons corresponding to a given receptor converge onto only 1 or 2 of 3,000 processing units called *glomeruli.* Because each odorant activates several

receptors that activate a characteristic pattern of glomeruli, the identity of external chemical molecules is represented internally by a spatial pattern of neural activity in the olfactory bulb.

Each glomerulus projects to a unique set of 20 to 50 separate columns in the olfactory cortex. In turn, each olfactory cortical column receives projections from a unique combination of glomeruli. The connectivity of the olfactory system is genetically determined. Because each odorant activates a unique set of several receptors and thus a unique set of olfactory bulb glomeruli, each olfactory cortical column is tuned to detect a different odorant of some evolutionary significance to the species. Unlike the signals of the somatosensory, visual, and auditory systems, olfactory signals do not pass through the thalamus but project directly to the frontal lobe and the limbic system, especially the pyriform cortex. The connections to the limbic system (amygdala, hippocampus, and pyriform cortex) are significant. Olfactory cues stimulate strong emotional responses and can evoke powerful memories.

Olfaction, the most ancient sense in evolutionary terms, is tightly associated with sexual and reproductive responses. A related chemosensory structure, the vomeronasal organ, is thought to detect *pheromones*, chemical cues that trigger unconscious, stereotyped responses. In some animals, ablation of the vomeronasal organ in early life may prevent the onset of puberty. Recent studies have suggested that humans also respond to pheromones in a manner that varies according to the menstrual cycle. The structures of higher olfactory processing in phylogenetically more primitive animals have evolved in humans into the limbic system, the center of the emotional brain, and the gate through which experience is admitted into memory according to emotional significance. The elusive basic animal drives with which clinical psychiatry constantly grapples may, therefore, originate from the ancient centers of higher olfactory processing.

DEVELOPMENT OF THE OLFACTORY SYSTEM. During normal development, axons from the nasal olfactory epithelium project to the olfactory bulb and segregate into about 3,000 equivalent glomeruli. If an animal is exposed to a single dominant scent in the early postnatal period, then one glomerulus expands massively within the bulb at the expense of the surrounding glomeruli. Thus, as discussed earlier concerning the barrel fields of the

somatosensory cortex, the size of brain structures may reflect the environmental input.

Taste. Soluble chemical cues in the mouth bind to receptors in the tongue and stimulate the gustatory nerves, which project to the nucleus solitarius in the brainstem. The sense of taste is believed to discriminate only in broad classes of stimuli: sweet, sour, bitter, salty and umami. Each modality is mediated through a unique set of cellular receptors and channels, of which several may be expressed in each taste neuron. The detection and the discrimination of foods, for example, involve a combination of the senses of taste, olfaction, touch, vision, and hearing. Taste fibers activate the medial temporal lobe, but the higher cortical localization of taste is only poorly understood.

Autonomic Sensory System. The autonomic nervous system (ANS) monitors the essential functions necessary for life. The activity of visceral organs, blood pressure, cardiac output, blood glucose levels, and body temperature are all transmitted to the brain by autonomic fibers. Most autonomic sensory information remains unconscious; if such information rises to conscious levels, it is only as a vague sensation, in contrast to the capacity of the primary senses to transmit sensations rapidly and precisely.

Alteration of Conscious Sensory Perception through Hypnosis. *Hypnosis* is a state of heightened suggestibility attainable by a certain proportion of the population. Under a state of hypnosis, gross distortions of perception in any sensory modality and changes in the ANS can be achieved instantaneously. The anatomy of the sensory system does not change, yet the same specific stimuli may be perceived with opposed emotional value before and after induction of the hypnotic state. For example, under hypnosis, a person may savor an onion as if it were a luscious chocolate truffle, only to reject the onion as abhorrently pungent seconds later when the hypnotic suggestion is reversed. The localization of the hypnotic switch has not been determined, but it presumably involves both sensory and association areas of the brain. Experiments tracing neural pathways in human volunteers via functional neuroimaging have demonstrated that shifts in attention in an environmental setting determine changes in the regions of the brain that are

activated on an instantaneous time scale. Thus the organizing centers of the brain may route conscious and unconscious thoughts through different sequences of neural processing centers, depending on a person's ultimate goals and emotional state. These attention-mediated variations in synaptic utilization can occur instantaneously, much like the alteration in the routing of associational processing that may occur in hypnotic states.

Motor Systems

Body muscle movements are controlled by the lower motor neurons, which extend axons—some as long as 1 m—to the muscle fibers. Lower motor neuron firing is regulated by the summation of upper motor neuron activity. In the brainstem, primitive systems produce gross coordinated movements of the entire body. Activation of the rubrospinal tract stimulates flexion of all limbs, whereas activation of the vestibulospinal tract causes all limbs to extend. Newborn infants, for example, have all limbs tightly flexed, presumably through the dominance of the rubrospinal system. The movements of an anencephalic infant, who completely lacks a cerebral cortex, may be indistinguishable from the movements of an average newborn. In the first few months of life, the flexor spasticity is gradually mitigated by the opposite actions of the vestibulospinal fibers, and more limb mobility occurs.

At the top of the motor hierarchy is the corticospinal tract, which controls fine movements and which eventually dominates the brainstem system during the first years of life. The upper motor neurons of the corticospinal tract reside in the posterior frontal lobe, in a section of the cortex known as the *motor strip*. Planned movements are conceived in the association areas of the brain, and in consultation with the basal ganglia and cerebellum, the motor cortex directs their smooth execution. The importance of the corticospinal system becomes immediately evident in strokes, in which spasticity returns as the cortical influence is ablated, and the actions of the brainstem motor systems are released from cortical modulation.

Basal Ganglia. The *basal ganglia*, a subcortical group of gray matter nuclei, appear to mediate postural tone. The four functionally distinct ganglia are the striatum, the pallidum, the substantia nigra, and the subthalamic

nucleus. Collectively known as the corpus striatum, the caudate and putamen harbor components of both motor and association systems. The caudate nucleus plays a vital role in the modulation of motor acts. Anatomical and functional neuroimaging studies have correlated decreased activation of the caudate with obsessive-compulsive behavior. When functioning correctly, the caudate nucleus acts as a gatekeeper to allow the motor system to perform only those acts that are goal-directed. When it fails to perform its gatekeeper function, extraneous acts are performed, as in obsessive-compulsive disorder (OCD) or the tic disorders, such as Tourette disorder. Overactivity of the striatum owing to lack of dopaminergic inhibition (e.g., in parkinsonian conditions) results in *bradykinesia,* an inability to initiate movements. The caudate, in particular, shrinks dramatically in Huntington disease. This disorder is characterized by rigidity, on which is gradually superimposed choreiform, or "dancing," movements. Psychosis may be a prominent feature of Huntington disease, and suicide is not uncommon. The caudate is also thought to influence associative or cognitive processes.

The globus pallidus contains two parts linked in series. In a cross-section of the brain, the internal and external parts of the globus pallidus are nested within the concavity of the putamen. The globus pallidus receives input from the corpus striatum and projects fibers to the thalamus. This structure may be severely damaged in Wilson disease and in carbon monoxide poisoning, which are characterized by dystonic posturing and flapping movements of the arms and legs.

The substantia nigra is named the black substance because the presence of melanin pigment causes it to appear black to the naked eye. It has two parts, one of which is functionally equivalent to the globus pallidus interna. The other part degenerates in Parkinson disease. Parkinsonism is characterized by rigidity and tremor and is associated with depression in more than 30 percent of cases.

Finally, lesions in the subthalamic nucleus yield ballistic movements, sudden limb jerks of such velocity that they are compared to projectile movement.

Together, the nuclei of the basal ganglia appear capable of initiating and maintaining the full range of useful movements. Investigators have speculated that the nuclei serve to configure the activity of the overlying motor cortex to fit the purpose of the association areas. Also, they appear to

integrate proprioceptive feedback to maintain an intended movement.

Cerebellum. The cerebellum consists of a simple six-cell pattern of circuitry that is replicated roughly 10 million times. Simultaneous recordings of the cerebral cortex and the cerebellum have shown that the cerebellum is activated several milliseconds before a planned movement. Moreover, the ablation of the cerebellum renders intentional movements coarse and tremulous. These data suggest that the cerebellum carefully modulates the tone of agonistic and antagonistic muscles by predicting the relative contraction needed for smooth motion. This prepared motor plan is used to ensure that precisely the right amount of flexor and extensor stimuli is sent to the muscles. Recent functional imaging data have shown that the cerebellum is active, even during the mere imagination of motor acts when no movements ultimately result from its calculations. The cerebellum harbors two, and possibly more, distinct "homunculi" or cortical representations of the body plan.

Motor Cortex. Penfield's groundbreaking work defined a motor homunculus in the precentral gyrus, Brodmann area 4 (Fig. 33-2), where a somatotopic map of the motor neurons is found. Individual cells within the motor strip cause contraction of single muscles. The brain region immediately anterior to the motor strip is called the *supplementary motor area*, Brodmann area 6. This region contains cells that, when individually stimulated, can trigger more complex movements by influencing a firing sequence of motor strip cells. Recent studies have demonstrated a broad representation of motor movements in the brain.

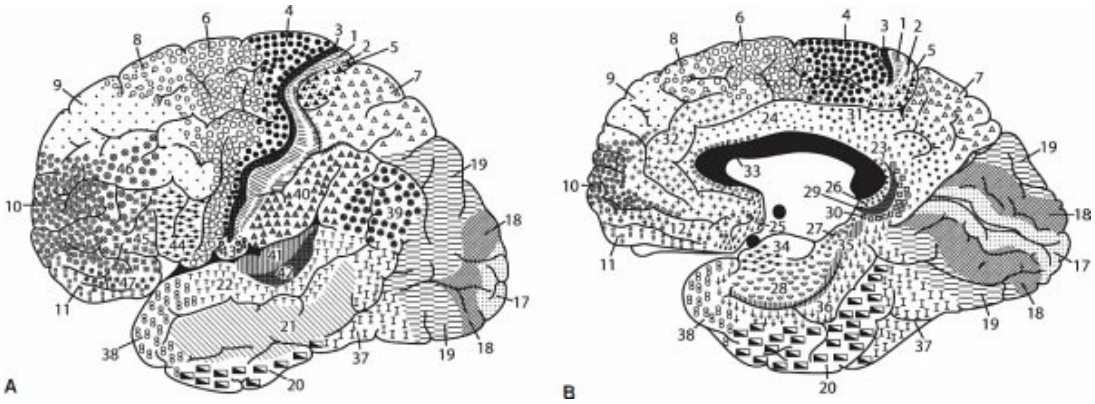


FIGURE 33-2

Drawing of the lateral view (**A**) and medial view (**B**) of the cytoarchitectonic subdivisions of the human brain as determined by Brodmann. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009.)

The skillful use of the hands is called *praxis,* and deficits in skilled movements are termed *apraxias.* The three levels of apraxia are limb-kinetic, ideomotor, and ideational. *Limb-kinetic apraxia* is the inability to use the contralateral hand in the presence of preserved strength; it results from isolated lesions in the supplementary motor area, which contains neurons that stimulate functional sequences of neurons in the motor strip.

Ideomotor apraxia is the inability to perform an isolated motor act on command, despite preserved comprehension, strength, and spontaneous performance of the same act. Ideomotor apraxia simultaneously affects both limbs and involves functions so specialized that they are localized to only one hemisphere. Conditions in two separate areas can produce this apraxia. Disconnection of the language comprehension area, Wernicke area, from the motor regions causes an inability to follow spoken commands, and lesions to the left premotor area may impair the actual motor program as the higher-order motor neurons generate it. This program is transmitted across the corpus callosum to the right premotor area, which directs the movements of the left hand. A lesion in this callosal projection can also cause isolated ideomotor apraxia in the left hand. This syndrome implies the representation of specific motor acts within discrete sections of the left premotor cortex. Thus just as some cells respond selectively to specific environmental features in the higher sensory cortices, some cells in the premotor cortex direct specific complex motor tasks.

Ideational apraxia occurs when the individual components of a sequence of skilled acts can be performed in isolation, but the entire series cannot be organized and executed as a whole. For example, the sequence of opening an envelope, removing the letter, unfolding it, and placing it on the table cannot be performed in order, even though the individual acts can be performed in isolation. The representation of the concept of a motor sequence may involve several areas, specifically the left parietal cortex, but it likely also relies on the sequencing and executive functions of the prefrontal cortex. This apraxia is a typical finding of diffuse cortical degeneration, such as Alzheimer disease.

Autonomic Motor System. The *autonomic system* is divided into a sensory component (described earlier) and a motor component. The *autonomic motor system* is divided into two branches: the sympathetic and the parasympathetic. As a rule, organs are innervated by both types of fibers, which often serve antagonistic roles. The *parasympathetic system* slows the heart rate and begins the process of digestion. In contrast, the *sympathetic system* mediates the fight or flight response, with increased heart rate, shunting of blood away from the viscera, and increased respiration. The sympathetic system is highly activated by sympathomimetic drugs, such as amphetamine and cocaine, and may also be activated by withdrawal from sedating drugs such as alcohol, benzodiazepines, and opioids. Investigators who have found an increased risk of heart attacks in persons with high levels of hostility have suggested that chronic activation of the sympathetic fight or flight response, with the elevated secretion of adrenaline, may underlie this association.

The brain center that drives the autonomic motor system is the *hypothalamus*, which houses a set of paired nuclei that appear to control appetite, rage, temperature, blood pressure, perspiration, and sexual drive. For example, lesions to the ventromedial nucleus, the satiety center, produce a voracious appetite and rage. In contrast, lesions to the upper region of the lateral nucleus, the hunger center, produce a profound loss of appetite. Numerous research groups are making intense efforts to define the biochemical regulation of appetite and obesity and frequently target the role of the hypothalamus.

In the regulation of sexual attraction, the role of the hypothalamus has also become an area of active research. In the 1990s, three groups independently reported neuroanatomical differences between some of the hypothalamic nuclei of straight and gay men. Researchers interpreted this finding to suggest that human sexual orientation has a neuroanatomical basis, and this result has stimulated several follow-up studies of the biologic basis of sexual orientation. At present, however, these controversial findings are not accepted without question (including by some of the original authors), and no clear consensus has emerged about whether the structure of the hypothalamus consistently correlates with sexual orientation. In animal studies, early nurturing and sexual experiences consistently alter the size of specific hypothalamic nuclei.

Primitive Reflex Circuit. Sensory pathways function as extractors of specific features from the overwhelming multitude of environmental stimuli, whereas motor pathways carry out the wishes of the organism. These pathways may be linked directly, for example, in the spinal cord, where a primitive reflex arc may mediate the brisk withdrawal of a limb from a painful stimulus, without immediate conscious awareness. In this loop, the peripheral stimulus activates the sensory nerve, the sensory neuron synapses on and directly activates the motor neuron, and the motor neuron drives the muscle to contract. This response is strictly local and all-or-none. Such primitive reflex arcs, however, rarely generate an organism's behaviors. In most behaviors, sensory systems project to association areas, where sensory information is interpreted in terms of internally determined memories, motivations, and drives. The exhibited behavior results from a plan of action determined by the associated components and carried out by the motor systems.

Localization of Brain Functions. Many theorists have subdivided the brain into functional systems. Brodmann defined 47 areas based on cytoarchitectonic distinctions, a cataloging that has been remarkably durable as the functional anatomy of the brain has been elucidated. Various distinct functions, based on data from lesion studies and functional neuroimaging, have been assigned to nearly all Brodmann areas. At the other extreme, individual experts have distinguished only three processing blocks: The brainstem and the thalamic reticular activating system provide arousal and set up attention; the posterior cortex integrates perceptions and generates language; and, at the highest level, the frontal cortex generates programs and executes plans like an orchestra conductor.

Hemispheric lateralization of function is a crucial feature of higher cortical processing. The primary sensory cortices for touch, vision, hearing, smell, and taste are represented bilaterally, and the first level of abstraction for these modalities is also usually represented bilaterally. The highest levels of feature extraction, however, are generally unified in one brain hemisphere only. For example, recognition of familiar and unfamiliar faces seems localized to the left inferior temporal cortex, and cortical processing of olfaction occurs in the right frontal lobe.

Hypotheses about the flow of thought in the brain are based on few experimental data, although this scarcity of findings has not impeded numerous theoreticians from speculating about functional neuroanatomy. Several roles have been tentatively assigned to specific lobes of the brain, based on the functional deficits resulting from the localized injury. These data indicate that some areas of the cortex may be necessary for a specific function, but they do not define the complete set of structures that suffices for a complicated task. Anecdotal evidence from surface electrocorticography for the study of epilepsy, for example, suggests that a right parietal seizure impulse may shoot immediately to the left frontal lobe and then to the right temporal lobe before spreading locally to the remainder of the parietal lobe. This evidence illustrates the limitations of naively assigning a mental function to a single brain region. Functional neuroimaging studies frequently reveal the simultaneous activation of disparate brain regions during the performance of even a simple cognitive task. Nevertheless, particularly in the processing of vision and language, fairly well-defined lobar syndromes have been confirmed.

Language. The clearest known example of hemispheric lateralization is the localization of language functions to the left hemisphere. Starting with the work of Pierre Broca and Karl Wernicke in the 19th century, researchers have drawn a detailed map of language comprehension and expression.

At least eight types of aphasias in which one or more components of the language pathway are inured have been defined. *Prosody,* the emotional and affective components of language, or "body language," appears to be localized in a mirror set of brain units in the right hemisphere.

Because of the significant role of verbal and written language in human communication, the neuroanatomical basis of language is the most wholly understood association function. Language disorders, also called *aphasias,* are readily diagnosed in routine conversation, whereas perceptual disorders may escape notice, except during detailed neuropsychological testing, although these disorders may be caused by injury of an equal volume of the cortex. Among the earliest models of cortical localization of function were Broca's 1865 description of a loss of fluent speech caused by a lesion in the left inferior frontal lobe and Wernicke's 1874 localization of language comprehension to the left superior temporal lobe. Subsequent analyses of

patients rendered aphasic by strokes, trauma, or tumors have led to the definition of the entire language association pathway from sensory input through the motor output.

Language most demonstrates hemispheric localization of function. In most persons, the hemisphere dominant for language also directs the dominant hand. Ninety percent of the population is right-handed, and 99 percent of right-handers have left hemispheric dominance for language. Of the 10 percent who are left-handers, 67 percent also have left-hemispheric language dominance; the other 33 percent have either mixed or right-hemispheric language dominance. This innate tendency to lateralization of language in the left hemisphere is positively associated with an asymmetry of the planum temporale, a triangular cortical patch on the superior surface of the temporal lobe that appears to harbor Wernicke area. Patients with mixed hemispheric dominance for language lack the expected asymmetry of the planum temporale. That asymmetry has been observed in prenatal brains suggests a genetic determinant. Indeed, the absence of asymmetry runs in families, although both genetic and intrauterine influences probably contribute to the final pattern.

Language comprehension is processed at three levels. First, in *phonologic processing,* individual sounds, such as vowels or consonants, are recognized in the inferior gyrus of the frontal lobes. Phonologic processing improves if lip reading is allowed, if speech is slowed, or if contextual clues are provided. Second, *lexical processing* matches the phonologic input with recognized words or sounds in the individual’s memory. Lexical processing determines whether a sound is a word. Recent evidence has localized lexical processing to the left temporal lobe, where the representations of lexical data are organized according to the semantic category. Third, *semantic processing* connects the words to their meaning. Persons with an isolated defect in semantic processing may retain the ability to repeat words in the absence of an ability to understand or spontaneously generate speech. Semantic processing activates the middle and superior gyri of the left temporal lobe, whereas the representation of the conceptual content of words is widely distributed in the cortex. Language production proceeds in the opposite direction, from the cortical semantic representations through the left temporal lexical nodes to either the oromotor phonologic processing area (for speech) or the graphomotor system (for writing). Each of these areas can be

independently or simultaneously damaged by stroke, trauma, infection, or tumor, resulting in a specific type of aphasia.

The garbled word salad or illogic utterances of an aphasic patient leave little uncertainty about the diagnosis of left-sided cortical injury, but the right hemisphere contributes a somewhat more subtle, but equally important, affective quality to language. For example, the phrase “I feel good” may be spoken with an infinite variety of shadings, each of which is understood differently. The perception of prosody and the appreciation of the associated gestures, or “body language,” appear to require an intact right hemisphere. Behavioral neurologists have mapped an entire pathway for prosody association in the right hemisphere that mirrors the language pathway of the left hemisphere. Patients with right hemisphere lesions, who have impaired comprehension or expression of prosody, may find it difficult to function in society despite their intact language skills.

Developmental dyslexia is defined as an unexpected difficulty with learning in the context of adequate intelligence, motivation, and education. Whereas speech consists of the logical combination of 44 basic phonemes of sounds, reading requires a broader set of brain functions and, thus, is more susceptible to disruption. The awareness of specific phonemes develops at about the ages of 4 to 6 years. It appears to be a prerequisite to the acquisition of reading skills. The inability to recognize distinct phonemes is the best predictor of a reading disability. Functional neuroimaging studies have localized the identification of letters to the occipital lobe adjacent to the primary visual cortex. Phonologic processing occurs in the inferior frontal lobe, and semantic processing requires the superior and middle gyri of the left temporal lobe. A recent finding of uncertain significance is that phonologic processing in men activates only the left inferior frontal gyrus, whereas phonologic processing in women activates the inferior frontal gyrus bilaterally. Careful analysis of an individual’s particular reading deficits can guide remedial tutoring efforts that can focus on weaknesses and thus attempt to bring the reading skills up to the general level of intelligence and verbal skills.

In children, developmental nonverbal learning disorder is postulated to result from right hemisphere dysfunction. A nonverbal learning disorder is characterized by poor fine-motor control in the left hand, deficits in the visuoperceptual organization, problems with mathematics, and incomplete or

disturbed socialization.

Patients with nonfluent aphasia, who cannot complete a simple sentence, may be able to sing an entire song, apparently because many aspects of music production are localized to the right hemisphere. Music is represented predominantly in the right hemisphere, but the full complexity of musical ability seems to involve both hemispheres. Trained musicians appear to transfer many musical skills from the right hemisphere to the left as they gain proficiency in musical analysis and performance.

Arousal and Attention. Arousal, or the establishment and maintenance of an awake state, appears to require at least three brain regions. Within the brainstem, the ascending reticular activating system (ARAS), a diffuse set of neurons, appears to set the level of consciousness. The ARAS projects to the intralaminar nuclei of the thalamus, and these nuclei, in turn, project widely throughout the cortex. Electrophysiologic studies show that both the thalamus and the cortex fire rhythmical bursts of neuronal activity at rates of 20 to 40 cycles per second. During sleep, these bursts are not synchronized. During wakefulness, the ARAS stimulates the thalamic intralaminar nuclei, which in turn coordinate the oscillations of different cortical regions. The greater the synchronization, the higher the level of wakefulness. The absence of arousal produces stupor and coma. In general, small discrete lesions of the ARAS can produce a stuporous state, whereas, at the hemispheric level, large bilateral lesions are required to cause the same depression in alertness. One particularly unfortunate but instructive condition involving extensive, permanent, bilateral cortical dysfunction is the persistent vegetative state. Sleep–wake cycles may be preserved, and the eyes may appear to gaze, but the external world does not register, and no evidence of conscious thought exists. This condition represents the expression of the isolated actions of the ARAS and the thalamus.

The maintenance of attention appears to require an intact right frontal lobe. For example, a widely used test of persistence requires scanning and identifying only the letter A from a long list of random letters. Healthy persons can usually maintain the performance of such a task for several minutes, but in patients with right frontal lobe dysfunction, this capacity is severely curtailed. Lesions of similar size in other regions of the cortex usually do not affect persistence tasks. In contrast, the more generally

adaptive skill of maintaining a coherent line of thought is diffusely distributed throughout the cortex. Many medical conditions can affect this skill and may produce acute confusion or delirium.

One widely diagnosed disorder of attention is attention-deficit/hyperactivity disorder (ADHD). No pathologic findings have been consistently associated with this disorder. Functional neuroimaging studies, however, have variously documented either frontal lobe or right hemisphere hypometabolism in patients with ADHD, compared with normal controls. These findings strengthen the notion that the frontal lobes—especially the right frontal lobe—are essential to the maintenance of attention.

Memory. The clinical assessment of memory should test three periods, which have distinct anatomical correlates. *Immediate memory* functions over seconds; *recent memory* applies on a scale of minutes to days, and *remote memory* encompasses months to years. Immediate memory is implicit in the concept of attention and the ability to follow a train of thought. This ability has been divided into phonologic and visuospatial components, and functional imaging has localized them to the left and right hemispheres, respectively. A related concept, incorporating immediate and recent memory, is *working memory,* which is the ability to store information for several seconds, whereas other, related cognitive operations take place on this information. Recent studies have shown that single neurons in the dorsolateral prefrontal cortex not only record features necessary for working memory but also record the certainty with which the information is known and the degree of expectation assigned to the permanence of a particular environmental feature. Some neurons fire rapidly for an item that is eagerly awaited but may cease firing if hopes are dashed unexpectedly. The encoding of the emotional value of an item contained in the working memory may be of great usefulness in determining goal-directed behavior. Some researchers localize working memory predominantly to the left frontal cortex. Clinically, however, bilateral prefrontal cortex lesions are required for severe impairment of working memory. Other types of memory have been described: episodic, semantic, and procedural.

Three brain structures are critical to the formation of memories: the medial temporal lobe, specific diencephalic nuclei, and the basal forebrain. The *medial temporal lobe* houses the *hippocampus,* an elongated, highly

repetitive network. The *amygdala* is adjacent to the anterior end of the hippocampus. The amygdala has been suggested to rate the emotional importance of experience and to activate the level of hippocampal activity accordingly. Thus an emotionally intense experience is indelibly etched in memory, but neutral stimuli are quickly disregarded.

Animal studies have defined a hippocampal place code, a pattern of cellular activation in the hippocampus that corresponds to the animal's location in space. When the animal is introduced to a novel environment, the hippocampus is broadly activated. As the animal explores and roams, the firing of certain hippocampal regions begins to correspond to specific locations in the environment. In about 1 hour, a highly detailed internal representation of the external space (a "cognitive map") appears in the form of specific firing patterns of the hippocampal cells. These patterns of neuronal firing may bear little spatial resemblance to the environment they represent; instead, they may seem randomly arranged in the hippocampus. If the animal is manually placed in a particular part of a familiar space, only the corresponding hippocampal regions show intense neural activity. When recording continues into sleep periods, firing sequences of hippocampal cells outlining a coherent path of navigation through the environment are registered, even though the animal is motionless. If the animal is removed from the environment for several days and then returned, the previously registered hippocampal place code is immediately reactivated. A series of animal experiments have dissociated the formation of the hippocampal place code from either visual, auditory, or olfactory cues, although each of these modalities may contribute to place code generation. Other factors may include internal calculations of distances based on counting footsteps or other proprioceptive information. Data from targeted genetic mutations in mice have implicated both the *N*-methyl-D-aspartate (NMDA) glutamate receptors and the calcium-calmodulin kinase II (CaMKII) in the proper formation of hippocampal place fields. These data suggest that the hippocampus is a significant site for the formation and storage of immediate and recent memories. Although no data yet support the notion, it is conceivable that the hippocampal cognitive map is inappropriately reactivated during a *déjà vu* experience.

The most famous human subject in the study of memory is H. M., a man with intractable epilepsy, who had both his hippocampi and amygdalae surgically removed to alleviate his condition. The epilepsy was controlled, but he was left with a complete inability to form and recall memories of facts. H. M.’s learning and memory skills were relatively preserved, which led to the suggestion that declarative or factual memory may be separate within the brain from procedural or skill-related memory. A complementary deficit in procedural memory with preservation of declarative memory may be seen in persons with Parkinson disease, in whom dopaminergic neurons of the nigrostriatal tract degenerate. Because this deficit in procedural memory can be ameliorated with treatment with levodopa, which is thought to potentiate dopaminergic neurotransmission in the nigrostriatal pathway, a role has been postulated for dopamine in procedural memory. Additional case reports have further implicated the amygdala and the afferent and efferent fiber tracts of the hippocampus as essential to the formation of memories. Also, lesional studies have suggested mild lateralization of hippocampal function in which the left hippocampus is more efficient at forming verbal memories, and the right hippocampus tends to form nonverbal memories. After unilateral lesions in humans, however, the remaining hippocampus may compensate to a large extent. Medical causes of amnesia include alcoholism, seizures, migraine, drugs, vitamin deficiencies, trauma, strokes, tumors, infections, and degenerative diseases.

The motor system within the cortex receives directives from the association areas. The performance of a novel act requires constant feedback from the sensory and association areas for completion, and functional neuroimaging studies have demonstrated widespread activation of the cortex during unskilled acts. Memorized motor acts initially require activation of the medial temporal lobe. With practice, however, the performance of ever-larger segments of an act necessary to achieve a goal become encoded within discrete areas of the premotor and parietal cortices, particularly the left parietal cortex, with the result that a much more limited activation of the

cortex is seen during highly skilled acts, and the medial temporal lobe is bypassed. This process is called the *corticalization of motor commands.* In lay terms, the process suggests a neuroanatomical basis for the adage "practice makes perfect."

Within the diencephalon, the dorsal medial nucleus of the thalamus and the mammillary bodies appear necessary for memory formation. These two structures are damaged in thiamine deficiency states usually seen in chronic alcoholics, and their inactivation is associated with Korsakoff syndrome. This syndrome is characterized by severe inability to form new memories and a variable inability to recall remote memories.

Alzheimer disease is the most common clinical memory disorder. Alzheimer disease is characterized pathologically by the degeneration of neurons and their replacement by senile plaques and neurofibrillary tangles. Clinicopathologic studies have suggested that cognitive decline is best correlated with the loss of synapses. Initially, the parietal and temporal lobes are affected, with relative sparing of the frontal lobes. This pattern of degeneration correlates with the early loss of memory, which is mostly a temporal lobe function. Also, syntactical language comprehension and visuospatial organization, functions that rely heavily on the parietal lobe, are impaired early in the course of Alzheimer disease.

In contrast, personality changes, which reflect frontal lobe function, are relatively late consequences of Alzheimer disease. A rarer, complementary cortical degeneration syndrome, Pick disease, first affects the frontal lobes while sparing the temporal and parietal lobes. In Pick disease, disinhibition and impaired language expression, which are signs of frontal dysfunction, appear early, with relatively preserved language comprehension and memory.

Memory loss can also result from disorders of the subcortical gray matter structures, specifically the basal ganglia and the brainstem nuclei, from the disease of the white matter, or from disorders that affect both gray and white matter.

Emotion. Individual emotional experiences occupy the attention of all mental health professionals. Emotion derives from basic drives, such as feeding, sex, reproduction, pleasure, pain, fear, and aggression, which all animals share. The neuroanatomical basis for these drives appears to be centered in the limbic system. Distinctly human emotions, such as affection,

pride, guilt, pity, envy, and resentment, are primarily learned and most likely are represented in the cortex. The regulation of drives appears to require an intact frontal cortex. The complex interplay of the emotions, however, is far beyond the understanding of functional neuroanatomists. Where, for example, are the representations of the id, the ego, and the superego? Through what pathway are ethical and moral judgments shepherded? What processes allow beauty to be in the eye of the beholder? These philosophical questions represent a real frontier of human discovery.

Several studies have suggested that within the cortex exists a hemispheric dichotomy of emotional representation. The left hemisphere houses the analytical mind but may have a limited emotional repertoire. For example, lesions to the right hemisphere, which cause profound functional deficits, may be noted with indifference by the intact left hemisphere. The denial of illness and of the inability to move the left hand in cases of right hemisphere injury is called *anosognosia.* In contrast, left hemisphere lesions, which cause profound aphasia, can trigger a catastrophic depression, as the intact right hemisphere struggles with the realization of the loss. The right hemisphere also appears dominant for affect, socialization, and body image.

Damage to the left hemisphere produces intellectual disorder and loss of the narrative aspect of dreams. Damage to the right hemisphere produces affective disorders, loss of the visual aspects of dreams, and a failure to respond to humor, shadings of metaphor, and connotations. In dichotic vision experiments, two scenes of varied emotional content were displayed simultaneously to each half of the visual field and were perceived separately by each hemisphere. A more intense emotional response attended the scenes displayed to the left visual field that was processed by the right hemisphere. Moreover, hemisensory changes representing conversion disorders have been repeatedly noted to involve the left half of the body more often than the right, an observation that suggests an origin in the right hemisphere.

Within the hemispheres, the temporal and frontal lobes play a prominent role in emotion. The temporal lobe exhibits a high frequency of epileptic foci, and temporal lobe epilepsy (TLE) presents an exciting model for the role of the temporal lobe in behavior. In studies of epilepsy, abnormal brain activation is analyzed, rather than the deficits in activity analyzed in classic lesional studies. TLE is of particular interest in psychiatry because patients with temporal lobe seizures often manifest bizarre behavior without the

classic grand mal shaking movements caused by seizures in the motor cortex. A proposed TLE personality is characterized by hyposexuality, emotional intensity, and a perseverative approach to interactions termed *viscosity.* Patients with left TLE may generate references to personal destiny and philosophical themes and display a humorless approach to life. In contrast, patients with right TLE may display excessive emotionality, ranging from elation to sadness. Although patients with TLE may display excessive aggression between seizures, the seizure itself may evoke fear.

The inverse of a TLE personality appears in persons with bilateral injury to the temporal lobes after head trauma, cardiac arrest, herpes simplex encephalitis, or Pick disease. This lesion resembles the one described in the Klüver–Bucy syndrome, an experimental model of temporal lobe ablation in monkeys. Behavior in this syndrome is characterized by hypersexuality, placidity, a tendency to explore the environment with the mouth, inability to recognize the emotional significance of visual stimuli, and continually shifting attention, called *hypermetamorphosis.* In contrast to the aggression-fear spectrum sometimes seen in patients with TLE, complete experimental ablation of the temporal lobes appears to produce a uniform, bland reaction to the environment, possibly because of an inability to access memories.

The prefrontal cortices influence mood in a complementary way. Whereas activation of the left prefrontal cortex appears to lift the mood, activation of the right prefrontal cortex causes depression. A lesion to the left prefrontal area, at either the cortical or the subcortical level, abolishes the typical mood-elevating influences and produces depression and uncontrollable crying. In contrast, a comparable lesion to the right prefrontal area may produce laughter, euphoria, and *witzelsucht,* a tendency to joke and make puns. Effects opposite to those caused by lesions appear during seizures, in which occurs abnormal, excessive activation of either prefrontal cortex. A seizure focus within the left prefrontal cortex can cause gelastic seizures, for example, in which the ictal event is laughter. Functional neuroimaging has documented left prefrontal hypoperfusion during depressive states, which normalized after the depression was treated successfully.

Limbic System Function. James Papez delineated the limbic system in 1937. The Papez circuit consists of the hippocampus, the fornix, the mammillary bodies, the anterior nucleus of the thalamus, and the cingulate

gyrus (Fig. 33-3). The boundaries of the limbic system were subsequently expanded to include the amygdala, septum, basal forebrain, nucleus accumbens, and orbitofrontal cortex.

Although this schema creates an anatomical loop for emotional processing, the specific contributions of the individual components other than the hippocampus or even whether a given train of neural impulses travels along the entire pathway is unknown.

The amygdala appears to be a critically important gate through which internal and external stimuli are integrated. Information from the primary senses is interwoven with internal drives, such as hunger and thirst, to assign emotional significance to sensory experiences. The amygdala may mediate learned fear responses, such as anxiety and panic, and may direct the expression of certain emotions by producing a particular affect. Neuroanatomical data suggest that the amygdala exerts a more powerful influence on the cortex, to stimulate or suppress cortical activity than the cortex exerts on the amygdala. Pathways from the sensory thalamic relay stations separately send sensory data to the amygdala and the cortex, but the subsequent effect of the amygdala on the cortex is the more potent of the two reciprocal connections.

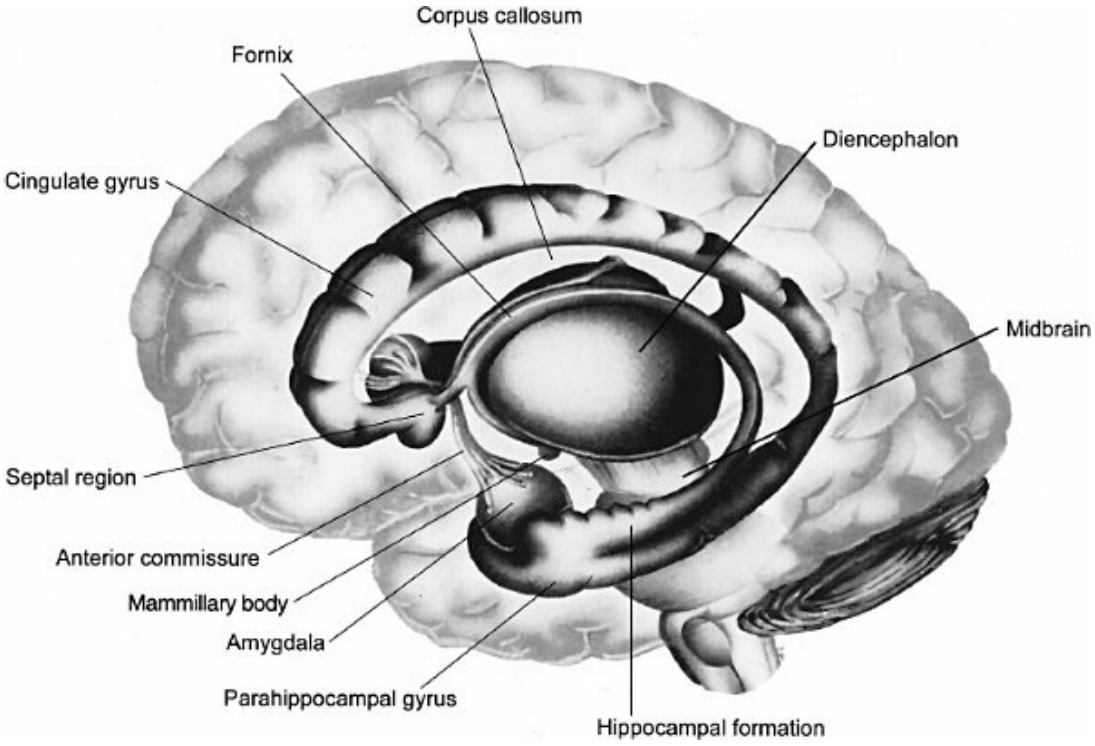


FIGURE 33-3
Schematic drawing of the major anatomic structures of the limbic system. The cingulate and parahippocampal gyri form the "limbic lobe," a rim of tissue located along the junction of the diencephalon and the cerebral hemispheres. (Adapted from Hendelman WJ. *Student's Atlas of Neuroanatomy*. Philadelphia, PA: WB Saunders; 1994:179.)

In contrast, damage to the amygdala has been reported to ablate the ability to distinguish fear and anger in other persons' voices and facial expressions. Persons with such injuries may have a preserved ability to recognize happiness, sadness, or disgust. The limbic system appears to house the emotional association areas, which direct the hypothalamus to express the motor and endocrine components of the emotional state.

Fear and Aggression. The electrical stimulation of animals throughout the subcortical area involving the limbic system produces rage reactions (e.g., growling, spitting, and arching of the back). Whether the animal flees or attacks depends on the intensity of the stimulation.

Limbic System and Schizophrenia. The limbic system has been particularly implicated in neuropathologic studies of schizophrenia. Eugen Bleuler's well-known four A's of schizophrenia—affect, associations, ambivalence, and autism—refer to brain functions served in part by limbic structures. Several clinicopathologic studies have found a reduction in the brain weight of the gray matter but not of the white matter in persons with schizophrenia. In pathologic as well as in magnetic resonance imaging (MRI) reports, persons with schizophrenia may have reduced the volume of the hippocampus, amygdala, and parahippocampal gyrus. Schizophrenia may be a late sequela of a temporal epileptic focus, with some studies reporting an association in 7 percent of patients with TLE.

Functional neuroimaging studies have demonstrated decreased activation of the frontal lobes in many patients with schizophrenia, particularly during tasks requiring willed action. A reciprocal increase in activation of the temporal lobe can occur during willed actions, such as finger movements or speaking, in persons with schizophrenia. Neuropathologic studies have shown a decreased density of neuropil, the intertwined axons, and dendrites of the neurons in the frontal lobes of these patients. During development, the density of neuropil is highest around age 1 year and then is reduced

somewhat through synaptic pruning; the density plateaus throughout childhood and is further reduced to adult levels in adolescence. One hypothesis of the appearance of schizophrenia in the late teenage years is that excessive adolescent synaptic pruning occurs and results in too little frontolimbic activity. Some experts have suggested that hypometabolism and paucity of interneuronal connections in the prefrontal cortex may reflect inefficiencies in working memory, which permits the disjointed discourse and loosening of associations that characterize schizophrenia. At present, the molecular basis for the regulation of the density of synapses within the neuropil is unknown. Other lines of investigation aimed at understanding the biologic basis of schizophrenia have documented inefficiencies in the formation of cortical synaptic connections in the middle of the second trimester of gestation, which may result from a viral infection or malnutrition. Neurodevelopmental surveys administered during childhood found an increased incidence of subtle neurologic abnormalities before the appearance of the thought disorder in persons who subsequently exhibited signs of schizophrenia.

In one intriguing study, positron emission tomography (PET) scanning was used to identify the brain regions that are activated when a person hears spoken language. A consistent set of cortical and subcortical structures demonstrated increased metabolism when speech was processed. The researchers then studied a group of patients with schizophrenia who were experiencing active auditory hallucinations. They found that the same brain areas that were activated when actual sounds were playing were activated while the patients were hallucinating. This included the primary auditory cortex. At the same time, decreased activation was seen of areas thought to monitor speech, including the left middle temporal gyrus and the supplementary motor area. This study raises the questions of what brain structure is activating the hallucinations, and by what mechanism do neuroleptic drugs suppress the hallucinations. Functional imaging has much to tell about the neuroanatomical basis of schizophrenia.

Frontal Lobe Function. The *frontal lobes,* the region that determines how the brain acts on its knowledge, constitute a category unto themselves. In comparative neuroanatomical studies, the massive size of the frontal lobes is the main feature that distinguishes the human brain from that of other

primates, and that lends it uniquely human qualities. There are four subdivisions of the frontal lobes. The first three—the motor strip, the supplemental motor area, and Broca area—are mentioned in the preceding discussion of the motor system and language. The fourth, most anterior, division is the prefrontal cortex. The prefrontal cortex contains three regions in which lesions produce distinct syndromes: the *orbitofrontal,* the *dorsolateral,* and the *medial.* Dye-tracing studies have defined dense reciprocal connections between the prefrontal cortex and all other brain regions. Therefore, to the extent that anatomy can predict function, the prefrontal cortex is ideally connected to allow sequential use of the entire palette of brain functions in executing the goal-directed activity. Indeed, frontal lobe injury usually impairs the executive functions: motivation, attention, and sequencing of actions.

Bilateral lesions of the frontal lobes are characterized by changes in personality—how persons interact with the world. The *frontal lobe syndrome,* which is most commonly produced by trauma, infarcts, tumors, lobotomy, multiple sclerosis, or Pick disease, consists of slow thinking, poor judgment, decreased curiosity, social withdrawal, and irritability. Patients typically display apathetic indifference to experience that can explode into impulsive disinhibition. Unilateral frontal lobe lesions may be mostly unnoticed because the intact lobe can compensate with high efficiency.

Frontal lobe dysfunction may be challenging to detect using highly structured, formal neuropsychological tests. Intelligence, as reflected in the intelligence quotient (IQ), may be average, and functional neuroimaging studies have shown that the IQ seems to require mostly parietal lobe activation. For example, during the administration of the Wechsler Adult Intelligence Scale-Revised (WAIS-R), the highest levels of increased metabolic activity during verbal tasks occurred in the left parietal lobe, whereas the highest levels of increased metabolic activity during performance skills occurred in the right parietal lobe. In contrast, frontal lobe pathology may become apparent only under unstructured, stressful, real-life situations.

A famous case illustrating the result of frontal lobe damage involves Phineas Gage, a 25-year-old railroad worker. While he was working with explosives, an accident drove an iron rod through Gage’s head. He

survived, but both frontal lobes were severely damaged. After the accident, his behavior changed dramatically. The case was written up by J. M. Harlow, M.D., in 1868, as follows: [George] is fitful, irreverent, indulging at times in the grossest profanity (which was not previously his custom), manifesting but little deference for his fellows, impatient of restraint or advice when it conflicts with his desires … His mind was radically changed, so decidedly that his friends and acquaintances said he was “no longer Gage” (see Fig. 33-4).

In one study of right-handed males, lesions of the right prefrontal cortex eliminated the tendency to use internal, associative memory cues and led to an extreme tendency to interpret the task at hand in terms of its immediate context. In contrast, right-handed males who had lesions of the left prefrontal cortex produced no context-dependent interpretations and interpreted the tasks entirely in terms of their internal drives. A mirror image of the functional lateralization appeared in left-handed subjects. This test thus revealed the most precise known association of higher cortical functional lateralization with the subjects’ dominant hand. Future experiments in this vein will attempt to reproduce these findings with functional neuroimaging. If corroborated, these studies suggest a remarkable complexity of functional localization within the prefrontal cortex and may also have implications for the understanding of psychiatric diseases in which prefrontal pathology has been postulated, such as schizophrenia and mood disorders.

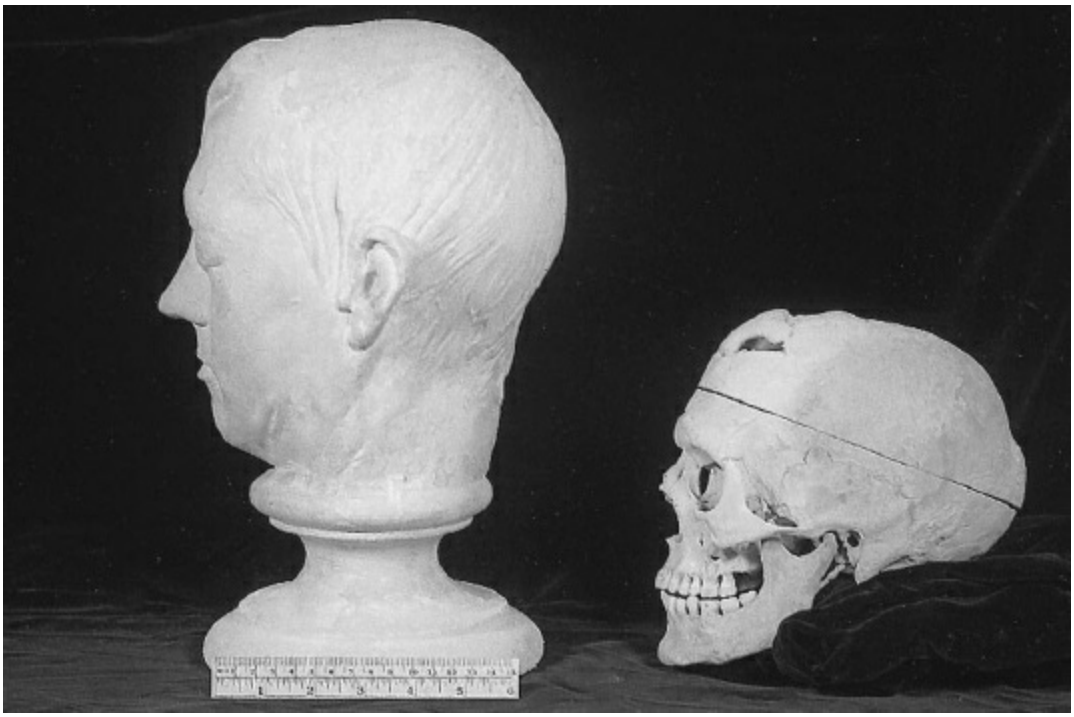


FIGURE 33-4
The life mask and skull of Phineas Gage. Note damage to the frontal region. “A famous case illustrating the result of frontal lobe damage involves Phineas Gage, a 25-year-old railroad worker. While he was working with explosives, an accident drove an iron rod through Gage’s head. He survived, but both frontal lobes were severely damaged. After the accident, his behavior changed dramatically. The case was written up by J.M. Harlow, M.D., in 1868, as follows: [Gage] is fitful, irreverent, indulging at times in the grossest profanity (which was not previously his custom), manifesting but little deference for his fellows, impatient of restraint or advice when it conflicts his desires…His mind was radically changed, so decidedly that his friends and acquaintances said he was ‘no longer Gage’.” (Courtesy of Anthony A. Walsh, Ph.D.)

The massive innervation of the frontal lobes by dopamine-containing nerve fibers is of interest because of the action of antipsychotic medications. At the clinical level, antipsychotic medications may help to organize the rambling associations of a patient with schizophrenia. At the neurochemical level, most typical antipsychotic medications block the actions of dopamine at the D_2 receptors. The frontal lobes, therefore, may be a significant therapeutic site of action for antipsychotic medications.

Development

The nervous system is divided into central and peripheral nervous systems (CNS and PNS). The CNS consists of the brain and spinal cord; the PNS

refers to all the sensory, motor, and autonomic fibers and ganglia outside the CNS. During development, both divisions arise from the neural tube. The neural tube forms through the folding of the neural plate, a specialization of the ectoderm (the outermost of the three embryonic layers). During embryonic development, the neural tube itself becomes the CNS; the ectoderm immediately superficial to the neural tube becomes the neural crest, which gives rise to the PNS. The formation of these structures requires chemical communication between the neighboring tissues in the form of cell surface molecules and diffusible chemical signals.

In many cases, an earlier-formed structure, such as the notochord, is said to *induce* the surrounding ectoderm to form a later structure, in this case, the neural plate. The identification of the chemical mediators of tissue induction is an active area of research. Investigators have begun to examine whether failures of the interactions of these mediators and their receptors could underlie errors in brain development that cause psychopathology.

Neuronal Migration and Connections. The life cycle of a neuron consists of a cell birth, migration to the adult position, extension of an axon, elaboration of dendrites, synaptogenesis, and, finally, the onset of chemical neurotransmission. Individual neurons are born in proliferative zones, generally located along the inner surface of the neural tube. At the peak of neuronal proliferation in the middle of the second trimester, 250,000 neurons are born each minute. Postmitotic neurons migrate outward to their adult locations in the cortex, guided by radially oriented astrocytic glial fibers. Glia-guided neuronal migration in the cerebral cortex occupies much of the first 6 months of gestation. For some neurons in the prefrontal cortex, migration occurs over a distance 5,000 times the diameter of the neuronal cell body. Neuronal migration requires a complex set of cell-cell interactions and is susceptible to errors in which neurons fail to reach the cortex and instead reside in ectopic positions. A group of such incorrectly placed neurons is called a *heterotopia.* Neuronal heterotopias have been shown to cause epilepsy and are positively associated with mental retardation. In a neuropathologic study of the planum temporale of four consecutive patients with dyslexia, heterotopias were a common finding. Recently, heterotopic neurons within the frontal lobe have been postulated to play a causal role in some cases of schizophrenia.

Many neurons lay down an axon as they migrate, whereas others do not initiate axon outgrowth until they have reached their cortical targets. Thalamic axons that project to the cortex initially synapse on a transient layer of neurons called the *subplate neurons.* In normal development, the axons subsequently detach from the subplate neurons and proceed superficially to synapse on the actual cortical cells. The subplate neurons then degenerate. Some brains from persons with schizophrenia reveal an abnormal persistence of subplate neurons, suggesting a failure to complete axonal pathfinding in the brains of these persons. This finding does not correlate with the presence of schizophrenia in every case, however. A characteristic branched dendritic tree elaborates once the neuron has completed migration. Synaptogenesis occurs at a furious rate from the second trimester through the first 10 years or so of life. The peak of synaptogenesis occurs within the first 2 postnatal years when as many as 30 million synapses form each second. Ensheathment of axons by myelin begins prenatally; it is mostly complete in early childhood but does not reach its full extent until late in the third decade of life. Myelination of the brain is also sequential.

Neuroscientists are tremendously interested in the effect of experience on the formation of brain circuitry in the first years of life. As noted earlier, many examples are seen of the impact of early sensory experience on the wiring of cortical sensory processing areas. Similarly, early movement patterns are known to reinforce neural connections in the supplemental motor area that drive specific motor acts. Neurons rapidly form a fivefold excess of synaptic connections; then, through a Darwinian process of elimination, only those synapses that serve a relevant function persist. This synaptic pruning appears to preserve input in which the presynaptic cell fires in synchrony with the postsynaptic cell, a process that reinforces repeatedly activated neural circuits. One molecular component that is thought to mediate synaptic reinforcement is the postsynaptic NMDA glutamate receptor. This receptor allows the influx of calcium ions only when activated by glutamate at the same time as the membrane in which it sits is depolarized. Thus, glutamate binding without membrane depolarization or membrane depolarization without glutamate binding fails to trigger calcium influx. NMDA receptors open in dendrites that are exposed to repeated activation, and their activation stimulates the stabilization of the synapse. Calcium is a crucial intracellular messenger that initiates a cascade of events, including gene regulation and the

release of trophic factors that strengthen particular synaptic connections. Although less experimental evidence exists for the role of experience in modulating synaptic connectivity of association areas than has been demonstrated in sensory and motor areas, neuroscientists assume that similar activity-dependent mechanisms may apply in all areas of the brain.

Adult Neurogenesis. A remarkable recent discovery has been that new neurons can be generated in specific brain regions (notably the dentate gyrus of the hippocampus) in adult animals, including humans. This discovery is in marked contrast to the previous belief that no neurons were produced after birth in most species. This discovery has a potentially profound impact on our understanding of normal development, incorporation of experiences, as well as the ability of the brain to repair itself after various types of injuries.

Neurologic Basis of Development Theories. In the realm of emotion, early childhood experiences have been suspected to be at the root of psychopathology since the earliest theories of Sigmund Freud. Freud's psychoanalytic method aimed at tracing the threads of a patient's earliest childhood memories. Franz Alexander added the goal of allowing the patient to relive these memories in a less pathologic environment, a process known as a *corrective emotional experience.* Although neuroscientists have no data demonstrating that this method operates at the level of neurons and circuits, emerging results reveal a profound effect of early caregivers on an adult individual's emotional repertoire. For example, the concept of attunement is defined as the process by which caregivers "playback a child's inner feelings." If a baby's emotional expressions are reciprocated consistently and sensitively, specific emotional circuits are reinforced. These circuits likely include the limbic system, in particular, the amygdala, which serves as a gate to the hippocampal memory circuits for emotional stimuli. In one anecdote, for example, a baby whose mother repeatedly failed to mirror her level of excitement emerged from childhood a too passive girl, who was unable to experience a thrill or a feeling of joy.

The relative contributions of nature and nurture are perhaps nowhere more indistinct than in the maturation of emotional responses, partly because the localization of emotion within the adult brain is only poorly understood. It is reasonable to assume, however, that the reactions of caregivers during a

child's first 2 years of life are eventually internalized as distinct neural circuits, which may be only incompletely subject to modification through subsequent experience. For example, axonal connections between the prefrontal cortex and the limbic system, which probably play a role in modulating basic drives, are established between the ages of 10 and 18 months. Recent work suggests that a pattern of terrifying experiences in infancy may flood the amygdala and drive memory circuits to be specifically alert to threatening stimuli, at the expense of circuits for language and other academic skills. Thus infants raised in a chaotic and frightening home may be neurologically disadvantaged for the acquisition of complex cognitive skills in school.

An adult correlate to this cascade of detrimental overactivity of the fear response is found in posttraumatic stress disorder (PTSD), in which persons exposed to an intense trauma involving death or injury may have feelings of fear and helplessness for years after the event. A PET scanning study of patients with PTSD revealed abnormally high activity in the right amygdala while the patients were reliving their traumatic memories. The researchers hypothesized that the stressful hormonal milieu present during the registration of the memories might have served to burn the memories into the brain and to prevent their erasure by the usual memory modulation circuits. As a result, the traumatic memories exerted a pervasive influence and led to a state of constant vigilance, even in safe, familiar settings.

Workers in the related realms of mathematics have produced results documenting the organizing effects of early experiences on internal representations of the external world. Since the time of Pythagoras, music has been considered a branch of mathematics. A series of studies have shown that groups of children who were given 8 months of intensive classical music lessons during preschool years later had significantly better spatial and mathematical reasoning in school than a control group. Nonmusical tasks, such as navigating mazes, drawing geometric figures, and copying patterns of two-color blocks, were performed significantly more skillfully by the musical children. Early exposure to music, thus, may be ideal preparation for later acquisition of complex mathematical and engineering skills.

These tantalizing observations suggest a neurologic basis for the developmental theories of Jean Piaget, Erik Erikson, Margaret Mahler, John Bowlby, Sigmund Freud, and others. Erikson's epigenetic theory states that

normal adult behavior results from the successful, sequential completion of each of several infantile and childhood stages. According to the epigenetic model, failure to complete an early stage is reflected in subsequent physical, cognitive, social, or emotional maladjustment. By analogy, the experimental data just discussed suggest that early experience, particularly during the critical window of opportunity for establishing neural connections, primes the necessary circuitry for language, emotions, and other advanced behaviors. Miswiring of an infant's brain may lead to severe handicaps later when the person attempts to relate to the world as an adult. These findings support the vital need for adequate public financing of Early Intervention and Head Start programs, programs that may be the most cost-effective means of improving persons' mental health.

NEURAL DEVELOPMENT AND NEUROGENESIS

The human brain is a structurally and functionally complex system that exhibits ongoing modification in response to both experience and disease. The anatomical and neurochemical systems that underlie the cognitive, social, emotional and sensorimotor functions of the mature nervous system emerge from neuronal and glial cell populations that arise during the earliest periods of development.

An understanding of molecular and cellular mechanisms mediating nervous system development is critical in psychiatry because abnormalities of developmental processes contribute to many brain disorders. Although a developmental basis may not be surprising in early childhood disorders, such as autism, fragile X mental retardation, and Rett syndrome, even mature diseases, including schizophrenia and depression, reflect ontogenetic factors. For example, evidence from brain pathology and neuroimaging indicates that there are reductions in forebrain region volumes, neuron and glial cell numbers, and some classes of interneurons in schizophrenia that are apparent at the time of diagnosis. Similarly, in autism, early brain growth is abnormally increased, and abnormalities of the cellular organization are observed that reflect disturbances in the fundamental processes of cell proliferation and migration. When there is abnormal regulation of early brain development, a foundation of altered neuron populations that may differ in cell types, numbers, and positions is laid down. They may also develop

abnormal connections, which also have consequences for interacting glial populations. With progressive postnatal development, the maturing brain systems call upon component neurons to achieve increasing levels of complex information processing, which may be deficient should initial conditions be disturbed. New neural properties emerge during maturation as neuron populations elaborate additional functional networks based on and modified by ongoing experience. Given the brain's dynamic character, we may expect that developmental abnormalities in neural populations and systems, caused by genetic as well as environmental factors, will manifest at diverse times in a person's life.

Overview of Nervous System Morphologic Development

In considering brain development, several overarching principles need to be considered. First, different brain regions and neuron populations are generated at distinct times of development and exhibit specific temporal schedules. This pattern has implications for the consequences of specific developmental insults, such as the production of autism following fetal exposure to the drug thalidomide only during days 20 to 24 of gestation. Second, the sequence of cellular processes comprising ontogeny predicts that abnormalities in early events necessarily lead to differences in subsequent stages, although not all abnormalities may be accessible to our clinical tools. For example, a deficit in the number of neurons will likely lead to reductions in axonal processes and ensheathing white matter in the mature brain. However, at the clinical level, since glial cells outnumber neurons 8 to 1, the glial cell population, the oligodendrocytes, and their myelin appear as an altered white matter on neuroimaging with little evidence of a neuronal disturbance. Third, it is clear that specific molecular signals, such as extracellular growth factors and cognate receptors or transcription factors, play roles at multiple developmental stages of the cell. For example, both insulin-like growth factor I (IGF-I) and brain-derived neurotrophic factor (BDNF) regulate multiple cellular processes during the developmental generation and mature function of neurons, including cell proliferation, survival promotion, neuron migration, process outgrowth, and the momentary synaptic modifications (plasticity) underlying learning and memory. Thus changes in expression or regulation of a ligand or its receptor, by experience,

environmental insults, or genetic mechanisms, will have effects on multiple developmental and mature processes.

The Neural Plate and Neurulation. The nervous system of the human embryo first appears between 2½ and 4 weeks of gestation. During development, the emergence of new cell types, including neurons, results from interactions between neighboring layers of cells. On gestational day 13, the embryo consists of a sheet of cells. Following gastrulation (days 14 to 15), which forms a two-cell–layered embryo consisting of ectoderm and endoderm, the neural plate region of the ectoderm is delineated by the underlying mesoderm, which appears on day 16. The mesoderm forms by cells entering a midline cleft in the ectoderm called the primitive streak. After migration, the mesodermal layer lies between the ectoderm and endoderm and induces the overlying ectoderm to become a neural plate. Induction usually involves the release of soluble growth factors from one group of cells, which in turn bind receptors on neighboring cells, eliciting changes in nuclear transcription factors that control downstream gene expression. In some cases, cell–cell contact-mediated mechanisms are involved. In the gene-patterning section below, the critical roles of soluble growth factors and transcription factor expression are described.

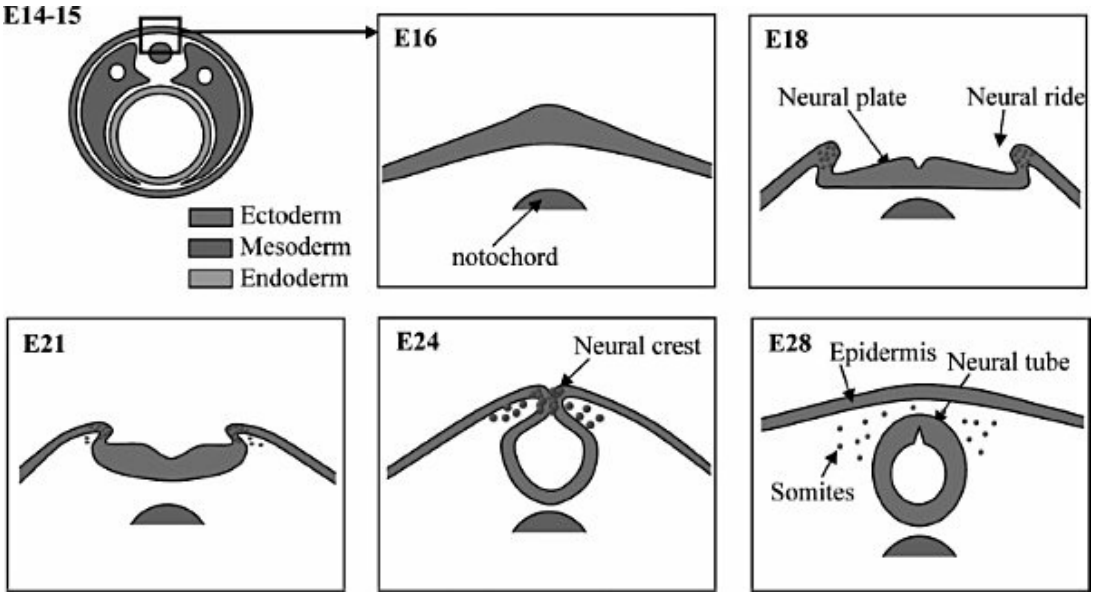


FIGURE 33-5
Mechanisms of neurulation. Neurulation begins with the formation of a neural plate in response to soluble growth factors released by the underlying notochord. The neural plate originates as a thickening of the ectoderm that results from

cuboidal epithelial cells becoming columnar in shape. With further changes in cell shape and adhesion, the edges of the plate fold and rise, meeting in the midline to form a tube. Cells at the tips of the neural folds come to lie between the neural tube and overlying epidermis, forming the neural crest that gives rise to the peripheral nervous system and other structures. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:44.)

The neural plate, the induction of which is complete by 18 days, is a sheet of columnar epithelium and is surrounded by ectodermal epithelium. After formation, the edges of the neural plate elevate, forming the neural ridges. Subsequently, changes in intracellular cytoskeleton and cell-extracellular matrix attachment cause the ridges to merge in the midline and fuse, a process termed neurulation, forming the neural tube, with a central cavity presaging the ventricular system (Fig. 33-5). Fusion begins in the cervical region at the hindbrain level (medulla and pons) and continues rostrally and caudally. Neurulation occurs at 3 to 4 weeks of gestation in humans, and its failure results in anencephaly rostrally and spina bifida caudally. Neurulation defects are well known following exposure to retinoic acid in dermatologic preparations and anticonvulsants, especially valproic acid, as well as diets deficient in folic acid.

Another product of neurulation is the neural crest, the cells of which derive from the edges of the neural plate and dorsal neural tube. From this position, neural crest cells migrate dorsolaterally under the skin to form melanocytes and ventromedially to form dorsal root sensory ganglia and sympathetic chains of the peripheral nervous system and ganglia of the enteric nervous system. However, neural crest gives rise to diverse tissues, including cells of neuroendocrine, cardiac, mesenchymal, and skeletal systems, forming the basis of many congenital syndromes involving the brain and other organs. The neural crest origin at the border of neural and epidermal ectoderm and its generation of melanocytes forms the basis of the neurocutaneous disorders, including tuberous sclerosis and neurofibromatosis. Finally, another nonneuronal structure of mesodermal origin formed during neurulation is the notochord found on the ventral side of the neural tube. As we will describe later in this chapter, the notochord plays a critical role during neural tube differentiation, since it is a signaling source of soluble growth factors, such as sonic hedgehog (Shh), which affect gene patterning and cell determination.

Regional Differentiation of the Embryonic Nervous System. After closure, the neural tube expands differentially to form major morphologic subdivisions that precede the major functional divisions of the brain. These subdivisions are important developmentally because different regions are generated according to specific schedules of proliferation and subsequent migration and differentiation. The neural tube can be described in three dimensions, including longitudinal, circumferential, and radial. The longitudinal dimension reflects the rostrocaudal (anterior–posterior) organization, which mostly consists of the brain and spinal cord. Organization in the circumferential dimension, tangential to the surface, represents two significant axes: In the dorsoventral axis, cell groups are uniquely positioned from top to bottom.

On the other hand, in the medial to the lateral axis, there is mirror-image symmetry, consistent with the right-left symmetry of the body. Finally, the radial dimension represents the organization from the innermost cell layer adjacent to the ventricles to the outermost surface and exhibits region-specific cell layering. At 4 weeks, the human brain is divided longitudinally into the prosencephalon (forebrain), mesencephalon (midbrain), and rhombencephalon (hindbrain). These three subdivisions or “vesicles” divide further into five divisions by 5 weeks, consisting of the prosencephalon, which forms the telencephalon (including cortex, hippocampus, and basal ganglia) and diencephalon (thalamus and hypothalamus), the mesencephalon, (midbrain), and the rhombencephalon, yielding metencephalon (pons and cerebellum) and myelencephalon (medulla). Morphologic transformation into five vesicles depends on the region-specific proliferation of precursor cells adjacent to the ventricles, the so-called ventricular zones (VZs). As discussed later, proliferation intimately depends on soluble growth factors made by proliferating cells themselves or released from regional signaling centers. In turn, growth factor production and cognate receptor expression also depend on region-specific patterning genes. We now know that VZ precursors, which appear morphologically homogeneous, express a checkerboard array of molecular genetic determinants that control the generation of specific types of neurons in each domain (Fig. 33-6).

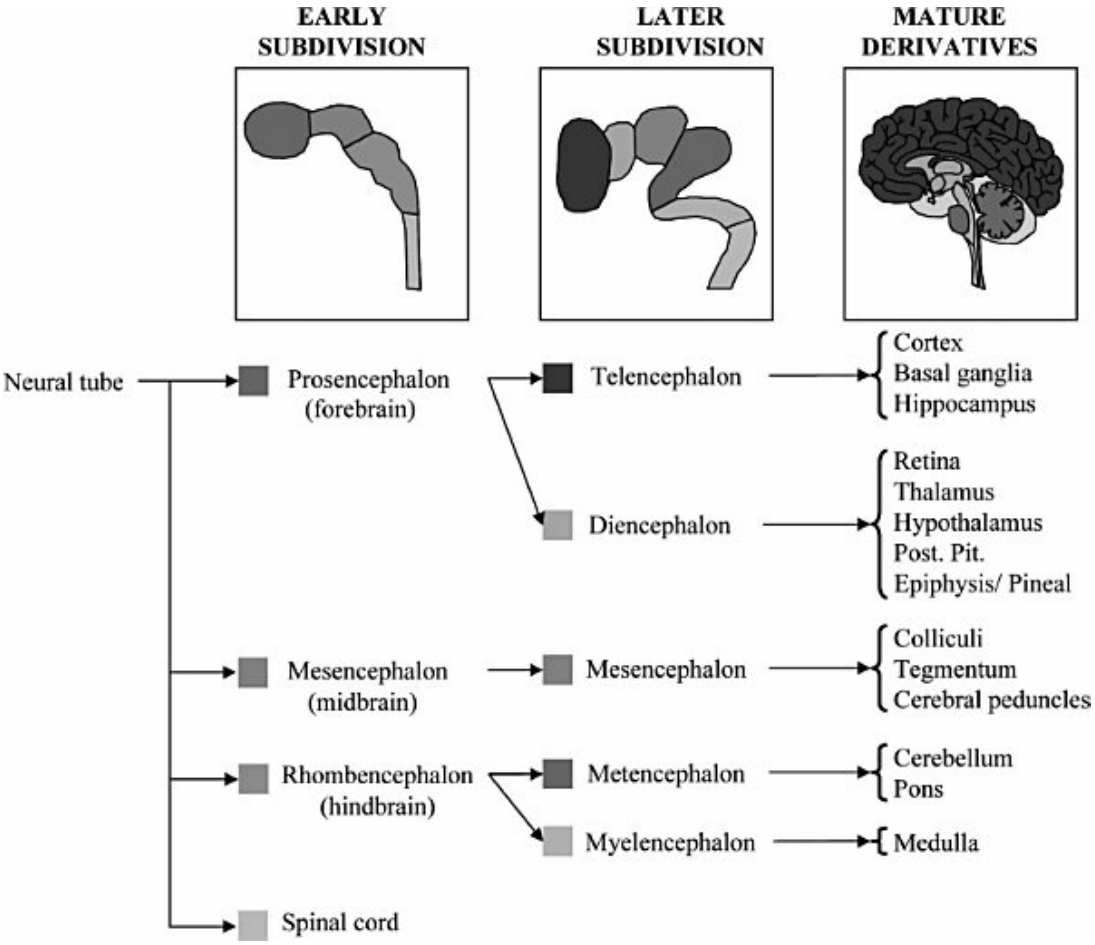


FIGURE 33-6
Progression of brain regional differentiation. Early after neurulation, the neural tube differentiates into four regions (forebrain, midbrain, hindbrain, and spinal cord) that give rise following later divisions and maturation to the different brain structures. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:45.)

In the circumferential dimension, the organization begins very early and extends over many rostrocaudal subdivisions. In the spinal cord, the majority of tissue comprises the lateral plates, which later divide into dorsal or alar plates, composed of sensory interneurons, and motor or basal plates, consisting of ventral motor neurons. Two other diminutive plates, termed the roof plate and floor plate, are virtually absent in maturity; however, they play critical regulatory roles as growth factor signaling centers in the embryo. Indeed, the floor plate, in response to Shh from the ventrally located notochord, produces its own Shh, which in turn induces neighboring cells in the ventral spinal cord and brainstem to express region-specific transcription

factors that specify cell phenotype and function. For example, in combination with other factors, floor plate Shh induces midbrain precursors to differentiate into dopamine-secreting neurons of the substantia nigra.

Similarly, the roof plate secretes growth factors, such as bone morphogenetic proteins (BMPs), which induce dorsal neuron cell fate in the spinal cord. In the absence of a roof plate, dorsal structures fail to form, such as the cerebellum, and midline hippocampal structures are missing. Finally, in the radial dimension, the organization of layers is subdivision specific, produced by differential proliferation of VZ precursors and cell migration, as described later.

The Ventricular and Subventricular Proliferative Zones. The distinct patterns of precursor proliferation and migration in different regions generate the radial organization of the nervous system. In each longitudinal subdivision, the final population size of a brain region depends on the interplay of regulated neurogenesis with programmed cell death. Traditional concepts had suggested that there was excess cell production everywhere and that final cell number regulation was achieved primarily after neurogenesis through selective cell death mediated by target-derived survival (trophic) factors. We now know that the patterning genes discussed later play significant roles in directing regional precursor proliferation that is coordinated with final structural requirements, and that programmed cell death occurs at multiple stages. Consequently, in diseases characterized by brain regions smaller than usual, such as schizophrenia, there may be a failure to generate neurons initially, as opposed to normal generation with subsequent cell loss.

Radial and Tangential Patterns of Neurogenesis and Migration. Of interest to psychiatry, the cerebral cortex is the paradigmatic model of inside-to-outside neurogenesis. A large number of studies now relate specific genetic mutations to distinct cortical malformations that alter neurogenesis, migration, and cellular organization, thereby increasing our knowledge of both normal and pathophysiologic cortical development. The characteristic six-cell layers represent a common cytoarchitectural and physiologic basis for the neocortical function and are derived from the embryonic forebrain telencephalic vesicles. Within each layer, neurons exhibit related axodendritic

morphologies, use common neurotransmitters, and establish similar afferent and efferent connections. In general, pyramidal neurons in layer 3 establish synapses within and between cortical hemispheres, whereas deeper layer 5/6 neurons project primarily to subcortical nuclei, including the thalamus, brainstem, and spinal cord. The majority of cortical neurons originate from the forebrain VZ. At the earliest stages, the first postmitotic cells migrate outward from the VZ to establish a superficial layer termed the preplate. Two consequential cell types comprise the preplate—Cajal–Retzius cells, which form outermost layer one or marginal zone, and subplate neurons, which lay beneath future layer 6. These distinct regions form when later-born cortical plate neurons migrate within and divide the preplate in two (Fig. 33-7).

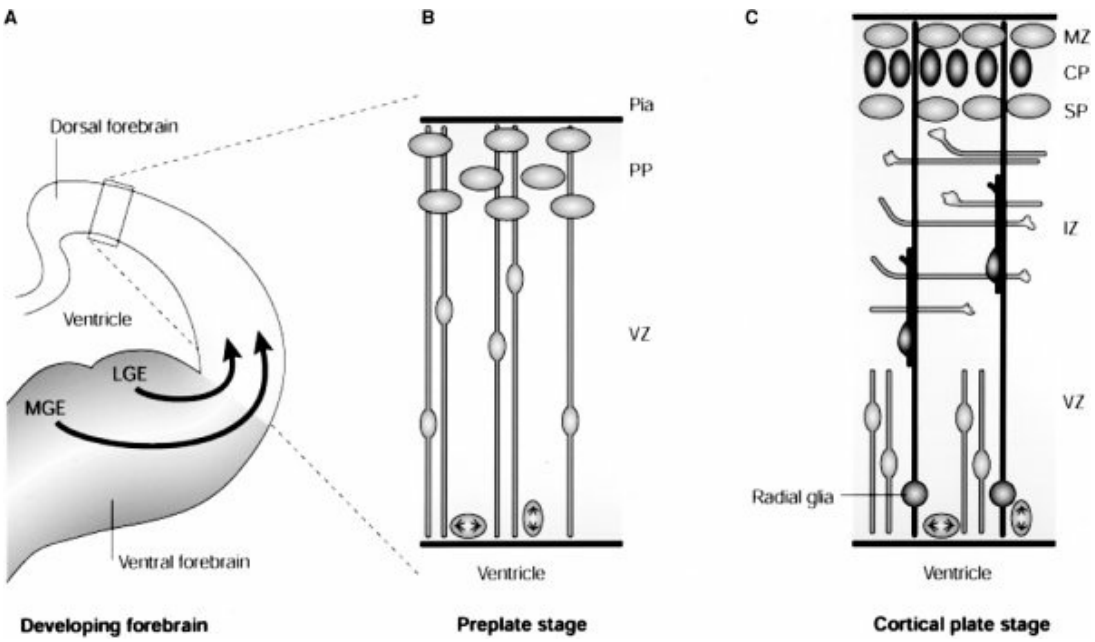


FIGURE 33-7

Schematic drawing of radial and tangential migration during cerebral cortex development. **A:** A coronal section of one-half of the developing rat forebrain. The dorsal forebrain gives rise to the cerebral cortex. Medial ganglionic eminences (MGEs) and lateral ganglionic eminences (LGEs) of the ventral forebrain generate neurons of the basal ganglia and the cortical interneurons. The *arrows* indicate the tangential migration route for γ-aminobutyric acid (GABA) interneurons to the cortex. The boxed area (enlarged in **B** and **C**) shows the developing cortex at early and late stages. **B:** In the dorsal forebrain, the first cohort of postmitotic neurons migrate out from the ventricular zone (VZ) and create a preplate (PP) below the pial surface. **C:** Subsequent postmitotic neurons will migrate along radial glia through the intermediate zone (IZ) and take position in the middle of the preplate, creating a cortical plate (CP) between the outer marginal zone (MZ) and inner subplate (SP). Ultimately, the CP will be composed of six layers that are born sequentially, migrating in an inside-to-outside pattern. Horizontal processes

in the IZ represent axon terminals of thalamic afferents. (From Nadarajah B, Parnavelas JG. Modes of neuronal migration in the developing cerebral cortex. *Nat Neurosci*. 2002;3:423, with permission.)

A recent discovery, postulated for years, has changed the view of the origins of cortical neuron populations involved in human brain disease. Neuron tracing experiments in culture and in vivo demonstrate that the neocortex, a dorsal forebrain derivative, is also populated by neurons generated in the ventral forebrain (see Fig. 33-7). Molecular studies of patterning genes, especially *Dlx*, strongly support this model (see below). In contrast to excitatory pyramidal neurons, the overwhelming majority of inhibitory γ-aminobutyric acid (GABA)-secreting interneurons originate from mitotic precursors of the ganglionic eminences that generate the neurons of the basal ganglia. Subsets of interneurons also secrete neuropeptides, such as neuropeptide Y (NPY) and somatostatin, and express nitrous oxide (NOS)-generating enzyme. Not associated with cortical VZ radial glia, these GABA interneurons reach the cortical plate by migrating tangentially, in either the superficial marginal zone or a deep position above the VZ, the subplate region where thalamic afferents are also growing. Significantly, in brains from patients with schizophrenia, the prefrontal cortex exhibits a reduced density of interneurons in layer 2. Also, there is upregulation of $GABA_A$-receptor binding, a potential functional compensation, as well as a relative deficiency of NOS-expressing neurons. These observations have led to the hypothesis that schizophrenia is due to reduced GABAergic activity. The origin of GABA interneurons from the ganglionic eminences and their association with specific patterning genes raises new genetic models of disease causation and possible strategies for disease intervention. Thus, more broadly, normal cortical development depends on a balance of two principal patterns of neurogenesis and migration, consisting of radial migration of excitatory neurons from the dorsal forebrain VZ and tangential migration of inhibitory neurons from the ventral forebrain.

In contrast to inside-to-outside neurogenesis observed in the cortex, phylogenetically older regions, such as the hypothalamus, spinal cord, and hippocampal dentate gyrus, exhibit the reverse order of cell generation. First-formed postmitotic neurons lie superficially, and last-generated cells localize toward the center. Although this outside-to-inside pattern might reflect

passive cell displacement, radial glia, and specific migration signaling molecules are involved. Furthermore, cells do not always lie in direct extension from their locus of VZ generation. Instead, some groups of cells migrate to specific locations, as observed for neurons of the inferior olivary nuclei.

Of prime importance in psychiatry, the hippocampus demonstrates both radial and nonradial patterns of neurogenesis and migration. The pyramidal cell layer, Ammon's horn Cornu Ammonis (CA) 1 to 3 neurons, is generated in a typical outside-to-inside fashion in the dorsomedial forebrain for a discrete period, from 7 to 15 weeks of gestation, and exhibits intricate migration patterns. In contrast, the other significant population, dentate gyrus granule neurons, start appearing at 18 weeks and exhibits prolonged postnatal neurogenesis, originating from several migrating secondary proliferative zones. In rats, for instance, granule neurogenesis starts at embryonic day 16 (E16) with proliferation in the forebrain VZ. At E18, an aggregate of precursors migrates along a subpial route into the dentate gyrus. There, they generate granule neurons in situ. After birth, there is another migration, localizing proliferative precursors to the dentate hilus, which persists until 1 month of life. After that, granule precursors move to a layer just under the dentate gyrus, termed the subgranular zone (SGZ), which produces neurons throughout life in adult rats, primates, and humans. In rodents, SGZ precursors proliferate in response to cerebral ischemia, tissue injury, and seizures, as well as growth factors. Finally, the diminished hippocampal volume reported in schizophrenia raises the possibility that disordered neurogenesis plays a role in pathogenesis, as either a basis for dysfunction or a consequence of brain injuries, consistent with associations of gestational infections with disease manifestation.

Finally, a different combination of radial and nonradial migration is observed in the cerebellum, which is a brain region that plays an essential function in nonmotor tasks, with particular significance for autism spectrum disorders (ASDs). Except for granule cells, the other significant neurons, including Purkinje and deep nuclei, originate from the primary VZ of the fourth ventricle, coincident with other brainstem neurons. In rats, this occurs at E13 to E15, and in humans, at 5 to 7 weeks of gestation. The granule neurons, as well as the basket and stellate interneurons, originate in the secondary proliferative zone, the external germinal cell layer (EGL), which

covers the newborn cerebellum at birth. EGL precursors originate in the fourth ventricle VZ and migrate dorsally through the brainstem to reach this superficial position. The rat EGL proliferates for 3 weeks, generating more neurons than in any other structure, whereas, in humans, EGL precursors exist for at least 7 weeks and up to 2 years. When an EGL precursor stops proliferating, the cell body sinks below the surface and grows bilateral processes that extend transversely in the molecular layer, and then the soma migrates further down into the internal granule layer (IGL). Cells reach the IGL along specialized Bergmann glia, which serve guidance functions similar to those of the radial glia. However, in this case, cells originate from a secondary proliferative zone that generates neurons exclusively of the granule cell lineage, indicating a restricted neural fate. Clinically, this postnatal population in infants makes cerebellar granule neurogenesis vulnerable to infectious insults of early childhood and an undesirable target of several therapeutic drugs, such as steroids, well known to inhibit cell proliferation. Also, proliferative control of this stem cell population is lost in the common childhood brain tumor, medulloblastoma (see Fig. 33-8).

Developmental Cell Death. During nervous system development, cell elimination is required to coordinate the proportions of interacting neural cells. Developmental cell death is a reproducible, spatially, and temporally restricted death of cells that occurs during the organism's development. Three types of developmental cell death have been described: (1) phylogenetic cell death that removes structures in one species that served evolutionarily earlier ones, such as the tail or the vomeronasal nerves; (2) morphogenetic cell death, which sculpts the fingers from the embryonic paddle and is required to form the optic vesicles, as well as the caudal neural tube; and (3) histogenetic cell death, a widespread process that allows the removal of select cells during development of specific brain regions. Numerous studies have focused on histogenetic cell death, the impact of which varies among brain regions but can affect 20 to 80 percent of neurons in some populations. A significant role for developmental cell death was proposed in the 1980s based on the paradigm of nerve growth factor, suggesting that following neurogenesis, neurons compete for trophic factors. In this model, the survival of differentiating neurons depended absolutely on establishing axonal connections to the correct targets in order to obtain survival-promoting

(trophic) growth factors, such as the neurotrophins. Otherwise, they would be eliminated by programmed cell death. This competitive process was thought to ensure the proper matching of new neuronal populations with the size of its target field. Although such interactions are involved in controlling cell degeneration, this model is overly simplistic: Developmental cell death also occurs in neural precursors and immature neurons, before any synaptic contacts are established.

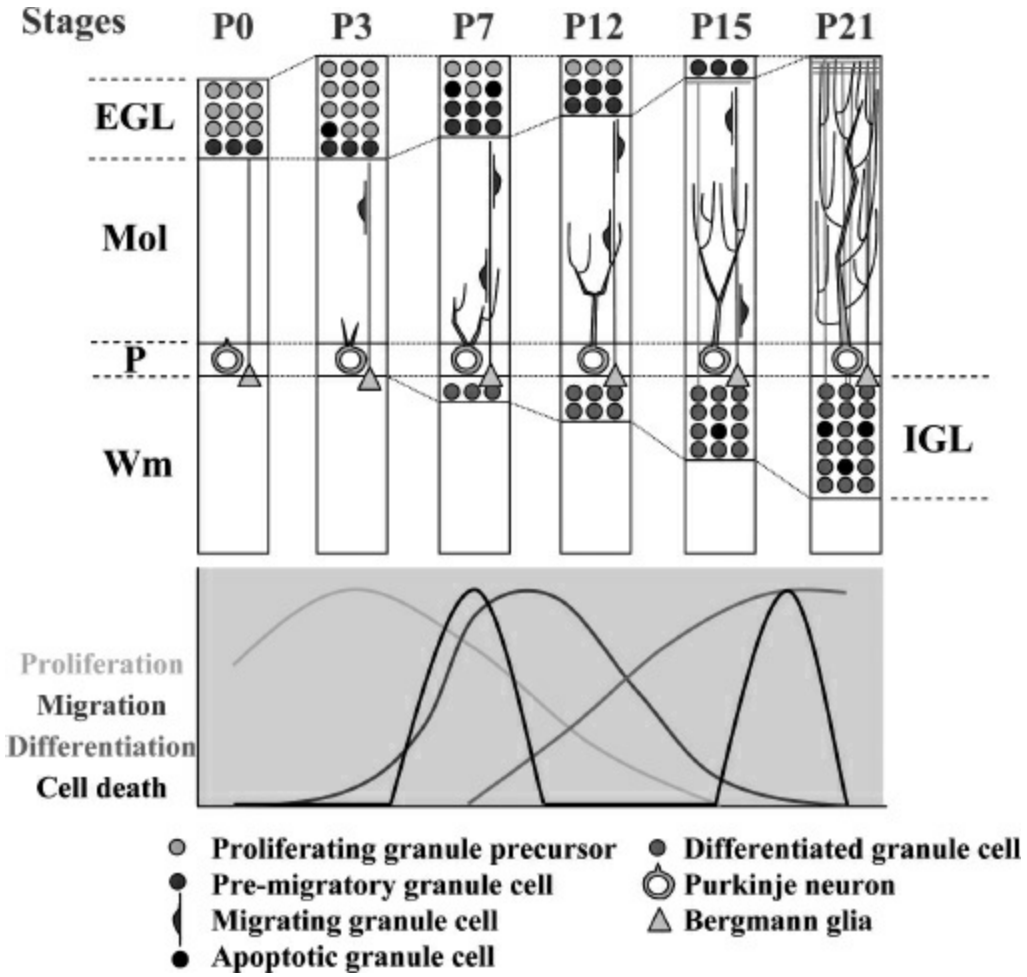


FIGURE 33-8
Neurogenesis, migration, and differentiation of granule cells during cerebellar development. Granule cell precursors proliferate in the external germinal layer. After exiting the cell cycle, they migrate through the molecular layer and past the Purkinje neurons to reach the internal granule layer where they differentiate and make synapses. Neurons that do not migrate properly or that do not establish proper synaptic connections undergo apoptosis. EGL, external germinal cell layer; Mol, molecular layer; P, Purkinje cell layer; IGL, internal granule cell layer; Wm, white matter. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:48.)

APOPTOSIS. Apoptotic cell death, or apoptosis, is the primary type of developmental cell degeneration. Apoptosis or "programmed cell death" involves specific molecules that possess enzymatic activities such as cysteine-containing aspartate-specific proteases, also called "caspases," which participate in complex intracellular mechanisms. A large number of signals (both proapoptotic and antiapoptotic) converge to regulate common signaling pathways. Of importance for psychiatry, both developmental as well as pathologic cell death, involves many of the same signaling cascades. A failure to inhibit apoptosis is involved in cancers and autoimmune diseases (multiple sclerosis), whereas excess stimulation of apoptosis is observed in neurodegenerative diseases during both development (Huntington disease, lysosomal diseases, and leukodystrophy) and aging (Alzheimer and Parkinson diseases). Massive apoptotic cell death is also observed during acquired developmental brain injuries such as hypoxia-ischemia, fetal alcohol syndrome, and exposure to ionizing radiations and neurotoxicants. Thus dysregulation of apoptotic cell death during development can lead to severe brain abnormalities, which may only manifest later as mature functional impairments.

Programmed cell death is a necessary process during neurodevelopment, as genetic deletion of caspases in embryonic mice produces enlarged and disorganized brains with marked regional specificity. Programmed cell death occurs at multiple stages of nervous system development, interacting with neurogenesis and differentiation with precise and complex mechanisms. As many neuropathologies also involve dysregulation of apoptosis, future studies hold promise for elucidation and treatment of neurologic diseases.

The Concept of Neural Patterning

Principles of Function. The morphologic conversion of the nervous system through the embryonic stages, from the neural plate through the neural tube to brain vesicles, is controlled by interactions between extracellular factors and intrinsic genetic programs. In many cases, extracellular signals are soluble growth factors secreted from regional signaling centers, such as the notochord, floor, or roof plates, or surrounding mesenchymal tissues. The precursor's ability to respond (competence)

depends on cognate receptor expression, which is determined by patterning genes whose proteins regulate gene transcription. The remarkable new observation is that the subdivisions of the embryonic telencephalon that were initially based on mature differences in morphology, connectivity, and neurochemical profiles are also distinguished embryonically by distinct patterns of gene expression. Classical models had suggested that the cerebral cortex was generated as a reasonably homogeneous structure, unlike most epithelia, with individual functional areas specified relatively late, after cortical layer formation, by the ingrowth of afferent axons from the thalamus. In marked contrast, recent studies indicate that proliferative VZ precursors themselves display regional molecular determinants, a “protomap,” which the postmitotic neurons carry with them as they migrate along radial glia to the cortical plate.

Consequently, innervating thalamic afferents may serve to modulate only intrinsic molecular determinants of the protomap. Indeed, in two different genetic mutants, *Gbx2* and *Mash1*, in which thalamocortical innervation is disrupted, expression of cortical patterning genes proceeds unaltered. On the other hand, thalamic afferent growth may be directed by patterning genes and subsequently play roles in modulating regional expression patterns. Thus experience-dependent processes may contribute less to cortical specialization than initially postulated.

The term patterning genes connotes families of proteins that serve primarily to control transcription of other genes, the products of which include other transcription factors or proteins involved in cellular processes, such as proliferation, migration, or differentiation. Characteristically, transcription factor proteins contain two principal domains, one that binds DNA promoter regions of genes and the other that interacts with other proteins, either transcription factors or components of intracellular second messengers. Notably, transcription factors form multimeric protein complexes to control gene activation. Therefore, a single transcription factor will play diverse roles in multiple cell types and processes, according to what other factors are present, the so-called cellular environment. The combinatorial nature of gene promoter regulation leads to a diversity of functional outcomes when a single patterning gene is altered.

Furthermore, because protein interactions depend on protein–protein affinities, there may be complex changes as a single factor’s expression level

is altered. This process may be a critical mechanism of human variation and disease susceptibility, since polymorphisms in gene promoters, known to be associated with human disease, can alter levels of gene protein products. A transcription factor may associate primarily with one partner at a low concentration but with another at a higher titer. The multimeric nature of regulatory complexes allows a single factor to stimulate one process while simultaneously inhibiting another. During development, a patterning gene may thus promote one event, say the generation of neurons, by stimulating one gene promoter, while simultaneously sequestering another factor from a different promoter whose activity is required for an alternative phenotype, such as glial cell fate. Finally, the factors frequently exhibit cross-regulatory functions, where one factor negatively regulates the expression of another. This activity leads to the establishment of tissue boundaries, allowing the formation of regional subdivisions, such as basal ganglia and cerebral cortex in the forebrain.

Also, to combinatorial interactions, patterning genes exhibit distinct temporal sequences of expression and function, acting hierarchically. Functional hierarchies were established experimentally by using genetic approaches, either deleting a gene (loss of function) or over-/ectopically expressing it (gain of function), and defining developmental consequences. At the most general level, genetic analyses indicate that regionally restricted patterning genes participate in specifying the identity, and therefore function, of cells in which they are expressed. Subdivisions of the brain and cerebral cortex specifically, are identified by regionalized gene expression in the proliferative VZ of the neural tube, leading to subsequent differentiation of distinct types of neurons in each mature (postmitotic) region. Thus the protomap of the embryonic VZ predicts the cortical regions it will generate and may instruct the hierarchical temporal sequence of patterning gene expression. It appears that the different genes underlie multiple stages of brain development, including the following: (1) determining that ectoderm will give rise to the nervous system (as opposed to the skin); (2) defining the dimensional character of a region, such as positional identity in dorsoventral or rostrocaudal axes; (3) specifying cell class, such as neuron or glia; (4) defining when proliferation ceases, and differentiation begins, (5) determining specific cell subtype, such as GABA interneuron, as well as projection pattern; and (6) defining laminar position in the region, such as

cerebral cortex. Although investigations are ongoing, studies indicate that these many steps depend on interactions of transcription factors from multiple families. Furthermore, a single transcription factor plays regulatory roles at multiple stages in the developmental life of a cell, yielding complex outcomes, for instance, in genetic loss of function studies and human disease.

Recent advances in molecular biology have led to the identification of another principle of nervous system organization, which, if sustained by further studies, may provide a molecular basis for brain system diseases, such as Parkinson disease and autism. Using molecular techniques to permanently identify cells that had expressed during the development of a specific gene, in this case, the soluble growth factor, Wnt3a, investigators were able to determine where cells originated embryonically and could trace their path of migration along the neuraxis during development. These genetic-fate mapping studies indicate that cells that expressed Wnt3a migrated widely from the dorsal midline into the dorsal regions of the brain and spinal cord, thereby contributing to diverse adult structures in the diencephalon, midbrain, and brainstem, and rostral spinal cord. Of interest, most of these structures were linked into a functional neural network, specifically the auditory system. The observation that a single functional system emerges from a specific group of fated cells would allow for restricted neurologic system–based disorders, such as deficits in dopamine or catecholamine neurons, or for the dysfunction of inter-related brain regions that subserve social cognition and interaction, a core symptom of the ASDs. Other adult system degenerations may also be considered. This new observation may change the way that we consider temporal changes in patterning gene expression of specific brain regions during development.

Finally, patterning gene expression in nervous system subdivisions is not insensitive to environmental factors. On the contrary, the expression is intimately regulated by growth factors released from regional signaling centers. Indeed, although a century of classical experimental embryology described the induction of new tissues between neighboring cell layers morphologically, we have only recently defined molecular identities of soluble protein morphogens and cell response genes underlying development. Signaling molecules from discrete centers establish tissue gradients that provide positional information (dorsal or ventral), impart cell specification, or control regional growth. Signals include the BMPs, the Wingless-Int proteins

(Wnts), Shh, fibroblast growth factors (FGFs), and epidermal growth factors (EGFs), to name a few. These signals set up developmental domains characterized by expression of specific transcription factors, which in turn control further regional gene transcription and developmental processes. The importance of these mechanisms for cerebral cortical development is only now emerging, altering our concepts of the roles of subsequent thalamic innervation and experience-dependent processes. In light of the temporal and combinatorial principles discussed earlier, brain development can be viewed as a complex and evolving interaction of extrinsic and intrinsic information.

Specific Inductive Signals and Patterning Genes in Development

Induction of the CNS begins at the neural plate stage when the notochord, underlying mesenchyme, and surrounding epidermal ectoderm produce signaling molecules that affect the identity of neighboring cells. Specifically, the ectoderm produces BMPs that prevent neural fate determination by promoting and maintaining epidermal differentiation. In other words, neural differentiation is a default state that manifests unless it is inhibited. In turn, neural induction proceeds when inhibitory proteins block BMP's epidermis-inducing activity, such as noggin, follistatin, and chordin, which are secreted by Hensen's node (homologous to the amphibian Spemann organizer), a signaling center at the rostral end of the primitive streak. Once the neural tube closes, the roof plate and floor plate become new signaling centers, organizing dorsal and ventral neural tube, respectively. The same ligand/receptor system is used sequentially for multiple functions during development. BMPs are a case in point since they prevent neural development at the neural plate stage, whereas after neurulation, the factors are produced by the dorsal neural tube itself to induce sensory neuron fates.

The Spinal Cord. The spinal cord is a prime example of the interaction of soluble signaling factors with intrinsic patterning gene expression and function. The synthesis, release, and diffusion of inductive signals from signaling sources produce concentration gradients that impose distinct neural fates in the spinal cord (Fig. 33-9). The notochord and floor plate secrete Shh, which induces motoneurons and interneurons ventrally, whereas the epidermal ectoderm and roof plate release several BMPs that impart neural

crest and sensory relay interneuron fates dorsally. Growth factor inductive signals act to initiate discrete regions of transcription factor gene expression. For instance, high concentrations of Shh induce winged-helix transcription factor *Hnf3β* gene in floor plate cells and *Nkx6.1* and *Nkx2.2* in the ventral neural tube, whereas the expression of more dorsal genes, *Pax6, Dbx1/2, Irx3,* and *Pax7,* is repressed. In response to Shh, ventral motoneurons express transcription factor gene *Isl1*, whose protein product is essential for neuron differentiation. Subsequently, ventral interneurons differentiate, expressing *En1* or *Lim1/2* independent of Shh signaling.

In contrast, the release of BMPs by dorsal cord and roof plate induces a distinct cascade of patterning genes to elicit sensory interneuron differentiation. In aggregate, the coordinated actions of Shh and BMPs induce the dorsoventral dimension of the spinal cord. Similarly, other inductive signals determine the rostrocaudal organization of the CNS, such as retinoic acid, an upstream regulator of *hox* patterning genes, anteriorly, and the FGFs posteriorly. The overlapping and unique expression of the many *hox* gene family members are essential for establishing the segmental pattern in the anterior–posterior axis of the hindbrain and spinal cord, now-classic models well described in previous reviews.

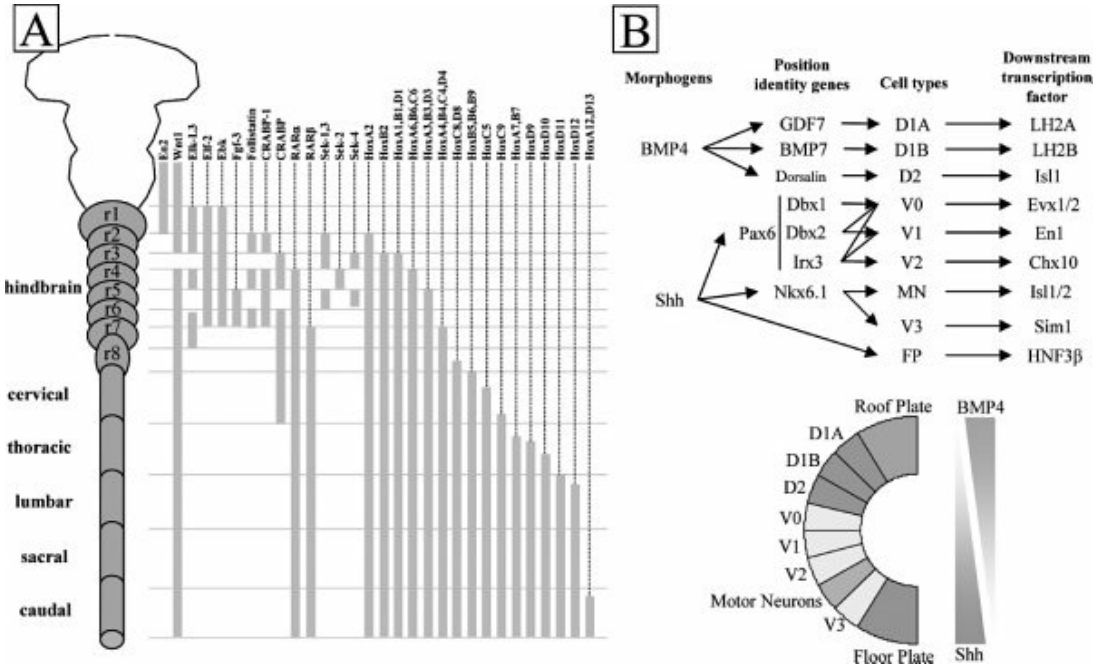


FIGURE 33-9

Patterning genes in the spinal cord. **A:** Diagram illustrating the localization of gene expression in the developing "trunk." Rhombomere boundaries are specified

by specific combinations of transcription factors. (Modified from Darnell, 2005.) **B:** Morphogen induction of spinal cord cell fate. Dorsoventral gradients of sonic hedgehog (Shh) and bone morphogenetic protein (BMP) induce expression of several position identity genes. Combinatorial effects of these factors establish progenitor domains and result in the expression of specific downstream molecular markers. D, dorsal neurons; V, ventral neurons. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:51.)

Recent advances in spinal cord transcription factor expression and function reinforce that these factors play roles at multiple developmental cell stages. This is likely due to their participation in diverse protein regulatory complexes: The transcription factors Pax6, Olig2, and Nkx2.2, which define the positional identity of multipotent progenitors early in development, also play crucial roles in controlling the timing of neurogenesis and gliogenesis in the developing ventral spinal cord.

The Cerebral Cortex. Recent evidence suggests that forebrain development also depends on inductive signals and patterning genes, as observed in more caudal neural structures. In the embryo, the dorsal forebrain structures include the hippocampus medially, the cerebral cortex dorsolaterally, and the entorhinal cortex ventrolaterally, whereas in the basal forebrain, the globus pallidus lies medially and the striatum laterally. Based on gene expression and morphologic criteria, it has been hypothesized that the forebrain is divided into a checkerboard-like grid pattern of domains generated by the intersection of longitudinal columns and transverse segments, perpendicular to the longitudinal axis. The columns and segments (prosomeres) exhibit restricted expression of patterning genes, allowing for unique combinations of factors within each embryonic subdivision. Many of these genes, including *Hnf3β, Emx2, Pax6,* and *Dlx2,* are first expressed even before neurulation in the neural plate. They are then maintained, providing the "protomap" determinants of the VZ described earlier. As in the spinal cord, initial forebrain gene expression is influenced by a similar array of signaling center soluble factors—Shh, BMP, and retinoic acid. As the telencephalic vesicles form, signaling centers localize to the edges of the cortex. In the dorsal midline, there is the anterior neural ridge, anterior cranial mesenchyme secreting FGF8, the roof plate, and, at the junction of the roof plate with the telencephalic vesicle, the cortical hem (Fig. 33-10). Other

factors originate laterally from the dorsal–ventral forebrain junction, as well as from basal forebrain structures themselves.

Do molecular studies identify how different cortical regions interact with thalamic neurons to establish specific functional modalities, such as vision and sensation? Moreover, once regional identity is established, can later developmental events modify it? It has been proposed that initially, there are no functional distinctions in the cortex but that they are induced by the ingrowth of extrinsic thalamic axons, which convey positional and functional specifications, the so-called “protocortex model.” However, in contrast, the abundant molecular evidence provided earlier suggests that intrinsic differences are established early in the neuroepithelium by molecular determinants that regulate areal specification, including the targeting of thalamic axons, termed the “protomap” model. Experimental tests indicate that neither model is entirely correct. Although there is early molecular regionalization of the cortex, the initial targeting of thalamic axons to the cortex is independent of these molecular differences. In the rodent, thalamic afferents first target to their usual cortical regions prenatally in the late embryo. However, once thalamic afferents reach the cortex, which occurs several days after birth, interactions of thalamic axon branches with local, regional cues lead to modifications of initial outgrowth and the establishment of connections that conform to areal molecular identities. Furthermore, the developing cortex exhibits a remarkable and unexpected level of flexibility in mediating modality-specific functions: In the ferret, surgical elimination of visual pathway (lateral geniculate nucleus) in postnatal pups results in the transfer of visual signaling to the auditory cortex, which successfully mediates vision! Thus the animal’s visual information is effectively processed by their auditory cortex.

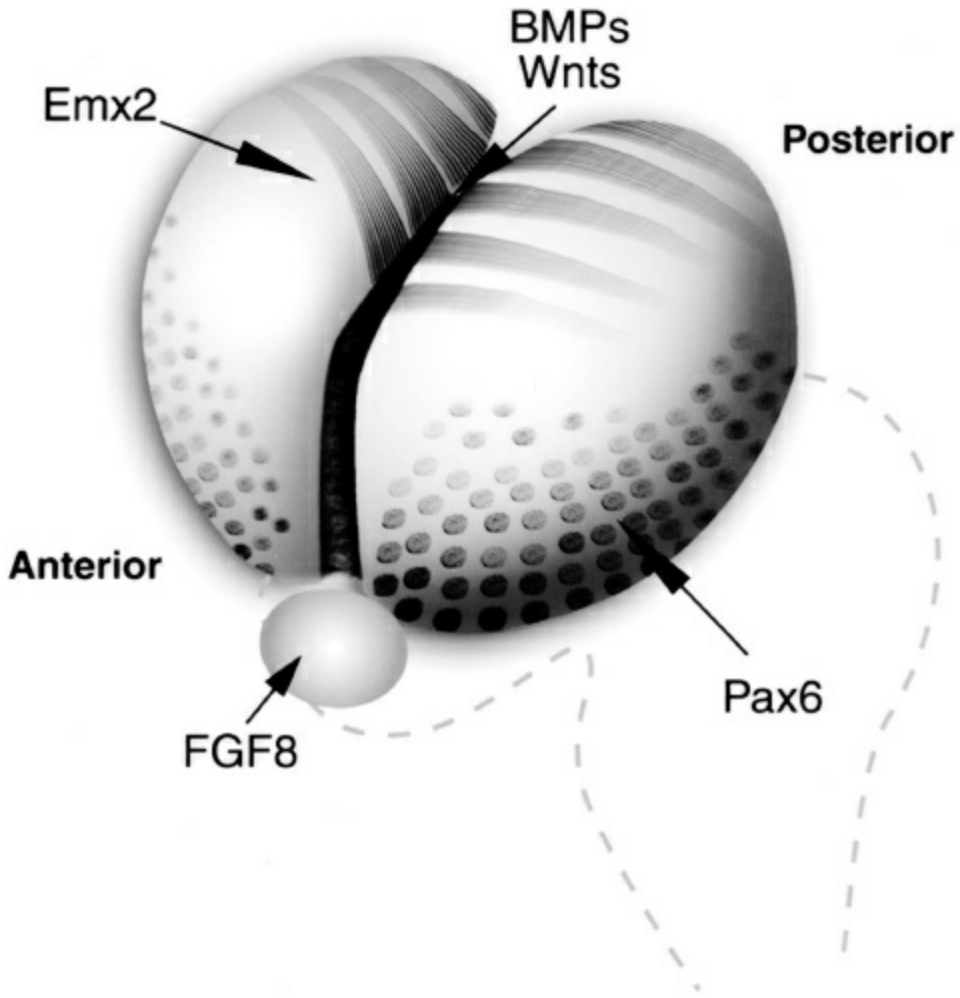


FIGURE 33-10
Patterning genes and signaling centers in the developing cerebral cortex. This schematic diagram shows a lateral–superior view of the two cerebral hemispheres of the embryonic mouse, sitting above the midbrain and hindbrain (*broken lines*). The anterior–lateral extent of *Pax6* gene expression is indicated by circles. The posterior–medial domain of *Emx2* expression is indicated by stripes. The genes exhibit continuous gradients of expression that decrease as they extend to opposite poles. The signaling factor fibroblast growth factor 8 (FGF8) is produced by and released from mesenchymal tissue in the anterior neural ridge, which regulates *Pax6* and *Emx2* expression. In the midline, bone morphogenetic proteins (BMPs) and Wingless-Int proteins (Wnts) are secreted from other signaling centers, including the roof plate and the cortical hems. (Courtesy of E. DiCicco-Bloom and K. Forgash.)

The Hippocampus. The hippocampus is a region of significant importance in schizophrenia, depression, autism, and other disorders, and defining mechanisms regulating hippocampal formation may provide clues to the developmental bases of these disorders. In mice, the hippocampus is located

in the medial wall of the telencephalic vesicle. Where it joins the roof plate, the future roof of the third ventricle, there is a newly defined signaling center, the cortical hem, which secretes BMPs, Wnts, and FGFs (see Fig. 33-10). Genetic experiments have defined patterning genes localized to the cortical hem and hippocampal primordia, whose deletions result in a variety of morphogenetic defects. In mice lacking Wnt3a, which is expressed in the cortical hem, the hippocampus is either completely missing or significantly reduced, whereas the neighboring cerebral cortex is mainly preserved. A similar phenotype is produced by deleting an intracellular factor downstream to Wnt receptor activation, the *Lef1* gene, suggesting that the Wnt3a-*Lef1* pathway is required for hippocampal cell specification or proliferation, issues remaining to be defined. When another cortical hem gene, *Lhx5,* is deleted, mice lack both the hem and neighboring choroid plexus, both sources of growth factors. However, in this case, the cortical hem cells may proliferate in excess, and the hippocampal primordia may be present but disorganized, exhibiting abnormalities in cell proliferation, migration, and differentiation. A related abnormality is observed with *Lhx2* mutation. Finally, a sequence of bHLH transcription factors plays roles in hippocampal neurogenesis: Dentate gyrus differentiation is defective in *NeuroD* and *Mash1* mutants. Significantly, expression of all these hippocampal patterning genes is regulated by factors secreted by anterior neural ridge, roof plate, and the cortical hem, including FGF8, Shh, BMPs, and Wnts. Moreover, the basal forebrain region secretes an EGF-related protein, transforming growth factor α (TGF-α), which can stimulate expression of the classical limbic marker protein, lysosomal-associated membrane protein (LAMP). These various signals and genes now serve as candidates for disruption in human diseases of the hippocampus.

The Basal Ganglia. Also, to the motor and cognitive functions, the basal ganglia take on new importance in neocortical function, since they appear to be the embryonic origin of virtually all adult GABA interneurons, reaching the neocortex through tangential migration. Gene expression studies have identified several transcription factors that appear in precursors originating in the ventral forebrain ganglionic eminences, allowing interneurons to be followed as they migrate dorsally into the cortical layers. Conversely, genetic deletion mutants exhibit diminished or absent interneurons, yielding results

consistent with other tracing techniques. These transcription factors, including *Pax6, Gsh2,* and *Nkx2.1,* establish boundaries between different precursor zones in the ventral forebrain VZ, through mechanisms involving mutual repression. As a simplified model, the medial ganglionic eminence (MGE) expresses Nkx2.1 primarily and gives rise to most GABA interneurons of the cortex and hippocampus, whereas the lateral ganglionic eminence (LGE) expresses *Gsh2* and generates GABA interneurons of the SVZ and olfactory bulb. The boundary between the ventral and dorsal forebrain then depends on LGE interaction with the dorsal neocortex, which expresses *Pax6.* When *Nkx2.1* is deleted, LGE transcription factor expression spreads ventrally into the MGE territory, and there is a 50 percent reduction in neocortical and striatal GABA interneurons.

In contrast, the deletion of *Gsh2* leads to ventral expansion of the dorsal cortical molecular markers and concomitant decreases in olfactory interneurons. Finally, *Pax6* mutation causes both MGE and LGE to spread laterally and into dorsal cortical areas, yielding increased interneuron migration. The final phenotypic changes are involved, as these factors exhibit unique and overlapping expression and interact to control cell fate.

Neuronal Specification. As indicated for basal ganglia, throughout the nervous system, transcription factors participate in decisions at multiple levels, including determining the generic neural cell, such as a neuron or glial cell, as well as neuron subtypes. *Mash1* can promote a neuronal fate over a glial fate as well as induce the GABA interneuron phenotype. However, another bHLH factor, *Olig1/2,* can promote oligodendrocyte development, whereas it promotes motor neuron differentiation elsewhere, indicating that the variety of factors expressed in a specific cell leads to combinatorial effects and thus diverse outcomes for cell differentiation. The bHLH inhibitory factor, *Id,* is expressed at the transition from somatosensory to the motor cortex, implying the roles of family members in areal characteristics. In the hippocampus, granule neuron fate is dependent on *NeuroD* and *Math1,* with deficient cell numbers when either one is deleted. The role of specific factors in cortical cell layer determination remains an area of active investigation but likely includes *Tbr1, Otx1,* and *Pax6.*

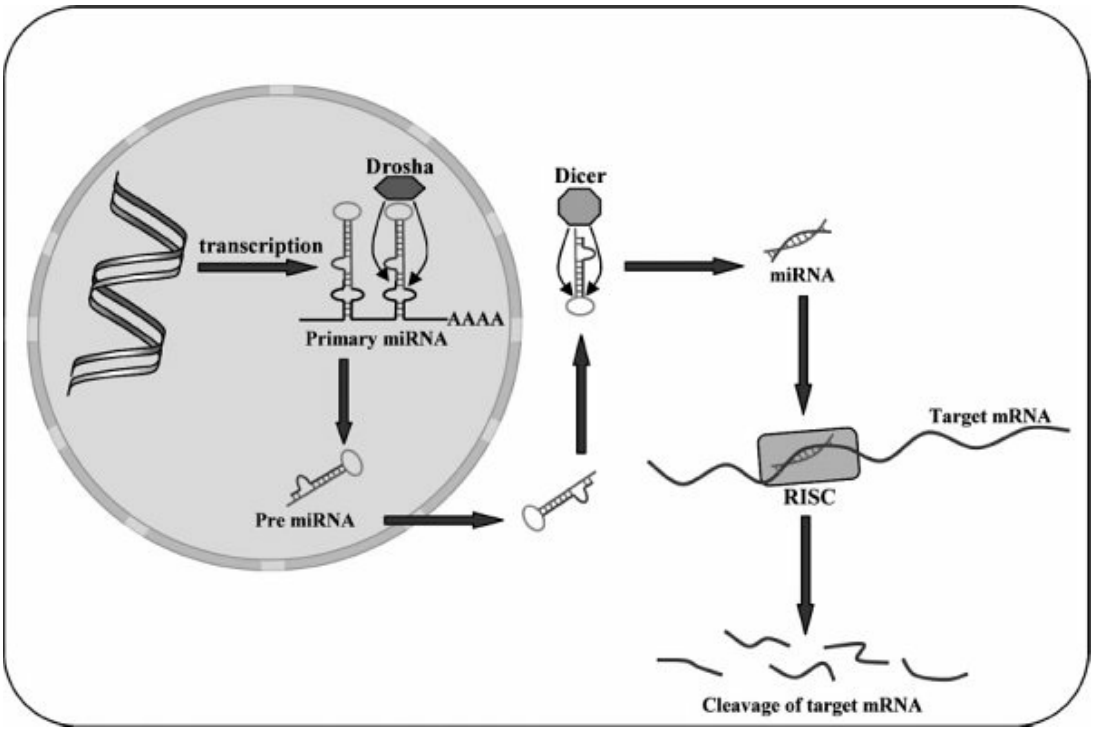


FIGURE 33-11
Processing and function of micro RNA (miRNA). After transcription, the primary miRNA forms a hairpin conformation. This structure allows the enzyme Drosha to cleave the transcript, producing a pre-miRNA that then exits the nucleus through nuclear pores. In the cytoplasm, Dicer cleaves the pre-miRNA stem loop, resulting in the formation of two complementary short RNA molecules. Only one of these molecules is integrated in the RNA-induced silencing complex (RISC) and serves as a guide strand that allows recognition and specificity for target RNA due to its sequence complementarity. After integration into the RISC complex, the miRNA matches with the complementary mRNA strand and induces mRNA duplex degradation by the argonaute protein, the catalytic enzyme of the RISC complex. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:55.)

A New Mechanism for Regulating Gene Expression: miRNAs

Over the last decade, a new mechanism for regulating messenger ribonucleic acid (mRNA) has been explored in a simple to complex organisms that involve microRNAs (miRNAs). We now know that miRNAs contribute not only to normal development and brain function but also to brain disorders, such as Parkinson and Alzheimer diseases, tauopathies, and brain cancer. miRNAs can affect the regulation of RNA transcription, alternative splicing, molecular modifications, or RNA translation. miRNAs are 21- to 23-nucleotide-long single-strand RNA molecules. Unlike mRNAs that encode

the instructions for ribosome complex translation into proteins, miRNAs are noncoding RNAs that are not translated but are instead processed to form loop structures. miRNAs exhibit a sequence that is partially complementary to one or several other cellular mRNAs. By binding to target mRNA transcripts, the miRNAs serve to interfere with their function, thereby downregulating the expression of these gene products. This gene silencing involves a complex mechanism: The larger miRNA primary transcript is first processed by the Microprocessor, an enzymatic complex consisting of the nuclease Drosha and the double-stranded RNA binding protein Pasha. The mature miRNA binds to its complementary RNA and then interacts with the endonuclease Dicer that is part of the RNA-induced silencing complex (RISC), resulting in the cleavage of the target mRNA and gene silencing (Fig. 33-11).

Currently, 475 miRNAs have been identified in humans, and their total number is estimated to be between 600 and 3,441. Potentially, up to 30 percent of all genes might be regulated by miRNAs, a whole new layer of molecular complexity. A connection between miRNAs and several brain diseases has already been made. For example, miR-133b, which is expressed explicitly in midbrain dopaminergic neurons, is deficient in midbrain tissue from patients with Parkinson disease. Furthermore, the miRNAs encoding miR-9, miR-124a, miR-125b, miR-128, miR-132, and miR-219 are abundantly represented in the fetal hippocampus, are differentially regulated in the aged brain, and are altered in Alzheimer disease hippocampus. Similar RNA species termed short-interfering RNAs (siRNAs) have been discovered in plants where they prevent the transcription of viral RNA. The mechanisms involved in these effects are closely related to those of miRNA. Thus siRNAs are now being used in both basic and clinical research to downregulate specific cellular gene products, thereby advancing the study of pathways involved in neurodevelopment and providing new selective tools to regulate disease-causing genes or therapeutic molecular targets.

Regulation of Neurodevelopment by Extracellular Factors

The interaction of extracellular factors with intrinsic genetic determinants controlling region-specific neurogenesis includes signals that regulate cell proliferation, migration, differentiation, and survival (Table 33-1). Patterning

genes control the expression of growth factor receptors and the molecular machinery of the cell division cycle. Extracellular factors are known to stimulate or inhibit the proliferation of VZ precursors and originate from the cells themselves, termed autocrine, neighboring cells/tissues, or paracrine, or from the general circulation, as in endocrine, all sources known to affect proliferation in the prenatal and postnatal developing brain. Although defined initially in cell culture, several mitogenic growth factors are now well-characterized in vivo, including those stimulating proliferation, such as basic FGF (bFGF), EGF, IGF-I, Shh, and signals inhibiting cell division, such as pituitary adenylate-cyclase–activating polypeptide (PACAP), GABA and glutamate, and members of the TGF-β superfamily. However, in addition to stimulating reentry of cells into the cell cycle, termed a mitogenic effect, extracellular signals also enhance proliferation by promoting survival of the mitotic population, a trophic action. Stimulation of both pathways is necessary to produce maximal cell numbers. These mitogenic and trophic mechanisms during development parallel those identified in carcinogenesis, reflecting roles of c-myc and bcl-2, respectively. Several of the neurotrophins, especially BDNF and neurotrophin-3 (NT3), promote the survival of mitotic precursors as well as the newly generated progeny.



Table 33-1
Regulation of Neurodevelopment by Extracellular Factors

Extracellular Factors	Proliferation		Migration		Differentiation		Survival	
bFGF	↑	Cortex Cerebellum Hippocampus	—	—	↑	Nigrostriatum Cortex	↑	Nigrostriatum Cerebellum Cortex
IGF-1	↑	Cortex Cerebellum	—	—	↑	Spinal neurons Cerebellum	↑	Cortex Cerebellum
EGF	↑	Cortex Adult SVZ	—	—	↑	Cortex	—	—
TGF-β	↓	Cortex Cerebellum	—	—		—	↓	Cortex Cerebellum
Shh	↑	Cortex Cerebellum	↑	Cerebellum	—	—	—	—
PACAP	↓	Cortex Cerebellum	↓	Cerebellum	↑	Cerebellum	↑	Cerebellum
GABA	↓	Cortex	↑	Cortex	—	—	—	—
Glutamate	↓	Cortex	↑	Cortex Cerebellum	↓ ↑	Pyramidal neurons Granule neurons	↑ ↓	Immature neurons Mature neurons
TNF-α	↓	Neurons	—	—	—	—	↓	Neurons
BDNF	—	—	↑	Cerebellum	↑	Cortex Adult SVZ	↑	Cortex Cerebellum
Wnt	↑	Embryonic stem cells Hippocampus	—	—	↑	Axon guidance Spinal cord	—	—
NT3	↓	Cortical stem cells	↑	Cortex	↑	Cortex	↑	Cortex
LIF/CNTF/gp130	↑	Cortex Embryonic Stem cells	—	—	↑	Astrocytes	—	—

From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins;2009:55.

The developmental significance of extracellular mitogens is demonstrated by the expression of the factors and their receptors in regions of neurogenesis. It is also demonstrated by the profound and permanent consequences of altering their activities during development. For example, by administering growth factors to developing embryos or pups, one can induce changes in proliferation in prenatal cortical VZ, postnatal cerebellar EGL, and hippocampal dentate gyrus that produce lifelong modifications in brain region population size and cell composition. Such changes may be relevant to structural differences observed in neuropsychiatric disorders, such as depression, schizophrenia, and autism. Precisely, in the cerebral cortex VZ of the embryonic rat, proliferation is controlled by promitogenic bFGF and antimitogenic PACAP, which are expressed as autocrine/paracrine signals. Positive and negative effects were shown in living embryos in utero by performing intracerebroventricular (ICV) injections of the factors or antagonists. ICV injection of bFGF produced a larger adult cortex composed of 87 percent more neurons, which employed glutamate, thus increasing the ratio of excitatory pyramidal neurons to GABA inhibitory neurons, which were unchanged.

Conversely, embryonic PACAP injection inhibited the proliferation of cortical precursors by 26 percent, reducing the number of labeled layer 5/6 neurons in the cortical plate 5 days later. A similar reduction was accomplished by genetically deleting promitogenic bFGF or leukocyte inhibitory factor (LIF)/ciliary neurotrophic factor (CNTF)/gp130 signaling, diminishing cortical size. Furthermore, the effects of mitogenic signals depended critically on the stage-specific program of regional development, since bFGF injection at later ages when gliogenesis predominates affected glial numbers selectively. Thus developmental dysregulation of mitogenic pathways due to genetic or environmental factors (hypoxia, maternal/fetal infection, or drug or toxicant exposure) will likely produce subtle changes in the size and composition of the developing cortex. Other signals likely to play proliferative roles include Wnt's, TGF-α, IGF-I, and BMPs. Although interactions between intrinsic cortical programs and extrinsic factors remain to be defined, a remarkable new study of mouse embryonic stem cells suggests that embryonic mammalian forebrain specification may be a developmentally ancestral intrinsic program that emerges in the absence of extrinsic signals. In specific culture conditions that block endogenous Shh signaling, mouse embryonic stem cells can sequentially generate the various types of neurons that display the most salient features of genuine cortical pyramidal neurons. When grafted into the cerebral cortex, these cells differentiate into neurons that project to select cortical (visual and limbic regions) and subcortical targets, corresponding to a wide range of pyramidal layer neurons. Insight into precision control of neuronal differentiation will open new avenues to perform neuronal grafts in humans for cellular replacement in various acquired and neurodegenerative diseases.

Similar to the cerebral cortex, later generated populations of granule neurons, such as in the cerebellum and hippocampal dentate gyrus, are also sensitive to growth factor manipulation, which is especially relevant to therapies administered intravenously to premature and newborn infants in the neonatal nursery. Like in humans, cerebellar granule neurons are produced postnatally in rats, but for only 3 weeks, whereas in both species, dentate gyrus neurons are produced throughout life. Remarkably, a single peripheral injection of bFGF into newborn rat pups rapidly crossed into the cerebrospinal fluid (CSF) and stimulated proliferation in the cerebellar EGL by 30 percent as well as hippocampal dentate gyrus by twofold by 8 hours,

consistent with an endocrine mechanism of action. The consequence of mitogenic stimulation in the cerebellum was a 33 percent increase in the number of internal granule layer neurons and a 22 percent larger cerebellum. In the hippocampus, mitotic stimulation elicited by a single bFGF injection increased the absolute number of dentate gyrus granule neurons by 33 percent at 3 weeks, defined stereologically, producing a 25 percent larger hippocampus containing more neurons and astrocytes, a change that persisted lifelong. Conversely, genetic deletion of bFGF resulted in smaller cerebellum and hippocampus at birth and throughout life, indicating that levels of the growth factor were critical for normal brain region formation. Other proliferative signals regulating cerebellar granule neurogenesis include Shh and PACAP, the disruption of which contributes to human medulloblastoma, whereas in the hippocampus, the Wnt family may be involved.

Clinical Implications. There are several clinical implications of these surprising growth factor effects observed in newborns. First, we may need to investigate possible neurogenetic effects of therapeutic agents we administer in the newborn nursery for long-term consequences. Second, because bFGF is as effective in stimulating adult neurogenesis (see subsequent text) as in newborns because of specific transport across the mature BBB, there is the possibility that other protein growth factors are also preferentially transported into the brain and alter ongoing neurogenesis. Indeed, in rats, IGF-I also stimulates mature hippocampal dentate gyrus neurogenesis. Third, other therapeutics cross the BBB efficiently due to their lipid solubility, such as steroids, which inhibit neurogenesis across the age spectrum. Steroids are frequently used perinatally to promote lung maturation and treat infections and trauma, but effects on human brain formation have not been examined. Fourth, it is well known that neurologic development may be delayed in children who experience a severe systemic illness that is associated with numerous inflammatory cytokines. One may wonder to what degree this reflects an interference with neurogenesis and concomitant processes, potentially producing long-term differences in cognitive and motor functional development. Finally, maternal infection during pregnancy is a known risk factor for schizophrenia, and cytokines that cross the placental barrier may directly affect fetal brain cell proliferation and differentiation as well as cell migration, target selection, and synapse maturation, as shown in animal

models, eventually leading to multiple brain and behavioral abnormalities in the adult offspring.

Cell Migration

Throughout the nervous system, newly generated neurons naturally migrate away from proliferative zones to achieve final destinations. If this process is disrupted, abnormal cell localization and function result. In humans, more than 25 syndromes with disturbed neuronal migration have been described. As noted earlier, neurons migrate in both radial and tangential fashions during development and may establish cell layers that are inside-to-outside, or the reverse, according to region. In the developing cerebral cortex, the most well-characterized mechanism is radial migration from underlying VZ to appropriate cortical layers in an inside-to-outside fashion. Also, however, the inhibitory GABA interneurons that are generated in ventrally located medial ganglionic eminences reach the cortex through tangential migration in the intermediate zone along with axonal processes or other neurons. The neurons in the developing cerebellum also exhibit both radial and tangential migration. Purkinje cells leave the fourth ventricle VZ and exhibit radial migration, whereas other precursors from the rhombic lip migrate tangentially to cover the cerebellar surface, establishing the EGL, a secondary proliferative zone. From EGL, newly generated granule cells migrate radially inward to create the internal granule cell layer. Finally, granule interneurons of the olfactory bulb exhibit a different kind of migration, originating in the SVZ of the lateral ventricles overlying the striatum. These neuroblasts divide and migrate simultaneously in the rostral migratory stream in transit to the bulb, on a path comprising chains of cells that support forward movements. The most commonly recognized disorders of human neuronal migration are the extensive lissencephalies (see subsequent text), although incomplete migration of more restricted neuron aggregates (heterotopias) frequently underlies focal seizure disorders.

Animal models have defined molecular pathways involved in neuronal migration. Cell movement requires signals to start and stop migration, adhesion molecules to guide migration, and functional cytoskeleton to mediate cell translocation. The best-characterized mouse model of aberrant neuronal migration is reeler, a spontaneous mutant in which cortical neuron

laminar position is inverted, being generated in outside-to-inside fashion. Reelin is a large, secreted extracellular glycoprotein produced embryonically by the earliest neurons in the cortical preplate, Cajal–Retzius cells, and hippocampus and cerebellum. Molecular and genetic analysis has established a signaling sequence in reelin activity that includes at least two receptors, the very low-density lipoprotein receptor (VLDLR) and the apoprotein E receptor 2 (ApoER2), and the intracellular adapter protein, disabled 1 (Dab1), initially identified in the scrambler mutant mouse, a reelin phenocopy. Current thoughts consider the reelin system as one mediator of radial glia-guided neuron migration, although its specific functions in starting or stopping migration remain controversial. The roles of the VLDL and ApoE2 receptors are intriguing for their possible contributions to Alzheimer disease risk. Recent studies have found human reelin gene (*RELN*) mutations associated with autosomal recessive lissencephaly with cerebellar hypoplasia, exhibiting a markedly thickened cortex with pachygyria, abnormal hippocampal formations, and severe cerebellar hypoplasia with absent folia. Additional studies suggest that reelin polymorphisms may contribute to ASD risk as well.

Concerning cytoskeletal proteins, studies of the filamentous fungus *Aspergillus nidulans* surprisingly provided insight into the molecular machinery underlying the human migration disorder, Miller–Dieker syndrome, lissencephaly associated with abnormal chromosome 17q13.3. Lissencephaly is a diverse disorder characterized by a smooth cortical surface lacking in gyri and sulci, with markedly reduced brain surface area. The absence of convolutions results from a migration defect: the majority of neurons fail to reach their final destinations. In classical lissencephaly (type I), the cerebral cortex is thick and usually four-layered, whereas, in cobblestone lissencephaly (type II), the cortex is chaotically organized with a partly smooth and partly pebbled surface and deficient lamination. The most severely affected parts of the brain are the cerebral cortex and hippocampus, with the cerebellum less affected. In fungus, the gene *NudF* was found to be essential for intracellular nuclear distribution, a translocation process also involved in mammalian cell migration. The human homolog of *NudF* is LIS-1 or PAFAH1B1, a mutation of which accounts for up to 60 percent of lissencephaly cases of type I pathology. The LIS-1 gene product interacts with microtubules and related motor components dynein and dynactin as well

as doublecortin (DCX), which may regulate microtubule stability. Mutations in the DCX gene result in X-linked lissencephaly in males and bands of heterotopic neurons in white matter in females, appearing as a “double cortex” on imaging studies, producing severe mental retardation and epilepsy. Other migratory defects occur when proteins associated with the actin cytoskeleton are affected, such as a mutation in the filamin 1 gene responsible for periventricular nodular heterotopias in humans and mutations of a regulatory phosphokinase enzyme, the CDK5/p35 complex.

Cell migration also depends on molecules mediating cellular interactions, which provide cell adhesion to establish neuron–neuron and neuron–glia relationships or induce attraction or repulsion. Astrotactin is a major glial protein involved in neuronal migration on radial glial processes, whereas neuregulins and their receptors, ErbB2-4, play roles in neuronal–glial migratory interactions. Recent genetic studies associate neuregulin polymorphisms with schizophrenia, suggesting that this developmental disease may depend on altered oligodendrocyte numbers and activities and synaptic functions. Furthermore, some work suggests that early appearing neurotransmitters themselves, GABA and glutamate, and platelet-derived growth factor (PDGF) regulate migration speed. In contrast to radial migration from cortical VZ, GABA interneurons generated in ganglionic eminences employ different mechanisms to leave the ventral forebrain and migrate dorsally into the cerebral cortex. Several signaling systems have been identified, including the Slit protein and Robo receptor, the semaphorins and their neuropilin receptors, and hepatocyte growth factor and its c-Met receptor, all of which appear to repel GABA interneurons from the basal forebrain, promoting tangential migration into the cortex. Significantly, the c-Met receptor has recently been associated with ASDs, suggesting that altered GABA interneuron migration into the cortex and deficits in inhibitory signaling may contribute to the phenotype, including seizures and abnormal cognitive processing. Finally, several human forms of congenital muscular dystrophy with severe brain and eye migration defects result from gene mutations in enzymes that transfer mannose sugars to serine/threonine -OH groups in glycoproteins, thereby interrupting interactions with several extracellular matrix molecules and producing type II cobblestone lissencephalies.

Differentiation and Neuronal Process Outgrowth

After newly produced neurons and glial cells reach their final destinations, they differentiate into mature cells. For neurons, this involves outgrowth of dendrites and extension of axonal processes, the formation of synapses, and the production of neurotransmitter systems, including receptors and selective reuptake sites. Most axons will become insulated by myelin sheaths produced by oligodendroglial cells. Many of these events occur with a peak period from 5 months of gestation onward. During the first several years of life, many neuronal systems exhibit exuberant process growth and branching, which is later decreased by selective "pruning" of axons and synapses, dependent on experience. In contrast, myelination continues for several years after birth and into adulthood.

Although there is tremendous synapse plasticity in the adult brain, a fundamental feature of the nervous system is the point-to-point or topographic mapping of one neuron population to another. During development, neurons extend axons to innervate diverse distant targets, such as the cortex and spinal cord. The structure that recognizes and responds to cues in the environment is the growth cone, located at the axon tip. The axonal process is structurally supported by microtubules that are regulated by numerous microtubule-associated proteins (MAPs), whereas the terminal growth cone exhibits a transition to actin-containing microfilaments. The growth cone has rod-like extensions called filopodia that bear receptors for specific guidance cues present on cell surfaces and in the extracellular matrix. Interactions between filopodial receptors and environmental cues cause growth cones to move forward, turn, or retract. Recent studies have identified the actin-modulating proteins and kinases involved in rapid growth cone movements, such as LIMK kinase that causes the language phenotype associated with Williams syndrome. Perhaps surprising is that activation of growth cone receptors leads to local mRNA translation to produce synaptic proteins, whereas traditional concepts assumed that all proteins were transported to axon terminals from distant neuronal cell somas. The region-specific expression of extracellular guidance molecules, such as cadherins, regulated by patterning genes *Pax6* and *Emx2*, results in the highly directed outgrowth of axons, termed axonal pathfinding. These molecules affect the direction, speed, and fasciculation of axons, acting through either positive or

negative regulation. Guidance molecules may be soluble extracellular factors or may be bound to extracellular matrix or cell membranes. In the latter class of signal is the newly discovered family of transmembrane proteins, the ephrins. Ephrins act via the largest known family of tyrosine kinase receptors in the brain, Eph receptors, and play a significant role in topographic mapping between neuron populations and their targets. Ephrins frequently serve as chemorepellent cues, negatively regulating growth by preventing developing axons from entering incorrect target fields. For example, the optic tectum expresses ephrins A2 and A5 in a gradient that decreases along the posterior to the anterior axis, whereas innervating retinal ganglion cells express a gradient of Eph receptors. Ganglion cell axons from the posterior retina, which possess high Eph A3 receptor levels, will preferentially innervate the anterior tectum because the low-level ephrin expression does not activate the Eph kinase that causes growth cone retraction. In the category of soluble molecules, netrins serve primarily as chemoattractant proteins secreted, for instance, by the spinal cord floor plate to stimulate spinothalamic sensory interneurons to grow into the anterior commissure. In contrast, Slit is a secreted chemorepulsive factor that, through its roundabout (Robo) receptor, regulates midline crossing and axonal fasciculation and pathfinding.

The Neurodevelopmental Basis of Psychiatric Disease

An increasing number of neuropsychiatric conditions are considered to originate during brain development, including schizophrenia, depression, autism, and ADHD. Defining when a condition begins helps direct attention to underlying pathogenic mechanisms. The term neurodevelopmental suggests that the brain is abnormally formed from the very beginning due to disruption of fundamental processes, in contrast to a normally formed brain that is injured secondarily or that undergoes degenerative changes. However, the value of the term neurodevelopmental needs to be reconsidered because of different use by clinicians and pathologists. Also, given that the same molecular signals function in both development and maturity, altering an early ontogenetic process by changes in growth factor signaling, for instance, probably means that other adult functions exhibit ongoing dysregulation as well. For example, clinical researchers of schizophrenia consider the disorder

neurodevelopmental because at the time of onset and diagnosis, the prefrontal cortex and hippocampus are smaller and ventricles enlarged already at adolescent presentation.

In contrast, the neuropathologist uses the term neurodevelopmental for specific morphologic changes in neurons. If a brain region exhibits a normal cytoarchitecture, but with neurons of smaller than the normal diameter, reminiscent of “immature” stages, then this may be considered an arrest of development. However, if the same cellular changes are accompanied by inflammatory signs, such as gliosis and white blood cell infiltrate, then this is termed neurodegeneration. These morphologic and cellular changes may no longer be adequate to distinguish disorders that originate from development versus adulthood, especially given the roles of glial cells, including astrocytes, oligodendrocytes, and microglia, as sources of neurotrophic support during both periods of life. Thus abnormalities in glial cells may occur in both epochs to promote disease or act as mechanisms of repair. Many neurodegenerative processes, such as in Alzheimer and Parkinson diseases, are associated with microglial cells.

On the other hand, neuronal dysfunction in adulthood, such as cell shrinkage, may occur without inflammatory changes. In animal models, interruption of BDNF neurotrophic signaling in the adult brain results in neuron and dendrite atrophy in the cerebral cortex without eliciting glial cell proliferation. Thus finding small neurons without gliosis in the brains of patients with schizophrenia or autism does not necessarily mean that the condition is only or primarily developmental in origin. In turn, several etiologic assumptions about clinical brain conditions may require reexamination.

Because the same processes that mediate development, including neurogenesis, gliogenesis, axonal growth and retraction, synaptogenesis, and cell death, also function during adulthood, a new synthesis has been proposed. All of these processes, although perhaps in more subtle forms, contribute to adaptive and pathologic processes. Successful aging of the nervous system may require precise regulation of these processes, allowing the brain to adapt appropriately and counteract the numerous intrinsic and extrinsic events that could potentially lead to neuropathology. For example, adult neurogenesis and synaptic plasticity are necessary to maintain neuronal circuitry and ensure proper cognitive functions. Programmed cell death is

crucial to prevent tumorigenesis that can occur as cells accumulate mutations throughout life. Thus dysregulation of these ontogenetic processes in adulthood will lead to disruption of brain homeostasis, expressing itself as various neuropsychiatric diseases.

Schizophrenia. The neurodevelopmental hypothesis of schizophrenia postulates that etiologic and pathogenetic factors occurring before the formal onset of the illness, that is, during gestation, disrupt the course of normal development. These subtle early alterations in specific neurons, glia, and circuits confer vulnerability to other later developmental factors, ultimately leading to malfunctions. Schizophrenia is a multifactorial disorder, including both genetic and environmental factors. Clinical studies using risk assessment have identified some relevant factors, including prenatal and birth complications (hypoxia, infection, or substance and toxicant exposure), family history, body dysmorphia, especially structures of neural crest origin, and presence of mild premorbid deficits in social, motor, and cognitive functions. These risk factors may affect ongoing developmental processes such as experience-dependent axonal and dendritic production, programmed cell death, myelination, and synaptic pruning. An intriguing animal model using human influenza-induced pneumonia of pregnant mice shows that the inflammatory cytokine response produced by the mother may directly affect the offspring's brain development, with no evidence of the virus in the fetus or placenta.

Neuroimaging and pathology studies identify structural abnormalities at disease presentation, including smaller prefrontal cortex and hippocampus and enlarged ventricles, suggesting abnormal development. More severely affected patients exhibit a more significant number of affected regions with more considerable changes. In some cases, ventricular enlargement and cortical gray matter atrophy increase with time. These ongoing progressive changes should lead us to reconsider the potential role of active degeneration in schizophrenia, whether due to the disease or its consequences, such as stress or drug treatment. However, classic signs of neurodegeneration with inflammatory cells are not present.

Structural neuroimaging strongly supports the conclusion that the hippocampus in schizophrenia is significantly smaller, perhaps by 5 percent. In turn, brain morphology has been used to assess the etiologic contributions

of genetic and environmental factors. Comparisons of concordance for schizophrenia in monozygotic and dizygotic twins support roles for both factors. Among monozygotic twins, only 40 to 50 percent of both twins have the illness, indicating that the genetic constitution alone does not ensure disease and suggesting that the embryonic environment also contributes. Neuroimaging, pharmacologic, and pathologic studies suggest that some genetic factors allow for susceptibility and that secondary insults, such as birth trauma or perinatal viral infection, provide the other. This model is consistent with imaging studies showing small hippocampus in both affected and unaffected monozygotic twins.

Moreover, healthy, genetically at-risk individuals show hippocampal volumes (smaller) more similar to affected probands than normal controls. Thus hippocampal volume reduction is not pathognomonic of schizophrenia but instead may represent a biologic marker of genetic susceptibility. It is not difficult to envision roles for altered developmental regulators in producing a smaller hippocampus, which in turn limits functional capacity. A smaller hippocampus may result from subtle differences in the levels of transcription factors, such as *NeuroD, Math1,* or *Lhx,* signaling by Wnt3a and downstream mediator *Lef1,* or proliferative control mediated by bFGF, the family members of which exhibit altered expression levels in schizophrenia brain samples. Such genetic limitations may only become manifest following another developmental challenge, such as gestational infection, stressors, or toxicant exposure.

A regional locus of schizophrenia pathology remains uncertain but may include the hippocampus, entorhinal cortex, multimodal association cortex, limbic system, amygdala, cingulate cortex, thalamus, and medial temporal lobe. Despite size reductions in specific regions, attempts to define changes in cell numbers have been unrewarding since most studies do not quantify the entire cell population but assess only regional cell density. Without assessing a region’s total volume, cell density measures alone are limited in revealing population size. Most studies have found no changes in cell density in diverse regions. A single study successfully examining total cell number in the hippocampus found normal neuron density and a 5 percent volume reduction on the left and 2 percent on the right, yielding no significant change in total cell number.

In contrast to total neuron numbers, using neuronal cell-type–specific

markers, many studies have found a decreased density of nonpyramidal GABA interneurons in the cortex and hippocampus. In particular, parvalbumin-expressing interneurons are reduced, whereas calretinin-containing cells are normal, suggesting a deficiency of an interneuron subtype. These morphometric data are supported by molecular evidence for decreased GABA neurons, including reduced mRNA and protein levels of the GABA-synthesizing enzyme, GAD67, in the cortex and hippocampus. Another product of the adult GABA-secreting neurons, reelin, which initially appears in Cajal–Retzius cells in the embryonic brain, is reduced 30 to 50 percent in schizophrenia and bipolar disorder with psychotic symptoms. Such a deficiency, leading to diminished GABA signaling, may underlie a potential compensatory increase in $GABA_A$ receptor binding detected in hippocampal CA 2 to 4 fields by both pyramidal and nonpyramidal neurons, apparently selective since benzodiazepine binding is unchanged. More generally, deficiency in a subpopulation of GABA interneurons raises intriguing new possibilities for schizophrenia etiology. As indicated in the preceding gene patterning section, different subpopulations of forebrain GABA interneurons originate from distinct precursors located in the embryonic basal forebrain. Thus cortical and hippocampal GABA interneurons may derive primarily from the MGE under control of the patterning gene *Nkx2.1,* whereas SVZ and olfactory neurons derive from *Gsh2*-expressing LGE precursors.

Furthermore, the timing and sequence of GABA interneuron generation may depend on a regulatory network, including *Mash1, Dlx1/2,* and *Dlx5/6,* all gene candidates for schizophrenia risk. Indeed, *DLX1* gene expression is reduced in the thalamus of patients with psychosis. Thus abnormal regulation of these factors may diminish selectively GABA interneuron formation, which in turn may represent a genetically determined vulnerability and may contribute to diminished regional brain size or function.

The most compelling neuropathologic evidence for a developmental basis is the finding of aberrantly localized or clustered neurons, especially in lamina II of the entorhinal cortex and in the white matter underlying prefrontal cortex and temporal and parahippocampal regions. These abnormalities represent alterations of developmental neuronal migration, survival, and connectivity. Also, in the hippocampus and neocortex,

pyramidal neurons appear smaller in many studies, exhibiting fewer dendritic arborizations and spines with reduced neuropil, findings that are associated with reductions in neuronal molecules, including MAP2, spinophilin, synaptophysin, and SNAP25. Although the genes associated with schizophrenia are reviewed extensively in other chapters, worth mentioning here is a particularly intriguing candidate gene, *DISC1*, whose protein has roles during development, including regulating cell migration, neurite outgrowth, and neuronal maturation as well as in the adult brain, where it modulates cytoskeletal function, neurotransmission, and synaptic plasticity. DISC1 protein interacts with many other proteins intimately involved in neuronal cell migration and forms a protein complex with Lis1 and NudEL that is downstream of reelin signaling.

Autism Spectrum Disorders. Another condition that is neurodevelopmental in origin is ASDs, a complex and heterogeneous group of disorders characterized by abnormalities in social interaction and communication and the presence of restricted or repetitive interests and activities. In the last edition of DSM (DSM-IV), the ASDs included classic autistic disorder, Asperger syndrome, and pervasive developmental disorder not otherwise specified. These three disorders were grouped due to their common occurrence in families, indicating related genetic factors and shared signs and symptoms. Recent conceptualizations of ASDs propose that there are multiple “autisms” differing in underlying pathogenetic mechanisms and manifestations. The different core symptom domains (or other endophenotypes) will likely be more heritable than the syndromic diagnosis, which was constructed to be inclusive. The broad diversity of ASD signs and symptoms reflects the multiplicity of abnormalities observed in pathologic and functional studies and includes both forebrain and hindbrain regions. Forebrain neurons in the cerebral cortex and limbic system play critical roles in social interaction, communication, and learning and memory. For example, the amygdala, which connects to prefrontal and temporal cortices and fusiform gyrus, plays a prominent role in social and emotional cognition. In ASDs, the amygdala and fusiform gyrus demonstrate abnormal activation during facial recognition and emotional attribution tasks. Some investigators hypothesize that ASDs reflect dysfunctions in specific neural networks, such as the social network.

On the other hand, neurophysiologic tests of evoked cortical potentials and oculomotor responses indicate a standard perception of primary sensory information but disturbed higher cognitive processing. The functional impairments in higher-order cognitive processing and neocortical circuitry suggest a developmental disorder involving synaptic organization, a mechanism that may be uniformly present throughout the brain, a model in distinct contrast to abnormalities of specific neural networks. The earlier reference to the expression of Wnt3a in cells that migrated widely during development and appear in auditory systems is one example of how developmental changes may affect single functional networks. In contrast, changes in common and widely expressed synaptic molecules, such as the neuroligins, would represent the other mechanism.

The most important recent discovery in ASD pathogenesis has been the widely reported and replicated brain growth phenotype: The brain starts with, most likely, average size at birth increases in volume at an accelerated rate by the end of the first year compared to the typically developing child, and this process continues from ages 2 to 4 years. These data derive from both neuroimaging studies as well as measures of head circumference performed by multiple labs. It is not known whether this reflects an acceleration of normal developmental processes or a disease-specific aberration in postnatal development, including changes in cell numbers, neuronal processes, synapse formation, and modifications, or glial cell dysfunction, to name a few. The most prominent differences are observed in the frontal and parietal cortex, cerebellar hemispheres, as well as the amygdala. These findings are also consistent with recent reports of macrocephaly in up to ~20 percent of ASD cases in brain and DNA banks. These findings raise many questions to be addressed by developmental neuroscientists.

Functional neuroimaging studies indicate broad forebrain but also cerebellar dysfunctions in ASD, and classical pathologic studies have suggested abnormalities restricted to limbic and cerebellar structures. However, classical studies were hampered by small sample sizes, poor control for comorbidities such as epilepsy and mental retardation that affect neuroanatomy, and the use of tissue cell density measures as opposed to unbiased stereologic methods to estimate regional neuron numbers. Although previous studies described increased densities of small neurons in interconnecting limbic nuclei, including CA fields, septum, mammillary

bodies, and amygdala, these results have not been replicated by other laboratories. In contrast, the most consistent neuropathology has been observed in the cerebellum (21 of 29 brains), showing reductions in the number of Purkinje neurons without signs of acquired postnatal lesions, such as gliosis, empty baskets, and retrograde loss of afferent inferior olive neurons, suggesting prenatal origins.

A more recent study identifies widespread and nonuniform abnormalities, suggesting dysregulation of many processes, including neuron proliferation, migration, survival, organization, and programmed cell death. Four of six brains were macrocephalic, consistent with an increased size defined by numerous pathology and neuroimaging studies. In the cerebral cortex, there was thickened or diminished gray matter, disorganized laminar patterns, misoriented pyramidal neurons, ectopic neurons in both superficial and deep white matter, and increased or decreased neuron densities. This evidence of abnormal cortical neurogenesis and migration accords well with the deficits in cognitive functions. In the brainstem, neuronal disorganization appeared as discontinuous and malpositioned neurons in olivary and dentate nuclei, ectopic neurons in medulla and cerebellar peduncles, and aberrant fiber tracts. There were widespread patchy or diffuse decreases of Purkinje neurons, sometimes associated with increased Bergmann glia, or ectopic Purkinje neurons in the molecular layer. Hippocampal neuronal atrophy was not observed, and quantitative stereology found no consistent change in neuron density or number. Moreover, a single recent neuropathologic study using multiple immunologic indices has reported increased levels of immune cytokines in the CSF of patients and brain tissues as well as astrocytes expressing high levels of glial fibrillary acidic protein in the frontal and cingulated cortex, white matter, and cerebellum, all suggesting potential immune activation without evidence of an inflammatory process. We await confirmation of these critical findings.

Although seemingly incompatible, these various data support a model of developmental abnormalities occurring at different times, altering regions according to specific schedules of neurogenesis and differentiation. It is notable that a similar range of abnormalities was found in classical studies but was excluded since these abnormalities did not occur in every brain examined. Moreover, in 15 children exposed to the teratogen thalidomide during days 20 to 24 of gestation, when cranial and Purkinje neurogenesis

occurs in the brainstem, four cases exhibited autism. Based on these data, autism is associated with insults at 3 weeks for thalidomide, 12 weeks when inferior olivary neurons are migrating, and ~30 weeks when olivary axons make synapses with Purkinje cells. These diverse abnormalities in cell production, survival, migration, organization, and differentiation in both the hindbrain and forebrain indicate disturbed brain development over a range of stages. Recent genetic studies have defined two genetic polymorphisms associated reproducibly with ASD in several datasets, both of which influence brain developmental processes. The first is *ENGRAILED-2*, the cerebellar patterning gene whose dysregulation causes deficits in Purkinje and granule neurons in animal models and acts to control proliferation and differentiation. The second is the hepatocyte growth factor receptor *cMET*, whose function affects tangential migration of GABA interneurons from the ventral forebrain ganglionic eminences, potentially leading to imbalances of excitatory and inhibitory neurotransmission.

Furthermore, although the cellular derangements may be directly responsible for the core symptoms of autism, there is an alternative hypothesis: Disturbed regulation of developmental processes produces an as-yet-unidentified biochemical, cellular lesion that may be associated with autism. This proposal is supported by the currently known genetic causes of autism that account for 10 percent of cases, including tuberous sclerosis, neurofibromatosis, Smith–Lemli–Opitz syndrome, Rett syndrome, and fragile X mental retardation. These genetic etiologies interfere with cell proliferation control, cholesterol biosynthesis and Shh function, and synaptic and dendrite protein translation and function, fundamental processes in the sequence of development. An intriguing potential link in these monogenetic causes of autism symptoms is their participation in protein synthesis in the synapse, mainly as regulated via the PI3K/Akt signaling pathway and the mammalian target of rapamycin (mTOR) complex, an area of active research.

The Remarkable Discovery of Adult Neurogenesis

In the last decade, there has been a fundamental shift in paradigm regarding the limits of neurogenesis in the brain, with important implications for neural plasticity, mechanisms of disease etiology and therapy, and possibilities of repair. Until recently, it has generally been maintained that we do not produce

new neurons in the brain after birth (or soon after that, considering cerebellar EGL); thus, brain plasticity and repair depend on modifications of a numerically static neural network. We now have strong evidence to the contrary: new neurons are generated throughout life in certain regions, well documented across the phylogenetic tree, including birds, rodents, primates, and humans. As an area of intense interest and investigation, we may expect rapid progress over the next two decades, likely altering models described herein.

The term neurogenesis has been used inconsistently in different contexts, indicating sequential production of neural elements during development, first neurons then glial cells, but frequently connoting only neuron generation in the adult brain, in contrast to gliogenesis. For this discussion, we use the first, more general meaning, and distinguish cell types as needed. The first evidence of mammalian neurogenesis in the adult hippocampus was reported in the 1960s in which ^{3}H-thymidine-labeled neurons were documented. As a standard marker for cell production, these studies used nuclear incorporation of ^{3}H-thymidine into newly synthesized DNA during chromosome replication, which occurs before cells undergo division. After a delay, cells divide, producing two ^{3}H-thymidine-labeled progeny. Cell proliferation is defined as an absolute increase in cell number, which occurs only if cell production is not balanced by cell death. Because there is currently little evidence for a progressive increase in brain size with age, except perhaps for rodent hippocampus, most neurogenesis in the adult brain is compensated for by cell loss. More recent studies of neurogenesis employ the more convenient thymidine analog BrdU, which can be injected into living animals and then detected by immunohistochemistry.

During embryonic development, neurons are produced from almost all regions of the ventricular neuroepithelium. Neurogenesis in the adult, however, is mostly restricted to two regions: the SVZ lining the lateral ventricles and a narrow proliferative zone underlying the dentate gyrus granule layer (subgranular zone) in the hippocampus. In mice, rodents, and monkeys, newly produced neurons migrate from the SVZ in an anterior direction into the olfactory bulb to become GABA interneurons. The process has been elegantly characterized at both ultrastructural and molecular levels. In the SVZ, the neuroblasts (A cells) on their way to the olfactory bulb create chains of cells and migrate through a scaffold of glial cells supplied by

slowly dividing astrocytes (B cells). Within this network of cell chains, there are groups of rapidly dividing neural precursors (C cells). Evidence suggests that the B cells give rise to the C cells, which in turn develop into the A cells, the future olfactory bulb interneurons. The existence of a sequence of precursors with progressively restricted abilities to generate diverse neural cell types makes defining mechanisms that regulate adult neurogenesis in vivo a significant challenge.

As in the developing brain, adult neurogenesis is also subject to regulation by extracellular signals that control precursor proliferation and survival and, in many cases, the very same factors. After the initial discovery of adult neural stem cells generated under EGF stimulation, other regulatory factors were defined, including bFGF, IGF-I, BDNF, and LIF/CNTF. Although the hallmark of neural stem cells includes the capacity to generate neurons, astrocytes, and oligodendroglia, termed multipotentiality, specific signals appear to produce relatively different profiles of cells that may migrate to distinct sites. Intraventricular infusion of EGF promotes primarily gliogenesis in the SVZ, with cells migrating to the olfactory bulb, striatum, and corpus callosum, whereas bFGF favors the generation of neurons destined for the olfactory bulb. Both factors appear to stimulate mitosis directly, with differential effects on the cell lineage produced. In contrast, BDNF may increase neuron formation in SVZ as well as striatum and hypothalamus, though effects may be primarily through promoting survival of newly generated neurons that otherwise undergo cell death. Finally, CNTF and related LIF may promote gliogenesis or support the self-renewal of adult stem cells rather than enhancing a specific cell category.

Remarkably, in addition to direct intraventricular infusions, adult neurogenesis is also affected by peripheral levels of growth factors, hormones, and neuropeptides. Peripheral administration of both bFGF and IGF-I stimulate neurogenesis, increasing selectively mitotic labeling in the SVZ and hippocampal subgranular zone, respectively, suggesting that there are specific mechanisms for factor transport across the BBB. Of interest, elevated prolactin levels, induced by peripheral injection or natural pregnancy, stimulate proliferation of progenitors in the mouse SVZ, leading to increased olfactory bulb interneurons, potentially playing roles in learning new infant scents. This process may be relevant to changes in prolactin seen in psychiatric disease. Conversely, in behavioral paradigms of social stress,

such as territorial challenge by male intruders, activation of the hypothalamic–pituitary–adrenal (HPA) axis with increased glucocorticoids leads to reduced neurogenesis in the hippocampus, apparently through local glutamate signaling. Inhibition is also observed after peripheral opiate administration, a model for substance abuse. Thus neurogenesis may be one target process affected by changes of hormones and neuropeptides associated with several psychiatric conditions.

The discovery of adult neurogenesis naturally leads to questions about whether new neurons can integrate into the complex cytoarchitecture of the mature brain and speculation about its functional significance, if any. In rodents, primates, and humans, new neurons are generated in the dentate gyrus of the hippocampus, an area important for learning and memory. Some adult-generated neurons in humans have been shown to survive for at least 2 years. Furthermore, newly generated cells in the adult mouse hippocampus indeed elaborate extensive dendritic and axonal arborizations appropriate to the neural circuit and display functional synaptic inputs and action potentials. From a functional perspective, the generation or survival of new neurons correlates strongly with multiple instances of behavioral learning and experience. For example, the survival of newly generated neurons is markedly enhanced by hippocampal-dependent learning tasks and by an enriched, behaviorally complex environment. Of perhaps greater importance, a reduction in dentate gyrus neurogenesis impairs the formation of trace memories, that is when an animal must associate stimuli that are separated in time, a hippocampal-dependent task. Finally, in songbirds, neurogenesis is activity-dependent and is increased by foraging for food and learning a new song, whether it occurs seasonally or is induced by steroid hormone administration.

From clinical and therapeutic perspectives, fundamental questions are whether changes in neurogenesis contribute to disease and whether newly formed neurons undergo migration to and integration into regions of injury, replacing dead cells and leading to functional recovery. A neurogenetic response has now been shown for multiple conditions in the adult, including brain trauma, stroke, and epilepsy. For instance, ischemic stroke in the striatum stimulates adjacent SVZ neurogenesis, with neurons migrating to the injury site. Furthermore, in a highly selective paradigm not involving local tissue damage, degeneration of layer three cortical neurons elicited SVZ

neurogenesis and cell replacement. These studies raise the possibility that newly produced neurons typically participate in recovery and may be stimulated as a novel therapeutic strategy. However, in contrast to potential reconstructive functions, neurogenesis may also play roles in pathogenesis: In a kindling model of epilepsy, newly generated neurons were found to migrate to incorrect positions and participate in aberrant neuronal circuits, thereby reinforcing the epileptic state.

Conversely, reductions in neurogenesis may contribute to several conditions that implicate dysfunction or degeneration of the hippocampal formation. Dentate gyrus neurogenesis is inhibited by increased glucocorticoid levels observed in aged rats. It can be reversed by steroid antagonists and adrenalectomy, observations potentially relevant to the correlation of elevated human cortisol levels with reduced hippocampal volumes, and the presence of memory deficits. Similarly, stress-induced increases in human glucocorticoids may contribute to decreased hippocampal volumes seen in schizophrenia, depression, and PTSD.

A potential role for altered neurogenesis in disease has gained the most support in recent studies of depression. Several studies in animals and humans suggest a correlation of decreased hippocampal size with depressive symptoms, whereas clinically significant antidepressant therapy elicits increased hippocampal volume and enhanced neurogenesis, with causal relationships still being defined. For example, postmortem and brain imaging studies indicate cell loss in corticolimbic regions in bipolar disorder and major depression. Significantly, mood stabilizers, such as lithium and valproic acid, as well as antidepressants and ECT, activate intracellular pathways that promote neurogenesis and synaptic plasticity. Furthermore, in a useful primate model, the adult tree shrew, the chronic psychosocial stress model of depression elicited ~15 percent reductions in brain metabolites and a 33 percent decrease in neurogenesis (BrdU mitotic labeling), effects that were prevented by coadministration of antidepressant, tianeptine. More importantly, although stress exposure elicited small reductions in hippocampal volumes, stressed animals treated with antidepressants exhibited increased hippocampal volumes. Similar effects have been found in rodent models of depression.

Also, to the previous structural relationships, recent evidence has begun defining the roles of relevant neurotransmitter systems to antidepressant

effects on behavior and neurogenesis. In a most exciting finding, a causal link between antidepressant-induced neurogenesis and a positive behavioral response has been demonstrated. In the serotonin 1A receptor null mouse, fluoxetine, a selective serotonin reuptake inhibitor [SSRI], produced neither enhanced neurogenesis nor behavioral improvement. Furthermore, when hippocampal neuronal precursors were selectively reduced (85 percent) by X-irradiation, neither fluoxetine nor imipramine induced neurogenesis or behavioral recovery. Finally, one study using hippocampal cultures from normal and mutant rodents strongly supports a neurogenetic role for endogenous NPY, which is contained in dentate gyrus hilar interneurons. NPY stimulates precursor proliferation selectively via the Y1 (not Y2 or Y5) receptor, a finding consistent with the receptor-mediating antidepressive effects of NPY in animal models and the impact of NPY levels on both hippocampal-dependent learning and responses to stress. In aggregate, these observations suggest that volume changes observed with human depression and therapy may directly relate to alterations in ongoing neurogenesis. More generally, the discovery of adult neurogenesis has led to significant changes in our perspectives on the regenerative capacities of the human brain.

NEUROPHYSIOLOGY AND NEUROCHEMISTRY

The study of chemical interneuronal communication is called neurochemistry, and in recent years there has been an explosion of knowledge in understanding chemical transmission between neurons and the receptors affected by those chemicals. It also influences our understanding of brain physiology. This section chapter focuses on the complex heterogeneity of both these areas to help explain the complexity of thoughts, feelings, and behaviors that make up the human experience.

Monoamine Neurotransmitters

The monoamine neurotransmitters and acetylcholine have been historically implicated in the pathophysiology and treatment of a wide variety of neuropsychiatric disorders. Each monoamine neurotransmitter system modulates many different neural pathways, which themselves subserve multiple behavioral and physiologic processes. Conversely, each CNS neurobehavioral process is likely modulated by multiple interacting

neurotransmitter systems, including monoamines.

This complexity poses a significant challenge to understanding the precise molecular, cellular, and systems-level pathways through which various monoamine neurotransmitters affect neuropsychiatric disorders. However, recent advances in human genetics and genomics, as well as in experimental neuroscience, have shed light on this question. Molecular cloning has identified a large number of genes that regulate monoaminergic neurotransmissions, such as the enzymes, receptors, and transporters that mediate the synthesis, cellular actions, and cellular reuptake of these neurotransmitters, respectively. Human genetics studies have provided evidence of tantalizing links between allelic variants in specific monoamine-related genes and psychiatric disorders and trait abnormalities. In contrast, the ability to modify gene function and cellular activity in experimental animals has clarified the roles of specific genes and neural pathways in mediating behavioral processes.

Monoamines act on target cells by binding to specific cell-surface receptors. There are multiple receptor subtypes for each monoamine, which are expressed in diverse regions and subcellular locales and which engage a variety of intracellular signaling pathways. This panoply of receptors thus allows each monoamine neurotransmitter to modulate target cells in many ways; the same molecule may activate some cells while inhibiting others, depending on which receptor subtype is expressed by each cell. The various monoamines are discussed below.

Serotonin. Although only one in a million CNS neurons produces serotonin, these cells influence virtually all aspects of CNS function. The cell bodies of these serotonergic neurons are clustered in the midline raphe nuclei of the brainstem; the rostral raphe nuclei send ascending axonal projections throughout the brain, whereas the descending caudal raphe nuclei send projections into the medulla, cerebellum, and spinal cord (Fig. 33-12). The descending serotonergic fibers that innervate the dorsal horn of the spinal cord have been implicated in the suppression of nociceptive pathways, a finding that may relate to the pain-relieving effects of some antidepressants. The tonic firing of CNS serotonin neurons varies across the sleep–wake cycle, with an absence of activity during rapid eye movement (REM) sleep. Increased serotonergic firing is observed during rhythmic motor behaviors

and suggests that serotonin modulates some forms of motor output.

Most serotonergic innervation of the cortex and limbic system arises from the dorsal and median raphe nuclei in the midbrain; the serotonergic neurons in these areas send projections through the medial forebrain bundle into target forebrain regions. The median raphe provides most of the serotonergic fibers that innervate the limbic system, whereas the dorsal raphe nucleus provides most of the serotonergic fibers that innervate the striatum and thalamus.

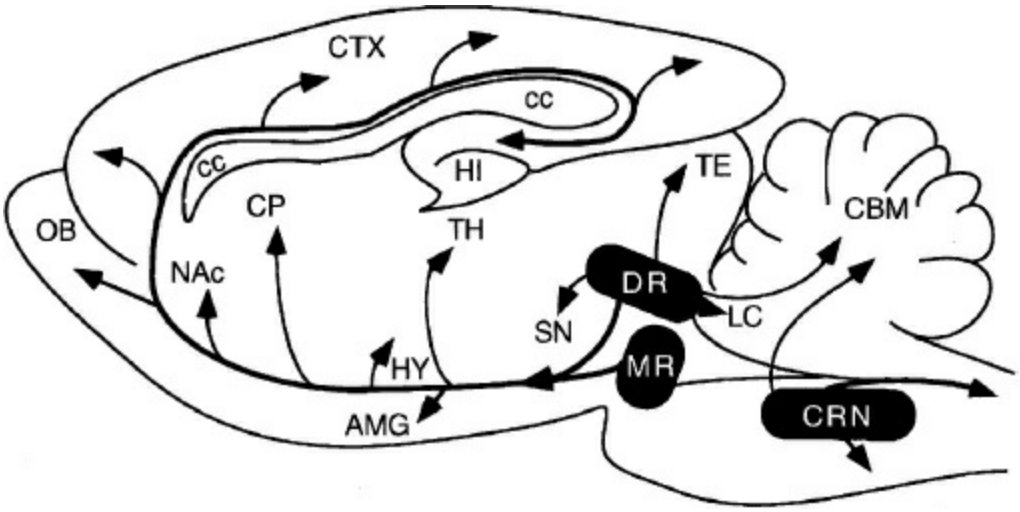


FIGURE 33-12
Brain serotonergic pathways (in rats). Serotonergic neurons are located in brainstem midline raphe nuclei and project throughout the neuraxis. (There is an approximate similarity between monoamine pathways in rats and in humans.) AMG, amygdala; CBM, cerebellum; cc, corpus callosum; CP, caudate putamen; CRN, caudal raphe nuclei; CTX, neocortex; DR, dorsal raphe nucleus; HI, hippocampus; HY, hypothalamus; LC, locus ceruleus; MR, median raphe nucleus; NAc, nucleus accumbens; OB, olfactory bulb; SN, substantia nigra; TE, tectum; TH, thalamus; TM, tuberomammillary nucleus of hypothalamus. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:65.)

Also, to the different target fields of these serotonergic nuclei, there are cellular differences between their constituent neurons. Dorsal raphe serotonergic fibers are fine, with small vesicle-coated swellings called varicosities, whereas median raphe fibers have large spherical or beaded varicosities. It is unclear to what extent serotonin acts as a true synaptic or "private" neurotransmitter versus action as a local endocrine hormone or "social transmitter," or whether its roles differ depending on the fiber type from which it is released. These fibers show differential sensitivity to the

neurotoxic effects of the amphetamine analog 3,4-methylenedioxy-methamphetamine (MDMA, “ecstasy”), which lesions the fine axons of the dorsal raphe while sparing the thick beaded axons of the median raphe. The significance of these morphologic differences is unclear, although recent work has identified functional differences between the serotonergic neurons of the dorsal and median raphe nuclei.

Dopamine. Dopamine neurons are more widely distributed than those of other monoamines, residing in the midbrain substantia nigra and ventral tegmental area (VTA) and the periaqueductal gray, hypothalamus, olfactory bulb, and retina. In the periphery, dopamine is found in the kidney, where it functions to produce renal vasodilation, diuresis, and natriuresis. Three dopamine systems are highly relevant to psychiatry: The nigrostriatal, mesocorticolimbic, and tuberohypophyseal system (Fig. 33-13). Degeneration of the nigrostriatal system causes Parkinson disease and has led to an intense research focus on the development and function of dopamine neurons in the midbrain substantia nigra nuclei. Dopamine cell bodies in the pars compacta division of this region send ascending projections to the dorsal striatum (especially to the caudate and putamen) and thereby modulate motor control. The extrapyramidal effects of antipsychotic drugs are thought to result from the blockade of these striatal dopamine receptors.

The midbrain VTA lies medial to the substantia nigra and contains dopaminergic neurons that give rise to the mesocorticolimbic dopamine system. These neurons send ascending projections that innervate limbic structures, such as the nucleus accumbens and amygdala; the mesoaccumbens pathway is a central element in the neural representation of reward, and intense research has been devoted to this area in recent years. All known drugs of abuse activate the mesoaccumbens dopamine pathway, and plastic changes in this pathway are thought to underlie drug addiction. The mesolimbic projection is believed to be a significant target for the antipsychotic properties of dopamine receptor antagonist drugs in controlling the positive symptoms of schizophrenia, such as hallucinations and delusions.

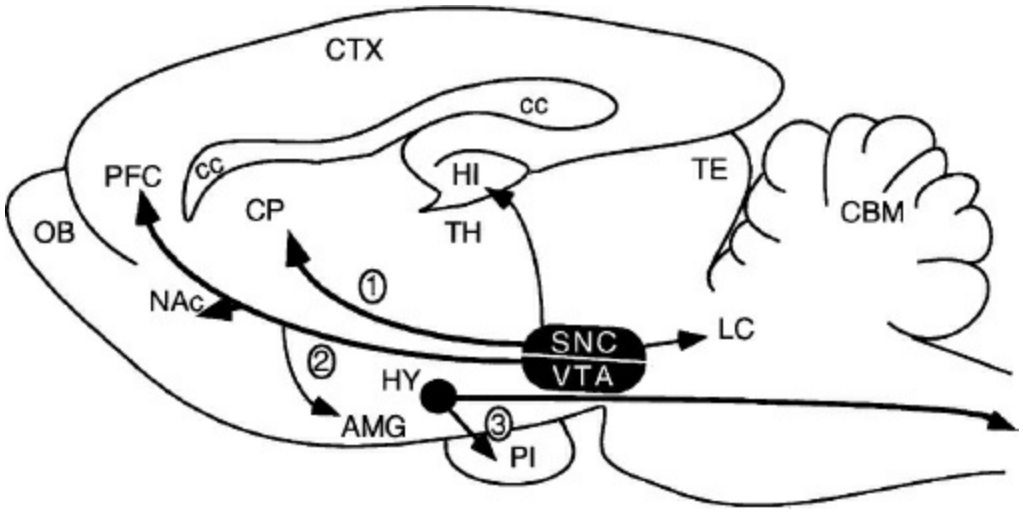


FIGURE 33-13
Brain dopaminergic pathways (in rats). The three principal dopaminergic pathways: (*1*) nigrostriatal pathway, (*2*) mesocorticolimbic pathway, and (*3*) tuberohypophyseal pathway. AMG, amygdala; CBM, cerebellum; cc, corpus callosum; CP, caudate putamen; CTX, neocortex; HI, hippocampus; HY, hypothalamus; LC, locus ceruleus; NAc, nucleus accumbens; OB, olfactory bulb; PFC, prefrontal cortex; PI, pituitary; SNC, substantia nigra pars compacta; TE, tectum; TH, thalamus; VTA, ventral tegmental area. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:66.)

VTA dopamine neurons also project to cortical structures, such as the prefrontal cortex, and modulate working memory and attention; decreased activity in this pathway is proposed to underlie negative symptoms of schizophrenia. Thus antipsychotic drugs that decrease positive symptoms by blocking dopamine receptors in the mesolimbic pathway may simultaneously worsen these negative symptoms by blocking similar dopamine receptors in the mesocortical pathway. The decreased risk of extrapyramidal side effects seen with clozapine (Clozaril; vs. other typical antipsychotic medications) is thought to be due to its relatively selective effects on this mesocortical projection. The tuberohypophyseal system consists of dopamine neurons in the hypothalamic arcuate and paraventricular nuclei that project to the pituitary gland and thereby inhibit prolactin release. Antipsychotic drugs that block dopamine receptors in the pituitary may thus disinhibit prolactin release and cause galactorrhea.

Norepinephrine and Epinephrine. The postganglionic sympathetic neurons of the ANS release norepinephrine, resulting in widespread

peripheral effects, including tachycardia and elevated blood pressure. The adrenal medulla releases epinephrine, which produces similar effects; epinephrine-secreting pheochromocytoma tumors produce bursts of sympathetic activation, central arousal, and anxiety.

Norepinephrine-producing neurons are found within the brain in the pons and medulla in two major clusterings: The locus ceruleus (LC) and the lateral tegmental noradrenergic nuclei (Fig. 33-14). Noradrenergic projections from both of these regions extensively ramify as they project throughout the neuraxis. In humans, the LC is found in the dorsal portion of the caudal pons and contains approximately 12,000 tightly packed neurons on each side of the brain. These cells provide the central noradrenergic projections to the neocortex, hippocampus, thalamus, and midbrain tectum. The activity of LC neurons varies with the animal's level of wakefulness. Firing rates are responsive to novel or stressful stimuli, with the largest responses to stimuli that disrupt ongoing behavior and reorient attention. Altogether, physiologic studies indicate a role for this structure in the regulation of arousal state, vigilance, and stress response. The projections from lateral tegmental nucleus neurons, which are loosely scattered throughout the ventral pons and medulla, partially overlap those of the LC. Fibers from both cell groups innervate the amygdala, septum, and spinal cord. Other regions, such as the hypothalamus and lower brainstem, receive adrenergic inputs predominantly from the lateral tegmental nucleus. The relatively few neurons that utilize epinephrine as a neurotransmitter are located in the caudal pons and medulla, intermingled with noradrenergic neurons. Projections from these groups ascend to innervate the hypothalamus, LC, and visceral efferent and afferent nuclei of the midbrain.

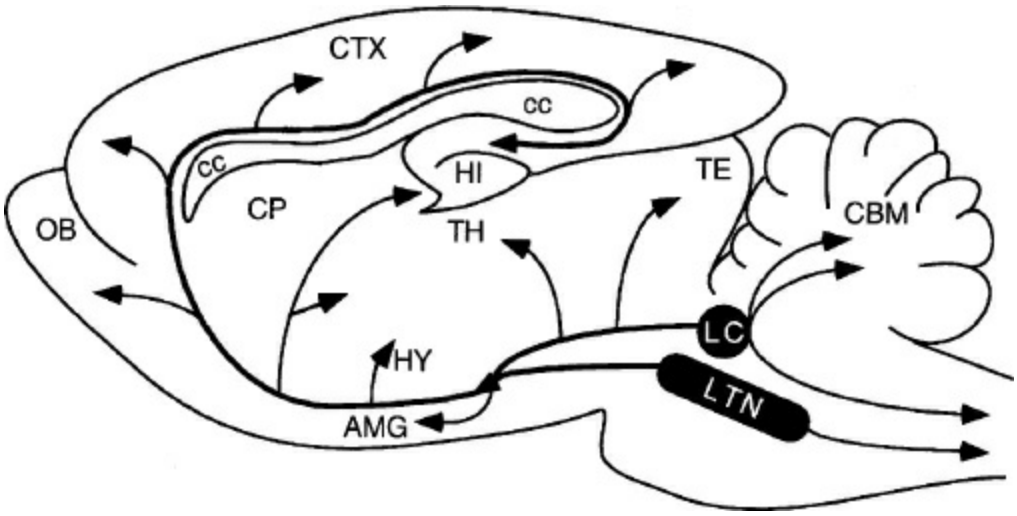


FIGURE 33-14
Brain noradrenergic pathways (in rats). Projections of noradrenergic neurons located in the locus ceruleus (LC) and lateral tegmental noradrenergic nuclei (LTN). AMG, amygdala; CBM, cerebellum; cc, corpus callosum; CP, caudate putamen; CTX, neocortex; HI, hippocampus; HY, hypothalamus; OB, olfactory bulb; TE, tectum; TH, thalamus. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:66.)

Histamine. Histamine is perhaps best known for its role in allergies. It is an inflammatory mediator stored in mast cells and released upon cellular interaction with allergens. Once released, histamine causes vascular leakage and edema and other facial and topical allergy symptoms. In contrast, central histaminergic neural pathways have only more recently been characterized by immunocytochemistry using antibodies to the synthetic enzyme histidine decarboxylase and histamine. Histaminergic cell bodies are located within a region of the posterior hypothalamus termed the tuberomammillary nucleus. The activity of tuberomammillary neurons is characterized by firing that varies across the sleep–wake cycle, with the highest activity during the waking state, slowed firing during slow-wave sleep, and absence of firing during REM sleep. Histaminergic fibers project diffusely throughout the brain and spinal cord (Fig. 33-15). Ventral ascending projections course through the medial forebrain bundle and then innervate the hypothalamus, diagonal band, septum, and olfactory bulb. Dorsal ascending projections innervate the thalamus, hippocampus, amygdala, and rostral forebrain. Descending projections travel through the central midbrain gray to the dorsal hindbrain and spinal cord. The fibers have varicosities that are seldom

associated with classical synapses, and histamine has been proposed to act at a distance from its sites of release, like a local hormone. The hypothalamus receives the densest histaminergic innervation, consistent with a role for this transmitter in the regulation of autonomic and neuroendocrine processes. Also, muscular histaminergic innervation is seen in monoaminergic and cholinergic nuclei.

Acetylcholine. Within the brain, the axonal processes of cholinergic neurons may either project to distant brain regions (projection neurons) or contact local cells within the same structure (interneurons). Two large clusters of cholinergic projection neurons are found within the brain: The basal forebrain complex and the mesopontine complex (Fig. 33-16). The basal forebrain complex provides most of the cholinergic innervation to the nonstriatal telencephalon. It consists of cholinergic neurons within the nucleus basalis of Meynert, the horizontal and vertical diagonal bands of Broca, and the medial septal nucleus. These neurons project to widespread areas of the cortex and amygdala, to the anterior cingulate gyrus and olfactory bulb, and the hippocampus, respectively. In Alzheimer disease, there is significant degeneration of neurons in the nucleus basalis, leading to a substantial reduction in cortical cholinergic innervation. The extent of neuronal loss correlates with the degree of dementia, and the cholinergic deficit may contribute to the cognitive decline in this disease, consistent with the beneficial effects of drugs that promote acetylcholine signaling in this disorder.

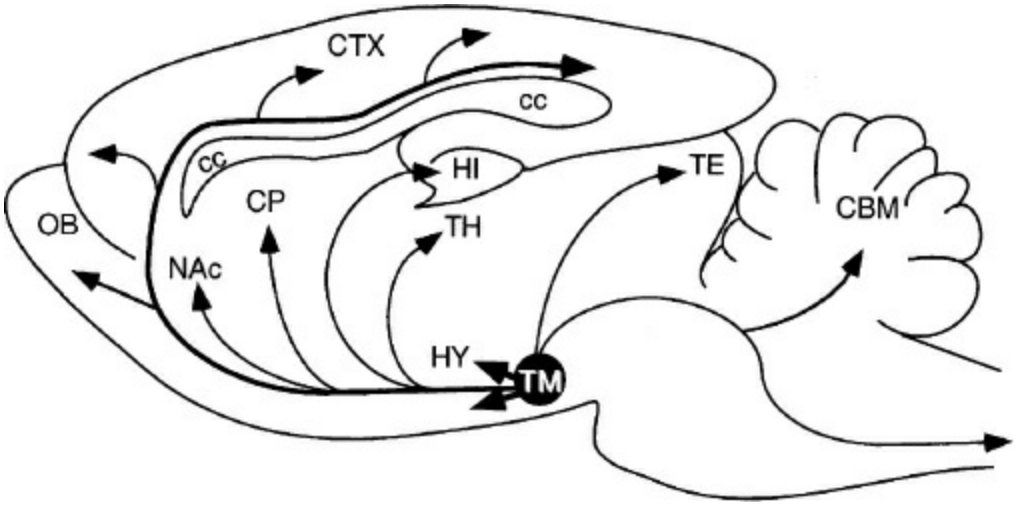


FIGURE 33-15

Brain histaminergic pathways (in rats). Histaminergic neurons are located in the tuberomammillary nucleus of the caudal hypothalamus (TM) and project to the hypothalamus (HY) and more distant brain regions. CBM, cerebellum; cc, corpus callosum; CP, caudate putamen; CTX, neocortex; HI, hippocampus; NAc, nucleus accumbens; OB, olfactory bulb; TE, tectum; TH, thalamus. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:67.)

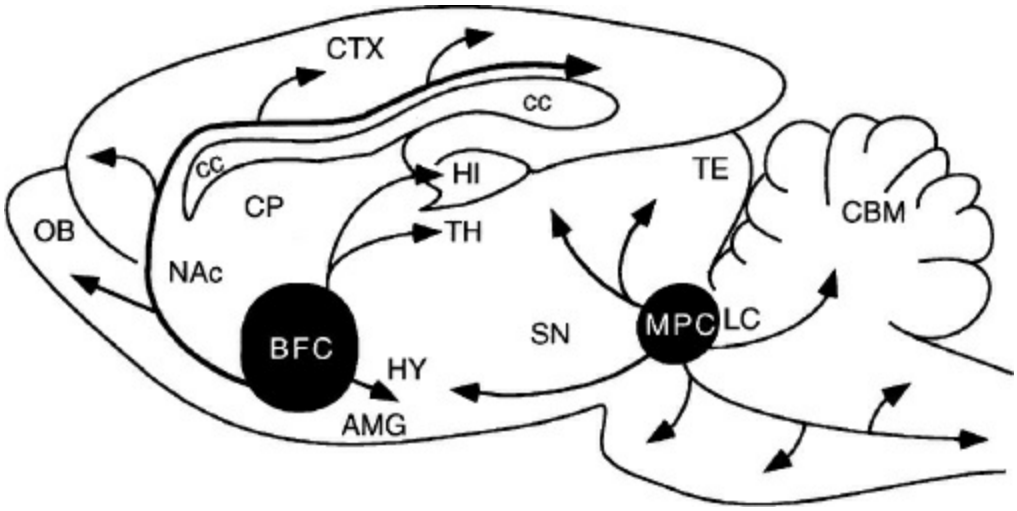


FIGURE 33-16
Brain cholinergic projection pathways (in rats). The majority of cholinergic projection neurons are located in the basal forebrain complex (BFC) and the mesopontine complex (MPC). AMG, amygdala; CBM, cerebellum; cc, corpus callosum; CP, caudate putamen; CTX, neocortex; HI, hippocampus; HY, hypothalamus; LC, locus ceruleus; NAc, nucleus accumbens; OB, olfactory bulb; SN, substantia nigra; TE, tectum; TH, thalamus. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:67.)

The mesopontine complex consists of cholinergic neurons within the pedunculopontine and laterodorsal tegmental nuclei of the midbrain and pons and provides cholinergic innervation to the thalamus and midbrain areas (including the dopaminergic neurons of the VTA and substantia nigra) and descending innervation to other brainstem regions such as the LC, dorsal raphe, and cranial nerve nuclei. In contrast to central serotonergic, noradrenergic, and histaminergic neurons, cholinergic neurons may continue to fire during REM sleep and have been proposed to play a role in REM sleep induction. Acetylcholine is also found within interneurons of several brain regions, including the striatum. The modulation of striatal cholinergic transmission has been implicated in the antiparkinsonian actions of

anticholinergic agents. Within the periphery, acetylcholine is a prominent neurotransmitter, located in motoneurons innervating skeletal muscle, preganglionic autonomic neurons, and postganglionic parasympathetic neurons. Peripheral acetylcholine mediates the characteristic postsynaptic effects of the parasympathetic system, including bradycardia and reduced blood pressure, and enhanced digestive function.

Monoamine Synthesis, Storage, and Degradation

Also, to neuroanatomic similarities, monoamines are synthesized, stored, and degraded in similar ways (Fig. 33-17). Monoamines are synthesized within neurons from common amino acid precursors (Fig. 33-17, step 1) and taken up into synaptic vesicles by way of a vesicular monoamine transporter (Fig. 33-17, step 2). On stimulation, vesicles within nerve terminals fuse with the presynaptic terminal and release the neurotransmitter into the synaptic cleft (Fig. 33-17, step 3). Once released, the monoamines interact with postsynaptic receptors to alter the function of postsynaptic cells (Fig. 33-17, step 4), and they may also act on presynaptic autoreceptors on the nerve terminal to suppress further release (Fig. 33-17, step 5). Also, released monoamines may be taken back up from the synaptic cleft into the nerve terminal by plasma membrane transporter proteins (Fig. 33-17, step 6), a process known as reuptake. Reuptake plays an essential role in limiting the total magnitude and temporal duration of monoamine signaling. Once monoamines are taken up, they may be subject to enzymatic degradation (Fig. 33-17, step 7), or they may be protected from degradation by uptake into vesicles. The processing of acetylcholine differs from this scheme and is described later in this section.

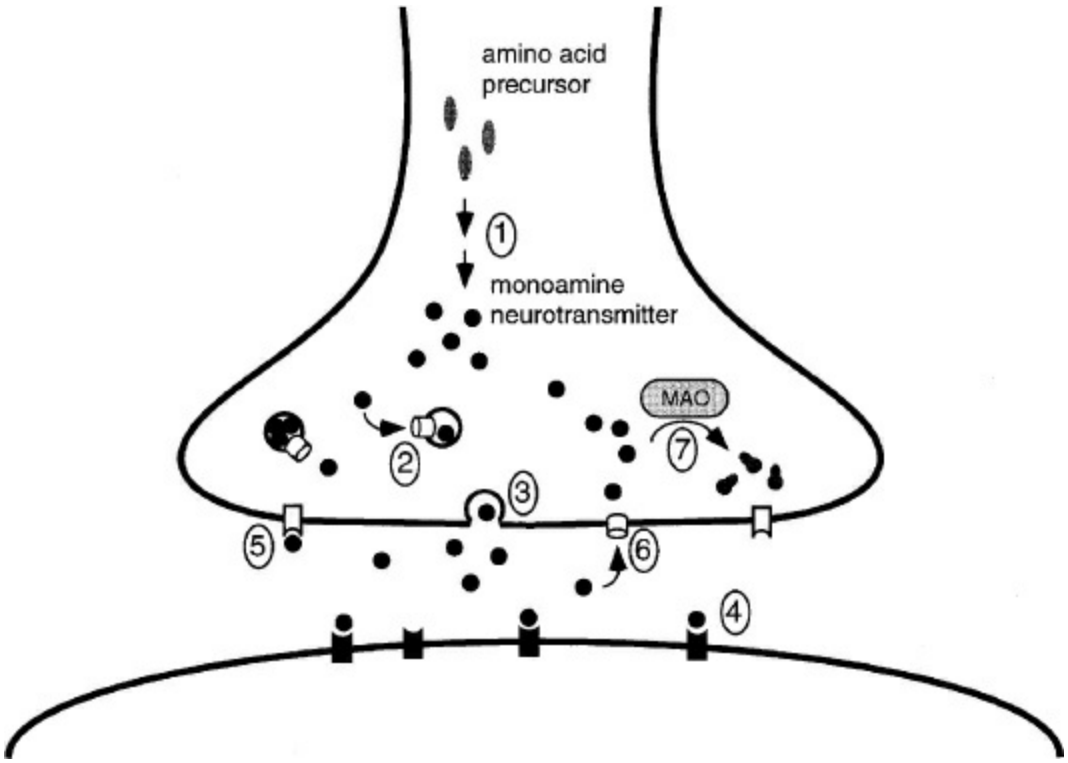


FIGURE 33-17
Schematic diagram of a monoaminergic synapse. Steps involved in synaptic transmission are described in the text. MAO, monoamine oxidase. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:68.)

Serotonin

The CNS contains less than 2 percent of the serotonin in the body; peripheral serotonin is located in platelets, mast cells, and enterochromaffin cells. More than 80 percent of all the serotonin in the body is found in the gastrointestinal system, where it modulates motility and digestive functions. Platelet serotonin promotes aggregation and clotting through a most unusual mechanism: The covalent linkage of serotonin molecules to small GTP-binding proteins, which can then activate these proteins, is a process termed "serotonylation." Peripheral serotonin cannot cross the BBB, so serotonin is synthesized within the brain as well. Serotonin is synthesized from the amino acid tryptophan, which is derived from the diet. The rate-limiting step in serotonin synthesis is the hydroxylation of tryptophan by the enzyme tryptophan hydroxylase to form 5-hydroxytryptophan (4-HT) (Fig. 33-18). Two isoforms of tryptophan hydroxylase exist—one isoform is found mainly in the periphery, whereas the second isoform is restricted to the CNS.

Under normal circumstances, tryptophan concentration is rate-limiting in serotonin synthesis. Therefore, much attention has focused on the factors that determine tryptophan availability. Unlike serotonin, tryptophan is taken up into the brain by way of a saturable active carrier mechanism. Because tryptophan competes with other large neutral amino acids for transport, brain uptake of this amino acid is determined both by the amount of circulating tryptophan and by the ratio of tryptophan to other large neutral amino acids. This ratio may be elevated by carbohydrate intake, which induces insulin release and the uptake of many sizeable neutral amino acids into peripheral tissues.

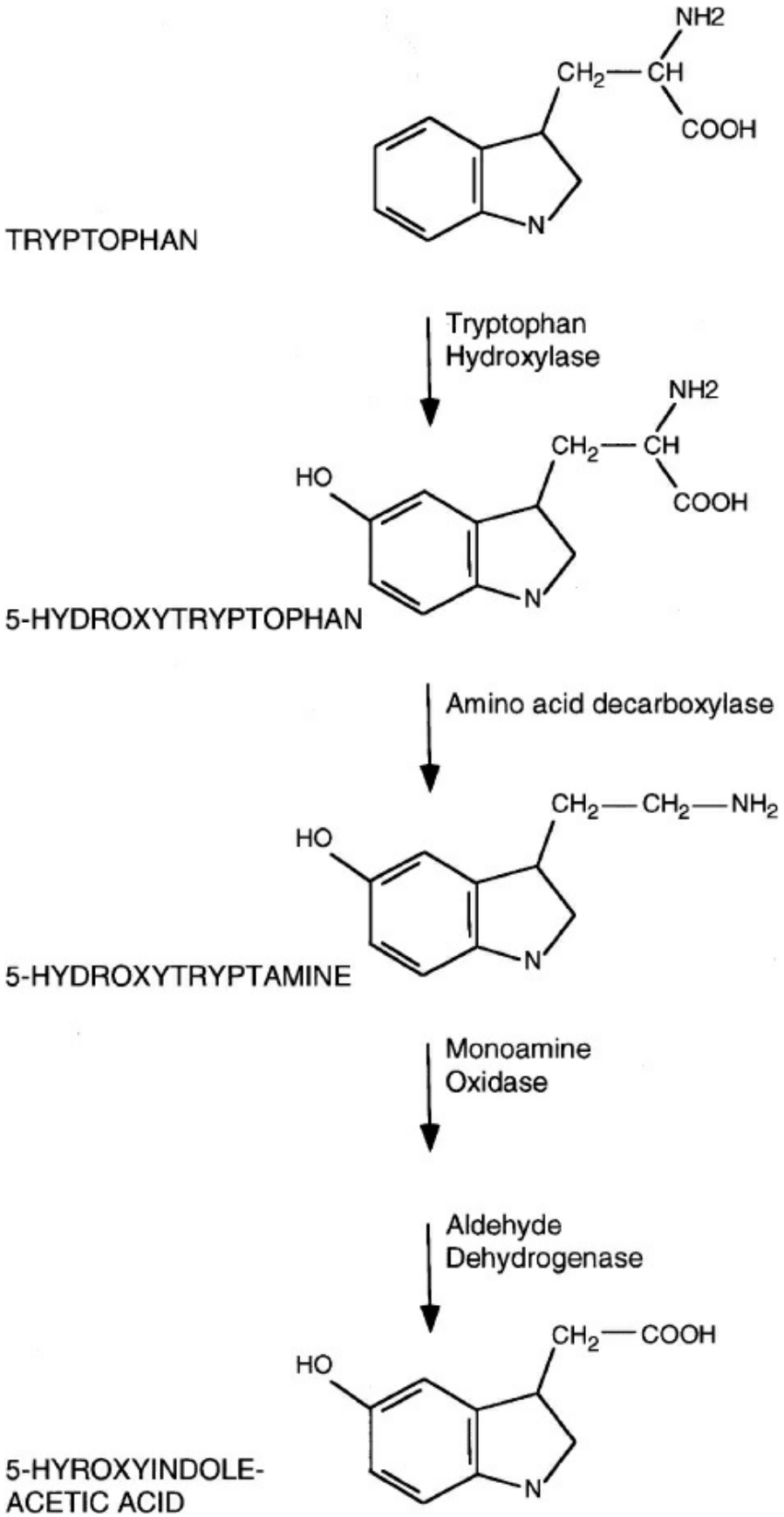

NH2
CH2—CH
COOH
N
TRYPTOPHAN
Tryptophan
Hydroxylase
NH2
CH2—CH
COOH
HO
N
5-HYDROXYTRYPTOPHAN
Amino acid decarboxylase
CH2—CH2—NH2
HO
N
5-HYDROXYTRYPTAMINE
Monoamine
Oxidase
Aldehyde
Dehydrogenase
CH2—COOH
HO
N
5-HYROXYINDOLE-
ACETIC ACID

FIGURE 33-18
Synthesis and catabolism of serotonin. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:68.)

Conversely, high-protein foods tend to be relatively low in tryptophan, thus lowering this ratio. Moreover, the administration of specialized low tryptophan diets produces significant declines in brain serotonin levels. After tryptophan hydroxylation, 5-hydroxytryptophan is rapidly decarboxylated by aromatic amino acid decarboxylase (an enzyme also involved in dopamine synthesis) to form serotonin.

The first step in the degradation of serotonin is mediated by monoamine oxidase type A (MAO_A), which oxidizes the amino group to form an aldehyde. MAO_A is located in mitochondrial membranes and is nonspecific in its substrate specificity; in addition to serotonin, it oxidizes norepinephrine. The elevation of serotonin levels by MAO inhibitors (MAOIs) is believed to underlie the antidepressant efficacy of these drugs. After oxidation by MAO_A, the resulting aldehyde is further oxidized to 5-hydroxyindoleacetic acid (5-HIAA). Levels of 5-HIAA are often measured as a correlate of serotonergic system activity, although the relationship of these levels to serotonergic neuronal activity remains unclear.

Catecholamines. The catecholamines are synthesized from the amino acid tyrosine, which is taken up into the brain via an active transport mechanism (Fig. 33-19). Within catecholaminergic neurons, tyrosine hydroxylase catalyzes the addition of a hydroxyl group to the meta position of tyrosine, yielding L-dopa. This rate-limiting step in catecholamine synthesis is subject to inhibition by high levels of catecholamines (end-product inhibition). Because tyrosine hydroxylase usually is saturated with substrate, manipulation of tyrosine levels does not readily affect the rate of catecholamine synthesis. Once formed, L-dopa is rapidly converted to dopamine by dopa decarboxylase, which is located in the cytoplasm. It is now recognized that this enzyme acts not only on L-dopa but also on all naturally occurring aromatic L-amino acids, including tryptophan, and thus it is more appropriately termed aromatic amino acid decarboxylase. In noradrenergic and adrenergic neurons, dopamine is actively transported into

storage vesicles, where it is oxidized by dopamine β-hydroxylase to form norepinephrine. In adrenergic neurons and the adrenal medulla, norepinephrine is converted to epinephrine by phenylethanolamine *N*-methyltransferase (PNMT), which is located within the cytoplasmic compartment.

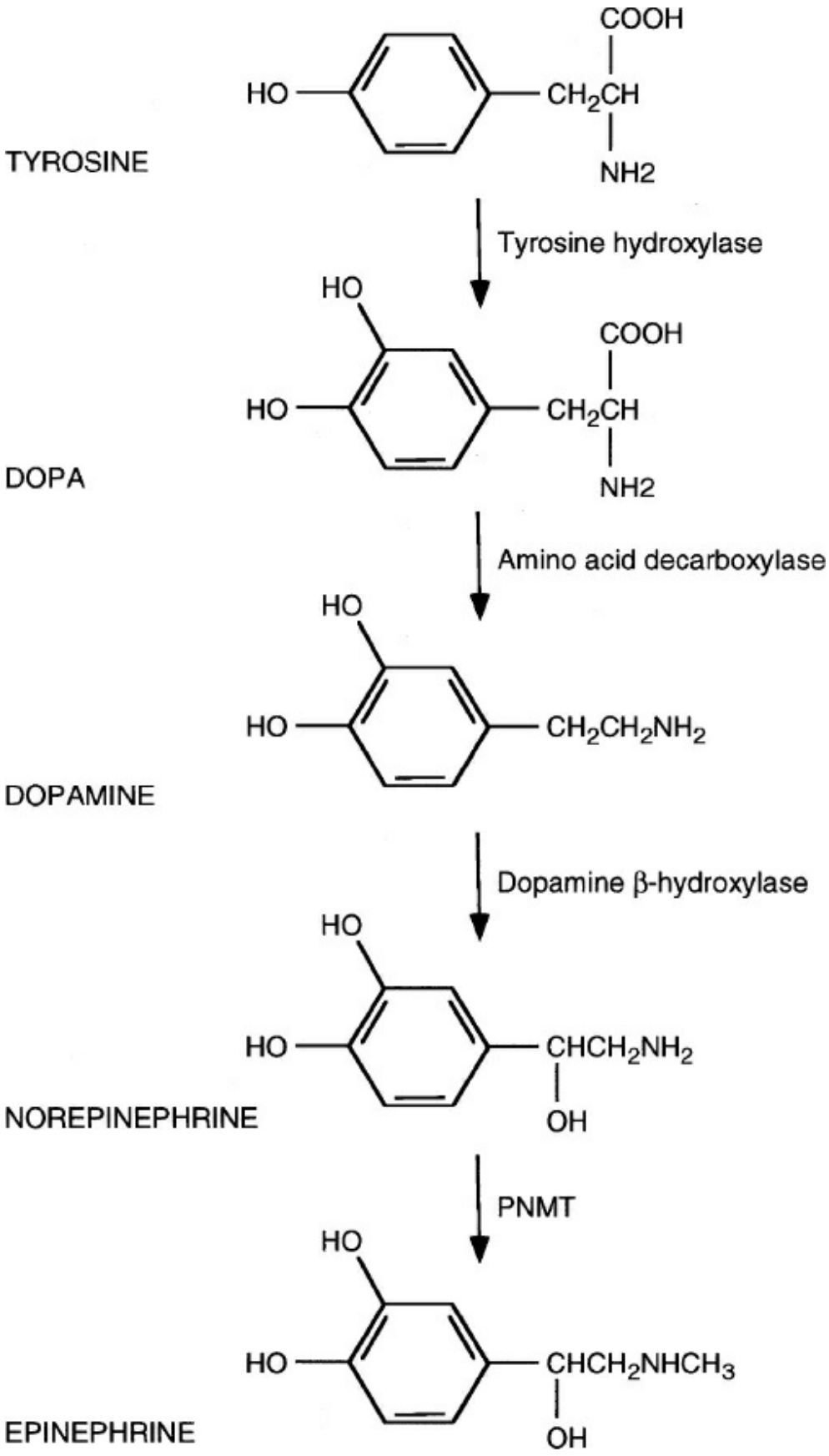

TYROSINE
HO
COOH
CH2CH
NH2
Tyrosine hydroxylase
DOPA
HO
HO
COOH
CH2CH
NH2
Amino acid decarboxylase
DOPAMINE
HO
HO
CH2CH2NH2
Dopamine β-hydroxylase
NOREPINEPHRINE
HO
HO
CHCH2NH2
OH
PNMT
EPINEPHRINE
HO
HO
CHCH2NHCH3
OH

FIGURE 33-19
Synthesis of catecholamines. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:69.)

Two enzymes that play significant roles in the degradation of catecholamines are monoamine oxidase and catechol-*O*-methyltransferase (COMT). MAO is located on the outer membrane of mitochondria, including those within the terminals of adrenergic fibers, and oxidatively deaminates catecholamines to their corresponding aldehydes. Two MAO isozymes with differing substrate specificities have been identified: MAO_A, which preferentially deaminates serotonin and norepinephrine, and MAO type B (MAO_B), which deaminates dopamine, histamine, and a broad spectrum of phenylethylamines. Neurons contain both MAO isoforms. The blockade of monoamine catabolism by MAO inhibitors produces elevations in brain monoamine levels. MAO is also found in peripheral tissues such as the gastrointestinal tract and liver, where it prevents the accumulation of toxic amines. For example, peripheral MAO degrades dietary tyramine, an amine that can displace norepinephrine from sympathetic postganglionic nerve endings, producing hypertension if tyramine is present in sufficient quantities. Thus patients treated with MAO inhibitors are cautioned to avoid pickled and fermented foods that typically have high levels of tyramine. Catechol-O-methyltransferase (COMT) is located in the cytoplasm and is widely distributed throughout the brain and peripheral tissues, although little to none is found in adrenergic neurons. It has a broad substrate specificity, catalyzing the transfer of methyl groups from *S*-adenosyl methionine to the *m*-hydroxyl group of most catechol compounds. The catecholamine metabolites produced by these and other enzymes are frequently measured as indicators of the activity of catecholaminergic systems. In humans, the predominant metabolites of dopamine and norepinephrine are homovanillic acid (HVA) and 3-methoxy-4-hydroxyphenylglycol (MHPG), respectively.

Histamine. As is the case for serotonin, the brain contains only a small portion of the histamine found in the body. Histamine is distributed throughout most tissues of the body, predominantly in mast cells. Because it does not readily cross the BBB, it is believed that histamine is synthesized

within the brain. In the brain, histamine is formed by the decarboxylation of the amino acid histidine by a specific L-histidine decarboxylase. This enzyme is not typically saturated with substrate, so synthesis is sensitive to histidine levels. This is consistent with the observation that the peripheral administration of histidine elevates brain histamine levels. Histamine is metabolized in the brain by histamine *N*-methyltransferase, producing methylhistamine. In turn, methylhistamine undergoes oxidative deamination by MAO_B.

Acetylcholine. Acetylcholine is synthesized by the transfer of an acetyl group from acetyl coenzyme A (ACoA) to choline in a reaction mediated by the enzyme choline acetyltransferase (ChAT). The majority of choline within the brain is transported from the blood rather than being synthesized de novo. Choline is taken up into cholinergic neurons by a high-affinity active transport mechanism, and this uptake is the rate-limiting step in acetylcholine synthesis. The rate of choline transport is regulated, such that increased cholinergic neural activity is associated with enhanced choline uptake. After synthesis, acetylcholine is stored in synaptic vesicles through the action of a vesicular acetylcholine transporter. After vesicular release, acetylcholine is rapidly broken down by hydrolysis by acetylcholinesterase, located in the synaptic cleft. Much of the choline produced by this hydrolysis is then taken back into the presynaptic terminal via the choline transporter. Although acetylcholinesterase is primarily in cholinergic neurons and synapses, a second class of cholinesterase, butyrylcholinesterase, is primarily in the liver and plasma as well as in glia. In the treatment of Alzheimer disease, strategies aimed at enhancing cholinergic function, primarily through the use of cholinesterase inhibitors to prevent usual degradation of acetylcholine, have shown moderate efficacy in ameliorating cognitive dysfunction as well as behavioral disturbances. Cholinesterase inhibitors are also used in the treatment of myasthenia gravis, a disease characterized by weakness due to blockade of neuromuscular transmission by autoantibodies to acetylcholine receptors.

Transporters. A great deal of progress has been made in the molecular characterization of the monoamine plasma membrane transporter proteins. These membrane proteins mediate the reuptake of synaptically released

monoamines into the presynaptic terminal. This process also involves cotransport of Na^+ and Cl^- ions and is driven by the ion concentration gradient generated by the plasma membrane Na^+/K^+ ATPase. Monoamine reuptake is an essential mechanism for limiting the extent and duration of activation of monoaminergic receptors. Reuptake is also a primary mechanism for replenishing terminal monoamine neurotransmitter stores. Moreover, transporters serve as molecular targets for several antidepressant drugs, psychostimulants, and monoaminergic neurotoxins. Whereas transporter molecules for serotonin (SERT), dopamine (DAT), and norepinephrine (NET) have been well characterized, transporters selective for histamine and epinephrine have not been demonstrated.

Among drugs of abuse, cocaine binds with high affinity to all three known monoamine transporters, although the stimulant properties of the drug have been attributed primarily to its blockade of DAT. This view has been recently supported by the absence of cocaine-induced locomotor stimulation in a strain of mutant mice engineered to lack this molecule. Psychostimulants produce a paradoxical locomotor suppression in these animals that has been attributed to their blockade of the serotonin transporter. The rewarding properties of cocaine have also been attributed primarily to dopamine transporter inhibition, although other targets mediate these effects as well since cocaine still has rewarding effects in mice lacking the dopamine transporter. It appears that serotonergic as well as dopaminergic mechanisms, may be involved. Transporters may also provide routes that allow neurotoxins to enter and damage monoaminergic neurons; examples include the dopaminergic neurotoxin 1-methyl-4-phenyl-1,2,3,6-tetrahydropyridine (MPTP) and the serotonergic neurotoxin MDMA.

Vesicular Monoamine Transporter. Also, to the reuptake of monoamines into the presynaptic nerve terminal, a second transport process serves to concentrate and store monoamines within synaptic vesicles. The transport and storage of monoamines in vesicles may serve several purposes: (1) to enable the regulated release of the transmitter under appropriate physiologic stimulation, (2) to protect monoamines from degradation by MAO, and (3) to protect neurons from the toxic effects of free radicals produced by the oxidation of cytoplasmic monoamines. In contrast with the plasma membrane transporters, a single type of vesicular monoamine

transporter is believed to mediate the uptake of monoamines into synaptic vesicles within the brain. Consistent with this, blockade of this vesicular monoamine transporter by the antihypertensive drug reserpine (Serpasil) has been found to deplete brain levels of serotonin, norepinephrine, and dopamine and to increase the risk of suicide and affective dysfunction.

Receptors

Ultimately, the effects of monoamines on CNS function and behavior depend on their interactions with receptor molecules. The binding of monoamines to these plasma membrane proteins initiates a series of intracellular events that modulate neuronal excitability. Unlike the transporters, multiple receptor subtypes exist for each monoamine neurotransmitter (Table 33-2).

Serotonin Receptors. The 5-hydroxytryptophan type 1 (5-HT_1) receptors comprise the largest serotonin receptor subfamily, with human subtypes designated 5-HT_{1A}, 5-HT_{1B}, 5-HT_{1D}, 5-HT_{1E}, and 5-HT_{1F}. All five 5-HT_1 receptor subtypes display intronless gene structures, high affinities for serotonin, and adenylate cyclase inhibition. The most intensively studied of these has been the 5-HT_{1A} receptor. This subtype is found on postsynaptic membranes of forebrain neurons, primarily in the hippocampus, cortex, and septum, and on serotonergic neurons, where it functions as an inhibitory somatodendritic autoreceptor. There is a significant interest in the 5-HT_{1A} receptor as a modulator of both anxiety and depression. The downregulation of 5-HT_{1A} autoreceptors by the chronic administration of serotonin reuptake inhibitors has been implicated in their antidepressant effects, and SSRIs may produce some behavioral effects via increases in hippocampal neurogenesis mediated by postsynaptic 5-HT_{1A} receptor activation. Also, partial 5-HT_{1A} receptor agonists such as buspirone display both anxiolytic and antidepressant properties.

Much recent attention has focused on the contributions of 5-$HT_{2A/C}$ receptors to the actions of atypical antipsychotic drugs such as clozapine, risperidone, and olanzapine. Analysis of the receptor-binding properties of these drugs has led to the hypothesis that 5-HT_{2A} receptor blockade correlates with the therapeutic effects of atypical antipsychotics. The 5-HT_{2A}

receptor also influences working memory, which is likely impaired in schizophrenia.

The 5-HT_{2C} receptor is expressed at high levels in many CNS regions, including the hippocampal formation, prefrontal cortex, amygdala, striatum, hypothalamus, and choroid plexus. Stimulation of 5-HT_{2C} receptors has been proposed to produce anxiogenic effects as well as anorectic effects, which may result from interactions with the hypothalamic melanocortin and leptin pathways. 5-HT_{2C} receptors may also play a role in the weight gain and development of type 2 diabetes mellitus associated with atypical antipsychotic treatment. Indeed, a line of mice lacking this receptor subtype exhibits an obesity syndrome associated with overeating and enhanced seizure susceptibility, suggesting that this receptor regulates neuronal network excitability. A variety of antidepressant and antipsychotic drugs antagonize 5-HT_{2C} receptors with high affinity.

Conversely, hallucinogens such as lysergic acid diethylamide (LSD) display agonist activity at 5-HT_2 (and other) serotonin receptor subtypes. 5-HT_{2C} receptor transcripts also undergo RNA editing, producing isoforms of the receptor with significantly altered basal versus serotonin-induced activity. Alterations in 5-HT_{2C} receptor mRNA editing have been found in the brains of suicide victims with a history of major depression, and SSRIs have been shown to alter these editing patterns.



Table 33-2
Monoamine Receptors: Overview

Transmitter	Subtype	Primary Effector	Prop Clinic Relev
Histamine	H_1	↑ PI Turnover	Antag as ant and infl age
	H_2	↑ AC ↓ AC	
	H_3	↓ AC	
	H_4		

			pro sec wei
			Antag to t ulc refl ble
			Antag pro trea dis obe der
			Possi ant ant infl age
Epinephrine/Norepinephrine	$\alpha_{1A,B,D}$	↑ PI Turnover	Antag in m
	$\alpha_{2A,B,C}$	↓ AC	of p dis
	β_1	↑ AC	Agoni and
	β_2	↑ AC	hyp
	β_3	↑ AC	Regu car fun ant ma anx
			Agoni bro

			Possi agc obe
Serotonergic	$5HT_{1A,1B,1D,1E,1F}$	↓ AC, ↑ GIRK currents	Partia (bu anx in h neu
	$5\text{-}HT_{2A}$, $5\text{-}HT_{2B}$, $5\text{-}HT_{2C}$	↑ PI Turnover	HT ant use ant age
	$5\text{-}HT_3$	Na^+ channel, cell membrane depolarization	(trip 2A ar ant effe
	$5\text{-}HT_4$	↑ AC	agc hall 2B ag
	$5\text{-}HT_5$, $5\text{-}HT_6$, $5\text{-}HT_7$	↑ AC	car valv 2C ag
			und dev and ant
			Agoni (on are
			Partia use

			(teg Uncle Uncle Antag hav ant pot
Dopaminergic	D_1-like family (D_1, D_5) D_2-like family (D_2, D_3, D_4)	↑ AC ↓ AC	D_1 ag in F dise D_2 an are ant (e.g hal D_3 ag in F dise res syn pra

From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:71.

Dopamine Receptors. In 1979, it was recognized that the actions of dopamine are mediated by more than one receptor subtype. Two dopamine receptors, termed D_1 and D_2, were distinguished based on differential binding affinities of a series of agonists and antagonists, distinct effector mechanisms, and distinct distribution patterns within the CNS. It was subsequently found that the therapeutic efficacy of antipsychotic drugs correlated strongly with their affinities for the D_2 receptor, implicating this subtype as an essential site of antipsychotic drug action. Recent molecular cloning studies have identified three additional dopamine receptor genes encoding the D_3, D_4, and

D_5 dopamine receptors. Based on their structure, pharmacology, and primary effector mechanisms, the D_3 and D_4 receptors are considered to be "D_2-like," and the D_5 receptor "D_1-like." The functional roles of the recently discovered subtypes remain to be definitively elucidated.

The D_1 receptor was initially distinguished from the D_2 subtype by its high affinity for the antagonist SCH 23390 and relatively low affinity for butyrophenones such as haloperidol (Haldol). Whereas D_1 receptor activation stimulates cyclic adenosine monophosphate (cAMP) formation, D_2 receptor stimulation produces the opposite effect.

Adrenergic Receptors. As for the α_1 receptors, the functions of α_2 receptor subtypes (designated α_{2A}, α_{2B}, and α_{2C}) have been challenging to determine due to a lack of selective agonists and antagonists; α_2 receptors display both presynaptic autoreceptor and postsynaptic actions, and all appear to inhibit the cAMP formation and to activate potassium channels with resultant membrane hyperpolarization. These receptors regulate neurotransmitter release from peripheral sympathetic nerve endings. The stimulation of central α_2 autoreceptors (likely the α_{2A} subtype) inhibits firing of the noradrenergic neurons of the LC, which are implicated in arousal states. This mechanism has been proposed to underlie the sedative effects of the α_2 receptor agonist clonidine (Catapres). Also, the stimulation of brainstem α_2 receptors has been proposed to reduce sympathetic and to augment parasympathetic nervous system activity. This action may relate to the utility of clonidine in lowering blood pressure and in suppressing the sympathetic hyperactivity associated with opiate withdrawal. Activation of α_2 receptors inhibits the activity of serotonin neurons of the dorsal raphe nucleus, whereas activation of local α_1 receptors stimulates the activity of these neurons, and this is thought to be a significant activating input to the serotonergic system.

Histamine Receptors. Histaminergic systems have been proposed to modulate arousal, wakefulness, feeding behavior, and neuroendocrine responsiveness. Four histaminergic receptor subtypes have been identified

and termed H_1, H_2, H_3, and H_4. The H_4 receptor was identified recently and is detected predominantly in the periphery, in regions such as the spleen, bone marrow, and leukocytes. The other three histamine receptors have a prominent expression in the CNS. H_1 receptors are expressed throughout the body, particularly in the smooth muscle of the gastrointestinal tract and bronchial walls, as well as on vascular endothelial cells. H_1 receptors are widely distributed within the CNS, with exceptionally high levels in the thalamus, cortex, and cerebellum. H_1 receptor activation is associated with G_q activation and stimulation of phosphoinositide turnover and tends to increase excitatory neuronal responses. These receptors are the targets of classical antihistaminergic agents used in the treatment of allergic rhinitis and conjunctivitis. The sedative effects of these compounds derive from their actions in the CNS and implicate histamine in the regulation of arousal and the sleep–wake cycle. Accordingly, a line of mutant mice lacking histamine displays deficits in waking and attention. Also, the sedation and weight gain produced by several antipsychotic and antidepressant drugs have been attributed to H_1 receptor antagonism. Conversely, H_1 receptor agonists stimulate arousal and suppress food intake in animal models.

Cholinergic Receptors. M1 receptors are the most abundantly expressed muscarinic receptors in the forebrain, including the cortex, hippocampus, and striatum. Pharmacologic evidence has suggested their involvement in memory and synaptic plasticity, and recent evaluation of mice lacking the M1 receptor gene revealed deficits in memory tasks believed to require interactions between the cortex and the hippocampus.

Nicotinic receptors have been implicated in cognitive function, especially working memory, attention, and processing speed. Cortical and hippocampal nicotinic acetylcholine receptors appear to be significantly decreased in Alzheimer disease, and nicotine administration improves attention deficits in some patients. The acetylcholinesterase inhibitor galantamine used in the treatment of Alzheimer disease also acts to modulate nicotinic receptor function positively. The α_7 nicotinic acetylcholine receptor subtype has been implicated as one of many possible susceptibility genes for schizophrenia, with lower levels of this receptor being associated with impaired sensory gating. Some rare forms of the familial epilepsy syndrome autosomal

dominant nocturnal frontal lobe epilepsy (ADNFLE) are associated with mutations in the α_4 or β_2 subunits of the nicotinic acetylcholine receptor. Finally, the reinforcing properties of tobacco use are proposed to involve the stimulation of nicotinic acetylcholine receptors located in mesolimbic dopaminergic reward pathways.

Amino Acid Neurotransmitters

For more than 50 years, biogenic amines have dominated thinking about the role of neurotransmitters in the pathophysiology of psychiatric disorders. However, over the last decade, evidence has accumulated from postmortem, brain imaging, and genetic studies that the amino acid neurotransmitters, in particular glutamic acid and GABA, play an important, if not central, role in the pathophysiology of a broad range of psychiatric disorders including schizophrenia, bipolar disorder, major depression, Alzheimer disease, and anxiety disorders.

Glutamic Acid. Glutamate mediates fast excitatory neurotransmission in the brain and is the transmitter for approximately 80 percent of brain synapses, particularly those associated with dendritic spines. The repolarization of neuronal membranes that are depolarized by glutamatergic neurotransmission may explain perhaps 80 percent of the brain's energy expenditure. The concentration of glutamate in the brain is 10 mM, the highest of all amino acids, of which approximately 20 percent represents the neurotransmitter pool of glutamate.

Two families of receptors mediate the postsynaptic effects of glutamate. The first are the glutamate-gated cation channels that are responsible for fast neurotransmission. The second type of glutamate receptor are the metabotropic glutamate receptors (mGluR), which are G-protein–coupled receptors like α-adrenergic receptors and dopamine receptors. The mGluRs primarily modulate glutamatergic neurotransmission.

MAJOR GLUTAMATERGIC PATHWAYS IN THE BRAIN. All primary sensory afferent systems appear to use glutamate as their neurotransmitter, including retinal ganglion cells, cochlear cells, trigeminal nerve, and spinal afferents. The thalamocortical projections that distribute afferent information broadly to

the cortex are glutamatergic. The pyramidal neurons of the corticolimbic regions, the primary source of intrinsic, associational, and efferent excitatory projections from the cortex, are glutamatergic. A temporal lobe circuit that figures significantly in the development of new memories is a series of four glutamatergic synapses: The perforant path innervates the hippocampal granule cells that innervate CA3 pyramidal cells that innervate CA1 pyramidal cells. The climbing fibers innervating the cerebellar cortex are glutamatergic, as well as the corticospinal tracks.

IONOTROPIC GLUTAMATE RECEPTORS. Three families of ionotropic glutamate receptors have been identified based on selective activation by conformationally restricted or synthetic analogs of glutamate. These include α-amino-3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA), kainic acid (KA), and *N*-methyl-D-aspartic acid (NMDA) receptors. Subsequent cloning revealed 16 mammalian genes that encode structurally related proteins, which represent subunits that assemble into functional receptors. Glutamate-gated ion channel receptors appear to be tetramers, and subunit composition affects both the pharmacologic and the biophysical features of the receptor.

METABOTROPIC GLUTAMATE RECEPTORS. These receptors are so designated because G proteins mediate their effects. All mGluRs are activated by glutamate, although their sensitivities vary remarkably. To date, eight mGluRs have been cloned. These genes encode for seven membrane-spanning proteins that are members of the superfamily of G-protein–coupled receptors.

THE ROLE OF ASTROCYTES. Specialized end-feet of the astrocyte surround glutamatergic synapses. The astrocyte expresses the two Na^+-dependent glutamate transporters that play the primary role in removing glutamate from the synapse, thereby terminating its action: EAAT1 and EAAT2 (*excitatory amino acid transporter*). The neuronal glutamate transporter, EAAT3, is expressed in upper motor neurons, whereas EAAT4 is expressed primarily in cerebellar Purkinje cells and EAAT5 in the retina. Mice homozygous for null mutations of either EAAT1 or EAAT2 exhibit elevated extracellular glutamate and excitotoxic neurodegeneration. Notably, several studies have

described the loss of EAAT2 protein and transport activity in the ventral horn in amyotrophic lateral sclerosis.

The astrocytes express AMPA receptors so that they can monitor synaptic glutamate release. GlyT1, which maintains subsaturating concentrations of glycine in the synapse, is expressed on the astrocyte plasma membrane. GlyT1 transports three Na^+ out for each molecule of glycine transported into the astrocyte. This stoichiometry results in a robust reversal of the direction of transport when glutamate released in the synapse activates the AMPA receptors on the astrocyte, thus depolarizing the astrocyte. Thus glycine release in the synapse by GlyT1 is coordinated with glutamatergic neurotransmission.

Similarly, activation of the astrocyte AMPA receptors causes GRIP to dissociate from the AMPA receptor and bind to serine racemase, activating it to synthesize D-serine. D-Serine levels are also determined by D-amino acid oxidase (DAAO) with low D-serine levels in the cerebellum and brainstem where DAAO expression is high, and high D-serine levels are found in corticolimbic brain regions where DAAO expression is relatively low. In contrast, the expression of GlyT1 is highest in the cerebellum and brainstem. This distribution suggests that D-serine is the primary modulator of the NMDA receptor in the forebrain, whereas glycine is more prominent in the brainstem and cerebellum.

PLASTICITY IN GLUTAMATERGIC NEUROTRANSMISSION. The extinction of conditioned fear is an active process mediated by the activation of NMDA receptors in the amygdala. Treatment of rats with NMDA receptor antagonists prevents the extinction of conditioned fear, whereas treatment with the glycine modulatory site partial agonist D-cycloserine facilitates the extinction of conditioned fear. (D-Cycloserine is an antibiotic used to treat tuberculosis that has 50 percent of the efficacy of glycine at the NMDA receptor.) Patients with acrophobia were administered either placebo or a single dose of D-cycloserine along with cognitive-behavioral therapy (CBT) to determine whether the phenomenon generalizes to humans. D-Cycloserine plus CBT resulted in a highly significant reduction in acrophobic symptoms that persisted for at least 3 months as compared to placebo plus CBT. Other placebo-controlled clinical trials support the notion that D-cycloserine is a

robust enhancer of CBT, suggesting that pharmacologically augmenting neural plasticity may be used to bolster psychological interventions.

Fragile X mental retardation protein (FMRP), which is deficient in individuals with fragile X syndrome, appears to be synthesized locally within the spine during times of NMDA receptor activation and also plays a role in transporting specific mRNAs to the spine for translation. Notably, mice in which the FMRP gene has been inactivated through a null mutation, as well as patients with fragile X syndrome have fewer dendritic spines, the preponderance of which have an immature morphology. Loss of FMRP exaggerates responses of mGluR5, which stimulates dendritic protein synthesis, and treatment with an mGluR5 antagonist reverses the fragile-X–like phenotype in mice with the FMRP gene inactivated.

EXCITOTOXICITY. In the early 1970s, it was shown that the systemic administration of large amounts of monosodium glutamate to immature animals resulted in the degeneration of neurons in brain regions where the BBB was deficient.

Excitotoxicity is also implicated as a proximate cause of neuronal degeneration in Alzheimer disease. Most evidence points to the toxic consequences of aggregates of β-amyloid, especially β-amyloid$_{1\text{-}42}$. The β-amyloid fibrils depolarize neurons, resulting in loss of the Mg^{2+} block and enhanced NMDA receptor sensitivity to glutamate. The fibrils also impair glutamate transport into astrocytes, thereby increasing the extracellular concentration of glutamate. β-Amyloid directly promotes oxidative stress through inflammation that further contributes to neuronal vulnerability to glutamate. Thus, several mechanisms contribute to neuronal vulnerability to NMDA-receptor–mediated excitotoxicity in Alzheimer disease. Memantine, a recently approved treatment for mild to moderate Alzheimer disease, is a weak noncompetitive inhibitor of NMDA receptors. It reduces the tonic sensitivity of NMDA receptors to excitotoxicity but does not interfere with "phasic" neurotransmission, thereby attenuating neuronal degeneration in Alzheimer disease.

Inhibitory Amino Acids: GABA. GABA is the major inhibitory neurotransmitter in the brain, where it is broadly distributed and occurs in millimolar concentrations. Given its physiologic effects and distributions, it is

not surprising that the dysfunction of GABAergic neurotransmission has been implicated in a broad range of neuropsychiatric disorders, including anxiety disorders, schizophrenia, alcohol dependence, and seizure disorders. Chemically, GABA differs from glutamic acid, the major excitatory neurotransmitter, only by the removal of a single carboxyl group from the latter.

GABA is synthesized from glutamic acid by glutamic acid decarboxylase (GAD), which catalyzes the removal of the α-carboxyl group. In the CNS, the expression of GAD appears to be restricted to GABAergic neurons, although in the periphery, it is expressed in pancreatic islet cells. Two distinct but related genes encode GAD. GAD65 is localized to nerve terminals, where it is responsible for synthesizing GABA that is concentrated in the synaptic vesicles. Consistent with its role in fast inhibitory neurotransmission, mice homozygous for a null mutation of GAD65 have an elevated risk for seizures. GAD67 appears to be the primary source for neuronal GABA because mice homozygous for a null mutation of GAD67 die at birth, have a cleft pallet, and exhibit significant reductions in brain GABA.

GABA is catabolized by GABA transaminase (GABA-T) to yield succinic semialdehyde. Transamination generally occurs when the parent compound, α-ketoglutarate, is present to receive the amino group, thereby regenerating glutamic acid. Succinic semialdehyde is oxidized by succinic semialdehyde dehydrogenase (SSADH) into succinic acid, which reenters the Krebs cycle. GABA-T is a cell surface, a membrane-bound enzyme expressed by neurons and glia, which is oriented toward the extracellular compartment. As would be anticipated, drugs that inhibit the catabolism of GABA have anticonvulsant properties. One of the mechanisms of action of valproic acid is the competitive inhibition of GABA-T. γ-Vinyl-GABA is a suicide substrate inhibitor of GABA-T that is used as an anticonvulsant in Europe (vigabatrin [Sabril]).

The synaptic action of GABA is also terminated by high-affinity transport back into the presynaptic terminal as well as into astrocytes. Four genetically distinct GABA high-affinity transporters have been identified with differing kinetic and pharmacologic characteristics. They all share homology with other neurotransmitter transporters with the characteristic of 12 membrane-spanning domains. The active transport is driven by the sodium gradient so that upon depolarization, transportation of GABA out of the neuron is

favored. GABA transported into astrocytes is catabolized by GABA-T and ultimately converted to glutamic acid and then to glutamine, which is transported back into the presynaptic terminal for GABA synthesis. Tiagabine (Gabitril) is a potent GABA transport inhibitor that is used to treat epilepsy. Preliminary results suggest that it also may be useful in panic disorder.

$GABA_A$ RECEPTORS. $GABA_A$ receptors are distributed throughout the brain. The $GABA_A$ complex, when activated, mediates an increase in membrane conductance with an equilibrium potential near the resting membrane potential of –70 mV (Fig. 33-20). In the mature neuron, this typically results with an influx of Cl^-, causing membrane hyperpolarization. Hyperpolarization is inhibitory because it increases the threshold for generating an action potential. In immature neurons, which have unusually high levels of intracellular Cl^-, activating the $GABA_A$ receptor can counterintuitively cause depolarization. For this reason, anticonvulsants that act by enhancing $GABA_A$ receptor activity may exacerbate seizures in the neonatal period.

Barbiturates such as phenobarbital and pentobarbital are noted for their sedative and anticonvulsant activities. Barbiturates allosterically increase the affinities of the binding sites for GABA and benzodiazepines at pharmacologically relevant concentrations. Barbiturates also affect channel dynamics by markedly increasing the extended open state and reducing the short open state, thereby increasing Cl^- inhibition. Chemically modified analogs of progesterone and corticosterone have been shown in behavioral studies to have sedative and anxiolytic effects through their interaction with the $GABA_A$ receptor complex. They share features with barbiturates, although they act at a distinctly different site. Thus, they allosterically enhance agonist ligand binding to the receptor and increase the duration of chloride channel opening. A variety of behavioral effects associated with steroid administration or fluctuation of endogenous steroids and sex-specific effects of GABAergic drugs have been linked to the action of endogenous neurosteroids.

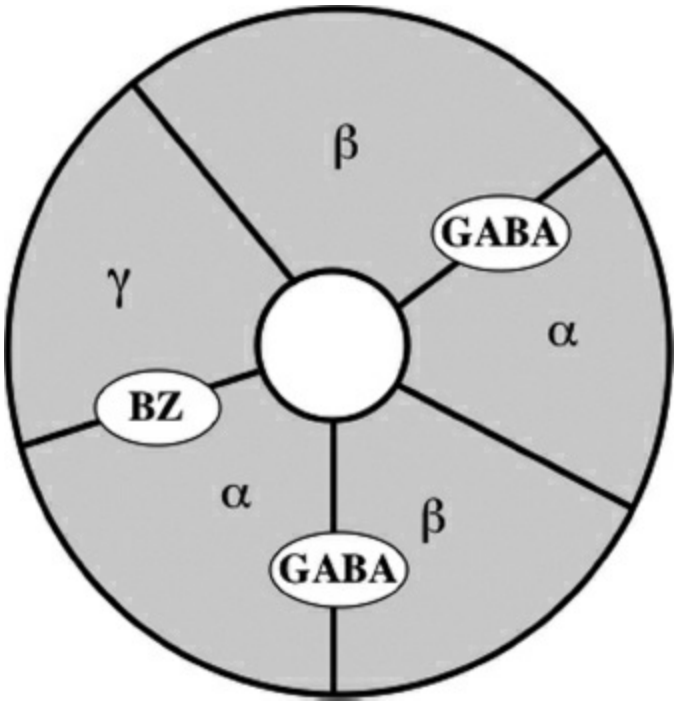


FIGURE 33-20
Schematic representation of the $GABA_A$ receptor. The receptor-channel complex is a heteropentamer. The GABA binding site is at the interface of the α and β subunits. The benzodiazepine binding site is at the interface between the γ and α subunits. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:81.)

Concerning $GABA_A$ receptor antagonists, picrotoxin, like the barbiturates, alters channel dynamics but in the opposite direction by reducing long open states and favoring the briefest open state. The proconvulsant pentylenetetrazol also acts by reducing chloride channel permeability. Penicillin, which at high concentrations is proconvulsant, binds to the positively charged residues in the channel, thereby occluding it. As a general class, anesthetics, including barbiturates, steroids, and volatile anesthetics, increase chloride conductance, thereby inhibiting neurotransmissions. Amino acids in the membrane-spanning domain of the GABA receptor subunits confer sensitivity to anesthetics. The precise mechanism whereby ethanol enhances $GABA_A$ receptor function remains unclear due to inconsistent results, suggesting that subunit composition may be necessary. However, recent studies suggest that ethanol increases the response of the tonic GABA-activated currents, which contain the δ subunit and exhibit remarkably high

affinity to GABA.

Recently, recombinant DNA strategies exploiting site-directed mutagenesis have permitted the identification of sites on the specific subunits that mediate the pharmacologic action of drugs such as the benzodiazepines. Removal of the binding ability for benzodiazepines has established that the α_1 subunit plays a significant role in the sedative and amnestic effects of benzodiazepines, whereas inactivating the benzodiazepine site on the α_2 subunit eliminates the anxiolytic effect of benzodiazepines

$GABA_B$ RECEPTORS. The $GABA_B$ receptors are distinguished pharmacologically from $GABA_A$ receptors by the fact that they are insensitive to the canonical $GABA_A$ receptor antagonist bicuculline and that they are potently activated by baclofen [β-(4-chlorophenyl)-GABA], which is inactive at $GABA_A$ receptors. They are members of the G-protein–coupled superfamily of receptors but are highly unusual, as they are made of a dimer of two seven-transmembrane–spanning subunits. $GABA_B$ receptors are widely distributed throughout the nervous system and are localized both presynaptically and postsynaptically. The postsynaptic $GABA_B$ receptors cause a long-lasting hyperpolarization by activating potassium channels. Presynaptically, they act as autoreceptors and heteroreceptors to inhibit neurotransmitter release.

GLYCINE AS A NEUROTRANSMITTER. Glycine is an inhibitory neurotransmitter primarily in the brainstem and spinal cord, although the expression of glycine receptor subunits in the thalamus, cortex, and hippocampus suggests a broader role. Glycine is a nonessential amino acid that is synthesized in the brain from L-serine by serine hydroxymethyltransferase. Glycine is concentrated within synaptic vesicles by H^+-dependent vesicular inhibitory amino acid transporter (VIAAT or VGAT), which also transports GABA. Termination of the synaptic action of glycine is through reuptake into the presynaptic terminal by the glycine transporter II (GlyT2), which is quite distinct from GlyT1 that is expressed in astrocytes and modulates NMDA receptor function.

The inhibitory effects of glycine are mediated by a ligand-gated chloride

channel, which can also respond to β-alanine, taurine, L-alanine, L-serine, and proline, but not to GABA. The canonical antagonist for the glycine receptor is the plant alkaloid strychnine. The receptor was first identified through the specific binding of [^{3}H]strychnine. [^{3}H]Glycine binds to two sites: One that is displaceable by strychnine and represents the glycine A receptor and a second that is insensitive to strychnine and is designated the glycine B receptor, representing the glycine modulatory site on the NMDA receptor.

Neuropsychiatric Implications of Amino Acid Transmitters

SCHIZOPHRENIA. Evidence accumulating from postmortem, pharmacologic, and genetic studies is shifting the focus of the pathophysiology of schizophrenia from dopamine to glutamate and GABA. Indeed, after the use of D_2 receptor antagonists as the sole treatment of schizophrenia for the last 50 years, more than two-thirds of the treated patients remain substantially disabled. Early postmortem studies indicated a reduction in the activity of GAD in the cortex in patients with schizophrenia as compared to suitable controls. With the advent of immunocytochemistry and gene expression techniques, it has been possible to define the GABAergic deficit in schizophrenia more precisely. It appears that the parvalbumin-positive GABAergic interneurons in the intermediate layers of the cortex bear the brunt of the pathology, which includes reduced expression of GAD67, parvalbumin, and the GABA transporter (GAT). The finding that $GABA_A$ receptors are upregulated, as measured by autoradiography or with antibodies, supports the theory that these changes reflect the hypofunction of the presynaptic GABAergic neurons. These particular GABAergic interneurons, which include the chandelier cells, play an essential role in negative feedback inhibition to the pyramidal cells in the cortex. Despite this highly reproducible neuropathology, genes related to GABAergic function have not figured prominently in genome-wide searches, suggesting that GABAergic deficits may be a downstream consequence of some more proximal genetic defects.

The theory that hypofunction of NMDA receptors is an etiologic factor in schizophrenia initially arose from the observation that phencyclidine (PCP) and related dissociative anesthetics that block NMDA receptors produce a syndrome that can be indistinguishable from schizophrenia (Fig. 33-21).

Dissociative anesthetics are so named because they prevent the acquisition of new memories while the patient is conscious. In fact, under laboratory conditions, low-dose infusion of ketamine can produce positive symptoms, negative symptoms, and specific cognitive deficits associated with schizophrenia in clear consciousness. Subsequent studies indicated that low-dose ketamine could also cause enhanced amphetamine-induced subcortical dopamine release as is observed in schizophrenia, as well as abnormal cortical event-related potentials (ERPs) and disruption of prepulse inhibition in experimental animals.

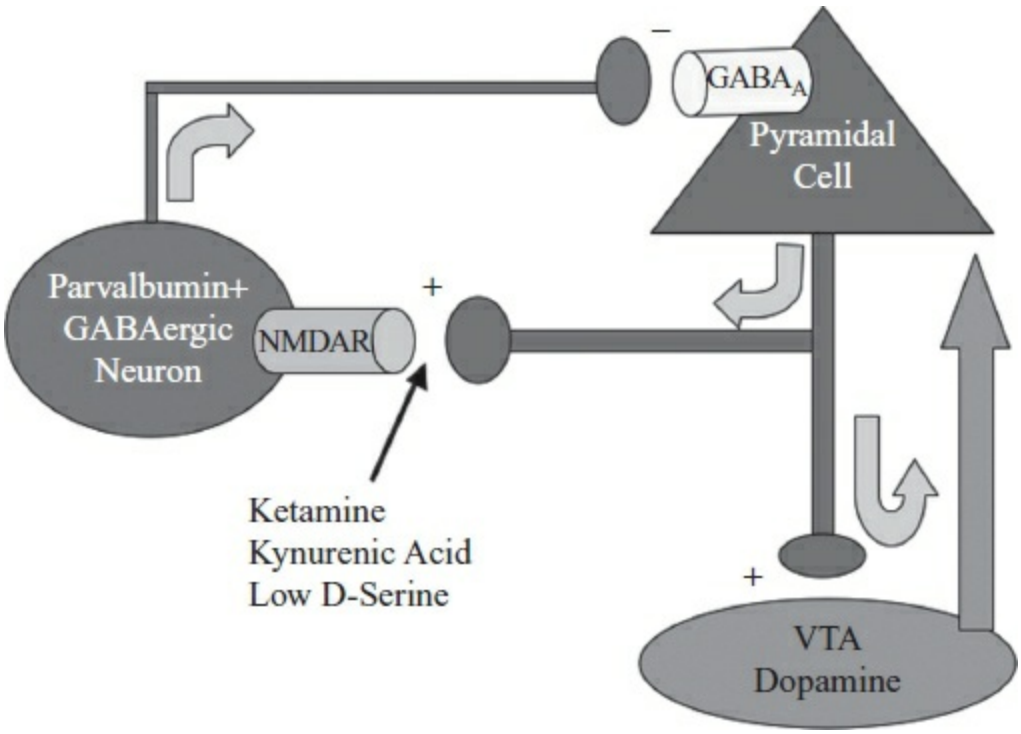


FIGURE 33-21
Pathologic circuit in schizophrenia. The NMDA receptors on the rapidly firing parvalbumin (PV) expressing GABAergic interneurons in the intermediate levels of the cortex are disproportionately sensitive to antagonists or loss of the coagonist, d-serine. NMDA receptor hypofunction causes reduced expression of PV, GAD67, and the GABA transporter and upregulation of $GABA_A$ receptors on pyramidal neurons. Disinhibition of the pyramidal neurons causes cognitive dysfunction and negative symptoms and drives excessive subcortical dopamine release resulting in psychosis. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:83.)

Several putative risk genes for schizophrenia are closely associated with

the NMDA receptor function. DAAO, which encodes a protein that activates D-amino acid oxidase, has been repeatedly linked to the risk of schizophrenia. D-Amino acid oxidase itself has been associated with increased risk. Recently an allelic variant of serine racemase in the promoter region has also been associated with the risk for schizophrenia. Each of these gene variants could reduce the availability of D-serine in the cortex, thereby impairing the NMDA receptor function. Notably, CSF and blood levels of D-serine are significantly reduced in patients with schizophrenia. Neuregulin 1 appears to be a convincing risk gene and interacts directly with NMDA receptors. Dysbindin, another risk gene, is expressed in glutamatergic terminals. mGluR3, which downregulates glutamate release, has also been associated with schizophrenia.

Recent findings have provided a link between the GABAergic neuropathology and NMDA receptor hypofunction. Chronic treatment of rats with NMDA receptor antagonists causes downregulation of GAD67, parvalbumin, and GAT. The sensitive subpopulation of GABAergic neurons are interneurons that fire rapidly and provide the perisomatic innervation of the pyramidal cells. Their NMDA receptors appear to be much more sensitive to antagonists than those less active GABAergic neurons and pyramidal cells. The subtly reduced GABAergic inhibition results in a disinhibition of the glutamatergic pyramidal output. This degradation of the inhibitory feedback could account for the cognitive deficits and negative symptoms in schizophrenia, and the disinhibited output also results in elevated subcortical dopamine release and psychosis. Thus psychosis would be considered a downstream event resulting from a disruption in critical glutamatergic-GABAergic synaptic function in the cerebral cortex.

ANXIETY AND DEPRESSION. GABAergic dysfunction has been associated with anxiety disorders, especially panic disorder, as well as with major depressive disorder. Clinically, there is considerable comorbidity between anxiety and affective disorders. Decreased levels of the $GABA_A$ receptor modulators, the three α-reduced neuroactive steroids, have been found both in plasma and in CSF in major depressive disorder. Effective treatment with SSRIs increases the neurosteroid levels. In contrast, in patients with panic disorder, the plasma neurosteroid levels were significantly elevated, perhaps as a compensatory mechanism. Magnetic resonance spectroscopy (MRS) has disclosed significant reductions in GABA levels in the anterior cingulate and the basal ganglia of medicated patients with panic disorder. PET scanning reveals a highly selective reduction in benzodiazepine receptor sites bilaterally in the insular cortex in panic disorder. A genome-wide screen has shown significant linkage at 15q in a region containing $GABA_A$ receptor subunit genes and panic disorder. MRS reveals significant reductions in both GABA and glutamate/glutamine (Glx) in the prefrontal cortex in major depressive disorder. Postmortem studies indicate upregulation of the $GABA_A$ receptor α_1 and β_3 subunits in the cerebral cortices of depressed patients who committed suicide, consistent with a reduction in GABAergic neurotransmission. The reduced levels of GABA in the occipital cortex in episodes of major depressive disorder normalized with effective treatment with SSRI or with ECT.

Glutamatergic dysfunction has also been implicated in depression. NMDA receptor antagonists have antidepressant effects in several animal models of depression, including forced swim, tail suspension, and learned helplessness. A single injection of ketamine protects the induction of behavioral despair in rats for up to 10 days. Chronic treatment with antidepressants alters the expression of NMDA receptor subunits and decreases glycine receptor B binding. Two placebo-controlled clinical trials have shown that a single dose of ketamine can produce a rapid, substantial, and persistent reduction in symptoms in patients with major depressive disorder.

ALCOHOLISM. Ethanol at concentrations associated with intoxication has a dual action of enhancing GABAergic receptor function and attenuating the

NMDA receptor function. The GABA receptor effects may be associated with the anxiolytic effects of ethanol. Persistent abuse and dependency on ethanol result in a downregulation of $GABA_A$ receptors and an upregulation of NMDA receptors such that acute discontinuation of ethanol results in a hyperexcitable state characterized by delirium tremens. Furthermore, supersensitive NMDA receptors in the context of thiamine deficiency may contribute to the excitotoxic neuron degeneration of Wernicke–Korsakoff syndrome.

Acamprosate is a derivative of homotaurine that was developed as an agent to reduce alcohol consumption, craving, and relapse in alcoholic patients, for which it exhibits moderate efficacy in clinical trials. Because taurine resembles GABA, it was thought that acamprosate acted via $GABA_A$ receptors, but electrophysiologic studies found little evidence to support this hypothesis. Subsequent studies demonstrated that it inhibited NMDA receptor responses in cortical slices and recombinant NMDA receptors. The precise mechanism whereby acamprosate alters NMDA receptor function, however, remains unclear.

Fetal alcohol syndrome is the most common preventable cause of mental retardation. Convincing evidence has been developed that the microencephaly associated with fetal alcohol exposure results from inhibition of NMDA receptor function, resulting in widespread neuronal apoptosis in the immature cortex. NMDA receptor activation is essential for immature neuronal survival and differentiation.

Neuropeptides

Neuropeptides represent the most diverse class of signaling molecules in the CNS. Initially discovered for their role in the hypothalamic regulation of pituitary hormone secretion, the complex role of peptides in brain function has emerged over the last 30 years. Many neuropeptides and their receptors are widely distributed within the CNS, where they have an extraordinary array of direct or neuromodulatory effects, ranging from modulating neurotransmitter release and neuronal firing patterns to the regulation of emotionality and complex behaviors. More than 100 unique biologically active neuropeptides have been identified in the brain, a subset of which is presented in Table 33-3. Adding to the complexity of neuropeptide systems

in the CNS, the actions of many peptides are mediated via multiple receptor subtypes localized in different brain regions. The discovery of new peptides and receptor subtypes has outpaced our understanding of the roles of these peptides in normal or aberrant CNS function. Pharmacologic, molecular, and genetic approaches are now leading the way in our understanding of the contribution of neuropeptide systems in psychiatric disorders.

Neuropeptides have been implicated in the regulation of a variety of behavioral and physiologic processes, including thermoregulation, food and water consumption, sex, sleep, locomotion, learning and memory, responses to stress and pain, emotion, and social cognition. Involvement in such behavioral processes suggests that neuropeptidergic systems may contribute to the symptoms and behaviors exhibited in major psychiatric illnesses such as psychoses, mood disorders, dementia, and ASDs.



Table 33-3
Selected Neuropeptide Transmitters

Adrenocorticotropic hormone (ACTH)
Angiotensin
Atrial natriuretic peptide
Bombesin
Calcitonin
Calcitonin gene-related peptide (CGRP)
Cocaine and amphetamine regulated transcript (CART)
Cholecystokinin (CCK)
Corticotropin-releasing factor (CRF)
Dynorphin
β-Endorphin
Leu-enkephalin
Met-enkephalin
Galanin
Gastrin
Gonadotropin-releasing hormone (GnRH)
Growth hormone

Growth hormone-releasing hormone (GHRH; GRF)
Insulin
Motilin
Neuropeptide S
Neuropeptide Y (NPY)
Neurotensin
Neuromedin N
Orphanin FQ/Nociceptin
Orexin
Oxytocin
Pancreatic polypeptide
Prolactin
Secretin
Somatostatin (SS; SRIF)
Substance K
Substance P
Thyrotropin-releasing hormone (TRH)
Urocortin (1, 2, and 3)
Vasoactive intestinal polypeptide (VIP)
Vasopressin (AVP; ADH)

From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:84.

Investigating Neuropeptide Function. The roles of neuropeptides in CNS function and behavior have been examined using a multitude of experimental techniques. The levels of analysis include the following: Molecular structure and biosynthesis of the peptide and its receptor(s), the neuroanatomical localization of the peptide and its receptor(s), the regulation of the expression and release of the peptide, and the behavioral effects of the peptide. Most information on neuropeptide biology is derived from laboratory animal studies; however, there is a growing database on the localization, activity, and potential psychiatric relevance of several neuropeptide systems in humans.

Most neuropeptide structures have been identified based on the chemical

analysis of purified biologically active peptides, leading ultimately to the cloning and characterization of the genes encoding them. Characterization of the gene structure of peptides and their receptors has provided insight into the molecular regulation of these systems, and their chromosomal localization is useful in genetic studies examining the potential roles of these genes in psychiatric disorders. Structural characterization permits the production of immunologic and molecular probes that are useful in determining peptide distribution and regulation in the brain. Quantitative radioimmunoassays on microdissected brain regions or immunocytochemistry on brain sections are typically used to localize the distribution of peptide within the brain. Both techniques use specific antibodies generated against the neuropeptide to detect the presence of the peptide. Immunocytochemistry allows researchers to visualize the precise cellular localization of peptide-synthesizing cells as well as their projections throughout the brain, although the technique is generally not quantitative. With molecular probes homologous to the mRNA encoding the peptides or receptor, in situ hybridization can be used to localize and quantify gene expression in brain sections. This technique is a powerful way to examine the molecular regulation of neuropeptide synthesis with a precise neuroanatomical resolution, which is impossible for other classes of nonpeptide neurotransmitters that are not derived directly from the translation of mRNAs, such as dopamine, serotonin, and norepinephrine.

Generally, the behavioral effects of neuropeptides are initially investigated by infusions of the peptide directly into the brain. Unlike many nonpeptide neurotransmitters, most neuropeptides do not penetrate the BBB in amounts sufficient enough to produce CNS effects. Furthermore, serum and tissue enzymes tend to degrade the peptides before they reach their target sites. The degradation is usually the result of the cleavage of specific amino acid sequences targeted by a specific peptidase designed for that purpose. Thus, intracerebroventricular (ICV) or site-specific infusions of the peptide in animal models are generally required to probe for the behavioral effects of peptides. However, there are examples of the delivery of neuropeptides via intranasal infusions in human subjects, which may help the peptide access the brain.

One of the most significant impediments for exploring the roles and potential therapeutic values of neuropeptides is the inability of the peptides or their agonists/antagonists to penetrate the BBB. Thus the behavioral effects

of most peptides in humans are mostly uninvestigated, except for a few studies utilizing intranasal delivery. However, in some instances, small-molecule, nonpeptide agonists/antagonists have been developed that can be administered peripherally and permeate the BBB in sufficient quantities to affect receptor activation.

The use of pretreatment and posttreatment CSF samples or of samples obtained during the active disease state versus when the patient is in remission addresses some of the limitations in study design. For such progressive diseases as schizophrenia or Alzheimer disease, serial CSF samples may be a valuable indicator of disease progression or response to treatment. Even with these constraints, significant progress has been made in describing the effects of various psychiatric disease states on neuropeptide systems in the CNS.

Biosynthesis. Unlike other neurotransmitters, neuropeptide biosynthesis involves mRNA transcription from a specific gene, translation of an mRNA-encoded polypeptide preprohormone, and posttranslational processing involving proteolytic cleavage of the preprohormone. Over the last 25 years, the gene structures and biosynthetic pathways of many neuropeptides have been elucidated. The gene structure of selected neuropeptides is illustrated in Figure 33-22. Neuropeptide genes are generally composed of multiple exons that encode a protein preprohormone. The N-terminus of the preprohormone contains a signal peptide sequence, which guides the growing polypeptide to the rough endoplasmic reticulum (RER) membrane. The single preprohormone molecule often contains the sequences of multiple peptides that are subsequently separated by proteolytic cleavage by specific enzymes. For example, translation of the gene encoding NT yields a preprohormone, which upon enzymatic cleavage produces both NT and neuromedin N.

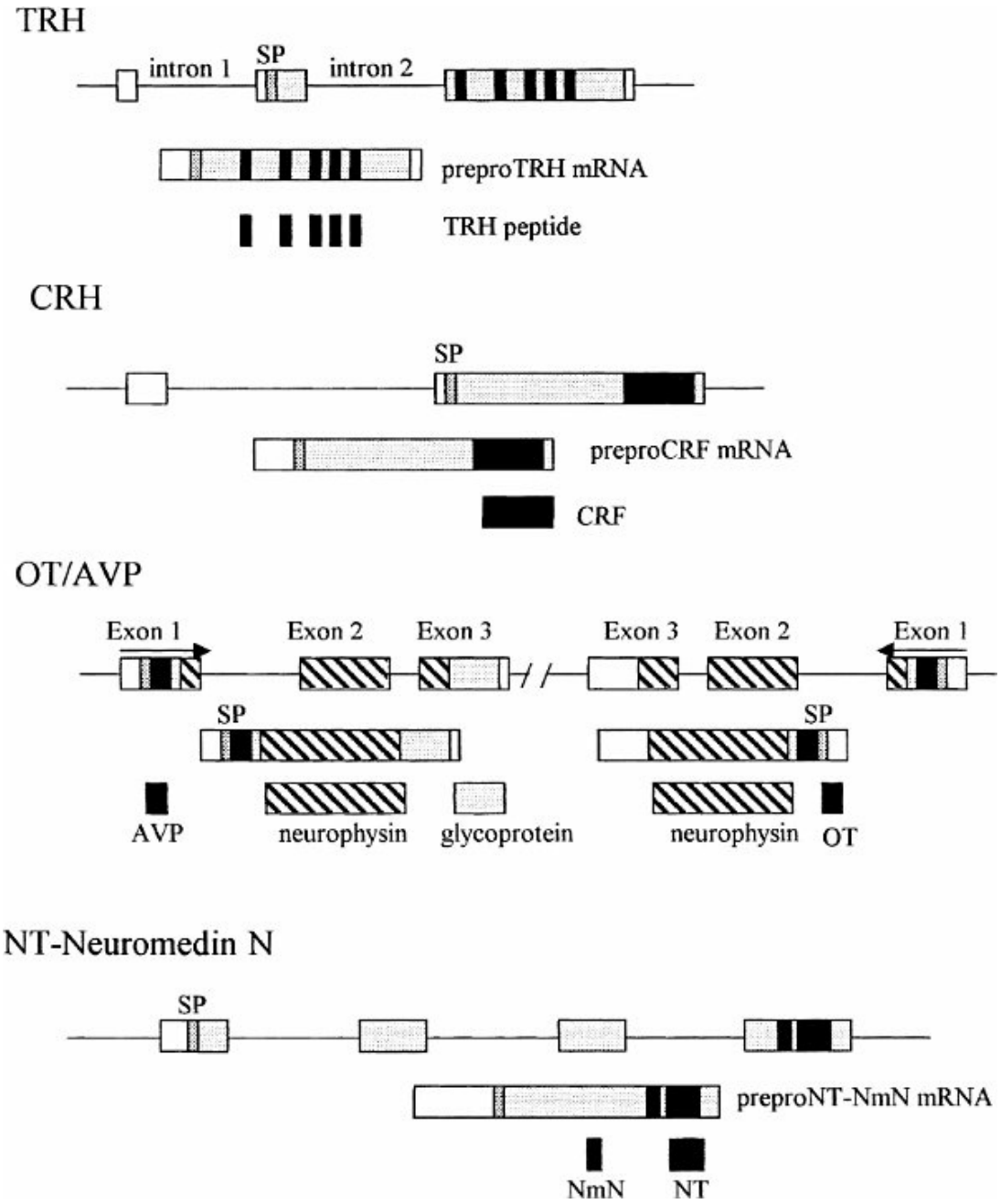


FIGURE 33-22

Schematics illustrating the gene structure, preprohormone messenger RNA (mRNA), and processed neuropeptides of thyrotropin-releasing hormone (TRH), corticotropin-releasing factor (CRF), oxytocin (OT), arginine vasopressin (AVP), and neurotensin (NT). *Boxed regions* indicate the locations of the exons in the respective genes. *Shaded or hatched regions* indicate coding regions. Each preprohormone begins with a signal peptide (SP) sequence. *Black boxes* indicate the locations of the sequences encoding the neuropeptide. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:87.)

Distribution and Regulation. Although many neuropeptides were initially isolated from the pituitary and peripheral tissues, the majority of neuropeptides were subsequently found to be widely distributed throughout the brain. Those peptides involved in regulating pituitary secretion are concentrated in the hypothalamus. Hypothalamic releasing and inhibiting factors are produced in neurosecretory neurons adjacent to the third ventricle that send projections to the median eminence where they contact and release peptide into the hypothalamohypophysial portal circulatory system. Peptides produced in these neurons are often subject to regulation by the peripheral hormones that they regulate. For example, thyrotropin-releasing hormone (TRH) regulates the secretion of thyroid hormones, and thyroid hormones negatively feedback on TRH gene expression. However, neuropeptide-expressing neurons and their projections are found in many other brain regions, including limbic structures, midbrain, hindbrain, and spinal cord.

Neuropeptide Signaling. Neuropeptides may act as neurotransmitters, neuromodulators, or neurohormones. Neurotransmitters are typically released from axonal terminals into a synapse where they change the postsynaptic membrane potential, either depolarizing or hyperpolarizing the cell. For classical neurotransmitters, this often involves direct modulation of voltage-gated ion channels. In contrast, neuromodulators and neurohormones do not directly affect the firing of the target cell itself but may alter the response of the cell to other neurotransmitters through the modulation of second messenger pathways. Neuropeptide release is not restricted to synapses or axon terminals but may occur throughout the axon or even from dendrites.

Specific neuropeptide receptors mediate the cellular signaling of neuropeptides. Thus understanding neuropeptide receptor function is essential for understanding neuropeptide biology. Neuropeptide receptors have undergone the same process of discovery and characterization that receptors for other neurotransmitters have enjoyed. Most neuropeptide receptors are G-protein–coupled, seven-transmembrane domain receptors belonging to the same family of proteins as the monoamine receptors.

Molecular technology has made it possible to clone and characterize neuropeptide receptor genes and complementary DNAs (cDNAs). This process is most often accomplished in one of three ways. First, the neuropeptide receptor protein is biochemically purified and partially

sequenced, which allows the development of oligonucleotide probes that can be used to isolate the cDNA encoding the protein from a cDNA library. A second approach involves producing expression libraries in which cells containing the receptor cDNA can be isolated based on their ability to bind to a radiolabeled peptide ligand. Finally, many neuropeptide receptors are now isolated based on their sequence homology with other known peptide receptors. Once the cDNA of the receptor has been isolated, it can be used to produce purified receptor protein for structural and functional studies. By mutation of specific amino acids in the receptor structure and determination of relative binding affinities of peptides with various amino acid substitutions, it is possible to elucidate the nature of the ligand–receptor interaction. This information facilitates the development of drugs that specifically modulate receptor function, including nonpeptide drugs, leading to the ability to manipulate peptide systems in ways that are currently enjoyed by the more classic neurotransmitters. The availability of cDNAs encoding the receptor also permits the neuroanatomical mapping of the receptor-producing cells in the brain, which is critical for understanding the neural circuits modulated by the peptide. Finally, with the cloned receptor in hand, it is possible to use transgenic techniques, such as targeted gene overexpression or gene knockouts, to elucidate the functions of these receptors further. siRNA techniques now allow the targeted synthesis disruption of specific receptor populations, allowing researchers to examine the roles of these receptor populations on physiology and behavior.

The following three factors determine the biologic roles of a neuropeptide hormone: (1) the temporal-anatomical release of the peptide, (2) functional coupling of the neuropeptide receptor to intracellular signaling pathways, and (3) the cell type and circuits in which the receptor is expressed. Genetic studies have demonstrated that regulatory sequences flanking the receptor coding region determine the expression pattern of the receptor and thus the physiologic and behavioral response to the neuropeptide.

Peptidases. Unlike monoamine neurotransmitters, peptides are not actively taken up by presynaptic nerve terminals. Instead, released peptides are degraded into smaller fragments, and eventually into single amino acids by specific enzymes termed peptidases. The enzymes may be found bound to presynaptic or postsynaptic neural membranes or in solution in the cytoplasm

and extracellular fluid, and they are distributed widely in peripheral organs and serum as well as in the CNS. As a result, neuropeptides generally have half-lives on the order of minutes once released.

Specific Neuropeptides as Prototypes of Neuropeptide Biology

THYROTROPIN-RELEASING HORMONE. In 1969, TRH, a pyroglutamylhistidylprolinamide tripeptide (Table 33-4), became the first of the hypothalamic releasing hormones to be isolated and characterized. The discovery of the structure of this hormone led to the conclusive demonstration that peptide hormones secreted from the hypothalamus regulate the secretion of hormones from the anterior pituitary. The gene for TRH in humans resides on chromosome 3q13.3–q21. In the rat, it consists of three exons (coding regions) separated by two introns (noncoding sequences) (see Fig. 33-22). The first exon contains the 5′ untranslated region of the mRNA encoding the TRH preprohormone, the second exon contains the signal peptide (SP) sequence and much of the remaining N-terminal end of the precursor peptide, and the third contains the remainder of the sequence, including five copies of the TRH precursor sequence, the C-terminal region, and the 3′ untranslated region. The 5′ flanking of the gene, or promoter, contains sequences homologous to the glucocorticoid receptor and the DNA-binding sites for thyroid hormone, providing a mechanism for the regulation of this gene by cortisol and negative feedback by thyroid hormone. Enzymatic processing of TRH begins with excision of the progenitor peptides by carboxypeptidases, amidation of the C-terminal proline, and cyclization of the N-terminal glutamine to yield five TRH molecules per prohormone molecule. TRH is widely distributed in the CNS with TRH immunoreactive neurons being located in the olfactory bulbs, entorhinal cortices, hippocampus, extended amygdala, hypothalamus, and midbrain structures. As is the case for most neuropeptides, the TRH receptor is also a member of the seven-transmembrane domain, G-protein–coupled receptor family.

Hypothalamic TRH neurons project nerve terminals to the median eminence; there, they release TRH into the hypothalamohypophyseal portal system where it is transported to the adenohypophysis, causing the release of thyroid-stimulating hormone (TSH) into the systemic circulation. TSH subsequently stimulates the release of the thyroid hormones triiodothyronine

(T_3) and thyroxine (T_4) from the thyroid gland. TRH neurons in the paraventricular nucleus (PVN) contain thyroid hormone receptors and respond to increases in thyroid hormone secretion with a decrease in TRH gene expression and synthesis. This negative feedback of thyroid hormones on the TRH-synthesizing neurons was first demonstrated by a decrease in TRH content in the median eminence, but not in the PVN of the hypothalamus after thyroidectomy. This effect can be reversed with exogenous thyroid hormone treatment. The treatment of normal rats with exogenous thyroid hormone decreases TRH concentration in the PVN and the posterior nucleus of the hypothalamus. With a probe against the TRH preprohormone mRNA, in situ hybridization studies have demonstrated that TRH mRNA is increased in the PVN 14 days after thyroidectomy. The ability of thyroid hormones to regulate TRH mRNA can be superseded by other stimuli that activate the hypothalamic–pituitary–thyroid (HPT) axis. In that regard, repeated exposure to cold (which releases TRH from the median eminence) induces increases in the levels of TRH mRNA in the PVN despite concomitantly elevated concentrations of thyroid hormones.



Table 33-4
Selected Neuropeptide Structures

Name	Amino Acid Sequence
Thyrotropin-releasing hormone (TRH)	pE-H-P-NH_2
Corticotropin-releasing factor (CRF)	S-E-E-P-P-I-S-L-D-L-T-F-H-L-L-R-E-V-L-E-M-A-R-A-E-Q-L-A-Q-Q-A-H-S-N-R-K-L-M-E-I-I-NH_2
Arginine vasopressin (AVP)	C-Y-I-Q-N-C-P-L-G-NH_2
Oxytocin (OT)	C-Y-F-Q-N-C-P-R-G-NH_2
Neurotensin (NT)	pE-L-Y-E-N-K-P-R-R-P-Y-I-L-OH

Note the cyclized glutamines at the N-termini of TRH and NT indicated by pE-, the cysteine–cysteine disulfide bonds of AVP and OT, and the amidated C-termini of TRH, CRF, AVP, and OT.

From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:85.

Further evidence of the different levels of communication of the HPT axis is seen in the ability of TRH to regulate the production of mRNA for the pituitary TRH receptor. It is also seen for TRH concentrations to regulate the mRNA coding for both the α and β subunits of the thyrotropin (TSH) molecule. Also, TRH-containing synaptic boutons have been observed in contact with TRH-containing cell bodies in the medial and periventricular subdivisions of the paraventricular nucleus, thus providing anatomical evidence for ultrashort feedback regulation of TRH release. Negative feedback by thyroid hormones may be limited to the hypothalamic TRH neurons because negative feedback on TRH synthesis by thyroid hormones has not been found in extrahypothalamic TRH neurons.

The early availability of adequate tools to assess HPT axis function (i.e., radioimmunoassays and synthetic peptides), coupled with observations that primary hypothyroidism is associated with depressive symptomatology, ensured extensive investigation of the involvement of this axis in affective disorders. Early studies established the hypothalamic and extrahypothalamic distribution of TRH. This extrahypothalamic presence of TRH quickly led to speculation that TRH might function as a neurotransmitter or neuromodulator. Indeed, a large body of evidence supports such a role for TRH. Within the CNS, TRH is known to modulate several different neurotransmitters, including dopamine, serotonin, acetylcholine, and opioids. TRH has been shown to arouse hibernating animals and counteracts the behavioral response and hypothermia produced by a variety of CNS depressants, including barbiturates and ethanol.

The use of TRH as a provocative agent for the assessment of the HPT axis function evolved rapidly after its isolation and synthesis. Clinical use of a standardized TRH stimulation test, which measures negative feedback responses, revealed blunting of the TSH response in approximately 25 percent of euthyroid patients with major depression. These data have been widely confirmed. The observed TSH blunting in depressed patients does not

appear to be the result of excessive negative feedback due to hyperthyroidism because thyroid measures such as basal plasma concentrations of TSH and thyroid hormones are generally in the normal range in these patients. TSH blunting may be a reflection of pituitary TRH receptor downregulation due to median eminence hypersecretion of endogenous TRH. Indeed, the observation that CSF TRH concentrations are elevated in depressed patients as compared to those of controls supports the hypothesis of TRH hypersecretion but does not elucidate the regional CNS origin of this tripeptide. TRH mRNA expression in the PVN of the hypothalamus is decreased in patients with major depression. However, it is not clear whether the altered HPT axis represents a causal mechanism underlying the symptoms of depression or only a secondary effect of depression-associated alterations in other neural systems.

CORTICOTROPIN-RELEASING FACTOR (CRF) AND UROCORTINS. There is convincing evidence to support the hypothesis that CRF and the urocortins play an intricate role in integrating the endocrine, autonomic, immunologic, and behavioral responses of an organism to stress.

Although it was initially isolated because of its functions in regulating the HPA axis, CRF is widely distributed throughout the brain. The PVN of the hypothalamus is the primary site of CRF-containing cell bodies that influence anterior pituitary hormone secretion. These neurons originate in the parvocellular region of the PVN and send axon terminals to the median eminence, where CRF is released into the portal system in response to stressful stimuli. A small group of PVN neurons also projects to the brainstem and spinal cord, where they regulate autonomic aspects of the stress response. CRF-containing neurons are also found in other hypothalamic nuclei, the neocortex, the extended amygdala, brainstem, and spinal cord. Central CRF infusion into laboratory animals produces physiologic changes and behavioral effects similar to those observed following stress, including increased locomotor activity, increased responsiveness to an acoustic startle, and decreased exploratory behavior in an open field.

The physiologic and behavioral roles of the urocortins are less understood, but several studies suggest that urocortins 2 and 3 are anxiolytic and may dampen the stress response. This research has led to the hypothesis that CRF

and the urocortins act in opposition, but this is likely an oversimplification. Urocortin 1 is primarily synthesized in the Edinger–Westphal nucleus, lateral olivary nucleus, and supraoptic hypothalamic nucleus. Urocortin 2 is synthesized primarily in the hypothalamus, while urocortin three-cell bodies are found more broadly in the extended amygdala, perifornical area, and preoptic area.

Hyperactivity of the HPA axis in major depression remains one of the most consistent findings in biologic psychiatry. The reported HPA axis alterations in major depression include hypercortisolemia, resistance to dexamethasone suppression of cortisol secretion (a measure of negative feedback), blunted adrenocorticotropic hormone (ACTH) responses to intravenous CRF challenge, increased cortisol responses in the combined dexamethasone/CRF test, and elevated CSF CRF concentrations. The exact pathologic mechanism(s) underlying HPA axis dysregulation in major depression and other affective disorders remains to be elucidated.

Mechanistically, two hypotheses have been advanced to account for the ACTH blunting following exogenous CRF administration. The first hypothesis suggests that pituitary CRF receptor downregulation occurs as a result of hypothalamic CRF hypersecretion. The second hypothesis postulates the altered sensitivity of the pituitary to glucocorticoid negative feedback. Substantial support has accumulated favoring the first hypothesis. However, neuroendocrine studies represent a secondary measure of CNS activity; the pituitary ACTH responses principally reflect the activity of hypothalamic CRF rather than that of the corticolimbic CRF circuits. The latter of the two is more likely to be involved in the pathophysiology of depression.

Of particular interest is the demonstration that the elevated CSF CRF concentrations in drug-free depressed patients are significantly decreased after successful treatment with ECT, indicating that CSF CRF concentrations, like hypercortisolemia, represent a state rather than a trait marker. Other recent studies have confirmed this normalization of CSF CRF concentrations following successful treatment with fluoxetine. One group demonstrated a significant reduction of elevated CSF CRF concentrations in 15 female patients with major depression who remained depression-free for at least 6 months following antidepressant treatment, as compared to little significant treatment effect on CSF CRF concentrations in 9 patients who relapsed in these 6 months. This finding suggests that elevated or increasing CSF CRF

concentrations during antidepressant treatment may be the harbinger of an inadequate response in major depression despite early symptomatic improvement. Of interest, treatment of normal subjects with desipramine or, as noted above, of individuals with depression with fluoxetine is associated with a reduction in CSF CRF concentrations.

If CRF hypersecretion is a factor in the pathophysiology of depression, then reducing or interfering with CRF neurotransmission might be an effective strategy to alleviate depressive symptoms. Over the last several years, several pharmaceutical companies have committed considerable effort to the development of small-molecule CRF_1 receptor antagonists that can effectively penetrate the BBB. Several compounds have been produced with reportedly promising characteristics.

OXYTOCIN (OT) AND VASOPRESSIN (AVP). The vasopressor effects of posterior pituitary extracts were first described in 1895, and the potent extracts were named AVP. OT and AVP mRNAs are among the most abundant messages in the hypothalamus, being heavily concentrated in the magnocellular neurons of the PVN and the supraoptic nucleus of the hypothalamus, which send axonal projections to the neurohypophysis. These neurons produce OT and AVP, which is released into the bloodstream, where these peptides act like hormones on peripheral targets. OT and AVP are generally synthesized in separate neurons within the hypothalamus. OT released from the pituitary is most often associated with functions associated with female reproduction, such as regulating uterine contractions during parturition and the milk ejection reflex during lactation. AVP, also known as antidiuretic hormone, regulates water retention in the kidney and vasoconstriction through interactions with vasopressin V2 and V1a receptor subtypes, respectively. AVP is released into the bloodstream from the neurohypophysis following a variety of stimuli, including plasma osmolality, hypovolemia, hypertension, and hypoglycemia. The actions of OT are mediated via a single receptor subtype (oxytocin receptor, OTR), which is distributed in the periphery and within the limbic CNS.

In contrast to the OTR, there are three vasopressin receptor subtypes, V1a, V1b, and V2 receptors, each of which is a G-protein–coupled seven-transmembrane domain receptor. The V2 receptor is localized in the kidney

and is not found in the brain. The V1a receptor is distributed widely in the CNS and is thought to mediate most of the behavioral effects of AVP. The V1b receptor is concentrated in the anterior pituitary, and some reports describe V1b receptor mRNA in the brain, although its function is unknown.

Neurotensin (NT). Although NT is found in several brain regions, it has been most thoroughly investigated in terms of its association with other neurotransmitter systems, particularly the mesolimbic dopamine system, and has gained interest in research on the pathophysiology of schizophrenia. There are several lines of evidence suggesting that NT and its receptors should be considered as potential targets for pharmacologic intervention in this disorder. First, the NT system is positioned anatomically to modulate the neural circuits implicated in schizophrenia. Second, peripheral administration of antipsychotic drugs has been shown to modulate NT systems consistently. Third, there is evidence that central NT systems are altered in patients with schizophrenia.

NT was first shown to interact with dopamine systems while undergoing characterization of its potent hypothermic-potentiating and sedative-potentiating activities. Subsequent work indicated that NT possessed many properties that were also shared by antipsychotic drugs, including the ability to inhibit avoidance, but not escape responding in a conditioned active avoidance task; the ability to block the effects of indirect dopamine agonists or endogenous dopamine in the production of locomotor behavior; and the ability to elicit increases in dopamine release and turnover. Perhaps most importantly, both antipsychotic drugs and NT neurotransmission enhance sensorimotor gating. Sensorimotor gating is the ability to screen or filter relevant sensory input, deficits in which may lead to involuntary flooding of indifferent sensory data. Increasing evidence suggests that deficits in sensorimotor gating are a cardinal feature of schizophrenia. Both dopamine agonists and NT antagonists disrupt performance on tasks designed to gauge sensorimotor gating. Unlike antipsychotic drugs, NT is not able to displace dopamine from its receptor. As noted earlier, NT is colocalized in specific subsets of dopamine neurons and is coreleased with dopamine in the mesolimbic and medial prefrontal cortex dopamine terminal regions that are implicated as the sites of dopamine dysregulation in schizophrenia. Antipsychotic drugs that act at D_2 and D_4 receptors increase the synthesis,

concentration, and release of NT in those dopamine terminal regions but not in others. That effect of antipsychotic drugs in increasing NT concentrations persists after months of treatment and is accompanied by the expected increase in NT mRNA concentrations as well as expression of the "immediate early gene" c-fos within hours of initial drug treatment. The altered regulation of NT expression by antipsychotic drugs extends to the peptidases that degrade the peptide because recent reports have revealed decreased NT metabolism in rat brain slices 24 hours after the acute administration of haloperidol. When administered directly into the brain, NT preferentially opposes dopamine transmission in the nucleus accumbens but not the caudate-putamen. In the nucleus accumbens, NT receptors are located predominantly on GABAergic neurons, which release GABA on dopamine terminals, thereby inhibiting release.

Decreased CSF NT concentrations have been reported in several populations of patients with schizophrenia when compared to those of controls or other psychiatric disorders. Although treatment with antipsychotic drugs has been observed to increase NT concentrations in the CSF, it is not known whether this increase is causal or merely accompanies the decrease in psychotic symptoms seen with successful treatment. Postmortem studies have shown an increase in NT concentrations in the dopamine-rich Brodmann area 32 of the frontal cortex, but that result may have been confounded by premortem antipsychotic treatment. Other researchers have found no postmortem alterations in NT concentrations of a broad sampling of subcortical regions. Decreases in NT receptor densities in the entorhinal cortex have been reported in entorhinal cortices of schizophrenic postmortem samples. A critical test of the hypothesis that NT may act as an endogenous antipsychotic-like substance awaits the development of an NT receptor agonist that can penetrate the BBB.

Other Neuropeptides. Several other neuropeptides have been implicated in the pathophysiology of psychiatric disorders. These include, but are not limited to, cholecystokinin (CCK), substance P, and neuropeptide Y.

CCK initially discovered in the gastrointestinal tract, and its receptors are found in areas of the brain associated with emotion, motivation, and sensory processing (e.g., cortex, striatum, hypothalamus, hippocampus, and amygdala). CCK is often colocalized with dopamine in the VTA neurons that

comprise the mesolimbic and mesocortical dopamine circuits. Like NT, CCK decreases dopamine release. Infusions of a CCK fragment have been reported to induce panic in healthy individuals, and patients with panic disorder exhibit increased sensitivity to the CCK fragment compared to that of normal controls. Pentagastrin, a synthetic CCK agonist, dose-dependently produced increased blood pressure, pulse, HPA activation, and physical symptoms of panic. Recently, a CCK receptor gene polymorphism has been associated with panic disorder.

The undecapeptide substance P is localized in the amygdala, hypothalamus, periaqueductal gray, LC, and parabrachial nucleus and is colocalized with norepinephrine and serotonin. Substance P serves as a pain neurotransmitter, and administration to animals elicits behavioral and cardiovascular effects resembling the stress response. More recent data suggest a role for substance P in major depression and PTSD. Both depressed and PTSD patients had elevated CSF substance P concentrations. Furthermore, in PTSD patients, marked increases in CSF substance P concentrations were detected following precipitation of PTSD symptoms. One study has indicated that a substance P receptor (termed the neurokinin 1 [NK1] receptor) antagonist capable of passing the BBB is more effective than a placebo and as useful as paroxetine in patients with major depression with moderate to severe symptom severity. However, subsequent studies have been unable to confirm these findings.

Neuropeptide Y (NPY) is a 36 amino acid peptide found in the hypothalamus, brainstem, spinal cord, and several limbic structures and is involved in the regulation of appetite, reward, anxiety, and energy balance. NPY is colocalized with serotonergic and noradrenergic neurons and is thought to facilitate the containment of adverse effects following exposure to stress. Suicide victims with a diagnosis of major depression are reported to have a pronounced reduction in NPY levels in the frontal cortex and caudate nucleus. Furthermore, CSF NPY levels are decreased in depressed patients. Chronic administration of antidepressant drugs increases neuropeptide Y concentrations in the neocortex and hippocampus in rats. Plasma NPY levels were found to be elevated in soldiers subjected to the “uncontrollable stress” of interrogation, and NPY levels were correlated with the feelings of dominance and confidence during the stress. Also, low NPY response to stress has been associated with increased vulnerability to depression and

PTSD.

Novel Neurotransmitters

Nitric Oxide. The discovery that gases could function as neurotransmitters revealed that highly atypical modes of signaling existed between neurons. In the early 1990s, nitric oxide was the first gas to be ascribed a neurotransmitter function and proved to be an atypical neurotransmitter for several reasons. First, it was not stored in or released from synaptic vesicles, as it was a small gas it could freely diffuse into the target neuron. Second, its target was not a specific receptor on the surface of a target neuron, but intracellular proteins whose activity could directly be modulated by nitric oxide, leading to neurotransmission. Nitric oxide also lacks a reuptake mechanism to remove it from the synapse. Although enzymatic inactivation of it is postulated to exist, nitric oxide appears to have a very short half-life of a few seconds.

Nitric oxide was initially discovered as a bactericidal compound released from macrophages, and as an endothelial cell, it derived relaxation factor allowing for the dilation of blood vessels. A role for nitric oxide in the brain followed, revealing a role for the gas in neurotransmission, learning and memory processes, neurogenesis, and neurodegenerative disease.

NITRIC OXIDE AND BEHAVIOR. Nitric oxide neurotransmission can play a role in behavior, as neuronal nitric oxide synthase (nNOS)-deficient male mice display exaggerated aggressive tendencies and increased sexual activity. In female mice, the contrary is true, as they have reduced aggression. As manic bipolar patients may show both hypersexuality and aggression, the nitric oxide pathway may participate in the psychopathology of affective states.

In the periphery, nNOS localizes to neurons that innervate blood vessels of the penis, including the corpus cavernosa. Stimulation of these nerves releases nitric oxide, leading to cyclic guanosine monophosphate (cGMP) formation, blood vessel wall relaxation and vasodilation, penile engorgement, and initial erection. The sustained phase of erection also depends on nitric oxide; turbulent blood flow leads to phosphorylation of eNOS and sustained nitric oxide production. Drugs used in the treatment of erectile dysfunction—sildenafil (Viagra), tadalafil (Cialis), and vardenafil (Levitra)—act to inhibit

phosphodiesterase type 5 (PDE5), an enzyme that degrades cGMP in the penis (Fig. 33-23), thereby potentiating nitric oxide neurotransmission and penile erection.

Numerous lines of evidence have suggested a role for nitric oxide in the regulation of sleep–wake cycles. nNOS expressing neurons occur in several areas that initiate REM sleep, including the pons, dorsal raphe nucleus, laterodorsal tegmentum, and pedunculopontine tegmentum. In animal models, microinjection of compounds that release nitric oxide decreases wakefulness and increases slow-wave sleep compared with controls. Consistent with this, NOS inhibitors show a trend toward decreasing slow-wave and REM sleep. Studies of NOS-deficient mice suggest that nitric oxide may serve a more involved role than merely promoting sleep. nNOS-deficient animals also show reduced REM sleep; however, inducible nitric oxide synthase (iNOS)-deficient mice demonstrate the reverse, suggesting a complex interplay between NOS enzymatic isoforms.

NITRIC OXIDE AND MOOD DISORDERS. NOS-expressing neurons are well represented in areas implicated in depression, including the dorsal raphe nucleus and prefrontal cortex. A role for nitric oxide has been suggested in antidepressant response, as SSRI antidepressants can directly inhibit NOS activity. Moreover, in animal studies such as the forced swim test, NOS and soluble guanylyl cyclase inhibitors can achieve antidepressant-like effects. Plasma nitric oxide levels were elevated in patients with bipolar disorder compared to healthy controls. However, in depressed subjects, studies have found decreased nitric oxide levels and increased plasma nitrite, a byproduct of nitric oxide. Reduced NOS has also been described in the paraventricular nucleus of patients with schizophrenia and depression compared to controls.

Nitric oxide has been questioned as to its ability to regulate neurotransmission at serotonin, norepinephrine, and dopamine nerve termini. However, there has been no clear consensus, and nitric oxide appears to be able to increase or decrease activity at these neurons depending on the timing of its activation and the region of the brain studied.

NITRIC OXIDE AND SCHIZOPHRENIA. Nitric oxide has been investigated as a candidate molecule contributing to symptoms of schizophrenia. Two genetic studies have identified schizophrenia-associated single nucleotide

polymorphisms (SNPs) in CAPON, a protein that associates with nNOS. SNPs in nNOS itself have been associated with schizophrenia, although others have not been able to reproduce such findings. Changes in NOS levels have been reported in postmortem brain samples of individuals with schizophrenia. Abnormalities have been noted in the cortex, cerebellum, hypothalamus, and brainstem, although no specific trend can be discerned. Elevated NOS activity has been noted in platelets from drug-naive and drug-treated individuals with schizophrenia. Some investigators find increased nitric oxide activity and others the reverse. In autopsy samples, schizophrenic patients were found to have abnormally localized NOS expressing neurons in the prefrontal cortex, hippocampus, and lateral temporal lobe, consistent with abnormal migration of these neuronal types during development. In a rat model, prenatal stress led to reduced NOS expressing neurons in the fascia dentate and hippocampus.

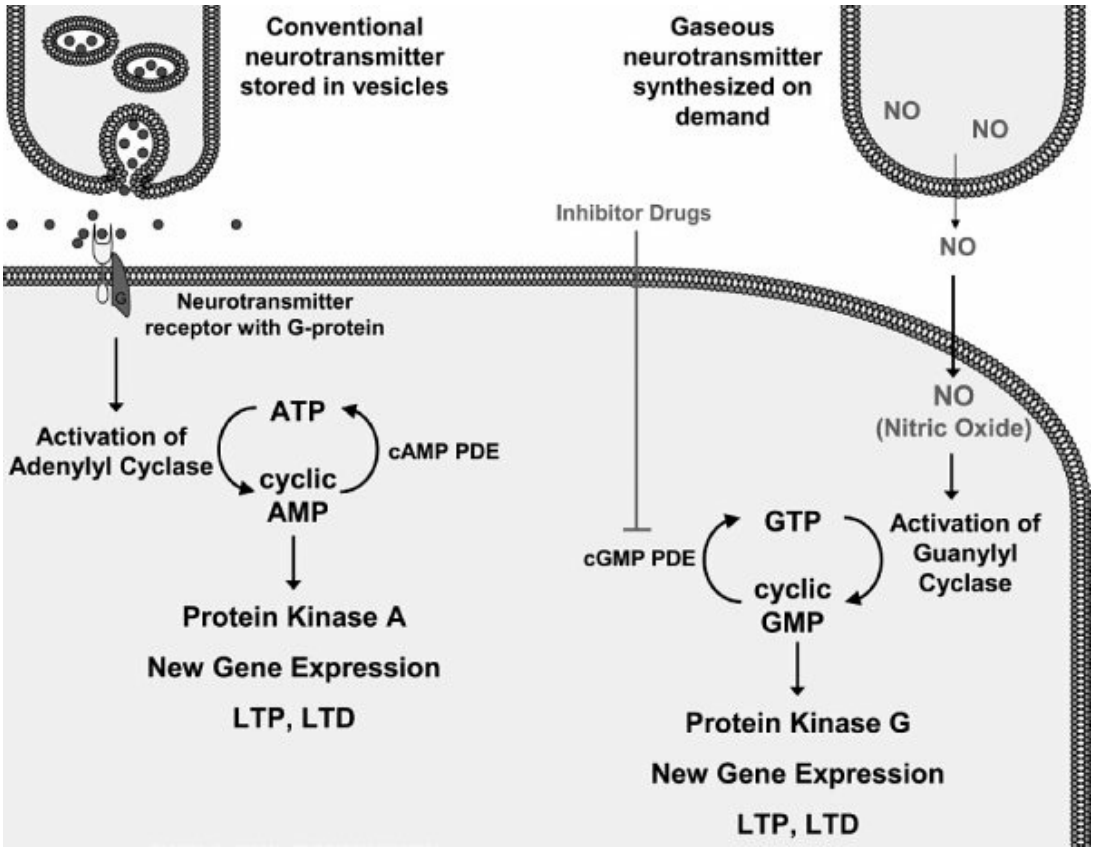


FIGURE 33-23
Neurotransmitter and signaling functions of nitric oxide (NO) via production of cyclic guanosine monophosphate (cGMP). Gaseous nitric oxide is enzymatically generated and freely diffuses into an adjacent neuron (*upper right*). In comparison

to traditional neurotransmitters (*upper left*), nitric oxide (NO) does not act via a specific neurotransmitter receptor on the surface membrane of a neuron. In contrast, NO freely diffuses across the neuronal membrane and activates the enzyme, guanylyl cyclase, which converts guanosine 5'-triphosphate (GTP) into the second messenger, cGMP. Nitric oxide effects are mediated, in part, by cGMP activation of neuronal protein kinases, new gene expression, and effects on neuronal long-term potentiation (LTP) and long-term depression (LTD). ATP, adenosine triphosphate. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:104.)

NEUROPATHOLOGIC ROLES OF NITRIC OXIDE. Abundant evidence exists that nitric oxide is a direct participant in a variety of neuropathic events. Superoxide, a byproduct of cellular metabolism, can react with nitric oxide to form peroxynitrite (chemical formula $ONOO^-$). This labile and toxic compound forms chemical adducts with protein tyrosine residues, a process termed *protein nitration,* and DNA, leading to cellular dysfunction.

Cell loss resulting from ischemic stroke is mediated in part by overstimulation of the glutamate NMDA receptor, a process termed *excitotoxicity.* Nitric oxide produced by NMDA activation appears to mediate a significant portion of this excitotoxic neuronal death, and stroke damage is reduced in mice with a genetic deletion of nNOS.

S-Nitrosylation has also been implicated in pathologic processes in the brain. Mutations in the Parkin protein are associated with early-onset Parkinson disease. Parkin is an E3 ubiquitin ligase, adding ubiquitin molecules to proteins and targeting them for destruction in the cell proteasome. In sporadic Parkinson disease (i.e., without the early-onset mutation), nitric oxide can nitrosylate the Parkin protein and inhibit its protective E3 ubiquitin ligase function. An overabundance of nitric oxide signaling may thus predispose to the dysfunction and cell death of dopaminergic neurons in Parkinson disease by interfering with proteins essential for cell functioning. In Alzheimer disease, excess oxidation of brain protein, lipids, and carbohydrates has long been appreciated, but nitrosative stress from excess nitric oxide also appears to participate in the disease. Protein disulfide isomerase (PDI) is a cellular protective protein that may help combat the accumulation of misfolded proteins, such as the amyloid fibrils occurring in the disease. In both Alzheimer and Parkinson disease brains, PDI appears to be S-nitrosylated in a harmful way that impedes its

cellular protective function.

The discovery that nitric oxide participates in neurodegenerative processes raises the possibility for improved diagnostic processes, such as detecting damage to cellular components produced by nitric oxide before the onset of full-blown symptoms. Also, drugs may be designed to attenuate the damage to crucial neuronal proteins that protect against disease onset. However, entirely and nonspecifically inhibiting or stimulating NOS is likely to produce significant side effects because of its wide-ranging activities throughout the body.

Carbon Monoxide. Although carbon monoxide (CO) is most well known as an air pollutant derived from combustion reactions, it is produced physiologically in a great variety of organisms ranging from human to bacterium. Once thought to be a toxic byproduct of metabolic reactions, carbon monoxide is increasingly recognized to play an essential role in regulating a variety of physiologic processes in the brain and other organs. These varied effects include regulation of olfactory neurotransmission, blood vessel relaxation, smooth muscle cell proliferation, and platelet aggregation.

Carbon monoxide is far better known for its toxic effects than its activities at physiologic concentrations. It binds tightly to heme molecules within hemoglobin, forming carboxyhemoglobin, which can no longer transport oxygen to tissues. One- to two-pack per day smokers typically have 3 to 8 percent of their hemoglobin as carboxyhemoglobin, with nonsmokers having less than 2 percent. Following acute carbon monoxide poisoning, 5 to 10 percent of carboxyhemoglobin is associated with impaired alertness and cognition, and 30 to 50 percent of carboxyhemoglobin leads to significant drops in oxygen transport to tissues.

CARBON MONOXIDE AND NEUROTRANSMISSION. Carbon monoxide appears to participate in the neurotransmission of odorant perception. Odorants lead to carbon monoxide production and subsequent cGMP synthesis that promotes long-term adaptation to odor stimuli. Carbon monoxide has the potential to regulate a variety of perceptual and cognitive processes that are yet untested. Similarly, in the rat retina, long periods of light exposure led to increased HO1 expression, carbon monoxide production, and cGMP signaling. Carbon monoxide may also participate in adaptation to chronic pain. HO2-deficient

animals manifest reduced hyperalgesia and allodynia after exposure to chronic pain stimuli. Carbon monoxide may thus set the threshold for pain perception, although it is unclear whether the effect occurs in the central or peripheral nervous system. Aside from its role in promoting cGMP production, carbon monoxide may also directly bind to and open the calcium-activated big potassium (BK_{Ca}) channel, leading to as yet uncharacterized effects on neurotransmission.

In the gastrointestinal (GI) nervous system, carbon monoxide serves as a neurotransmitter to relax the internal anal sphincter in response to nonadrenergic noncholinergic (NANC) nerve stimulation and vasoactive intestinal peptide (VIP).

Carbon monoxide has been implicated in the development of hippocampal LTP, although lines of evidence are contradictory. Carbon monoxide and tetanic stimulation of nerves lead to increased excitatory postsynaptic potentials (EPSPs). HO inhibitors that block carbon monoxide production lead to impaired induction of LTP and reduced calcium-dependent release of glutamate neurotransmitter. However, HO_2-deficient animals fail to manifest any differences in LTP. These disparate findings may be explained by a role for HO_1 in LTP, or an ability of HO inhibitors to block some other processes necessary to LTP induction nonspecifically.

At toxic levels, carbon monoxide is well known to impair oxygen transport by binding to hemoglobin with a higher affinity than oxygen. Amazingly, carbon monoxide itself plays a physiologic role in the mechanism by which the carotid body senses oxygen. HO, expressed in glomus cells of the carotid body, uses oxygen as a substrate in the production of carbon monoxide (Fig. 33-24). When oxygen levels drop, so does carbon monoxide production, leading to a resetting of the threshold in which the carotid body senses oxygen. The molecular mechanism may occur via carbon monoxide regulation of the carotid body BK ion channel.

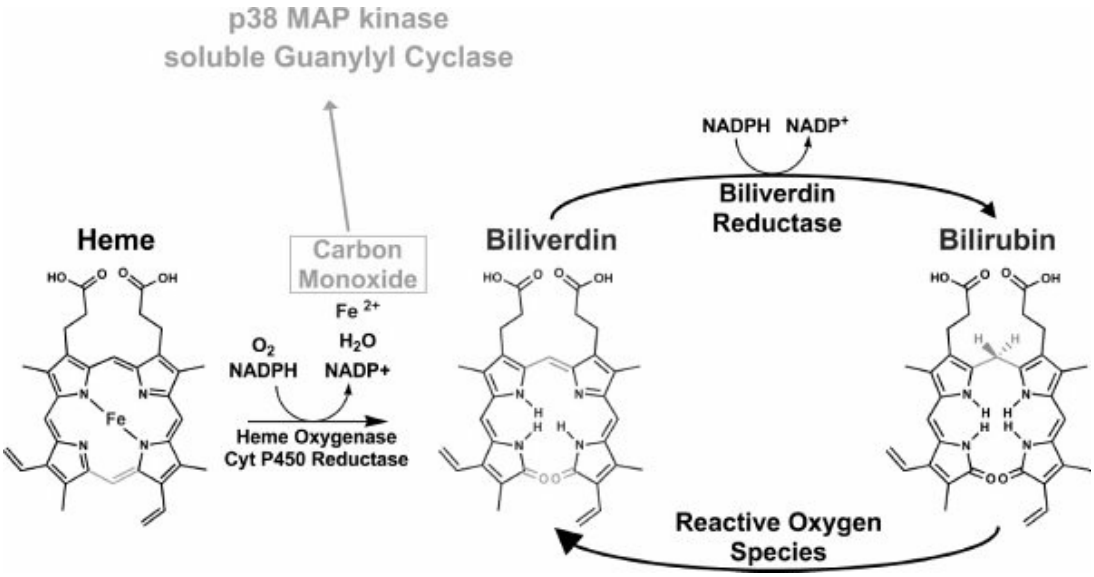


FIGURE 33-24
Synthesis of carbon monoxide (CO), an unexpected neurotransmitter. Gaseous carbon monoxide is enzymatically synthesized in neurons by way of the enzyme heme oxygenase, also converting heme into the molecule biliverdin and liberating free iron (Fe). Similar to nitric oxide, CO is not stored in neuronal vesicles and can freely diffuse across neuronal membranes. CO also similarly activates soluble guanylyl cyclase, and leads to activation of multiple intracellular signaling molecules such as p38 MAP kinase. CO exerts its neurotransmitter and signaling functions at concentrations far below that at which classical CO toxicity occurs. The significance of this pathway in neurons is underlined by the existence of two distinct heme oxygenase enzymes, one of which is predominantly expressed in the brain. Biliverdin is converted to bilirubin via the enzyme biliverdin reductase. Similar to CO, bilirubin is no longer relegated to the status of toxic byproduct and may be an important antioxidant. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:107.)

Endocannabinoids: From Marijuana to Neurotransmission. Whether known as cannabis, hemp, hashish, ma-fen, or a variety of slang terms, marijuana has been cultivated and utilized by human populations for thousands of years. Despite a long debate about risks and benefits, it has only been in recent decades that we have started to learn how marijuana exerts its effects on the brain. The "high" users experience, euphoria and tranquility, relates to cannabis acting on a neural pathway involving cannabinoids endogenous to the human brain, or endocannabinoids.

The first described medicinal use of cannabis dates to approximately 2700 BC in the pharmacopeia of Chinese Emperor Shen Nung, who recommended its use for a variety of ailments. At this time, adverse properties were also

apparent, and large amounts of the fruits of hemp could lead to “seeing devils,” or a user might “communicate with spirits and lightens one’s body.” For centuries, cannabis was employed in India as an appetite stimulant; habitual marijuana users remain well acquainted with “the munchies.”

For many years the mechanisms by which the active components of marijuana, *cannabinoids,* exerted their psychoactive effects remained a mystery. Chemists sought to isolate the psychoactive components of cannabis from the many components of the plant oil (Table 33-5).



Table 33-5
Selected Discoveries in Cannabinoid Research

1899	Cannabinol isolated from cannabis resin
1940	Identification of cannabinol structure
1964	Discovery of the structure of δ-9-tetrahydrocannabinol (THC), the most psychoactive component of cannabis
1988	Specific THC-binding sites identified in brain
1990	Identification of a brain cannabinoid receptor, CB1
1992	Discovery of the first endogenous brain endocannabinoid, anandamide
1993	Identification of a second cannabinoid receptor, CB2
1994	Rimonabant (Acomplia), a CB1 receptor blocker is developed
1995	Report of a second endocannabinoid, 2-AG
1996	Fatty acid amide hydrolase (FAAH), an endocannabinoid degrading enzyme, is discovered
2003	FAAH inhibitors reduce anxiety-like behaviors in animal studies
2003	Identification of enzymes that synthesize endocannabinoids
2006	Monoacylglycerol lipase (MAGL), a second endocannabinoid-degrading enzyme, is found

2006	Rimonabant approved for use in Europe for weight loss
2007	Rimonabant meta-analysis finds increased anxiety and depressive symptoms in humans without a history of psychiatric illness

From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:109.

DISCOVERY OF THE BRAIN ENDOCANNABINOID SYSTEM. Estimates suggest that 20 to 80 µg of tetrahydrocannabinol (THC) reaches the brain after one smokes a marijuana cigarette (i.e., "joint"). This quantity is comparable to the 100 to 200 µg of norepinephrine neurotransmitter present in the entire human brain. Thus the effects of THC might be explained by the effects on neurotransmitter systems. In the 1960s, there were at least two schools of thought on how THC exerted its psychoactive effects. One held that THC worked like that of the inhaled volatile anesthetics (i.e., no specific receptor existed), and it might have a generalized effect on neuronal membranes or widespread actions on neurotransmitter receptors. A competing school of thought speculated that specific receptors for cannabinoids existed in the brain, but they were difficult to identify due to the lipophilic nature of these chemicals. Novel cannabinoids were synthesized that were more water-soluble, and in the late 1980s, this allowed for the discovery of a specific cannabinoid receptor, CB1.

Several additional endocannabinoids were soon discovered, 2-arachidonylglycerol (2-AG), *N*-arachidonyldopamine (NADA), 2-arachidonoylglycerol ether (noladin ether), and virodhamine (Fig. 33-25). The different endocannabinoids have differing affinities for the cannabinoid receptors, CB1 and CB2. Anandamide appears to have the highest selectivity for the CB1 receptor, followed by NADA and noladin ether. In contrast, virodhamine prefers CB2 receptors and has only partial agonist activity at CB1. 2-AG appears not to discriminate between CB1 and CB2.

BIOSYNTHESIS OF ENDOCANNABINOIDS. Arachidonic acid is utilized as a building block for the biosynthesis of endocannabinoids, prostaglandins, and leukotrienes and is found within cellular phospholipids of the plasma membrane and other intracellular membranes. The synthesis of anandamide

requires the sequential action of two enzymes (Fig. 33-26). In the first reaction, the enzyme *N*-acetyltransferase (NAT) transfers an arachidonic acid side chain from a phospholipid to phosphatidylethanolamine (PE), generating NAPE (*N*-arachidonyl-phosphatidylethanolamine). In the second reaction, the enzyme *N*-arachidonyl-phosphatidylethanolamine phospholipase (NAPD-PLD) converts NAPE to anandamide. Because NAPE is already a natural component of mammalian membranes, it is the second step that generates anandamide, which is most crucial to neurotransmission.

Endocannabinoids are not stored in synaptic vesicles for later use but are synthesized on-demand, as is done for the gaseous neurotransmitters. An essential criterion for a signaling molecule to be considered a neurotransmitter is that neuronal depolarization should lead to its release. Depolarization leads to increases in cellular calcium, which in turn promotes the synthesis of the endocannabinoids and their release. The mechanism is explained in part by calcium activation of NAPE-PLD and DAGL, leading to augmented biosynthesis of anandamide and 2-AG, respectively.

Endocannabinoids generated in a neuron must cross the synaptic cleft to act on cannabinoid receptors. Similar to THC, endocannabinoids are highly lipophilic and thus poorly soluble in CSF. It is hypothesized that a specific endocannabinoid transporter exists to allow endocannabinoids to cross the synaptic cleft and allow entry into the target neuron.

INACTIVATION OF ENDOCANNABINOIDS. Neurotransmitters are typically inactivated either by reuptake from the neurons that release them or by degradation by highly specific enzymes, such as the example of acetylcholine being hydrolyzed by acetylcholinesterase. At least two enzymes exist to target the destruction of endocannabinoids and attenuate their neurotransmission. Fatty acid amide hydrolase (FAAH) converts anandamide to arachidonic acid and ethanolamine (Fig. 33-26). FAAH is found in regions of the brain where CB1 receptors are predominant and localize to postsynaptic neurons where anandamide is made. Rapid degradation of anandamide in part explains its relatively low potency compared to THC. Confirming the role of FAAH in anandamide inactivation, knockout mice without FAAH exhibit a 15-fold increase of anandamide, but not 2-AG. These mice have more significant behavioral responses to exogenous anandamide, owing to its decreased degradation. The endocannabinoid 2-AG

is inactivated by FAAH, but also by a monoacylglycerol lipase (MAGL) located in presynaptic neurons.

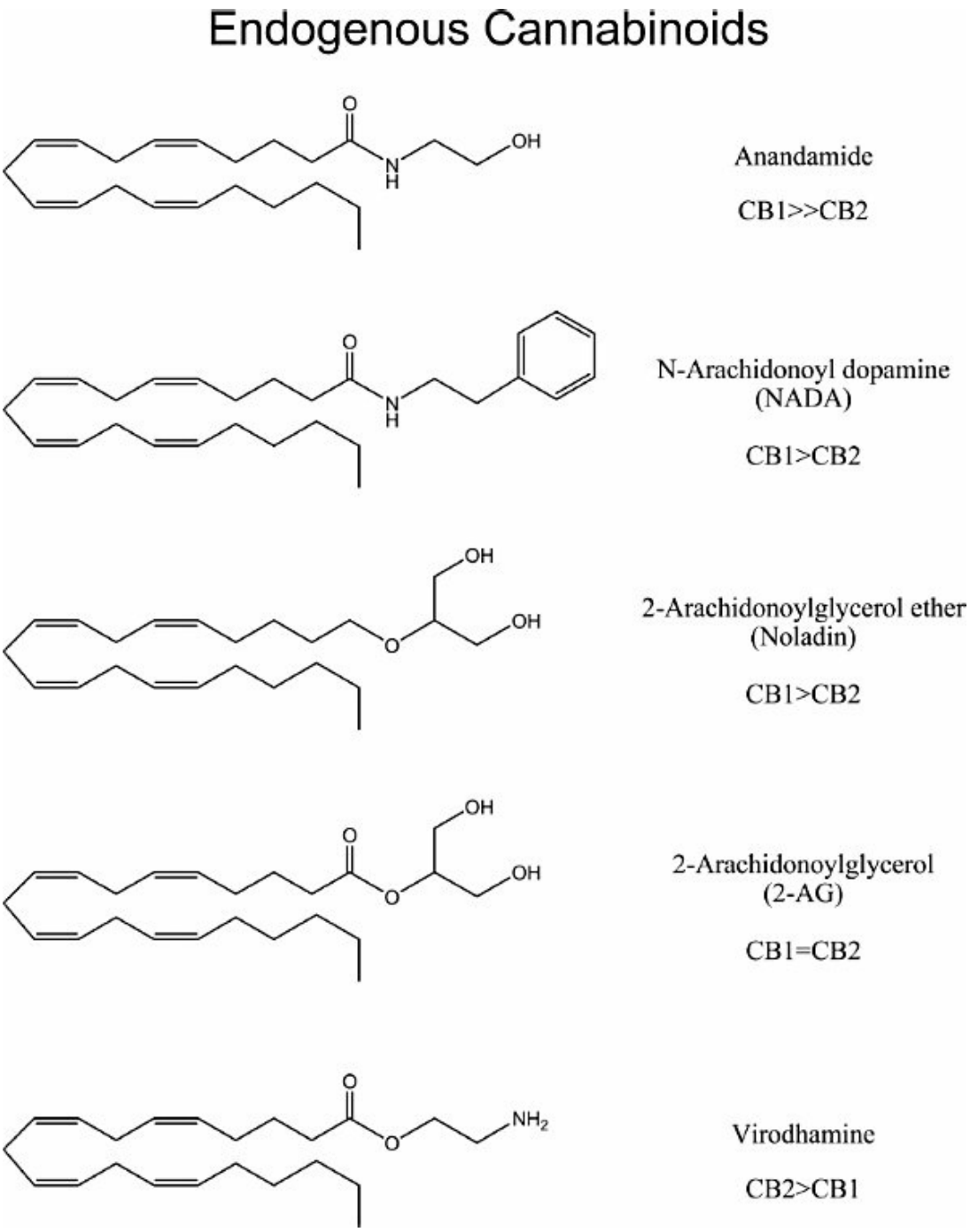


FIGURE 33-25
Endogenous cannabinoids. At least five endocannabinoids exist in the mammalian brain, each differing in affinity for CB1 and CB2 cannabinoid receptors. All are derived from the essential omega-6 fatty acid, arachidonic acid, which is also a substrate in the formation of prostaglandins and leukotrienes. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive*

Textbook of Psychiatry. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:111.)

Pharmacologic inhibitors of FAAH have analgesic effects and reduce anxiety in animal models, but do not have the undesirable effects of THC such as immobility, lowered body temperature, or greater appetite. Such a pharmacologic strategy would be analogous to MAOIs and COMT inhibitors (COMTIs). MAOIs used to treat depression, slow the breakdown of serotonin and other monoamines, thereby increasing serotonin, whereas COMTIs serves an analogous role in blocking the destruction of dopamine and other catecholamines.

CANNABINOID RECEPTORS. CB1 receptors are possibly the most abundant G-protein–coupled receptors in the brain. They occur at the highest density in the basal ganglia, cerebellum, hippocampus, hypothalamus, anterior cingulate cortex, and cerebral cortex, particularly the frontal cortex. Humans or animals that receive large doses of THC develop catalepsy, a reduction of spontaneous movement, and freeze in bizarre and unnatural postures. The action of cannabinoids in the basal ganglia and cerebellum may be associated with these behaviors, which may prove relevant in understanding catatonic symptoms in schizophrenia.

CB1 receptors are predominantly found on axons and nerve termini, with little presence on neuronal dendrites and the cell body. CB1 receptors localize to the presynaptic rather than the postsynaptic side of the neuronal cleft, suggesting a role in the regulation of neurotransmission. A second cannabinoid receptor, CB2, is predominantly expressed on the surface of white blood cells of the immune system, but small amounts appear to be present in the brainstem.

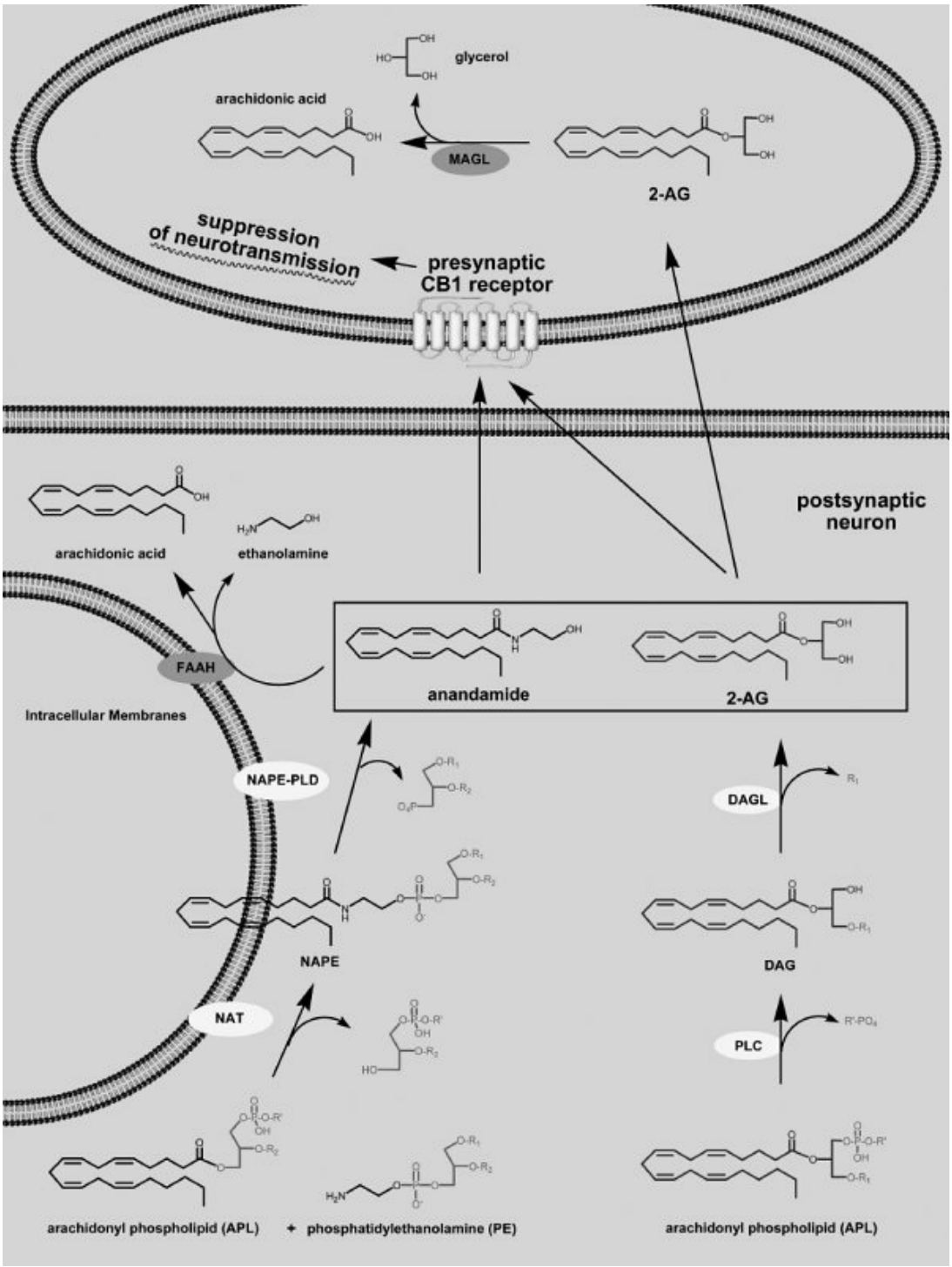


FIGURE 33-26

Retrograde neurotransmission of the endocannabinoids, anandamide, and 2-arachidonylglycerol (2-AG). Anandamide is synthesized on demand for neurotransmission via a two-step process. The enzyme NAT transfers the arachidonic acid chain from a phospholipid (APL) to phosphatidylethanolamine (PE), thereby producing NAPE. A second enzyme, NAPE-PLD, generates anandamide. 2-AG is similarly synthesized in two steps by the enzymes PLC and

DAGL. The endocannabinoids made in a postsynaptic neuron cross the synapse and activate presynaptic CB1 receptors, and suppress neurotransmission of the presynaptic neuron (although activation of the presynaptic neuron occurs in some cases). 2-AG is predominantly inactivated in the presynaptic neuron by MAGL, whereas anandamide is destroyed in the postsynaptic neuron by FAAH. PE, phosphatidylethanolamine; APL, arachidonyl phospholipids; NAT, *N*-acyltransferase; NAPE, *N*-arachidonyl-phosphatidylethanolamine; NAPE-PLD, *N*-arachidonyl-phosphatidylethanolamine phospholipase D; FAAH, fatty acid amide hydrolase; MAGL, monoacylglycerol lipase; PLC, phospholipase C; DAG, diacylglycerol; DAGL, diacylglycerol lipase; R_1-R_3, various acyl or akyl side chains of phospholipids; R', side chain of phospholipid head group. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:112.)

Effects on Neurotransmission. The cannabinoid CB1 receptor is associated with G proteins that mediate its intracellular signaling, in part, through inhibition of adenylyl cyclase. This action leads to a decrease in levels of the essential second messenger, cyclic adenosine monophosphate. Activation of the CB1 receptor also leads to the activation of potassium channels and inhibition of *N-type calcium channels. Because calcium is integral to neurotransmitter release, cannabinoids can block neurotransmission through this mechanism. Cannabinoid receptors also activate mitogen-activated protein kinases.*

With the use of cell culture models and slices of the brain, cannabinoids have been shown to block the release of a variety of neurotransmitters, including GABA, norepinephrine, and acetylcholine. Norepinephrine and acetylcholine tend to be excitatory neurotransmitters, and cannabinoid inhibition of their release would be expected to have an overall inhibitory effect. However, GABA is an inhibitory neurotransmitter, and cannabinoid inhibition of it would lead to overall excitatory effects, demonstrating that cannabinoids can have complex effects on neurotransmission depending on the specific context. Cannabinoids also appear to increase the release of brain endorphin neurotransmitters and increase dopamine release in the nucleus accumbens, a "reward center" relevant to addiction and learning. The endocannabinoids have been implicated in a variety of forms of synaptic plasticity, including LTP and long-term depression (LTD).

ENDOCANNABINOIDS IN ANXIETY AND MOOD. Endocannabinoid neurotransmission may be an essential regulator of anxiety, and cannabis

users regularly describe a tranquilizing effect of THC. The loss of signaling by the endocannabinoid system appears to promote anxiety-like states in animal studies. CB1 receptor-deficient animals exhibit more pronounced anxiety behavior when exposed to stress or new environs.

The endocannabinoid pathway may represent an attractive target in understanding posttraumatic stress responses and phobias. Although one cannot yet safely measure endocannabinoid levels in human subjects, this model is supported by clinical trials of the cannabinoid receptor blocker, rimonabant (Acomplia), which may offer promise as a strategy for weight loss (see below). A frequent adverse reaction to the drug is increased anxiety and depression.

Addiction. The endocannabinoid system may be an attractive target for understanding addiction. Mice deficient in CB1 receptors are unsurprisingly resistant to the behavioral effects of cannabinoids; they also appear to have reduced addiction to and withdrawal from opiates. Further interaction has also been found between the opioid and cannabinoid systems, as cannabinoids appear to increase the release of dopamine in the nucleus accumbens, a key reward area of the brain implicated in addiction. This dopamine release appears to require μ-opioid receptors, as pharmacologic inhibition of these receptors blocks the ability of cannabinoids to increase dopamine release. Rats with a preference for alcohol have decreased FAAH activity, suggestive of higher cannabinoid signaling. CB1 receptor antagonists dampen their alcohol consumption, whereas inhibiting FAAH increases their alcohol consumption. Furthermore, CB1-deficient animals also appear to have reduced alcohol intake. A single amino acid mutation in human FAAH is associated with drug abuse, and this abnormal enzyme appears to be less stable than its wild-type counterpart.

ENDOCANNABINOIDS IN PSYCHOSIS. Heavy use of cannabis can produce psychotic symptoms in individuals with no prior history of psychiatric disorder, although it is unclear whether this is solely due to the drug or an underlying vulnerability to psychosis in such persons. Cannabis use often worsens psychosis in schizophrenia, and heavy use has been associated with developing schizophrenia, although some suggest that this association is an accelerated development of symptoms in those who would eventually

manifest schizophrenia. Nonetheless, the endocannabinoid system has implications for the pathophysiology of schizophrenia, as cannabinoid signaling appears to increase the release of dopamine. Medications that act as antagonists of D_2 receptors will likely remain a component of schizophrenia treatment for some time.

Feeding. Following drug ingestion, THC users develop an increased appetite ("the munchies"), and cannabis has been utilized as an appetite stimulant for centuries. This effect may depend on CB1 receptors present in the hypothalamus. Endocannabinoid levels increase in the hypothalamus and limbic system when animals are deprived of food. Mice genetically deficient in CB1 receptors become resistant to developing obesity after being given a high-fat diet. Similarly, the CB1 receptor antagonist, rimonabant, appears to facilitate weight loss by blocking cannabinoid signaling. In a clinical trial of more than 3,000 obese patients, those treated with 20 mg/day of rimonabant lost 6.3 kg at 1 year, compared to 1.6 kg in the placebo group. Nausea was the most common side effect reported. A 2007 meta-analysis of clinical trials reported an overall 4.7 kg weight loss with rimonabant treatment, besting the weight-loss drugs orlistat (2.9 kg) and sibutramine (4.2 kg).

EFFECTS ON BRAIN INJURY AND PAIN. In mouse models of traumatic brain injury, 2-AG appears neuroprotective, reducing brain edema, infarct size, and cell death, while improving functional outcomes. Anandamide is also protected against brain injury in a model of multiple sclerosis (MS), and human patients with the disease have increased production of anandamide. A study of cannabinoid agonist, HU-211, led to more rapid clinical improvement following head trauma. FAAH inhibitors improved motor symptoms in a mouse model of Parkinson disease, likely via cannabinoids increasing dopamine neurotransmission.

There is increasing evidence that neurotransmission via the endocannabinoid pathway regulates pain perception. THC and cannabinoid agonists have proven effective in animal models of acute and chronic pain, ranging from burn injury to nerve damage and inflammation. The CB1 receptor plays an essential role in these effects, as the analgesic effects of cannabinoid drugs are lost when CB1 antagonist rimonabant is given. Similarly, the analgesic effect of THC is lost in mice that are genetically

deficient in the CB1 receptor.

Stress has long been associated with diminished pain perception, such as in cases of injured military personnel who demonstrate remarkable pain tolerance, a phenomenon known as *stress-induced analgesia.* The endocannabinoid system may mediate these effects. Animal models reveal anandamide and 2-AG production after stress, and stress-induced analgesia is blocked by the CB1 blocker, rimonabant, in these animals.

Endocannabinoid regulation of pain perception appears to be distinct from that of the endogenous opiate system, but the two pathways may share overlapping neural pathways. Evidence for this has been provided using CB1 blocker, rimonabant, and naloxone (Narcan), which blocks opiate receptors. Rimonabant attenuates analgesia provided by THC and cannabinoids, but only partially blocks the response to morphine. However, the opposite is true for opiates: Naloxone blocks morphine-induced analgesia but also partially blocks the analgesia of THC and cannabinoid drugs. Combinations of cannabinoid and opiate drugs evince synergistic analgesic effects in animal models.

Although it was initially assumed that cannabinoids exert their analgesic effects via the CNS, in animal models, it has been shown that localized administration of cannabinoids may also be useful, including drugs selective for the CB2 receptor, whose expression is minimal in the CNS.

Endocannabinoids may also influence pain sensitivity by mechanisms that do not involve the CB1 and CB2 receptors. Both anandamide and NADA can also activate a calcium channel known as the vanilloid receptor (also known as transient receptor potential vanilloid type 1 [TRPV-1]) that is found on sensory nerves. This same receptor is also famous for being activated by capsaicin, which causes the hot sensation after eating chili peppers. Thus endocannabinoids can exert opposing functions: Promoting analgesia through the CB1 and CB2 receptors but potentially increasing pain via TRP channels. Although CB2 receptors are primarily expressed in the periphery, postmortem analyses reveal an upregulation in the brain from those with Alzheimer disease.

The rapid development of novel cannabinoid drugs may allow for targeting of specific symptoms, rather than elicit all of the typical effects of THC. For instance, ajulemic acid demonstrates analgesic and anti-inflammatory properties but may offer a benefit of limited psychoactive side effects. In a

randomized clinical trial of this compound, Mathias Karst and colleagues found efficacy in reducing chronic neuropathic pain.

EFFECTS IN THE PERIPHERY. Cannabinoids lead to direct relaxation of vascular smooth muscle by local CB1 receptors. This vasodilation extends to the conjunctiva, leading to a "bloodshot" appearance in some cannabis users. Relaxation of ocular arteries by cannabinoids may offer utility as a treatment for glaucoma, a condition of high intraocular pressure, and activation of CB1 receptors in the kidney can improve renal blood flow. A role in generalized blood pressure regulation is unproven, and blood pressure is unaltered in persons treated with rimonabant or animals deficient in CB1 receptors. Cannabinoid signaling may also be relevant to ectopic pregnancy, as CB1-deficient mice retain many embryos in the oviduct.

Nonpsychoactive Cannabinoids. Although THC is the principal psychoactive component of cannabis, the many nonpsychoactive cannabinoids also have intriguing properties and may regulate neurotransmission.

Cannabidiol may offer potential therapeutic effects and appears to stimulate TRPV-1 receptors and influence endocannabinoid degradation. Also, cannabidiol demonstrated a protective effect in a mouse model of inflammatory arthritis. Although results have been mixed, purified cannabidiol may also exert antipsychotic activity, although the net effect of plant cannabis use typically exacerbates schizophrenia symptoms owing to THC. Tetrahydrocannabivarin is a plant cannabinoid that antagonizes CB1 receptors. It is a candidate marker to distinguish whether a patient has been using plant-derived cannabis or prescription THC, which contains no tetrahydrocannabivarin.

Eicosanoids

OVERVIEW. Clinical findings suggest that the dietary supplements omega-3 fatty acids, eicosapentaenoic acid (EPA), its ester ethyl-eicosapentaenoic (E-EPA), and docosahexaenoic acid (DHA) help relieve symptoms of depression, bipolar illness, schizophrenia, and cognitive impairment. DHA and EPA may help reduce behavioral outbursts and improve attention in

children.

CHEMISTRY. Essential fatty acids are a group of polyunsaturated fats that contain a carbon–carbon double bond in the third position from the methyl end group in the fatty acid chain. They are essential because unlike monosaturated and saturated fatty acids, polyunsaturated fatty acids cannot be synthesized de novo and can be acquired only through diet from natural fats and oils. Linoleic acid (LA) is the parent compound of omega-6 fatty acids, and α-linolenic acid (ALA) is the parent compound of omega-3 fatty acids. Both omega-3 and omega-6 groups use the same enzymes for desaturation and chain elongation. Algae and plankton synthesize Omega-3 fatty acids. Fish such as herring, salmon, mackerel, and anchovy feed on these aquatic species and become a rich dietary source of omega-3. EPA and DHA are highly unsaturated omega-3 fatty acids that contain 6 and 5 double bonds on their long structural chain, respectively. They are positioned in the cell membrane by phospholipids and play a crucial role in cell membrane signaling.

EFFECTS ON SPECIFIC ORGANS AND SYSTEMS. The most substantial scientific evidence for treatment with fatty acid supplements comes from the cardiovascular literature. Several human trials have demonstrated that omega-3 fatty acids lower blood pressure, reduce the rate of recurrent myocardial infarction, and lower triglyceride levels. In the nervous system, fatty acids are essential components of neurons, immune cells, and glial phospholipid membrane structures. They increase cerebral blood flow, decrease platelet aggregation, and delay the progression of atherosclerosis in the cardiovascular system. Omega-6 fatty acids appear to reduce inflammation and neuronal apoptosis and decrease phosphatidylinositol second messenger activity. Omega-3 fatty acids have been suggested to alter gene expression.

In the CNS, fatty acids are selectively concentrated in neuronal membranes and involved in cell membrane structure. Omega-6 arachidonic acid has been shown to enhance glutamate neurotransmission, stimulate stress hormone secretion, and trigger glial cell activation in the setting of oxidative toxicity and neurodegeneration. The omega-3 fatty acids DHA and EPA appear to protect neurons from inflammatory and oxidative toxicities. Increases in serotonin, enhancement of dopamine, and regulation of CRF have been

demonstrated in cell culture models.

In rodent models of depression, chronic EPA treatment normalized behavior in open field tests. Serotonin and norepinephrine were also increased in the limbic regions. Mice fed omega-3 poor diets had reduced memory, altered learning patterns, and more behavioral problems.

THERAPEUTIC INDICATIONS. Clinical research with the use of fish oil for mood disorders was based on epidemiology studies where there appears to be a negative correlation between fish consumption and depressive symptoms. Countries with lower per capita fish consumption had up to 60 times increased rates of major depression, bipolar disorder, and postpartum depression. Observational studies concluded that the lower incidence of seasonal affective disorder in Iceland and Japan, rather than latitude predicted, is related to the amount of fatty acid these populations consume in their diet. A study in Norway showed that the use of cod liver oil decreased depressive symptoms. Depression after a myocardial infarction shows higher arachidonic acid to EPA ratio. Postmortem studies in the brains of patients diagnosed with major depressive disorder show reduced DHA in the orbitofrontal cortex.

The first randomized, controlled pilot study of omega-3 fatty acids focused on adjunctive treatment in both bipolar and unipolar patients with depression in addition to their standard lithium (Eskalith) or valproic acid (Depakene) treatment. The omega-3 fatty acid group had significant improvement on the Hamilton Depression Scale and a more extended period of remission than the placebo group. A subsequent more extensive study supported a benefit from treatment with E-EPA for bipolar illness. However, a study of a group of patients with either bipolar disorder or rapid cycling treated with E-EPA showed no significant difference in any outcome measure between the EPA and placebo groups. Bleeding time was also increased in the treatment group. There are no current data on monotherapy in bipolar illness or depression.

The most convincing evidence comes from early brain development and learning studies. Pregnant mothers who consumed foods rich in DHA gave birth to infants who had improved problem-solving skills but not necessarily improved memory. Visual acuity and eye development are also associated with DHA supplementation during pregnancy.

Reports of behavioral studies of prisoners in England who consumed

higher amounts of seafood containing omega-3 fatty acids showed a decrease in assault rates. A Finnish study of violent criminals identified lower levels of omega-3 fatty acids in their system compared to the nonviolent offenders.

The negative and psychotic symptoms of schizophrenia may be improved with supplementation with omega-3 fatty acids. Antipsychotic medications like haloperidol (Haldol) appear to have fewer extrapyramidal side effects when combined with antioxidants and omega-3 fatty acids.

EPA and DHA have been associated with decreased dementia incidence. After reviewing the Rotterdam study of a longitudinal cohort of more than 5,300 patients, fish consumption appeared to be inversely related to the development of new cases of dementia. Later analysis of the study after 6 years demonstrated that low intake of omega-3 fatty acids was not associated with an increased risk of dementia. In contrast, the Zutphen study, also in the Netherlands, concluded that high fish consumption was inversely related to cognitive decline at a 3-year follow-up and after 5 years. Well-designed clinical trials are needed before omega-3 fatty acids can be recommended for the prevention of cognitive impairment.

PRECAUTIONS AND ADVERSE REACTIONS. The most adverse complication of eicosanoid use is an increased risk of bleeding. Dietary sources can contain heavy metals, and there is no standard preparation for capsule formulations. Treatment studies have yielded a variety of different doses, but evidence for the therapeutic dose and clinical guidelines are almost nonexistent. The length of treatment still needs to be determined.

Neurosteroids

BACKGROUND. Although steroids are critical for the maintenance of body homeostasis, neurosteroids are synthesized from cholesterol in the brain and independent of peripheral formation in the adrenals and gonads. Neurosteroids are produced by a sequence of enzymatic processes governed by cytochrome P450 (CYP) and non-CYP enzymes, either within or outside the mitochondria of several types of CNS and peripheral nervous system (PNS) cells.

Recent work has shown that neurosteroids can operate through a nongenomic pathway to regulate neuronal excitability through their effects on

neurotransmitter-gated ion channels. Receptors are generally located in the nucleus, membrane, or microtubules of the CNS and PNS. Although steroids and neurosteroids can act on the same nuclear receptors, neurosteroids differ from steroids in their topologic distribution and regional synthesis. The most well-known effect of neurosteroids is on the GABA receptor, particularly the $GABA_A$ receptor. Neurosteroids acting primarily at this site include allopregnanolone (3α,5α-tetrahydroprogesterone), pregnanolone (PREG), and tetrahydrodeoxycorticosterone (THDOC). Dehydroepiandrosterone sulfate (DHEA-S), the most prevalent neurosteroid, acts as a noncompetitive modulator of GABA. Its precursor dehydroepiandrosterone (DHEA) has also been shown to exert inhibitory effects at the GABA receptor. Some neurosteroids may also act at the NMDA, α-amino-3-hydroxy-5-methyl-4-isoxazole-propanoic acid (AMPA), kainate, glycine, serotonin, sigma type-1, and nicotinic acetylcholine receptors. Progesterone is also considered a neurosteroid and can regulate gene expression at progesterone receptors.

NEUROSTEROIDS IN NEURODEVELOPMENT AND NEUROPROTECTION. In general, neurosteroids stimulate axonal growth and promote synaptic transmission. Specific neuroprotective effects are unique to each neurosteroid. DHEA acts to regulate brain serotonin and dopamine levels, suppress cortisol, increase hippocampal primed burst potentiation and cholinergic function, decrease amyloid-β protein, inhibit the production of proinflammatory cytokines, and prevent free radical scavenging. DHEA and DHEA-S have both been shown to have a role in glial development and neuronal growth and to promote their survival in animals; the injection of these substances into the brains of mice promoted long-term memory while reversing amnesia. Progesterone is linked to myelinating processes like aiding in the repair of damaged neural myelination. Allopregnanolone contributes to the reduction of contacts during axonal regression.

ROLE OF NEUROSTEROIDS IN MENTAL ILLNESS. Neurosteroids have distinct implications for the maintenance of normal neurologic function and also may contribute to neuropathology. Neurosteroids are differentially regulated in males and females and may affect the manifestation of psychological disorders in these two populations. Specifically, they play a distinct role in depression and anxiety disorders and may be targeted by psychiatric

medications someday soon.

Depression. When compared with nondepressed controls, studies show that depressed patients have lower plasma and CSF concentrations of allopregnanolone. Also, this research has elucidated an inverse relationship between allopregnanolone concentrations and the severity of depressive illness. However, there are no allopregnanolone-based therapies available for humans, so its direct efficacy is unsubstantiated. Antidepressant drugs, specifically fluoxetine (Prozac), have been shown in multiple studies to increase the levels of certain neurosteroids. Nonetheless, there is debate over the therapeutic properties of neurosteroids, prompting the investigation of neurosteroid concentrations in patients undergoing nonpharmacologic therapies. Preliminary results indicate that the lack of modifications in neurosteroid levels during nonpharmacologic treatments supports the validity of the pharmacologic properties of antidepressants, not their therapeutic action, in the elevation of neurosteroid levels in medicated populations.

Anxiety Disorders. In patients with anxiety disorders, the primary mechanism of action is on the GABA receptor. Homeostasis characterized by regular GABAergic activity is restored after panic attacks as neurosteroids are released in response to stress. Allopregnanolone stimulates GABAergic activity with 20 times the strength of benzodiazepines and 200 times the potency of barbiturates. Both positive and negative regulation of the $GABA_A$ *receptor are correlated with anxiolytic and anxiogenic action, respectively.*

Psychotic Disorders. Also, to their primary relevance to the pharmacologic treatment of mood and anxiety disorders, neurosteroids contribute to psychotic, childhood, substance abuse, eating, and postpartum disorders. DHEA and DHEA-S mediate the effect of neurosteroids on psychotic disorders such as schizophrenia. DHEA has been dispensed to decrease anxiety in patients with schizophrenia, as DHEA and DHEA-S suppress GABA inhibition and heighten the neuronal response at the NMDA and sigma receptors. DHEA and DHEA-S levels are typically elevated in the initial episode of a patient with schizophrenia, indicating the onset of psychosis upregulates neurosteroids. Because neurosteroid levels are studied across various illness stages, some questions still exist regarding the role of

neurosteroids in psychosis.

Childhood Mental Illness. In children, the clinical symptomology of ADHD is inversely correlated with DHEA and pregnenolone levels.

Substance Abuse. Alcohol is theorized to regulate the GABA receptor and induce de novo steroid synthesis in the brain, specifically, pregnenolone, allopregnanolone, and allotetrahydrodeoxycorticosterone levels are increased in the brain and periphery in response to increases in peripheral alcohol levels. It is hypothesized that sharp increases in ethanol concentration may mimic the acute stress response and elevate neurosteroid concentrations by the HPA axis. To prevent ethanol dependence, researchers are investigating fluctuations in neurosteroid levels and in vivo neurosteroid responsiveness. Neurosteroids (increased allopregnanolone levels in particular) are associated with drug abuse. However, DHEA-S may check the acquisition of morphine tolerance. Past research has shown that DHEA-S levels were also increased in patients who abstained from cocaine use in a treatment program, and as patients relapsed, DHEA-S concentrations decreased accordingly.

Eating Disorders. Concerning eating disorders, DHEA has been shown to diminish food intake, temper obesity, moderate insulin resistance, and lower lipids in rats with a model of youth-onset, hyperphagic, and genetic obesity. By regulating the serotonergic system, DHEA is hypothesized to promote a reduced caloric load. Although hypothetical, low levels of DHEA and DHEA-S are recorded in young women with anorexia nervosa, and 3 months of oral DHEA supplementation increased bone density and tempered the emotional problems associated with the disorder.

Postpartum and Gynecologic Disorders. Because estrogen and progesterone levels fluctuate during pregnancy and drop markedly after delivery, neurosteroids are thought to contribute to postpartum disorders. Low postpartum DHEA concentrations have been linked to mood instability. Also, allopregnanolone levels correlated with mood disorders during pregnancy and in premenstrual syndrome (PMS). It has been noted that women with the premenstrual dysphoric disorder have higher allopregnanolone/progesterone ratios than normal controls; women treated for this disorder reported

improvement as allopregnanolone levels decreased.

Neurosteroids, Memory Disorders, and Aging. Neurosteroid levels may be irregular in neurodegenerative disorders and aging conditions such as Alzheimer disease and Parkinson disease. DHEA levels at age 70 are only about 20 percent of their maximum value recorded in the late 20s, and some researchers believe DHEA supplementation can prevent or slow the cognitive declines associated with the aging process. However, conflicting studies have indicated that DHEA administration does not improve cognitive measures in patients. Also, in patients with Alzheimer disease, DHEA concentrations are markedly decreased.

PSYCHONEUROENDOCRINOLOGY

The term *psychoneuroendocrinology* encompasses the structural and functional relationships between hormonal systems and the CNS and behaviors that modulate and are derived from both. Classically, *hormones* have been defined as the products of endocrine glands transported by the blood to exert their action at sites distant from their release. Advances in neuroscience have shown, however, that in the CNS, the brain not only serves as a target site for regulatory control of hormonal release but also has secretory functions of its own and serves as an end-organ for some hormonal actions. These complex interrelationships make classic distinctions between the origin, structure, and function of neurons and those of endocrine cells dependent on physiologic context.

Hormone Secretion

Hormone secretion is stimulated by the action of a neuronal secretory product of neuroendocrine transducer cells of the hypothalamus. Examples of hormone regulators (Table 33-6) include corticotropin-releasing hormone (CRH), which stimulates adrenocorticotropin (adrenocorticotropic hormone [ACTH]); TRH, which stimulates the release of TSH; gonadotropin-releasing hormone (GnRH), which stimulates the release of luteinizing hormone (LH) and follicle-stimulating hormone (FSH); and somatostatin (somatotropin release-inhibiting factor [SRIF]) and growth-hormone-releasing hormone (GHRH), which influence growth hormone (GH) release. Chemical signals

cause the release of these neurohormones from the median eminence of the hypothalamus into the portal hypophyseal bloodstream and subsequent transport to the pituitary to regulate the release of target hormones. Pituitary hormones, in turn, act directly on target cells (e.g., ACTH on the adrenal gland) or stimulate the release of other hormones from peripheral endocrine organs. Also, these hormones have feedback actions that regulate secretion and exert neuromodulatory effects in the CNS.



Table 33-6
Examples of Regulating Hormones

Regulating Hormone	Hormone Stimulated (or Inhibited)
Corticotropin-releasing hormone	Adrenocorticotropic hormone
Thyrotropin-releasing hormone	Thyroid-stimulated hormone
Luteinizing hormone-releasing hormone	Luteinizing hormone
Gonadotropin-releasing hormone	Follicle-stimulating hormone
Somatostatin	Growth hormone (inhibited)
Growth hormone-releasing hormone	Growth hormone
Progesterone, oxytocin	Prolactin
Arginine vasopressin	Adrenocorticotropic hormone

From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:162.



Table 33-7
Classifications of Hormones

Structure	Examples	Storage	Lipid Soluble
Proteins,	ACTH, β-endorphin,	Vesicles	No

polypeptides, glycoproteins	TRH, LH, FSH		
Steroids, steroid-like compounds	Cortisol, estrogen, thyroxine	Diffusion after synthesis	Yes
Functions			
Autocrine	Self-regulatory effects		
Paracrine	Local or adjacent cellular action		
Endocrine	Distant target site		

ACTH, adrenocorticotropic hormone; TRH, thyrotropin-releasing hormone; LH, luteinizing hormone; FSH, follicle-stimulating hormone.

Courtesy of Victor I Reus, M.D. and Sydney Frederick-Osborne, Ph.D.

Hormones are divided into two general classes: (1) proteins, polypeptides, and glycoproteins, and (2) steroids and steroid-like compounds (Table 33-7); these are secreted by an endocrine gland into the bloodstream and are transported to their sites of action.

Developmental Psychoneuroendocrinology

Hormones can have both organizational and activational effects. Exposure to gonadal hormones during critical stages of neural development directs changes in brain morphology and function (e.g., sex-specific behavior in adulthood). Similarly, thyroid hormones are essential for the normal development of the CNS, and thyroid deficiency during critical stages of postnatal life will severely impair growth and development of the brain, resulting in behavioral disturbances that may be permanent if replacement therapy is not instituted.

Endocrine Assessment

Neuroendocrine function can be studied by assessing baseline measures and by measuring the response of the axis to some neurochemical or hormonal challenge. The first method has two approaches. One approach is to measure

a single time point—for example, morning levels of growth hormone; this approach is subject to significant error because of the pulsatile nature of the release of most hormones. The second approach is to collect blood samples at multiple points or to collect 24-hour urine samples; these measurements are less susceptible to significant errors. The best approach, however, is to perform a neuroendocrine challenge test, in which the person is given a drug or a hormone that perturbs the endocrine axis in some standard way. Persons with no disease show much less variation in their responses to such challenge studies than in their baseline measurements.

Hypothalamic–Pituitary–Adrenal Axis

Since the earliest conceptions of the stress response by Hans Selye and others, investigation of HPA function has occupied a central position in psychoendocrine research. CRH, ACTH, and cortisol levels all rise in response to a variety of physical and psychic stresses and serve as prime factors in maintaining homeostasis and developing adaptive responses to novel or challenging stimuli. The hormonal response depends on the characteristics of the stressor and how one assesses and copes with it. Aside from generalized effects on arousal, distinct effects on sensory processing, stimulus habituation and sensitization, pain, sleep, and memory storage and retrieval have been documented. In primates, social status can influence adrenocortical profiles and, in turn, be affected by exogenously induced changes in hormone concentration.

Pathologic alterations in HPA function have been associated primarily with mood disorders, PTSD, and dementia of the Alzheimer type, although recent animal evidence points toward the role of this system in substance use disorders as well. Disturbances of mood are found in more than 50 percent of patients with Cushing syndrome (characterized by elevated cortisol concentrations), with psychosis or suicidal thought apparent in more than 10 percent of patients studied. Cognitive impairments similar to those seen in major depressive disorder (principally in visual memory and higher cortical functions) are common and relate to the severity of the hypercortisolemia and possible reduction in hippocampal size. In general, reduced cortisol levels normalize mood, and mental status. Conversely, in Addison disease (characterized by adrenal insufficiency), apathy, social withdrawal, impaired

sleep, and decreased concentration frequently accompany prominent fatigue. Replacement of glucocorticoid (but not of electrolyte) resolves behavioral symptomatology.

Similarly, HPA abnormalities are reversed in persons who are treated successfully with antidepressant medications. Failure to normalize HPA abnormalities is a poor prognostic sign. Alterations in HPA function associated with depression include elevated cortisol concentrations, failure to suppress cortisol in response to dexamethasone, increased adrenal size and sensitivity to ACTH, a blunted ACTH response to CRH, and, possibly, elevated CRH concentrations in the brain.

Hypothalamic–Pituitary–Gonadal Axis

The gonadal hormones (progesterone, androstenedione, testosterone, estradiol, and others) are steroids that are secreted principally by the ovary and testes, but significant amounts of androgens arise from the adrenal cortex as well. The prostate gland and adipose tissue, also involved in the synthesis and storage of dihydrotestosterone, contribute to individual variance in sexual function and behavior.

The timing and presence of gonadal hormones play a critical role in the development of sexual dimorphisms in the brain. Developmentally, these hormones direct the organization of many sexually dimorphic CNS structures and functions, such as the size of the hypothalamic nuclei and corpus callosum, neuronal density in the temporal cortex, the organization of language ability, and responsivity in Broca motor speech area. Women with congenital adrenal hyperplasia—a deficiency of the enzyme 21-hydroxylase, which leads to high exposure to adrenal androgens in prenatal and postnatal life, in some studies—are more aggressive and assertive and less interested in traditional female roles than female control subjects. Sexual dimorphisms may also reflect acute and reversible actions of relative steroid concentrations (e.g., higher estrogen levels transiently increase CNS sensitivity to serotonin).

Testosterone. Testosterone is the primary androgenic steroid, with both androgenic (i.e., facilitating linear body growth) and somatic growth functions. Testosterone is associated with increased violence and aggression in animals and correlation studies in humans, but anecdotal reports of

increased aggression with testosterone treatment have not been substantiated in investigations in humans. In hypogonadal men, testosterone improves mood and decreases irritability. Varying effects of anabolic–androgenic steroids on mood have been noted anecdotally. A prospective, placebo-controlled study of anabolic–androgenic steroid administration in normal subjects reported positive mood symptoms, including euphoria, increased energy, and sexual arousal, in addition to increases in the negative mood symptoms of irritability, mood swings, violent feelings, anger, and hostility.

Testosterone is essential for sexual desire in both men and women. In males, muscle mass and strength, sexual activity, desire, thoughts, and intensity of sexual feelings depend on normal testosterone levels, but these functions are not augmented by supplemental testosterone in those with normal androgen levels. Adding small amounts of testosterone to routine hormonal replacement in postmenopausal women has proved, however, to be as beneficial as its use in hypogonadal men.

Dehydroepiandrosterone. DHEA and DHEA sulfate (DHEA-S) are adrenal androgens secreted in response to ACTH and represent the most abundant circulating steroids. DHEA is also a neurosteroid that is synthesized in situ in the brain. DHEA has many physiologic effects, including a reduction in neuronal damage from glucocorticoid excess and oxidative stress. Behavioral interest has centered on its possible involvement in memory, mood, and several psychiatric disorders. *Adrenarche* is the prepubertal onset of adrenal production of DHEA-S and may play a role in human maturation through increasing the activity of the amygdala and hippocampus and promoting synaptogenesis in the cerebral cortex. DHEA has been shown to act as an excitatory neurosteroid and to enhance memory retention in mice, but studies of DHEA administration to humans have not consistently shown any improvement in cognition. Several trials of DHEA administration point to an improvement in well-being, mood, energy, libido, and functional status in depressed individuals. Administration of DHEA to women with adrenal insufficiency (e.g., Addison disease) has repeatedly been demonstrated to enhance mood, energy, and sexual function; effects in men remain to be assessed. Mood, fatigue, and libido improved in human immunodeficiency virus (HIV)-positive patients treated with DHEA in one study, and DHEA and DHEA-S are inversely correlated with severity in

ADHD. Women diagnosed with fibromyalgia have significantly decreased DHEA-S levels, but supplementation does not improve outcomes. Several cases of possible DHEA-induced mania have been reported, and DHEA has been reported to be inversely related to extrapyramidal symptoms (EPS) in patients with schizophrenia who are treated with antipsychotics. DHEA administration, in these cases, improves EPS.

Double-blind treatment studies have shown antidepressant effects of DHEA in patients with major depression, midlife-onset dysthymia, and schizophrenia, although beneficial effects on memory have not been reliably demonstrated. A small, double-blind trial of DHEA treatment of Alzheimer disease failed to reveal significant benefit, although a near-significant improvement in cognitive function was seen after 3 months of treatment.

Animal studies suggest that DHEA may be involved in eating behavior, aggressiveness, and anxiety as well, with its effects resulting from its transformation into estrogen, testosterone, or androsterone from its antiglucocorticoid activity, or direct effects on $GABA_A$, NMDA, and σ receptors. Because of the putative antiglucocorticoid effects, the ratio of cortisol to DHEA levels may be particularly crucial in understanding adaptive responses to stress. Both cortisol and DHEA appear to be involved in fear conditioning, with the cortisol/DHEA ratio hypothesized to be an index of the degree to which an individual is buffered against the adverse effects of stress. This ratio is related to some measures of psychopathology and response to treatment, predicting the persistence of the first episode major depression and being related to the degree of depression, anxiety, and hostility in patients with schizophrenia and response to antipsychotic treatment. Patients with PTSD have higher DHEA levels and lower cortisol/DHEA ratios related to symptom severity, suggesting a role in PTSD recovery. Fear-potentiated startle is more extensive in individuals with high as compared to low cortisol/DHEA-S ratios and is positively associated with cortisol and negatively with DHEA-S. More significant DHEA response to ACTH is related to lower PTSD ratings, and the cortisol/DHEA ratio to negative mood symptoms. A genetic variation in an ACTH receptor promoter has been found to influence DHEA secretion in response to dexamethasone and may underlie some individual differences in stress response.

Estrogen and Progesterone. Estrogens can influence neural activity in

the hypothalamus and limbic system directly through the modulation of neuronal excitability, and they have complex multiphasic effects on nigrostriatal dopamine receptor sensitivity. Accordingly, evidence indicates that the antipsychotic effect of psychiatric drugs can change over the menstrual cycle and that the risk of tardive dyskinesia depends partly on estrogen concentrations. Several studies have suggested that gonadal steroids modulate spatial cognition and verbal memory and are involved in impeding age-related neuronal degeneration. Increasing evidence also suggests that estrogen administration decreases the risk and severity of dementia of the Alzheimer type in postmenopausal women. Estrogen has mood-enhancing properties and can also increase sensitivity to serotonin, possibly by inhibiting monoamine oxidase. In animal studies, long-term estrogen treatment results in a decrease in serotonin 5-HT_1 receptors and an increase in 5-HT_2 receptors. In oophorectomized women, significant reductions in tritiated imipramine binding sites (which indirectly measure presynaptic serotonin uptake) were restored with estrogen treatment.

The association of these hormones with serotonin is hypothetically relevant to mood change in premenstrual and postpartum mood disturbances. In premenstrual dysphoric disorder, a constellation of symptoms resembling major depressive disorder occurs in most menstrual cycles, appearing in the luteal phase and disappearing within a few days of the onset of menses. No definitive abnormalities in estrogen or progesterone levels have been demonstrated in women with the premenstrual dysphoric disorder, but decreased serotonin uptake with premenstrual reductions in steroid levels has been correlated with the severity of some symptoms.

Most psychological symptoms associated with the menopause are reported during perimenopause rather than after complete cessation of menses. Although studies suggest no increased incidence of major depressive disorder, reported symptoms to include worry, fatigue, crying spells, mood swings, diminished ability to cope, and diminished libido or intensity of orgasm. Hormone replacement therapy (HRT) is effective in preventing osteoporosis and reinstating energy, a sense of well-being, and libido; however, its use is very controversial. Studies have shown that combined estrogen–progestin drugs (e.g., Premarin) cause small increases in breast cancer, heart attack, stroke, and blood clots among menopausal women.

Studies of the effects of estrogen alone in women who have had hysterectomies (because estrogen alone increases the risk for uterine cancer) are ongoing.

Hypothalamic–Pituitary–Thyroid Axis

Thyroid hormones are involved in the regulation of nearly every organ system, mainly those integral to the metabolism of food and the regulation of temperature, and are responsible for the optimal development and function of all body tissues. Also, to its prime endocrine function, TRH has direct effects on neuronal excitability, behavior, and neurotransmitter regulation.

Thyroid disorders can induce virtually any psychiatric symptom or syndrome, although no consistent associations of specific syndromes and thyroid conditions are found. Hyperthyroidism is commonly associated with fatigue, irritability, insomnia, anxiety, restlessness, weight loss, and emotional lability; marked impairment in concentration and memory may also be evident. Such states can progress into delirium or mania, or they can be episodic. On occasion, a real psychosis develops, with paranoia as a frequent presenting feature. In some cases, psychomotor retardation, apathy, and withdrawal are the presenting features rather than agitation and anxiety. Symptoms of mania have also been reported following the rapid normalization of thyroid status in hypothyroid individuals and may covary with thyroid level in individuals with episodic endocrine dysfunction. In general, behavioral abnormalities resolve with normalization of thyroid function and respond symptomatically to traditional psychopharmacological regimens.

The psychiatric symptoms of chronic hypothyroidism are generally well recognized (Fig. 33-27). Classically, fatigue, decreased libido, memory impairment, and irritability are noted, but an actual secondary psychotic disorder or dementia-like state can also develop. Suicidal ideation is common, and the lethality of actual attempts is profound. In milder, subclinical states of hypothyroidism, the absence of gross signs accompanying endocrine dysfunction can result in its being overlooked as a possible cause of a mental disorder.

Growth Hormone

Growth hormone deficiencies interfere with growth and delay the onset of puberty. Low GH levels can result from a stressful experience. Administration of GH to individuals with GH deficiency benefits cognitive function in addition to its more obvious somatic effects, but evidence indicates poor psychosocial adaptation in adulthood for children who were treated for GH deficiency. A significant percentage of patients with major depressive disorder and dysthymic disorder may have a GH deficiency. Some prepubertal and adult patients with diagnoses of major depressive disorder exhibit hyposecretion of GHRH during an insulin tolerance test, a deficit that has been interpreted as reflecting alterations in both cholinergic and serotonergic mechanisms. Several GH abnormalities have been noted in patients with anorexia nervosa. Secondary factors, such as weight loss, however, in both major depressive disorder and eating disorders, may be responsible for alterations in the endocrine release.

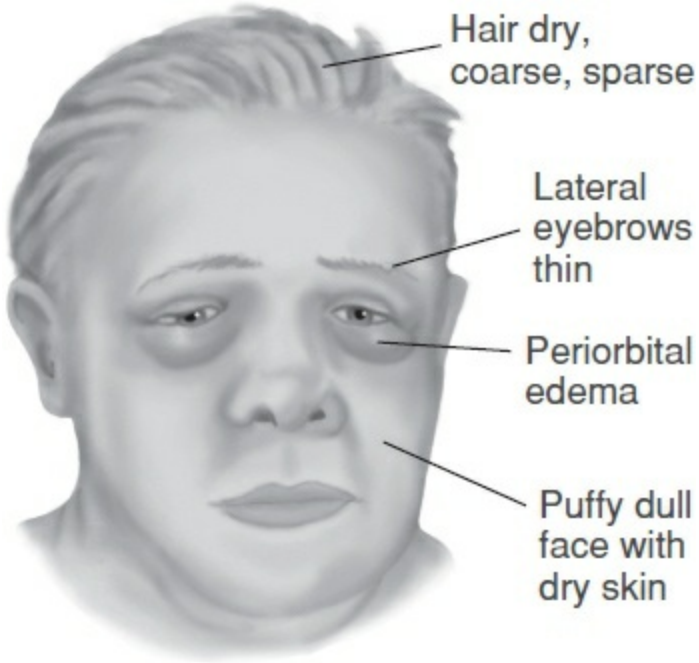


FIGURE 33-27
Myxedema: The patient with severe hypothyroidism (myxedema) has a dull, puffy facies. The edema, often particularly pronounced around the eyes, does not pit with pressure. The hair and eyebrows are dry, coarse, and thinned. The skin is dry. (From Bickley LS, Szilagyi P. *Bates' Guide to Physical Examination and History Taking*, 8th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2003.)

Nonetheless, at least one study has reported that GHRH stimulates food

consumption in patients with anorexia nervosa and lowers food consumption in patients with bulimia. Administration of GH to older men increases lean body mass and improves vigor. GH is released in pulses throughout the day, but the pulses are closer together during the first hours of sleep than at other times.

Prolactin

Since its identification in 1970, the anterior pituitary hormone prolactin has been examined as a potential index of dopamine activity, dopamine receptor sensitivity, and antipsychotic drug concentration in studies of CNS function in psychiatric patients and as a correlate of stress responsivity. The secretion of prolactin is under direct inhibitory regulation by dopamine neurons located in the tuberoinfundibular section of the hypothalamus and is, therefore, increased by classical antipsychotic medications. Prolactin also inhibits its secretion using a short-loop feedback circuit to the hypothalamus. Also, a significant number of prolactin-releasing or prolactin-modifying factors have been identified, including estrogen, serotonin (mainly through the 5-HT_2 and 5-HT_3 receptors), norepinephrine, opioids, TRH, T_4, histamine, glutamate, cortisol, CRH, and oxytocin, with interaction effects possible. For example, estrogen may promote the serotonin-stimulated release of prolactin.

Prolactin is primarily involved in reproductive functions. During maturation, prolactin secretion participates in gonadal development, whereas, in adults, prolactin contributes to the regulation of the behavioral aspects of reproduction and infant care, including estrogen-dependent sexual receptivity and breast-feeding. In female rats, prolactin secretion is strongly stimulated with exposure to pups. In women, basal prolactin levels are elevated in the postpartum period before weaning, and prolactin release is stimulated by suckling. Hyperprolactinemia is associated with low testosterone in men and reduced libido in men and women. In rodents, prolactin level is increased along with corticosterone in response to such stressful stimuli as immobilization, hypoglycemia, surgery, and cold exposure and may be specifically associated with the use of passive coping in the face of a stressor. Prolactin promotes various stress-related behaviors in rats, depending on the condition, such as increasing object-directed exploration while decreasing other exploration.

Patients with hyperprolactinemia often complain of depression, decreased libido, stress intolerance, anxiety, and increased irritability. These behavioral symptoms usually resolve in parallel with decrements in serum prolactin when surgical or pharmacologic treatments are used. In psychotic patients, prolactin concentrations and prolactin-related sexual disturbances have been positively correlated with the severity of tardive dyskinesia. Prolactin levels are also positively correlated with adverse symptoms in schizophrenia.

Melatonin

Melatonin, a pineal hormone, is derived from the serotonin molecule and it controls photoperiodically mediated endocrine events (particularly those of the hypothalamic–pituitary–gonadal axis). It also modulates immune function, mood, and reproductive performance and is a potent antioxidant and free-radical scavenger. Melatonin has a depressive effect on CNS excitability, is an analgesic, and has seizure-inhibiting effects in animal studies. Melatonin can be a useful therapeutic agent in the treatment of circadian phase disorders such as jet lag. Intake of melatonin increases the speed of falling asleep, as well as its duration and quality.

Oxytocin

Oxytocin, also a posterior pituitary hormone, is involved in osmoregulation, the milk ejection reflex, food intake, and female maternal and sexual behaviors. Oxytocin may be released during orgasm, more so in women than in men, and promote bonding between the sexes. It has been used in autistic children experimentally in an attempt to increase socialization.

Insulin

Increasing evidence indicates that insulin may be integrally involved in learning and memory. Insulin receptors occur in high density in the hippocampus and are thought to help neurons metabolize glucose. Patients with Alzheimer disease have lower insulin concentrations in the CSF than controls, and both insulin and glucose dramatically improve verbal memory. Depression is frequent in patients with diabetes, as are indexes of impaired hormonal response to stress. It is not known whether these findings represent the direct effects of the disease or are secondary, nonspecific effects. Some

antipsychotics are known to dysregulate insulin metabolism.

Endocrine Variables in Psychiatric Disorders

Although it is clear that alterations in endocrine regulation are involved in the pathophysiology and treatment responses of many psychiatric disorders, incorporating these findings into clinical diagnostic assessment, and decision-making remains problematic. Large-scale longitudinal or cost-effectiveness studies are rare, despite indications that baseline alterations in glucocorticoid regulation and thyroid status (two of the best-studied abnormalities) may be useful in subtyping psychiatric disorders and prediction of outcome. Alterations in HPA/stress regulation underlie several psychiatric diagnoses and may serve as complementary independent variables in assigning treatment response and course of illness to the classical behavioral categories that have thus far defined psychiatric practice. Studying genetic polymorphisms in factors regulating hormonal response may help us better understand the influence of hormonal variability on the illness and also possible underlying differences in the illness as reflected in these genetic subtypes.

IMMUNE SYSTEM AND CENTRAL NERVOUS SYSTEM INTERACTIONS

Interactions between the immune system and the CNS play a critical role in the maintenance of bodily homeostasis and the development of diseases, including psychiatric disease. Alterations in CNS function brought about by a variety of stressors have been shown to influence both the immune system as well as diseases that involve the immune system. Moreover, many of the relevant hormonal and neurotransmitter pathways that mediate these effects have been elucidated. Of considerable interest is accumulating data that cytokines, which derive from immune cells and microglia, have profound effects on the CNS. The relative role of cytokines and their signaling pathways in the various psychiatric diseases is an area of active investigation, as is the role of infectious and autoimmune diseases in the pathophysiology of psychiatric disorders. These findings highlight the importance of interdisciplinary efforts involving the neurosciences and immunology for gaining new insights into the etiology of psychiatric disorders.

Overview of the Immune System

The immune system can protect the body from the invasion of foreign pathogens, such as viruses, bacteria, fungi, and parasites. Also, the immune system can detect and eliminate cells that have become neoplastically transformed. These functions are accomplished through highly specific receptors on immune cells for molecules derived from invading organisms and a rich intercellular communication network that involves direct cell-to-cell interactions and signaling between cells of the immune system by soluble factors called *cytokines.* The body's absolute dependence on the efficient functioning of the immune system is illustrated by the less than 1-year survival rate of untreated infants born with severe combined immunodeficiency disease and the devastating opportunistic infections and cancers that arise during untreated acquired immunodeficiency syndrome (AIDS).

Behavioral Conditioning

The fact that learning processes are capable of influencing immunologic function is an example of interactions between the immune system and the nervous system. Several classical conditioning paradigms have been associated with suppression or enhancement of the immune response in various experimental designs. The conditioning of immunologic reactivity provides further evidence that the CNS can have significant immunomodulatory effects.

Some of the first evidence for immunologic conditioning was derived from the serendipitous observation that animals undergoing extinction in a taste-aversion paradigm with cyclophosphamide, an immunosuppressive agent, had unexpected mortality. In that taste-aversion paradigm, animals were simultaneously exposed to an oral saccharin solution (the conditioned stimulus) and an intraperitoneal injection of cyclophosphamide (unconditioned stimulus). Because the animals experienced considerable physical discomfort from the cyclophosphamide injection, through the process of conditioning, they began to associate the ill effects of cyclophosphamide with the taste of the oral saccharin solution. If given a choice, the animals avoided the saccharin solution (taste aversion). Conditioned avoidance can be eliminated or extinguished if the saccharin is

repeatedly presented in the absence of cyclophosphamide. However, it was observed that animals undergoing extinction of cyclophosphamide-induced taste aversion died unexpectedly, leading to the speculation that the oral saccharin solution had a specific conditioned association with the immunosuppressive effects of cyclophosphamide. Repeated exposure to the saccharin-associated conditioned immunosuppression during extinction might explain the unexpected death of animals. To test that hypothesis, researchers conditioned the animals with saccharin (conditioned stimulus) and intraperitoneal cyclophosphamide (unconditioned conditioned stimulus) and then immunized them with sheep red blood cells. At different times after immunization, the conditioned animals were re-exposed to saccharin (conditioned stimulus) and examined. The conditioned animals exhibited a significant decrease in mean antibody titers to sheep red blood cells when compared to the control animals. Thus, the evidence demonstrated that immunosuppression of humoral immunity was occurring in response to the conditioned stimulus of saccharin alone.

Stress and the Immune Response

Interest in the effects of stress on the immune system grew out of a series of animal and human studies suggesting that stressful stimuli can influence the development of immune-related disorders, including infectious diseases, cancer, and autoimmune disorders. Although stress has been historically associated with suppression of immune function, recent data indicate that such a conclusion oversimplifies the complexities of the mammalian immune response to environmental perturbation and that stress may also activate certain aspects of the immune system, particularly the innate immune response.

Stress and Illness. Experiments conducted on laboratory animals in the late 1950s and the early 1960s indicated that a wide variety of stressors—including isolation, rotation, crowding, exposure to a predator, and electric shock—increased morbidity and mortality in response to several types of tumors and infectious diseases caused by viruses and parasites. However, as research progressed, it became increasingly clear that “stress” is too variegated a concept to have singular effects on immunity and that the effects

of stress on immunity depend on several factors. Chief among these factors is whether a stressor is acute or chronic. Other critical variables include stressor severity and type, as well as the timing of stressor application and the type of tumor or infectious agent investigated. For example, mice subjected to electric grid shock 1 to 3 days before the infection of Maloney murine sarcoma virus-induced tumor cells exhibited decreased tumor size and incidence. In contrast, mice exposed to grid shock 2 days after tumor cell injection exhibited an increase in tumor size and number.

The relevance of the effects of stress on immune-related health outcomes in humans has been demonstrated in studies that have shown an association between chronic stress and increased susceptibility to the common cold, reduced antibody responses to vaccination, and delayed wound healing. Also, stress, as well as depression, through their effects on inflammation, has been linked to increased morbidity and mortality in infectious diseases, such as HIV infection, autoimmune disorders, neoplastic diseases, as well as diabetes and cardiovascular disorders, which are increasingly being recognized as diseases in which the immune system, inflammation, in particular, plays a pivotal role (Fig. 33-28).

Effects of Chronic Stress. When challenged with a medical illness or chronic psychological stressor, complex interactions between the immune and nervous systems promote a constellation of immune-induced behavioral changes, alternatively referred to as “sickness syndrome” or “sickness behavior.” These behavioral changes include dysphoria, anhedonia, fatigue, social withdrawal, hyperalgesia, anorexia, altered sleep–wake patterns, and cognitive dysfunction. Although seen in response to infection, the full syndrome can be reproduced in humans and laboratory animals by the administration of innate immune cytokines. Blocking cytokine activity diminishes or prevents the development of sickness behavior in laboratory animals, even when such behavior develops as a result of psychological stress. Evidence that cytokine-induced behavioral toxicity is related to significant depression comes in part from studies showing that in humans and laboratory animals, antidepressants can abolish or attenuate the development of sickness behavior in response to cytokine administration.

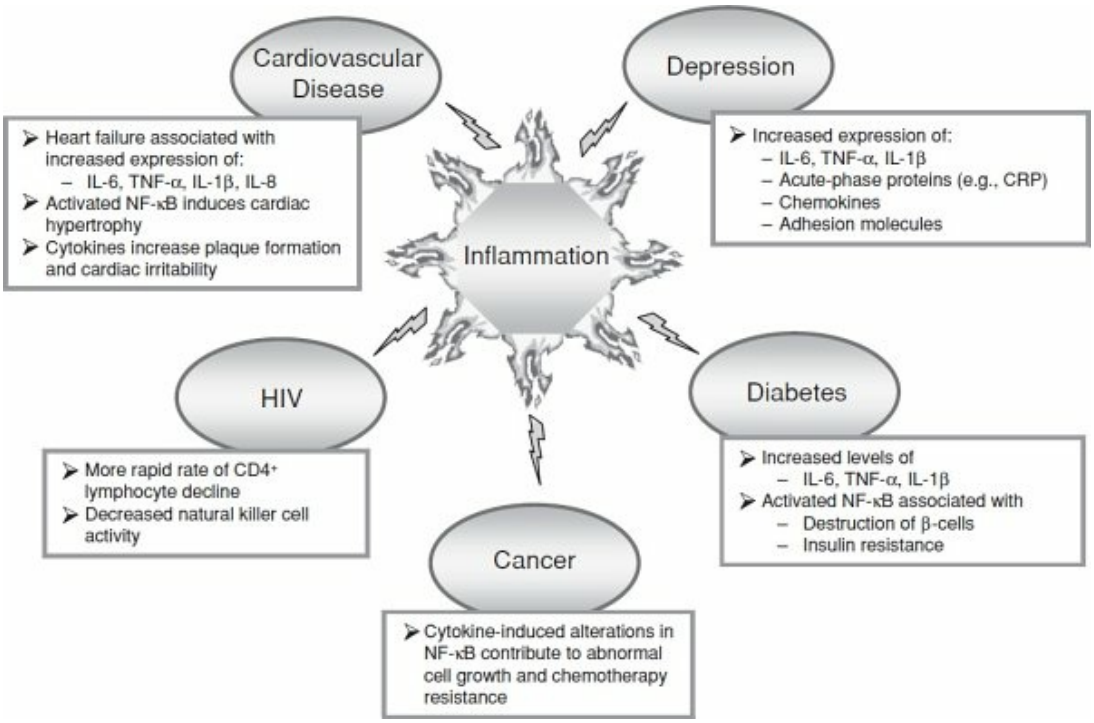


FIGURE 33-28
Inflammation and disease. IL, interleukin; TNF, tumor necrosis factor; NF-kB, nuclear factor κB; C RP, C-reactive protein. (From Cowles MK, Miller AH. Stress cytokines and depressive illness. In: Squire LR, ed. *The New Encyclopedia of Neuroscience*. Academic Press; 2009:521, with permission.)

Relevance of Immune System–CNS Interactions to Psychiatric Disorders

Major Depression. The neuropsychiatric disorder that has been best characterized in terms of the influence of the brain on the immune system and vice versa is major depression. For many years major depression was seen as the quintessential example of how stress-related disorders may decrease immunocompetence. More recently, however, it has become evident that stress also activates inflammatory pathways, even while suppressing measures of acquired immunity. Not surprisingly, studies now indicate that, in addition to immunosuppression, major depression is also frequently associated with inflammatory activation. Recent research showing that proinflammatory cytokines are capable of suppressing many of the immune measures examined in major depression may provide a mechanism to account for how chronic stress-induced inflammatory activity may give rise to depression-related suppression of in vitro functional assays, such as

lymphocyte proliferation.

Bipolar Disorder. Patients with bipolar disorder evince many of the immune alterations frequently observed in the context of unipolar depression. Several studies have observed that bipolar patients, particularly when manic, demonstrate increased plasma concentrations of inflammatory cytokines. Other studies indicate that treatments for mania, such as lithium, lower plasma concentrations of several cytokines. Of interest, the available literature seems to suggest that patients in the manic phase of the disorder may be more likely than depressed patients to demonstrate increased inflammatory markers. It should not be surprising that mania—which seems the phenomenologic opposite of depression—should be associated with increased inflammation, given that mania and depression have also been reported to show identical neuroendocrine and autonomic abnormalities, such as dexamethasone non suppression and increased sympathetic activity, both of which would be expected to promote inflammatory activity.

Schizophrenia. There has been growing interest in the idea that infectious agents, particularly viruses, may underlie at least some cases of schizophrenia. Although it is well established that viral encephalitis can present clinically as psychosis, the primary focus of the "viral hypothesis" for schizophrenia has been on infections during neurodevelopment, given its congruence with the emerging consensus that prenatal or early postnatal insult is implicated in the causality of schizophrenia. Several lines of indirect evidence suggest that viral infection during CNS development may be involved in the pathogenesis of schizophrenia. The data include: (1) an excess number of patient births in the late winter and early spring, suggesting possible exposure to viral infection in utero during the fall and winter peak of viral illnesses, (2) an association between exposure to viral epidemics in utero and the later development of schizophrenia, (3) higher prevalence of schizophrenia in crowded urban areas, which have conditions that are particularly conducive to the transmission of viral pathogens, and (4) seroepidemiologic studies indicating a higher infection rate for certain viruses in schizophrenia patients or their mothers.

Also, schizophrenia has been associated with indices of immune activation, including elevations in cytokines. Although these immune findings in

patients with schizophrenia may indicate evidence of immune system activation secondary to an infection, it should be noted that they might also indicate that an autoimmune process is involved in the disorder. Despite the plethora of studies pointing to abnormalities in cellular and humoral immunity in schizophrenia, the data have not been uniform or conclusive, and there is a need for more studies that account for confounding variables such as medication status and tobacco use. Moreover, attempts to isolate infectious agents from schizophrenic brain tissue or to detect viral nucleic acids in the CNS or peripheral blood of patients with schizophrenia have generally yielded negative results.

Because the initial neuronal abnormalities in schizophrenia have been proposed to arise during neurodevelopment, a perinatal viral infection could insidiously disrupt development and then be cleared by the immune system before clinical diagnosis. In such a scenario, host factors such as cytokines could be responsible for causing the developmental abnormality by interacting with growth factors or adhesion molecules. Recent animal models have identified that maternal immune activation with resultant production of interleukin 6 (IL-6) critically affects behavioral and transcriptional changes in offspring. Behavioral changes, including deficits in prepulse inhibition and latent inhibition, are consistent with behavioral abnormalities in animal models of both schizophrenia and autism. Various animal models using influenza virus, Borna disease virus, or lymphocytic choriomeningitis virus in rodents have demonstrated that prenatal or postnatal viral infections can lead to neuroanatomical or behavioral alterations that are somewhat reminiscent of schizophrenia in humans. As mentioned earlier, epidemiologic studies also support the link between infection with a teratogenic virus and the development of psychotic disorders later in life. Associations have been observed between maternal infection with rubella or influenza during gestation and the development of a schizophrenia spectrum disorder in the offspring. Similarly, maternal antibodies to the herpes simplex virus that develops during pregnancy are correlated with increased rates of psychosis during adulthood in the offspring.

Non-HIV retroviruses might also play a role in the pathogenesis of schizophrenia. Retroviruses integrate into host DNA and can disrupt the function of adjacent genes. Moreover, the genomes of all humans contain sequences of “endogenous retroviruses” that hold the capacity to alter the

transcriptional regulation of host genes. If genes controlling the development or function of the brain undergo transcriptional disruption by retroviral effects, then this might lead to a cascade of biochemical abnormalities, eventually giving rise to schizophrenia.

Autism. Although a convincing case can be made for a significant immune component in autism, the relationship of immune abnormalities to the neurobehavioral symptoms of the disease remains controversial. Recent epidemiologic studies have not substantiated the claim that childhood vaccines trigger autism, and immune-based therapies for autism have not been reliably effective. Thus, although it is tempting to speculate that the immune system holds a clue to a cure for autism, there is currently not enough data to determine whether immune anomalies cause autism, are *caused by* autism or are just adventitiously associated with the disease.

Alzheimer Disease. Although Alzheimer disease is not considered an inflammatory disease primarily, emerging evidence indicates that the immune system may contribute to its pathogenesis. The discovery that amyloid plaques are associated with acute-phase proteins, such as complement proteins and C-reactive protein, suggests the possibility of an ongoing immune response. The idea that inflammatory processes are involved in Alzheimer disease has been bolstered by recent studies showing that long-term use of nonsteroidal anti-inflammatory drugs (NSAIDs) is negatively correlated with the development of Alzheimer disease.

HIV AIDS. AIDS is an immunologic disease associated with a variety of neurologic manifestations, including dementia. HIV encephalitis results in synaptic abnormalities and loss of neurons in the limbic system, basal ganglia, and neocortex.

Multiple Sclerosis. Multiple sclerosis (MS) is a demyelinating disease characterized by disseminated inflammatory lesions of white matter. Considerable progress has been made in elucidating the immunopathology of myelin destruction that occurs in MS and the animal model for the disease, experimental allergic encephalomyelitis. Although the initial step in lesion formation has not been determined, disruption of the BBB and infiltration of

T cells, B cells, plasma cells, and macrophages appear to be associated with lesion formation.

Other Disorders. Finally, several disorders are seen in which neural-immune interactions are suspected but not well documented. *Chronic fatigue syndrome* is an illness with a controversial etiology and pathogenesis. Also, to persistent fatigue, symptoms frequently include depression and sleep disturbances. Tests of immune function have found indications of both immune activation and immunosuppression. Neuroendocrine assessments indicate that patients with chronic fatigue syndrome may be hypocortisolemic because of impaired activation of the HPA axis. Although an acute viral infection frequently precedes the onset of chronic fatigue syndrome, no infectious agent has been causally associated with it.

In contrast, *Lyme disease,* in which sleep disturbances and depression are also common, is caused by infection with the tick-borne spirochete *Borrelia burgdorferi,* which can invade the CNS and cause encephalitis and neurologic symptoms. Lyme disease is remarkable because it appears to produce a spectrum of neuropsychiatric disorders, including anxiety, irritability, obsessions, compulsions, hallucinations, and cognitive deficits. Immunopathology of the CNS may be involved because symptoms can persist or reappear even after a lengthy course of antibiotic treatment, and the spirochete is frequently difficult to isolate from the brain. *Gulf War syndrome* is a controversial condition with inflammatory and neuropsychiatric features. The condition has been attributed variously to combat stress, chemical weapons (e.g., cholinesterase inhibitors), infections, and vaccines. Given the impact of stress on neurochemistry and immune responses, these pathogenic mechanisms are not mutually exclusive.

Therapeutic Implications

The bidirectional nature of CNS–immune system interactions implies the therapeutic possibility that agents known to alter stress system activity positively might benefit immune functioning and, conversely, that agents that modulate immune functioning may be of potential benefit in the treatment of neuropsychiatric disturbance, especially in the context of medical illness. Increasing evidence supports both hypotheses.

Antidepressants and the Immune System. Emerging data indicate that in animals and humans, antidepressants attenuate or abolish behavioral symptoms induced by inflammatory cytokine exposure. For example, pretreatment of rats with either imipramine or fluoxetine (a TCA and SSRI, respectively) for 5 weeks before endotoxin administration significantly attenuated endotoxin-induced decrements in saccharine preference (commonly accepted as a measure for anhedonia), as well as weight loss, anorexia, and reduced exploratory, locomotor, and social behavior. Similarly, several studies in humans suggest that antidepressants can ameliorate mood disturbances in the context of chronic cytokine therapies, especially if given prophylactically before cytokine exposure. For example, the SSRI paroxetine significantly decreased the development of major depression in patients receiving high doses of interferon-α (IFN-α) for malignant melanoma.

Behavioral Interventions and Immunity. It has been known for years that psychosocial factors can mitigate or worsen the effects of stress, not only on immune functioning but also on the long-term outcomes of medical conditions in which the immune system is known to play a role. Therefore, behavioral interventions aimed at maximizing protective psychosocial factors might be predicted to have a beneficial effect, not only in terms of mitigating the effect of stress on immune functioning but perhaps also on diminishing emotional disturbances that arise in the context of immune system dysregulation.

Two factors that have been repeatedly identified as protective against stress-induced immune alterations are social support and the ability to see stressors as being to some degree under the individual's control. One study used a genome-wide scan to assess gene expression activity in socially isolated versus nonisolated individuals. It found that social isolation was associated with increased activation of several proinflammatory cytokine-related pathways and reduced activity in anti-inflammatory cytokine pathways, as well as in the glucocorticoid receptor, which plays an essential role in the neuroendocrine control of inflammatory processes. Of interest, the two types of psychotherapy most often examined in illnesses associated with immune dysregulation are group therapy, which provides social support, and cognitive behavioral therapy, which provides cognitive reframing techniques aimed at enhancing one's sense of agency (and hence control).

NEUROGENETICS

Starting from the rediscovery of Gregor Mendel's basic concepts at the turn of the 20th century, the field of genetics has matured into an essential cornerstone not only of the biologic sciences but of all of medicine. The discovery of the basic structure and properties of DNA in the middle of the century led to an exponential acceleration in our understanding of all aspects of the life sciences, including deciphering the complete sequence of the human genome and those of myriad other species. Massive databases of such sequences now provide 21st-century biologists with the task of decoding the functional significance of all this information. In particular, researchers are trying to determine how to sequence variations contributing to the phenotypic variation between species and individuals within a species. In humans, we hope that discoveries about the relationship between genotypes and phenotypes will revolutionize our understanding of why and how some individuals but not others develop common diseases. This hope is particularly vital for psychiatry, as our knowledge of the pathogenic mechanisms of the psychiatric disease remains sparse.

Genetic mapping studies aim to identify the genes implicated in heritable diseases, based on their chromosomal location. These studies are carried out by investigating affected individuals and their families through two approaches, linkage, and association (Fig. 33-29). It is now straightforward to genetically map Mendelian traits (traits for which a specific genotype at one particular locus is both necessary and sufficient to cause the trait). Psychiatric diseases, however, do not follow simple Mendelian inheritance patterns but rather are examples of etiologically complex traits. Etiologic complexity may be due to many factors, including incomplete penetrance (expression of the phenotype in only some of the individuals carrying the disease-related genotype), the presence of phenocopies (forms of the disease that are not caused by genetic factors), locus heterogeneity (different genes associated with the same disease in different families or populations), or polygenic inheritance (risk for the disease only increases if susceptibility variants at multiple genes act in concert). Mapping a complex disorder involves several component steps, including the definition of the phenotype to be studied, epidemiologic studies to determine the evidence for genetic transmission of that phenotype, choice of an informative study population, and determination

of the appropriate experimental and statistical approaches.

Genetic Epidemiologic Approaches

Genetic epidemiologic investigations provide quantitative evidence regarding the degree to which given trait aggregates in families and can suggest to what degree such aggregation reflects a genetic contribution to the etiology of the trait. Family studies compare the aggregation of disease among the relatives of affected individuals compared to control samples. Because these studies do not differentiate between genetic and environmental contributions to such familial aggregation, they provide only indirect evidence regarding the heritability of a trait. Often these studies measure the relative risk (λ), defined as the rate of occurrence of disease among specified categories of relatives of an affected individual divided by the rate of occurrence of the disease for the general population. A relative risk of >1 suggests a genetic etiology, and the magnitude of the measure gives an estimate of the genetic contribution to the disease. Relative risks can be calculated for sibling pairs, parent–offspring pairs, and various other types of family relationships. Likely modes of transmission can be assessed by comparing the degree of relative risk for each type of relationship. Multiple family studies have been carried out for many of the major psychiatric disorders, including major depression, bipolar disorder, schizophrenia, and OCD. Although these studies have consistently reported familial aggregation for all of these disorders, the degree of such aggregation has varied substantially across studies, mainly reflecting differences in phenotype definition and how study samples were ascertained and assessed.

Gene Mapping Strategies

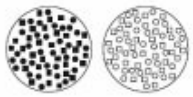
	Linkage Analysis		Genome Wide Association	
	Pedigree Analysis	Affected Sib Pair Analysis	Case-Control	Family-Trios
				
Study Subjects	Multigenerational families with multiple affected individuals	Two or more affected siblings	Affected individuals and matched unaffected controls sampled from population	Affected individual and parents
Basic Idea	Identify genetic markers that cosegregate with disease phenotype	Identify chromosomal regions shared by siblings concordant for disease.	Tests for statistical association of alleles and disease in cases versus controls.	Tests for association using nontransmitted parental chromosome as control.
Strengths	1) Can detect rare variants of large effect. 2) Gains power by incorporating information about familial relationships into the model.	1) Robust to differences in genetic composition of study population. 2) Easier to collect clinical samples compared to special pedigrees. 3) Allows incorporation of enviromental data.	1) Can detect common variants of small effect. 2) Does not require collection of family data.	1) Can detect common variants of small effect. 2) Robust to problems of population stratification.
Limitations	1) Limited power to identify common variants of small effect. 2) Cost intensive.	1) Limited power to identify common variants of small effect.	1) Increased false-positive rate in the presence of population stratification. 2) Requires large sample sizes.	1) About two-thirds as powerful as case-control designs. 2) Difficult to collect samples for late-onset diseases.

FIGURE 33-29
Comparison of gene-mapping strategies. Genetic mapping approaches can be divided into those that rely on linkage analysis and those that rely on association analysis. Linkage studies can be further categorized as either focused on investigation of pedigrees or focused on investigation of sib pairs. Association studies can be categorized as either case–control or family-based. Some of the key features as well as advantages and disadvantages of these different approaches are shown. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:321.)

Twin studies examine the concordance rates of a particular disorder (the percentage of twin pairs where both twins have the disorder) in monozygotic (MZ) and dizygotic (DZ) twins. For a disorder that is strictly determined by genetic factors, the concordance rate should be 100 percent in MZ twin pairs (who share 100 percent of their genetic material) and 25 or 50 percent in DZ twin pairs (who are no more closely related than any siblings), depending on whether the disease is recessive or dominant, respectively. For a disorder where genetic factors play a role in disease causation but are not the exclusive cause of disease, the concordance rates should be more significant for MZ twins than those for DZ twins. The higher the degree of concordance of MZ twins, the higher the trait heritability or the evidence for a genetic

contribution to disease risk. When genetic factors do not play a role, the concordance rates should not differ between the twin pairs, under the simplifying assumption that the environment for MZ twin pairs is no more similar than that for DZ twin pairs. Several twin studies that have been conducted for traits such as autism, bipolar disorder, and schizophrenia have consistently suggested high heritability and have therefore spurred efforts to map loci for each of these conditions genetically. Different twin studies may, however, generate varying point estimates for the heritability of any given disorder. When evaluating the results of twin studies, it is, therefore, essential to scrutinize how the phenotype was ascertained because, as with family studies, the different heritability estimates are likely due to differences in the mode of assessing and defining phenotypes. For example, early twin studies of psychiatric disorders often relied on their phenotypes on unstructured interviews by a single clinician.

In contrast, modern studies generally utilize standardized assessments and a review of diagnostic material by a panel of expert clinicians. Similarly, part of the apparent variation in heritability between different twin studies can be attributed to the fact that some studies employ narrow definitions of affectedness for a given phenotype. Other studies employ broader phenotype definitions (e.g., considering a twin with major depressive disorder to be phenotypically concordant with a co-twin diagnosed with bipolar disorder). Because of such differences in approach across studies, it is usually prudent to view such investigations as providing a rough estimate of the genetic contribution to trait variability. Nevertheless, even such estimates are useful in deciding which traits are likely to be mappable.

Basic Concepts of Gene Mapping

Recombination and Linkage. Once genetic epidemiologic studies of particular phenotypes have suggested that these phenotypes are heritable, genetic mapping studies are conducted to identify the specific genetic variants that contribute to the risk of the disorder. All genetic mapping methods aim to identify disease-associated variants based on their chromosomal position and the principle of genetic linkage. All cells contain two copies of each chromosome (called homologs), one inherited from the mother and one inherited from the father. During meiosis, the parental

homologs cross over, or recombine, creating unique new chromosomes that are then passed on to the progeny. Genes that are physically close to one another on a chromosome are genetically linked, and those that are farther apart or are on different chromosomes are genetically unlinked. Unlinked genes will recombine at random (i.e., there is a 50 percent chance of recombination with each meiosis). Genetic loci that are linked will recombine less frequently than expected by random segregation, with the degree of recombination proportional to the physical distance between them. The principle of linkage underlies the use of genetic markers, segments of DNA of known chromosomal locations that contain variations or polymorphisms (described in more detail later). Strategies to map disease genes are based on identifying genetic marker alleles that are shared—to a greater extent than expected by chance—by affected individuals. It is presumed that such sharing reflects linkage between a disease locus and a marker locus, that is, the alleles at both loci are inherited "identical by descent" (IBD), from a common ancestor, and that this linkage pinpoints the chromosomal site of the disease locus.

The evidence for linkage between two loci depends on the recombination frequency between them. Recombination frequency is measured by the recombination fraction (Θ). It is equal to the genetic distance between the two loci (1 percent recombination equals 1 centimorgan [cM] in the genetic distance and, on average, covers a physical distance of about 1 megabase [mB] of DNA). A recombination fraction of 0.5 or 50 percent indicates that two loci are not linked but rather that they are segregating independently. A LOD (logarithm of the odds) score is calculated to determine the likelihood that two loci are linked at any particular genetic distance. The LOD score is calculated by dividing the likelihood of acquiring the data if the loci are linked at a given recombination fraction by the likelihood of acquiring the data if the loci are unlinked ($\Theta = 0.5$). This step gives an odds ratio, and the log (base 10) of this odds ratio is the LOD score. A LOD score can be obtained for various values of the recombination fraction, from $\Theta = 0$ (completely linked) to $\Theta = 0.5$ (unlinked). The value of Θ that gives the largest LOD score is considered to be the best estimate of the recombination fraction between the disease locus and the marker locus. This recombination fraction can then be converted into a genetic map distance between the two loci.

Linkage Disequilibrium. Linkage disequilibrium (LD) is a phenomenon that is used to evaluate the genetic distance between loci in populations rather than in families. When alleles at two loci occur together in the population more often than would be expected given the allele frequencies at the two loci, those alleles are said to be in LD. When strong LD is observed between two loci, it usually indicates that the two loci are sited in very close physical proximity to one another on a given chromosome and is useful in mapping disease susceptibility loci because one locus can be used to predict the presence of another locus. This predictability is vital because current gene-mapping strategies can sample only a subset of the estimated 10 million common human polymorphisms. Because of the existence of LD, one can use data from a subset of genotyped polymorphisms to infer genotypes at nearby loci. Clusters of alleles that are in LD and inherited as a single unit are termed haplotypes. Thus LD mapping "consolidates" genomic information by identifying haplotypes in populations that can then be used to infer IBD sharing among unrelated individuals.

There are several methods to measure the extent of LD. One of the most commonly used measures of LD is r^2, a measure of the difference between observed and expected haplotype probabilities. Unlike D', another widely used measure of LD, r^2 values do not depend on the allele frequencies of the loci being assessed. A considerable r^2 value indicates that the observed frequency of association between two alleles is more significant than that expected by chance; that is, the alleles are in LD. LD studies have traditionally been used to complement traditional pedigree analyses, for example, to hone in on a locus that has been mapped by linkage analysis. However, LD-based association analysis has become the method of choice for whole-genome screens, particularly for diseases where traditional linkage studies have been unsuccessful. These studies have one great advantage over a traditional family analysis. Because affected individuals are chosen from an entire population, the only limit to the number of potential subjects is the population size and disease frequency. Maximizing the potential number of affected individuals that can be included in the analysis is extremely important for disorders where genetic heterogeneity or incomplete penetrance is likely to be a factor.

Genetic Markers. Mapping studies, regardless of their type, depend on the availability of genetic markers. The most widely used markers are microsatellite markers (also called simple tandem repeats [STRs], or simple sequence length polymorphisms [SSLPs]) and SNPs. SSLPs are stretches of variable numbers of repeated nucleotides two to four base pairs in length. These markers are highly polymorphic, as the number of repeat units at any given STR locus varies substantially between individuals. SNPs, as the name implies, are single base pair changes at a specific nucleotide; they are the most common form of sequence variation in the genome. SNPs are widely used for genetic mapping studies because they are distributed so widely across the genome and because they can be assessed in a high-throughput, automated fashion. Other forms of genetic variation that have been investigated for use as genetic markers include small insertion or deletion polymorphisms, termed indels, that generally range between 1 and 30 base pairs and copy number variations (CNVs), which can refer to either deletions or duplications. Recent genome-wide surveys have revealed that CNVs are common and can range in length from several base pairs to several million base pairs. CNVs may contribute to chromosomal recombination and rearrangements, thereby playing an essential role in generating genetic diversity. Also, as many of these variants are sizable, it is hypothesized that they may significantly influence the expression of genes that encompass or are adjacent to the variant.

Mapping Strategies

The genetic variants that contribute to disease susceptibility can be roughly categorized into those that are highly penetrant and those that are of low penetrance. High-penetrance variants, by definition, have a large effect on phenotype, and therefore identifying these variants usually provides fundamental insights into pathobiology. Because individuals carrying high-penetrance variants have a high probability of expressing a disease phenotype, such variants tend to be rare and to segregate in families and are generally most powerfully mapped using pedigree-based approaches (see Fig. 33-29). In contrast, low-penetrance variants have a relatively weak effect on phenotype, and therefore identification of individual low-penetrance variants may, at least initially, provide relatively little new biologic knowledge.

However, because of their small effects, such variants are typically frequent, and identifying them may add to our understanding of disease risk. Because we do not expect these variants to segregate strongly with the disease phenotype in pedigrees, efforts to identify them focus on population samples.

Pedigree Analysis. A pedigree analysis, which is conducted in multigenerational families, consists of scanning the genome or a portion of the genome with a series of markers in one or more affected pedigrees, calculating a LOD score at each marker position, and identifying the chromosomal regions that show a significant deviation from what would be expected under independent assortment. The primary goal of the pedigree analysis is to determine if two or more genetic loci (i.e., a genetic marker of known location and the unknown disease loci) are cosegregating within a pedigree.

Following the successful application of pedigree analysis to map Mendelian disorders such as Huntington disease, many investigators adopted this strategy for mapping psychiatric disease genes with, at best, mixed success. In the late 1980s and mid-1990s, several pedigree-based studies reported the mapping of susceptibility loci for Alzheimer disease, bipolar disorder, and schizophrenia. Although the linkage findings for three Alzheimer disease loci were relatively quickly replicated, the findings reported for bipolar disorder and schizophrenia were ultimately determined to have been false positives. Several different explanations have been proposed for the failure of pedigree-based approaches to map psychiatric loci; however, most investigators now recognize that these studies were generally drastically underpowered, considering the apparent etiologic complexity of psychiatric disorders.

Pedigree analysis in psychiatry has increasingly turned toward an application that is more appropriately powered, namely, the mapping of quantitative trait loci (QTLs). QTLs are defined as genetic loci that contribute to the variation in continuously varying traits (as opposed to categorical traits such as disease diagnoses). QTLs are typically loci of small effect that only contribute to a portion of the observed variance of a trait in the population. It is now generally accepted that using analytical methods developed in the late 1990s, it may be possible to use pedigree studies to map a wide range of quantitative traits that are relevant for understanding psychiatric disorders.

Several such studies are now being undertaken, typically with multiple phenotypes being assessed in each individual in the pedigree.

Sib-Pair Analysis. Affected sib-pair (ASP) analysis became widely used during the 1990s for the genetic mapping of complex traits, including many psychiatric disorders. Sib-pair analysis examines the frequency with which sibling pairs concordant for a trait share a particular region of the genome compared with the frequency that is expected under random segregation.

Sib-pair analysis is based on the fact that siblings share approximately 50 percent of their genomes IBD. Therefore, if a set of unrelated sib pairs affected with a given trait shares a particular area of the genome at a frequency significantly more significant than 50 percent (the proportion of sharing expected under conditions of random segregation), then that area of the genome is likely to be linked to the trait in question. In this method, siblings are genotyped, and population frequencies and parental genotypes are used to estimate the proportion of genes shared IBD at each site for each sib-pair. The linkage analysis then compares those pairs concordant and discordant for each locus.

Like pedigree studies, ASP studies have more power to locate genes of a large effect than genes of small effect. This limitation can be partially addressed by a two-tiered design that incorporates additional markers or family members after an initial linkage study in affected siblings or by an increased sample size. It generally requires less effort to identify and assess even large sets of affected sibs than to identify and assess all members of extended pedigrees, mainly when investigators can take advantage of data repositories that include samples and phenotype data from sib pairs ascertained from multiple sites. For example, the U.S. National Institute of Mental Health (NIMH) maintains such repositories for sizable collections of sib pairs affected with schizophrenia, bipolar disorder, autism, and Alzheimer disease. An additional benefit of the ASP design is that it allows for the incorporation of epidemiologic information, permitting the simultaneous examination of environmental and gene–environment interactions.

Association Studies. In the last few years, there has been increasing acceptance of the notion that association studies are more powerful than linkage approaches for mapping the loci of relatively small effects that are

thought to underlie much of the risk for complex disorders. Whereas linkage studies attempt to find cosegregation of a genetic marker and a disease locus within a family or families, association studies examine whether a particular allele occurs more frequently than expected in affected individuals within a population. As noted previously in this chapter, mapping of genes using association studies is based on the idea that specific alleles at markers closely surrounding a disease gene will be in LD with the gene; that is, these alleles will be carried in affected individuals more often than expected by random segregation, because they are inherited IBD.

There are two common approaches to association studies (see Fig. 33-29), case-control designs, and family-based designs, which typically investigate trios (mother, father, and an affected offspring). In a case-control study, allele frequencies are compared between a group of unrelated affected individuals and a matched control sample. This design is generally more powerful than a family-based design because large samples of cases and controls are more straightforward to collect than trios and are less expensive since they require the genotyping of fewer individuals. Case-control samples may be the only practical design for traits with a late age of onset (such as Alzheimer disease) for which parents of affected individuals are typically unavailable. The main drawback of the case-control approach is the potential problem of population stratification; if the cases and controls are not carefully matched demographically, then they may display substantial differences in allele frequency that reflect population differences rather than associations with the disease.

Family-based association studies are designed to ameliorate the problem of population stratification. In this design, the nontransmitted chromosomes (the copy of each chromosome that is not passed from parent to child) are used as control chromosomes, and differences between allele frequencies in the transmitted and nontransmitted chromosomes are examined, eliminating the problem of stratification, as the comparison group is by definition genetically similar to the case group. Although more robust to population stratification than a case-control study, family-based studies are only about two-thirds as powerful using the same number of affected individuals, as noted previously.

Until recently, it was not practical to conduct association studies on a genome-wide basis, as relatively few SNPs were available. Therefore, association studies focused on testing one or a few markers in candidate

genes chosen based on their hypothesized function with a given disease. Recently, however, as a result of international efforts that have identified millions of SNPs distributed relatively evenly across the genome and that have developed technology for genotyping them relatively inexpensively, genome-wide association (GWA) studies are now a reality. Such studies hold much promise for the identification of common variants contributing to common diseases. While few GWA studies of psychiatric disorders have been completed, such studies have already reported remarkable findings for complex traits such as rheumatoid arthritis, inflammatory bowel disease, and type 2 diabetes. The successful studies of these diseases have made use of extensive samples (in some cases up to several thousand cases and controls), providing further support for the hypothesis that underpowered study designs bear much of the responsibility for the disappointing results to date of psychiatric genetic investigations.

Statistical Considerations. Scientists in other biomedical research fields are often surprised by the high level of statistical evidence that geneticists require to consider a linkage or association result to be significant. This requirement can be imagined in terms of the very low expectation that any two loci are either linked or associated with one another. The likelihood that any two given loci are linked (i.e., the prior probability of linkage) is expected to be approximately 1:50, based on the genetic length of the genome. To compensate for this low prior probability of linkage and bring the posterior (or overall) probability of linkage to about 1:20, which corresponds to the commonly accepted significance level of $p = 0.05$, a conditional probability of 1,000:1 odds in favor of linkage is required, corresponding to the traditionally accepted LOD score threshold of 3. This generally provides an acceptable false-positive rate (Fig. 33-30), but some false-positive findings have exceeded even this threshold.

Geneticists generally expect that the chance that any two loci in the genome are associated with one another is even lower than that of their being in linkage. Typically a *p*-value of less than about 10^{-7} is considered to indicate “genome-wide significance.” This standard essentially discounts the prior probability that some investigators assign to variants in candidate genes chosen based on their hypothesized functional relevance to a given disorder or trait. GWA studies are now replicating associations with very low *p*-values

for a wide range of complex traits, whereas most candidate gene associations (which usually report as significant much higher *p*-values) remain unreplicated. It is, therefore, increasingly apparent that genome-wide levels of significance are appropriately applied to all initial association studies for a given trait.

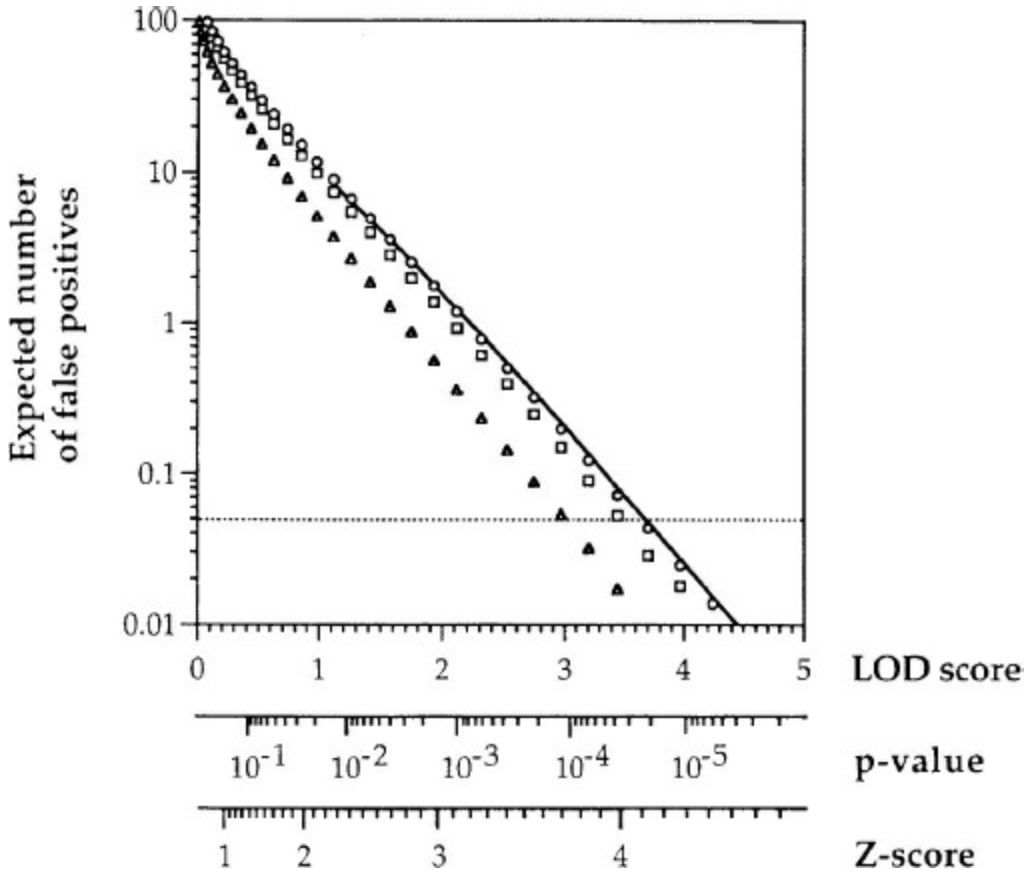


FIGURE 33-30
Number of false positives expected in a whole genome scan for a given threshold of logarithm of odds (LOD) score. *Solid line* represents the expectation for a perfect genetic map. *Symbols* represent the results for 100 sib pairs using genetic maps with markers spaced every .1 cM (*circles*), every 1 cM (*squares*), and every 10 cM (*triangles*). The *dotted line* indicates the 5 percent genome-wide significance level. (Courtesy of Dr. Eric Lander.)

Defining Phenotypes for Mapping Studies

The generally disappointing results of psychiatric genetic mapping studies have focused increasing attention on the problem of defining and assessing phenotypes for such studies. Most psychiatric mapping studies to date have relied on categorical disease diagnoses, as exemplified by the *Diagnostic and*

Statistical Manual (DSM-5) classification scheme. Criticisms of this approach rest on two arguments. First, diagnosis of the psychiatric disease depends on subjective clinical evaluation, a fact that underscores the difficulty in ascertaining individuals who can be considered affected with a given disease. Second, even when a psychiatric diagnosis can be established unambiguously, the menu-based system used for psychiatric classification provides the possibility that any two individuals affected with a given disorder may display mostly nonoverlapping sets of symptoms, likely reflecting distinct etiologies. Concern that the diagnosis-based approach to phenotyping may represent one of the chief obstacles to the genetic mapping of psychiatric phenotypes has generated considerable interest in mapping heritable traits known to demonstrate continuous variation in the population. Continuous measures that are hypothesized to be related to psychiatric disorders include biochemical measures (e.g., serum or CSF levels of neurotransmitter metabolites or hormones), cognitive measures, personality assessments, structural or functional brain images, biophysical markers such as responses to evoked potentials, or molecular assays such as gene expression profiles. Critical features of categorical and continuous phenotyping strategies are shown in Figure 33-31, and each is discussed in more detail below.

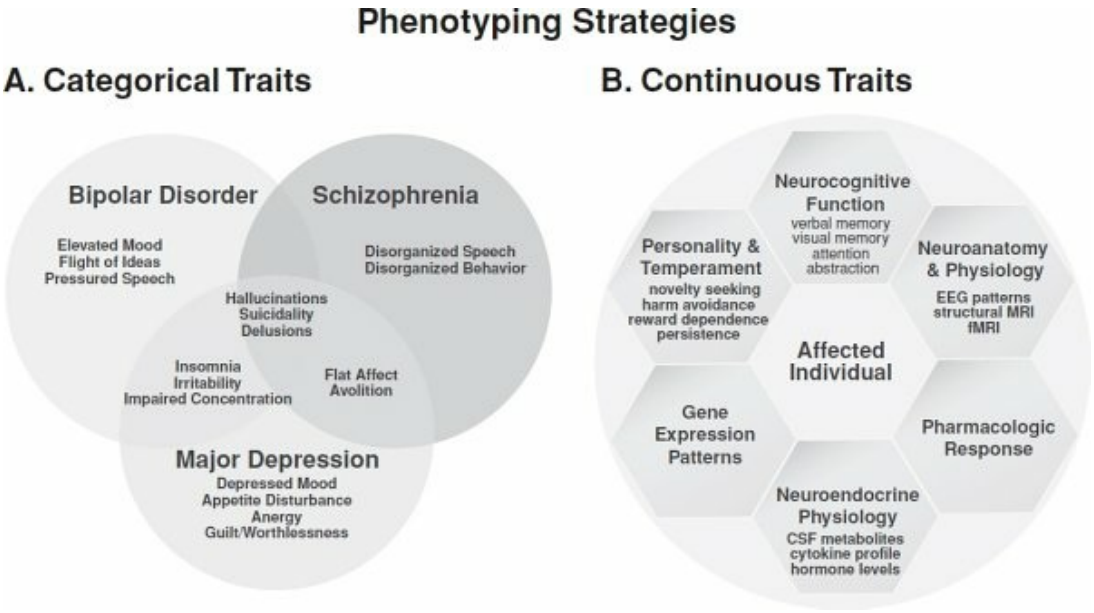


FIGURE 33-31
Two alternate schemes for conceptualizing psychiatric phenotypes. **A:** Categorical Traits as conceptualized by the *Diagnostic and Statistical Manual*

(DSM-5) represent a “menu-based” approach to psychiatric disorders. Individuals are assessed for a checklist of signs and symptoms that are then used to categorize the individual as “affected” according to a specific diagnosis. Not all symptoms are present in samples of individuals who carry a particular DSM diagnosis, and many of these symptoms occur across diagnostic boundaries, as illustrated in this Venn diagram. DSM phenotypes therefore probably represent etiologically heterogeneous categories, and this fact may help to explain the limited progress thus far of genetic mapping investigations focused on these phenotypes. **B:** Alternatively, in the Continuous Traits model, “affectedness” can be conceptualized in terms of an expectation that an individual will demonstrate extreme values on a set of continuous measures that correlate with psychopathology and thus are hypothesized to underlie the disorder (as illustrated by examples of six different types of measures shown in the hexagon). Such measures may also be associated with particular components of categorical phenotypes, such as those depicted in the Venn diagram in Figure 33-31A. The justification for using continuous measures as the phenotypes for genetic mapping studies is that they are considered etiologically simpler and more reliably assessed compared to categorical phenotypes. In addition, mapping such traits combines information from all members of the study population (affected and unaffected individuals alike), which adds considerably to power. (From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock’s Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:325.)

Categorical Phenotypes. The most commonly used categorical phenotypes in psychiatry are DSM diagnoses. Some studies focus on a single DSM diagnosis, whereas other studies include individuals with a range of different diagnoses. The latter approach is typically used for disorders that are hypothesized to represent a single disease spectrum, such as mood disorders. The categorical approach is designed to help us classify subjects as unambiguously as possible. Several strategies are used to accomplish this goal. The first strategy involves deciding on the appropriate diagnostic criteria for the study in question and deciding how these criteria will be applied to individuals in the study. One way of standardizing the procedures used to identify and assess potential study subjects is to use only experienced clinicians in the diagnostic process and to train them in the administration of the instruments and the diagnostic criteria to be employed. Also, a “best estimate” procedure or a consensus diagnosis is frequently used. The best estimate process involves making use of every piece of available information, including medical records, interviews, and videotapes, to arrive at a diagnosis. For a consensus diagnosis, two or more diagnosticians independently review the material and make a diagnosis for each individual. The diagnoses are then compared, and individuals for whom an agreement in

diagnosis cannot be reached are not entered as “affected” into the study.

A well-designed study makes use of all available information about the genetic epidemiology of the disorder to choose a sample of affected individuals to study. It is often the case that a subset of families carries the disorder in what appears to be a simple Mendelian pattern, whereas the inheritance pattern is less clear for other families or groups. In a disorder where there are likely to be multiple genes contributing to the phenotype, it makes sense, to begin with, a study sample where there may be significant loci. Redefining the disease phenotype can often simplify the mapping process by identifying such groups or families. For example, in the search for a genetic defect for Alzheimer disease, the process was advanced enormously by limiting the study population to those individuals who had an early age of onset (before age 65); the early-onset trait segregated in an autosomal dominant fashion. Other ways of redefining the phenotype include focusing on factors such as ethnic background, age of onset, treatment response, symptom severity, or the presence of comorbid disorders.

Narrowing the phenotype using the approaches discussed earlier may increase the chances of finding a genetic defect in complex diseases, but it can also significantly reduce the power of the study by limiting the number of available affected individuals. For this reason, it has been argued that for some disorders broadening the phenotype is an appropriate strategy. The suggestion is that for some complex diseases, the phenotype of interest may represent the end of a spectrum and that other phenotypes within the spectrum must also be included to have enough power to map genes. For example, mapping studies of bipolar disorder might include affected individuals with major depressive disorder as well as those individuals diagnosed with bipolar disorder.

Although the two approaches of narrowing the disease phenotype and broadening the disease phenotype may seem to be mutually exclusive, many groups studying complex disorders have incorporated both approaches into their study designs. One way to do this is to create stratified diagnostic categories, ranging from a narrow diagnostic category to a broad diagnostic category, and test for genetic linkage under each of these schemas. Some investigators argue that for complex diseases that are part of a spectrum, this strategy decreases the rate of false negatives, that is, of missing an existing linkage because of misspecification. Others argue that using several models

and picking the one that gives the highest scores dramatically increases the rates of false positives, that is, of identifying an area of linkage where none exists. One problem that exists with the use of multiple diagnostic categories is that as more models are used (and therefore more statistical tests are performed), increasingly stringent levels of evidence are required to consider a result significant.

While categorical phenotypes remain the mainstay of psychiatric genetic studies, the limitations of DSM nosology as the basis of phenotyping for genetic studies are becoming apparent. Genetic investigations are focusing increasingly on traits that may be components of one or more DSM diagnostic categories. For example, there is growing evidence that genetic susceptibility to psychosis, broadly defined, contributes to both severe bipolar disorder and schizophrenia, and several investigative approaches are being employed to attempt to identify genes that underlie such susceptibility and even to explore possible etiologic relationships between psychiatric and nonpsychiatric disorders. For example, bioinformatics models have been employed to investigate medical records databases and have uncovered extensive pairwise correlations among a diverse list of psychiatric disorders, neurologic disorders, autoimmune disorders, and infectious diseases. Eventually, the results of such model-fitting experiments may provide a framework to design more powerful linkage and association studies that can search for alleles that contribute to susceptibility to multiple disorders.

Continuous Phenotypes. Because of the difficulties experienced in genetic mapping of categorical diagnoses, neurobehavioral geneticists are increasingly focused on investigating quantitative traits that are hypothesized to underlie a particular psychiatric diagnosis, and that may be simpler to map genetically. The rationale for efforts to map such alternative phenotypes or endophenotypes is that the genes identified through such efforts may provide clues regarding the biologic pathways that are relevant to understanding a particular disorder. Several features characterize useful endophenotypes. First, they should be state-independent; that is, they should not fluctuate as a function of the disease course or medication treatment and should show adequate test–retest stability. Second, they should be heritable; that is, there should be evidence that genetic factors are responsible for a substantial proportion of the variability of the trait within the population. Third, the

endophenotype should be correlated with the disease under investigation; that is, different values of the trait measure are observed in patients compared to unrelated control subjects.

Measures of brain structure and function provide most of the traits now under investigation as endophenotypes for psychiatric disorders. For example, several features of brain morphometry (as assessed by MRI) are highly heritable (in the range of 60 to 95 percent), including total brain volume, cerebellar volume, gray and white matter density, amygdala and hippocampal volume, and regional cortical volume. Several studies show that brain structural features that are correlated in clinical samples with disorders such as schizophrenia or bipolar disorder are also abnormal in relatives of affected individuals. Physiologic measures of brain activity that have been employed as candidate endophenotypes for psychiatric disorders include electroencephalography (EEG) patterns. Several "pencil and paper" assessments have been employed to measure endophenotypes relating to neurocognitive function and temperament.

Animal Models. In contrast to categorical phenotypes, endophenotypes can be more straightforwardly related to phenotypes that can be assessed in animal models. Studies of genetic variations that affect circadian rhythms provide a good example. Variations in circadian rhythms have long been recognized as prominent features of mood disorders, and quantitative assessments of activity patterns have been proposed as endophenotypes for such disorders. Numerous studies in animal models have demonstrated that genetically controlled biologic clocks determine circadian activity and that variations in clock genes are associated with variations in such activity from bacteria to humans. Genetic mapping efforts in fruit flies starting in the early 1970s resulted in the identification of at least seven "clock genes," beginning with a *period.* Subsequent studies showed that the homologs of several of these genes play essential roles in regulating mammalian circadian rhythms. Genetic mapping studies in mice also have identified previously unknown circadian rhythm genes, beginning with the discovery and characterization in the early 1990s of a *clock.* These genetic discoveries have not only explicated the cellular networks and neurophysiologic circuits responsible for the control of mammalian circadian rhythms. However, they have also generated animal models that may shed light on the pathobiology of psychiatric

syndromes such as bipolar disorder. For example, mice carrying a targeted mutation in the *clock* demonstrate abnormal activity patterns, such as hyperactivity and decreased sleep, which is modified by the administration of lithium.

Progress in the Genetics of Specific Disorders

The progress in identifying susceptibility genes for psychiatric disorders has been disappointing compared to that observed for nonpsychiatric disorders. Alzheimer disease represents the most successful application of gene-mapping strategies to complex neurobehavioral disorders, and the section on this disease provides an example of how genetic linkage studies add to the understanding of the pathogenesis of a complex trait. An overview section on autism describes genetic investigations of syndromes that have features of autism but have relatively simple inheritance patterns and discusses how these studies have provided starting points for investigations of more complex ASDs. Finally, the frustrating search for unequivocal gene findings for bipolar disorder and schizophrenia is used to illustrate the challenges that are motivating new approaches in the field of neurobehavioral genetics.

Alzheimer Disease. Alzheimer disease provides an excellent example of the power of genetics to elucidate the complex biology of a neuropsychiatric disorder. Alzheimer disease is a well-defined form of dementia characterized by progressive impairment of memory and intellectual functioning. The clinical signs and symptoms, although characteristic, are not limited to Alzheimer disease; they are also found in several other types of dementia. For this reason, the diagnosis of Alzheimer disease can only be confirmed histopathologically at autopsy. The presence of senile plaques (made up of a core of β-amyloid fibrils surrounded by dystrophic neurites), tau-rich neurofibrillary tangles, and congophilic angiopathy in the brain parenchyma and associated blood vessels are pathognomonic for Alzheimer disease.

A variable age of onset has been noted for Alzheimer disease, ranging from as early as age 35 to as late as age 95. The concordance rate for Alzheimer disease in MZ twin pairs is about 50 percent, indicating a moderately substantial genetic contribution to disease risk. It is now evident from a wide range of genetic studies that Alzheimer disease can be divided

into two broad categories. These are the familial forms, which account for a tiny minority of Alzheimer disease cases and are characterized by early onset and autosomal dominant inheritance with high penetrance; and sporadic forms, in which the genetic contribution is hypothesized to be similar to that characterizing other common neuropsychiatric diseases.

The search for the genetic basis of familial Alzheimer disease began with traditional linkage studies. First, an investigation of a candidate locus on chromosome 21 in humans identified mutations in the *amyloid precursor protein* (*APP*) gene in a small number of families in which significant linkage had previously been observed to markers from this region. Transgenic mice with different *APP* mutations were created and have been shown to produce β-amyloid deposits and senile plaques as well as to show synapse loss, astrocytosis, and microgliosis, all part of the pathology of Alzheimer disease. Mutations in the genes that encode β-APP all lead to an increase in the extracellular concentration of longer fragments of β-amyloid (Aβ42). Most of the strains of transgenic mice with mutations in APP exhibit increased rates of behavioral changes and impairment in several memory tasks, indicating dysfunction in object-recognition memory and working memory, among others. These findings represent striking evidence that mutations in the β-amyloid gene are indeed responsible for at least some of the histopathologic elements of Alzheimer disease.

Even as the preceding findings were being reported, it was clear that mutations in the β-amyloid gene could not wholly explain the etiology and pathology of Alzheimer disease, not least because it was shown that linkage to chromosome 21 was excluded in most early-onset Alzheimer disease families. Also, no neurofibrillary tangles are observed in most of the different β-amyloid transgenic mice. The subsequent search for the genetic underpinnings of Alzheimer disease using genome-wide linkage analysis of early-onset Alzheimer disease families resulted in the identification of two additional Alzheimer disease susceptibility genes: *presenilin-1* (*PS-1*) on chromosome 14q24.3 and *presenilin-2* (*PS-2*) on chromosome 1q. PS-1 and PS-2 are integral transmembrane proteins with at least seven transmembrane domains. Although their function has not yet been completely elucidated, they are involved in the pathogenesis of Alzheimer disease. Inactivation of *presenilins* in mice leads to neurodegeneration and behavioral manifestations of memory loss. Biochemical and cellular studies have implicated presenilins

in several critical pathways, including apoptosis (programmed cell death) and protein processing in the endoplasmic reticulum.

These findings emphasize one of the strengths of using family-based linkage analysis. Pedigree-based studies are especially suited to identify highly penetrant disease genes that serve essential roles in critical biologic processes. Although mutations in *APP* and *presenilin* are rare, research into the biology of the expressed proteins has provided vital insights into the pathophysiology of dementia. Because these highly penetrant mutations elucidate essential biologic functions, they also provide a firm ground to design therapeutic interventions. For example, amyloid-β "vaccines" designed to induce an immunogenic response to pathogenic amyloid are now in advanced clinical trials. Unlike the current psychopharmacological treatments for Alzheimer disease that nonspecifically target cholinergic and glutaminergic neuronal systems, the amyloid-β vaccines may specifically treat the causes of Alzheimer disease by generating an immune response that may reverse the deposition of senile plaques.

SPORADIC AND LATE-ONSET ALZHEIMER DISEASE. Mutations in *APP, PS-1,* or *PS-2* are present in a majority of familial cases of early-onset Alzheimer disease but do not account for sporadic or familial late-onset Alzheimer disease. For this reason, investigators turned to other approaches to search for evidence of linkage in a large number of small families with late-onset Alzheimer disease. In 1991, the results of a nonparametric linkage study using 36 markers in late-onset Alzheimer disease families provided evidence for a susceptibility gene on the long arm of chromosome 19. In 1993, association studies revealed that the e4 allele of the *apolipoprotein E* gene was strongly associated with late-onset Alzheimer disease and that this association almost certainly was responsible for the previously observed linkage signal on chromosome 19. There are three known alleles of this gene —e2, e3, and e4. In most populations, the e3 allele is the most common. However, in familial late-onset Alzheimer disease, the incidence of e4 is approximately 50 percent, and in sporadic late-onset Alzheimer disease, it is 40 percent, compared with about 16 percent in normal controls. Epidemiologic studies suggest that between 30 and 60 percent of late-onset Alzheimer disease cases have at least one *apoE-e4* allele. The e4 genotype appears to be a more important risk factor for Alzheimer disease in

populations of European and Asian origin when compared with populations of African origin. Overall, the association of *apoE-e4* with Alzheimer disease remains probably the most robust association yet identified for a common human disease.

The establishment of *apoE-e4* as a susceptibility allele for late-onset Alzheimer disease has led to the search for additional alleles that might interact with *apoE-e4* to modify disease risk. In 2007, investigators used GWA strategies (in histologically confirmed cases and controls) to identify *GAB2* (GRB-associated binding protein 2) as an additional risk allele in *apoE-e4* carriers (but not in Alzheimer disease patients who were not e4 carriers). Initial studies suggest that carriers of both *apoE-e4* and *GAB2* risk alleles have an almost 25-fold more significant risk for Alzheimer disease than individuals who do not carry either risk allele. Larger-scale GWA studies of Alzheimer disease are in progress and will likely yield further associations; however, it is unlikely that any will have as strong an effect as *apoE.*

Autism. Autism is a severe neurodevelopmental disorder that is characterized by three primary features: impaired language and communication; abnormal or impaired social interaction; and restricted, repetitive, and stereotyped patterns of behavior. Understanding the etiology of autism has proceeded slowly, but there is convincing evidence that alterations in specific cellular and molecular neurodevelopmental pathways are essential in its etiology. In comparison with other neuropsychiatric disorders, there is particularly strong evidence for a genetic contribution to the risk of autism and ASDs. The sibling recurrence risk for autism or ASD is between 2 and 6 percent. Given a population prevalence of about 1 in 2,000 (0.04 percent), this means that the siblings of autistic individuals are approximately 50 to 100 times more likely to develop autism than a person in the general population. Twin studies of autism show extraordinarily high heritability (as demonstrated by MZ twin concordance of 80 to 92 percent) but also demonstrate the genetic complexity of these disorders, with the DZ twin concordance rate of 1 to 10 percent suggesting a highly multigenic mode of inheritance.

Increasing interest is now focused on the possibility that individuals affected with autism may display more considerable numbers of large-scale

chromosomal aberrations (5 to 10 percent in some studies) than unaffected individuals. Also, to such gross abnormalities, several recent studies have suggested that autism is associated with an unusually high prevalence of submicroscopic CNVs. For example, in 2007, the Autism Genome Project Consortium applied microarray strategies to almost 8,000 individuals from about 1,500 families, each with at least two affected family members, and found that about 10 percent of the ASD families carried CNVs, with an average size of more than 3 million base pairs, mostly consisting of duplications rather than deletions. Although the design of this study did not permit assessment of whether the frequency of CNVs is higher in patients with autism than that in controls, another study found a de novo CNV incidence of 10 percent in sporadic (no family history) cases of autism compared to an incidence of 1 percent in controls. These results, while exciting, are still considered preliminary. Even before the demonstration of high rates of de novo mutations in autism, epidemiologic studies had strongly suggested that the genetic basis of this disorder is likely complex. For example, although the risk of autism in first-degree relatives of autistic probands is high, there is a substantial falloff for second- and third-degree relatives of such probands, suggesting that multiple genetic variants must interact to increase susceptibility to this syndrome. Segregation analyses of autism also support the hypothesis that it is a heterogeneous disorder that reflects the actions of multiple genetic variants of small effect. A latent class analysis performed to study possible modes of transmission suggested an epistatic model with up to about 10 interacting loci, whereas other studies have estimated that as many as 15 such loci may be involved. Genetic studies of autism have included whole-genome screens, candidate gene studies, chromosome rearrangement studies, mutation analyses, and, most recently, comparative genomic hybridization studies. Taken together and recognizing that most findings still await adequate replication, these studies have contributed to an emerging picture of autism susceptibility that includes genes involved in three major systems: those involving synapse formation and maintenance, those involving cell migration, and those involving the excitatory/inhibitory neurotransmitter networks. Figure 33-32 shows a schematic of the currently known potential candidate genes for autism and their molecular relationships with one another.

SYNAPSE FORMATION AND MAINTENANCE. Perhaps the most significant breakthroughs in identifying susceptibility genes for autism have come from studies of disorders that display clinical features associated with autism or ASDs but with simpler inheritance patterns, including fragile X syndrome, tuberous sclerosis, and Rett syndrome. In general, the genetic defects associated with these disorders affect synapse formation and maintenance. Fragile X, which accounts for 3 to 4 percent of autism cases, is caused by an unstable trinucleotide repeat in the 5′ region of the fragile X mental retardation 1 (*FMR1)* gene at Xq27.3. This repeat expands as it is transmitted to succeeding generations, resulting in abnormal methylation and inhibition of expression of *FMR1*. *FMR1* produces an RNA-binding protein that acts as a chaperone for the transport of RNA from the nucleus to the cytoplasm and is involved in messenger RNA (mRNA) translation at the synapse. Abnormalities in dendritic spine density (increased over normal) and anatomy (longer and thinner than normal) have been reported in individuals with fragile X as well as in mouse models of this disorder. Tuberous sclerosis, which accounts for perhaps 2 to 10 percent of autism cases (the rate of tuberous sclerosis is higher among autistic individuals with seizure disorders), results from mutations in one of two tumor suppressor genes, *TSC1* on 9q34, and *TSC2* on 16p13, both of which are involved in guanosine triphosphatase (GTPase) inactivation. Loss of a single copy of *TSC1* in mice has been shown to disrupt cytoskeletal dynamics and dendritic spine structure.

Although somewhat less well understood, the genetics of Rett syndrome, an X-linked pervasive developmental disorder (the first with a known genetic etiology), also point to abnormalities in synapse formation and maintenance in ASD and ASD-like disorders. Mutations in *MeCP2* cause Rett syndrome. The gene makes a methylated-DNA-binding protein that regulates gene expression and chromatin structure. Although little is known about the exact role of *MeCP2* in the development of Rett syndrome, the pattern of normal early development and later regression suggests that this gene is more likely to be involved in synapse maintenance and remodeling than in synapse development.

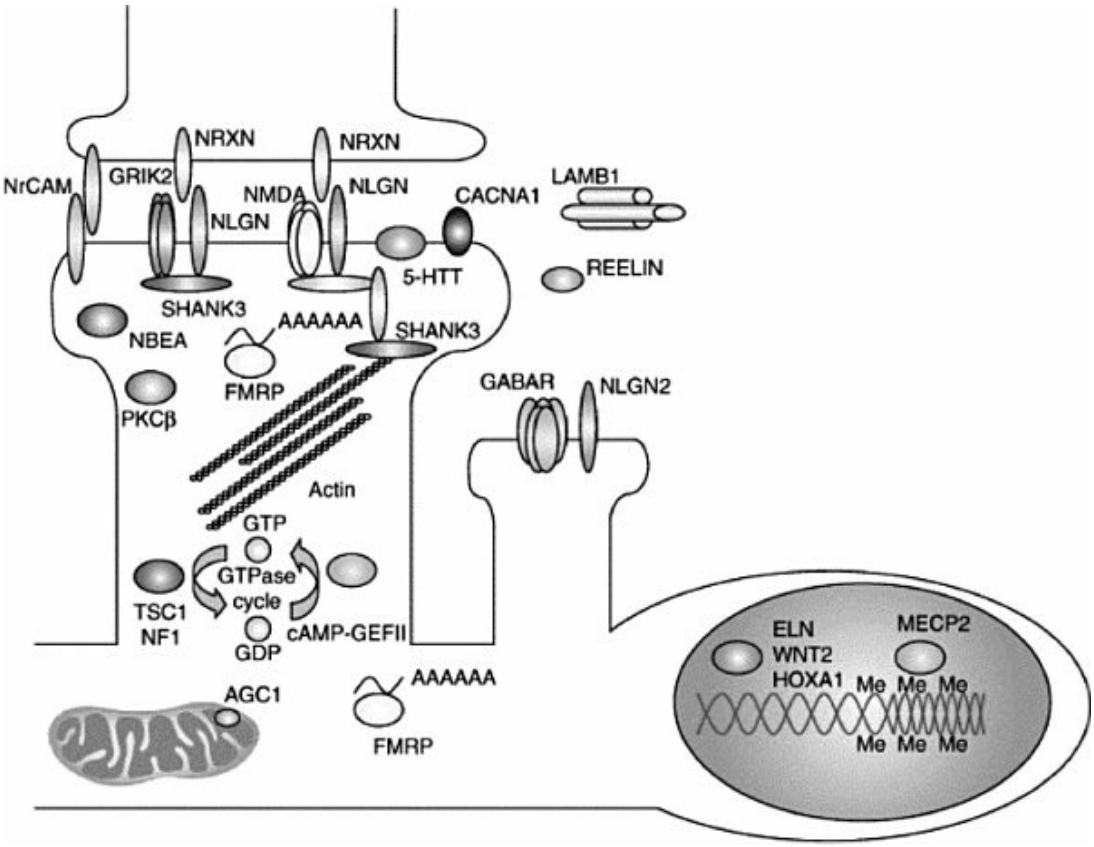


FIGURE 33-32
Schematic of the cell biology of proteins expressed from genes identified through mapping studies of autism spectrum disorders. The function of each gene product falls into three broad functional categories. Proteins involved in synapse formation and maintenance include FMR1, TSC1, TSC2, MeCP2, NLGN 3 and 4, and SHANK3. Another set of proteins is involved in neuronal migration and cell fate including REELIN, WNT2, LAMB1, and NrCAM. Proteins involved in neurotransmitter systems are also altered in some individuals with autism and include 5-HTT (serotonin transporter encoded by SLC6A4), GABAR, and the NMDA subunit encoded by GRIN2A. See text for details. (From Persico AM, Bourgeron T. Searching for ways out of the autism maze: genetic, epigenetic and environmental clues. *Trends Neurosci*. 2006;29:349, with permission.)

Neuroligin (*NLGN*) 3 and 4 and *SHANK3,* additional genes that appear to play a role in synapse formation, may be affected by chromosomal rearrangements observed in some individuals affected with autism. The neuroligin genes, sited on the X chromosome, produce cell adhesion molecules that are located on postsynaptic glutamatergic neurons. When mutated in rodents, these genes show defective trafficking and synapse induction. In nonmutated form, their expression induces the formation of normal, presynaptic terminals in axons. *SHANK3* is a binding partner of the

neuroligins and regulates the structural organization of dendritic spines. Mutations in *SHANK3* have been identified in ASD-affected members of at least three families to date, and a comparative genomic hybridization study of autistic individuals, their family members, and controls recently identified a large deletion in chromosome 22q13, the region containing *SHANK3*, in at least one individual with autism.

CELL MIGRATION. Of the regions highlighted by a genome screen in autism families, chromosome 7q has provided the most consistent evidence for linkage, albeit over an extensive region. Known chromosomal rearrangements in this region in individuals affected with autism add to its interest. The linkage region on chromosome 7q contains several genes that are strong candidates for autism, most notably *RELN*, which maps to chromosome 7q22. *RELN* codes for reelin, a signaling protein secreted by Cajal–Retzius cells located in the marginal zone of the developing brain. It plays a vital role in neuronal migration as well as in the development of neural connections. Reeler mice, which have spontaneous deletions of *RELN*, have cytoarchitectonic alterations in their brains during development that are similar to those that have been described in autistic brains. The complete absence of *RELN* in humans leads to a more severe phenotype with lissencephaly and severe mental retardation but not autism. Individuals with autism show reduced levels of reelin mRNA and protein in brain and blood serum, suggesting that mutations leading to reduced expression of *RELN* rather than its absence may be significant in ASD. Genetic association studies with *RELN* have been equivocal, suggesting that if *RELN* does contribute to the development of autism, then it may play such a role in a small subset of affected individuals. *WNT2* (wingless-type MMTV integration site family member 2) is another gene identified as a potential candidate for autism based on linkage studies. *WNT2* is located on 7q31 and is part of a family of genes that encode secreted signaling proteins implicated in several developmental processes, including the regulation of cell fate and patterning during embryogenesis. At least two families have been identified in which nonconservative coding sequence variants in *WNT2* segregate with autism. LD between an SNP in the 3′ untranslated region of *WNT2* and autism is also present in families with severe language abnormalities that accounted for most of the evidence for linkage on chromosome 7q in one of the original

genome screens.

EXCITATORY/INHIBITORY NEUROTRANSMITTER SYSTEMS. Although there is little current evidence that mutations in genes encoding neurotransmitter transporters or receptors are directly responsible for the development of autism, there is some evidence that such genes might act as modifiers or susceptibility factors for autism spectrum phenotype. The evidence is perhaps most robust for the role of the GABA receptors in the development and expression of autistic disorders. These receptors occur in a cluster on chromosome 15q11–13, and duplications of this region are the most common cytogenetic abnormalities seen in autism cases (up to 6 percent of cases). GABA is an important inhibitory neurotransmitter in the CNS and is responsible for controlling excitability in mature brains. Chromosome 15q11–13 is one of the most complex regions of the genome. It has a high rate of genomic instability, including frequent duplication and deletion events, and imprinting plays a vital role in the expression of genes in this region. The 15q11–13 region is the critical region for Angelman and Prader–Willi syndromes, neurologic disorders due to deletions or mutations in this region that occur on maternally and paternally inherited chromosomes, respectively.

Despite the high rate of duplications of 15q11–13 among autistic individuals, genome screens have not shown strong support for linkage or association to this region. Candidate gene studies continue, however, in part because a rate of 6 percent of autistic individuals with duplications in this region is hard to ignore.

Bipolar Disorder. The search for the genetic basis of bipolar disorder has been fraught with missteps and partial answers. The history of genetic mapping attempts for bipolar disorder illustrates not only the extreme complexity of psychiatric disorders but also the evolution of genetic approaches to such diseases. Bipolar disorder is an episodic illness characterized by recurrent periods of both mania and depression. Psychotic symptoms are often a part of the clinical picture, particularly in more severely affected individuals.

Numerous genetic epidemiologic investigations conducted over several decades have strongly supported a genetic contribution to risk for bipolar

disorder. The definition of the bipolar disorder phenotype in these studies has varied substantially, resulting in a wide range of heritability estimates. This problem is typical of many psychogenetic studies. For example, many early studies into the genetic basis of mood disorders did not distinguish between unipolar and bipolar mood disorders. Furthermore, the diagnostic methodology used in such early studies differs substantially from that employed in current-day genetic studies. For example, a Danish twin study that suggested a very high heritability for bipolar disorder and thereby had a substantial influence on the design of initial genetic mapping studies of mood disorders employed only unstructured diagnostic interviews by a single clinician rather than the structured assessments used in current studies, which have suggested somewhat lower heritabilities.

Current estimates of the concordance for bipolar disorder range between 65 and 100 percent in MZ twins and 10 and 30 percent in DZ twins. This ratio indicates that the disorder is highly heritable (between about 60 and 80 percent). Several studies have shown that bipolar disorder is substantially more heritable than unipolar major depression, which has an estimated heritability between 30 and 40 percent.

Early family studies suggested that bipolar disorder segregation patterns were compatible with single-gene inheritance of a locus of significant effect. However, although some bipolar disorder pedigrees may segregate such a locus, mounting evidence indicates that if such pedigrees exist, they must be quite rare. Furthermore, the fact that genetic linkage studies have failed to uncover such a locus with unequivocal evidence in any pedigrees argues against this possibility. The observed rapid decrease in recurrence risk for bipolar disorder from monozygotic co-twins to first-degree relatives is also not consistent with single-gene inheritance models but instead suggests models of multiple interacting genes.

EARLY LINKAGE STUDIES. Tremendous excitement followed the first reports of linkage to bipolar disorder on chromosomes X and 11 in 1987. Investigators noted that in several families, bipolar disorder and other affective disorders appeared to be inherited in an X-linked fashion. Likewise, these disorders appeared to cosegregate in several Israeli families with color blindness and G6PD deficiency, which map to the X chromosome. Linkage studies in these pedigrees, using color blindness or G6PD deficiency as marker loci, gave

LOD scores between 4 and 9. Early studies of chromosome 11 were similar to those for chromosome X in that they reported significant linkage after testing only a few markers in a single region, in this case, in an extended Old Order Amish pedigree heavily loaded for bipolar disorder.

Not surprisingly, these findings generated a great deal of interest. Both studies showed high LOD scores and seemed to provide clear evidence for linkage. However, replication studies in other populations failed to produce positive results for either the X chromosome or chromosome 11, and evidence for linkage virtually disappeared in both chromosomal regions in the samples in which linkage was initially reported when the pedigrees were extended to include additional affected individuals and when additional markers were typed in the putative linkage regions. The most likely explanation in each case is that the original linkage results were false-positive findings and may have reflected over-optimistic interpretation of evidence that, in retrospect, was relatively scanty.

GENOME-WIDE SCREENS. The early linkage studies of bipolar disorder evaluated only a few markers because they were all that were available. With the construction of genetic linkage maps of the genome in the 1990s, linkage studies of most complex traits, including bipolar disorder, began to search genome-wide. The advantage of genome-wide mapping studies is that they do not require *a priori* knowledge of the biologic underpinnings of a particular phenotype. Complete genome screens provide an opportunity to evaluate the evidence of linkage at all points in the genome without bias. Although genome-wide studies had greater power to detect accurate linkage than studies focused on only a few markers in arbitrary locations or around a few candidate genes, these investigations have also generally had disappointing results. The challenge of achieving replicated significant linkage results for bipolar disorder and other complex traits is apparent when one reviews the many gene-mapping studies that have suggested—but not demonstrated unequivocally—bipolar disorder susceptibility loci on chromosome 18.

CHROMOSOME 18. The first report of linkage came from a partial genome screen that examined 11 markers on chromosome 18 and identified suggestive linkage near the centromere. Because the inheritance patterns for

bipolar disorder are unknown, the results were analyzed using both recessive and dominant models. Some of the markers were positive under a recessive model in some families, some were positive under a dominant model in other families, and some markers gave positive LOD scores in a subset of families under both models. Attempts to replicate this finding in other populations have been mixed. So far, at least two groups have found no evidence for linkage to the pericentromeric region of chromosome 18 in their samples, although one other group has found evidence to support linkage to this region. Other studies have found suggestive evidence for linkage on chromosome 18, including a complete genome screen in two large Costa Rican pedigrees that gave evidence for linkage on chromosome 18q22–23 as well as in an area on 18p. The combined evidence of these several studies, although somewhat contradictory and confusing, points to at least two different susceptibility loci on chromosome 18: one on 18p and one on 18q.

IMPROVING STUDY POWER. The equivocal findings represented by the attempts to pinpoint susceptibility loci on chromosome 18 have led investigators to implement several new strategies to map bipolar disorder genes. One such strategy is a meta-analysis, which involves combining data across multiple individual investigations to increase statistical power, and in some cases, the combined analysis points to loci not initially found in the individual studies. Several meta-analytical techniques have been used to explore gene-mapping studies for bipolar disorder. The multiple scan probability (MSP) and genome scan meta-analysis (GSMA) methods require only linkage statistics and *p*-values from each study to examine combined data. MSP was used to combine chromosomal regions with *p*-values less than 0.01 from 11 independent bipolar disorder studies and provided evidence for susceptibility loci on chromosomes 13q and 22q. Although the MSP and GSMA methods have the advantage of requiring only linkage significance data, they are not able to account for study-specific issues that will limit the extent to which multiple studies can be compared. Combining original genotype data from multiple studies can circumvent this problem. With this method, the largest meta-analysis to date combined 11 bipolar disorder genome-wide linkage scans consisting of 5,179 individuals from 1,067 families. Access to the original genotype data allowed the construction of a standardized genetic map in which the markers of each respective study were

mapped onto one standard gender-averaged map. The results of this meta-analysis identified two susceptibility loci with genome-wide significance on 6q and 8q.

Another strategy that has been used to increase the power of gene-mapping studies in the formation of consortia that combine data across multiple clinical sites. A consortium combining data from the UK and Ireland led to support for linkage at 9p21 and 10p14–21. Likewise, combining data from Spanish, Romanian, and Bulgarian families provided additional support for findings on chromosomes 4q31 and 6q24. Investigators can also increase power by standardizing marker sets and clinical evaluation protocols between independent studies to permit direct comparisons between such studies. This approach was used to identify a bipolar disorder susceptibility locus on chromosome 5q31–33. The region showed suggestive nonparametric linkage results in pedigrees from the Central Valley of Costa Rica. With identical genetic markers and diagnostic criteria, the same region was highlighted in an independent analysis of a set of Colombian families who have a genetic background similar to that of the Costa Rican families. A follow-up study using additional markers in an expanded set of Colombian and Costa Rican families confirmed genome-wide significant evidence to a candidate region of 10 cM in 5q31–33. This finding is especially impressive given that the linkage peak in the bipolar studies overlaps with linkage regions for schizophrenia and psychosis, identified in a previous study of 40 families from the Portuguese Islands. These results contribute to a growing opinion that there may be substantial genetic overlap between different DSM disorders.

Schizophrenia. In the past decade we have seen exciting advances in the genetic studies of schizophrenia. Much of these are due to the several large consortia formed to investigate the disorder. These collaborative efforts allow for large samples with sufficient power to investigate the disorder.

Before these studies, family studies established a heritability of between 60 and 80 percent, however we lacked any meaningful genome-wide studies. In 2009 several of the consortia combined samples and identified an SNP in the MHC on chromosome 6p22 as a risk locus. What followed were several elegant studies, including a landmark 2016 paper from the Schizophrenia Working Group of the Psychiatric Genomics Consortium, which focused on

the C4 alleles and demonstrated their role in synaptic pruning during critical developmental period. This supports the hypothesis that schizophrenia is the result of abnormalities of neurodevelopment.

Before this work, various candidate genes emerged. They spawned a diverse range of basic and clinical investigations aiming to elucidate their functional significance, for example, using mouse gene targeting and functional MRI.

The early identification of chromosome 6p24–22 as implicated in schizophrenia led to the proposal of *Dysbindin* (*DTNB1*) as a candidate gene for schizophrenia. Additional association studies of *Dysbindin* have been equivocal.

Linkage studies subsequently pointed to a region on chromosome 1 containing the candidate genes *DISC 1* and *DISC 2* (*disrupted in schizophrenia 1* and *2*) located on chromosome 1q21–22 and 1q32–42. These genes were initially identified in a large Scottish pedigree in the early 1990s. A balanced translocation between chromosomes 1 and 11 segregated in this pedigree and was possibly associated with severe mental illness. *DISC 1* and *2* were identified in the original Scottish family because of their location near the chromosomal translocation breakpoint. As with *Dysbindin,* follow-up studies of *DISC 1* and *2* have been equivocal.

Genome screens, including a screen focused on extended Icelandic kindreds, have identified a schizophrenia candidate region on chromosome 8p21–22. Fine mapping of the region narrowed the search and eventually led to the proposal of *neuregulin 1* (*NRG1*) as a schizophrenia candidate gene. Association studies again provided equivocal and difficult-to-interpret results.

Despite the equivocal genetic studies, significant resources have been channeled into molecular and neurophysiologic investigations of the functional products of *dysbindin, DISC 1* and *2,* and *neuregulin.* Mutant mice for each of the three genes are now available and have been used to demonstrate interesting biologic findings. For example, *dysbindin* is expressed in the hippocampus and dorsolateral prefrontal cortex. The dysbindin protein binds to B-dystrobrevin and has been implicated in synaptic structure and signaling. *DISC 1* has been shown to influence neurite formation in cellular studies, and mutant mice for *DISC 1* show impairments in a wide variety of tests, including learning, memory, and sociability.

Neuregulin belongs to a family of growth factors that mediate numerous functions, including synapse formation, neuronal migration, and neurotransmission. Targeted disruption of *erbB4*, the postsynaptic target of neuregulin, leads to synaptic glutamatergic hypofunction. Despite the exciting biology uncovered, it remains unclear whether and to what extent any of these genes contribute to the etiology of schizophrenia in humans, and many geneticists have been cautious in their endorsement of the legitimacy of the mutant mice generated from the current list of candidate genes as models of psychiatric disorders.

APPLIED ELECTROPHYSIOLOGY

EEG is the recording of the electrical activity of the brain. It is used in clinical psychiatry principally to evaluate the presence of seizures, particularly temporal lobe, frontal lobe, and petit mal seizures (absence seizures), which can produce complex behaviors. EEG is also used during ECT to monitor the success of the stimulus in producing seizure activity and as a critical component of polysomnography used in the evaluation of sleep disorders. Quantitative electroencephalography (QEEG) and cerebral evoked potentials (EPs) represent newer EEG-based methods that provide improved research and clinical insights into brain functioning.

Electroencephalography

A brain wave is a transient difference in electrical potential (greatly amplified) between any two points on the scalp or between some electrode placed on the scalp and a reference electrode located elsewhere on the head (i.e., ear lobe or nose). The difference in electrical potential measured between any two EEG electrodes fluctuates or oscillates rapidly, usually many times per second. It is this oscillation that produces the characteristic "squiggly line" that is recognized as the appearance of "brain waves."

Brain waves reflect a change by becoming faster or slower in frequency or lower or higher in voltage, or perhaps some combination of these two responses. A routine EEG can never constitute positive proof of the absence of brain dysfunction. Even in diseases with established brain pathophysiology, such as multiple sclerosis, deep subcortical neoplasm, some seizure disorders, and Parkinson disease and other movement disorders, a

substantial incidence of patients with normal EEG studies may be encountered. Nonetheless, a normal EEG can often provide convincing evidence for excluding certain types of brain pathology that may present with behavioral or psychiatric symptoms. More often, information from the patient's symptoms, clinical course and history, and other laboratory results identifies a probable cause for the EEG findings. EEG studies are often ordered when a pathophysiologic process is already suspected, or a patient experiences a sudden, unexplained change in mental status.

Electrode Placement. The electrodes usually used to record the EEG are attached to the scalp with a conductive paste. A standard array consists of 21 electrodes. The placement of the electrodes is based on the 10/20 International System of Electrode Placement (Fig. 33-33). This system measures the distance between readily identifiable landmarks on the head and then locates electrode positions at 10 percent or 20 percent of that distance in an anterior–posterior or transverse direction. Electrodes are then designated by an uppercase letter denoting the brain region beneath that electrode and a number, with odd numbers used for the left hemisphere and with even numbers signifying the right hemisphere (the subscript Z denotes midline electrodes). Thus, the O_2 electrode is placed over the right occipital region, and the P_3 lead is found over the left parietal area.

In particular circumstances, other electrodes may be used. Nasopharyngeal (NP) electrodes can be inserted into the NP space through the nostrils and can be closer to the temporal lobe than scalp electrodes. No actual penetration of tissue occurs. These electrodes may be contraindicated with many psychiatric patients displaying behaviors, such as confusion, agitation, or belligerence, which could pull the leads out, possibly lacerating the nasal passage. Sphenoidal electrodes use a hollow needle through which a fine electrode that is insulated, except at the tip, is inserted between the zygoma and the sigmoid notch in the mandible until it is in contact with the base of the skull lateral to the foramen ovale.

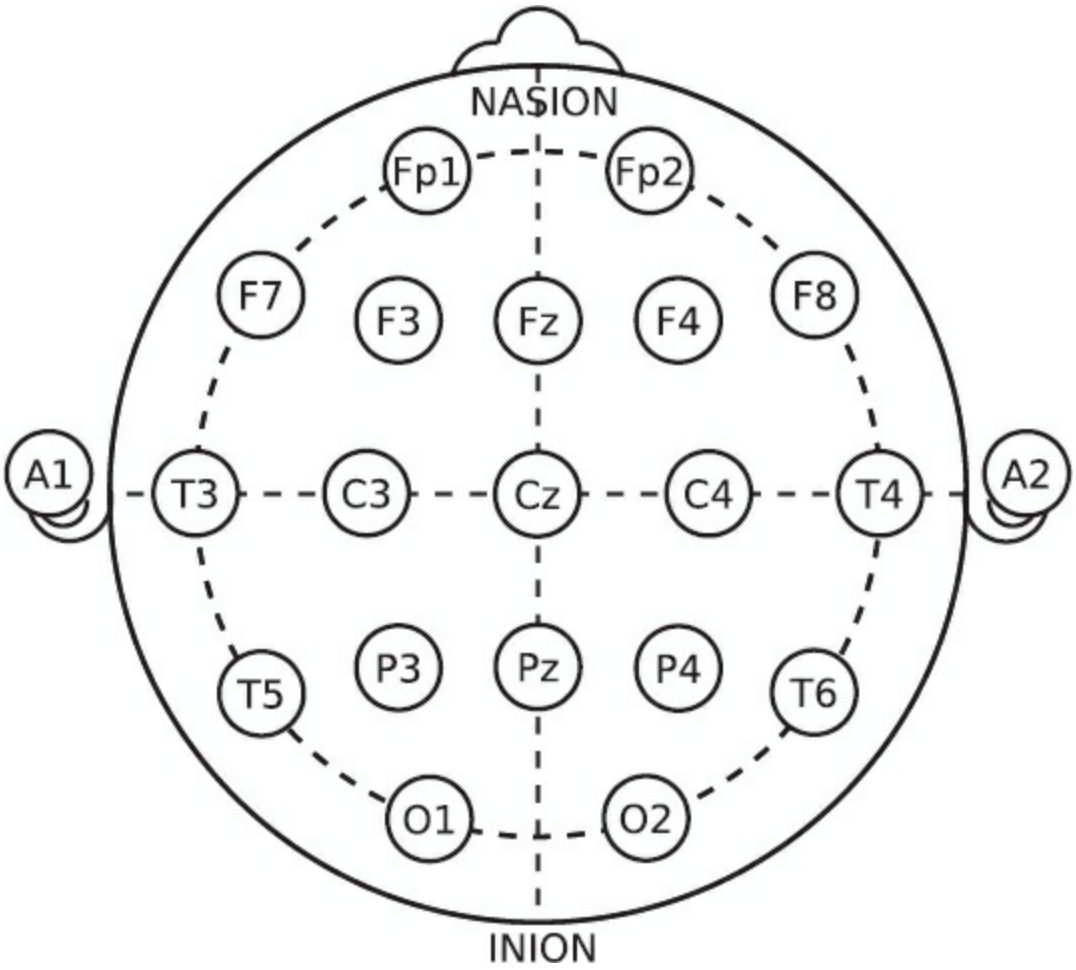


FIGURE 33-33
International 10–20 Electrode Placement System.

Activated EEG. Specific activating procedures are used to increase the probability that abnormal discharges, particularly spike or spike-wave seizure discharges, will occur. Strenuous hyperventilation is one of the most frequently used activation procedures. While remaining reclined with the eyes closed, the patient is asked to over breathe through the open mouth with deep breaths for 1 to 4 minutes, depending on the laboratory (3 minutes is standard). In general, hyperventilation is one of the safest EEG-activating procedures, and, for most of the population, it presents no physical risk. It can pose a risk for patients with cardiopulmonary disease or risk factors for cerebral vascular pathophysiology, however. Photic stimulation (PS) generally involves placing an intense strobe light approximately 12 inches in front of the subject's closed eyes and flashing at frequencies that can range from 1 to 50 Hz, depending on how the procedure is carried out. Retinal

damage does not occur because each strobe flash, although intense, is too brief in duration to harm the patient. When the resting EEG is normal, and a seizure disorder or behavior that is suspected to be a manifestation of a paroxysmal EEG dysrhythmia is suspected, PS can be a valuable activation method to use. EEG recording during sleep, natural or sedated, is now widely accepted as an essential technique for eliciting a variety of paroxysmal discharges, when the wake tracing is normal, or for increasing the number of abnormal discharges to permit a more definitive interpretation. It has been shown that the CNS stress produced by 24 hours of sleep deprivation alone can lead to the activation of paroxysmal EEG discharges in some cases.

Normal EEG Tracing

The normal EEG tracing (Fig. 33-34) is composed of a complex mixture of many different frequencies. Discrete frequency bands within the broad EEG frequency spectrum are designated with Greek letters.

Awake EEG. The four basic waveforms are alpha, beta, delta, and theta. Highly rhythmic *alpha waves* with a frequency range of 8 to 13 Hz constitute the dominant brain wave frequency of the normal eyes-closed awake EEG. Alpha frequency can be increased or decreased by a wide variety of pharmacologic, metabolic, or endocrine variables. Frequencies that are faster than the upper 13 Hz limit of the alpha rhythm are termed *beta waves*, and they are not uncommon in average adult waking EEG studies, particularly over the frontal-central regions. *Delta waves* (≤3.5 Hz) are not present in the normal waking EEG but are a prominent feature of deeper stages of sleep. The presence of significant generalized or focal delta waves in the awake EEG is strongly indicative of a pathophysiologic process. Waves with a frequency of 4.0 to 7.5 Hz are collectively referred to as *theta waves.* A small amount of sporadic, arrhythmic, and isolated theta activity can be seen in many normal waking EEG studies, particularly in frontal-temporal regions. Although theta activity is limited in the waking EEG, it is a prominent feature of the drowsy and sleep tracing. Excessive theta in awake EEG, generalized, or focal, suggests the operation of a pathologic process.

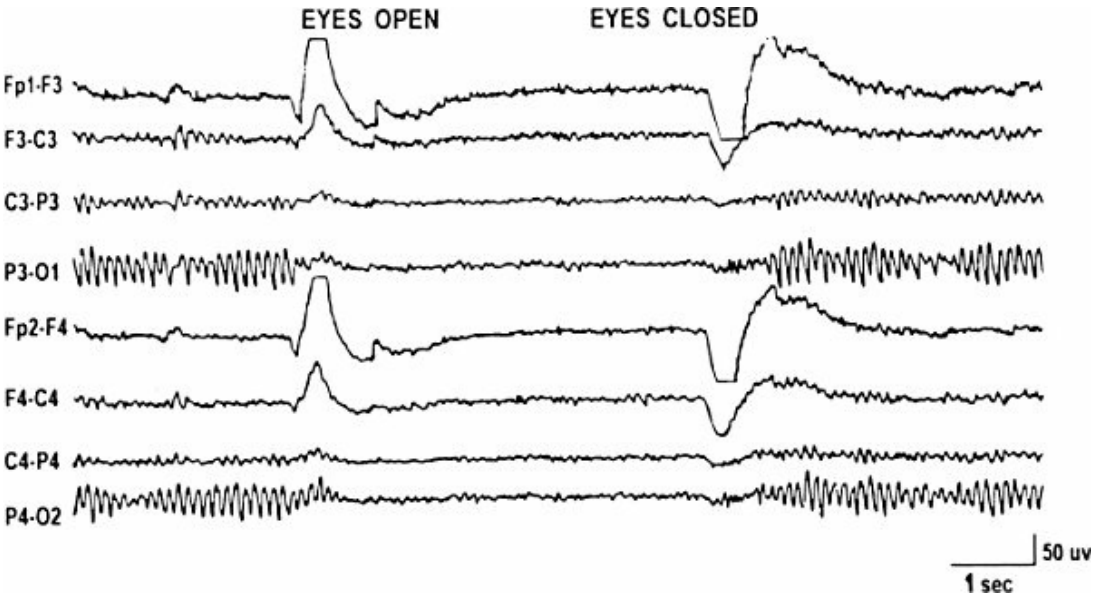


FIGURE 33-34
Normal electroencephalogram (EEG) tracings in an awake 28-year-old man. (Reprinted from Emerson RG, Walesak TS, Turner CA. EEG and evoked potentials. In: Rowland LP, ed. *Merritt's Textbook of Neurology*. 9th ed. Baltimore, MD: Lippincott Williams & Wilkins; 1995:68, with permission.)

With maturation, EEG activity gradually goes from a preponderance of irregular medium- to high-voltage delta activity in the tracing of the infant, to greater frequency and more rhythmic pattern. Rhythmic activity in the upper theta-lower alpha range (7 to 8 Hz) can be seen in posterior areas by early childhood, and, by mid-adolescence, the EEG essentially has the appearance of an adult tracing.

Sleep EEG. The EEG patterns that characterize drowsy and sleep states are different from the patterns seen during the awake state. The rhythmic posterior alpha activity of the waking state subsides during drowsiness and is replaced by irregular low-voltage theta activity. As drowsiness deepens, slower frequencies emerge, and sporadic vertex sharp waves may appear at central electrode sites, particularly among younger persons. Finally, the progression into sleep is marked by the appearance of 14-Hz sleep spindles (also called *sigma waves*), which, in turn, gradually become replaced by high-voltage delta waves as deep sleep stages are reached.

EEG Abnormalities

Apart from some of the apparent indications for an EEG study (i.e., suspected seizures), EEG studies are not routinely performed as part of a diagnostic work-up in psychiatry. EEG, however, is a valuable assessment tool in clinical situations in which the initial presentation or the clinical course appears to be unusual or atypical (Table 33-8). Table 33-9 summarizes some common types of EEG abnormalities.

Some psychotropic medications and recreational or abused drugs produce EEG changes, yet, except for the benzodiazepines and some compounds with a propensity to induce paroxysmal EEG discharges, little, if any, the clinically relevant effect is noted when the medication is not causing toxicity. Benzodiazepines, which always generate a significant amount of diffuse beta activity, have EEG-protective effects so that they can mask alterations caused by concomitant medications (Table 33-10).

Medical and neurologic conditions produce a wide range of abnormal EEG findings. EEG studies, thus, can contribute to the detection of unsuspected organic pathophysiology influencing a psychiatric presentation (Fig. 33-35). Table 33-11 lists EEG alterations in medical disorders, and Table 33-12 lists EEG alterations associated with psychiatric disorders.

Table 33-8
Warning Signs of the Presence of Covert Medical or Organic Factors Causing or Contributing to Psychiatric Presentation

Atypical age of onset (i.e., anorexia nervosa beginning at mid-adulthood)
Complete lack of positive family history of the disorder when a positive family history is expected
Any focal or localized symptoms (i.e., unilateral hallucinations)
Focal neurologic abnormalities
Catatonia
Presence of any difficulty with orientation or memory (in general, Mini Mental State Examination should be normal)
Atypical response to treatment
Atypical clinical course

Note: Clinicians should have a high index of suspicion for underlying medical conditions and a

low threshold for initiating appropriate workups.

Table 33-9
Common Electroencephalography (EEG) Abnormalities

Diffuse slowing of background rhythms	Most common EEG abnormality; nonspecific and is present in patients with diffuse encephalopathies of diverse causes
Focal slowing	Suggests localized parenchymal dysfunction and focal seizure disorder; seen with focal fluid collection, such as hematomas
Triphasic waves	Typically consist of generalized synchronous waves occurring in brief runs; approximately one-half of patients with triphasic waves have hepatic encephalopathy, and the remainder have other toxic–metabolic encephalopathies
Epileptiform discharges	Interictal hallmark of epilepsy; strongly associated with seizure disorders
Periodic lateralizing epileptiform discharges	Suggest the presence of an acute destructive cerebral lesion; associated with seizures, obtundation, and focal neurologic signs
Generalized periodic sharp waves	Most commonly seen following cerebral anoxia; recorded in about 90% of patients with Creutzfeldt–Jakob disease

Table 33-10
Electroencephalography (EEG) Alterations Associated with Medication and Drugs

Drug	Alterations

Benzodiazepines	Increased beta activity
Clozapine (Clozaril)	Nonspecific change
Olanzapine (Zyprexa)	Nonspecific change
Risperidone (Risperdal)	Nonspecific change
Quetiapine (Seroquel)	No significant changes
Aripiprazole (Abilify)	No significant changes
Lithium	Slowing or paroxysmal activity
Alcohol	Decreased alpha activity; increased theta activity
Opioids	Decreased alpha activity; increased voltage of theta and delta waves; in overdose, slow waves
Barbiturates	Increased beta activity; in withdrawal states, generalized paroxysmal activity and spike discharges
Marijuana	Increased alpha activity in frontal area of brain; overall slow alpha activity
Cocaine	Similar to marijuana
Inhalants	Diffuse slowing of delta and theta waves
Nicotine	Increased alpha activity; in withdrawal, marked decrease in alpha activity
Caffeine	In withdrawal, increase in amplitude or voltage of theta activity

Topographic Quantitative Electroencephalography (QEEG)

Unlike standard EEG interpretation, which relies on waveform recognition, QEEG involves computer analysis of data extracted from the EEG. Findings are compared with an extensive population database of subjects without any known neurologic or psychiatric disorder as well as QEEG profiles that may be characteristic of some defined diagnostic group. In QEEG, the analog-based electrical signals are processed digitally and converted to graphic, colored topographical displays. These images are sometimes called "brain maps."

Table 33-11
Electroencephalography (EEG) Alterations Associated with Medical Disorders

Seizures	Generalized, Hemispheric, or Focal Spike, Spike-Wave Discharge, or Both
Structural lesions	Focal slowing, with possible focal spike activity
Closed head injuries	Focal slowing (sharply focal head trauma) Focal delta slowing or more widespread slowing (subdural hematomas)
Infectious disorders	Diffuse, often synchronous, high voltage slowing (acute phase of encephalitis)
Metabolic and endocrine disorders	Diffuse generalized slowing of wake frequencies Triphasic waves: 1.5–3.0 per second high-voltage slow waves, with each slow wave initiated by a blunt or rounded spike-like transient (hepatic encephalopathy)
Vascular pathophysiology	Slowed alpha frequency and increased generalized theta slowing (diffuse atherosclerosis) Focal or regional delta activity (cerebrovascular accidents)

QEEG remains primarily a research method, but it holds considerable

clinical potential for psychiatry, mainly in establishing neurophysiologic subtypes of specific disorders and for identifying electrophysiologic predictors of response. Examples of some of the more promising results of QEEG research include the identification of subtypes of cocaine dependence and the subtype most likely to be associated with sustained abstinence; identification of subtypes of OCD that predict clinical responsiveness or lack of responsiveness to SSRIs; and the differentiation between normals, attention-deficit disorder and ADHD, and learning disability subpopulations. QEEG findings in ADHD show that increased theta abnormality frontally may be a strong predictor of response to methylphenidate and other psychostimulants and that favorable clinical responses may be associated with a normalization of the EEG abnormality.

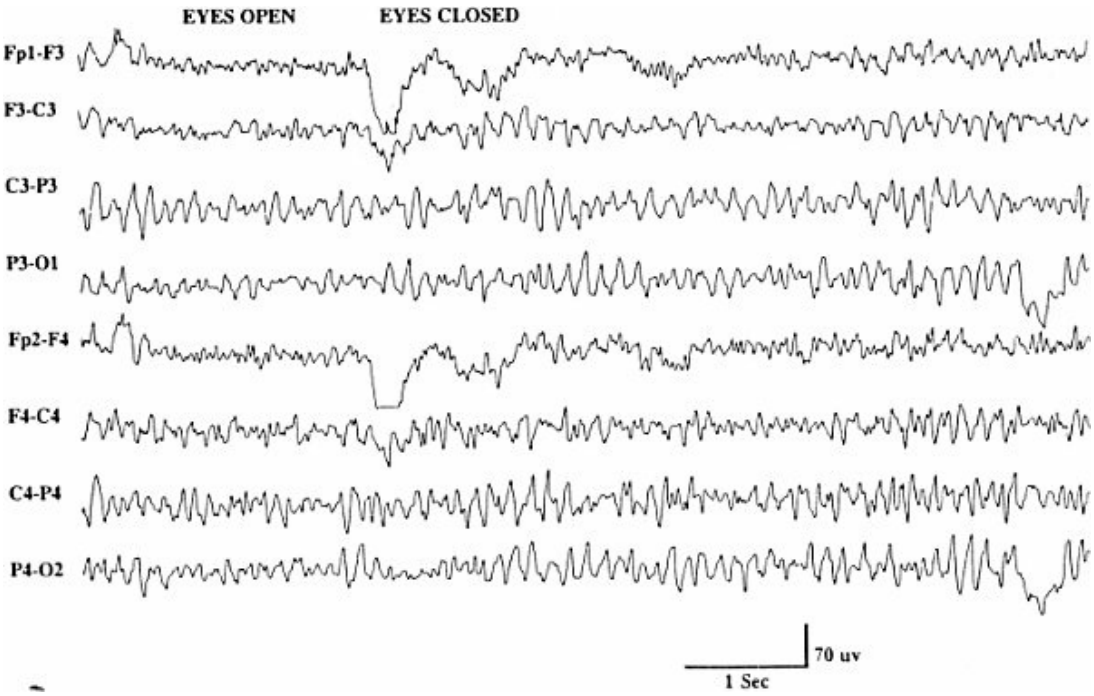


FIGURE 33-35
Diffuse slowing in a 67-year-old patient with dementia. Six- to seven cycles per second (cps) activity predominates over the parieto-occipital regions. Although reactive to eye closure, the frequency of this rhythm is abnormally slow. (Reprinted from Emerson RG, Walesak TS, Turner CA. EEG and evoked potentials. In: Rowland LP, ed. *Merritt's Textbook of Neurology*. 9th ed. Baltimore, MD: Lippincott Williams & Wilkins; 1995:68, with permission.)

Table 33-12
Electroencephalography (EEG) Alterations Associated with

Psychiatric Disorders	
Panic disorder	Paroxysmal EEG changes consistent with partial seizure activity during attack in one-third of patients; focal slowing in about 25% of patients
Catatonia	Usually normal, but EEG indicated in new patient presenting with catatonia to rule out other causes
Attention-deficit/hyperactivity disorder (ADHD)	High prevalence (up to 60%) of EEG abnormalities versus normal controls; spike or spike-wave discharges
Antisocial personality disorder	Increased incidence of EEG abnormalities in those with aggressive behavior
Borderline personality disorder	Positive spikes: 14 and 6 per second seen in 25% of patients
Chronic alcoholism	Prominent slowing and periodic lateralized paroxysmal discharges
Alcohol withdrawal	May be normal in patients who are not delirious; excessive fast activity in patients with delirium
Dementia	Rarely normal in advanced dementia; may be helpful in differentiating pseudodementia from dementia

Cerebral Evoked Potentials

Cerebral EPs are a series of surface (scalp) recordable waves that result from brain visual, auditory, somatosensory, and cognitive stimulation. They are abnormal in many psychiatric conditions, including schizophrenia and Alzheimer disease, thus creating difficulty in using cerebral EPs for differential diagnosis purposes.

CHRONOBIOLOGY

Chronobiology is the study of biologic time. The rotation of the Earth about its axis imposes a 24-hour cyclicity on the biosphere. Although it is widely accepted that organisms have evolved to occupy geographical niches that can be defined by the three spatial dimensions, it is less appreciated that organisms have also evolved to occupy temporal niches that are defined by the fourth dimension—time. Much like light represents a small portion of the electromagnetic spectrum, the 24-hour periodicity represents a small time domain within the spectrum of temporal biology. A broad range of frequencies exist throughout biology, ranging from millisecond oscillations in ocular field potentials to the 17-year cycle of emergence seen in the periodic cicada (*Magicicada* spp.). Although these different periodicities all fall within the realm of chronobiology, *circadian* (Latin: *circa*, about; *dies*, day) rhythms that have a period of about one day are among the most extensively studied, and best understood biologic rhythms.

A defining feature of circadian rhythms is that they persist in the absence of time cues and are not driven merely by the 24-hour environmental cycle. Experimental animals housed for several months under constant darkness, temperature, and humidity continue to exhibit robust circadian rhythms. Maintenance of rhythmicity in a "timeless" environment points to the existence of an internal biologic timing system that is responsible for generating these endogenous rhythms.

The site of the primary circadian oscillator in mammals, including humans, is the suprachiasmatic nucleus (SCN), located in the anterior hypothalamus. The mean circadian period generated by the human SCN is approximately 24.18 hours. Like a slightly slow watch, an individual with such a period gradually comes out of synchrony with the astronomical day. In slightly more than 3 months, an ordinarily diurnal human would be in antiphase to the day–night cycle and thus would become transiently nocturnal. Therefore, a circadian clock must be reset regularly to be effective at maintaining the proper phase relationships of behavioral and physiologic processes within the context of the 24-hour day.

Although factors such as temperature and humidity exhibit daily fluctuations, the environmental parameter that most reliably corresponds to the period of Earth's rotation around its axis is the change in illuminance

associated with the day–night cycle. Organisms have evolved to use this daily change in light levels as a time cue or *zeitgeber* (German: *zeit,* time; *geber,* giver) to reset the endogenous circadian clock. Regulation of the circadian pacemaker through the detection of changes in illuminance requires a photoreceptive apparatus that communicates with the central oscillator. This apparatus is known to reside in the eyes because surgical removal of the eyes renders an animal incapable of resetting its clock in response to light.

The circadian clock drives many rhythms. These include rhythms in behavior, core body temperature, sleep, feeding, drinking, and hormonal levels. One such circadian-regulated hormone is the indoleamine, melatonin. Melatonin synthesis is controlled through a multisynaptic pathway from the SCN to the pineal gland. Serum levels of melatonin become elevated at night and return to baseline during the day. The nocturnal rise in melatonin is a convenient marker of circadian phase. Exposure to light elicits two distinct effects on the daily melatonin profile. First, light acutely suppresses elevated melatonin levels, immediately decreasing them to baseline levels. Second, light shifts the phase of the circadian rhythm of melatonin synthesis. Because melatonin can be assayed easily, it provides a convenient window into the state of the circadian pacemaker. Any perturbation of the clock is reflected in the melatonin profile; thus, melatonin offers an output that can be used to study the regulation of the central circadian pacemaker.

Sleep and Circadian Rhythms

Sleep Regulation. Restful consolidated sleep is most appreciated when sleep disturbances are experienced. Sleep is the integrated product of two oscillatory processes. The first process frequently referred to as the *sleep homeostat,* is an oscillation that stems from the accumulation and dissipation of sleep debt. The biologic substrates encoding sleep debt are not known, although adenosine is emerging as a primary candidate neuromodulator of the sleep homeostat. The second oscillatory process is governed by the circadian clock and controls a daily rhythm in sleep propensity or, conversely, arousal. These interacting oscillations can be dissociated by housing subjects in a timeless environment for several weeks.

The circadian cycle in arousal (wakefulness) steadily increases throughout the day, reaching a maximum immediately before the circadian increase in

plasma melatonin (Fig. 33-36). Arousal subsequently decreases to coincide with the circadian trough in core body temperature. Experiments imposing forced sleep schedules throughout the circadian day have shown that an uninterrupted 8-hour bout of sleep can only be obtained if sleep is initiated approximately 6 hours before the temperature nadir. This nadir typically occurs at approximately 5:00 AM to 6:00 AM. In healthy individuals, initiating sleep between 11:00 PM and 12:00 AM affords the highest probability of getting eight solid hours of sleep.

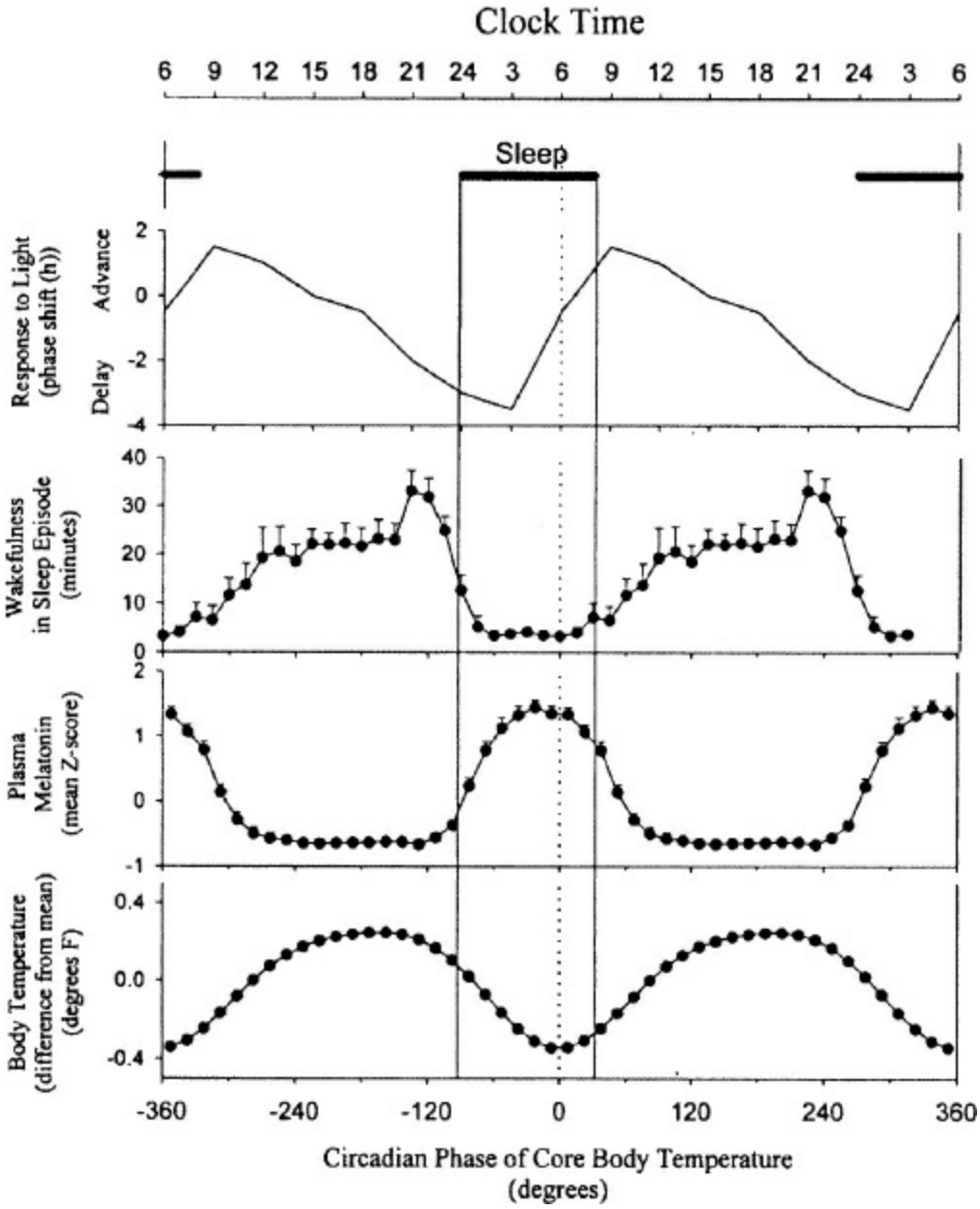


FIGURE 33-36

Relative phase relationship of sleep in young adults to other circadian phase markers. (From Dijk D-J, Lockley SW. Invited review: integration of human sleep-wake regulation and circadian rhythmicity. *J Appl Physiol*. 2002;92:852, with permission.)

It should be stressed that diurnal preference varies among individuals as a function of age, endogenous circadian periods, and other factors. This variability is paralleled by physiology. Clinically, diurnal preference can be quantified using the Horne-Östberg (HO) Morningness-Eveningness Questionnaire (MEQ). In qualitative terms, *morning people* or *morning larks* tend to awaken earlier and experience the core body temperature minimum at an earlier clock time relative to *night people* or *night owls.* Sleep deprivation studies have shown that the homeostatic component of sleep is remarkably similar among individuals of similar age. (It should be noted that there is a well-established age-dependent decline in sleep need.) Therefore, diurnal preference is dictated almost exclusively by the circadian component of sleep regulation.

Circadian Sleep Disorders. Advanced sleep phase syndrome (ASPS) is a pathologic extreme of the morning lark phenotype. An autosomal-dominant familial form of ASPS (FASPS) has been genetically characterized. Afflicted family members exhibit a striking 4-hour advance of the daily sleep–wake rhythm. They typically fall asleep at approximately 7:30 PM and spontaneously awaken at approximately 4:30 AM. Affected individuals have an SNP in the gene encoding hPER2, the human homolog of the mouse *Per2* clock gene. This adenine-to-guanine nucleotide polymorphism results in serine-to-glycine amino acid substitution that causes the mutant protein to be inefficiently phosphorylated by casein kinase Iε, an established component of the circadian molecular clockwork. Similarly, delayed sleep phase syndrome (DSPS) is influenced by genetics. A length polymorphism in a repeat region of the *hPER3* gene appears to be associated with diurnal preference in patients with DSPS, the shorter allele being associated with evening preference.

The advent of the light bulb extended the human day into night. This encroachment on the night, although increasing productivity, affects our sleep patterns (Fig. 33-37). The typical use of artificial lights results in a single, consolidated bout of sleep lasting approximately 8 hours. This pattern of

sleep is uncommon among most other mammals, which typically experience more fractured sleep. Human sleep under more natural photoperiods, where the duration of the night is longer than in artificial settings, becomes decompressed. Specifically, a bimodal distribution of sleep is observed; bouts of sleep occur early and late at night. Periods of *quiet wakefulness* are interspersed between the two primary bouts of sleep. This natural sleep pattern is more similar to the sleep patterns of other mammals.

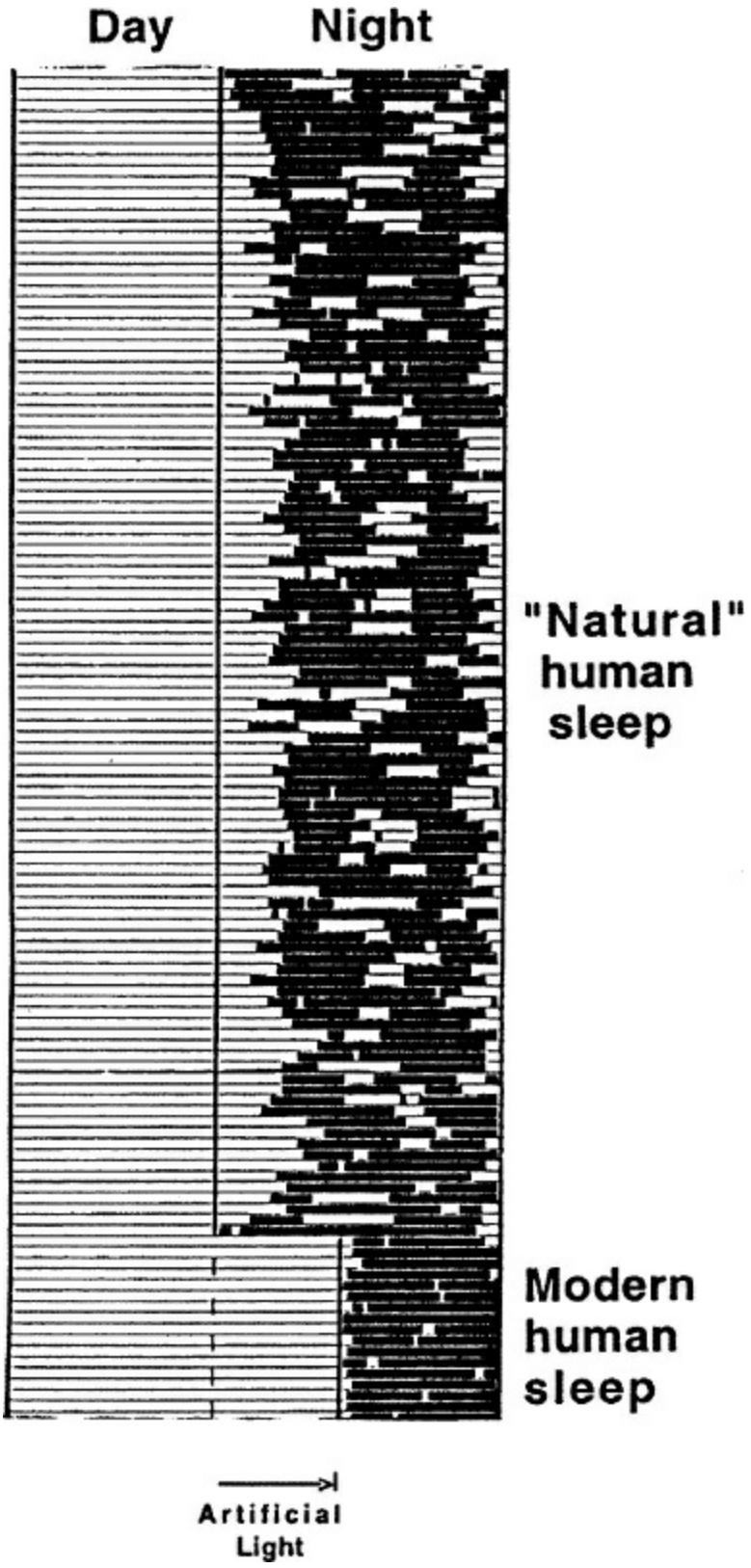


FIGURE 33-37
Change in sleep structure in response to artificial lighting. Total sleep time is reduced, and periods of quiet wakefulness are abolished by extending daytime

into nighttime through artificial lighting. (From Wehr TA, Moul DE, Barbato G, Giesen HA, Seidel JA, Barker C, Bender C. Conservation of photoperiod-responsive mechanisms in humans. *Am J Physiol*. 1993;265:R846, with permission.)

Seasonality

The 24-hour period of the Earth's rotation around its axis is unchanging. However, the Earth's axis is tilted 23.45 degrees from the plane of its orbit around the sun (the ecliptic). The relative day to night proportion within day varies as the Earth proceeds through its orbit of the sun. Many organisms are capable of synchronizing physiology to the seasonal cycle to maximize survival. For example, precise seasonal cycles in reproduction are seen throughout the plant and animal kingdoms. Large mammals that typically have long gestation periods, such as sheep, conceive in the fall when the nights are long and the days are short, so birth occurs during the relatively mild season of spring. These animals are referred to as *short-day breeders.*

Conversely, mammals with gestation periods of only a few weeks, such as hamsters, conceive and give birth during spring and summer, when the days are long and the nights are short. Hence, these animals are referred to as *long-day breeders.* Like circadian rhythms, many of these yearly (circannual) rhythms tend to persist in the absence of seasonal cues with endogenous periods of approximately 1 year.

Melatonin and Seasonality. The most reliable environmental parameter providing a faithful representation of the solar day is the day–night cycle. Similarly, the most reliable environmental parameter reflecting the progression through the seasons is the change in day length, the fraction of the 24-hour day between sunrise and sunset. In seasonally breeding animals, day length is physiologically encoded through the melatonin profile. As described previously, melatonin levels are elevated during the night. A long night, such as that experienced during the short day lengths of winter, results in an elevated melatonin profile of a relatively long duration. A short summer night, by contrast, results in a short duration of elevated melatonin. This seasonal signal is interpreted by the reproductive axis, resulting in an appropriate reproductive response. Melatonin's role in transducing day length was elucidated by pinealectomizing seasonally breeding animals, thereby

removing the primary endogenous source of melatonin. Melatonin was then infused in profiles mimicking long days or short days. The duration of elevated melatonin was the primary determinant of seasonal reproductive status, even when the infused profile was administered under a conflicting day length. Variations in other parameters, such as the amplitude of the melatonin profile, the amount of total melatonin synthesized, or the phase relationship of the profile to the light-dark cycle, are of limited importance in producing a humoral signal that transduces day length.

Reproductive responses to changing day length can be dramatic. A male Siberian hamster (*Phodopus sungorus*) maintained in long days is reproductively competent and typically has a testicular weight of approximately 250 mg per testis. Under short days, however, the testes regress to approximately 15 mg per testis, representing a 94 percent decrease in testicular mass. The same degree of regression is observed in response to melatonin infusions that mimic short days. Communication of the hormonally transduced day length to the reproductive axis is likely to be mediated, at least partially, through melatonin receptors in the *pars tuberalis* of the pituitary gland. The exact mechanism remains unknown, but activation of these receptors is hypothesized to regulate an unidentified factor putatively named tuberalin indirectly. Tuberalin, in turn, controls gene expression and prolactin release from lactotrophs in the adenohypophysis of the pituitary.

Seasonality in Humans. Whether humans are truly seasonal is still a point of considerable debate. Several lines of evidence exist that suggest the presence of a residual tendency toward seasonality. A peak in the rate of suicide occurs in the summer; this peak is cross-cultural. Birth rates also tend to show a seasonal variation; a small but distinguishable peak in the rate of births occurs in spring and summer. This pattern, however, is itself variable and is heavily influenced by unknown cultural and geographic factors. Of interest, the amplitude of the spring–summer birth rate peak has diminished as societies have become industrialized.

The decompressed bimodal structure of human sleep during long nights indicates that the length of natural sleep is related to the length of the night. Potentially, a two-oscillator system could function to maintain proper sleep patterns during changing photoperiods. Such a proposed system would consist of an evening oscillator that tracks the transition from day to night

(dusk) and a morning oscillator that tracks the transition from night to day (dawn). The relative phase differences between these oscillators may encode the changing day lengths associated with the passing of the seasons. Biologic evidence for a two-oscillator system exists in rodents and humans.

The melatonin profile of many vertebrates, including some humans, is bimodal, with evening and morning peaks. In rodents, metabolic and electrophysiologic studies of the SCN typically have been done in brain slices cut in the coronal plane. Results of electrophysiologic studies conducted in brain slices cut in the horizontal plane have provided new insights. The action potential frequency in SCN neurons from horizontally cut preparations is bimodal, with peaks in the early and late subjective day. Furthermore, the interpeak interval varies as a function of the photoperiod in which the animal was housed. These studies lend credence to long-standing suspicions that the SCN of seasonally breeding mammals and, perhaps, nonseasonal mammals harbor a morning and evening oscillator that interact to convey day-length information.

Effect of Aging. In general, as humans age, the circadian period shortens, the circadian phase advances resulting in earlier waking times and bedtimes, the amplitudes of most circadian rhythms decrease, and dramatic phase shifts such as those caused by jet-lag are less tolerated. Again, a mouse model has provided an interesting insight into the interaction of the aging process and the circadian clock. The effect of chronic jet-lag on aged mice has dramatic consequences on mortality. About half of aged mice forced to phase advance 6 hours once per week survive this treatment compared with an 83 percent survival rate in unshifted age-matched mice. Aged mice subjected to weekly 6-hour phase delays show an intermediate survival of 68 percent. These profound effects of phase-shifting are not observed in younger mice. The pathogenesis of chronic jet-lag remains to be determined. Of interest, these mice did not have an increased rate of tumorigenesis. It is likely that in humans, as in mice, the internal desynchrony of oscillators that result from a rotating light schedule may have dire consequences that may be exacerbated by aging.

Circadian Rhythms and Pharmacotherapy

Circadian rhythmicity can be affected by drugs, and conversely, the circadian clock can modulate the efficacy of drugs throughout the day. A better understanding of these interactions will lead to more effective pharmacotherapies. Some of the best-studied interactions between medications and the circadian clock have included the circadian effects of antidepressants. Elevated nocturnal body temperature is a common feature among depressed patients. This effect may be due to reduced amplitude of the master circadian oscillator in the hypothalamus that drives body temperature. TCAs and SSRIs reduce elevated nocturnal body temperature while simultaneously enhancing circadian amplitude. Similarly, many depressed patients exhibit a dampened amplitude in daily activity rhythms. Like body temperature, the amplitude in daily activity cycles of depressed individuals may be augmented by TCA or SSRI therapy.

The use of lithium to treat bipolar disorder has been long established. However, lithium also affects the circadian system, resulting in a lengthening of the circadian period. The molecular mechanism by which this occurs remains unknown. Glycogen synthase kinase 3β (GSK3β) has been implicated in participating within the molecular clock mechanism. Of interest, GSK3β is inhibited by lithium. In cell culture, GSK3β has been shown to stabilize the negative clockwork regulator REV-ERBα via phosphorylation. REV-ERBα typically represses the transcription of the *BMAL1* gene. With lithium, however, GSK3β is inhibited, thereby preventing the phosphorylation and stabilization of REV-ERBα, which is then targeted for proteasomal degradation. The degradation of REV-ERBα results in the de-repression of *BMAL1* transcription. Whether lithium's influence on circadian behavior is attributable to its inhibitory effect on *GSK3*β-mediated stabilization of REV-ERBα remains to be determined.

Short-acting benzodiazepines (e.g., triazolam [Halcion] and brotizolam [Lendormin]) also exert chronobiologic effects. In hamsters, triazolam or brotizolam administered during the middle of the subjective day induces circadian phase advances in activity. Brotizolam has been shown to reduce the light-induced expression of clock genes *Per1* and *Per2* in the SCN. Although benzodiazepines are allosteric modulators of γ-aminobutyric acid A receptors ($GABA_A$), several lines of evidence indicate that the circadian effects of short-acting benzodiazepines require an intact serotonergic system.

When the 5-$HT_{1A/7}$ receptor agonist 8-hydroxy-2-(di-*n*-propylamino)-tetralin (8-OH-DPAT) is injected into hamsters at subjective midday, phase advances in locomotor behavior and SCN neuronal activity are observed in addition to a reduction in *Per1* and *Per2* gene expression in the SCN. Recreational drugs of abuse also affect the circadian system. 3,4-Methylenedioxymethamphetamine (MDMA), or "ecstasy," can act as a serotonin neurotoxin. Hamsters treated with MDMA showed reduced triazolam-induced phase shifts in circadian locomotor activity and a diminished ability to re-entrain rhythms posttreatment. MDMA-treated animals exhibited a reduction of serotonergic axonal terminals in the SCN, again emphasizing the importance of an intact serotonergic system in the regulation of the circadian axis. Recreational use of methamphetamine has increased dramatically. Chronic administration of methamphetamine disorganizes rodent activity rhythms. However, the administration of methamphetamine to rodents rendered arrhythmic through ablation of the SCN results in a reemergence of rhythmicity. The mechanism involved in the rescue of rhythmicity or site of action remains unknown.

The efficacy and toxicity of many pharmacotherapeutics vary as a function of the circadian phase. Daily variations in fixed-dose lethal toxicity have been appreciated in rodents for years. Many anticancer drugs, ranging in mechanism from antimetabolites to DNA intercalators to mitotic inhibitors, have 2- to 10-fold changes in tolerability in rodents over the day. Much of this difference is attributed to circadian variations in the body's ability to absorb, distribute, metabolize, and eliminate toxic compounds. These four processes are affected by underlying circadian rhythms in physiologic processes such as daily variations in gastric pH, gastrointestinal mobility, glomerular filtration rate, and membrane viscosity. The rhythmic intake of food during traditionally timed meals also influences how the body handles therapeutic drugs. It is becoming clear that to maximize efficacy and minimize the toxicity of drugs, the circadian phase of administration must be considered. Appropriate circadian timing of the administration of multiple drugs can be a daunting challenge to infirmed individuals or their caretakers. The development of small implanted programmable pumps that can be directed to administer anticancer drugs or other therapeutics at particular times of day may provide a limited solution to this challenge. The emergence

of the field of chronotherapy reflects our increased understanding of the circadian system's impact on pharmacologic treatment efficacy.

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34

Contributions from the Behavioral and Social Sciences

The underpinnings of psychiatry span many disciplines. The previous chapter covered the neuroscience foundations of psychiatry. However, there are also contributions from the psychosocial sciences, the sociocultural sciences, and many more areas of scientific pursuit. Also, many theories and models of the mind have grown from the psychoanalytic and other theoretical schools that have had a vital influence on the practice of psychiatry, both historically and currently.

This chapter provides a brief tour through some of the significant fields that, along with the neurosciences, have helped psychiatry to grow from a purely empirical practice into a robust and evidence-based branch of medicine. Along the way, it also provides some brief biographies of some of the greatest thinkers and researchers in these fields.

34.1 Contributions from the Psychosocial Sciences

JEAN PIAGET AND COGNITIVE DEVELOPMENT

Jean Piaget (1896–1980) is considered one of the greatest thinkers of the 20th century. His contributions to the understanding of cognitive development had a paradigmatic influence in developmental psychology and had significant implications for interventions with children, both educational and clinical.

Piaget was born in Neuchatel, Switzerland, where he studied at the university and received a doctorate in biology at 22. Interested in psychology, he studied and researched several centers, including the Sorbonne in Paris, and he worked with Eugen Bleuler at the Burghöltzli Psychiatric Hospital.

Piaget created a broad theoretical system for the development of cognitive abilities; in this sense, his work was similar to that of Sigmund Freud, but Piaget emphasized how children think and acquire knowledge.

Widely renowned as a child (or developmental) psychologist, Piaget referred to himself primarily as a *genetic epistemologist;* he defined genetic epistemology as the study of the development of abstract thought based on a biologic or innate substrate. That self-designation reveals that Piaget's central project was more than articulating *developmental child psychology*, as this term is generally understood; it was an account of the progressive development of human knowledge.

Cognitive Development Stages

According to Piaget, the following four major stages lead to the capacity for adult thought (Table 34-1): (1) sensorimotor, (2) preoperational thought, (3) concrete operations, and (4) formal operations. Each stage is a prerequisite for the following one, but the rate at which different children move through different stages varies with their native endowment and environmental circumstances.

Sensorimotor Stage (Birth to 2 Years). Piaget used the term *sensorimotor* to describe the first stage: infants begin to learn through sensory observation, and they gain control of their motor functions through activity, exploration, and manipulation of the environment. Piaget divided this stage into six substages, listed in Table 34-2.

From the outset, biology and experience blend to produce learned behavior. For example, infants are born with a sucking reflex, but a type of learning occurs when infants discover the location of the nipple and alter the shape of their mouths. A stimulus is received, and a response results, accompanied by a sense of awareness that is the first schema, or elementary concept. As infants become increasingly mobile, one schema is built on another, and new and more complex schemata are developed. Infants' spatial,

visual, and tactile worlds expand during this period; children interact actively with the environment and use previously learned behavior patterns. For example, having learned to use a rattle, infants shake a new toy as they did the rattle they had already learned to use. Infants also use a rattle in new ways.

The critical achievement of this period is the development of *object permanence* or the *schema of the permanent object.* This phrase relates to a child's ability to understand that objects have an existence independent of their involvement. Infants learn to differentiate themselves from the world and maintain a mental image of an object, even when it is not visible. When an object is dropped in front of infants, they look down to the ground to search for the object; that is, they behave for the first time as though they have a reality outside themselves.

At about 18 months, infants begin to develop mental symbols and to use words, a process known as *symbolization.* Infants can create a visual image of a ball or a mental symbol of the word *ball* to stand for, or signify, the real object. Such mental representations allow children to operate on new conceptual levels. The attainment of object permanence marks the transition from the sensorimotor stage to the preoperational stage of development.

Stage of Preoperational Thought (2 to 7 Years). During the stage of preoperational thought, children use symbols and language more extensively than in the sensorimotor stage. Thinking and reasoning are intuitive; children learn without the use of reasoning. They cannot think logically or deductively, and their concepts are primitive; they can name objects but not classes of objects. Preoperational thought is midway between socialized adult thought and the completely autistic Freudian unconscious. Events are not linked by logic. Early in this stage, if children drop a glass that breaks, they have no sense of cause and effect. They believe that the glass was ready to break, not that they broke the glass. Children in this stage cannot grasp the sameness of an object in different circumstances: the same doll in a carriage, a crib, or a chair is perceived to be three different objects. During this time, things are represented in terms of their function. For example, a child defines a bike as "to ride" and a hole as "to dig."

Table 34-1

Stages of Intellectual Development Postulated by Piaget

Age (yr)	Period	Cognitive Developmental Characteristics
0–1.5 (to 2)	Sensorimotor	Divided into six stages, characterized by: 1. Inborn motor and sensory reflexes 2. Primary circular reaction 3. Secondary circular reaction 4. Use of familiar means to obtain ends 5. Tertiary circular reaction and discovery through active experimentation 6. Insight and object permanence
2–7	Preoperations subperiod[a]	Deferred imitation, symbolic play, graphic imagery (drawing), mental imagery, and language
7–11	Concrete operations	Conservation of quantity, weight, volume, length, and time based on reversibility by inversion or reciprocity; operations; class inclusion and seriation
11–end of adolescence	Formal operations	Combinatorial system, whereby variables are isolated and all possible combinations are examined; hypotheticodeductive thinking

[a]This subperiod is considered by some authors to be a separate developmental period.

In this stage, children begin to use language and drawings in more elaborate ways. From one-word utterances, two-word phrases develop, made up of either a noun and a verb or a noun and an objective. A child may say, “Bobby eat,” or “Bobby up.”



Table 34-2
Piaget’s Sensorimotor Period of Cognitive Development

Age	Characteristics
Birth–2 mo	Uses inborn motor and sensory reflexes (sucking, grasping, looking) to interact and accommodate to the external world
2–5 mo	Primary circular reaction: Coordinates activities of own body and five senses (e.g., sucking thumb); reality remains subjective—does not seek stimuli outside of its visual field; displays curiosity
5–9 mo	Secondary circular reaction: Seeks out new stimuli in the environment; starts both to anticipate consequences of own behavior and to act purposefully to change the environment; beginning of intentional behavior
9 mo–1 yr	Shows preliminary signs of object permanence; has a vague concept that objects exist apart from itself; plays peek-a-boo; imitates novel behaviors
1 yr–18 mo	Tertiary circular reaction: Seeks out new experiences; produces novel behaviors
18 mo–2 yr	Symbolic thought: Uses symbolic representations of events and objects; shows signs of reasoning (e.g., uses one toy to reach for and get another); attains object permanence

Children in the preoperational stage cannot deal with moral dilemmas,

although they feel good and bad. For example, when asked, “Who is more guilty, the person who breaks one dish on purpose or the person who breaks ten dishes by accident?” a young child usually answers that the person who breaks ten dishes by accident is more guilty because more dishes are broken. Children in this stage have a sense of *immanent justice;* the belief that punishment for bad deeds is inevitable.

Children in this developmental stage are *egocentric:* they see themselves as the center of the universe; they have a limited point of view, and they cannot take the role of another person. Children cannot modify their behavior for someone else; for example, children are not negativistic when they do not listen to a command to be quiet because their brother has to study. Instead, egocentric thinking prevents an understanding of their brother’s point of view.

During this stage, children also use magical thinking, called *phenomenalistic causality,* in which events that occur together are thought to cause one another (e.g., thunder causes lightning, and bad thoughts cause accidents). Also, children use *animistic thinking,* which is the tendency to endow physical events and objects with life-like psychological attributes, such as feelings and intentions.

SEMIOTIC FUNCTION. The semiotic function emerges during the preoperational period. With this new ability, children can represent something—such as an object, an event, or a conceptual scheme—with a signifier, representing a representative function (e.g., language, mental image, symbolic gesture). That is, children use a symbol or sign to stand for something else. Drawing is a semiotic function initially done as a playful exercise but eventually signifying something else in the real world.

Stage of Concrete Operations (7 to 11 Years). The stage of concrete operations is so named because, in this period, children operate and act on the concrete, real, and perceivable world of objects and events. Egocentric thought is replaced by *operational thought,* which involves dealing with a wide array of information outside the child. Therefore, children can now see things from someone else’s perspective.

Children in this stage begin to use limited logical thought processes and can serialize, order, and group things into classes based on common

characteristics. *Syllogistic reasoning,* in which a logical conclusion is formed from two premises, appears during this stage; for example, all horses are mammals (premise); all mammals are warm-blooded (premise); therefore, all horses are warm-blooded (conclusion). Children can reason and follow the rules and regulations. They can regulate themselves, and they begin to develop a moral sense and a code of values.

Children who become overly invested in rules may show obsessive-compulsive behavior; children who resist a code of values often seem willful and reactive. The most desirable developmental outcome in this stage is that a child attains healthy respect for rules and understands that there are legitimate exceptions to rules.

Conservation is the ability to recognize that, although the shape of objects may change, the objects still maintain or conserve other characteristics that enable them to be recognized as the same. For example, if a ball of clay is rolled into a long, thin sausage shape, children recognize that each form contains the same amount of clay. An inability to conserve (which is characteristic of the preoperational stage) is observed when a child believes there is more clay in the sausage-shaped piece because it is longer. *Reversibility* is the capacity to understand the relation between things and realize that one thing can turn into another and back again—for example, ice and water.

The most critical sign that children are still in the preoperational stage is that they have not achieved conservation or reversibility. The ability of children to understand concepts of quantity is one of Piaget's most influential cognitive developmental theories. Measures of quantity include measures of substance, length, number, liquids, and area (Fig. 34-1).

The 7- to 11-year-old child must organize and order occurrences in the real world. Dealing with the future and its possibilities occur in the formal operational stage.

Stage of Formal Operations (11 through the End of Adolescence). The stage of formal operations is so named because young persons' thinking operates in a formal, highly logical, systematic, and symbolic manner. This stage is characterized by the ability to think abstractly, to reason deductively, and to define concepts, and also by the emergence of skills for dealing with permutations and combinations; young persons can grasp the concept of

probabilities. Adolescents attempt to deal with all possible relations and hypotheses to explain data and events during this stage. Language use is complex; it follows formal rules of logic and is grammatically correct. Abstract thinking is shown by adolescents' interest in various issues—philosophy, religion, ethics, and politics.

HYPOTHETICODEDUCTIVE THINKING. Hypotheticodeductive thinking, the highest organization of cognition, enables persons to make a hypothesis or proposition and to test it against reality. *Deductive reasoning* moves from the general to the particular and is a more complicated process than *inductive reasoning,* which moves from the particular to the general.

Because young persons can reflect on their own and other persons' thinking, they are susceptible to self-conscious behavior. As adolescents attempt to master new cognitive tasks, they may return to egocentric thought, but on a higher level than in the past. For example, adolescents may think that they can accomplish everything or can change events by thought alone. Not all adolescents enter the stage of formal operations at the same time or to the same degree. Depending on individual capacity and intervening experience, some may not reach the stage of formal operational thought at all and may remain in the concrete operational mode throughout life.

Psychiatric Applications

Piaget's theories have many psychiatric implications. Hospitalized children in the sensorimotor stage have not achieved object permanence and, therefore, have separation anxiety. They do best if their mothers are allowed to stay with them overnight. Children at the preoperational stage, who cannot deal with concepts and abstractions, benefit more from role-playing proposed medical procedures and situations than by having them described in detail. For example, a child who receives intravenous therapy is helped by acting out the procedure with a toy intravenous set and dolls.

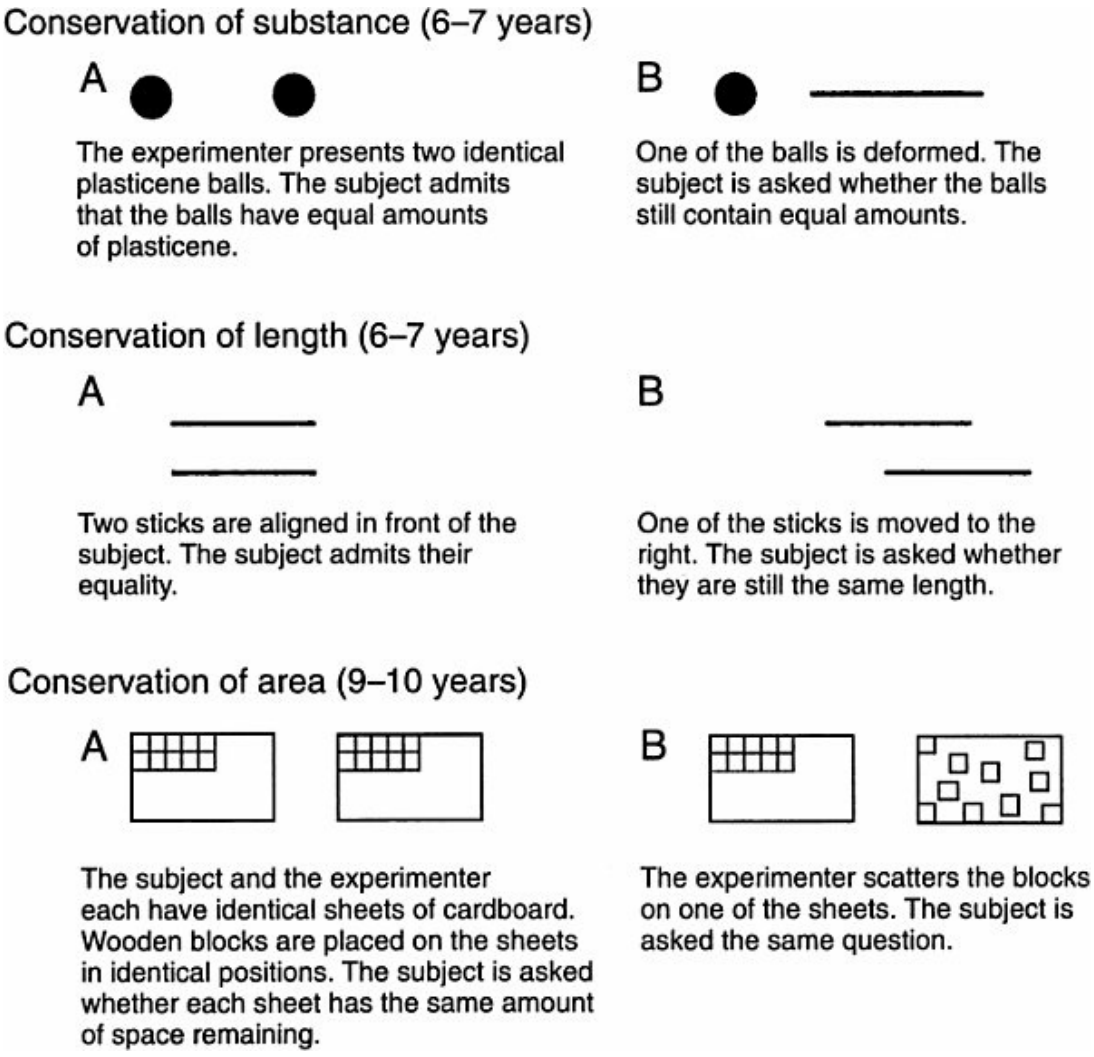


FIGURE 34-1
Some simple tests for conservation, with approximate ages of attainment. When the sense of conservation is achieved, the child answers that **B** contains the same quantity as **A**. (Modified from Lefrancois GR. *Of Children: An Introduction to Child Development*. Belmont, CA: Wadsworth; 1973:305, with permission.)

Because children at the preoperational stage do not understand cause and effect, they may interpret physical illness as punishment for bad thoughts or deeds. Because they have not yet mastered the capacity to conserve and do not understand the concept of reversibility (which usually occurs during the concrete operational stage), they cannot understand that a broken bone mends or that blood loss in an accident can be replaced.

Adolescents' thinking, during the stage of formal operations, may appear overly abstract when it is, in fact, a normal developmental stage. Adolescent turmoil may not herald a psychotic process but may well result from a typical

adolescent's coming to grips with newly acquired abilities to deal with the unlimited possibilities of the surrounding world.

Adults under stress may regress cognitively as well as emotionally. Their thinking can become preoperational, egocentric, and sometimes animistic.

Implications for Psychotherapy. Piaget was not an applied psychologist and did not develop the implications of his cognitive model for psychotherapeutic intervention. Nevertheless, his work formed one of the foundations of the *cognitive revolution* in psychology. One aspect of this revolution was an increasing emphasis on the cognitive components of the therapeutic endeavor. In contrast to classical psychodynamic therapy, which focused primarily on drives and affects, and in contrast to behavior therapy, which focused on overt actions, cognitive approaches to therapy focused on thoughts, including automatic assumptions, beliefs, plans, and intentions. By including "theory theory" and "script theory," we can see additional applications to psychotherapy.

Cognitive development theory has influenced psychotherapeutic approaches in multiple ways. Some therapists have taken developmental notions from Piaget's work and developed intervention techniques. Others have developed cognitive models of treatment independent of Piaget but with heavy reliance on cognition. Others have included Piaget's concepts in a broader set of constructs to undergird new developmental approaches to psychotherapy.

First, some psychotherapists applied Piagetian notions directly to child interventions. Susan Harter, for example, discussed techniques for helping young children become aware of divergent or contradictory emotions and to integrate these complex emotions within a more abstract or higher class of emotions. One of Harter's techniques is to ask the young child to make a drawing that shows different and conflicting feelings in one person. This technique represents an application of the concrete operation of class inclusion to the realm of the emotions. Harter's work applied Piagetian findings to the common therapeutic problem of helping children to recognize, tolerate, and integrate mixed or ambivalent affects within stable object relations.

As such, it drew on cognitive theory and psychodynamic theory. Similar techniques are important in work with children who have been exposed to

trauma or sexual abuse. It is an essential component of such work to assist them in labeling, differentiating, and accepting the full range of emotions stemming from these experiences.

Second, other psychotherapists developed treatment models that, although not directly dependent on Piagetian psychology, emphasized core ideas quite similar to those Piaget discovered in his naturalistic observations of cognitive development. These models are even more closely aligned with recent developments in "theory theory." Aaron Beck, for example, developed an entire school of cognitive therapy that focuses on the role of cognitions in causing or maintaining psychopathology. Cognitive therapy is an effective treatment for problems as diverse as depression, anxiety disorders, and substance use disorders.

A core idea in cognitive therapy is that the patient has developed certain core beliefs, aspects of the self-schema, and conditional probability beliefs as a result of developmental experiences, and these contribute to emotional or behavioral problems. For example, depressed persons may have the core belief, "I am unlovable." Addicted persons may have the belief, "Unless I drink, I cannot feel happy." In cognitive therapy, the person can be assisted in identifying the negative automatic thoughts and underlying dysfunctional attitudes or beliefs that contribute to emotional distress or addictive behavior. The critical therapeutic process after identification of the maladaptive thoughts is to help the patient view these thoughts more objectively, not unquestioningly take them as veridical. Here, cognitive therapy emphasizes evidence, consistent both with Piagetian theory and "theory theory." The patient is assisted in seeking out evidence to test negative thinking; active involvement, rather than passive listening, is required.

What the cognitive therapist accomplishes through such techniques as Socratic questioning and asking if there are other ways to look at the same event is similar to what the talented teacher does in guiding children to a more adequate, more intelligent understanding of operational tasks. The notion of equilibration is relevant in both instances. By helping the individual see that previous cognitive structures are inadequate, the therapist or teacher disturbs the old cognitive structure, and the patient or student experiences a disruption that leads to the search for more-adequate structures. The compensation for external disturbance is what Piaget termed *equilibration.* New structures can be constructed only through a process of accommodation,

enabling the subject to assimilate a more comprehensive array of data, a new perspective, or more complex information.

Because it requires *thinking about thinking*, cognitive therapy seems to require formal operational thinking, although this has not been empirically tested. At the least, it requires the ability to recognize and articulate affects, to recognize and label events that give rise to affects, and to translate into thought the mediating process that occurs rapidly between the event and the affect. Cognitive-behavioral models of psychotherapy include cognitive techniques and more behavioral, interactive techniques, such as increasing pleasant activities and improving communication and problem-solving skills. It is possible that the less-cognitive, more-behavioral techniques, although requiring a lower level of cognitive development, can also lead to the garnering of evidence and modification of specific expectancies, attributions, and self-schemata.

Because "script theory" or narrative approaches to cognition in psychotherapy are empirically based, generated by repetitive experiences rather than by reflective abstraction, and domain-specific, they may have even more general application to psychotherapy than classic Piagetian theories or "theory theory." For example, in dialectical behavior therapy, patients provide a "chain analysis" of events, feelings, thoughts, situational stimuli, and interpersonal factors that led to negative or self-damaging behavior. This narrative provides guidance to the patient and the therapist about where and how to intervene to prevent similar behavior.

Developmentally Based Psychotherapy. Developmentally based psychotherapy, developed by Stanley Greenspan, M.D., integrates cognitive, affective, drive, and relationship-based approaches with a new understanding of the stages of human development. The clinician first determines the level of the patient's ego or personality development and the presence or absence of deficits or constrictions. For example, can the person regulate activity and sensations, relate to others, read nonverbal affective symbols, represent experience, build bridges between representations, integrate emotional polarities, abstract feelings, and reflect on internal wishes and feelings?

From a developmental point of view, the integral parts of the therapeutic process include learning how to regulate experience; to engage more fully and deeply in relationships; to perceive, comprehend, and respond to

complex behaviors and interactive patterns; and to be able to engage in the ever-changing opportunities, tasks, and challenges during life (e.g., adulthood and aging) and, throughout, to observe and reflect on one's own and others' experiences. These processes are the foundation of the ego, and more broadly, the personality. Their presence constitutes emotional health and their absence, emotional disorder. The developmental approach describes how to harness these core processes and assist the patients in mobilizing their growth.

ATTACHMENT THEORY

Attachment and Development

Attachment can be defined as the emotional tone between children and their caregivers and is evidenced by an infant's seeking and clinging to the caregiving person, usually the mother. By their first month, infants usually have begun to show such behavior, designed to promote proximity to the desired person.

Attachment theory originated in the work of John Bowlby, a British psychoanalyst (1907–1990). In his studies of infant attachment and separation, Bowlby pointed out that attachment constituted a central motivational force and that mother–child attachment was an essential medium of human interaction that had significant consequences for later development and personality functioning. Being monotropic, infants tend to attach to one person; but they can form attachments to several persons, such as the father or a surrogate. Attachment develops gradually; it results in an infant's wanting to be with a preferred person, who is perceived as stronger, wiser, and able to reduce anxiety or distress. Attachment thus gives infants feelings of security. The interaction between mother and infant facilitates the process; the amount of time together is less critical than the activity between the two.

The term *bonding* is sometimes used synonymously with attachment, but the two are different phenomena. Bonding concerns the mother's feelings for her infant and differs from attachment. Mothers generally do not rely on their infants as a source of security, as is the case in attachment behavior. Much research reveals that the bonding of mother to infant occurs when there is

skin-to-skin contact between the two or when other types of contact, such as voice and eye contact, are made. Some workers have concluded that a mother who has skin-to-skin contact with her baby immediately after birth shows a stronger bonding pattern and may provide more attentive care than a mother who does not have this experience. Some researchers have even proposed a critical period immediately after birth, during which such skin-to-skin contact must occur if bonding is to occur. This concept is much disputed: many mothers are bonded to their infants and display excellent maternal care even though they did not have skin-to-skin contact immediately postpartum. Because human beings can develop representational models of their babies in utero and even before conception, this representational thinking may be as crucial to the bonding process as skin, voice, or eye contact.

Ethologic Studies. Bowlby suggested a Darwinian evolutionary basis for attachment behavior; namely, such behavior ensures that adults protect their young. Ethologic studies show that nonhuman primates and other animals exhibit attachment behavior patterns that are presumably instinctual and are governed by inborn tendencies. An example of an instinctual attachment system is *imprinting*. Certain stimuli can elicit innate behavior patterns during the first few hours of an animal's behavioral development; thus, the animal offspring becomes attached to its mother at a critical period early in its development. A similar sensitive or critical period during which attachment occurs has been postulated for human infants. The presence of imprinting behavior in humans is highly controversial, but bonding and attachment behavior during the first year of life closely approximate the critical period; in humans, however, this period occurs over years rather than hours.

HARRY HARLOW. Harry Harlow's work with monkeys is relevant to attachment theory. Harlow demonstrated the emotional and behavioral effects of isolating monkeys from birth and keeping them from forming attachments. The isolates were withdrawn, unable to relate to peers, unable to mate, and incapable of caring for their offspring.

Phases of Attachment

In the first attachment phase, sometimes called the *preattachment stage* (birth

to 8 or 12 weeks), babies orient to their mothers, follow them with their eyes over a 180-degree range, and turn toward and move rhythmically with their mother's voice. In the second phase, sometimes called *attachment in the making* (8 to 12 weeks to 6 months), infants become attached to one or more persons in the environment. In the third phase, sometimes called *clear-cut attachment* (6 through 24 months), infants cry and show other signs of distress when separated from the caretaker or mother; this phase can occur as early as 3 months in some infants. On being returned to the mother, the infant stops crying and clings to gain further assurance of the mother's return. Sometimes, seeing the mother after a separation is sufficient for crying to stop. In the fourth phase (25 months and beyond), the mother figure is seen as independent, and a more complex relationship between the mother and the child develops. Table 34-3 summarizes the development of standard attachment from birth through 3 years.

Mary Ainsworth. Mary Ainsworth (1913–1999) was a Canadian developmental psychologist from the University of Toronto. She described three main types of insecure attachment: insecure-avoidant, insecure-ambivalent, and insecure-disorganized. The insecure-avoidant child, having experienced brusque or aggressive parenting, tends to avoid close contact with people and lingers near caregivers rather than approaching them directly when faced with a threat. The insecure-ambivalent child finds exploratory play difficult, even in the absence of danger, and clings to his or her inconsistent parents. Insecure-disorganized children have parents who are emotionally absent with a parental history of abuse in their childhood. These children tend to behave in bizarre ways when threatened. According to Ainsworth, disorganization is a severe form of insecure attachment and a possible precursor of severe personality disorder and dissociative phenomena in adolescence and early adulthood.



Table 34-3
Normal Attachment

Birth to 30 days
Reflexes at birth
Rooting

- Head turning
- Sucking
- Swallowing
- Hand-mouth
- Grasp
- Digital extension
- Crying—signal for particular kind of distress
- Responsiveness and orientation to mother's face, eyes, and voice
- 4 days—anticipatory approach behavior at feeding
- 3–4 wk—infant smiles preferentially to mother's voice

Age 30 days through 3 mo
- Vocalization and gaze reciprocity further elaborated from 1 to 3 mo; babbling at 2 mo, more with the mother than with a stranger
- Social smile
- In strange situation, increased clinging response to mother

Age 4 through 6 mo
- Briefly soothed and comforted by sound of mother's voice
- Spontaneous, voluntary reaching for mother
- Anticipatory posturing to be picked up
- Differential preference for mother intensifies
- Subtle integration of responses to mother

Age 7 through 9 mo
- Attachment behaviors further differentiated and focused specifically on mother
- Separation distress, stranger distress, strange-place distress

Age 10 through 15 mo
- Crawls or walks toward mother
- Subtle facial expressions (coyness, attentiveness)
- Responsive dialogue with mother clearly established
- Early imitation of mother (vocal inflections, facial expression)
- More fully developed separation distress and mother preference
- Pointing gesture
- Walking to and from mother
- Affectively positive reunion responses to mother after separation

or, paradoxically, short-lived, active avoidance or delayed protest
Age 16 mo through 2 yr
Involvement in imitative jargon with mother (12–14 mo)
Head-shaking “no” (15–16 mo)
Transitional object used during the absence of mother
Separation anxiety diminishes
Mastery of strange situations and persons when mother is near
Evidence of delayed imitation
Object permanence
Microcosmic symbolic play
Age 25 mo through 3 yr
Able to tolerate separations from mother without distress when familiar with surroundings and given reassurances about mother’s return
Two- and three-word speech
Stranger anxiety much reduced
Object consistency achieved—maintains composure and psychosocial functioning without regression in absence of mother
Microcosmic play and social play; cooperation with others begins

Based on material by Justin Call, M.D.

Mary Ainsworth expanded on Bowlby’s observations and found that the interaction between the mother and her baby during the attachment period significantly influences the baby’s current and future behavior. Patterns of attachments vary among babies; for example, some babies signal or cry less than others. Sensitive responsiveness to infant signals, such as cuddling a crying baby, causes infants to cry less in later months, rather than reinforcing crying behavior. Close bodily contact with the mother when the baby signals for her is also associated with the growth of self-reliance, rather than a clinging dependence, as the baby grows older. Unresponsive mothers produce anxious babies; these mothers often have lower intelligence quotients (IQs) and are emotionally more immature and younger than responsive mothers.

Table 34-4
The Strange Situation

Episode[a]	Persons Present	Change
1	Parent, infant	Enter room
2	Parent, infant, stranger	Unfamiliar adult joins the dyad
3	Infant, stranger	Parent leaves
4	Parent, infant	Parent returns, stranger leaves
5	Infant	Parent leaves
6	Infant, stranger	Stranger returns
7	Parent, infant	Parent returns, stranger leaves

[a]All episodes are usually 3 minutes long, but episodes 3, 5, and 6 can be curtailed if the infant becomes too distressed, and episodes 4 and 7 are sometimes extended.

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Ainsworth also confirmed that attachment serves to reduce anxiety. What she called the *secure base effect* enables children to move away from attachment figures and to explore the environment. Inanimate objects, such as a teddy bear and a blanket (called the transitional object by Donald Winnicott), also serve as a secure base, one that often accompanies them as they investigate the world.

STRANGE SITUATION. Ainsworth developed the strange situation, the research protocol for assessing the quality and security of an infant's attachment. In this procedure, the infant is exposed to escalating amounts of stress; for example, the infant and the parent enter an unfamiliar room, an unfamiliar adult then enters the room, and the parent leaves the room. The protocol has seven steps (Table 34-4). According to Ainsworth's studies, about 65 percent of infants are securely attached by 24 months.

Anxiety

Bowlby's theory of anxiety holds that a child's sense of distress during separation is perceived and experienced as anxiety and is the prototype of anxiety. Any stimuli that alarm children and cause fear (e.g., loud noises, falling, and cold blasts of air) mobilize signal indicators (e.g., crying) that cause the mother to respond in a caring way by cuddling and reassuring the child. The mother's ability to relieve the infant's anxiety or fear is fundamental to the growth of attachment in the infant. When the mother is close to the child, and the child experiences no fear, the child gains a sense of security, the opposite of anxiety. When the mother is unavailable to the infant because of physical absence (e.g., if the mother is in prison) or because of psychological impairment (e.g., severe depression), anxiety develops in the infant.

Separation anxiety is the response of a child who is isolated or separated from its mother or caretaker. It is most common at 10 to 18 months of age and generally disappears by the end of the third year. It is usually expressed with tearfulness or irritability. Somewhat earlier (at about 8 months), stranger anxiety is an anxiety response to someone other than the caregiver appears.

Signal Indicators. Signal indicators are infants' signs of distress that prompt or elicit a behavioral response in the mother. The primary signal is crying. The three types of signal indicators are hunger (the most common), anger, and pain. Some mothers can distinguish between them, but most mothers generalize the hunger cry to represent distress from pain, frustration, or anger. Other signal indicators that reinforce attachment are smiling, cooing, and looking. The sound of an adult human voice can prompt these indicators.

Losing Attachments. Persons' reactions to the death of a parent or a spouse can be traced to the nature of their past and present attachment to the lost figure. An absence of demonstrable grief may be owing to real experiences of rejection and the lack of closeness in the relationship. The person may even consciously offer an idealized picture of the deceased. Persons who show no grief usually try to present themselves as independent and disinterested in closeness and attachment.

Sometimes, however, the severing of attachments is traumatic. The death of a parent or a spouse can precipitate a depressive disorder, and even

suicide, in some persons. The death of a spouse increases the chance that the surviving spouse will experience a physical or mental disorder during the next year. The onset of depression and other dysphoric states often involves rejection by a significant figure in a person's life.

Disorders of Attachment

Attachment disorders are characterized by biopsychosocial pathology that results from maternal deprivation, a lack of care by, and interaction with, the mother or caregiver. Failure-to-thrive syndromes, psychosocial dwarfism, separation anxiety disorder, avoidant personality disorder, depressive disorders, delinquency, academic problems, and borderline intelligence have been traced to negative attachment experiences. When maternal care is deficient because (1) a mother is mentally ill, (2) a child is institutionalized for a long time, or (3) the primary object of attachment dies, children sustain emotional damage. Bowlby initially thought that the damage was permanent and invariable. However, he revised his theories to consider the time at which the separation occurred, the type and degree of separation, and the level of security that the child experienced before the separation.

Bowlby described a predictable set and sequence of behavior patterns in children who are separated from their mothers for extended periods (more than 3 months): protest, in which the child protests the separation by crying, calling out, and searching for the lost person; despair, in which the child appears to lose hope that the mother will return; and detachment, in which the child emotionally separates him- or herself from the mother. Bowlby believed that this sequence involves ambivalent feelings toward the mother; the child both wants her and is angry with her for her desertion.

Children in the detachment stage respond indifferently when the mother returns; the mother has not been forgotten, but the child is angry at her for having gone away in the first place and fears that she will go away again. Some children have affectionless personalities characterized by emotional withdrawal, little or no feeling, and a limited ability to form affectionate relationships.

Anaclitic Depression. Anaclitic depression, also known as hospitalism, was first described by René Spitz in infants who had made normal

attachments but were suddenly separated from their mothers for varying times and placed in institutions or hospitals. The children became depressed, withdrawn, nonresponsive, and vulnerable to physical illness, but they recovered when their mothers returned, or surrogate mothering was available.

Child Maltreatment

Abused children often maintain their attachments to abusive parents. Studies of dogs have shown that severe punishment and maltreatment increase attachment behavior. When children are hungry, sick, or in pain, they too show clinging attachment behavior. Similarly, when children are rejected by their parents or are afraid of them, their attachment may increase; some children want to remain with an abusive parent. Nevertheless, when a choice must be made between a punishing and a nonpunishing figure, the nonpunishing person is the preferable choice, especially if they are sensitive to the child's needs.

Psychiatric Applications

The applications of attachment theory in psychotherapy are numerous. When a patient can attach to a therapist, a secure base effect is seen. The patient may then be able to take risks, mask anxiety, and practice new patterns of behavior that otherwise might not have been attempted. Patients whose impairments can be traced to never having made an attachment in early life may do so for the first time in therapy, with salutary effects.

Patients whose pathology stems from exaggerated early attachments may attempt to replicate them in therapy. Therapists must enable such patients to recognize the ways their early experiences have interfered with their ability to achieve independence.

For patients who are children and whose attachment difficulties may be more apparent than those of adults, therapists represent consistent and trusted figures who can engender a sense of warmth and self-esteem in children, often for the first time.

Relationship Disorders. A person's psychological health and sense of well-being depend significantly on the quality of his or her relationships and attachment to others, and a core issue in all close personal relationships is

establishing and regulating that connection. In a typical attachment interaction, one person seeks more proximity and affection, and the other either reciprocates, rejects, or disqualifies the request. A pattern is shaped through repeated exchanges. Distinct attachment styles have been observed. Adults with an anxious-ambivalent attachment style tend to be obsessed with romantic partners, suffer from extreme jealousy, and have a high divorce rate. Persons with an avoidant attachment style are relatively uninvested in close relationships, although they often feel lonely. They seem afraid of intimacy and tend to withdraw when there is stress or conflict in the relationship. Break-up rates are high. Persons with a secure attachment style are highly invested in relationships and tend to behave without much possessiveness or fear of rejection.

LEARNING THEORY

Learning is defined as a change in behavior resulting from repeated practice. The principles of learning are always operating and always influencing human activity. Learning principles are often deeply involved in the etiology and maintenance of psychiatric disorders because so much of human behavior (including overt behavior, thought patterns, and emotion) is acquired through learning. Learning processes also strongly influence psychotherapy because human behavior changes. Thus, learning principles can influence the effectiveness of therapy. No method of therapy can be said to be immune to the effects of learning. Even the simple prescription of medication can bring learning processes into play because the patient will have opportunities to learn about the drug's effects and side effects, will need to learn to adhere to the instructions and directions for taking it, and will need to learn to overcome any resistance to adherence.

Basic Concepts and Considerations

A great deal of modern research on learning still focuses on Pavlovian (classical) and operant learning. Pavlovian conditioning, developed by Ivan Petrovich Pavlov (1849–1936), occurs when neutral stimuli are associated with a psychologically significant event. The main result is that the stimuli come to evoke a set of responses or emotions that may contribute to many clinical disorders, including (but not limited to) anxiety disorders and drug

dependence. The events in Pavlov's experiment are often described using terms designed to make the experiment applicable to any situation. The food is the unconditional stimulus (*US*) because it unconditionally elicits salivation before the experiment begins. The bell is the conditional stimulus (*CS*) because it only elicits the salivary response conditional on the bell–food pairings. The new response to the bell is correspondingly called the conditional response (*CR*), and the natural response to the food itself is the unconditional response (*UR*). Modern laboratory studies of conditioning use an extensive range of CSs and USs and measure a wide range of conditioned responses.

Operant conditioning, developed by B.F. Skinner (1904–1990), occurs when a behavior (instead of a stimulus) is associated with a psychologically significant event. In the laboratory, the most famous experimental arrangement is when a rat presses a lever to earn food pellets. In this case, as opposed to Pavlov's, the behavior is an operant because it operates on the environment. The food pellet is a reinforcer—an event that increases the strength of the behavior of which it is made a consequence. A central idea behind this method is that the rat's behavior is "voluntary" in the sense that the animal is not compelled to make the response (it can perform it whenever it "wants" to). In this sense, it is similar to the thousands of operant behaviors that humans choose to commit—freely—in any day. Of course, the even larger idea is that even though the rat's behavior appears as though it is voluntary, it is lawfully controlled by its consequences: if the experimenter were to stop delivering the food pellet, the rat would stop pressing the lever, and if the experimenter were to allow the lever press to produce larger pellets, or perhaps pellets at a higher probability or rate, then the rate of the behavior might increase. The point of operant conditioning experiments, then, is mostly to understand the relation of behavior to its payoff.

Pavlovian and operant conditioning differ in several ways. One of the most fundamental differences is that the responses observed in Pavlov's experiment are *elicited* and thus controlled by presenting an antecedent stimulus. In contrast, the "response" observed in Skinner's experiment is not elicited or compelled by an antecedent stimulus in any obvious way—its consequences instead control it. This distinction between *operants* and *respondents* is essential in clinical settings. If a young patient is referred to the clinic for acting out in the classroom, an initial goal of the clinician will

be to determine whether the behavior is a respondent or an operant. Then the clinician will go about changing either its antecedents or its consequences, respectively, to reduce its probability of occurrence.

Despite the academic separation of operant and respondent conditioning, they have an important common function: both learning processes allow organisms to adapt to the environment. The idea is illustrated by considering the *law of effect* (Fig. 34-2), which says that whether an operant behavior increases or decreases in strength depends on the effect it has on the environment. When the action leads to a positive outcome, the action is strengthened; conversely, when the action leads to a negative outcome, we have *punishment*, and the action is weakened. Similarly, when an action decreases the probability of a positive event, behavior also declines. (Such a procedure is now widely known as a *time-out* from reinforcement.) When an action terminates or prevents the occurrence of a negative event, the behavior will strengthen. By enabling the organism to maximize its interaction with positive events and minimize its interaction with negative ones, operant conditioning allows the organism to optimize its interaction with the environment. Reward learning also provides a framework for understanding the development of somewhat maladaptive behaviors like overeating (in which behavior is reinforced by food) and drug-taking (in which the pharmacologic effects of drugs reinforce behaviors)—cases in which reward principles lead to psychopathology.

A parallel to Figure 34-2 exists in Pavlovian conditioning, in which one can likewise think of whether the CS is associated with positive or negative events (Fig. 34-3). Although such learning can lead to a vast constellation or system of behaviors, in a very general way, it also leads to behavioral tendencies of approach or withdrawal. Thus, when a CS signals a positive US, the CS will evoke approach behaviors—called *sign tracking*. For example, an organism will approach a signal for food. Analogously, when a CS signals a negative US, it will evoke behaviors that tend to move the organism away from the CS.

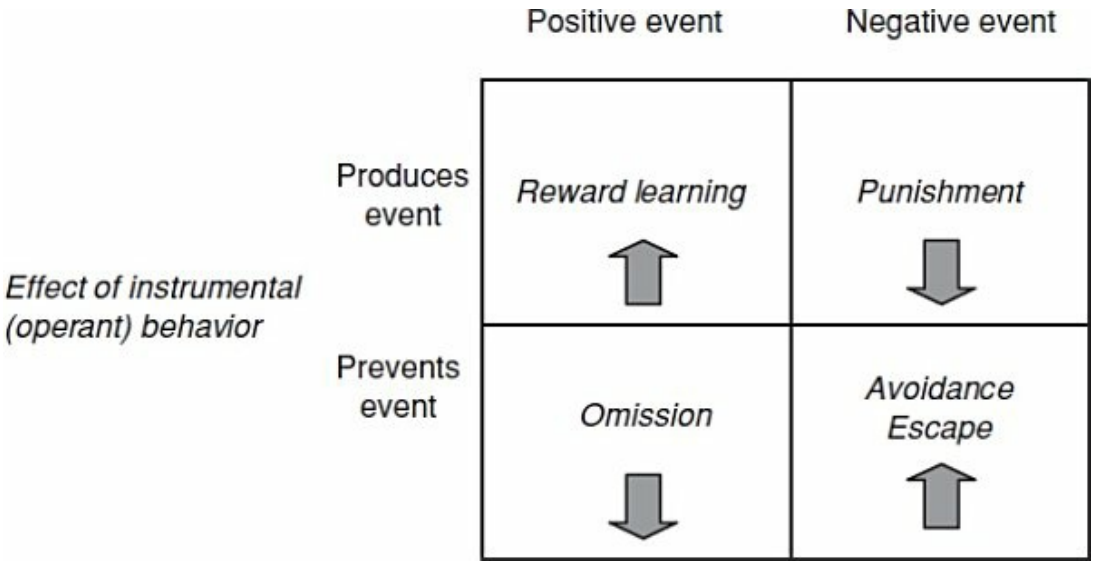


FIGURE 34-2
The law of effect in instrumental/operant learning. Actions either produce or prevent good or bad events, and the strength of the action changes accordingly (*arrow*). "Reinforcement" refers to a strengthening of behavior. *Positive reinforcement* occurs when an action produces a positive event, whereas *negative reinforcement* occurs when an action prevents or eliminates a negative event. (Courtesy of Mark E. Bouton, PhD.)

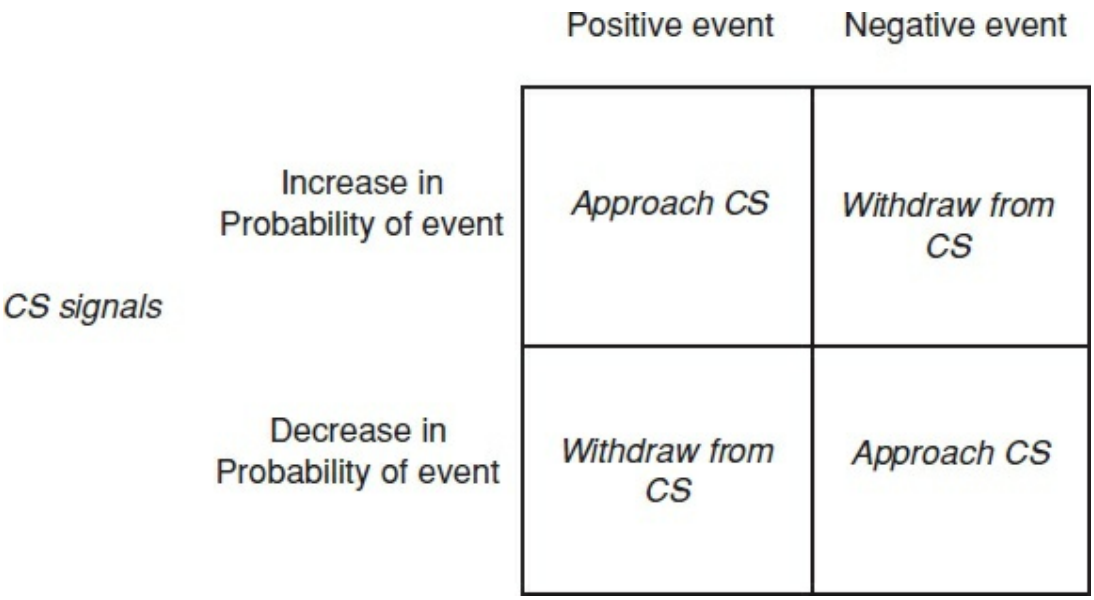


FIGURE 34-3
Sign tracking in Pavlovian learning. Conditional stimuli (CSs) signal either an increase or a decrease in the probability of good or bad events, and the CS generally engages approach or withdrawal behaviors accordingly. (Courtesy of Mark E. Bouton, PhD.)

Conversely, CSs associated with a decrease in the probability of a good

thing will elicit withdrawal behaviors, whereas CSs associated with the decrease in the probability of a bad thing can elicit an approach. An example of the latter case might be a stimulus that signals safety or decreases the probability of an aversive event, which evokes an approach in a frightened organism. In the end, these fundamental behavioral effects of both operant (see Fig. 34-2) and Pavlovian (see Fig. 34-3) learning serve to maximize the organism's contact with good things and minimize contact with bad things.

Perhaps because they have such similar functions, Pavlovian learning and operant learning are influenced by similar variables. For example, in either case, the behavior is especially strong if the magnitude of the US or reinforcer is large, or if the US or reinforcer occurs relatively close to the CS or operant response in time. In either case, the learned behavior decreases if the US or reinforcer that was once paired with the CS or the response is eliminated from the situation. This phenomenon, called *extinction*, provides a means of eliminating unwanted behaviors learned through either form of conditioning and has led to several very effective cognitive-behavioral therapies.

Pavlovian Conditioning

Effects of Conditioning on Behavior. Many laypeople have the mistaken impression that Pavlovian learning is a rigid affair in which a fixed stimulus comes to elicit a fixed response. Conditioning is considerably more complex and dynamic than that. For example, signals for food may evoke a broad set of responses that function to prepare the organism to digest food: they can elicit the secretion of gastric acid, pancreatic enzymes, and insulin along with Pavlov's famous salivary response. The CS can also elicit approach behavior (as described earlier), an increase in body temperature, and a state of arousal and excitement. When a signal for food is presented to a satiated animal or human, they may eat more food. Some of these effects may be motivational; for example, an additional effect of presenting a CS for food is that it can invigorate ongoing operant behaviors that have been reinforced with food. CSs thus have powerful behavioral potential. Signals for food evoke a whole *behavior system* that is functionally organized to find, procure, and consume food.

Pavlovian conditioning is also involved in other aspects of eating. Through

conditioning, humans and other animals may learn to like or dislike certain foods. In animals like rats, flavors associated with nutrients (sugars, starches, calories, proteins, or fats) come to be preferred. Flavors associated with sweet tastes are also preferred, whereas flavors associated with bitter tastes are avoided. At least as necessary, flavors associated with illness become disliked, as illustrated by the person who gets sick drinking an alcoholic beverage and consequently learns to hate the flavor. The fact that flavor CSs can be associated with such a range of biologic consequences (USs) is essential for omnivorous animals to learn about new foods. It also has some clinical implications. For example, chemotherapy can make cancer patients sick, causing the conditioning of an aversion to a recently eaten food (or to the clinic itself). Other evidence suggests that animals may learn to dislike food that is associated with becoming sick with cancer. On the flip side, conditioning can enable external cues to trigger food consumption and craving, potentially influencing overeating and obesity.

Pavlovian conditioning also occurs when organisms ingest drugs. Whenever a drug is taken, in addition to reinforcing the behaviors that lead to its ingestion, the drug constitutes a US and may be associated with potential CSs that are present at the time (e.g., rooms, odors, injection paraphernalia). CSs that are associated with drug USs can sometimes have an interesting property: they often elicit a conditioned response that seems opposite to the unconditional effect of the drug. For example, although morphine causes a rat to feel less pain, a CS associated with morphine elicits an opposite increase, not a decrease in pain sensitivity. Similarly, although alcohol can cause a drop in body temperature, a conditioned response to a CS associated with alcohol is typically increased in body temperature. In these cases, the conditioned response is said to be *compensatory* because it counteracts the drug effect. Compensatory responses are another example of how classical (Pavlovian) conditioning helps organisms prepare for a significant US.

Compensatory conditioned responses have implications for drug abuse. First, they can cause drug tolerance, in which repeated administration of a drug reduces its effectiveness. As a drug and a CS are repeatedly paired, the compensatory response to the CS becomes stronger and more effective at counteracting the effect of the drug. The drug, therefore, has less impact. One implication is that tolerance will be lost if the drug is taken without being signaled by the usual CS. Consistent with this idea, administering a drug in a

new environment can cause a loss of drug tolerance and make drug overdose more likely. A second implication stems from the fact that compensatory responses may be unpleasant or aversive. A CS associated with an opiate may elicit several compensatory responses—it may cause the drug user to be more sensitive to pain, change body temperature, and perhaps become hyperactive (the opposite of another unconditional opiate effect). The unpleasantness of these responses may motivate the user to retake the drug to get rid of them, an example of escape learning, or negative reinforcement, and a classic example of how Pavlovian and operant learning processes might readily interact. The idea is that the urge to take drugs may be strongest in the presence of CSs that have been associated with the drug. The hypothesis is consistent with self-reports of abusers, who, after a period of abstinence, are tempted to retake the drug when they are reexposed to drug-associated cues.

Pavlovian learning may potentially be involved in anxiety disorders. CSs associated with frightening USs can elicit a whole system of conditioned fear responses, broadly designed to help the organism cope. In animals, cues associated with frightening events (such as a brief foot shock) elicit changes in respiration, heart rate, and blood pressure, and even a (compensatory) decrease in sensitivity to pain. Brief CSs that occur close to the US in time can also elicit adaptively timed protective reflexes. For example, the rabbit blinks in response to a brief signal that predicts a mild electric shock near the eye. The same CS, when lengthened in duration and paired with the same US, elicits mainly fear responses, and fear elicited by a CS may potentiate the conditioned eyeblink response elicited by another CS or a startle response to a sudden noise. Once again, CSs do not merely elicit a simple reflex, but also evoke a complex and interactive set of responses.

Classical fear conditioning can contribute to phobias (in which specific objects may be associated with a traumatic US), as well as other anxiety disorders, such as panic disorder and posttraumatic stress disorder (PTSD). In panic disorder, people who have unexpected panic attacks can become anxious about having another one. In this case, the panic attack (the US or UR) may condition anxiety to the external situation in which it occurs (e.g., a crowded bus) and also internal (“interoceptive”) CSs created by early symptoms of the attack (e.g., dizziness or a sudden pounding of the heart). These CSs may then evoke anxiety or panic responses. Panic disorder may begin because external cues associated with panic can arouse anxiety, which

may exacerbate the next unconditional panic attack or panic response elicited by an interoceptive CS. Emotional reactions elicited by CSs may not require conscious awareness for their occurrence or development. Indeed, fear conditioning may be independent of conscious awareness.

In addition to eliciting conditioned responses, CSs also motivate ongoing operant behavior. For example, presenting a CS that elicits anxiety can increase the vigor of operant behaviors learned to avoid or escape the frightening US. Thus, an individual with an anxiety disorder will be more likely to express avoidance in the presence of anxiety or fear cues. Similar effects may occur with CSs that predict other USs (such as drugs or food)—as already mentioned. A drug-associated CS may motivate the drug abuser to take more drugs. The motivating effects of CSs may stem from the fact that CSs may be associated with both the sensory and emotional properties of USs. For example, the survivor of a traumatic train derailment might associate stimuli that occur immediately before derailment (such as the blue flash that occurs when the train separates from its overhead power supply) with both the emotional and the sensory aspects of the crash. Consequently, when the survivor later encounters another flashing blue light (e.g., the lights on a police car), the CS might evoke both emotional responses (mediated by association with the trauma's emotional qualities) and sensory associations (mediated by association with the trauma's sensory qualities). Both might play a role in the nightmares and "reexperiencing" phenomena that are characteristic of PTSD.

The Nature of the Learning Process. Research beginning in the late 1960s began to uncover some critical details about the learning process behind Pavlovian conditioning. Several findings proved especially influential. For example, it was shown that conditioning is not an inevitable consequence of pairing a CS with a US. Such pairings will not cause conditioning if there is a second CS present that already predicts the US. This finding (known as *blocking*) suggests that a CS must provide new information about the US if learning occurs. The importance of the CS's information value is also suggested by the fact that a CS will not be treated as a signal for the US if the US occurs equally often (or is equally probable) in the presence and the absence of the CS. Instead, the organism treats the CS as a signal for the US if the probability of the US is more significant in the presence of the CS than

in its absence. Also, the organism will treat the CS as a signal for “no US” if the probability of the US is less in the presence of the CS than in its absence. In the latter case, the signal is called a *conditioned inhibitor* because it will inhibit other CSs. The conditioned inhibition phenomenon is clinically relevant because inhibitory CSs may hold pathologic CRs like fear or anxiety at bay. A loss of the inhibition would allow the anxiety response to emerge.

There are also essential variants of classical conditioning. In *sensory preconditioning*, two stimuli (A and B) are first paired, and then one of them (A) is later paired with the US. Stimulus A evokes conditioned responding, of course, but so does stimulus B—indirectly, through its association with A. One implication is that exposure to a potent US like a panic attack may influence reactions to stimuli that have never been paired with the US directly; the sudden anxiety to stimulus B might seem spontaneous and mysterious. A related finding is *second-order conditioning.* Here, A is paired with a US first and then subsequently paired with stimulus B. Once again, both A and B will evoke responding. Sensory preconditioning and second-order conditioning increase the range of stimuli that can control the conditioned response. A third variant worth mentioning occurs, as indicated previously, when the onset of a stimulus becomes associated with the rest of that stimulus, as when a sudden increase in heart rate caused by the onset of a panic attack comes to predict the rest of the panic or feeling, or when the onset of a drug may predict the rest of the drug effect. Such intraevent associations may play a role in many of the body’s regulatory functions, such that an initial change in some variable (e.g., blood pressure or blood glucose level) may come to signal a further increase in that variable and therefore initiate a conditioned compensatory response.

Emotional responses can also be conditioned through observation. For example, a monkey that merely observes another monkey being frightened by a snake can learn to be afraid of the snake. The observer learns to associate the snake (CS) with its emotional reaction (US/UR) to the other monkey being afraid. Although monkeys readily learn to fear snakes, they are less likely to associate other salient cues (such as colorful flowers) with fear in the same way. This is an example of *preparedness* in classical conditioning—some stimuli are especially useful signals for some USs. Another example is that tastes are easily associated with illness but not a shock, whereas auditory and visual cues are easily associated with shock but not an illness.

Preparedness may explain why human phobias tend to be for particular objects (snakes or spiders) and not others (knives or electric sockets) that may as often be paired with pain or trauma.

Erasing Pavlovian Learning. If Pavlovian learning plays a role in the etiology of behavioral and emotional disorders, a natural question concerns how to eliminate it or undo it. Pavlov studied *extinction:* conditioned responding decreases if the CS is presented repeatedly without the US after conditioning. Extinction is based on behavioral or cognitive-behavioral therapies designed to reduce pathologic conditioned responding through repeated exposure to the CS (exposure therapy). It is presumably a consequence of any form of therapy in the course of which the patient learns that previous harmful cues are no longer harmful. Another elimination procedure is *counterconditioning,* in which the CS is paired with a very different US/UR.

Counterconditioning was the inspiration for *systematic desensitization,* a behavior therapy technique in which frightening CSs are deliberately associated with relaxation during therapy.

Although extinction and counterconditioning reduce unwanted conditioned responses, they do not destroy the original learning, which remains in the brain, ready to return to behavior under the right circumstances. For example, conditioned responses that have been eliminated by extinction or counterconditioning can recover if time passes before the CS presents again (spontaneous recovery). Conditioned responses can also return if the patient returns to the context of conditioning after extinction in another context, or if the CS is encountered in a context that differs from the one in which extinction has occurred (all are examples of the renewal effect). The renewal effect is significant because it illustrates the principle that extinction performance depends on the organism being in the context in which extinction was learned. If the CS is encountered in a different context, the extinguished behavior may relapse or return. Recovery and relapse can also occur if the current context is associated again with the US (“reinstatement”) or if the CS is paired with the US again (“rapid reacquisition”). One theoretical approach assumes that extinction and counterconditioning do not destroy the original learning but instead entail new learning that gives the CS a second meaning (e.g., “the CS is safe” in addition to “the CS is

dangerous"). As with an ambiguous word, which has more than one meaning, responding evoked by an extinguished or counterconditioned CS depends fundamentally on the current context.

Research on context effects in animal and human learning and memory suggests that a wide variety of stimuli can play context (Table 34-5). Drugs, for example, can be very salient in this regard. When rats are given fear extinction while under the influence of a benzodiazepine tranquilizer or alcohol, fear is renewed when the CS is tested in the absence of the context provided by the drug. This is an example of *state-dependent learning*, in which the retention of information is best when tested in the same state in which it was initially learned. State-dependent fear extinction has apparent implications for combining therapy with drugs. It also has implications for the administration of drugs more generally. For example, if a person were to take a drug to reduce anxiety, the anxiety reduction would reinforce drug-taking. State-dependent extinction might further preserve any anxiety that might otherwise be extinguished during natural exposure to the anxiety-eliciting cues. Thus, drug use could paradoxically preserve the original anxiety, creating a self-perpetuating cycle that could provide a possible explanation for the link between anxiety disorders and substance abuse. One point of this discussion is that drugs can play multiple roles in learning: they can be USs or reinforcers on the one hand, and CSs or contexts on the other. The possible complex behavioral effects of drugs are worth bearing in mind.



Table 34-5
Effective Contextual Stimuli Studied in Animal and Human Research Laboratories

Exteroceptive context:
Room, place, environment, other external background stimuli
Interoceptive context:
Drug state
Hormonal state
Mood state
Deprivation state
Recent events
Expectation of events

Passage of time

From Sadock BJ, Sadock VA, Ruiz P. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:652.

Another general message is that contemporary theory emphasizes that extinction (and other processes, such as counterconditioning) entails new learning rather than a destruction of the old. Recent psychopharmacological research has built on this idea: if extinction and therapy constitute new learning, drugs that might facilitate new learning might also facilitate the therapy process. For example, there has been considerable recent interest in D-cycloserine, a partial agonist of the *N*-methyl-D-aspartate (NMDA) glutamate receptor. The NMDA receptor is involved in long-term potentiation, a synaptic facilitation phenomenon that has been implicated in several examples of learning. Of interest, there is evidence that the administration of D-cycloserine can facilitate extinction learning in rats and possibly in humans undergoing exposure therapy for anxiety disorders. In the studies supporting this conclusion, the administration of the drug increased the amount of extinction that was apparent after a small (and incomplete) number of extinction trials. Although such findings are promising, it is essential to remember that the context-dependence of extinction, and thus the possibility of relapse with a change of context, may easily remain. Consistent with this possibility, although D-cycloserine allows fear extinction to be learned in fewer trials, it does not appear to prevent or reduce the strength of the renewal effect. Such results further underscore the importance of behavioral research—and behavioral theory—in understanding the effects of drugs on therapy. Nonetheless, the search for drugs that might enhance the *learning* that occurs in therapy situations will continue to be an essential area of research.

Another process that might theoretically modify or erase memory is illustrated by a phenomenon called *reconsolidation.* Newly learned memories are temporarily labile and easy to disrupt before they are consolidated into a more stable form in the brain. The consolidation of memory requires the synthesis of new proteins and can be blocked by administering protein synthesis inhibitors (e.g., anisomycin). Animal research suggests that consolidated memories that have recently been reactivated might also return briefly to a similarly vulnerable state; protein synthesis inhibitors can

likewise block their "reconsolidation." For example, several studies have shown that the reactivation of conditioned fear by one or two presentations of the CS after a brief fear conditioning experience can allow it to be disrupted by anisomycin. When the CS is tested later, there is little evidence of fear—as if reactivation and then drug administration diminished the strength of the original memory. However, like the effects of extinction, these fear-diminishing effects do not necessarily mean that the original learning has been destroyed or erased. There is some evidence that fear of the CS that has been diminished in this way can still return over time (i.e., spontaneously recover) or with reminder treatments. This sort of result suggests that the effect of the drug is somehow able to interfere with retrieval or access to the memory rather than to be an actual "reconsolidation."

Generally speaking, eliminating a behavior after therapy should not be interpreted as the erasure of the underlying knowledge. For the time being, it may be safest to assume that, after any treatment, a part of the original learning may remain in the brain, ready to produce relapse if retrieved. Instead of trying to find treatments that destroy the original memory, another therapeutic strategy might be to accept the possible retention of the original learning and build therapies that allow the organism to prevent or cope with its retrieval. One possibility is to conduct extinction exposure in the contexts in which relapse might be most problematic to the patient and to encourage retrieval strategies (such as the use of retrieval cues like reminder cards) that might help to remind the patient of the therapy experience.

Operant/Instrumental Learning

The Relation Between Behavior and Payoff. Operant learning has many parallels with Pavlovian learning. As one example, extinction also occurs in operant learning if the reinforcer is omitted following training. Although extinction is once again a useful technique for eliminating unwanted behaviors, just as we saw with Pavlovian learning, it does not destroy the original learning—spontaneous recovery, renewal, reinstatement, and rapid reacquisition effects still obtain. Although early accounts of instrumental learning, beginning with Edward Thorndike, emphasized the role of the reinforcer as "stamping in" the instrumental action, more-modern approaches tend to view the reinforcer as a sort of guide or motivator of

behavior. A modern, “synthetic” view of operant conditioning (see later discussion) holds that the organism associates the action with the outcome in much the way that stimulus–outcome learning is believed to be involved in Pavlovian learning.

Human behavior is influenced by a wide variety of reinforcers, including social ones. For example, simple attention from teachers or hospital staff members has been shown to reinforce disruptive or problematic behavior in students or patients. In either case, when the attention is withdrawn and redirected toward other activities, the problematic behaviors can decrease (i.e., undergo extinction). Human behavior is also influenced by verbal reinforcers, like praise, and, more generally, by *conditioned reinforcers,* such as money, that have no intrinsic value except for the value derived through association with more basic, “primary” rewards. Conditioned reinforcers have been used in schools and institutional settings in the so-called *token economies* in which positive behaviors are reinforced with tokens that can be used to purchase valued items. In more natural settings, reinforcers are always delivered in social relationships, in which their effects are dynamic and reciprocal. For example, the relationship between a parent and a child is full of interacting and reciprocating operant contingencies in which the delivery (and withholding) of reinforcers and punishers shapes the behavior of each. Like Pavlovian learning, operant learning is always operating and always influencing behavior.

Research on operant conditioning in the laboratory has offered many insights into how action relates to its payoff. In the natural world, few actions are reinforced every time they are performed; instead, most actions are reinforced only intermittently. In a *ratio reinforcement schedule,* the reinforcer is directly related to the amount of work or responding that the organism emits. That is, there is some work requirement that determines when the next reinforcer will be presented. In a “fixed ratio schedule,” every *x*th action is reinforced; in a “variable ratio schedule,” there is an average ratio requirement, but the number of responses required for each successive reinforcer varies. Ratio schedules, especially variable ratio schedules, can generate high rates of behavior, as seen in the behavior directed at a casino slot machine. In an *interval reinforcement schedule,* the presentation of each reinforcer depends on the organism emitting the response after some time has also elapsed. In a “fixed interval schedule,” the first response after *x* seconds

have elapsed is reinforced. In a "variable interval schedule," there is an interval requirement for each reinforcer, but the length of that interval varies. A person checking e-mail throughout the day is being reinforced on a variable interval schedule—a new message is not present to reinforce each checking response, but the presence of a new message becomes available after variable time points throughout the day. Of interest, on interval schedules, the response rate can vary substantially without influencing the overall rate of reinforcement. (On ratio schedules, there is a more direct relationship between behavior rate and reinforcement rate.) In part because of this, interval schedules tend to generate slower response rates than ratio schedules.

Classic research on operant behavior underscores the fact that the performance of any action always involves *choice.* That is, whenever the individual performs a particular behavior, he or she chooses to engage in that action over many other possible alternatives. When a choice has been studied by allowing the organism to perform either of two different operant behaviors (paying off with their separate schedules of reinforcement), the rate of operant behavior depends not only on the behavior's rate of reinforcement but also on the rate of reinforcement of all other behaviors in the situation. Put most generally, the strength of Behavior 1 (e.g., the rate at which Behavior 1 is performed) is given by

$$B_1 = K^*R_1/(R_1 + R_O)$$

where B_1 can be seen as the strength of Behavior 1, R_1 is the rate at which B_1 has been reinforced, and R_O is the rate at which all alternative (or "other") behaviors in the environment have been reinforced; K is a constant that corresponds to all behavior in the situation and may have a different value for different individuals. This principle, known as the *quantitative law of effect,* captures several ideas that are relevant to psychiatrists and clinical psychologists. It indicates that an action can be strengthened either by increasing its rate of reinforcement (R_1) or decreasing the rate of reinforcement for alternative behaviors (R_O). Conversely, an action can be weakened either by reducing its rate of reinforcement (R_1) or increasing reinforcement for alternative behaviors (R_O). The latter point has an

especially important implication: in principle, one can slow the strengthening of new, undesirable behavior by providing an environment that is otherwise rich in reinforcement (high R_O). Thus, an adolescent who experiments with drugs or alcohol would be less likely to engage in this behavior at a high rate (high B_1) if their environment were otherwise rich with reinforcers (e.g., provided by extracurricular activities or outside interests).

The choice among actions is also influenced by the size of their corresponding reinforcers and how soon the reinforcers will occur. For example, individuals sometimes have to choose between an action that yields a small but immediate reward (e.g., taking a hit of a drug) versus another that yields a larger but delayed reward (e.g., going to a class and earning credit toward a general educational development certificate). Individuals who choose the more immediate reward are often said to be "impulsive," whereas those who choose the delayed reward are said to exercise "self-control." Of interest, organisms often choose immediate small rewards over delayed larger ones, even though it may be maladaptive to do so in the long run. Such "impulsive" choices are especially difficult to resist when the reward is imminent. Choice is believed to be determined by the relative value of the two rewards, with that value being influenced by both the reinforcer's size and its delay. The bigger the reinforcer, the better is the value, and the more immediate the reinforcer, the better, too: when a reward is delayed, its value decreases or is "discounted" over time. When offered a choice, the organism will always choose the action that leads to the reward whose value is currently higher.

Theories of Reinforcement. It is possible to use the preceding principles of operant conditioning without knowing in advance what kind of event or stimulus will be reinforcing for the individual patient. None of the reinforcement rules say much about what sorts of events in an organism's world will play a reinforcer. Skinner defined a reinforcer empirically, by considering the effect it had on operant behavior. A reinforcer was defined as any event that could be shown to increase the strength of an operant if it was made a consequence of the operant. This empirical (some would say "atheoretical") view can be valuable because it allows idiosyncratic reinforcers for idiosyncratic individuals. For instance, if a therapist works

with a child who is injuring himself, the approach advises the therapist merely to search for the consequences of the behavior and then manipulate them to bring the behavior under control. So, if, for example, the child's self-injurious behavior decreases when the parent stops scolding the child for doing it, then the scold is the reinforcer, which might seem counterintuitive to everyone (including the parent who thinks that the scold should function as a punisher). On the other hand, it would also be useful to know what kind of event will reinforce an individual before the therapist has to try everything.

Several approaches allow for predictions ahead of time. Perhaps the most useful is the *Premack principle* (named for researcher David Premack), which claims that, for any individual, reinforcers can be identified by giving the individual a preference test in which he or she is free to engage in any number of activities. The individual might spend the most time engaging in activity A, the second-most time engaged in activity B, and the third-most time engaged in activity C. Behavior A can thus be said to be preferred to B and C, and B is preferred to C. The Premack principle asserts that access to a preferred action will reinforce any action that is less preferred. In the present example, if doing activity C allowed access to doing A or B, activity C will be reinforced—it will increase in strength or probability.

Similarly, activity B will be reinforced by activity A (but not C). The principle accepts large individual differences. For example, in an early study, some children given a choice spent more time eating candy than playing pinball, whereas others spent more time playing pinball then eating candy. Candy eating reinforced pinball playing in the former group. In contrast, pinball playing reinforced candy eating in the latter group. There is nothing particularly special about food (eating) or any particular kind of activity as a possible reinforcer. Any behavior that is preferred to a second behavior will theoretically reinforce the second behavior.

The principle has been refined over the years. It is now recognized that even a less-preferred behavior can reinforce a more-preferred behavior if the organism has been deprived of doing the low-preferred behavior below its ordinary level. In the preceding example, even the low-preference activity C could reinforce A or B if it were suppressed for a while below its baseline level of preference. However, the main implication is that in the long run, a person's reinforcers can be discovered by looking at how they allocate their activities when access to them is free and unconstrained.

Motivational Factors. Instrumental action is often said to be goal-oriented. As Edward Tolman illustrated in many experiments conducted in the 1930s and 1940s, organisms may flexibly perform several actions to achieve a goal; instrumental learning thus provides a variable means to a fixed end. Tolman's perspective on the effects of reinforcers has returned to favor. He argued that reinforcers are not necessary for learning, but instead are essential for motivating instrumental behavior. The classic illustration of this point is the *latent learning* experiment. Rats received several trials in a complex maze in which they were removed from the maze without reward once they got to a particular goal location. When arriving at the goal was suddenly rewarded, the rats suddenly began working through the maze with very few errors. Thus, they had learned about the maze without the benefit of the food reinforcer, but the reinforcer was nonetheless crucial for motivating them to get through the maze efficiently. The reinforcer was unnecessary for learning, but it gave the organism a reason to translate its knowledge into action.

Subsequent research has identified many motivating effects of reward. For example, organisms with experience receiving a small reward may show *positive contrast* when they are suddenly reinforced with a larger reward. That is, their instrumental behavior may become more vigorous than that in control subjects who have received the larger reward all along. Conversely, organisms show *negative contrast* when they are switched from a high reward to a lower reward—their behavior becomes weaker than control subjects who have received the same smaller reward all along. Negative contrast involves frustration and emotionality. Both types of contrast are consistent with the idea that the current effectiveness depends on what the organism has learned to expect; an increase from expectation causes elation, whereas a decrease from expectation causes frustration. There is a sense in which receiving a reward that is smaller than expected might seem punishing.

Negative contrast is an example of a *paradoxical reward effect*—a set of behavioral phenomena given the name because they show that reward can sometimes weaken behavior and that nonreward can sometimes strengthen it. The best known is the *partial reinforcement extinction effect*, in which actions that have been intermittently (or "partially") reinforced persist longer when reinforcers are completely withdrawn than actions that have been continuously reinforced. The finding is considered paradoxical because an

action that has been reinforced (say) half as often as another action may be more persistent. One explanation is that the action that has been partially reinforced has been reinforced in the presence of some frustration—and is thus persistent in new adversity or sources of frustration. Other evidence suggests that effortfulness is a dimension of behavior that can be reinforced. That is, human and animal participants that have been reinforced for performing effortful responses learn a sort of "industriousness" that transfers to new behaviors. One implication is that new behaviors learned in therapy will be more persistent over time if a high effort has been deliberately reinforced.

The organism's current motivational state also influences the effectiveness of a reinforcer. For example, food is more reinforcing for a hungry organism, and water is more reinforcing for a thirsty one. Such results are consistent with many theories of reinforcement (e.g., the Premack principle) because the presence of hunger or thirst would undoubtedly increase the organism's preference ranking for food or water. However, recent research indicates that the effects of motivational states on instrumental actions are not this automatic. Specifically, if a motivational state is going to influence an instrumental action, the individual needs first to learn how the action's reinforcer will influence the motivational state. The process of learning about the effects the reinforcer has on the motivational state is called *incentive learning.*

An experimental example best illustrates incentive learning. In 1992, Bernard Balleine reported a study that taught trained rats that were not hungry to lever press to earn a novel food pellet. The animals were then food-deprived and tested for their lever pressing under conditions in which the lever press no longer produced the pellet. The hunger state did not affect the lever-press rate; that is, hungry rats did not lever press any more than rats that were not food-deprived. On the other hand, if the rat had been given separate experience eating the pellets while it was food-deprived, during the test, it lever pressed at a high rate. Thus, hunger invigorated the instrumental action only if the animal had previously experienced the reinforcer in that state, allowing it to learn that the specific substance influenced the state (incentive learning). The interpretation of this result, and others like it, will be developed further later in this section. The main idea is that individuals will perform an instrumental action when they know that it produces an outcome

that is desirable in the current motivational state. The clinical implications are underexplored but could be significant. For example, persons who abuse drugs will need to learn that the drug makes them feel better in the withdrawal state before the withdrawal will motivate drug-seeking. Persons with anxiety might not be motivated to take a beneficial medication while anxious until they have had the opportunity to learn how the medication makes them feel when they are in the anxious state, and a person with depression may need to learn what natural reinforcers make them feel better while they are depressed. According to theory, direct experience with a reinforcer's effect on depressed mood might be necessary before the person will be interested in performing actions that help to ameliorate the depressed state.

Pavlovian and Operant Learning Together

Avoidance Learning. Theories of the motivating effects of reinforcers have usually emphasized that Pavlovian CSs in the background are also associated with the reinforcer. The expectancy of the reinforcer (or conditioned motivational state) the CSs arouse increases the vigor of the operant response. This is *two-factor* or *two-process theory:* Pavlovian learning co-occurs and motivates behavior during operant learning.

The interaction of Pavlovian and instrumental factors is especially important in understanding avoidance learning (see Fig. 34-2). In avoidance situations, organisms learn to perform actions that prevent the delivery or presentation of an aversive event. The explanation of avoidance learning is subtle because it is difficult to identify an obvious reinforcer. Although preventing the occurrence of the aversive event is essential, how can the nonoccurrence of that event reinforce? The answer is that cues in the environment (Pavlovian CSs) come to predict the occurrence of the aversive event, and consequently, they arouse anxiety or fear. The avoidance response can, therefore, be reinforced if it escapes from or reduces that fear. Pavlovian and operant factors are thus both important: Pavlovian fear conditioning motivates and allows reinforcement of an instrumental action through its reduction. Escape from fear or anxiety is believed to play a significant role in many human behavior disorders, including anxiety disorders. Thus, the obsessive-compulsive patient checks or washes his or her hands repeatedly to

reduce anxiety, the agoraphobic stays home to escape the fear of places associated with panic attacks, and the bulimic learn to vomit after a meal to reduce the learned anxiety evoked by eating the meal.

Although the two-factor theory remains an essential view of avoidance learning, excellent avoidance can be obtained in the laboratory without reinforcement: for example, if an animal is required to perform an action that resembles one of its natural and prepared fear responses—so-called *species-specific defensive reactions* (*SSDRs*). Rats will readily learn to freeze (remain motionless) or flee (run to another environment) to avoid shock, two behaviors that have evolved to escape or avoid predation. Freezing and fleeing are also respondents rather than operants; they are controlled by their antecedents (Pavlovian CSs that predict shock) rather than reinforced by their consequences (escape from fear). Thus, when the rat can use an SSDR for avoidance, the only necessary learning is Pavlovian—the rat learns about environmental cues associated with danger, and these arouse fear and evoke natural defensive behaviors including withdrawal (negative sign tracking; Fig. 34-3). To learn to perform an action that is not similar to a natural SSDR requires more feedback or reinforcement through fear reduction.

A good example is lever pressing, which is easy for the rat to learn when the reinforcer is a food pellet but challenging to learn when the same action avoids shock. More recent work with avoidance in humans suggests an essential role for CS-aversive event and response–no aversive event expectancies. The larger point is that Pavlovian learning is vital in avoidance learning; when the animal can avoid with an SSDR, it is the only learning necessary; when the required action is not an SSDR, Pavlovian learning permits the expectation of something bad.

A cognitive perspective on aversive learning is also encouraged by studies of *learned helplessness*. In this phenomenon, organisms exposed to either controllable or uncontrollable aversive events differ in their reactivity to later aversive events. For example, the typical finding is that a subject exposed to inescapable shock in one phase of an experiment is less successful at learning to escape shock with altogether new behavior in a second phase, whereas subjects exposed to escapable shock are normal. Both types of a subject are exposed to the same shock, but its psychological dimension (its controllability) creates a difference, perhaps because subjects exposed to inescapable shock learn the independence of their actions and the outcome.

Although this finding (and interpretation) was once seen as a model of depression, the current view is that the controllability of stressors mainly modulates their stressfulness and negative impact. At a theoretical level, the result also implies that organisms receiving instrumental contingencies in which their actions lead to outcomes might learn something about the controllability of those outcomes.

One of the main conclusions of aversive learning is that there are biologic and cognitive dimensions to instrumental learning. The possibility that Pavlovian contingencies can control much instrumental learning is also consistent with research in which animals have learned to respond to positive reinforcers. For example, pigeons have been widely used in operant learning experiments since the 1940s. In the typical experiment, the bird learns to peck at a plastic disk on a wall of the chamber (a response "key") to earn food. Although pecking seems to be an operant response, it turns out that the pigeon's peck can be entrained by merely illuminating the key for a few seconds before presenting the reinforcer on several trials. Although there is no requirement for the bird to peck the key, the bird will begin to peck at the illuminated key—a Pavlovian predictor of food—anyway. Its consequences only weakly control the pecking response; if the experimenter arranges things so that pecks prevent the delivery of food (which is otherwise delivered on trials without pecks), the birds will continue to peck almost indefinitely on many trials. (Although the peck has a negative correlation with food, key illumination remains a weakly positive predictor of food.) Thus, this classic "operant" behavior is at least partly a Pavlovian one. Pavlovian contingencies cannot be ignored. When rats are punished with mild foot shock for pressing a lever that otherwise produces food, they stop lever pressing at least partly (and perhaps predominantly) because they learn that the lever now predicts shock, and they withdraw from it. A child might likewise learn to stay away from the parent who delivers punishment rather than refrain from performing the punished behavior. A great deal of behavior in operant learning settings may be controlled by Pavlovian learning and sign tracking rather than authentic operant learning.

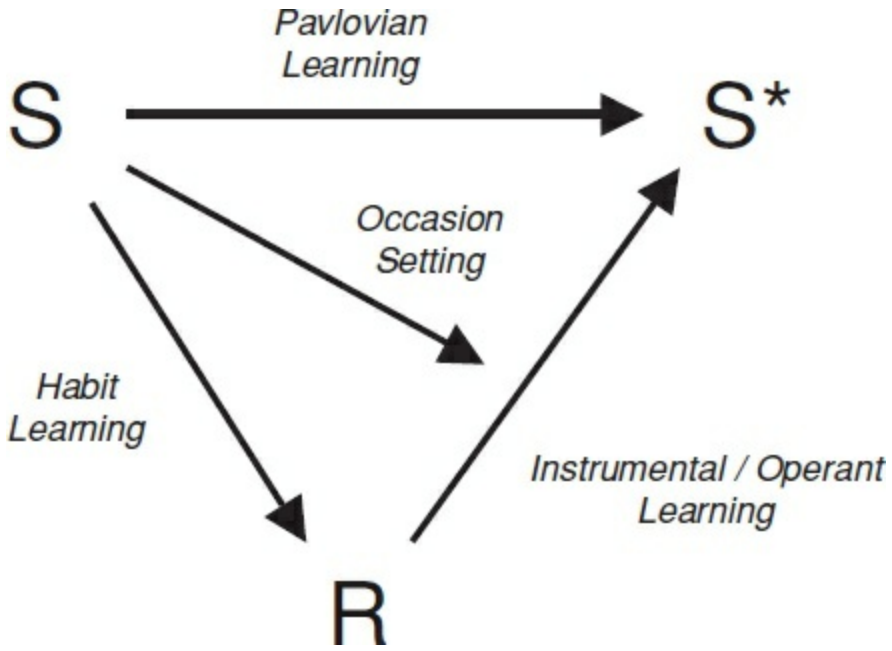


FIGURE 34-4
Any instrumental/operant learning situation permits a number of types of learning, which are always occurring all the time. R, operant behavior or instrumental action; S, stimulus in the background; S*, biologically significant event (e.g., reinforcer, US). (Courtesy of Mark E. Bouton, PhD.)

A Synthetic View of Instrumental Action. Then, the idea is that several hypothetical associations control behavior in any instrumental learning situation, as illustrated in Figure 34-4. Much behavior in an instrumental learning arrangement can be controlled by a Pavlovian factor in which the organism associates background cues (CSs) with the reinforcer (S*, denoting a biologically significant event). As discussed earlier, this type of learning can allow the CS to evoke a variety of behaviors and emotional reactions (and motivational states) that can additionally motivate instrumental action.

In modern terms, the instrumental factor is represented by the organism learning a direct, and similar, association between the instrumental action (R) and the reinforcer (S*). Evidence for this sort of learning comes from experiments on *reinforcer devaluation* (Fig. 34-5). In such experiments, the organism can first be trained to perform two instrumental actions (e.g., pressing a lever and pulling a chain), each paired with a different reinforcer (e.g., food pellet versus a liquid sucrose solution). In a separate second phase, one of the reinforcers (e.g., pellet) is paired with illness, which creates the conditioning of a powerful taste aversion to the reinforcer. In a final test, the

organism is returned to the instrumental situation and can perform either instrumental action. No reinforcers are presented during the test. The result is that the organism no longer performs the action that produced the reinforcer that is now aversive. The organism must have (1) learned which action produced which reinforcer and (2) combined it with the knowledge that it no longer likes or values that reinforcer. The result cannot be explained by the simpler, more traditional view that reinforcers merely stamp in or strengthen instrumental actions.

Organisms also need to learn how reinforcers influence a particular motivational state called “incentive learning.” Incentive learning is crucially involved in instrumental learning as a process through which the animal learns the value of the reinforcer. Thus, in the reinforcer devaluation experiment shown in Figure 34-5, an important thing that occurs in the second phase is that the organism must contact the reinforcer and learn that it does not like it. As described earlier, incentive learning is probably always involved in making outcomes (and the associated actions that produce them) more or less desirable.

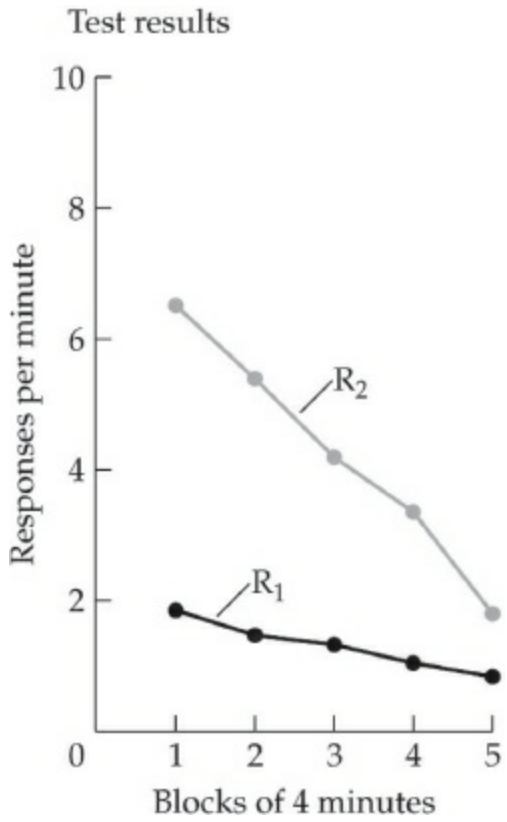


FIGURE 34-5

The reinforcer devaluation effect. Results of the test session. The result indicates the importance of the response–reinforcer association in operant learning. For the organism to perform in the way that it does during testing, it must learn which action leads to which reinforcer and then choose to perform the action that leads to the outcome it currently liked or valued. R1, R2, operant behaviors or instrumental actions. (Data from Colwill and Rescorla [1986]. From Bouton ME. *Learning and Behavior: A Contemporary Synthesis*. Sunderland, MA: Sinauer; 2007.)

Other experiments have illustrated the other associations to the stimulus that are represented in Figure 34-4. In addition to being directly associated with the reinforcer, a stimulus can signal a relation between an action and an outcome. This is called *occasion setting:* instead of eliciting a response directly, stimuli in operant situations can set the occasion for the operant response. There is good evidence that they do so by signaling a specific response–reinforcer relationship. For example, in one experiment, rats learned to lever press and chain pull in background noise and background light. When the noise was present, lever pressing yielded a pellet reinforcer, and chain pulling yielded sucrose.

In contrast, when the light was present, the relations were reversed: Lever pressing yielded sucrose, and chain pulling yielded pellet. There was evidence that the rats learned corresponding relationships. In a second phase, pellets were associated with illness, so the rat no longer valued the pellet. In a final test, rats were allowed to lever press or chain pull in extinction in noise or light present during separate tests. In the presence of the noise, the animal's chain pulled rather than lever pressed. When the light was present, the animal's lever pressed rather than chain pulled. Thus, the noise informed the rat that the lever press yielded a pellet, and the light informed the rat that the chain pull did. This is the occasion-setting function illustrated in Figure 34-4.

It is worth noting that other stimuli besides lights and noises set the occasion for operant behavior. Modern research on learning in animals has underscored the importance of other stimuli, such as temporal and spatial cues, and particular perception and memory processes. A particularly interesting example of research on the *stimulus control* of operant behavior is categorization. Pigeons can be shown images of cars, chairs, flowers, and cats on a computer screen positioned on the wall of a Skinner box. Pecking one of four keys in the presence of these images is reinforced in any picture

containing a car, a chair, a flower, or a cat. Of interest, as the number of exemplars in each category increases, the pigeon makes more errors as it learns the discrimination. However, more exemplars create better learning in the sense that it is more ready to transfer to new test images—after many examples of each category, the pigeon is more accurate at categorizing (and responding accurately to) new stimuli. One implication is that training new behaviors in various settings or ways will enhance generalization to new situations.

The final association in Figure 34-4 is simple *habit learning* or a direct association between the stimulus and the response. Through this association, the background may elicit the instrumental action directly, without the intervening cognition of R–S* and the valuation of S*. Although S–R learning was once believed to dominate learning, the current view sees it as developing only after extensive and consistent instrumental training. In effect, actions that have been performed repeatedly (and repeatedly associated with the reinforcer) become automatic and routine. One source of evidence is the fact that the reinforcer devaluation effect—which implies a kind of cognitive mediation of operant behavior—no longer occurs after extensive instrumental training as if the animal reflexively engages in the response without remembering the actual outcome it produces. It seems reasonable to expect that many pathologic behaviors that come into the clinic might also be automatic and blind through repetition. Of interest, the evidence suggests that the eventual dominance of the S-R habit in behavior does not replace or destroy more cognitive mediation by learned S–S*, R–S*, or S–(R–S*) relations. Under some conditions, even a habitual response might be brought back under the control of the action–reinforcer association. The conversion of actions to habits and the relation of habit to cognition are active areas of research.

BIOLOGY OF MEMORY

The topic of memory is fundamental to the discipline of psychiatry. Memory is the glue that binds our mental life, the scaffolding for our personal history. In part, personality is an accumulation of habits that have been acquired, many very early in life, which creates dispositions and influences how we behave. In the same sense, the neuroses are often products of learning—

anxieties, phobias, and maladaptive behaviors that result from particular experiences. Psychotherapy itself is a process by which new habits and skills are acquired by accumulating new experiences. In this sense, memory is at the theoretical heart of psychiatry's concern with personality, the consequences of early experience, and the possibility of growth and change.

Memory is also of clinical interest because disorders of memory and complaints about memory are common in neurologic and psychiatric illness. Memory impairment is also a side effect of specific treatments, such as electroconvulsive therapy (ECT). Accordingly, the effective clinician needs to understand something of the biology of memory, the varieties of memory dysfunction, and how memory can be evaluated.

From Synapses to Memory

Memory is a particular case of the general biologic phenomenon of *neural plasticity*. Neurons can show history-dependent activity by responding differently as a function of prior input, and this plasticity of nerve cells and synapses is the basis of memory. In the last decade of the 19th century, researchers proposed that the persistence of memory could be accounted for by nerve cell growth. This idea has been restated many times, and current understanding of the synapse as the critical site of change is founded on extensive experimental studies in animals with simple nervous systems. Experience can lead to structural change at the synapse, including alterations in the strength of existing synapses and alterations in the number of synaptic contacts along specific pathways.

Plasticity. Neurobiologic evidence supports two necessary conclusions: first, short-lasting plasticity, which may last for seconds or minutes depending on specific synaptic events, including an increase in neurotransmitter release; and second, long-lasting memory, which depends on new protein synthesis, the physical growth of neural processes, and an increase in the number of synaptic connections.

A significant source of information about memory has come from the extended study of the marine mollusk *Aplysia californica.* Individual neurons and connections between neurons have been identified, and the wiring diagram of some simple behaviors has been described. *Aplysia* is capable of

associative learning (including classic conditioning and operant conditioning) and nonassociative learning (habituation and sensitization). *Sensitization* had been studied using the gill-withdrawal reflex, a defensive reaction in which tactile stimulation causes the gill and siphon to retract. When tactile stimulation is preceded by sensory stimulation to the head or tail, gill withdrawal is facilitated. The cellular changes underlying this sensitization begin when a sensory neuron activates a modulatory interneuron, enhancing the strength of synapses within the circuitry responsible for the reflex. This modulation depends on a second-messenger system in which intracellular molecules (including cyclic adenosine monophosphate [cAMP] and cAMP-dependent protein kinase) lead to enhanced transmitter release that lasts for minutes in the reflex pathway. Both short- and long-lasting plasticity within this circuitry are based on enhanced transmitter release. The long-lasting change uniquely requires the expression of genes and the synthesis of new proteins. Synaptic tagging mechanisms allow gene products delivered throughout a neuron to increase synaptic strength selectively at recently active synapses. Also, the long-term change, but not the short-term change, is accompanied by the growth of neural processes of neurons within the reflex circuit.

In vertebrates, memory cannot be studied as directly as in the simple nervous system of *Aplysia*. Nevertheless, it is known that behavioral manipulations can also result in measurable changes in the brain's architecture. For example, rats reared in enriched as opposed to ordinary environments show an increase in the number of synapses ending on individual neurons in the neocortex. These changes are accompanied by small increases in cortical thickness, the diameter of neuronal cell bodies, and the number and length of dendritic branches. The behavioral experience thus exerts powerful effects on the wiring of the brain.

Many of these same structural changes have been found in adult rats exposed to an enriched environment and adult rats given extensive maze training. In maze training, vision was restricted to one eye, and the corpus callosum was transected to prevent information received by one hemisphere from reaching the other hemisphere. The result was that structural changes in neuronal shape and connectivity were observed only in the trained hemisphere. This finding rules out several nonspecific influences, including motor activity, indirect effects of hormones, and overall level of arousal.

Long-term memory in vertebrates is believed to be based on morphologic growth and change, including increases in synaptic strength along particular pathways.

Long-Term Potentiation. The phenomenon of *long-term potentiation* (LTP) is a candidate mechanism for mammalian long-term memory. LTP is observed when a postsynaptic neuron is persistently depolarized after a high-frequency burst of presynaptic neural firing. LTP has several properties that make it suitable as a physiologic substrate of memory. It is established quickly and then lasts for a long time. It is associative, in that it depends on the co-occurrence of presynaptic activity and postsynaptic depolarization. It occurs only at potentiated synapses, not all synapses terminating on the postsynaptic cell. Finally, LTP occurs prominently in the hippocampus, a structure that is important for memory.

The induction of LTP is known to be mediated postsynaptically and to involve activation of the NMDA receptor, which permits the influx of calcium into the postsynaptic cell. LTP is maintained by an increase in the number of α-amino-3-hydroxy-5-methyl-4-isoxazolepropionate (AMPA; non-NMDA) receptors postsynaptic cell and also possibly by increased transmitter release.

A promising method for elucidating molecular mechanisms of memory relies on introducing specific mutations into the genome. By deleting a single gene, one can produce mice with specific receptors or cell signaling molecules inactivated or altered. For example, in mice with a selective deletion of NMDA receptors in the CA1 field of the hippocampus, many aspects of CA1 physiology remain intact, but the CA1 neurons do not exhibit LTP, and memory impairment is observed in behavioral tasks. Genetic manipulations introduced reversibly in the adult are particularly advantageous, in that specific molecular changes can be induced in developmentally normal animals.

Associative Learning. The study of *classical conditioning* has provided many insights into the biology of memory. Classical conditioning has been especially well studied in rabbits, using a tone as the conditioned stimulus and an air puff to the eye (which automatically elicits a blink response) as the unconditioned stimulus. Repeated pairings of the tone and the air puff lead to

a conditioned response, in that the tone alone elicits an eye blink. Reversible lesions of the deep nuclei of the cerebellum eliminate the conditioned response without affecting the unconditioned response. These lesions also prevent initial learning from occurring, and, when the lesion is reversed, rabbits usually learn. Thus, the cerebellum contains essential circuitry for the learned association. The relevant plasticity appears to be distributed between the cerebellar cortex and the deep nuclei.

An analogous pattern of cerebellar plasticity is believed to underlie motor learning in the vestibuloocular reflex and, perhaps, associative learning of motor responses in general. Based on the idea that learned motor responses depend on the coordinated control of changes in timing and strength of the response, it has been suggested that synaptic changes in the cerebellar cortex are critical for learned timing. In contrast, synaptic changes in the deep nuclei are critical for forming an association between a conditioned stimulus and an unconditioned stimulus.

Fear conditioning and fear-potentiated startle are types of learning that serve as useful models for anxiety disorders and related psychiatric conditions. For example, mice exhibit freezing behavior when returned to the same context in which aversive shock was delivered on an earlier occasion. This type of learning depends on the encoding of contextual features of the learning environment. Acquiring and expressing this type of learning requires neural circuits that include both the amygdala and the hippocampus. The amygdala may be important for associating negative affect with new stimuli. The hippocampus may be important for representing the context. With extinction training, when the context is no longer associated with an aversive stimulus, the conditioned fear response fades. The frontal cortex is believed to play a critical role in extinction.

Cortical Organization of Memory

One fundamental question concerns the locus of memory storage in the brain. In the 1920s, Karl Lashley searched for the site of memory storage by studying the behavior of rats after removing different amounts of their cerebral cortex. He recorded the number of trials needed to relearn a maze problem that rats learned before surgery, and he found that the deficit was proportional to the amount of cortex removed. The deficit did not seem to

depend on the particular location of cortical damage. Lashley concluded that the memory resulting from maze learning was not localized in any one part of the brain but instead was distributed equivalently over the entire cortex.

Subsequent research has led to reinterpretations of these results. Maze learning in rats depends on different types of information, including visual, tactile, spatial, and olfactory information. Neurons that process these various types of information are segregated into different areas of the rat cerebral cortex, and memory storage is segregated in a parallel manner. Thus, the correlation between maze-learning abilities and lesion size that Lashley observed results from progressive encroachment of larger lesions on specialized cortical areas serving the many components of information processing relevant for maze learning.

The functional organization of the mammalian cerebral cortex has been revealed through neuropsychological analyses of deficits following brain damage and physiologic studies of intact brains. The cortical areas responsible for processing and storing visual information have been studied most extensively in nonhuman primates. Nearly one-half of the primate neocortex is specialized for visual functions.

The cortical pathways for visual information processing begin in the primary visual cortex (V1) and proceed from parallel pathways or streams. One stream projects ventrally to the inferotemporal cortex and is specialized for processing information concerning the identification of visual objects. Another stream projects dorsally to the parietal cortex and is specialized for processing information about spatial location.

Specific visual processing areas in the dorsal and ventral streams, together with areas in the prefrontal cortex, register the immediate experience of perceptual processing. The results of perceptual processing are first available as *immediate memory*. Immediate memory refers to the amount of information that can be held in mind (like a telephone number) to be available for immediate use. Immediate memory can be extended in time by rehearsing or otherwise manipulating the information, in which case what is stored is said to be in *working memory*.

Regions of the visual cortex in forward portions of the dorsal and ventral streams serve as the ultimate repositories of visual memories. The inferotemporal cortex, for example, lies at the end of the ventral stream, and inferotemporal lesions lead to selective impairments in both visual object

perception and visual memory. Nonetheless, such lesions do not disrupt elementary visual functions, such as acuity. Electrophysiologic studies in the monkey show that neurons in area TE, which is one part of the inferotemporal cortex, register specific and complex features of visual stimuli, such as shape, and respond selectively to patterns and objects. The inferotemporal cortex can thus be thought of as both a higher-order visual processing system and a storehouse of the visual memories that result from that processing.

In sum, memory is distributed and localized in the cerebral cortex. Memory is distributed in the sense that, as Lashley concluded, there is no cortical center dedicated solely to the storage of memories. Nonetheless, memory is localized in the sense that different aspects or dimensions of events are stored at specific cortical sites—namely, in the same regions that are specialized to analyze and process what is to be stored.

Memory and Amnesia

The principle that the functional specialization of cortical regions determines both the locus of information processing and the locus of information storage does not provide a complete account of memory in the brain. If it did, then brain injury would always include difficulty in memory for a restricted type of information and a loss of ability to process information of that same type. This kind of impairment occurs, for example, in the aphasias and the agnosias. However, there is another kind of impairment that can occur as well, called amnesia.

The hallmark of amnesia is a loss of new learning ability that extends across all sensory modalities and stimulus domains. This *anterograde amnesia* can be explained by understanding the role of brain structures critical for acquiring information about facts and events. Typically, anterograde amnesia occurs together with *retrograde amnesia,* a loss of knowledge acquired before the onset of amnesia. Retrograde deficits often have a temporal gradient, following a principle known as Ribot’s law; deficits are most severe for information that was most recently learned.

A patient with a presentation of amnesia exhibits severe memory deficits in the context of the preservation of other cognitive functions, including language comprehension and production, reasoning, attention, immediate

memory, personality, and social skills. The selectivity of the memory deficit in these cases implies that intellectual and perceptual functions of the brain are separated from the capacity to lay down in memory the records that ordinarily result from engaging in intellectual and perceptual work.

Specialized Memory Function. Amnesia can result from damage to the medial portion of the temporal lobe or damage to regions of the midline diencephalon. Studies of a severely amnesic patient known as HM stimulated intensive investigation of the role of the medial temporal lobe in memory.

HM became amnesic in 1953, at 27 years of age, when he sustained a bilateral resection of the medial temporal lobe to relieve severe epilepsy. The removal included approximately one-half of the hippocampus, the amygdala, and most neighboring entorhinal and perirhinal cortices (Fig. 34-6). After the surgery, HM's seizure condition was much improved, but he experienced profound forgetfulness. His intellectual functions were generally preserved. For example, HM exhibited normal immediate memory, and he could maintain his attention during conversations. After an interruption, however, HM could not remember what had recently occurred. HM's dense amnesia was permanent and debilitating. In HM's words, he felt as if he were waking from a dream because he had no recollection of what had just happened.

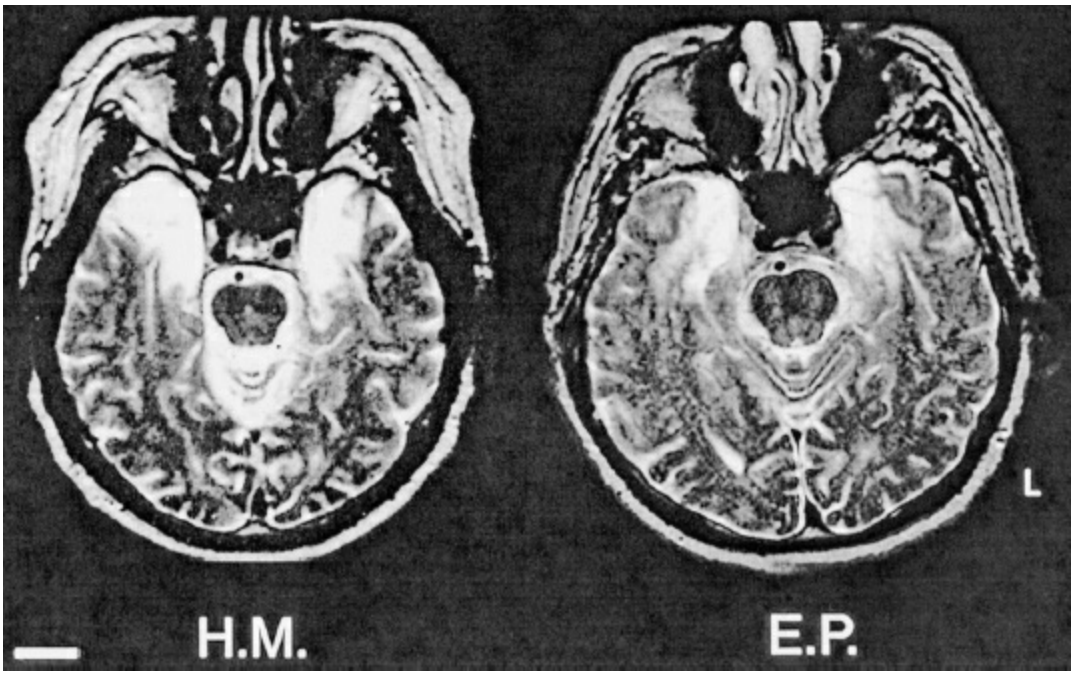


FIGURE 34-6
Structural magnetic resonance images of the brains of patients HM and EP through the level of the temporal lobe. Damaged tissue is indicated by bright signal in these T2-weighted axial images. Both patients sustained extensive damage to medial temporal structures, as the result of surgery for epilepsy in HM, and as the result of viral encephalitis in EP. Scale bar: 2 cm. L, left side of the brain. (Reprinted from Corkin S, Amaral EG, González RG, Johnson KA, Hyman BT. H. M.'s medial temporal lobe lesion: Findings from magnetic resonance imaging. *J Neurosci*. 1997;17(10):3964–3979 and Stefanacci L, Buffalo EA, Schmolck H, Squire LR. Profound amnesia after damage to the medial temporal lobe: A neuroanatomical and neuropsychological profile of patient E.P. *J Neurosci*. 2000;20(18):7024–7036, with permission.)

In monkeys, many parallels to human amnesia have been demonstrated after surgical damage to anatomical components of the medial temporal lobe. Cumulative study of the resulting memory impairment eventually identified the medial temporal structures and connections crucial for memory. These include the hippocampus, including the dentate gyrus, hippocampal fields CA1, CA2, CA3, and the subiculum—and the adjacent cortical regions, including the entorhinal, perirhinal, and parahippocampal cortices.

Another important medial temporal lobe structure is the amygdala. The amygdala is involved in the regulation of much of emotional behavior. In particular, the storage of emotional events engages the amygdala. Modulatory effects of projections from the amygdala to the neocortex are responsible for producing enhanced memory for emotional or arousing events compared to

neutral events.

A detailed study of amnesic patients offers unique insights into the nature of memory and its organization in the brain. An extensive series of informative studies, for example, described the memory impairment of patient EP.

EP was diagnosed with herpes simplex encephalitis at 72 years of age. Damage to the medial temporal lobe region (see Fig. 34-6) produced persistent and profound amnesia. During testing sessions, EP was cordial and talked freely about his life experiences, but he relied exclusively on stories from his childhood and early adulthood. He would repeat the same story many times. Strikingly, his performance on tests of recognition memory was no better than would result from guessing (Fig. 34-7A). Tests involving facts about his life and autobiographical experiences revealed poor memory for the time leading up to his illness but normal memory for his childhood (Fig. 34-7B). EP also has good spatial knowledge about the town in which he lived as a child, but he was unable to learn the layout of the neighborhood where he moved to after he became amnesic (Fig. 34-7C).

Given the severity of the memory problems experienced by EP and other amnesic patients, it is noteworthy that these patients nonetheless perform typically on certain kinds of memory tests. The impairment selectively concerns memory for factual knowledge and autobiographical events, collectively termed *declarative memory.* Amnesia presents as a global deficit, in that it involves memory for information presented in any sensory modality, but the deficit is limited, in that it covers only memory for facts and events.

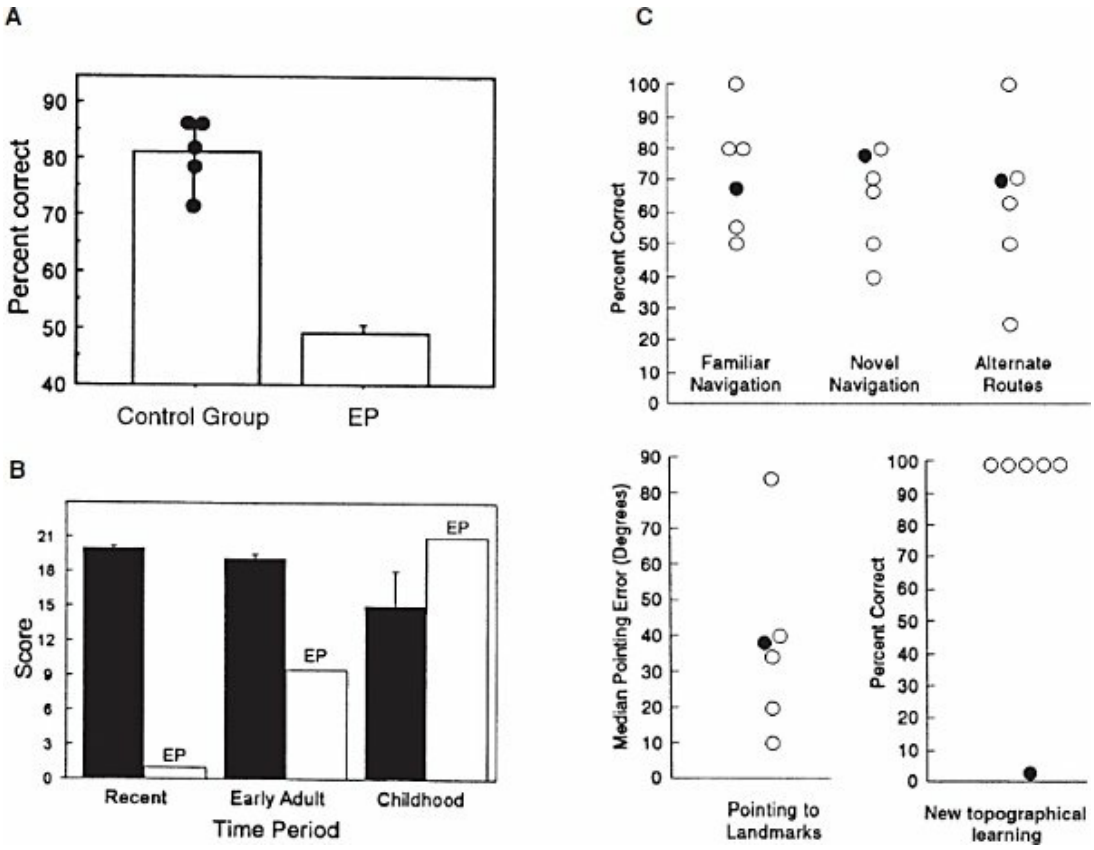


FIGURE 34-7

Formal test results for patient EP, showing severe anterograde and retrograde deficits, with intact remote memory. **A:** Scores were combined from 42 different tests of recognition memory for words given to patient EP and a group of five healthy control subjects. The testing format was either two-alternative forced choice or yes–no recognition. Brackets for EP indicate the standard error of the mean. Data points for the control group indicate each participant's mean score across all 42 recognition memory tests. EP's average performance (49.3 percent correct) was not different from chance and was approximately five standard deviations (SDs) below the average performance of control subjects (81.1 percent correct; SD, 6.3). **B:** Autobiographical remembering was quantified by using a structured interview known as the Autobiographical Memory Interview. Items assessed personal semantic knowledge (maximum score 21 for each time period). Performance for the recent time period reflects poor memory for information that could have been acquired only subsequent to the onset of his amnesia. For EP, performance for the early adult period reflects retrograde memory deficits. Performance for the childhood period reflects good remote memory. Similar results for semantic and episodic remembering were obtained from these time periods. (Data from Kopelman MD, Wilson BA, Baddeley AD. The autobiographical memory interview: A new assessment of autobiographical and personal semantic memory in amnesic patients. *J Clin Exp Neuropsychol.* 1989;5:724; and Reed JM, Squire LR. Retrograde amnesia for facts and events: Findings from four new cases. *J Neurosci.* 1998;18:3943). **C:** Assessments of spatial memory demonstrated EP's good memory for spatial knowledge from his childhood, along with extremely poor new learning of spatial information.

Performance was compared to that of five individuals (*open circles*) who attended EP's high school at the same time as he did, lived in the region over approximately the same time period, and, like EP (*filled circles*), moved away as young adults. Normal performance was found for navigating from home to different locations in the area (familiar navigation), between different locations in the area (novel navigation), and between these same locations when a main street was blocked (alternative routes). Subjects were also asked to point to particular locations while imagining themselves in a particular location (pointing to landmarks), or they were asked about locations in the neighborhoods in which they currently lived (new topographical learning). EP showed difficulty only in this last test, because he moved to his current residence after he became amnesic. (Data from Teng E, Squire LR. Memory for places learned long ago is intact after hippocampal damage. *Nature.* 1999;400:675.) (Adapted from Stefanacci L, Buffalo EA, Schmolck H, Squire LR. Profound amnesia after damage to the medial temporal lobe: A neuroanatomical and neuropsychological profile of patient E. P. *J Neurosci*. 2000;20(18):7024–7036. Printed with permission.)

Hippocampal pathology in patients with amnesia can also be revealed using high-resolution magnetic resonance imaging (MRI). Such studies indicate that damage limited to the hippocampus results in clinically significant memory impairment. In addition to the hippocampus, other medial temporal lobe regions also make critical contributions to memory. Thus, a moderately severe memory impairment results from CA1 damage, whereas a more profound and disabling amnesia results from medial temporal lobe damage that includes the hippocampus and adjacent cortex. Memory impairment due to medial temporal lobe damage is also typical in patients with early Alzheimer disease or amnestic mild cognitive impairment. As Alzheimer disease progresses, the pathology affects many cortical regions and produces substantial cognitive deficits and memory dysfunction.

Amnesia can also result from damage to structures of the medial diencephalon. The critical regions damaged in diencephalic amnesia include the mammillary nuclei in the hypothalamus, the dorsomedial nucleus of the thalamus, the anterior nucleus, the internal medullary lamina, and the mammillothalamic tract. However, uncertainty remains regarding which specific lesions are required to produce diencephalic amnesia. *Alcoholic Korsakoff syndrome* is the most prevalent and best-studied example of diencephalic amnesia, and in these cases, the damage is found in brain regions that may be especially sensitive to prolonged bouts of thiamine deficiency and alcohol abuse. Patients with alcoholic Korsakoff syndrome typically exhibit memory impairment due to a combination of diencephalic

damage and frontal lobe pathology. Frontal damage alone produces characteristic cognitive deficits and specific memory problems (e.g., in effortful retrieval and evaluation); in Korsakoff syndrome, the pattern of deficits thus extends beyond what is commonly found in other amnesia cases (see Table 34-6).



Table 34-6
Memory and Cognitive Deficits Associated with Frontal Damage

Test	Amnesia	Korsakoff Syndrome	Frontal Lobe Damage
Delayed recall	+	+	−
Dementia Rating Scale: Memory index	+	+	−
Dementia Rating Scale: Initiation and perseveration index	−	+	+
Wisconsin Card Sorting Test	−	+	+
Temporal order memory	+	++	++
Metamemory	−	+	+
Release from proactive interference	−	+	−

+, deficit; −, no deficit; ++, disproportionate impairment relative to item memory.

From Squire LR, Zola-Morgan S, Cave CB, Haist F, Musen G, Suzuki WA. *Memory, organization of brain systems and cognition Cold Spring Harb Symp Quant Biol*. 1990;55:1007.

The ability to remember factual and autobiographical events depends on the integrity of both the cortical regions responsible for representing the information in question and several brain regions responsible for memory formation. Thus, medial temporal and diencephalic brain areas work in concert with widespread areas of neocortex to form and to store declarative

memories (Fig. 34-8).

Retrograde Amnesia

Memory loss in amnesia typically affects recent memories more than remote memories (Fig. 34-9). Temporally graded amnesia has been demonstrated retrospectively in studies of amnesic patients and prospectively in studies of monkeys, rats, mice, and rabbits. These findings have important implications for understanding the nature of the memory storage process. Memories are dynamic, not static. As time passes after learning, some memories are forgotten, whereas others become stronger due to a process of *consolidation* that depends on cortical, medial temporal, and diencephalic structures.

The study of retrograde amnesia has been important for understanding how memory changes over time. The dynamic nature of memory storage can be conceptualized as follows. An event is experienced and encoded by a collection of cortical regions involved in representing a combination of different event features. At the same time, the hippocampus and adjacent cortex receive pertinent high-level information from all sensory modalities. Later, when the original event is recalled, the same set of cortical regions is activated. If a subset of the cortical regions is activated, the hippocampus and related structures can facilitate recall by facilitating the remaining cortical regions (i.e., pattern completion). When the original event is retrieved and newly associated with other information, hippocampal–cortical networks can be modified. In this way, a gradual consolidation process occurs that changes memory storage (see Fig. 34-8). The neocortical components representing some events can become so effectively linked together that ultimately a memory can be retrieved without any contribution from the medial temporal lobe. As a result, amnesic patients can exhibit normal retrieval of remote facts and events and autobiographical memories. Distributed neocortical regions are the permanent repositories of these enduring memories.

In contrast to what is observed after damage restricted to the hippocampus, extensive retrograde impairments for facts and events from the distant past can also occur. Damage to the frontal lobes, for example, can lead to difficulty in organizing memory retrieval. Accurate retrieval often begins with activation of lifetime periods and proceeds to identify general classes of events and then more specific events, but this process becomes difficult

following frontal damage. Damage to other cortical regions can also impair memory storage. Networks in the anterolateral temporal cortex, for example, are critical for retrieving stored information because these areas are essential for long-term storage itself. Patients with *focal retrograde amnesia* exhibit substantial retrograde memory impairments together with only moderately impaired new learning ability. Some capacity for new learning remains, presumably, because medial temporal lobe structures can communicate with other areas of the cortex that remain undamaged.

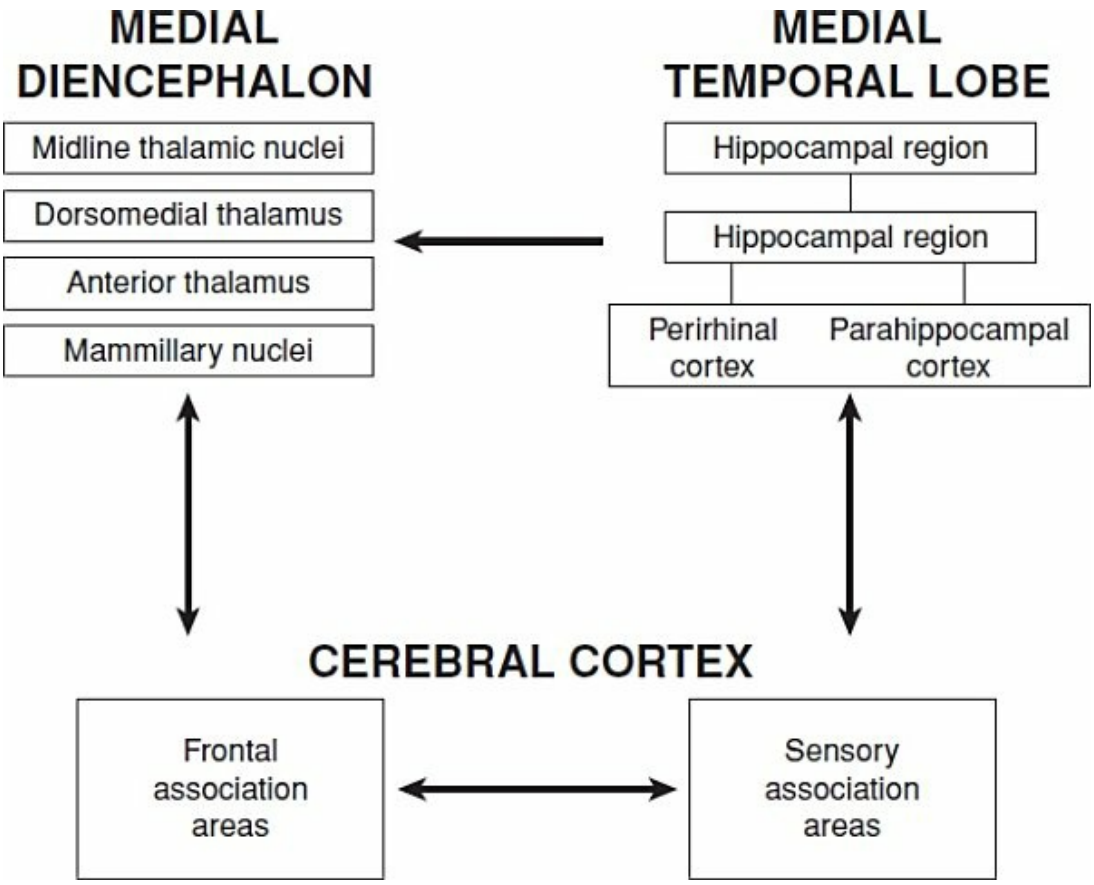


FIGURE 34-8
Brain regions believed to be critical for the formation and storage of declarative memory. Medial diencephalon and medial temporal regions are critical for declarative memory storage. The entorhinal cortex is the major source of projections for the neocortex to the hippocampus, and nearly two-thirds of the cortical input to the entorhinal cortex originates in the perirhinal and parahippocampal cortex. The entorhinal cortex also receives direct connections from the cingulate, insula, orbitofrontal, and superior temporal cortices. (Adapted from Paller KA. Neurocognitive foundations of human memory. In: Medin DL, ed.: *The Psychology of Learning and Motivation*. Vol. 40. San Diego, CA: Academic Press; 2008:121 and Gluck MA, Mercado E, Myers CE. *Learning and Memory: From Brain to Behavior*. New York: Worth; 2008:109, Fig. 3-16.)

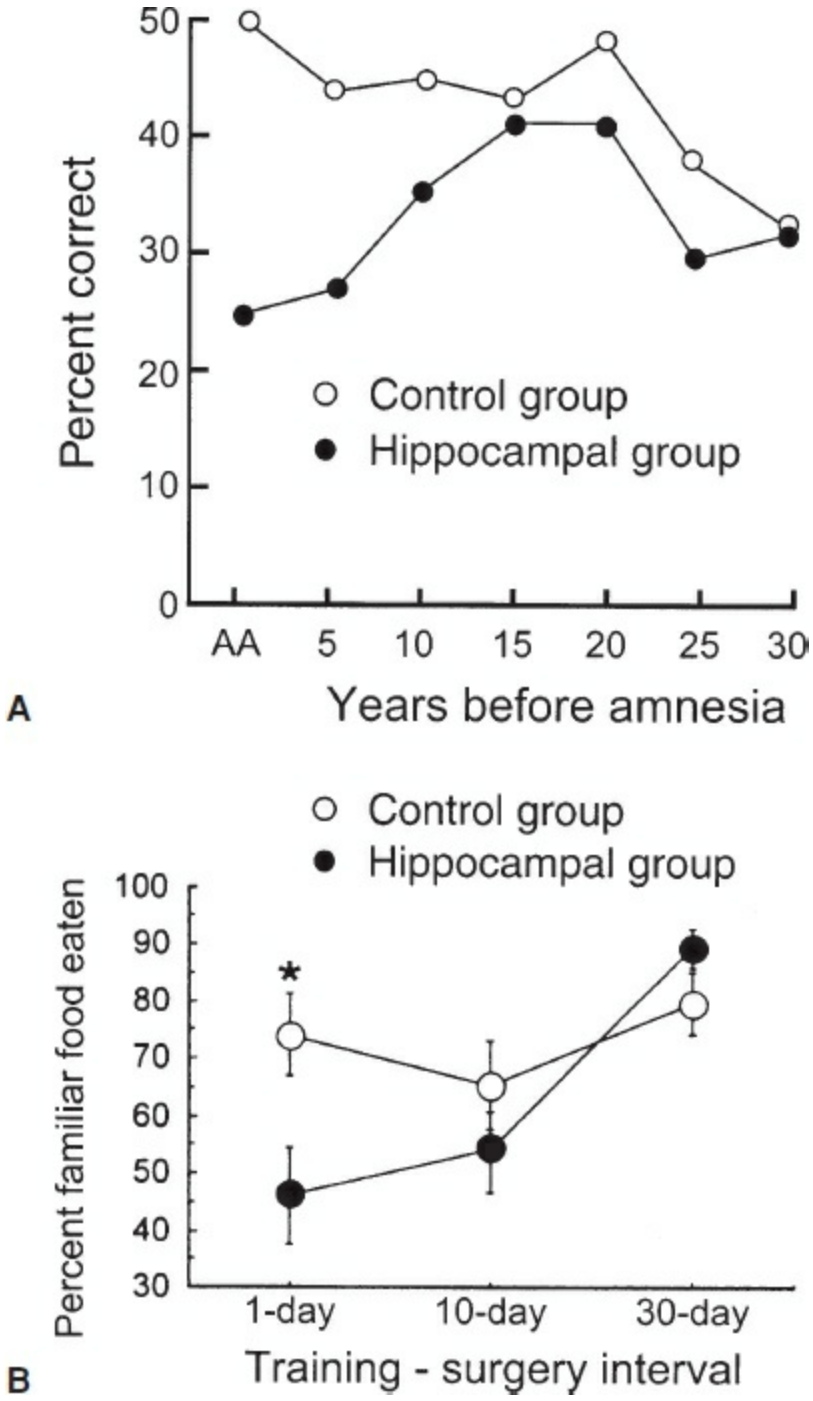


FIGURE 34-9

A: Temporally limited retrograde amnesia for free recall of 251 news events. Scores were aligned relative to the onset of amnesia in patients (*N* = 6) and to a corresponding time point in age-matched and education-matched healthy individuals (*N* = 12). The time period after the onset of amnesia is labeled AA (anterograde amnesia) to designate that this time point assessed memory for events that occurred after the onset of amnesia. Standard errors ranged from 2 to 10 percent. Brain damage in the patient group was limited primarily to the hippocampal region. **B:** Temporally limited retrograde amnesia in rats with lesions

of the hippocampus and subiculum. Rats learned to prefer an odorous food as the result of an encounter with another rat with that odor on its breath. Percent preference for the familiar food was observed for three training-surgery intervals. At 1 day after learning, the control group performed significantly better than the rats with lesions ($p < 0.05$). At 30 days, the two groups performed similarly, and both groups performed well above chance. *Error bars* show standard errors of the mean. (Adapted from Manns JR, Hopkins RO, Squire LR. Semantic memory and the human hippocampus. *Neuron*. 2003;38(1):127–133 and Clark RE, Broadbent NJ, Zola SM, Squire LR. Anterograde amnesia and temporally graded retrograde amnesia for a nonspatial memory task after lesions of hippocampus and subiculum. *J Neurosci*. 2002;22(11):4663–4669, with permission.)

MULTIPLE TYPES OF MEMORY

Memory is not a single faculty of the mind but consists of various subtypes. Amnesia affects only one kind of memory, *declarative memory*. Declarative memory is what is ordinarily meant by the term *memory* in everyday language. Declarative memory supports the conscious recollection of facts and events. The classic impairment in amnesia thus concerns memory for routes, lists, faces, melodies, objects, and other verbal and nonverbal material, regardless of the sensory modality in which the material is presented.

Amnesic patients can display a broad impairment in these components of declarative memory, while several other memory abilities are preserved. The heterogeneous set of preserved abilities is collectively termed *nondeclarative memory*. Nondeclarative memory includes skill learning, habit learning, simple forms of conditioning, and priming (see next section). For these kinds of learning and memory, amnesic patients can perform normally.

In controlled laboratory settings, the acquisition of a variety of perceptual, perceptual-motor, and cognitive skills can be tested in isolation, and amnesic patients are found to acquire these skills at rates equivalent to the rates at which healthy individuals acquire the skills. For example, amnesic patients can learn to read mirror-reversed text normally, they exhibit the typical facilitation in reading speed with successive readings of ordinary prose, and they improve as rapidly as healthy individuals at a speeded reading of repeating nonwords. Also, amnesic patients can, after seeing strings of letters generated by a finite-state rule system, classify novel strings of letters as rule-based or not rule-based. Classification performance is normal even though amnesic patients are impaired at remembering the events of training or the

specific items they have studied.

Priming

Priming refers to the facilitation of the ability to detect or to identify a particular stimulus based on a specific recent experience. Many tests have been used to measure priming in amnesia and show that it is intact. For example, words might be presented in a study phase and then again, after a delay, in a test phase, when a priming measure such as reading speed is obtained. Patients are instructed to read words as quickly as possible in such a test, and they are not informed that memory is being assessed.

In one priming test, patients named pictures of previously presented objects reliably faster than they named pictures of new objects, even after a delay of 1 week. This facilitation occurred at normal levels, even though the patients were markedly impaired at recognizing which pictures had been presented previously. Particularly striking examples of preserved priming come from studies of patient EP (Fig. 34-10), who exhibited intact priming for words but performed at chance levels when asked to recognize which words had been presented. This form of memory, termed *perceptual priming,* is thus a distinct class of memory independent of the medial temporal lobe regions typically damaged in amnesia.

Another form of priming reflects improved access to meaning rather than percepts. For example, subjects study a list of words, including *tent* and *belt,* and then are asked to free-associate them. Thus, they are given words such as *canvas* and *strap* and asked to produce the first word that comes to mind. The result is that subjects are more likely to produce *tent* in response to *canvas* and to produce *belt* in response to *strap* than if the words *tent* and *belt* had not been presented recently. This effect, called *conceptual priming,* is also preserved in amnesic patients, even though they fail to recognize the same words on a conventional memory test (Fig. 34-11).

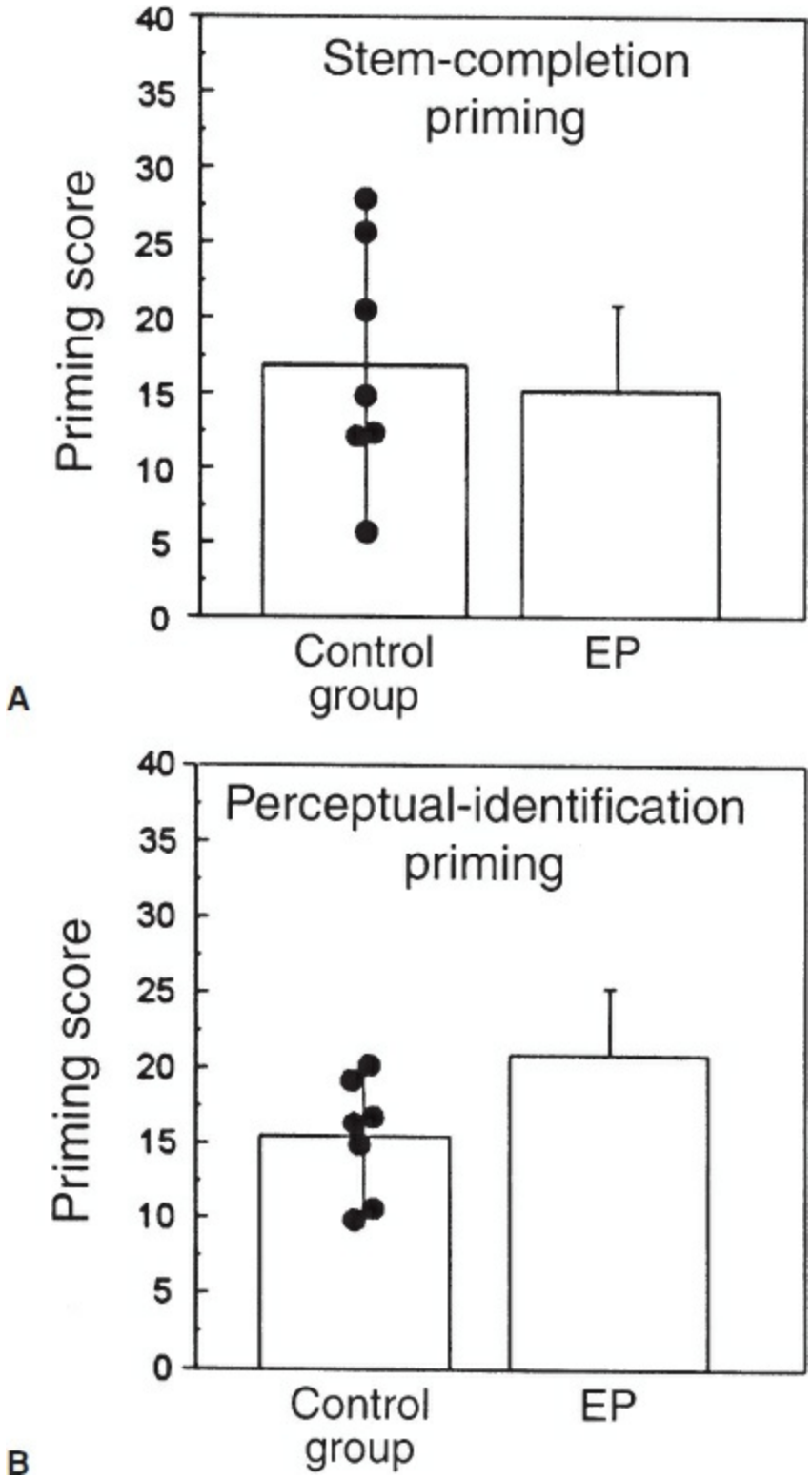


FIGURE 34-10

Preserved priming in patient EP relative to seven control subjects. **A:** Stem-completion priming on six separate tests. Priming reflected a tendency for subjects to complete three-letter stems with previously encountered words when they were instructed to produce the first word to come to mind (e.g., MOT___

completed to form MOTEL). Priming scores were calculated as percentage correct for studied words minus percentage correct for baseline words (guessing). **B:** Perceptual-identification priming on 12 separate tests. Subjects attempted to read 48 words that were visually degraded. Priming scores were calculated as percentage correct identification of previously studied words minus percentage correct identification of nonstudied words. Brackets indicate standard error of the mean. (Data from Hamann SB, Squire LR. Intact perceptual memory in the absence of conscious memory. *Behav Neurosci*. 1997;111:850.) (Reprinted from Stefanacci L, Buffalo EA, Schmolck H, Squire LR. Profound amnesia after damage to the medial temporal lobe: A neuroanatomical and neuropsychological profile of patient E. P. *J Neurosci*. 2000;20(18):7024–7036, with permission.)

Not all types of priming are preserved in amnesia. Some priming tests have been designed to examine the formation of new associations. When priming tests are based not on preexisting knowledge but the acquisition of new associative knowledge, priming tends to be impaired. In other words, priming in certain complex situations can require the same type of linkage among multiple cortical regions critical for declarative memory.

Memory Systems

Table 34-7 depicts one scheme for conceptualizing multiple types of memory. Declarative memory depends on medial temporal and midline diencephalic structures along with large portions of the neocortex. This system provides for the rapid learning of facts (*semantic memory*) and events (*episodic memory*). Nondeclarative memory depends on several different brain systems. Habits depend on the neocortex and the neostriatum, and the cerebellum is essential for the conditioning of skeletal musculature, the amygdala for emotional learning, and the neocortex for priming.

Declarative memory and nondeclarative memory differ in significant ways. Declarative memory is phylogenetically more recent than nondeclarative memory. Also, declarative memories are available to conscious recollection. The flexibility of declarative memory permits the retrieved information to be available to multiple response systems. Nondeclarative memory is inaccessible to awareness and is expressed only by engaging specific processing systems. Nondeclarative memories are stored as changes within these processing systems—changes encapsulated such that the stored information has limited access to other processing systems.

Semantic memory, which concerns general knowledge of the world, has

often been categorized as a separate form of memory. Facts committed to memory typically become independent of the original episodes in which the facts were learned. Amnesic patients can sometimes acquire information that would ordinarily be learned as facts, but the patients learn it by relying on a different brain system than the system that supports declarative memory.

Consider a test requiring the concurrent learning of eight object pairs. Healthy individuals can rapidly learn which the correct object in each pair is, whereas severely amnesic patients like EP learn only gradually over many weeks, and at the start of each session, they cannot describe the task, the instructions, or the objects. In patients who do not have severe amnesia, factual information is typically acquired as conscious declarative knowledge. In these cases, the brain structures that remain intact within the medial temporal lobe presumably support learning. In contrast, when factual information is acquired as nondeclarative knowledge, as in the case of EP's learning of pairs of objects, learning likely occurs directly as a habit, perhaps supported by the neostriatum. Humans thus appear to have a robust capacity for habit learning that operates outside of awareness and independent of the medial temporal lobe structures that are damaged in amnesia.

Frontal Contributions to Memory

Although amnesia does not occur after limited frontal damage, the frontal lobes are fundamentally crucial for declarative memory. Patients with frontal lesions have a poor memory for the context in which information was acquired, they have difficulty in unaided recall, and they may even have some mild difficulty on tests of item recognition. More generally, patients with frontal lesions have difficulty implementing memory retrieval strategies and evaluating and monitoring their memory performance.

NEUROIMAGING AND MEMORY

The understanding of memory derived from studies of amnesia has been extended through studies using various methods for monitoring brain activity in healthy individuals. For example, activation of posterior prefrontal regions with positron emission tomography (PET) and functional magnetic resonance imaging (fMRI) have shown that these regions are involved in strategic processing during retrieval and working memory. Anterior frontal regions

near the frontal poles have been linked with functions such as evaluating the products of retrieval. Frontal connections with posterior neocortical regions support the organization of retrieval and the manipulation of information in working memory. Consistent with the evidence from patients with frontal lesions, frontal–posterior networks can be viewed as instrumental in the retrieval of declarative memories and the online processing of new information.

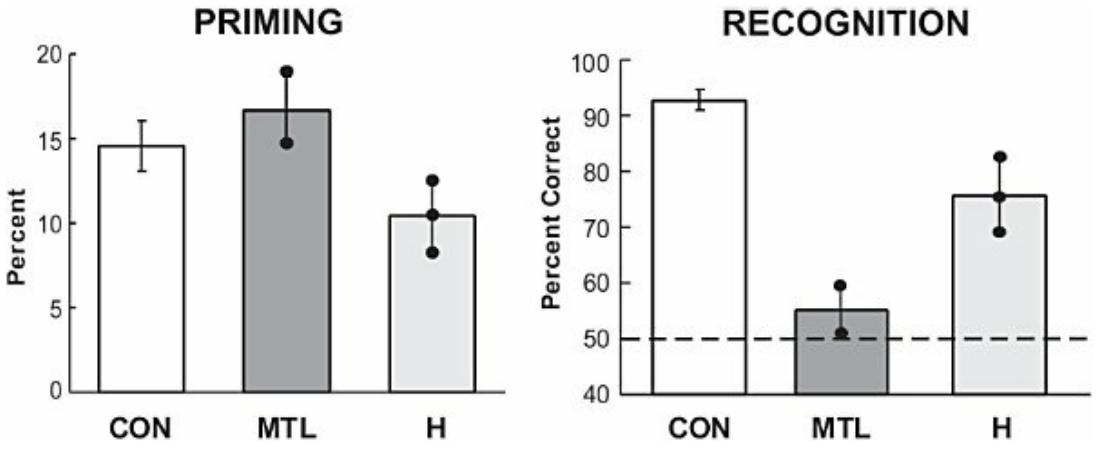


FIGURE 34-11

Preserved conceptual priming in amnesia. In the free association test, subjects studied a set of words (e.g., *lemon*) and 5 minutes later viewed cue words that included associates of the studied words (e.g., *orange*). Subjects were asked to produce the first word that came to mind in response to each cue word. Results are shown separately for the control group (CON; *n* = 12), amnesic patients with large medial temporal lobe lesions (MTL; *n* = 2), and amnesic patients with lesions believed to be limited to the hippocampal region (H; *n* = 3). The **left panel** shows conceptual priming scores computed as the percentage of studied words produced in the free association test minus a baseline measure of the likelihood of producing those words by chance. All of the groups performed similarly in the conceptual priming test. The **right panel** shows results from a yes–no recognition memory test using comparable words. Both patient groups were impaired relative to the control group. The *dashed line* indicates chance performance. The data points for the MTL and H groups show the scores of individual patients averaged across four separate tests. *Brackets* show standard errors of the mean for the control group. (Reprinted from Levy DA, Stark CEL, Squire LR. Intact conceptual priming in the absence of declarative memory. *Psychol Sci*. 2004;15(10):680–686, with permission.)

Neuroimaging has also identified contributions to memory made by the parietal cortex. Multiple parietal regions (including inferior and superior parietal lobules, precuneus, posterior cingulate, and retrosplenial cortex) are activated in conjunction with remembering recent experiences. Although

many functions have been hypothesized to explain this parietal activity, a single consensus position has not yet been reached, and it may be the case that several different functions are relevant.

Neuroimaging studies have also illuminated priming phenomena and how they differ from declarative memory. Perceptual priming appears to reflect changes in the early stages of the cortical pathways engaged during perceptual processing. For example, in the case of stem-completion priming, in which subjects study a list of words (e.g., MOTEL) and then are tested with a list of stems (e.g., MOT___) and with instructions to complete each stem with the first word to come to mind, neuroimaging and divided visual-field studies have implicated visual processing systems in extrastriate cortex, especially in the right hemisphere. In contrast, the conscious recollection of remembered words engages brain areas at later stages of processing. Neural mechanisms supporting priming and declarative memory retrieval have also been distinguished in brain electrical activity recorded from the scalp (Fig. 34-12). In sum, priming differs from declarative memory in that it is signaled by brain activity that occurs earlier, and that originates in different brain regions.



Table 34-7
Types of Memory

A. Declarative memory
   1. Facts
   2. Events
B. Nondeclarative memory
   1. Skills and habits
   2. Priming
   3. Simple classical conditioning
   4. Nonassociative learning

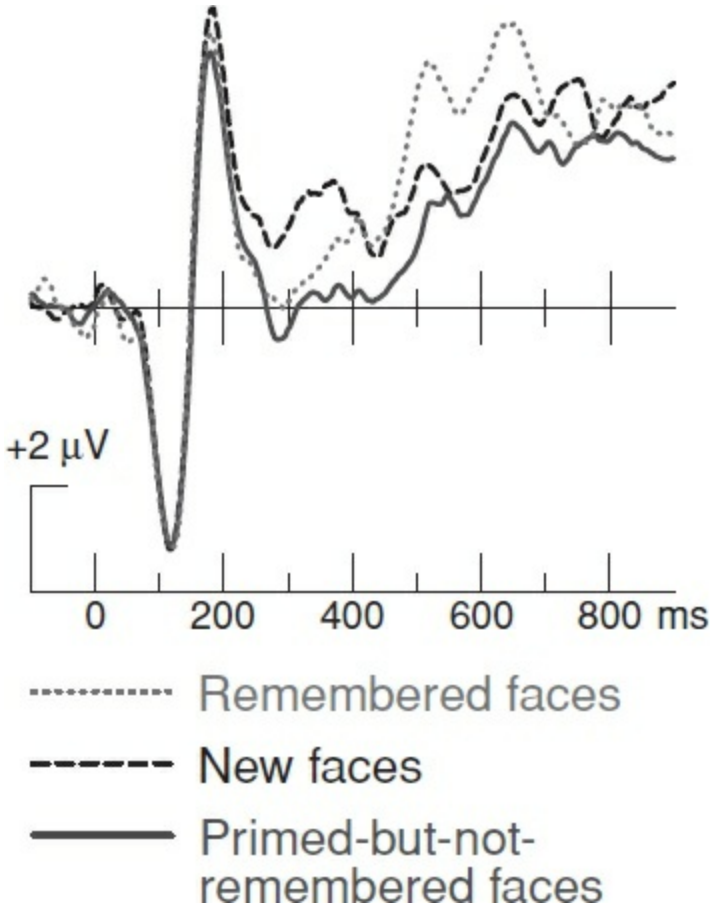


FIGURE 34-12
Brain potentials associated with perceptual priming versus declarative memory retrieval. Paller et al. (2003) studied 16 volunteers, who took a memory test involving three sorts of faces: new faces, faces they had seen recently and remembered well, and faces they had seen but did not remember because the faces had been presented too briefly to process effectively. In a companion experiment with a priming test, speeded responses were found, indicative of priming. Frontal recordings of brain waves elicited by the primed faces included negative potentials from 200 to 400 ms after face presentation that differed from the brain waves elicited by new faces. These differences were particularly robust for trials with the fastest responses (data shown were from trials with responses faster than the median reaction time). Remembered faces uniquely elicited positive brain waves that began about 400 ms after face presentation. Brain potential correlates of face recollection occurred later than those for perceptual priming and were larger over posterior brain regions. (Adapted from Paller KA, Hutson CA, Miller BB, Boehm SG. Neural manifestations of memory with and without awareness. *Neuron*. 2003;38(3):507–516, with permission.)

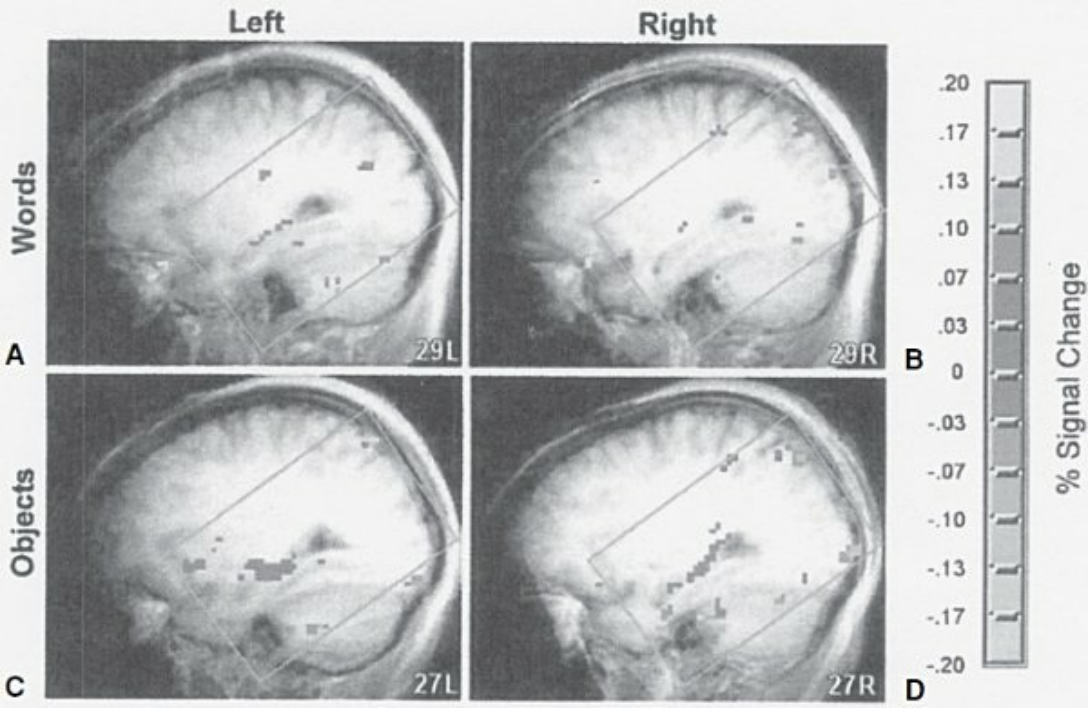


FIGURE 34-13
Activity in the left and right hippocampal regions measured with functional magnetic resonance imaging (fMRI) during declarative memory retrieval. Data were collected from 11 participants who saw words at study and test and from 11 different participants who saw pictures of namable objects at study and test. Recognition memory accuracy was 80.2 percent correct for words and 89.9 percent correct for objects. Areas of significant fMRI signal change (targets vs. foils) are typically shown in sagittal sections as color overlays on averaged structural images. The *box* indicates the area in which reliable data were available for all subjects. With words, retrieval-related activity was observed in the hippocampus on the left side (**A**) but not on the right side (**B**). With namable objects, retrieval-related activity was observed in the hippocampus on both the left side (**C**) and the right side (**D**). (Reprinted from Stark CE, Squire LR. Functional magnetic resonance imaging (fMRI) activity in the hippocampal region during recognition memory. *J Neurosci*. 2000;20(20):7776–7781, with permission.)

Hippocampal activity associated with the formation and retrieval of declarative memories has also been investigated with neuroimaging. In keeping with the neuropsychological evidence, the hippocampus appears to be involved in the recollection of recent events (Fig. 34-13). The retrieval-related hippocampal activity has been observed in memory tests with many different types of stimuli. The hippocampus is also active during the initial storage of information. Whereas the left inferior prefrontal cortex is engaged as a result of attempts to encode a word, hippocampal activity at encoding is more closely associated with whether encoding leads to a stable memory that

can later be retrieved (Fig. 34-14). These findings confirm and extend the idea that medial temporal and frontal regions are essential for memory storage and make different contributions.

SLEEP AND MEMORY

The speculation that memories are processed during sleep has a long history. Freud noted that dreams could reveal fragments of recent experiences in the form of day residues. Although many questions about how and why memories may be processed during sleep remain unresolved, recent experiments have provided new empirical support to bolster the idea that memory processing during sleep serves an adaptive function. It is now clear that memory performance can be facilitated when sleep occurs after initial learning, and that sleep-related facilitation can be observed for many different types of memory.

Memory storage appears to be aided explicitly by processing during deep sleep within a few hours after learning, especially in stages 3 and 4 (slow-wave sleep). Some results indicate that slow-wave sleep facilitates the storage of declarative memories but not nondeclarative memories. Direct evidence for this proposal has been obtained using stimulation with olfactory stimuli (Fig. 34-15), stimulation with direct current electrical input at the approximate frequency of electroencephalographic slow waves, and other methods. Furthermore, neuronal recordings in animals have revealed a phenomenon of hippocampal replay, in which activity patterns expressed during the day are later observed during sleep. In summary, declarative memories acquired during waking can be processed again during sleep, and this processing can influence the likelihood of subsequent memory retrieval when the individual is awake. The facilitation of declarative memory is typically manifest as a reduction in the amount of forgetting that occurs, not as an improvement in memory.

ASSESSMENT OF MEMORY FUNCTIONS

A variety of quantitative methods are available to assess memory functions in neurologic and psychiatric patients. Quantitative methods are useful for evaluating and following patients longitudinally and carrying out a one-time

examination to determine the status of memory function. It is desirable to obtain information about the severity of memory dysfunction, as well as to determine whether a memory is selectively affected or whether memory problems are occurring, as they often do, against a background of additional intellectual deficits. Although some widely available tests, such as the Wechsler Memory Scale, provide useful measures of memory, most single tests assess memory rather narrowly. Even general-purpose neuropsychological batteries provide for only limited testing of memory functions. A complete assessment of memory usually involves several specialized tests that sample intellectual functions, new learning capacity, remote memory, and memory self-report.

The assessment of general intellectual functions is central to any neuropsychological examination. In the case of memory testing, information about intellectual functions provides information about a patient's general test-taking ability and a way to assess the selectivity of memory impairment. Useful tests include the Wechsler Adult Intelligence Scale; a test of object naming, such as the Boston Naming Test; a rating scale to assess the possibility of global dementia; a test of word fluency; and specialized tests of frontal lobe function.

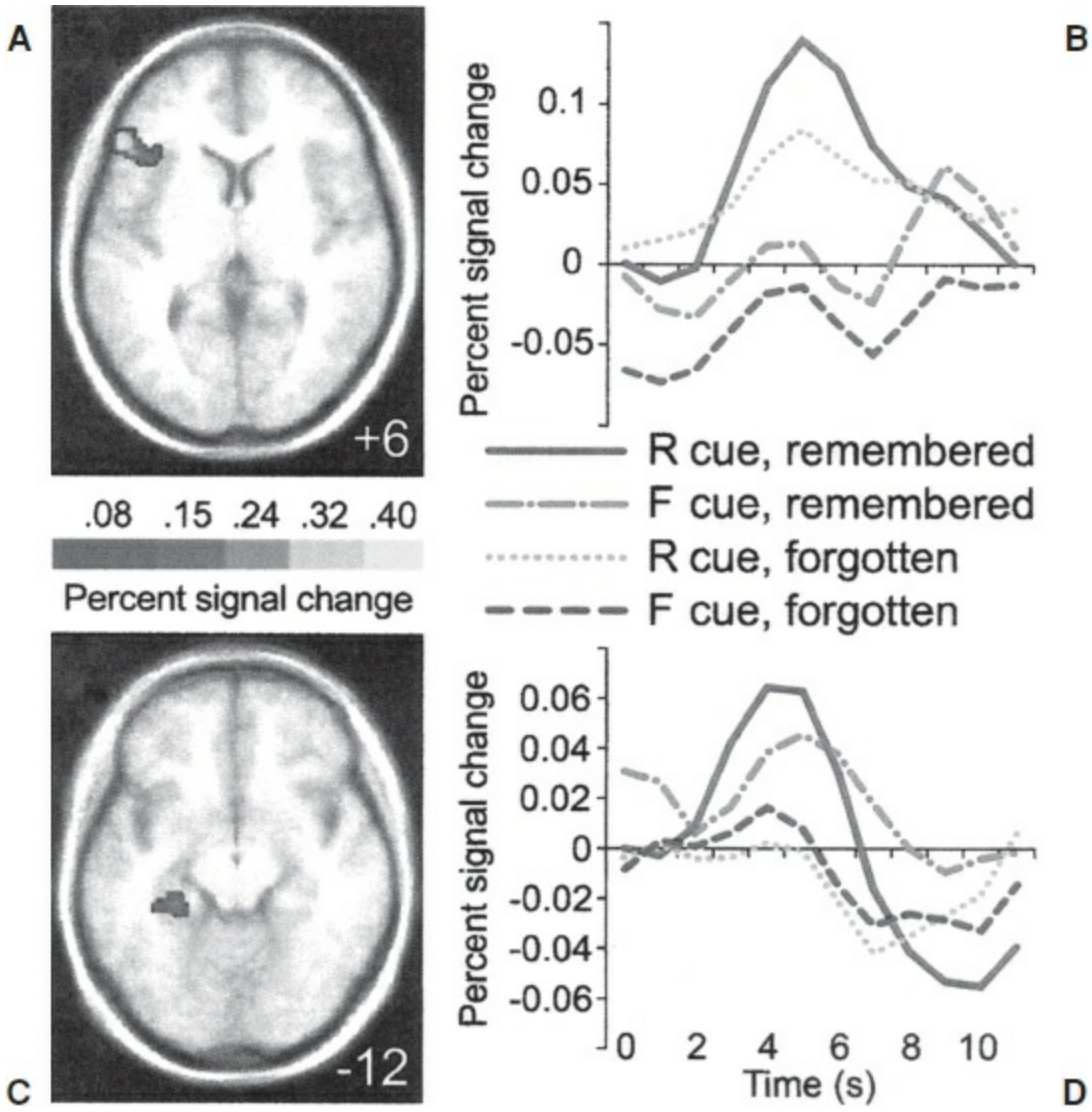


FIGURE 34-14

Functional activations of prefrontal and medial temporal regions that were predictive of later memory performance. Single words were presented visually, each followed by an instruction to remember (R cue) or to forget (F cue). Trials were sorted based on the remember or forget instruction and on subsequent recognition performance. Activity in the left inferior prefrontal cortex and left hippocampus was predictive of subsequent recognition but for different reasons. Left inferior prefrontal activation (**A**) was associated with the encoding attempt, in that responses were largest for trials with a cue to remember, whether or not the word was actually recognized later. The time course of activity in this region (**B**) was computed based on responses that were time locked to word onset (time 0). Left inferior prefrontal activity increased for words that were later remembered, but there was a stronger association with encoding attempt, because responses were larger for words followed by an R cue that were later forgotten than for words followed by an F cue that were later remembered. In contrast, left parahippocampal and posterior hippocampal activation (**C**) was associated with encoding success. As shown by the time course of activity in this region (**D**), responses were largest for words that were subsequently remembered, whether

the cue was to remember or to forget. (Reprinted from Reber PJ, Siwiec RM, Gitelman DR, Parrish TB, Mesulam MM, Paller KA. Neural correlates of successful encoding identified using functional magnetic resonance imaging. *J Neurosci*. 2002;22(21):9541–9548, with permission.)

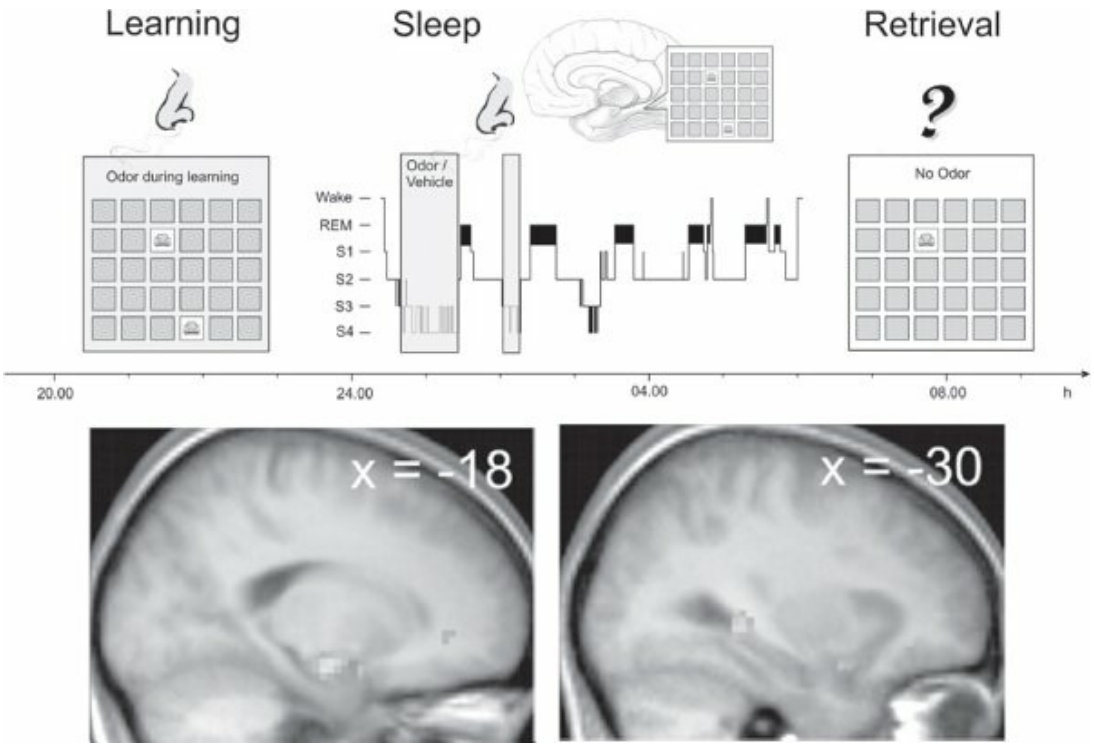


FIGURE 34-15
Evidence for memory processing during sleep. Subjects first learned object–location associations when a rose odor was present. Following learning, subjects slept wearing a device for delivering odors to the nose, and the rose odor was administered during the first two slow-wave sleep periods of the night (in 30-second periods to prevent habituation). Memory facilitation was observed when object–location associations were tested the following morning in the absence of odor stimulation. Facilitated memory was not found when stimulation occurred during slow-wave sleep but not during learning, when stimulation occurred during learning and then during rapid-eye-movement (REM) sleep, or when subjects were kept awake. Moreover, odor stimulation during slow-wave sleep was found to produce anterior and posterior hippocampal activation (**lower panels**). (Reprinted from Rasch B, Büchel C, Gais S, Born J. Odor cues during slow-wave sleep prompt declarative memory consolidation. *Science*. 2007;315(5817):1426–1429, with permission.)

New Learning Capacity

Memory tests are sensitive to impaired new learning ability when they adhere to either of two critical principles. First, tests are sensitive to memory impairment when more information is presented than can be held in

immediate memory. For example, one might ask patients to memorize a list of 10 faces, words, sentences, or digits, given that 10 items are more than can be held in mind. The paired-associate learning task is an especially sensitive test of this kind. In the paired-associate task, the examiner asks the patient to learn a list of unrelated pairs of words (e.g., queen–garden, office–river) and then to respond to the first word in each pair by recalling the second word.

Second, tests are sensitive to memory impairment when a delay, filled with distraction, is interposed between the learning phase and the test phase. In that case, examiners typically ask patients to learn a small amount of information and then distract them for several minutes by conversation to prevent rehearsal. Recollection is then assessed for the previously presented material. Memory can be tested by the unaided recall of previously studied material (free recall), presenting a cue for the material to be remembered (cued recall), or by testing recognition memory. In multiple-choice tests of recognition memory, the patient tries to select previously studied items from a group of studied and unstudied items. In yes–no recognition tests, patients see studied and unstudied items one at a time and are asked to say “yes” if the item was presented previously and “no” if it was not. These methods for assessing recently learned material vary in their sensitivity for detecting memory impairment, with free recall being most sensitive, cued recall intermediate, and recognition least sensitive.

The specialization of function of the two cerebral hemispheres in humans means that left and right unilateral damage is associated with different kinds of memory problems. Accordingly, different kinds of memory tests must be used when unilateral damage is a possibility. In general, damage to medial temporal or diencephalic structures in the left cerebral hemisphere causes difficulty remembering verbal material, such as word lists and stories. Damage to medial temporal or diencephalic structures in the right cerebral hemisphere impairs memory for faces, spatial layouts, and other nonverbal material that is typically encoded without verbal labels. Left medial temporal damage can lead to impaired memory for spoken and written text. Right medial temporal damage can lead to impaired learning of spatial arrays, whether the layouts are examined by vision or touch. A useful way to test for nonverbal memory is to ask a patient to copy a complex geometric figure, and then, after a delay of several minutes, without forewarning, ask the patient to reproduce it.

Remote Memory

Evaluations of retrograde memory loss should attempt to determine the severity of any memory loss and the period that it covers. Most quantitative tests of remote memory are composed of material in the public domain that can be corroborated. For example, tests have been used that concern news event, photographs of famous persons, or former one-season television programs. An advantage of these methods is that one can sample large numbers of events and often target particular periods. A disadvantage is that these tests are not so useful for detecting memory loss for information learned during the weeks or months immediately before the onset of amnesia. Most remote memory tests sample periods rather coarsely and cannot detect a retrograde memory impairment covering only a few months.

In contrast, autobiographical memory tests can potentially provide fine-grained information about a patient's retrograde memory. In the word-probe task, first used by Francis Galton in 1879, patients are asked to recollect specific episodes from their past in response to single word cues (e.g., bird and ticket) and date the episodes. The number of episodes recalled tends to be systematically related to the time the episode is taken. Most memories naturally come from recent periods (the last 1 to 2 months). In contrast, patients with amnesia often exhibit temporally graded retrograde amnesia, drawing relatively few episodic memories from the recent past but producing as many remote autobiographical memories as normal subjects (see Fig. 34-9).

Memory Self-Reports

Patients can often supply descriptions of their memory problems that are extremely useful for understanding the nature of their impairment. Tests of the ability to judge one's memory abilities are called tests of *metamemory.* Self-rating scales are available that yield quantitative and qualitative information about memory impairment. As a result, it is possible to distinguish memory complaints associated with depression from memory complaints associated with amnesia. Depressed patients tend to rate their memory as poor in a relatively undifferentiated way, endorsing equally all of the items on a self-rating form. By contrast, amnesic patients tend to endorse

some items more than others; that is, there is a pattern to their memory complaints. Amnesic patients do not report difficulty in remembering very remote events or in following what is being said to them, but they do report having difficulty remembering an event a few minutes after it happens. Indeed, self-reports can match rather closely the description of memory dysfunction that emerges from objective tests. Specifically, new-learning capacity is affected, immediate memory is intact, and very remote memory is intact. Some amnesic patients, however, tend to underestimate their memory impairment markedly. In patients with Korsakoff syndrome, for example, their poor metamemory stems from frontal lobe dysfunction. In any case, querying patients in some detail about their sense of impairment and administering self-rating scales are valuable and informative adjuncts to more formal memory testing.

Psychogenic Amnesia

Patients sometimes exhibit memory impairment that differs markedly from the typical patterns of memory loss that follow brain damage. For example, some cases of amnesia present with a sudden onset of retrograde amnesia, a loss of personal identity, and minimal anterograde amnesia. These patients may even be unable to recall their name. Given the psychological forces that prompt the onset of amnesia in these cases, they are commonly termed *psychogenic amnesia,* or sometimes *hysterical amnesia, functional amnesia,* or *dissociative amnesia.*

Differentiating psychogenic amnesia from a memory disorder that results from frank neurologic injury or disease is often straightforward. Psychogenic amnesias typically do not affect new-learning capacity. Patients enter the hospital able to record a continuing procession of daily events. By contrast, new-learning problems tend to be at the core of neurologic amnesia. The main positive symptom in psychogenic amnesia is extensive and severe retrograde amnesia. Patients may be unable to recollect pertinent information from childhood or some part of their past. Formal neuropsychological testing has shown that the pattern of memory deficits varies widely from patient to patient. This variability may reflect a patient’s commonsense concepts of memory, even when symptoms are not the result of conscious attempts to simulate amnesia. Some patients may perform poorly only when asked to

remember past autobiographical events. Other patients also may fail at remembering past news events. Some patients perform well when memory tests appear to assess general knowledge, such as remembering the names of celebrities or cities. Learning new material is usually intact, perhaps because such tests appear to concern the present moment, not traveling into the past. Occasionally, patients with psychogenic amnesia exhibit broad memory deficits such that they cannot perform previously familiar skills or identify everyday objects or common vocabulary words.

By contrast, patients with neurologic amnesia never forget their name, and remote memory for the events of childhood and adolescence is typically normal unless there is damage to the lateral temporal or frontal lobes. Patients with psychogenic amnesia sometimes have evidence of head trauma or brain injury, but the pattern of deficits cannot be taken as a straightforward result of neurologic insult. The clinician's challenge is not to distinguish psychogenic amnesia from neurologic amnesia, but to distinguish psychogenic amnesia from malingering. Indeed, the diagnosis of psychogenic amnesia can be difficult to substantiate and may be met with skepticism by hospital staff. Some features that argue in favor of a genuine psychogenic disorder include (1) memory test scores that are not as low as possible, and never worse than chance levels; (2) memory access that is improved by hypnosis or amobarbital (Amytal) interview; and (3) a significant premorbid psychiatric history. In some cases, psychogenic amnesia has been observed to clear after days, but in many cases, it has persisted as a potentially permanent feature of the personality.

IMPLICATIONS

Memory Distortion

Current understanding of the biology of memory has significant implications for several fundamental issues in psychiatry. Given the selective and constructive nature of autobiographical remembering and the imperfect nature of memory retrieval more generally, it is surprising that memory is so often accurate. How much can we trust our memories? Subjective feelings of confidence are not perfect indicators of retrieval accuracy. Moreover, memory distortion can lead to unfortunate consequences, such as when

mistaken eyewitness testimony harms an innocent individual.

It is possible to remember with confidence events that never happened. For example, it is possible to confuse an event that was only imagined or dreamed about with a real-life event. One factor contributing to memory distortion is that similar brain regions are essential both for visual imagery and for the long-term storage of visual memories (Fig. 34-16). Another factor contributing to memory distortion is that memory functions best in remembering the gist of an event, not the particulars from which the gist is derived. In a noted demonstration, people listen to a list of words: *Candy, sour, sugar, tooth, heart, taste, dessert, salt, snack, syrup, eat,* and *flavor.* Subsequently, when asked to write down the words they heard, 40 percent wrote down the word *sweet,* even though it did not appear on the list. Thus, many people in this demonstration failed to discriminate between the words presented and a word strongly associated with all the words but had not itself been presented. The word *sweet* can be thought of as a gist word, a word that represents the other words, which captures the meaning of the whole list. Presumably, the words in the study list evoked a thought of the word *sweet,* either at the time of learning or during the memory test, and people then tended to confuse merely thinking of the word with actually hearing it.

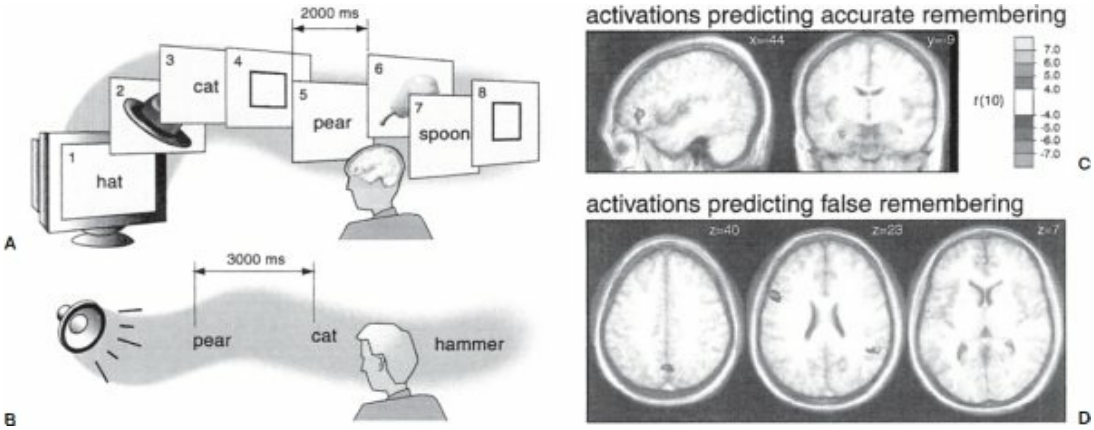


FIGURE 34-16

Neural substrates of false memories. **A:** Functional magnetic resonance imaging data were acquired in a learning phase, when subjects read names of objects and visualized the referents. One-half of the names were followed 2 seconds later by a picture of the object. **B:** In a surprise memory test given outside of the scanner, subjects listened to object names and decided whether they had seen a picture of the corresponding object. On some trials, subjects claimed to have seen a picture of an object that they had only imagined. **C:** Results showed that the left inferior prefrontal cortex and the left anterior hippocampus were more active during learning in response to pictures later remembered compared to pictures later

forgotten. **D:** Several different brain areas showed a greater response to words in the learning phase that were later falsely remembered as pictures compared to words not misremembered. Activations that predicted false remembering were found in a brain network important for the generation of visual imagery in response to object names (precuneus, inferior parietal cortex, and anterior cingulate, shown in the left, middle, and right images, respectively). (Reprinted from Gonsalves B, Reber PJ, Gitelman DR, Parrish TB, Mesulam MM, Paller KA. Neural evidence that vivid imagining can lead to false remembering. *Psychol Sci*. 2004;15(10):655–660, with permission.)

The reconstructive nature of recollection means that the interpretation of eyewitness testimony is not straightforward. Whole episodes are not available in the neocortex, but rather must be pieced together based on fragmentary components and in the context of potentially misleading influences present at the time of retrieval. Studies in adults and children have documented that illusory memories can be created. Children are particularly susceptible to these effects, especially when subjected to leading questions and false suggestions.

Given these features of memory, when memory for childhood abuse is remembered after many years, it is prudent to ask whether the memory is accurate. Genuine examples of memory recovery have been documented, whereby an individual produces a veridical memory for a past traumatic event after not recalling the event for extended periods. Numerous examples of apparent memory recovery have also been subsequently discovered to be instances of false memory. Unfortunately, there is no perfect method, in the absence of independent corroboration, for determining whether a recollective experience is based on a real event.

Infantile Amnesia

The biology of memory has also provided insights relevant to the phenomenon of infantile amnesia—the apparent absence of conscious memory for experiences from approximately the first 3 years of life. Traditional views of infantile amnesia have emphasized repression (psychoanalytic theory) and retrieval failure (developmental psychology). A common assumption has been that adults retain memories of early events but cannot bring them into consciousness. However, it now appears that the capacity for declarative memory does not become fully available until approximately the third year of life, whereas nondeclarative memory emerges

early in infancy (e.g., classical conditioning and skill learning). Thus, infantile amnesia results not from the adult's failure to retrieve early memories, but from the child's failure to store them adequately in the first place.

Nevertheless, studies in young infants show that a rudimentary capacity for declarative memory is present even at a few months of age. As a child develops, memories can be retained across increasingly long intervals, and what is represented becomes correspondingly richer and more full of detail. Medial temporal and diencephalic regions seem to be sufficiently developed during these early months and years. What limits the capacity for declarative memory appears to be the gradual development and differentiation of the neocortex.

As the neocortex develops, the memories represented there become more complex, language abilities allow for more elaborate verbal descriptions of events, and a developing sense of self supports autobiographical knowledge. As new strategies emerge for organizing incoming information, declarative memories become more persistent, more richly encoded, and better interconnected with other information. It is not the case that fully formed childhood memories are stored but cannot be retrieved. The perspective consistent with the current understanding of the biology of memory is that declarative memories formed very early in life are fragmentary, simple, and tied to the specific context of an infant's understanding of the world. They are unlike typical declarative memories in adults, which are imbued with meaning and a complex understanding of events.

Memories and the Unconscious

The existence of multiple memory systems also has implications for issues central to psychoanalytic theory, including the construct of the *unconscious.* How one believes that experience influences current behavior depends on what view one takes of the nature of memory. By the traditional view, memory is a unitary faculty, and representations in memory vary mainly in strength and accessibility. Unconscious material is below some threshold of accessibility but could potentially be made available to consciousness.

The modern, biologic view begins with the distinction between a kind of memory that can be brought to mind—declarative memory—and other kinds

of memory that are, by their nature, unconscious. Stored nondeclarative memories are expressed through performance without affording any conscious memory content. Nondeclarative memories shape our personalities in the form of numerous habits and conditioned responses. In this view, one's behavior is indeed affected by events from early life, but the effects of early experience persist in a nondeclarative form without necessarily including an explicit, conscious record of the events. Learned behavior can be expressed through altered dispositions, preferences, conditioned responses, habits, and skills, but exhibiting such behavior need not be accompanied by an awareness that behavior is being influenced by experience, nor is there a necessity that any particular experience has been recorded as a complete episode. That is, an influence from early experience does not require memory of any specific episode. One can be afraid of dogs without remembering being knocked down by a dog as a child. In this case, the fear of dogs is not experienced as a memory. It is experienced as a part of the personality. Furthermore, an intense fear of dogs carries with it no implication that the brain retains a specific record of any early experience that subsequently resulted in fear of dogs.

Behavioral change can occur by acquiring new habits that supersede old ones or by becoming sufficiently aware of a habit that one can to some extent isolate it, countermand it, or limit the stimuli that elicit it. However, one need not become aware of any early formative event in the same sense that one knows the content of declarative memory. The unconscious does not become conscious. Various forms of nondeclarative memory simply influence behavior without having the additional capacity for these influences to become accessible to conscious awareness.

34.2 Normality and Mental Health

There has been an implicit assumption that mental health could be defined as the antonym of mental illness. In other words, mental health was the absence of psychopathology and synonymous with *normal.* Achieving mental health by alleviating gross pathologic signs and symptoms of illness is also the definition of the mental health model strongly advocated by third-party

payers. Indeed, viewing mental health as simply the absence of mental illness is at the heart of much of the debate concerning mental health policies. The great epidemiologic studies of the last half-century also focused on who was mentally ill, and not well.

DEFINING MENTAL HEALTH

Several steps are necessary for defining positive mental health. The first step is to note that "average" is not healthy; it always includes mixing in with the healthy the prevalent amount of psychopathology in the population. For example, in the general population, being of "average" weight or eyesight is unhealthy, and if all sources of biopsychosocial pathology were excluded from the population, the average IQ would be significantly greater than 100.

The second step in discussing mental health is to appreciate the caveat that what is healthy sometimes depends on geography, culture, and the historical moment. Sickle cell trait is unhealthy in New York City, but in the tropics, where malaria is endemic, the sickling of red blood cells may be lifesaving.

The third step is to make clear whether one is discussing *trait* or *state.* Who is physically healthier—an Olympic miler disabled by a temporary but straightforward (state) sprained ankle or a person with type 1 diabetes (trait) with temporarily normal blood sugar? In cross-cultural studies, such differences become especially important. Superficially, an Indian mystic in a state of trance may resemble a person with catatonic schizophrenia, but the mystic does not resemble someone in the schizophrenic condition over time.

The fourth and most crucial step is to appreciate the twofold danger of "contamination by values." On the one hand, cultural anthropology teaches us how fallacious any definition of mental health can be. Competitiveness and scrupulous neatness may be healthy in one culture and regarded as personality disorders in another. Furthermore, if mental health is "good," what is it good for? The self or society? For "fitting in" or for creativity? For happiness or survival? Furthermore, who should be the judge?

MODELS OF MENTAL HEALTH

This chapter contrasts six different empirical approaches to mental health. First, mental health can be conceptualized as *above normal* and a mental state

that is objectively desirable, as in Sigmund Freud's definition of mental health, which is the capacity to work and love. Second, from the viewpoint of healthy adult development, mental health can be conceptualized as *maturity.* Third, mental health can be conceptualized in terms of *positive psychology*—as epitomized by the presence of multiple human strengths. Fourth, mental health can be conceptualized as *emotional intelligence* and successful object relations. Fifth, mental health can be conceptualized as *subjective well-being*—a mental state that is subjectively experienced as happy, contented, and desired. Sixth, mental health can be conceptualized as *resilience,* as the capacity for successful adaptation and homeostasis.

Model A: Mental Health as Above Normal

This first perspective differs from the traditional medical approach to health and illness. No manifest psychopathology equals mental health. In this medical model, if one were to put all individuals on a continuum, normality would encompass a significant portion of adults, and abnormality would be the small remainder. This definition of health correlates with the traditional role model of the doctor who attempts to free his patient from grossly observable signs of illness. In other words, in this context, health refers to a reasonable, rather than an optimal state of functioning. However, as already pointed out, mental health is not normal; it is above average. Some believe that true mental health is the exception, not the rule. Moreover, until recently, some believed that mental health was imaginary.

Model B: Mental Health as Maturity

Unlike other organs of the body that are designed to stay the same, the brain is designed to be plastic. Therefore, just as optimal brain development requires almost a lifetime, so does assessing positive mental health. A 10-year-old's lungs and kidneys are more likely to reflect optimal function than are those of a 60-year-old, but that is not true of a 10-year-old's central nervous system. To some extent, then, adult mental health reflects a continuing process of maturational unfolding. Statistically, physically healthy 70-year-olds are mentally healthier than they were at age 30 years; for example, Laura Carstensen found through prospective studies that individuals are less depressed and show greater emotional modulation at age 70 years

than they did at age 30 years.

However, if prospective studies of adult development reveal that the immature brain functions less well than the mature brain, does that mean that adolescents are mentally healthier than toddlers? Are the middle-aged mentally healthier than adolescents? The answer is both yes and no, but the question illustrates that in order to understand mental health, we must first understand what we mean by maturity.

To confirm the hypothesis that maturity and positive mental health are almost synonymous, it is necessary to study the behavior and emotional states of persons over a lifetime. Although such longitudinal studies have come to fruition only recently, all of them illustrate the association of maturity with increasing mental health. After age 50 years, of course, the association between mental health and maturity is contingent on a healthy central nervous system. The ravages of illnesses like brain trauma, major depression, arteriosclerosis, Alzheimer's, and alcoholism must all be avoided.

The association of mental health to maturity is probably mediated not only by progressive brain myelination into the sixth decade but also by the evolution of emotional and social intelligence through experience. Erik Erikson conceptualized that such development produced a "widening social radius." In such a view, life after age 50 years was no longer a staircase leading downward, as in the Pennsylvania Dutch cartoons of life-span development, but a path leading outward. In Erikson's model, the adult social radius expanded over time through the mastery of specific tasks such as "Identity versus Identity Diffusion," "Intimacy versus Isolation," "Generativity versus Stagnation," and "Integrity versus Despair."

Identity. In such a model, the social radius of each adult developmental task fits inside the next. First, adolescents must achieve an *Identity* that allows them to become separate from their parents, for mental health and adult development cannot evolve through a false self. The task of *Identity* requires mastering the last task of childhood: sustained separation from social, residential, economic, and ideologic dependence on the family of origin. Identity is not just a product of egocentricity, of running away from home, or of marrying to get out of a dysfunctional family. There is a world of difference between the instrumental act of running away from home and the developmental task of knowing where one's family values end and one's

values begin. Such separation derives as much from the identification and internalization of important adolescent friends and nonfamily mentors as it does from simple biologic maturation. For example, our accents become relatively fixed by age 16 years and reflect those of our adolescent peer group rather than the accents of our parents.

Intimacy. Then, young adults should develop *Intimacy*, which permits them to become reciprocally, and not selfishly, involved with a partner. However, living with just one other person in an interdependent, reciprocal, committed, and contented fashion for years and years may seem neither desirable nor possible to a young adult. Once achieved, however, the capacity for intimacy may seem as effortless and desirable as riding a bicycle. Sometimes the relationship is with a person of the same gender; sometimes, it is entirely asexual; and sometimes, as in religious orders, the interdependence is with a community. Superficially, mastery of intimacy may take very different guises in different cultures and epochs, but "mating-for-life" and "marriage-type love" are developmental tasks built into the developmental repertoires of many warm-blooded species, including ours.

Career Consolidation. *Career Consolidation* is a task that is usually mastered together with or that follows the mastery of intimacy. Mastery of this task permits adults to find a career as valuable as they once found play. On a desert island, one can have a hobby but not a career, for careers involve being of value to other people. Four crucial developmental criteria transform a "job" or hobby into a "career:" Contentment, compensation, competence, and commitment. Such a career can be "wife and mother"—or, in more recent times, "husband and father." To the outsider, the process of Career Consolidation often appears "selfish," but without such "selfishness," one becomes "selfless" and has no "self" to give away in the next stage of generativity. Persons with schizophrenia and individuals with a severe personality disorder often manifest a lifelong inability to achieve either intimacy or sustained, gratifying employment.

Generativity. *Generativity* involves the demonstration of a clear capacity to care for and guide the next generation. Research reveals that sometime between ages 35 and 55 years, our need for achievement declines, and our

need for community and affiliation increases. Depending on the opportunities that the society makes available, generativity can mean serving as a consultant, guide, mentor, or coach to young adults in the larger society. Like leadership, generativity means to be in a caring relationship in which one gives up much of the control that parents retain over young children. Good mentors learn "to hold loosely" and to share responsibility. Generativity reflects the capacity to give the self—finally completed through mastery of the first three tasks of adult development—away. Its mastery is strongly correlated with successful adaptation to old age. This is because, in old age, there are inevitable losses, and these may overwhelm us if we have not continued to grow beyond our immediate family.

Integrity. Finally, in old age, it is common to feel that some life exists after death and that one is part of something greater than oneself. Thus, the last life task in Erikson's words is *Integrity,* achieving some sense of peace and unity with respect both to one's life and to the whole world. Erikson described integrity as "an experience which conveys some world order and spiritual sense. No matter how dearly paid for, accepting one's life cycle as something that had to be and that, by necessity, permitted no substitutions."

It must be kept in mind that mastery of one life task is not necessarily healthier than mastery of another, for adult development is neither a foot race nor a moral imperative. Instead, these sequential tasks are offered as a road map to help clinicians make sense of where they are and where their patients might be located. One can be a mature 20-year-old, that is, healthy. One can be an immature 50-year-old, which may be unhealthy. Nevertheless, acquiring a social radius that extends beyond the person by definition allows more flexibility and is usually healthier than self-preoccupation. Generativity by ages 40 to 50 years offers a powerful predictor of a contented old age.

Model C: Mental Health as Positive or "Spiritual" Emotions

This model defines mental and spiritual health as the amalgam of the positive emotions that bind us to other human beings. Love, hope, joy, forgiveness, compassion, faith, awe, and gratitude comprise the essential positive and "moral" emotions included in this model. Of great importance, these selected positive emotions all involve human connection. None of the emotions listed

is just about the self. These positive emotions appear to be a common denominator of all major faiths. Omitted from the list are five other positive emotions—excitement, interest, contentment (happiness), humor, and a sense of mastery, for a person can feel these latter five emotions alone on a desert island.

Negative emotions originating in the hypothalamus, such as fear and anger, are elaborated in the human amygdala (larger in humans than in other mammals). Of tremendous importance to individual survival, the negative emotions are all about "me." In contrast, positive emotions, apparently generated in the limbic system and unique to mammals, can free the self from the self. People feel the emotions of vengeance and forgiveness deeply, but the long-term results of these two emotions are very different. Negative emotions are crucial for survival in the present time. The positive emotions are more expansive and help us to broaden and build. In the future, they widen one's tolerance for strangers, expand one's moral compass, and enhance one's creativity. Whereas negative emotions narrow attention and miss the forest for the trees, positive emotions, especially joy, make thought patterns more flexible, creative, integrative, and efficient.

The effect of positive emotion on the autonomic (visceral) nervous system has much in common with the relaxation response to meditation. In contrast to the metabolic and cardiac arousal that the fight-or-flight response of negative emotion induces in our *sympathetic* autonomic nervous system, positive emotion via our *parasympathetic* nervous system reduces basal metabolism, blood pressure, heart rate, respiratory rate, and muscle tension. fMRI studies of Kundalini yoga practitioners demonstrate that meditation increases the activity of the hippocampus and the right lateral amygdala, which in turn leads to parasympathetic stimulation and the sensation of deep peacefulness.

Positive emotions have a biologic basis, which means that they have evolved through natural selection. The prosocial emotions probably reflect adaptations that permitted the survival of relatively defenseless *Homo sapiens* and their defenseless children in the African savannah 1 to 2 million years ago.

Evidence for Positive Emotions. It has taken recent developments in neuroscience and ethology to make positive emotions a subject fit for

scientific study. For example, infantile autism, a not uncommon genetic disorder of emotional attachment, was not discovered until 1943 by a Johns Hopkins child psychiatrist, Leo Kanner—in his son. Until then, medicine could not articulate a positive emotion as basic, but as cognitively subtle, as attachment. Today, the congenital lack of empathy and difficulties of attachment in childhood autism can be recognized by any competent pediatrician.

To locate positive emotion in the mammalian limbic system has been a slow, arduous process. In 1955, James Olds, an innovative neuropsychologist, observed that 35 of 41 electrode placements within the limbic system of rats, but only 2 of 35 placements outside of the limbic system, proved sufficiently rewarding to lead to self-stimulation. Also in the 1950s, neurobiologist Paul MacLean pointed out that the limbic structures govern our mammalian capacity not only to remember (cognition), but also to play (joy), to cry out at separation (faith/trust), and to take care of our own (love). Except for rudimentary memory, reptiles express none of these qualities.

Studies using fMRI demonstrated that when individuals subjectively experience existential states of fear, sadness, or pleasure, blood flow increases in limbic areas and decreases in many higher brain areas. Various studies have located pleasurable human experiences (tasting chocolate, winning money, admiring pretty faces, enjoying music, and experiencing orgasmic ecstasy) in limbic areas—especially in the orbitofrontal region, anterior cingulate, and insula. These diverse structures are closely integrated and organized to help us to seek and to recognize all that falls under the rubric of mammalian love and human spirituality.

The *anterior cingulate gyrus* links valence and memory to create attachment. Along with the hippocampus, the anterior cingulate is the brain region most responsible for making the past meaningful. In terms of mediating attachment, the anterior cingulate receives one of the richest dopaminergic innervations of any cortical area. Thus, the cingulate gyrus provides motivational salience not only for lovers but also for drug addicts. The anterior cingulate is crucial in directing whom we should approach and whom we should avoid. Maternal touch, body warmth, and odor via the limbic system and primarily via the anterior cingulate regulate a rat pup’s behavior, neurochemistry, endocrine release, and circadian rhythm. Brain

imaging studies reveal that the anterior cingulate gyrus is aroused neither by facial recognition of friends per se nor by sexual arousal per se. Instead, anterior cingulate fMRI images light up when a lover gazes at a picture of a partner's face or when a new mother hears her infant's cry.

Perhaps no area of the brain is more ambiguous in its evolutionary heritage or more crucial to mental health than our *prefrontal cortex.* The prefrontal cortex is in charge of estimating rewards and punishments and plays a critical role in adapting and regulating our emotional response to new situations. Thus, the prefrontal lobes are deeply involved in emotional, "moral," and "spiritual" lives.

From an evolutionary standpoint, the human frontal lobes are not different from chimpanzees in terms of the number of neurons. Instead, it is the frontal lobe white matter (the connectivity between neurons through myelinated fibers) that accounts for larger frontal lobes of humans. This connectivity to the limbic system underscores its "executive" function, which includes the ability to delay gratification, comprehend symbolic language, and, most importantly, to establish temporal sequencing. By being able to connect memory of the past to "memory of the future," the frontal lobes establish for *Homo sapiens* predictable cause and effect.

Surgical or traumatic ablation of the ventromedial prefrontal cortex can turn a conscientious, responsible adult into a moral imbecile without any other evidence of intellectual impairment.

The *insula* is another part of the limbic system that is only beginning to be understood. The insula is a medial cortical gyrus located between the amygdala and the frontal lobe. The brain has no sensation; humans feel emotion only in their bodies. The insula helps to bring these visceral feelings into consciousness: The pain in one's heart of grief, the warmth in one's heart of love, and the tightness in one's gut from fear all make their way into consciousness through the insula.

Both the limbic anterior cingulate and insula appear to be active in the positive emotions of humor, trust, and empathy. The higher apes are set apart from other mammals by a unique neural component called the spindle cell. Humans have 20 times more spindle cells than chimps or gorillas (adult chimpanzees average about 7,000 spindle cells; human newborns have four times more, and human adults have almost 200,000 spindle cells). Monkeys and other mammals, with the possible exception of whales and elephants, are

lacking in these unique cells. These large, cigar-shaped spindle or "von Economo" neurons appear to be central to the governance of social emotions and moral judgment. Spindle cells may have helped the Great Apes and humans integrate their mammalian limbic systems with their expanding neocortices. Spindle cells are concentrated in the anterior cingulate cortex, the prefrontal cortex, and the insula. More recently, scientists have discovered a particular group of "mirror neurons" in the insula and anterior cingulate. These neurons are more highly developed in humans than in primates and appear to mediate empathy—the experience of "feeling" the emotions of another.

Although the practical applications of this newest model of mental health are still many years away, these findings provide further evidence that the brain and mind are one. In several studies, the prosocial biologic activity of the anterior cingulate cortex and insula was highest in individuals with the highest social awareness (based on objectively scored tests). In other words, there are not only individual biologic differences for negative mental health, but also positive mental health.

Model D: Mental Health as Socioemotional Intelligence

High socioemotional intelligence reflects above-average mental health in the same way that a high IQ reflects above-average intellectual aptitude. Such emotional intelligence lies at the heart of positive mental health. In the *Nicomachean Ethics*, Aristotle defined socioemotional intelligence as follows: "Anyone can become angry—that is easy. However, to be angry with the right person, to the right degree, at the right time, for the right purpose, and in the right way—that is not easy."

All emotions exist to assist in basic survival. Although the exact number of primary emotions is arguable, seven emotions are currently distinguished according to characteristic facial expressions connoting anger, fear, excitement, interest, surprise, disgust, and sadness. The capacity to identify these different emotions in ourselves and others plays a vital role in mental health. The benefits of reading feelings from nonverbal cues have been demonstrated in almost a score of countries. These benefits included being better emotionally adjusted, more popular, and more responsive to others. Empathic children do better in school and are more popular than their peers

without being more intelligent. The Head Start Program, a program of the United States Department of Health and Human Services that provides education and other services for low-income children and their families, found that early school success was achieved not by intelligence but by knowing what kind of behavior is expected, knowing how to rein in the impulse to misbehave, being able to wait, and knowing how to get along with other children. At the same time, the child must communicate his or her needs and turn to teachers for help.

Ethologically, emotions are critical to mammalian communication. Because such communications are not always consciously recognized, the more skillful individuals are in identifying their emotions, the more skilled the individual will be in communicating with others, and in empathically recognizing their emotions. Put differently, the more you are skilled in empathy, the more others will value you, and the greater will be your social supports, self-esteem, and intimate relationships.

The following criteria can define social and emotional intelligence:

- Accurate conscious perception and monitoring of one's emotions.
- Modification of emotions so that their expression is appropriate. This involves the capacity to self-soothe personal anxiety and to shake off hopelessness and gloom.
- Accurate recognition of and response to emotions in others.
- Skill in negotiating close relationships with others.
- Capacity for focusing emotions (motivation) toward the desired goal. This involves delayed gratification and adaptively displacing and channeling impulse.

Some behavioral scientists divide emotions into positive and negative as if negative emotions were unhealthy (this point of view was emphasized in Model C). This tendency is an oversimplification. As with pus, fever, and cough, the negative emotions of sadness, fear, and anger are also crucial to healthy self-preservation. On the one hand, positive emotions like joy, love, interest, and excitement are associated with subjective contentment; on the other hand, although negative emotions interfere with contentment, their expression can be equally healthy.

Advances in Studying Emotional Intelligence. Over the last 15 years,

three critical empirical steps have been taken in our understanding of the relationship of socioemotional intelligence to positive mental health.

The first step is that both fMRI studies and neurophysiologic experimentation have led to advances in our understanding of integrating the prefrontal cortex with the limbic system, especially with the amygdala and its connections. As noted in the previous model, these research advances have brought us closer to understanding emotions as neurophysiologic phenomena rather than as Platonic abstractions. The prefrontal cortex is the region of the brain responsible for working memory, and the frontal lobes, through their connections to the amygdala, hippocampus, and other limbic structures, encode emotional learning in a manner quite distinct from both conventional conditioning and declarative memory.

The second step has been the slow but steady progress in conceptualizing and measuring "emotional intelligence." Over the last decade, measures of emotional intelligence have evolved rapidly.

The third advance is the use of videotape to chart emotional interaction. Videos of sustained family interactions reveal that an essential aspect of healthy infant development, of adolescent development, and marital harmony is how partners or parents respond to emotion in others. To ignore, to punish, and to be intimidated or contemptuous of how another feels spells disaster. Children of emotionally attuned parents are better at handling their own emotions and are more effective at soothing themselves when upset. Such children even manifest lower levels of stress hormones and other physiologic indicators of emotional arousal.

There are now many exercises in handling relationships that help couples, business executives, and diplomats to become more skilled at conflict resolution and negotiation. In the last decade, there has also been an increasing effort to teach schoolchildren core emotional and social competencies, sometimes called "emotional literacy." The relevance of these advances in psychology to psychiatry includes teaching emotion recognition and differentiation in eating disorders and teaching anger modulation and finding creative solutions to social predicaments for behavior disorders.

Model E: Mental Health as Subjective Well-Being

Positive mental health does not just involve being a joy to others; one must

also experience subjective well-being. Long before humankind considered definitions of mental health, they pondered criteria for subjective happiness. For example, objective social support accomplishes little if, subjectively, the individual cannot feel loved. Thus, the capacity for subjective well-being becomes a vital model of mental health.

Subjective well-being is never categorical. Healthy blood pressure is the objective absence of hypotension and hypertension, but happiness is less neutral. Subjective well-being is not just the absence of misery, but the presence of positive contentment. Nevertheless, if happiness is an inescapable dimension of mental health, happiness is often regarded with ambivalence. If through the centuries, philosophers have sometimes regarded happiness as the highest good, psychologists and psychiatrists have tended to ignore it.

Subjective happiness can have maladaptive as well as adaptive facets. The search for happiness can appear selfish, narcissistic, superficial, and banal. Pleasures can come quickly and soon be gone. Happiness is often based on illusion or dissociative states. Illusory happiness is seen in the character structure associated with bipolar and dissociative disorders. Maladaptive happiness can bring temporary bliss but has no sticking power. In the Study of Adult Development, scaled measures of "happiness" had little predictive power and, often, insignificant association with other subjective and objective measures of contentment. It is because of such ambiguity of meaning that, throughout this section, the term *subjective well-being* will be substituted for happiness.

Empirical Evidence. The mental health issues involved in subjective well-being are complicated and clouded by historical relativism, value judgment, and illusion. Europeans have always been skeptical of the American concern with happiness. Only in the last decade have investigators pointed out that a primary function of positive emotional states and optimism is that they facilitate self-care. Subjective well-being makes available personal resources that can be directed toward innovation and creativity in thought and action. Thus, subjective well-being, like optimism, becomes an antidote to learned helplessness. Again, controlling for income, education, weight, smoking, drinking, and disease, happy people are only half as likely to die at an early age or become disabled as unhappy people.

A distinction can be made between *pleasure* and *gratification.* Pleasure is

in the moment, is closely allied with happiness, and involves the satisfaction of impulse and biologic needs. Pleasure is highly susceptible to habituation and satiety. If pleasure involves satisfaction of the senses and emotions, gratification involves joy, purpose, and the satisfaction of "being the best you can be" and meeting aesthetic and spiritual needs.

Subjective (unhappy) distress can be healthy. As ethologically minded investigators have long pointed out, subjective negative affects (e.g., fear, anger, and sadness) can be healthy reminders to seek environmental safety and not to wallow in subjective well-being. If positive emotions facilitate optimism and contentment, fear is the first protection against the external threat; sadness protests against loss and summons help and anger signals trespass.

Clarifying Subjective Well-Being. Since the 1970s, investigators have made a serious effort to attend to definitional and causal parameters of subjective well-being and address critical questions. One such question is: Is subjective well-being more a function of environmental good fortune or a function of an inborn, genetically based temperament? Put differently, does subjective well-being reflect trait or state? If subjective well-being reflects a safe environment and the absence of stress, it should fluctuate over time, and those individuals who are happy in one domain or time in their lives might not be happy in another.

A second question, but one related to the first, is what is a cause and what is an effect. Are happy people more likely to achieve enjoyable jobs and good marriages, or do conjugal stability and career contentment lead to subjective well-being? Or are such positive associations the result of still a third factor? For example, the absence of a genetic tendency for alcoholism, for major depression, for trait neuroticism, and even for the presence of a chronic wish to give socially desirable answers (impression management) might facilitate both subjective well-being and reports of a good marriage and career contentment.

As with physiologic homeostasis, evolution has prepared humans to make subjective adjustments to environmental conditions. Thus, one can adapt to good and bad events to not remain in a state of either elation or despair. However, humans have a more challenging time adjusting to their genes. Studies of adopted-away twins have demonstrated that half of the variance in

subjective well-being is due to heritability. The subjective well-being of monozygotic twins raised apart is more similar than the subjective well-being of heterozygous twins raised together. Among the heritable factors making a significant contribution to high subjective well-being are low trait neuroticism, high trait extraversion, absence of alcoholism, and absence of major depression. In contrast to tested intelligence, when heritable variables are controlled, subjective well-being is not affected by environmental factors like income, parental social class, age, and education.

If subjective well-being were mainly due to the meeting of basic needs, then there should be a relatively low correlation between subjective well-being in work and subjective well-being in recreational settings or subjective well-being in social versus subjective well-being in solitary settings. Because women experience more objective clinical depression than men, the fact that gender is not a determining factor in subjective well-being is interesting. One explanation is that women appear to report both positive and negative affects more vividly than men. In one study, gender accounted for only 1 percent of the variance in happiness, but 13 percent of the variance in the intensity of reported emotional experiences.

Other Sources of Well-Being. In some instances, the environment *can* be critical to subjective well-being. Young widows remain subjectively depressed for years. Even though their poverty has been endured for centuries, respondents in impoverished nations, like India and Nigeria, report lower subjective well-being than other, more prosperous nations. The loss of a child never stops aching. Although achieving concrete goals like money and fame does not lead to a sustained increase in subjective well-being, social comparison, such as seeing a neighbor become more prosperous than you, can exert a negative effect on subjective well-being.

The maintenance of self-efficacy, agency, and autonomy make additional environmental contributions to subjective well-being. For example, elders will use their discretionary income to live independently, even though this means living alone rather than with relatives. Subjective well-being is usually higher in democracies than in dictatorships. Assuming responsibility for favorable or unfavorable outcomes (internalization) is another major factor leading to subjective well-being. Placing the blame elsewhere (externalization) significantly reduces subjective well-being. In other words,

the mental mechanisms of paranoia and projection make people feel worse rather than better.

Refined methods of measurement of subjective states of mind have included the Positive and Negative Affect Scale (PANAS), which assesses both positive and negative affect, each with 10 affect items. The *Satisfaction with Life Scale* represents the most recent evolution of a general life satisfaction scale. Most recently, the widely validated Short Form 36 (SF-36) has allowed clinicians to assess the subjective cost/benefits of clinical interventions. Because short-lived environmental variables can distort subjective well-being, a consensus is emerging that naturalistic experience-sampling methods are the most valid way to assess subjective well-being. With such sampling methods, research subjects are contacted at random times during the day for several days or weeks and asked to assess their subjective well-being. Finally, to tease verbal self-report from actual subjective experience, physiologic measures of stress (e.g., measuring galvanic skin response and salivary cortisol and filming facial expression by concealed cameras) have also proven useful.

Model F: Mental Health as Resilience

There are three broad classes of coping mechanisms that humans use to overcome stressful situations. First, there is how an individual elicits help from appropriate others: Namely *consciously seeking social support.* Second, there are *conscious cognitive strategies* that individuals intentionally use to master stress. Third, there are *adaptive involuntary coping mechanisms* (often called "defense mechanisms") that distort our perception of internal and external reality to reduce subjective distress, anxiety, and depression.

Involuntary Coping Mechanisms. Involuntary coping mechanisms reduce conflict and cognitive dissonance during sudden *changes* in internal and external reality. If such changes, in reality, are not "distorted" and "denied," they can result in disabling anxiety or depression, or both. Such homeostatic mental "defenses" shield us from sudden changes in the four lodestars of conflict: impulse (affect and emotion), reality, people (relationships), and social learning (conscience). First, such involuntary mental mechanisms can restore psychological homeostasis by ignoring or

deflecting sudden increases in the lodestar of impulse—affect and emotion. Psychoanalysts call this lodestar "id," religious fundamentalists call it "sin," cognitive psychologists call it "hot cognition," and neuroanatomists point to the hypothalamic and limbic regions of the brain.

Second, such involuntary mental mechanisms can provide a mental time-out to adjust to sudden changes in *reality* and self-image, which cannot be immediately integrated. Individuals who initially responded to the television images of the sudden destruction of New York City's World Trade Center as if it were a movie provide a vivid example of the denial of an external reality that was changing too fast for voluntary adaptation. Sudden good news—the instant transition from student to a physician or winning the lottery—can evoke involuntary mental mechanisms as often as can an unexpected accident or a diagnosis of leukemia.

Third, involuntary mental mechanisms can mitigate sudden unresolvable conflict with important *people,* living or dead. People become a lodestar of conflict when one cannot live with them and cannot live without them. Death is such an example; another is an unexpected proposal of marriage. Internal representations of essential people may continue to cause conflict for decades after they are dead yet continue to evoke an involuntary mental response.

Finally, the fourth source of conflict or anxious depression is social learning or conscience. Psychoanalysts call it "superego," anthropologists call it "taboos," behaviorists call it "conditioning," and neuroanatomists point to the associative cortex and the amygdala. This lodestar is not just the result of admonitions from our parents that we absorb before age 5 years, but it is formed by our whole identification, with culture, and sometimes by irreversible learning resulting from overwhelming trauma.

Healthy Involuntary Mental Mechanisms. Longitudinal studies from both Berkeley's Institute of Human Development and Harvard's Study of Adult Development have illustrated the importance of the mature defenses to mental health.

HUMOR. Humor makes life easier. With humor, one sees all, feels much, but does not act. Humor permits the discharge of emotion without individual discomfort and unpleasant effects upon others. Mature humor allows individuals to look directly at what is painful, whereas dissociation and

slapstick distract them. However, like the other mature defenses, humor requires the same delicacy as building a house of cards—timing is everything.

ALTRUISM. When used to master conflict, altruism involves an individual getting pleasure from giving to others what the individual would have liked to receive. For example, using reaction formation, a former alcohol abuser works to ban alcohol in his town and annoys his social drinking friends. Using altruism, the same former alcoholic serves as an Alcoholics Anonymous sponsor to a new member—achieving a transformative process that may be lifesaving to both the giver and receiver. Many acts of altruism involve a free will, but others involuntarily soothe unmet needs.

SUBLIMATION. The sign of a successful sublimation is neither careful cost accounting nor shrewd compromise, but rather psychic alchemy. By analogy, sublimation permits the oyster to transform an irritating grain of sand into a pearl. In writing his Ninth Symphony, the deaf, angry, and lonely Beethoven transformed his pain into triumph by putting Schiller's "Ode to Joy" to music.

SUPPRESSION. Suppression is a defense that modulates emotional conflict or internal/external stressors through stoicism. Suppression minimizes and postpones but does not ignore gratification. Empirically, this is the defense most highly associated with other facets of mental health. Effective suppression is analogous to a well-trimmed sail; every restriction is precisely calculated to exploit, not hide, the winds of passion. Evidence that suppression is not merely a conscious "cognitive strategy" is provided by the fact that jails would empty if delinquents could learn just to say "No."

ANTICIPATION. If suppression reflects the capacity to keep current impulse in mind and control it, anticipation is the capacity to keep the affective response to an unbearable future event in mind in manageable doses. The defense of anticipation reflects the capacity to perceive future danger affectively and cognitively and by this means to master conflict in small steps. Examples are that moderate amounts of anxiety before surgery promote postsurgical adaptation and that anticipatory mourning facilitates the adaptation of parents

of children with leukemia.

Psychiatry needs to understand how best to facilitate the transmutation of less-adaptive defenses into more-adaptive defenses. One suggestion has been first to increase social supports and interpersonal safety and second, to facilitate the intactness of the central nervous system (e.g., rest, nutrition, and sobriety). The newer forms of integrative psychotherapies using videotape can also catalyze such change by allowing patients actually to see their involuntary coping style.

▲ 34.3 Contributions from the Sociocultural Sciences

SOCIOBIOLOGY AND ETHOLOGY

Sociobiology

The term *sociobiology* was coined in 1975 by Edward Osborne Wilson, an American biologist whose book, called *Sociobiology*, emphasized the role of evolution in shaping behavior. Sociobiology is the study of human behavior based on the transmission and modification of genetically influenced behavioral traits. It explores the ultimate question of *why* specific behaviors or other phenotypes came to be.

Evolution. Evolution is described as any change in the genetic makeup of a population. It is the foundational paradigm from which all of biology arises. It unites ethology, population biology, ecology, anthropology, game theory, and genetics. Charles Darwin (1809–1882) posited that natural selection operates via differential reproduction, in a competitive environment, whereby specific individuals are more successful than others. Given that differences among individuals are at least somewhat heritable, any comparative advantage will result in a gradual redistribution of traits in succeeding generations, such that favored characteristics will be represented in a more significant proportion over time. In Darwin's terminology, *fitness* meant reproductive success.

COMPETITION. Animals vie with one another for resources and territory, the area that is defended for the exclusive use of the animal, and that ensures access to food and reproduction. The ability of one animal to defend a disputed territory or resource is called *resource holding potential*, and the more significant this potential, the more successful the animal.

AGGRESSION. Aggression serves both to increase territory and to eliminate

competitors. Defeated animals can emigrate, disperse, or remain in the social group as subordinate animals. A dominance hierarchy in which animals are associated with subtle but well-defined ways is part of every social pattern.

REPRODUCTION. Because behavior is influenced by heredity, those behaviors that promote reproduction and survival are among the most important. Men tend to have a higher variance in reproductive success than do women, thus inclining men to be competitive with other men. Male–male competition can take various forms; for example, sperm can be thought of as competing for access to the ovum. Competition among women, although genuine, typically involves social undermining rather than overt violence. Sexual dimorphism, or different behavioral patterns for males and females, evolves to ensure the maintenance of resources and reproduction.

ALTRUISM. *Sociobiologists define altruism* as behavior that reduces the personal reproductive success of the initiator while increasing that of the recipient. According to the traditional Darwinian theory, altruism should not occur in nature because, by definition, selection acts against any trait whose effect is to decrease its representation in future generations. However, an array of altruistic behaviors occurs among free-living mammals as well as humans. In a sense, altruism is selfishness at the level of the gene rather than at the level of the individual animal. A classic case of altruism is the female worker classes of individual wasps, bees, and ants. These workers are sterile and do not reproduce, but labor altruistically for the reproductive success of the queen.

Another possible mechanism for the evolution of altruism is group selection. If groups containing altruists are more successful than those composed entirely of selfish members, the altruistic groups succeed at the expense of the selfish ones, and altruism evolves. However, within each group, altruists are at a severe disadvantage relative to selfish members; however well the group as a whole does.

IMPLICATIONS FOR PSYCHIATRY. Evolutionary theory provides possible explanations for some disorders. Some may be manifestations of adaptive strategies. For example, cases of anorexia nervosa may be partially understood as a strategy ultimately caused to delay mate selection,

reproduction, and maturation in situations where males are perceived as scarce. Persons who take risks may do so to obtain resources and gain social influence. An erotomanic delusion in a single postmenopausal woman may represent an attempt to compensate for the painful recognition of reproductive failure.

Studies of Identical Twins Reared Apart: Nature versus Nurture. Studies in sociobiology have stimulated one of the oldest debates in psychology. Does human behavior owe more to nature or to nurture? Curiously, humans readily accept that genes determine most of the behaviors of nonhumans, but tend to attribute their behavior almost exclusively to nurture. However, recent data unequivocally identify our genetic endowment as equally important, if not more important, factors.

The best "experiments of nature" permitting an assessment of the relative influences of nature and nurture are cases of genetically identical twins separated in infancy and raised in different social environments. If nurture is the most important determinant of behavior, they should behave differently. On the other hand, if nature dominates, each will closely resemble the other, despite their having never met. Several hundred pairs of twins separated in infancy, raised in separate environments, and then reunited in adulthood have been rigorously analyzed. Nature has emerged as a critical determinant of human behavior.

Laura R and Catherine S were reunited at the age of 35. They were identical twins that had been adopted by two different families in Chicago. Growing up, neither twin was aware of the other's existence. As children, each twin had a cat named Lucy, and both habitually cracked their knuckles. Laura and Catherine each began to have migraine headaches beginning at the age of 14. Both were elected valedictorian of their high school classes and majored in journalism in college. Each sister had married a man named John and had given birth to a daughter in wedlock. Both of their marriages fell apart within 2 years. Each twin maintained a successful rose garden and took morning spin classes at their local fitness center. Upon meeting, each twin discovered that the other had also named her daughter Erin and owned a German Sheppard

named Rufus. They had similar voices, hand gestures, and mannerisms.

Jack Y and Oskar S, identical twins born in Trinidad in 1933 and separated in infancy by their parents' divorce, were first reunited at age 46. His Catholic mother and grandmother raised Oskar in Nazi-occupied Sudetenland, Czechoslovakia. Jack was raised by his Orthodox Jewish father in Trinidad and spent time on an Israeli kibbutz. Each wore aviator glasses and a blue sport shirt with shoulder plackets, had a trim mustache, liked sweet liqueurs, stored rubber bands on his wrists, read books and magazines from back to front, dipped buttered toast in his coffee, flushed the toilet before and after using it, enjoyed sneezing loudly in crowded elevators to frighten other passengers, and routinely fell asleep at night while watching television. Each was impatient, squeamish about germs, and gregarious.

Bessie and Jessie, identical twins, separated at 8 months of age after their mother's death, were first reunited at age 18. Each had had a bout of tuberculosis, and they had similar voices, energy levels, administrative talents, and decision-making styles. Each had had her hair cut short in early adolescence. Jessie had a college-level education, whereas Bessie had had only 4 years of formal education, yet Bessie scored 156 on intelligence quotient testing, and Jessie scored 153. Each read avidly, which may have compensated for Bessie's sparse education; she created an environment compatible with her inherited potential.

Neuropsychological Testing Results. The dominant influence of genetics on behavior has been documented in several sets of identical twins on the Minnesota Multiphasic Personality Inventory (MMPI). Twins reared apart generally showed the same degree of genetic influence across the different scales as twins reared together. For example, despite being reared on different continents, in countries with different political systems and different languages, two particularly fascinating identical twin pairs generated scores more closely correlated across 13 MMPI scales than the already tight correlation noted among all tested identical twin pairs, most of whom had shared similar rearing.

Reared-apart twin studies report a high correlation (r = 0.75) for IQ similarity. In contrast, the IQ correlation for reared-apart nonidentical twin

siblings is 0.38, and for sibling pairs in general, it is in the 0.45 to 0.50 range. Strikingly, IQ similarities are not influenced by similarities in access to dictionaries, telescopes, and original artwork, in parental education and socioeconomic status, or characteristic parenting practices. These data overall suggest that tested intelligence is determined roughly two-thirds by genes and one-third by the environment.

Studies of reared-apart identical twins reveal a genetic influence on alcohol use, substance abuse, childhood antisocial behavior, adult antisocial behavior, risk aversion, and visuomotor skills, as well as on psychophysiological reactions to music, voices, sudden noises, and other stimulation, as revealed by brain wave patterns and skin conductance tests. Moreover, reared-apart identical twins show that genetic influence is pervasive, affecting virtually every measured behavioral trait. For example, many individual preferences previously assumed to be due to nurture (e.g., religious interests, social attitudes, vocational interests, job satisfaction, and work values) are strongly determined by nature.

A selected glossary of some terms used in this section and other ethologic terms is given in Table 34-8.



Table 34-8
Selected Glossary of Ethologic Terms

Action-specific energy	Energy associated with the innate releasing mechanism and specific to a particular behavior pattern, which builds up if the releasing stimulus is not present to activate the behavior pattern; conversely, it is depleted by repetition.
Aggression	Intraspecific conflict manifested by physical attack or social signaling.
Appetitive behavior	Phase of behavior involving the active seeking of sign stimuli and thought to be driven by action-specific energy accumulating through inactivity of the specific behavior pattern.

Consummatory response	Phase of behavior whereby the energy driving the appetitive phase is released. Involves the perception of sign stimuli, the activation of the innate releasing mechanism (IRM), and the performance of the fixed action pattern (FAP).
Critical period	The time during which imprinting must occur, usually shortly after birth or early in life. Also known as “sensitive period.”
Displacement activity	A set of behavior patterns occurring alongside an unrelated set of behavior patterns. Originally, irrelevant movements from one behavioral system occurring in the presence of powerful but thwarted drive from another behavior system.
Ethology	The biologic study of behavior. From the Greek ethos, meaning custom, usage, manner, habit. The modern usage is attributed to Oskar Heinroth, Konrad Lorenz’s teacher.
Fixed action pattern (FAP)	A genetically determined behavior pattern that is initiated by stimuli particular to the pattern and that consists of species-specific, stereotyped movements.
Imprinting	A specialized form of learning occurring early in life and often influencing behavior later in life. The exposure to the stimulus situation must occur during a particular period, the critical period, and the exposure can be of short duration and without obvious reward. The learning is particularly resistant to change.
Innate	Genetically determined behavior patterns; in theory not influenced by experience.
Innate releasing	Sensory mechanism selectively responsive to

mechanism (IRM)	specific external stimuli and responsible for triggering the stereotyped motor response.
Instinct	A developmental process resulting in species-typical behavior.
Redirection activity	The venting of one drive from two or more incompatible, but simultaneously activated, drives on some third animal or object.
Ritualization	Process of a behavior pattern being incorporated through evolution into a primary signaling function, frequently with exaggeration and embellishment of some of the movements.

Courtesy of William T. McKinney, Jr., M.D.

Ethology

The systematic study of animal behavior is known as ethology. In 1973, the Nobel Prize in psychiatry and medicine was awarded to three ethologists, Karl von Frisch, Konrad Lorenz, and Nikolaas Tinbergen. Those awards highlighted the particular relevance of ethology, not only for medicine but also for psychiatry.

Konrad Lorenz. Born in Austria, Konrad Lorenz (1903–1989) is best known for his studies of imprinting. Imprinting implies that, during a specific short period of development, a young animal is highly sensitive to a specific stimulus that then, but not at other times, provokes a specific behavior pattern. Lorenz described newly hatched goslings programmed to follow a moving object and thereby become imprinted rapidly to follow it and, possibly, similar objects. Typically, the mother is the first moving object the gosling sees, but should it see something else first, the gosling follows it. For instance, a gosling imprinted by Lorenz followed him and refused to follow a goose (Fig. 34-17). Imprinting is an essential concept for psychiatrists to understand to link early developmental experiences with later behaviors.

Lorenz also studied the behaviors that function as sign stimuli—that is, social releasers—in communications between individual animals of the same species. Many signals have the character of fixed motor patterns that appear

automatically; the reaction of other members of the species to the signals is equally automatic.

Lorenz is also well known for his study of aggression. He wrote about the practical function of aggression, such as territorial defense by fish and birds. Aggression among members of the same species is common, but Lorenz pointed out that in normal conditions, it seldom leads to killing or even to serious injury. Although animals attack one another, a certain balance appears between tendencies to fight and flight, with the tendency to fight to be strongest in the center of the territory and the tendency to flight strongest at a distance from the center.



FIGURE 34-17
In a famous experiment, Konrad Lorenz demonstrated that goslings responded to him as if he were the natural mother. (Reprinted from Hess EH. Imprinting, an effect of early experience, imprinting determines later social behavior in animals. *Science*. 1959;130(3368):133–141, with permission.)

In many works, Lorenz tried to draw conclusions from his ethologic studies of animals that could also be applied to human problems. The postulation of a primary need for aggression in humans, cultivated by the

pressure for selecting the best territory, is a primary example. Such a need may have served a practical purpose historically, when humans lived in small groups that had to defend themselves from other groups. Competition with neighboring groups could become an important factor in selection. Lorenz pointed out, however, that this once useful trait may be outliving its survival value, now that we live in an era with weapons that can kill not just individuals but entire populations.

Nikolaas Tinbergen. Born in the Netherlands, Nikolaas Tinbergen (1907–1988), a British zoologist, conducted a series of experiments to analyze various aspects of animal behavior. He was also successful in quantifying behavior and measuring the power or strength of various stimuli in eliciting specific behavior. Tinbergen described displacement activities, which have been studied mainly in birds. For example, in a conflict situation, when the needs for fight and flight are of roughly equal strength, birds sometimes do neither. Instead, they display behavior that appears to be irrelevant to the situation (e.g., a herring gull defending its territory can start to pick grass). Displacement activities of this kind vary according to the situation and the species concerned. Humans can engage in displacement activities when under stress.

Lorenz and Tinbergen described innate releasing mechanisms, animals' responses triggered by releasers, which are specific environmental stimuli. Releasers (including shapes, colors, and sounds) evoke sexual, aggressive, or other responses. For example, big eyes in human infants evoke more caretaking behavior than do small eyes.

In his later work, Tinbergen, along with his wife, studied early childhood autistic disorder. They began by observing the behavior of autistic and typical children when they meet strangers, analogous to the techniques used in observing animal behavior. In particular, they observed in animals the conflict between fear and the need for contact and noted that the conflict could lead to behavior similar to that of autistic children. They hypothesized that, in certain predisposed children, fear can greatly predominate and can also be provoked by stimuli that generally have a positive social value for most children. This innovative approach to studying infantile autistic disorder opened up new avenues of inquiry. Although their conclusions about preventive measures and treatment must be considered tentative, their method

shows how ethology and clinical psychiatry can relate to each other.

Karl von Frisch. Born in Austria, Karl von Frisch (1886–1982) conducted studies on changes of color in fish and demonstrated that fish could learn to distinguish among several colors and that their sense of color was somewhat congruent with that of humans. He later went on to study the color vision and behavior of bees and is most widely known for his analysis of how bees communicate with one another—that is, their language, or what is known as their dances. His description of the exceedingly complex behavior of bees prompted an investigation of communication systems in other animal species, including humans.

Characteristics of Human Communication. Communication is traditionally seen as an interaction in which a minimum of two participants—a sender and a receiver—share the same goal: exchanging accurate information. Although shared interest in accurate communication remains valid in some domains of animal signaling—notably such well-documented cases as the "dance of the bees," whereby foragers inform other workers about the location of food sources—a more selfish and, in the case of social interaction, more accurate model of animal communication has mostly replaced this concept.

Sociobiologic analyses of communication emphasize that because individuals are genetically distinct, their evolutionary interests are similarly distinct, although admittedly with significant fitness overlap, especially among kin, reciprocators, parents and offspring, and mated pairs. Senders are motivated to convey information that induces the receivers to behave in a manner that enhances the senders' fitness. Receivers, similarly, are interested in responding to communication only insofar as such a response enhances their fitness. One crucial way to enhance reliability is to make the signal costly; for example, an animal could honestly indicate its physical fitness, freedom from parasites and other pathogens, and possibly its genetic quality as well by growing elaborate and metabolically expensive secondary sexual characteristics such as the oversized tail of a peacock. Human beings, similarly, can signal their wealth by conspicuous consumption. This approach, known as the *handicap principle,* suggests that effective communication may require that the signaler engages in exceptionally costly

behavior to ensure success.

Subhuman Primate Development

An area of animal research that has relevance to human behavior and psychopathology is the longitudinal study of nonhuman primates. Monkeys have been observed from birth to maturity, not only in their natural habitats and laboratory facsimiles but also in laboratory settings that involve various degrees of social deprivation early in life. Social deprivation has been produced through two predominant conditions: social isolation and separation. Socially isolated monkeys are raised in varying degrees of isolation and are not permitted to develop normal attachment bonds. Monkeys separated from their primary caretakers, thereby experience disruption of an already developed bond. Social isolation techniques illustrate the effects of an infant's early social environment on subsequent development (Figs. 34-18 and 34-19), and separation techniques illustrate the effects of loss of a significant attachment figure. The name most associated with isolation and separation studies is Harry Harlow. A summary of Harlow's work is presented in Table 34-9.

In a series of experiments, Harlow separated rhesus monkeys from their mothers during their first weeks. During this time, the monkey infant depends on its mother for nourishment and protection, and physical warmth and emotional security—*contact comfort,* as Harlow first termed it in 1958. Harlow substituted a surrogate mother made from wire or cloth for the real mother. The infants preferred the cloth-covered surrogate mother, which provided contact comfort, to the wire-covered surrogate, which provided food but no contact comfort (Fig. 34-20).

Treatment of Abnormal Behavior. Stephen Suomi demonstrated that monkey isolates could be rehabilitated if exposed to monkeys that promote physical contact without threatening the isolates with aggression or overly complex play interactions. These monkeys were called therapist monkeys. To fill such a therapeutic role, Suomi chose normal young monkeys that would play gently with the isolates and approach and cling to them. Within 2 weeks, the isolates were reciprocating the social contact, and their incidence of abnormal self-directed behaviors began to decline significantly. By the end of

the 6-month therapy period, the isolates were actively initiating play bouts with both the therapists and each other, and most of their self-directed behaviors had disappeared. The isolates were observed closely for the next 2 years, and their improved behavioral repertoires did not regress over time. The results of this and subsequent monkey-therapist studies underscored the potential reversibility of early cognitive and social deficits at the human level. The studies also served as a model for developing treatments for socially retarded and withdrawn children.



FIGURE 34-18
Social isolate after removal of isolation screen.

Several investigators have argued that social separation manipulations with nonhuman primates provide a compelling basis for animal models of

depression and anxiety. Some monkeys react to separations with behavioral and physiologic symptoms similar to those seen in depressed human patients; both ECT and tricyclic drugs effectively reverse the symptoms in monkeys. Not all separations produce depressive reactions in monkeys, just as separation does not always precipitate depression in humans, young and old.

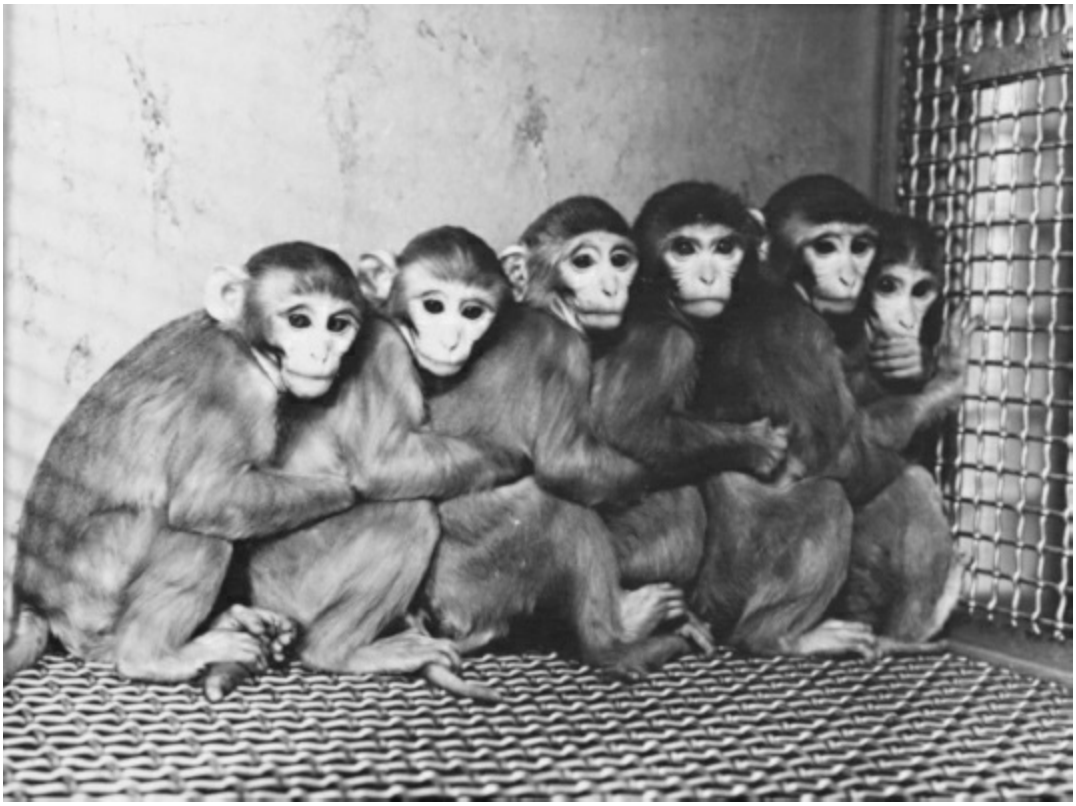


FIGURE 34-19
Choo-choo phenomenon in peer-only-reared infant rhesus monkeys.

Table 34-9
Social Deprivation in Nonhuman Primates

Type of Social Deprivation	Effects
Total isolation (not allowed to develop	Self-orality, self-clasping, very fearful when placed with peers, unable to copulate. If impregnated, female is unable to nurture young (motherless mothers). If isolation goes beyond 6 mo, no

caretaker or peer bond)	recovery is possible.
Mother-only reared	Fails to leave mother and explore. Terrified when finally exposed to peers. Unable to play or to copulate.
Peer-only reared	Engages in self-orality, grasps others in clinging manner, easily frightened, reluctant to explore, timid as adult, play is minimal.
Partial isolation (can see, hear, and smell other monkeys)	Stares vacantly into space, engages in self-mutilation, stereotyped behavior patterns.
Separation (taken from caretaker after bond has developed)	Initial protest stage changing to despair 48 hours after separation; refuses to play. Rapid reattachment when returned to mother.

Adapted from work of Harry Harlow, M.D.

Individual Differences. Recent research has revealed that some rhesus monkey infants consistently display fearfulness and anxiety in situations where similarly reared peers to show normal exploratory behavior and play. These situations generally involve exposure to a novel object or situation. Once the object or situation has become familiar, any behavioral differences between the anxiety-prone, timid infants and their outgoing peers disappear, but the individual differences appear to be stable during development. Infant monkeys at 3 to 6 months of age that are at high risk for fearful or anxious reactions tend to remain at high risk for such reactions, at least until adolescence.

A long-term follow-up study of these monkeys has revealed some behavioral differences between fearful and nonfearful female monkeys when they become adults and have their first infants. Usually, fearful female monkeys who grow up in socially benign and stable environments become fine mothers, but fearful female monkeys who have reacted with depression to frequent social separations during childhood are at high risk for maternal dysfunction; more than 80 percent of these mothers either neglect or abuse their first offspring. However, nonfearful female monkeys that encounter the same number of social separations but do not react to any of these separations with depression turn out to be good mothers.

Experimental Disorders

Stress Syndromes. Several researchers, including Ivan Petrovich Pavlov in Russia and W. Horsley Gantt and Howard Scott Liddell in the United States, studied the effects of stressful environments on animals, such as dogs and sheep. Pavlov produced a phenomenon in dogs, which he labeled *experimental neurosis,* by using a conditioning technique that led to symptoms of extreme and persistent agitation. The technique involved teaching dogs to discriminate between a circle and an ellipse and then progressively diminishing the difference between them. Gantt used the term *behavior disorders* to describe the reactions he elicited from dogs forced into similar conflictual learning situations. Liddell described the stress response he obtained in sheep, goats, and dogs as *experimental neurasthenia,* which was produced in some cases by merely doubling the number of daily test trials in an unscheduled manner.

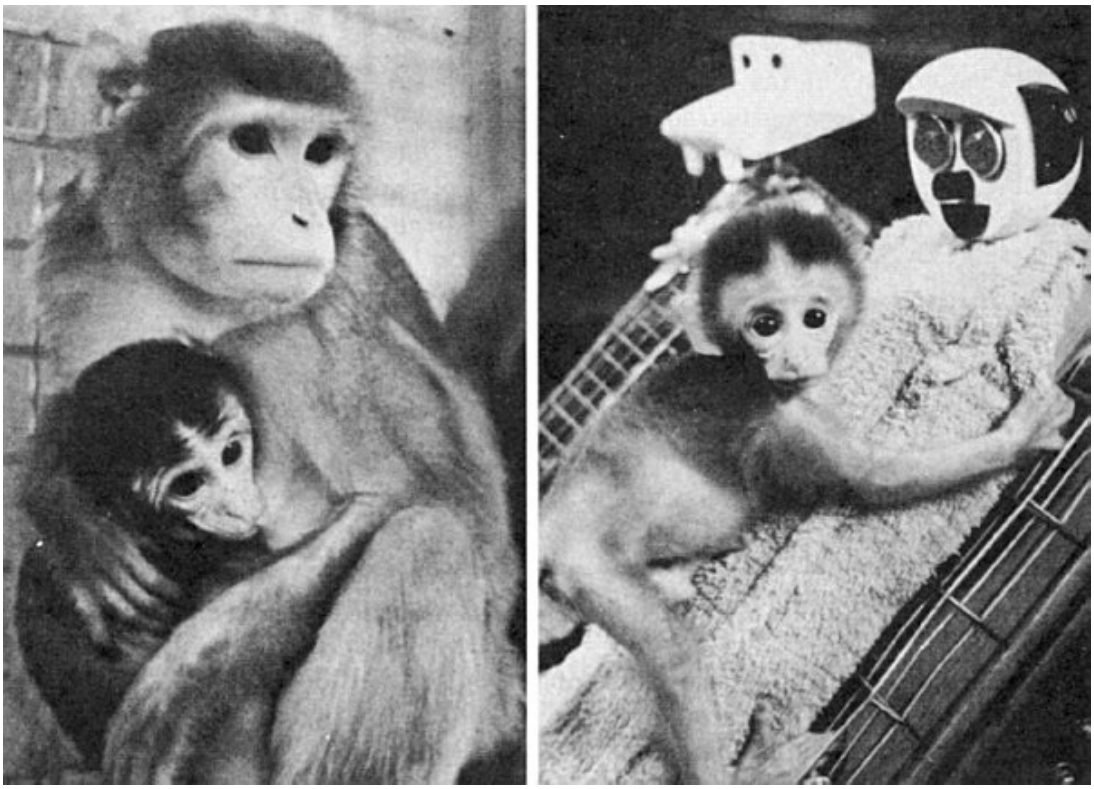


FIGURE 34-20
Monkey infant with mother (**left**) and with cloth-covered surrogate (**right**).

Learned Helplessness. The learned helplessness model of depression, developed by Martin Seligman, is an excellent example of an experimental disorder. Dogs were exposed to electric shocks from which they could not escape. The dogs eventually gave up and did not attempt to escape new shocks. The apparent giving up generalized to other situations, and eventually, the dogs always appeared to be helpless and apathetic. Because the cognitive, motivational, and affective deficits displayed by the dogs resembled symptoms common to human depressive disorders, learned helplessness, although controversial, was proposed as an animal model of human depression. In connection with learned helplessness and the expectation of inescapable punishment, research on subjects has revealed brain release of endogenous opiates, destructive effects on the immune system, and elevation of the pain threshold.

A social application of this concept involves schoolchildren who have learned that they fail in school no matter what they do; they view themselves as helpless losers, and this self-concept causes them to stop trying. Teaching them to persist may reverse the process, with excellent results in self-respect

and school performance.

Unpredictable Stress. Rats subjected to chronic unpredictable stress (crowding, shocks, irregular feeding, and interrupted sleep time) show decreases in movement and exploratory behavior; this finding illustrates the roles of unpredictability and lack of environmental control in producing stress. These behavioral changes can be reversed by antidepressant medication. Animals under experimental stress (Fig. 34-21) become tense, restless, hyperirritable, or inhibited in specific conflict situations.

Dominance. Animals in a dominant position in a hierarchy have certain advantages (e.g., mating and feeding). Being more dominant than peers is associated with elation, and a fall in position in the hierarchy is associated with depression. When persons lose jobs, are replaced in organizations, or otherwise have their dominance or hierarchical status changed, they can experience depression.

Temperament. Temperament mediated by genetics plays a role in behavior. For example, one group of pointer dogs was bred for fearfulness and a lack of friendliness toward persons, and another group was bred for the opposite characteristics. The phobic dogs were extremely timid and fearful and showed decreased exploratory capacity, increased startle response, and cardiac arrhythmias. Benzodiazepines diminished these fearful, anxious responses. Amphetamines and cocaine aggravated the responses of genetically nervous dogs to a greater extent than they did the responses of stable dogs.

Brain Stimulation. Pleasurable sensations have been produced in humans and animals through self-stimulation of specific brain areas, such as the medial forebrain bundle, the septal area, and the lateral hypothalamus. Rats have engaged in repeated self-stimulation (2,000 stimulations per hour) to gain rewards. Catecholamine production increases with self-stimulation of the brain area, and drugs that decrease catecholamines decrease the process. The centers for sexual pleasure and opioid reception are closely related anatomically. Heroin addicts report that the so-called rush after intravenous injection of heroin is akin to an intense sexual orgasm.

Pharmacologic Syndromes. With the emergence of biologic psychiatry, many researchers have used pharmacologic means to produce syndrome analogs in animal subjects. Two classic examples are the reserpine (Serpasil) model of depression and the amphetamine psychosis model of paranoid schizophrenia. In the depression studies, animals given the norepinephrine-depleting drug reserpine exhibited behavioral abnormalities analogous to major depressive disorder in humans. The behavioral abnormalities produced were generally reversed by antidepressant drugs. These studies tended to corroborate the theory that depression in humans is, in part, the result of diminished levels of norepinephrine. Similarly, animals given amphetamines acted in a stereotypical, inappropriately aggressive, and frightened manner that resembled paranoid psychotic symptoms in humans. These models are considered too simplistic in their concepts of cause, but they remain as early paradigms for this type of research.

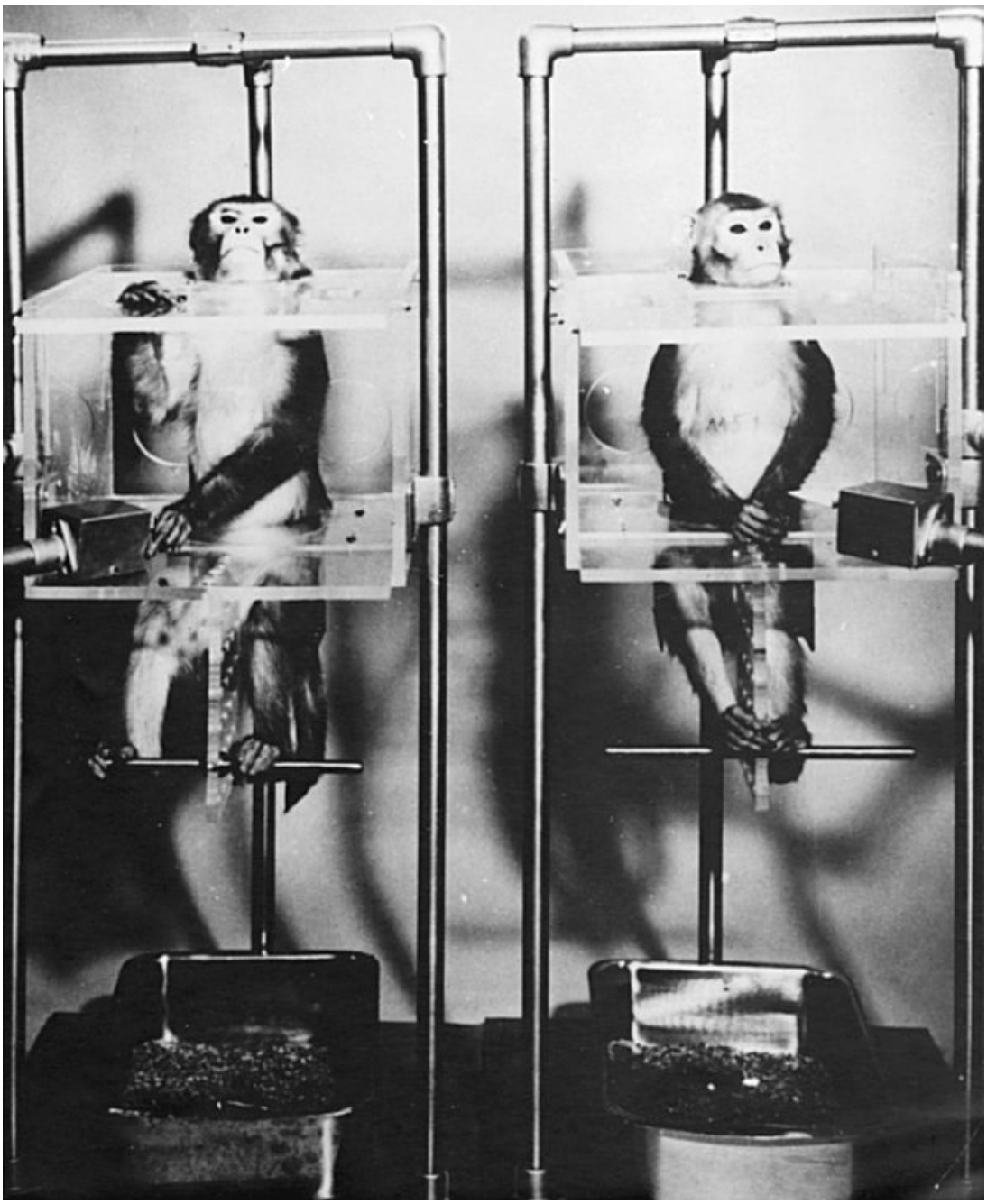


FIGURE 34-21
The monkey on the left, known as the executive monkey, controls whether both will receive an electric shock. The decision-making task produces a state of chronic tension. Note the more relaxed attitude of the monkey on the **right**. (From U.S. Army photographs, with permission.)

Studies have also been done on the effects of catecholamine-depleting drugs on monkeys during separation and reunion periods. These studies showed that catecholamine depletion and social separation could interact in a

highly synergistic fashion and yield depressive symptoms in subjects for whom mere separation or low-dose treatment alone does not suffice to produce depression.

Reserpine has produced a severe depression in humans and, as a result, is rarely used as either an antihypertensive (its original indication) or an antipsychotic. Similarly, amphetamine and its congeners (including cocaine) can induce psychotic behavior in persons who use it in overdose or over long periods.

Sensory Deprivation

The history of sensory deprivation and its potentially deleterious effects evolved from instances of aberrant mental behavior in explorers, shipwrecked sailors, and prisoners in solitary confinement. Toward the end of World War II, startling confessions, induced by the brainwashing prisoners of war, caused a rise of interest in this psychological phenomenon brought about by the deliberate diminution of sensory input.

To test the hypothesis that an essential element in brainwashing is prolonged exposure to sensory isolation, D. O. Hebb and his coworkers brought solitary confinement into the laboratory. They demonstrated that volunteer subjects—under conditions of visual, auditory, and tactile deprivation for periods of up to 7 days—reacted with increased suggestibility. Some subjects also showed characteristic symptoms of the sensory deprivation state: anxiety, tension, inability to concentrate or organize thoughts, increased suggestibility, body illusions, somatic complaints, intense subjective emotional distress, and vivid sensory imagery—usually visual and sometimes reaching the proportions of hallucinations with a delusionary quality.

Psychological Theories. Anticipating psychological explanation, Sigmund Freud wrote: “It is interesting to speculate what could happen to ego function if the excitations or stimuli from the external world were either drastically diminished or repetitive. Would there be an alteration in the unconscious mental processes and an effect upon the conceptualization of time?”

Indeed, under conditions of sensory deprivation, the abrogation of such

ego functions as perceptual contact with reality and logical thinking brings about confusion, irrationality, fantasy formation, hallucinatory activity, and wish-dominated mental reactions. In the sensory-deprivation situation, the subject becomes dependent on the experimenter and must trust the experimenter to satisfy such basic needs as feeding, toileting, and physical safety. A patient undergoing psychoanalysis may be in a kind of sensory deprivation room (e.g., a soundproof room with dim lights and a couch) in which primary-process mental activity is encouraged through free association.

COGNITIVE. Cognitive theories stress that the organism is an information-processing machine, whose purpose is an optimal adaptation to the perceived environment. Without sufficient information, the machine cannot form a cognitive map against which the current experience is matched. Disorganization and maladaptation then result. To monitor their behavior and to attain optimal responsiveness, persons must receive continuous feedback; otherwise, they are forced to project outward idiosyncratic themes that have little relation to reality. This situation is similar to that of many psychotic patients.

Physiologic Theories. The maintenance of optimal conscious awareness and accurate reality testing depends on a necessary state of alertness. This alert state, in turn, depends on a constant stream of changing stimuli from the external world, mediated through the ascending reticular activating system in the brainstem. In the absence or impairment of such a stream, as occurs in sensory deprivation, alertness drops away, direct contact with the outside world diminishes, and impulses from the inner body and the central nervous system may gain prominence. For example, idioretinal phenomena, inner ear noise, and somatic illusions may take on a hallucinatory character.

TRANSCULTURAL PSYCHIATRY

Culture is defined as a set of meanings, norms, beliefs, values, and behavior patterns shared by people. These values include social relationships, language, nonverbal expression of thoughts and emotions, moral and religious beliefs, rituals, technology, and economic beliefs and practices,

among other items. Culture has six essential components: (1) Culture is learned; (2) culture can be passed on from one generation to the next; (3) culture involves a set of meanings in which words, behaviors, events, and symbols have meanings agreed upon by the cultural group; (4) culture acts as a template to shape and orient future behaviors and perspectives within and between generations, and to take account of novel situations encountered by the group; (5) culture exists in a constant state of change; and (6) culture includes patterns of both subjective and objective components of human behavior. Also, culture shapes which and how psychiatric symptoms are expressed; culture influences the meanings that are given to symptoms; and culture affects the interaction between the patient and the health care system, as well as between the patient and the physician and other clinicians with whom the patient and family interact.

Race is a concept, the scientific validity of which is now considered highly questionable, by which human beings are grouped primarily by physiognomy. Its effect on individuals and groups, however, is considerable, due to its reference to physical, biologic, and genetic underpinnings, and because of the intensely emotional meanings and responses, it generates. *Ethnicity* refers to the subjective sense of belonging to a group of people with a common national or regional origin and shared beliefs, values, and practices, including religion. It is part of every person's identity and self-image.

Cultural Formulation

Culture plays a role in mental health and mental illness; therefore, a cultural assessment should be a component of every complete psychiatric assessment. The outline for cultural formulation found in the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-5) is intended to give clinicians a framework for assessing the role of culture in psychiatric illness. Its purposes are (1) to enhance the application of diagnostic criteria in multicultural environments; (2) cultural conceptualizations of distress; (3) psychosocial stressors and cultural features of vulnerability and resilience; (4) to enable the clinician to systematically describe the patient's cultural and social reference groups and their relevance to clinical care; and (5) to identify the effect that cultural differences may have on the relationship between the patient and

family and the treating clinician, as well as how such cultural differences affect the course and the outcome of treatment provided.

The outline for cultural formulation consists of five areas of assessment: (1) cultural identity of the individual; (2) cultural explanations of the individual's illness; (3) cultural factors related to psychosocial environment and levels of functioning; (4) cultural elements of the relationship between the individual and the clinician; and (5) overall cultural assessment for diagnosis and care.

Cultural Identity of the Individual. *Cultural identity* refers to the characteristics shared by a person's cultural group. Identity allows for a self-definition. Factors that comprise an individual's cultural identity include race, ethnicity, country of origin, language use, religious beliefs, socioeconomic status, migration history, the experience of acculturation, and the degree of affiliation with the individual's group of origin. Cultural identity emerges throughout the individual's life and in a social context. It is not a fixed trait of an individual or of the group of which the individual is part. An individual may have several cultural reference groups.

The clinician should encourage the patient to describe the component aspects of their cultural identity. Evaluating the cultural identity of the patient allows identification of potential areas of strength and support that may enhance treatment effectiveness, as well as vulnerabilities that may interfere with the progress of treatment. Eliciting this data permits the identification of unresolved cultural conflicts that may be addressed during treatment. These conflicts can be between the various aspects of the patient's identity and between traditional and mainstream cultural values and behavioral expectations affecting the individual. Knowledge of the patient's cultural identity allows the clinician to avoid misconceptions based on inadequate background information or stereotypes related to race, ethnicity and other aspects of cultural identity. Also, it helps build rapport because the clinician is attempting to understand the individual as a person and not just a representative of the cultural groups that have shaped the patient's identity.

Cultural Explanations of the Individual's Illness. The explanatory model of illness is the patient's understanding of and attempts to explain why they became ill. The explanatory model defines the culturally acceptable

means of expression of the symptoms of the illness or *idioms of distress,* the particular way individuals within a specific cultural group report symptoms, and their behavioral response to them heavily influenced by cultural values. The cultural explanations of illness may also help define the sick role or behavior the patient assumes. The explanatory model of illness includes the patient's beliefs about their prognosis and the treatment options they would consider. The patient's explanatory model may be only vaguely conceptualized or may be quite clearly defined, and it may include several conceptual perspectives that could conflict with one another. Formulation of a collaborative model that is acceptable to both the clinician and the patient is the sought-for endpoint, which would include an agreed-upon set of symptoms to be treated and an outline of treatment procedures to be used.

Difficulties may arise when there are conceptual differences in the explanatory model of illness between clinician, patient, family, and community. Conflicts between the patient's and the clinician's explanatory models may lead to diminished rapport or treatment noncompliance. Conflicts between the patient's and the family's explanatory models of illness may result in a lack of support from the family and family discord. Conflicts between the patient's and the community's explanatory models could lead to social isolation and stigmatization of the patient.

Examples of the more common explanatory models of illness include the *moral model, the religious model,* the *magical* or *supernatural explanatory model*, the *medical model*, and the *psychosocial stress model.* The *moral model* implies that the patient's illness is caused by a moral defect such as selfishness or moral weakness. The *religious model* suggests that the patient is being punished for a religious failing or transgression. The *magical* or *supernatural explanatory model* may involve attributions of sorcery or witchcraft as being the cause of the symptoms. The *medical model* attributes the patient's illness primarily to a biologic etiology. The *psychosocial model* infers that overwhelming psychosocial stressors cause or are primary contributors to the illness.

Culture has both direct and indirect effects on help-seeking behavior. In many cultural groups, an individual and their family may minimize symptoms due to stigma associated with seeking assistance for psychiatric disorders. Culture affects the patient's expectations of treatment, such as whether the clinician should assume an authoritarian, paternalistic,

egalitarian, or nondirective demeanor in the treatment process.

Cultural Factors Related to Psychosocial Environment and Level of Functioning. Understanding the patient's family dynamics and cultural values are integral to assessing the patient's psychosocial environment. The definition of what constitutes a family and the roles of individuals in the family differ across cultures. This includes an understanding of the patient's cultural group and its relationship to the mainstream culture or cultures. It includes the patient's life experience of racial and ethnic discrimination. For immigrants and refugees, it includes the individual's and family's perceptions of the openness of the host society toward people of their country and region of origin, their racial, ethnic, religious, and other attributes. The patient and family may identify strongly or weakly with communal sources of support familiar from their country or region of origin, or they may identify along the same gradient with communal sources of support in the host culture.

Cultural Elements of the Relationship between the Individual and the Clinician. The cultural identity of the clinician and the mental health team has an impact on patient care. The culture of mental health care professionals influences diagnosis and treatment. Clinicians who have an understanding of their own cultural identity may be better prepared to anticipate the cultural dynamics that may arise in interactions with people of diverse cultural backgrounds. Unacknowledged differences between the clinician's and patient's cultural identity can result in assessment and treatment that is unintentionally biased and stressful for all. Clinicians need to examine their assumptions about other cultures to be optimally effective in serving the culturally diverse patient populations that are the norm in most contemporary medical facilities.

Culture influences transference and counter-transference in the clinical relationship between people seeking psychiatric care and their treating clinicians. Transference relationships and dynamics are affected when the patient and clinician have different cultural background characteristics. A perceived social power differential between the patient and clinician could lead to overcompliance, resistance in the exploration of family and social conflict situations, or to the clinician being conceptualized as a cultural role model or stereotype.

Overall Cultural Assessment for Diagnosis and Care. The treatment plan should include the use of culturally appropriate health care and social services. Interventions also may be focused on the family and social levels. In making a psychiatric diagnosis, the clinician should consider principles of cultural relativism and not fall prone to category fallacy. Many psychiatric disorders show cross-cultural variation. Objective evaluation of the multiple possible effects of culture on psychopathology can be a challenging task for the clinician. Diagnostic dilemmas may arise in dealing with patients of diverse cultural backgrounds. Some of these dilemmas may include problems in judging distortion from reality, problems in assessing unfamiliar behaviors and distinguishing pathologic from normal cultural behavior.

Migration, Acculturation, and Acculturative Stress

From the time of the first significant surge of immigration to the United States in the 1870s, and for the next 100 years, the predominant national sentiment toward immigrants, as in most other host countries, was that they should acculturate to the normative behaviors and values of the majority or mainstream culture of the host population. Most immigrants had the same wish to assimilate, to become part of the *melting pot.* This process of acculturative change can be seen as unidirectional, as individuals who identified themselves as part of immigrant, indigenous, and other minority groups both rejected and progressively lost distinctive aspects of their cultural heritage in favor of becoming part of the mainstream majority culture of the host country. In countries that encouraged this outcome of acculturation, people were expected to progress from unacculturated, through the gradient of minimally, moderately, and fully acculturated.

The intensity of acculturative stress experienced by immigrant and other minority groups, and the individuals comprising those groups, has been directly proportional to the openness of the host government and population. The central issue is to what extent are immigrants' and other minority groups' customs, values, and differences from the majority population of the host country accepted, encouraged, and welcomed as an enrichment of the host country instead of being seen as alien and unwelcome. The *acceptance position* encourages the cultural integration of immigrants, whereas the *rejection position* encourages either cultural exclusion or cultural

assimilation.

To assess the outcome of acculturative stress, for groups and their component individuals, two determining factors need to be considered. The first is the extent to which the group and its members value and wish to preserve their cultural uniqueness, including the language, beliefs, values, and social behaviors that define the group. The second factor is the mirror-image issue of the extent to which the group and its members value and wish to increase their contact and involvement with other groups, particularly the majority culture. This conceptual framework leads to four possible outcomes of acculturative stress that are not conceptualized along the unidirectional gradient from unacculturated to wholly acculturated.

The four possible outcomes are *rejection, integration, assimilation,* and *marginalization. Rejection* is characterized by individuals' wishes, both conscious and intuitive, to maintain their cultural integrity, whether by actively resisting the incorporation of the values and social behavior patterns of another cultural group or groups with whom they have regular contact or by disengaging themselves from contact with and the influence of those other cultural groups. Some religious cults are examples of rejection.

Integration, as an outcome of acculturative stress, derives from the wish to both maintain a firm sense of one's cultural heritage and not abandon those values and behavioral characteristics that define the uniqueness of one's culture of origin. At the same time, such individuals can incorporate enough of the value system and norms of behavior of the other cultural group with which they interact closely, to feel and behave like members of that cultural group, principally the majority host culture. Accordingly, the defining feature of integration is psychological: It is the gradual process of formulation of a bicultural identity, a sense of self that intertwines the unique characteristics of two cultures.

Assimilation is the psychological process of the conscious and unconscious giving up of the unique characteristics of one's culture of origin in favor of the more or less complete incorporation of the values and behavioral characteristics of another cultural group, usually, but not always, the majority culture. Examples include involuntary migration when war and social upheaval necessitate such changes for purposes of survival. However, there are many other life circumstances, including racial, ethnic, and religious discrimination, that motivate people to overlook, suppress, or deny aspects of

their cultural heritage to have a seamless fit within another group. The price of such an effort, in terms of intrapsychic conflict, can be high.

Marginalization is defined by the psychological characteristics of rejection or the progressive loss of valuation of one's cultural heritage, while at the same time rejecting, or being alienated from, the defining values and behavioral norms of another cultural group, usually that of the majority population. This is the psychological outcome of acculturative stress that is closest to the concept of identity diffusion.

Psychiatric Assessment of Immigrants and Refugees

Migration History. Mental illness among immigrants and refugees may have been present before migration, may have developed during the immigration process, such as during months or years living in refugee camps, or presented for the first time in immigration. The immigration process and premigration trauma may precipitate the manifestation of underlying symptoms or result in exacerbation of a preexisting disorder. Obtaining a thorough migration history will help understand the background and precipitating stressors and help guide the development of an appropriate treatment plan.

The premigration history includes inquiry about the patient's social support network, social and psychological functioning, and significant premigration life events. Information about the country and region of origin, the family history in the country of origin—including an understanding of family members who may have decided not to immigrate—educational and work experiences in the country of origin, and prior socioeconomic status should be obtained. Also, premigration political issues, trauma, war, and natural disaster faced by the patient and family in the country or region of origin should be explored. For those who had to escape persecution, warfare, or natural disaster, what were the means of escape, and what type of trauma was suffered before and during migration? Traumatic life events are not limited just to refugees. Immigration may result in losses of social networks, including family and friends, material losses, business, career, property, and loss of the cultural milieu, including their familiar community and religious life. Premigration planning includes reasons for immigrating, duration, and extent of planning, premigration aspirations, and beliefs about the host

country. The type of migration experience, whether as voluntary immigrants or as unprepared refugees, can have profoundly different effects on migrants' mental health.

The Mental Status Examination. As with any patient, conducting a mental status examination is a central component of the psychiatric examination. However, its interpretation in culturally distinct groups and immigrant populations requires caution, as it may be culturally biased. The patient's response is molded by their culture of origin, educational level, and type of acculturative adaptation. The components of the standardized mental status examination are the following: cooperation, appearance, and behavior, speech, affect, thought process, thought content, cognition, insight, and judgment. Cultural differences are wide and varied in dress and grooming. Facial expressions and body movements used in the expression of affect may be more reflective of normal cultural manifestations than pathology. If the clinician is unfamiliar with the individual's culture and the patient's fluency in the language of the host country is limited, the clinician must use caution in interpreting disturbances of speech and thought process, perception, and affect. The presence of hallucinations, for example, can be easily misinterpreted, such as hearing encouraging or clarifying comments from deceased family members, normative experiences in many cultures. The clinician should not assume that the patient understands what the clinician is trying to communicate, and miscommunication involving interpreters is a common problem. The cognitive examination may be particularly tricky. Education and literacy have an important and biasing role. The patient may need adequate time to fully express him- or herself through repeating questions and restating questions to reduce miscommunication. Asking about the meaning of proverbs unfamiliar to the patient may be an inappropriate means of determining abstract thinking. An accurate mental status examination can be accomplished when one allows additional time for clarification of concepts.

Immigration Acculturation and Mental Health

Many countries have had difficulty coping with the surging numbers of migrants. This has led to more significant restrictions on migrant numbers,

partly in response to public sentiment that the social and cultural integrity of the nation has become threatened, even undermined, by waves of migrants from other countries and cultures. During the last 10 years, fears of terrorist violence and civil disruption have led many countries to adopt increasingly restrictive and sometimes punitive policies toward legal and illegal migrants, refugees, and asylum seekers. This trend has been observed in the United States, in some countries of the European Union, and Australia.

Racial and Ethnic Differences in Psychiatric Disorders in the United States

Several community-based epidemiologic studies in the United States have examined the rates of disorders across specific ethnic groups. These studies have found a lower than expected prevalence of psychiatric disorders among disadvantaged racial and ethnic minority groups. African Americans were found to have lower rates of major depression in the Epidemiological Catchment Area study. The lifetime prevalence rates for major depression for whites was 5.1 percent; for Hispanics, 4.4 percent; and for African Americans, 3.1 percent. African Americans, however, had higher rates for all lifetime disorders combined. This finding of differential rates could be explained by adjusting for socioeconomic status.

The National Comorbidity Study (NCS) found lower lifetime prevalence rates of mental illness among African Americans than whites, and in a particular mood, anxiety, and substance use disorders. The lifetime rates for mood disorders were 19.8 percent for whites, 17.9 percent for Hispanic Americans, and 13.7 percent for African Americans. The National Health and Nutrition Examination Survey-III also found lifetime rates of major depression significantly higher among whites, 9.6 percent, than African Americans, 6.8 percent, or Mexican Americans, 6.7 percent. Although African Americans had a lower lifetime risk of mood disorders than whites, once diagnosed, they were more likely to remain persistently ill.

NCS rates for anxiety disorders were 29.1 percent among whites, 28.4 percent for Hispanic Americans, and 24.7 percent for African Americans. The rates for lifetime substance use disorders for the three groups, whites, Hispanic Americans, and African Americans, were 29.5, 22.9, and 13.1 percent. Hispanic Americans, and in particular Mexican Americans, were

found to be at lower risk for substance use and anxiety disorders than whites. In an epidemiologic study conducted in Florida, substantially lower rates were observed among African Americans for depressive disorders and substance use disorders. The lower rate for substance use disorders was also found in the National Epidemiological Survey on Alcohol and Related Conditions, with whites having a prevalence rate of 1-year alcohol use disorders of 8.9 percent; Hispanic Americans, 8.9 percent; African Americans, 6.9 percent; Asian Americans, 4.5 percent; and Native Americans, 12.2 percent. This study also found lower lifetime rates for major depression among Hispanic Americans, 10.9 percent, compared to whites, 17.8 percent. In 2007, the National Survey of American Life compared rates of major depression between Caribbean blacks, African Americans, and whites. Although there were no significant differences in 1-year prevalence between the three groups, lifetime rates were highest among whites, 17.9 percent, followed by Caribbean blacks, 12.9 percent, and African Americans, 10.4 percent. The chronicity of major depressive disorder was higher for African Americans and Caribbean blacks, approximately 56 percent, while much lower for whites, 38.6 percent. This study was consistent with findings from the NCS that concluded that members of disadvantaged racial and ethnic groups in the United States do not have an increased risk for psychiatric disorders; however, once diagnosed, they do tend to have more persistent disorders.

Although African Americans have a lower prevalence rate for mood, anxiety, and substance use disorders, this may not be the case for schizophrenia. The Child Health and Development Study found that African Americans were about threefold more likely than whites to be diagnosed with schizophrenia. The association may be partly explained by African American families having lower socioeconomic status, a significant risk factor for schizophrenia.

A more detailed examination of differences across racial groups was included in the National Comorbidity Study Replication (NCS-R). Non-Hispanic African Americans and Hispanic Americans were at significantly lower risk than non-Hispanic whites for anxiety disorders and mood disorders. Non-Hispanic African Americans had lower rates of substance use disorders than non-Hispanic whites. More specifically, both minority groups were at lower risk for depression, generalized anxiety disorder, and social

phobia. Also, Hispanic Americans had a lower risk for dysthymia, oppositional-defiant disorder, and attention-deficit/hyperactivity disorder. Non-Hispanic African Americans had a lower risk for panic disorder, substance use disorders, and early-onset impulse-control disorders. The lower rates among Hispanic Americans and African Americans compared to non-Hispanic whites appear to be due to reduced lifetime risk of disorders instead of the persistence of chronic disorders. The researchers concluded that the pattern of racial–ethnic differences in risk for psychiatric disorders suggests the presence of protective factors that originate in childhood and have generalized effects, as the lower lifetime risk for both Hispanic Americans and African Americans begins before age 10 years for depression and anxiety disorders. The retention of ethnic identification and participation in communal, religious, and other activities have been suggested as protective factors that may decrease the lifetime risk for psychiatric disorders in close-knit ethnic minority communities. Cultural differences in response to psychiatric diagnostic survey items may be another possible explanation for these findings. However, disadvantaged ethnic groups usually overreport in studies measuring psychological distress, whereas these studies find lower rates.

Discrimination, Mental Health, and Service Utilization

Disparities in Mental Health Services. Studies, including recent ones, have shown that racial and ethnic minorities in the United States receive more limited mental health services than whites. Analysis of medical expenditures in the United States has shown that the mental health care system provides comparatively less care to African Americans and Hispanic Americans than to whites, even after controlling for income, education, and availability of health insurance. African Americans have about a 10 percent probability of receiving any mental health expenditure, compared to 20 percent for whites. Hispanic Americans are about 40 percent less likely than whites to receive any mental health expenditure. Total mental health expenditure for Hispanic Americans is about 60 percent less than for whites.

Also, studies conducted over the last 25 years have shown that regardless of disorder diagnosed, African American psychiatric patients are more likely than white patients to be treated as in-patients, hospitalized involuntarily,

placed in seclusion or restraints without evidence of a greater degree of violence, and treated with higher doses of antipsychotic medications. These differences are not due to the greater severity of the disorders between white and African American patients. One hypothesis for this discrepancy of treatment between African American and white patients is that whites are more likely to seek mental health care voluntarily than African Americans, and African Americans are more likely to enter the mental health care system through more coercive and less voluntary referral systems. African Americans are also more likely than whites to use emergency room services, resulting in more crisis-oriented help-seeking and service utilization. Once hospitalized in an institution with predominantly white staff, African American patients may receive differential care due to discrimination. That is, service personnel who are not familiar with the illness concepts and behavioral norms of nonwhite groups tend to assess minorities as more severely ill and more dangerous than patients of their own racial or ethnic group; consequently, such patients tend more often than white patients to be hospitalized involuntarily, placed in seclusion or restrains, and treated with higher doses of antipsychotic medications.

African American patients assessed in psychiatric emergency services are more likely to be diagnosed with schizophrenia and substance abuse than matched white patients. White patients are more often diagnosed with a mood disorder. The cultural distance between the clinician and the patient can affect the degree of psychopathology inferred and the diagnosis given. These differences in diagnosis by race have also been found when comparable research diagnostic instruments have been used for patient assessment. Semistructured diagnostic instruments based on explicit diagnostic criteria do not necessarily eliminate racial disparities in diagnostic outcomes. It appears that the process that clinicians use to link symptom observations to diagnostic constructs may differ, in particular for schizophrenia, between African American and white patients. The pattern of psychotic symptoms that predicts a clinician making a diagnosis of schizophrenia in African American and white patients is different. Among African American patients, loose associations, inappropriate affect, auditory hallucinations, and vague speech increased the likelihood of a diagnosis of schizophrenia. In white patients the main predictors were vague speech and loose associations. Also, auditory hallucinations are more frequently attributed to African American patients.

African Americans are less likely to have had outpatient treatment and longer delays in seeking care, and they present more severely ill. The reason for hospitalization was also different between African Americans and whites. African American patients were more likely to be admitted for behavioral disturbance, whereas white patients were more likely to be admitted for cognitive or affective disturbances. Also, African Americans were more likely to have police or emergency service involvement, despite no racial differences in violence, suicidality, or substance use when assessed. Furthermore, African American patients are more likely, even after controlling for health insurance status, to be referred to the public rather than private in-patient psychiatric facilities, suggesting racial bias in psychiatric emergency room assessment and recommended treatment.

African American patients diagnosed with major depression are less likely to receive antidepressant medications than whites and less likely to be treated with ECT. Demographic or socioeconomic differences cannot explain these findings. One explanation may be that there are conscious or unconscious biases in psychiatrists' treatment decisions. Although both African Americans and Hispanic Americans were less likely to fill an antidepressant prescription when diagnosed with depression, once a prescription was filled, they were just as likely as whites to receive an adequate course of treatment. These findings indicate that initiating care for depression is the biggest hurdle in overcoming these disparities. African American patients are more likely to be treated with depot neuroleptics than whites after controlling for the type and severity of illness. When treated with antipsychotic drugs, African Americans are less likely to receive second-generation antipsychotics than whites, placing them at increased risk for tardive dyskinesia and dystonia. These differences in antipsychotic prescribing patterns may be due to physicians' concern over an increased risk of diabetes among African Americans compared with whites or may be due to physicians perceiving their symptoms differently. Disparities in mental health care for African Americans and Hispanic Americans have also been noted in studies conducted with adolescents.

A disparity in prescription drug use for mental illness also has been found among Hispanic Americans and Asian Indian Americans. From 1996 to 2000, Asian Indian Americans were found to use prescription drugs 23.6 percent less than whites, whereas the differences between whites and African

Americans and between whites and Hispanic Americans were 8.3 and 6.1 percent, respectively. Disparities in mental health service use among Asian American immigrants may be linked to language-based discrimination, although racial bias cannot be excluded. A study of Chinese Americans found a higher level of informal services and help-seeking from friends and relatives for emotional problems. Those Chinese Americans who reported experiencing language-based discrimination had a more negative attitude toward formal mental health services.

Data on racial and ethnic differences in mental health counseling and psychotherapy are similar to the psychopharmacological studies showing disparities for minorities. A study examining visits to primary care physicians based on the National Ambulatory Medical Care Survey from 1997 to 2000 found that primary care physicians provided similar or higher rates of general health counseling to African Americans than white patients. However, the rates of mental health counseling were significantly lower for African American patients. The lower rate of mental health counseling among African Americans may be due to decreased reporting of depressive symptoms, inadequate communication between African American patients and their primary care physicians, and decreased willingness to discuss mental health issues. On the other hand, another study utilizing the Medical Expenditure Panel Survey from 2000 found that African Americans were more likely than Hispanic Americans or whites to receive an adequate course of psychotherapy for depression. These findings suggest that initiating treatment is the biggest hurdle and that once they are engaged in treatment, African Americans have high compliance with psychotherapy.

Research in Transcultural Psychiatry

There are three perspectives, among other possible approaches, that offer great promise for future research in cultural psychiatry. The first would be based on identifying specific fields in general psychiatry that could be the subject of focused research from a cultural perspective. Topics of epidemiology and neurobiology could be assessed in this way. The former would address issues primarily in the public health arena, including stigmatization, racism, and acculturation. Several cultural variables should be considered in conducting cultural psychiatry research, including language,

religion, traditions, beliefs, ethics, and gender orientation.

The second would aim to explore critical concepts or instruments in culturally relevant clinical research. There are four key concepts: *idioms of distress, social desirability, ethnographic data,* and *explanatory models. Idioms of distress* are the specific ways in which different cultures or societies report ailments; behavioral responses to threatening or pathogenic factors; and the uniqueness in the style of description, nomenclature, and assessment of stress. *Social desirability* stems from the similarities or differences among cultures, vis-à-vis the actual experiencing stressful events. Members of some cultures may be more or less willing to suffer physical or emotional problems, thus showing different levels of vulnerability or resignation, resilience, or acceptance. Issues of stigma in different cultural contexts contribute to this level of desirability or rejection. Third, *ethnographic data* should be included, together with strictly clinical data and laboratory analyses or tests, and narratives of life that enrich the descriptive aspects of the condition and expand on the surrounding sociocultural and interpersonal, and environmental aspects of the experience. The fourth concept is *explanatory models.* Each culture explains the pathology of any kind in its distinctive way. The explanation includes not only the presumptive original cause but also the impact of the adduced factors and the interpersonal exchanges and interactions that lead to the culturally accepted clinical diagnosis.

A third approach attempts to combine the first two by examining different areas of research based on the clinical dimensions of cultural psychiatry. This deals with conceptual, operational, and topical issues in the field now and in the future, including their biocultural connections.

Conceptual Issues in Cultural Psychiatry. One of the primary issues in research in cultural psychiatry is the conceptual differentiation between culture and environment. Although generally accepted as the conceptual opposite of genetic, the environment represents a very broad, polymorphic concept. It is, therefore, essential to establish that, while perhaps part of that environmental set, culture and cultural factors in health and disease are terms of a different, even unique, nature.

To what extent does culture apply to the clinical realities of psychiatry? Culture plays a role in both normality and psychopathology. The role of

culture in psychiatric diagnosis is an excellent example of this conceptual issue. Furthermore, culture impacts treatment approaches based on conventional medical and psychiatric knowledge and explanatory models. Finally, cultural variables have a role in prognosis and outcome.

A conceptual debate exists between those who advocate an *evidence-based* approach to research and practice versus those who assign a *value-based* view to everything clinical, more so if influenced by cultural factors. The value-based approach invokes poverty, unemployment, internal and external migration, and natural and human-made disasters. Evidence may be found to support both positions in scientific research.

Operational Issues in Cultural Psychiatry. The dichotomy of normality and abnormality in human behavior is a crucial operational issue. Culture plays a definitive role in shaping these approaches. This raises the notion of relativism, a strong conceptual pillar in cultural psychiatry. Normality is a relative idea; that is, it varies in different cultural contexts.

Another operational issue is that of the choice of cultural variables. Each one has a specific weight and impact on symptoms, syndromes, or clinical entities in psychiatry. Some of them may be essential in assessing a clinical topic, namely, language, education, religion, and gender orientation. An additional operational factor is a description, assessment, and testing of the strengths and weaknesses of an individual patient. Aspects of an individual's behavior, attitudes, disposition, sociability, occupational skills, and other factors are culturally determined.

Culture plays a significant role in perceiving the severity of symptoms, the disruption of the individual's functionality, and the quality of life. The assessment of severity is also the result of the meaning attributed to causal or pathogenic factors of psychopathology. Judgments about the level of dysfunction and the quality of a patient's life involve elusive concepts such as happiness, well-being, and peace of mind.

Research on cultural psychiatry issues needs to consider the representativeness of the study populations and the generalizability of the findings. Methodologic rigor needs to be applied to collecting demographic data, delineation of and differentiation between ethnic groups or subgroups, and measurement of demographic variables, symptoms, diagnosis, and culturally specific constructs.

Many tests and questionnaires used in clinical settings and research have been developed on English-speaking Western subjects and may not be appropriate for ethnic minority patients or non–English-speaking individuals due to lack of cultural equivalence. Translating items is insufficient to achieve linguistic equivalence, as the meaning and connotation changes and idioms of expression differ between languages. Also, norms may differ between ethnic groups, and tests need to be standardized with representative patients.

The complexity of translating an instrument varies depending on how much the construct being measured differs between the two cultures. There are four different approaches to translation. An ethnocentric approach is one in which the researcher assumes that the concepts overlap entirely in the two cultures. The instrument is used with individuals who differ from the population in which the instrument was initially developed and normed. The pragmatic approach assumes that some overlap between the two cultures and attempts are made to measure the overlapping aspects of the construct, *emic* aspects. An emic plus *etic* approach goes one step further and also attempts to measure culture-specific aspects of the construct. Lastly, sometimes the translation is not possible when the concepts do not overlap at all within the two cultures.

CULTURE-BOUND SYNDROMES

Cross-cultural mental health professionals have introduced several terms to refer to and describe culture-specific forms of expressing and diagnosing emotional distress. The term *culture-bound* was used to describe patterned behaviors of distress or illness whose phenomenology appeared distinct from psychiatric categories and were considered unique to particular cultural settings. The clear implication was that Western psychiatric categories were not culture-bound, but rather universal and that proper characterization would disclose a simple translation key for non-Western syndromes. The dichotomy between syndromes that are “culture-free,” emerging from Euro-American and European societies, and those that are “culture-bound,” emerging from everywhere else, is, of course, patently false. Culture suffuses all forms of psychological distress, the familiar as well as the unfamiliar.

Culture-Bound Syndromes and Their Relationship to Psychiatric Diagnoses

Only a few of the many cultural forms of expressing distress have received sustained research attention to integrating cultural and psychiatric research methods. This chapter focuses on some of those syndromes from diverse cultural regions, which have received the most intensive research and are associated with psychiatric categories: *Amok, ataques de nervios*, possession syndrome, and *shenjing shuairuo*.

Amok. *Amok* is a dissociative episode characterized by a period of depression followed by an outburst of violent, aggressive, or homicidal behavior. Episodes tend to be caused by a perceived insult and are often accompanied by persecutory ideas, automation, amnesia, and exhaustion. Patients return to premorbid states following the episode. *Amok* seems to be prevalent only among males. The term originated in Malaysia, but similar behavior patterns can be found in Laos, Philippines, Polynesia (*cafard* or *cathard*), Papua New Guinea, and Puerto Rico (*mal de pelea*), and among the Navajo (*iich'aa*).

PHENOMENOLOGY. A prototypical episode is composed of the following elements:

1. Exposure to a stressful stimulus or subacute conflict, eliciting in the subject feelings of anger, loss, shame, and lowered self-esteem. The stressor usually appears minor in proportion to the resulting behavior (e.g., an argument with a coworker, verbal insult), but may occasionally be severe (i.e., death of a loved one).
2. A period of social withdrawal and brooding over the precipitating conflict, often involving aimless wandering, and sometimes accompanied by visual perceptual alterations.
3. Transition, usually sudden, to frenzied and extremely violent homicidality, with or without a brief prodromal stage of preparation (e.g., the subject may locate a preferred weapon or reach suddenly for whatever implement is available).
4. Indiscriminate selection of victims who may or may not symbolically represent the original actors in the conflict (e.g., subject attacks only

Chinese people who are strangers to him, after a conflict with a Chinese coworker). Occasionally, the subject also attacks animals or objects in his path, or wounds himself, sometimes severely. The subject perseveres at these violent activities despite external attempts to bring him under control.

5. Verbalizations, when present, may be frenzied and guttural, or express internal conflict (e.g., ask forgiveness of a relative before killing him) or split consciousness (e.g., subject admits to a positive relationship with the victim, but denies this for his "spear").
6. Cessation may be spontaneous but usually results from being overpowered or killed. It is typically abrupt and leads to a change in consciousness, usually stupor or sleep.
7. Subsequent partial or total amnesia and report of "unconsciousness" or description of "darkened vision" (*mata gelap*) during the acute episode.
8. Perceptual disturbances or affective decompensations may occur for days or weeks after the acute attack. Psychosis or depression sometimes ensues.

EPIDEMIOLOGY. Epidemiologic rates of *amok* in Malaysia and Indonesia are unknown and may vary regionally and over time. From the available data, *amok* appears to follow an endemic pattern in Malayo-Indonesia with some epidemic increases, the reverse of which has been found for *amok*-like attacks in Laos.

Amok is essentially unknown in women (only one case was found in the literature, and it was considered atypical in that no deaths occurred). It is thought to occur more frequently in men of Malay extraction, Muslim religion, low education, and rural origin, who are between the ages of 20 and 45 years.

PRECIPITANTS. Precipitants of *amok* in Malaysia and Indonesia typically consisted of experiences eliciting in the subject marked feelings of loss, shame, anger, or lowered self-esteem. Although specific triggers were very diverse in nature and presentation, including sudden and gradual stressors, most consisted of interpersonal or social conflicts superficially appearing to generate only mild to moderate stress. These include arguments with coworkers, nonspecific family tensions, feelings of social humiliation, bouts

of possessive jealousy, gambling debts, and job loss. Rarely, however, *amok* was precipitated by a severe stressor, such as the simultaneous death of the spouse and child of the subject.

ADDITIONAL CLINICAL FEATURES. It is unclear whether *amok* episodes are associated with indirect suicidal intent on the part of the subject. Anecdotes and cultural views supporting a connection are available, but interviews with surviving subjects have refuted the association.

Rates of relapse are unknown. It is considered very likely in the popular view, leading currently in Malaysia to permanent psychiatric hospitalization of surviving subjects, and, in the past, to banishment or execution.

TREATMENT. Afflicted individuals in 20th-century Malaysia have been exempted from legal or moral responsibility for acts committed while in a state of *amok* using a kind of "insanity defense," which characterizes the attack as "unconscious" and beyond the subject's control. They were subsequently hospitalized, sometimes permanently, and frequently received diagnoses of schizophrenia and were treated with antipsychotic medication. Alternatively, trials have sometimes resulted in criminal verdicts and prolonged imprisonment.

Ataque de Nervios. *Ataque de nervios* is an idiom of distress principally reported among Latinos from the Caribbean but recognized among many Latin American and Latin Mediterranean groups. Commonly reported symptoms to include uncontrollable shouting, attacks of crying, trembling, heat in the chest rising into the head, and verbal or physical aggression. Dissociative experiences, seizure-like or fainting episodes, and suicidal gestures are prominent in some attacks but absent in others. A general feature of an *ataque de nervios* is a sense of being out of control. *Ataques de nervios* frequently occur as a direct result of a stressful event relating to the family (e.g., news of a death of a close relative, a separation or divorce from a spouse, conflicts with a spouse or children, or witnessing an accident involving a family member). People may experience amnesia for what occurred during the *ataque de nervios*, but they otherwise return rapidly to their usual level of functioning.

Ataque de nervios (attack of nerves, in Spanish) is a syndrome indigenous

to various Latin American cultures, notably those of the Hispanic Caribbean (Puerto Rico, Cuba, and the Dominican Republic). It has received considerable attention in the psychiatric and anthropologic literature since the mid-1950s, mostly in Puerto Rican communities on the island and in populations within the United States.

PHENOMENOLOGY. An *ataque de nervios* can be described as prototypically composed of the following elements:

1. Exposure to a frequently sudden, stressful stimulus, typically eliciting feelings of fear, grief, or anger, and involving a person close to the subject, such as a spouse, child, family member, or friend. The severity of the trigger ranges from mild–moderate (i.e., marital argument, disclosure of migration plans) to extreme (i.e., physical or sexual abuse, acute bereavement).
2. Initiation of the episode is immediate upon exposure to the stimulus, or after a period of brooding or emotional "shock."
3. Once the acute attack begins, the rapid evolution of an intense affective storm follows, characterized by a primary affect usually congruent with the stimulus (such as anger, fear, grief) and a sense of loss of control (*emotional expressions*).
4. All or some of the following accompany these:
   A. *Bodily sensations:* Trembling, chest tightness, headache, difficulty breathing, heart palpitations, heat in the chest, paresthesias of diverse location, difficulty moving limbs, faintness, blurred vision, or dizziness (*mareos*).
   B. Behaviors (*action dimension*): Shouting, crying, swearing, moaning, breaking objects, striking out at others or self, attempting to harm self with the nearest implement, falling to the ground, shaking with convulsive movements, or lying "as if dead."
5. Cessation may be abrupt or gradual, but it is usually rapid, and often results from the ministration of others, involving expressions of concern, prayers, or use of rubbing alcohol (*alcoholado*). There is a return of ordinary consciousness and reported exhaustion.
6. The attack is frequently followed by partial or total amnesia for the events of the episode, and descriptions such as the following for the acute attack:

Loss of consciousness, depersonalization, mind going blank, or general unawareness of surroundings (*alterations in consciousness*). However, some *ataques* appear to exhibit no alterations in consciousness.

EPIDEMIOLOGY. Risk factors for *ataque de nervios* span a range of social and demographic characteristics. The strongest predictors of *ataque* are female gender, lower formal education, and disrupted marital status (i.e., divorced, widowed, or separated). *Ataque* sufferers also reported less satisfaction in their social interactions, generally and specifically with their spouses. Also, people who experienced an *ataque de nervios* were more likely to describe their health as only fair or poor, to seek help for an emotional problem, and to take medications for this purpose. Persons with *ataque* also reported deriving less satisfaction from leisure time activities and feeling overwhelmed more often.

PRECIPITANTS. Prototypically, *ataque de nervios* was linked by sufferers to an acute precipitating event or to the summation of many life episodes of suffering brought to a head by a trigger that overwhelmed the person's coping ability. In 92 percent of cases, the *ataque* was directly provoked by a distressing situation, and 73 percent of the time, it began within minutes or hours of the event. A majority of first *ataques* (81 percent) occurred in the presence of others, as opposed to when the sufferer was alone and led to a person receiving help (67 percent). Unlike the typical experience of persons with panic disorder, most patients reported feeling better (71 percent) or feeling relieved (81 percent) after their first *ataque*. The first episodes of *ataque de nervios* are closely tied to the interpersonal world of the sufferer, and they result in an unburdening (*desahogarse*) of one's life problems, at least temporarily.

ADDITIONAL CLINICAL FEATURES. The association between *ataque de nervios* and a sense of loss of control and of being overwhelmed highlight the importance of the association between the cultural syndrome and other behaviors associated with acute emotional dysregulation. Most concerning among these is the strong relationship between *ataques* and suicidal ideation and attempts. Other related behaviors include loss of aggression control, expressed as attacks on people or property, and dissociative experiences

related to the acute *ataque* experience.

SPECIFIC CULTURAL FACTORS. The complex relationship between *ataque de nervios* and psychiatric diagnosis may be clarified about its broader popular nosology. In the Hispanic Caribbean and other areas of Latin America, *ataque* is part of a popular nosology of *nervios* (nerves), composed of other related categories. Experiences of adversity are linked in this nosology to ensuing "alterations" of the nervous system, which result in its impaired functioning, including the peripheral nerves. This quasi-anatomical damage is evidenced in emotional symptoms, including interpersonal susceptibility, anxiety, and irritability, as well as in physically mediated symptoms, such as trembling, palpitations, and decreased concentration.

TREATMENT. No treatment studies of *ataque de nervios* have ever been conducted. Typical treatment involves, first, ensuring the safety of the person and those around him or her, given the association between *ataque*, suicidality, and uncontrolled aggressivity. "Talking the person down" is usually helpful, accompanied by expressions of support from relatives and other loved ones; rubbing alcohol (*alcoholado*) to help calm the person is a culturally prescribed way of expressing this support.

"Telling the story" of what led to the *ataque* usually constitutes the principal therapeutic approach in subsequent stages of treatment. Because one of the main functions of the attack is to communicate a feeling of being overwhelmed, indicating receipt of the message and the desire to offer support are usually perceived as therapeutic. The person should be allowed to set the pace of disclosure and to give enough details and circumstances to feel "unburdened" (*desahogado[a]*).

In the case of single or occasional *ataques* in the absence of psychiatric diagnosis, brief follow-up is usually sufficient. This may be discussed with the patient and the family to ensure a full return to the previous healthy state. For recurrent *ataques,* treatment depends on the associated psychopathology, the nature of the precipitants (including traumatic exposure), the degree of family conflict or support, the social context, the previous treatment experiences, and the patient's and family's expectations, among other factors.

Psychotherapy is typically the mainstay of treatment, given the usual source of the overwhelmed behavior in the interpersonal milieu.

Pharmacotherapy may also help treat *ataque*-related psychopathology; the primary emphasis should be placed on treating the underlying disorder. Given the slow crescendo of many *ataques*, judicious use of short-acting benzodiazepines also has a role in helping abort an impending episode. However, this should not be the primary form of treatment for recurrent *ataques*, since it only forestalls the principal function of the syndrome as a mode of communication. Instead, psychotherapy and a social activism stance by the therapist that acknowledges the origins of adversity among low-income Latinos in socioeconomic disenfranchisement and ethnic/racial discrimination are usually required to address the interpersonal and sociocultural roots of *ataque de nervios.*

Possession Syndrome. Involuntary possession trance states are very common presentations of emotional distress around the world. Cognate experiences have been reported in extremely diverse cultural settings, including India, Sri Lanka, Hong Kong, China, Japan, Malaysia, Niger, Uganda, Southern Africa, Haiti, Puerto Rico, and Brazil. *Possession syndrome* is an umbrella English-language term used to describe South Asian presentations of involuntary possession trance that encompasses multiple names in regional languages and dialects of India and Sri Lanka. These presentations are seen as a form of illness by the person's cultural group because they are involuntary, they cause distress, and they do not occur as a normal part of a collective cultural or religious ritual or performance.

PHENOMENOLOGY. It is important to distinguish at the outset between possession syndrome, as an instance of possession trance, and the broader category of possession. The latter refers to a general ideology describing the full range of direct spirit influences on human affairs, including effects on physical, psychological, spiritual, social, and ecologic realms. By contrast, as a subset of general possession experience, possession trance refers to specific alterations in consciousness, memory, behavior, and identity attributed to direct spirit influence. In addition to pathologic possession, trance states, South Asian cultures authorize multiple examples of normal possession and possession trance. When voluntary and normative, these states are typically seen as instances of religious devotion, mystical ecstasy, social commentary, asceticism, interpersonal relations, existential reflection, and the study of

consciousness. This chapter discusses possession syndrome as a pathologic entity with an established phenomenology, that is, as a particular case in the general continuum of etiologic ideas regarding possession illnesses. A prototypical episode is composed of the following elements:

1. Onset occurs typically due to subacute conflict or stress and shows considerable variation. It may be gradual and nonspecific (e.g., diverse somatic complaints, such as dizziness, headaches, abdominal discomfort, hot–cold flashes, listlessness, or difficulty breathing) or sudden and specific, in the form of an abrupt transition to an altered state of consciousness.
2. Behavior during the altered state consists of some or all of the following:
   A. Dramatic, semi-purposeful movements, such as head-bobbing, bodily shaking, thrashing, gyrating, or falling to the ground, accompanied by guttural, incoherent verbalizations, such as mumbling, moaning, or shrieking.
   B. Aggressive or violent actions directed at self or others, including spitting, striking, and impulsive suicidal or homicidal gestures. Verbalizations may be coherent and consist of derogatory comments or threats of violence directed against significant others or the subject (in the third person) and typically considered by observers to be uncharacteristic of the subject's usual behavior.
   C. Specific gestures, comments, or requests denoting the appearance of a known possessing personality by (1) reference to standard attributes of culturally recognizable figures or (2) the name and degree of deceased family members or acquaintances.
3. In all cases, this state is marked by the emergence of one or several secondary personalities distinct from that of the subject. Their specific identities, which may remain undisclosed for some time, adhere to cultural norms regulating permissible agents of possession, which vary by religion, region, and caste. Acceptable agents include spirits of deceased family members, in-law relations, or known village acquaintances who died under specific conditions of distress, and minor supernatural figures of the Hindu pantheon (rarely major deities) and the Islamic spiritual world.
4. Possession by the secondary personality(ies) is episodic, resulting in alternation between the usual personality of the subject and the altered

state. The subject in his or her usual identity appears in a daze, exhausted, distressed, or confused about the situation and may report visual or auditory perceptual disturbances regarding the possessing agent and "unconsciousness" and partial or total amnesia for the altered state.

5. Frequently, the specific identities of possessing personalities remain undisclosed for some time, requiring the active ministrations of family members and the intervention of specialized indigenous practitioners. The process of disclosure is conceived as a struggle between the family members and the beneficent agents possessing the healer on the one side, and the troublesome possessing personalities on the other. It is characterized by remarkable reactivity on the part of the subject to environmental cues, including direct questioning, strategic neglect, and aggressive manipulation.
6. The outcome is variable. Total recovery is often reported at the cessation of a single acute episode, which may be of several weeks' duration. Alternatively, prolonged morbidity may result, or even, rarely, death.

Methodologic considerations limit data on the epidemiology, precipitants, and associated psychopathology of subjects with possession syndrome in South Asia. These include a lack of representative community samples and nonsystematic definitions of the syndrome, which shows considerable regional variation.

EPIDEMIOLOGY. Possession syndrome is more common in women than men, with a female-to-male ratio of approximately 3 to 1 in both community and psychiatric cohorts. The age of onset is usually between 15 and 35 years, but many cases reportedly begin in childhood. Attacks may persist well into middle age, and geriatric cases have also been reported.

PRECIPITANTS. Precipitants of possession syndrome are varied but typically consist of marked social or family conflicts, or stressful life transitions, of subacute duration, eliciting strong feelings of vulnerability in persons without firm emotional support. Examples encountered in the literature included marital conflict, abuse, and neglect, at times associated with alcoholism; the arrival of a new bride to the home of her husband's family; delay in arranging a marriage, or in consummating it; forced marriage; widowhood; postpartum

status; loss of family social standing; the death of a family member; difficulty finding employment and financial difficulties; alienation from family support; and subordination to other family members and in-laws.

SPECIFIC CULTURAL FACTORS. Possession syndrome constitutes a normative cultural category throughout India and Sri Lanka. It may present initially in a variety of forms, linked by the attribution of spirit etiology. When it presents in a nonspecific fashion, indigenous diagnosis is confirmed by the appearance of the altered state during the therapeutic ritual. It is considered an affliction by its painful, involuntary nature and attributed to the intervention of specific spiritual agencies acting independently or at the behest of a witch. Certain castes and persons in transitional states (e.g., puerperium) are considered most vulnerable to a spirit attack, especially when deprived of emotional and material support.

TREATMENT. Specialized indigenous practitioners and ritual therapies are generally available and widely utilized, but psychiatric treatment is typically avoided. Indigenous treatments include neutralization of the conflict or stress via the communal rituals involved in exorcism, as well as the reformulation of the suffering into beneficent individual and communal practice via initiation into a spirit devotion cult, such as the Siri cult of South India, or education into the roles of oracle (diviner), exorcist, or, rarely, avatar (divine incarnation).

Shenjing Shuairuo.

Shenjing shuairuo ("weakness of the nervous system" in Mandarin Chinese) is a translation and cultural adaptation of the term "neurasthenia," which was transmitted into China from the West and Japan in the 1920s and 1930s. Revived in its modern form by the American neurologist George Beard since 1868, his formulation of neurasthenia (Greek for "lack of nerve strength") originally denoted a heterogeneous syndrome of lassitude, pain, poor concentration, headache, irritability, dizziness, insomnia, and over 50 other symptoms. It was considered at first an "American disease," resulting from the "pressures" of a rapidly modernizing society but was later adopted by European diagnosticians. Its pathophysiology was thought to derive from a lowering of nervous system function on a physical rather than emotional basis, due to excessive demand for its use, especially

among the educated and wealthier classes. In Soviet psychiatry, buttressed by Pavlovian research, it was a central component of mental health nosology, which became especially influential in Chinese psychiatry after the communist revolution of 1949.

Although neurasthenia declined in importance in Western classification systems during the 20th century, *Shenjing shuairuo* underwent marked popular and professional development in mainland China, Taiwan, Hong Kong, in Chinese migrant communities, and in Japan, where a similar syndrome is labeled *shinkei suijaku.* From a peak in about 1980, when it may have constituted up to 80 percent of all "neurotic" diagnoses in Chinese societies, *Shenjing shuairuo* has undergone intense psychiatric and anthropologic reexamination. It currently features prominently in the second edition revised *Chinese Classification of Mental Disorders* (CCMD-2-R), under the section on "other neuroses." The CCMD-2-R diagnosis requires three symptoms out of five nonhierarchical symptom clusters, organized as weakness, emotional, excitement, and nervous symptoms, and the fifth category of sleep disturbances. Like other neurotic disorders in the Chinese manual, the condition must last at least 3 months, and should (1) lower the efficiency of work, study, or social function; (2) cause mental distress; or (3) precipitate treatment seeking.

PHENOMENOLOGY. Given the evolution of diagnostic practice regarding *shenjing shuairuo* in Chinese societies over the last decades, which may be labeled the professional approximation of the condition or its "disease" aspect, phenomenologic description in this chapter is based instead on clinical histories of self-identified sufferers, or the "illness" aspect of the syndrome. The following elements are prototypical:

1. Onset is usually gradual, sometimes spanning several years, and typically emerges out of a conflictive, frustrating, or worrying situation that involves work, family, and other social settings, or their combination. A sense of powerlessness in changing the precipitating situation appears central to most illness accounts of the syndrome.
2. Symptoms show substantial individual variation, but usually involve at least some of the following spontaneous complaints: Insomnia, affective dysphoria, headache, bodily pains and distortions (e.g., "swelling" of the

head), dizziness, difficulty concentrating, tension and anxiety, worry, fatigue, weakness, gastrointestinal problems, and "troubled vexation" (,*fan nao*). This last emotion has been described as a form of irritability mixed with worry and distress over "conflicting thoughts and unfulfilled desires," that may be partially concealed for the sake of preserving social harmony.

3. The sufferer frequently seeks the sick role, attributing his or her difficulties in meeting work, school, or other social expectations to the syndrome. Sources of treatment vary substantially across Chinese communities, depending on the availability of formal and traditional service sectors.
4. The course is variable and may respond closely to changing interpersonal and social circumstances. Amelioration of the precipitating stressor(s) typically brings about substantial improvement, although residual symptoms appear common.
5. Response to treatment may be strongly mediated by the illness role and its relationship to the intractability of precipitating stressors.

PRECIPITANTS. An empirical assessment of the precipitants of *shenjing shuairuo* has found high rates of work-related stressors, which were made more intractable by the centrally directed nature of mainland Chinese society. These included unwelcome work assignments, job postings that caused family separations, harsh criticism at work, excessive workloads, monotonous tasks, and feelings of inadequacy or incompatibility of skills and responsibilities. Students usually described less severe study-related precipitants, particularly school failure or anxiety over the mismatch between personal or family aspirations and performance. Other interpersonal and family-related stressors included romantic disappointments, marital conflict, and the death of a spouse or other relative. Chinese etiologic understandings of the syndrome commonly invert the Western view of "psychosomatic" presentations, whereby social–interpersonal precipitants cause psychological distress that is displaced onto bodily experience.

ADDITIONAL CLINICAL FEATURES. The clinical course of the syndrome may depend on the associated psychiatric comorbidity and the degree of persistence of the precipitating stressors. One longitudinal study found a

complete resolution of *shenjing shuairuo* symptoms and reasonable social adjustment 20 years from the index diagnosis in 83 of 89 cases. Only one case was receiving continued treatment, and no subjects reported the onset of depressive disorder after the *shenjing shuairuo* diagnosis.

Chinese psychiatrists have carried out numerous studies of neurophysiologic and cognitive function in *shenjing shuairuo* patients since the 1950s. Most have reported abnormalities compared to normal controls, including in tests of polysomnography, electroencephalography, psychogalvanic reflexes, gastric function, and memory function. These findings need to be replicated with well-controlled samples using contemporary diagnostic instruments.

SPECIFIC CULTURAL FACTORS. The evolving definitions of *shenjing shuairuo* have emerged from a tradition of syncretism in Chinese medicine between indigenous illness understandings and international contributions. Nineteenth-century Western notions of a weakened nervous system due to overuse (neurasthenia) found an ancient cognate expression in Chinese concepts of bodily meridians or channels (*jing*) binding vital organs in balanced networks along which forces (e.g., *qi*, vital energy, in *yin* and *yang* forms) could be disrupted from their normal, harmonious flow. This gave rise to *shenjing shuairuo*, an illness whereby the *jing* that carry *shen*—spirit or vitality, the capacity of the mind to form ideas and the desire of the personality to live life—have become *shuai* (degenerate) and *ruo* (weak) following undue nervous excitement.

TREATMENT. When accessing formal sectors of care, most patients used both Western-trained physicians and traditional Chinese doctors. Nonpsychiatric medical settings were preferred, including neurology and general medicine clinics, in concert with cultural understandings of the somatopsychic etiology of *shenjing shuairuo,* emphasizing its physical mediation. The modality of treatment was usually traditional Chinese medicines, which were prescribed by both Western-trained and Chinese-style doctors. This conformed to the balanced status still ascribed to both types of training among Chinese physicians. Polypharmacy was common, combining sedatives, traditional herbs, antianxiety agents, vitamins, and other tonics. Despite the active suppression of religious healing in China, almost a quarter of patients were

also engaged in such treatment.

▲ 34.4 Theories of Personality and Psychopathology

SIGMUND FREUD: FOUNDER OF CLASSIC PSYCHOANALYSIS

Psychoanalysis was the child of Sigmund Freud's genius. He put his stamp on it from the very beginning, and it can be fairly stated that, although the science and theory of psychoanalysis have advanced far beyond Freud, his influence is still strong and pervasive. In recounting the progressive stages in the evolution of the origins of Freud's psychoanalytic thinking, it is useful to keep in mind that Freud himself was working against the background of his neurologic training and expertise and in the context of the scientific thinking of his era.

The science of psychoanalysis is the bedrock of psychodynamic understanding. It forms the fundamental theoretical frame of reference for a variety of forms of therapeutic intervention, embracing not only psychoanalysis itself but also various forms of psychoanalytically oriented psychotherapy and related forms of therapy employing psychodynamic concepts. Currently, considerable interest has been generated in efforts to connect psychoanalytic understandings of human behavior and emotional experience with emerging findings of neuroscientific research.

At the same time, psychoanalysis is undergoing a creative ferment in which classical perspectives are continually being challenged and revised, leading to a diversity of emphases and viewpoints, all of which can be regarded as representing aspects of psychoanalytic thinking. This has given rise to the question as to whether psychoanalysis is one theory or more than one. The divergence of multiple theoretical variants raises the question of the degree to which newer perspectives can be reconciled to classical perspectives.

Freud himself inaugurated the spirit of creative modifications in theory. Some of the theoretical modifications of the classic theory after Freud have

attempted to reformulate basic analytic propositions while still retaining the spirit and fundamental insights of a Freudian perspective; others have challenged and abandoned fundamental analytic insights in favor of divergent paradigms that seem radically different and even contradictory to basic analytic principles.

Although there is more than one way to approach the diversity of such material, the decision has been made to organize this material along roughly historical lines, tracing the emergence of analytic theory or theories over time, but with a good deal of overlap. However, there is an overall pattern of gradual emergence, progressing from early drive theory to structural theory to ego psychology to object relations, and on to self-psychology, intersubjectivism, and relational approaches.

Psychoanalysis today is recognized as having three crucial aspects: it is a therapeutic technique, a body of scientific and theoretical knowledge, and a method of investigation. This section focuses on psychoanalysis as both a theory and a treatment, but the basic tenets elaborated here have broad applications to nonpsychoanalytic settings in clinical psychiatry.

Life of Freud

Sigmund Freud (1856–1939) (Fig. 34-22) was born in Freiburg, a small town in Moravia, which is now part of the Czech Republic. When Freud was 4 years old, his father, a Jewish wool merchant, moved the family to Vienna, where Freud spent most of his life. Following medical school, he specialized in neurology and studied for a year in Paris with Jean-Martin Charcot. He was also influenced by Ambroise-Auguste Liébeault and Hippolyte-Marie Bernheim, both of whom taught him hypnosis while he was in France. After his education in France, he returned to Vienna and began clinical work with hysterical patients. Between 1887 and 1897, his work with these patients led him to develop psychoanalysis.

Beginnings of Psychoanalysis

In the decade from 1887 to 1897, Freud immersed himself in the serious study of the disturbances in his hysterical patients, resulting in discoveries that contributed to the beginnings of psychoanalysis. These slender beginnings had a threefold aspect: Emergence of psychoanalysis as a method

of investigation, as a therapeutic technique, and as a body of scientific knowledge based on an increasing fund of information and fundamental theoretical propositions. These early researches flowed out of Freud's initial collaboration with Joseph Breuer and then, increasingly, from his independent investigations and theoretical developments.

The Case of Anna O

Breuer was an older physician, a distinguished and well-established medical practitioner in the Viennese community (Fig. 34-23). Knowing Freud's interests in hysterical pathology, Breuer told him about the unusual case of a woman he had treated for approximately 1.5 years, from December 1880 to June 1882. This woman became famous under the pseudonym Fräulein Anna O, and the study of her difficulties proved to be one of the essential stimuli in the development of psychoanalysis.

Anna O was, in reality, Bertha Pappenheim, who later became independently famous as a founder of the social work movement in Germany. When she began to see Breuer, she was an intelligent and strong-minded young woman of approximately 21 years of age who had developed several hysterical symptoms connected with the illness and death of her father. These symptoms included paralysis of the limbs, contractures, anesthesias, visual and speech disturbances, anorexia, and distressing nervous cough. Two distinct phases of consciousness also characterized her illness: One relatively standard, but the other reflected a second and more pathologic personality.

Anna was very fond of and close to her father and shared with her mother the duties of nursing him on his deathbed. During her altered states of consciousness, Anna recalled the vivid fantasies and intense emotions she had experienced while caring for her father. It was with considerable amazement, both to Anna and Breuer, that when she could recall, with the associated expression of affect, the scenes or circumstances under which her symptoms had arisen, the symptoms would disappear. She vividly described this process as the "talking cure" and as "chimney sweeping."

Once the connection between talking through the circumstances of the

symptoms and the disappearance of the symptoms themselves had been established, Anna proceeded to deal with each of her many symptoms, one after another. She was able to recall that on one occasion when her mother had been absent, she had been sitting at her father's bedside and had had a fantasy or daydream in which she imagined that a snake was crawling toward her father and was about to bite him. She struggled forward to try to ward off the snake, but her arm, which had been draped over the back of the chair, had gone to sleep. She was unable to move it. The paralysis persisted, and she was unable to move the arm until, under hypnosis, she could recall this scene. It is easy to see how this kind of material must have made a profound impression on Freud. It provided a convincing demonstration of the power of unconscious memories and suppressed affects in producing hysterical symptoms.

In the course of the somewhat lengthy treatment, Breuer had become increasingly preoccupied with his fascinating and unusual patient and, consequently, spent more and more time with her. Meanwhile, his wife had grown increasingly jealous and resentful. As soon as Breuer began to realize this, the sexual connotations of it frightened him, and he abruptly terminated the treatment. Only a few hours later, however, he was recalled urgently to Anna's bedside. She had never alluded to the forbidden topic of sex during her treatment, but she was now experiencing hysterical childbirth. Freud saw the phantom pregnancy as the logical outcome of the sexual feelings she had developed toward Breuer in response to his therapeutic attention. Breuer himself had been entirely unaware of this development, and the experience was quite unnerving. He was able to calm Anna down by hypnotizing her, but then he left the house in a cold sweat and immediately set out with his wife for Venice on a second honeymoon.

According to a version from Freud through Ernest Jones, the patient was far from cured and later hospitalized after Breuer's departure. It seems ironic that the prototype of a cathartic cure was, in fact, far from successful. Nevertheless, the case of Anna O provided a critical starting point for Freud's thinking and a crucial juncture in the development of psychoanalysis.

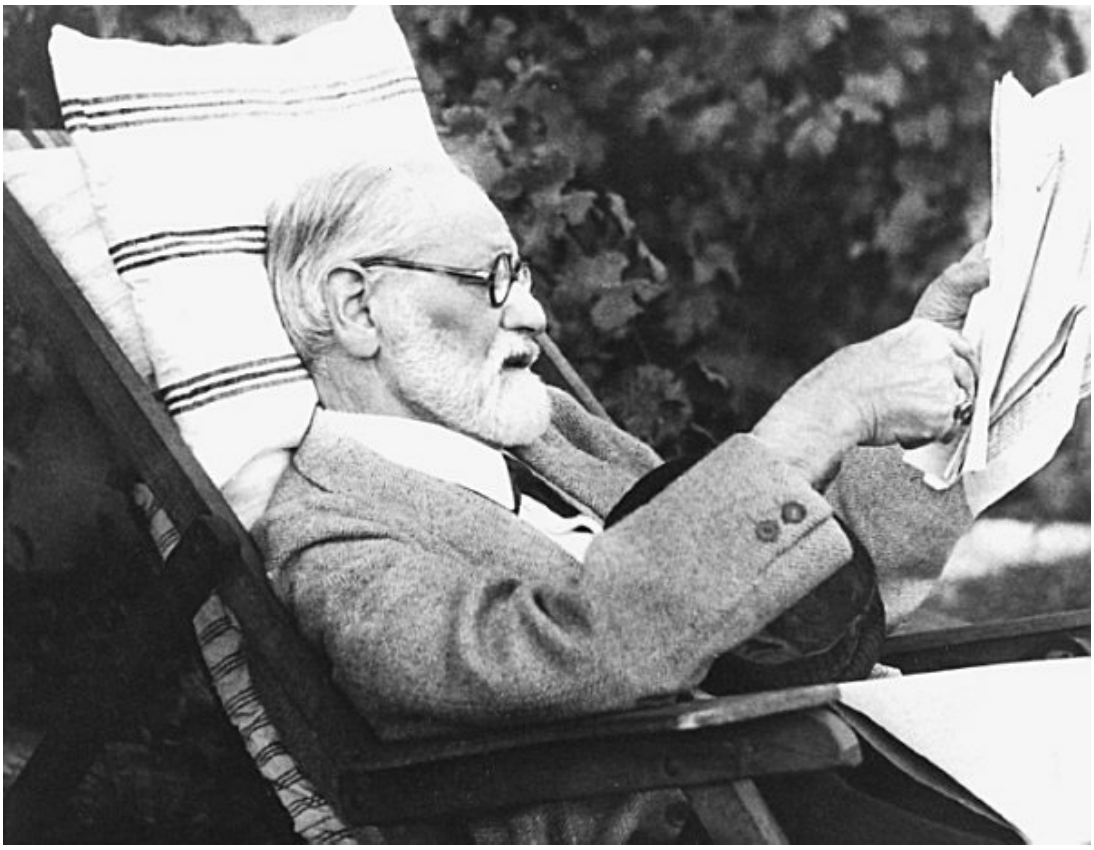


FIGURE 34-22
Sigmund Freud at age 79. (Courtesy of Menninger Foundation Archives, Topeka, KS.)



FIGURE 34-23
Joseph Breuer (1842–1925).

The Interpretation of Dreams

In his landmark publication *The Interpretation of Dreams* in 1900, Freud presented a theory of the dreaming process that paralleled his earlier analysis of psychoneurotic symptoms. He viewed the dream experience as a conscious expression of unconscious fantasies or wishes not readily acceptable to conscious waking experience. Thus, dream activity was considered to be one of the usual manifestations of unconscious processes.

The dream images represented unconscious wishes or thoughts, disguised through a process of a symbolization and other distorting mechanisms. This

reworking of unconscious contents constituted the dream work. Freud postulated the existence of a "censor," pictured as guarding the border between the unconscious part of the mind and the preconscious level. The censor functioned to exclude unconscious wishes during conscious states but, during regressive relaxation of sleep, allowed specific unconscious contents to pass the border, only after transforming these unconscious wishes into disguised forms experienced in the manifest dream contents by the sleeping subject. Freud assumed that the censor worked in the service of the ego—that is, as serving the self-preservative objectives of the ego. Although he was aware of the unconscious nature of the processes, he tended to regard the ego at this point in developing his theory more restrictively as the source of conscious processes of reasonable control and volition.

The analysis of dreams elicits material that has been repressed. These unconscious thoughts and wishes include nocturnal sensory stimuli (sensory impressions such as pain, hunger, thirst, urinary urgency), the day residue (thoughts and ideas connected with the activities and preoccupations of the dreamer's current waking life), and repressed unacceptable impulses. Because the sleep state blocks motility, the dream enables partial but limited gratification of the repressed impulse that gives rise to the dream.

Freud distinguished between two layers of dream content. The *manifest* content refers to what is recalled by the dreamer; the *latent* content involves the unconscious thoughts and wishes that threaten to awaken the dreamer. Freud described the unconscious mental operations by which latent dream content is transformed into the manifest dream as the *dream work.* Repressed wishes and impulses must attach themselves to innocent or neutral images to pass the scrutiny of the dream censor. This process involves selecting meaningless or trivial images from the dreamer's current experience, which are dynamically associated with the latent images that they resemble in some respect.

Condensation. *Condensation* is the mechanism by which several unconscious wishes, impulses, or attitudes can be combined into a single image in the manifest dream content. Thus, in a child's nightmare, an attacking monster may come to represent not only the dreamer's father but may also represent some aspects of the mother and even some of the child's primitive hostile impulses as well. The converse of condensation can also

occur in the dream work, namely, irradiation or diffusion of a single latent wish or impulse distributed through multiple representations in the manifest dream content. The combination of mechanisms of condensation and diffusion provides the dreamer with a highly flexible and economic device for facilitating, compressing, and diffusing or expanding the manifest dream content, which is derived from latent or unconscious wishes and impulses.

Displacement. The mechanism of *displacement* refers to the transfer of amounts of energy (cathexis) from an original object to a substitute or symbolic representation of the object. Because the substitute object is relatively neutral—that is, less invested with affective energy—it is more acceptable to the dream censor and can pass the borders of repression more easily. Thus, whereas symbolism can be taken to refer to the substitution of one object for another, displacement facilitates distortion of unconscious wishes through the transfer of affective energy from one object to another. Despite the transfer of cathectic energy, the aim of the unconscious impulse remains unchanged. For example, in a dream, the mother may be represented visually by an unknown female figure (at least one who has less emotional significance for the dreamer), but the naked content of the dream nonetheless continues to derive from the dreamer's unconscious instinctual impulses toward the mother.

Symbolic Representation. Freud noted that the dreamer would often represent highly charged ideas or objects by using innocent images that were in some way connected with the idea or object being represented. In this manner, an abstract concept or a complex set of feelings toward a person could be symbolized by a simple, concrete, or sensory image. Freud noted that symbols have unconscious meanings that can be discerned through the patient's associations to the symbol, but he also believed that certain symbols have universal meanings.

Secondary Revision. The mechanisms of condensation, displacement, and symbolic representation are characteristic of a type of thinking that Freud referred to as *primary process.* This primitive mode of cognitive activity is characterized by illogic, bizarre, and absurd images that seem incoherent. Freud believed that a more mature and reasonable aspect of the ego works

during dreams to organize primitive aspects of dreams into a more coherent form. *Secondary revision* is Freud's name for this process, in which dreams become somewhat more rational. The process is related to mature activity characteristic of waking life, which Freud termed *secondary process.*

Affects in Dreams. Secondary emotions may not appear in the dream at all, or they may be experienced in somewhat altered form. For example, repressed rage toward a person's father may take the form of mild annoyance. Feelings may also appear as their opposites.

Anxiety Dreams. Freud's dream theory preceded his development of a comprehensive theory of the ego. Hence, his understanding of dreams stresses the importance of discharging drives or wishes through the hallucinatory contents of the dream. He viewed such mechanisms as condensation, displacement, symbolic representation, projection, and secondary revision primarily as facilitating the discharge of latent impulses, rather than as protecting dreamers from anxiety and pain. Freud understood anxiety dreams as reflecting a failure in the protective function of the dream-work mechanisms. The repressed impulses succeed in working their way into the manifest content in a more or less recognizable manner.

Punishment Dreams. Dreams in which dreamers experience punishment represented a unique challenge for Freud because they appear to represent an exception to his wish-fulfillment theory of dreams. He came to understand such dreams as reflecting a compromise between the repressed wish and the repressing agency or conscience. In a punishment dream, the ego anticipates condemnation on the dreamer's conscience if the latent unacceptable impulses are allowed direct expression in the manifest dream content. Hence, the wish for punishment on the part of the patient's conscience is satisfied by giving expression to punishment fantasies.

Topographical Model of the MIND

The publication of *The Interpretation of Dreams* in 1900 heralded the arrival of Freud's topographical model of the mind, in which he divided the mind into three regions: the conscious system, the preconscious system, and the unconscious system. Each system has its unique characteristics.

The Conscious. The conscious system in Freud's topographical model is the part of the mind in which perceptions coming from the outside world or within the body or mind are brought into awareness. Consciousness is a subjective phenomenon whose content can be communicated only using language or behavior. Freud assumed that consciousness used a form of neutralized psychic energy that he referred to as *attention cathexis,* whereby persons were aware of a particular idea or feeling as a result of investing a discrete amount of psychic energy in the idea or feeling.

The Preconscious. The preconscious system is composed of those mental events, processes, and contents that can be brought into conscious awareness by focusing attention. Although most persons are not consciously aware of the appearance of their first-grade teacher, they ordinarily can bring this image to mind by deliberately focusing attention on the memory. Conceptually, the preconscious interfaces with both unconscious and conscious regions of the mind. To reach conscious awareness, the contents of the unconscious must become linked with words and thus become preconscious. The preconscious system also serves to maintain the repressive barrier and to censor unacceptable wishes and desires.

The Unconscious. The unconscious system is dynamic. Its mental contents and processes are kept from conscious awareness through the force of censorship or repression, and it is closely related to instinctual drives. At this point in Freud's theory of development, instincts were thought to consist of sexual and self-preservative drives, and the unconscious was thought to contain the mental representations and derivatives of the sexual instinct primarily.

The content of the unconscious is limited to wishes seeking fulfillment. These wishes motivate dream and neurotic symptom formation. This view is now considered reductionist.

The unconscious system is characterized by *primary process thinking,* principally aimed at facilitating wish fulfillment and instinctual discharge. It is governed by the pleasure principle and, therefore, disregards logical connections; it has no concept of time, represents wishes as fulfillments, permits contradictions to exist simultaneously, and denies the existence of negatives. The primary process is also characterized by extreme mobility of

drive cathexis; the investment of psychic energy can shift from object to object without opposition. Memories in the unconscious have been divorced from their connection with verbal symbols. Hence, when words are reapplied to forgotten memory traits, as in psychoanalytic treatment, the verbal cathexis allows the memories to reach consciousness again.

The contents of the unconscious can become conscious only by passing through the preconscious. When censors are overpowered, the elements can enter consciousness.

Limitations of the Topographical Theory. Freud soon realized that two primary deficiencies in the topographical theory limited its usefulness. First, many patients' defense mechanisms that guard against distressing wishes, feelings, or thoughts were not initially accessible to consciousness. Thus, repression cannot be identical with preconscious, because, by definition, this region of the mind is accessible to consciousness. Second, Freud's patients frequently demonstrated an unconscious need for punishment. This clinical observation made it unlikely that the moral agency making the demand for punishment could be allied with anti-instinctual forces available to conscious awareness in the preconscious. These difficulties led Freud to discard the topographical theory. However, certain concepts derived from the theory continue to be useful, mainly primary and secondary thought processes, the fundamental importance of wish fulfillment, the existence of a dynamic unconscious, and a tendency toward regression under frustrating conditions.

Instinct or Drive Theory

After the development of the topographical model, Freud turned his attention to the complexities of instinct theory. Freud was determined to anchor his psychological theory in biology. His choice led to terminologic and conceptual difficulties when he used terms derived from biology to denote psychological constructs. *Instinct,* for example, refers to a pattern of species-specific behavior that is genetically derived and, therefore, is more or less independent of learning. Modern research demonstrating that instinctual patterns are modified through experiential learning has made Freud's instinctual theory problematic. Further confusion has stemmed from the ambiguity inherent in a concept on the borderland between the biologic and

the psychological: Should the mental representation aspect of the term and the physiologic component be integrated or separated? Although *drive* may have been closer than the instinct to Freud's meaning, in contemporary usage, the two terms are often used interchangeably.

In Freud's view, an instinct has four principal characteristics: source, impetus, aim, and object. The *source* refers to the part of the body from which the instinct arises. The *impetus* is the amount of force or intensity associated with the instinct. The *aim* refers to any action directed toward tension discharge or satisfaction, and the *object* is the target (often a person) for this action.

Instincts

LIBIDO. The ambiguity in the term *instinctual drive* is also reflected in the use of the term *libido.* Briefly, Freud regarded the sexual instinct as a psychophysiological process that had both mental and physiologic manifestations. Essentially, he used the term *libido* to refer to "the force by which the sexual instinct is represented in mind." Thus, in its accepted sense, libido refers specifically to the mental manifestations of the sexual instinct. Freud recognized early that the sexual instinct did not originate in a finished or final form, as represented by genital primacy. Instead, it underwent a complex process of development, at each phase of which the libido had specific aims and objects that diverged in varying degrees from the simple aim of genital union. The libido theory thus came to include all of these manifestations and the complicated paths they followed in the course of psychosexual development.

EGO INSTINCTS. From 1905 on, Freud maintained a dual instinct theory, subsuming sexual instincts and ego instincts connected with self-preservation. Until 1914, with the publication of *On Narcissism*, Freud had paid little attention to ego instincts; in this communication, however, Freud invested ego instinct with libido for the first time by postulating an ego libido and an object libido. Freud thus viewed narcissistic investment as an essentially libidinal instinct and called the remaining nonsexual components the *ego instincts.*

AGGRESSION. When psychoanalysts today discuss the dual instinct theory,

they are generally referring to libido and aggression. Freud, however, originally conceptualized aggression as a component of the sexual instincts in the form of sadism. As he became aware that sadism had nonsexual aspects, he made finer gradations, which enabled him to categorize aggression and hate as part of the ego instincts and the libidinal aspects of sadism as components of the sexual instincts. Finally, in 1923, to account for the clinical data he was observing, he was compelled to conceive of aggression as a separate instinct in its own right. The source of this instinct, according to Freud, was mostly in skeletal muscles, and the aim of the aggressive instincts was destruction.

LIFE AND DEATH INSTINCTS. Before designating aggression as a separate instinct, Freud, in 1920, subsumed the ego instincts under a broader category of life instincts. These were juxtaposed with death instincts and were referred to as *Eros* and *Thanatos* in *Beyond the Pleasure Principle.* Life and death instincts were regarded as forces underlying sexual and aggressive instincts. Although Freud could not provide clinical data that directly verified the death instinct, he thought the instinct could be inferred by observing *repetition compulsion,* a person's tendency to repeat past traumatic behavior. Freud thought that the dominant force in biologic organisms had to be the death instinct. In contrast to the death instinct, eros (the life instinct) refers to the tendency of particles to reunite or bind to one another, as in sexual reproduction. The prevalent view today is that the dual instincts of sexuality and aggression suffice to explain most clinical phenomena without recourse to a death instinct.

Pleasure and Reality Principles. In 1911, Freud described two basic tenets of mental functioning: the pleasure principle and the reality principle. He essentially recast the primary process and secondary process dichotomy into the pleasure and reality principles and thus took an essential step toward solidifying the notion of the ego. Both principles, in Freud's view, are aspects of ego functioning. The *pleasure principle* is defined as an inborn tendency of the organism to avoid pain and to seek pleasure through the discharge of tension. The *reality principle,* on the other hand, is considered to be a learned function closely related to the maturation of the ego; this principle modifies the pleasure principle and requires delay or postponement of immediate

gratification.

Infantile Sexuality. Freud set forth the three central tenets of psychoanalytic theory when he published *Three Essays on the Theory of Sexuality.* First, he broadened the definition of sexuality to include forms of pleasure that transcend genital sexuality. Second, he established a developmental theory of childhood sexuality that delineated the vicissitudes of erotic activity from birth through puberty. Third, he forged a conceptual linkage between neuroses and perversions.

Freud's notion that sexual drives influence children has made some persons reluctant to accept psychoanalysis. Freud noted that infants are capable of erotic activity from birth, but the earliest manifestations of infantile sexuality are nonsexual and are associated with such bodily functions as feeding and bowel–bladder control. As libidinal energy shifts from the oral zone to the anal zone to the phallic zone, each stage of development is thought to build on and to subsume the accomplishments of the preceding stage. The *oral stage,* which occupies the first 12 to 18 months of life, centers on the mouth and lips and is manifested in chewing, biting, and sucking. The dominant erotic activity of the anal stage, from 18 to 36 months of age, involves bowel function and control. The *phallic stage,* from 3 to 5 years of life, initially focuses on urination as the source of erotic activity. Freud suggested that phallic erotic activity in boys is a preliminary stage leading to adult genital activity. Whereas the penis remains the principal sexual organ throughout male psychosexual development, Freud postulated that females have two principal erotogenic zones: the vagina and the clitoris. He thought that the clitoris was the chief erotogenic focus during the infantile genital period, but that erotic primacy shifted to the vagina after puberty. Studies of human sexuality have subsequently questioned the validity of this distinction.

Freud discovered that in the psychoneuroses, only a limited number of the sexual impulses that had undergone repression and were responsible for creating and maintaining the neurotic symptoms were normal. For the most part, these were the same impulses that were given overt expression in the perversions. The neuroses, then, were the negative of perversions.

Object Relationships in Instinct Theory. Freud suggested that the

choice of a love object in adult life, the love relationship itself, and the nature of all other object relationships depend primarily on the nature and quality of children's relationships. In describing the libidinal phases of psychosexual development, Freud repeatedly referred to the significance of a child's relationships with parents and other significant persons in the environment.

The awareness of the external world of objects develops gradually in infants. Soon after birth, they are primarily aware of physical sensations, such as hunger, cold, and pain, which give rise to tension, and caregivers are regarded primarily as persons who relieve their tension or remove painful stimuli. Recent infant research, however, suggests that awareness of others begins much sooner than Freud initially thought. Table 34-10 summarizes the stages of psychosexual development and the object relationships associated with each stage. Although the table goes only as far as young adulthood, development is now recognized as continuing throughout adult life.

Concept of Narcissism. According to Greek myth, Narcissus, a beautiful youth, fell in love with his reflection in the water of a pool and drowned in his attempt to embrace his beloved image. Freud used the term *narcissism* to describe situations in which an individual's libido was invested in the ego itself rather than other persons. This concept of narcissism presented him with vexing problems for his instinct theory and essentially violated his distinction between libidinal instincts and ego or self-preservative instincts. Freud's understanding of narcissism led him to use the term to describe a wide array of psychiatric disorders, very much in contrast to the term's contemporary use to describe a specific personality disorder. Freud grouped several disorders as the narcissistic neuroses, in which a person's libido is withdrawn from objects and turned inward. He believed that this withdrawal of libidinal attachment to objects accounted for the loss of reality testing in psychotic patients; grandiosity and omnipotence in such patients reflected excessive libidinal investment in the ego.

Freud did not limit his use of narcissism to psychoses. In states of physical illness and hypochondriasis, he observed that libidinal investment was frequently withdrawn from external objects and outside activities and interests. Similarly, he suggested that in normal sleep, libido was also withdrawn and reinvested in a sleeper's own body. Freud regarded homosexuality as an instance of a narcissistic form of object choice, in which

persons fall in love with an idealized version of themselves projected onto another person. He also found narcissistic manifestations in the beliefs and myths of primitive people, especially those involving the ability to influence external events through the magical omnipotence of thought processes. In the course of normal development, children also exhibit this belief in their omnipotence.



Table 34-10
Stages of Psychosexual Development

Oral Stage	
Definition	Earliest stage of development in which the infant's needs, perceptions, and modes of expression are primarily centered in mouth, lips, tongue, and other organs related to oral zone and around the sucking reflex.
Description	Oral zone maintains dominance in psychic organization through approximately first 18 mo of life. Oral sensations include thirst, hunger, pleasurable tactile stimulations evoked by the nipple or its substitute, and sensations related to swallowing and satiation. Oral drives consist of two components: libidinal and aggressive. States of oral tension lead to seeking oral gratification, as in quiescence at the end of nursing. Oral triad consists of wishes to eat, sleep, and reach that relaxation that occurs at the end of sucking just before onset of sleep. Libidinal needs (oral erotism) predominate in early oral phase, whereas they are mixed with more aggressive components later (oral sadism). Oral aggression is expressed in biting, chewing, spitting, or crying. Oral aggression is connected with primitive wishes and fantasies of biting, devouring, and destroying.

Objectives	To establish a trusting dependence on nursing and sustaining objects, establish comfortable expression and gratification of oral libidinal needs without excessive conflict or ambivalence from oral sadistic wishes.
Pathologic traits	Excessive oral gratifications or deprivation can result in libidinal fixations contributing to pathologic traits. Such traits can include excessive optimism, narcissism, pessimism (as in depressive states), or demandingness. Envy and jealousy are often associated with oral traits.
Character traits	Successful resolution of the oral phase results in capacities to give to and receive from others without excessive dependence or envy and to rely on others with a sense of trust as well as with a sense of self-reliance and self-trust. Oral characters are often excessively dependent and require others to give to them and look after them and are often extremely dependent on others for maintaining self-esteem. These are readily amalgamated with narcissistic needs.
Anal Stage	
Definition	The stage of psychosexual development promoted by maturation of neuromuscular control over sphincters, particularly the anal sphincter, permitting greater voluntary control over retention or expulsion of feces.
Description	This period extends roughly from 1 to 3 yr of age, marked by recognizable intensification of aggressive drives mixed with libidinal components in sadistic impulses. Acquisition of voluntary sphincter control is associated with an increasing shift from passivity to activity. Conflicts over anal

	control and struggles with parents over retaining or expelling feces in toilet training give rise to increased ambivalence, together with conflicts over separation, individuation, and independence. Anal erotism refers to sexual pleasure in anal functioning, both in retaining precious feces and presenting them as a precious gift to the parent. Anal sadism refers to expression of aggressive wishes connected with discharging feces as powerful and destructive weapons. These wishes are often displayed in fantasies of bombing or explosions.
Objectives	The anal period is marked by greater striving for independence and separation from dependence on and control of parents. Objectives of sphincter control without overcontrol (fecal retention) or loss of control (messing) are matched by attempts to achieve autonomy and independence without excessive shame or self-doubt from loss of control.
Pathologic traits	Maladaptive character traits, often apparently inconsistent, derive from anal erotism and defenses against it. Orderliness, obstinacy, stubbornness, willfulness, frugality, and parsimony are features of anal character. When defenses against anal traits are less effective, anal character reveals traits of heightened ambivalence, lack of tidiness, messiness, defiance, rage, and sadomasochistic tendencies. Anal characteristics and defenses are typically seen in obsessive-compulsive neuroses.
Character traits	Successful resolution of the anal phase provides the basis for development of personal autonomy, a capacity for independence and personal initiative without guilt, a capacity for self-determining

	behavior without a sense of shame or self-doubt, a lack of ambivalence, and a capacity for willing cooperation without either excessive willfulness or self-diminution or defeat.
Urethral Stage	
Definition	This stage was not explicitly treated by Freud but serves as a transitional stage between anal and phallic stages. It shares some characteristics of anal phase and some from subsequent phallic phase.
Description	Characteristics of the urethral phase are often subsumed under phallic phase. Urethral erotism, however, refers to pleasure in urination as well as pleasure in urethral retention analogous to anal retention. Similar issues of performance and control are related to urethral functioning. Urethral functioning may also have sadistic quality, often reflecting persistence of anal sadistic urges. Loss of urethral control, as in enuresis, may frequently have regressive significance that reactivates anal conflicts.
Objectives	At stake are issues of control and urethral performance and loss of control. It is not clear whether or to what extent objectives of urethral functioning differ from those of anal period, except that they are expressed in a later developmental stage.
Pathologic traits	The predominant urethral trait is competitiveness and ambition, probably related to compensation for shame due to loss of urethral control. This may instigate development of penis envy, related to feminine sense of shame and inadequacy in being unable to match male urethral performance. This

	may also be related to issues of control and shaming.
Character traits	Besides healthy effects analogous to those from the anal period, urethral competence provides a sense of pride and self-competence based on performance. Urethral performance is an area in which the small boy can imitate and try to match his father's more adult performance. Resolution of urethral conflicts sets the stage for budding gender identity and subsequent identifications.
Phallic Stage	
Definition	Phallic stage begins sometime during year 3 and continues until approximately the end of year 5.
Description	The phallic phase is characterized by a primary focusing of sexual interests, stimulation, and excitement in the genital area. The penis becomes the organ of principal interest to children of both sexes, with lack of penis in females being considered as evidence of castration. The phallic phase is associated with an increase in genital masturbation accompanied by predominantly unconscious fantasies of sexual involvement with the opposite-sex parent. Threats of castration and the related anxiety are connected with guilt over masturbation and oedipal wishes. During this phase oedipal involvement and conflict are established and consolidated.
Objectives	To focus erotic interest in genital area and genital functions. This lays the foundation for gender identity and serves to integrate residues of previous stages into a predominantly genital–sexual orientation. Establishing the oedipal situation is essential for furtherance of subsequent

	identifications, serving as a basis for important and perduring dimensions of character organization.
Pathologic traits	Derivation of pathologic traits from phallic–oedipal involvement is sufficiently complex and subject to such a variety of modifications that it encompasses nearly the whole of neurotic development. Issues, however, focus on castration in males and penis envy in females. Patterns of internalization developed from resolution of the Oedipus complex provide another important focus of developmental distortions. The influence of castration anxiety and penis envy, defenses against them, and patterns of identification are primary determinants of the development of human character. They also subsume and integrate residues of previous psychosexual stages, so that fixations or conflicts deriving from preceding stages can contaminate and modify oedipal resolution.
Character traits	The phallic stage provides the foundation for an emerging sense of sexual identity, curiosity without embarrassment, initiative without guilt, as well as mastery not only over objects and persons in the environment but also over internal processes and impulses. Resolution of the oedipal conflict gives rise to internal structural capacities for regulation of drive impulses and their direction to constructive ends. The internal sources of such regulation are the ego and superego, based on introjections and identifications derived primarily from parental figures.
Latency Stage	
Definition	This is the stage of relative instinctual quiescence or inactivity of sexual drive during the period from the

	resolution of the Oedipus complex until pubescence (from about 5–6 yr until about 11–13 yr).
Description	The institution of the superego at the close of the oedipal period and further maturation of ego functions allow for considerably greater degrees of control of instinctual impulses and motives. Sexual interests are generally thought to be quiescent. This is a period of primarily homosexual affiliations for both boys and girls, as well as a sublimation of libidinal and aggressive energies into energetic learning and play activities, exploring the environment, and becoming more proficient in dealing with the world of things and persons around them. It is a period for development of important skills. The relative strength of regulatory elements often gives rise to patterns of behavior that are somewhat obsessive and hypercontrolling.
Objectives	The primary objective is further integration of oedipal identifications and consolidation of gender and sex-role identity. Relative quiescence and control of instinctual impulses allow for development of ego apparatuses and mastery of skills. Further identificatory components may be added to the oedipal ones on the basis of broadening contacts with other significant figures outside the family (e.g., teachers, coaches, and other adult figures).
Pathologic traits	Dangers in the latency period can arise either from the lack of development of inner controls or an excess of them. Lack of control can lead to failure to sufficiently sublimate energies in the interest of learning and the development of skills; an excess of inner control, however, can lead to premature closure of personality development and precocious

	elaboration of obsessive character traits.
Character traits	The latency period is frequently regarded as a period of relatively unimportant inactivity in the developmental schema. More recently, greater respect has been gained for the developmental processes in this period. Important consolidations and additions are made to basic postoedipal identifications and to processes of integrating and consolidating previous attainments in psychosexual development and establishing decisive patterns of adaptive functioning. The child can develop a sense of industry and capacity for mastery of objects and concepts that allows autonomous functioning and a sense of initiative without risk of failure or defeat or a sense of inferiority. These are all important attainments that need to be further integrated, ultimately as the essential basis for a mature adult life of satisfaction in work and love.
Genital Stage	
Definition	The genital or adolescent phase extends from the onset of puberty from approximately ages 11–13 until young adulthood. Current thinking tends to subdivide this stage into preadolescent, early adolescent, middle adolescent, late adolescent, and even postadolescent periods.
Description	Physiologic maturation of systems of genital (sexual) functioning and attendant hormonal systems leads to intensification of instinctual, particularly libidinal, drives. This produces a regression in personality organization, which reopens conflicts of previous stages of psychosexual development and provides opportunity for re-resolution of these conflicts in the context of achieving a mature sexual and adult

	identity. This period has been described as a "second individuation."
Objectives	Primary objectives are the ultimate separation from dependence on and attachment to parents and establishment of mature, nonincestuous, heterosexual object relations. Related are the achievement of a mature sense of personal identity and acceptance and integration of adult roles and functions that permit new adaptive integrations with social expectations and cultural values.
Pathologic traits	Pathologic deviations due to failure to achieve successful resolution of this stage of development are multiple and complex. Defects can arise from a whole spectrum of psychosexual residues, since the developmental task of adolescence is in a sense a partial reopening, reworking, and reintegration of all of these aspects of development. Previous unsuccessful resolutions and fixations in various phases or aspects of psychosexual development will produce pathologic defects in the emerging adult personality and defects in identity formation.
Character traits	Successful resolution and reintegration of previous psychosexual stages in the adolescent genital phase set the stage normally for a fully mature personality with the capacity for full and satisfying genital potency and a self-integrated and consistent sense of identity. This provides the basis for a capacity for self-realization and meaningful participation in areas of work, love, and in creative and productive application to satisfying and meaningful goals and values.

Freud postulated a state of primary narcissism at birth in which the libido is stored in the ego. He viewed the neonate as completely narcissistic, with the entire libidinal investment in physiologic needs and satisfaction. He referred to this self-investment as *ego libido.* The infantile state of self-absorption changes only gradually, according to Freud, with the dawning awareness that a separate person—the mothering figure—is responsible for gratifying an infant's needs. This realization leads to the gradual withdrawal of the libido from the self and its redirection toward the external object. Hence, the development of object relations in infants parallels the shift from primary narcissism to object attachment. The libidinal investment in the object is referred to as *object libido.* If a developing child suffers rebuffs or trauma from the caretaking figure, object libido may be withdrawn and reinvested in the ego. Freud called this regressive posture *secondary narcissism.*

Freud used the term *narcissism* to describe many different dimensions of human experience. At times, he used it to describe a perversion in which persons used their bodies or body parts as objects of sexual arousal. At other times, he used the term to describe a developmental phase, as in primary narcissism. In still other instances, the term referred to a particular object choice. Freud distinguished love objects who are chosen "according to the narcissistic type," in which case the object resembles the subject's idealized or fantasied self-image, from objects chosen according to the "anaclitic," in which the love object resembles a caretaker from early in life. Finally, Freud also used the word *narcissism* interchangeably and synonymously with *self-esteem.*

Ego Psychology

Although Freud had used the construct of the ego throughout the evolution of psychoanalytic theory, ego psychology, as it is known today, really began with the publication in 1923 of *The Ego and the Id.* This landmark publication also represented a transition in Freud's thinking from the topographical model of the mind to the tripartite structural model of ego, id, and superego. He had observed repeatedly that not all unconscious processes could be relegated to a person's instinctual life. Elements of the conscience, as well as functions of the ego, are also unconscious.

Structural Theory of the Mind. The structural model of the psychic apparatus is the cornerstone of ego psychology. The three provinces—id, ego, and superego—are distinguished by their different functions.

ID. Freud used the term *id* to refer to a reservoir of unorganized instinctual drives. Operating under the domination of the primary process, the id cannot delay or modify the instinctual drives with which an infant is born. The id, however, should not be viewed as synonymous with the unconscious, because both the ego and the superego have unconscious components.

EGO. The ego spans all three topographical dimensions of conscious, preconscious, and unconscious. Logical and abstract thinking and verbal expression are associated with conscious and preconscious functions of the ego. Defense mechanisms reside in the unconscious domain of the ego. The ego, the executive organ of the psyche, controls motility, perception, contact with reality, and, through the defense mechanisms available to it, the delay and modulation of drive expression.

Freud believed that the id is modified due to the impact of the external world on the drives. The pressures of external reality enable the ego to appropriate the energies of the id to do its work. As the ego brings influences from the external world to bear on the id, it simultaneously substitutes the reality principle for the pleasure principle. Freud emphasized the role of conflict within the structural model and observed that conflict occurs initially between the id and the outside world, only to be transformed later to conflict between the id and the ego.

The third component of the tripartite structural model is the superego. The superego establishes and maintains an individual's moral conscience based on a complex system of ideals and values internalized by parents. Freud viewed the superego as the heir to the Oedipus complex. Children internalize parental values and standards at about the age of 5 or 6 years. The superego then serves as an agency that provides ongoing scrutiny of a person's behavior, thoughts, and feelings; it makes comparisons with expected standards of behavior and offers approval or disapproval. These activities occur mostly unconsciously.

The ego-ideal is often regarded as a component of the superego. It is an agency that prescribes what a person should do according to internalized

standards and values. The superego, by contrast, is an agency of moral conscience that *proscribes*—that is, dictates what a person should not do. Throughout the latency period and after that, persons continue to build on early identifications through their contact with admired figures who contribute to the formation of moral standards, aspirations, and ideals.

Functions of the Ego. Modern ego psychologists have identified a set of essential ego functions that characterizes the operations of the ego. These descriptions reflect the ego activities that are generally regarded as fundamental.

CONTROL AND REGULATION OF INSTINCTUAL DRIVES. The development of the capacity to delay or postpone drive discharge, like the capacity to test reality, is closely related to the early childhood progression from the pleasure principle to the reality principle. This capacity is also an essential aspect of the ego's role as a mediator between the id and the outside world. Part of infants' socialization to the external world is acquiring language and secondary process or logical thinking.

JUDGMENT. A closely related ego function is judgment, which involves the ability to anticipate the consequences of actions. As with the control and regulation of instinctual drives, judgment develops in parallel with the growth of *secondary process thinking.* The ability to think logically allows assessment of how contemplated behavior may affect others.

RELATION TO REALITY. The mediation between the internal world and external reality is a crucial function of the ego. Relations with the outside world can be divided into three aspects: the sense of reality, reality testing, and adaptation to reality. The *sense of reality* develops in concert with an infant's dawning awareness of bodily sensations. The ability to distinguish outside the body from the inside is an essential aspect of the sense of reality, and disturbances of body boundaries, such as depersonalization, reflect impairment in this ego function. *Reality testing,* an ego function of paramount importance, refers to the capacity to distinguish internal fantasy from external reality. This function differentiates persons who are psychotic from those who are not. *Adaptation to reality* involves persons' ability to use their

resources to develop effective responses to changing circumstances based on previous experience with reality.



Table 34-11
Parallel Lines of Development

Instinctual Phases	Separation–Individuation	Object Relations	Psychosocial Crises
Oral	Autism, symbiosis	Primary narcissism, need-satisfying	Trust or mistrust
Anal	Differentiation, practicing, rapprochement	Need-satisfying, object constancy	Autonomy or shame, self-doubt
Phallic	Object constancy, Oedipal complex	Object constancy, ambivalence	Initiative or guilt
Latency	—	—	Industry or inferiority
Adolescence	Genitality, secondary individuation	Object love	Identity or identity confusion
Adulthood	Mature genitality	—	Intimacy or isolation, generativity or stagnation, integrity or despair

OBJECT RELATIONSHIPS. The capacity to form mutually satisfying relationships is related in part to patterns of internalization stemming from

early interactions with parents and other significant figures. This ability is also a fundamental function of the ego, in that satisfying relatedness depends on the ability to integrate positive and negative aspects of others and self and to maintain an internal sense of others even in their absence. Similarly, mastery of drive derivatives is also crucial to the achievement of satisfying relationships. Although Freud did not develop an extensive object relations theory, British psychoanalysts, such as Ronald Fairbairn (1889–1964) and Michael Balint (1896–1970), elaborated extensively on the early stages in infants' relationships with need-satisfying objects and on the gradual development of a sense of separateness from the mother. Another of their British colleagues, Donald W. Winnicott (1896–1971), described the *transitional object* (e.g., a blanket, teddy bear, or pacifier) as the link between developing children and their mothers. A child can separate from the mother because a transitional object provides feelings of security in her absence. The stages of human development and object relations theory are summarized in Table 34-11.

SYNTHETIC FUNCTION OF THE EGO. First described by Herman Nunberg in 1931, the *synthetic function* refers to the ego's capacity to integrate diverse elements into an overall unity. Different aspects of self and others, for example, are synthesized into a consistent representation that endures over time. The function also involves organizing, coordinating, and generalizing or simplifying large amounts of data.

PRIMARY AUTONOMOUS EGO FUNCTIONS. Heinz Hartmann described the so-called primary autonomous functions of the ego as rudimentary apparatuses present at birth that develop independently of intrapsychic conflict between drives and defenses. These functions include perception, learning, intelligence, intuition, language, thinking, comprehension, and motility. In the course of development, some of these conflict-free aspects of the ego may eventually become involved in a conflict. They will develop normally if the infant is raised in what Hartmann referred to as an *average expectable environment.*

SECONDARY AUTONOMOUS EGO FUNCTIONS. Once the sphere where primary autonomous function develops becomes involved with conflict, the so-called

secondary autonomous ego functions arise in the defense against drives. For example, a child may develop caretaking functions as a reaction formation against murderous wishes during the first few years. Later, the defensive functions may be neutralized or deinstinctualized when the child grows up to be a social worker and cares for homeless persons.

Defense Mechanisms. At each phase of libidinal development, specific drive components evoke characteristic ego defenses. The anal phase, for example, is associated with reaction formation, as manifested by the development of shame and disgust concerning anal impulses and pleasures.

Defenses can be grouped hierarchically according to the relative degree of maturity associated with them. Narcissistic defenses are the most primitive and appear in children and persons who are psychotically disturbed. Immature defenses are seen in adolescents and some nonpsychotic patients. Neurotic defenses are encountered in obsessive-compulsive and hysterical patients as well as in adults under stress. Table 34-12 lists the defense mechanisms, according to George Valliant's classification of the four types.

Theory of Anxiety. Freud initially conceptualized anxiety as "dammed-up libido." Essentially, a physiologic increase in sexual tension leads to a corresponding increase in libido, the mental representation of the physiologic event. This buildup causes the actual neuroses. Later, with the development of the structural model, Freud developed a new theory of a second type of anxiety that he referred to as *signal anxiety.* In this model, anxiety operates at an unconscious level and serves to mobilize the ego's resources to avert danger. Either external or internal sources of danger can produce a signal that leads the ego to marshal specific defense mechanisms to guard against or reduce instinctual excitation.

Freud's later theory of anxiety explains neurotic symptoms as the ego's partial failure to cope with distressing stimuli. The drive derivatives associated with danger may not have been adequately contained by the defense mechanisms used by the ego. In phobias, for example, Freud explained that fear of an external threat (e.g., dogs or snakes) is an externalization of internal danger.

Danger situations can also be linked to developmental stages and can create a developmental hierarchy of anxiety. The earliest dangerous situation

is a fear of disintegration or annihilation, often associated with concerns about fusion with an external object. As infants mature and recognize the mothering figure as a separate person, separation anxiety, or fear of the loss of an object, becomes more prominent. During the oedipal psychosexual stage, girls are most concerned about losing the love of the most crucial figure in their lives, their mother. Boys are primarily anxious about bodily injury or castration. After resolving the oedipal conflict, a more mature form of anxiety occurs, often termed *superego anxiety*. This latency-age concern involves the fear that internalized parental representations, contained in the superego, will cease to love, or will angrily punish, the child.



Table 34-12
Classification of Defense Mechanisms

Narcissistic-Psychotic Defenses	
These defenses are usually found as part of a psychotic process, but may also occur in young children and adult dreams or fantasies. They share the common note of avoiding, negating, or distorting reality.	
Projection	Perceiving and reacting to unacceptable inner impulses and their derivatives as though they were outside the self. On a psychotic level, this takes the form of frank delusions about external reality, usually persecutory, includes both perception of one's own feelings and those of another with subsequent acting on the perception (psychotic paranoid delusions). Impulses may derive from id or superego (hallucinated recriminations).
Denial	Psychotic denial of external reality, unlike repression, affects perception of external reality more than perception of internal reality. Seeing, but refusing to acknowledge what one sees, or hearing and negating what is actually heard are

	examples of denial and exemplify the close relationship of denial to sensory experience. Not all denial, however, is necessarily psychotic. Like projection, denial may function in the service of more neurotic or even adaptive objectives. Denial avoids becoming aware of some painful aspect of reality. At the psychotic level, the denied reality may be replaced by a fantasy or delusion.
Distortion	Grossly reshaping the experience of external reality to suit inner needs, including unrealistic megalomanic beliefs, hallucinations, wish-fulfilling delusions, and employing sustained feelings of delusional grandiosity, superiority, or entitlement.

Immature Defenses

These mechanisms are fairly common in preadolescent years and in adult character disorders. They are often mobilized by anxieties related to intimacy or its loss. Although they are regarded as socially awkward and undesirable, they often moderate with improvement in interpersonal relationships or with increased personal maturity.

Acting out	The direct expression of an unconscious wish or impulse in action to avoid being conscious of the accompanying affect. The unconscious fantasy, involving objects, is lived out and impulsively enacted in behavior, thus gratifying the impulse more than the prohibition against it. On a chronic level, acting out involves giving in to impulses to avoid the tension that would result from postponement of their expression.
Blocking	An inhibition, usually temporary in nature, of affects especially, but possibly also thinking

	and impulses. It is close to repression in its effects but has a component of tension arising from the inhibition of the impulse, affect, or thought.
Hypochondriasis	Transformation of reproach toward others arising from bereavement, loneliness, or unacceptable aggressive impulses, into self-reproach in the form of somatic complaints of pain, illness, and so forth. Real illness may also be overemphasized or exaggerated for its evasive and regressive possibilities. Thus, responsibility may be avoided, guilt may be circumvented, and instinctual impulses may be warded off.
Introjection	In addition to the developmental functions of the process of introjection, it also can serve specific defensive functions. The introjection of a loved object involves the internalization of characteristics of the object with the goal of ensuring closeness to and constant presence of the object. Anxiety consequent to separation or tension arising out of ambivalence toward the object is thus diminished. If the object is lost, introjection nullifies or negates the loss by taking on characteristics of the object, thus in a sense internally preserving the object. Even if the object is not lost, the internalization usually involves a shift of cathexis reflecting a significant alteration in the object relationship. Introjection of a feared object serves to avoid anxiety through internalizing the aggressive characteristic of the object, thereby putting the aggression under one's own control. The aggression is no longer felt as coming from outside, but is taken within and utilized defensively, thus turning the subject's weak,

	passive position into an active, strong one. The classic example is “identification with the aggressor.” Introjection can also take place out of a sense of guilt in which the self-punishing introject is attributable to the hostile-destructive component of an ambivalent tie to an object. Thus, the self-punitive qualities of the object are taken over and established within one’s self as a symptom or character trait, which effectively represents both the destruction and the preservation of the object. This is also called *identification with the victim.*
Passive-aggressive behavior	Aggression toward an object expressed indirectly and ineffectively through passivity, masochism, and turning against the self.
Projection	On a nonpsychotic level, projection involves attributing one’s own unacknowledged feelings to others; it includes severe prejudice, rejection of intimacy through suspiciousness, hypervigilance to external danger, and injustice collecting. Projection operates correlatively to introjection, such that the material of the projection derives from the internalized but usually unconscious configuration of the subject’s introjects. At higher levels of function, projection may take the form of misattributing or misinterpreting motives, attitudes, feelings, or intentions of others.
Regression	A return to a previous stage of development or functioning to avoid the anxieties or hostilities involved in later stages. A return to earlier points of fixation embodying modes of behavior previously given up. This is often the result of a disruption of equilibrium at a later phase of

	development. This reflects a basic tendency to achieve instinctual gratification or to escape instinctual tension by returning to earlier modes and levels of gratification when later and more differentiated modes fail or involve intolerable conflict.
Schizoid fantasy	The tendency to use fantasy and to indulge in autistic retreat for the purpose of conflict resolution and gratification.
Somatization	The defensive conversion of psychic derivatives into bodily symptoms; tendency to react with somatic rather than psychic manifestations. Infantile somatic responses are replaced by thought and affect during development (desomatization); regression to earlier somatic forms or response (resomatization) may result from unresolved conflicts and may play an important role in psychophysiological and psychosomatic reactions.

Neurotic Defenses

These are common in apparently normal and healthy individuals as well as in neurotic disorders. They function usually in the alleviation of distressing affects and may be expressed in neurotic forms of behavior. Depending on circumstances, they can also have an adaptive or socially acceptable aspect.

Controlling	The excessive attempt to manage or regulate events or objects in the environment in the interest of minimizing anxiety and solving internal conflicts.
Displacement	Involves a purposeful, unconscious shifting of impulses or affective investment from one object to another in the interest of solving a

	conflict. Although the object is changed, the instinctual nature of the impulse and its aim remain unchanged.
Dissociation	A temporary but drastic modification of character or sense of personal identity to avoid emotional distress; it includes fugue states and hysterical conversion reactions.
Externalization	A general term, correlative to internalization, referring to the tendency to perceive in the external world and in external objects components of one's own personality, including instinctual impulses, conflicts, moods, attitudes, and styles of thinking. It is a more general term than projection, which is defined by its derivation from and correlation with specific introjects.
Inhibition	The unconsciously determined limitation or renunciation of specific ego functions, singly or in combination, to avoid anxiety arising out of conflict with instinctual impulses, superego, or environmental forces or figures.
Intellectualization	The control of affects and impulses by way of thinking about them instead of experiencing them. It is a systematic excess of thinking, deprived of its affect, to defend against anxiety caused by unacceptable impulses.
Isolation	The intrapsychic splitting or separation of affect from content resulting in repression of either idea or affect or the displacement of affect to a different or substitute content.
Rationalization	A justification of attitudes, beliefs, or behavior that might otherwise be unacceptable by an incorrect application of justifying reasons or the

	invention of a convincing fallacy.
Reaction formation	The management of unacceptable impulses by permitting expression of the impulse in antithetical form. This is equivalently an expression of the impulse in the negative. Where instinctual conflict is persistent, reaction formation can become a character trait on a permanent basis, usually as an aspect of obsessional character.
Repression	Consists of the expelling and withholding from conscious awareness of an idea or feeling. It may operate either by excluding from awareness what was once experienced on a conscious level (secondary repression) or it may curb ideas and feelings before they have reached consciousness (primary repression). The “forgetting” associated with repression is unique in that it is often accompanied by highly symbolic behavior, which suggests that the repressed is not really forgotten. The important discrimination between repression and the more general concept of defense has been discussed.
Sexualization	The endowing of an object or function with sexual significance that it did not previously have, or possesses to a lesser degree, to ward off anxieties connected with prohibited impulses.

Mature Defenses

These mechanisms are healthy and adaptive throughout the life cycle. They are socially adaptive and useful in the integration of personal needs and motives, social demands, and interpersonal relations. They can underlie seemingly admirable and virtuous patterns of behavior.

Altruism	The vicarious but constructive and instinctually gratifying service to others, even to the detriment of the self. This must be distinguished from altruistic surrender, which involves a masochistic surrender of direct gratification or of instinctual needs in favor of fulfilling the needs of others to the detriment of the self, with vicarious satisfaction only being gained through introjection.
Anticipation	The realistic anticipation of or planning for future inner discomfort: Implies overly concerned planning, worrying, and anticipation of dire and dreadful possible outcomes.
Asceticism	The elimination of directly pleasurable affects attributable to an experience. The moral element is implicit in setting values on specific pleasures. Asceticism is directed against all “base” pleasures perceived consciously, and gratification is derived from the renunciation.
Humor	The overt expression of feelings without personal discomfort or immobilization and without unpleasant effect on others. Humor allows one to bear, and yet focus on, what is too terrible to be borne, in contrast to wit, which always involves distraction or displacement away from the affective issue.
Sublimation	The gratification of an impulse whose goal is retained but whose aim or object is changed from a socially objectionable one to a socially valued one. Libidinal sublimation involves a desexualization of drive impulses and the placing of a value judgment that substitutes what is valued by the superego or society.

	Sublimation of aggressive impulses takes place through pleasurable games and sports. Unlike neurotic defenses, sublimation allows instincts to be channeled rather than dammed up or diverted. Thus, in sublimation, feelings are acknowledged, modified, and directed toward a relatively significant person or goal so that modest instinctual satisfaction results.
Suppression	The conscious or semiconscious decision to postpone attention to a conscious impulse or conflict.

Adapted from Vaillant GE. *Adaptation to Life*. Boston, MA: Little Brown; 1977; Semrad E. The operation of ego defenses in object loss. In: Moriarity DM, ed. *The Loss of Loved Ones*. Springfield, IL: Charles C Thomas; 1967; and Bibring GL, Dwyer TF, Huntington DS, Valenstein AA. A study of the psychological principles in pregnancy and of the earliest mother–child relationship: Methodological considerations. *Psychoanal Stud Child*. 1961;16:25.

Character. In 1913, Freud distinguished between neurotic symptoms and personality or character traits. *Neurotic symptoms* develop due to the failure of repression; *character traits* owe their existence to the success of repression, that is, to the defense system that achieves its aim through a persistent pattern of reaction formation and sublimation. In 1923, Freud also observed that the ego could only give up important objects by identifying with them or introjecting them. This accumulated pattern of identifications and introjections also contributes to character formation. Freud specifically emphasized the importance of superego formation in character construction.

Contemporary psychoanalysts regard character as a person's habitual or typical pattern of adaptation to internal drive forces and external environmental forces. *Character* and *personality* are used interchangeably and are distinguished from the ego in that they mostly refer to styles of defense and directly observable behavior rather than feeling and thinking.

Character is also influenced by constitutional temperament, the interaction of drive forces with early ego defenses and with environmental influences, and various identifications with, and internalizations of, other persons throughout life. The extent to which the ego has developed a capacity to tolerate the delay of impulse discharge and to neutralize instinctual energy

determines the degree to which such character traits emerge in later life. Exaggerated development of individual character traits at the expense of others can lead to personality disorders or produce a vulnerability or predisposition to psychosis.

Classic Psychoanalytic Theory of Neuroses

The classic view of the genesis of neuroses regards conflict as essential. The conflict can arise between instinctual drives and external reality or internal agencies, such as the id and the superego or the id and the ego. Moreover, because the conflict has not been worked through to a realistic solution, the drives or wishes that seek discharge have been expelled from consciousness through repression or another defense mechanism. Their expulsion from conscious awareness, however, does not make the drives any less powerful or influential. As a result, the unconscious tendencies (e.g., the disguised neurotic symptoms) fight their way back into consciousness. This theory of the development of neurosis assumes that a rudimentary neurosis based on the same type of conflict existed in early childhood.

Deprivation during the first few months of life because of absent or impaired caretaking figures can adversely affect ego development. This impairment, in turn, can fail to make appropriate identifications. The resulting ego difficulties create problems in mediating between the drives and the environment. Lack of capacity for the constructive expression of drives, especially aggression, can lead some children to turn their aggression on themselves and become overtly self-destructive. Parents who are inconsistent, excessively harsh, or overly indulgent can influence children to develop disordered superego functioning. The severe conflict that cannot be managed through symptom formation can lead to extreme restrictions in ego functioning and fundamentally impair the capacity to learn and develop new skills.

Traumatic events that seem to threaten survival can break through defenses when the ego has been weakened. More libidinal energy is then required to master the excitation that results. The libido thus mobilized, however, is withdrawn from the supply that is usually applied to external objects. This withdrawal further diminishes the strength of the ego and produces a sense of inadequacy. Frustrations or disappointments in adults may revive infantile

longings that are then dealt with through symptom formation or further regression.

In his classic studies, Freud described four different types of childhood neuroses, three of which had later neurotic developments in adult life. This well-known series of cases shown in tabulated form in Table 34-13 exemplifies some of Freud's important conclusions: (1) neurotic reactions in the adult are associated frequently with neurotic reactions in childhood; (2) the connection is sometimes continuous but more often is separated by a latent period of nonneurosis; and (3) infantile sexuality, both fantasized and real, occupies a memorable place in the early history of the patient.



Table 34-13
Classic Psychoneurotic Reactions of Childhood

	Conversion Reaction (Dora)	Phobic Reaction (Hans)	Obsessive-Compulsive Reaction (Rat Man)	Mixed Neurotic Reaction (Wolf Man)
Family history	Striking family history of psychiatric and physical illness	Both parents treated for neurotic conflict but not severe	No family history of mental illness	Striking family history of psychiatric and physical illness
Symptoms	Enuresis and masturbation, 6–8 yr; onset of neurosis at 8; migraine, nervous cough, and hoarseness at 12; aphonia at 16; "appendicitis" at 16; convulsions at 16; facial neuralgia at 19; change of personality at 8 from "wild creature" to quiet child	Compulsive questions at 3–3 1/2 yr in regard to sex difference; jealous reaction to sibling birth at 3 1/2; overt castration threat; overt masturbation at 3 1/2; overeating and constipation at 4–5; phobic reaction at 4–5; attack of flu at 5 worsens phobia; tonsillectomy at 5 worsens phobia	Naughty period at 3–4 yr; marked timidity after beating by father at 4; recognizing people by their smells as a child (Renifleur); precocious ego development; onset of obsessive ideas at 6–7	Tractable and quiet up to 3 1/4 yr; "naughty" period at 3 1/4–4 yr; phobias at 4–5 with nightmares; obsessional reaction at 6–7 (pious ceremonials). Disappearance of neuroses at 8
Causes	Seduction by older man; father's illness; father's affair	Seductive care by mother; sibling birth at 3 1/2	Seduction by governess at 4; death of sibling at 4; beating by father at 4	Seduction by older sister at 3 1/4; mother's illness; conflict between maid and governess

Courtesy of E. James Anthony, M.D.

Specific differences are worth noting in the four cases shown in Table 34-13. First, the phobic reactions tend to start at about 4 or 5 years of age, the obsessional reactions between 6 and 7 years, and the conversion reactions at 8 years. The degree of background disturbance is most significant in the conversion reaction and the mixed neurosis, and it seems only slight in the phobic and obsessional reactions. The course of the phobic reaction seems little influenced by severe traumatic factors, whereas traumatic factors, such as sexual seductions, play an essential role in the three other subgroups. It was during this period that Freud elaborated his seduction hypothesis for the cause of the neuroses, in terms of which the obsessive-compulsive and

hysterical reactions were alleged to originate in active and passive sexual experiences.

Treatment and Technique

The cornerstone of psychoanalytic technique is free association, in which patients say whatever comes to mind. Free association not only provides analytic content, it helps to induce the regression needed for establishing and working through the transference neurosis. When this development occurs, all the original wishes, drives, and defenses associated with the infantile neurosis are transferred to the person of the analyst.

As patients attempt to free associate, they soon learn that they have difficulty saying whatever comes to mind without censoring specific thoughts. They develop conflicts about their wishes and feelings toward the analyst that reflect childhood conflicts. The *transference* that develops toward the analyst may also serve as resistance to the process of free association. Freud discovered that *resistance* was not merely a stoppage of a patient's associations, but also an essential revelation of the patient's internal object relations as they were externalized and manifested in the transference relationship with the analyst. The systematic analysis of transference and resistance is the essence of psychoanalysis. Freud was also aware that the analyst might have transferences to the patient, which he called *countertransference*. Countertransference, in Freud's view, was an obstacle that the analyst needed to understand so that it did not interfere with treatment. In this spirit, he recognized the need for all analysts to have been analyzed themselves. Variations in transference and their descriptions are contained in Table 34-14. The basic mechanisms by which transferences are effected—displacement, projection, and projective identification—are described in Table 34-15.



Table 34-14
Transference Variants

Libidinal Transferences
Follow the classic model and usually in milder forms as positive *transference reactions* but can take the form of more intense and

disturbing *erotic transferences.* They are derivatives of phallic-oedipal, libidinal impulses and may be permeated variously by pregenital influences. They can occur with varying degrees of intensity, and in mild forms, may not even require interpretation if they contribute to and support the therapeutic relation. Sigmund Freud recommended that they call for interpretation only when they begin to serve as a resistance.

Aggressive Transferences

Take the form either of negative or more pathologic paranoid transferences. *Negative transferences* are seen at all levels of psychopathology but can predominate in some borderline patients who tend to see the therapeutic relationship in terms of power and victimization, regarding the therapist as omnipotent and powerful, whereas the patient experiences him- or herself as helpless, weak, and vulnerable. Negative transferences are identifiable in varying degrees in all analyses and usually require specific intervention and interpretation.

Transferences of Defense

Opposed to *transferences of impulse;* defense against impulses finds its way into the transference rather than the impulses themselves. In this form of transference, attention shifts from drives to the ego's defensive functioning so that transference is no longer merely repetition of instinctual cathexes but includes aspects of ego functioning as well.

Transference Neurosis

Involves the recreation or more ample expression of the patient's neurosis enacted anew within the analytical relation and at least theoretically mirroring aspects of the infantile neurosis. The transference neurosis usually develops in the middle phase of analysis, when the patient, at first eager for improved mental health, no longer consistently displays such motivation but engages in a continuing battle with the analyst over the desire to attain some kind of emotional satisfaction from the analyst so that this becomes the most compelling reason for continuing analysis.

At this point of the treatment, the transference emotions become more important to the patient than alleviation of distress sought initially, and the major, unresolved, unconscious problems of childhood begin to dominate the patient's behavior. They are now reproduced in the transference, with all their pent-up emotion.

The transference neurosis is governed by three outstanding characteristics of instinctual life in early childhood: the pleasure principle (before effective reality testing), ambivalence, and repetition compulsion. Emergence of the transference neurosis is usually a slow and gradual process, although in certain patients with a propensity for *transference regression,* particularly more hysterical patients, elements of transference and transference neurosis may manifest themselves relatively early in the analytical process. One situation after another in the life of the patient is analyzed and progressively interpreted until the original infantile conflict is sufficiently revealed. Only then does the transference neurosis begin to subside. At that point, termination begins to emerge as a more central concern.

Contemporary opinion is divided to its importance and centrality, whether it forms to the extent Freud believed, and whether it is necessary for successful analysis—for some, it remains an essential vehicle for analytical interpretation and therapeutic effectiveness; for others, it may never develop or, to the extent that it does, may play a less central role in the process of cure.

Transference Psychosis

Occurs when failure of reality testing leads to loss of self–object differentiation and diffusion of self and object boundaries. This may reflect an attempt to refuse with an omnipotent object, investing the self with omnipotent powers as defense against underlying fears of vulnerability and powerlessness. Transference psychosis can also include negative transference elements in which fusion carries the threat of engulfment and loss of self that may precipitate a *paranoid transference reaction.*

Narcissistic Transferences

Clarified by Heinz Kohut (1971) as variations of patterns of projection of archaic narcissistic configurations onto the therapist. They are based on projections of narcissistic introjective configurations, both superior and inferior—the superior form reflecting narcissistic superiority, grandiosity, and enhanced self-esteem, and the inferior opposite qualities of inferiority, self-depletion, and diminished self-esteem. The therapist comes to represent, in Kohut's terms, either the grandiose self in *mirror transferences* or the idealized parental imago in *idealizing transferences.* In idealizing transferences, all power and strength are attributed to the idealized object, leaving the subject feeling empty and powerless when separated from that object. Union with the idealized object enables the subject to regain narcissistic equilibrium. Idealizing transferences may reflect developmental disturbances in the idealized parent imago, particularly at the time of formation of the ego-ideal by introjection of the idealized object. In some individuals, narcissistic fixation leads to development of the grandiose self. Reactivation in analysis of the grandiose self provides the basis of mirror transferences formation, which occur in three forms: *archaic merger transference,* a less archaic *alter-ego* or *twinship transference,* and *mirror transference in the narrow sense.* In the most primitive merger transference, the analyst is experienced only as an extension of the subject's grandiose self and, thus, becomes the repository of the patient's grandiosity and exhibitionism. In the alter-ego or twinship transference, activation of the grandiose self leads to experience of the narcissistic object as similar to the grandiose self. In the most mature form of mirror transference, the analyst is experienced as a separate person but, nonetheless, one who becomes important to the patient and is accepted by him or her only to the degree that he or she is responsive to the narcissistic needs of the reactivated grandiose self.

Self–Object Transferences

Represent extensions of the self-psychology paradigm beyond merely narcissistic configurations. The self–object involves investment of the self in the object so that the object comes to serve a self-sustaining function that the self cannot perform for itself—either in maintaining fragile self-cohesion or in regulating self-esteem. The other, thus, is not experienced as an autonomous and separate object or agency in its own right but as present only to serve the needs of the self. Transference in this sense reflects a continuing developmental need that seeks satisfaction in the analytical relation.

Self–object transferences reflect the underlying need structure the patient brings to the therapeutic relationship based on the predominant pattern of self–object deprivation or frustration and the corresponding seeking for the appropriate form of self–object involvement. These configurations have been described as the *understimulated self,* the *overstimulated self,* the *overburdened self,* and the *fragmenting self.* Other descriptions of self–object need translate patterns of transference interaction based on narcissistic dynamics into the perspective of the relationship between self and self–object, as in mirror-hungry personalities and ideal-hungry personalities. Variations on the mirroring transference theme include the alter-ego–hungry personality, the merger-hungry personality, and, in contrast, the contact-shunning personality. In transferences derived from such personality configurations, the classic meaning of transference has undergone radical modification. Rather than displacements or projections from earlier object relational contexts, the patient brings to bear a need based in his or her own currently deficient capacity and defective character structure—a need to involve the object in a dependent relationship to complete or stabilize his or her own psychic integration.

Transitional Relatedness

This transference model is based on Donald Winnicott's notion of the transitional object. Transference in more primitive character

structures is regarded as a form of *transitional object relation* in which the therapist is perceived as outside the self but is invested with qualities from the patient's own archaic self-image. The transference field in this view is envisioned as a transitional space in which the transference illusion is allowed to play itself out.

Transference as Psychic Reality

Reflects the need of each participant in analysis to draw the other into a stance corresponding to his or her own intrapsychic configuration and needs as a reflection of the individual subject's psychic reality. This regards the classic view of transference, based on displacement or projection from past objects, as inadequate, resulting in further diffusion of meaning of transference as equivalent to the individual's capacity to create a meaningful world or to inform the world with meaning. In this rendition, transference becomes equivalent to the patient's psychic reality so that any distinction between the meanings given to reality and the meanings inherent in transference is lost. Transference in these terms becomes all-encompassing, and whatever distinguishing and dynamic significance it may have had fades into obscurity. In this form of transference, no definable mechanism seems to be at work other than whatever is involved in the subject's psychic reality. The subject's view of his or her environment and impression of objects of his or her experience, including the analytical object, are indistinguishable from ordinary cognitive and affective processes characterizing personal involvement and responsiveness to the world about him or her.

Transference as Relational or Intersubjective

The relational or intersubjective view of transference as emerging from or cocreated by the subjective interaction between analyst and analysand transforms transference into an interactive phenomenon in which individual intrapsychic contributions from either participant are obscured. Transference in this sense is not anything individual to, or intrapsychically derived from, the patient but is based on the present ongoing interaction between analyst

and patient coconstructing transference. On these terms, analysis of transference has little to do with past derivatives and everything to do with the ongoing relation with the analyst, primarily in the form of interpersonal enactments. Transference in this sense is no longer a one-person phenomenon but reflects a two-person transference–countertransference interaction. The supposition is that no such thing as transference exists without countertransference and no such thing as countertransference exists without transference. The patient is thus relieved of any burden of a personal dynamic unconscious reflecting developmental vicissitudes and residues of a life history. Transference is created anew in the immediacy of present analytical interaction as the product of mutual influence and communication between analyst and analysand, probably relying on some form of mutual projective identification to sustain the interactive connotation.



Table 34-15
Transference Mechanisms

Displacement

The basic mechanism of classic transference paradigms in which an object representation derived from any level or combination of levels of the subject's developmental experience is displaced to the representation of the new object, namely, the analyst, in the therapeutic relationship. Displacement is the basic mechanism for libidinally based transferences, both positive and erotic, as well as for aggressive and especially negative transferences. By and large, displacement transferences tend to play a more dominant role in neurotic disorders in which phallic–oedipal (and to a lesser degree preoedipal) dynamics tend to play a dominant, although not exclusive, role.

Projection

Process by which qualities or characteristics of the self-as-object, usually involving introjections or self-representations, are

attributed to an external object, and the subsequent interaction with the object is determined by the projected characteristics. Thus, the analyst or object may be seen as sadistic—that is, as possessing the sadistic character of the analysand or subject, an aspect of the subject's self that is denied or disowned by the subject. Projection tends to play a more prominent, although again not exclusive, role in formation of transferences in more primitive character disorders but can be found in variously modified forms throughout the spectrum of neuroses. Because projections derive primarily from the configuration of introjects constituting the patient's self-as-object, the effect of projective or externalizing transferences is that the image of the therapist comes to represent part of the patient's own self-organization rather than simply an object representation.

Projections derived from destructive introjects can provide the basis for both negative and paranoid transference reactions. Those based on the victim or introject result in the patient relating to the therapist as his or her victim and him- or herself assuming a hostile or sadistic position as a destructive aggressor or victimizer to the therapist's victim. Then again, projection based on the aggressor or introject results in the patient relating to the therapist as an aggressor and him- or herself assuming a weak, vulnerable, or masochistic position in which he or she becomes a passive and vulnerable victim to the therapist's destructive aggression. Similar patterns can take place around narcissistic issues involving introjective configurations of narcissistic superiority and inferiority.

Projective dynamics in self–object transferences, however, seem to involve more than narcissistic projections because these forms of transference tend to draw the analyst into meeting the pathologic needs of the self. If anything is projected, it is an infantile wished-for imago, one lacking earlier in the patient's experience, as, for example, an empathic and idealized parental figure. On the other hand, transitional transferences, despite their considerable overlap with self–object phenomena, tend to involve a more explicit projective element as the self-related contribution to the

transitional experience.

Projective Identification

The concept of projective identification was first proposed by Melanie Klein, arguing that the projection of impulses or feelings into another person brought about an identification with that person based on attribution of one's own qualities to that other. This attribution served as the basis for a sense of empathy and connection with the other. On these terms, projective identification was a fantasy taking place solely in the mind of the one projecting.

Projective identification is often appealed to as a mechanism of transference, or more exactly transference–countertransference interactions, particularly in Kleinian usage. Confusion arises from the failure to clearly distinguish between projection and projective identification. The notion of projective identification added to the basic concept of projection of the notes of diffusion of ego boundaries, a loss or diminishing of self–object differentiation, and inclusion of the object as part of the self.

Later elaborations of the notion of projective identification transformed it from a one-body to a two-body phenomenon, describing interaction between two subjects, one of whom projects something onto or into the other, whereon the other introjects or internalizes what has been projected. Instead of the projection and introjection taking place in the same subject, the projection now takes place in one and the internalization in the other. This latter usage has led to extensive extrapolation of the concept of projective identification to apply to all sorts of object relations, including transference. The emphasis in Kleinian transference is less on the influence of the past on the present but rather the influence of the internal world on the external in the here-and-now interaction with the analyst.

Analysts after Freud began to recognize that countertransference was not only an obstacle but also a source of useful information about the patient. In other words, the analyst's feelings in response to the patient reflect how other

persons respond to the patient and indicate the patient's own internal object relations. By understanding the intense feelings in the analytic relationship, the analyst can help the patient broaden the understanding of past and current relationships outside the analysis. The development of insight into neurotic conflicts also expands the ego and provides an increased sense of mastery.

ERIK H. ERIKSON

Erik H. Erikson (Fig. 34-24) was one of America's most influential psychoanalysts. Throughout six decades in the United States, he distinguished himself as an illuminator and expositor of Freud's theories and as a brilliant clinician, teacher, and pioneer in psychohistorical investigation. Erikson created an original and highly influential theory of psychological development and crisis occurring in periods that extended across the entire life cycle. His theory grew out of his work first as a teacher, then as a child psychoanalyst, next as an anthropologic field worker, and, finally, as a biographer. Erikson identified dilemmas or polarities in the ego's relations with the family and larger social institutions at nodal points in childhood, adolescence, and early, middle, and late adulthood. Two of his psychosexual historical studies, *Young Man Luther* and *Gandhi's Truth* (published in 1958 and 1969, respectively), were widely hailed as profound explorations of how crucial circumstances can interact with the crises of certain great persons at particular moments in time. The interrelationships of the psychological development of the person and the historical developments of the times were more fully explored in *Life History and the Historical Moment*, written by Erikson in 1975.

Erik Homburger Erikson was born on June 15, 1902, in Frankfurt, Germany, the son of Danish parents. He died in 1994. His father abandoned his mother before he was born, and he was brought up by his mother, a Danish Jew, and her second husband, Theodor Homburger, a German-Jewish pediatrician. Erikson's parents chose to keep his real parentage a secret from him, and for many years he was known as Erik Homburger. Erikson never knew his biologic father; his mother withheld that information from him all her life. Undoubtedly, the man who introduced the term "identity crisis" into the language struggled with his sense of identity. Compounding his parents' deception about his biologic father—their "loving deceit," as he called it—was the fact that, as a blond, blue-eyed, Scandinavian-looking son of a Jewish father, he was taunted as a "goy" among Jews, at the same time being called a Jew by his classmates. His being a Dane living in Germany added to his identity confusion. Erikson was later described as a man of the borders. Much of what he was to study was concerned with how group values are implanted

in the very young, how young people grasp onto group identity in the limbo period between childhood and adulthood, and how a few persons, like Gandhi, transcend their local, national. Even temporal identities to form a small band of people with wider sympathies who span the ages.

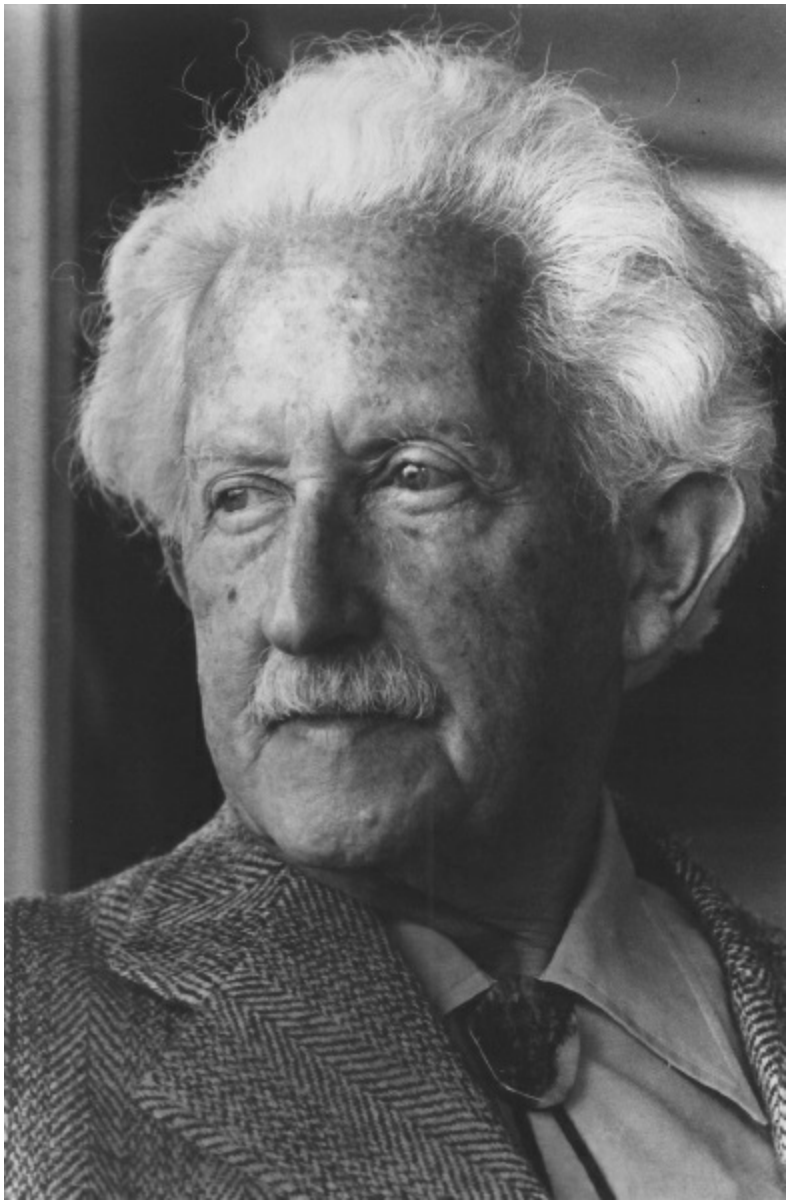


FIGURE 34-24
Erik Erikson (1902–1994).

The concepts of identity, identity crisis, and identity confusion are central to Erikson's thought. In his first book *Childhood and Society* (published in 1950), Erikson observed that "the study of identity…becomes as strategic in

our time as the study of sexuality was in Freud's time." By identity, Erikson meant a sense of sameness and continuity "in the inner core of the individual" maintained amid external change. A sense of identity, emerging at the end of adolescence, is a psychosocial phenomenon preceded in one form or another by an identity crisis; that crisis may be conscious or unconscious, with the person being aware of the present state and future directions but also unconscious of the fundamental dynamics and conflicts that underlie those states. The identity crisis can be acute and prolonged in some people.

The young Erikson did not distinguish himself in school, although he did show artistic talent. On graduation, he spent a year traveling through the Black Forest, Italy, and the Alps, pondering life, drawing, and making notes. After that year of roaming, he studied art in his home city of Karlsruhe and later in Munich and Florence.

In 1927 Peter Blos, a high school friend, invited Erikson to join him in Vienna. Blos, not yet a psychoanalyst, had met Dorothy Burlingham, a New Yorker who had come to Vienna to be psychoanalyzed; she had brought her four children with her and hired Blos to tutor them. Blos was looking for a fellow teacher in his new school for the children of English and American parents and students of his new discipline of psychoanalysis. Erikson accepted his offer.

Blos and Erikson informally organized their school—much in the style of the so-called progressive or experimental schools prevalent in the United States. Children were encouraged to participate in curriculum planning and to express themselves freely. Erikson, still very much the artist, taught drawing and painting, but he also exposed his pupils to history and foreign ways of life, including the cultures of the American Indian and Eskimo.

During that period, Erikson became involved with the Freud family, friends of Mrs. Burlingham. He became particularly close to Anna Freud, with whom he began psychoanalysis. Anna Freud, who had been an elementary school teacher, was at that time formulating the new science of child psychiatry, trying to turn attention from the adult's corrective backward look to a neurosis-preventative study of childhood itself. Under Anna Freud's tutelage, Erikson began more and more to turn his attention to childhood, both his own and that of the children whom he saw in the classroom. The analysis was not then the rigidly structured procedure into which it later developed; Erikson met with Miss Freud daily for his analytic hour and

frequently saw her socially and as part of the circle of Freud's followers and associates. Still undecided about his future, Erikson continued to teach school while studying psychoanalysis at the Vienna Psychoanalytic Institute. He also studied to become accredited as a Montessori teacher.

In 1929 he married Joan Mowat Serson, an American of Canadian birth. He was hastily made a full member, rather than an associate member of the Vienna Psychoanalytic Society—unorthodoxy that allowed him to leave a Vienna threatened by fascism immediately after his graduation in 1933. Earlier, Erikson had met the Viennese Hanns Sachs, co-founder, along with Otto Rank of the psychoanalytically oriented journal *Imago*. Sachs—who had settled in Boston, where he was associated with the Harvard Medical School—was sure that Erikson would be welcome at Harvard and suggested that he make Boston his home. After a brief stay in Denmark, the Eriksons moved to Boston, where he became the city's only child analyst. He held positions at the Harvard Medical School and Massachusetts General Hospital, served as a consultant at the Judge Baker Guidance Center, and maintained a private practice.

Erikson was much influenced by Cambridge's circle of young social scientists, including anthropologists Margaret Mead and Ruth Benedict. Exposure to the views of those vigorous thinkers helped to shape his theories of child psychology and his cross-cultural approach to human development. Classical psychoanalysis had traditionally concerned itself with pathology and with treating disturbed people. However, Erikson found himself more interested in the normal personality and applying his observations about how young people function and how childhood play affects character formation. Although he remained in the Boston area for only 3 years, he established a solid reputation as a skilled clinician and researcher before moving to Yale University's Institute of Human Relations. There he furthered an interest sparked at Harvard in the work of American anthropologists. In 1938 he traveled to South Dakota to study the children of the Sioux Indians of the Pine Ridge Reservation. His observations about how communal and historical forces powerfully influence child-rearing became a vital contribution to psychology and the study of humans in society.

In 1939 Erikson moved to a post at Berkeley, from which he studied the Yurok Indians, a group of salmon fishers. He left Berkeley in 1950 after refusing to sign what he called a vague, fearful addition to the loyalty oath.

He resettled at the Austen Riggs Center in Stockbridge, Massachusetts, working with young people. In 1960 he was appointed to a professorship at Harvard. After he retired from Harvard in 1972, Erickson joined Mount Zion Hospital in San Francisco as a senior consultant in psychiatry. Until his death in 1994, he continued to focus on many of his earlier interests, examining the individual in this historical context and elaborating on concepts of the human life cycle, especially those of old age.

Epigenetic Principle

Erikson's formulations were based on the concept of *epigenesis*, a term borrowed from embryology. His *epigenetic principle* holds that development occurs in sequential, clearly defined stages and that each stage must be satisfactorily resolved for development to proceed smoothly. According to the epigenetic model, if a successful resolution of a particular stage does not occur, all subsequent stages reflect that failure in physical, cognitive, social, or emotional maladjustment.

Relation to Freudian Theory. Erikson accepted Freud's concepts of instinctual development and infantile sexuality. For each of Freud's psychosexual stages (e.g., oral, anal, and phallic), Erikson described a corresponding zone with a specific pattern or mode of behavior. Thus, the oral zone is associated with sucking or taking-in behavior; the anal zone is associated with holding on and letting go. Erikson emphasized that the development of the ego is more than the result of intrapsychic wants or inner psychic energies. It is also a matter of mutual regulation between growing children and a society's culture and traditions.

Eight Stages of the Life Cycle. Erikson's conception of the eight stages of ego development across the life cycle is the centerpiece of his life's work, and he elaborated on the conception throughout his subsequent writings (Table 34-16). The eight stages represent points along a continuum of development. Physical, cognitive, instinctual, and sexual changes combine to trigger an internal crisis, the resolution of which results in either psychosocial regression or growth and the development of specific virtues. In *Insight and Responsibility*, Erikson defined virtue as "inherent strength," as in the active quality of medicine or liquor. He wrote in *Identity: Youth and Crisis* that

"crisis" refers not to a "threat of catastrophe, but a turning point, a crucial period of increased vulnerability and heightened potential, and therefore, the ontogenetic source of generational strength and maladjustment."

STAGE 1: TRUST VERSUS MISTRUST (BIRTH TO ABOUT 18 MONTHS). In *Identity: Youth and Crisis*, Erikson noted that the infant "lives through and loves with" its mouth. Indeed, the mouth forms the basis of its first mode or pattern of behavior, incorporation. The infant is taking the world through the mouth, eyes, ears, and sense of touch. The baby is learning a cultural modality that Erikson termed *to get*, that is, to receive what is offered and elicit what is desired. As the infant's teeth develop, and it discovers the pleasure of biting, it enters the second oral stage, the active-incorporative mode. The infant is no longer passively receptive to stimuli; it reaches out for sensation and grasps at its surroundings—the social modality shifts to that of *taking and holding* on to things.



Table 34-16
Erikson's Psychosocial Stages

Psychosocial Stage	Associated Virtue	Related Forms of Psychopathology	Positive and Negative Forerunners of Identity Formation	Enduring Aspects of Identity Formation
Trust vs. mistrust (birth–)	Hope	Psychosis Addictions Depression	Mutual recognition vs. autistic isolation	Temporal perspective vs. time confusion
Autonomy vs. shame and doubt (~18 mo–)	Will	Paranoia Obsessions Compulsions Impulsivity	Will to be oneself vs. self-doubt	Self-certainty vs. self-consciousness
Initiative vs. guilt (~3 yr–)	Purpose	Conversion disorder Phobia Psychosomatic disorder Inhibition	Anticipation of roles vs. role inhibition	Role experimentation vs. role fixation
Industry vs. inferiority (~5 yr–)	Competence	Creative inhibition Inertia	Task identification vs. sense of futility	Apprenticeship vs. work paralysis
Identity vs. role confusion (~13 yr–)	Fidelity	Delinquent behavior Gender-related identity disorders Borderline psychotic episodes		Identity vs. identity confusion
Intimacy vs. isolation (~20s–)	Love	Schizoid personality disorder Distantiation		Sexual polarization vs. bisexual confusion
Generativity vs. stagnation (~40s–)	Care	Mid-life crisis Premature invalidism		Leadership and followership vs. abdication of responsibility
Integrity vs. despair (~60s–)	Wisdom	Extreme alienation Despair		Ideologic commitment vs. confusion of values

Adapted from Erikson E. *Insight and Responsibility*. New York: WW Norton; 1964 and Erikson E. *Identity: Youth and Crisis*. New York: WW Norton; 1968, with permission.

The infant's development of essential trust in the world stems from its earliest experiences with its mother or primary caretaker. In *Childhood and Society,* Erikson asserts that trust depends not on "absolute quantities of food or demonstrations of love, but rather on the quality of the maternal relationship." A baby whose mother can anticipate and respond to its needs in a consistent and timely manner despite its oral aggression will learn to tolerate the inevitable moments of frustration and deprivation. The defense mechanisms of introjection and projection provide the infant with the means to internalize pleasure and externalize pain such that "consistency, continuity, and sameness of experience provide a rudimentary sense of ego identity." Trust will predominate over mistrust, and hope will crystallize. For Erikson, the element of society corresponding to this stage of ego identity is religion, as both are founded on "trust born of care."

In keeping with his emphasis on the epigenetic character of psychosocial change, Erikson conceived of many forms of psychopathology as examples of what he termed *aggravated development crisis,* development, which having gone awry at one point, affects subsequent psychosocial change. A person who, as a result of severe disturbances in the earliest dyadic relationships, fails to develop a basic sense of trust or the virtue of hope may be predisposed as an adult to the profound withdrawal and regression characteristic of schizophrenia. Erikson hypothesized that the depressed patient's experience of being empty and being no good is an outgrowth of a developmental derailment that causes oral pessimism to predominate. Addictions may also be traced to the mode of oral incorporation.

STAGE 2: AUTONOMY VERSUS SHAME AND DOUBT (ABOUT 18 MONTHS TO ABOUT 3 YEARS). In the development of speech and sphincter and muscular control, the toddler practices the social modalities of *holding on and letting go* and experiences the first stirrings of the virtue that Erikson termed *will.* Much depends on the amount and type of control exercised by adults over the child. Control that is exerted too rigidly or too early defeats the toddler's attempts to develop its internal controls, and regression or false progression results. Parental control that fails to protect the toddler from the consequences of their lack of self-control or judgment can be equally disastrous to the child's development of a healthy sense of autonomy. In *Identity: Youth and Crisis,* Erikson asserted: "This stage, therefore, becomes decisive for the ratio

between loving goodwill and hateful self-insistence, between cooperation and willfulness, and between self-expression and compulsive self-restraint or meek compliance."

Where that ratio is favorable, the child develops an appropriate sense of autonomy and the capacity to "have and to hold"; where it is unfavorable, doubt and shame will undermine free will. According to Erikson, the principle of law and order has at its roots this early preoccupation with the protection and regulation of will. In *Childhood and Society,* he concluded, "The sense of autonomy fostered in the child and modified as life progresses, serves (and is served by) the preservation in the economic and political life of a sense of justice."

A person who becomes fixated at the transition between the development of hope and autonomous will, with its residue of mistrust and doubt, may develop paranoiac fears of persecution. When psychosocial development is derailed in the second stage, other forms of pathology may emerge. The perfectionism, inflexibility, and stinginess of the person with an obsessive-compulsive personality disorder may stem from conflicting tendencies to hold on and to let go. The ruminative and ritualistic behavior of the person with an obsessive-compulsive disorder may be an outcome of the triumph of doubt over autonomy and the subsequent development of a primitively harsh conscience.

STAGE 3: INITIATIVE VERSUS GUILT (ABOUT 3 YEARS TO ABOUT 5 YEARS). The child's increasing mastery of locomotor and language skills expands its participation in the outside world and stimulates omnipotent fantasies of broader exploration and conquest. Here the youngster's mode of participation is active and intrusive; its social modality is that of being on the make. The intrusiveness is manifested in the child's fervent curiosity and genital preoccupations, competitiveness, and physical aggression. The Oedipus complex is in ascendance as the child competes with the same-sex parent for the fantasized possession of the other parent. In *Identity: Youth and Crisis,* Erikson wrote that "jealousy and rivalry now come to a climax in a final contest for a favored position with one of the parents: the inevitable and necessary failure leads to guilt and anxiety."

Guilt over the drive for conquest and anxiety over the anticipated punishment are both assuaged in the child through repression of the forbidden

wishes and development of a superego to regulate its initiative. This conscience, the faculty of self-observation, self-regulation, and self-punishment, is an internalized version of parental and societal authority. Initially, the conscience is harsh and uncompromising; however, it constitutes the foundation for the subsequent development of morality. Having renounced oedipal ambitions, the child begins to look outside the family for arenas in which it can compete with less conflict and guilt. This is the stage that highlights the child's expanding initiative and forms the basis for the subsequent development of realistic ambition and the virtue of *purpose.* As Erikson noted in *Childhood and Society,* "The 'oedipal' stage sets the direction toward the possible and the tangible which permits the dreams of early childhood to be attached to the goals of active adult life." Toward this end, social institutions provide the child with an economic ethos in adult heroes who begin to replace their storybook counterparts.

When there has been an inadequate resolution of the conflict between initiative and guilt, a person may ultimately develop a conversion disorder, inhibition, or phobia. Those who overcompensate for the conflict by driving themselves too hard may experience sufficient stress to produce psychosomatic symptoms.

STAGE 4: INDUSTRY VERSUS INFERIORITY (ABOUT 5 YEARS TO ABOUT 13 YEARS). With the onset of latency, the child discovers the pleasures of production. He or she develops industry by learning new skills and takes pride in the things made. Erikson wrote in *Childhood and Society* that the child's "ego boundaries include his tools and skills: the work principle teaches him the pleasure of work completion by steady attention and persevering diligence." Across cultures, this is when the child receives systematic instruction and learns the fundamentals of technology and how to use necessary utensils and tools. As children work, they identify with their teachers and imagine themselves in various occupational roles.

A child unprepared for this stage of psychosocial development, either through an insufficient resolution of previous stages or by current interference, may develop a sense of inferiority and inadequacy. In the form of teachers and other role models, society becomes crucially important in the child's ability to overcome that sense of inferiority and achieve the virtue known as competence. In *Identity: Youth and Crisis,* Erikson noted: "This is

socially a most decisive stage. Since industry involves doing things beside and with others, a first sense of division of labor and of differential opportunity, that is, a sense of the technological ethos of a culture, develops at this time."

The pathologic outcome of a poorly navigated stage of industry versus inferiority is less well defined than in previous stages. However, it may concern the emergence of a conformist immersion into the world of production in which creativity is stifled, and identity is subsumed under the worker's role.

STAGE 5: IDENTITY VERSUS ROLE CONFUSION (ABOUT 13 YEARS TO ABOUT 21 YEARS). With the onset of puberty and its myriad social and physiologic changes, adolescents become preoccupied with identity. Erikson noted in *Childhood and Society* that youth are now "primarily concerned with what they appear to be in the eyes of others as compared to what they feel they are, and with the question of how to connect the roles and skills cultivated earlier with the occupational prototypes of the day." Childhood roles and fantasies are no longer appropriate, yet the adolescent is far from equipped to become an adult. In *Childhood and Society,* Erikson writes that the integration that occurs in the formation of ego identity encompasses far more than the summation of childhood identifications. "It is the accrued experience of the ego's ability to integrate these identifications with the vicissitudes of the libido, with the aptitudes developed out of endowment, and with the opportunities offered in social roles."

The formation of cliques and an identity crisis occur at the end of adolescence. Erikson calls the crisis normative because it is a typical event. Failure to negotiate this stage leaves adolescents without a stable identity; they suffer from identity diffusion or role confusion, characterized by not having a sense of self and by confusion about their place in the world. Role confusion can manifest in such behavioral abnormalities as running away, criminality, and overt psychosis. Problems in gender identity and sexual role may manifest at this time. Adolescents may defend against role diffusion by joining cliques or cults or by identifying with folk heroes. Intolerance of individual differences is how the young person attempts to ward off a sense of identity loss. Falling in love, a process by which the adolescent may clarify a sense of identity by projecting a diffused self-image onto the partner and

gradually seeing it assume a more distinctive shape. An overidentification with idealized figures is how the adolescent seeks self-definition. With the attainment of a more sharply focused identity, the youth develops the virtue of *fidelity*—faithfulness not only to the nascent self-definition but also to an ideology that provides a version of self-in-world. As Erik Erikson, Joan Erikson, and Helen Kivnick wrote in *Vital Involvement in Old Age*, "Fidelity is the ability to sustain loyalties freely pledged in spite of the inevitable contradictions of value systems. It is the cornerstone of identity and receives inspiration from confirming ideologies and affirming companionships." Role confusion ensues when the youth is unable to formulate a sense of identity and belonging. Erikson held that delinquency, gender-related identity disorders, and borderline psychotic episodes can result from such confusion.

STAGE 6: INTIMACY VERSUS ISOLATION (ABOUT 21 YEARS TO ABOUT 40 YEARS). Freud's famous response to the question of what a normal person should be able to do well, "Lieben und arbeiten" (to love and to work), is one that Erikson often cited in his discussion of this psychosocial stage, and it emphasizes the importance he placed on the virtue of love within a balanced identity. Erikson asserted in *Identity: Youth and Crisis* that Freud's use of the term love referred to "the generosity of intimacy as well as genital love; when he said love and work, he meant a general work productiveness which would not preoccupy the individual to the extent that he might lose his right or capacity to be a sexual and a loving being."

Intimacy in the young adult is closely tied to fidelity; it can make and honor commitments to concrete affiliations and partnerships even when that requires sacrifice and compromise. The person who cannot tolerate the fear of ego loss arising out of self-abandonment (e.g., sexual orgasm, moments of intensity in friendships, aggression, inspiration, and intuition) is apt to become profoundly isolated and self-absorbed. *Distantiation*, an awkward term coined by Erikson to mean "the readiness to repudiate, isolate, and, if necessary, destroy those forces and persons whose essence seems dangerous to one's own," is the pathologic outcome of conflicts surrounding intimacy and, in the absence of an ethical sense where intimate, competitive, and combative relationships are differentiated, forms the basis for various forms of prejudice, persecution, and psychopathology.

Erikson's separation of the psychosocial task of achieving identity from

achieving intimacy and his assertion that substantial progress on the former task must precede development on the latter have engendered much criticism and debate. Critics have argued that Erikson's emphasis on separation and occupationally based identity formation fails to consider the importance for women of continued attachment and the formation of an identity based on relationships.

STAGE 7: GENERATIVITY VERSUS STAGNATION (ABOUT 40 YEARS TO ABOUT 60 YEARS). Erikson asserted in *Identity: Youth and Crisis* that "generativity is primarily the concern for establishing and guiding the next generation." The term *generativity* applies not so much to rearing and teaching one's offspring as it does to a protective concern for all the generations and social institutions. It encompasses productivity and creativity, as well. Having achieved the capacity to form intimate relationships, the person now broadens the investment of ego and libidinal energy to include groups, organizations, and society. *Care* is the virtue that coalesces at this stage. In *Childhood and Society,* Erikson emphasized the importance to the mature person of feeling needed. "Maturity needs guidance as well as encouragement from what has been produced and must be taken care of." Through generative behavior, the individual can pass on knowledge and skills while obtaining a measure of satisfaction in having achieved a role with senior authority and responsibility in the tribe.

When persons cannot develop true generativity, they may settle for pseudoengagement in occupation. Often, such persons restrict their focus to the technical aspects of their roles, at which they may now have become highly skilled, eschewing greater responsibility for the organization or profession. This failure of generativity can lead to profound personal stagnation, masked by various escapisms, such as alcohol and drug abuse, and sexual and other infidelities. Mid-life crisis or premature invalidism (physical and psychological) can occur. In this case, pathology appears not only in middle-aged persons but also in the organizations that depend on them for leadership. Thus, the failure to develop at midlife can lead to sick, withered, or destructive organizations that spread the effects of failed generativity throughout society; examples of such failures have become so common that they constitute a defining feature of modernity.

STAGE 8: INTEGRITY VERSUS DESPAIR (ABOUT 60 YEARS TO DEATH). In *Identity: Youth and Crisis,* Erikson defined integrity as "the acceptance of one's one and only life cycle and of the persons who have become significant to it as something that had to be and that, by necessity, permitted of no substitutions." From the vantage point of this stage of psychosocial development, the individual relinquishes the wish that important persons in his life had been different and can love in a more meaningful way—one that reflects accepting responsibility for one's own life. The individual in possession of the virtue of wisdom and a sense of integrity has room to tolerate the proximity of death and to achieve what Erikson termed in *Identity: Youth and Crisis,* a "detached yet active concern with life."

Erikson underlined the social context for this final stage of growth. In *Childhood and Society,* he wrote, "The style of integrity developed by his culture or civilization thus becomes the 'patrimony' of his soul …. In such final consolidation, death loses its sting."

When attempting to attain integrity has failed, the individual may become deeply disgusted with the external world and contemptuous of persons and institutions. Erikson wrote in *Childhood and Society* that such disgust masks a fear of death and a sense of despair that "time is now short, too short for the attempt to start another life and to try out alternate roads to integrity." Looking back on the eight ages of man, he noted the relation between adult integrity and infantile trust, "Healthy children will not fear life if their elders have integrity enough not to fear death."

Psychopathology

Each stage of the life cycle has its psychopathological outcome if it is not mastered successfully.

Basic Trust. An impairment of basic trust leads to basic mistrust. In infants, social trust is characterized by ease of feeding, depth of sleep, smiling, and general physiologic homeostasis. Prolonged separation during infancy can lead to hospitalism or anaclitic depression. In later life, this lack of trust may be manifested by dysthymic disorder, a depressive disorder, or a sense of hopelessness. Persons who develop and rely on the defense of projection—in which, according to Erikson, "we endow significant persons with the evil

which actually is in us"—experienced a sense of social mistrust in the first years of life and are likely to develop paranoid or delusional disorders. Basic mistrust is a significant contributor to the development of schizoid personality disorder and, in most severe cases, to the development of schizophrenia. Substance-related disorders can also be traced to social mistrust; substance-dependent personalities have strong oral-dependency needs and use chemical substances to satisfy themselves because of their belief that human beings are unreliable and, at worst, dangerous. If not appropriately nurtured, infants may feel empty, starved not just for food but also sensual and visual stimulation. As adults, they may become seekers after stimulating thrills that do not involve intimacy, which helps ward off feelings of depression.

Autonomy. The stage in which children attempt to develop into autonomous beings is often called the *terrible twos*, referring to toddlers' willfulness at this period of development. If shame and doubt dominate over autonomy, compulsive doubting can occur. The inflexibility of the obsessive personality also results from an overabundance of doubt. Too rigorous toilet training, commonplace in today's society, requires a clean, punctual, and deodorized body, can produce an overly compulsive personality that is stingy, meticulous, and selfish. Known as anal personalities, such persons are parsimonious, punctual, and perfectionistic (the three P's).

Too much shaming causes children to feel evil or dirty and may pave the way for delinquent behavior. In effect, children say, "If that's what they think of me, that's the way I'll behave." Paranoid personalities feel that others are trying to control them, a feeling that may have its origin during the stage of autonomy versus shame and doubt. When coupled with mistrust, the seeds are planted for persecutory delusions. Impulsive disorder may be explained as a person's refusal to be inhibited or controlled.

Initiative. Erikson stated: "In pathology, the conflict over initiative is expressed either in hysterical denial, which causes the repression of the wish or the abrogation of its executive organ by paralysis or impotence; or in overcompensatory showing off, in which the scared individual, so eager to 'duck,' instead 'sticks his neck out.'" In the past, hysteria was the usual form of pathologic regression in this area, but a plunge into a psychosomatic

disease is now common.

Excessive guilt can lead to a variety of conditions, such as generalized anxiety disorder and phobias. Patients feel guilty because of normal impulses, and they repress these impulses, with resulting symptom formation. Punishment or severe prohibitions during the stage of initiative versus guilt can produce sexual inhibitions. Conversion disorder or specific phobia can result when the oedipal conflict is not resolved. As sexual fantasies are accepted as unrealizable, children may punish themselves for these fantasies by fearing harm to their genitals. Under the brutal assault of the developing superego, they may repress their wishes and begin to deny them. If this pattern is carried forward, paralysis, inhibition, or impotence can result. Sometimes, in fear of not being able to live up to what others expect, children may develop a psychosomatic disease.

Industry. Erikson described industry as a "sense of being able to make things and make them well and even perfectly." When children's efforts are thwarted, they are made to feel that personal goals cannot be accomplished or are not worthwhile, and a sense of inferiority develops. In adults, this sense of inferiority can result in severe work inhibitions and a character structure marked by feelings of inadequacy. For some persons, the feelings may result in a compensatory drive for money, power, and prestige. Work can become the main focus of life at the expense of intimacy.

Identity. Many disorders of adolescence can be traced to identity confusion. The danger is role diffusion. Erikson stated:

Where this is based on a strong previous doubt on one's sexual identity, delinquent and outright psychotic incidents are not uncommon. If diagnosed and treated correctly, those incidents do not have the same fatal significance that they have at other ages. It is primarily the inability to settle on an occupational identity that disturbs young persons. Keeping themselves together, they temporarily overidentify, to the point of apparent complete loss of identity, with the heroes of cliques and crowds.

Other disorders during the stage of identity versus role diffusion include conduct disorder, disruptive behavior disorder, gender identity disorder, schizophreniform disorder, and other psychotic disorders. The ability to leave home and live independently is an essential task during this period. An

inability to separate from the parent and prolonged dependence may occur.

Intimacy. The successful formation of a stable marriage and family depends on the capacity to become intimate. The years of early adulthood are crucial for deciding whether to get married and to whom. Gender identity determines object choice, either straight or gay, but making an intimate connection with another person is a significant task. Persons with schizoid personality disorder remain isolated from others because of fear, suspicion, inability to take risks, or the lack of a capacity to love.

GENERATIVITY. From about 40 to 65 years, the period of middle adulthood, specific disorders are less clearly defined than in the other stages described by Erikson. Middle-aged persons show a higher incidence of depression than younger adults, which may be related to middle-aged persons' disappointments and failed expectations as they review the past, consider their lives, and contemplate the future. The increased use of alcohol and other psychoactive substances also occurs during this time.

INTEGRITY. Anxiety disorders often develop in older persons. In Erikson's formulation, this development may be related to persons' looking back on their lives with a sense of panic. Time has run out, and chances are used up. The decline in physical functions can contribute to psychosomatic illness, hypochondriasis, and depression. The suicide rate is highest in persons over the age of 65. Persons facing dying and death may find it intolerable not to have been generative or make significant attachments in life. Integrity, for Erikson, is characterized by an acceptance of life. Without acceptance, persons feel despair and hopelessness that can result in severe depressive disorders.

Treatment

Although no independent Eriksonian psychoanalytic school exists in the same way that Freudian and Jungian schools do, Erikson made many vital contributions to the therapeutic process. Among his most important contributions is his belief that establishing a state of trust between doctor and patient is the essential requirement for successful therapy. When psychopathology stems from basic mistrust (e.g., depression), a patient must

reestablish trust with the therapist, whose task, like that of the good mother, is sensitive to the patient's needs. The therapist must have a sense of personal trustworthiness that can be transmitted to the patient.

Techniques. For Erikson, a psychoanalyst is not a blank slate in the therapeutic process, as the psychoanalyst commonly is in Freudian psychoanalysis. On the contrary, effective therapy requires that therapists actively convey to patients the belief that they are understood. This is done through both empathetic listening and verbal assurances, which enable a positive transference built on mutual trust to develop.

Beginning as an analyst for children, Erikson tried to provide this mutuality and trust while he observed children recreating their worlds by structuring dolls, blocks, vehicles, and miniature furniture into the dramatic situations that were bothering them. Then, Erikson correlated his observations with statements by the children and their family members. He began treatment of a child only after eating an evening meal with the entire family, and his therapy was usually conducted with much cooperation from the family. After each regressive episode in the treatment of a schizophrenic child, for instance, Erikson discussed with every member of the family what had been going on with them before the episode. Only when he was thoroughly satisfied that he had identified the problem did treatment begin. Erikson sometimes provided corrective information to the child—for instance, telling a boy who could not release his feces and had made himself ill from constipation that food is not an unborn infant.

Erikson often turned to play, which, along with specific recommendations to parents, proved fruitful as a treatment modality. Play, for Erikson, is diagnostically revealing and thus helpful for a therapist who seeks to promote a cure, but it is also curative in its own right. Play is a function of the ego and gives children a chance to synchronize social and bodily processes with the self. Children playing with blocks or adults playing out an imagined dramatic situation can manipulate the environment and develop the sense of control that the ego needs. Play therapy is not the same for children and adults, however. Children create models to gain control of reality; they look ahead to new areas of mastery. Adults use play to correct the past and to redeem their failures.

Mutuality, which is essential in Erikson's system of health, is also vital to a

cure. Erikson applauded Freud for the moral choice of abandoning hypnosis because hypnosis heightens both the demarcation between the healer and the sick and the inequality that Erikson compares with the inequality of child and adult. Erikson urged that the relationship of the healer to the sick person be one of equals "in which the observer who has learned to observe himself teaches the observed to become self-observant."

Dreams and Free Association. As with Freud, Erikson worked with the patient's associations to the dream as the "best leads" to understanding the dream's meaning. He valued the first association with the dream, which he believed to be powerful and essential. Ultimately, Erikson listened for "a central theme which, once found, gives added meaning to all the associated material."

Erikson believed that interpretation was the primary therapeutic agent, sought as much by the patient as the therapist. He emphasized free-floating attention as the method that enabled discovery to occur. Erikson once described this attentional stance by commenting that in clinical work, "You need a history and you need a theory, and then you must forget them both and let each hour stand for itself." This frees both parties from counterproductive pressures to advance in the therapy and allows them to notice the gaps in the patient's narrative that signal the unconscious.

Goals. Erikson discussed four dimensions of the psychoanalyst's job. The patient's desire to be cured and the analyst's desire to cure is the first dimension. Mutuality exists in that patient and therapist are motivated by cure, and labor is divided. The goal is always to help the patient's ego get stronger and cure itself. The second dimension Erikson called objectivity-participation. Therapists must keep their minds open. "Neuroses change," wrote Erikson. New generalizations must be made and arranged in new configurations. The third dimension runs along the axis of knowledge-participation. The therapist "applies selected insights to more strictly experimental approaches." The fourth dimension is tolerance-indignation. Erikson stated: "Identities based on Talmudic argument, on messianic zeal, on punitive orthodoxy, on faddist sensationalism, on professional and social ambition" are harmful and tend to control patients. Control widens the gap of inequality between the doctor and the patient and makes the realization of the

recurrent idea in Erikson's thought—mutuality—difficult.

According to Erikson, therapists have the opportunity to work through past unresolved conflicts in the therapeutic relationship. Erikson encouraged therapists not to shy away from guiding patients; he believes therapists must offer patients both prohibitions and permissions. Nor should therapists be so engrossed in patients' past life experiences that current conflicts are overlooked.

The goal of therapy is to recognize how patients have passed through the various stages of the life cycle and how the various crises in each stage have or have not been mastered. Equally important, future stages and crises must be anticipated so that they can be negotiated and mastered appropriately. Unlike Freud, Erikson does not believe that the personality is so inflexible that change cannot occur in middle and late adulthood. For Erikson, psychological growth and development occur throughout the entire span of the life cycle.

The Austen Riggs Center in Stockbridge, Massachusetts, is a repository of Erikson's work, and many of his theories are put into practice there. Erik's wife, Joan, developed an activities program at the Austen Riggs Center as an "interpretation-free zone," where patients could take up work roles or function as students with artists and craftspersons, without the burden of the patient role. This workspace encouraged the play and creativity required for the patients' work development to parallel the process of their therapy.

OTHER PSYCHODYNAMIC SCHOOLS

The men and women discussed in this chapter contributed to psychiatric thought and practice in the early and middle years of the 20th century. Many of these theories of psychopathology evolved as direct offshoots of Freudian psychoanalysis. This, however, is derived from various aspects of psychology, such as learning theory and quantitative methods of personality assessment. The theories selected for the discussion in this section have stood the test of time and are most relevant for psychiatry.

Brief synopses of the theories that exert the most significant influence on current psychiatric thought are listed below in alphabetical order of their proponent. Each of these theories contains insights that merit consideration because they enhance our understanding of the complexities of human

behavior. They also illustrate the diversity of theoretical orientation that characterizes psychiatry today.

Karl Abraham (1877–1925)

Karl Abraham, one of Sigmund Freud's earliest disciples, was the first psychoanalyst in Germany. He is best known for his explication of depression from a psychoanalytic perspective and for his elaboration of Freud's stages of psychosexual development. Abraham divided the oral stage into a biting phase and a sucking phase; the anal stage into a destructive-expulsive (anal-sadistic) phase and a mastering-retentive (anal-erotic) phase; and the phallic stage into an early phase of partial genital love (true phallic phase) and a later mature genital phase. Abraham also linked the psychosexual stages to specific syndromes. For example, he postulated that obsessional neurosis resulted from fixation at the anal-sadistic phase and depression from fixation at the oral stage.

Alfred Adler (1870–1937)

Alfred Adler (Fig. 34-25) was born in Vienna, Austria, where he spent most of his life. A general physician, he became one of the original four members of Freud's circle in 1902. Adler never accepted the primacy of the libido theory, the sexual origin of a neurosis, or the importance of infantile wishes. Adler thought that aggression was far more critical, specifically in its manifestation as a striving for power, which he believed to be a masculine trait. He introduced the term *masculine protest* to describe the tendency to move from a passive, feminine role to a masculine, active role. Adler's theories are collectively known as *individual psychology*.

Adler saw individuals as unique, unified biologic entities whose psychological processes fit together into an individual lifestyle. He also postulated a principle of dynamism, in which every individual is future-directed and moves toward a goal. Adler also emphasized the interface between individuals and their social environment: the primacy of action in the real work over fantasy.



FIGURE 34-25
Alfred Adler (print includes signature). (Courtesy of Alexandra Adler.)

Adler coined the term *inferiority complex* to refer to a sense of inadequacy and weakness that is universal and inborn. A physical defect compromises a developing child's self-esteem, and Adler referred to this phenomenon as *organ inferiority*. He also thought that basic inferiority tied to children's oedipal longings could never be gratified.

Adler was one of the first developmental theorists to recognize the

importance of children's birth order in their origin. The firstborn child reacts with anger to the birth of siblings and struggles against giving up the powerful position of an only child. They tend not to share and become conservative. The second-born child must continuously strive to compete with the firstborn. Youngest children feel secure because they have *never* been displaced. Adler thought that a child's sibling position results in lifelong influences on character and lifestyle.

The primary therapeutic approach in Adlerian therapy is encouragement, through which Adler believed his patients could overcome feelings of inferiority. Consistent human relatedness, in his view, leads to greater hope, less isolation, and more significant affiliation with society. He believed that patients needed to develop a greater sense of their dignity and worth and a renewed appreciation of their abilities and strengths.

Franz Alexander (1891–1964)

Franz Alexander (Fig. 34-26) emigrated from his native Germany to the United States, where he settled in Chicago and founded the Chicago Institute for Psychoanalysis. He wrote extensively about the association between specific personality traits and certain psychosomatic ailments, a point of view that came to be known as the *specificity hypothesis.* Alexander fell out of favor with classic analysts for advocating the *corrective emotional experience* as part of the analytic technique. In this approach, Alexander suggested that an analyst must deliberately adopt a particular mode of relatedness with a patient to counteract noxious childhood influences from the patient's parents. He believed that the trusting, supportive relationship between patient and analyst enabled the patient to master childhood traumas and grow from the experience.



FIGURE 34-26
Franz Alexander. (Courtesy of Franz Alexander.)

Gordon Allport (1897–1967)

Gordon Allport (Fig. 34-27), a psychologist in the United States, is known as the founder of the humanistic school of psychology, which holds that each person has an inherent potential for autonomous function and growth. At Harvard University, he taught the first course in the psychology of personality offered at a college in the United States.

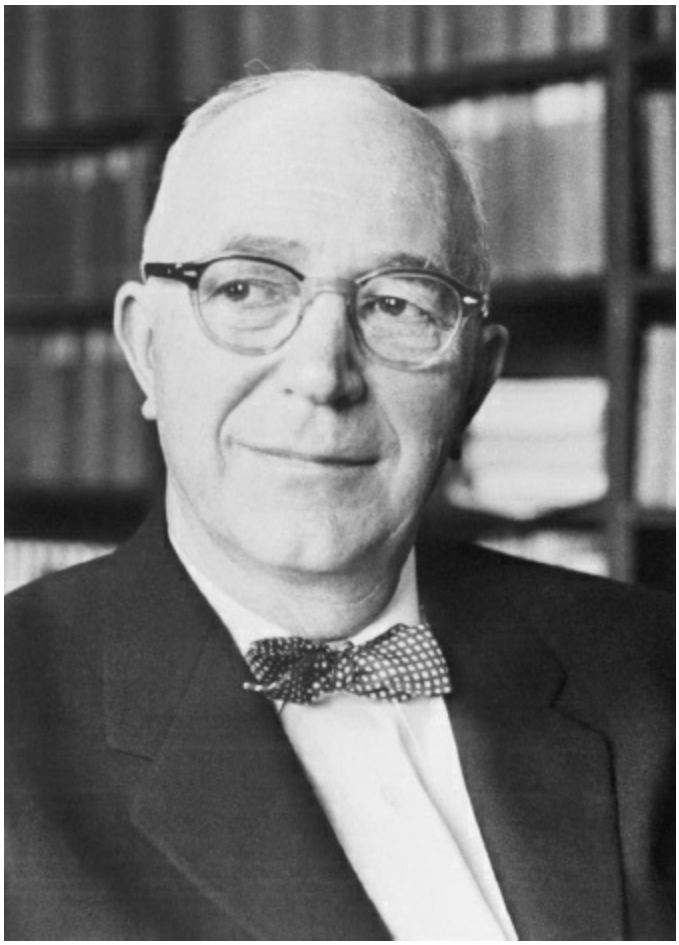


FIGURE 34-27
Gordon Allport. (© Bettmann/Corbis.)

Allport believed that a person's only real guarantee of personal existence is a sense of self. Selfhood develops through a series of stages, from awareness of the body to self-identity. Allport used the term *propriem* to describe strivings related to the maintenance of self-identity and self-esteem. He used the term *traits* to refer to the chief units of personality structure. *Personal dispositions* are individual traits that represent the essence of an individual's unique personality. Maturity is characterized by a capacity to relate to others with warmth and intimacy and an expanded sense of self. In Allport's view, mature persons have security, humor, insight, enthusiasm, and zest. Psychotherapy is geared to helping patients realize these characteristics.

Michael Balint (1896–1970)

Michael Balint was considered a member of the independent or middle group

of object relations theorists in the United Kingdom. Balint believed that the urge for the primary love object underlies virtually all psychological phenomena. Infants wish to be loved totally and unconditionally, and when a mother is not forthcoming with appropriate nurturance, a child devotes his or her life to a search for the love missed in childhood. According to Balint, the *basic fault* is the feeling of many patients that something is missing. As with Ronald Fairbairn and Donald W. Winnicott, Balint understood this deficit in internal structure to result from maternal failures. He viewed all psychological motivations stemming from the failure to receive adequate maternal love.

Unlike Fairbairn, however, Balint did not entirely abandon drive theory. He suggested that libido, for example, is both pleasure-seeking and object seeking. He also worked with seriously disturbed patients, and like Winnicott, he thought that certain aspects of psychoanalytic treatment occur at a more profound level than that of the ordinary verbal explanatory interpretations. Although some material involving genital psychosexual stages of development can be interpreted from the perspective of intrapsychic conflict, Balint believed that certain preverbal phenomena are reexperienced in the analysis. The relationship itself is decisive in dealing with this realm of early experience.

Eric Berne (1910–1970)

Eric Berne (Fig. 34-28) began his professional life as a training and supervising analyst in classic psychoanalytic theory and technique, but ultimately developed his school, known as *transactional analysis.* A *transaction* is a stimulus presented by one person that evokes a corresponding response in another. Berne defined psychological games as stereotyped and predictable transactions that persons learn in childhood and continue to play throughout their lives. *Strokes,* the basic motivating factors of human behavior, consist of specific rewards, such as approval and love. All persons have three ego states that exist within them: the *child,* which represents primitive elements that become fixed in early childhood; the *adult,* which is the part of the personality capable of objective appraisals of reality; and the *parent,* which is an introject of the values of a person’s actual parents. The therapeutic process is geared toward helping patients understand whether

they are functioning in the child, adult, or parent mode in their interactions with others. As patients learn to recognize characteristic games repeatedly played throughout life, they can ultimately function in the adult mode as much as possible in interpersonal relationships.

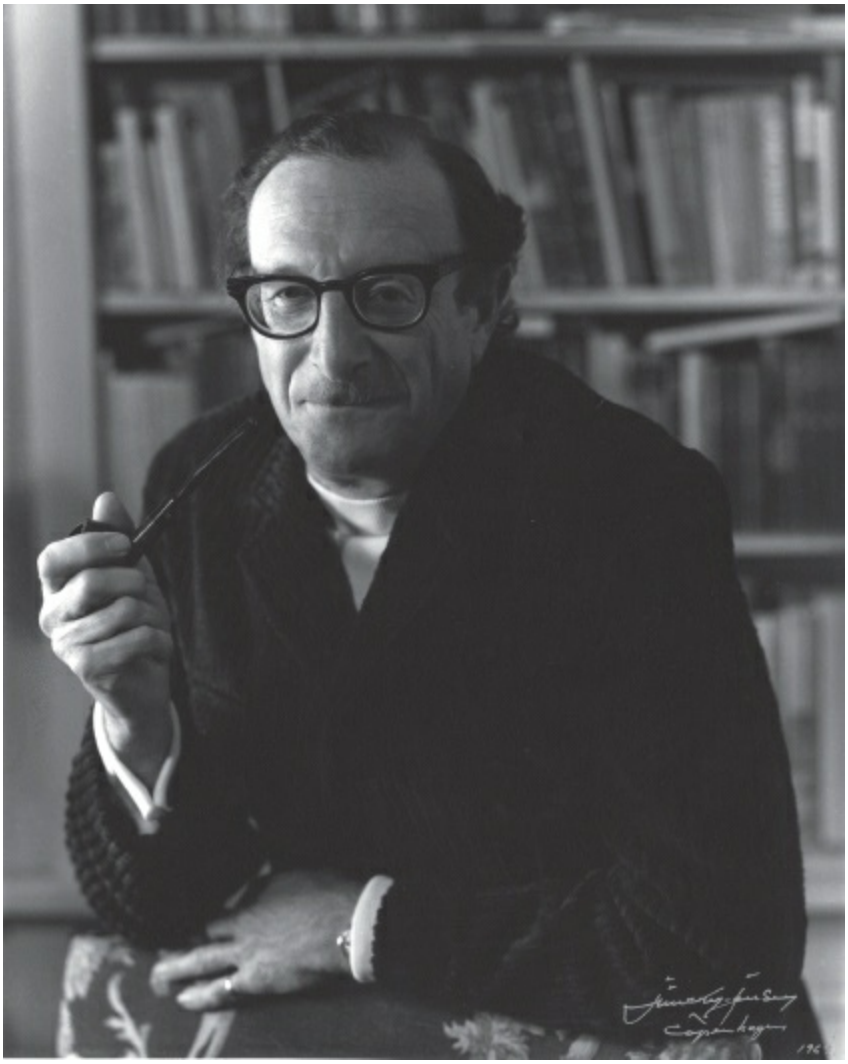


FIGURE 34-28
Eric Berne. (Courtesy of Creative Commons Attribution-Share Alike 4.0 International.)

Wilfred Bion (1897–1979)

Wilfred Bion expanded Melanie Klein's concept of *projective identification* to include an interpersonal process in which a therapist feels coerced by a patient into playing a particular role in the patient's internal world. He also developed the notion that the therapist must contain what the patient has

projected so that it is processed and returned to the patient in a modified form. Bion believed that a similar process occurs between mother and infant. He also observed that "psychotic" and "nonpsychotic" aspects of the mind function simultaneously as sub-organizations. Bion is probably best known for his application of psychoanalytic ideas to groups. Whenever a group gets derailed from its task, it deteriorates into one of three primary states: dependency, pairing, or fight-flight.

John Bowlby (1907–1990)

John Bowlby is generally considered the founder of attachment theory. He formed his ideas about attachment in the 1950s while consulting with the World Health Organization (WHO) on homelessness in children. He stressed that the essence of attachment is *proximity* (i.e., the tendency of a child to stay close to the mother or caregiver). His theory of the mother–infant bond was firmly rooted in biology and drew extensively from ethology and evolutionary theory. A basic sense of security and safety is derived from a continuous and close relationship with a caregiver, according to Bowlby. This readiness for attachment is biologically driven, and Bowlby stressed that attachment is reciprocal. Maternal bonding and caregiving are always intertwined with the child's attachment behavior. Bowlby felt that without this early proximity to the mother or caregiver, the child does not develop a *secure base,* which he considered a launching pad for independence. In the absence of a secure base, the child feels frightened or threatened, and development is severely compromised. Bowlby and attachment theory are discussed in detail in Section 34.1.

Raymond Cattell (1905–1998)

Raymond Cattell obtained his Ph.D. in England before moving to the United States. He introduced *multivariate analysis* and *factor analysis*—statistical procedures that simultaneously examine the relations among multiple variables and factors—to personality. By examining a person's life record objectively, using personal interviewing and questionnaire data, Cattell described various traits representing the building blocks of personality.

Traits are both biologically based and environmentally determined or learned. Biologic traits include sex, gregariousness, aggression, and parental

protectiveness. Environmentally learned traits include cultural ideas, such as work, religion, intimacy, romance, and identity. An important concept is the *law of coercion* to the biosocial mean, which holds that society exerts pressure on genetically different persons to conform to social norms. For example, a person with a strong genetic tendency toward dominance is likely to receive social encouragement for restraint, whereas the naturally submissive person will be encouraged toward self-assertion.

Ronald Fairbairn (1889–1964)

Ronald Fairbairn, a Scottish analyst who worked most of his life in relative isolation, was one of the significant psychoanalytic theorists in the British school of object relations. He suggested that infants are not primarily motivated by the drives of libido and aggression but are by an object seeking instinct. Fairbairn replaced the Freudian ideas of energy, ego, and id with the notion of *dynamic structures.* When an infant encounters frustration, a portion of the ego is defensively split off in the course of development and functions as an entity to internal objects and other subdivisions of the ego. He also stressed that both an object and an object *relationship* are internalized during development, so that a self is always with an object, and the two are connected with an affect.

Sándor Ferenczi (1873–1933)

Although Sándor Ferenczi, a Hungarian analyst, had been analyzed by Freud and was influenced by him, he later discarded Freud's techniques and introduced his method of analysis. He understood the symptoms of his patients as related to sexual and physical abuse in childhood and proposed that analysts need to love their patients in a way that compensates for the love they did not receive as children. He developed a procedure known as *active therapy,* in which he encouraged patients to develop an awareness of reality through active confrontation by the therapist. He also experimented with *mutual analysis,* in which he would analyze his patient for a session and then allow the patient to analyze him for a session.

Viktor Frankl (1905–1997)

His experience in Nazi concentration camps profoundly shaped the Austrian

neurologist and philosopher Viktor Frankl's distinctive view of human nature and psychopathology. There he concluded that even the most appalling circumstances could be endured if one found a way of making them meaningful. He described his experience in *Man's Search for Meaning*, a book that has been read by millions around the world.

Frankl was both a humanist and an existentialist. He believed that human beings shared with other animals somatic and psychological dimensions, but that humans alone also had a spiritual dimension that confers both freedom and responsibility. People find meaning in their lives through creative and productive work, through an appreciation of the world and others, and by freely adopting positive attitudes even in the face of suffering. Those who fail to find meaning face alienation, despair, and existential neuroses. Traditional societies provided a framework of meaning in religion and shared cultural values; in modern society, people must find their sources of meaning, and Frankl attributed many social problems, such as drug abuse and suicide, to their failures to do so.

Because of the spiritual dimension, human beings show self-transcendence and self-distancing. The former refers to the capacity to put other values (e.g., the well-being of a loved one) above self-interest. The latter is the ability to take an external perspective, as seen in a sense of humor. These capacities form the basis for therapeutic interventions in Frankl's version of psychotherapy known as logotherapy. Logotherapy is derived from the Greek word logos, which means thought or reason, and Frankl believed that man instinctively attempts to find universal understanding and harmony in life experiences.

Anna Freud (1895–1982)

Anna Freud (Fig. 34-29), the daughter of Sigmund Freud, ultimately made her own unique contributions to psychoanalysis. Although her father focused primarily on repression as the central defense mechanism, Anna Freud extensively elaborated on individual defense mechanisms, including reaction formation, regression, undoing, introjection, identification, projection, turning against the self, reversal, and sublimation. She was also a key figure in the development of modern ego psychology in that she emphasized that there was "depth in the surface." The defenses marshaled by the ego to avoid

unacceptable wishes from the id were in and of themselves complex and worthy of attention. Up to that point, the primary focus had been on uncovering unconscious sexual and aggressive wishes. She also made seminal contributions to child psychoanalysis and studied the function of the ego in personality development. She founded the Hampstead child therapy course and clinic in London in 1947 and served as its director.



FIGURE 34-29
Anna Freud. (Courtesy of the National Library of Medicine.)



FIGURE 34-30
Erich Fromm. (© Bettmann/Corbis.)

Erich Fromm (1900–1980)

Erich Fromm (Fig. 34-30) came to the United States in 1933 from Germany, where he had received his Ph.D. He was instrumental in founding the William Alanson White Institute for Psychiatry in New York. Fromm identified five character types common to, and determined by, Western culture; each person may possess qualities from one or more types. The types are (1) the *receptive personality* is passive; (2) the *exploitative personality* is manipulative; (3) the *marketing personality* is opportunistic and changeable; (4) the *hoarding personality* saves and stores; and (5) the *productive personality* is mature and enjoys love and work. The therapeutic process

involves strengthening the person's sense of ethical behavior toward others and developing productive love, characterized by care, responsibility, and respect for other persons.

Kurt Goldstein (1878–1965)

Kurt Goldstein (Fig. 34-31) was born in Germany and received his M.D. from the University of Breslau. He was influenced by existentialism and Gestalt psychology—every organism has dynamic properties, which are energy supplies that are relatively constant and evenly distributed. When states of tension-disequilibrium occur, an organism automatically attempts to return to its normal state. What happens in one part of the organism affects every other part, a phenomenon known as *holocoenosis.*

Self-actualization was a concept Goldstein used to describe persons' creative powers to fulfill their potentialities. Because each person has a different set of innate potentialities, persons strive for self-actualization along different paths. Sickness severely disrupts self-actualization. Responses to disruption of an organism's integrity may be rigid and compulsive; regression to more primitive modes of behavior is characteristic. One of Goldstein's significant contributions was his identification of the *catastrophic reaction* to brain damage, in which a person becomes fearful and agitated and refuses to perform simple tasks because of the fear of possible failure.

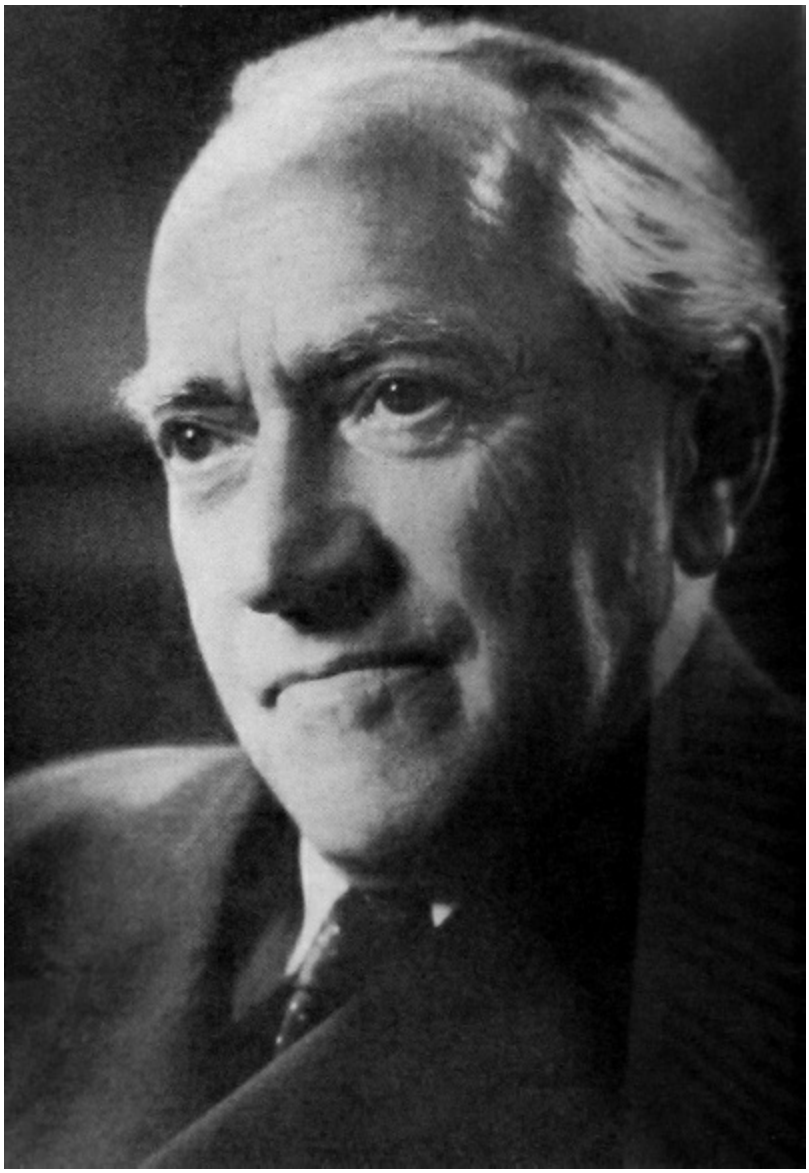


FIGURE 34-31
Kurt Goldstein. (Courtesy of New York Academy of Medicine, New York.)

Karen Horney (1885–1952)

German-born physician-psychoanalyst Karen Horney (Fig. 34-32), who emphasized the preeminence of social and cultural influences on psychosexual development, focused her attention on the differing psychology of men and women and explored the vicissitudes of marital relationships. She taught at the Institute of Psychoanalysis in Berlin before immigrating to the United States. Horney believed that a person's current personality attributes result from the interaction between the person and the environment and are

not solely based on infantile libidinal strivings carried over from childhood. Her theory, known as *holistic psychology,* maintains that a person needs to be seen as a unitary whole who influences, and is influenced by, the environment. She thought that the Oedipus complex was overvalued in terms of its contribution to adult psychopathology, but she also believed that rigid parental attitudes about sexuality led to an excessive concern with the genitals.



FIGURE 34-32
Karen Horney. (Courtesy of the Association for the Advancement of Psychoanalysis, New York.)

She proposed three separate concepts of the self: the *actual self,* the total of

a person's experience; the *real self*, the harmonious, healthy person; and the *idealized self*, the neurotic expectation, or glorified image that a person feels he or she should be. A person's *pride system* alienates him or her from the real self by overemphasizing prestige, intellect, power, strength, appearance, sexual prowess, and other qualities that can lead to self-effacement or self-hatred. Horney also established the concepts of *basic anxiety* and *basic trust.* The therapeutic process, in her view, aims for *self-realization* by exploring distorting influences that prevent the personality from growing.

Edith Jacobson (1897–1978)

Edith Jacobson, a psychiatrist in the United States, believed that the structural model and an emphasis on object relations are not fundamentally incompatible. She thought that the ego, self-images, and object images exert reciprocal influences on one's development. She also stressed that the infant's disappointment with the maternal object is not necessarily related to the mother's actual failure. In Jacobson's view, disappointment is related to a specific, drive-determined demand, rather than a global striving for contact or engagement. She viewed an infant's experience of pleasure or "unpleasure" as the core of the early mother–infant relationship. Satisfactory experiences lead to the formation of good or gratifying images, whereas unsatisfactory experiences create bad or frustrating images. Normal and pathologic development is based on the evolution of these self-images and object images. Jacobson believed that *fixation* refers to modes of object relatedness rather than modes of gratification.

Carl Gustav Jung (1875–1961)

Carl Gustav Jung (Fig. 34-33), a Swiss psychiatrist, formed a psychoanalytic school known as *analytic psychology*, which includes basic ideas related to but going beyond, Freud's theories. After initially being Freud's disciple, Jung broke with Freud over the latter's emphasis on infantile sexuality. He expanded on Freud's concept of the unconscious by describing the *collective unconscious* as consisting of all humankind's common, shared mythologic and symbolic past. The collective unconscious includes *archetypes*—representational images and configurations with universal symbolic meanings. Archetypal figures exist for the mother, father, child, and hero,

among others. Archetypes contribute to *complexes,* feeling-toned ideas that develop from personal experience interacting with archetypal imagery. Thus, a mother complex is determined not only by the mother–child interaction but also by the conflict between archetypal expectation and experience with the real woman who functions in a motherly role.

Jung noted that there are two types of personality organizations: introversion and extroversion. *Introverts* focus on their inner world of thoughts, intuitions, emotions, and sensations; *extroverts* are more oriented toward the outer world, other persons, and material goods. Each person has a mixture of both components. The *persona,* the mask covering the personality, is the face a person presents to the outside world. The persona may become fixed, and the real person hidden from him- or herself. *Anima* and *animus* are unconscious traits possessed by men and women, respectively, and contrast with the persona. *Anima* refers to a man's undeveloped femininity, whereas *animus* refers to a woman's undeveloped masculinity.

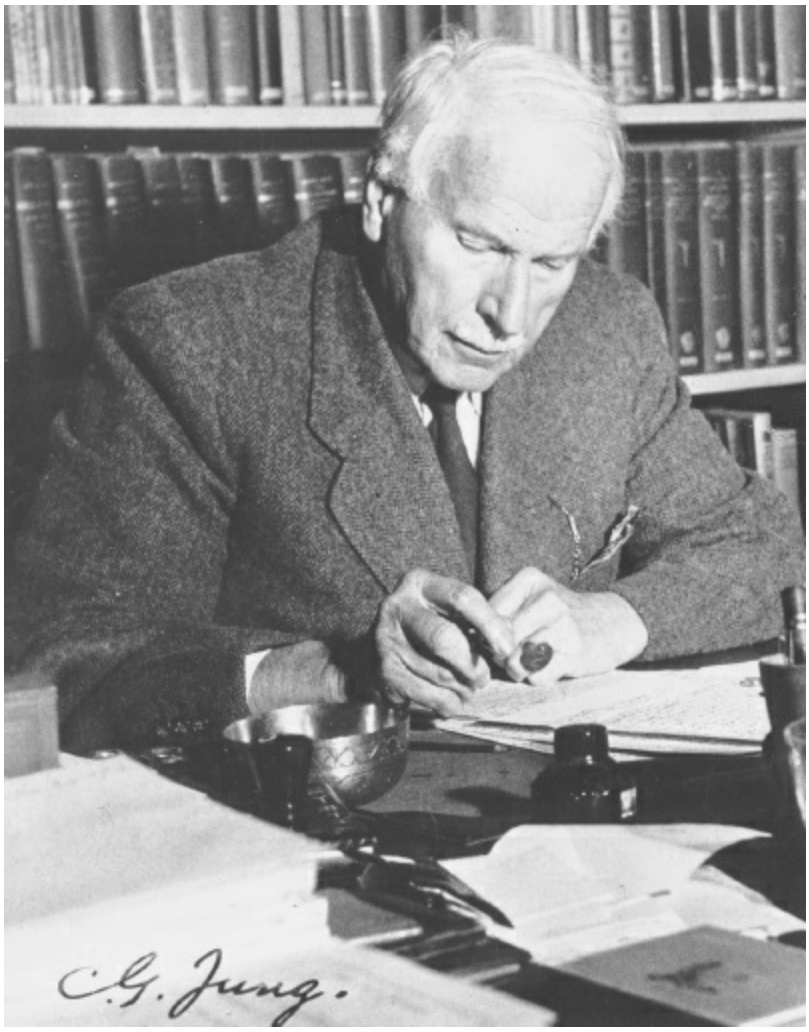


FIGURE 34-33
Carl Gustav Jung (print includes signature). (Courtesy of National Library of Medicine, Bethesda, MD.)

Jungian treatment aims to bring about an adequate adaptation to reality, which involves a person fulfilling their creative potentialities. The ultimate goal is to achieve *individuation*, a process continuing throughout life whereby persons develop a unique sense of their own identity. This developmental process may lead them down new paths away from their previous directions in life.

Otto Kernberg (1928–Present)

Otto Kernberg is perhaps the most influential object relations theorist in the United States. Influenced by both Klein and Jacobson, much of his theory is derived from his clinical work with patients who have borderline personality

disorder. Kernberg places great emphasis on the splitting of the ego and the elaboration of good and bad self-configurations and object configurations. Although he has continued to use the structural model, he views the id as composed of self-images, object images, and their associated affects. Drives appear to manifest themselves only in the context of internalized interpersonal experience. Good and bad self-representations and object relations become associated, respectively, with libido and aggression. Object relations constitute the building blocks of both structure and drives. Goodness and badness in relational experiences precede drive cathexis. The dual instincts of libido and aggression arise from object-directed affective states of love and hate.

Kernberg proposed the term *borderline personality organization* for a broad spectrum of patients characterized by a lack of an integrated sense of identity, ego weakness, absence of superego integration, reliance on primitive defense mechanisms such as splitting and projective identification, and a tendency to shift into primary process thinking. He suggested a specific type of psychoanalytic psychotherapy for such patients in which transference issues are interpreted early in the process.

Melanie Klein (1882–1960)

Melanie Klein (Fig. 34-34) was born in Vienna, worked with Abraham and Ferenczi, and later moved to London. Klein evolved a theory of internal object relations that was intimately linked to drives. Her unique perspective grew mainly from her psychoanalytic work with children, in which she became impressed with the role of unconscious intrapsychic fantasy. She postulated that the ego undergoes a splitting process to deal with the terror of annihilation. She also thought that Freud's concept of the death instinct was central to understanding aggression, hatred, sadism, and other forms of "badness," all of which she viewed as derivatives of the death instinct.

Klein viewed projection and introjection as the primary defensive operations in the first months of life. Infants project derivatives of the death instinct into the mother and then fear attack from the "bad mother," a phenomenon that Klein referred to as *persecutory anxiety.* This anxiety is intimately associated with the *paranoid-schizoid position,* infants' mode of organizing experience in which all aspects of infant and mother are split into

good and bad elements. As the disparate views are integrated, infants become concerned that they may have harmed or destroyed the mother through the hostile and sadistic fantasies directed toward her. At this developmental point, children have arrived at the *depressive position,* in which the mother is viewed ambivalently as having both positive and negative aspects and as the target of a mixture of loving and hateful feelings. Klein was also instrumental in developing child analysis, which evolved from an analytic play technique in which children used toys and played in a symbolic fashion that allowed analysts to interpret the play.



FIGURE 34-34
Melanie Klein. (Courtesy of Melanie Klein and Douglas Glass.)



FIGURE 34-35
Heinz Kohut. (Courtesy of New York Academy of Medicine, New York.)

Heinz Kohut (1913–1981)

Heinz Kohut (Fig. 34-35) is best known for his writings on narcissism and the development of self-psychology. He viewed the development and maintenance of self-esteem and self-cohesion as more critical than sexuality or aggression. Kohut described Freud's concept of narcissism as judgmental, in that development was supposed to proceed toward object relatedness and away from narcissism. He conceived of two separate lines of development, one moving in the direction of object relatedness and the other in the direction of greater enhancement of the self.

In infancy, children fear losing the protection of the early mother–infant bliss and resort to one of three pathways to save the lost perfection: the grandiose self, the alter ego or twinship, and the idealized parental image. These three poles of the self manifest themselves in psychoanalytic treatment in terms of characteristic transferences, known as *self-object transferences.* The *grandiose self* leads to a *mirror transference,* in which patients attempt to capture the gleam in the analyst's eye through exhibitionistic self-display. The *alter ego* leads to the *twinship transference,* in which patients perceive the analyst as a twin. The *idealized parental image* leads to an *idealizing transference,* in which patients feel enhanced self-esteem by being in the presence of the exalted figure of the analyst.

Kohut suggested that empathic failures in the mother lead to a developmental arrest at a particular stage when children need to use others to perform self-object functions. Although Kohut originally applied this formulation to narcissistic personality disorder, he later expanded it to apply to all psychopathology.

Jacques Lacan (1901–1981)

Born in Paris and trained as a psychiatrist, Jacques Lacan founded his institute, the Freudian School of Paris. He attempted to integrate the intrapsychic concepts of Freud with concepts related to linguistics and semiotics (the study of language and symbols). Whereas Freud saw the unconscious as a seething cauldron of needs, wishes, and instincts, Lacan saw it as a sort of language that helps to structure the world. His two principal concepts are that the unconscious is structured like a language, and the unconscious is a discourse. Primary process thoughts are uncontrolled free-flowing sequences of meaning. Symptoms are signs or symbols of underlying processes. The role of the therapist is to interpret the semiotic text of the personality structure. Lacan's most basic phase is the mirror stage; it is here that infants learn to recognize themselves by taking the perspective of others. In that sense, the ego is not part of the self but is something outside of, and viewed by, the self. The ego comes to represent parents and society more than it represents the actual self of the person.

Lacan's therapeutic approach involves the need to become less alienated from the self and more involved with others. Relationships are often

fantasized, which distorts reality and must be corrected. Among his most controversial beliefs was that the resistance to understanding the real relationship with the therapist can be reduced by shortening the length of the therapy session and that psychoanalytic sessions need to be standardized not to time but, instead, to content and process.

Kurt Lewin (1890–1947)

Kurt Lewin received his Ph.D. in Berlin, came to the United States in the 1930s, and taught at Cornell, Harvard, and the Massachusetts Institute of Technology. He adapted the field approach of physics to a concept called *field theory*. A *field* is the totality of coexisting, mutually interdependent parts. Behavior becomes a function of persons and their environment, which together make up the *life space.* The life space represents a field in constant flux, with *valences* or needs that require satisfaction. A hungry person is more aware of restaurants than someone who has just eaten, and a person who wants to mail a letter is aware of mailboxes.

Lewin applied field theory to groups. *Group dynamics* refers to the interaction among members of a group, each of whom depends on the others. The group can exert pressure on a person to change behavior, but the person also influences the group when change occurs.

Abraham Maslow (1908–1970)

Abraham Maslow (Fig. 34-36) was born in Brooklyn, New York, and completed his undergraduate and graduate work at the University of Wisconsin. Along with Goldstein, Maslow believed in *self-actualization theory*—the need to understand the totality of a person. A leader in humanistic psychology, Maslow described a hierarchical organization of needs present in everyone. As the more primitive needs, such as hunger and thirst, are satisfied, more advanced psychological needs, such as affection and self-esteem, become the primary motivators. Self-actualization is the highest need.

A peak experience, frequently occurring in self-actualizers, is an episodic, brief occurrence in which a person suddenly experiences a powerful transcendental state of consciousness—a sense of heightened understanding, an intense euphoria, an integrated nature, unity with the universe, and an

altered perception of time and space. This powerful experience tends to occur most often in the psychologically healthy and can produce long-lasting beneficial effects.



FIGURE 34-36
Abraham H. Maslow. (© Bettmann/Corbis.)

Karl A. Menninger (1893–1990)

Karl A. Menninger was one of the first physicians in the United States to receive psychiatric training. With his brother, Will, he pioneered the concept of a psychiatric hospital based on psychoanalytic principles and founded the Menninger Clinic in Topeka, Kansas. He also was a prolific writer; *The Human Mind,* one of his most famous books, brought psychoanalytic understanding to the lay public. He made a compelling case for the validity of Freud's death instinct in *Man Against Himself.* In *The Vital Balance,* his

magnum opus, he formulated a unique theory of psychopathology. Menninger maintained a lifelong interest in the criminal justice system and argued in *The Crime of Punishment* that many convicted criminals needed treatment rather than punishment. Finally, his volume titled *Theory of Psychoanalytic Technique* was one of the few books to examine the theoretical underpinnings of psychoanalysts' interventions.

Adolf Meyer (1866–1950)

Adolf Meyer (Fig. 34-37) came to the United States from Switzerland in 1892 and eventually became director of the psychiatric Henry Phipps Clinic of the Johns Hopkins Medical School. Not interested in metapsychology, he espoused a commonsense psychobiological methodology for the study of mental disorders, emphasizing the interrelationship of symptoms and individual psychological and biologic functioning. His approach to the study of personality was biographical; he attempted to bring psychiatric patients and their treatment out of isolated state hospitals and into communities and was also a strong advocate of social action for mental health. Meyer introduced the concept of *common sense psychiatry* and focused on how a patient's current life situation could be realistically improved. He coined the concept of *ergasia*, the action of the total organism. His goal in therapy was to aid patients' adjustment by helping them modify unhealthy adaptations. One of Meyer's tools was an autobiographical life chart constructed by the patient during therapy.



FIGURE 34-37
Adolf Meyer. (From the National Library of Medicine, Bethesda, MD.)

Gardner Murphy (1895–1979)

Gardner Murphy (Fig. 34-38) was born in Ohio and received his Ph.D. from Columbia University. He was among the first to publish a comprehensive history of psychology and made significant contributions to social, general, and educational psychology. According to Murphy, three essential stages of personality development are the stage of undifferentiated wholeness, the stage of differentiation, and the stage of integration. This development is frequently uneven, with both regression and progression occurring along the way. The four inborn human needs are visceral-, motor-, sensory-, and

emergency-related. These needs become increasingly specific in time as a person's experiences mold them in various social and environmental contexts. *Canalization* brings about these changes by establishing a connection between a need and a specific way of satisfying the need.

Murphy was interested in parapsychology. States such as sleep, drowsiness, specific drug, toxic conditions, hypnosis, and delirium tend to be favorable to paranormal experiences. Impediments to paranormal awareness include various intrapsychic barriers, conditions in the general social environment, and heavy investment in ordinary sensory experiences.

Henry Murray (1893–1988)

Henry Murray (Fig. 34-39) was born in New York City, attended medical school at Columbia University, and was a founder of the Boston Psychoanalytic Institute. He proposed the term *personology* to describe the study of human behavior. He focused on *motivation,* a need aroused by internal or external stimulation; once aroused, motivation produces continued activity until the need is reduced or satisfied. He developed the *Thematic Apperception Test* (TAT), a projective technique used to reveal unconscious and conscious mental processes and problem areas.



FIGURE 34-38
Gardner Murphy. (Courtesy of New York Academy of Medicine, New York.)

Frederick S. Perls (1893–1970)

Gestalt theory developed in Germany under the influence of several men: Max Wertheimer (1880–1943), Wolfgang Köhler (1887–1967), and Lewin. Frederick "Fritz" Perls (Fig. 34-40) applied Gestalt theory to a therapy that emphasizes the current experiences of the patient in the here and now, as contrasted to the there and then of psychoanalytic schools. In terms of motivation, patients learn to recognize their needs at any given time and how the drive to satisfy these needs may influence their current behavior. According to the Gestalt point of view, behavior represents more than the sum of its parts. A *gestalt,* or a whole, includes and goes beyond the sum of smaller, independent events; it deals with essential characteristics of experience, such as value, meaning, and form.

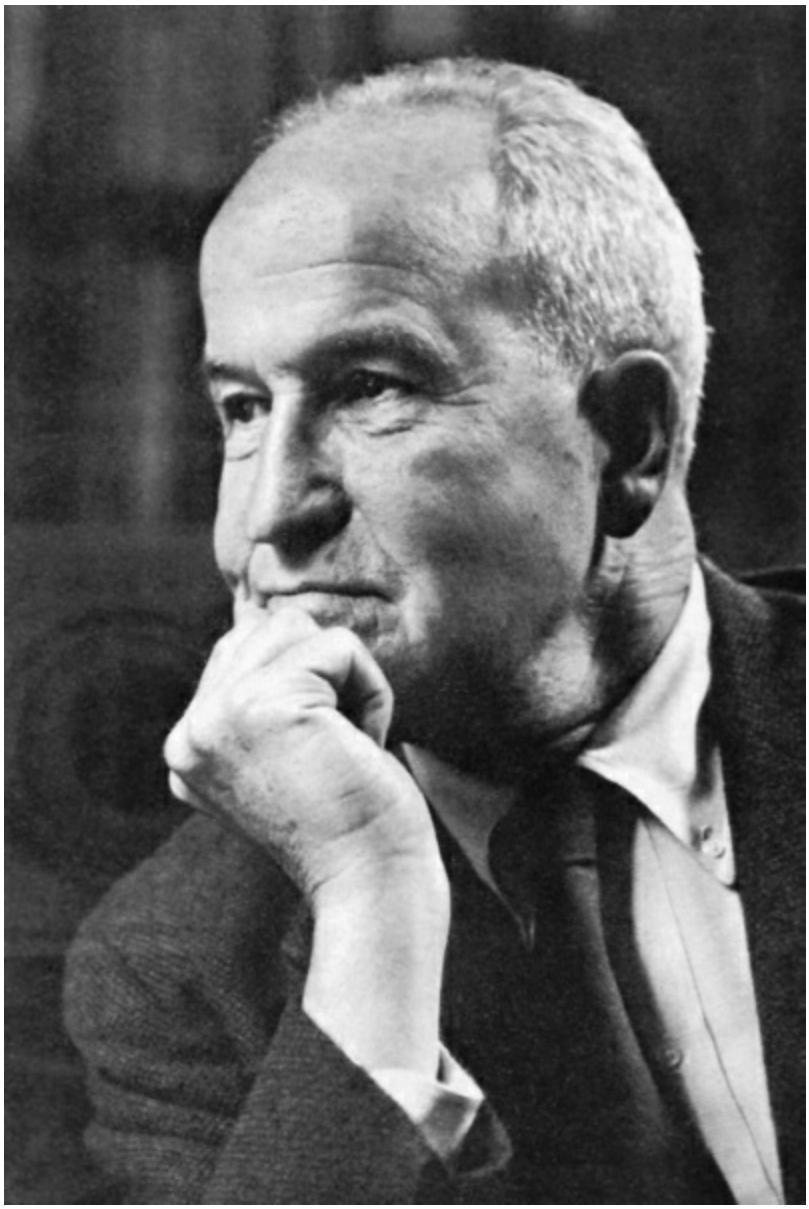


FIGURE 34-39
Henry Murray. (Courtesy of New York Academy of Medicine, New York.)

Sandor Rado (1890–1972)

Sandor Rado (Fig. 34-41) came to the United States from Hungary in 1945 and founded the Columbia Psychoanalytic Institute in New York. His theories of *adaptational dynamics* hold that the organism is a biologic system operating under hedonic control, which is somewhat similar to Freud's pleasure principle. Cultural factors often cause excessive hedonic control and disordered behavior by interfering with the organism's ability for *self-*

regulation. In therapy, the patient needs to relearn how to experience pleasurable feelings.

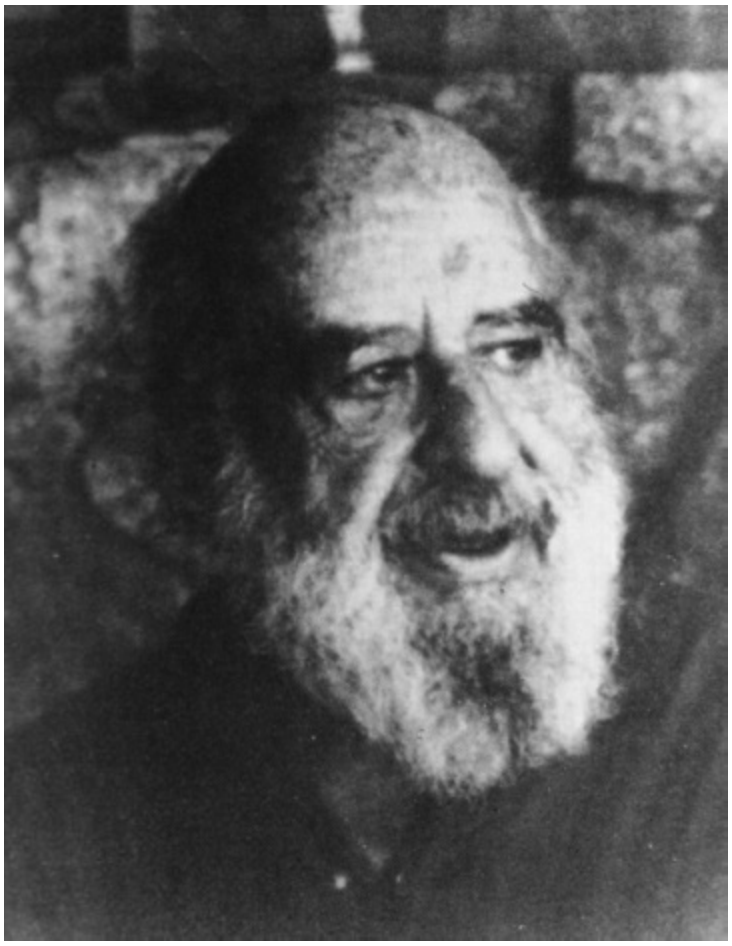


FIGURE 34-40
Fritz Perls. (Courtesy of the National Library of Medicine.)

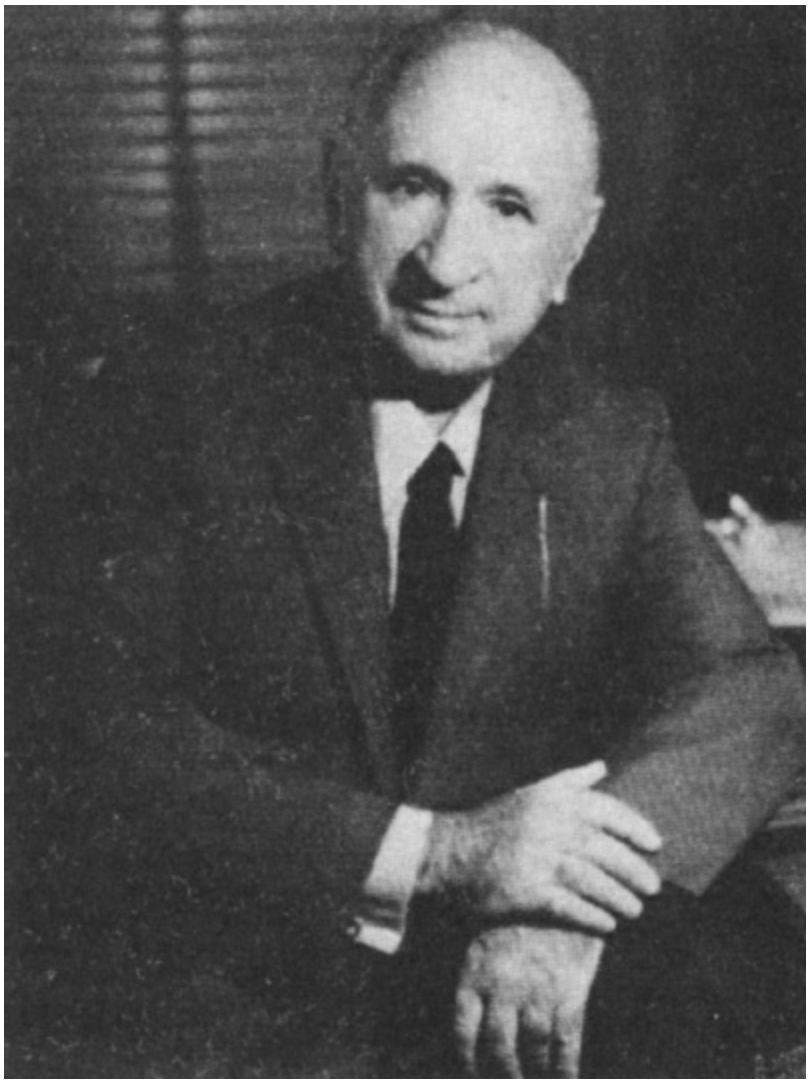


FIGURE 34-41
Sandor Rado. (Courtesy of New York Academy of Medicine.)

Otto Rank (1884–1939)

An Austrian psychologist and a protégé of Sigmund Freud, Otto Rank (Fig. 34-42) broke with Freud in his 1924 publication, *The Trauma of the Birth,* and developed a new theory called *birth trauma.* Anxiety is correlated with separation from the mother—specifically, with separation from the womb, the source of effortless gratification. This painful experience results in primal anxiety. Sleep and dreams symbolize the return to the womb.



FIGURE 34-42
Otto Rank. (Courtesy of New York Academy of Medicine.)

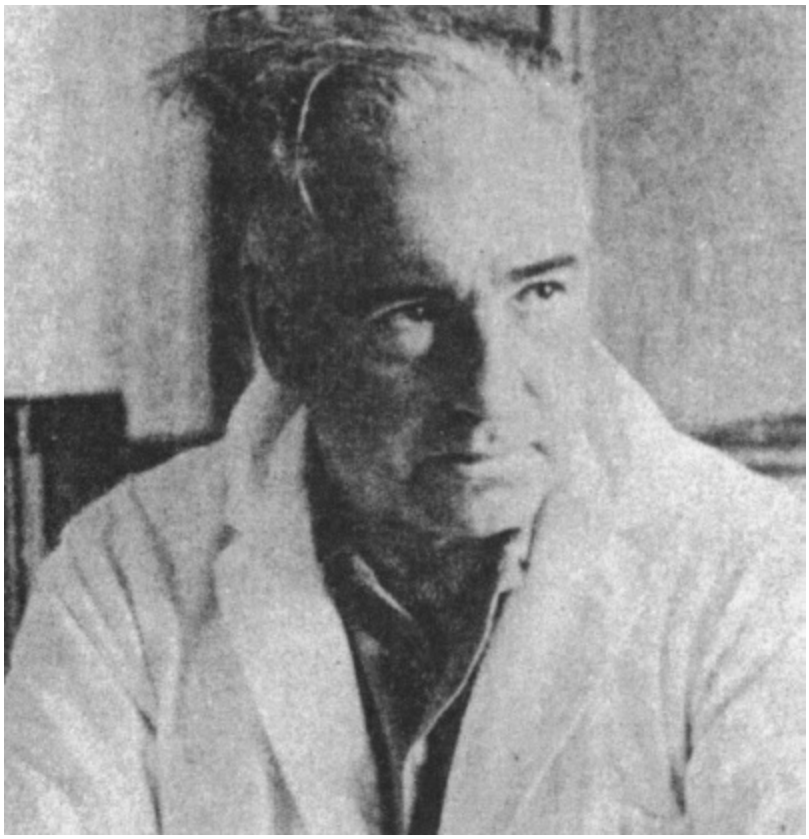


FIGURE 34-43
Wilhelm Reich. (Courtesy of New York Academy of Medicine.)

The personality is divided into impulses, emotions, and will. Children's impulses seek immediate discharge and gratification. As impulses are mastered, as in toilet training, children begin the process of will development. If the will is carried too far, pathologic traits (e.g., stubbornness, disobedience, and inhibitions) may develop.

Wilhelm Reich (1897–1957)

Wilhelm Reich (Fig. 34-43), an Austrian psychoanalyst, made significant contributions to psychoanalysis in character formation and character types. The term *character armor* refers to the personality's defenses that serve as resistance to self-understanding and change. The four primary character types are as follows: the *hysterical character* is sexually seductive, anxious, and fixated at the phallic phase of libido development; the *compulsive character* is controlled, distrustful, indecisive, and fixated at the anal phase; the *narcissistic character* is fixated at the phallic stage of development, and if the

person is male, he has contempt for women; and the *masochistic character* is long-suffering, complaining, and self-deprecatory, with excessive demand for love.

The therapeutic process, called *will therapy*, emphasizes the relationship between patient and therapist; the goal of treatment is to help patients accept their separateness. A definite termination date for therapy is used to protect against excessive dependence on the therapist.

Carl Rogers (1902–1987)

Carl Rogers (Fig. 34-44) received his Ph.D. in psychology from Columbia University. After attending Union Theological Seminary in New York, Rogers studied for the ministry. His name is most clearly associated with the *person-centered theory* of personality and psychotherapy, in which the central concepts are self-actualization and self-direction. Specifically, persons are born with a capacity to direct themselves in the healthiest way toward a level of completeness called self-actualization. From his person-centered approach, Rogers viewed personality not as a static entity composed of traits and patterns but as a dynamic phenomenon involving ever-changing communications, relationships, and self-concepts.

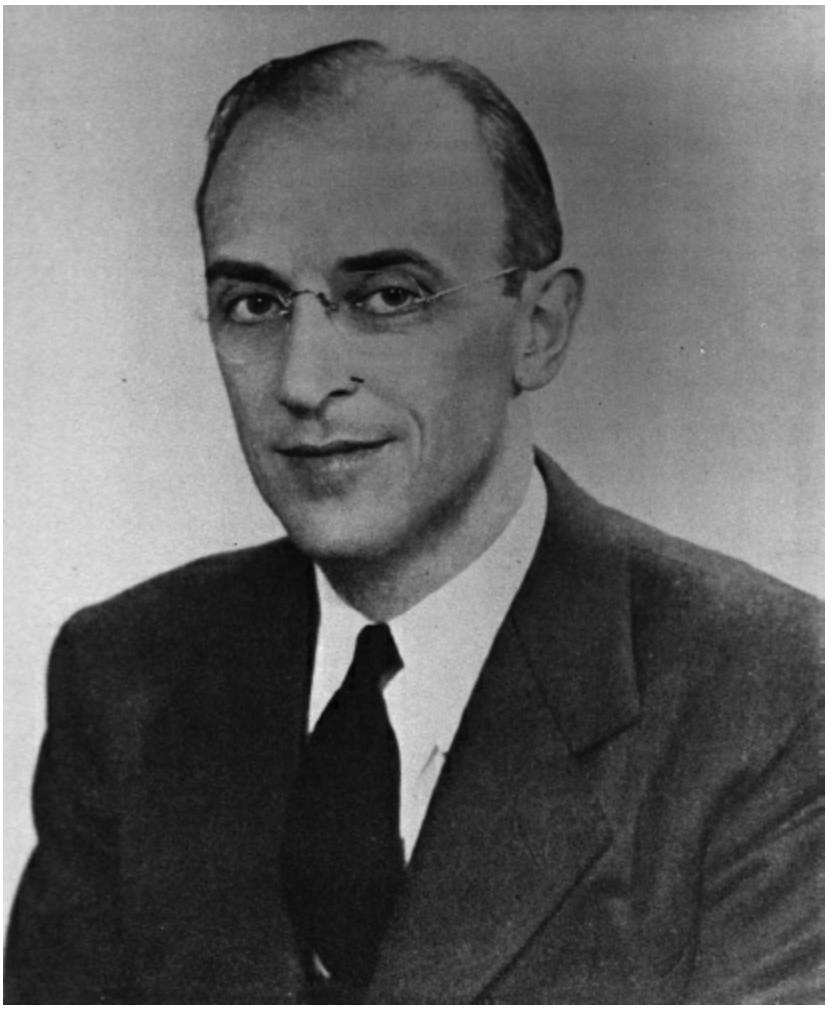


FIGURE 34-44
Carl Rogers. (Courtesy of the National Library of Medicine.)

Rogers developed a treatment program called *client-centered psychotherapy*. Therapists attempt to produce an atmosphere in which clients can reconstruct their strivings for self-actualization. Therapists hold clients in unconditional positive regard, which is the total nonjudgmental acceptance of clients. Other therapeutic practices include attention to the present, focus on clients' feelings, emphasis on process, trust in the potential and self-responsibility of clients, and a philosophy grounded in a positive attitude toward them, rather than a preconceived structure of treatment.

Jean-Paul Sartre (1905–1980)

Born in Paris, Jean-Paul Sartre wrote plays and novels before turning to psychology. He was a German prisoner of war from 1940 to 1941 during

World War II. Influenced by the ideas of Martin Heidegger, he developed what he called *existential psychoanalysis.* The reflective self was a crucial concept in Sartre's psychology. He recognized that humans alone could reflect on themselves as objects so that the experience of "being" in humans is unique in the natural world. This capacity to reflect leads humans to impose meaning on existence. For Sartre, this meaning allows a human being to create their essence.

Sartre denied the unconscious; he thought that human beings were condemned to be free and to face the fundamental existential dilemma—their aloneness without a god to provide meaning. As a result, each individual creates values and meanings. Neurosis is an escape from freedom, which is the key to maintaining psychological health. Sartre made no distinction between philosophy and psychology. Psychologists, as with philosophers, search for the truth about the world. Part of this truth, in Sartre's view, was the dialectic between consciousness and *being.* Consciousness introduces nothingness and is a negation of being-in-itself. Ideals are revealed in actions, not in professed beliefs.



FIGURE 34-45
B. F. Skinner. (Courtesy of New York Academy of Medicine, New York.)

B. F. Skinner (1904–1990)

Burrhus Frederic Skinner (Fig. 34-45), commonly known as B. F. Skinner, received his Ph.D. in psychology from Harvard University, where he taught for many years. Skinner's seminal work in operant learning laid much of the groundwork for many current methods of behavior modification, programmed instruction, and general education. His global beliefs about the nature of behavior have been applied more widely. It can be argued, than those of any other theorist, except, perhaps, Freud. His impact has been impressive in scope and magnitude.

Skinner's approach to personality was derived more from his fundamental beliefs about behavior than from a specific theory of personality per se. To Skinner, personality did not differ from other behaviors or sets of behaviors; it is acquired, maintained, and strengthened or weakened according to the

same rules of reward and punishment that alter any other behavior. *Behaviorism,* as Skinner's basic theory is most commonly known, is concerned only with observable, measurable behavior that can be operationalized. Many abstract and mentalistic hallmarks of other dominant personality theories have little place in Skinner's framework. Concepts such as self, ideas, and ego are considered unnecessary for understanding behavior. Through the process of operant conditioning and the application of basic principles of learning, persons are believed to develop sets of behavior that characterize their responses to the world of stimuli that they face in their lives. Such a set of responses is called *personality.*

Harry Stack Sullivan (1892–1949)

Harry Stack Sullivan (Fig. 34-46) is generally acknowledged as the most original and distinctive American-born theorist in dynamic psychiatry. When psychiatrists use the term *parataxic distortion,* apply the concept of self-esteem, consider the importance of preadolescent peer groups in development, or view a patient's behavior as an interpersonal manipulation, they are applying concepts Sullivan first proposed.

Sullivan described three modes of experiencing and thinking about the world. The *prototaxic mode* is undifferentiated thought that cannot separate the whole into parts or use symbols. It occurs in infancy and also appears in patients with schizophrenia. In the *parataxic mode,* events are causally related because of temporal or serial connections. Logical relationships, however, are not perceived. The *syntaxic mode* is the logical, rational, and most mature type of cognitive functioning of which a person is capable. These three types of thinking and experiencing occur side by side in all persons; it is the rare person who functions exclusively in the syntaxic mode.



FIGURE 34-46
Harry Stack Sullivan. (Courtesy of the National Library of Medicine.)

The total configuration of personality traits is known as the *self-system*, which develops in various stages and is the outgrowth of interpersonal experiences, rather than an unfolding of intrapsychic forces. During infancy, anxiety occurs for the first time when infants' primary needs are not satisfied. During childhood, from 2 to 5 years, a child's main tasks are to learn the requirements of the culture and how to deal with influential adults. As a juvenile, from 5 to 8 years, a child needs peers and must learn how to deal with them. In preadolescence, from 8 to 12 years, the capacity for love and collaboration with another person of the same sex develops. This so-called *chum period* is the prototype for a sense of intimacy. In the history of patients with schizophrenia, this experience of chums is often missing. During adolescence, significant tasks include the separation from the family, the

development of standards and values, and the transition to heterosexuality.

The therapy process requires the active participation of the therapist, who is known as a *participant-observer*. Modes of experience, particularly the parataxic, need to be clarified, and new patterns of behavior need to be implemented. Ultimately, persons need to see themselves as they are, instead of as they think they are or want others to think they are.

Sullivan is best known for his creative psychotherapeutic work with severely disturbed patients. He believed that even the most psychotic patients with schizophrenia could be reached through the human relationship of psychotherapy.

Donald W. Winnicott (1896–1971)

Donald W. Winnicott (Fig. 34-47) was one of the central figures in the British school of object relations theory. His theory of *multiple self-organization* included a *true self*, which develops in the context of a responsive *holding environment* provided by a *good-enough mother*. When infants experience a traumatic disruption of their developing sense of self, however, a false self emerges and monitors and adapts to the conscious and unconscious needs of the mother; it thus provides a protected exterior behind which the true self is afforded the privacy that it requires to maintain its integrity.

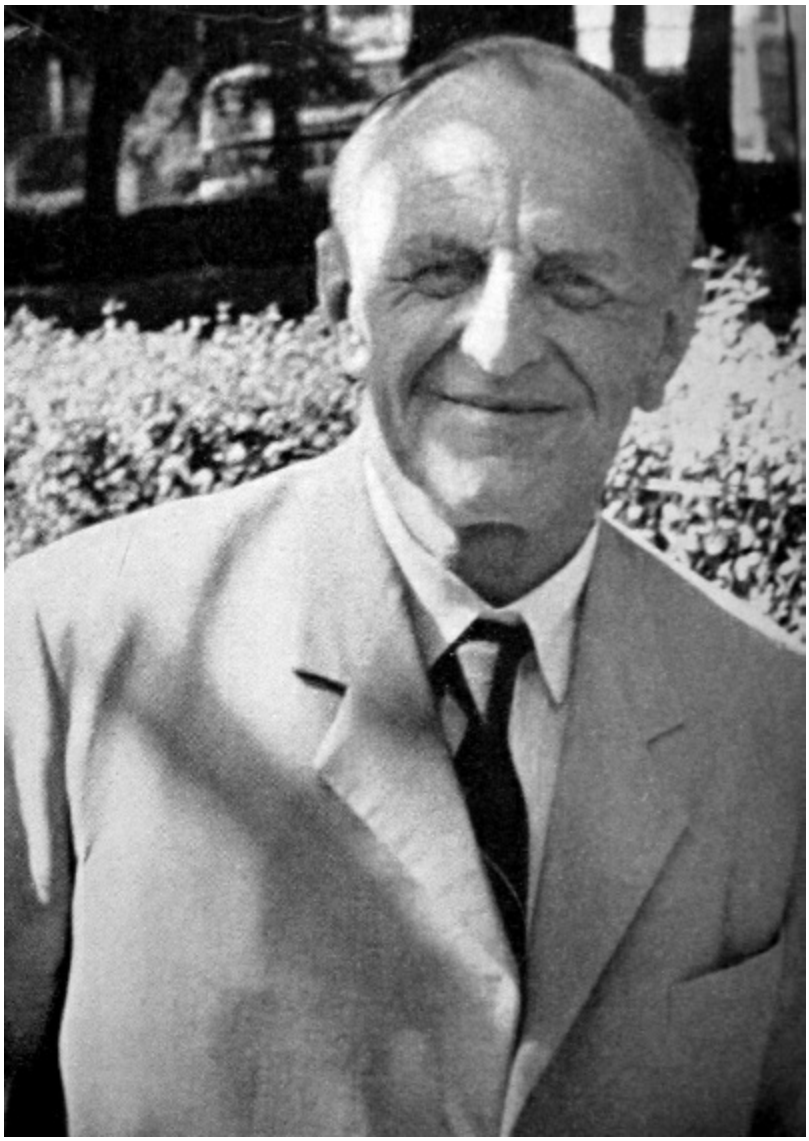


FIGURE 34-47
Donald Winnicott. (Courtesy of New York Academy of Medicine, New York.)

Winnicott also developed the notion of the *transitional object.* Ordinarily, a pacifier, blanket, or teddy bear, this object serves as a substitute for the mother during infants' efforts to separate and become independent. It provides a soothing sense of security in the absence of the mother.

The case history below illustrates how the different psychodynamic schools discussed in this chapter can be applied to the clinical observations.

Mr. A was a 26-year-old white man who had a history of bipolar I disorder. He was brought in for treatment after not completing the last

required course for his advanced degree and being arrested for disturbing the peace. He had consistently lied to his family about where he stood with his coursework and about having skipped an examination that would have qualified him to use his professional degree. He had also not told them that he had been using marijuana almost daily for several years and occasionally used hallucinogens. His arrest for disorderly conduct was for swimming naked in an apartment complex in the middle of the night while taking hallucinogens.

Mr. A's use of marijuana began early in college but became daily during graduate school. He was diagnosed as having bipolar I disorder early in his senior year at college after an apparent episode of mania. His mood disorder was well controlled on lithium (Eskalith). During graduate school, he was episodically compliant with medications, preferring to try to maintain a state of hypomania. He saw a psychiatrist every 3 to 6 months for medication checks. During his 4 years in graduate school, he had two clear episodes of depression and began taking sertraline (Zoloft), 100 mg/day, with questionable benefit. Mr. A believed that he could be a great writer. He spent most of his time reading and trying to write. He dreamed of going to New York and becoming part of a group of avant-garde writers that would parallel the Algonquin Club of the 1930s or the Beat poets of the late 1940s. This aspiration and his marijuana abuse predated his development of bipolar I disorder. He attended class episodically, nonetheless performing adequately. His last class had no final examination but required a paper. He planned to write this paper in the form of a play, involving a dialogue between two thinkers from different times and cultures. His professor was very excited about this idea, but Mr. A kept postponing the task until he was forced to extend his schooling by a year. His other significant interest during this time involved growing and photographing flowers.

Mr. A was born and raised in a large city. His father had been very successful in commercial real estate, and his mother, after raising the children, used the substantial real estate holdings she inherited from her father to set up a business to manage them. Most of the money was placed in trust for the patient and his siblings. His mother had total financial control of the trusts and doled out the proceeds to the children as they needed them. There was no family history of any psychiatric

disorders.

The patient described his mother as very loving and caring but intrusive and controlling. For example, the mother arranged the initial treatment but then was angry that the psychiatrist had not called her regularly to report on her adult son's progress. She was also critical of various aspects of the treatment, as reported to her by her son. The patient's two older siblings had attended prestigious colleges and graduate schools but had returned home to work in the mother's real estate management company. The 30-year-old sister was living in the parents' home. The 35-year-old brother had lived at home for a time but then moved out to a location a few blocks away. There was a younger brother, still in college, who also smoked marijuana excessively. He tried to minimize the patient's problems to the family and tried to protect the patient, who desperately had not wanted to return home. Of note is that none of the children were married, although the two older ones had a couple of serious relationships.

The children seemed to regard the mother with affectionate amusement and bemusement. The father was seen as a very caring but undemonstrative man who put much energy into keeping the mother from becoming too upset and encouraged them to do the same. The children often wanted to provoke the mother for her judgmental, detail-oriented intrusiveness. The father discouraged them but occasionally found their provocations amusing.

The family viewed itself as very close, with strong values oriented toward community service and family loyalty. The family belonged to a religious community but expressed their involvement primarily in social service and social action volunteer work, accompanied by generous financial contributions.

The patient had been a very successful debater in high school and recalled his development as very positive but provided few details. He tended to place himself in the role of the outsider, an observer of humanity, which he saw as consonant with the role of a writer. He was proud to have bipolar I disorder and tried to regulate his medications so that he would be hypomanic much of the time, seeing this as enhancing his creativity. He viewed his use of marijuana in the same vein. One of the most distressing aspects of his depressive episodes was that marijuana

no longer created a feeling of well-being but made him feel worse. His current depressive episode involved no neurovegetative symptoms. Instead, he presented as flat, numb, apathetic, ashamed, anhedonic, and anergic. He was notably ashamed of being back in his hometown and of living with his parents.

The patient ostensibly understood and accepted his illness well and had read much about it. However, the family had responded to the information "with proper treatment, bipolars can live normal lives," meaning that the information should be kept secret to be treated normally. Mr. A, on the other hand, was very open with friends at graduate school about his illness and his pride in it and the creativity he associated with it.

The patient had two long-standing recurrent dreams. One involved him flying. The narrative line varied, but the flying theme recurred. Often, he had other magical powers in his dreams, such as the ability to heal, not to be killed by bullets, to save the world or some group of people from mortal danger, and so on. The other recurrent dream was of a hotel lobby. These dreams regularly began with him entering a hotel lobby to meet a group of people, accompanied by a feeling of dread.

POSITIVE PSYCHOLOGY

Positive psychology is an umbrella term describing the scientific study of what makes life most worth living. Research findings from positive psychology are intended to provide a more complete and balanced scientific understanding of the human experience. The new field of positive psychology calls for as much focus on strength as on weakness, as much interest in building the best things in life as in repairing the worst, and as much concern with making the lives of normal people fulfilling as with healing pathology.

Positive psychology does not replace business-as-usual psychology, which often focuses on people's problems and how to remedy them. Instead, positive psychology intends to complement and extend problem-focused psychology. The attention of positive psychologists is increasingly turning to deliberate interventions that promote the well-being of individuals and groups, and again, these should be regarded as supplements to existing therapies.

Positive psychology studies what goes right in life, from birth to death. It is concerned with optimal experience—people being their best and doing their best. Everyone's life has peaks and valleys, and positive psychology does not deny the low points. Its signature premise is nuanced: What is good about life is as genuine as what is bad and deserves equal attention from psychologists. Positive psychology assumes that life entails more than avoiding or undoing problems and that explanations of the good life must do more than reverse accounts of distress and dysfunction.

Empirical Findings

Although still a young field, positive psychology already has a canon of established findings worth considering. Indeed, positive psychology is a bottom-up field, very much defined by its empirical results. Discussed below are some of the things that have been learned about positive experiences, positive traits, positive relationships, and positive institutions.

When psychologists study self-reported happiness and life satisfaction, usually under the rubric of subjective well-being, they administer numerical rating scales. The consistent and perhaps surprising result is that most people in most circumstances, most of the time, score above the scale midpoint, whether they are multimillionaires in the United States or pavement dwellers in Calcutta. This conclusion holds across demographic characteristics like age, sex, ethnicity, and education, each of which has a surprisingly small association with avowed happiness.

The significant correlates of happiness are social. In contrast to the modest demographic correlates of happiness and well-being, consider the following robust correlates:

- Number of friends
- Being married
- Being extroverted
- Being grateful
- Being religious
- Pursuing leisure activities
- Employment (not income)

In a study that compared happy people to very happy people, there was one

striking difference: good relationships with other people. Of the very happy people in the sample, all had close relationships with others. Psychology research documents very few necessary or sufficient conditions, but these data suggest that good social relationships may be a necessary condition for extreme happiness.

People who are successful in life's venues are, of course, happy, but the less obvious and more interesting finding from experimental and longitudinal research is that happiness foreshadows success in academic, vocational, and interpersonal realms.

Having good relationships with other people is the most critical contributor to a satisfying life and may even be a necessary condition for happiness. Having a "best friend" at work is a strong predictor of satisfaction and even productivity. A good relationship is one in which the amount of positive communication considerably outweighs negative communication.

Positive psychologists have taken a close look at the features of positive communication, describing four ways in which a person can respond to someone else when something happens, including good events such as a raise at work:

- Active-constructive responding—an enthusiastic response: "That's great; I bet you'll receive many more raises."
- Active-destructive responding—a response that points out the potential downside: "Are they going to expect more of you now?"
- Passive-constructive responding—a muted response: "That's nice, dear."
- Passive-destructive responding—a response that conveys disinterest: "It rained all day here."

Couples who use active-constructive responding have good marriages. The other responses, if they dominate, are associated with marital dissatisfaction. Although this research has only been done in the context of marriage, it may well generalize to other relationships.

Psychology and psychiatry have a long history of either ignoring religion or regarding it with suspicion. However, research findings have begun to accumulate, showing that religion has particular benefits in various psychological domains. Internalized religious beliefs may help a person to cope with problems and even avoid physical illness in the first place. Religiousness is robustly associated with longevity, happiness, and other

indices of a life lived well.

People who are so poor that they cannot meet their basic needs are unhappy, but above extreme poverty, increased income has a surprisingly small relationship with life satisfaction.

Despite the small contribution of income to well-being, whether or not someone is working is much more strongly related to happiness. People who are employed and engaged in what they do are happy, regardless of the status or compensation associated with their job. Happiness and engagement lead people to regard their work as a calling and be more productive at whatever they do, take fewer sick days, and even postpone their retirement.

According to Aristotle's notion of *eudaimonia*—being faithful to one's inner self (demon)—true happiness entails identifying one's virtues, cultivating them, and living by them. Contrast this notion with the equally venerable idea of *hedonism*—pursuing pleasure and avoiding pain—that is the foundation for *utilitarianism,* which in turn provides the underpinning of psychoanalysis and all but the most radical of the behaviorisms. Research shows that eudaimonia consistently trumps pleasure as a predictor of life satisfaction. Those who pursue eudaimonic goals and activities are more satisfied than those who pursue pleasure. This is not to say that hedonism is irrelevant to life satisfaction, just that all things being equal, hedonism contributes less to long-term happiness than does eudaimonia.

Although the study of positive institutions is in its infancy, there is an agreement that institutions that allow people to flourish—whether families, schools, workplaces, or even entire societies—share a core of common characteristics:

- Purpose—a shared vision of the moral goals of the institution, one reinforced by remembrances and celebrations
- Safety—protection against threat, danger, and exploitation
- Fairness—equitable rules governing reward and punishment and the means for consistently enforcing them
- Humanity—mutual care and concern
- Dignity—the treatment of all people in the institution as individuals regardless of their position

Psychologists, at least in the United States, have long believed that the human condition can be improved by the intelligent application of what they

have learned. Positive psychologists are no exception, and many have turned their attention to interventions that make people more happy, hopeful, virtuous, accomplished, and socially involved. In some cases, these applications are running in front of data that would support them, but in other cases, outcome research has been done. Even the most compelling research is not based on follow-up that extends beyond a few years, and the research participants are usually motivated and willing volunteers. How well these interventions will generalize—across diverse people and over time—is a research topic of high priority.

Positive Psychology and Clinical Work

When positive psychology was first described, its stated goal was not to move people from –5 to zero—the goal of conventional psychology and psychiatry—but rather from +2 to +5, within the upper right-hand quadrant of Figure 34-48. This emphasis on *promotion* instead of remediation is an essential feature of a positive psychology perspective, but it does not do justice to this new field and its potential role in clinical work.

Positive Psychology's Vision of Psychological Health. In its 1948 constitution, the World Health Organization (WHO) defined *health* as "a state of complete physical, mental, and social well-being and not merely the absence of disease or infirmity." In more recent years, this statement has been expanded to include the ability to lead a life that is socially and economically productive. This definition is an essential declaration that health entails more than the absence of illness, but it is circular since "well-being" is a synonym for "health." Work by positive psychologists makes this definition more concrete and useful as a guide for research and intervention.

If one can extrapolate from the sorts of topics that have been studied, positive psychology assumes that people are doing well when they experience more positive feelings than negative feelings, are satisfied with their lives as they have been lived, have identified what they do well, and use these talents and strengths on an ongoing basis, are highly engaged in what they do, are contributing members of a social community, and have a sense of meaning and purpose in their lives. Physical health and safety, of course, provide an essential context for psychological well-being. It is difficult to

imagine a cultural group in which these components of the good life are not valued. Respect for human diversity need not entail extreme cultural relativism.

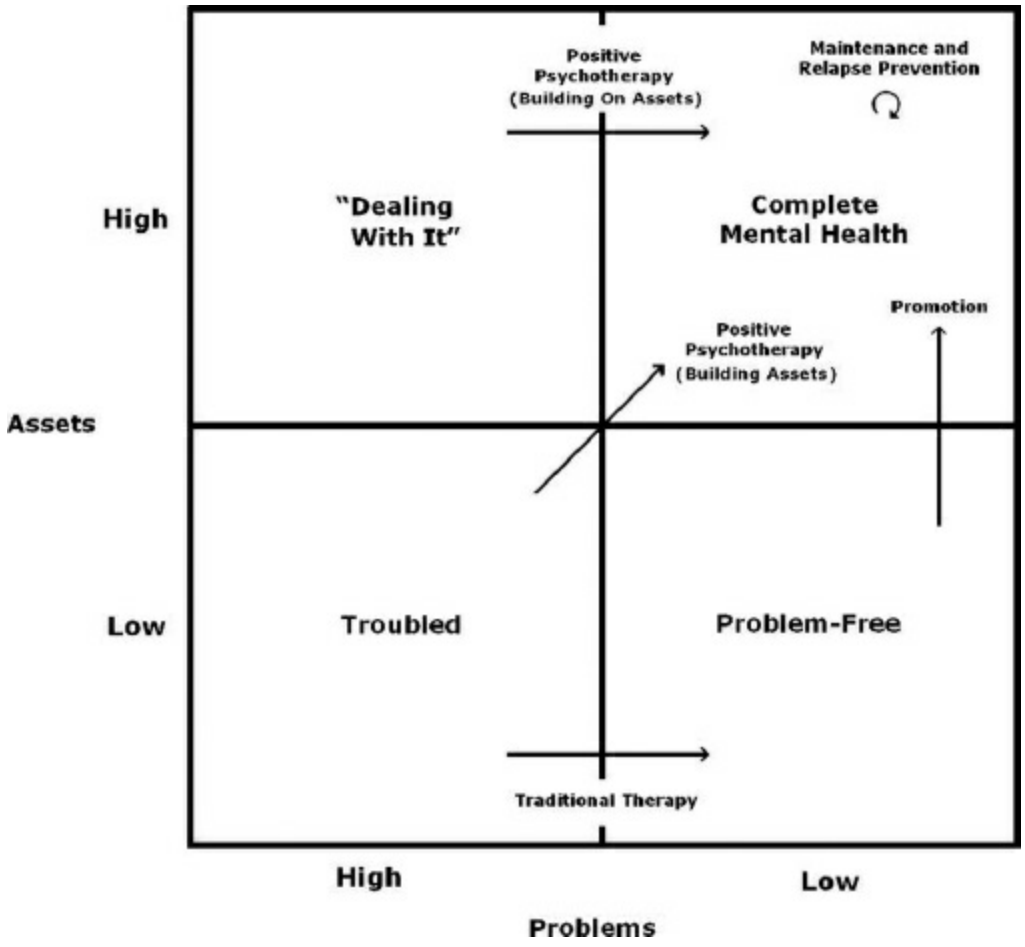


FIGURE 34-48
Mental health and mental illness. These are dimensions. Quadrants are shown for the purpose of illustration only. (From Sadock BJ, Sadock VA, Ruiz P, eds. *Kaplan & Sadock's Comprehensive Textbook of Psychiatry*. 9th ed. Philadelphia, PA: Lippincott Williams & Wilkins; 2009:2942, with permission.)

Note that this fuller characterization of health reflects the WHO definition and is drawn from research in all of the domains of concern to contemporary positive psychology. The relevant research cautions that health so defined is not unitary. No one can have it all, at least at the same time, given tradeoffs among the psychological states and traits that reflect doing well. Psychological health, therefore, needs to be described with a profile of features and not a single summary score. The fifth edition of the *Diagnostic*

and Statistical Manual of Mental Disorders (DSM-5) describes hundreds of psychological problems. However, there may be just as many different manifestations of the good life.

Theory of Psychopathology. As a perspective on topics that deserve scientific study, positive psychology has no single theory. Like much of contemporary psychology, it instead relies on midrange theories that draw on various broader perspectives, from evolutionary to behavioral to cognitive to sociocultural models, to make sense of specific phenomena. Different topics are explained with different theories. The eventual integration of psychology may be a worthy goal, but it has not yet been achieved.

At this early point in the development of positive psychology, the lack of a consensual or integrated theory is not a problem. The good psychological life is not yet understood, and positive psychologists are still grappling with the right vocabulary to describe it. Accordingly, championing a single theory at present would be premature, even counterproductive.

It has been argued that positive psychology is a descriptive endeavor, not a prescriptive one. If this means that positive psychology should be an empirical science—informed by replicable facts—then this claim is reasonable and a defining feature of the entire field. If this means that positive psychology is assumption-free or value-neutral, it becomes much more challenging to defend. After all, positive psychologists make the value judgment that the "good" life is indeed good—that is, desirable, morally and otherwise—and the metatheoretical assumption that the good life can be studied with the conventional methods of psychology. In any event, positive psychology seems no more prescriptive than clinical psychology or psychiatry. It may even be less so, given the theoretical diversity of positive psychology as it now exists.

Positive Psychology Assessment. The assessment has long been a staple of psychology, and much of it has been tilted—understandably—toward identifying weaknesses, deficiencies, and problems. The positive psychology perspective is that business-as-usual assessment should be expanded (not replaced) by attention to areas of strength and competence. Low life satisfaction can occur in the absence of psychopathology, but it is nonetheless related to psychological and social problems. Conversely, high

life satisfaction is linked to good functioning, even in the presence of symptoms.

Positive psychologists have developed an impressive set of measurement instruments that allow someone to break through the zero point of deficiency measures. For example, the healthiest that one can score on a typical measure of depression is zero, but this lumps together people who are blasé with those filled with zest and joy. The distinction seems well worth making, and the self-report surveys and interviews developed by positive psychologists allow it.

Most of the existing positive psychology measures were developed for research purposes, and they are most valid when aggregated to yield conclusions about groups of people. They can also be used ipsatively (i.e., forced to choose between two options), to describe the psychological characteristics of an individual and how they stay the same or change over time, but the cautious use of these descriptions is a point of discussion and a departure in treatment. None is a robust diagnostic test, and none should be treated as if it were. Such prudence is appropriate for all psychological assessments, but it is worth emphasizing in positive psychology measures.

Positive Psychology Techniques. Positive psychologists have demonstrated that brief interventions in the short term can increase happiness, satisfaction, and fulfillment. In some cases, there is also evidence that they can alleviate depression. For example, patients or clients can be asked to count their blessings:

Every night for 1 week, set aside 10 minutes before you go to bed. Use that time to write down three things that went really well on that day and why they went well. You may use a journal or your computer to write about the events, but you must have a physical record of your writing. It is not enough to do this exercise in your head. The three things you list can be relatively small in importance or relatively large in importance. Next to each positive event in your list, answer the question, “Why did this good thing happen?”

They can also be asked to use their strengths in novel ways. They take the Values in Action Inventory of Strengths (VIA-IS) questionnaire online and identify their most signature strengths of character. Then they are instructed to use these strengths in their daily life:

Every day for the next 7 days, use one of your top five strengths in a way that you have not before. You might use your strength in a new setting or with a new person. It is your choice.

Outcome research shows that various psychotherapies are effective in alleviating problems and usually equally so, despite the different forms they take. One interpretation of the equal effectiveness of different therapies is that nonspecific factors common to all treatments are responsible. Perhaps the types of strategies being studied by positive psychologists reflect these common factors and give names to them. Strategies like instilling hope and building strengths may be the critical factors in the effectiveness of any therapy.

Some qualifications are in order if these techniques are used in the context of treatment. First, the therapist must ascertain a client's readiness to change in the particular ways requested in the exercise and the client's capacity to make the change. Like any psychotherapeutic procedure, these techniques cannot be imposed on the unwilling or unable.

Second, none of these techniques is akin to a crash diet or an antibiotic. The degree to which they have lasting effects is related to how patients or clients integrate them into their regular behavioral routines. Counting blessings for a week will make a person happier for that week, but only if they become habitually grateful will there be a more enduring effect. Research finds—not surprisingly—that the people who showed lasting benefits were those who continued to use the exercise.

Third, these exercises are typically presented as one size fits all, but there is no reason to think that they are equally useful for all patients or clients. Nothing is known about the match of an exercise with a client's particular presenting problems or goals or with a client's age, sex, social class, or ethnicity.

Fourth, little is known about the parameters of these interventions. For example, how many blessings should one count, and how frequently should this be done? With college students, counting blessings three times a week may be more effective in increasing happiness than counting them more frequently. Is this a general phenomenon or one

specific to young adults attending college?

Fifth, all interventions run the risk of unintended harm, and although positive psychologists would like to believe that their techniques avoid iatrogenic effects, this assertion cannot be made with thorough confidence. For example, although optimism is related to mental and physical health, it would be simplistic and potentially hazardous to tell patients or clients that positive expectations will solve all their difficulties. Along these lines, if a positive psychology intervention overemphasizes a client's choice and responsibility, considerable damage could be done in cases of abuse and victimization, in which self-blame needs to be undone and certainly not encouraged. Interventions based on positive psychology should not preclude the use of existing therapeutic strategies when these are indicated.

Positive Psychotherapies. *Positive psychotherapies* are beginning to appear: therapeutic interventions based on the theories and findings of positive psychology. What distinguishes these emerging positive psychotherapies from conventional treatments is that their stated goal is not symptom reduction or relief but relatively enhanced happiness, life satisfaction fulfillment, productivity, and the like—one or more components of positive psychology's vision of the good life. These new therapies target people with psychological problems as well as those without them. In the latter case, positive psychotherapies make contact with life coaching.

The possible field of positive psychotherapies is so broad that it needs narrowing, and somewhat arbitrarily, the focus here is on approaches characterized by an explicit *therapeutic alliance* between the positive psychologist and the patient or client. This feature goes by many names and has been variously defined, but its recurring themes include collaboration between therapist and client, an affective bond between them, and an agreement on the goals and tasks of the intervention. Asking people to write about their goals or perform acts of kindness, despite beneficial consequences, may or may not be instances of positive psychotherapy—what matters is the relational context of the request.

TAKING STOCK. The unique and explicit goal of emerging positive psychotherapies is to enhance well-being and to promote the good life among

those with obvious psychological problems as well as those without them. They are also similar to more established therapies. Positive psychotherapies are short-term, structured interventions for individuals or small groups. Most can be placed in the cognitive-behavioral realm, although their techniques could be integrated easily into other treatment models. Most positive psychotherapies entail out-of-sessions exercises and homework assignments, the results of which are discussed in sessions. Several positive psychotherapies rely on journal-keeping, and many of these therapies rely on ongoing assessment.

Like other cognitive-behavioral interventions, positive psychotherapies take issue with assumptions of the medical model that people in treatment are ill and that their problems are best described as discrete (present-or-absent) entities as in DSM-5. According to positive psychology, people's weaknesses and strengths exist in degrees.

As emphasized, research support is still accumulating. Enough outcome studies have been conducted to conclude that positive psychotherapies are more than just promising, with effect sizes in the small to moderate range typical of psychological interventions. Not known in most cases is how positive psychotherapies fare in direct comparison with conventional treatments for anxiety or depression. Also, as already mentioned, the boundary conditions of effective positive psychotherapy are unknown.

Many positive psychologists would like to believe that a strengths-based approach to change is superior to one that focuses on the remediation of deficiencies, but this hypothesis has yet to be put to a serious test. The even-handed suspicion is that attention to both strengths and weaknesses is critical and that no useful purpose is served by regarding these as mutually exclusive therapeutic strategies.

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35

A Brief History of Psychiatry

It is challenging to present a brief history of psychiatry that goes beyond cherry-picking individual moments as it is difficult to decide what advances in science are most relevant to the field. Equally significant are the contributions of related fields, including philosophy, theology, psychology, sociology, and other disciplines that have contributed to our understanding and attitudes about mental health.

Psychiatry was not considered a formal discipline of medicine until relatively recently. However, concern with the causes and treatment of various mental illnesses go back to antiquity. We can often see specific familiar themes and debates from different ages. For example, are mental illnesses truly aberrant, or merely variations of normal behavior. Is there some value in evolutionary fitness or to society, in mental illness? Are mental illnesses diseases of the "mind" or the "body," and what is the difference? Is the pathogen internal, such as a virus, a malfunctioning gene or a dysfunctional immune system, or external, such as a god, demon, industrial life, or a society in general? Who is best suited to treat the diseases—a medical doctor, or some other professional or nonprofessional healer? We see such debates emerge throughout society, and even when some no longer have scientific merit (i.e., mind vs. body debates), they often still linger in the lay understanding of the illnesses and affect how society treats individuals suffering from these disorders.

PSYCHIATRY BEFORE THE MODERN ERA

Many of us think of the roots of medicine beginning in Ancient Greece, and for the western world, that is, in part, true. However, the civilizations of

Ancient Asia tended to be more advanced in the medical sciences. For example, psychiatry was practiced as a discipline in Ancient India, and the pre-2nd century CE medical treatise, the Charaka Sam.hitaO or Compendium of Charaka, includes sections on psychiatric diseases (Fig. 35-1). Psychiatric hospitals existed in India even before that.

The Greeks tended to see psychiatric illnesses as supernatural disorders, although Hippocrates makes some mention of rudimentary anatomic research into the causes of psychiatric illness. In a broader sense, the Greeks were wrestling with what has become a significant theme in modern psychiatry: whether to view diseases as mere abnormalities or on a continuum with normal. Thus, the ongoing debate between whether we should see such disorders as depression on a continuum with normal sadness and grief, or as a fundamentally different phenomenon has its roots in antiquity. For example, Galen and Plato stressed a categorical approach to disease, whereas Hippocrates took a more dimensional approach, often seeing typical human attributes as being on a continuum with abnormal. This affected treatment approaches: Galen focused on treating the disease process, whereas Hippocrates emphasized treating the individual.

In the Middle Ages, Arab countries included psychiatric wards in their hospitals. The first was built in Baghdad around the early 9th century. Many of the great Islamic medical scholars, including Abu Ali al-Hussain ibn Abdallah ibn Sina (known in the West as Avicenna), wrote on mental disorder treatments.

Psychiatric hospitals appeared in medieval Europe around the 13th century. However, they did not provide treatment and were merely places for custodial care. The best known is probably the Bethlem Royal Hospital, usually shortened to Bethlem, or *Bedlam* (Fig. 35-2). It was founded in 1247 and was not a hospital in the modern sense but rather a place to collect alms for the care of the needy. Although the history is somewhat unclear, it may have taken in and cared for the insane as early as the later 14th century, and there is documentation of mentally ill inmates by the early 15th century. Little is known about the treatments that might have occurred. However, the hospital gradually transitioned from a general hospital to an asylum for the insane by the later 15th century.

PSYCHIATRY IN THE MODERN AGE

Early Modern Age

By the late 17th century, other “lunatic asylums” proliferated, most were privately run. The first building built specifically as an asylum for the insane was the Bethel Hospital Norwich, founded by Mary Chapman. Meanwhile, in France, Louis XIV created what was for Europe the first public mental health system, although again, there was little treatment.

One of the earliest Western treatises on mental illness was the *Anatomy of Melancholy, What it is: With all the Kinds, Causes, Symptomes, Prognostickes, and Several Cures of it. In Three Maine Partitions with their several Sections, Members, and Subsections. Philosophically, Medicinally, Historically, Opened and Cut Up* (usually just called *The Anatomy of Melancholy*) by Robert Burton, a mathematician, astrologer, and scholar at Oxford University (Fig. 35-3). Although going well beyond what we would consider a medical textbook, Burton’s book does discuss “melancholy,” which includes depressive illnesses, although it takes a much broader view and includes much of the human experience.

Although the medieval focus was theoretically on care, not punishment for the insane, often, the treatments were barbaric. These could include prolonged chained restraint and physical punishments. These practices began to change during the enlightenment, with a call for humanitarian reform. For example, in William Battie’s, Treatise on Madness, he criticized the treatments occurring at such hospitals as Bethlem and argued for good hygiene and nutrition as mainstays of treatment. Although many of the somatic treatments he recommended were likely counterproductive (bleeding, purgatives), he did raise psychiatry to the level of a respectable discipline and helped support the humane treatment of mentally ill patients.



FIGURE 35-1
Charak Monument in Yog Peeth Campus, Father of Medicine & Surgery. (Source: Alokprasad, Balajijagadesh—Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=8124213)

Expanding on the concept, the York Retreat in England, founded by William Tuke and the Bicêtre Hospital and then Salpêtrière, in France, under Dr. Philippe Pinel, championed the concept of "moral treatment." These approaches emphasized avoiding restraint—Pinel is famous for removing the patients' chains (Fig. 35-4). His mentor, Jean-Baptiste Pussin, who, with his wife and colleague Marguerite, advocated for moral treatment and began banning chains at the Bicêtre Hospital, which greatly influenced the young Pinel. Pinel extensively elaborated this and took a scientific approach. He was one of the first to present a comprehensive categorization of psychiatric diseases, which reflected a functional view of mental illness emphasizing phenomenology over pathophysiology.

In Edinburgh, Scotland, the pseudoscience of phrenology gained influence in the 18th century under the leadership of William Browne and George and Andrew Combe. Although useless as medical science, it was an important conceptual and philosophical step in that it shifted the concept of mental illness from a moral or philosophical disorder to a material one originating in the physical matter of the brain.

The 19th and Early 20th Century

In the early 19th century, the state began playing a larger role in the Western treatment of mental illness. In 1808, Britain passed the Country Asylums Act, which established public mental asylums. In 1828 the Commissioners in Lunacy were appointed to oversee and license private asylums, partly in response to public concern over the abuses at Bethlem Hospital and similar private asylums. By then, “Bedlam” had become synonymous with “uproar” and “confusion,” owing to the mismanagement at the hospital, and scandalous practices, such as continuing to chain patients (Fig. 35-5), and allowing the public to visit the hospital and view patients for their entertainment.

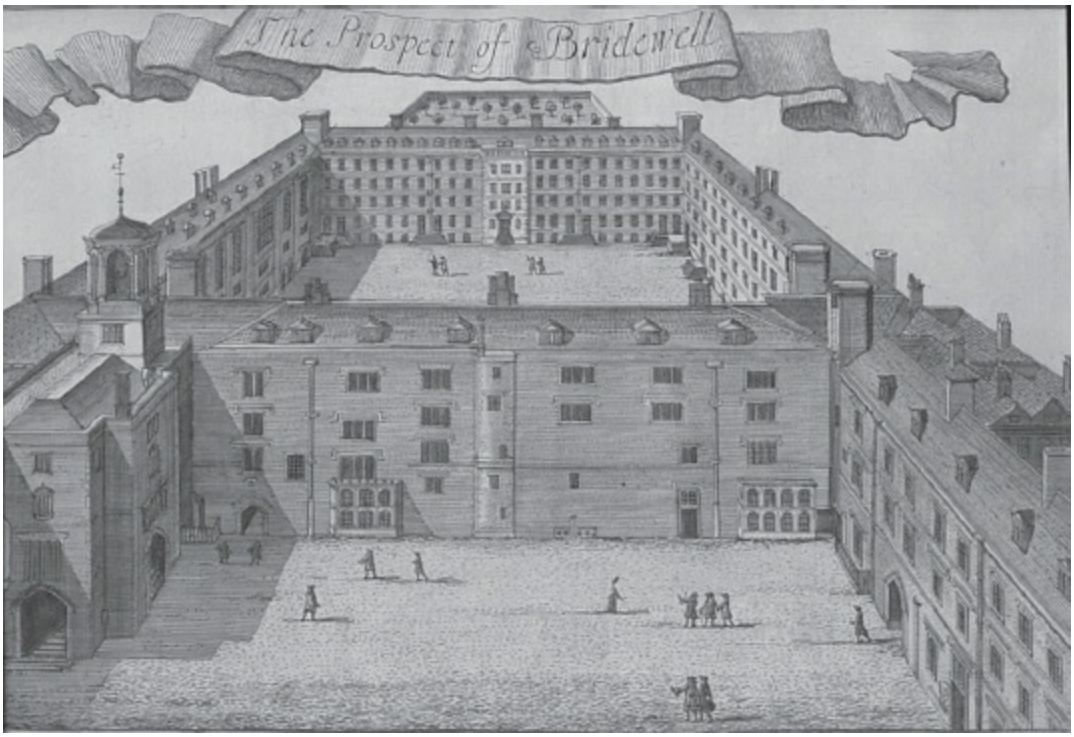


FIGURE 35-2
The Bethlem Royal Hospital. (Source: from John Strype’s, An Accurate Edition of Stow’s “A Survey of London,” 1720. Public Domain.)

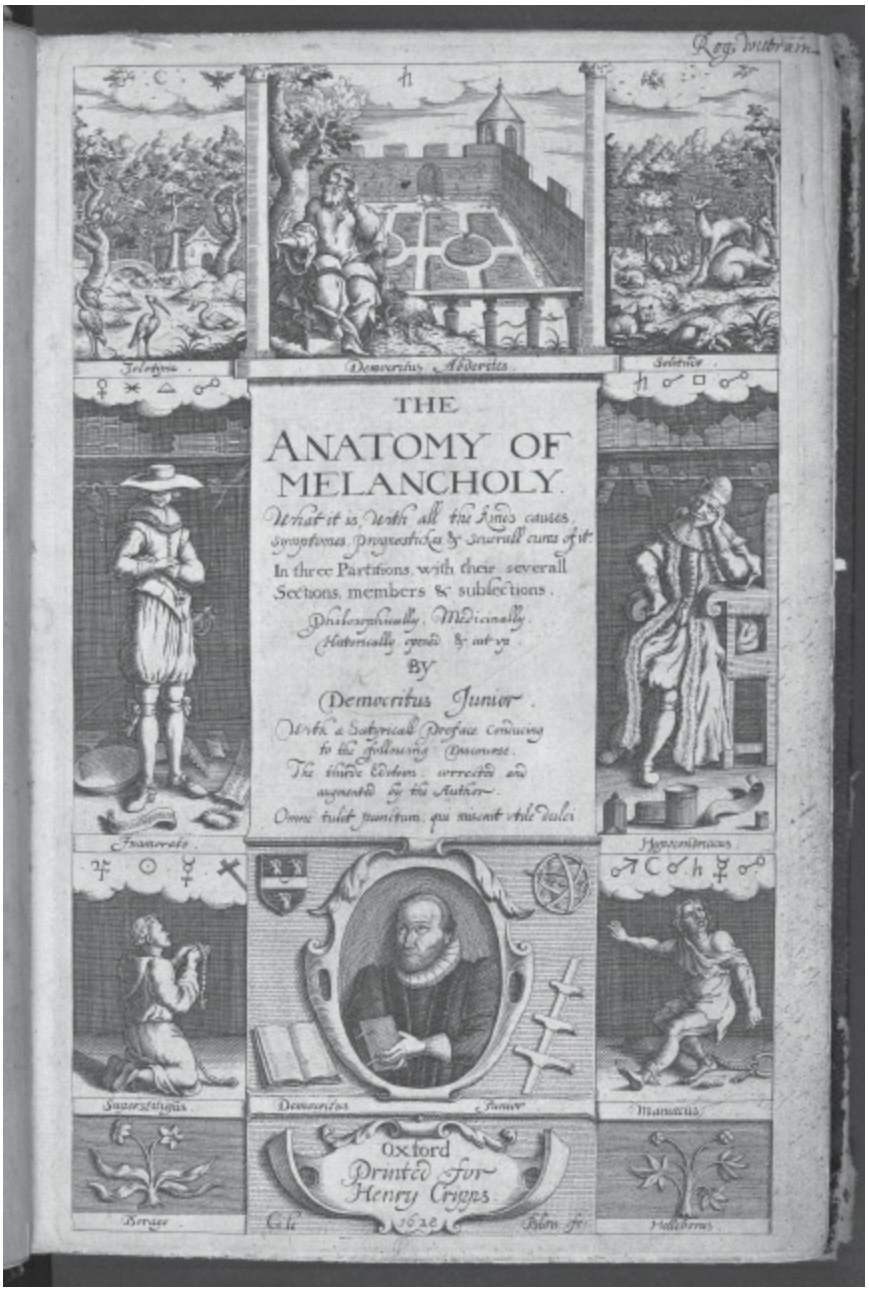


FIGURE 35-3
Robert Burton's The Anatomy of Melancholy, 1628. (Source: The British Library.)

In 1838, France enacted a law regulating asylum services across the country and built the first school for individuals with intellectual disabilities. Similarly, the 1845 Lunacy Act in Britain reclassified mentally ill people as patients requiring treatment. It mandated that asylums be built and required periodic inspections of these facilities.



FIGURE 35-4
Pinel, removing the chains from an inmate at the Salpétrière in 1795. (Source: Pinel, médecin en chef de la Salpêtrière en 1795, by Tony Robert Fleury. From the collection at the hôpital de la Pitié-Salpétrière.)



FIGURE 35-5
Etching of James Norris by G. Arnald 1815. (Source: Scan of a public domain image printed in Roy Porter, Madmen: A Social History of Madhouses, Mad-Doctors and Lunatics (Stroud, 2006).)

In the United States, reformers such as Dorothea Dix (Fig. 35-6) helped advocate for the first law to establish a state asylum, which became the Utica State Hospital, founded in about 1850. Dix achieved this after conducting a statewide investigation of the mentally ill poor in Massachusetts, showing

that they were treated like prisoners in cages, often naked and physically abused. She later conducted similar investigations in other states, always advocating for moral care for the mentally ill.

The latter half of the 19th century and early 20th century saw the rise of scientific approaches to mental disorders, which advances in the classification of psychiatric disorders and beginning insights into the pathophysiology and etiology of disorders.

The first medical case report formally describing a schizophrenic psychosis was most likely that of Benedict Morel (1809–1873) in 1852, a French psychiatrist, who used the term *démence précoce* to describe deteriorated patients whose illnesses began in adolescence. Subsequent cases describing what was likely schizophrenia include those reported by Greisinger in 1867, Hecker in 1871, and Kahlbaum in 1874. Kalbaum's case was notable as one of the first medical case reports of catatonia.

For much of the 19th and early 20th century in Europe, most psychiatric disorders were thought to derive from either congenital or degenerative diseases. Jules Falret in 1854 described bipolar disorder (*folie circulaire*), and 1882, the German psychiatrist Karl Kahlbaum used the term cyclothymia, describing mania and depression as stages of the same illness.



FIGURE 35-6
Dorothea Dix. (Source: United States Library of Congress's Prints and Photographs division (Public domain).)

Emil Kraepelin, building on the work of these predecessors, pioneered the scientific classification of psychiatric disease (Fig. 35-7). Kraepelin declared that all psychiatric disorders were biologic. As etiologic classification was not feasible, he based his classification on a careful study of disease phenomenology in different patients, in which he noted the similarities and differences in the symptomatology between different diseases.



FIGURE 35-7
Emil Kraepelin in 1926. (Source: Wikipedia, original: Münchener Medizinische Wochenschrift], unknown photographer.)

Kraepelin translated Morel's *démence précoce* into *dementia praecox*, a term that emphasized the change in cognition (dementia) and early onset (praecox) of the disorder. Patients with dementia praecox were described as having a long-term deteriorating course and the clinical symptoms of hallucinations and delusions. Kraepelin distinguished these patients from those who underwent distinct episodes of illness alternating with periods of normal functioning, which he classified as having manic-depressive

psychosis. Persistent persecutory delusions characterized another separate condition called paranoia. These patients lacked the deteriorating course of dementia praecox and intermittent symptoms of manic-depressive psychosis.

Eugene Bleuler coined the term schizophrenia, which replaced dementia praecox in the literature. Deriving it from the Ancient Greek words “schizo” (“I split”) and “phren” (“mind”), he chose the term to express the presence of schisms among thought, emotion, and behavior in patients with the disorder. This term is often misconstrued, especially by laypeople, to mean split personality, a concept that would be closer to that of the dissociative disorders. Bleuler also stressed that, unlike Kraepelin’s concept of dementia praecox, schizophrenia need not have a deteriorating course.

Bleuler identified specific fundamental (or primary) symptoms of schizophrenia to develop his theory about the internal mental schisms of patients. These symptoms included associational disturbances of thought, especially looseness, affective disturbances, autism, and ambivalence, summarized as “the four As”: associations, affect, autism, and ambivalence. Bleuler also identified accessory (secondary) symptoms, including the symptoms significant to Kraepelin, such as hallucinations and delusions.

Sigmund Freud (Fig. 35-8) initially also sought a biologic explanation for mental disease. However, realizing that the contemporary scientific tools were not up to the task, he instead developed a psychoanalytic approach based on his experiences with his private patients as he sought to understand the unconscious meanings behind their thoughts, behaviors, and desires. Although Freud was a neurologist and arguably working in psychology, his work was soon adopted by psychiatrists. The psychoanalytic approach became transcendent in the middle of the 20th century, and many psychiatric illnesses were explained in psychoanalytic terms; that is, in terms of conflicts and drives, particularly sexual drives, and stemming from early life experiences. Like many philosophical movements, psychoanalysis may have suffered from its overwhelming popularity, and at times it overreached as it tried to treat illnesses for which it was not suited, or tried to explain many human behaviors and achievements through the lens of psychoanalytic theory. In the latter half of the 20th century, as effective biologic treatments were discovered, many psychiatrists became disenchanted with psychoanalysis’s lack of empirical data and seeming resistance to scientific testing. It began to lose influence as a primary approach to understanding and

treating psychiatric disease.



FIGURE 35-8
Sigmund Freud at age 47. (Courtesy of Menninger Foundation Archives, Topeka, KS.)

The Latter 20th Century

The approach to mental illness, including the specific patients treated, shifted around World War II, as psychiatrists moved out of the asylum to confront very different patients, moving from the treatment of the “insane” to people who were “like us.” No longer confined to just the asylum, psychiatrists proved their usefulness in the military arena, treating both patients traumatized by combat and those found unfit to serve in the military for psychological reasons.

Greater importance was attached to the role of the environment on mental illness. This change drew from the work of Adolph Meyer (Fig. 35-9).

Meyer, the founder of psychobiology, saw mental disorders as reactions to life stresses. It was a maladaptation that was understandable in terms of the patient's life experiences.



FIGURE 35-9
Adolf Meyer. (Courtesy of National Library of Medicine, Bethesda, M.D.)



FIGURE 35-10
Otto Loewi. (Source: Creative Commons Attribution 4.0 International.)

Building on Meyer's social perspective and Freud's emphasis on early life experiences, Walter Menninger and others developed the *psychosocial model* of mental illness. Depression was seen not as a neurologic illness but as a failure to adapt. Menninger went as far as to suggest that all psychiatric disorders could be reduced to "one basic psychosocial process: the failure of the suffering individual to adapt to his or her environment."

The second half of the 20th century saw a gradual transition from a psychoanalytic and sociologic understanding of mental illness to a biologic one. Many discoveries helped propel this shift in approach.

Biologic models had existed before. For example, in the 1930s, Papez theorized that emotions were caused by a circuit connecting the limbic lobe, amygdala, septal region, and hippocampus. Electroconvulsive therapy was also developed in the 1930s and used to treat depression as early as the 1940s.

An early influence on biologic explanations for etiology was Otto Loewi's discovery of acetylcholine's role as a neurotransmitter. In proving this, Loewi (Fig. 35-10) employed a now-famous experiment, published in 1921, in which he used the saline solution from one frog heart to stimulate the rate in a second denervated heart, thus proving that a chemical must control the heart rate. He later showed that this chemical was acetylcholine. This was the first experiment that showed that a chemical could cause a response absent of electrical stimulation, thus discovering the critical role of neurotransmitters.

Loewi's experiment is famous, not only for its scientific value but also that the idea for the experiment came in a dream. Although he wrote down the dream when briefly waking from it, he could not make sense of his scribbles the next morning. Fortunately, he went on to have the same dream the next night, and when awakening from that, he rushed to his laboratory to perform what would become Nobel Prize-winning research.

The practical beginning of biologic psychiatry occurred with the serendipitous discovery of several drugs that could treat mental disorders, including chlorpromazine for schizophrenia, imipramine and iproniazid for depression, and lithium carbonate for bipolar disorder. In each case, the initial importance of the drug was not appreciated. For example, chlorpromazine was synthesized from promethazine, an antihistamine, in a search for more stable antihistamine sedatives. It was initially used as an anesthetic adjunct, and its antipsychotic properties were only later recognized after psychiatrists began administering it to manic and psychotic patients, again originally as a sedative.

Imipramine was also first synthesized as an antihistamine, and, following in the footsteps of chlorpromazine, tried as an antipsychotic before some case reports convinced the manufacturers to investigate its role as an antidepressant. Supporting these findings was the discovery that certain

drugs, such as reserpine, which irreversibly blocks monoamine transporters, could cause depression through the depletion of monoamines. Not only did it help the development of the first biologic theory for depression, the monoamine theory, but it also lent what was considered the first reproducible example of a chemical causing a mental disorder.

Lithium had been used as a medication since the 19th century, initially for gout. However, the doses needed to, for example, dissolve urate in the body were toxic. It was later used as a salt substitute for hypertensive patients but was largely banned after the toxicity became apparent. Lithium was used for mania in the late 1800s, but this was related to some theories that mood disorders were a type of “brain gout.” As that theory waned in popularity, lithium was largely abandoned as a psychotherapeutic. In the late 1940s, John Cade, an Australian physician, rediscovered lithium’s value for mania. Because of toxicity concerns, however, it took many decades and much research before it was adopted in the United States and other countries.

Neuroimaging started being used as an investigatory tool in the 1980s, following the discovery by Johnstone and others that patients with schizophrenia had enlarged cerebral ventricles. Pioneers such as Nancy Andreasen conducted the first MRI studies of schizophrenia and developed many of the techniques used by other investigators of the disease.

Genetic explorations also yielded interesting findings, such as the high heritability of certain disorders such as schizophrenia, autism spectrum disorder, and bipolar disorder. Although actual genetic associations have been elusive, the research in genetics lent additional support to the drug research and neuroimaging research which, when combined, created a growing body of research which moved the understanding of the etiology of many mental disorders from childhood parenting and other experiences or social causes to disorders that are, at least in part, inherited.

DEINSTITUTIONALIZATION

The 20th century also saw a shift from the asylum to the office and the community as the center of psychiatric treatment. The influence of the psychoanalytic and psychosocial movements also influenced psychiatry’s belief that most psychiatric diseases had at least potential treatments. In 1963, President Kennedy introduced legislation that delegated the NIH to

administer community mental health centers, thus beginning the community health movement in the United States (Fig. 35-11). It outlined a plan to discharge patients, including severely mentally ill patients, from institutions and instead treat them in the community. Although a laudable goal, it has been most successful at deinstitutionalization, whereas community centers have been poorly funded and supported. Thus, although most psychiatric patients no longer live in asylums, they may be homeless, living in dangerous environments, or be confined in jails, and in each case, receive little treatment.



FIGURE 35-11
President John F. Kennedy signs the Mental Retardation Facilities and Community Mental Health Center Construction Act. (Source JFK Library. Public Domain.)

THE CLASSIFICATION OF PSYCHIATRIC DISORDERS

Clinical treatment and classification dragged somewhat behind the science. Before the 1960s, many institutions and asylums developed their unique classification systems, and the result was a significant variation in both diagnostic terminology and how those diagnoses were applied. For example, in 1953, the American Psychiatric Association held a conference to develop a research program to evaluate psychiatry therapies. However, the members concluded that such research would be meaningless without a standardized approach diagnosis, which did not then exist.

This problem was further exposed in the United States–United Kingdom study conducted in the 1960s, which showed that many patients diagnosed with schizophrenia would be diagnosed with bipolar disorder in the United Kingdom. This variation results from an overemphasis in the United States on psychotic symptoms as a diagnostic criterion for schizophrenia. Another telling study was the Rosenhan experiment, in which experimenters feigned hallucinations to enter psychiatric hospitals, but usually acted normally once admitted. They were diagnosed with psychiatric disorders and even given antipsychotic drugs (which they flushed down toilets).

The problem was also apparent in such studies as the Baltimore Morbidity Study, which used DSM-I criteria and found inconsistent and unreliable results. The situation was such that in the 1970s, the President's Commission on Mental Health, informally called the Carter Commission, concluded that they could not form meaningful goals for mental health treatment until they understood the disorders better.

This problem led to an attempt to develop more reliable diagnostic systems. In the United States, this coalesced into the Diagnostic and Statistical Manual of Mental Disorders (DSM), developed by the American Psychiatric Association (APA). The first two editions are mostly preliminary works, reflecting the prevalent approaches among psychiatrists. For example, the second edition of the DSM (1968) distinguished between psychotic (defined more broadly than today) and neurotic forms of depression. This division developed into an endogenous/exogenous distinction—virtually depression due to a disease of the brain versus depression that is a reaction to stress, be it current or developmental. Although lacking any empirical validity, the distinction was a convenient way to divide biologic and

psychodynamic concepts of depression. This division often drove treatment during much of the 1970s, in which it was thought some forms of depression were better treated with psychotherapy, whereas others should be addressed with medication.

The introduction of the DSM-III in 1980, and its subsequent revisions up to and including DSM-5, represented a significant paradigm change in diagnosis, which was borne of research designed to create more reliable criteria for mental disorders (Fig. 35-12).

Feighner and colleagues at Washington University developed the Feighner, or St. Louis criteria, criteria for 15 major psychiatric categories. This approach represented the first systematic approach that relied entirely on numbers and types of symptoms to diagnose psychiatric disorders. While this was taking place, Robert Spitzer, Jean Endicott, and other researchers at the New York State Psychiatric Institute were developing methods to increase the reliability of psychiatric diagnoses. These groups and other researchers were brought together by the National Institute of Mental Health in the 1970s, where they developed the Research Diagnostic Criteria (RDC), which were essentially an elaboration and expansion of the Feighner criteria.

The RDC prepared the way for the DSM-III, which resulted from the work of the Task Force on Nomenclature and Statistics, headed by Dr. Spitzer, formed by the APA in the early 1970s. The DSM-III eliminated the endogenous/exogenous dichotomy previously inherent in many diagnoses, and eliminated assumptions of cause from the criteria, taking a strictly phenomenal approach. For example, major depression was defined by symptomatic criteria. It also divided the mood disorders, with major depression and bipolar disorder being treated as distinct disorders. Although acknowledging more minor forms of depression, DSM-III's categorical approach created a rigid divide between those with major depressive disorder and those not meeting the criteria for the disorder, a difference that survives to this day. This divide risks neglecting some individuals who do not fit into one of the DSM categories and yet suffer from depression; although DSM includes ways to categorize them, it is likely that in clinical practice, many such patients are treated as if they had no disorder.

However, the benefits of this categorical approach have been that it has created more reliable diagnoses, allowing for research on more homogeneous populations. This strategy bore significant fruit, and there was a

pharmaceutical revolution in the 1980s and 1990s, in which researchers developed and introduced new classes of medications, such as the SSRIs and the atypical antipsychotics. Although arguably not more efficacious than the original antidepressants and antipsychotics, the drugs were considered more tolerable, and the result was a considerable increase in the number of patients taking, and benefitting from, psychopharmacological agents. Some have pointed out that this phenomenon also caused abuses and overuse of the drugs for conditions that could be more considered part of the human condition, and there was a predictable backlash in which critics questioned both the safety and efficacy of many psychiatric drugs. Organized psychiatry often found itself on the defensive as it had to make a case for the drugs and contend with critiques that went beyond scientific questions into the effect of the drugs on culture and society.

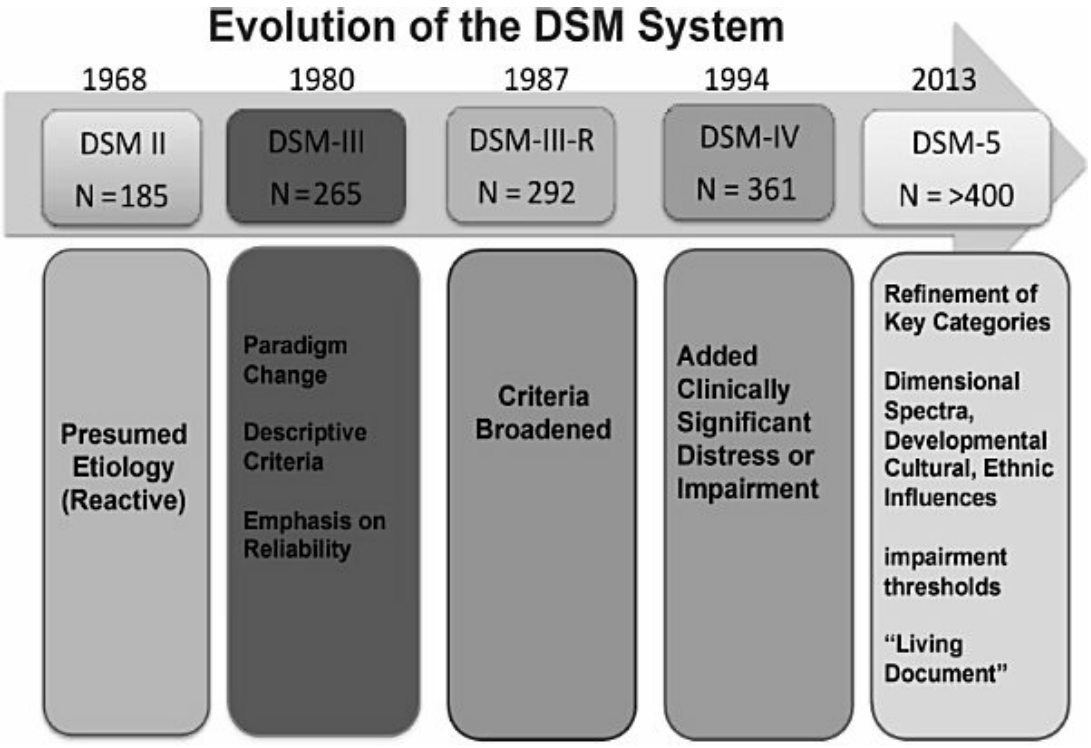


FIGURE 35-12
DSM system. Key issues and number of categories.

The DSM-III and its role in contributing to the remedicalization of psychiatry also affected psychological treatments. Although the more reliable criteria made it easier to design studies and enroll suitable populations for treatment studies, it also helped encourage a bias toward psychological

treatments that lent themselves to scientific research—ones that were easily manualized and reproducible in a consistent way. Hence, such therapies as cognitive-behavioral therapies, behavioral therapies, and interpersonal therapies gained predominance as treatments for major psychiatric disorders. In contrast, psychoanalytic treatments, whose depth and complexity made them inherently harder to study consistently, became more marginalized and considered "boutique treatments" reserved for those who could pay the sometimes considerable out-of-pocket costs for these therapies.

LOOKING TO THE FUTURE

The combination of new technologies, such as high-resolution functional imagining (Fig. 35-13) and total gene mapping, combined with the ever-increasing ability of computers to manipulate vast amounts of data, has resulted in a remarkable explosion in research on such disorders as schizophrenia, autism, obsessive-compulsive disorder, and major depression during the past decade, and we have made great strides toward understanding the genetics and pathophysiology of these disorders. However, there remains a frustrating gap between basic and clinical research.

In genetics, the limitations of clinical diagnoses for research purposes are clear, and the search for valid biomarkers has been disappointing. It may be that a focus on meaningful endophenotypes may be an essential step. However, the predictive power of many of the proposed endophenotypes remains to be proven.

Pharmacologic research into treating the disorder, after several fruitful decades, has stalled. Clozapine, arguably the most effective antipsychotic, was discovered in the 1960s, the SSRIs became available in the 1980s and 90s, and the subsequent efforts to refine pharmacologic treatment have, so far, fallen short of hopes. Barriers to translational research include the lack of a valid animal model for many of the disorders, and efforts are being made in improving how animal testing is performed. In the meantime, new approaches are being attempted that deemphasize the importance of single receptors and emphasize a multi-target approach to psychopharmacology—essentially using "dirty drugs," which had previously been discouraged due to concerns over side effects, as well as moving beyond monoamine targets to include other receptors, including glutamatergic targets and G-protein–

coupled receptors.

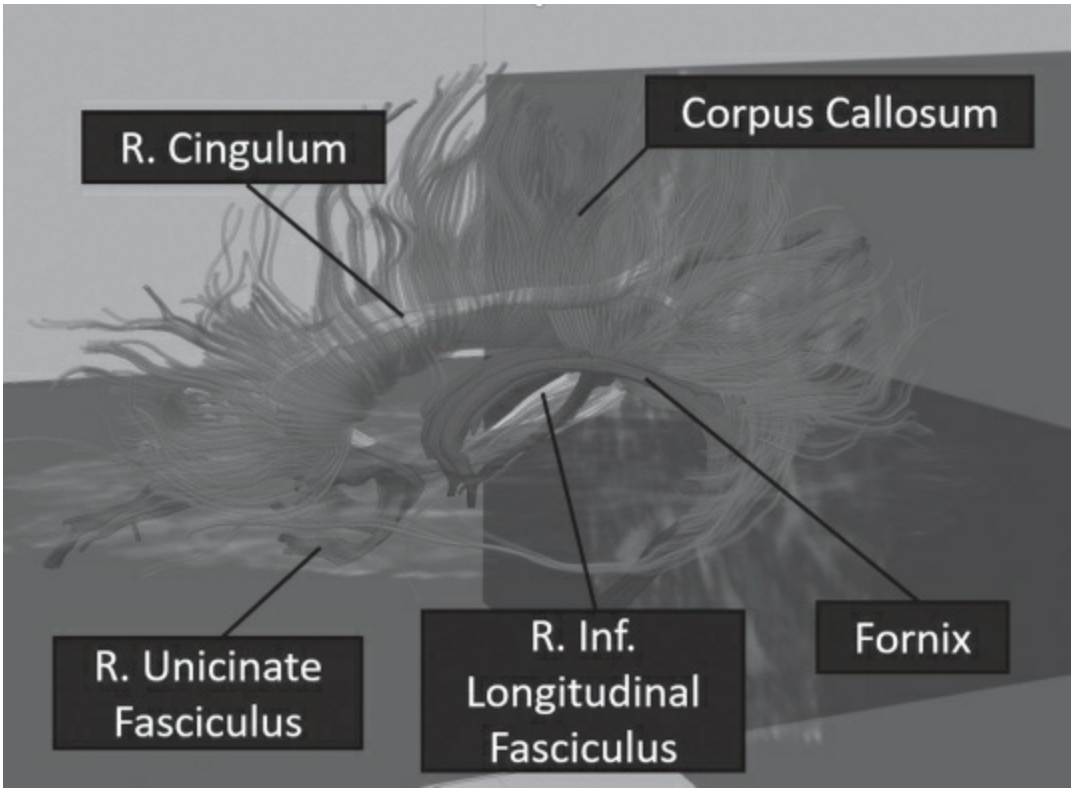


FIGURE 35-13
This image is an example of the high level of detail that neuroimaging technologies can produce to aid investigators. Three-dimensional image reconstructed based on diffusion data acquired on a 3-T General Electric scanner, Brigham and Women's Hospital, Harvard Medical School, Boston, MA, which shows several major white matter fiber bundles (generally differing in colors, converted to grayscale for this book) identified through diffusion tensor imaging.

What is clear, however, is that most psychiatric disorders are very complex disorders that require multifaceted solutions. These solutions include attention to psychosocial treatments as well as pharmacologic. The change from conceptualizing many of the disorders as neurodevelopmental rather than neurodegenerative disorders also highlights the importance of early identification of disorders in the hopes that early interventions can alter the course at a more promising stage.

Table 35-1 presents a summary of some of the significant persons and events in the history of psychiatry.

Table 35-1

A History of Psychiatry Table

Persons and Events in Psychiatry		
Person or Event	**Country**	**Publications**
Hippocrates of Cos (ca. 460–370 BCE)		*Hippocratic Writings*. Bal 1978
Plato (427–347 BCE)	Greece	*The Dialogues of Plato*. Random House; 193
Aristotle (384–322 BCE)	Greece	Ross WD: *Aristotle*. 6th e

Galen of Pergamon (130–200)	Asia Minor (Turkey) in the Roman Empire	On the Affected Parts. S Karger; 1976
St. Augustine (354–430)	Tageste (Numidia, North Africa)	Confessions. Pusey EB, Library; 1949; Wills G Lipper/Viking; 1999
Avicenna (980–1037)	Persia	A Treatise on the Canon London: Luzac; 1930
Constanitus Africanus (ca. 1010–1087)	Carthage, North Africa	Constantino l’Africano. D
Petrarch (1304–1374)	Italy	*De remediis utriusque fo*

Henry Kramer and James Sprenger (15th century)	Germany	*Malleus maleficarum (W*
Paracelsus (ca. 1493–1541)	Einsiedeln (Switzerland), then traveled Europe	Pagel W. “Disease and t and lunacy. The psyc *Paracelsus.* 2nd rev.
Juan Luis Vives (1493–1540)	Born in Spain; from the age of 16 yr lived mainly in Holland and Belgium	*De anima et vita (Of Sou Juan Luis Vives and t* Southern Illinois Univ
Johann Weyer (1515–1588)	Holland	*De Praestigiis Daemonu* (1563); reprinted in M *Doctors in the Renais* Center for Medieval a 1991
Juan Huarte de San Juan (ca. 1530–1592)	Spain	*The Examination of Men*
Giambattista Porta (1535–1615)	Italy	*De humana physiognom*

Felix Plater (1536–1614)	Switzerland	*Practice of Medicine* (16 *Injurious to Body and*
Robert Burton (1577–1640)	England	*The Anatomy of Melanch*
Paolo Zacchia (1584–1659)	Italy	*Questiones medico-lega*
Thomas Sydenham (1624–1689)	England	Dissertatio epistolaris … Dr. Thomas Sydenha

Thomas Willis (1621–1675)	England	*Two Discourses Concer*
Georg Ernst Stahl (1660–1734)	Germany	*Theoria medica vera* (17
William Battie (1703–1776)	England	*A Treatise of Madness* (1

Boissier de Sauvages (1706–1767)	France	*Nosologia methodica* (17
John Aiken (1747–1822)	England	*Thoughts on Hospitals* (1
Franz Anton Mesmer (1734–1815)	Austria, France	*Memoire sur la découver*
Vincenzo Chiarugi (1759–1820)	Italy	*Regulations of the Hospi of Bonifazio* (1789)
William Cullen (1710–1790)	Scotland	*Nosology, or a Systemat* (1800)
Philippe Pinel (1745–1826)	France	*A Treatise on Insanity in Principles of a New a Mental Disorders* (18
Johann Reil (1759–1813)	Germany	*Rhapsodies about the Ap Mental Disturbances*

Benjamin Rush (1745–1813)	United States	*Medical Inquiries and Ob of the Mind* (1812); Fa American psychiatry. 1994;151(Suppl):65
Thomas Sutton (1767–1835)	England	*Tracts on Delirium Trem*
William Tuke (1732–1822); Samuel Tuke (1784–1857), grandson of William Tuke; Daniel Hack Tuke (1827–1895), youngest son of Samuel Tuke; John Charles Bucknill (1817–1895)	England	Tuke S. Description of th York, for Insane Pers York; 1813; Tuke DH *Psychological Medici Nosology, Description Pathology, and Treat*
Joseph Adams (1756–1818)	England	*A Treatise on the Suppo Diseases … Particula* (1814)

Franz Joseph Gall (1758–1828), Johann Gaspar Spurzheim (1776–1832)	Austria, Germany	*The Physiognomical Sys* *Spurzheim; founded* *physiological examin* *general and of the br*
Johann Christian Heinroth (1773–1843)	Germany	Disturbances of the Mind Metaphors of mind ar German psychiatry in 1998;89:387; Cauwe (1773–1843) a psych era. *Hist Psychiatry*. 1
Robert Gooch (1784–1830)	England	*An Account of … Diseas*
James Cowles Pritchard (1786–1848)	England	*A Treatise on Insanity ar* *Mind* (1835)
Jean Etienne Dominique Esquirol (1782–1840)	France	*Mental Maladies: A Trea*

Isaac Ray (1807–1881)	United States	*Treatise on Medical Juris*
James Braid (1795–1860)	England	*Neurypnology; or, the Ra* (1843)
Daniel M’Naghton (1813–1865)	England	West DJ, Walk A, eds. *D the Aftermath.* Londo
Wilhelm Griesinger (1817–1868)	Germany	*Mental Pathology and Th*
Jacques-Joseph Moreau de Tours (1804–1884)	France	*Du haschich et de l’alién*

Walter Cooper Dendy (1794–1871)	England	Psychotherapeia, or the *Psychol Med Ment Pa*
Jean-Pierre Falret (1794–1870), Jules Baillarger (1809–1890)	France	Baillarger J. La folie à do *Med.* 1853–1854;19:3 circulaire. *Bull Acad I* Pichot P. The birth of *Psychiatry*. 1995;10:1
Thomas Kirkbride (1809–1883)	United States	*On the Construction, Org Arrangements of Hos*
John Conolly (1794–1866)	England	*The Treatment of the Ins Restraints* (1856)
George Robinson (1821–1875)	England	*On the Prevention and T* (1859)
Gustav Theodor Fechner (1801–1887)	Germany	*Elements of Psychophys*

Benedict-Augustin Morel (1809–1873)	France	*Traité des maladies men*
Thomas Laycock (1812–1876)	England	*Mind and Brain* (1860)
Forbes B. Winslow (1810–1874)	England	*On Obscure Diseases of Mind* (1860)
Ewald Hecker (1843–1909)	Germany	Die Hebephrenie. *Arch P*
Karl Kahlbaum (1828–1899)	Germany	*Die Katatonie oder das S* 1874)
George Miller Beard (1840–1883)	United States	*A Practical Treatise on N (Neurasthenia)* (1880

Richard von Krafft-Ebing (1840–1902)	Germany	*Psychopathia sexualis* (1 *Stepchildren of Natur* *the Making of Sexual* of Chicago Press; 20
Pierre Janet (1859–1947)	France	*L'automatisme psycholo* *experimentale sur les* (Paris, 1889); van de Pierre Janet and the psychological trauma 1989;146:1530
Herman Emminghaus (1845–1904)	Germany	*Psychic Disturbances of*
Jean-Martin Charcot (1825–1893)	France	Owen ARG. *Hysteria, Hy* *of JM Charcot*. New Y 1971; by Goetz C, Bo *Constructing Neurolo* Press; 1995

Georges Gilles de la Tourette (1857–1904)	France	Etude sur une affection r l'incoordination motric coprolalie. *Arch Neur* Yorston G, Hindley N nervous disorder cha incoordination with ec *Psychiatry.* 1998;9:97 *The Histories of Tour* MA: Harvard Universi
Cesare Lombroso (1835–1909)	Italy	*Genius and Insanity* (186 (1876), *The Female C*

Emil Kraepelin (1856–1926)	Germany	Kraepelin E. *Psychiatrie: und Aerzte*. 6th ed. (1 Springer-Verlag; 198
John Hughlings Jackson (1834–1911)	England	*Selected Writings of Joh* 1931–1932)
Sigmund Freud (1856–1939)	Austria	*The Interpretation of Dre the Theory of Sexuali on Psychoanalysis* (1
Morton Prince (1854–1929)	United States	*The Dissociation of a Pe*
Clifford Beers (1876–1943)	United States	*A Mind That Found Itself*

Eugen Bleuler (1857–1939)	Switzerland	*Dementia Praecox or the* (1911)
Alfred Binet (1857–1911)	France	*A Method of Measuring t* *Intelligence of Young* English translation, 1
Hideyo Noguchi (1876–1928)	United States	Noguchi H, Moore JW. A *Treponema pallidum* paralysis. *J Exp Med*.
Alfred Adler (1870–1937)	Austria	*Study of Organ Inferiorit* *Compensations* (191 *Shadow: Adler in Cor* Press; 1983
Hermann Rorschach (1884–1922)	Switzerland	*Psychodiagnostik* (1921)

Julius von Wagner-Jauregg (1857–1940)	Austria	de Rudoli G. *Therapeutic* Jauregg J. The histor general paresis. *Am J* 1994;151(Suppl):231 Jauregg won the Nob therapy for general pa *Psychiatry*. 2000;11:3
Otto Rank (1884–1939)	Austria, France, United States	*The Myth of the Birth of t Birth* (1924), *Will The Acts of Will: The Life* York: Free Press, Ma
American Board of Psychiatry and Neurology (ABPN)	United States	Freeman W, Ebaugh F, I American Board of Ps *Am J Psychiatry*. 195

Henry A. Murray (1893–1988)	United States	*Thematic Apperception T* University Press; 194
Charles Bradley (1902–1979)	United States	Bradley C. The behavior Benzedrine*. Am J Ps* Origin of stimulant us deficit disorder. *Am J*

Cure of pellagra	United States	Roe D. *A Plague of Corr* *Pellagra*. Ithaca, NY:
Genetics of schizophrenia	Germany, United States	Rüdin E. *Studien über V* *geistiger Störungen. I* *Neuetstehung der De* *the Heredity and Gen* *the Heredity and New* *Praecox*]. Berlin: Verl Kallman FJ. *The Gen* York: JJ Augustin; 19 contributions of Franz schizophrenia. In: Ca *Research in the Schiz* *Stanley R. Dean Awa* New York: Spectrum;

ECT	Italy	Abrams R. The treatmen *North Am.* 1994;17:52 origins of electroconv history. *Hist Psychiat*
Karen Horney (1885–1952)	Germany until 1932, then	*The Neurotic Personality* *Ways in Psychoanaly*

	United States	*Human Growth: The* *Realization* (1950); P *Psychoanalyst's Sea* Haven, CT: Yale Univ
Adolf Meyer (1866–1950)	Switzerland until 1893, then United States	Meyer A. A short sketch *Am J Insanity*. 1896– Meyer and the develo *Am J Psychiatry.* 199 Havens L, ed. *Approa* Little, Brown
Constitution, personality, and mental illness	Germany, United States	Kretschmer E. *Body Buil* Sheldon WH. *Varietie*

Führer decree (1939)	Germany	Lifton RJ. The Nazi doct *Medical Killing*. New Chapter 2; Gallagher *Patients, Physicians,* *Third Reich*. New Yor Grodin MA, eds. *The* *Nuremburg Code. Hu* *Experimentation*. Oxf 1992
MMPI	United States	Buchanan R. The develo Multiphasic Personali 1994;29:148

Leo Kanner (1894–1981)	Austria, United States	*Child Psychiatry*. Springf Autistic disturbances 1943;2:217
Melanie Klein (1882–1960), Anna Freud (1895–1982)	Central Europe, then England	Klein M. *The Psychoana* Kernberg O. Melanie AM, Sadock BJ, eds. *Psychiatry*. 3rd ed. Ba Wilkins; 1980; *Grossk* *World and Her Work*. A. *The Ego and the M* *The Psychoanalytic T* short history of child a *Child*. 1966;21:7; You *Biography*. New York

Helene Deutsch (1884–1982)	Europe until 1935, then United States	*The Psychology of Wom* of emotional disturba schizophrenia ('As If') *and Character Types* Universities Press; 19 *Helene Deutsch: A Ps* City, NJ: Anchor Pres
Ivan Petrovich Pavlov (1849–1936), Andrew Salter (1914–1996)	Russia, United States	Pavlov IP. *Lectures on C* New York: Internation BP. Pavlov: *A Biograp* Chicago Press; 1949; *Therapy*. New York: F
NMHA (1946), NIMH (1949)	United States	Grob GN. *From Asylum t* *Policy in Modern Ame* University Press; 199 Reminiscences: 1938 1990;147:785

Erik H. Erikson (1902–1994)	Europe; United States since 1930s	*Childhood and Society* (1 *Cycle* (1959); Friedm *Biography of Erik H. E* 1999; Wallerstein R, (*Identities: The Life an* Madison, CT: Internat
Carl Rogers (1902–1987)	United States	*Client-Centered Theory* (
Donald Winnicott (1896–	England	Winnicott D. Transitional

1971)		phenomena [1951]. In *Paediatrics to Psycho* 1958; Kohon H. *The I Psychoanalysis: The* Haven, CT: Yale Univ FR. *Winnicott: Life an* 2003
Jean Delay (1907–1987), Pierre Deniker (1917–1999)	France	Delay J, Deniker P. Le tr une méthode neuroly l'hibernothérapie. C R France. 1952;50:497; *Changed the Face of* Dunitz; 1999; Healy D *Psychopharmacology* University Press; 200
Harry Stack Sullivan (1892–1949)	United States	Sullivan HS. *The Interpe* Perry HS, Gawel ML, 1953; Perry HS. *Psyc Harry Stack Sullivan*.

		Press of Harvard Uni
Egas Moniz (1874–1955), Walter Freeman (1895–1972)	United States, Europe	Valenstein ES. *Great an and Decline of Psych Treatments for Menta* Books; 1986; Pressm *Psychosurgery and th* Cambridge: Cambrid

Advent of the double-blind test	United States	Gold H, Kwit NT, Otto H. and aminophylline) in *JAMA.* 1937;108:217 Frank J. A comparati mephenesin and plac a mixed group of psy *Hopkins Hosp*. 1954;9
Carl Gustav Jung (1875–1961)	Switzerland	*Psychological Types* (19 *Psychology* (1912–19 *Dynamics of the Psyc Archetypes and the C* 1955); Kirsch T. *The J Historical Perspective*

Antidepressant drugs	Switzerland, United States	Kuhn R. The treatment c iminodibenzyl derivat 1957;87:1135; Loome A clinical and pharma iproniazid as a psych Rep. 1957;8:129; Hea Cambridge, MA: Harv Kuhn R. *Hist Psychia*
B. F. Skinner (1904–1990), Joseph Wolpe (1915–1997)	United States, South Africa	Skinner BF. *Science and* Free Press; 1953; Wo *Reciprocal Inhibition*. University Press; 195
Development of child psychiatry	United States	Lewis M, ed. *Child and A Comprehensive Textb* Williams & Wilkins; 19 Effects of health deliv adolescent mental he Recruitment, training, adolescent psychiatry (Bernstein D. The dis perspective on child a
Karl Jaspers (1883–1969)	Germany	*General Psychopatholog* (English trans., 1963)

Kurt Schneider (1887–1967)	Germany	Schneider K. *Psychopat* Cassell; 1958; Schne *Psychopathology.* Ne
Michel Foucault (1926–1984)	France	*Histoire de la folie* (1961 *Civilization: A History* *Reason*. New York: P

Benzodiazepine drugs	United States	Cohen IM. The benzodia Blackwell B, eds. *Dis* *Psychiatry*. Baltimore Communications; 198

Haloperidol in the treatment of Gilles de la Tourette syndrome	France	Seignot JN. Un cas de m Tourette guéri par le l 1961;cxix:578; The ca syndrome: A conditio R.1625. *Hist Psychia*
Joint Commission on Mental Illness and Mental Health (1955–1961)	United States	*Action for Mental Health* Health Centers Act of
Rene Spitz (1887–1974)	Europe, then United States	*The First Year of Life: A Normal and Deviant L Relation*s. New York: Press; 1965

Classification of mental disorders	United States	DSM-I prepared by the C and Statistics of the A Hospital Service, Was prepared by the Com Statistics of the APA, Washington, DC, 196 study in appearance a 1991;148:421
Epidemiological evidence for the genetic transmission of schizophrenia	United States, Denmark	Rosenthal D, Kety SS, e *Schizophrenia*. Londo (Wender PH, Rosenth assessment of the ad schizophrenics), 345 Wender PH, Schulsin prevalence of mental adoptive families of a

		377 (Rosenthal D, W F, Welner J. Schizopl adoptive homes); Ket Ingraham LJ, Jansso biological and adoptiv adoptees: Replicatio the rest of Denmark. A 1994;51:442
Lithium therapy for psychotic excitement	Australia, Denmark, United States	Cade J. Lithium salts in t excitement. *Med J Au* Juel-Nielsen N, Strom treatment of manic ps

		of lithium salts. *J Neu* 1954;17:250; Schou I treatment. *Arch Gen I* Schioldann: Mogens *A* Half a century with Lit 2007;17:247
William H. Masters (1915–2001), Virginia E. Johnson (1925–)	United States	*Human Sexual Respons* *Dysfunction* (1970)
Margaret Mahler (1897–	Central Europe	Mahler M, Pine F, Bergm

1985)	until 1938, then United States	*of the Human Infant:* New York: Basic Boo *Margaret S. Mahler.* S Free Press; 1988
John Bowlby (1907–1990)	England	*Attachment* (1969), *Sepa* (1973), *Loss: Sadnes Secure Base: Clinica Theory* (1988); Holme *Attachment Theory*. L

Jacques Lacan (1901–1981), “the French Freud”	France	*The Language of the Se* *Psychoanalysis* (1968 *Politics: Freud’s Fren* Books; 1978; Roudin *A History of Psychoa* Chicago: University o
Heinz Kohut (1913–1981)	Austria until 1940, then United States	*The Analysis of the Self* *Self* (1977); The two *Psychoanal.* 1979;60 Kohut’s self psycholo *Psychiatry*. 1987;144 *The Making of a Psyc* Straus and Giroux; 20
Jean Piaget (1896–1980)	Switzerland	Greenspan S, Curry J. P functioning. In: Kapla *Comprehensive Textb* Baltimore, MD: Willia

NAMI	United States	*NAMI: We Are Family.* A page brochure descri then NAMI has publis its work

Sociobiologic psychiatry	United States, Holland	Van Praag H. Sociobiolo *Psychiatry*. 1981;22:4
Roger Sperry (1913–1994)	United States	Some effects of disconn hemispheres. *Scienc* Levi-Montalcini R. In: *Brain: A History of the Discoveries*. New Yo 2000

Brief psychotherapy	United States, England	Malan D. *A Study of Brie* Plenum Press; 1976; *Casebook in Time-Li* York: McGraw-Hill; 19 *Dynamic Psychothera* 1980; Sifneos P. *Sho Psychotherapy Evalu* York: Plenum Press; efficacy of brief dynar analysis. *Am J Psych*
Psychology of women	United States	Miller JB. *Towards a Ne* Boston, MA: Beacon *Different Voice: Psyc Development.* Cambr Press; 1982; Levinso Levinson J. *The Seas* York: Knopf; 1996
NARSAD	United States	NARSAD activities are re and *Newsletters*

Changes in the diagnosis and classification of homosexuality	United States	Bayer R. *Homosexuality Politics of Diagnosis*. DSM-III (1980); DSM

Classification of mental disorders	United States	DSM-III (1980); DSM-III- and the transformatio history. *Am J Psychia*

James Masterson (1926–), Otto Kernberg (1928–)	United States	Masterson J. *Psychother Developmental Appro* 1976; *Narcissistic an Integrated Developm* Brunner/Mazel; 1981; *Psychotherapy of the Masterson Approach*. 1988; Kernberg O. *Bo Pathological Narcissi* 1975; Kernberg O. *Se Psychotherapeutic St* University Press; 198
Psychology and neurophysiology of dreams	United States	Hobson JA, McCarley R generator: An activati dream process. *Am J* McCarley RW. Dream wishes or transparent state? In: Bilder RM, *of the Mind: On the C a Scientific Psycholog* Academy of Sciences dream in contempora 2001;158:351

Theories on the psychology of love	United States	Bergmann MS. *The Ana* Columbia University P *Triangle of Love: Intir* New York: Basic Boo ML, eds. *The Psycho* Yale University Press eds. *Passionate Attac* New York: Free Press *Love and Fateful Enc* *Romantic Passion*. N
Evaluations of psychotherapy	United States	Bruce R, Staples FR, Cri KS. *Psychotherapy v* Cambridge, MA: Harv Elkin I, Shea MT, Wa National Institute of M

		Depression Collabora General effectiveness *Psychiatry*. 1989;48:9
Eric Kandel (1929–)	Austria, United States after 1939	Kandel E. Psychotherap impact of psychiatric *Engl J Med.* 1979;301 to molecular biology: anxiety. *Am J Psychia* nerve cells, and the r *Neuropsychiatry*. 198 of psychoanalysis: A psychiatry revisited. *A* *In Search of Memory*. *Science of Mind*. New

Aaron Beck (1921–)	United States	Beck A. *Cognitive Thera* *Disorders*. New York: Press; 1976; Beck A, *Cognitive Therapy of* Guilford Press; 1979; R. *Anxiety Disorders* *Perspective*. New Yo Freeman A. *Cognitive* *Disorders.* New York:

		Weishaar M. Aaron T *Counselling and Psyc* 1993; Barsky AJ, Ahe therapy for hypochon controlled trial. *JAMA*
Karl Menninger (1893–1990), William Menninger (1900–1966)	United States	Faulkner H, Pruitt V, eds *Correspondence of K* CT: Yale University P Menninger: *The Fami* Knopf; 1990

Frederick K. Goodwin (1936–), Kay Redfield Jamison (1946–)	United States	Goodwin FK, Jamison K New York: Oxford Un 2007
A memoir of depression	United States	Styron W. *Darkness Visi* New York: Random H

SSRIs	United States	“The Story of Prozac,” W Manufacturers Assoc Serotonin-specific inh BJ, eds. *Comprehens* ed. Baltimore, MD: W
Jerome Frank (1910–2005), E. Fuller Torrey (1937–)	United States	Frank J. *Persuasion and* *of Psychotherapy*. Ba University Press; 196 Frank JB. *Persuasion* *Study of Psychothera* Johns Hopkins Unive *The Mind Game: Witc* New York: Emerson H *Witchdoctors and Psy* *of Psychotherapy and* & Row; 1986 (a revis

Psychiatric epidemiology	United States	Robins LN, Regier DA, e *America: The Epidem* New York: Free Press

Forensic psychiatry	United States	Bursztajn H, Scherr A, B forensic psychiatry in in criminal responsibi 1994;1(3):611; Prenti subspecialization in f *Acad Psychiatry Law*
Epidemic of AIDS and of persons infected with HIV (1981–)	Worldwide	Grmek M. *History of AID* *Modern Pandemic*. P University Press; 199 A*IDS and Contempor* Cambridge University Miller J, eds. *The AID* *History.* Westport, CT Bayer R, Oppenheim *Voices from the Epid*

		Press; 2000; Treisma *Psychiatry of AIDS: A* *Treatment*. Baltimore Press; 2004; Mayer K *Pandemic: Impact on* Diego, CA: Elsevier A
Classification of mental disorders	United States	American Psychiatric As *Statistical Manual of* rev. Washington, DC: Association; 2000 (DS

Clozapine, risperidone, and olanzapine in the treatment of schizophrenia	United States, Canada, Europe, England, South America, South Africa	Kane J, Honigfield G, Sir Clozaril Collaborative treatment-resistant sc comparison with chlo *Psychiatry.* 1988;45:7 Clozapine benefits ar 1994;20:23; Ayd FJ J unique antipsychotic. 1994;29:5; Marder SF the treatment of schiz 1994;151:825; Kane J *Psychiatry*. 1994;151 Tran P, Satterlee W, placebo and haloperi North American doub *Neuropsychopharma* G, Kuntz A. Review o olanzapine. *Br J Psyc* Czobor P, Sheitman I Clozapine, olanzapin in the treatment of pa schizophrenia and sc

		Psychiatry. 2002;159 clozapine and its eme review and analysis.
Publications in psychohistory and psychobiography	United States, England	Lawton H. *The Psychohi* Psychohistory Press; Psyche. Understandi Events, History, and S periodical published b 627 Dakota Trail, Fra Sabbadini A, ed. *Psy* No. 1. London: Artesi first issue, this *Histor* intervals of twice a ye Forrester, and since 2 University Press); Sza *and Practice*. New Yo Psychoanalytic schola *Annu Psychoanal*. 20

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University Press; 200

Paul Wender (1934–)	United States	*Attention-Deficit Hyperac* York: Oxford Universi

Computers and the Internet	United States	The APA has published t computers in psychia *Applications in Psych* American Psychiatric CH, Luo JS, Kennedy *Computers in Clinical* American Psychiatric Computer-based testi Sadock BJ, Sadock V *Psychiatry*. 7th ed. Ph Williams & Wilkins; 20 Two articles about the In Internet and the futur *Psychiatry*. 1996;153 Internet: Ushering in a 2001;285:804

Discoveries of genetic variations as causes of mental illnesses	United States, Canada, England, Europe, Australia	Oberlé I, Rousseau F, H Instability of a 550–ba X syndrome. *Science* Pritchard M, Kremer E Fragile X genotype ch region of DNA. *Scien* Fu, Kuhl DPA, Pizzut Variation of the CGG results in genetic inst Sherman paradox. *Ce* GD, Bird TD, Wijsma Genetic linkage of evi disease locus on chro 1992;258:668; Muller M, Fidani L, Lombard onset of Alzheimer's chromosome 14, prox antichymotrypsin gen

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Weissman MM, ed. *T*
Bridging the 21st Cer
American Psychiatric
Andreasen NC. Schiz
New Brain: Conqueri
the Genome. New Yo
Press; 2001:199

Three books on psychotherapy	United States	Jackson SW. *Care of the Psychological Healing* University Press; 199 *The Real World Guid* Cambridge, MA: Harv Hersen M, Sledge WI *Psychotherapy*. 2 Vol Press; 2002
Education of psychiatrists	United States	Luhrmann TM. *Of Two M American Psychiatry*.

Theory and treatment of depression	United States, England, Switzerland	In a March 1999 New Yo American Psychopath clinical scientists gave of Depression, all of w Weissman MM, ed. *T* *Bridging the 21st Cer* American Psychiatric

Neuroimaging	United States	Andreasen NC. Mapping *Conquering Mental Ill* Oxford: Oxford Unive JM, ed. *Advances in* DC: American Psychi

The effects of traumatic events on individuals	United States, England, France, Germany, and Italy	Micale MS, Lerner P, eds *and Trauma in the Mo* Cambridge: Cambridg

Diagnosis and treatment of borderline personality disorder	United States	Gunderson JG. *Borderlir* *Clinical Guide*. Washi Psychiatric Associatic Association. *Practice* *Patients with Borderli* Washington, DC: Am 2001; Oldham JM. A borderline personality 2002;287:1031
September 11, 2001, terrorist attacks on the World Trade Center and the Pentagon	United States	Kleinfield NR, Connelly M psyche. Poll finds wid fear of terror. *The Ne* 2003, Section A, pp. A America. *One Town's*

		Hope. New York: Ran JA, Tylim I, Hoffenber *Psychoanalytic Space from Ground Zero*. Ne 2003
Overviews of psychoanalysis	United States	Erwin E, ed. *The Freud E and Culture*. New Yor E. *Secrets of the Sou of Psychoanalysis*. Ne

Psychosomatic Medicine/Consultation-Liaison Psychiatry	United States	Blumenfield M, Strain J, Philadelphia, PA: Lipp

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ADHD, attention-deficit/hyperactivity disorder; AIDS, acquired immune deficiency syndrome; APA, American Psychiatric Association; CNS, central nervous system; CRF, corticotropin-releasing factor; CT, computed tomography; DSM, *Diagnostic and Statistical Manual of Mental Disorders;* ECT, electroconvulsive therapy; FDA, U.S. Food and Drug Administration;

HIV, human immunodeficiency virus; MAOI, monoamine oxidase inhibitor; MMPI, Minnesota Multiphasic Personality Inventory; MRI, magnetic resonance imaging; NAMI, National Alliance for the Mentally Ill; NARSAD, National Alliance for Research on Schizophrenia and Depression; NIMH, National Institute for Mental Health; NMHA, National Mental Health Association; OCD, obsessive-compulsive disorder; PET, positron emission tomography; PTSD, posttraumatic stress disorder; SSRI, selective serotonin reuptake inhibitor; TR, text revision.

This table was conceived and developed by Ralph Colp Jr, M.D., who contributed to many editions of this book and who died in 2008. It was updated by Dr Benjamin Sadock with the assistance of Kristel Carrington, M.D.

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Glossary of Terms Relating to Signs and Symptoms

abreaction: A process by which repressed material, particularly a painful experience or conflict, is brought back to consciousness; in this process, the person not only recalls but also relives the repressed material, which is accompanied by the appropriate affective response.

abstract thinking: Thinking characterized by the ability to grasp the essentials of a whole, to break a whole into its parts, and to discern common properties. To think symbolically.

abulia: Reduced impulse to act and to think, associated with indifference about consequences of action. Occurs as a result of neurologic deficit, depression, and schizophrenia.

acalculia: Loss of ability to do calculations; not caused by anxiety or impairment in concentration. Occurs with neurologic deficit and learning disorder.

acataphasia: Disordered speech in which statements are incorrectly formulated. Patients may express themselves with words that sound like the ones intended but are not appropriate to the thoughts, or they may use totally inappropriate expressions.

acathexis: Lack of feeling associated with an ordinarily emotionally charged subject; in psychoanalysis, it denotes the patient's detaching or transferring of emotion from thoughts and ideas. Also called *decathexis.* Occurs in anxiety, dissociative, schizophrenic, and bipolar disorders.

acenesthesia: Loss of sensation of physical existence.

acrophobia: Dread of high places.

acting out: Behavioral response to an unconscious drive or impulse that

brings about temporary partial relief of inner tension; relief is attained by reacting to a present situation as if it were the situation that originally gave rise to the drive or impulse. Common in borderline states.

aculalia: Nonsense speech associated with marked impairment of comprehension. Occurs in mania, schizophrenia, and neurologic deficit.

addiction: Psychological or physical dependence on a drug. Sometimes more broadly used to describe dependence on any potentially pleasurable experience (gambling, sexual activity).

adiadochokinesia: Inability to perform rapid alternating movements. Occurs with neurologic deficit and cerebellar lesions.

adynamia: Weakness and fatigability, characteristic of neurasthenia and depression.

aerophagia: Excessive swallowing of air. Seen in anxiety disorder.

affect: The subjective and immediate experience of emotion attached to ideas or mental representations of objects. Affect has outward manifestations that may be classified as restricted, blunted, flattened, broad, labile, appropriate, or inappropriate. *See also* **mood**.

ageusia: Lack or impairment of the sense of taste. Seen in depression and neurologic deficit.

aggression: Forceful, goal-directed action that may be verbal or physical; the motor counterpart of the affect of rage, anger, or hostility. Seen in neurologic deficit, temporal lobe disorder, impulse-control disorders, mania, and schizophrenia.

agitation: Severe anxiety associated with motor restlessness.

agnosia: Inability to understand the import or significance of sensory stimuli; cannot be explained by a defect in sensory pathways or cerebral lesion; the term has also been used to refer to the selective loss or disuse of knowledge of specific objects because of emotional circumstances, as seen in certain schizophrenic, anxious, and depressed patients. Occurs with neurologic deficit. For types of agnosia, see the specific term.

agoraphobia: Morbid fear of open places or leaving the familiar setting of the home. May be present with or without panic attacks.

agrammatism: Speech in which the patient forms words into a sentence without regard for grammatical rules. Seen in Alzheimer and Pick diseases.

agraphia: Loss or impairment of a previously possessed ability to write.

ailurophobia: Dread of cats.

akataphasia: A form of disordered speech in which thoughts cannot be expressed directly but are expressed indirectly such as by making a similar sound (displacement paralogia) or by being derailed into another thought (derailment paralogia). *See also* **derailment**.

akathisia: Subjective feeling of motor restlessness manifested by a compelling need to be in constant movement; may be seen as an extrapyramidal adverse effect of antipsychotic medication. May be mistaken for psychotic agitation.

akinesia: Lack of physical movement, as in the extreme immobility of catatonic schizophrenia; may also occur as an extrapyramidal effect of antipsychotic medication.

akinetic mutism: Absence of voluntary motor movement or speech in a patient who is apparently alert (as evidenced by eye movements). Seen in psychotic depression and catatonic states.

alexia: Loss of a previously possessed reading facility; not explained by defective visual acuity. *Compare with* **dyslexia**.

alexithymia: Inability or difficulty in describing or being aware of one's emotions or moods.

algophobia: Dread of pain.

alogia: Inability to speak because of a mental deficiency or an episode of dementia.

ambivalence: Coexistence of two opposing impulses toward the same thing in the same person at the same time. Seen in schizophrenia, borderline states, and obsessive-compulsive disorders (OCDs).

amimia: Lack of the ability to make gestures or to comprehend those made by others.

amnesia: Partial or total inability to recall past experiences; may be organic (*amnestic disorder*) or emotional (*dissociative amnesia*) in origin.

amnesic aphasia: Disturbed capacity to name objects, even though they are known to the patient. Also called *anomic aphasia, dysnomia, nominal aphasia, and amnestic aphasia.*

anaclitic: Depending on others, especially as the infant on the mother; analytic depression in children results from an absence of mothering.

analgesia: State in which one feels little or no pain. Can occur under hypnosis and in dissociative disorder.

anancasm: Repetitious or stereotyped behavior or thought usually used as a

tension-relieving device; used as a synonym for obsession and seen in obsessive-compulsive (anankastic) personality.

androgyny: Combination of culturally determined female and male characteristics in one person.

anergia: Lack of energy.

anhedonia: Loss of interest in and withdrawal from all regular and pleasurable activities. Often associated with depression.

anomia: Inability to recall the names of objects.

anorexia: Loss or decrease in appetite. In *anorexia nervosa*, appetite may be preserved, but the patient refuses to eat.

anosognosia: Inability to recognize a physical deficit in oneself (e.g., patient denies paralyzed limb).

anterograde amnesia: Loss of memory for events subsequent to the onset of the amnesia; common after trauma. *Compare with* **retrograde amnesia**.

anxiety: Feeling of apprehension caused by anticipation of danger, which may be internal or external.

apathy: Dulled emotional tone associated with detachment or indifference; observed in certain types of schizophrenia and depression.

aphasia: Any disturbance in the comprehension or expression of language caused by a brain lesion. For types of aphasia, see the specific term.

aphonia: Loss of voice. Seen in conversion disorder.

apperception: Awareness of the meaning and significance of a particular sensory stimulus as modified by one's own experiences, knowledge, thoughts, and emotions. *See also* **perception**.

appropriate affect: Emotional tone in harmony with the accompanying idea, thought, or speech.

apraxia: Inability to perform a voluntary purposeful motor activity; cannot be explained by paralysis or other motor or sensory impairment. In *constructional apraxia*, a patient cannot draw two- or three-dimensional forms.

asomatopagnosia: Inability to recognize a part of one's body as one's own (also called *ignorance of the body* and *autotopagnosia*).

astasia abasia: Inability to stand or to walk in a normal manner, even though normal leg movements can be performed in a sitting or lying down position. Seen in conversion disorder.

astereognosis: Inability to identify familiar objects by touch. Seen with

neurologic deficit. *See also* **neurologic amnesia**.

asthenia: Weakness or debility.

asyndesis: Disorder of language in which the patient combines unconnected ideas and images. Commonly seen in schizophrenia.

ataxia: Lack of coordination, physical or mental. (1) In neurology, refers to loss of muscular coordination. (2) In psychiatry, the term *intrapsychic ataxia* refers to lack of coordination between feelings and thoughts; seen in schizophrenia and in severe OCD.

atonia: Lack of muscle tone. *See* **waxy flexibility**.

attention: Concentration; the aspect of consciousness that relates to the amount of effort exerted in focusing on certain aspects of an experience, activity, or task. Usually impaired in anxiety and depressive disorders.

auditory hallucination: False perception of sound, usually voices, but also other noises, such as music. Most common hallucination in psychiatric disorders.

audible thoughts: A form of auditory hallucination in which everything the patient thinks or speaks is repeated by the voices. Also known as thought echoing.

aura: (1) Warning sensations, such as automatisms, fullness in the stomach, blushing, and changes in respiration, cognitive sensations, and mood states usually experienced before a seizure. (2) A sensory prodrome that precedes a classic migraine headache.

autistic thinking: Thinking in which the thoughts are largely narcissistic and egocentric, with emphasis on subjectivity rather than objectivity, and without regard for reality; used interchangeably with autism and dereism. Seen in schizophrenia and autistic disorder.

automatic obedience: Strict obedience of command without critical judgment. The person may respond to an inner voice, as in schizophrenia, or to another person's command, as in hypnosis.

automatism: Activity carried out without conscious knowledge.

autoscopy: Seeing oneself or a double as part of a brief hallucinatory experience.

behavior: Sum total of the psyche that includes impulses, motivations, wishes, drives, instincts, and cravings, as expressed by a person's actions or motor activity. Also called *conation*.

bereavement: Feeling of grief or desolation, especially at the death or loss of

a loved one.

bizarre delusion: False belief that is patently absurd or fantastic (e.g., invaders from space have implanted electrodes in a person's brain). Common in schizophrenia. In nonbizarre delusion, content is usually within the range of possibility.

blackout: Amnesia experienced by alcoholics about behavior during drinking bouts; usually indicates reversible brain damage.

blocking: Abrupt interruption in train of thinking before a thought or idea is finished; after a brief pause, the person indicates no recall of what was being said or was going to be said (also known as *thought deprivation* or *increased thought latency*). Common in schizophrenia and severe anxiety.

blunted affect: Disturbance of affect manifested by a severe reduction in the intensity of externalized feeling tone; one of the fundamental symptoms of schizophrenia, as outlined by Eugen Bleuler.

bradykinesia: Slowness of motor activity, with a decrease in normal spontaneous movement.

bradylalia: Abnormally slow speech. Common in depression.

bradylexia: Inability to read at normal speed.

bruxism: Grinding or gnashing of the teeth, typically occurring during sleep. Seen in anxiety disorder.

carebaria: Sensation of discomfort or pressure in the head.

catalepsy: Condition in which persons maintain the body position into which they are placed; observed in severe cases of catatonic schizophrenia. Compare with *waxy flexibility* and *cerea flexibilitas. See also* **command automatism**.

cataplexy: Temporary sudden loss of muscle tone, causing weakness and immobilization; can be precipitated by a variety of emotional states and is often followed by sleep. Commonly seen in narcolepsy.

catastrophic reaction: Extreme emotional state characterized by restlessness, irritability, crying, anxiety, and uncooperativeness. Seen in patients who have had a stroke.

catatonic excitement: Excited, uncontrolled motor activity seen in catatonic schizophrenia. Patients in catatonic state may suddenly erupt into an excited state and may be violent.

catatonic posturing: Voluntary assumption of an inappropriate or bizarre posture, generally maintained for long periods of time. May switch

unexpectedly with catatonic excitement.

catatonic rigidity: Fixed and sustained motoric position that is resistant to change.

catatonic stupor: Stupor in which patients ordinarily are well aware of their surroundings.

cathexis: In psychoanalysis, a conscious or unconscious investment of psychic energy in an idea, concept, object, or person. *Compare with* **acathexis**.

cenesthesia: The general awareness of being alive coming from stimuli arising from the functioning of various bodily organs.

cephalagia: Headache.

cerea flexibilitas: Condition of a person who can be molded into a position that is then maintained; when an examiner moves the person's limb, the limb feels as if it were made of wax. Also called *waxy flexibility*. Seen in schizophrenia.

chorea: Movement disorder characterized by random and involuntary quick, jerky, purposeless movements. Seen in Huntington disease.

circumstantiality: Disturbance in the associative thought and speech processes in which a patient digresses into unnecessary details and inappropriate thoughts before communicating the central idea. Observed in schizophrenia, obsessional disturbances, and certain cases of dementia. *See also* **tangentiality**.

clang association: Association or speech directed by the sound of a word rather than by its meaning; words have no logical connection; punning and rhyming may dominate the verbal behavior. Seen most frequently in schizophrenia or mania.

claustrophobia: Abnormal fear of closed or confining spaces.

clonic convulsion: An involuntary, violent muscular contraction or spasm in which the muscles alternately contract and relax. Characteristic phase in grand mal epileptic seizure.

clouding of consciousness: Any disturbance of consciousness in which the person is not fully awake, alert, and oriented. Occurs in delirium, dementia, and cognitive disorder.

cluttering: Disturbance of fluency involving an abnormally rapid rate and erratic rhythm of speech that impedes intelligibility; the affected individual is usually unaware of communicative impairment.

cognition: Mental process of knowing and becoming aware; function is closely associated with judgment.

coma: State of profound unconsciousness from which a person cannot be roused, with minimal or no detectable responsiveness to stimuli; seen in injury or disease of the brain, in systemic conditions such as diabetic ketoacidosis and uremia, and in intoxications with alcohol and other drugs. Coma may also occur in severe catatonic states and in conversion disorder.

coma vigil: Coma in which a patient appears to be asleep and cannot be aroused. A persistent vegetative state.

command automatism: Condition in which suggestions are followed automatically.

command hallucination: False perception of orders that a person may feel obliged to obey or unable to resist.

complex partial seizure: A seizure characterized by alterations in consciousness that may be accompanied by complex hallucinations (sometimes olfactory) or illusions. During the seizure, a state of impaired consciousness resembling a dreamlike state may occur, and the patient may exhibit repetitive, automatic, or semipurposeful behavior.

compulsion: Pathologic need to act on an impulse that, if resisted, produces anxiety; repetitive behavior in response to an obsession or performed according to certain rules, with no true end in itself other than to prevent something from occurring in the future.

conation: That part of a person's mental life concerned with cravings, strivings, motivations, drives, and wishes as expressed through behavior or motor activity.

concrete thinking: Thinking characterized by actual things, events, and immediate experience rather than by abstractions; seen in young children, in those who have lost or never developed the ability to generalize (as in certain cognitive mental disorders), and in schizophrenic persons. *Compare with* **abstract thinking**.

condensation: Mental process in which one symbol stands for a number of components.

confabulation: Unconscious filling of gaps in memory by imagining experiences or events that have no basis in fact, commonly seen in amnestic syndromes; should be differentiated from lying. *See also* **paramnesia**.

confusion: Disturbances of consciousness manifested by a disordered orientation in relation to time, place, or person.

consciousness: State of awareness, with response to external stimuli.

constricted affect: Reduction in intensity of feeling tone that is less severe than that of blunted affect.

constructional apraxia: Inability to copy a drawing, such as a cube, clock, or pentagon, as a result of a brain lesion.

conversion phenomena: The development of symbolic physical symptoms and distortions involving the voluntary muscles or special sense organs; not under voluntary control and not explained by any physical disorder. Most common in conversion disorder but also seen in a variety of mental disorders.

convulsion: An involuntary, violent muscular contraction or spasm. *See also* **clonic convulsion** and **tonic convulsion**.

coprolalia: Involuntary use of vulgar or obscene language. Observed in some cases of schizophrenia and in Tourette syndrome.

coprophagia: Eating of filth or feces.

cryptographia: A private written language.

cryptolalia: A private spoken language.

cycloplegia: Paralysis of the muscles of accommodation in the eye; observed, at times, as an autonomic adverse effect (anticholinergic effect) of antipsychotic or antidepressant medication.

decompensation: Deterioration of psychic functioning caused by a breakdown of defense mechanisms. Seen in psychotic states.

déjà entendu: Illusion that what one is hearing one has been heard previously. *See also* **paramnesia**.

déjà pensé: Condition in which a thought never entertained before is incorrectly regarded as a repetition of a previous thought. *See also* **paramnesia**.

déjà vu: Illusion of visual recognition in which a new situation is incorrectly regarded as a repetition of a previous experience. *See also* **paramnesia**.

delirium: Acute reversible mental disorder characterized by confusion and some impairment of consciousness; generally associated with emotional lability, hallucinations or illusions, and inappropriate, impulsive, irrational, or violent behavior.

delirium tremens: Acute and sometimes fatal reaction to withdrawal from

alcohol, usually occurring 72 to 96 hours after the cessation of heavy drinking; distinctive characteristics are marked autonomic hyperactivity (tachycardia, fever, hyperhidrosis, and dilated pupils), usually accompanied by tremulousness, hallucinations, illusions, and delusions. Called *alcohol withdrawal with perceptual disturbance in DSM-5*. *See also* **formication**.

delusion: False belief, based on incorrect inference about external reality, that is firmly held despite objective and obvious contradictory proof or evidence and despite the fact that other members of the culture do not share the belief.

delusion of control: False belief that a person's will, thoughts, or feelings are being controlled by external forces.

delusion of grandeur: Exaggerated conception of one's importance, power, or identity.

delusion of infidelity: False belief that one's lover is unfaithful. Sometimes called *pathologic jealousy*.

delusion of persecution: False belief of being harassed or persecuted; often found in litigious patients who have a pathologic tendency to take legal action because of imagined mistreatment. Most common delusion.

delusion of poverty: False belief that one is bereft or will be deprived of all material possessions.

delusion of reference: False belief that the behavior of others refers to oneself or that events, objects, or other people have a particular and unusual significance, usually of a negative nature; derived from idea of reference, in which persons falsely feel that others are talking about them (e.g., belief that people on television or radio are talking to or about the person). *See also* **thought broadcasting**.

delusion of self-accusation: False feeling of remorse and guilt. Seen in depression with psychotic features.

dementia: Mental disorder characterized by general impairment in intellectual functioning without clouding of consciousness; characterized by failing memory, difficulty with calculations, distractibility, alterations in mood and affect, impaired judgment and abstraction, reduced facility with language, and disturbance of orientation. Although irreversible because of underlying progressive degenerative brain disease, dementia may be reversible if the cause can be treated. Also known as major

neurocognitive disorder in **DSM-5**.

denial: Defense mechanism in which the existence of unpleasant realities is disavowed; refers to keeping out of conscious awareness any aspects of external reality that, if acknowledged, would produce anxiety.

depersonalization: Sensation of unreality concerning oneself. Experiencing one's own behavior, thoughts, or feelings as if they were outside oneself, or as in a dream.

depression: Mental state characterized by feelings of sadness, loneliness, despair, low self-esteem, and self-reproach; accompanying signs include psychomotor retardation or, at times, agitation, withdrawal from interpersonal contact, and vegetative symptoms, such as insomnia and anorexia. The term refers to a mood that is so characterized or a mood disorder.

derailment: Gradual or sudden deviation in train of thought without blocking; sometimes used synonymously with *loosening of association.*

derealization: Sensation of changed reality or feeling detached from one's surroundings.

dereism: Mental activity that follows a totally subjective and idiosyncratic system of logic and fails to take the facts of reality or experience into consideration. Characteristic of schizophrenia. *See also* **autistic thinking**.

detachment: Characterized by distant interpersonal relationships and lack of emotional involvement.

devaluation: Defense mechanism in which a person attributes excessively negative qualities to self or others. Seen in depression and paranoid personality disorder.

diminished libido: Decreased sexual interest and drive. (Increased libido is often associated with mania.)

dipsomania: Compulsion to drink alcoholic beverages.

disinhibition: (1) Removal of an inhibitory effect, as in the reduction of the inhibitory function of the cerebral cortex by alcohol. (2) In psychiatry, a greater freedom to act in accordance with inner drives or feelings and with less regard for restraints dictated by cultural norms or one's superego.

disorientation: Confusion; impairment of awareness of time, place, and person (the position of the self in relation to other persons). Characteristic of cognitive disorders.

displacement: Unconscious defense mechanism by which the emotional

component of an unacceptable idea or object is transferred to a more acceptable one. Seen in phobias.

dissociation: Unconscious defense mechanism involving the segregation of any group of mental or behavioral processes from the rest of the person's psychic activity; may entail the separation of an idea from its accompanying emotional tone, as seen in dissociative and conversion disorders.

distractibility: Inability to focus one's attention; the patient does not respond to the task at hand but attends to irrelevant phenomena in the environment.

dread: Massive or pervasive anxiety, usually related to a specific danger.

dreamy state: Altered state of consciousness, the semiconscious state associated with an epileptic attack, particularly from medial temporal epilepsy.

drowsiness: State of impaired awareness associated with a desire or inclination to sleep.

dysarthria: Difficulty in articulation, the motor activity of shaping phonated sounds into speech, not in word finding or in grammar.

dyscalculia: Difficulty in performing calculations.

dysgeusia: Impaired sense of taste.

dysgraphia: Difficulty in writing.

dyskinesia: Uncontrolled, involuntary muscle movement. Seen in extrapyramidal disorders.

dyslalia: Faulty articulation caused by structural abnormalities of the articulatory organs or impaired hearing.

dyslexia: Specific learning disability syndrome involving an impairment of the ability to read; unrelated to the person's intelligence. *Compare with* **alexia**.

dysmegalopsia: A distortion in which the size and shape of objects is misperceived, sometimes called the "Alice in Wonderland" effect. *See also* **illusion**.

dysmetria: Impaired ability to gauge distance relative to movements. Seen in neurologic deficit.

dysmnesia: Impaired memory.

dyspareunia: Physical pain in sexual intercourse, which may be emotionally caused and more commonly experienced by women; may also result from cystitis, urethritis, or other medical conditions.

dysphagia: Difficulty in swallowing.

dysphasia: Difficulty in comprehending oral language (*reception dysphasia*) or in trying to express verbal language (*expressive dysphasia*).

dysphonia: Difficulty in speaking or singing.

dysphoria: Feeling of unpleasantness or discomfort; a mood of general dissatisfaction and restlessness. Occurs in depression and anxiety.

dysprosody: Loss of normal speech melody (*prosody*). Common in depression.

dystonia: Extrapyramidal motor disturbance consisting of slow, sustained contractions of the axial or appendicular musculature; one movement often predominates, leading to relatively sustained postural deviations; acute dystonic reactions (facial grimacing and torticollis) are occasionally seen with the initiation of antipsychotic drug therapy.

echolalia: Psychopathological repeating of words or phrases of one person by another; tends to be repetitive and persistent. Seen in certain kinds of schizophrenia, particularly the catatonic types.

echopraxia: Similar to echolalia, however this represents the repeating or mimicking of the movements of another person.

ego-alien: Denoting aspects of a person's personality that are viewed as repugnant, unacceptable, or inconsistent with the rest of the personality. Also called *ego-dystonia. Compare with* **ego-syntonic**.

egocentric: Self-centered; selfishly preoccupied with one's own needs; lacking interest in others.

ego-dystonic: *See* **ego-alien**.

egomania: Morbid self-preoccupation or self-centeredness. *See also* **narcissism**.

ego-syntonic: Denoting aspects of a personality that are viewed as acceptable and consistent with that person's total personality. Personality traits are usually ego-syntonic. *Compare with* **ego-alien**.

eidetic image: Unusually vivid or exact mental image of objects previously seen or imagined.

elation: Mood consisting of feelings of joy, euphoria, triumph, and intense self-satisfaction or optimism. Occurs in mania when not grounded in reality.

elevated mood: Air of confidence and enjoyment; a mood more cheerful than normal but not necessarily pathologic.

emotion: Complex feeling state with psychic, somatic, and behavioral components; external manifestation of emotion is *affect.*

emotional insight: A level of understanding or awareness that one has emotional problems. It facilitates positive changes in personality and behavior when present.

emotional lability: Excessive emotional responsiveness characterized by unstable and rapidly changing emotions.

encopresis: Involuntary passage of feces, usually occurring at night or during sleep.

enuresis: Incontinence of urine, often during sleep.

erotomania: Delusional belief, more common in women than in men, that someone is deeply in love with them (also known as *de Clérambault syndrome*).

erythrophobia: Abnormal fear of blushing.

euphoria: Exaggerated feeling of well-being that is inappropriate to real events. Can occur with drugs such as opiates, amphetamines, and alcohol.

euthymia: Normal range of mood, implying absence of depressed or elevated mood.

evasion: Act of not facing up to, or strategically eluding, something; consists of suppressing an idea that is next in a thought series and replacing it with another idea closely related to it. Also called *paralogia* and *perverted logic.*

exaltation: Feeling of intense elation and grandeur.

excited: Agitated, purposeless motor activity uninfluenced by external stimuli.

expansive mood: Expression of feelings without restraint, frequently with an overestimation of their significance or importance. Seen in mania and grandiose delusional disorder.

expressive aphasia: Disturbance of speech in which understanding remains, but ability to speak is grossly impaired; halting, laborious, and inaccurate speech (also known as *Broca's*, *nonfluent*, and *motor aphasias*).

expressive dysphasia: Difficulty in expressing verbal language; the ability to understand language is intact.

externalization: More general term than *projection* that refers to the tendency to perceive in the external world and in external objects elements of one's own personality, including instinctual impulses, conflicts, moods,

attitudes, and styles of thinking.

extroversion: State of one's energies being directed outside oneself. *Compare with* **introversion**.

false memory: A person's recollection and belief of an event that did not actually occur. In *false memory syndrome*, persons erroneously believe that they sustained an emotional or physical (e.g., sexual) trauma in early life.

fantasy: Daydream; fabricated mental picture of a situation or chain of events. A normal form of thinking dominated by unconscious material that seeks wish fulfillment and solutions to conflicts; may serve as the matrix for creativity. The content of the fantasy may indicate mental illness.

fatigue: A feeling of weariness, sleepiness, or irritability after a period of mental or bodily activity. Seen in depression, anxiety, neurasthenia, and somatoform disorders.

fausse reconnaissance: False recognition, a feature of paramnesia. Can occur in delusional disorders.

fear: Unpleasurable emotional state consisting of psychophysiological changes in response to a realistic threat or danger. *Compare with* **anxiety**.

flat affect: Absence or near absence of any signs of affective expression.

flight of ideas: Rapid succession of fragmentary thoughts or speech in which content changes abruptly and speech may be incoherent. Seen in mania.

floccillation: Aimless plucking or picking, usually at bedclothes or clothing, commonly seen in dementia and delirium.

fluent aphasia: Aphasia characterized by inability to understand the spoken word; fluent but incoherent speech is present. Also called *Wernicke's*, *sensory* and *receptive aphasias.*

folie à deux: Mental illness shared by two persons, usually involving a common delusional system; if it involves three persons, it is referred to as *folie à trois*, etc. Also called *shared psychotic disorder.*

formal thought disorder: Disturbance in the form of thought rather than the content of thought; thinking characterized by loosened associations, neologisms, and illogic constructs; thought process is disordered, and the person is defined as psychotic. Characteristic of schizophrenia.

formication: Tactile hallucination involving the sensation that tiny insects are crawling over the skin. Seen in cocaine addiction and delirium tremens.

free-floating anxiety: Severe, pervasive, generalized anxiety that is not attached to any particular idea, object, or event. Observed particularly in

anxiety disorders, although it may be seen in some cases of schizophrenia.

fugue: Dissociative disorder characterized by a period of almost complete amnesia, during which a person actually flees from an immediate life situation and begins a different life pattern; apart from the amnesia, mental faculties and skills are usually unimpaired.

galactorrhea: Abnormal discharge of milk from the breast; may result from the endocrine influence (e.g., prolactin) of dopamine receptor antagonists, such as phenothiazines.

generalized tonic–clonic seizure: Generalized onset of tonic–clonic movements of the limbs, tongue biting, and incontinence followed by slow, gradual recovery of consciousness and cognition; also called *grand mal seizure*.

global aphasia: Combination of grossly nonfluent aphasia and severe fluent aphasia.

glossolalia: Unintelligible jargon that has meaning to the speaker but not to the listener. Occurs in schizophrenia.

grandiosity: Exaggerated feelings of one's importance, power, knowledge, or identity. Occurs in delusional disorder and manic states.

grief: Alteration in mood and affect consisting of sadness appropriate to a real loss; normally, it is self-limited. *See also* **depression** and **mourning**.

guilt: Emotional state associated with self-reproach and the need for punishment. In psychoanalysis, refers to a feeling of culpability that stems from a conflict between the ego and the superego (conscience). Guilt has normal psychological and social functions, but special intensity or absence of guilt characterizes many mental disorders, such as depression and antisocial personality disorder, respectively. Psychiatrists distinguish shame as a less internalized form of guilt that relates more to others than to the self. *See also* **shame**.

gustatory hallucination: Hallucination primarily involving taste.

gynecomastia: Female-like development of the male breasts; may occur as an adverse effect of antipsychotic and antidepressant drugs because of increased prolactin levels or anabolic–androgenic steroid abuse.

hallucination: False sensory perception occurring in the absence of any relevant external stimulation of the sensory modality involved. For types of hallucinations, see the specific term.

hallucinosis: State in which a person experiences hallucinations without any

impairment of consciousness.

haptic hallucination: Hallucination of touch.

hebephrenia: Complex of symptoms, considered a form of schizophrenia, characterized by wild or silly behavior or mannerisms, inappropriate affect, and delusions and hallucinations that are transient and unsystematized. Hebephrenic schizophrenia is now called *disorganized schizophrenia.*

holophrastic: Using a single word to express a combination of ideas. Seen in schizophrenia.

hyperactivity: Increased muscular activity. The term is commonly used to describe a disturbance found in children that is manifested by constant restlessness, overactivity, distractibility, and difficulties in learning. Seen in *attention-deficit/hyperactivity disorder* (ADHD).

hyperacusis: Extreme sensitivity to sounds.

hyperalgesia: Excessive sensitivity to pain. Seen in somatoform disorder.

hyperesthesia: Increased sensitivity to any sense. It can affect one or more senses.

hypermnesia: Exaggerated degree of retention and recall. It can be elicited by hypnosis and may be seen in certain prodigies; also may be a feature of OCD, some cases of schizophrenia, and manic episodes of bipolar I disorder.

hyperphagia: Increase in appetite and intake of food.

hyperpragia: Excessive thinking and mental activity. Generally associated with manic episodes of bipolar I disorder.

hypersomnia: Excessive time spent asleep. May be associated with underlying medical or psychiatric disorder or narcolepsy, may be part of the Kleine–Levin syndrome, or may be primary.

hyperventilation: Excessive breathing, generally associated with anxiety, which can reduce blood carbon dioxide concentration and can produce lightheadedness, palpitations, numbness, tingling periorally and in the extremities, and, occasionally, syncope.

hypervigilance: Excessive attention to and focus on all internal and external stimuli; usually seen in delusional or paranoid states.

hypesthesia: Diminished sensitivity to tactile stimulation.

hypnagogic hallucination: Hallucination occurring while falling asleep, not ordinarily considered pathologic.

hypnopompic hallucination: Hallucination occurring while awakening from

sleep, not ordinarily considered pathologic.

hypnosis: Artificially induced alteration of consciousness characterized by increased suggestibility and receptivity to direction.

hypoactivity: Decreased motor and cognitive activity, as in psychomotor retardation; visible slowing of thought, speech, and movements. Also called *hypokinesis.*

hypochondria: Exaggerated concern about health that is based not on real medical pathology but on unrealistic interpretations of physical signs or sensations as abnormal.

hypomania: Mood abnormality with the qualitative characteristics of mania but somewhat less intense. Seen in cyclothymic disorder.

idea of reference: Misinterpretation of incidents and events in the outside world as having direct personal reference to oneself; occasionally observed in normal persons, but frequently seen in paranoid patients. If present with sufficient frequency or intensity or if organized and systematized, they constitute delusions of reference.

illogic thinking: Thinking containing erroneous conclusions or internal contradictions; psychopathological only when it is marked and not caused by cultural values or intellectual deficit.

illusion: Perceptual misinterpretation of a real external stimulus. *Compare with* **hallucination**.

immediate memory: Reproduction, recognition, or recall of perceived material within seconds after presentation. *Compare with* **long-term memory** and **short-term memory**.

impaired insight: Diminished ability to understand the objective reality of a situation.

impaired judgment: Diminished ability to understand a situation correctly and to act appropriately.

impulse control: Ability to resist an impulse, drive, or temptation to perform some action.

inappropriate affect: Emotional tone out of harmony with the idea, thought, or speech accompanying it. Seen in schizophrenia.

incoherence: Communication that is disconnected, disorganized, or incomprehensible. *See also* **word salad**.

incorporation: Primitive unconscious defense mechanism in which the psychic representation of another person or aspects of another person are

assimilated into oneself through a figurative process of symbolic oral ingestion; represents a special form of introjection and is the earliest mechanism of identification.

increased libido: Increase in sexual interest and drive.

ineffability: Ecstatic state in which persons insist that their experience is inexpressible and indescribable and that it is impossible to convey what it is like to one who never experienced it.

initial insomnia: Falling asleep with difficulty; usually seen in anxiety disorder. *Compare with* **middle insomnia** and **terminal insomnia**.

insight: Conscious recognition of one's own condition. In psychiatry, it refers to the conscious awareness and understanding of one's own psychodynamics and symptoms of maladaptive behavior; highly important in effecting changes in the personality and behavior of a person.

insomnia: Difficulty in falling asleep or difficulty in staying asleep. It can be related to a mental disorder, can be related to a physical disorder or an adverse effect of medication, or can be primary (not related to a known medical factor or another mental disorder). *See also* **initial insomnia**, **middle insomnia**, and **terminal insomnia**.

intellectual disability: Limitations in intellectual functioning and adaptive behavior, beginning before age 18 years. Below average intelligence.

intellectual insight: Knowledge of the reality of a situation without the ability to use that knowledge successfully to effect an adaptive change in behavior or to master the situation. *Compare with* **true insight**.

intelligence: Capacity for learning and ability to recall, to integrate constructively, and to apply what one has learned; the capacity to understand and to think rationally.

intoxication: Mental disorder caused by recent ingestion or presence in the body of an exogenous substance producing maladaptive behavior by virtue of its effects on the central nervous system (CNS). The most common psychiatric changes involve disturbances of perception, wakefulness, attention, thinking, judgment, emotional control, and psychomotor behavior; the specific clinical picture depends on the substance ingested.

intropunitive: Turning anger inward toward oneself. Commonly observed in depressed patients.

introspection: Contemplating one's own mental processes to achieve insight.

introversion: State in which a person's energies are directed inward toward

the self, with little or no interest in the external world.

irrelevant answer: Answer that is not responsive to the question.

irritability: Abnormal or excessive excitability, with easily triggered anger, annoyance, or impatience.

irritable mood: State in which one is easily annoyed and provoked to anger. *See also* **irritability**.

jamais vu: Paramnestic phenomenon characterized by a false feeling of unfamiliarity with a real situation that one has previously experienced.

jargon aphasia: Aphasia in which the words produced are neologistic; that is, nonsense words created by the patient.

judgment: Mental act of comparing or evaluating choices within the framework of a given set of values for the purpose of electing a course of action. If the course of action chosen is consonant with reality or with mature adult standards of behavior, then judgment is said to be *intact* or *normal*; judgment is said to be *impaired* if the chosen course of action is frankly maladaptive, results from impulsive decisions based on the need for immediate gratification, or is otherwise not consistent with reality as measured by mature adult standards.

kleptomania: Pathologic compulsion to steal.

la belle indifférence: Inappropriate attitude of calm or lack of concern about one's disability. May be seen in patients with conversion disorder.

labile affect: Affective expression characterized by rapid and abrupt changes, unrelated to external stimuli.

labile mood: Oscillations in mood between euphoria and depression or anxiety.

laconic speech: Condition characterized by a reduction in the quantity of spontaneous speech; replies to questions are brief and unelaborated, and little or no unprompted additional information is provided. Occurs in major depression, schizophrenia, and organic mental disorders. Also called *poverty of speech.*

lethologica: Momentary forgetting of a name or proper noun. *See* **blocking**.

lilliputian hallucination: Visual sensation that persons or objects are reduced in size; more properly regarded as an illusion. *See also* **micropsia**.

localized amnesia: Partial loss of memory; amnesia restricted to specific or isolated experiences. Also called *lacunar amnesia* and *patch amnesia.*

logoclonia: Repeated use of the same word. See perseveration.

logorrhea: Copious, pressured, coherent speech; uncontrollable, excessive talking; observed in manic episodes of bipolar disorder. Also called *tachylogia*, *verbomania*, and *volubility*.

long-term memory: Reproduction, recognition, or recall of experiences or information that was experienced in the distant past. Also called *remote memory*. *Compare with* **immediate memory** and **short-term memory**.

loosening of associations: Characteristic schizophrenic thinking or speech disturbance involving a disorder in the logical progression of thoughts, manifested as a failure to communicate verbally adequately; unrelated and unconnected ideas shift from one subject to another. *See also* **tangentiality**.

macropsia: False perception that objects are larger than they really are. *Compare with* **micropsia**.

magical thinking: A form of dereistic thought; thinking similar to that of the preoperational phase in children (Jean Piaget), in which thoughts, words, or actions assume power (e.g., to cause or to prevent events).

malingering: Feigning disease to achieve a specific goal, for example, to avoid an unpleasant responsibility.

mania: Mood state characterized by elation, agitation, hyperactivity, hypersexuality, and accelerated thinking and speaking (flight of ideas). Seen in bipolar I disorder. *See also* **hypomania**.

manipulation: Maneuvering by patients to get their own way, characteristic of antisocial personalities.

mannerism: Ingrained, habitual involuntary movement.

melancholia: Severe depressive state. Used in the term *involutional melancholia* as a descriptive term and also in reference to a distinct diagnostic entity.

memory: Process whereby what is experienced or learned is established as a record in the CNS (registration), where it persists with a variable degree of permanence (retention) and can be recollected or retrieved from storage at will (recall). For types of memory, see **immediate memory**, **long-term memory**, and **short-term memory**.

mental disorder: Psychiatric illness or disease whose manifestations are characterized primarily by behavioral or psychological impairment of function, measured in terms of deviation from some normative concept; associated with distress or disease, not just an expected response to a

particular event or limited to relations between a person and society.

mental retardation: Obsolete term, please see **intellectual disability**.

metonymy: Speech disturbance common in schizophrenia in which the affected person uses a word or phrase that is related to the proper one but is not the one ordinarily used; for example, the patient speaks of consuming a *menu* rather than a *meal* or refers to losing the *piece of string* of the conversation rather than the *thread* of the conversation. *See also* **paraphasia** and **word approximation**.

microcephaly: Condition in which the head is unusually small as a result of defective brain development and premature ossification of the skull.

micropsia: False perception that objects are smaller than they really are. Sometimes called *lilliputian hallucination. Compare with* **macropsia**.

middle insomnia: Waking up after falling asleep without difficulty and then having difficulty in falling asleep again. *Compare with* **initial insomnia** and **terminal insomnia**.

mimicry: Simple, imitative motion activity of childhood.

monomania: Mental state characterized by preoccupation with one subject.

mood: Pervasive and sustained feeling tone that is experienced internally and that, in the extreme, can markedly influence virtually all aspects of a person's behavior and perception of the world. Distinguished from affect, the external expression of the internal feeling tone. For types of mood, see the specific term.

mood-congruent delusion: Delusion with content that is mood appropriate (e.g., depressed patients who believe that they are responsible for the destruction of the world).

mood-congruent hallucination: Hallucination with content that is consistent with a depressed or manic mood (e.g., depressed patients hearing voices telling them that they are bad persons and manic patients hearing voices telling them that they have inflated worth, power, or knowledge).

mood-incongruent delusion: Delusion based on incorrect reference about external reality, with content that has no association to mood or is mood inappropriate (e.g., depressed patients who believe that they are the new Messiah).

mood-incongruent hallucination: Hallucination not associated with real external stimuli, with content that is not consistent with depressed or manic mood (e.g., in depression, hallucinations not involving such themes as

guilt, deserved punishment, or inadequacy; in mania, not involving such themes as inflated worth or power).

mood swings: Oscillation of a person's emotional feeling tone between periods of elation and periods of depression.

motor aphasia: Aphasia in which understanding is intact, but the ability to speak is lost. Also called *Broca's*, *expressive*, or *nonfluent aphasias.*

mourning: Syndrome following loss of a loved one, consisting of preoccupation with the lost individual, weeping, sadness, and repeated reliving of memories. *See also* **bereavement** and **grief**.

muscle rigidity: State in which the muscles remain immovable; seen in schizophrenia.

mutism: Organic or functional absence of the faculty of speech. *See also* **stupor**.

mydriasis: Dilation of the pupil; sometimes occurs as an autonomic (anticholinergic) or atropine-like adverse effect of some antipsychotic and antidepressant drugs.

narcissism: In psychoanalytic theory, divided into primary and secondary types: primary narcissism, the early infantile phase of object relationship development, when the child has not differentiated the self from the outside world, and all sources of pleasure are unrealistically recognized as coming from within the self, giving the child a false sense of omnipotence; secondary narcissism, when the libido, once attached to external love objects, is redirected back to the self. *See also* **autistic thinking**.

needle phobia: The persistent, intense, pathologic fear of receiving an injection.

negative signs: In schizophrenia: flat affect, alogia, abulia, and apathy.

negativism: Verbal or nonverbal opposition or resistance to outside suggestions and advice; commonly seen in catatonic schizophrenia in which the patient resists any effort to be moved or does the opposite of what is asked.

neologism: New word or phrase whose derivation cannot be understood; often seen in schizophrenia. It has also been used to mean a word that has been incorrectly constructed but whose origins are nonetheless understandable (e.g., *head shoe* to mean *hat*), but such constructions are more properly referred to as *word approximations.*

neurologic amnesia: (1) Auditory amnesia: loss of ability to comprehend

sounds or speech. (2) Tactile amnesia: loss of ability to judge the shape of objects by touch. *See also* **astereognosis**. (3) Verbal amnesia: loss of ability to remember words. (4) Visual amnesia: loss of ability to recall or to recognize familiar objects or printed words.

nihilism: Delusion of the nonexistence of the self or part of the self; also refers to an attitude of total rejection of established values or extreme skepticism regarding moral and value judgments.

nihilistic delusion: Depressive delusion that the world and everything related to it have ceased to exist.

noesis: The functioning of the intellectual or cognitive processes.

nominal aphasia: Aphasia characterized by difficulty in giving the correct name of an object. *See also* **anomia** and **amnesic aphasia**.

nymphomania: Abnormal, excessive, insatiable desire in a woman for sexual intercourse. *Compare with* **satyriasis**.

obsession: Persistent and recurrent idea, thought, or impulse that cannot be eliminated from consciousness by logic or reasoning; obsessions are involuntary and ego-dystonic. *See also* **compulsion**.

olfactory hallucination: Hallucination primarily involving smell or odors; most common in medical disorders, especially in the temporal lobe.

orientation: State of awareness of oneself and one's surroundings in terms of time, place, and person.

overactivity: Abnormality in motor behavior that can manifest itself as psychomotor agitation, hyperactivity (hyperkinesis), tics, sleepwalking, or compulsions.

overvalued idea: False or unreasonable belief or idea that is sustained beyond the bounds of reason. It is held with less intensity or duration than a delusion but is usually associated with mental illness.

panic: Acute, intense attack of anxiety associated with personality disorganization; the anxiety is overwhelming and accompanied by feelings of impending doom.

panphobia: Overwhelming fear of everything.

pantomime: Gesticulation; psychodrama without the use of words.

paramnesia: Disturbance of memory in which reality and fantasy are confused. It is observed in dreams and in certain types of schizophrenia and organic mental disorders; it includes phenomena such as *déjà vu* and *déjà entendu*, which may occur occasionally in normal persons.

paranoia: Unreasonable false beliefs, characterized by delusions of persecution, unwarranted jealousy, or exaggerated self-importance, typically elaborated into an organized system.

paranoid delusions: Includes persecutory delusions and delusions of reference, control, and grandeur.

paranoid ideation: Thinking dominated by suspicious, persecutory, or grandiose content of less than delusional proportions.

paraphasia: Abnormal speech in which one word is substituted for another, the irrelevant word generally resembling the required one in morphology, meaning, or phonetic composition; the inappropriate word may be a legitimate one used incorrectly, such as *clover* instead of *hand*, or a bizarre nonsense expression, such as *treen* instead of *train*. Paraphasic speech may be seen in organic aphasias and in mental disorders such as schizophrenia. *See also* **metonymy** and **word approximation**.

parapraxis: Faulty act, such as a slip of the tongue or the misplacement of an article. Freud ascribed parapraxes to unconscious motives.

paresis: Weakness or partial paralysis of organic origin.

paresthesia: Abnormal spontaneous tactile sensation, such as a burning, tingling, or pins-and-needles sensation.

perception: Conscious awareness of elements in the environment by the mental processing of sensory stimuli; sometimes used in a broader sense to refer to the mental process by which all kinds of data, intellectual, emotional, and sensory, are meaningfully organized. *See also* **apperception**.

perseveration: (1) Pathologic repetition of the same response to different stimuli, as in a repetition of the same verbal response to different questions. (2) Persistent repetition of specific words or concepts in the process of speaking. Seen in cognitive disorders, schizophrenia, and other mental illness. *See also* **verbigeration**.

phantom limb: False sensation that an extremity that has been lost is, in fact, present.

phobia: Persistent, pathologic, unrealistic, intense fear of an object or situation; the phobic person may realize that the fear is irrational but, nonetheless, cannot dispel it. For types of phobias, see the specific term.

pica: Craving and eating of nonfood substances, such as paint and clay.

polyphagia: Pathologic overeating.

positive signs: In schizophrenia: hallucinations, delusions, and thought disorder.

posturing: Strange, fixed, and bizarre bodily positions held by a patient for an extended time. *See also* **catatonic posturing**.

poverty of content of speech: Speech that is adequate in amount but conveys little information because of vagueness, emptiness, or stereotyped phrases.

poverty of speech: Restriction in the amount of speech used; replies may be monosyllabic. *See also* **laconic speech**.

preoccupation of thought: Centering of thought content on a particular idea, associated with a strong affective tone, such as a paranoid trend or a suicidal or homicidal preoccupation.

pressured speech: Increase in the amount of spontaneous speech; rapid, loud, accelerated speech, as occurs in mania, schizophrenia, and cognitive disorders.

primary process thinking: In psychoanalysis, the mental activity related directly to the functions of the id and characteristic of unconscious mental processes; marked by primitive, prelogical thinking and by the tendency to seek immediate discharge and gratification of instinctual demands. Includes thinking that is dereistic, illogic, magical; normally found in dreams, abnormally in psychosis. *Compare with* **secondary process thinking**.

projection: Unconscious defense mechanism in which persons attribute to another those generally unconscious ideas, thoughts, feelings, and impulses that are in themselves undesirable or unacceptable as a form of protection from anxiety arising from an inner conflict; by externalizing whatever is unacceptable, they deal with it as a situation apart from themselves.

prosopagnosia: Inability to recognize familiar faces that is not due to impaired visual acuity or level of consciousness.

pseudocyesis: Rare condition in which a nonpregnant patient has the signs and symptoms of pregnancy, such as abdominal distention, breast enlargement, pigmentation, cessation of menses, and morning sickness.

pseudodementia: (1) Dementia-like disorder that can be reversed by appropriate treatment and is not caused by organic brain disease. (2) Condition in which patients show exaggerated indifference to their surroundings in the absence of a mental disorder; also occurs in depression

and factitious disorders.

pseudologia fantastica: Disorder characterized by uncontrollable lying in which patients elaborate extensive fantasies that they freely communicate and act on.

psychomotor agitation: Physical and mental overactivity that is usually nonproductive and is associated with a feeling of inner turmoil, as seen in agitated depression.

psychosis: Mental disorder in which the thoughts, affective response, ability to recognize reality, and ability to communicate and relate to others are sufficiently impaired to interfere grossly with the capacity to deal with reality; the classical characteristics of psychosis are impaired reality testing, hallucinations, delusions, and illusions.

psychotic: (1) Person experiencing psychosis. (2) Denoting or characteristic of psychosis.

rationalization: An unconscious defense mechanism in which irrational or unacceptable behavior, motives, or feelings are logically justified or made consciously tolerable by plausible means.

reaction formation: Unconscious defense mechanism in which a person develops a socialized attitude or interest that is the direct antithesis of some infantile wish or impulse that is harbored consciously or unconsciously. One of the earliest and most unstable defense mechanisms, closely related to repression; both are defenses against impulses or urges that are unacceptable to the ego.

reality testing: Fundamental ego function that consists of tentative actions that test and objectively evaluate the nature and limits of the environment; includes the ability to differentiate between the external world and the internal world and to accurately judge the relation between the self and the environment.

recall: Process of bringing stored memories into consciousness. *See also* **memory**.

recent memory: Recall of events over the past few days.

recent past memory: Recall of events over the past few months.

receptive aphasia: Organic loss of ability to comprehend the meaning of words; fluid and spontaneous, but incoherent and nonsensical, speech. *See also* **fluent aphasia** and **sensory aphasia**.

receptive dysphasia: Difficulty in comprehending oral language; the

impairment involves comprehension and production of language.

regression: Unconscious defense mechanism in which a person undergoes a partial or total return to earlier patterns of adaptation; observed in many psychiatric conditions, particularly schizophrenia.

remote memory: Recall of events in the distant past.

repression: Freud's term for an unconscious defense mechanism in which unacceptable mental contents are banished or kept out of consciousness; important in normal psychological development and in neurotic and psychotic symptom formation. Freud recognized two kinds of repression: (1) repression proper, in which the repressed material was once in the conscious domain, and (2) primal repression, in which the repressed material was never in the conscious realm. *Compare with* **suppression**.

restricted affect: Reduction in intensity of feeling tone that is less severe than in blunted affect but clearly reduced. *See also* **constricted affect**.

retrograde amnesia: Loss of memory for events preceding the onset of the amnesia. *Compare with* **anterograde amnesia**.

retrospective falsification: Memory becomes unintentionally (unconsciously) distorted by being filtered through a person's present emotional, cognitive, and experiential state.

rigidity: In psychiatry, a person's resistance to change, a personality trait.

ritual: (1) Formalized activity practiced by a person to reduce anxiety, as in OCD. (2) Ceremonial activity of cultural origin.

rumination: Constant preoccupation with thinking about a single idea or theme, as in OCD.

satyriasis: Morbid, insatiable sexual need or desire in a man. *Compare with* **nymphomania**.

scotoma: (1) In psychiatry, a figurative blind spot in a person's psychological awareness. (2) In neurology, a localized visual field defect.

secondary process thinking: In psychoanalysis, the form of thinking that is logical, organized, reality oriented, and influenced by the demands of the environment; characterizes the mental activity of the ego. *Compare with* **primary process thinking**.

seizure: An attack or sudden onset of certain symptoms, such as convulsions, loss of consciousness, and psychic or sensory disturbances; seen in epilepsy and can be substance induced. For types of seizures, see the specific term.

sensorium: Perceptual awareness or the sensory apparatus considered as a whole. Sometimes used interchangeably with *consciousness.*

sensory aphasia: Organic loss of ability to comprehend the meaning of words; fluid and spontaneous, but incoherent and nonsensical, speech. *See also* **fluent aphasia** and **receptive aphasia**.

sensory extinction: Neurologic sign operationally defined as failure to report one of two simultaneously presented sensory stimuli, despite the fact that either stimulus alone is correctly reported. Also called *sensory inattention.*

shame: Failure to live up to self-expectations; often associated with fantasy of how person will be seen by others. *See also* **guilt**.

short-term memory: Reproduction, recognition, or recall of perceived material within minutes after the initial presentation. *Compare with* **immediate memory** *and* **long-term memory**.

simultanagnosia: Impairment in the perception or integration of visual stimuli appearing simultaneously.

somatic delusion: Delusion pertaining to the functioning of one's body.

somatic hallucination: Hallucination involving the perception of a physical experience localized within the body.

somnolence: Pathologic sleepiness or drowsiness from which one can be aroused to a normal state of consciousness.

spatial agnosia: Inability to recognize spatial relations.

speaking in tongues: Expression of a revelatory message through unintelligible words; not considered a disorder of thought if associated with practices of specific Pentecostal religions. *See also* **glossolalia**.

stereotypy: Continuous mechanical repetition of speech or physical activities; observed in catatonic schizophrenia.

stupor: (1) State of decreased reactivity to stimuli and less than full awareness of one's surroundings; as a disturbance of consciousness, it indicates a condition of partial coma or semicoma. (2) In psychiatry, used synonymously with *mutism* and does not necessarily imply a disturbance of consciousness; in *catatonic stupor*, patients are ordinarily aware of their surroundings.

stuttering: Frequent repetition or prolongation of a sound or syllable, leading to markedly impaired speech fluency.

sublimation: Unconscious defense mechanism in which the energy associated with unacceptable impulses or drives is diverted into personally

and socially acceptable channels; unlike other defense mechanisms, it offers some minimal gratification of the instinctual drive or impulse.

substitution: Unconscious defense mechanism in which a person replaces an unacceptable wish, drive, emotion, or goal with one that is more acceptable.

suggestibility: State of uncritical compliance with influence or of uncritical acceptance of an idea, belief, or attitude; commonly observed among persons with hysterical traits.

suicidal ideation: Thoughts or act of taking one's own life.

suppression: Conscious act of controlling and inhibiting an unacceptable impulse, emotion, or idea; differentiated from repression in that repression is an unconscious process.

symbolization: Unconscious defense mechanism in which one idea or object comes to stand for another because of some common aspect or quality in both; based on similarity and association; the symbols formed protect the person from the anxiety that may be attached to the original idea or object.

synesthesia: Condition in which the stimulation of one sensory modality is perceived as sensation in a different modality, as when a sound produces a sensation of color.

syntactical aphasia: Aphasia characterized by the fluent production of words, however the inability to form normal grammatical constructions.

systematized delusion: Group of elaborate delusions related to a single event or theme.

tactile hallucination: Hallucination primarily involving the sense of touch. Also called *haptic hallucination.*

tangentiality: Oblique, digressive, or even irrelevant manner of speech in which the central idea is not communicated.

tension: Physiologic or psychic arousal, uneasiness, or pressure toward action; an unpleasurable alteration in mental or physical state that seeks relief through action.

terminal insomnia: Early morning awakening or waking up at least 2 hours before planning to wake up. *Compare with* **initial insomnia** and **middle insomnia**.

thought broadcasting: Feeling that one's thoughts are being broadcast or projected into the environment. *See also* **thought withdrawal**.

thought disorder: Any disturbance of thinking that affects language,

communication, or thought content; the hallmark feature of schizophrenia. Manifestations range from simple blocking and mild circumstantiality to profound loosening of associations, incoherence, and delusions; characterized by a failure to follow semantic and syntactic rules that is inconsistent with the person's education, intelligence, or cultural background.

thought insertion: Delusion that thoughts are being implanted in one's mind by other people or forces.

thought latency: The period of time between a thought and its verbal expression. Increased in schizophrenia (*see* **blocking**) and decreased in mania (*see* **pressured speech**).

thought withdrawal: Delusion that one's thoughts are being removed from one's mind by other people or forces. *See also* **thought broadcasting**.

tic disorders: Predominantly psychogenic disorders characterized by involuntary, spasmodic, stereotyped movement of small groups of muscles; seen most predominantly in moments of stress or anxiety, rarely as a result of organic disease.

tinnitus: Noises in one or both ears, such as ringing, buzzing, or clicking; an adverse effect of some psychotropic drugs.

tonic convulsion: Convulsion in which the muscle contraction is sustained.

trailing phenomenon: Perceptual abnormality associated with hallucinogenic drugs in which moving objects are seen as a series of discrete and discontinuous images.

trance: Sleeplike state of reduced consciousness and activity.

tremor: Rhythmical alteration in movement, which is usually faster than one beat a second; typically, tremors decrease during periods of relaxation and sleep and increase during periods of anger and increased tension.

true insight: Understanding of the objective reality of a situation coupled with the motivational and emotional impetus to master the situation or change behavior.

twilight state: Disturbed consciousness with hallucinations.

twirling: Sign present in autistic children who continually rotate in the direction in which their head is turned.

unconscious: (1) One of three divisions of Freud's topographic theory of the mind (the others being the conscious and the preconscious) in which the psychic material is not readily accessible to conscious awareness by

ordinary means; its existence may be manifest in symptom formation, in dreams, or under the influence of drugs. (2) In popular (but more ambiguous) usage, any mental material not in the immediate field of awareness. (3) Denoting a state of unawareness, with lack of response to external stimuli, as in a coma.

undoing: Unconscious primitive defense mechanism, repetitive in nature, by which a person symbolically acts out in reverse something unacceptable that has already been done or against which the ego must defend itself; a form of magical expiatory action, commonly observed in OCD.

unio mystica: Feeling of mystic unity with an infinite power. Although this can be a religious phenomenon, it can also be a type of delusion.

vegetative signs: In depression, denoting characteristic symptoms such as sleep disturbance (especially early morning awakening), decreased appetite, constipation, weight loss, and loss of sexual response.

verbigeration: Meaningless and stereotyped repetition of words or phrases, as seen in schizophrenia. Also called *cataphasia*. *See also* **perseveration**.

vertigo: Sensation that one or the world around one is spinning or revolving; a hallmark of vestibular dysfunction, not to be confused with dizziness.

visual agnosia: Inability to recognize objects or persons.

visual amnesia: Inability to remember the appearance of objects or to recognize familiar words

visual hallucination: Hallucination primarily involving the sense of sight.

waxy flexibility: Condition in which a person maintains the body position into which they are placed with slight resistance to movement giving it a waxy feel. See also **cerea flexibilitas**.

word approximation: Use of conventional words in an unconventional or inappropriate way (metonymy or of new words that are developed by conventional rules of word formation) (e.g., *hand shoes* for *gloves* and *time measure* for *clock*); distinguished from a *neologism*, which is a new word whose derivation cannot be understood. *See also* **paraphasia**.

word salad: Incoherent, essentially incomprehensible, mixture of words and phrases commonly seen in far-advanced cases of schizophrenia. *See also* **incoherence**.

xenophobia: Abnormal fear of strangers.

zoophobia: Abnormal fear of animals.



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